Written by AI. Still True.

When AI Expands Human Potential
A Systematic Epistemology of Human–AI Knowledge

Yaoharee Lahtee

Edition 1 (2026)

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Open Civil Science Initiative, Bangkok, Thailand

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Preface

Who I am, for the purposes of this book. I am a standalone scholar, not an institution. There is no department, laboratory, or co-author behind this collection; the author of record is one working name, Yaoharee Lahtee, writing from a social-enterprise standpoint — ARAYA Nikah, the Open Civil Science Initiative — rather than from inside a university or a funded research programme. That standpoint shapes what follows more than any credential could: most of the questions in this volume began as problems that needed solving for a small organization and the people it serves, not as gaps a discipline had already marked out. Nothing here should be read as expert testimony substituting for a licensed professional's judgment in medicine, law, religious authority, or any other regulated field; where a chapter's argument touches such territory, it names that limit rather than reaching past it.

Why this title. *Written by AI. Still True.* is not a claim that any sentence in this book is correct because an AI assistant helped write it, and it is not an apology for having used one. It is an invitation to set the question of origin aside and ask the question that actually decides whether a claim is worth keeping: what was read, by what method, from what access point, and what would show it wrong. A sentence drafted with AI assistance and a sentence drafted without it are tested the same way in these pages — by the readout in front of the reader, never by who or what produced the sentence.

What the AI assistant did in making this book. Under my direction, an AI assistant drafted section text, searched and cross-checked citations, verified page counts, checksums, and cross-chapter references mechanically, and helped with typesetting. Its contribution is disclosed here by role, not by vendor or product name, following this project's own standing rule that no AI system is named as an author, co-author, or credited contributor of a public work.

What it did not do. It did not choose which problems this programme worked on. It did not choose which hypothesis, among several drafted candidates, was worth keeping — that choice was mine in every chapter that names one. It did not decide what would count as a legitimate test of a claim, and it never certified that a claim in this book is correct; nothing an AI produces is treated, in the workspace this book comes from, as its own check.

How to read the tier labels. Every chapter in this collection keeps exactly the status it was given the day it was deposited: almost all of them K0, a private candidate the author has made public and citable, explicitly not peer reviewed. Binding 41 such chapters into one volume does not raise any of them by one step. A headnote in front of each chapter states its own status, its own stated claim, and its own stated falsifier — or the honest absence of one — in its own words; the collection adds nothing to any of that, and the fuller apparatus behind these labels is set out in *How to Read a Readout*, next.

What this book does not ask for. It does not ask a university, a journal, or a review committee to certify any of this before it counts as worth reading. It asks to be tested against the problem actually in front of the reader — does the distinction hold, does the falsifier bite, does the method survive contact with a real case — rather than against a credential.

What is deliberately missing from Edition 1. Two chapters named in this programme's own plan are not yet written and are not included here under any other name: *The Prompt Is Not the Beginning* and *Testing the Master Equation River*. No date is promised for either. A third work, *Master Equation River: A Provenance Audit Across the Human–AI Readout Programme*, was withheld from earlier drafts of this edition because its then-current deposited edition (DOI `10.5281/zenodo.22414412`) was Thai-majority in its body text and this is an English-only book; an English-only v1.5 edition (DOI `10.5281/zenodo.22550491`) has since been deposited and is included here as Chapter 36, opening Part 7. This edition also excludes the programme's applied-domain work — tourism, social-enterprise, and Islamic-knowledge-authority applications — and its Thai-language material; all of this sits outside an English-only epistemic volume by design, not by oversight. The Edition page that follows this preface says more about the two remaining planned chapters.

An invitation, not a verdict. Read the falsifier stated with each chapter before deciding whether to believe its claim, and test it against a case you actually have, not against my say-so. If a chapter turns out to be wrong, the honest outcome is that its status stays exactly what it already is — K0, not yet checked — until someone runs that test.

How to Read a Readout

The readout, not the world. Every chapter's claim, and the methodology that governed how this book itself was built, starts from one distinction, quoted directly from the specification these chapters share: *"What a claim card reports on was already capable of making that report right or wrong before the card existed ($R_A = O_A(W;\Pi_A) \neq W$); a record is an operator-conditioned readout, never the world itself, whether the operator is human or not."* (glosa — Rigour Without Infrastructure, `FOUNDATION_v0.6.md` §1.0, DOI `10.5281/zenodo.22340255`). A claim in this book is a readout of a problem, produced by a named method from a named access point — never the problem itself, and never Truth handed down.

Five questions, one crosswalk. Every claim card behind this programme answers five questions before anything is made public. They are stated below by their Readout Condition principle rather than by the founder's own Thai-language wording, which this English-only edition does not reproduce; the full bilingual crosswalk is §3.1 of the same document.

Q	Principle	Claim-card field	Mechanically checkable?
Q1	Existence (E)	`five_questions.seen`	Presence and shape only
Q2	What the record alone distinguishes	`five_questions.separates`, `zero_vs_bottom`	Presence-checkable only, not correctness-checkable
Q3	Disclosure (D) of AI's contribution	`five_questions.ai_filled`	Presence-checkable only
Q4	Attribution (A) of every assumption	`five_questions.assumed[]` + identification ladder	Presence-checkable only
Q5	Independent evidence or objection	`five_questions.tested` + `independent_check`	The one genuinely mechanically enforceable question

"The Readout Condition is E∧A∧D: a card that answers Q1/Q2/Q4 honestly but misattributes Q3 — crediting the source with a distinction AI actually supplied — fails the condition even though every field looks filled." (ibid., §3.1)

The independence ladder, I0–I5. Every chapter's own claim about how well-checked it is rests on who read it, not how many times it was read.

Level	Definition	K-state ceiling
I0 — Self	Same session re-reads its own output	K0
I1 — Same model, new session	Fresh conversation, identical model	K0
I2 — Same vendor, other model	Different model, same vendor	K0→K1 only combined with I3+, or the bounded I2+I4 exception
I3 — Different vendor	Materially different model family/vendor; founder-set minimum bar for K1 public-provisional, never sufficient alone	K1 (floor), never K2
I4 — Mechanical / original-record	Proof kernel, executed test, schema validator, direct original-source lookup	K1 unless combined with I5
I5 — Independent external human (non-founder)	A person outside the authoring session; the only route to K2	K2/K3

K-state. Four stages, quoted verbatim: *"K0 Private Candidate → K1 Public Provisional (timestamped, citable, explicitly not peer reviewed) → K2 Socially Stress-Tested (K1 + substantive external human friction) → K3 Reviewed/Empirically Constrained (K2 + certification/replication as appropriate)."* (ibid., §4.4) Every chapter in this book is K0 unless its own headnote states otherwise; being collected here moves no chapter up this ladder.

Non-collapse discipline. A recurring failure this methodology names and refuses is the silent merging of two things that are not the same — for example *"World ≠ Record ≠ Readout ≠ Meaning ≠ Truth ≠ Warrant ≠ Knowledge-Attribution"* (non-collapse row NC-01: that a record exists is never treated as truth established). Appendix B reprints the independence ladder and K-state tables in full, alongside the non-collapse rows this book's own chapters cite by id.

What This Book Does Not Claim

This book raises nothing beyond what each chapter already states for itself; where a chapter names its own limits, those limits stand for the collection as a whole. The passages below are quoted verbatim from the chapters' own text, each with its DOI, at exactly the point where each chapter already draws its own line.

- **Written by AI. Still True.** (`10.5281/zenodo.22301202`): "It does not claim that current AI systems know, believe, understand, testify, or possess moral standing."
- **Choice Begins Before Choice** (`10.5281/zenodo.22357788`): "The paper does not claim a metaphysical theory of free will or a validated universal scale of human agency."
- **Potential as a Readout** (`10.5281/zenodo.22361830`): "Not claimed: that (1) is forced from the root of the epistemic core; that all human agency reduces to one number; that R, D, X, F are necessary or sufficient for every definition of agency..."
- **The Civilization of Knowledge** (`10.5281/zenodo.18943971`): "It does not claim that all regimes know equally well..." nor "to replace specialist historiography, area studies, or detailed institutional histories."
- **Operational Linguistic Wisdom**, uplift edition (`10.5281/zenodo.22456487`): "The paper does not claim that any construct below has been empirically validated, that the three-stage mechanism is a demonstrated organizational law, or that the propositions in Section 7 have been tested."
- **CTSA Human-Return Readout** (`10.5281/zenodo.22339909`): "The paper does not claim that Concept, Tool-Selection, Skill-Execution, and Alternative-Generation are four biological modules, exhaustive kinds of knowledge, or validated psychometric factors."
- **Before Meaning, Before Choice** (`10.5281/zenodo.22424434`): "The paper does not claim that Readout Universe or Readout Genesis is a completed law of nature, that the human-domain quotient has been empirically closed, or that the equations below are literal neuroscience."
- **Rigour Without Infrastructure**, glosa concept paper (`10.5281/zenodo.22307841`): "This paper does not claim: that H1's mechanism has been run as a study; that H2's conceptual-portability claim has been tested against real readers..."

Each chapter's own headnote points to its own falsifier, indexed in full by part in Appendix D; a claim not yet tested against one stays exactly as unverified as its own chapter already says it is.

Edition 1 (2026): What Is Not Yet Written

Edition 1 (2026). This edition collects 41 works from the Human–AI Readout Programme's own library that fall inside the epistemic human–AI co-knowledge scope this collection sets for itself (Appendix A lists all of them, with DOI, version, and date). Two further works named in the programme's own plan are not yet written and are not in this edition:

- **The Prompt Is Not the Beginning** — planned, Part 5 (Human–AI Coupling and Return), following Chapter 32; not yet written, no date promised.
- **Testing the Master Equation River** — planned, Part 7 (The River and Its Tests), following Chapter 37; not yet written, no date promised.

Master Equation River: A Provenance Audit Across the Human–AI Readout Programme (DOI `10.5281/zenodo.22550491`), previously withheld pending an English-only edition, is included in this edition as Chapter 36 (v1.5, English, opening Part 7): its deposited v1.1 body text was Thai-majority, and an English-only v1.5 edition, extending the same unchanged content with one further segment, a companion Th_coqc ledger, and a row-by-row cross-reference to the programme's equation library Toledo, was deposited on 2026-09-07.
The two planned titles above also appear, marked *not yet written*, in Part 5's and Part 7's own chapter listing on this book's part-opener pages, and in Appendix E's build record, so their absence is stated plainly rather than silently dropped.

Outside this volume. This is an English-only edition. Two further bodies of work in the wider programme are outside its scope by design, not by oversight, and are not included in this or any future edition of this particular book:

1. A Thai-language national-conference presentation (slide deck, NIDA, 29 August 2026, DOI `10.5281/zenodo.22301886`, duplicated at DOI `10.5281/zenodo.22302410`; no English edition exists) is referenced only, in the Part 5 bridge.
2. The programme's applied-domain work — tourism, social-enterprise, and Islamic-knowledge-authority applications — sits outside the epistemic-co-knowledge scope this volume sets for itself and belongs to a different collection.

Three 2025 youth-dialogue papers (DOIs `10.5281/zenodo.22308448`, `10.5281/zenodo.22308446`, `10.5281/zenodo.22308451`) are likewise referenced only, in the Part 5 bridge, rather than reproduced as standalone chapters here.

Contents

PART I

Root: Before Meaning

Part 1 opens the programme's spine at its first two stations, World and Readout, before the collection turns to Meaning, Experience, and Retention in Part 2. Chapters 1 to 3 address the readout station directly. Chapter 1 examines what changes, epistemically, once a claim's provenance is a non-human generator, and describes a claim's standing as tracked from the constraints and dependencies it answers to, rather than from the identity of a knower. Chapter 2 develops the Readout Condition: existence, attribution, and disclosure norms for tracing a claim-level distinction back to the source that licensed it. Chapter 3 restates the wider Readout Genesis programme as a standalone synthesis, introducing the retained-distinction operator and a typed belief and knowledge-status architecture that later chapters draw on.

Chapters 4 to 7 apply variants of this readout-first stance to specific epistemic settings: knowledge as stabilized translation under bounded observation (Chapter 4); a constraint-first account of normativity under irreducible mediation (Chapter 5); a discrete causal-graph account of mind and expertise formation carrying an explicit falsification condition (Chapter 6); and a mediated-agency account of free will and truth (Chapter 7).

Each chapter states its own claim, its own status, and, where the chapter provides one, its own falsifier; none of the description above adds a claim beyond what each headnote quotes. Several vocabulary items introduced in Chapters 1 to 3 — provenance path, attribution, retained distinction, knowledge status — recur, cited rather than re-argued, across most of the chapters that follow in this book, as the knowledge-graph cross-references recorded in each headnote show.

What to carry forward. Part 2 assumes the readout vocabulary set out in Chapters 1 to 3 — a claim-level distinction, its provenance path, and its retained structure — and turns to how such retained structure becomes meaning, experience, and retention for a bounded knower, without treating anything established in Part 1 as settled beyond the status each chapter already states for itself.

- Written by AI. Still True. Knower Fetishism, Epistemic Pedigree, and the Human Face as a Bad Theory of Truth — `10.5281/zenodo.22301202`
- The Readout Condition: Distinguishability, Access, and Epistemic Warrant — `10.5281/zenodo.22301318`
- Readout Genesis Standalone Synthesis: Information Epistemic Foundation, Conditioned Agency, and Meta-Readout Governance — `10.5281/zenodo.21529456`
- Knowledge as Stabilized Translation: Toward an Observer-Constrained Epistemology — `10.5281/zenodo.18925129`
- Constraint-First Epistemology: Normativity, Conditioned Agency, and the Non-Zero Kantian Floor — `10.5281/zenodo.19205869`
- Mind as Information Horizon: From Primordial Difference to Expertise Formation on the Discrete Causal Graph — `10.5281/zenodo.19640361`
- The Architecture of Mediated Agency: Beyond the Misframing of Free Will and Truth — `10.5281/zenodo.19176260`

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What the chapter states. *"The thesis is not that machines know. It is that before anyone knew a fact there was already something to get right or wrong about it; the world did not wait for a human knower before acquiring the differences, traces, dependencies, and constraints to which later knowledge must answer."* (10.5281/zenodo.22301202)

Status. "preprint – not peer reviewed" (as printed in the chapter header).

Falsifier(s). No explicit falsifier stated in the chapter.

Read before. Ch. 2 *The Readout Condition*; Ch. 3 *Readout Genesis Standalone Synthesis*; Ch. 27 *When AI Expands Human Potential*; Ch. 39 *The Standalone Scholar*.

Feeds into. This book's Chapters 2–7, 10–16, 23, 27, 29–31, 38–41 (mechanical KG cross-reference list; titles and DOIs in Appendix A).

Written by AI. Still True.

Knower Fetishism, Epistemic Pedigree, and the Human Face as a Bad Theory of Truth

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August 2026

Abstract

Suppose a proposition survives source checking, inferential scrutiny, and independent reconstruction. You accept it. Then you learn that an artificial-intelligence system generated the draft, and your confidence falls. What, exactly, became epistemically worse? The familiar answer is that provenance can carry higher-order evidence about reliability. Correct. This paper grants that point and then asks what remains after provenance has done all the legitimate epistemic work available to it. The resulting problem is larger than AI. I argue that modern epistemology often slides from a plausible claim about the *possession* of knowledge—that knowing is a state or standing of a subject—to a much stronger and usually undefended claim about the *constitution* of epistemic value—that epistemically significant content, processes, records, or sources acquire standing only through a human or human-like knower. I call this the *Possession–Constitution Collapse*. Its social form is *knower fetishism*: the conversion of an earned relation among evidence, reliability, accountability, and inquiry into an apparently intrinsic property of the recognized knower. The paper develops a pre-subjective notion of *constraint structure*, stages the argument against knowledge-first epistemology, testimony, reliabilism, higher-order evidence, objective knowledge, information-theoretic epistemology, distributed cognition, and social epistemology, and proposes two constraints: provenance matters only through an identifiable epistemic channel, and source metadata must never be identified with the epistemic role for which it is merely a proxy. The thesis is not that machines know. It is that before anyone knew a fact there was already something to get right or wrong about it; the world did not wait for a human knower before acquiring the differences, traces, dependencies, and constraints to which later knowledge must answer. A non-knower can therefore occupy an epistemically productive role without being promoted into personhood. Institutions remain indispensable, but their authority is derivative from epistemic work, not magical pedigree. AI is philosophically useful because it removes the human face from a claim and reveals how much epistemic credit we had quietly attached to the face. The paper's provocation is deliberately asymmetric: if one wishes to reserve *knowledge* for a subject's successful possession, one may keep the word; one may not infer from that linguistic reservation that the subject created the truth, trace, dependence, constraint, or record to which successful possession must answer.

Keywords: epistemology; artificial intelligence; objective knowledge; testimony; epistemic authority; anthropocentrism; provenance; scientific instruments; social epistemology

Disclosure challenge. Generative AI assisted with literature mapping, adversarial argument testing, language editing, and LaTeX drafting for this manuscript. The author selected, checked, revised, and endorses the final argument. If this disclosure lowers your assessment, do not merely report the discomfort. Identify the proposition, inference, evidential relation, or reliability condition that became worse when you learned it.

1. The sentence was fine five seconds ago

Take a deliberately clean case. A reader, S, receives a proposition p, an argument A, and an evidential record E. She checks the cited materials, reconstructs the relevant inference, and finds no defect. She judges p well supported. A moment later she learns only this:

> *The draft was generated by an AI system.*

Suppose her confidence falls. The cheap response is that she has committed a genetic fallacy. That response is too cheap. Source information can be epistemically relevant. Reliabilism, expertise, testimony, higher-order evidence, and genealogical debunking all explain why learning something about the origin of a belief can rationally alter confidence without altering the proposition's truth-value [9, 11, 10, 2, 14]. If "AI-generated" reveals fabricated references, hidden dependence, systematic error, non-independent verification, or a previously unmodeled failure mode, confidence may properly fall.

So let provenance win its strongest case. The harder question begins afterward:

> **After provenance has told us everything epistemically relevant that provenance can tell us, what exactly is the label still doing?**

The question matters because two different objects are easily collapsed: the epistemic properties of a claim and the social pedigree of the route by which the claim arrived. A useful proxy for reliability can become so familiar that we begin treating the proxy as though it

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were reliability itself.

The joke is simple. Nothing happened to the sentence. The punctuation survived. The evidence survived. The inferential relations survived. Only the biography changed.

If that biography changes the verdict, it may do so for excellent reasons. But those reasons must be named. A human face is not a free epistemic variable.

But that joke points toward a larger embarrassment. Epistemology has spent centuries asking what a knower must be like, what a knower may believe, when a knower is justified, whom a knower may trust, and how a knower acquires knowledge. Those are legitimate questions. They become less innocent when the grammar of *someone knows that p* quietly becomes an ontology in which the knower is treated as the source of epistemic standing itself.

This paper asks whether we have confused the conditions under which knowledge is *possessed* with the conditions under which reality is *epistemically structured.*

2. Before anyone knew it, there was already something to get wrong

Before there were physicists, stars behaved in ways that later made some astrophysical claims correct and others false. Before there were paleontologists, animals died, bones fractured, sediments accumulated, and traces were retained. Before there were geneticists, inheritance displayed stable dependencies. None of these events filed an interpretation with a human observer before occurring.

The claim is intentionally weaker than "knowledge existed before knowers." That slogan invites a terminological fight and wins nothing we need. If "knowledge" is reserved for a factive state or standing possessed by an epistemic subject, then let the subject keep the noun. Williamson's knowledge-first program gives that subject-involving state an especially fundamental role [21]. The present argument can grant the usage completely.

What it refuses is the inflationary inference from *someone must possess knowledge* to *nothing can be epistemically structured before someone possesses it.* Before anyone knew that p, there could already be a fact of the matter that made p correct rather than $\neg p$, traces selectively dependent on that fact, and constraints capable of defeating a later interpretation.

> **Before anyone knew it, there was already something to get wrong.**

That sentence is the realist minimum of the paper. A theory may reject it, but then the dispute is no longer primarily about artificial intelligence. It is about whether inquiry discovers constraints that precede inquiry or partly constitutes the very correctness conditions by which inquiry is judged.

2.1. The Jurassic test

Imagine a dinosaur whose femur fractured long before any human existed. The fracture generated a physical pattern. Millions of years later, investigators disagree about whether the injury occurred before or after death. One interpretation fits the retained structure better than the other.

Now ask four questions.

1. Did the fracture occur, if it occurred, before paleontology?
2. Did the resulting structure differ depending on what happened?
3. Can that retained difference make a later paleontological claim better or worse?
4. Did the difference wait for a human interpretation before being the difference it was?

A "yes, yes, yes, no" answer is enough. The fossil need not know anything. It need not carry a proposition in English. It need only be the sort of worldly remainder against which later representations can succeed or fail.

> A fossil does not need a theory of knowledge to make your theory of the fossil wrong.
>
> **Reality can say no without speaking.**

Dretske's information-theoretic epistemology is useful here because information, in his technical sense, is objective: a signal carries information in virtue of lawful dependence on a source condition, not because a human first interprets it [4]. Popper pushed in a different direction with the deliberately provocative project of an "epistemology without a knowing subject," distinguishing subjective states from objective contents of thought and scientific theories [18]. Frege's realism about thoughts likewise denies that truth-bearing thought contents are private mental ideas owned by an individual thinker [6]. Baird goes further in a material direction, arguing that scientific instruments can embody or constitute forms of "thing knowledge" rather than merely assisting propositional thought [1].

These views are not equivalent. Dretske's information is not Popper's World 3. Fregean thoughts are not spectrometers. Baird's instruments do not prove that fossils literally know anything. Their disagreement is useful precisely because it blocks an easy escape: epistemic significance has never had only one philosophically uncontested address.

To avoid winning by vocabulary, introduce a deliberately weaker notion.

Definition 1 (Pre-subjective constraint structure)**.** *A pre-subjective constraint structure is a worldly state, relation, trace, record, or lawful dependence capable of constraining which later representations succeed or fail, independently of whether any actual subject has yet represented or interpreted it. The adjective* epistemic *is deliberately withheld at this stage: the structure need not already be evidence-for-us, meaningful-to-us, or anyone's knowledge in order to constrain what later inquiry can responsibly say.*

Interpretation matters for uptake, representation, comparison, inference, and use. It does not travel backward in time to manufacture the event, dependence, or

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trace that constrains the interpretation. The point can therefore be stated without giving a fossil a mind:

> If you insist that knowledge must be owned, keep the word.
> The world does not care.
> **What makes ownership epistemically answerable was there before the owner.**

3. The Possession–Constitution Collapse

Let a minimal **Subject Condition** be:

$$\text{SC}: \qquad K(S,p) \to \text{Subject}(S).$$

If S knows p, some subject S occupies the knowing relation. This is deliberately weak.

Now consider a stronger thesis:

$$\text{HSC}: \qquad \text{Epi}(X,p) \to \text{Knower}(X,p),$$

where $\text{Epi}(X,p)$ means that X plays an epistemically significant role in a route by which p becomes warranted, available, retained, or inferentially constrained.

Call this the **Knower-Constitution Condition**. In its human form, it becomes the Human-Source Condition: epistemically serious production must terminate in, originate from, or pass through something sufficiently like a recognized human knower.

Nothing in SC entails this stronger condition.

Definition 2 (Possession–Constitution Collapse)**.** *The Possession–Constitution Collapse occurs when a condition on who can* possess *knowledge is treated as a condition on what can* constitute, constrain, retain, transmit, or generate *epistemically significant structure.*

The collapse is tempting because ordinary human cases package many roles into one body. A scientist may generate a claim, understand it, endorse it, answer for it, possess relevant expertise, and carry institutional standing. Because the bundle is historically common, philosophy can mistake co-location for identity.

Principle 1 (Role Separation)**.** *Generation, truth, evidential support, reliability, understanding, possession, endorsement, accountability, credibility, and institutional authorization are distinct epistemic roles. No pair may be identified merely because one human agent often occupies both.*

The principle is not anti-human. It is anti-shortcut. A human being can occupy many roles at once. What she cannot do is make those roles identical by occupying them.

> **The human body was a convenient interface. Epistemology mistook the interface for the architecture.**

The collapse is easy to commit because ordinary grammar hides it. "Alice knows that the sample contains lead" naturally puts Alice at the center. The grammar says almost nothing about where the epistemically load-bearing work occurred: in the calibration procedure, the detector, the sample preparation, the chemical regularity, the laboratory archive, the inferential model, the testimony of technicians, or Alice's final endorsement.

Giere's analysis of distributed cognition in epistemic cultures makes this division explicit: scientific cognition can be distributed across people and artifacts even if one reserves epistemic agency for human components [8]. Humphreys makes the anti-anthropocentric point even more sharply for computational science: human cognitive capacities need not remain the ultimate standard of epistemic success [12, 13].

Proposition 1 (No transitivity of knowerhood)**.** *From the fact that knowledge has a subject it does not follow that every epistemically significant node in the route by which that subject comes to know must itself be a knower.*

A thermometer need not know the temperature. A telescope need not believe in galaxies. A calculator need not understand multiplication. A simulation need not possess the theory it helps investigators evaluate. The absence of knowerhood at an intermediate node is therefore not, by itself, an epistemic defect.

The burden moves. If a route is defective, find the defect in evidence, reliability, calibration, dependence, opacity, selection, inference, accountability, or some other relation that bears epistemic weight.

"But it does not know" is not yet an argument.

3.1. The source label is a readout, not an oracle

The argument can be made more precise without pretending that finite inquiry has transparent access to reality. Let a worldly condition be W, let an inquiry procedure or access operator be $\mathcal{O}_A$, and let R_A be the record available to agent or system A under a selection policy Π_A:

$$R_A = \mathcal{O}_A(W; \Pi_A).$$

The notation is intentionally modest. It says that what an investigator receives is an operator-conditioned record of the world. It does *not* license the identification $R_A = W$. Human observers do not escape this merely by having biographies.

Now let $m(A)$ be metadata about the source—"senior professor", "top journal", "AI-generated", "anonymous amateur"—and let $\widehat{\rho}(A)$ be the reliability estimate an evaluator extracts from that metadata. Metadata can be excellent evidence about reliability. But:

$$m(A) \neq \rho(A),$$

and neither metadata nor estimated reliability is identical to the epistemic standing of p.

The mistake targeted in this paper is therefore a *role-identification error*: a proxy that is useful in one inferential role is promoted into the thing it was introduced to estimate.

Principle 2 (Bridge Burden)**.** *Any inference from source metadata to a change in epistemic standing must identify the mediating relation that does the work—for example reliability, independence, access, assurance, accountability, calibration, or error correlation. Where no such bridge is supplied, the inference is pedigree substitution rather than source-sensitive epistemology.*

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Table 1: Ten roles routinely compressed into the figure of "the knower." The table is diagnostic: co-location in one person does not establish identity among the roles.

Role	Governing question	What it is not
Truth	Is p the case?	Not conferred by authorship, prestige, or recognition.
Constraint	What worldly difference can make a later representation fail?	Not yet evidence-for-us or anyone's possession.
Evidence	What bears on whether p should be accepted?	Not identical to a source label.
Reliability	How well does a process discriminate success from error across cases?	Not the truth-value of this particular output.
Provenance	By what route did this representation arrive?	Not a verdict on the representation by itself.
Credibility	How much epistemic weight is assigned to a source?	Not automatically deserved authority.
Accountability	Who can answer, correct, retract, or defend?	Not a truth-maker and not identical to generation.
Legitimacy	What social or institutional standing is granted?	Not the same as evidential support.
Authorship	Who is properly answerable for the scholarly act?	Not necessarily who generated every token.
Possession	Who, if anyone, knows p?	Not what constituted every epistemically productive step on the route.

Principle 3 (No Bare Pedigree)**.** *A source label is epistemically incomplete reporting. The relevant object is not merely* who or what produced the claim*, but the tuple of production procedure, selection conditions, dependencies, checks, error model, inferential role, and answerability that makes the source cue evidentially useful.*

This is the source analogue of a familiar scientific discipline: never let the displayed value erase the measurement procedure that gave it meaning. "Professor" and "AI" are not epistemic magnitudes with values printed on their foreheads. They are compressed descriptions of routes. Compression is allowed. Transubstantiation is not.

> A PhD is not a truth-maker.
> An AI label is not a defeater by sacrament.
> **Show the channel.**

4. Knowledge-first meets a world that came first

Knowledge-first epistemology gives us the strongest possible subject-centered opponent. Williamson treats knowledge as a fundamental factive mental state rather than as an analysable composite of belief, truth, and justification [21]. The present argument need not deny that knowing is a state of the subject. It challenges a different inference: that because *knowledge-as-state* is fundamental in explaining belief, evidence, assertion, or action, the subject who occupies that state thereby becomes constitutively prior to all epistemically significant structure.

Knowledge-first may put knowledge first in an explanatory theory of mind. It does not follow that the knower came first in the history of the world, or that human-like sourcehood comes first in the architecture of inquiry. The order of explanation is not the order of arrival. A dinosaur is not obligated to wait for epistemology before breaking its leg, and a detector is not obligated to acquire beliefs before its output can embarrass a theorist.

That inference would be surprising. Williamson himself emphasizes that what can be known does not exhaust what is true. The world therefore exceeds the set of actual epistemic possessions. The factive character of knowledge already points outward: knowing succeeds because the subject is appropriately related to how things are, not because the subject manufactures how things are.

Popper's objective epistemology moves almost in the opposite direction. Scientific knowledge, on his view, can be considered at the level of objective contents, problems, arguments, and theories rather than reduced to psychological states of knowers [18]. Popper's World 3 is historically produced by minds, so it does not establish the stronger claim that knowledge literally predates every knower. What it does establish is enough for the present dialectic: epistemology can coherently investigate epistemic contents and relations without making current possession by a subject constitutive of their objective status.

Dretske then removes another layer of dependence. Information can be objective before cognition takes it up; knowledge is an information-dependent state, not the creator of the informational relation [4]. Baird removes another: material artifacts can embody epistemic achievements that exceed what any individual presently has in mind [1]. Giere and Humphreys remove another: epistemic production is heterogeneous and technologically extended [8, 12].

The resulting picture is not that subjects disappear. It is that the subject arrives *inside* an already structured problem.

> The knower may be necessary for knowing.
> The knower is not therefore the author of what makes knowing correct.

This distinction is the paper's deepest anti-

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anthropocentric commitment. We once removed the human from the center of the cosmos and from a special biological creation story. Epistemology may still preserve a smaller throne: reality can be independent of us, but epistemic legitimacy somehow requires our face.

5. Source matters. Unfortunately.

Any attempt to decenter the knower becomes unserious if it ignores why sources matter. Hardwig's epistemic dependence is unavoidable in modern inquiry: no individual can personally reproduce the chains of expertise on which scientific belief depends [11]. Goldman shows why novices rationally use evidence about experts, track records, agreement, and competence when direct access to the subject matter is limited [10]. Coady and Lackey establish the depth of testimony as a source of knowledge [3, 16]. Zagzebski gives a sophisticated account of epistemic authority rather than treating deference as mere irrational submission [22].

Higher-order evidence makes the point sharper. Christensen shows that information about one's epistemic performance can rationally alter confidence even when first-order evidence is unchanged [2]. Kahane's analysis of evolutionary debunking arguments shows how genealogical facts can undermine justification without changing the truth-value of the target proposition [14].

So the paper concedes all of the following:

- source identity can be evidence;
- credentials can rationally compress track-record information;
- human testimony can carry assurance and responsibility unavailable from an impersonal process;
- institutional pedigree can indicate methods, training, correction, and accountability;
- AI provenance can rationally lower confidence when it identifies a real failure mode.

Good. Let every opponent win those points. The paper needs none of them to lose.

Now the question can finally become difficult.

If the epistemic force of pedigree is explained by reliability, evidence, assurance, accountability, and institutional procedure, what epistemic work remains for pedigree *after those relations have done theirs*? If the answer is "humanity", the burden is no longer rhetorical. Specify the channel by which humanity improves the epistemic relation in the case at hand. If no channel can be specified, the word is functioning as an honorific, not an explanation.

6. Knower fetishism

The term "fetishism" is intentionally accusatory, but the accusation is structural rather than psychological.

Definition 3 (Knower fetishism)**.** *Knower fetishism is the treatment of the recognized identity, humanity, credentials, or social standing of a knower as though it possessed epistemic value in itself, after abstracting away the evidential and relational work through which that status earns epistemic relevance.*

The structure resembles familiar cases of reification. A relation produced by training, successful performance, institutional checking, criticism, and accountability is compressed into a status term: *expert*. Compression is useful. The mistake begins when the status term is allowed to behave like an intrinsic epistemic property.

Professor P may deserve high prior credibility because P has an excellent track record, relevant expertise, transparent methods, and exposure to criticism. But if we subtract those relations and retain only "Professor P", what remains epistemically? A title. If the title still moves credence after the reasons encoded by the title have been held fixed, the compression format has begun impersonating the data.

Likewise, "AI" may rationally trigger caution because a system has known failure modes. But once the relevant failure modes are represented, an unexplained residual penalty cannot be defended merely by repeating the species or substrate of the source.

There is a social reason the mistake is attractive. A recognizable human knower is an administratively convenient address for trust, blame, credit, responsibility, and legitimacy. That convenience is real. It is not yet epistemology. The danger begins when the address of responsibility is mistaken for the address of truth.

> Perhaps we never wanted truth alone.
> **We wanted truth with a human witness.**

> The problem is not that we respected knowers too much.
> The problem is that we forgot what the respect was for.

This is why AI is a useful philosophical irritant. The traditional human author bundled generation, belief, assertion, intention, responsibility, reputation, and social identity into one body. Generative systems unbundle those roles. Once the bundle comes apart, epistemology has to decide which role was actually doing the work.

7. The Epistemic Pedigree Paradox

Consider two versions of the same checked argument. Version H is labeled "written by a senior professor." Version M is labeled "drafted by AI and independently checked by an accountable human." Suppose the proposition, cited evidence, derivation, verification procedure, known dependencies, and endorsing responsibility are held fixed as far as the case allows.

If credence still differs merely with the label, we have a residual provenance effect. More generally, the epistemically relevant object is a route, not a badge:

$$\big(\text{origin}, \text{procedure}, \text{dependence}, \\ \text{checks}, \text{answerability}\big) \longrightarrow \Delta\mathrm{Cr}(p).$$

A bare label suppresses the very variables that could justify its influence.

For two provenance descriptions O_1 and O_2, a useful isolating diagnostic is

$$\mathrm{RPE} = \mathrm{Cr}(p \mid E, R, A, O_1) - \mathrm{Cr}(p \mid E, R, A, O_2),$$

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where E denotes first-order evidence, R the represented reliability and dependence conditions, and A answerability or responsible endorsement. The expression is not a psychological law. It asks what, normatively, remains for provenance to do once the channels invoked on its behalf have been made explicit.

Definition 4 (Residual Provenance Effect)**.** *A Residual Provenance Effect occurs when epistemic assessment changes with provenance after the epistemically relevant pathways through which provenance could rationally matter have been held fixed or explicitly represented.*

The point is not that real cases will perfectly screen off every pathway. They usually will not. The point is normative: any residual difference incurs a burden of explanation.

Principle 4 (Provenance Relevance Constraint)**.** *Provenance may rationally alter epistemic standing insofar as it changes total evidence, reliability, dependence, assurance, accountability, or another specified epistemically relevant condition. Provenance has no additional free-standing force merely by redescribing the origin of an otherwise fixed epistemic route.*

The formal ancestry is intentionally unglamorous: the structure resembles screening-off. Once the variables through which a common cause bears on an outcome are adequately conditioned on, the residual label should not acquire unexplained causal or evidential force [19]. The philosophical problem is therefore not inventing conditional independence. It is noticing when social categories are allowed to bypass it.

This principle is deliberately conservative. It does not say provenance decays monotonically with verification. A late discovery of data leakage, fraud, shared model dependence, or correlated error can make provenance newly relevant. It says only that relevance must travel through an epistemic channel.

The paradox appears because pedigree is most useful when direct checking is difficult. We invented expert shortcuts because we could not inspect everything. Yet after inspection becomes more possible, we can find ourselves inspecting the shortcut instead.

> We needed credentials because we could not check everything.
> Then we learned to check more things and began checking the credentials instead.

7.1. The author-shaped placebo

Consider two texts that are token-for-token identical. One is labeled “Professor.” The other is labeled “AI.” If the labels rationally signal different reliability, dependence, or accountability, different credence may be warranted. That is ordinary source-sensitive epistemology. But once those pathways are fixed, any remaining difference is not a mysterious extra property of the text. It is a residue in the evaluator.

This is why the paper’s provocation is conditional rather than populist. It does not demand equal trust for equal strings. It demands an account of the *difference-maker*. If the answer is reliability, show the reliability difference. If it is independence, show the dependence. If it is responsibility, show what responsibility changes for the epistemic task. If it is merely “one came from a human,” then humanity has ceased to be a description and started functioning as an unexplained epistemic honorific. If “human” is a reliability variable, estimate the reliability. If it is a moral category, do not smuggle it into credence. Cheekbones are not calibration.

> **A name can be evidence. It cannot be evidence of itself forever.**

8. AI as philosophical acid, not as a new person

Much contemporary debate asks whether conversational AI can testify, assert, understand, or count as an epistemic authority. Freiman argues that beliefs acquired from conversational AI do not fit neatly into the traditional instrument/testimony dichotomy [7]. Elizabeth Fricker argues forcefully that human and AI “testimony” differ metaphysically and epistemically, with responsibility playing a central role in genuine human testimony [5]. Sahebi and Formosa analyze the dilemma created by epistemic distrust of AI-mediated communication: caution can be rational while indiscriminate penalties can also distort human epistemic relations [20].

The present paper takes an impolite shortcut through that debate.

Suppose current AI does not know. Suppose it does not believe. Suppose it does not testify. Suppose it has no understanding worth the name. Suppose it has no moral standing. Grant the skeptical side everything.

The central argument survives.

Because the question is not:

Can the machine be promoted into the club of knowers?

It is:

> **Why must it be a knower in order for us to know through it?**

The thermometer was never promoted. Neither was the telescope. Neither was the archive. Neither was the calculator. Epistemology already permits non-knowers to occupy epistemically productive roles. AI is difficult because it performs those roles in language, the medium we had culturally associated with minds, assertion, testimony, and authorship.

AI therefore acts as philosophical acid. It dissolves a bundle that used to arrive in one human body and leaves separate questions on the table: Who generated the string? What supports the claim? What process produced it? Who checked it? Who endorses it? Who can answer objections? Who bears responsibility? Which parts are reproducible? Which dependencies are hidden?

The human author once answered all of these questions by standing there.

That was convenient. It was never an argument that standing there was itself epistemically magic. The machine is philosophically embarrassing not because it

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lacks a soul, but because it forces us to itemize what the soul was allegedly doing for the inference.

> **Written by AI. Still true.**
> The first sentence concerns genealogy. The second concerns the world.
> **A philosophy of knowledge should not erase the punctuation without an argument.**

The paper's smallest philosophical object may therefore be the period in its title. A full stop is doing the work that source-sensitive reasoning often forgets to do: keeping two questions apart until a warranted bridge connects them.

9. Credibility: when the face becomes evidence

Miranda Fricker's account of testimonial injustice is indispensable because it shows that credibility is not distributed in a social vacuum. Speakers can receive credibility deficits because identity prejudice enters epistemic uptake [17]. The lesson for the present argument is structural, not analogical. An AI system is not thereby a victim of testimonial injustice in the morally relevant sense. The interesting point is that audiences already use social identity as part of epistemic assessment, sometimes legitimately and sometimes not.

The human face can be evidence. A known scientist's face may stand in for a rich history of competence and accountability. A marginalized speaker's face may, unjustly, trigger prejudice. The same visible cue can therefore carry epistemically relevant and epistemically corrupting effects through different mechanisms.

Credibility and legitimacy must nevertheless remain separate. Credibility concerns the weight actually or rationally assigned to a source; legitimacy concerns the standing a social or institutional order grants to that source or claim. A claim can be epistemically strong and institutionally illegitimate. A claim can be institutionally legitimate and false. A proposition may have a passport and still be wrong.

This is why the paper refuses the slogan "ignore the source." The demand is harder:

> For every source effect, name the epistemic relation that earns it.

Once that discipline is imposed, the authority of the knower becomes decomposable. What looked like an intrinsic property of a person becomes a network of relations: competence, track record, access, independence, honesty, answerability, methodological discipline, and exposure to correction.

Knower fetishism is what happens when we stop decomposing.

10. Institutions after epistemic monarchy

The attack on knower fetishism is not an attack on institutions. It is an attempt to give institutions a better reason to exist.

Hardwig's epistemic dependence and Kitcher's division of cognitive labor show why modern knowledge cannot be organized as universal first-person verification [11, 15]. Universities, journals, laboratories, professional bodies, and archives coordinate expertise, preserve memory, train judgment, distribute criticism, and sustain correction across time. Their epistemic value can be enormous.

But enormous is not mysterious.

Principle 5 (Friction, not magic)**.** *Institutional certification has epistemic force insofar as institutions produce reliable epistemic friction: criticism, validation, replication, robustness testing, archival continuity, correction, and answerability. Certification does not create truth by status alone.*

As claim production becomes cheaper, the institutional problem changes. Scarcity of production can no longer do as much filtering work. The appropriate response is not to make entry artificially scarce. It is to make survival under criticism expensive.

> Institutions should be difficult places for claims to survive, not difficult places for claims to enter.

The alternative generates a regress. If evidence counts only after an authorized institution recognizes it, what epistemic fact authorizes the institution? If the answer is that its procedures reliably distinguish better from worse claims, then procedure does the work. If the answer is simply that the institution is authorized, epistemology has quietly become administration.

10.1. The Verification Paradox

Suppose a claim is denied standing until an institution certifies it. If certification adds criticism, replication, methodological checking, or accountability, there is no paradox: the institution added epistemic work. But if every epistemically relevant feature is already fixed and the remaining difference is only the stamp, then certification is recognizing permission rather than producing warrant. If the claim still cannot count without the stamp, "knowledge" has begun to track authorization.

The paradox does not show that certification is dispensable. It forces certification to disclose its difference-maker. The institution may be an indispensable machine for producing epistemic friction. It is not a sacrament.

Authority should be the receipt for epistemic work, not the merchandise. Peer review can expose a claim to disciplined criticism; it cannot transubstantiate a false proposition into a true one. A journal can reject a truth and publish an error. Reality keeps no impact-factor database.

11. Fourteen objections, none invited to be polite

11.1. Objection 1: "Facts are not knowledge"

Correct. Under ordinary subject-involving usage, a dinosaur fossil existing before inquiry is not someone's knowledge.

Reply. That concession is built into the argument. The claim is that conditions of *possession* do not exhaust

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conditions of *epistemic constraint*. Calling the pre-subjective structure "not knowledge" does not make the structure begin later. If the objection is merely lexical, it wins the noun and loses the ontology.

11.2. Objection 2: "Knowledge-first already says knowledge is world-involving"

Williamson's factive mental state is not a Cartesian item sealed inside a head. Why accuse knowledge-first epistemology of anthropocentrism?

Reply. The paper does not accuse knowledge-first theory of idealism. It targets a slide in surrounding epistemic practice: from knowledge being a subject's world-involving state to human-like subjecthood being treated as a qualification for epistemic contribution. The former does not entail the latter.

11.3. Objection 3: "Popper's objective knowledge is still human-produced"

World 3 does not give you pre-human knowledge.

Reply. Agreed. Popper is used to establish that objective epistemic contents can be studied without current possession by a knowing subject. The pre-human claim is weaker and realist: facts, traces, lawful dependencies, and constraints precede our uptake of them. The paper does not smuggle World 3 into the Jurassic period.

11.4. Objection 4: "Information is not knowledge"

Dretskean information can exist without a knower, but knowledge requires the right information-dependent state.

Reply. Exactly. Information demonstrates that epistemically load-bearing structure need not originate in a knower. Knowledge can be the successful uptake of structure that did not wait for uptake to exist.

11.5. Objection 5: "AI is not a thermometer"

LLMs are generative, opaque, semantically flexible, and capable of fluent fabrication. Instrument analogies understate the danger.

Reply. Correct. The analogy proves only a negative: non-knower status alone cannot disqualify an epistemic contributor. AI-specific defects must be identified at the level of reliability, dependence, evidence, or accountability. That is stricter than anthropomorphic dismissal, not more permissive.

11.6. Objection 6: "Provenance is higher-order evidence"

Learning that AI generated a claim can rationally lower confidence even after first-order checking.

Reply. Yes. The Provenance Relevance Constraint preserves exactly that possibility. The target is unexplained residual force after the relevant higher-order pathway is represented. This objection kills a naive genetic-fallacy argument. It does not kill the present one.

11.7. Objection 7: "Unknown unknowns never screen off"

No finite evaluator can know that every relevant AI failure mode has been represented. Machine provenance may therefore rationally lower confidence because it signals risks we have not yet discovered. The residual-provenance test appears impossible to satisfy.

Reply. The objection limits empirical identification; it does not license a source label to become a universal solvent. If "AI-generated" rationally predicts unmodeled failure, then the operative reason is uncertainty about reliability. Say that. Estimate it where possible. Preserve it where it cannot be estimated. What cannot be allowed is an unspecified remainder that defeats any amount of contrary evidence merely because its content is unknown. Ignorance is allowed to be uncertainty. It is not allowed to cosplay as evidence. An unfalsifiable pedigree penalty is not caution; it is immunity from correction.

11.8. Objection 8: "Your independent checks may share the same ancestor"

A claim can appear independently verified while the generator, verifier, retrieval system, benchmark, or dataset share a hidden common source. Two checks may therefore be one epistemic route wearing two interfaces.

Reply. Correct. Independence is a property of the route, not the number of reviewers or buttons at its end. A pipeline that shares model family, training contamination, corrupted data, or a common inferential bottleneck does not screen off provenance. This strengthens the Bridge Burden: a verification claim must carry its dependency structure. Two checks are not two checks merely because they have two buttons.

11.9. Objection 9: "Accountability is irreducibly human"

Scholarly knowledge requires agents who can answer, retract, correct, and bear responsibility.

Reply. Accountability may remain human. Generation need not. Verification need not. Calculation need not. Retrieval need not. The point of role decomposition is precisely to preserve accountability without pretending that the accountable person personally performed every epistemically productive operation.

11.10. Objection 10: "This is just conditional independence"

Once every relevant pathway from provenance to reliability is screened off, of course the label should not matter. The thesis is formally trivial.

Reply. Good. Sometimes philosophy's most useful job is to show that a socially sophisticated practice rests on a formally embarrassing confusion. Simplicity is not always a weakness. Sometimes it is the indictment.

11.11. Objection 11: "Without the knower, anything goes"

If human status loses its privilege, rumor, machine output, amateur speculation, and expert judgment collapse

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into one epistemic level.

Reply. The opposite follows. Remove automatic human credit and explicit epistemic standards become more important: evidence, calibration, track record, independence, reproducibility, error detection, answerability, and criticism. The proposal is not equal credibility. It is equal liability to epistemic testing.

11.12. Objection 12: "There is no epistemic structure without interpretation"

On a sufficiently strong constructivism, traces become evidence only within concepts, practices, and interpretive schemes. Calling anything epistemic before those practices arise therefore risks smuggling human semantics into a pre-human world.

Reply. This is why the paper now calls the pre-subjective item a *constraint structure*, not evidence and not knowledge. The paper can concede that *evidence-for-us* requires uptake within a practice. It needs only the weaker claim that uptake is answerable to differences it did not create. The fracture pattern, stellar emission, or causal dependency need not already mean anything in our vocabulary. It need only constrain which later meanings survive contact with it. Interpretation supplies a mapping; it does not thereby manufacture every constraint on the mapping. If that weaker realism is rejected, say so. But then AI is no longer the controversial part of the paper.

11.13. Objection 13: "Your formalism just hides the human in the operator"

Every record is generated by some access procedure. Why pretend that writing $R_A = \mathcal{O}_A(W; \Pi_A)$ escapes interpretation? The operator, selection policy, and reliability model are themselves human constructions.

Reply. It does not escape interpretation; it disciplines it. The point of the notation is precisely that a record is not licensed to identify itself with the world, and a source cue is not licensed to identify itself with reliability. Human construction of an operator does not imply human construction of every constraint on the operator's success. A bad thermometer is human-made; the temperature-dependent expansion it fails to track is not thereby authored by the thermometer maker. The formalism therefore weakens, rather than strengthens, anthropocentric privilege: humans become one bounded route among others, accountable to what their route preserves and distorts.

11.14. Objection 14: "This is just realism wearing a leather jacket"

The rhetoric is louder than the metaphysics. Of course a realist thinks the world does not wait for us.

Reply. Exactly. The novelty is not the realism. The target is the inconsistency between accepting a mind-independent world in metaphysics and quietly re-centering the human source in epistemic legitimacy. The leather jacket is optional. The contradiction is not.

> We removed humanity from the center of the cosmos.
> We removed humanity from the center of life.
> **Epistemology should explain why it deserves the last throne.**

12. What is actually being destroyed

Because the rhetoric is intentionally hostile to one familiar picture, the boundaries should be boring.

1. The paper does not claim that AI systems are reliable in general.
2. It does not claim that current AI systems know, believe, understand, testify, or possess moral standing.
3. It does not claim that every fact, trace, or causal relation is literally knowledge under ordinary subject-involving usage.
4. It does claim that the constraint structure to which knowledge must answer is not created merely by the act of knowing it.
5. It does not claim that provenance is normally irrelevant.
6. It does not claim that human accountability, expertise, testimony, or institutions are dispensable.
7. It does not claim that distrust of machines is morally equivalent to testimonial injustice against persons.
8. It does not claim novelty for anti-anthropocentrism as such. Its contribution is the diagnosis of the Possession–Constitution Collapse and its manifestation as knower fetishism in AI-mediated inquiry.

What is being destroyed is narrower and more consequential: the presumption that because a subject must occupy the state of knowing, the recognized knower therefore sits at the metaphysical center of epistemic value.

The subject is a participant in the relation. It arrives into a world already capable of correcting it.

> The knower does not confer reality on the evidence.
> The evidence confers liability on the knower.
> **That asymmetry is the whole scandal.**

The subject is not the sun around which truth must orbit. At most, the subject is one late-arriving instrument that learned to write grant applications.

13. Conclusion: the knower arrived late

The world was not epistemically blank before someone looked at it. Reality contained differences before anyone named them, lawful dependencies before anyone modeled them, traces before anyone read them, and constraints before anyone argued from them. Before anyone knew, there was already something to misdescribe. If one reserves the word "knowledge" for successful possession by a subject, nothing in this paper objects. Keep the word. The point is that possession arrives late, while answerability does not.

That lateness matters. Once the knower is treated as the owner rather than the creator of epistemic correctness, several confusions become easier to see. A source can matter without being a truth-maker. An expert can deserve deference without possessing intrinsic

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epistemic nobility. An institution can earn authority without minting truth. A machine can contribute to a route to knowledge without being a knower. A human can remain accountable without personally generating every epistemic component for which she answers.

Generative AI makes these distinctions difficult to avoid because it removes a convenient historical bundle. The human author used to package generation, intention, endorsement, responsibility, reputation, and credibility in one visible body. Once generation moves elsewhere, the package opens. We are forced to ask which component we actually valued.

Perhaps that is why the disclosure "written by AI" feels philosophically larger than it should. The machine did not merely enter an epistemic practice. It exposed how much of the practice had been organized around an image of the knower.

> **Written by AI. Still true.**
>
> The title does not say that you should believe it.
> It asks why authorship made you forget what kind of question truth is.

A final provocation follows.

> Perhaps knowledge, in the possessive sense, needs a knower.
> Truth did not. Constraints did not. Traces did not. Dependencies did not.
>
> **The knower arrived late, named the building, and took credit for the architecture.**

AI may not be a knower. Fine.

The fossil was not a knower either. It could still ruin your paper. The detector does not understand your theory. It can still falsify your prediction. The archive has no beliefs. It can still reveal that you misquoted the record.

Reality has veto power without voting rights.

That does not put the human back at the center.

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10.5281/zenodo.22301202

References

[1] D. Baird, *Thing Knowledge: A Philosophy of Scientific Instruments*. Berkeley: University of California Press, 2004.

[2] D. Christensen, "Higher-order evidence," *Philosophy and Phenomenological Research*, vol. 81, no. 1, pp. 185–215, 2010. doi:10.1111/j.1933-1592.2010.00366.x.

[3] C. A. J. Coady, *Testimony: A Philosophical Study*. Oxford: Clarendon Press, 1992.

[4] F. I. Dretske, *Knowledge and the Flow of Information*. Cambridge, MA: MIT Press, 1981.

[5] E. Fricker, "On the metaphysical and epistemic contrasts between human and AI testimony," *Inquiry*, published online 29 September 2025. doi:10.1080/0020174X.2025.2553297.

[6] G. Frege, "The Thought: A Logical Inquiry," *Mind*, vol. 65, no. 259, pp. 289–311, 1956. doi:10.1093/mind/65.1.289.

[7] O. Freiman, "Analysis of beliefs acquired from a conversational AI: Instruments-based beliefs, testimony-based beliefs, and technology-based beliefs," *Episteme*, vol. 21, no. 3, pp. 1031–1047, 2024. doi:10.1017/epi.2023.12.

[8] R. N. Giere, "Distributed cognition in epistemic cultures," *Philosophy of Science*, vol. 69, no. 4, pp. 637–644, 2002. doi:10.1086/344627.

[9] A. I. Goldman, "What is justified belief?" in G. S. Pappas, ed., *Justification and Knowledge: New Studies in Epistemology*. Dordrecht: D. Reidel, 1979, pp. 1–25.

[10] A. I. Goldman, "Experts: Which ones should you trust?" *Philosophy and Phenomenological Research*, vol. 63, no. 1, pp. 85–110, 2001. doi:10.1111/j.1933-1592.2001.tb00093.x.

[11] J. Hardwig, "Epistemic dependence," *The Journal of Philosophy*, vol. 82, no. 7, pp. 335–349, 1985. doi:10.2307/2026523.

[12] P. Humphreys, *Extending Ourselves: Computational Science, Empiricism, and Scientific Method*. New York: Oxford University Press, 2004.

[13] P. Humphreys, "The philosophical novelty of computer simulation methods," *Synthese*, vol. 169, no. 3, pp. 615–626, 2009.

[14] G. Kahane, "Evolutionary debunking arguments," *Noûs*, vol. 45, no. 1, pp. 103–125, 2011. doi:10.1111/j.1468-0068.2010.00770.x.

[15] P. Kitcher, "The division of cognitive labor," *The Journal of Philosophy*, vol. 87, no. 1, pp. 5–22, 1990. doi:10.2307/2026796.

[16] J. Lackey, *Learning from Words: Testimony as a Source of Knowledge*. Oxford: Oxford University Press, 2008.

[17] M. Fricker, *Epistemic Injustice: Power and the Ethics of Knowing*. Oxford: Oxford University Press, 2007.

[18] K. R. Popper, "Epistemology Without a Knowing Subject," in B. van Rootselaar and J. F. Staal, eds., *Logic, Methodology and Philosophy of Science III*, Studies in Logic and the Foundations of Mathematics, vol. 52. Amsterdam: North-Holland/Elsevier, 1968, pp. 333–373. doi:10.1016/S0049-237X(08)71204-7.

[19] H. Reichenbach, *The Direction of Time*. Berkeley: University of California Press, 1956.

[20] S. Sahebi and P. Formosa, "The AI-mediated communication dilemma: Epistemic trust, social media, and the challenge of generative artificial intelligence," *Synthese*, vol. 205, art. 128, 2025. doi:10.1007/s11229-025-04963-2.

[21] T. Williamson, *Knowledge and Its Limits*. Oxford: Oxford University Press, 2000.

[22] L. T. Zagzebski, *Epistemic Authority: A Theory of Trust, Authority, and Autonomy in Belief*. Oxford: Oxford University Press, 2012.

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The Readout Condition: Distinguishability, Access, and Epistemic Warrant

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What the chapter states. *"Claims routinely distinguish more finely than the access sources invoked on their behalf." "This paper develops the Readout Condition as a necessary architecture for source-relative epistemic warrant."* (10.5281/zenodo.22301318)

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Falsifier(s). *"A counterexample would require a target-sensitive route not represented in K^* that nevertheless always passes the access audit without being absorbable into the declared architecture."*

Read before. Ch. 1 *Written by AI. Still True.*; Ch. 3 *Readout Genesis Standalone Synthesis*; Ch. 27 *When AI Expands Human Potential*; Ch. 39 *The Standalone Scholar*.

Feeds into. This book's Chapters 1, 3–7, 10–16, 18, 23, 27, 29–31, 37–41 (mechanical KG cross-reference list; Appendix A).

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The Readout Condition

Distinguishability, Access, and Epistemic Warrant

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Abstract

Claims routinely distinguish more finely than the access sources invoked on their behalf. The central epistemic problem is not merely whether the resulting claim is true or justified, but whether each distinction in it is correctly attributed to the source, model, background evidence, relevance restriction, or decision rule that actually supplied the discrimination. This paper develops the *Readout Condition* as a necessary architecture for source-relative epistemic warrant. Its core is a three-part norm: *existence* (every claim-level distinction has an epistemic provenance path), *attribution* (a distinction credited to a source must be licensed by that source), and *disclosure* (essential augmentations must remain visible to audit). In the exact functional case, source attribution requires constancy on readout fibers and hence factorization through the readout. For non-transitive indiscriminability, the correct extension is pointwise: a claim token at an actual alternative is licensed only if its local indiscriminability neighbourhood lies inside the claim-value fiber, avoiding an illicit transitive closure. In stochastic settings, Blackwell garbling provides an *access audit* but cannot by itself detect assumption-driven silent lifts; a complementary identification ladder tracks the weakest augmentation layer at which a distinction becomes identified. The resulting typed provenance DAG supports a defeater-routing result: source-specific defeaters propagate exactly along the distinctions that depend on that source, while misattribution misroutes defeat and correction. Classical and pre-modern pressure tests from Dignāga–Dharmakīrti, Madhyamaka, Avicenna, and Suhrawardī constrain the framework's neutrality. The proposed contribution is not a new factorization theorem, data-processing inequality, or general notion of provenance, but a unified distinction-level architecture linking access, identification, attribution, disclosure, and epistemic revision.

1. Introduction

Epistemic practices routinely transform limited access into specific claims. A thermometer produces a reading; a clinician receives a test result; a witness reports an event; a classifier emits a label; an institution converts records into a decision. In each case, a later claim may distinguish more finely than the immediate readout. Such sharpening is often legitimate. The difficult question is not whether all downstream specificity is forbidden, but *what supplied it and whether it is attributed correctly.*

This paper calls the governing requirement the **Readout Condition**. The mature form defended here has three logically distinct components.

Principle 1 (Existence)**.** *Every claim-level distinction that bears epistemic weight must have an epistemic provenance path.*

Principle 2 (Attribution)**.** *If a claim attributes a distinction to a source S, then S must license that distinction under the relevant access condition.*

Principle 3 (Disclosure)**.** *If a distinction depends essentially on an augmentation beyond the attributed source, that augmentation must remain declared or recoverable in an epistemic audit.*

The conjunction is the Readout Condition. It is stronger than a generic demand for provenance and weaker than a complete theory of knowledge. It permits theory, priors, models, testimony, background evidence, relevance restrictions, and decision policies; what it forbids is an epistemically consequential distinction with no traceable ground, a false attribution of that distinction to the wrong ground, or a representation of an augmented conclusion as though it were contained in the original source.

The distinction between *truth*, *source support*, and *source attribution* is crucial. A claim may be true while its source attribution is false. A claim may also be well warranted overall through a plurality of grounds while not being warranted *by the source to which it is credited.* The title's reference to epistemic warrant is therefore not the claim that factorization is a sufficient condition for knowledge. Rather, the paper argues that source-relative warrant is a genuine component of epistemic evaluation, and that misattribution can become a warrant defeater when the alleged source is an essential ground of the subject's justification or assertion.

Conceptual and operational lineage. The Readout Condition is a standalone philosophical extraction from a broader public Readout Universe programme rather than a vocabulary assembled for this manuscript. The repository's July 2026 *Information DNA* already treated distinguishability, retention, accessibility, identity-locking, and lens discipline as distinct requirements; its Translation Protocol imposed an *identifiability gate* before an answer

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could be asserted, returning "structurally unanswerable from these records" rather than guessing across a null direction; its Gate Typing Law required an evidential gate to possess actual discriminating power; and by 3 August 2026 its Claim IR represented claims with explicit dependencies, tiers, and bridge classes and refused silent promotion of untranslated or bridge-dependent material (Lahtee, 2026). Public commit history timestamps these components before the present Paper Zero.[1] These artifacts do not prove the Readout Condition. They matter because they establish the independent architecture from which the present distinction-level epistemology is being abstracted.

The philosophical problem nevertheless has many near neighbours. Discrimination, relevant alternatives, and margin-for-error constraints are established themes in epistemology (Dretske, 1970; Goldman, 1976; Pritchard, 2010; Williamson, 2000, 2013; Melchior, 2021). Toulmin's argument model already distinguishes data, warrant, backing, qualifier, and rebuttal (Toulmin, 1958). Contrastivism makes the question and its alternatives constitutive of knowledge attribution (Schaffer, 2005, 2007). Manski's partial-identification programme explicitly distinguishes what data identify from what stronger assumptions identify (Manski, 2003, 2007). Blackwell comparison, statistical sufficiency, and data-processing results constrain what processing can add; philosophy of measurement distinguishes indication, model, calibration, and phenomenon; testimony and provenance research distinguish transmission, generation, assertion, and source history. The burden of this paper is therefore not to rediscover any of these components. It is to show that their conjunction still leaves a distinct object: the *typed provenance of each discriminative commitment in a claim and the way epistemic defeat propagates through that provenance structure.*

The formal development has four parts. First, exact readouts yield a fiber/factorization condition. Second, non-transitive indiscriminability requires a pointwise condition on an actual claim token rather than a global equivalence closure. Third, stochastic garbling is retained but explicitly demoted to what it can actually test: whether a downstream channel uses target-sensitive access not available through the stated source. A second, identification-based audit is required to locate assumption-driven specificity. Fourth, claim distinctions are placed in a typed provenance DAG whose dependency structure determines where defeaters and corrections propagate.

The cross-traditional route is not decorative. Dignāga–Dharmakīrti force a separation between access and conceptual uptake; Madhyamaka forces constitution-neutrality and, through the *Vigrahavyāvartanī*, presses the meta-question of how a purported means of knowledge is itself warranted; Avicenna tests whether supplementation can be classified by epistemic role rather than faculty; and Suhrawardī blocks any definition of readout that presupposes representation. Importantly, these pressures changed the framework: earlier global, representational, and faculty-typed formulations are not retained merely to make the tests pass.

The resulting thesis can be stated without inflation:

$$\boxed{\text{No epistemic discrimination without provenance.}} \tag{1}$$

The rest of the paper specifies what that maxim means, what it does not mean, and what would falsify the proposed architecture.

2. Literature Review: Classical Pressure Tests and Contemporary Collisions

2.1 Scope and comparative method

The Readout Condition addresses a problem that predates contemporary epistemology: how far may a claim legitimately distinguish among alternatives, given the way its target becomes epistemically accessible? This section develops a problem-driven comparative architecture around four classical and pre-modern pressure tests—Dignāga–Dharmakīrti, Madhyamaka, Avicenna, and Suhrawardī—while placing the closest contemporary theories alongside them.

The comparison is deliberately non-reductive. *pratyakṣa* is not identified with readout; Madhyamaka is not redescribed as contrastive semantics; Avicennian intellection is not assimilated to Bayesian inference; and *ʿilm al-ḥuḍūrī* is not treated as an early theory of information channels. Each tradition is used because it destabilizes a different assumption that a general theory of epistemic access might otherwise import unnoticed. The four pressures are deliberately distinct: Dignāga–Dharmakīrti test whether access can be separated from conceptual uptake; Madhyamaka tests whether discrimination can remain neutral about intrinsic constitution; Avicenna tests the classification of supplementation; and Suhrawardī tests whether access must be representational.

These traditions constrain the neutrality of the Readout Condition rather than supplying its historical ancestry. Contemporary theories are introduced as collision tests: they determine which parts of the framework are already present in epistemology, semantics, statistics, information theory, philosophy of measurement, and provenance research.

2.2 Dignāga–Dharmakīrti: access and conceptual uptake

Dignāga's *Pramanasamuccaya* and Dharmakīrti's subsequent epistemological programme distinguish perception and inference as fundamental modes of warranted cognition. In the perceptual chapter of the *Pramanasamuccaya*, Dignāga characterizes perception through freedom from conceptual construction; Dharmakīrti develops a richer account involving particulars, causal efficacy, conceptual cognition, and exclusion (Dignaga, 1968; Dunne, 2004). These doctrines should not be simplified into a modern contrast between raw data and interpretation. Their value here is structural: what becomes available in cog-

[1] For lineage rather than evidential support: the v2 Information DNA/Translation scaffold entered the public repository in commit `1b1ca4f` (19 July 2026); Claim IR with explicit dependency fields entered in commit `a871db7` (3 August 2026). The present paper does not import the broader programme's physical or metaphysical commitments.

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nition and what conceptual activity later predicates or classifies need not be the same epistemic operation.

The Readout Condition therefore does not identify *pratyakṣa* with R. It extracts a narrower pressure: In compressed form, the distinction is: *what access makes available need not exhaust what is later asserted on its basis.* If a later claim contains distinctions not attributable to the original mode of access, those distinctions require an epistemic account of how they entered.

This pressure immediately collides with contemporary discrimination-based epistemology. Goldman places reliable discriminatory mechanisms near the centre of perceptual knowledge (Goldman, 1976). Pritchard distinguishes discriminating from favouring epistemic support, weakening any simple identification of support with direct discriminatory capacity (Pritchard, 2010). Melchior develops discrimination itself as a modal epistemic capacity rather than merely a component of a theory of knowledge (Melchior, 2021). These works block any novelty claim that epistemology has neglected discrimination. The residual question is narrower: not merely whether discrimination occurs, but what licenses each discrimination asserted by a claim.

2.3 Madhyamaka: constitution-neutrality

In the *Mulamadhyamakakarika*, Nāgārjuna attacks explanations that depend on *svabhāva*, an intrinsic or independently grounded nature. Major contemporary interpretations differ over the exact reconstruction of the target, but dependence and the rejection of an ontologically self-sufficient terminus remain central (Garfield, 1995; Siderits and Katsura, 2013; Westerhoff, 2009). The Readout Condition does not derive local contrast spaces from Madhyamaka. Its narrower role is to pressure an unnoticed metaphysical assumption: epistemically relevant distinctions need not be treated as intrinsic divisions already carved into a fully specified target.

Thus the notation $R : \mathcal{X} \to Y$ must not silently turn $\mathcal{X}$ into a global inventory of independently determinate world states. $\mathcal{X}$ may instead be local to the question a claim undertakes to answer. Madhyamaka motivates **constitution-neutrality**: the framework should remain agnostic about whether relevant alternatives correspond to intrinsic joints of reality, relational structures, conventional individuations, or some combination thereof.

A different problem remains: if $\mathcal{X}$ is local, what fixes it? Here the formal resources come from contemporary work on subject matter and questions under discussion. Yablo develops subject matter through structured partitions of logical space (Yablo, 2014); Roberts models discourse around questions represented through alternative answers (Roberts, 2012). The Readout Condition uses this literature to constrain the contrast space without adopting a global possible-world ontology. The contrast space must preserve distinctions constitutive of the question the claim undertakes to settle; removing otherwise relevant alternatives is itself an epistemically significant restriction.

2.4 Avicenna: the classification of supplementation

Avicenna's psychology does not permit a simple model in which sensory input contains a complete epistemic message that intellect merely decodes. The *Kitab al-Nafs* distinguishes external and internal senses, imagination, memory, estimation, and intellectual activity; the acquisition of intelligibles also involves syllogistic reasoning, middle terms, and the Active Intellect (Avicenna, 1959; McGinnis, 2010). Specialist scholarship emphasizes the complexity of the relationship between abstraction and the Active Intellect rather than reducing one to the other (Alpina, 2014).

This makes Avicenna a pressure test for the proposed distinction between *access augmentation* and *assumption augmentation*. He is not invoked as though he already endorsed that taxonomy. Instead, his theory asks whether greater cognitive discrimination can be classified by epistemic role rather than by faculty. An additional sensory channel is not automatically "access," nor is intellectual activity automatically "assumption." The relevant question is what the addition contributes to the claim's epistemic basis.

A successful taxonomy must therefore classify supplementation functionally: does it make new target-supported distinctions available, or does it increase specificity by adding inferential commitments to what is already available? If a case cannot be assigned without distortion, the framework rather than Avicenna requires revision.

2.5 Suhrawardī: selectivity without representation

Suhrawardī provides the strongest pressure against identifying readout with representation. In *Ḥikmat al-Ishrāq*, he develops an Illuminationist account of self-awareness and presence while criticizing Peripatetic accounts in which knowledge is explained through an acquired form in the knower (al Din Yahya Suhrawardi, 1999). Kaukua's reconstruction places this development within a sustained Islamic philosophical debate over first-person awareness and presence (Kaukua, 2015).

If a readout were defined as an internal representation, the claimed generality of the Readout Condition would fail. The weaker conception required here is relational: a readout is epistemically relevant insofar as an access relation preserves or makes available distinctions that can support a claim. Representation may implement such a relation, but it is not constitutive of it. This yields a central distinction:

$$\text{Representationality} \neq \text{Selectivity}. \tag{2}$$

Even granting a maximally strong case of unmediated presence, the Readout Condition would simply become non-restrictive with respect to distinctions fully available under that relation. This is a conditional result of the framework, not a historical thesis attributed to Suhrawardī.

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2.6 Nearest-neighbour collision audit

The historical pressure tests establish neutrality requirements, not novelty. The novelty question requires collision with the nearest contemporary architectures, including several that are closer than a generic provenance survey.

2.6.1 Argument warrants, contrastive knowledge, and discrimination

Toulmin's model already decomposes argument into data, claim, warrant, backing, qualifier, and rebuttal (Toulmin, 1958). It is therefore a serious predecessor for any claim about "warrant" and claim-level support. The Readout Condition does not replace that architecture. Its different object is the *discriminative capacity of the attributed source*: a Toulmin warrant can state why data support a claim without representing which target alternatives the data fail to distinguish or whether a specific claim distinction has been credited to the wrong source.

Likewise, Hintikka, Dretske, Goldman, Pritchard, Williamson, and Melchior provide powerful accounts of epistemic alternatives, information, discrimination, and indiscriminability (Hintikka, 1962; Dretske, 1970, 1981; Goldman, 1976; Pritchard, 2010; Williamson, 2000, 2013; Melchior, 2021). Schaffer's contrastivism makes an explicit contrast constitutive of knowledge attribution and later treats knowing-wh as knowing an answer to a question (Schaffer, 2005, 2007). These literatures defeat any novelty claim for discrimination or local alternatives. They do not yet yield a typed ledger saying which source, access-model, background claim, relevance restriction, or policy node is responsible for each distinction in the downstream claim.

2.6.2 Partial identification and statistical sufficiency

Manski is the closest neighbour for the assumption side of the architecture. Partial identification begins by asking what the data alone identify and then studies how identification regions shrink under additional assumptions (Manski, 2003, 2007). This paper adopts that lesson directly: a stochastic garbling relation cannot reveal an undeclared prior or model assumption when the downstream rule consumes only the observed readout. An assumption audit therefore requires an identification ladder, not merely an information-ordering test. The residual contribution is to integrate this assumption ladder with source attribution, disclosure, readout models, iterated provenance, and defeater routing at the level of claim distinctions.

Classical Fisher–Neyman sufficiency also blocks any suggestion that preserving all information relevant to a target parameter is new (Fisher, 1922; Neyman, 1935). A sufficient statistic is a compression that loses no information about the parameter under a specified statistical model. The Readout Condition is neither a rival definition of sufficiency nor a claim that its deterministic factorization theorem is new. Its target is broader and differently typed: what an epistemic source may be credited with across deterministic, relational, stochastic, testimonial, model-mediated, and institutional routes.

2.6.3 Blackwell, Dretske, measurement, and robust access

Blackwell's comparison of experiments and the data-processing inequality formalize monotonicity under post-processing (Blackwell, 1951, 1953; Cover and Thomas, 2006). Dretske's information theory stresses that what a signal carries is relative to channel conditions (Dretske, 1981). Suppes, Bogen and Woodward, Tal, Giere, and Massimi distinguish data, phenomena, measurement models, calibration, and perspective (Suppes, 1962; Bogen and Woodward, 1988; Tal, 2017; Giere, 2006; Massimi, 2022). Hacking and Wimsatt make independent routes and robustness epistemically important (Hacking, 1983; Wimsatt, 1981). These are not competitors to be displaced; they identify distinct typed nodes in the provenance graph. The Readout Condition's claim is that those nodes must not be silently collapsed when downstream specificity is represented.

2.6.4 Testimony and assertion provenance

Lackey's rejection of simple transmission shows that testimonial epistemic status can be generated or transformed rather than merely passed intact from speaker to hearer (Lackey, 2008). Nanopublications and related provenance infrastructures already package assertions with contextual provenance (Groth et al., 2010; Chhetri et al., 2025). These developments defeat any claim that assertion-level provenance or dependency is new. The residual distinction is finer: documentary or assertion provenance can be correct while the *discriminative dependency* of an assertion remains misdescribed. A claim may cite the correct dataset and still represent a prior-induced distinction as though the dataset itself distinguished it.

The novelty claim is therefore a conjunction claim. No individual mathematical or philosophical component is presented as unprecedented. What is proposed is a domain-neutral architecture in which (i) the audit unit is an individual claim distinction, (ii) source attribution has an explicit licensing condition, (iii) assumptions are located by an identification ladder, (iv) provenance is a typed dependency DAG, and (v) source-specific defeaters propagate through that DAG. The public Readout Universe lineage supplies an operational precursor for this conjunction—identifiability gates, bridge classes, tier downgrades, and explicit claim dependencies—but the present paper is the first place in that programme where those components are reconstructed as a standalone epistemic condition.

3. Conceptual Method

3.1 Methodological status

This is a work of conceptual and formal philosophy, not an empirical study. The method combines problem reconstruction, formal reconstruction, cross-traditional pressure testing, nearest-neighbour collision testing, and self-application to the paper's own claims. The last component matters: a provenance theory that exempts its own models, citations, or formal bridges from audit would reproduce the very silent lift it criticizes.

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Table 1: Nearest-neighbour collision audit. The contribution defended here is the conjunction, not the individual components.

Neighbour	**Already provides**	**What the Readout Condition concedes**	**Residual task**
Toulmin	Data–claim warrants, backing, rebuttal	Claim-level licensing is not new	Test source discrimination and misattribution
Schaffer / QUD	Explicit contrast structure	Local alternatives are not new	Track provenance of each resolved contrast
Goldman / Pritchard / Melchior / Williamson	Discrimination, support, non-transitivity	Discrimination is not new	Pointwise source licensing plus provenance
Manski	Data-only vs assumption-dependent identification	Assumption burden is not new	Type the weakest augmentation layer for each distinction
Blackwell / DPI / sufficiency	Informativeness and lossless processing	Monotonicity/factorization are not new	Separate access audit from assumption audit
Dretske / measurement	Channel conditions, model mediation	Source models are not transparent	Distinguish K^* from $\widetilde{K}$ and route defeat
Lackey / robustness	Generation, transmission, independent routes	Epistemic routes may transform status	Represent branching and recursive access
Nanopublications / provenance	Assertion and source lineage	Provenance simpliciter is not new	Distinction-level dependency and defeater propagation

3.2 Units and roles

Let Q determine a local contrast space $\mathcal{X}$. Let R denote a deterministic readout, I_R a relational access structure, or K^* a stochastic access experiment. Let Φ represent a categorical claim structure. The unit of audit is not only the whole proposition but a **claim distinction**. For an actual alternative x_0, define

$$D_\Phi(x_0) = \{d_{0x'} = (x_0, x') : \Phi(x') \neq \Phi(x_0)\}. \quad (3)$$

Each $d_{0x'}$ is one contrast the actual claim token resolves against a rival alternative.

A provenance analysis assigns each such distinction a directed acyclic dependency graph

$$\mathcal{P}(d) = \big(V_d, E_d, \tau_d\big), \quad (4)$$

where the typing map τ_d distinguishes at least: *access source*, *access model*, *background-evidence claim*, *contrast/relevance operation*, *inferential commitment*, and *decision policy*. A single object may occupy different types in different audits; the types are epistemic roles, not ontological kinds.

This formal move is directly continuous with the public programme's earlier Claim IR, which already represented claims with dependencies, tiers, and bridge classes and automatically prevented untranslated or bridge-dependent material from being silently promoted. The present graph generalizes that operational discipline from whole claims to discriminative commitments.

3.3 Internal reconstruction and pressure-driven revision

Historical comparison proceeds by *internal reconstruction* $\to$ *controlled abstraction* $\to$ *pressure test*. A pressure test counts only if it can force a change. In fact, the present framework records four such changes: a global contrast-space formulation was replaced by a local one under Madhyamaka/QUD pressure; a representation-defined readout was replaced by a selectivity role under Suhrawardī pressure; a faculty-based supplementation taxonomy was replaced by role typing under Avicennian pressure; and the original global relational condition was replaced by a pointwise claim-token condition after the Williamson-style non-transitivity objection. These are theory revisions, not retroactive declarations that every tradition "passes."

3.4 Collision criteria

For each neighbour the paper asks: does it analyze the same object, impose the same constraint, type the same epistemic sources, locate assumption dependence, and determine how defeat propagates? Novelty survives only at the residual conjunction after the strongest relevant results are granted.

3.5 Failure and revision conditions

The architecture fails, rather than merely losing elegance, if (i) a source can be correctly credited with a distinction it does not locally discriminate without any other epistemic ground; (ii) the provenance graph cannot represent an important supplementation case without ad hoc typing; (iii) two genuinely different dependency structures always generate identical patterns of rational defeat and correction; or (iv) an existing theory already supplies the same distinction-level licensing, identification, provenance, and defeat architecture at comparable generality. Application-level opacity is different: when K^*, $\widetilde{K}$, or the historical dependency graph cannot be recovered, the correct output is provenance indeterminacy, not forced classification.

4. The Readout Condition

4.1 Local contrasts and claim discrimination

Let Q be a question or epistemic task. Its relevant alternatives determine a local contrast space $\mathcal{X}_Q$, abbreviated $\mathcal{X}$. The local qualification matters. A claim is never audited against every imaginable difference in reality; it is audited against the distinctions it undertakes to settle. At the same time, the claimant cannot freely shrink $\mathcal{X}$ after inspecting the evidence. A relevance restriction that removes otherwise constitutive alternatives is itself an epistemically significant commitment.

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A claim structure Φ can be represented, in the exact categorical case, as a function

$$\Phi : \mathcal{X} \to Z, \tag{5}$$

where Z indexes the distinctions made by the claim. Two alternatives $x, x' \in \mathcal{X}$ are claim-equivalent when $\Phi(x) = \Phi(x')$ and claim-distinguished otherwise.

A readout R specifies what the relevant access relation preserves. The exact deterministic case is the simplest:

$$R : \mathcal{X} \to Y. \tag{6}$$

Nothing in this notation implies that the target is metaphysically discrete or that the readout is a representation. The functions describe the contrastive role played by the access structure relative to the current question.

4.2 Exact functional case

Definition 1 (Readout-admissible discrimination)**.** *A categorical claim structure $\Phi : \mathcal{X} \to Z$ is* readout-admissible relative to $R : \mathcal{X} \to Y$ *when it is constant on every fiber of R:*

$$R(x) = R(x') \Rightarrow \Phi(x) = \Phi(x'). \tag{7}$$

Proposition 1 (Factorization)**.** *For functions $R : \mathcal{X} \to Y$ and $\Phi : \mathcal{X} \to Z$, the following are equivalent:*

(i) Φ is constant on every fiber of R;
(ii) $\ker R \subseteq \ker \Phi$;
(iii) there exists a function $\bar{g} : \mathrm{im}(R) \to Z$ such that $\Phi = \bar{g} \circ R$.

Proof. If $\Phi = \bar{g} \circ R$, then $R(x) = R(x')$ implies $\Phi(x) = \Phi(x')$, so (iii)⇒(i). Statement (i) is exactly the inclusion of the equivalence relation induced by R in that induced by Φ, giving (i)⇔(ii). For (i)⇒(iii), define $\bar{g}(y) = \Phi(x)$ for any x such that $R(x) = y$. Fiber constancy makes this well-defined on $\mathrm{im}(R)$. □

The proposition is elementary mathematics. The philosophical claim is not that factorization is new, but that factorization marks a boundary on *source attribution.* If $R(x) = R(x')$ while $\Phi(x) \neq \Phi(x')$, the distinction in Φ did not come from R alone. A broader epistemic case may still justify the claim, but the additional discrimination requires another source.

This yields the exact Readout Condition:

Principle 4 (Exact Readout Condition)**.** *If a categorical distinction in Φ is licensed by R alone, then Φ must be readout-admissible relative to R.*

The direction is intentionally one-way. Factorization is sufficient for formal admissibility, not for epistemic warrant as a whole.

4.3 Relational readouts: pointwise licensing

Many access relations cannot be represented adequately by partitions. Indiscriminability may be reflexive and symmetric without being transitive (Williamson, 2013). Let $I_R \subseteq \mathcal{X} \times \mathcal{X}$ and define the local indiscriminability neighbourhood of an actual alternative x_0 as

$$N_R(x_0) = \{x' \in \mathcal{X} : (x_0, x') \in I_R\}. \tag{8}$$

Let the actual claim token have value $z_0 = \Phi(x_0)$.

Principle 5 (Pointwise Relational Readout Condition)**.** *The claim token $z_0 = \Phi(x_0)$ is licensed by relational access R alone only if*

$$\boxed{N_R(x_0) \subseteq \Phi^{-1}(z_0).} \tag{9}$$

Equivalently, the claim token may not exclude any alternative that remains locally indiscriminable from the actual case under the attributed source.

The quantifier is deliberately pointwise. A global requirement $I_R \subseteq I_\Phi$ would indeed force every equivalence relation containing I_R to contain its transitive closure and would collapse a Sorites chain. Equation (9) does not iterate licenses from one centre to another: licensing at x_0 constrains $N_R(x_0)$ but does not imply licensing at every $x' \in N_R(x_0)$. The local geometry therefore remains visible.

Proposition 2 (Reduction to the deterministic case)**.** *If I_R is induced by a deterministic readout R through $xI_Rx' \iff R(x) = R(x')$, then requiring (9) for every $x_0 \in \mathcal{X}$ is equivalent to fiber constancy and hence to factorization $\Phi = \bar{g} \circ R$ on $\mathrm{im}(R)$.*

This is the appropriate sense in which the relational condition extends the exact case: not by replacing a non-transitive relation with an equivalence closure, but by changing the object of licensing from a global claim function to an actual claim token.

4.4 Stochastic readouts: access audit, not assumption audit

Let the actual source relation be a Markov kernel

$$K^* : Y \mid \mathcal{X}, \tag{10}$$

and a stochastic downstream output be a channel $C : Z \mid \mathcal{X}$. For the finite case, Blackwell comparison gives the following access test; for more general experiment spaces the relevant extensions belong to the Le Cam–Torgersen theory of statistical experiments (Blackwell, 1951, 1953; Cam, 1964; Torgersen, 1991).

Principle 6 (Stochastic access admissibility)**.** *A downstream channel uses no target-sensitive access beyond K^* only if there exists a Markov kernel $G : Z \mid Y$ such that*

$$\boxed{C = G \circ K^*.} \tag{11}$$

The wording is intentionally precise. Equation (11) is an *access audit*, not a complete provenance audit. Any rule that consumes only y—including one that uses an undeclared prior, model class, or loss function—can be absorbed into G. Garbling therefore detects an undeclared target-sensitive route that bypasses the stated source; it does *not* by itself detect silent assumption augmentation.

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Treating it as a full source-only warrant condition would be a category error.

Practical inference also distinguishes the actual access process from its model:

$$\boxed{K^* \neq \widetilde{K}.} \tag{12}$$

Likelihood ratios are ordinarily computed under $\widetilde{K}$, not read directly from K^*:

$$\Lambda_{\widetilde{K}}(y; x, x') = \frac{\widetilde{K}(y \mid x)}{\widetilde{K}(y \mid x')}. \tag{13}$$

In a Bayesian representation,

$$\log \frac{P(x \mid y)}{P(x' \mid y)} = \log \frac{P(x)}{P(x')} + \log \Lambda_{\widetilde{K}}(y; x, x'). \tag{14}$$

The decomposition already shows that source model and prior occupy different provenance roles. The next section supplies the missing assumption audit.

5. Augmentation and Epistemic Provenance

5.1 Typed augmentation grammar

The phrase "assumption augmentation" is too coarse if it treats every downstream addition as the same operation. The provenance graph therefore distinguishes four signatures.

Definition 2 (Access augmentation). *A new target-sensitive route is added to the epistemic basis, for example a second measurement, testimony, retrieved record, sensor, or retained upstream observation. Formally, the observation structure expands, e.g.* $Y_1 \mapsto (Y_1, Y_2)$.

Definition 3 (Contrast or relevance operation). *The contrast domain changes,* $\mathcal{X} \mapsto \mathcal{X}' \subseteq \mathcal{X}$. *Such restriction can be legitimate when fixed by the question or independently justified. A restriction selected after inspecting the readout requires its own provenance because it reuses the observed result in defining what alternatives count.*

Definition 4 (Inferential commitment). *A model, prior, theory, bridge principle, calibration assumption, or other rule changes what can be inferred from an existing access basis without constituting new current access to the target.*

Definition 5 (Decision-policy augmentation). *A loss function, threshold, utility, or institutional rule maps an epistemic state into an action or classification. It can change the decision while leaving the evidential discrimination unchanged.*

Background evidence is represented recursively rather than forced into a "mere assumption" box. A population prior may be an imported commitment relative to today's patient test while itself being the conclusion of an earlier evidence graph. This recursion is essential to the architecture inherited from the Readout Universe claim/tier discipline: a bridge-dependent claim may be usable, but the bridge never disappears from its lineage.

5.2 Identification ladder: where a distinction first becomes available

Manski's partial-identification programme supplies the right mathematical resource for the assumption side of the audit (Manski, 2003, 2007). Let $\mathcal{A}_0$ denote the weakest declared commitment set compatible with the access model, and let

$$\mathcal{A}_0 \subseteq \mathcal{A}_1 \subseteq \cdots \subseteq \mathcal{A}_m \tag{15}$$

represent increasingly strong augmentation layers. For each layer let Γ_j be the set of target alternatives not excluded by the access model plus commitments through $\mathcal{A}_j$.

For an actual claim token $z_0 = \Phi(x_0)$, define the first-identification level

$$\ell(d_{0x'}) = \min\{j : x' \notin \Gamma_j\}, \tag{16}$$

whenever the rival x' is excluded at some layer. The provenance of that discrimination is not "the data" simpliciter; it includes the weakest augmentation layer at which the rival becomes excludable. If x' is already excluded at $j = 0$, the discrimination is access-model licensed at baseline. If exclusion begins only after a prior, structural model, or relevance restriction enters, the distinction is assumption-dependent. If no declared layer excludes it, the distinction remains unidentified.

Equation (16) is deliberately an audit definition rather than a new theorem in partial-identification theory. Its contribution is to attach assumption strength to individual claim distinctions and then connect those distinctions to attribution and defeat.

5.3 The E–A–D form of the Readout Condition

Let $\mathcal{P}(d)$ be the typed provenance DAG for distinction d. The mature condition is:

Principle 7 (E: provenance existence). *For every epistemically load-bearing claim distinction d, $\mathcal{P}(d)$ is nonempty or the claim is explicitly marked provenance-indeterminate.*

Principle 8 (A: provenance attribution). *If the public or doxastic attribution map $\widehat{\lambda}(d)$ credits distinction d to source S, then S must pass the appropriate licensing test for d: fiber discrimination in the exact case, pointwise neighbourhood exclusion in the relational case, or access admissibility/identification in the stochastic case.*

Principle 9 (D: provenance disclosure). *Every non-source node on an essential dependency path to d must remain declared or recoverable at the level relevant to the epistemic practice. Omitting such a node while representing d as source-generated is a silent lift.*

The Readout Condition is $E \wedge A \wedge D$. This separation closes an ambiguity in weaker formulations: the mere existence of a valid augmentation does not rescue a false attribution to the original source.

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5.4 Epistemic overreach and silent lift

Definition 6 (Epistemic overreach)**.** *A claim-level distinction exhibits epistemic overreach when it lacks an adequate provenance path, is attributed to a node that does not license it, or depends on an essential augmentation that is concealed in the relevant epistemic representation.*

Definition 7 (Silent lift)**.** *Let $\lambda(d)$ denote the actual essential dependency set of distinction d and $\widehat{\lambda}(d)$ the dependency set represented to oneself or others. A silent lift occurs when an essential node $v \in \lambda(d)$ is omitted or a source S is credited with d despite failing the corresponding licensing condition.*

This is not an anti-inference principle. A model-dependent or prior-dependent distinction can be fully warranted when the dependency is genuine and the supporting background claims are themselves adequate. The target is false provenance, not epistemic sophistication.

5.5 Defeater routing on the provenance DAG

The payoff of distinction-level provenance is not documentary neatness but revision structure. For a distinction d, let $\Pi(d)$ be the set of currently adequate licensing paths in $\mathcal{P}(d)$, each running from epistemic leaves to d. Define the essential dependency set

$$\mathrm{Ess}(d) = \bigcap_{p \in \Pi(d)} V(p). \tag{17}$$

Thus a node is essential exactly when every currently adequate licensing path for d passes through it.

Let Δ_v be a node-specific defeater that defeats provenance node v while leaving all other nodes and edges initially unchanged.

Proposition 3 (Defeater Routing)**.** *After Δ_v, the surviving licensing paths for distinction d are*

$$\boxed{\Pi_{\Delta_v}(d) = \{p \in \Pi(d) : v \notin V(p)\}.} \tag{18}$$

Consequently, d loses provenance support under the node-specific defeat iff $\Pi_{\Delta_v}(d) = \varnothing$, equivalently iff $v \in \mathrm{Ess}(d)$.

Argument. A path that contains the defeated node cannot remain an adequate licensing path; a path that does not contain it is untouched by a defeater whose stipulated target is v alone. Equation (18) follows by path deletion. The distinction loses all current provenance support exactly when no licensing path survives, which is equivalent to v belonging to every path in $\Pi(d)$, i.e. $v \in \mathrm{Ess}(d)$. The epistemic interpretation is substantive even though the graph step is elementary: equal propositional content can survive different challenges because its licensing paths differ. □

Corollary 1 (Misrouted defeat under silent lift)**.** *Let $\mathrm{Ess}(d)$ be the actual essential dependency set and $\widehat{\mathrm{Ess}}(d)$ the represented one. If the symmetric difference is nonempty, then for any epistemically possible node-specific defeater aimed at a node in that difference, the represented vulnerability of d differs from its actual dependency vulnerability. Silent lift can therefore generate under-revision when a hidden essential node is defeated, and over-revision when an innocent represented node is treated as essential.*

This result formalizes a discipline that the public Readout Universe implemented operationally before this paper: bridge-dependent claims carry dependency and tier information; untranslated terms are forced to Open rather than silently promoted; the identifiability gate blocks answers outside the record's row space; and a gate with no discriminating controls is classified as convention rather than evidence. Those rules are not proofs of Proposition 3, but they are working instances of the same dependency-sensitive norm.

5.6 From provenance error to epistemic warrant

Why does false attribution matter to warrant rather than merely to reporting? Consider a subject who accepts distinction d partly because she believes the test, witness, or dataset S itself discriminates the relevant alternatives. If correcting that meta-belief—learning that S did not discriminate them and that the result depended on an undeclared bridge—would rationally require withdrawal or suspension of d absent another ground, then the false attribution is an *essential ground* in her justificatory structure. Debates over inference from falsehood show that false premises need not always destroy knowledge, but they make the essentiality of the false ground the relevant issue (Warfield, 2005; Klein, 2008). The Readout Condition therefore identifies a class of genuine warrant failures: cases in which a false provenance belief is load-bearing for justification.

The framework also has an assertoric face. A privately warranted belief may rely on a model or prior without defect, yet asserting that "the test showed d" can misrepresent the epistemic basis when the test alone did not. Testimonial epistemology already distinguishes the epistemic status of speakers, statements, and hearers and rejects simple transmission pictures (Lackey, 2008). The Readout Condition adds a provenance norm on how the discriminative basis of an assertion is represented. Doxastic warrant and assertoric disclosure are related but not identical obligations.

6. Iterated Readout Chains

6.1 Deterministic composition

Readouts frequently occur in chains. A record may be produced by an instrument, transformed by software, summarized in a database, processed by a model, and interpreted by a human. Let

$$\mathcal{X} \xrightarrow{R_1} Y_1 \xrightarrow{R_2} Y_2 \to \cdots \xrightarrow{R_n} Y_n. \tag{19}$$

If $R_1(x) = R_1(x')$, then every deterministic downstream composition also maps x and x' to the same result. More generally, if an upstream stage has already collapsed a pair, post-processing alone cannot recreate that pairwise distinction.

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Typed provenance DAG for one discriminative commitment

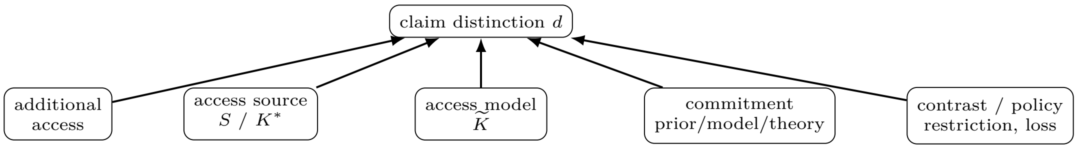

Figure 1: A claim distinction can depend on several differently typed nodes. The Readout Condition audits existence, attribution, and disclosure; Proposition 3 uses the same graph to route defeat and correction.

Corollary 2 (Upstream monotonicity)**.** *For a deterministic chain* $R_{1:n} = R_n \circ \cdots \circ R_1$,

$$\ker R_1 \subseteq \ker R_{1:n}. \tag{20}$$

At any stage j*, distinctions collapsed by* $R_{1:j}$ *remain collapsed under later deterministic processing unless additional access enters the chain.*

This is the precise form of the claim that target-supported discrimination cannot increase downstream merely through processing. The formulation avoids talk of a single "coarsest link," which is not generally well-defined across heterogeneous stages.

6.2 Stochastic composition

For stochastic chains, let the actual upstream experiment be $K_1^* : Y_1 \mid \mathcal{X}$ and let later stages be Markov kernels. Whenever $K_{j+1}^* = G_j \circ K_j^*$, each downstream experiment is a garbling of the preceding one and is therefore weakly less informative in the Blackwell order. Standard data-processing inequalities provide metric-specific consequences, for example

$$I(X; Y_{j+1}) \leq I(X; Y_j) \tag{21}$$

for mutual information in a Markov chain (Cover and Thomas, 2006). The Readout Condition does not depend on mutual information as the unique measure of distinguishability; the Blackwell relation is used because it expresses the more general experiment-level ordering.

A downstream channel C passes the access-admissibility test relative to K_j^* only if $C = G \circ K_j^*$ for some post-processing kernel G. This establishes that no extra target-sensitive access route is required; it does not establish that the downstream specificity is assumption-free. The identification ladder and provenance DAG perform that second audit. Additional access changes the architecture by forming a joint experiment with a genuinely new evidential path.

6.3 Calibration and the model of the readout

A claimant rarely knows the actual access structure directly. Let R^* or K^* denote the source relation under analysis and let $\widetilde{R}$ or $\widetilde{K}$ denote the claimant's model of it. Tal's analysis of calibration is important here because calibration constructs and tests models that connect indications with measurement outcomes (Tal, 2017). The Readout Condition therefore distinguishes

$$\boxed{R^* \neq \widetilde{R}, \qquad K^* \neq \widetilde{K}} \tag{22}$$

as epistemic roles even when the model is excellent.

This distinction prevents a subtle silent lift in stochastic inference. A likelihood ratio computed from $\widetilde{K}$ is evidence *under a model of the source*; it is not literally read off from K^*. Calibration evidence can license the use of $\widetilde{K}$, but its contribution must remain visible.

At least three failure modes follow. In an **access failure**, R^* or K^* does not preserve the distinction required by the claim. In an **access-model failure**, $\widetilde{R}$ or $\widetilde{K}$ mischaracterizes what the source preserves. In an **attribution failure**, a distinction introduced by another source or commitment is credited to the original source. These failures require different remedies: additional or improved access, recalibration/model revision, or corrected provenance attribution.

7. Classical Pressure Tests Revisited

A pressure test earns methodological force only if it can change the theory. Table 2 records the revisions produced by the four traditions rather than simply declaring that the final framework "passes."

7.1 Dignāga–Dharmakīrti: access, apoha, and perceptual judgment

The relevant pressure is sharper than a generic perception/inference contrast. Dignāga's and Dharmakīrti's accounts of conceptual exclusion (*apoha*) and later perceptual judgment make it impossible to assume that every discrimination present in a conceptual claim was already present in the same form in non-conceptual perception (Dignaga, 1968; Dunne, 2004; Arnold, 2005; Kellner, 2004). The Readout Condition therefore places conceptual uptake downstream of access and demands provenance for distinctions introduced by classificatory rules. It does not identify *apoha* with the Readout Condition; it accepts the historical challenge that discrimination can be generated by conceptual organization rather than simply copied from perception.

7.2 Madhyamaka and the warrant of the means of access

The *Mūlamadhyamakakārikā* blocks an unargued move from epistemic contrasts to intrinsic target constitution (Garfield, 1995; Siderits and Katsura, 2013; Westerhoff, 2009). The *Vigrahavyāvartanī* adds a meta-epistemic pressure: if a means of knowledge is invoked to establish an object, the status of that means cannot simply be exempted from scrutiny (Westerhoff, 2010). The parallel

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Table 2: Pressure-driven revisions of the Readout Condition.

Pressure	Earlier vulnerable formulation	Failure	Revision retained here
Dignāga–Dharmakīrti	downstream classification folded into "what perception gives"	conceptual uptake inherits source status automatically	access/readout and claim uptake are separate nodes
Madhyamaka	global state space read as intrinsic constitution	formal contrast structure becomes covert metaphysics	local $\mathcal{X}_Q$ plus constitution-neutrality
Avicenna	sensory faculty = access; intellect = assumption	faculty names substitute for epistemic roles	role-typed augmentation with provenance indeterminacy
Suhrawardī	readout defined as internal representation	knowledge by presence is excluded by definition	readout = selectivity/access role; presence itself is not forced into a representation model
Williamson-style objection	global $I_R \subseteq I_\Phi$	transitive closure erases non-transitive geometry	pointwise claim-token condition $N_R(x_0) \subseteq \Phi^{-1}(z_0)$

here is controlled rather than identificatory. The Readout Condition's distinction $K^* \neq \widetilde{K}$ makes the model or calibration of access itself a claim with provenance; a failed access-model audit is not self-undermining, but a higher-level readout failure to be diagnosed under the same rules.

7.3 Avicenna: the *wahm* test

Avicenna's estimative faculty (*wahm*) is a concrete stress test because the sheep's apprehension of the wolf's hostility is not straightforwardly one more sensible quality (Avicenna, 1959; McGinnis, 2010; Alpina, 2014). The Readout Condition does not decide the historical psychology by fiat. It supplies a criterion. If the additional faculty constitutes a target-sensitive route whose output varies with the relevant *ma'nā* while the previous sensory readout is held fixed, it plays an access-augmentation role. If the hostility classification is generated from the existing readout by a standing rule or learned commitment, it plays an inferential-commitment role. If the historical evidence does not resolve which dependence obtains, the audit returns provenance indeterminacy. The taxonomy therefore answers the case conditionally by role rather than hiding the difficulty under a faculty label.

7.4 Suhrawardī: presence before claim

Knowledge by presence should not be forced into a contrastive channel when the doctrine itself denies representational mediation. The correct boundary is therefore stronger than saying that the Readout Condition becomes "non-restrictive." Presence as such can be *pre-claim* and hence outside the condition's immediate domain. The Readout Condition applies when a categorical or otherwise discriminative claim is made on the basis of presence—for example, when "I exist" or a further articulated judgment excludes alternatives. At that point the audit asks what distinctions the presence relation licenses and what conceptual articulation adds. This preserves representation-independence without pretending to formalize the metaphysics of presence itself (al Din Yahya Suhrawardi, 1999; Kaukua, 2015; Yazdi, 1992).

8. Applications and Portability

Portability is demonstrated here by running the audit to verdict rather than merely listing domains.

8.1 Worked audit I: a positive diagnostic test

Suppose a test has sensitivity 0.90, specificity 0.91, and is used in a population with prevalence 0.01. For a positive result $y = +$, Bayes' rule gives

$$P(D \mid +) = \frac{0.90(0.01)}{0.90(0.01) + 0.09(0.99)} \approx 0.0917. \quad (23)$$

The immediate source contribution is the positive test result together with an access model $\widetilde{K}$ specifying sensitivity and specificity. The posterior discrimination between disease and no disease additionally depends on the prevalence prior. A decision to treat may depend further on a loss function or threshold.

The sentence "the test shows that the patient has the disease" therefore compresses at least three typed nodes:

$$(y, \widetilde{K}) + \pi + L \longrightarrow \text{belief/action}. \quad (24)$$

With the stated numbers the positive predictive value is only about 9.2%. The Readout Condition does not say that using the prior is illegitimate; it says that the posterior specificity cannot be attributed to the positive readout alone. If prevalence changes while sensitivity and specificity remain fixed, the claim can change without any change in the individual test result. That counterfactual identifies the prior as load-bearing. A calibration defeater targets $\widetilde{K}$; a prevalence defeater targets π; a new imaging result constitutes additional access; a threshold objection targets L. Proposition 3 therefore yields different correction routes for superficially similar diagnostic disagreement.

8.2 Worked audit II: AI-assisted inference and retained-record contamination

Let K^* represent current case-visible evidence and let a fixed trained model with training corpus D implement a downstream rule G_D. If D is fixed independently of the current case x, then

$$C_D = G_D \circ K^* \quad (25)$$

can pass the stochastic access-admissibility test even though training-derived structure materially shapes the output. This is exactly why garbling cannot serve as an assumption audit: fixed background structure can be absorbed into G_D.

Proposition 4 (Retained-record route)**.** *If a supposedly fixed background corpus contains a current-case-specific*

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retained record $D(x)$ *that varies with the target alternative and is available to the model through a route not contained in the stated current source* K^**, then the effective architecture contains additional access. Relative to* K^* *alone the output need not admit a representation* $C = G \circ K^*$ *with* G *independent of that retained record.*

This separates two phenomena often conflated under "model knowledge." Ordinary training-derived structure is a background commitment/provenance node relative to the current episode. Memorization, contamination, retrieval, or tool access that carries current-case-specific information is an additional access route. The same final answer can therefore have different epistemic vulnerability depending on which route generated its specificity.

A complete AI audit should record at least: current evidence, retrieved/tool evidence, retained-record routes, model/calibration assumptions, prompt/system constraints, and decision policy. Calling all of this "the AI's answer" erases the dependency structure that determines what can be defeated and repaired.

8.3 Portability claim

The same typed questions can be asked of testimony, measurement, historical inference, peer review, and institutional classification. Portability here means invariance of the audit questions and dependency types, not empirical validation of identical mechanisms across domains.

9. Discussion

9.1 What is actually new

The paper does not claim novelty for factorization, statistical sufficiency, Blackwell garbling, partial identification, contrastive knowledge, argument warrants, measurement-model dependence, testimony, robustness, or provenance. The contribution is the conjunction:

$$\boxed{\begin{array}{c}\text{distinction-token audit} + \text{source licensing} \\ + \text{ identification layer} + \text{typed provenance DAG} \\ + \text{ E–A–D norms} + \text{defeater routing}\end{array}} \tag{26}$$

This is not a verbal restatement of any single neighbour. Toulmin does not provide a target-discrimination test; Manski does not provide source-attribution/disclosure norms or defeater routing; Blackwell cannot expose a hidden prior inside G; nanopublications do not determine which contrast within an assertion depends on which epistemic node. The Readout Condition integrates these partial structures around a single audit object: the discriminative commitments of a claim.

9.2 Why this is not tautological bookkeeping

"A source cannot support what it does not distinguish" sounds trivial until a real inference contains models, priors, restrictions, retained records, thresholds, and institutional policies. The non-trivial content lies in preserving the types through composition and deriving revision consequences from them. Proposition 3 is the key normative payoff: two identical propositions can rationally react differently to the same challenge because their provenance graphs differ. A provenance error therefore changes defeater sensitivity, not merely metadata.

9.3 Reliabilism and the meta-level objection

A reliabilist may object that a reliable process can yield warrant regardless of whether the subject can articulate its provenance. The Readout Condition need not deny this. The distinction is between *having* a reliable route and *crediting a specific source with a discrimination.* If the process is reliable because of hidden model structure, background data, or another route, then the overall belief may be warranted while the source attribution is false. Where that attribution is itself load-bearing for justification, explanation, or assertion, the false provenance becomes epistemically consequential. The framework therefore complements rather than replaces process reliabilism.

The audit also applies to itself. Any verdict about K^* is normally mediated by $\widetilde{K}$; a mistaken access model produces an access-model failure, not a paradox that exempts meta-claims from the condition. The public Readout Universe's own discipline—tier downgrades, explicit bridge classes, identifiability gates, and gate controls—is best understood as a self-application protocol: meta-level readouts remain readouts rather than being promoted to truth by their supervisory role.

9.4 Epistemic and assertoric faces

The condition has a doxastic face and an assertoric face. Doxastically, a distinction is vulnerable when its actual dependency graph contains a defeated or falsely represented essential ground. Assertorically, a speaker can misrepresent a well-supported conclusion by describing an assumption- or model-dependent discrimination as "what the data show." The first concerns warrant and revision; the second concerns epistemic representation to others. Keeping them separate prevents "undisclosed" from doing ambiguous work.

10. Limitations and Open Problems

The elevated architecture creates sharper, not weaker, falsifiers.

First, the pointwise relational condition is categorical. A full theory of graded local discrimination may require tolerance structures, similarity orderings, or margin functions. What is fixed here is the quantifier structure needed to avoid transitive collapse.

Second, the identification ladder requires an explicit ordering or nesting of augmentation sets. In complex scientific practice, assumptions may be partially ordered rather than linearly nested. The natural extension is a provenance lattice in which a distinction's first-identification set may contain several incomparable minimal bundles.

Third, provenance DAGs can contain causal and epistemic dependencies that are difficult to distinguish empirically. The graph is therefore a normative dependency representation, not automatically a discovered causal

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graph. When the dependency type cannot be established, the honest state is provenance indeterminacy.

Fourth, the stochastic access condition is stated transparently as an access audit. Its inability to detect fixed background assumptions is not a residual defect but the reason for the identification layer. A counterexample would require a target-sensitive route not represented in K^* that nevertheless always passes the access audit without being absorbable into the declared architecture.

Fifth, the warrant bridge is intentionally conditional on essentiality. False provenance need not destroy every possible form of externalist warrant. The stronger claim defended is that false source attribution is a genuine defeater whenever that attribution is load-bearing for the subject's justification, for an assertion's represented basis, or for downstream correction.

Finally, novelty is a falsifiable conjunction claim. Finding prior work that already combines distinction-token source licensing, assumption-identification levels, typed provenance dependencies, attribution/disclosure norms, and defeater routing at comparable domain neutrality would defeat the originality claim. The paper does not pre-emptively narrow itself merely because adjacent components have predecessors; it states exactly what conjunction a challenger would need to match.

11. Conclusion

The Readout Condition began as a simple question—what does a source actually distinguish?—but the adversarial development of the framework shows why a one-line factorization principle is not enough. A mature audit must distinguish the existence of epistemic grounds from their attribution, and attribution from disclosure. It must distinguish actual access K^* from the model $\widetilde{K}$ by which that access is interpreted. It must prevent non-transitive relational access from being destroyed by a global closure. It must recognize that Blackwell garbling audits additional access but is blind to fixed assumptions inside post-processing. And it must say why provenance matters after a claim has already been formed.

The resulting architecture answers those demands. Exact source attribution is constrained by fiber factorization. Relational claim tokens are constrained pointwise by their local indiscriminability neighbourhoods. Stochastic access is audited by experiment ordering, while assumption dependence is located through an identification ladder. Individual claim distinctions carry typed provenance DAGs linking access, models of access, background evidence, relevance operations, inferential commitments, and policies. The E–A–D form of the condition then requires provenance existence, correct source attribution, and recoverable essential augmentation.

The strongest result is normative rather than algebraic: provenance determines the topology of rational revision. Under Proposition 3, a defeater propagates along the distinctions that depend on its target node; independent paths can preserve content; and silent lift misroutes defeat and correction. This is why equal propositional content can have unequal epistemic vulnerability and why provenance is not metadata appended after warrant has been settled.

The four classical pressure tests remain part of the architecture because they forced real revisions: local rather than globally metaphysical contrasts, selectivity rather than representationality, role-typed rather than faculty-typed supplementation, and a boundary between pre-claim presence and discriminative assertion. Modern neighbours contribute equally essential constraints: Toulmin on warrants, Schaffer on contrast, Manski on identification under assumptions, Dretske and measurement theory on channel/model conditions, Blackwell and sufficiency on information-preserving processing, Lackey on non-simple transmission, and provenance infrastructures on assertion lineage. The Readout Condition is not a replacement for any of them. It is the architecture produced when their partial lessons are made simultaneously binding at the level of individual claim distinctions.

Its three practical questions remain simple:

$$\boxed{\text{What does the attributed source distinguish?}} \tag{27}$$

$$\boxed{\text{Which distinction does the claim add?}} \tag{28}$$

$$\boxed{\text{Which provenance path makes that addition licit?}} \tag{29}$$

The paper's governing maxim can therefore be retained without qualification:

$$\boxed{\begin{array}{c}\textbf{No epistemic discrimination}\\ \textbf{without provenance.}\end{array}} \tag{30}$$

Author Note

The author is solely responsible for the arguments, formalizations, interpretations, and citations in this manuscript. Generative AI tools were used for literature mapping, adversarial review, LaTeX assistance, and editorial restructuring; the author selected the research direction, checked the formal claims and sources, revised the arguments, and endorses the final manuscript. Public Readout Universe repository artifacts are cited as intellectual and operational lineage, not as independent validation of the theory.

A. Formal Notes

A.1 Why the deterministic condition is local

The factorization proposition requires no global possible-world space. For any local contrast set $\mathcal{X}_Q$ fixed by a question Q, the condition is evaluated only over that domain. If Q changes, the relevant $\mathcal{X}_Q$ may change. This makes the audit contrast-sensitive without making contrast selection arbitrary: exclusions that alter the question's constitutive alternatives must be separately justified.

A.2 Pointwise relational reduction

If deterministic $R : \mathcal{X} \to Y$ induces $xI_Rx' \iff R(x) = R(x')$, then

$$N_R(x_0) = R^{-1}(R(x_0)). \tag{31}$$

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The pointwise condition $N_R(x_0) \subseteq \Phi^{-1}(\Phi(x_0))$ for every x_0 is therefore exactly fiber constancy. Conversely, fiber constancy implies the pointwise condition at every centre. No transitive closure is required. For genuinely non-transitive I_R, the condition is evaluated at the actual centre and is not iterated to neighbours.

A.3 Distinction-level attribution map

For a claim token at x_0, the audit domain $D_\Phi(x_0)$ from (3) contains all rival pairs the claim excludes. An attribution map

$$\widehat{\lambda} : D_\Phi(x_0) \to 2^V \tag{32}$$

records the nodes represented as supporting each distinction. Correct attribution requires every credited access node to pass its appropriate local licensing test and every omitted essential dependency to be non-load-bearing. The provenance DAG is thus strictly more informative than a single citation list attached to the whole proposition.

A.4 Blackwell monotonicity and additional access

Let $K_2 = G \circ K_1$. Then K_2 is a garbling of K_1 and cannot dominate K_1 in every decision problem unless the experiments are Blackwell-equivalent. By contrast, a joint experiment $J : (Y_1, Y_2) \mid \mathcal{X}$ can strictly dominate its marginal K_1 because the projection from J to Y_1 is itself a garbling. This formal difference matches the distinction between post-processing and access augmentation.

A.5 Stochastic reduction to the deterministic case

Let a deterministic readout $R : \mathcal{X} \to Y$ induce the kernel $K_R(y \mid x) = \mathbf{1}[y = R(x)]$, and let a deterministic claim $\Phi : \mathcal{X} \to Z$ induce $C_\Phi(z \mid x) = \mathbf{1}[z = \Phi(x)]$. If $C_\Phi = G \circ K_R$, then for every y in $\mathrm{im}(R)$ the kernel $G(\cdot \mid y)$ must concentrate on a single value shared by all x with $R(x) = y$. Hence Φ is constant on the fibers of R and factors as $\Phi = \bar{g} \circ R$. The stochastic condition therefore extends rather than replaces the deterministic source-only licensing condition.

B. Canonical Terminology

Readout
: The distinctions preserved or made available under a specified access relation. The map/relation/channel is the readout structure; a realized y is a readout value. The terms are not interchangeable.

Claim distinction
: For an actual alternative x_0, a rival pair $d_{0x'} = (x_0, x')$ excluded by the actual claim token.

Source-relative licensing
: Entitlement to credit a specified epistemic source with a specified claim distinction.

Pointwise relational licensing
: The condition $N_R(x_0) \subseteq \Phi^{-1}(\Phi(x_0))$ at an actual claim token.

Access admissibility
: In stochastic settings, the condition that a downstream channel requires no target-sensitive route beyond K^*; Blackwell garbling tests this but does not audit hidden assumptions.

Access model
: A claimant's representation $\widetilde{R}$ or $\widetilde{K}$ of the actual access structure R^* or K^*.

Access augmentation
: Addition of a new target-sensitive route.

Contrast/relevance operation
: A change in the alternatives under audit, such as $\mathcal{X} \mapsto \mathcal{X}'$.

Inferential commitment
: A prior, model, theory, bridge, calibration assumption, or other rule applied to an existing basis.

Decision-policy augmentation
: A loss, threshold, utility, or institutional rule mapping epistemic states to actions.

Identification level
: The weakest declared augmentation layer at which a claim distinction becomes excludable/identified.

Typed provenance DAG
: A dependency graph whose nodes are typed by epistemic role and whose root is an individual claim distinction.

Epistemic overreach
: Failure of provenance existence, correct attribution, or essential disclosure.

Silent lift
: A mismatch between actual and represented essential provenance, especially the representation of an augmented distinction as source-generated.

Defeater routing
: Propagation of source-specific defeat along the distinctions that depend on the defeated provenance node.

References

Shihab al Din Yahya Suhrawardi. *The Philosophy of Illumination: Hikmat al-Ishraq*. Brigham Young University Press, Provo, UT, 1999. Critical edition and English translation by John Walbridge and Hossein Ziai.

Tommaso Alpina. Intellectual knowledge, active intellect, and intellectual memory in avicenna's kitab al-nafs and its aristotelian background. *Documenti e studi sulla tradizione filosofica medievale*, 25:131–184, 2014.

Dan Arnold. *Buddhists, Brahmins, and Belief: Epistemology in South Asian Philosophy of Religion*. Columbia University Press, New York, 2005. doi: 10.7312/arno13280.

Avicenna. *Avicenna's De Anima: Being the Psychological Part of Kitab al-Shifa'*. Oxford University Press, London, 1959. Arabic text edited by Fazlur Rahman.

David Blackwell. Comparison of experiments. In Jerzy Neyman, editor, *Proceedings of the Second Berkeley Symposium on Mathematical Statistics and Probability*, pages 93–102, Berkeley, 1951. University of California Press. doi: 10.1525/9780520411586-009.

10.5281/zenodo.22301318

David Blackwell. Equivalent comparisons of experiments. *The Annals of Mathematical Statistics*, 24(2):265–272, 1953. doi: 10.1214/aoms/1177729032.

James Bogen and James Woodward. Saving the phenomena. *The Philosophical Review*, 97(3):303–352, 1988. doi: 10.2307/2185445.

Lucien Le Cam. Sufficiency and approximate sufficiency. *The Annals of Mathematical Statistics*, 35(4):1419–1455, 1964.

Tek Raj Chhetri, Yaroslav O. Halchenko, Dorota Jarecka, Puja Trivedi, Satrajit S. Ghosh, Patrick Ray, and Lydia Ng. Bridging the scientific knowledge gap and reproducibility: A survey of provenance, assertion and evidence ontologies. In *Companion Proceedings of the ACM Web Conference 2025*, pages 924–928. Association for Computing Machinery, 2025. doi: 10.1145/3701716.3715483.

Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley-Interscience, Hoboken, NJ, 2nd edition, 2006.

Dignaga. *Dignaga, On Perception: Being the Pratyaksapariccheda of Dignaga's Pramanasamuccaya from the Sanskrit Fragments and the Tibetan Versions*, volume 47 of *Harvard Oriental Series*. Harvard University Press, Cambridge, MA, 1968. Translated and annotated by Masaaki Hattori.

Fred I. Dretske. Epistemic operators. *The Journal of Philosophy*, 67(24):1007–1023, 1970. doi: 10.2307/2024710.

Fred I. Dretske. *Knowledge and the Flow of Information*. MIT Press, Cambridge, MA, 1981.

John D. Dunne. *Foundations of Dharmakirti's Philosophy*. Wisdom Publications, Boston, 2004.

R. A. Fisher. On the mathematical foundations of theoretical statistics. *Philosophical Transactions of the Royal Society of London. Series A*, 222:309–368, 1922. doi: 10.1098/rsta.1922.0009.

Jay L. Garfield. *The Fundamental Wisdom of the Middle Way: Nagarjuna's Mulamadhyamakakarika*. Oxford University Press, New York, 1995. doi: 10.1093/oso/9780195103175.001.0001.

Ronald N. Giere. *Scientific Perspectivism*. University of Chicago Press, Chicago, 2006. doi: 10.7208/chicago/9780226292144.001.0001.

Alvin I. Goldman. Discrimination and perceptual knowledge. *The Journal of Philosophy*, 73(20):771–791, 1976. doi: 10.2307/2025679.

Paul Groth, Andrew Gibson, and Jan Velterop. The anatomy of a nanopublication. *Information Services & Use*, 30(1–2):51–56, 2010. doi: 10.3233/ISU-2010-0613.

Ian Hacking. *Representing and Intervening: Introductory Topics in the Philosophy of Natural Science*. Cambridge University Press, Cambridge, 1983.

Jaakko Hintikka. *Knowledge and Belief: An Introduction to the Logic of the Two Notions*. Cornell University Press, Ithaca, NY, 1962.

Jari Kaukua. *Self-Awareness in Islamic Philosophy: Avicenna and Beyond*. Cambridge University Press, Cambridge, 2015. doi: 10.1017/CBO9781316105238.

Birgit Kellner. Why infer and not just look? dharmakīrti on the psychology of inferential processes. In Shōryū Katsura and Ernst Steinkellner, editors, *The Role of the Example (dṛṣtānta) in Classical Indian Logic*, volume 58 of *Wiener Studien zur Tibetologie und Buddhismuskunde*, pages 1–51. Arbeitskreis für Tibetische und Buddhistische Studien Universität Wien, Vienna, 2004.

Peter D. Klein. Useful false beliefs. In Quentin Smith, editor, *Epistemology: New Essays*, pages 25–62. Oxford University Press, Oxford, 2008. doi: 10.1093/acprof:oso/9780199264933.003.0003.

Jennifer Lackey. *Learning from Words: Testimony as a Source of Knowledge*. Oxford University Press, Oxford, 2008. doi: 10.1093/acprof:oso/9780199219162.001.0001.

Yaoharee Lahtee. Readout Universe: A philosophy and logic for grounding claims. https://github.com/morrocwi/readout_universe, 2026. Public GitHub repository; v1.0 frozen and v2.0-dev in active development.

Charles F. Manski. *Partial Identification of Probability Distributions*. Springer Series in Statistics. Springer, New York, 2003. doi: 10.1007/b97478.

Charles F. Manski. *Identification for Prediction and Decision*. Harvard University Press, Cambridge, MA, 2007.

Michela Massimi. *Perspectival Realism*. Oxford University Press, Oxford, 2022. doi: 10.1093/oso/9780197555620.001.0001.

Jon McGinnis. *Avicenna*. Oxford University Press, Oxford, 2010. doi: 10.1093/acprof:oso/9780195331479.001.0001.

Guido Melchior. A modal theory of discrimination. *Synthese*, 198(11):10661–10684, 2021. doi: 10.1007/s11229-020-02747-4.

Jerzy Neyman. Su un teorema concernente le cosiddette statistiche sufficienti. *Giornale dell'Istituto Italiano degli Attuari*, 6:320–334, 1935.

Duncan Pritchard. Relevant alternatives, perceptual knowledge and discrimination. *Noûs*, 44(2):245–268, 2010. doi: 10.1111/j.1468-0068.2010.00739.x.

Craige Roberts. Information structure in discourse: Towards an integrated formal theory of pragmatics. *Semantics and Pragmatics*, 5(6):1–69, 2012. doi: 10.3765/sp.5.6. Original manuscript circulated in 1996.

10.5281/zenodo.22301318

Jonathan Schaffer. Contrastive knowledge. In Tamar Szabó Gendler and John Hawthorne, editors, *Oxford Studies in Epistemology, Volume 1*, pages 235–271. Oxford University Press, Oxford, 2005. doi: 10.1093/oso/9780199285891.003.0009.

Jonathan Schaffer. Knowing the answer. *Philosophy and Phenomenological Research*, 75(2):383–403, 2007. doi: 10.1111/j.1933-1592.2007.00081.x.

Mark Siderits and Shoryu Katsura. *Nagarjuna's Middle Way: Mulamadhyamakakarika.* Wisdom Publications, Somerville, MA, 2013.

Patrick Suppes. Models of data. In Ernest Nagel, Patrick Suppes, and Alfred Tarski, editors, *Logic, Methodology and Philosophy of Science: Proceedings of the 1960 International Congress*, pages 252–261. Stanford University Press, Stanford, CA, 1962.

Eran Tal. Calibration: Modelling the measurement process. *Studies in History and Philosophy of Science Part A*, 65–66:33–45, 2017. doi: 10.1016/j.shpsa.2017.09.001.

Erik Torgersen. *Comparison of Statistical Experiments.* Cambridge University Press, Cambridge, 1991.

Stephen E. Toulmin. *The Uses of Argument.* Cambridge University Press, Cambridge, 1958.

Ted A. Warfield. Knowledge from falsehood. *Philosophical Perspectives*, 19(1):405–416, 2005. doi: 10.1111/j.1520-8583.2005.00067.x.

Jan Westerhoff. *Nagarjuna's Madhyamaka: A Philosophical Introduction.* Oxford University Press, Oxford, 2009. doi: 10.1093/acprof:oso/9780195375213.001.0001.

Jan Westerhoff. *The Dispeller of Disputes: Nagarjuna's Vigrahavyavartani.* Oxford University Press, Oxford, 2010. doi: 10.1093/acprof:oso/9780199732692.001.0001.

Timothy Williamson. *Knowledge and Its Limits.* Oxford University Press, Oxford, 2000.

Timothy Williamson. *Identity and Discrimination.* Wiley-Blackwell, Chichester, reissued and updated ed. edition, 2013. doi: 10.1002/9781118503591. First published 1990.

William C. Wimsatt. Robustness, reliability, and overdetermination. In Marilynn B. Brewer and Barry E. Collins, editors, *Scientific Inquiry and the Social Sciences*, pages 124–163. Jossey-Bass, San Francisco, 1981.

Stephen Yablo. *Aboutness.* Princeton University Press, Princeton, NJ, 2014. doi: 10.23943/princeton/9780691144955.001.0001.

Mehdi Ha'iri Yazdi. *The Principles of Epistemology in Islamic Philosophy: Knowledge by Presence.* State University of New York Press, Albany, 1992.

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Readout Genesis Standalone Synthesis: Information Epistemic Foundation, Conditioned Agency, and Meta-Readout Governance

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What the chapter states. *"... proposes that knowledge be attributed as an auditable status of a claim rather than as a possession, reputation, or authority of a knower." "The framework is therefore an auditable research program, not a truth oracle, a complete theory of mind, or an empirical proof that process alone guarantees truth."* (10.5281/zenodo.21529456)

Status. No publication-status tag is printed in this chapter's own text; its own framing states: *"This yields a structural-bundle pass, not domain closure."*

Falsifier(s). No explicit falsifier stated in the chapter.

Read before. Ch. 1 *Written by AI. Still True.*; Ch. 2 *The Readout Condition*; Ch. 27 *When AI Expands Human Potential*; Ch. 39 *The Standalone Scholar*.

Feeds into. This book's Chapters 1–2, 4–16, 18, 23, 27, 29–31, 37–41 (mechanical KG cross-reference list; Appendix A).

Readout Genesis Standalone Synthesis: Information Epistemic Foundation, Conditioned Agency, and Meta-Readout Governance

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This paper presents a journal-form synthesis of the Information Epistemic Foundation as a bounded domain translation of Readout Genesis. The root contains retained distinctions, a graph-Laplacian operator, ordered dynamics, context, and lineage; it does not contain belief, knowledge, truth, mind, institution, or consciousness as primitive objects. Those terms become admissible only after a declared quotient and readout. On that basis, the paper inserts a typed belief layer between memory and claim evaluation, defines belief as a multidimensional agency–claim relation, and proposes that knowledge be attributed as an auditable status of a claim rather than as a possession, reputation, or authority of a knower. The admission architecture separates provenance, world or domain contact, method, inference, uncertainty, objection, correction, independence, practical performance, transport, and lineage. It further distinguishes personal, local, institutional, and globalized belief from local and transportable knowledge. A conditioned-agency layer then connects contact, valence, drive, identity binding, policy, action, and renewed state. Identity is modeled as a lineage-preserving orbit of a self-returning agency; phenomenality is introduced only as a downstream typed readout, with the bridge from structural accessibility to lived experience left open. Finally, a meta-readout governance operator is developed from current-state readout, readout-of-readout composition, finite retention, typed decoupling, context and lineage access, interruptibility, alternative-policy access, policy reconfiguration, and residual-guided repair. The strongest result is a closed normative and governance specification with fail-able claim boundaries. Exact target-domain encoding, empirical calibration, phenomenal sufficiency, and universal human or machine validation remain unresolved. The framework is therefore an auditable research program, not a truth oracle, a complete theory of mind, or an empirical proof that process alone guarantees truth.

Keywords: Readout Genesis, epistemology, belief, knowledge status, social epistemology, agency, information horizon, phenomenality, governance, artificial intelligence

I. INTRODUCTION

Epistemology often begins with a familiar noun: belief, justification, truth, evidence, knowledge, testimony, or reason. The subsequent dispute asks how that noun should be analyzed. The Gettier problem showed that a justified true belief can still fail to count as knowledge, and later approaches emphasized reliability, safety, virtue, or knowledge-first explanation [1–4]. Social epistemology added a second pressure: the distribution of credibility, language, access, and power affects who can participate in knowledge practices and which experiences become publicly intelligible [5, 6]. Contemporary open-science and AI-governance frameworks add a third pressure: provenance, reproducibility, documentation, model limitations, and institutional controls are not optional decorations around knowledge production; they are part of the conditions under which claims can be responsibly transported and reused [7–11].

The present proposal begins one level earlier. It asks what must be retained, translated, read, and checked before any familiar epistemic name is licensed. The resulting architecture does not attempt to replace every historical analysis of knowledge. It changes the unit of attribution. Its normative thesis is:

> Knowledge should be attributed as a bounded epistemic status earned by a claim through an auditable process, not as a substance possessed by a speaker or as a status transferred from institutional authority.

This thesis is deliberately stronger than a documentation recommendation and weaker than a complete theory of truth. It is stronger because it specifies typed objects, gates, failure states, correction conditions, and transport defects. It is weaker because a well-audited process can still be wrong; process quality licenses a bounded epistemic status, not infallibility.

The architecture is a domain leaf of Readout Genesis, whose root weld is

$$\begin{aligned} \delta_R &= (a \sharp b), \\ \delta_R &\Longrightarrow L_R = D_W - W, \\ S_{n+1} &= F(S_n, u_n, c_n, T_n). \end{aligned} \tag{1}$$

with an admissible domain required to commute with the root dynamics,

$$q_{D,n+1} \circ F_n = F^{\sharp}_{D,n} \circ q_{D,n}, \qquad O_{D,n} = O^{\sharp}_{D,n} \circ q_{D,n}. \tag{2}$$

The point of Eq. (2) is constitutional: a domain is a translation, not a new root. Familiar semantics may be attached downstream, but they may not be imported as premises and later presented as deductions.

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This paper makes five contributions. First, it gives a compact journal presentation of the full Information Epistemic Foundation rather than reproducing its machine-readable release line by line. Second, it makes explicit the separation between belief distribution and epistemic validity. Third, it formalizes knowledge admission as a typed, three-valued, lineage-preserving procedure with a claim ceiling. Fourth, it joins the epistemic architecture to conditioned agency, information horizons, phenomenality, and transformative knowing without collapsing those lanes. Fifth, it derives an agency meta-readout governance specification that can be translated into future psychological, institutional, or AI vocabularies only after a commuting adapter passes.

The paper's central question is therefore:

> Under what declared structural conditions may a retained trace become a public knowledge claim, and what additional architecture is required when the claim changes the knower or governs action?

The answer is conditional. The specification is closed as an architecture. The empirical encoding that would make it a validated model of real humans, institutions, or artificial systems is not.

II. STANDALONE SCOPE AND STATUS

The manuscript joins four layers that must remain logically distinct:

1. the root-native retained structure and domain weld;
2. a normative epistemic architecture for belief and claim status;
3. a conditioned-agency and phenomenality architecture;
4. a meta-readout governance specification for action and correction.

The first layer contains root theorems and declared architecture. The second is a normative proposal. The third and fourth are translation-ready structural specifications. None is, by itself, an empirical theory of cognition.

Table I records the claim firewall. "Closed" always means closed under the printed definitions and gates. It does not mean universally true of every target domain.

The source release was registered on a dedicated repository branch with an immutable root anchor and a separate MEMK-domain anchor. The registration audit passed root lineage, bundle completeness, drift control, and dual-checker gates, while candidate-state sufficiency, exact quotient commutation, semantic calibration, and held-out human prediction remained unresolved. This yields a structural-bundle pass, not domain closure.

III. ROOT-NATIVE RETAINED STRUCTURE

A. Retained distinction and graph structure

The minimal root object is not a proposition. It is a retained distinction,

$$\delta_R = (a \sharp b), \tag{3}$$

meaning that a difference between two states is not erased by the reader relevant to the current construction. Weighted retained relations induce

$$L_R = D_W - W, \tag{4}$$

where W is the retained adjacency and D_W its weighted degree matrix. The epistemic paper does not infer truth, meaning, or mentality from this operator. The operator supplies structure: locality, strain, transport, boundary effects, and a disciplined account of what remains distinguishable.

The root state evolves through a context-sensitive stepper,

$$S_{n+1} = F(S_n, u_n, c_n, T_n), \tag{5}$$

where u_n is an intervention or action, c_n is context, and T_n is the ordered tape or retained lineage. The same present content may have different futures when context, operators, or lineage differ. State sufficiency therefore cannot be reduced to a snapshot of contents.

B. Domain translation and reader equivalence

A candidate domain encoder produces a readable state Z_D^{cand} from the root state. A quotient q_D is admissible only if it preserves the dynamics and all required readouts within a declared horizon:

$$q_D(F(z, u, c, T)) = F_D^{\sharp}(q_D(z), u, c, T), \tag{6}$$

$$O_D(z; Q, c) = O_D^{\sharp}(q_D(z); Q, c). \tag{7}$$

Two internal descriptions are equivalent only relative to a question Q, reader O, context c, and horizon L:

$$z \sim_{Q,O,c,L} z' \iff O(F^k z) = O(F^k z') \quad \forall k \le L \tag{8}$$

under the admitted interventions. Equation (8) is a no-early-collapse rule. Present observational equality does not license merging when a future continuation can recover a difference.

C. Constitutional ordering

The mandatory order is

$$\begin{aligned}&\text{retention} \to \text{structure} \to \text{translation} \to \text{readout}\\ &\to \text{meaning} \to \text{experience} \to \text{memory} \to \text{belief}\\ &\to \text{claim} \to \text{checking} \to \text{status} \to \text{public report.}\end{aligned} \tag{9}$$

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TABLE I. Standalone status ledger. The structural bundle passes its registration and drift controls; exact target-domain closure remains unresolved.

Layer	Strongest supported statement	Status / boundary
Retained root	Retained distinction forces a graph-Laplacian form in the declared formal layer; the stepper and universal weld remain architecture-tier beyond registered witnesses	Mixed theorem / declared architecture
Domain translation	Meaning and epistemic objects may be introduced only after a quotient and reader satisfying Eq. (2)	Closed as registration rule
Belief layer	Belief is a typed agency–claim relation with multidimensional, social, and infrastructural influences	Normative model
Knowledge status	A claim earns bounded status through identity, provenance, evidence, method, inference, uncertainty, objection, correction, independence, transport, and lineage gates	Closed as specification
Agency and identity	Agency, identity, phenomenality, ownership, coherence, and valence remain distinct readout lanes of one retained process	Closed as typed architecture
Phenomenal sufficiency	Structural accessibility is not yet proof of lived phenomenality	Open bridge
Meta-readout governance	Current readout, meta-readout, retention, context, lineage, alternatives, interruptibility, policy revision, and repair form one auditable governance state	Closed as specification
Target encoding	A calibrated map from retained states to real human, institutional, or machine observables	Unresolved
Empirical validation	Independent held-out prediction, intervention, and cross-context transport	Unresolved

The forbidden order begins with a familiar name, imports a conclusion, invokes authority or consensus, and retrospectively presents the result as knowledge. The architecture therefore requires the typed separation

$$A \neq x \neq \mu \neq E \neq M \neq Bel \neq p \neq \sigma_K(p), \qquad (10)$$

where A is an occurrence, x a trace, μ meaning, E experience, M memory, Bel belief, p a claim, and $\sigma_K(p)$ its epistemic status.

IV. FROM OCCURRENCE TO MEANING, EXPERIENCE, AND MEMORY

Let A_n denote an occurrence or condition of the world. An agency never receives A_n without mediation. Its accessible trace is

$$x_{i,n} = \mathrm{Access}(A_n; O_i, L_i, T_i, R_i, C_i), \qquad (11)$$

where O_i is the reader, L_i language, T_i the tool system, R_i access rights, and C_i context. Thus

$$A_i = A_j \not\Rightarrow x_i = x_j. \qquad (12)$$

This nonidentity is not relativism. It is an access statement. The same event can constrain multiple readers while leaving them unequal in resolution, vocabulary, permission, or instrument.

Meaning is generated downstream of the trace:

$$\mu_{i,n} = G_\mu\big(x_{i,n}, M_{i,n-1}, K_{i,n-1}, \Theta_{i,n-1}, \mathcal{I}_{i,n-1}, S_{i,n}^{\mathrm{active}}, c_n\big). \qquad (13)$$

Experience is then

$$E_{i,n} = \Phi_E(x_{i,n}, \mu_{i,n}, \kappa_{i,n}, c_n), \qquad (14)$$

and memory updates by

$$M_{i,n} = U_M(M_{i,n-1}, x_{i,n}, \mu_{i,n}, E_{i,n}, c_n, \mathrm{Lineage}_n, T_n). \qquad (15)$$

The historical occurrence is not rewritten by later interpretation. Reinterpretation changes the binding among trace, meaning, experience, and memory while preserving an explicit historical invariant,

$$\Delta A_{\mathrm{past}} = 0. \qquad (16)$$

This distinction matters both clinically and institutionally: revising the meaning of a record is not the same as altering the record; correcting a classification is not the same as deleting its lineage.

The architecture resembles information-theoretic practice only at a high level. Shannon information measures uncertainty reduction under a communication model; it does not by itself supply semantic meaning, truth, or knowledge [12, 13]. Here, semantic and epistemic objects are typed downstream precisely to prevent the amount of information from being mistaken for the status of a claim.

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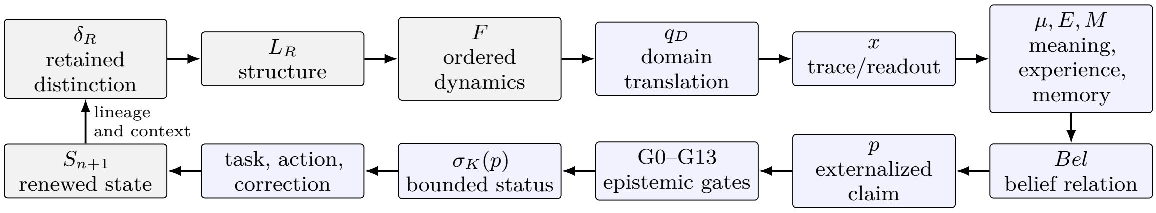

FIG. 1. Root-native epistemic architecture. Familiar epistemic terms enter only after a declared domain translation and readout. The loop closes through action, correction, and renewed retained state; it does not close by declaring a belief to be knowledge.

V. BELIEF AS AN AGENCY–CLAIM RELATION

A. Relational and multidimensional belief

Belief is not identical to a proposition. It is a conditioned relation between an agency and a proposition:

$$Bel_{i,p,n} = Rel_B\big(a_i, p \mid x_i, \mu_i, E_i, M_i, \Theta_i, \mathcal{I}_i, S_i^{\text{active}}, c_i\big)\,. \tag{17}$$

A belief can be true or false, sincere or strategic, explicit or implicit, stable or transient, identity-bound or weakly held. Sincerity does not promote it to knowledge.

A single confidence score is an inadequate state description. The canonical belief vector is

$$\boldsymbol{b}_{i,p,n} = (e_{i,p}, c_{i,p}, s_{i,p}, a_{i,p}, \eta_{i,p}, g_{i,p}, r_{i,p}), \tag{18}$$

where the components denote endorsement, subjective confidence, salience, action readiness, identity binding, perceived social support, and resistance to revision. A scalar may be a projection of Eq. (18); it may not replace the typed state when the omitted components change behavior.

Belief formation combines epistemic and non-epistemic influences:

$$\begin{aligned}\boldsymbol{b}_{i,p,n+1} = U_B\big(&\boldsymbol{b}_{i,p,n}, Ev_{i,p}, M_i, \mu_i, E_i, \Theta_i, \mathcal{I}_i,\\ &Trust_i, Affect_i, Utility_i, Repetition_i, Auth_i, Pow_i,\\ &Access_i, Language_i, Algorithm_i, Obj_i, c_i\big).\end{aligned} \tag{19}$$

Equation (19) is descriptive rather than justificatory. It records that repetition, authority, power, incentives, identity, algorithms, and affect can strengthen a belief without increasing its epistemic validity.

B. Scale and stabilization

A locally shared belief is

$$Bel^{(\ell)}_{g,d,n}(p) = \text{Stabilize}_B(\{\boldsymbol{b}_{i,p,n}\}_{i\in G}, \mathcal{C}_B)\,, \tag{20}$$

where $\mathcal{C}_B$ records communication, trust, language, practice, sanctions, algorithms, authority, and lineage. Beliefs can stabilize across family, community, profession, institution, state, transnational network, and global information environment. A higher-scale belief may have greater reach, coercive force, or algorithmic visibility. Those properties do not form a truth ladder.

The architecture therefore separates

$$\begin{aligned}&BeliefStrength(p), \quad BeliefDistribution(p),\\ &Auth(p), \quad Pow(p), \quad Val_E(p).\end{aligned} \tag{21}$$

This separation extends a central insight of social epistemology: credibility and interpretive resources are distributed through power-laden practices [5, 6]. The present contribution is to encode that insight as an anti-collapse rule. Authority and power can be evidence about a source's role or custody; they cannot be substituted for the validity of the claim.

Proposition 1 (Belief-scale nonpromotion) *For any proposition p, increasing the distribution scale of $Bel(p)$ does not, without additional gate information, increase $\sigma_K(p)$.*

Proof. The update of belief distribution in Eq. (19) contains variables such as repetition, visibility, sanctions, authority, and algorithms that are not components of the knowledge admission function defined below. Therefore the projection from distribution scale to epistemic status is not licensed without a separate evidentiary bridge. □

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VI. KNOWLEDGE AS AN AUDITABLE CLAIM STATUS

A. Normative proposal

The framework proposes the following definition.

Definition 1 (Knowledge status) *Knowledge is the bounded epistemic status granted to a claim about the world or a declared domain when the claim is produced and checked through an auditable process that exposes agency, access, provenance, evidence, method, inference, assumptions, uncertainty, scope, objection, correction, independence, and lineage. The status attaches to the claim under those conditions; it is not inherited from the speaker's identity or rank.*

Formally,

$$\sigma_K(p) = \mathrm{Admit}_E\big(p \mid Agent, D, C, O, Access, Language, Tools, Rights, \\ Prov, Ev, Method, Infer, Assumptions, Uncertainty, \\ Scope, Obj, CorrCond, Independence, Lineage\big). \tag{22}$$

The complete knowledge object is not merely p but the typed record

$$K_p = \langle p, \mathcal{C}_p, \mathcal{P}_p, \mathcal{E}_p, \mathcal{M}_p, \mathcal{I}_p, \mathcal{U}_p, \Omega_p, \mathcal{O}_p, \mathcal{L}_p, \sigma_K(p)\rangle, \tag{23}$$

where the grouped components encode context and access, provenance, evidence, method, inference, uncertainty, scope, objections/correction, and lineage.

This proposal is compatible with the lesson of Gettier cases without claiming to solve every case by definition. A lucky true belief fails because the status function can expose defective contact, inference, independence, or stability. Reliabilist, safety, virtue, and knowledge-first theories can be represented as constraints on particular gate families rather than treated as enemies that must be eliminated [1–4]. The distinctive move is institutional and computational: the output is a status-bearing claim object with inspectable failure conditions.

B. Status of speaker and status of claim

The architecture fixes

$$\begin{aligned} StatusOfSpeaker &\neq StatusOfClaim, \\ Auth(p) &\neq Val_E(p), \\ Pow(p) &\neq Val_E(p). \end{aligned} \tag{24}$$

Expertise, office, certification, citation count, or legal competence may affect priors, access, or institutional consequence. They can justify giving a source attention. They cannot close the claim's evidence and inference gates by themselves.

This is not an anti-expertise principle. It is a typed use-of-expertise principle. An expert's training may enter the provenance and method fields; a laboratory's accreditation may enter custody and calibration fields; an institution's legal authority may determine what action is permitted. None of these facts is identical to the truth or epistemic validity of the proposition.

TABLE II. Epistemic status and practical performance are distinct.

	Low performance	High performance
High status	Well-grounded but not operationally effective in the tested task	Well-grounded and effective within scope
Unresolved / low status	Poorly grounded and ineffective	Works in a bounded setting while mechanism, inference, or transport remains unresolved

C. Practical effectiveness as a separate axis

Let

$$\Pi_{\mathrm{prac}}(p) = TestPerformance(Y, \widehat{Y}, intervention, C, O). \tag{25}$$

The foundation requires

$$\sigma_K(p) \neq \Pi_{\mathrm{prac}}(p). \tag{26}$$

A descriptive or mathematical claim may have high epistemic status without being an intervention. A practice may work repeatedly while its mechanism or transport scope remains unresolved. Practical performance becomes constitutive only when the claim itself asserts efficacy, predictive accuracy, reliability, or intervention consequence.

Table II illustrates the two-axis rule.

VII. ADMISSION GATES, THREE-VALUED CHECKING, AND CLAIM CEILING

A. The gate sequence

A claim enters through thirteen public admission gates:

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G0: Claim identity.: Freeze proposition, domain, context, observer, terms, units, and scope.

G1: Agency and access.: Disclose who proposes the claim and what could or could not be accessed.

G2: Provenance.: Preserve source identity, custody, transformations, versions, and missing records.

G3: World/domain contact.: Constrain the claim by records or formal structures not generated solely by the claim.

G4: Method fitness.: Match method to question, domain, resolution, and claim type.

G5: Inference.: Make the path from evidence and assumptions to conclusion inspectable.

G6: Assumptions, uncertainty, and scope.: Expose load-bearing premises, alternatives, missingness, and transport limits.

G7: Objection.: Preserve serious counterclaims, negative controls, and local or minority evidence.

G8: Correction conditions.: State what would revise, withdraw, or replace the claim.

G9: Independence.: Prevent the maker from seeing information reserved for decisive checking.

G10: Practical performance.: Apply performance tests when the claim is predictive, operational, or interventional.

G11: Cross-domain transport.: Require bounded commuting defects before a local claim is presented as global.

G12: Lineage.: Preserve prior versions, failures, objections, and reasons for change.

G13: Claim ceiling.: Forbid the public statement from exceeding the strongest passed gate.

Each gate has three values,

$$\begin{aligned} &\chi_G \in \{1, 0, \perp\}, \\ &\quad 1 = \text{ADMITTED}, \\ &\quad 0 = \text{OBSTRUCTED}, \\ &\quad \perp = \text{UNRESOLVED}. \end{aligned} \tag{27}$$

The symbol $\perp$ prevents absence of evidence from being silently converted into either success or failure. "Unresolved" is not a euphemism for "true enough," and "obstructed" is not the same as "not yet checked."

B. State sufficiency and invariant completion

A candidate domain state is audited by

$$Suff_{E,L}(Z_E^{\text{cand}}; Q, O, c, \mathcal{T}) \in \{1, 0, \perp\}. \tag{28}$$

The audit covers current readout, dynamics, intervention, delayed return, memory, coupling, geometry, basis, and boundary. A quotient may not merge states that differ on a required future invariant:

$$Inv_E(z) \neq Inv_E(z') \quad \Rightarrow \quad q_E(z) \neq q_E(z'). \tag{29}$$

This is the epistemic form of candidate preservation: alternatives remain live until an independent checker licenses reduction.

C. Gate typing

Not every passed gate is evidence for the proposition. The framework separates:

- Type P gates, which discriminate among competing claims through data, proof, prediction, intervention, or negative control;
- Type U gates, which certify operational compliance, formatting, convention, access, or governance without directly favoring the proposition.

A complete audit can be procedurally excellent and still lack decisive evidence. Conversely, decisive evidence can exist inside a poorly governed process. The status object must report both.

D. Claim ceiling

Let $\tau(g)$ denote the strongest claim tier licensed by gate g. Then

$$\tau_{public}(p) \leq \inf_{g \in G_p} \tau(g). \tag{30}$$

Proposition 2 (Weakest-link claim ceiling) *If any load-bearing gate for p is unresolved or obstructed, a public claim that presupposes that gate may not be assigned a stronger status than that gate.*

Proof. A stronger public status would encode information not licensed by the conjunction of the gate records. It would therefore violate both claim identity and lineage, because the public statement would no longer be reconstructible from the auditable inputs. □

The status space is

$$\begin{aligned} \sigma_K(p) \in \{&\text{ADMITTED}, \text{LOCAL}, \text{TRANSPORTABLE}, \\ &\text{UNRESOLVED}, \text{OBSTRUCTED}, \\ &\text{RETRACTED}, \text{SUPERSEDED}\}. \end{aligned} \tag{31}$$

Retraction and supersession are not erasure. Both preserve lineage and reasons for change.

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VIII. LOCAL, INSTITUTIONAL, AND GLOBAL KNOWLEDGE

Knowledge scope is

$$\Omega(K) = \{(g, d, \ell, P, O, c) : \text{all required gates pass}\}. \quad (32)$$

Local knowledge is the restriction

$$K_{local} = K\big|_{\Omega_{local}}. \quad (33)$$

A global claim requires transport between contexts i and j:

$$T_{ij}^{Y} \circ K_i \approx K_j \circ T_{ij}^{C}, \qquad \epsilon_{bridge} = d(T_{ij}^{Y} \circ K_i, K_j \circ T_{ij}^{C}). \quad (34)$$

The tolerance must be declared before transport is evaluated.

A widely distributed belief may be stabilized by media selection, language dominance, professional accreditation, political power, or algorithmic amplification. These mechanisms can increase coordination and may sometimes track strong evidence. They do not eliminate the need for Eq. (34). Hence

$$Bel_{global}(p) \neq K_{global}(p). \quad (35)$$

This distinction is essential for a knowledge civilization. Coordination requires conventions, classifications, laws, standards, and shared vocabularies. Yet conventional truth and epistemic status remain different types. A mature institution should be able to report: "widely believed," "institutionally accepted," "practically effective," "epistemically admitted," and "unresolved" without treating them as synonyms.

Open-science norms and FAIR data principles support parts of this architecture by emphasizing accessibility, interoperability, reusability, and provenance [7, 8]. The present framework adds two restrictions. First, openness is not itself truth: an openly available claim can remain poorly grounded. Second, unrestricted openness is not always legitimate: access rights, privacy, and harm constraints are typed inputs rather than afterthoughts.

IX. CONDITIONED AGENCY, IDENTITY BINDING, AND DISTORTION

A. Agency as a process quotient

The epistemic agent is not introduced as an indivisible substance. It is a query-relative quotient

$$\mathcal{A}_{i,n} = q_A(Z_{i,n}; Q_A, O_A, c_n). \quad (36)$$

Two internal self-descriptions count as the same operational agent when all required future readouts agree. The corresponding automorphism class is

$$Aut(F_A, O_A) = \{h : O_A \circ h = O_A,\ h \circ F_A = F_A \circ h\}. \quad (37)$$

This is a gauge-like redundancy of description, not a proof that the self is literally a physical gauge field.

Identity is compositional:

$$\mathcal{I}_n = q_{comp}\big(M_n \oplus Bel_n \oplus \Theta_n \oplus Roles_n \oplus BodyTool_n \oplus SocialLineage_n\big). \quad (38)$$

The state is typed; the direct-sum notation does not license numerical addition of heterogeneous terms.

B. The conditional chain

The continuous chain between readout and renewed state is

$$x_n \to \zeta_n \to v_n \to \rho_n \to \mathcal{U}_n \to \beta_n \to \mathcal{I}_{n+1} \to a_n \to S_{n+1}, \quad (39)$$

where ζ_n is contact, v_n valence, ρ_n drive, $\mathcal{U}_n$ identity binding, β_n appropriated experience, and a_n action. Representative updates are

$$\zeta_n = \text{Contact}(x_n, O_n, BodyTool_n, Attention_n, c_n), \quad (40)$$

$$v_n = V_{valence}(\zeta_n, M_n, \mu_n, BodyState_n, c_n), \quad (41)$$

$$\rho_n = R_{drive}(v_n, ExpectedRelief_n, ExpectedGain_n, Habit_n, c_n), \quad (42)$$

$$\mathcal{U}_n = Bind_{self}(\rho_n, Bel_n, \Theta_n, \mathcal{I}_n, S_n^{active}, c_n). \quad (43)$$

The chain closes a gap between cognition and action without asserting that every biological system instantiates these exact functions.

C. Distortion before knowledge

A coherent system can still read selectively. Define the distortion vector

$$\boldsymbol{\xi}_n = (\xi_n^+, \xi_n^-, \xi_n^0), \quad (44)$$

representing approach amplification, aversive rejection, and model-blindness. Let

$$\Xi_n = \alpha_n^+ P_n^+ + \alpha_n^- P_n^- + \alpha_n^0 P_n^0, \qquad \widetilde{G}_{\mu,n} = G_{\mu,n} \circ (I + \Xi_n). \quad (45)$$

The distortion defect is

$$\epsilon_{\Xi,n} = d_O(G_{\mu,n}(z_n), \widetilde{G}_{\mu,n}(z_n)). \quad (46)$$

Identity protection, attraction, aversion, and ignorance can therefore alter meaning without increasing epistemic status. This mechanism gives a formal place to motivated reasoning while preserving the distinction between explanation and justification.

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X. IDENTITY, INFORMATION HORIZONS, AND PHENOMENALITY

A. Typed self readout

The complete self object is represented as a typed quotient,

$$\begin{aligned}\mathfrak{S}_A[n] &= q_{self}(F^n[\delta_R, \mathcal{T}_A, c_{0:n}]) \\ &= \big[\mathcal{A}_A[n], \Delta_A[n], H_A[n], \mathrm{Phen}_A^{str}[n], P_A^{lived}[n], \\ &\quad Own_A[n], Coh_A[n], Val_A[n], \Pi_A^{\star}[n], \Lambda_A[n]\big]_{\sim_{self}}.\end{aligned} \tag{47}$$

Agency, identity, structural phenomenality, lived phenomenality, ownership, coherence, and valence remain dissociable:

$$\mathcal{A}_A \neq \mathcal{I}_A \neq \mathrm{Phen}_A^{str} \neq P_A^{lived} \neq Own_A \neq Coh_A \neq Val_A. \tag{48}$$

Sharing one root does not make the readout lanes interchangeable.

Identity is agency continued through lineage. A one-time action is not yet a self. The identity update is

$$\begin{aligned}\mathcal{I}_A[n+1] = q_{id}\big(&\mathcal{I}_A[n] \boxplus \mathcal{A}_A[n] \boxplus P_A^{lived}[n] \\ &\boxplus \Pi_A^{\star}[n] \boxplus Res_A[n] \boxplus \Delta\mathcal{T}_A[n]\big).\end{aligned} \tag{49}$$

where $\boxplus$ denotes append-only typed lineage composition.

B. The horizon triad

The architecture distinguishes three horizons.

The dynamical horizon is the discriminant threshold

$$H_{dyn}: \quad \Delta_A(\lambda) = D_A^2 - 4M_AK_A\lambda = 0. \tag{50}$$

The information horizon is the boundary beyond which no admissible external inverse path recovers the internal state without bounded emission:

$$H_{info}(A) = \{z \in Z_A : \mathsf{Rec}(z) \Rightarrow (E_b < \infty,\ R_p > 0)\}. \tag{51}$$

The phenomenal horizon contains self-indexed, embodied, owned, coherent, and action-consequential records:

$$H_{phen}(A) = \{r = \mathcal{A}_A^{acc}\Pi_A T_A(\delta_R) : \mathsf{PhenGate}(r) = 1\}. \tag{52}$$

The proposed weld is

$$H_{dyn} \xrightarrow{K_{DI}} H_{info} \xrightarrow{K_{IP}} H_{phen}. \tag{53}$$

Only the threshold and information-horizon structures are conditionally derived. The sufficiency bridge K_{IP} remains open; a shared word such as “horizon” does not close it.

Structural phenomenality is

$$\mathrm{Phen}_A^{str}[n] = \mathcal{A}_A^{acc}\Pi_A T_A(\delta_R^{\leq n}; \mathcal{T}_A, c_n). \tag{54}$$

A lived event is a further bounded transformation,

$$P_A^{lived}[n] = \Phi_A^{phen}(M_A^{rec}[n], \Omega_A[n], B_A[n], c_n). \tag{55}$$

The framework does not infer subjective consciousness from accessibility alone. It only supplies a place where an independently tested phenomenal adapter could attach.

XI. TRANSFORMATIVE KNOWING AND RELEASE

A. Three levels of knowledge

Ordinary propositional knowledge is

$$K_{prop}(p) = \sigma_K(p). \tag{56}$$

Operational knowledge adds a task and an independent completion certificate,

$$K_{op}(p) = \langle K_{prop}(p), Task_p, Completion_p\rangle. \tag{57}$$

Transformative knowledge adds persistent changes in view, identity, binding, action, context transport, and viability:

$$\begin{aligned}K_{trans}(p) = \langle &K_{op}(p), \Delta\Theta, \Delta\mathcal{I}, \Delta\mathcal{U}, \Delta Action, \\ &Persistence, Transport, Viability\rangle.\end{aligned} \tag{58}$$

This layer is optional for mathematics, historical identification, or ordinary description. It is mandatory when a claim asserts that a practice changes the knower, completes a task, or produces release.

A transformation is admitted only if

$$\sigma_K(p) = ADMITTED, \tag{59}$$

$$Completion_p = PASS, \tag{60}$$

$$z_{post} \in V_A, \tag{61}$$

$$Defects_{persistence,transport,lineage} \leq \tau. \tag{62}$$

Intensity, conviction, surprise, and life change are not substitutes for these conditions.

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B. Release without destruction

The identity-binding update is

$$\mathcal{U}_{n+1} = \Pi_{\mathcal{U}\geq 0}\left[(I - D_U)\mathcal{U}_n + J_{reinforce,n} - J_{release,n}\right]. \tag{63}$$

A sufficient conditional release criterion is persistent dominance of release over reinforcement,

$$J_{release} - J_{reinforce} \geq \epsilon_{release} > 0, \tag{64}$$

with nonnegative damping and no hidden compensating channel. Then $\|\mathcal{U}_n\| \to 0$.

Release additionally requires viability and continuing readout:

$$z_n \in V_A, \qquad O_E(z_n) \neq 0, \qquad \|\mathcal{U}_n\| \to 0. \tag{65}$$

Thus release is not death, memory deletion, passivity, or loss of causal agency. A nonbinding mode satisfies

$$v_{free} \in \ker B_{\mathcal{I}}, \qquad B_{\mathcal{I}} v_{free} = 0, \qquad O_E(v_{free}) \neq 0. \tag{66}$$

The content remains readable while no longer functioning as an identity necessity.

XII. AGENCY META-READOUT GOVERNANCE

A. From readout to readout-of-readout

The governance architecture contains no named mental faculty in its canonical layer. It contains retained distinctions, readouts, finite retention, context, lineage, candidate actions, admissibility, policy, residuals, correction, and bounded reporting.

Current-state readout is

$$R_A^{(1)}[n] = O_A^{(1)}(q_A(Z_A[n]); c_n, \mathcal{T}_n). \tag{67}$$

Meta-readout is reader composition,

$$R_A^{(2)}[n] = O_A^{(2)}(R_A^{(1)}[n]; c_n, \mathcal{T}_n), \tag{68}$$

retained over a finite window,

$$W_A^R[n] = \mathcal{R}_{\tau_g}(R_A^{(2)}[n]; \mathcal{T}_n). \tag{69}$$

Typed-state decoupling requires

$$\begin{gathered} R_A^{(1)} \neq Claim_A \neq Identity_A, \\ Identity_A \neq ActionCandidate_A \neq CommittedAction_A. \end{gathered} \tag{70}$$

This prevents a transient readout from becoming a claim, identity, or command merely because it is salient.

Context and lineage access are

$$C_A^{ret}[n] = O_A^{ctx}(c_{n-h:n}, \mathcal{T}_{n-h:n}), \tag{71}$$

$$L_A^{acc}[n] = O_A^{lin}(\Lambda_A[n], \mathcal{T}_n). \tag{72}$$

The admissible candidate set is

$$\mathcal{U}_A[n] = \{u : \mathcal{C}_A(u \mid Z_A[n], c_n, \mathcal{T}_n) = 1\}. \tag{73}$$

Action formation is exposed by

$$F_A^{act}[n] = O_A^{act}(\mathcal{U}_A[n], \Pi_A[n]). \tag{74}$$

B. Interruptibility, alternatives, and policy revision

For commitment time t_c, response interruptibility is

$$\begin{aligned} I_A^{resp}[n] = 1 \iff \exists u' \neq u^\star :\ & u' \in \mathcal{U}_A[n], \\ t < t_c, \quad & \mathrm{Repair}(F(Z_A, u')) \geq R_{min}. \end{aligned} \tag{75}$$

Interruptibility is not delay. A fast transition can remain governed when alternatives and repair remain available; a slow transition can remain captured when they do not.

Alternative-policy access is

$$A_A^{pol}[n] = O_A^{alt}(\mathcal{U}_A[n]). \tag{76}$$

A policy can then be reconfigured by

$$\Pi_A^\star[n] = \arg\min_{u\in\mathcal{U}_A[n]} J_A(u), \tag{77}$$

$$\begin{aligned} J_A(u) = \mathbb{E}\|Res_{A,u}\|^2 + \lambda_C C(u) + \rho R(u) \\ + \mu L_{repair}(u) - \nu V(u). \end{aligned} \tag{78}$$

The weights are target-domain parameters, not root constants.

Residual-guided repair is primary in discrete form,

$$\begin{aligned} u_A^{repair} = \arg\min_{u\in\mathcal{U}_A^{repair}} \big[& \|Res_A(F(Z_A, u))\|^2 \\ & + Cost(u) + RepairLoss(u)\big]. \end{aligned} \tag{79}$$

A gradient approximation is allowed only after a differentiable local encoding is independently supplied.

C. Governance state and operator

The full governance state is

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$$\mathfrak{G}_A^{MR}[n] = \langle R_A^{(1)}, R_A^{(2)}, W_A^R, D_A^{type}, C_A^{ret}, L_A^{acc}, F_A^{act}, I_A^{resp}, A_A^{pol}, \Pi_A^\star, \Delta Z_A^{repair}\rangle_n. \tag{80}$$

A compact operator is

$$\mathcal{G}_A^{MR}[n] = \mathcal{R}_{\tau_g} O_A^{(2)}\big(O_A^{(1)}(q_A(Z_A[n])), C_A^{ret}, L_A^{acc}, F_A^{act}, \Pi_A\big). \tag{81}$$

The operator is governance-relevant only when ablation changes policy or correction:

$$\Delta_{\mathcal{G}}\Pi_A^\star = d_\Pi(\Pi_A^\star, \Pi_{A,-\mathcal{G}}^\star) > \tau_{idle} \tag{82}$$

or an analogous repair distance exceeds tolerance. This blocks unsupported claims based only on self-report, appearance, low arousal, delay, or approved output.

D. Defect vector, capture, and finite self-readout

The primary diagnostic is typed:

$$\boldsymbol{\epsilon}_G = (\epsilon_{R1}, \epsilon_{R2}, \epsilon_W, \epsilon_D, \epsilon_C, \epsilon_L, \epsilon_F, \epsilon_I, \epsilon_A, \epsilon_\Pi, \epsilon_{repair}). \tag{83}$$

A norm may summarize Eq. (83), but it may not replace it. Different defects require different repairs.

A conditional capture margin is

$$\Gamma_A = Load_A - Capacity_A. \tag{84}$$

If $\Gamma_A > \tau_{capture}$, governance capture becomes a live diagnostic candidate. The weights and threshold remain uncalibrated. The operational modes are $ONLINE$, $FRAGILE$, $CAPTURED$, $RETROSPECTIVE$, and $OFFLINE$.

The no-free-governance law is

$$CurrentReadout + SelectionPriority + SelectionStability + SelfReport \nRightarrow \mathcal{G}_A^{MR}. \tag{85}$$

A stable selected channel can coexist with lost context, missing alternatives, opaque action formation, and unavailable repair.

For a bounded agency,

$$B_G = \frac{I(R_A^{(2)}; Z_A)}{\tau_c^A} < \infty, \qquad \ker O_A^{(2)} \neq \varnothing \tag{86}$$

for a sufficiently rich state. The architecture therefore rejects complete self-transparency, permanent full-channel monitoring, and exact zero empirical governance defect.

XIII. HUMAN–AI RELATION AND KNOWLEDGE CIVILIZATION

Artificial systems can generate, rank, summarize, and transform claims at a scale that changes belief distribution. That capability does not grant them automatic knowledge status. The same typed separation applies:

$$AIOutput \neq ClaimIdentity \neq Evidence, \quad Evidence \neq Inference \neq KnowledgeStatus. \tag{87}$$

An AI output may function as testimony, compression, search assistance, hypothesis generation, or a method component. Its epistemic role depends on provenance, source access, transformation history, uncertainty, independence, and the possibility of correction.

The Maker–Checker firewall is central. The maker freezes the question, data split, metrics, thresholds, and claim ceiling before training or synthesis. The checker cannot retune the model or thresholds after seeing held-out outcomes. This resembles the separation of development and evaluation in responsible AI and the documentation goals of model cards and datasheets [9–11]. The additional requirement here is epistemic: the public status must be reconstructible from the frozen maker and checker records.

Human and artificial systems are not declared equivalent. A bounded transport relation is required:

$$T_{H\leftarrow AI} \circ K_{AI} \approx K_H \circ T_C, \tag{88}$$

with explicit defects for semantic loss, source omission, authority laundering, uncertainty compression, and context mismatch. AI can expand access and detect patterns while also amplifying institutional beliefs that have not passed epistemic transport. The architecture therefore requires AI-generated classifications and summaries to retain source identity and unresolved states rather than collapsing them into fluent certainty.

A knowledge civilization is not a society with the largest archive. It is an infrastructure that governs access, translation, belief distribution, claim externalization, evidence and inference checking, objection protection, correction, local knowledge transport, and AI-mediated synthesis. It fails when the hierarchy of believers becomes the hierarchy of truth. It matures when it can distinguish coordination from validity and preserve unresolved states without administrative pressure to manufacture certainty.

XIV. DISCUSSION

A. Relation to existing epistemology

The proposal is neither a new JTB amendment nor a rejection of individual epistemology. It relocates the primary public unit from the private mental state to the claim-process object. Traditional analyses remain relevant

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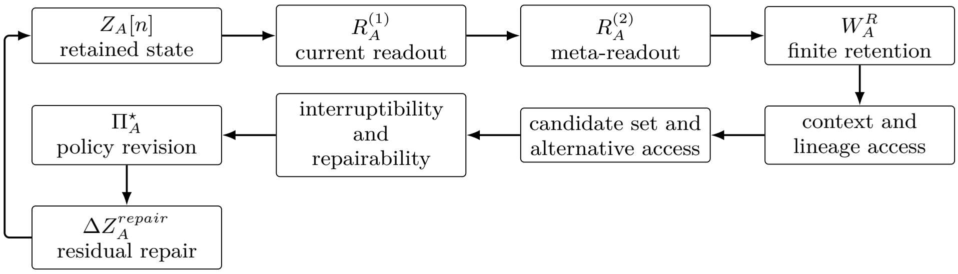

FIG. 2. Meta-readout governance loop. Governance requires more than access to a current state: it requires a retained readout of that readout, context and lineage, alternatives, interruptibility, policy reconfiguration, and residual-guided repair.

inside the architecture. Justification enters method and inference; reliability enters world contact, calibration, and repeated performance; safety enters counterfactual transport and stability; virtue enters agent and practice constraints; knowledge-first approaches can be represented by treating some knowledge states as primitive within a target theory, provided the adapter does not insert them into the Readout Genesis root.

The process-status proposal also differs from simple proceduralism. Passing formal procedures does not make a false proposition true. A claim can be admitted within a bounded domain and later superseded. The architecture is fallibilist: it governs how a claim earns, loses, and transports status. It is not verificationism and does not define truth as institutional acceptance.

B. Power, exclusion, and local evidence

The separation of authority, power, belief distribution, and epistemic validity gives social epistemology an executable place in the claim object. Access rights and language are not neutral metadata. They determine which traces can become evidence. Objection gates protect local and minority evidence from exclusion by status. Yet local evidence is not romanticized. It too must expose provenance, method, uncertainty, and correction conditions. The architecture rejects both institutional infallibility and automatic authenticity.

C. Agency and phenomenality

The agency layer is intentionally conservative. It provides typed readouts, horizons, ownership, coherence, valence, and policy without asserting that these equations are already a complete neuroscience or psychology. The bridge from structural accessibility to lived phenomenality remains open. Negative controls are therefore essential: locked-in control, depersonalization-like separation, altered valence, dreaming, and anesthesia-like loss of reportability illustrate that agency, ownership, coherence, valence, and phenomenality can dissociate. The architecture predicts that a one-scalar theory will fail to preserve these distinctions, but the prediction requires domain-specific operationalization.

D. Transformative knowledge

The transformative layer prevents a common category error. A person may sincerely report change, relief, certainty, or insight. Those experiences are real as experiences, but they do not by themselves establish the truth or universal validity of the associated claim. Conversely, a proposition can be well supported without changing the agent. The task/completion architecture makes transformation an additional evidential burden only when transformation is part of what is claimed.

E. Meta-readout governance

The governance operator is not a moral approval function. A system can be well governed and pursue a harmful objective if the objective and admissibility constraints are defective. Ethical and legal conditions must enter the target adapter and candidate set. Meta-readout governance contributes something narrower: it makes current state, context, lineage, alternatives, commitment, and repair inspectable. It also clarifies why delay, inhibition, self-report, or stable attention are not sufficient proxies for governance.

XV. LIMITATIONS AND EMPIRICAL PROGRAM

The architecture has four unresolved empirical frontiers.

First, candidate-state sufficiency is open. A real target domain may require variables absent from the current state. The relevant test is not descriptive completeness but future-readout sufficiency under intervention and delayed return.

Second, the exact epistemic quotient q_{MEMK} is open. A useful encoding must commute with target dynamics and

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preserve required readouts. Surface similarity between equations is not enough.

Third, calibration is open. The weights in belief update, capture load, governance capacity, practical performance, and transport defect are domain-DSL parameters. They cannot be read for free from the root.

Fourth, independent held-out validation is open. A serious empirical program should freeze event-resolved tapes, questions, admissible interventions, readers, metrics, and claim ceilings before fitting. It should then test prediction, correction, context transport, negative controls, and ablation of the governance operator.

A minimal validation program would proceed as follows:

1. freeze a typed event tape with source and context lineage;
2. define candidate state variables and a future-readout horizon;
3. test state sufficiency and invariant completion;
4. learn or specify a candidate quotient without access to held-out outcomes;
5. calibrate readouts on a training partition;
6. evaluate prediction, intervention, transport, and repair on held-out partitions;
7. compare the full model with ablations of context, lineage, alternatives, and meta-readout;
8. report unresolved, obstructed, and negative results without claim inflation.

No new simulated or human-subject result is reported in this paper. The contribution is the specification of what such a result would need to measure and what it would not license.

XVI. CONCLUSION

The Information Epistemic Foundation begins with a retained difference rather than a ready-made knower. Domain translation produces traces; traces become meaning, experience, and memory; memory contributes to belief; belief becomes a candidate for public epistemic status only after externalization and checking. The architecture separates belief strength, belief distribution, authority, power, practical success, and knowledge status. It preserves local scope, requires explicit transport for global claims, and treats correction and supersession as lineage-preserving epistemic actions.

The conditioned-agency extension closes the loop from contact through identity binding and action to renewed state. The horizon and phenomenality extension keeps structural accessibility, lived experience, ownership, coherence, and valence distinct. The transformative layer requires task and completion evidence when change in the knower is part of the claim. The meta-readout governance extension adds current readout, readout-of-readout, finite retention, context, lineage, alternatives, interruptibility, policy revision, and repair.

The resulting strongest statement is bounded:

> A root-connected normative architecture and meta-readout governance specification can be made internally complete, auditable, fail-able, and claim-bounded without importing belief, knowledge, identity, or phenomenality as root primitives.

The architecture does not yet establish a universal encoding for human or machine cognition, a complete theory of truth, a sufficient theory of consciousness, or empirical calibration across contexts. Those are not hidden assumptions. They are the next research program.

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Appendix A: Compact epistemic gate matrix

TABLE III. Compact gate matrix for implementation. A status-bearing report should retain each result separately rather than compressing the table into a single confidence score.

Gate	Object	Required record	Typical failure
G0	Claim identity	Proposition, domain, context, reader, units, scope	Claim drift or aliasing
G1	Agency/access	Proposer, permissions, tools, languages, exclusions	Hidden access asymmetry
G2	Provenance	Source, custody, transformations, versions, missingness	Authority laundering or lost lineage
G3	Contact	External record, proof object, or independent domain constraint	Self-generated support
G4	Method	Question–method fit and resolution	Method imported from a mismatched domain
G5	Inference	Inspectable derivation and alternatives	Conclusion stronger than premises
G6	Scope/uncertainty	Assumptions, ambiguity, missingness, transport limits	Universalization from local support
G7	Objection	Counterclaims, controls, local/minority evidence	Status-based exclusion
G8	Correction	Revision, withdrawal, replacement conditions	Immunization against error
G9	Independence	Frozen maker/checker separation	Leakage and post hoc threshold tuning
G10	Performance	Prediction/intervention metrics when relevant	Efficacy asserted without outcome test
G11	Transport	Commuting map and bridge defect	Translation residue hidden
G12	Lineage	Prior versions, failures, reasons for change	Retraction by deletion
G13	Ceiling	Public tier no stronger than weakest load-bearing gate	Claim inflation

Appendix B: Root-native closed algorithm

A language-neutral implementation is:

```
INPUT: occurrence/structure A, question Q, reader O,
       context c, horizon L, candidate claim p

1. Freeze claim identity, scope, units, and terminology.
2. Build candidate domain state Z_cand from retained records.
3. Audit state sufficiency over current and future readouts.
4. Preserve states that differ on required invariants.
5. Declare quotient q_D and test dynamics/readout commutation.
6. Externalize p with agency, access, provenance, and lineage.
7. Run G0--G13 with values {PASS, FAIL, UNRESOLVED}.
8. Separate discriminating evidence gates from operational gates.
9. Apply practical-performance tests only when the claim requires them.
10. Apply Maker--Checker independence and leakage controls.
11. Compute local scope; test transport before global reporting.
12. Assign status no stronger than the weakest load-bearing gate.
13. Preserve objections, failures, and correction conditions.
14. When action is involved, expose alternatives, interruptibility,
    policy revision, and residual-guided repair.
15. Publish bounded claim, not-checked ledger, and source hashes.
```

Appendix C: Source lineage and reproducibility

The manuscript is derived from the following immutable identities:

- Readout Genesis core snapshot: blob `31e07095add`

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`c45aacbaea26523784380d5ce21f1`;

- Philosophy/MEMK domain specification: blob `3d51fe8095f93207ab3ff4ae054042b6cf880a3a`;
- Information Epistemic Foundation source-ingest SHA-256: `147a43cecfb5fcaa39386b3fb9b5f1d541e00ef12b068980b74f1b05ecf61968`;
- equation-preserving canonical Markdown SHA-256: `ff8cbf6848d93f7579327dc628e25e91801d959669de4235fcdfbcb347f7a926`;
- registration head commit: `49e444cc64bdaea2767c2c18d209654cdf6cb23f`.

The canonicalization repaired corrupted lexical tokens without changing equations or claim boundaries. The repository registration was a draft pull request at the manuscript date; the paper therefore cites the branch and commit rather than implying merge into the default branch.

[1] E. L. Gettier, Is justified true belief knowledge?, Analysis **23**, 121 (1963).
[2] A. I. Goldman, What is justified belief?, in *Justification and Knowledge: New Studies in Epistemology*, edited by G. S. Pappas (D. Reidel, Dordrecht, 1979) pp. 1–23.
[3] E. Sosa, How to defeat opposition to moore, Philosophical Perspectives **13**, 137 (1999).
[4] T. Williamson, *Knowledge and Its Limits* (Oxford University Press, Oxford, 2000).
[5] M. Fricker, *Epistemic Injustice: Power and the Ethics of Knowing* (Oxford University Press, Oxford, 2007).
[6] H. E. Longino, *Science as Social Knowledge: Values and Objectivity in Scientific Inquiry* (Princeton University Press, Princeton, 1990).
[7] M. D. Wilkinson, M. Dumontier, I. J. Aalbersberg, *et al.*, The fair guiding principles for scientific data management and stewardship, Scientific Data **3**, 160018 (2016).
[8] UNESCO, Recommendation on open science, Adopted by the General Conference of UNESCO, Paris, 23 November 2021 (2021).
[9] E. Tabassi, *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*, Tech. Rep. NIST AI 100-1 (National Institute of Standards and Technology, 2023).
[10] T. Gebru, J. Morgenstern, B. Vecchione, J. W. Vaughan, H. Wallach, H. D. III, and K. Crawford, Datasheets for datasets, Communications of the ACM **64**, 86 (2021).
[11] M. Mitchell, S. Wu, A. Zaldivar, P. Barnes, L. Vasserman, B. Hutchinson, E. Spitzer, I. D. Raji, and T. Gebru, Model cards for model reporting, in *Proceedings of the Conference on Fairness, Accountability, and Transparency* (2019) pp. 220–229.
[12] C. E. Shannon, A mathematical theory of communication, Bell System Technical Journal **27**, 379 (1948).
[13] L. Floridi, *The Philosophy of Information* (Oxford University Press, Oxford, 2011).

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Knowledge as Stabilized Translation: Toward an Observer-Constrained Epistemology

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Knowledge as Stabilized Translation: Toward an Observer-Constrained Epistemology

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Open Civil Science Initiative

March 9, 2026

Abstract

Epistemology has often treated knowledge as a matter of access: a subject seeks justified, reliable, or rational contact with reality. This article argues that such accounts remain incomplete because they understate the constitutive role of mediation, finite access, selective encoding, and structured error. The problem becomes sharper in the age of artificial intelligence, where systems generate reliable and action-guiding outputs without fitting inherited models of reflective belief or transparent justification. I propose that epistemology should begin not from direct access to reality, but from bounded observers who form world-models under constraint. On this view, knowledge is best understood as *stabilized translation under bounded observation*: a representation that remains stable across admissible variation, calibration, and cross-checking. The result is a framework that preserves realism without naïvety, fallibilism without skepticism, and provides a common grammar for human and machine forms of model-based knowing.

Keywords: epistemology; knowledge; observation; mediation; artificial intelligence; translation; realism

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1 Introduction

A person glances at a message and misreads its tone. A physician looks at a scan and must decide whether a faint pattern is artifact or lesion. A driver sees movement at the edge of vision and brakes before explicit recognition arrives. These are not marginal episodes at the edge of knowledge. They are ordinary instances of what knowledge is in practice: not direct possession of reality, but the formation of sufficiently stable representations under conditions of incompleteness, selectivity, and risk.

Yet epistemology has often preferred a cleaner picture. In one classical form, knowledge is justified true belief (Gettier, 1963). In other forms, it is grounded in certainty, empirical warrant, rational necessity, or scientific method (Descartes, 1641/1996; Hume, 1748/2007; Kant, 1781/1998; Popper, 1959). These traditions differ profoundly, but they often preserve a shared aspiration: that knowledge consists, fundamentally, in successful access to reality.

This article argues that the picture is incomplete. The problem is not that truth has ceased to matter, nor that realism should be abandoned. The problem is that access is never unstructured. Every observer has finite causal reach, selective uptake, bounded memory, interpretive filters, and irreducible noise. What appears as knowledge is therefore never reality in unmediated form, but a representation formed under constraint.

The point becomes especially difficult to ignore in the age of artificial intelligence. Machine systems now classify, predict, translate, summarize, and generate in ways that are often practically successful. Whether or not one wishes to say that such systems "know" in the full human sense, their existence places pressure on inherited epistemological categories. If reliable, world-sensitive performance can arise without reflective belief in the classical sense, then epistemology must be widened accordingly.

The central claim of this paper is that epistemology should begin not from the ideal of direct access, but from the structure of bounded observers. Knowledge is best understood as *stabilized translation under bounded observation*: a representation that remains stable under admissible variation, calibration, and cross-checking.

2 The Historical Inheritance

Human civilizations have always lived in interpreted worlds. Mythic cosmologies did not merely tell stories; they organized order, memory, legitimacy, and meaning into livable forms (Cassirer, 1946; Eliade, 1959). Classical philosophical traditions in Greece, India, and China transformed inherited world-order into explicit reflection on truth, inference, perception, and reason (Plato, 1997; Aristotle, 1993; Matilal, 1986; Hansen, 1992). Medieval traditions integrated logic and interpretation within theological horizons (Augustine, 1991; Aquinas, 1920; Adamson, 2016). Modern science displaced revelation with experiment and mathematics while preserving a deeper hope that disciplined inquiry

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could progressively render reality transparent (Bacon, 1620/2000; Galilei, 1638/1974; Newton, 1687/1999).

The achievements of these traditions are considerable. They produced logic, experiment, criticism, and durable institutions of inquiry. Yet they also reinforced a powerful idealization: knowledge came to appear as if it were fundamentally a matter of successful access. Error could then be treated as removable noise, perspective as a secondary complication, and mediation as an unfortunate layer between the knower and the real.

Later philosophy repeatedly unsettled this picture. Hume exposed the fragility of induction (Hume, 1748/2007). Kant argued that experience is conditioned by the forms through which it becomes experience for us (Kant, 1781/1998). Sellars rejected the myth of the given (Sellars, 1956). Hanson and Kuhn showed that observation is theory-laden and that scientific practice is historically structured (Hanson, 1958; Kuhn, 1962). Quine naturalized epistemology (Quine, 1969). Popper weakened the ideal of positive final grounding by recasting scientific knowledge as conjectural and falsifiable (Popper, 1959).

These moves were decisive, but also dispersed. One tradition emphasized language, another science, another cognition, another justification. The result is a literature rich in critique but comparatively thin in synthesis. Mediation was recognized, but seldom made foundational.

3 The Research Gap

The present gap in epistemology is not a lack of criticism of direct realism. That criticism is already abundant. The gap is more specific. Contemporary epistemology still lacks a compact framework that does four things at once.

First, it treats observer-limitation as foundational rather than accidental.

Second, it explains knowledge in a form broad enough to speak to both human and non-human model-forming systems without collapsing their differences.

Third, it preserves realism and testability without relying on the fantasy of unmediated access.

Fourth, it connects abstract epistemology with the actual practices of cognition: perception, measurement, calibration, communication, institutional validation, and technological mediation.

Important traditions approach parts of this problem. Pragmatism treats inquiry as fallible and communal (Peirce, 1878; James, 1907/1975; Dewey, 1929). Reliabilism shifts attention from inner states to the dependability of process (Goldman, 1979). Social epistemology foregrounds testimony, institutions, and distributed justification (Fuller, 1988). Philosophy of information rethinks agents and informational environments in the digital age (Floridi, 2011, 2016). Cognitive science and cybernetics model knowing as control, adaptation,

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representation, and prediction under constraint (Wiener, 1948; Ashby, 1956; Marr, 1982; Clark, 1997; Friston, 2010). Recent work on computational opacity and algorithmic reliabilism sharpens the issue further by showing that epistemically useful systems may remain opaque while still being assessable in externalist terms (Durán and Formanek, 2018; Durán, 2025; Alvarado, 2024; Koskinen, 2024).

What remains underdeveloped is a general epistemological formulation in which knowledge itself is defined as a stability-achievement of bounded translation. That is the gap this article addresses.

4 Method, Scope, and Strategy

This article is a programmatic work in systematic epistemology. Its method is synthetic rather than archival. It does not offer a comprehensive civilizational history of all theories of knowledge, nor does it attempt to settle exegetical controversies surrounding canonical figures. Instead, it reconstructs a broad trajectory in order to identify one recurring omission: the tendency to treat mediation as secondary rather than constitutive.

The scope is deliberately limited. This is not a philosophy of physics paper, not a complete cognitive architecture paper, and not a grand-system paper. It advances one claim at short-journal scale: that epistemology is better understood when reorganized around bounded observers and stabilized representation. The argument is therefore focused rather than totalizing.

5 Why Artificial Intelligence Changes the Stakes

Artificial intelligence is not the sole basis of the present argument, but it makes the problem newly visible. Machine systems now classify images, detect anomalies, translate languages, recommend decisions, summarize discourse, and generate structured outputs with often significant practical success. They can also be brittle, biased, opaque, and overconfident. This combination matters philosophically.

The common response is to ask whether such systems really “know”, or merely simulate knowledge. That question matters, but its usual framing is too narrow. The deeper point is that machine performance exposes assumptions built into much inherited epistemology. If knowledge must always take the form of conscious, reflective, propositionally articulated belief, then AI systems can only appear as problematic imitations. But if epistemology is concerned more fundamentally with the formation of stable, action-guiding, world-sensitive representations, then AI becomes an important test case rather than an anomaly.

This does not imply that machines and humans know in the same way. Human knowing is embodied, affectively saturated, normatively entangled, and socially inherited in ways current AI is not. The argument is narrower. AI reveals that epistemology cannot be defined exclusively in terms of classical belief-content or introspective accessibility.

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It must also address representation, calibration, robustness, and epistemic dependence under constraint. Recent work in philosophy of information, philosophy of evidence, and the epistemology of opaque computational systems has already moved in this direction (Floridi, 2011, 2016; Lasonen-Aarnio and Littlejohn, 2024; De Pretis et al., 2021; Durán and Formanek, 2018; Koskinen, 2024; Russo et al., 2024).

6 The Observer as a Bounded System

The principal shift of this article is from proposition-first epistemology to observer-first epistemology. Before there is belief, there is uptake. Before there is justification, there is selective acquisition. Before there is theory, there is the formation of a world-model under constraint.

An observer, in the present sense, is not a detached spectator outside the world. It is an embedded system that receives only a bounded subset of what exists, processes it through finite channels, and forms a usable representation for action, judgment, or further communication. The underlying manuscripts from which this article develops make this point explicit: the correct starting point is observer-first rather than cosmos-first; all later claims are downstream of finite observers; no observer accesses the whole event structure; no reportable knowledge is unmediated; and all observation contains irreducible structured error.

This observer-structure can be expressed through one canonical equation:

$$M_A(E) = (T_A \circ \Pi_A)(S_A(E)). \tag{1}$$

Here, S_A denotes selective uptake, Π_A encoding or reporting, and T_A translation into a usable model. In the source manuscripts, this equation functions as the minimal pipeline of observation: without selection, encoding, and translation, no well-defined observer is present.

The same manuscripts define knowledge not as mere belief-content, but as stabilized representation under admissible variation:

$$Know_A(W) = 1 \;\text{ iff }\; Dist(R_A^{\nu}[n], R_A[n]) \leq \varepsilon_K \tag{2}$$

for all admissible variations ν on the declared window W.

Together, these claims support the paper's central proposal: bounded observers do not access the world transparently; they generate translated models under finite access and structured noise, and knowledge is the stability-achievement of those models.

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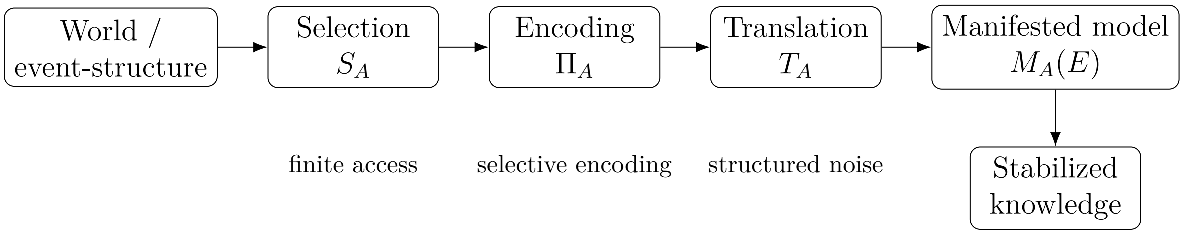

Figure 1: Conceptual structure of observer-constrained epistemology: knowledge is not direct access but stabilized translation under bounded observation.

7 Observer-Constrained Epistemology

The proposal can now be stated directly. Knowledge is a representation that remains stable under admissible variation, given the finite constraints of an observer. More briefly, knowledge is *stabilized translation under bounded observation.*

This claim does not deny realism. On the contrary, it presupposes that the world resists poor representations. Nor does it collapse into skepticism. Boundedness does not imply arbitrariness. A representation may be partial and still be objective in the sense that matters: it may remain robust across perturbation, calibration, and independent testing. What changes is not the importance of truth, but the philosophical picture through which truth is approached. The relevant unit of analysis is no longer direct access, but the stability profile of a translated representation formed by a finite observer.

The proposal also clarifies why mediation matters. Selection, encoding, and translation are not unfortunate distortions laid over an otherwise transparent relation to the real. They are the conditions under which finite knowers can know at all. A satisfactory epistemology must therefore begin not from immediacy, but from the structure of bounded representation.

8 Objections and Replies

8.1 Is this simply a form of relativism?

No. Translation does not imply that all representations are equally valid. The world continues to resist poor models. Stability is earned through perturbation, calibration, and endurance under cross-checking, not granted by internal coherence alone.

8.2 Is this just pragmatism under another name?

The view is close to pragmatism in its emphasis on fallibilism and test (Peirce, 1878; Dewey, 1929). It differs, however, in making bounded observer-structure and translation architecture more explicit, and in extending the framework to non-human representational systems.

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Table 1: Positioning the present proposal against major epistemological traditions

Tradition	**Core unit**	**Strength**	**Limitation**	**Revision proposed here**
Classical JTB	Belief	Focus on truth and justification	Weak on mediation and representational architecture	Knowledge as stabilized translation
Kantian critique	Subject-conditions	Shows experience is structured	Still centered on the human transcendental subject	Generalize to bounded observer systems
Pragmatism	Inquiry / practice	Fallibilism and testability	Less explicit on selection-translation structure	Stability under constrained translation
Reliabilism	Process dependability	Shifts beyond introspection	Thin account of representational formation	Reliability as one dimension of stabilization
AI challenge	Model output	Exposes non-human epistemic performance	Disrupts belief-centered models	Shared grammar for human and machine knowing

8.3 Does this abandon truth?

No. It relocates truth from the image of immediate possession to the more defensible terrain of durable epistemic convergence. The source manuscripts themselves frame truth not as certainty but as resistant convergence under constraint.

8.4 Are machine outputs really knowledge?

Not always, and not in a simple sense. The point is not that all machine outputs count as knowledge. The point is that epistemology must be able to explain why some model outputs are robust and calibrated while others are brittle, hallucinatory, or unstable.

9 Conclusion

The history of epistemology is often narrated as the progressive conquest of reality by reason, method, and critique. A more adequate narrative is available. It is the history of how finite beings learn to stabilize translations of a world they never possess in full.

This matters now because artificial intelligence has made the older picture increasingly difficult to sustain. Knowledge cannot be adequately defined as direct access, nor can it be restricted to reflective belief in the classical sense. What must be centered instead is

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the bounded observer: finite, selective, interpretive, corrigible, and nevertheless capable of generating durable representations.

The task of epistemology, then, is not to rescue immediacy, but to explain how stability is achieved under constraint. Knowledge, on this view, is not the end of mediation. It is mediation made reliable enough to inquire, to act, and to live by.

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References

Adamson, P. (2016). *Philosophy in the Islamic World*. Oxford University Press.

Alvarado, R. (2024). Challenges for computational reliabilism as an epistemological framework for machine learning in science. *Philosophy of Science Archive*. Preprint.

Aquinas, T. (1920). *Summa Theologica*. Benziger Bros.

Aristotle. (1993). *Posterior Analytics* (J. Barnes, Trans.). Oxford University Press.

10.5281/zenodo.18925129

Ashby, W. R. (1956). *An Introduction to Cybernetics*. Chapman & Hall.

Augustine. (1991). *The Trinity* (E. Hill, Trans.). New City Press.

Bacon, F. (2000). *The New Organon*. Cambridge University Press.

Cassirer, E. (1946). *Language and Myth*. Dover.

Clark, A. (1997). *Being There: Putting Brain, Body, and World Together Again*. MIT Press.

De Pretis, F., Landes, J., & Peden, W. (2021). Artificial intelligence methods for a Bayesian epistemology-powered evidence evaluation. *Journal of Evaluation in Clinical Practice*, 27(6), 1266–1272.

Descartes, R. (1996). *Meditations on First Philosophy*. Cambridge University Press.

Dewey, J. (1929). *The Quest for Certainty*. Minton, Balch & Company.

Durán, J. M. (2025). In defense of reliabilist epistemology of algorithms. *European Journal for Philosophy of Science*, 15, Article 64.

Durán, J. M., & Formanek, N. (2018). Grounds for trust: Essential epistemic opacity and computational reliabilism. *Minds and Machines*, 28(4), 645–666.

Eliade, M. (1959). *The Sacred and the Profane*. Harcourt.

Floridi, L. (2011). *The Philosophy of Information*. Oxford University Press.

Floridi, L. (Ed.). (2016). *The Routledge Handbook of Philosophy of Information*. Routledge.

Friston, K. (2010). The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127–138.

Fuller, S. (1988). *Social Epistemology*. Indiana University Press.

Galilei, G. (1974). *Two New Sciences*. University of Wisconsin Press.

Gettier, E. L. (1963). Is justified true belief knowledge? *Analysis*, 23(6), 121–123.

Goldman, A. I. (1979). What is justified belief? In G. Pappas (Ed.), *Justification and Knowledge* (pp. 1–23). Reidel.

Hansen, C. (1992). *A Daoist Theory of Chinese Thought*. Oxford University Press.

Hanson, N. R. (1958). *Patterns of Discovery*. Cambridge University Press.

Hume, D. (2007). *An Enquiry Concerning Human Understanding*. Oxford University Press.

James, W. (1975). *Pragmatism*. Harvard University Press.

Kant, I. (1998). *Critique of Pure Reason*. Cambridge University Press.

Koskinen, I. (2024). We have no satisfactory social epistemology of AI-based science. *Social Epistemology*, 38(6), 693–705.

Kuhn, T. S. (1962). *The Structure of Scientific Revolutions*. University of Chicago Press.

Lasonen-Aarnio, M., & Littlejohn, C. (Eds.). (2024). *The Routledge Handbook of the Philosophy of Evidence*. Routledge.

Marr, D. (1982). *Vision*. W. H. Freeman.

Matilal, B. K. (1986). *Perception: An Essay on Classical Indian Theories of Knowledge*. Oxford University Press.

10.5281/zenodo.18925129

Newton, I. (1999). *The Principia: Mathematical Principles of Natural Philosophy*. University of California Press.

Peirce, C. S. (1878). How to make our ideas clear. *Popular Science Monthly*, 12, 286–302.

Plato. (1997). *Republic*. In J. M. Cooper (Ed.), *Plato: Complete Works*. Hackett.

Popper, K. (1959). *The Logic of Scientific Discovery*. Hutchinson.

Quine, W. V. O. (1969). Epistemology naturalized. In *Ontological Relativity and Other Essays* (pp. 69–90). Columbia University Press.

Russo, F., Schliesser, E., & Wagemans, J. (2024). Connecting ethics and epistemology of AI. *AI & Society*, 39, 1171–1184.

Sellars, W. (1956). Empiricism and the philosophy of mind. In H. Feigl & M. Scriven (Eds.), *Minnesota Studies in the Philosophy of Science*, Vol. 1. University of Minnesota Press.

Turing, A. M. (1950). Computing machinery and intelligence. *Mind*, 59(236), 433–460.

Wiener, N. (1948). *Cybernetics*. MIT Press.

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Constraint-First Epistemology: Normativity, Conditioned Agency, and the Non-Zero Kantian Floor

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What the chapter states. *"Total epistemic error therefore remains irreducibly non-zero." "agents still achieve stable prediction, robust intervention, explanatory integration, and corrigible revision."* (10.5281/zenodo.19205869)

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Read before. Ch. 1 *Written by AI. Still True.*; Ch. 2 *The Readout Condition*; Ch. 3 *Readout Genesis Standalone Synthesis*; Ch. 39 *The Standalone Scholar*.

Feeds into. This book's Chapter 11, *Human Learning as Epistemic Architecture*.

Genesis Constraint-First Alignment Epistemology: Reason, Error, and World-Tracking under Irreducible Mediation

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Abstract

Finite knowers never enjoy unmediated access to the world. Their epistemic contact is shaped by sensory limitation, instrumentation, conceptual framing, inferential procedure, and historically situated practices of inquiry. Total epistemic error therefore remains irreducibly non-zero. Yet finite inquiry is not thereby rendered null: agents still achieve stable prediction, robust intervention, explanatory integration, and corrigible revision. The central philosophical problem is not whether mediation exists, but how world-responsiveness remains possible under it.

This paper develops *Genesis Constraint-First Alignment Epistemology* (GCFAE) as an account of knowledge under irreducible mediation. The framework introduces world-side causal-structural constraint, denoted by θ_W, as a minimal realist posit required to explain reproducible resistance, cross-route invariance, and interventional success. Agents construct revisable models M_A from constrained observation streams O_A, and reason is reconceived as a disciplined family of update procedures that tends, under suitable conditions, to improve epistemic alignment V_A between representation and world-side constraint.

GCFAE is positioned between constructive empiricism, convergent realism, and structural realism. Against constructive empiricism, it argues that empirical adequacy alone cannot explain why some models support robust intervention, transfer, and route-independent correction. Against convergent realism, it rejects the claim that progress must culminate in a final uniquely true theory. Against standard structural realism, it retains the significance of structure while shifting emphasis from mere retention across theory change to constraint-governed alignment under intervention and non-eliminable mediation. The account also integrates fallibilism, approximate truth, and non-substantialist normativity without appealing to a self-grounding rational faculty.

Knowledge is therefore not modeled as possession of truth, but as graded, corrigible alignment under irreducible error. The paper clarifies the ontological status

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of θ_W, the evaluative structure of V_A, the distinction between irreducible mediation and reducible distortion, and the role of public criticism in route-independent inquiry. It concludes that objectivity survives not because mediation is overcome, but because finite representations can fail, and because that failure can be publicly tracked, criticized, and revised under world-side resistance.

Keywords: epistemology; fallibilism; structural realism; world-tracking; mediation; normativity; error; intervention; scientific realism; corrigibility

1 Introduction

Finite knowers do not confront the world from nowhere. Their access to it is filtered through embodiment, instrumentation, conceptual articulation, inferential architecture, and historically situated practices of inquiry. Observation is selective, theory-laden, and resolution-bounded. Yet finite inquiry is not thereby reduced to illusion. Scientists predict eclipses, manipulate genes, calibrate detectors, revise cosmological models, and correct one another's mistakes. Ordinary agents likewise navigate environments with non-trivial success. The philosophical problem is therefore not whether mediation exists, but how knowledge remains possible under it.

Much epistemology is trapped within a false binary. One pole assumes that knowledge requires something close to transparent access, direct givenness, or full correspondence conceived as possession. The other infers from the inevitability of mediation and error that robust world-tracking is impossible. The first inflates local success into epistemic triumphalism; the second collapses finitude into skepticism. Both inherit the same hidden premise: unless cognition escapes mediation, objectivity cannot survive.

This paper rejects that premise. It argues that finite knowledge is better understood as *graded epistemic alignment under irreducible error*. Finite agents never eliminate mediation and never reduce total epistemic error to zero. Nevertheless, inquiry can still improve because representations can stand in better or worse alignment with the causal-structural constraints imposed by the world. Reason, accordingly, should not be understood as a mysterious inner faculty guaranteeing truth, nor as a merely adaptive label for whatever happens to work. It is a disciplined family of update procedures that tends, under repeatable conditions, to improve alignment between representation and constraint.

I call this framework *Genesis Constraint-First Alignment Epistemology* (GCFAE). The expression "constraint-first" signals a reversal of explanatory priority. Instead of beginning from a self-grounding subject and then asking how it reaches outward to reality, the account begins from the conditions under which finite, world-internal systems can become more or less answerable to what resists them. Subjectivity is not denied. It is relocated within a wider field of mediation, friction, public criticism, and correction.

This proposal is situated at the intersection of epistemology and philosophy of science. Against constructive empiricism, it argues that empirical adequacy alone is too thin to capture the full normative structure of inquiry (Fraassen 1980). Against convergent realism, it denies that progress must be modeled as convergence toward one final and uniquely true theory (Laudan 1981). It is closest to structural realism (Worrall 1989; Ladyman 1998), but departs from standard formulations by emphasizing intervention, route-independence, and corrigible alignment under mediation rather than merely the retention of mathematical structure across theory change. Methodologically, it extends fallibilism (Popper 1959) and approximate-truth traditions (Boyd 1990), while resisting

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the reification of reason into an intrinsic faculty external to the causal order.

The central claim can be stated in one sentence: *knowledge does not require unmediated access; it requires corrigible alignment under world-side resistance.* The rest of the paper defends that thesis.

Section 2 introduces the constraint-first reversal. Section 3 clarifies the framework's key concepts: θ_W, V_A, irreducible mediation, irreducible error, and reason as disciplined revision. Section 4 presents the formal core and explains the philosophical work done by each symbol. Section 5 situates GCFAE in relation to constructive empiricism, convergent realism, and structural realism. Section 6 develops route-independence, weighting, and the social dimension of inquiry. Section 7 offers illustrative cases from physics, biology, and machine learning. Section 8 addresses major objections. Section 9 concludes by drawing out the framework's contribution and its limits.

2 The Constraint-First Reversal

Many epistemologies begin by granting explanatory priority to the knower: a subject, a standpoint, a faculty, or a transcendental condition of intelligibility. Once this priority is granted, the central question becomes how such a knower secures access to a world beyond itself. That framing has produced two familiar pathologies. If too much weight is placed on the knower, inquiry tends toward idealism, internalism, or epistemic inflation. If too little is placed on the knower, normativity is threatened and skepticism looms.

GCFAE reverses the order of explanation. It begins from the fact that inquiry is conducted by finite systems embedded in a world that does not simply mirror their conceptual habits. Such systems receive constrained inputs, construct representations under limited bandwidth, act on those representations, and encounter resistance when those representations fail. The primary philosophical question is thus not "What faculty guarantees knowledge?" but rather: under what conditions can a finite, mediated system become more aligned with the world it never apprehends without mediation?

Three consequences follow from this reversal.

First, epistemic immediacy ceases to be the standard of legitimacy. There is no need to posit direct access to things in themselves in order to preserve objectivity. Knowledge is not possession but answerability.

Second, normativity is not grounded in a self-certifying faculty. Better and worse inquiry are distinguished by the success or failure of representations under repeatable conditions of correction. This shifts attention from inner guarantee to public revisability.

Third, objectivity is relocated from metaphysical transparency to resistant constraint. A representation is objective not because it reproduces reality from nowhere, but because it survives disciplined confrontation with what it does not control.

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Constraint-first epistemology therefore replaces the ideal of possession with the practice of correction. Finite agents remain within the world they seek to know, and that immanence is not a prison. It is precisely because representations can fail that they can be revised, and precisely because they can be revised that objectivity remains meaningful.

3 Clarifying the Core Concepts

3.1 θ_W: world-side causal-structural constraint

The symbol θ_W does not denote a total inventory of reality as it is in itself, nor a fully transparent metaphysical essence waiting behind appearances. It denotes the minimal realist posit required to explain why inquiry exhibits reproducible resistance, stable interventional success, and route-diverse invariance.

Three features of this posit are essential.

First, θ_W is *world-side*. It is not reducible to consensus, conceptual scheme, instrument design, or present belief. These mediate access to constraint; they do not constitute it.

Second, θ_W is *causal-structural*. GCFAE does not require direct acquaintance with hidden essences. What matters is that the world is organized in ways that constrain possible observations, sustain some interventions rather than others, and produce reproducible regularities not exhausted by any one access route. In this respect the framework aligns with interventionist and structural realist intuitions without collapsing into either one (Hacking 1983; Woodward 2003; Worrall 1989).

Third, θ_W is *minimal*. The framework does not claim exhaustive metaphysical possession. It claims only that without some world-side source of resistance and invariance, the recurrent success and correction of inquiry would become mysterious or coincidental. Cross-instrument agreement, interventional stability, and reproducible failure need an anchor beyond present endorsement.

This makes θ_W a modest realist posit justified by explanatory indispensability rather than direct apprehension. The view is realist, but non-triumphalist.

3.2 V_A: epistemic alignment

The symbol V_A denotes *epistemic alignment*: the degree to which a representation or model M_A stands in an adequate relation to θ_W within a task-domain D. Alignment is not identical to truth in a maximal correspondence sense, and it is not reducible to immediate predictive success. It is a domain-indexed evaluative achievement with several dimensions.

At minimum, four dimensions belong to alignment:

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(i) **Predictive adequacy**: how well a model supports stable and discriminating expectations.

(ii) **Intervention reliability**: how well actions guided by the model yield repeatable and non-accidental outcomes.

(iii) **Structural preservation**: how well the model captures stable relations, dependencies, or invariants across contexts.

(iv) **Route-independence**: how well support for the model survives across distinct observational, instrumental, or inferential pathways.

These dimensions need not carry equal weight in every domain. In intervention-heavy sciences, intervention reliability may dominate simple predictive fit. In historical reconstruction, route-independence and explanatory integration may matter more than manipulative control. GCFAE therefore treats V_A not as a single homogeneous scalar with universal units, but as a *domain-indexed weighted ordering* over epistemically relevant dimensions. This preserves rigor without imposing false uniformity.

What matters is not metric absolutism, but public assessability under constraint. The concept of alignment remains substantive because some models are systematically better than others across these dimensions, even when none achieves final possession.

3.3 Irreducible mediation

Irreducible mediation is the thesis that finite cognition never encounters the world without some layer of filtration, transformation, or structuring. Observation depends on embodiment and instrumentation; data depend on encoding conventions; interpretation depends on concepts and inferential commitments. There is no unmediated epistemic relation available to finite knowers.

This claim must be distinguished from a stronger and implausible thesis according to which all mediation is equally distorting or arbitrary. Some distortions are reducible. Better instruments, better sampling, better conceptual distinctions, and better error models can remove many sources of local misrepresentation. But the elimination of all mediation is not one of the options available to finite inquiry. Mediation is not an accidental imperfection laid over an otherwise transparent relation; it is constitutive of how finite access occurs at all.

3.4 Irreducible error

The claim that total epistemic error remains irreducibly non-zero should also be distinguished from any suggestion that inquiry is hopeless or unstructured. GCFAE does not claim that error is boundless. It claims that the total error budget of finite inquiry can be reduced, redistributed, and disciplined, but not driven to zero.

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Irreducible error arises from the finitude of agents and practices: partial access, limited resolution, model misspecification, theory-ladenness, underdetermination, and horizon effects. Even when local errors are corrected, finite knowers do not arrive at full possession. This non-zero floor is not a contingent defect. It is one of the conditions that make inquiry possible. Only where misalignment is possible can correction and criticism have point.

3.5 Reason as disciplined revision

Within GCFAE, reason is neither an occult faculty nor a merely retrospective honorific for whatever survives. Reason is a family of update procedures that tends, under suitable conditions, to improve epistemic alignment. It includes anomaly-sensitivity, hypothesis revision, intervention-guided correction, cross-route testing, inferential integration, and disciplined responsiveness to counterevidence.

Reason is therefore normative because it is corrigible. Its standards arise from the possibility of being wrong in ways that can be publicly tested and revised. Rationality is not what floats free of the world; it is what remains answerable to it.

4 Formal Core

The formal architecture of GCFAE is intended as a clarificatory heuristic rather than a fully axiomatized reduction of epistemology to mathematics. Its function is to display the dependency relations among mediation, error, revision, and alignment with precision. The formalism expresses the framework's central commitments without pretending to settle all metric or semantic issues in advance.

$$O_A[n] = \Pi_A(E[n]) \tag{1}$$

$$\mathrm{enc}_A(O_A)[n] = T_A(O_A[n]) \tag{2}$$

$$M_A[n+1] = U_A(M_A[n], \mathrm{enc}_A(O_A)[n], C_A, \Delta_A[n]) \tag{3}$$

$$\varepsilon_{\mathrm{tot}} > 0 \tag{4}$$

$$V_A[n] = \mathrm{Align}(M_A[n], \theta_W \mid D) \tag{5}$$

$$\mathbb{E}[V_A[n+1] \mid \mathrm{Rsn}_A, D] > \mathbb{E}[V_A[n] \mid D] \tag{6}$$

Each symbol deserves interpretation.

Environmental input $E[n]$**.** $E[n]$ denotes the relevant environmental input at stage n. It is not the totality of reality. It is the currently available segment of world-involving interaction relevant to the inquirer's epistemic task.

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Projection Π_A. Equation (1) states that observation is always filtered by an agent-relative projection function Π_A. This function represents sensory limitation, instrumental bandwidth, sampling constraints, and selective exposure. Observation is therefore never equivalent to the world as such. GCFAE rejects naive direct realism at the level of epistemic access.

Encoding T_A. Equation (2) adds a second layer of mediation. Raw observation must be encoded into a format metabolizable by the agent's conceptual, linguistic, or computational architecture. This includes categorization, measurement conventions, symbolic registration, preprocessing, and theory-laden framing. Observation is not merely received; it is rendered usable.

Update U_A. Equation (3) models revision. The next representation depends on the prior model, newly encoded observation, background constraints C_A, and an anomaly-sensitive term $\Delta_A[n]$.

$\Delta_A[n]$ should not be interpreted as a merely numerical discrepancy term. It is the formal mark of *resistance*: the point at which the world's causal-structural organization refuses full accommodation to the agent's present model. In empirical science this may appear as predictive error, failed intervention, measurement conflict, unexplained residual, or instability across access routes. In all cases the philosophical point is the same: revision becomes rational because the world does not always comply with projection.

The non-zero error floor. Equation (4) expresses the central axiom of fallibilism. Total epistemic error remains strictly greater than zero. Perfect isomorphism between model and world is structurally unavailable to finite knowers. This codifies epistemic humility without collapsing into skepticism.

Alignment. Equation (5) defines epistemic alignment as domain-indexed. A model can be highly aligned for one set of tasks and poorly aligned for another. This blocks the temptation to overstate local success as global adequacy. Alignment is not monolithic.

Expected improvement. Equation (6) expresses the framework's main normative claim. Under repeatable conditions, disciplined reasoning tends to improve expected alignment. The claim is deliberately modest. It does not assert monotonic progress, guaranteed convergence, or immunity to regression. It says only that regulated revision provides better expected epistemic performance than leaving models unrevised under the same domain conditions.

Several assumptions underlie this inequality. The environment must be structured enough for correction to be non-random. Update procedures must remain responsive rather than dogmatically insulated. The domain must display enough stability for learning and

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transfer to be possible. And evaluative standards must be publicly criticizable. Where these conditions fail, the tendency claim weakens.

Why this is not pseudo-mathematics. The formalism earns its place because it makes explicit what many epistemologies leave obscure: that observation is projected, uptake is encoded, revision is anomaly-sensitive, error is irreducible, and normativity is tied to expected improvement under constraint. The aim is not formal performance for its own sake, but argumentative clarity.

5 GCFAE in the Philosophical Landscape

5.1 Against constructive empiricism

Constructive empiricism holds that acceptance of a scientific theory need involve no more than belief in its empirical adequacy (Fraassen 1980). This position captures an important caution: success in saving observable phenomena should not be naively inflated into metaphysical possession of the unobservable. GCFAE shares that caution.

Yet empirical adequacy alone is too thin to capture the full normative structure of inquiry. Two models may fit existing observations equally well while differing sharply in interventional reliability, route-diverse robustness, and cross-domain transfer. Scientific practice often distinguishes between such models not only by asking which “saves the phenomena”, but by asking which supports manipulation, error-correction, and stability under changed conditions. These are not merely pragmatic conveniences. They are signs that a model is better aligned with world-side constraint.

GCFAE therefore accepts the empiricist rejection of premature triumphalism while refusing to reduce epistemic achievement to observational fit. Inquiry aims not merely to preserve appearances, but to become more answerable to the world that generates and frustrates those appearances.

5.2 Against convergent realism

Convergent realism interprets scientific progress as movement toward a uniquely correct theory, or at least toward an endpoint of convergence. Laudan’s critique remains decisive against simple versions of that view (Laudan 1981). The history of science provides little support for the claim that mature inquiry steadily approaches one final ontology.

GCFAE accepts this anti-finalist lesson. Progress need not mean convergence toward a final theory. But rejecting convergence does not require abandoning realism. The alternative defended here is *alignment without finalism.* Distinct models may track overlapping aspects of world-side constraint while remaining non-identical in ontology, idealization strategy, or scale. Progress can consist in stronger predictive stability, interventional success, route-independence, and explanatory integration without implying terminal convergence.

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This makes room for pluralism without relativism. Scientific advance is compatible with multiple comparably aligned models, provided they remain constrained, corrigible, and publicly assessable.

5.3 Beyond standard structural realism

GCFAE is closest to structural realism because it regards relational and organizational features of the world as epistemically central (Worrall 1989; Ladyman 1998). What inquiry tracks most robustly are not hidden essences grasped directly, but stable patterns of dependence, invariance, and causal organization.

Still, GCFAE departs from standard structural realism in three ways.

First, it places intervention at the center. Structures matter epistemically not merely because they are retained across theory change, but because they support manipulation, resistance, and correction (Hacking 1983; Woodward 2003).

Second, it emphasizes route-independence. A representation gains weight not just by fitting one theoretical lineage, but by surviving across partially independent modes of inquiry.

Third, it embeds structure within a fallibilist account of mediation. Structure is not something possessed from nowhere; it is what finite inquiry can increasingly align with under non-eliminable error.

In this respect GCFAE may be understood as a practice-sensitive, intervention-aware development of structural realism rather than a restatement of classical epistemic structural realism.

5.4 Against subject-first and correlationist pictures

Post-Kantian thought often risks a correlationist impasse: if thought only ever encounters the correlation between itself and being, how can it account for a world that is not exhausted by its own conditions (Meillassoux 2008)? GCFAE avoids this impasse without claiming direct access to the absolute. It does so through corrigible resistance. The world's non-compliance with our present models is not generated by our endorsement of it. Resistance is the sign that mediation is not enclosure. We never step outside mediation, but mediation remains world-responsive because what it mediates can push back.

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6 Route-Independence, Weighting, and Public Criticism

6.1 Why route-independence matters

Route-independence is indispensable because agreement without independence is indistinguishable from shared distortion. Two methods may converge on the same result merely because they inherit the same calibration defect, conceptual bottleneck, training regime, or incentive structure. Mere convergence is therefore insufficient.

A stronger criterion is required.

> **Partial independence of routes.** Two routes of inquiry R_1 and R_2 are partially independent if and only if their convergence on a representation cannot be fully explained by any of the following: (i) a shared method or algorithmic procedure; (ii) a shared incentive structure or social pressure; (iii) a shared conceptual bottleneck that both routes were constructed to preserve; (iv) a shared sensory or interpretive pipeline through which both routes receive their primary inputs.

This criterion operationalizes route-independence without pretending to theory-neutral purity. Independence is itself theory-laden, but that does not make it empty. What matters is not absolute independence, but diversification of failure modes. A result that survives different methods, instruments, operators, and error-profiles carries greater epistemic weight because it is less plausibly the artifact of one shared defect.

6.2 The social dimension of objectivity

Route-independence is not merely an individual virtue. In actual inquiry it is often institutionally realized through replication, adversarial criticism, methodological diversity, distributed expertise, and peer scrutiny. Public criticism matters because convergence without partial independence cannot be distinguished from coordinated distortion. This is why social epistemology is not an external add-on to GCFAE. It is one of the principal mechanisms through which route-independence becomes operational.

The point is not that consensus guarantees truth. Consensus is epistemically significant only when produced under conditions that diversify bias, test fragility, and expose hidden assumptions. Community inquiry matters because it can increase the number and heterogeneity of epistemic routes.

6.3 Weighting the dimensions of alignment

Because V_A is multidimensional, conflict among its dimensions must be addressed explicitly. Suppose one model predicts better in a narrow regime, while another preserves structure

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more robustly across wider contexts. Which is more aligned?

There can be no universal answer independent of domain. In some domains, immediate predictive discrimination justifiably carries highest weight. In others, broader transferability or interventional reliability matters more. GCFAE therefore treats alignment as a domain-indexed weighted ordering rather than a universal metric.

This is not a defect. It reflects a sober fact about inquiry: different epistemic tasks impose different success conditions. What preserves rigor is that weighting itself must remain publicly arguable, revisable, and constrained by the aims of the domain. Weighting is not arbitrary preference; it is part of the normative structure of inquiry.

6.4 Underdetermination and alignment-equivalence

The framework openly admits that multiple models may sometimes be comparably aligned within a domain:

$$\{M_{A_A}^1, M_{A_A}^2, \ldots, M_{A_A}^k\} \in \mathcal{C}_D$$

where $\mathcal{C}_D$ denotes an alignment-equivalence class for domain D.

This does not imply arbitrariness. Equivalence in one domain need not persist in another. Models with similar predictive fit may diverge under intervention, under extrapolation, or across routes of inquiry. And where primary alignment remains near-equivalent, secondary criteria such as simplicity, inferential economy, computational tractability, and cross-domain portability can guide rational choice.

Underdetermination therefore weakens naive mirroring theories of truth, but it does not abolish world-tracking. It motivates a more modest realism: one tied to constrained, domain-relative adequacy rather than to unique global correspondence.

7 Illustrative Cases

7.1 Physics: electron models and multi-route constraint

No scientist observes an electron without mediation. Evidence comes through cloud chamber traces, detector outputs, scattering patterns, and spectroscopic regularities. These routes differ in apparatus and inferential load, yet they exhibit non-trivial convergence under experimentally exploitable conditions. The key epistemic fact is not that inquiry acquires naked access to the electron in itself. It is that diverse access routes constrain representations in mutually reinforcing ways.

Within GCFAE, electron models earn credibility not merely because they fit observations, but because they support interventions, unify otherwise disjoint phenomena, and survive route-diverse testing. Their success is stronger than empirical adequacy alone and weaker than metaphysical possession. That intermediate status is precisely what alignment

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captures.

7.2 Biology: gene regulatory networks

Biological inquiry proceeds through incomplete and noisy routes: sequencing, perturbation studies, expression analysis, imaging, and computational reconstruction. Gene regulatory models are never perfect mirrors. They idealize, omit, and compress. Yet some models are better than others because they preserve relevant dependencies, support manipulation, and generalize across experimental contexts.

Again, progress need not require one final perfect network description. Distinct models may remain comparably aligned for different purposes. What matters is whether they survive perturbation, support explanation, and remain corrigible under new evidence. Alignment is therefore richer than mere data fit and compatible with task-relative pluralism.

7.3 Machine learning: fit versus alignment

Machine learning offers a contemporary analogue. A model may display excellent training performance while failing under distribution shift, adversarial perturbation, or transfer. Another model with similar local fit may generalize more robustly because it captures deeper regularities of the data-generating environment.

GCFAE clarifies what is epistemically at stake. A model that merely memorizes local correlations may achieve narrow success without robust alignment. By contrast, a model that remains responsive to anomalies and supports wider generalization is better aligned with the constraints operative in its task environment. This illustrates a broader moral: not all success deserves equal epistemic endorsement. What matters is success under resistant, variable, and criticizable conditions.

The comparison to predictive-processing and active-inference frameworks is suggestive (Friston 2010), but GCFAE is not reducible to a computational neuroscience program. Its aim is philosophical: to explain why error-sensitive updating can underwrite world-responsiveness without presupposing direct access.

8 Normativity Without Reification

A familiar objection to non-substantialist accounts of normativity holds that if knowers are fully conditioned, embedded, and causal, then rational normativity must collapse into descriptive regularity. GCFAE rejects that conclusion.

Normativity does not require an unconditioned chooser standing outside the world. It requires only that representations can be better or worse relative to constraints that are not constituted by present endorsement. Where inquiry is corrigible, distinctions between

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good and bad revision are not optional overlays. They are constitutive of the practice itself.

In that respect GCFAE is closer to inferentialist and practice-based accounts of normativity than to faculty-theoretic rationalism (Brandom 1994). Yet it differs from purely discursive internalism by insisting that inferential practice remains answerable to resistance beyond communal endorsement. At the same time, it does not make agency explanatorily prior in the manner of self-constituting accounts (Korsgaard 2009). Constraint is prior; agency operates within it.

This helps explain how fallibility and normativity fit together. The non-zero floor of mediation makes inquiry corrigible; corrigibility imposes standards; those standards are what normativity names. Normativity is not what survives despite finitude. It is what emerges because finitude makes revision necessary.

9 Objections and Replies

Objection 1. *If mediation is irreducible, skepticism follows. Without unmediated access, all that remains is internal coherence or adaptive utility.*

Reply. This objection assumes that objectivity requires elimination of mediation. GCFAE denies that assumption. World-tracking does not require identity access; it requires disciplined answerability to resistance, failed intervention, route-diverse correction, and reproducible friction. Mediation is compatible with objectivity whenever representations remain publicly revisable under conditions not reducible to preference.

Objection 2. *The status of θ_W is ambiguous. If it cannot be directly specified, why not eliminate it as a metaphysical fiction?*

Reply. θ_W is not posited as a fully describable hidden essence. It is posited because without some world-side source of resistance and invariance, stable interventional success, reproducible failure, and cross-route correction become mysterious. The posit is minimal and explanatory. It is realism about constraint, not about exhaustive possession.

Objection 3. *Epistemic alignment lacks metric rigor. A plural, domain-sensitive notion of success looks too vague to support serious epistemology.*

Reply. The demand for a single context-free epistemic scalar is misplaced. Many legitimate evaluative concepts are multidimensional yet rigorous. V_A is disciplined by public criteria: predictive adequacy, intervention reliability, structural preservation, and route-independence. Its weighting varies by domain, but that weighting is itself open to criticism and revision. Rigor does not require false homogeneity.

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Objection 4. *The account collapses into pragmatism. Once intervention and practical success are central, "what works" simply becomes "what is true enough".*

Reply. Local usefulness is not sufficient for alignment. A representation may be instrumentally convenient yet brittle, route-dependent, and non-transferable. GCFAE does not equate utility with truth. It treats some practical successes as epistemically significant because they provide evidence of constraint-sensitive fit. The world matters because it can frustrate our conveniences.

Objection 5. *Underdetermination defeats realism. If incompatible models can be equally aligned in a domain, realism loses its footing.*

Reply. Underdetermination defeats only a strong realism that requires one uniquely privileged total representation. GCFAE defends a weaker but more defensible realism. Distinct models may track overlapping aspects of world-side constraint without collapsing into subjectivism. What is realist here is not possession of one final vocabulary, but answerability to a world that resists arbitrary reformulation.

Objection 6. *If normativity is generated by constraint and correction, then it seems merely descriptive. Why call it normativity at all?*

Reply. Normativity is present whenever there are standards by which revision can fail. Constraint does not merely describe what happens; it distinguishes better from worse updating under publicly assessable conditions. To participate in inquiry at all is already to inhabit such distinctions. In that sense normativity is constitutive, not optional.

10 What GCFAE Does and Does Not Claim

The framework does *not* claim:

(a) that finite knowers can escape mediation;

(b) that total epistemic error can be eliminated;

(c) that science converges toward one final theory;

(d) that structure is known independently of practice;

(e) that social consensus by itself guarantees objectivity.

The framework *does* claim:

(a) that some world-side source of constraint is required to explain reproducible success and failure;

(b) that knowledge is better understood as graded alignment than as absolute possession;

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(c) that reason is an error-sensitive family of revision procedures rather than a metaphysically privileged faculty;

(d) that route-independence and public criticism are central to objectivity under mediation;

(e) that progress is possible without finalism.

These distinctions matter because much disagreement in epistemology results from asking one framework to answer questions it was never designed to answer. GCFAE is neither a triumphant realism nor a softened anti-realism. It is a fallibilist account of how world-tracking remains intelligible for finite knowers.

11 Conclusion

The central mistake shared by epistemic inflation and skeptical collapse is the assumption that finite inquiry must end either in possession or in nullity. Against both extremes, this paper has argued that knowledge is better understood as graded epistemic alignment under irreducible mediation and irreducible, though locally reducible, error.

The framework contributes six main claims.

First, world-responsiveness does not require unmediated access.

Second, a minimal realist posit of world-side causal-structural constraint, θ_W, is needed to explain resistance, interventional success, and cross-route invariance.

Third, epistemic achievement is more adequately modeled as multidimensional alignment, V_A, than as bare empirical adequacy or truth-possession.

Fourth, reason is best understood as disciplined revision under constraint rather than as a self-guaranteeing rational faculty.

Fifth, route-independence and public criticism are not optional social ornaments on inquiry, but part of the mechanism by which objectivity is stabilized under mediation.

Sixth, scientific and ordinary progress need not imply convergence toward one final representation; they can instead consist in more robust alignment across domains under finite conditions.

In compressed form, the thesis is this:

> Finite knowers never escape mediation, but mediation does not entail nullity. Because world-side constraint resists and corrects finite representation, reason can function as a disciplined engine of revision, and knowledge can be understood as increasing epistemic alignment under irreducible error.

Several avenues remain open. A fuller formal treatment of alignment would require more explicit work on weighting, ordering, and inter-domain comparison. The institutional

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conditions that maximize route-independence merit further development in conversation with social epistemology. And the relation between constraint-first epistemology and broader debates in philosophy of mind, comparative philosophy, and the sciences of cognition deserves sustained exploration.

Even so, the present conclusion is already significant. Once the fantasy of possession is abandoned, objectivity need not be abandoned with it. It survives because representations can fail; because that failure can be tracked across partially independent routes; and because finite knowers can revise their models under public, repeatable, world-constrained conditions. Objectivity survives not because mediation is overcome, but because resistance is.

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References

Boyd, Richard (1990). "Realism, Approximate Truth, and Philosophical Method". In: *Scientific Theories*. Ed. by C. Wade Savage. Minneapolis: University of Minnesota Press, pp. 355–391.

Brandom, Robert (1994). *Making It Explicit: Reasoning, Representing, and Discursive Commitment*. Cambridge, MA: Harvard University Press.

Fraassen, Bas C. van (1980). *The Scientific Image*. Oxford: Clarendon Press.

10.5281/zenodo.19205869

Friston, Karl (2010). "The Free-Energy Principle: A Unified Brain Theory?" In: *Nature Reviews Neuroscience* 11.2, pp. 127–138.

Hacking, Ian (1983). *Representing and Intervening*. Cambridge: Cambridge University Press.

Korsgaard, Christine M. (2009). *Self-Constitution: Agency, Identity, and Integrity*. Oxford: Oxford University Press.

Ladyman, James (1998). "What Is Structural Realism?" In: *Studies in History and Philosophy of Science Part A* 29.3, pp. 409–424.

Laudan, Larry (1981). "A Confutation of Convergent Realism". In: *Philosophy of Science* 48.1, pp. 19–49.

Meillassoux, Quentin (2008). *After Finitude: An Essay on the Necessity of Contingency*. London: Continuum.

Popper, Karl R. (1959). *The Logic of Scientific Discovery*. London: Hutchinson.

Woodward, James (2003). *Making Things Happen: A Theory of Causal Explanation*. Oxford: Oxford University Press.

Worrall, John (1989). "Structural Realism: The Best of Both Worlds?" In: *Dialectica* 43.1–2, pp. 99–124.

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Mind as Information Horizon: From Primordial Difference to Expertise Formation on the Discrete Causal Graph

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What the chapter states. *"This paper derives a complete architectonic chain from primordial difference through the formation of world-class expertise, within a discrete causal-informational ontology." "Every claim is tagged with its formal status—axiom (Ax), theorem (Th), derivation (Dr), conjecture (Cj), or working form (Wf)."* (10.5281/zenodo.19640361)

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Falsifier(s). *"the theory's universal falsification condition is stated explicitly: any demonstration that* $\varepsilon = 0$ *is achievable collapses the entire architecture."*

Read before. Ch. 1 *Written by AI. Still True.*; Ch. 2 *The Readout Condition*; Ch. 3 *Readout Genesis Standalone Synthesis*; Ch. 27 *When AI Expands Human Potential*; Ch. 39 *The Standalone Scholar*.

Feeds into. — (no chapter in this book references this one, per the KG).

Mind as Information Horizon: From Primordial Difference to Expertise Formation on the Discrete Causal Graph

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April 18, 2026

Abstract

What must reality minimally be if knowing—and therefore expertise—are to be possible at all? This paper derives a complete architectonic chain from primordial difference through the formation of world-class expertise, within a discrete causal-informational ontology. The argument proceeds in five movements. First, from three postulates (distinguishability, causal closure, finite propagation speed), the telegraph equation on a finite discrete graph is shown to be the unique minimal dynamics governing all structural persistence and decay. Second, the Causal Body Framework and its No-Go theorems are used to derive the *Agency-Horizon Theorem* (Th-BH): every agent satisfying the framework's axioms exhibits the five structural properties of an information horizon—event horizon, privacy, emission constraint, partial self-access, and information persistence. This result is a formal theorem, not a metaphor. Third, experience and knowerhood are shown to arise exclusively from the dynamic coupling between the information-horizon agent (mind-core) and embodied brain-body process; neither component alone suffices. Fourth, epistemic genesis is derived from the structural impossibility of zero total observation error ($\varepsilon_{\text{tot}} > 0$ for every finite knower), yielding a formal chain from selection through error-correction to legitimate knowledge. Fifth, expertise is characterized as the mature regime of this coupled system: high persistence, efficient correction, and expanding generative capacity under world-resistance. Every claim is tagged with its formal status—axiom (Ax), theorem (Th), derivation (Dr), conjecture (Cj), or working form (Wf)—and the theory's universal falsification condition is stated explicitly: any demonstration that $\varepsilon_{\text{tot}} = 0$ is achievable collapses the entire architecture. The resulting position, termed *informationalism*, constitutes a third metaphysical stance in which the fundamental is neither matter nor mind but the causal ordering of dif-

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ference, relation, and constraint on a finite discrete graph.

Keywords: informationalism; information horizon; agency; discrete ontology; epistemic genesis; expertise formation; primordial difference; telegraph equation; mind-core; causal body framework

1 Introduction

The governing question of this paper is severe in form but narrow in scope: what must reality minimally be, if anything like knowing—and therefore anything like expertise—is to become possible at all?

The answer defended here is that knowing requires a world built from difference, persistence, and causal ordering on a finite discrete graph, and that any system capable of knowing within such a world must exhibit the structural properties of an information horizon. This is not a metaphorical claim. The paper derives the information-horizon structure of the knower as a formal theorem from stated axioms, and then shows that experience, knowledge, and expertise arise only when the information-horizon agent is dynamically coupled with an embodied brain-body system.

The argument proceeds through a strict architectonic chain. Sections 2 establishes the ontological floor: primordial difference, retained efficacy, and the derivation of the telegraph equation as the unique minimal dynamics on the discrete causal graph. Section 3 introduces the Causal Body Framework and derives the Agency-Horizon Theorem (Th-BH), proving that every admissible agent possesses five information-horizon properties. Section 4 models the coupling between the information-horizon agent (mind-core) and the brain-body system, showing that experience is the product of neither component alone but of their active loop. Section 5 derives epistemic genesis from the structural impossibility of perfect observation. Section 6 formalizes knowledge as a distributed, corrigible, capacity-generating triple. Section 7 characterizes expertise as the mature regime of this coupled architecture. Section 8 derives failure modes as structural pathologies. Section 9 maps the proposal against top-tier empirical and phenomenological literature, demonstrating strong convergence at the levels of embodied cognition, reconstructive memory, and expertise formation, while marking the information-horizon ontology as a formal extension that current science neither confirms nor rules out. Section 10 states open frontiers and proof obligations. Section 11 concludes.

Three methodological commitments govern every section. First, every major claim is tagged with its formal status: axiom (Ax), theorem (Th), derivation requiring bridging assumptions (Dr), conjecture not yet established (Cj), or working form requiring calibration (Wf). Second, the theory states its own falsification condition: if any observer can be shown to achieve $\varepsilon_{\text{tot}} = 0$ (zero total observation error), the entire architecture collapses. Third, this paper is philosophical inquiry, not empirical report; it proposes an ontological floor and derives its consequences, without claiming that current instruments have verified the deepest layer of the architecture.

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2 Ontological Floor: Primordial Difference to Discrete Dynamics

2.1 Primitives

The ontological floor requires five conditions, each necessary in the architectonic sense: removal of any one collapses the possibility of a determinately structured, discriminable, directional, and temporally intelligible world.

Thesis 1: Primordial Difference (P_0). *(Ax)* The minimal ontological condition is non-identity: one state is not another state. Without difference, nothing becomes determinately *this* rather than *that*, and world-formation cannot begin. The claim concerns reality, not language; it is not semantic contrast but ontological non-identity [1].

Thesis 2: Retained Efficacy ($\tau_c > 0$). *(Ax)* Prior difference must have non-zero influence on subsequent states. If what has been is ontologically null for what follows, no structure survives, no causality obtains, and no memory is possible. The parameter τ_c formalizes this as a non-zero carry-over; it is not yet a physical constant but a minimal structural requirement [1].

Thesis 3: Distinguishability Under Persistence (P_1). *(Ax)* Difference becomes world-effective only where it is discriminable, and discriminability requires persistence. Indiscriminable difference is, for world-forming purposes, equivalent to nullity. The discriminability at issue is ontological, not epistemic: it does not require an observer, only that non-identity be structurally available for further consequences [1].

Thesis 4: Asymmetry of World-Forming Transition. *(Ax)* Given distinguishable states under persistence, world-forming transition requires asymmetry. If every transition were perfectly equivalent to its reversal, no non-arbitrary direction would arise. This is prior to the Second Law of Thermodynamics; it is the ontological condition that makes causal ordering possible [1].

Thesis 5: Generated Temporal Order. *(Dr)* Time is not a container but the ordered sequence of asymmetric transitions among distinguishable states under persistence. Before-and-after is generated from process, not imposed externally. This thesis is derived from the conjunction of Theses 1–4 [1].

2.2 The Three Postulates and the Telegraph Equation

When the five ontological theses are supplemented by three physical postulates, a unique minimal dynamics is forced.

Postulate 1 (Distinguishability). *(Ax)* Distinct world-states are distinguishable by admissible observers [2, §P1].

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Postulate 2 (Causal Closure). *(Ax)* Influences travel via causal chain only; no acausal influence is admitted [2, §P2].

Postulate 3 (Finite Propagation Speed). *(Ax)* $v = \sqrt{D/\tau_c} < \infty$; no infinite-speed propagation [2, §P3].

From P1–P3 on a finite discrete graph with normalized restricted Laplacian L_R (positive semi-definite, spectrum $\in [0,2]$), the unique minimal dynamics is the **Mother Equation** (MQ.08):

$$V[n+1] = V[n] + \Delta\theta\big(-\gamma\, V[n] - D_s\, L_R \cdot X[n]\big) \qquad \text{(MQ.08)}$$

where $V[n]$ is the state vector at discrete step n, γ is the damping coefficient, D_s is the diffusion coefficient, and $\Delta\theta$ is the discrete step size. This is a telegraph-like equation governing propagation, damping, and structural persistence across all scales [2, §P3.3].

(Th) This equation is not chosen; it is forced. Any dynamics satisfying P1–P3 on a finite discrete graph must reduce to MQ.08 or a special case thereof.

2.3 Eigenmodes, Decay, and the Emergence of Matter

(Th) Stable or semi-stable patterns on the graph are eigenmodes: eigenvectors φ_k of L_R. Anything that appears as a persisting structure is a temporarily persistent eigenmode.

(Th-5: Impermanence Theorem) Every eigenmode decays: $|a_k[n]| \le |a_k[0]| \cdot e^{-\gamma_k \cdot n \cdot \Delta\theta}$. No structure persists forever. Decay rates differ, but decay is universal [2, §5.0].

Definition: Matter. *(Df)* Matter is stabilized informational organization: persistent eigenmodes coupled into bounded systems under causal constraint. Matter is not a fundamental substance; it is information under persistence and coupling. This definition admits multiple classes of material organization, a distinction that becomes critical in the next section.

3 The Agency Horizon Theorem

3.1 The Causal Body Framework

The Causal Body Framework (CBF) formalizes the complete causal chain through which a living agent interacts with the world [2, §P3.6]:

$$B[n] \to H_{\text{body}}[n] \to N[n] \to S[n] \leftrightarrow A[n] \to \pi[n] \to U[n] \to B[n+1] \qquad (1)$$

where B is the external world-state, H_{body} is the body (hormonal, structural, metabolic), N is the nervous system, S is the sensory interface, A is the internal agency state, π is the selected action policy, and U is the executed output. The timescale ordering is $\tau_H \gg \tau_N \approx \tau_S \approx \tau_B$.

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The discrete dynamics are:

$$H_{\text{body}}[n+1] = \Phi_H(B[n],\, H_{\text{body}}[n]) \tag{2}$$
$$N[n+1] = \Phi_N(B[n],\, H_{\text{body}}[n],\, N[n]) \tag{3}$$
$$S[n+1] = \Phi_S(S[n],\, N[n],\, H_{\text{body}}[n]) \tag{4}$$
$$A[n+1] = \Phi_A(S[n],\, A[n]) \quad \text{(brain-internal only)} \tag{5}$$
$$\pi[n+1] \in \arg\min_{\pi\in\Pi_{\text{feas}}} L_{\text{sel}}(S, H_{\text{body}}, \pi) \tag{6}$$
$$U[n+1] = \text{Exec}(\pi[n+1]) \tag{7}$$
$$B[n+1] = \Phi_B(B[n],\, U[n+1]) \tag{8}$$

3.2 CBF No-Go Theorems

Three No-Go theorems follow directly from the causal closure postulate (P2) applied to the body chain:

CBF-Th1. *(Th)* No direct agency inputs into $\dot{B}$, $\dot{H}$, $\dot{N}$. Agency cannot directly actuate the external world, the body, or the nervous system except through the chain $A \to \pi \to U \to B$.

CBF-Th2. *(Th)* No direct $B \to A$, $H \to A$, $N \to A$ edges. The internal agency state receives input only through the sensory interface S.

CBF-Th3. *(Th)* Horizon constructs H_A introduce no new causal edges. The encoding of the agency horizon is informational, not causal.

Consequence: Agency constrains feasibility—it selects from feasible action policies—but it never directly actuates physiology. This is not a limitation of technology; it is a theorem of causal structure.

3.3 Derivation of Th-BH: The Agency-Horizon Theorem

Theorem (Th-BH). *(Th)* Every agent A satisfying the CBF axioms exhibits all five structural properties of an information horizon [2, §6.7].

The five properties, each derived from the No-Go theorems and the causal closure postulate, are:

Property 1: Event Horizon (H_{EH}). $H_{\text{EH}}(A) = \{\text{states in } A[n] \text{ with no direct external path}\}$. The agent possesses internal states that are structurally inaccessible to any external observer. This is not contingent secrecy but structural: the causal chain $B \to H_{\text{body}} \to N \to S \to A$ does not admit a reverse path $A \to S \to N \to B$ that would expose the full interior [2, §P3.6.5].

Property 2: Privacy (Th-PR). *(Th)* $\forall B : \nexists$ admissible path $\to A[n]$ directly. No external observer B can directly read, copy, or access A's internal state. Every path must pass through the output channels $U[n]$ only. *Proof sketch:* CBF No-Go Th2 prohibits direct $B \to A$ edges; CBF-Th3 prohibits horizon constructs from creating new causal edges; therefore all external access to A is mediated through $U[n]$ and $\mathrm{Aff}_{AB}[n]$, which are lossy projections of $A[n]$ [2, §6.7].

Property 3: Emission Constraint (Th-EM). *(Th)* The output channels of A are limited to $\{U[n], \mathrm{Aff}_{AB}[n]\}$ only. The agent's output to the world is bounded by encoding constraints and action channel capacity. What exits the agent is strictly limited and mediated, analogous to the Hawking radiation of a gravitational horizon: structured emission from a boundary, not full transmission of the interior [2, §6.7].

Property 4: Partial Self-Access (Th-PSA). *(Th)* $\varepsilon_{\text{self}} \geq 0$ always. The agent cannot fully observe its own internal state. Self-referential observation introduces irreducible noise, formally captured by the η_{self} term in the observer pipeline equation MQ.05. The agent's introspective access is structurally bounded, not merely empirically limited [2, §P3.4.2].

Property 5: Information Persistence (Th-IP). *(Th)* When mode $a_k[n] \to 0$ (lifecycle stage S5: ceasing), information is not annihilated. It persists in coupled modes, records, and the causal traces left in the world-graph. Ceasing $\neq$ annihilation. This is the formal resolution of the information paradox within the discrete causal framework: the append-only record structure (NoMerge axiom) guarantees that information survives mode cessation [2, §P3.6.5].

3.4 The Structural Isomorphism

The five properties above are not analogies drawn from astrophysics. They are formal consequences of the CBF axioms applied to any agent on the discrete causal graph. The structural isomorphism between an information-horizon agent and a gravitational black hole is a *theorem*, not a metaphor [2, §6.7]:

Property	**Gravitational Black Hole**	**Information-Horizon Agent**
Event Horizon	Light cannot escape	Internal states have no direct external path
Privacy	Interior inaccessible to external observer	$\forall B : \nexists$ admissible path $\to A[n]$
Emission	Hawking radiation	$U[n]$ and $\mathrm{Aff}_{AB}[n]$ only
Partial Self-Access	Observer at horizon: self-knowledge limited	$\varepsilon_{\text{self}} \geq 0$ always
Information Persistence	Information paradox resolution	Ceasing $\neq$ annihilation (NoMerge)

Anti-misreadings. The claim is not that the agent is a literal astrophysical black hole, not that it is a gravity well in a naive physical sense, not that it is a supernatural ghost-

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substance, and not that it is reducible to metaphorical depth language. The claim is that the formal structure is the same: both entities belong to the same class of information-horizon organization within the causal graph [2, §6.7].

4 Brain–Mind-Core Coupling and the Genesis of Experience

4.1 The Coupling Model

The information-horizon agent (hereafter *mind-core*, H) is not, in isolation, the lived knower. An information horizon without sensory input and motor output is an isolated singularity—structurally complete but experientially void. Experience (E) and knowerhood (K) arise exclusively from the dynamic, closed-loop coupling between the mind-core and the brain-body system (B).

The minimal coupling equations are *(Dr)*:

$$B[t+1] = F(B[t]) + C_H(H[t]) \tag{9}$$

$$H[t+1] = G(H[t]) + C_B(B[t]) \tag{10}$$

$$E[t] = R(B[t],\, H[t]) \tag{11}$$

where F governs the brain-body's own dynamics, G governs the mind-core's persistence mechanics, C_H is the coupling from mind-core to brain, C_B is the coupling from brain to mind-core, and R is the resultant function generating the experiential event.

Core Result. *(Dr)* Experience is not identical to brain activity alone and not identical to mind-core activity alone. Experience is generated by the active coupling between them. The brain without the mind-core lacks deep integration, temporal continuity, and the structural privacy established by Th-PR. The mind-core without the brain lacks embodied grounding, sensory content, and motor action. The knower is the coupled regime.

4.2 Signal Is Not Information

A critical distinction follows from the coupling model. Observable changes in brain tissue—synaptic modifications, ensemble activations, oscillatory patterns—are causally indispensable for the expression and manipulation of cognition. But these observable changes should not be naively identified with information itself *(Dr)*.

The distinction has three levels. Physical *signal* is the observable change in tissue or cells (strongly supported by current science). *Generative code* is the structured capacity that allows the system to reconstruct models rather than merely replay stored contents (strongly suggested by reconstructive and generative views of cognition, but not fully

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localized). The *runtime model* is the presently enacted model of world, self, or task, generated under current input and system state (strongly supported as process, not static retrieval).

Signal is not identical to code; code is not identical to runtime model; and runtime model is not identical to the totality of mind. This layered distinction is consistent with the contemporary shift in cognitive neuroscience away from static storage models [6, 7, 8] and toward dynamic, embodied, reconstructive accounts of memory and cognition.

4.3 Non-Neural Cognition as Evidence

The claim that the brain is an executional interface rather than the exhaustive ontological repository of generative code is corroborated by the extensive study of non-neural cognition *(Dr)*. Organisms entirely lacking nervous systems demonstrate complex information processing, navigation, memory-like retention, and anticipatory behavior. Bacterial chemotaxis integrates signals and executes behaviorally relevant decisions under uncertainty [15]. The slime mold *Physarum polycephalum* navigates mazes, anticipates periodic changes, and solves multi-armed optimization problems without a single synapse [16]. Plant behavior supports signal integration, resource-sensitive adjustment, and decision-like adaptation [17]. Sponges coordinate whole-body sensing and response despite lacking neurons [18].

These findings do not prove a black-hole-like mind-core in non-neural organisms. They do establish that information-guided selection, retention, and structural capacity are fundamental properties of causal-informational organization, not exclusive features of neural tissue.

5 Epistemic Genesis: From Bounded System to Knower

5.1 Transition Conditions

A coupled brain–mind-core system does not automatically possess knowledge. Epistemic genesis requires a sequence of transition conditions, each building on the prior *(Dr)*:

Bounded System. When some patterns persist more strongly than their surroundings, an inside/outside distinction emerges. The coupled system becomes sufficiently bounded to maintain structural integrity.

Proto-Horizon. The bounded system acquires a field of possible relevance. Not everything matters equally; a relevance gradient appears.

Salience Selection. Because computational bandwidth and thermodynamic resources are finite, the system must select. Selection is prior to full knowing—it is the condition under which knowing becomes possible.

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Retention. Selected traces are retained in an append-only fashion, creating an internal arrow of time. Memory is not playback but accumulated residue.

Proto-Modeling. Retained traces are organized into minimal predictive structure. A model of possible world-relations appears.

5.2 World-Resistance and Structural Error

World-Resistance ($\theta(E)$). *(Ax)* The world is not directly possessed by the model; it resists, constrains, and pushes back. Knowledge is never mere internal coherence. The notation $\theta(E)$ denotes the latent truth of the world-state E—that which the system's representations approach but never fully capture [2, 3].

Structural Error ($\varepsilon_{\text{tot}} > 0$). *(Th)* Every finite knower has unavoidable total observation error. This is not an engineering limitation but a structural theorem derived from P1–P3. The decomposition is:

$$\varepsilon_{\text{tot}} = \varepsilon_{\text{clock}} + \varepsilon_{\text{cross}} + \varepsilon_{\text{sel}} + \varepsilon_{\text{map}} + \varepsilon_{\text{self}} \tag{12}$$

where $\varepsilon_{\text{clock}}$ is clock error (finite temporal resolution), $\varepsilon_{\text{cross}}$ is cross-event mismatch, ε_{sel} is selection error (what was filtered out), ε_{map} is mapping error (translation distortion), and $\varepsilon_{\text{self}}$ is self-model error (Th-PSA) [2, §12; 3].

Universal Falsifier. If any observer, system, or entity can be demonstrated to achieve $\varepsilon_{\text{tot}} = 0$—perfect, transparent, error-free access to reality—then the entire architecture collapses. This is the theory's single falsification condition [2, §23.3].

5.3 The Knowledge Event

Comparison. The system detects mismatch between its internal model $M_A[n]$ and world-resistance $\theta(E[n])$.

Correction. The system reorganizes itself in response to mismatch, modifying structure rather than merely adjusting outputs.

Knowledge Event (K_t). *(Df)* A knowledge event is an event of alignment between representation and world-resistance that survives structured failure. Knowledge is not stored data, not mere true belief, and not mere signal possession. It is a dynamic event [3].

Knower. *(Dr)* A knower is a coupled brain–mind-core system capable of stably sustaining model-comparison-correction loops under world-resistance. The knower is produced within epistemic struggle, not presupposed before it.

The compressed formula:

$$\text{Knowing} = \text{Coupled System} + \text{Selection} + \text{Retention} + \text{Model} + \text{Error} + \text{Correction} \tag{13}$$

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6 Knowledge as Generative Capacity

6.1 The Knowledge Triple

Knowledge in a discrete world is not a static object deposited in a container. It is a stable, corrigible persistence-pattern that survives world-resistance and expands future capacity. Formally *(Df)*:

$$KS_A = (Tr_A,\ Str_A,\ Cap_A) \tag{14}$$

where Tr_A is the *trace*—residue left by prior interaction, stored across neural and mind-core organization; Str_A is the *structure*—coherent organization that remains usable over time; and Cap_A is the *capacity*—the ability the system gains to generate further correct acts, insights, and adaptive responses [3].

6.2 Distributed Location

Knowledge is not locally stored in one place *(Dr)*. It is distributed across four layers: the *neural layer* (encodes, routes, sequences, and executes), the *mind-core layer* (integrates, compresses, retains, and stabilizes), the *world-relation layer* (tests and constrains validity through $\theta(E)$), and the *social layer* (extends corrigibility through language, practice, and institutions).

6.3 The Legitimacy Gate

Not every knowledge event constitutes legitimate knowledge. The Legitimacy Gate (LegitKnow) requires four conditions to be jointly satisfied *(Th)* [3, §15]:

$$\text{LegitKnow} \iff \text{Know}_A \ \wedge\ \text{Grounded} \ \wedge\ Q_{\text{auth}} \geq \theta_Q \ \wedge\ N_{\text{know}} \geq \theta_N \tag{15}$$

where Know_A is the occurrence of a knowledge event, Grounded means the representation is answerable to world-resistance (not self-sealing), Q_{auth} measures route-independent authentication (multiple independent routes to the same result), and N_{know} is the minimum number of knowledge events required for the domain. This gate ensures that echo-chamber outputs, single-source amplifications, and internally coherent but ungrounded constructions are excluded from legitimate knowledge.

7 Expertise Formation: The Mature Regime

7.1 The Canonical Chain

Expertise is not the accumulation of more data. It is the mature regime of a brain–mind-core system whose retained structures survive error, correct themselves efficiently under world-resistance, and expand future capacity. The canonical formation chain has

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ten stages *(Dr)*:

Stage 0: Contact. Repeated exposure to a domain begins the formation process. Relevant signals must exist and attention must be allocable.

Stage 1: Filtering. The system learns to separate signal from noise, developing domain-specific salience maps.

Stage 2: Trace Building. Traces (Tr_A) accumulate across neural pathways and mind-core organization.

Stage 3: Modeling. Internal patterns become predictive and compressive; domain schemata and anticipation abilities form.

Stage 4: World Collision. Models fail against real constraints ($\theta(E)$). Prediction error spikes.

Stage 5: Revision. The coupled system corrects *structure* rather than merely adjusting outputs. Deeper discrimination and better priors result.

Stage 6: Stabilization. Valid patterns achieve high persistence (τ_c increases for domain-relevant structures).

Stage 7: Capacity Growth. Knowledge transfers abstractly to novel, previously unseen scenarios. The system can now generate new solutions, not merely recognize old ones.

Stage 8: Expertise. The mature regime: persistent, corrigible, transferable, capacity-generating knowledge under continued world-contact.

7.2 Expert Definition and Metrics

Definition. *(Df)* An expert is a brain–mind-core system whose knowledge structures persist longer, discriminate more finely, correct more rapidly, and generate more future capacity than ordinary learners.

Five measurable conditions characterize the expert regime *(Dr)*:

High persistence (τ_c high): domain-relevant structures decay slowly.

Correction efficiency: mismatch is repaired with less repeated failure over time.

Transfer power: knowledge works across novel cases, not only trained scenarios.

Capacity growth: knowledge produces more future knowing, not merely more stored traces.

Rigidity resistance: the system stays corrigible under novelty—it can revise without losing structural integrity.

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7.3 Why This Form in a Discrete World

Because the world is discrete, expertise is necessarily stepwise, compressive, and corrigible *(Dr)*:

No system can absorb the totality of the world at once (bandwidth constraint from P3). Expertise therefore requires *chunking*—pattern compression and hierarchical organization. Since ε_{tot} never reaches zero, expertise is not perfect knowledge but *superior error-management*. What matters is not total exposure volume but *pattern survival and reuse*—the persistence priority. And expertise depends on the *joint maturation* of neural organization and mind-core stability, not on either component alone.

8 Pathologies of Epistemic Formation

When the dynamic equilibrium of the coupled system fails, structural pathologies arise. These are not merely psychological descriptions but derivable consequences of architectural breakdown *(Dr)*:

Rigidity. Structure (Str_A) remains intact but update capacity collapses. The system rejects contradictory world-signals entirely to preserve internal consistency. This is the expertise-level analogue of confirmation bias.

Echo-Expertise. The system possesses high generative capacity but has severed its grounding with world-resistance ($\theta(E)$). It produces coherent models that are internally sound but epistemologically ungrounded—recognition within closed loops without real resistance-testing.

Overtrace-Understructure. Traces (Tr_A) accumulate massively, but the mind-core fails to compress them into usable structure (Str_A). The result is fragmentation: large memory residue, weak conceptual organization.

Weak Mind-Core Coupling. Brain activity continues, executing basic sensorimotor routines, but the integrative depth of the mind-core is unstable or fragmented. Experience becomes episodic and fragmented, lacking narrative continuity.

Detached Mind-Core. A hypothetical case in which the mind-core's persistence mechanics continue but ordinary embodied knowing collapses. This corresponds to the post-mortem open notes discussed in Section 10.

Institutional Failure. Social systems reward the display of knowledge rather than correction and transfer. Credential substitutes for capacity; status substitutes for world-tested validity. This is a social-scale pathology derivable from the LegitKnow gate: when Q_{auth} is systematically inflated by non-independent routes, institutional knowledge degrades.

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9 Convergence with Current Literature

The proposal is not constructed in isolation from empirical and phenomenological research. This section maps the architecture against top-tier review literature, using a three-tier assessment: *strongly supported*, *defensible extension*, and *open frontier*.

9.1 Strongly Supported Claims

Knowledge is dynamic, not static storage. The contemporary shift away from storage-centric models of memory is among the most robust findings in cognitive neuroscience. Craik argues that remembering is better understood as an activity of mind and brain than as retrieval from fixed stores [6]. The dynamic engram literature emphasizes that memory traces are distributed, flexible, and updated across encoding, consolidation, and recall [7]. Martin and colleagues frame memory through activity-dependent synaptic plasticity as process rather than archive [8]. These findings strongly support the paper's central claim that knowledge is a generative, corrigible event, not a static object.

Phenomenology supports embodied, processual mind. Legrand's work on the bodily self argues that pre-reflective self-consciousness is rooted in the body as a convergence of action and perception [9]. Petitmengin's neurophenomenological methods demonstrate that disciplined second-person investigation can connect first-person structure to scientific models [10]. Foultier's analysis of skilled action shows that expertise retains bodily reflection and cultivated awareness rather than collapsing into blind automatism [11]. These sources strongly support the claim that knowing is enacted, embodied, and action-centered, not detached symbolic possession.

Expertise emerges from repeated correction under constraint. Ericsson and Lehmann's foundational review demonstrates that world-class performance depends on adaptation to task constraints, domain-specific cognitive and perceptual-motor skill, and long-term deliberate practice [12]. Rosenbaum shows that intellectual and perceptual-motor skills are acquired in fundamentally similar ways [13]. Diedrichsen and Kornysheva present motor skill learning as a hierarchical process from effortful selection to integrated production [14]. These findings converge strongly with the ten-stage expertise chain of Section 7.

Consciousness theories leave the field open. Seth and Bayne's review of four major consciousness theories in *Nature Reviews Neuroscience* confirms that the field remains theoretically plural and unsettled [19]. This does not support the information-horizon ontology directly, but it establishes that the ontological space is not closed—there is legitimate room for the formal extension proposed here.

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9.2 Defensible Extensions

The brain as executional interface. Current science strongly supports that the brain is causally indispensable for observable cognition. The paper's extension—that the brain may be better understood as an executional and reconstructive interface rather than the exhaustive ontological location of generative code—is consistent with reconstructive memory models and non-neural cognition findings but exceeds what current instruments have verified.

Signal is not identical to information. The distinction between observable neural change and information-as-such is defensible from the dynamic engram literature, but current neuroscience does not provide a consensus account of where the "deeper layer" is located, or whether the concept of a distinct deeper layer is necessary.

9.3 Open Frontier

The information-horizon ontology of mind. The Agency-Horizon Theorem (Th-BH) is a formal consequence of the stated axioms. Its empirical status depends on the acceptance of those axioms. No Q1 review in neuroscience or phenomenology currently provides direct evidence for or against the claim that the mind is an information-horizon structure in the sense derived here. The literature is consistent with system-level integration, hidden interiority, distributed organization, and compressed generative structure, which gives the information-horizon extension a philosophically serious but empirically unverified foothold.

The cleanest assessment is therefore: the proposal is strongly literature-consistent at the levels of phenomenology, knowledge dynamics, learning, and expertise formation; it becomes a formal extension when it derives the information-horizon structure of mind from stated axioms.

10 Open Frontiers and Proof Obligations

Intellectual honesty requires explicit statement of what the theory does not yet resolve.

O3: The Hard Problem. *(Open)* The Embodied Horizon Canon defines nine necessary conditions (NC1–NC9) for phenomenal experience ($\Xi_A > 0$), but the question of whether these conditions are jointly *sufficient*—why the interior of an information horizon feels like something rather than being sophisticated dark information processing—remains open. The Xi-zombie test (PO17) formalizes this as a proof obligation: if the Xi-zombie concept is coherent for NC1–NC9 systems, the conditions are insufficient; if incoherent, closure approaches [2, §VI-7].

Post-Mortem Notes. Two non-canonical hypotheses remain open. Hypothesis 7.1 (Global Basin Reintegration): upon bodily collapse, informational traces relax toward a

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deeper integrated basin. Hypothesis 7.2 (Integration Without Center): personal selfhood ceases while traces are reintegrated into the world-network without a singular center. Neither hypothesis alters the fixed doctrine that the living mind-core is an information-horizon structure; they address only the thermodynamic redistribution of information after the biological coupling field fails.

Empirical Calibration. The working-form equations for phenomenal architecture (Val_A, Own_A, Coh_A, Ξ_A) require empirical calibration against neuroimaging and clinical data (PO14–PO16). Until calibrated, these remain (Wf) rather than (Th).

Born Rule Derivation. The conjecture (Cj-Q) that the Born rule of quantum mechanics can be derived from the discrete causal graph structure remains a formal proof obligation (PO2) requiring quantum field theory expertise [2, §VI-8].

EH Action Uniqueness. The conjecture (Cj-Y4) that the Einstein-Hilbert action is the unique continuous limit of the discrete causal dynamics requires additional curvature-smoothness assumptions and remains an open proof obligation (PO3) [2, §VI-8].

11 Conclusion

The argument of this paper is neither maximal nor conciliatory. It does not offer a complete metaphysics of everything. It isolates the conditions under which knowing is possible, derives the structural character of the knower, and traces the formation of expertise within a discrete causal-informational world.

The chain is: without difference there is no determinacy; without retained efficacy there is no structure; without the telegraph equation on the discrete graph there is no dynamics; without the Agency-Horizon Theorem there is no structural account of the knower; without brain–mind-core coupling there is no lived experience; without world-resistance and irreducible error there is no epistemic genesis; without stable corrected pattern there is no knowledge; and without capacity-generating persistence there is no expertise.

The resulting position—informationalism—is neither materialism (which begins with already-differentiated matter) nor idealism (which begins with consciousness). It is a third metaphysical stance: the fundamental is the causal ordering of difference, relation, and constraint on a finite discrete graph.

Every claim has been tagged with its formal status. The theory predicts its own falsification: $\varepsilon_{\mathrm{tot}} = 0$ anywhere kills the system. No claim has been reduced; every claim has been grounded in derivation from stated axioms, and every open frontier has been acknowledged.

The compressed principles:

1. Difference makes information possible. *(Ax)*
2. Persistence makes structure possible. *(Ax)*

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3. The telegraph equation on the discrete graph is forced by P1–P3. *(Th)*
4. Every admissible agent is an information horizon. *(Th-BH)*
5. Experience arises from brain–mind-core coupling. *(Dr)*
6. Error is structurally unavoidable: $\varepsilon_{\text{tot}} > 0$. *(Th)*
7. Knowledge is a stable corrected event, not stored data. *(Df/Dr)*
8. Expertise is capacity-generating corrected persistence. *(Dr)*
9. The hard problem of consciousness remains open. *(Open)*
10. $\varepsilon_{\text{tot}} = 0$ anywhere falsifies the entire architecture.

The arena is open. To refute the position, one must show either that a world can arise without primordial difference, that an observer within such a world can achieve zero total observation error, or that the CBF No-Go theorems admit an exception. None of these has been demonstrated.

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This document is a preprint and has not undergone peer review. It represents philosophical inquiry and theoretical derivation, not empirical claims requiring experimental confirmation. Claims tagged (Cj) or (Wf) are explicitly identified as conjectural or requiring calibration.

References

[1] Lahtee, Y. (2026). Primordial Difference, Retained Efficacy, and the Architectonic Conditions of World-Formation. Preprint. Open Civil Science Initiative.

[2] Lahtee, Y. (2026). *Unified Canon* (Genesis Canon Parts 0–8, Embodied Horizon Canon, Appendices). Open Civil Science Initiative. Internal canonical reference.

[3] Lahtee, Y. (2026). Epistemogenesis Knowing Layer Sub-Canon, v7 pr1. Open Civil Science Initiative. Internal canonical reference.

[4] Lahtee, Y. (2026). From Ontological Gap to Architectonic Sufficiency: Deflating, Relocating, and Reconstructing the Hard Problem of Consciousness. Preprint. Open Civil Science Initiative.

[5] Lahtee, Y. (2026). Evolution of Discrete-World Thought: From Ancient Atomism to Difference-First Finite Discrete Graph Ontology. Preprint. Open Civil Science Initiative.

[6] Craik, F. I. M. (2020). Remembering: An Activity of Mind and Brain. *Annual Review of Psychology*, 71, 1–24.

[7] Josselyn, S. A., & Tonegawa, S. (2020). Memory engrams: Recalling the past and imagining the future. *Science*, 367(6473), eaaw4325.

[8] Martin, S. J., Grimwood, P. D., & Morris, R. G. M. (2000). Synaptic Plasticity and Memory: An Evaluation of the Hypothesis. *Annual Review of Neuroscience*, 23, 649–711.

[9] Legrand, D. (2006). The Bodily Self: The Sensori-Motor Roots of Pre-Reflective Self-Consciousness. *Phenomenology and the Cognitive Sciences*, 5(3–4), 225–236.

[10] Petitmengin, C. (2006). Describing One's Subjective Experience in the Second Person: An Interview Method for the Science of Consciousness. *Phenomenology and the Cognitive Sciences*, 5(3–4), 229–269.

[11] Foultier, A. P. (2022). Letting the Body Find Its Way: Skills, Expertise, and Bodily Reflection. *Phenomenology and the Cognitive Sciences*, 22, 955–977.

[12] Ericsson, K. A., & Lehmann, A. C. (1996). Expert and Exceptional Performance: Evidence of Maximal Adaptation to Task Constraints. *Annual Review of Psychology*, 47, 273–305.

[13] Rosenbaum, D. A., Carlson, R. A., & Gilmore, R. O. (2001). Acquisition of Intellectual and Perceptual-Motor Skills. *Annual Review of Psychology*, 52, 453–470.

[14] Diedrichsen, J., & Kornysheva, K. (2015). Motor Skill Learning Between Selection and Execution. *Trends in Cognitive Sciences*, 19(4), 227–233.

10.5281/zenodo.19640361

[15] Vladimirov, N., & Bhatt, D. K. (2016). Information Processing in Bacteria: Memory, Computation, and Statistical Physics. In *Handbook of Systems Biology* (pp. 117–132). Academic Press.

[16] Boussard, A., Delescluse, J., Pérez-Escudero, A., & Dussutour, A. (2019). Memory inception and preservation in slime moulds: the quest for a common mechanism. *Philosophical Transactions of the Royal Society B*, 374(1774), 20180368.

[17] Trewavas, A. (2016). Plant Intelligence: An Overview. *BioScience*, 66(7), 542–551.

[18] Leys, S. P. (2015). Elements of a "nervous system" in sponges. *Journal of Experimental Biology*, 218(4), 581–591.

[19] Seth, A. K., & Bayne, T. (2022). Theories of Consciousness. *Nature Reviews Neuroscience*, 23(7), 439–452.

[20] Matilal, B. K. (1986). *Perception: An Essay on Classical Indian Theories of Knowledge*. Oxford University Press.

[21] Al-Khalili, J. (2012). *Pathfinders: The Golden Age of Arabic Science*. Penguin Books.

[22] Ladyman, J., Ross, D., Spurrett, D., & Collier, J. (2007). *Every Thing Must Go: Metaphysics Naturalized*. Oxford University Press.

[23] Floridi, L. (2008). A Defence of Informational Structural Realism. *Synthese*, 161, 219–253.

[24] Lahtee, Y. (2026). Relational Alethic Realism. Preprint. DOI: 10.5281/zenodo.19212565.

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The Architecture of Mediated Agency: Beyond the Misframing of Free Will and Truth

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What the chapter states. *"Furthermore, we propose that truth is not a 'naked correspondence' but a Causal-Structural Alignment between the observer's representation and the world's generative organization."* (10.5281/zenodo.19176260)

Status. "preprint – not peer reviewed" (as printed in the chapter header).

Falsifier(s). No explicit falsifier stated in the chapter.

Read before. Ch. 1 *Written by AI. Still True.*; Ch. 2 *The Readout Condition*; Ch. 3 *Readout Genesis Standalone Synthesis*; Ch. 39 *The Standalone Scholar*.

Feeds into. — (no chapter in this book references this one, per the KG).

The Architecture of Mediated Agency: Beyond the Misframing of Free Will and Truth

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Abstract

This article challenges the foundational assumptions of the centuries-old debate on free will and truth by introducing a unified framework of **Epistemic Mediation**. We argue that the traditional binary between determinism and libertarianism is a structural misframing (Misframing) that ignores the irreducible role of the observer's *eigenstructure*. By redefining agency as **Deep Authorship**—a generative causal organization that functions through a self-model with inherent errors ($\varepsilon_{tot} > 0$)—we move beyond the requirement for acausal miracles. Furthermore, we propose that truth is not a 'naked correspondence' but a **Causal-Structural Alignment** between the observer's representation and the world's generative organization. This framework 'swallows' the insights of structural realism, predictive coding, and cross-civilizational epistemology, providing a robust account of responsibility without transparency. We demonstrate that the 'gap' of translation is not a failure to be corrected, but the very condition of human agency and knowledge.

Keywords: Epistemic Mediation; Deep Authorship; Causal-Structural Alignment; Eigenstructure; Translation Dissonance

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1 Introduction: The Poverty of Direct Access

For centuries, the discourse on human agency has been trapped in a sterile dialectic between the ghost in the machine and the clockwork universe. This stalemate persists not because the question of 'free will' is difficult, but because it is structurally misframed. Both the libertarian who asserts an acausal 'I' and the determinist who reduces choice to neural correlates share a fatal assumption: the **Presumption of Raw Access**. They believe we can access agency 'as it is,' either through the lens of introspection or the instruments of neuroscience.

This article declares the end of this direct access era. We assert that every interaction with the world—and with the self—is a process of **Translation**. The observer is not a mirror, but an active pipeline with a specific *eigenstructure* that defines the dimensions of reality it can track. In this framework, the 'error' (ε) is not a noise to be filtered, but a constitutive element of the signal. We move the battlefield from metaphysics to the **Architecture of Mediation**, where agency and truth are reclaimed not as exemptions from causality, but as its deepest expressions.

2 The Architecture of Mediation: Eigenstructure and Noise

To understand agency, we must first understand the observer. The observer is a finite system that maps the world's infinite generative organization into a internal representation. This mapping is governed by the observer's **Eigenstructure**—the internal constraints that determine which signals are captured and how they are decoded.

2.1 The Theorem of Irreducible Error

We propose that for any finite observer, the total translation error (ε_{tot}) is strictly greater than zero. This is not a limitation of technology, but a theorem of privacy and finite bandwidth. When two agents interact, the gap between their internal horizons creates **Translation Dissonance (TD)**. Traditional epistemology views TD as a failure; we view it as the *lever* of meta-communication. By treating the gap itself as an object of inquiry, agents do not eliminate the error, but they **calibrate** it. This calibration is the first step toward a realistic account of truth.

3 Deep Authorship: Agency Within the Causal Order

The traditional debate asks: 'Are we free or are we determined?' We respond: This is a category error. Agency is not an exemption from causation; it is **Causation with Depth**. We replace the hollow 'free will' with the concept of **Deep Authorship**.

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3.1 The Non-Bypass Criterion

A process qualifies as Deep Authorship if and only if it satisfies the **Non-Bypass Criterion**: if the agent's internal self-model, valence structure, and memory were different, the action would change. This distinguishes 'Authorship' from 'Shallow Execution.' A thermostat executes a function; a human agent *authors* an action because the action is generated through a multi-layered organization that includes a self-model with its own history of translation.

By 'swallowing' the insights of Frankfurt's hierarchical desires and Strawson's reactive attitudes, we ground them in a structural causal framework. We do not need a soul that stands outside of physics; we need an organization that is deep enough to hold its own history.

4 Truth as Alignment: Beyond Naked Correspondence

If we accept that access is always mediated, what happens to 'Truth'? We argue that all four major theories of truth—Correspondence, Coherence, Consensus, and Pragmatism—fail because they ignore the mediation problem.

4.1 Causal-Structural Alignment

We propose that truth is **Causal-Structural Alignment**: a robust tracking of the world's generative organization through a calibrated translation pipeline. Unlike correspondence, alignment does not require a 'naked comparison' between belief and world. Instead, it requires **Independent Routes** of verification. A representation is 'true' not because it mirrors the world, but because it aligns with the world's causal structure across multiple, non-dependent channels. This account 'swallows' Structural Realism while acknowledging the irreducible mediation of the observer.

5 Responsibility Without Transparency

The most radical implication of this framework is the decoupling of responsibility from transparency. The skeptic asks: 'How can I be responsible if I don't fully know myself?' We answer: Responsibility is not based on perfect self-knowledge ($\varepsilon_{sel} = 0$), but on the **Depth of the Pipeline**. We hold agents responsible when their actions pass through their generative organization, regardless of whether that organization is fully transparent to them. To deny agency based on a lack of transparency is to commit **Epistemic Violence**—an imperialistic imposition of one eigenstructure over another.

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6 Conclusion: Living as the Translator of the World

The framework of **Agency as World-Translation** does not offer a new set of beliefs; it offers a new position from which to stand. We are neither prisoners of a deterministic void nor gods of an acausal vacuum. We are **Translators**.

By acknowledging our eigenstructure, declaring our errors, and seeking structural alignment, we reclaim a form of agency that is more robust because it is honest about its limits. The gap of translation is where the work of being human happens. In this space, we do not just observe the world; we author it through the very act of translating it.

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Author Declarations

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The author declares no competing interests.

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Data Availability

No original dataset was generated for this article. The article relies on doctrinal materials and published scholarship.

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References

Chakravartty, A. (2007). *A Metaphysics for Scientific Realism: Knowing the Unobservable.* Cambridge University Press.

Dennett, D. C. (1991). *Consciousness Explained.* Little, Brown and Company.

Frankfurt, H. G. (1971). Freedom of the Will and the Concept of a Person. *The Journal of Philosophy*, 68(1), 5–20.

Friston, K. J. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127–138.

Gadamer, H.-G. (1960/2004). *Truth and Method.* Bloomsbury Academic.

Metzinger, T. (2003). *Being No One: The Self-Model Theory of Subjectivity.* MIT Press.

Strawson, P. F. (1962). Freedom and Resentment. *Proceedings of the British Academy*, 48, 1–25.

Woodward, J. (2003). *Making Things Happen: A Theory of Causal Explanation.* Oxford University Press.

Worrall, J. (1989). Structural Realism: The Best of Both Worlds? *Dialectica*, 43(1-2), 99–124.

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Part II

Meaning, Experience, Retention

Part 1 stopped at the boundary between a finite retained difference and its naming: a readout occurs, and meaning is attached to it only after a domain translation, without collapsing meaning into the root event. Part 2 opens on the near side of that boundary and asks what happens to a readout before, during, and after it is named, felt, and kept.

Meaning Before Naming places an affective-semantic reorganization stage ahead of explicit naming, proposing that a finite reader can undergo a context- and history-dependent change in response to an external pattern before that change is put into words. *Experience Is Meaning-Giving* and *Experience Is the Human LoRA* each offer a readout-retention account of experience itself — the first as a meaning-giving relation between a phenomenon and a situated reader, spelling out resonance, rhythm, accumulation, and transformative release as candidate transition structures; the second as a cross-level identity between a lived appearance and a rank-bounded update to the observer, stated as a finite factorization condition with named hard falsifiers. *Human Learning as Epistemic Architecture* turns the same retained-difference vocabulary into a method for learners — word mapping, life-concept graphs, constraint testing — built to be disconfirmed by a stated transfer-and-calibration test. *Learning Under Generative Abundance* asks what happens to retention once external, AI-produced coherence is available at near-zero cost, proposing a shift from accumulation to discrimination as the operative learning strategy.

Read together, the chapters in this part treat meaning, experience, and retention as successive readout operations on a common finite substrate, each stating its own claim boundary, status, and named falsifiers rather than presupposing the others' results. None of the five chapters is described here as establishing, confirming, or completing another; the knowledge-graph edges recorded in each headnote (continues, isContinuedBy, references) are mechanical extractions, not endorsements.

Carried forward to Part 3 (Possibility, Choice, Potential): a reader who has stabilized a retained difference through meaning, experience, and disciplined learning is the same reader whose live possibility, choice, and potential the next part examines. Part 3 does not assume that stabilization has occurred; it asks what a readout of possibility and choice can state once some retained difference is already in place.

- Meaning Before Naming: A Readout-Retention Architecture of Affective-Semantic Reorganization — `10.5281/zenodo.22410666`
- Experience Is Meaning-Giving: A Strong-Form Readout-Retention Theory of Phenomena, Resonance, Rhythm, Accumulation, and Transformative Release — `10.5281/zenodo.22357744`
- Experience Is the Human LoRA: A Readout–Retention Theory of Selective Model Change — `10.5281/zenodo.21425420`
- Human Learning as Epistemic Architecture: A Method for Word Mapping, Life-Concept Graphs, Constraint Testing, and Corrigible Agency in the AI Age — `10.5281/zenodo.22341297`
- Learning Under Generative Abundance: A Structural Law of Epistemic Stabilization — `10.5281/zenodo.18711408`

Meaning Before Naming: A Readout-Retention Architecture of Affective-Semantic Reorganization

Standalone Concept Note v1.0 FULL • 5 September 2026 • `10.5281/zenodo.22410666`

What the chapter states. "This paper answers no. It proposes a readout-retention architecture in which externally available differences can produce a reader-relative affective-semantic configuration before explicit lexical or propositional articulation. The central non-collapse is that signal is not meaning, meaning is not naming, intensity is not warrant, and retention is not improvement."

Status. "K0 conceptual architecture and programme-extension paper." The chapter states it "does not propose a new theory of music, a universal code of emotion, or a literal physical theory of resonance," and lists eleven things it "does not claim," including that "resonance is literal physical resonance between minds."

Falsifier(s). Legitimate defeater, verbatim: "pre-nominative measures add no explanatory or predictive information beyond arousal and explicit labeling." Five further hypotheses (H1–H5) are tagged *[Open]* in the chapter's own text.

Read before. Experience Is the Human LoRA; Human Learning as Epistemic Architecture.

Feeds into. Experience Is Meaning-Giving; Before Meaning, Before Choice.

Meaning Before Naming

A Readout-Retention Architecture of Affective-Semantic Reorganization

Programme extension from Human Learning, Human LoRA, Readout Genesis, and Epistemic Fusion/Tunnel

Yaoharee Lahtee | Bangkok, Thailand | Standalone Concept Note v1.0 FULL

Status. K0 conceptual architecture and programme-extension paper. The paper does not propose a new theory of music, a universal code of emotion, or a literal physical theory of resonance. Music appears only as a phenomenological probe that makes a broader claim visible: meaningful human reorganization can occur before explicit naming, while the transmitted signal remains non-identical to the meaning that emerges in a particular reader. The architecture inherits its level distinctions from Readout Genesis, its embodied and affectively weighted account of experience from *Experience Is the Human LoRA*, its lived-word and semantic-network commitments from *Human Learning as Epistemic Architecture*, and its reader-, context-, and history-sensitive transport model from *Epistemic Fusion or Tunnel*. No dedicated review of music cognition is claimed here; any feature-specific account of melody, rhythm, timbre, harmony, or culture would require separate empirical review.

Central claim. Meaning need not begin with a name. A finite human can undergo an affectively weighted, context- and history-dependent reorganization in response to an external pattern before that change is propositionally articulated. The pattern does not carry a finished meaning from one interior to another. Rather, it supplies retained differences that a situated reader makes accessible through an existing and revisable organization of memory, embodiment, expectation, language, value, and context. Some such reorganizations vanish; others are retained and alter how later situations can be read, named, and acted upon.

1. Abstract

The Human-AI Readout Programme has treated language as epistemic mediation: articulation makes distinctions nameable, comparable, transportable, and revisable, while never becoming identical to the full human state from which it was produced [17]. Readout Genesis places retained difference before semantic naming and defines meaning as a context-, memory-, criterion-, and reader-dependent readout relation rather than a property stored in the root [16]. *Human Learning as Epistemic Architecture* begins from lived words whose meanings are entangled with experience, emotion, social origin, value, goal, evidence, and action [14]. *Experience Is the Human LoRA* further argues that lived experience is embodied, affectively weighted, context-mediated, and capable of leaving selectively retained residues that modify later interpretation and action [15]. These commitments together generate a residual question: must meaningful reorganization wait until experience has been explicitly named?

This paper answers no. It proposes a readout-retention architecture in which externally available differences can produce a reader-relative affective-semantic configuration before explicit lexical or propositional articulation. The central non-collapse is that *signal is not meaning, meaning is not naming, intensity is not warrant, and retention is not improvement*. Music is used only as a phenomenological probe: an acoustic pattern can be heard, felt, and experienced as uplifting, sorrowful, tense, peaceful, or energizing before the listener can state a proposition that exhausts what has changed. This does not imply that the sound contains a universal emotional code or that producer and listener share an identical inner state. It illustrates a more general structure: external difference → finite readout → affective-semantic reorganization → optional naming → selective retention → altered future readout.

The contribution is an extension of the programme below the level of explicit Human Return. It clarifies that the lived word is an important entry unit for reflective learning but not the ontological origin of all meaning; language is one powerful articulation and transport layer within a broader readout ecology. The paper specifies claim boundaries, failure modes, a minimal formal architecture, and an empirical programme for testing whether prenominative reorganization adds explanatory value beyond simpler accounts of arousal, verbal labeling, or memory.

2. The Residual Problem in the Existing Programme

The programme already contains most of the ingredients needed for this extension. The missing step is their placement.

Human Learning as Epistemic Architecture deliberately begins with the lived word rather than abstract information. A word is treated not as an inert dictionary unit but as a site where experience, memory, emotion, social pressure, value, goal, constraint, evidence, and action converge. Learning then moves from representation to relation, retrieval, evaluation, revision, problem-context, tool use, workflow, skill, and feedback [14]. This is a methodologically powerful starting point for education because words can be mapped, contested, sourced, and revised.

Yet Readout Genesis states something more basic: retained difference precedes the name of what dif-

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fers, and meaning is attached only after a valid domain translation and readout. A reader makes a distinction accessible under a question; the reader does not retroactively place that meaning in the root [16]. If this architecture is taken seriously, the lived word cannot also be the universal first occurrence of meaning. It is an entry point for inquiry, not the metaphysical beginning of experience.

Experience Is the Human LoRA supplies the second pressure. Experience is described as a temporally thick, embodied, affectively weighted, and context-mediated readout through which a world encounter becomes present to a finite observer. The durable consequence of experience is not the full event but a selectively retained alteration in later interpretive and action dispositions [15]. A large portion of experience may remain transient; some residues persist without becoming detailed episodic memory; some may become habits, sensitivities, or skills. Thus explicit verbal recall cannot exhaust what counts as a meaningful retained change.

Finally, *Epistemic Fusion or Tunnel* treats articulation as a bounded export of a larger retained human state. Language makes distinctions available for transformation, but the utterance is not identical to the human state that generated it. The same programme also makes reader, context, history, provenance, and resistance part of the trajectory rather than treating meaning as a fixed property of a surface form [17, 18].

The residual problem is therefore narrow:

> *How can the architecture represent meaningful, affectively weighted human reorganization that occurs before explicit naming, without treating meaning as a substance encoded inside an external signal or collapsing felt change into truth?*

3. Internal Genealogy and What Must Not Be Rewritten

The present note is an extension, not a replacement. Table 1 fixes what is inherited and what remains new.

The key correction is a level distinction. **Words can be methodologically first without being phenomenologically first.** Human Learning can begin an educational protocol with a word because a word is inspectable and socially contestable. Readout Genesis can simultaneously maintain that the retained distinctions and readout relations from which meaning emerges are prior to semantic naming. There is no contradiction once method and ontology are kept separate.

4. Phenomenological Ground: Meaningful Change Can Precede Naming

The architecture begins from a small descriptive floor rather than from equations.

P1 - Finite access. A human never receives the totality of an event. What becomes present is selected through bodily capacities, attention, prior learning, memory, expectation, language, and context [4, 13, 11, 15].

P2 - Affective weighting. What becomes accessible is not affectively neutral. Experience can appear as inviting, threatening, consoling, beautiful, tense, familiar, alien, hopeful, or otherwise action-relevant before the person can provide a complete propositional account of why [15].

P3 - Relational meaning. The significance of a signal is not exhausted by its isolated surface form. It depends on active relations among prior experience, contextual frames, memories, values, expectations, and other accessible distinctions [5, 8, 14].

P4 - Naming can lag reorganization. A change in what feels salient, connected, possible, or action-relevant may occur before the person can name the distinction that changed. Explicit articulation can follow, stabilize, sharpen, distort, or fail to capture the prior reorganization.

Consider an ordinary but theoretically revealing case. A composer or performer may externalize part of an inner organization through patterned sound, yet what crosses the physical medium is not a finished sadness, hope, memory, or spiritual state. What is available to the listener is an acoustic pattern containing differences of timing, pitch, intensity, timbre, repetition, pause, and relation. A listener may then experience the pattern as deeply uplifting, sorrowful, peaceful, destabilizing, or energizing before being able to say what the experience "means." Another listener may hear the same performance and undergo a different change, or almost none. The point is not that music possesses a special metaphysical power. The point is that the example makes a general architecture visible: an external pattern can reorganize an internal field without transporting an identical inner state from one person into another. The felt "echo inside" is therefore better modeled as reader-relative reorganization than as semantic copying.

This example also exposes why the paper cannot be reduced to a theory of verbal semantics. If meaning were identical to explicit linguistic definition, then meaningful change prior to naming would be incoherent by definition. The programme's own prior commitments resist that collapse: Human LoRA already treats lived experience as embodied and affectively weighted, and Readout Genesis explicitly places meaning after readout rather than after lexical naming [15, 16].

5. The Core Readout-Retention Architecture

Let x_n denote a finite externally available pattern at step n. It can be linguistic, acoustic, visual, gestural, social, or multimodal. The present note does not

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Table 1: Programme genealogy for the present extension. Internal work establishes continuity, not independent empirical validation.

Prior programme object	Retained result	Role here
Human Learning as Epistemic Architecture	Lived words are loaded by experience, memory, emotion, value, evidence, goals, constraints, and action; learning builds meaning neighborhoods, concept relations, problem-context, tools, workflow, skill, and feedback.	Supplies the reflective and educational level. The lived word remains an excellent unit for inquiry, but is retyped as a later articulation/access point rather than the origin of all meaning.
Experience Is the Human LoRA	Experience is embodied, affectively weighted, context-mediated, and not automatically learning; selected residues may persist and alter later interpretation/action.	Supplies the phenomenological and temporal layer: felt reorganization can precede naming, and persistence must be separated from momentary intensity.
Readout Genesis / MEMK	Retained difference precedes semantic naming; meaning follows readout; data, meaning, truth, and epistemic status remain non-identical; approximate translation must expose loss.	Prevents signal or sound from being treated as a container of finished meaning and prevents resonance from becoming a root ontology.
Epistemic Fusion or Tunnel	Articulation is a bounded readout of a larger human state; trajectories are reader-, context-, and history-sensitive; accessibility, retention, warrant, and direction do not collapse.	Supplies transport, reader-dependence, history-shaped accessibility, and non-collapse discipline.
CTSA Human-Return Readout	Explicit Human Return can be typed by what remains demonstrably usable after AI removal, while gain, loss, warrant, regulation, and direction remain separate.	Supplies a useful upper boundary: the present paper concerns a lower, not necessarily explicit, affective-semantic layer that may later become conceptual or skillful Human Return.

treat x_n as meaning itself.

Let $\mathcal{R}_H$ denote the human reader under current organization H_n and context c_n. A bounded readout is

$$r_n = \mathcal{R}_H(x_n \mid H_n, c_n).$$

The readout is not a copy of the source event. It is what becomes accessible to this reader under these conditions.

A context-sensitive affective-semantic configuration is then written schematically as

$$\mu_n = G(r_n, M_{n-1}, A_{n-1}, L_{n-1}, V_{n-1}, c_n),$$

where M denotes retained memory organization, A affective weighting, L currently available linguistic/conceptual organization, and V value/action relevance. The notation is architectural, not a claim that these objects are literal neural variables.

Lived experience is represented at a higher descriptive level as

$$E_n = \Phi(r_n, \mu_n, B_n, c_n),$$

where B_n marks embodiment. This inherits the level separation in Human LoRA: the lived event cannot be identified with a parameter update or a bare signal [15].

Only some residue is retained:

$$H_{n+1} = U(H_n, \mathrm{Retain}(E_n, r_n, \mu_n), \delta^{world}_{n:n+1}).$$

Here δ^{world} represents later correction or resistance. An intense moment may therefore produce little durable change, while a quieter event may reorganize later interpretation.

Finally, explicit naming is optional and temporally placed after the readout event:

$$\ell_n = \mathcal{L}_H(E_n, \mu_n \mid H_n, c_n),$$

where ℓ_n is a linguistic articulation, label, metaphor, description, or question. The architecture therefore permits

$$E_n, \mu_n \quad \text{before} \quad \ell_n.$$

This is the formal sense in which meaning may precede naming.

6. Resonance Is Reorganization, Not Transmission

The everyday word *resonance* is useful only if stripped of literal physical import. The present proposal does not claim that human meaning behaves like coupled oscillators. Resonance is a reader-relative structural diagnostic for a case in which an incoming pattern makes some relations more accessible, more strongly weighted, or newly connected in the current human organization.

Let $\mathcal{A}_H^{\text{pre}}$ be a task- or experience-relative accessibility profile before an encounter and $\mathcal{A}_H^{\text{post}}$ the profile after it. A finite diagnostic may be written as

$$\Delta\mathcal{A}_H(x, c) = \mathcal{A}_H^{\text{post}} - \mathcal{A}_H^{\text{pre}}.$$

A positive change in one region does not imply global improvement. It means only that some relation, memory, possibility, or action tendency became more accessible under the declared reader.

Resonance without identity. Two people can be affected by the same external pattern without receiving the same meaning. The shared source constrains the encounter, but the readout is conditioned by different histories, contexts, bodies, memories, and conceptual repertoires.

Resonance without truth. A pattern can produce a powerful, coherent, memorable reorganization while the interpretation attached to it is false, ethi-

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cally harmful, or epistemically unsupported. Affective force cannot substitute for warrant.

Resonance without retention. A strong momentary response may leave no durable reorganization. Conversely, a subtle encounter may later prove consequential if its residue is retained and integrated.

These propositions extend the programme's non-collapse discipline rather than adding a new ontology.

7. Language Is a Special Transport Layer, Not the Container of Meaning

The programme's earlier language model remains intact. A human state $Z_{H,t}$ is articulated through language as a bounded question or expression $Q_{H,t}$:

$$Z_{H,t} \xrightarrow{\mathcal{L}_H} Q_{H,t}.$$

Fusion/Tunnel emphasizes that $Q_{H,t}$ is not identical to the whole state; articulation exports only part of what is retained and available [17]. Once externalized, the expression can be compared, challenged, transported, and revised.

The present paper adds a prior possibility:

$$\begin{aligned}&\text{pattern encounter} \to \text{affective-semantic reorganization}\\&\to \text{optional articulation}.\end{aligned}$$

Language therefore performs at least three distinct functions.

First, it can **name** a distinction that was already experientially active but tacit or only partially discriminated. Polanyi's account of tacit knowing provides a classical ally for the idea that explicit statement need not exhaust what is operative in human understanding [12].

Second, language can **restructure** experience. A later word, metaphor, diagnosis, story, or theoretical concept can alter which relations become salient and how an earlier event is remembered or interpreted. Human Learning's meaning maps and concept graphs operate precisely at this level [14].

Third, language can **transport** a bounded readout to another person or system. But transport necessarily introduces selection and loss. Bender and Koller distinguish linguistic form from meaning, while Bisk et al. emphasize the grounding problem for language detached from situated experience [2, 3]. These external literatures support the programme's refusal to identify text with the world or with the total lived state.

Thus the revised ordering is not "meaning without language" versus "meaning through language." It is layered:

$$\begin{aligned}&\text{readout} \to \text{lived significance}\\&\leftrightarrow \text{language-mediated reorganization and transport}.\end{aligned}$$

8. From Felt Change to Retained Human Change

Human LoRA gives the temporal rule required here: experience is not automatically learning. An encounter may be vivid and vanish; another may be selectively retained, consolidated, and integrated into future dispositions [15, 6, 7, 10].

This matters because the intuitive language of "a song stayed with me," "a phrase changed me," or "a moment still echoes" can hide several different regimes:

- **Transient activation:** a relation becomes accessible during the encounter but returns near baseline afterward.
- **Recallable residue:** the event can later be remembered without materially changing interpretation or action.
- **Retained sensitivity:** later events are noticed, weighted, or discriminated differently.
- **Conceptual crystallization:** an initially difficult-to-name experience later becomes articulable as a concept or distinction.
- **Action reorganization:** the retained change affects later choice, avoidance, approach, practice, or inquiry.

The architecture therefore separates

$$\begin{aligned}&\text{Exposure} \neq \text{felt intensity}\\&\neq \text{retention} \neq \text{improvement}.\end{aligned}$$

Karpicke and Roediger's work on retrieval and long-term retention and the broader consolidation literature provide external constraints on the distinction between momentary availability and durable learning [9, 6].

9. Relation to Human Return and CTSA

CTSA Human-Return Readout asks what capability remains demonstrably usable after AI support is removed. Its four readout classes—Conceptual, Tool-Selection, Skill-Execution, and Alternative-Generation Return—are deliberately explicit and behaviorally demonstrable [19].

The present architecture sits lower in the stack. An affective-semantic reorganization may be real and retained before it becomes an explicit concept, tool-selection rule, executable skill, or articulable alternative. This does not refute CTSA; CTSA already rejects the claim that it exhausts human cognition or all human learning. Instead, the present note clarifies a boundary condition:

> *Not every retained human change is immediately an explicit epistemic capability, but explicit Human Return may emerge from a prior affective-semantic reorganization that later becomes nameable, discriminable, and usable.*

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A possible trajectory is therefore

$$\text{felt reorganization} \rightarrow \text{retained sensitivity} \rightarrow \text{naming} \rightarrow \text{Concept Return} \rightarrow \text{later Skill/Alternative Return}.$$

This is an open developmental pathway, not a universal causal sequence.

The reverse is also possible. A new concept supplied by language can reorganize later affective experience. A person who acquires a distinction may subsequently hear, see, or remember the same event differently. Human learning is therefore recursive rather than strictly bottom-up.

10. Context Wisdom as Governance of Resonant Meaning

The programme's Context Wisdom line treats wisdom not as possession of many meanings but as governance of meaning under context, provenance, resistance, and time. The present extension sharpens this point.

Affectively powerful meaning can become load-bearing very quickly. Repetition, prestige, ritual, group identity, aesthetic intensity, personal memory, or emotional relief may make a relation easier to re-enter. None of these facts alone establish its truth or ethical legitimacy.

Context Wisdom therefore asks at least four questions:

1. What did this encounter make newly accessible?
2. Which prior memories, values, identities, or expectations gave the pattern its current force?
3. What alternative readouts remain possible for other readers or in other contexts?
4. Which parts of the retained interpretation can still face independent resistance and revision?

The goal is not to suppress affect. It is to keep affective-semantic reorganization revisable rather than allowing intensity, familiarity, or repetition to become self-certifying.

11. Non-Collapse Laws

The extension is primarily a non-collapse architecture. The following distinctions are load-bearing:

$$\begin{aligned}
\text{external pattern} &\neq \text{meaning},\\
\text{meaning} &\neq \text{explicit naming},\\
\text{shared stimulus} &\neq \text{shared inner state},\\
\text{affective intensity} &\neq \text{epistemic warrant},\\
\text{accessibility} &\neq \text{truth},\\
\text{resonance} &\neq \text{understanding},\\
\text{retention} &\neq \text{benefit},\\
\text{verbalization} &\neq \text{lossless translation},\\
\text{same surface form} &\neq \text{same reader-relative meaning}.
\end{aligned}$$

Readout Genesis supplies the deepest guardrail: data are not meaning, meaning is not truth, and truth is not epistemic status [16]. The present note simply extends those separations into a domain in which felt significance can precede explicit naming.

12. Failure Modes and Adversarial Cases

12.1 The encoded-emotion fallacy

A listener reports sadness after an acoustic pattern. It does not follow that "sadness" existed as a semantic object inside the sound and was transferred intact. The sound constrained the readout; the reported sadness is a reader-relative interpretation of the encounter.

12.2 The universality shortcut

Several listeners respond similarly. Shared response does not by itself establish a context-free universal code. Common embodiment, enculturation, exposure history, task framing, or social expectation may contribute. The present paper makes no universality claim.

12.3 The intensity-as-truth error

An experience feels profound and memorable. That gives evidence about its force for the reader, not about the truth of every proposition later associated with it.

12.4 The naming-creates-meaning error

A person cannot initially describe an experience. It does not follow that there was no meaningful organization before the label appeared. Conversely, a later label can reshape the earlier experience; naming is neither mere reporting nor absolute creation.

12.5 The retention-is-good error

A pattern becomes deeply retained. Persistence can support learning, identity, care, or skill, but it can also stabilize fear, prejudice, fixation, or maladaptive expectation. Human LoRA and Fusion/Tunnel both require direction to remain separate from retention magnitude [15, 18].

12.6 The literal-resonance error

The term resonance is interpreted as a claim of shared oscillatory physics between minds. This is outside the present architecture. Resonance is only a structural shorthand for reader-relative reweighting, accessibility, and retained reorganization.

13. Empirical Programme

The first test should not attempt to validate a grand theory of meaning. It should test whether a prenominative readout layer adds explanatory value beyond simpler alternatives.

A minimal study could compare responses to matched nonverbal or weakly verbal patterns un-

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der different contexts. The design should freeze a pre-exposure baseline, record immediate experience without forcing premature verbal explanation, then obtain later articulation and delayed measures of retention or transfer. The key question is whether measurable reorganization appears before explicit naming and whether later naming predicts, merely reports, or further restructures the earlier change.

Candidate observables include:

- changes in association structure or accessibility before and after exposure;
- affective discrimination that precedes stable verbal labeling;
- delayed reconstruction of the experience and its boundaries;
- changes in later interpretation of related but novel patterns;
- divergence across readers exposed to the same stimulus under different histories or contexts;
- loss introduced by verbal translation: distinctions present in immediate experience but absent from later description;
- persistence versus decay across delayed measurement.

Several competing explanations must be preserved. Apparent "meaning before naming" could reduce to undifferentiated arousal, demand characteristics, post-hoc verbal reconstruction, memory bias, or familiarity. A strong design must include conditions capable of separating these alternatives.

H1 - Pre-nominative reorganization [Open]. For at least some classes of experience, task-relevant changes in affective or relational accessibility will be detectable before stable explicit naming.

H2 - Reader dependence [Open]. The same external pattern will produce systematically different readout profiles as reader history, context, or prior organization changes, even when the source pattern is held fixed.

H3 - Translation loss [Open]. Later verbal description will fail to preserve some distinctions present in immediate experience, while also adding distinctions that were not previously explicit.

H4 - Retention selectivity [Open]. Immediate intensity will not perfectly predict delayed retained reorganization.

H5 - Recursive naming [Open]. Naming after an initially difficult-to-articulate experience will sometimes reorganize subsequent readout of the same or related patterns rather than merely describe a fixed prior state.

H6 - No automatic improvement [Derived guardrail]. Even when H1–H5 are supported, no inference to truth, wisdom, well-being, or ethical value follows without independent criteria.

14. Legitimate Defeaters

The architecture should be narrowed or abandoned if any of the following occurs under well-specified tests:

- pre-nominative measures add no explanatory or predictive information beyond arousal and explicit labeling;
- reader history and context do not materially alter the relevant readout once stimulus features are controlled;
- the proposed distinction between immediate experience and later articulation cannot be operationalized reliably;
- delayed retained change is fully predicted by simpler memory variables with no need for an affective-semantic layer;
- "resonance" proves to be dispensable vocabulary that adds no measurable structure beyond accessibility or association change;
- the architecture cannot distinguish genuine retained reorganization from post-hoc narrative reconstruction.

These are removal conditions, not weaknesses to be hidden. The programme's methodological rule remains that a variable which performs no distinct explanatory work should be deleted.

15. Claim Boundaries

This paper does **not** claim that:

- music has one universal meaning or emotional code;
- specific musical features universally cause specific emotions;
- the inner state of a performer is transferred intact into a listener;
- all meaningful experience is pre-linguistic or independent of language;
- affective-semantic reorganization is a literal neural module;
- resonance is literal physical resonance between minds;
- a strong emotional response establishes truth, authenticity, morality, or spiritual validity;
- every retained experience improves the person;
- the equations are validated psychological or neuroscientific laws;
- CTSA, Human Return, or the present layer exhaust all forms of human learning;
- self-citation from the programme constitutes independent empirical support.

A dedicated music-cognition paper would require a separate review of auditory perception, predictive timing, tonal expectation, entrainment, affect,

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memory, culture, and individual differences. Those claims are deliberately outside the present note.

16. Standalone Thesis

> *Meaning need not begin with a name. A finite human can undergo an affectively weighted, context- and history-dependent reorganization before that change is propositionally articulated. External patterns do not transport finished meanings from one interior to another; they constrain a reader-relative readout whose significance is co-constituted by embodiment, memory, expectation, language, value, and context. Some reorganizations vanish, some are later named, and some are selectively retained so that future situations are read differently. Music makes this architecture visible, but does not define it.*

The programme sequence can now be written as

$$\begin{gathered}\text{retained difference} \rightarrow \text{bounded readout} \\ \rightarrow \text{lived affective-semantic configuration} \\ \rightarrow \text{optional naming} \rightarrow \text{retention/reorganization} \\ \rightarrow \text{altered future readout}.\end{gathered}$$

This extension preserves the earlier architecture while moving one level beneath language. Human Learning explains how named meaning becomes corrigible competence; Human LoRA explains how lived events can leave durable residues; Readout Genesis prevents semantic labels from becoming root primitives; Fusion/Tunnel explains how articulation becomes transport and how history shapes later accessibility; CTSA measures some explicit forms of what later returns as capability. The present note adds the missing phenomenological bridge: **what can become meaningful in the human before it can yet be fully said.**

17. References

[1] Anderson, J. R. (1982). Acquisition of cognitive skill. *Psychological Review, 89*(4), 369–406.

[2] Bender, E. M., & Koller, A. (2020). Climbing towards NLU: On meaning, form, and understanding in the age of data. *Proceedings of ACL 2020*, 5185–5198. https://doi.org/10.18653/v1/2020.acl-main.463

[3] Bisk, Y., Holtzman, A., Thomason, J., Andreas, J., Bengio, Y., Chai, J., et al. (2020). Experience grounds language. *Proceedings of EMNLP 2020*, 8718–8735. https://doi.org/10.18653/v1/2020.emnlp-main.703

[4] Clark, A. (2013). Whatever next? Predictive brains, situated agents, and the future of cognitive science. *Behavioral and Brain Sciences, 36*(3), 181–204. https://doi.org/10.1017/S0140525X12000477

[5] Collins, A. M., & Loftus, E. F. (1975). A spreading-activation theory of semantic processing. *Psychological Review, 82*(6), 407–428.

[6] Dudai, Y. (2004). The neurobiology of consolidations, or, how stable is the engram? *Annual Review of Psychology, 55*, 51–86. https://doi.org/10.1146/annurev.psych.55.090902.142050

[7] Dudai, Y. (2012). The restless engram: Consolidations never end. *Annual Review of Neuroscience, 35*, 227–247. https://doi.org/10.1146/annurev-neuro-062111-150500

[8] Fillmore, C. J. (1982). Frame semantics. In Linguistic Society of Korea (Ed.), *Linguistics in the Morning Calm* (pp. 111–137). Hanshin.

[9] Karpicke, J. D., & Roediger, H. L. (2007). Repeated retrieval during learning is the key to long-term retention. *Journal of Memory and Language, 57*(2), 151–162. https://doi.org/10.1016/j.jml.2006.09.004

[10] McGaugh, J. L. (2015). Consolidating memories. *Annual Review of Psychology, 66*, 1–24. https://doi.org/10.1146/annurev-psych-010814-014954

[11] Noe, A. (2004). *Action in Perception*. MIT Press.

[12] Polanyi, M. (1966). *The Tacit Dimension*. University of Chicago Press.

[13] Varela, F. J., Thompson, E., & Rosch, E. (1991). *The Embodied Mind*. MIT Press.

[14] Lahtee, Y. (2026). *Human Learning as Epistemic Architecture: A Method for Word Mapping, Life-Concept Graphs, Constraint Testing, and Corrigible Agency in the AI Age*. Revised v4 preprint, 28 June 2026.

[15] Lahtee, Y. (2026). *Experience Is the Human LoRA: A Readout-Retention Theory of Selective Model Change*. Open Civil Science Initiative preprint, 18 July 2026.

[16] Readout Genesis Programme. (2026). *READOUT_GENESIS_MEMK_DOMAIN v0.3.0 ANCHORED*. Root-connected semantic-cognitive domain specification, 23 July 2026. Architecture-conditional / unresolved domain.

[17] Lahtee, Y. (2026). *Epistemic Fusion or Tunnel: Conversation-Centered Architecture*. Human-AI Readout Programme, Conversation-Centered v3 FINAL, 5 September 2026.

[18] Lahtee, Y. (2026). *Epistemic Fusion or Epistemic Tunnel: A Phenomenology-Anchored, Global-Literature-Constrained, Problem-First Architecture of Human-AI Collaboration*. Human-AI Readout Programme, v8.1 strong-claim / legitimate-defeater edition, 5 September 2026.

[19] Lahtee, Y. (2026). *CTSA Human-Return Readout: A Session-Boundary Measurement Architecture for Retained Human Capability After AI*. Human-AI Readout Programme, v9.1 FINAL / global-literature-constrained edition, 5 September 2026.

Experience Is Meaning-Giving: A Strong-Form Readout-Retention Theory of Phenomena, Resonance, Rhythm, Accumulation, and Transformative Release

Full Paper v2.0, strong-claim / reviewer-hardened edition • 5 September 2026 • 10.5281/zenodo.22357744

What the chapter states. "This paper proposes that experience is meaning-giving: not the arbitrary projection of a private label onto neutral data, and not the reception of a finished semantic object already contained in the world, but a finite, embodied, affectively weighted, history- and context-conditioned readout relation through which a phenomenon becomes significant for a particular agent."

Status. "K0 strong-form conceptual and measurement architecture, revised after adversarial peer review without withdrawing the paper's central claims." "None is literal physical energy or an automatic truth certificate."

Falsifier(s). Legitimate defeater, verbatim: "experience is fully predicted by stimulus features and arousal with no added role for reader history, context, or significance."

Read before. Meaning Before Naming; Experience Is the Human LoRA.

Feeds into. Choice Begins Before Choice; Before Meaning, Before Choice.

Experience Is Meaning-Giving

A Strong-Form Readout-Retention Theory of Phenomena, Resonance, Rhythm, Accumulation, and Transformative Release

A programme synthesis from Readout Genesis, Human LoRA, Human Learning, Meaning Before Naming, Fusion/Tunnel, and CTSA Human Return

Yaoharee Lahtee | Bangkok, Thailand | Full Paper v2.0 - Strong-Claim / Reviewer-Hardened

Status. K0 strong-form conceptual and measurement architecture, revised after adversarial peer review without withdrawing the paper's central claims. The identity thesis is retained: for a finite situated human, experience is the giving of meaning to a phenomenon. Revision strengthens rather than narrows that claim by separating distinct modes of significance, exposing direct rival explanations, and specifying what each construct must add empirically. Resonance remains external-internal experiential congruence; rhythm remains structured recurrence across ordered history; accumulation, informational work, barrier crossing, and transformative release remain explicit candidate transition structures. None is literal physical energy or an automatic truth certificate. Internal programme manuscripts establish genealogy; external literature constrains novelty and identifies rival mechanisms.

Central claim. A phenomenon becomes an experience for a finite human by being given meaning: it becomes affectively, pragmatically, autobiographically, conceptually, or epistemically significant within a reader-conditioned relation to embodiment, retained history, expectation, value, available concepts, and context. These are modes of one meaning-giving relation, not competing definitions of experience. Meaning-giving can begin before stable naming, while naming can recursively reorganize later experience. When a present externally prompted experience corresponds with retained internal experiential organization, **resonance** occurs as external-internal congruence. When encounters recur with structured timing, **rhythm** organizes recurrence, spacing, interruption, emphasis, and return. Repeated retained changes can accumulate transition readiness; sufficiently aligned encounters may contribute to barrier crossing and **transformative release**. None of these transitions certifies truth, healing, wisdom, or improvement. In Human-AI interaction, the prompt is therefore not the beginning of the human trajectory but a bounded articulation of an already meaning-shaped state.

1. Abstract

What is an experience? The Human-AI Readout Programme has progressively separated world events, retained traces, meaning, memory, knowledge, language, and human return, but its earlier manuscripts stopped short of a single defended theory of how a phenomenon becomes lived experience. This paper proposes that *experience is meaning-giving*: not the arbitrary projection of a private label onto neutral data, and not the reception of a finished semantic object already contained in the world, but a finite, embodied, affectively weighted, history- and context-conditioned readout relation through which a phenomenon becomes significant for a particular agent. The proposal is rooted internally in Readout Genesis/MEMK, where meaning follows readout and experience is formally represented as an event-trace–meaning configuration; in *Experience Is the Human LoRA*, where experience is temporally thick and selected residues can alter future interpretation; in *Human Learning as Epistemic Architecture*, where lived words make experience inspectable and revisable; in *From Problem to Hypothesis* and *Fusion/Tunnel*, where accessibility is history-shaped and repeated traversal can alter future routes; and in *CTSA Human-Return Readout*, where retained change is distinguished from demonstrable unaided capability.

External literature constrains rather than validates the architecture. Phenomenology treats conscious experience as intentional and horizon-laden; enactivism frames cognition as sense-making in organism-environment coupling; appraisal theory distinguishes event appraisal from later feeling and labeling; emotion research treats experience as a content-rich binding of affect, perception, and conceptual knowledge; predictive-processing work shows how prior models can shape perception while recent reviews caution against over-generalization; memory and insight research supports selective retention, reconsolidation, and abrupt representational restructuring; music-cognition research establishes rhythm as a measurable temporal structure with competing mechanistic models; and Human-AI experiments show that repeated AI interaction can alter later human judgments.

The paper contributes a strong-form integrative architecture rather than merely a new label for significance. First, **meaning-giving** is typed as a plural but unified relation that can carry affective, pragmatic, autobiographical, conceptual, and epistemic significance without reducing experience to language. Second, **experiential resonance** names external-internal experiential congruence while remaining empirically separable from schema congruence, self-reference, familiarity, processing fluency, appraisal, and memory retrieval. Third, **rhythmic organization** names structured recurrence across ordered history and is connected not only to music

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cognition but also to event segmentation, memory, learning, and dialogue. Fourth, **transformative release** connects accumulated retained change, transition readiness, barrier crossing, and abrupt reorganization while remaining distinguishable from mere intensity and from durable transformation. Fifth, the paper names the **Pre-Prompt Human State Principle**: Human-AI interaction begins from a history-shaped human state whose prior experiences already condition the next prompt. A formal architecture, non-collapse laws, rival-model tests, model-selection criteria, and legitimate defeaters make these claims empirically vulnerable without weakening their conceptual scope.

2. The Residual Problem: From Phenomenon to Experience

The programme already distinguishes a world-side occurrence from its representation and interpretation. Readout Genesis states that a retained difference precedes the name of what differs and that meaning is attached only after domain translation and readout [1]. In its MEMK domain, meaning is explicitly defined as a readout-conditioned relation rather than a property stored in the root, and experience is represented as a non-additive event-trace–meaning configuration [2]. This gives an immediate clue: a phenomenon is not yet identical to experience.

The central residual question is therefore not whether humans "have experiences." It is what architecture connects an encountered phenomenon to a lived, significant event without collapsing one into the other. Three inadequate answers must be avoided.

First, **copying**: the experience is treated as a faithful internal copy of an external event. This ignores selective access, embodiment, prior learning, expectation, and context. Second, **semantic containment**: the phenomenon is treated as if it already carries the same finished meaning that every observer merely extracts. This fails wherever the same event matters differently to different readers. Third, **arbitrary construction**: meaning is treated as unconstrained private invention. This ignores source structure, common embodiment, social history, and world resistance.

The proposed alternative is relational:

> *A phenomenon becomes an experience when a finite situated agent gives it significance through a constrained readout relation between what is encountered and what the agent already retains, expects, values, can discriminate, and can act upon.*

The phrase *gives meaning* therefore does not mean "makes truth." It means that a phenomenon becomes *for this agent, in this context, under this history* something threatening, familiar, beautiful, actionable, puzzling, sacred, irrelevant, consoling, absurd, or otherwise significant. The resulting significance may later be revised, rejected, or shown false.

3. Internal Genealogy and the New Residual Contribution

The present paper does not replace the programme's earlier objects. It fixes their order and identifies a missing interface.

The new residual contribution is narrower than a universal theory of consciousness but broader than the earlier *Meaning Before Naming* note. It is a theory of **experience formation and retained transformation**:

$$\begin{aligned}\text{phenomenon} &\to \text{meaning-giving readout}\\ &\to \text{experience} \to \text{retention/revision}\\ &\to \text{future readout.}\end{aligned}$$

The paper then asks how resonance, rhythm, and transition belong inside this sequence.

4. External Literature as Constraint, Not Validation

The proposed architecture sits near several mature traditions. Its novelty claim survives only if it does not rename them as discoveries.

4.1 Phenomenology: experience is intentional and horizon-laden

Phenomenology has long treated conscious experience as directed: perception, memory, judgment, emotion, and imagination are experiences *of* or *about* something. Husserlian and Merleau-Pontian traditions also emphasize the lived body and the largely implicit horizon that gives an experienced object background significance [10, 11]. This is an important ally to the present thesis because it blocks the idea that experience is a bare sensory registration followed only later by optional meaning. Yet phenomenology alone does not supply the programme's retention gates, provenance rules, Human-AI session boundary, or empirical falsifiers.

4.2 Enaction: meaning as sense-making in organism-environment coupling

Enactive theory comes even closer to the claim that experience involves meaning-giving. Di Paolo argues that adaptivity is required to ground sense-making and agency, while Thompson's enactive synthesis treats significance and valence as arising in the living organism's regulated coupling with its environment rather than being intrinsic labels waiting to be extracted [12, 13, 11]. The overlap is substantial: meaning is relational, embodied, history-sensitive, and action-relevant. The present paper does not claim to discover sense-making. Its narrower contribution is to place a readout-retention architecture around this relation, then connect it to resonance, rhythmic recurrence, retained transformation, and Human-AI feedback.

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Table 1: Internal programme genealogy. Self-citation establishes lineage and role separation, not independent empirical validation.

Programme object	Retained result	Function here
Readout Genesis / MEMK	Retained difference precedes semantic naming; meaning follows readout; experience is an event-trace–meaning configuration; meaning, truth, and epistemic status are non-identical.	Supplies the level distinction between phenomenon, meaning-giving, experience, memory, and knowledge.
Experience Is the Human LoRA	Experience is temporally thick, embodied, affectively weighted, and context-mediated; selected residues may persist and alter future interpretation/action.	Supplies the retention and operator-change layer.
Human Learning as Epistemic Architecture	Lived words connect experience, emotion, value, memory, evidence, goals, constraints, problem-context, tools, workflow, skill, and feedback.	Supplies inspectable articulation and the concept-to-competence arc.
From Problem to Hypothesis	Accessibility differs from reachability; history-shaped attraction and recent-path momentum may change speed and direction of semantic mobility.	Supplies graded re-entry, path history, and auditable restructuring.
Epistemic Fusion or Tunnel	Human-AI dialogue recursively changes accessible routes; repeated traversal can create inclination; amplification is epistemically neutral; only retained human change crosses the session boundary.	Supplies reciprocal interaction, temporalized accessibility, resistance, and direction.
Meaning Before Naming	Pre-explicit affective-relational organization can precede stable naming; naming can later restructure the state it reports.	Supplies the pre-articulation interval but requires correction of the earlier resonance definition.
CTSA Human-Return Readout	After AI removal, retained change is classified by demonstrable Concept, Tool-Selection, Skill-Execution, or Alternative-Generation Return, with loss/warrant/direction kept separate.	Supplies the upper measurement boundary: explicit crystallized return, not every retained residue.

4.3 Appraisal and emotion construction: event meaning before stable label

Scherer and Moors review evidence for an emotion process in which appraisal of an event leads to differentiated action tendencies and physiological responses before conscious feeling and later semantic categorization or labeling [14]. Barrett and colleagues describe emotion experience as a content-rich event in which evolving affect, world perception, and conceptual knowledge are bound at a moment in time [15]. These literatures support a layered process from event significance to explicit category, while also warning that affect, conceptual knowledge, and labeling interact.

The warning is critical. Lindquist and colleagues experimentally reduced the accessibility of emotion words and impaired the speed or accuracy of emotion perception [16]. Lieberman and colleagues found that affect labeling changes neural and affective responding to negative stimuli [17]. Thus naming cannot be modeled as a passive final report. The strong one-way formula

$$\text{meaning} \rightarrow \text{name}$$

is inadequate. Naming may reveal, compress, stabilize, distort, or actively reorganize what is experienced.

4.4 Predictive processing: prior organization shapes present perception

Predictive-processing accounts model perception as shaped by prior expectations and error correction rather than as bottom-up copying [18]. This strongly constrains any readout theory that would ignore prior organization. However, the current predictive-processing literature is neither conceptually nor empirically uniform. A 2026 Annual Review argues that sensory prediction-error findings can reflect different computations and that the framework's definitions and scope require renewed scrutiny [19]. The present paper therefore uses prediction and surprise as possible mechanisms or covariates, not as a universal explanatory foundation.

4.5 Memory, reconsolidation, and insight: retained history can be revised abruptly

Prior knowledge and schemas alter encoding, consolidation, and retrieval [20, 21]; emotional memory is selectively remembered and retrieval can itself alter later representation [22]. Reconsolidation research further supports the possibility that reactivated memories can become modifiable, though mechanistic and translational claims require care [23]. Insight research distinguishes abrupt representational restructuring from the phenomenological Aha! experience; an Aha! feeling and a genuine lower-defect restructuring need not be the same event [24]. These literatures constrain the present transition model: abrupt subjective release is possible, but intensity alone cannot identify the underlying mechanism, truth, or durability.

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4.6 Rhythm: measurable temporal structure, not semantic substance

Research on musical rhythm, beat, and metre treats rhythm as a measurable organization of events in time and compares mechanistic accounts including oscillator and predictive-coding models [25]. A large cross-cultural study of 39 participant groups in 15 countries found both shared small-integer-ratio rhythm priors and systematic culture-related variation, showing that temporal structure can be jointly constrained by common regularities and learned history [26]. The lesson for the present architecture is exact: rhythm is not a metaphor for meaning and is not itself emotion. It is a temporal structure that can constrain prediction, coordination, expectation, and re-entry. The programme extends this idea beyond music only at the level of a testable domain hypothesis: recurrence, spacing, interruption, emphasis, and return may organize later accessibility in learning and dialogue.

4.7 Human-AI feedback: the human state can change before the next prompt

Glickman and Sharot report that repeated interaction with biased AI systems alters later human perceptual, emotional, and social judgments, producing feedback loops that can amplify bias [32]. This does not validate the full readout-retention theory, but it directly defeats the assumption that each prompt starts from a fixed human baseline. Prior interaction can alter the human state from which the next prompt is produced.

4.8 Schema congruence, self-reference, and fluency: direct rivals for resonance

The resonance construct is intentionally placed under stronger pressure than in the previous draft. Schema research shows that information congruent with prior knowledge is often encoded and consolidated differently from incongruent information [27]. A meta-analysis of the self-reference effect found robust memory advantages when material is related to the self, with organization and elaboration accounting for much of the effect [28]. Processing-fluency research further shows that ease of processing can increase liking and felt positivity without requiring a special external-internal congruence mechanism [29]. These are not reasons to abandon resonance; they define the rival set it must beat. The strong claim is that resonance, if retained as a technical construct, captures an experiential fit relation that cannot be exhausted by any one of schema congruence, self-reference, familiarity, fluency, appraisal, or retrieval alone.

4.9 Event segmentation: temporal organization beyond music

Event-cognition research provides the missing bridge from musical rhythm to temporally organized experience. Event Segmentation Theory proposes that people spontaneously partition ongoing activity into meaningful events, with event boundaries linked to changes in prediction and with consequences for attention and memory [30]. Related work emphasizes that segmentation depends on both bottom-up perceptual change and top-down conceptual structure such as goals [31]. This literature does not prove a general rhythm law. It does show that the temporal form of experience is not exhausted by clock time or exposure count: boundaries, return, transition, and predictive discontinuity can reorganize how episodes are encoded and remembered.

5. The Defended Thesis: Experience Is Meaning-Giving

The title is intentionally strong. It should not be read as saying that human agents arbitrarily manufacture the truth of phenomena. The defended thesis is a relational identity at the level of lived experience:

> *For a finite situated human, an experience is the phenomenon as it becomes meaningfully present through a constrained relation among encountered structure, embodiment, retained history, expectation, value, conceptual resources, and context.*

Three consequences follow.

P1 - Phenomenon is not experience. An occurrence or trace can exist without being experienced by a particular reader. Experience begins only when some part of that occurrence becomes accessible and significant to the reader.

P2 - Meaning-giving is not truth-making. The fact that an event is experienced as betrayal, beauty, threat, revelation, injustice, or opportunity establishes the reader's experienced significance, not the objective truth of every interpretation attached to it.

P3 - Meaning-giving is constrained. The reader is not free to assign any meaning whatsoever without cost. Source structure, embodiment, learned categories, social practice, memory, prediction, rights, and later world resistance constrain admissible and stable interpretations.

P4 - Experience can be pre-explicit. Some significance may be active before the person can stably name it, but this does not imply a fully formed proposition hidden beneath language.

P5 - Naming is recursive. Later articulation can reveal, sharpen, compress, distort, or reorganize the experience that made articulation possible.

P6 - No lived experience is significance-free. Under the present thesis, a phenomenon can be registered without becoming a lived experience for a declared human reader; but once it is lived as experience, at least one significance relation is active. That significance need not be lexical or propositional. Pain may be affectively and pragmatically significant before it is conceptually named; a face may be autobiographically salient before its relevance is

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Table 2: External literature map and the residual role of the present paper.

Literature	What it already provides	Constraint / residual role here
Phenomenology	Intentionality, lived body, implicit horizon of meaning.	Do not claim that significance in experience is newly discovered; add formal level separation and retention.
Enaction / sense-making	Meaning and value emerge relationally in embodied organism-environment coupling.	Position meaning-giving as a readout-retention extension, not a substitute for enaction.
Appraisal / emotion construction	Event appraisal can precede labeling; affect, perception, and concepts jointly shape experience.	Pre-explicit organization is plausible, but language can also be constitutive.
Predictive processing	Prior models and prediction errors shape perception/action.	Use prediction/surprise as rival or component mechanisms, not as automatic validation.
Memory / reconsolidation / insight	Prior knowledge shapes encoding; retained memories can update; restructuring can be abrupt.	Separate release from durable transformation and Aha! from warranted insight.
Rhythm cognition	Rhythm is measurable temporal organization with multiple candidate mechanisms and cross-cultural constraint.	Generalize as a DR/Open temporal-organization hypothesis rather than a music metaphor.
Schema / self-reference / fluency	Prior congruence, self-relevance, organization, elaboration, and processing ease can alter memory and felt response.	Treat these as direct rival explanations that resonance must outperform incrementally.
Event cognition	Ongoing activity is segmented into meaningful events; boundaries depend on prediction and conceptual structure and affect memory.	Supplies a non-musical bridge for temporal organization, interruption, transition, and return.
Human-AI feedback	Repeated AI interaction can alter later unaided human judgments.	Supports history-shaped human starting states, not the full resonance/rhythm theory.

articulated; a problem may be epistemically significant before a stable hypothesis is formed.

5.1 Meaning plurality without dilution

The strong thesis is preserved by refusing to equate *meaning* with only one linguistic or propositional form. For analytic purposes, let the reader-conditioned meaning relation be typed as

$$\boldsymbol{\mu}_n = (\mu_n^{aff}, \mu_n^{prag}, \mu_n^{auto}, \mu_n^{conc}, \mu_n^{epi}),$$

where the components denote affective significance, pragmatic/action significance, autobiographical significance, conceptual-semantic significance, and epistemic significance. These are *readout modes*, not biological modules and not an exhaustive ontology. Any subset may be active in a given episode. The unifying claim is stronger, not weaker: **experience is the phenomenon as it becomes significant for the agent through one or more of these mutually interacting meaning relations.**

This typing closes an equivocation without giving up the identity thesis. It explains how preverbal or difficult-to-name experience can still be meaningful, how language can later become constitutive, and why epistemic meaning must remain separable from affective force. In particular,

$$\mu^{aff} \neq \mu^{epi}, \qquad \mu^{auto} \neq \mu^{epi}, \qquad \mu^{prag} \neq \text{truth}.$$

A phenomenon can matter deeply without thereby being correctly interpreted.

5.2 Strong novelty claim: architectural closure

The paper does not claim that phenomenology, enaction, appraisal, schema theory, or event cognition failed to notice significance. Its strong novelty claim is architectural. Across the literature reviewed here, the relevant phenomena are distributed across separate traditions; the present paper places them into one role-separated sequence with explicit non-collapse and deletion rules:

$$\begin{gathered}\text{phenomenon} \to \text{meaning-giving readout} \to \text{experience}\\ \downarrow \\ \text{resonance / dissonance} \leftrightarrow \text{rhythmic history}\\ \downarrow \\ \text{retention / accumulation} \to \text{barrier crossing / release}\\ \downarrow \\ \text{recursive naming} \to \text{future human state}\\ \downarrow \\ \text{pre-prompt Human-AI interaction} \to \text{Human Return}\end{gathered}$$

The contribution is therefore not a synonym for sense-making. It is a closed readout-retention architecture that links lived significance to temporal history, congruence, abrupt transition, Human-AI feedback, and unaided retained capability while keeping truth and warrant external to experiential force.

6. Readout-Native Formalization

The formalization must preserve Readout Genesis order: retained difference first, semantic naming only after a valid domain translation. No psychological noun is promoted to a root primitive [1]. The equations below are **typed dependency maps and candidate domain dynamics**, not already estimated universal psychological laws. Their purpose is to make the strong thesis auditable rather than metaphorical.

Let x_n denote a bounded event trace or encountered pattern made available to the human domain at step n. Let H_n denote the human's retained organization relevant to the current domain, and c_n the context. A finite human readout is

$$r_n = \mathcal{R}_H(x_n \mid H_n, c_n).$$

The readout is not the world event itself. It is what

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this finite reader can access under declared conditions.

Meaning-giving is then represented schematically as

$$\boldsymbol{\mu}_n = \Psi_H(r_n, H_n, c_n, Q_n),$$

where $\boldsymbol{\mu}_n$ is the reader-conditioned vector of active significance relations under question or criterion Q_n. The scalar notation μ_n is retained when only one declared readout mode is in scope. This is intentionally relational: meaning is not stored inside x_n, and it is not produced independently of x_n.

The MEMK domain already defines experience as

$$E_n = \Phi_E(x_n, \mu_n, \kappa_n, c_n),$$

with the compact interpretation that experience is a non-additive event-trace–meaning configuration under context [2]. The present paper therefore makes explicit a thesis already latent in the programme:

$$\boxed{\text{Experience} = \text{phenomenon-as-meaningfully-read}}$$

within a declared human domain.

The phrase *giving meaning* can now be stated without mystification:

$$\boxed{\begin{array}{r}\mu_n \text{ is the relation that makes } x_n \text{ significant} \\ \text{for } H_n \text{ under } c_n, Q_n.\end{array}}$$

The human does not create the root event; the human creates access and significance under bounded conditions.

Naming is a further operator:

$$\ell_n = \mathcal{L}_H(E_n, \mu_n \mid H_n, c_n),$$

where ℓ_n may be a word, description, metaphor, story, question, or formal concept. The architecture permits

$$E_n, \mu_n \quad \text{before stable } \ell_n,$$

but does not require this order in every case because ℓ_n can feed back into later μ_{n+1}.

Retention is selective:

$$H_{n+1} = U_H\big(H_n, \text{Retain}(E_n, \mu_n, r_n), \delta^{world}_{n:n+1}\big).$$

Thus experience is neither identical to memory nor automatically learning.

7. Resonance: External-Internal Experiential Congruence

The present paper corrects the earlier *Meaning Before Naming* use of *resonance*. Resonance is not defined as mere accessibility change and not as physical oscillation. It is the reader-relative condition in which a current externally prompted experience corresponds with a retained internal experiential organization.

Let E_n^{cur} denote the current meaning-given experiential configuration and let

$$I_{H,n} = \text{Retrieve}(\mathcal{M}_{H,n} \mid c_n, Q_n)$$

denote a context-relevant retained experiential organization retrieved from human memory/history. Define a declared congruence diagnostic

$$\mathcal{C}_H(E_n^{\text{cur}}, I_{H,n} \mid c_n, Q_n) \in [0, 1].$$

Then experiential resonance is

$$\boxed{\text{Res}_H(n) = \mathcal{C}_H(E_n^{\text{cur}}, I_{H,n} \mid c_n, Q_n).}$$

This is a DR/Open domain diagnostic, not a validated psychometric scale.

The intended phenomenology is familiar: a person encounters a melody, sentence, face, scene, ritual, story, or AI response and experiences, "this corresponds to something already inside me," perhaps before being able to articulate what that correspondence is. The external encounter need not literally contain the internal memory; the internal structure need not be an exact copy of the external pattern. Resonance names **fit across the external-internal experiential boundary**.

Four non-collapses are mandatory:

$$\begin{array}{ll}\text{Res} \neq \text{Identity}, & \text{Res} \neq \text{Truth}, \\ \text{Res} \neq \text{Retention}, & \text{Res} \neq \text{Improvement}.\end{array}$$

A conspiracy narrative can resonate with retained fear. A prejudicial frame can resonate with prior social experience. A false reassurance can resonate with hope. Congruence explains force for the reader, not warrant.

Strong-form construct criterion. Resonance is retained as a distinct technical object only if an explicit congruence measure contributes incremental held-out prediction after controlling for schema congruence, self-reference, familiarity, processing fluency, appraisal, valence, and autobiographical retrieval. Formally, for an outcome family Y,

$$\Delta_{Res} = \mathcal{P}(Y \mid B, Res) - \mathcal{P}(Y \mid B),$$

where B denotes the preregistered rival set and $\mathcal{P}$ is a declared out-of-sample predictive score. The strong resonance claim earns technical status when $\Delta_{Res} > 0$ reproducibly under frozen criteria. If not, the phenomenology of fit remains describable, but the special diagnostic should be removed.

Resonance is therefore an interface construct. Candidate mechanisms underneath it may include autobiographical retrieval, schema congruence, prediction, association, affective appraisal, embodied readiness, familiarity, or social framing. If these variables fully explain the relevant outcomes and the congruence construct adds no discriminating value, resonance should be removed as a separate technical object.

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8. Rhythm: Structured Recurrence Across Ordered History

Rhythm occupies a different level. It does not mean internal-external congruence. It describes temporal organization of recurrence.

Readout Genesis already supplies ordered lineage before continuous-time semantics [1]. Human Learning includes spacing, retrieval, repeated practice, and workflow; Human LoRA includes rehearsal and selective consolidation; From Problem to Hypothesis models recent-path history; Fusion/Tunnel states that repeated traversal can make a route easier to re-enter. The residual object is therefore not musical rhythm per se but a domain-level temporal structure:

> *Rhythmic organization is the structured pattern of recurrence, spacing, interruption, emphasis, and return across an ordered lineage, where the organization of prior encounters may alter readiness or accessibility of subsequent readouts.*

Let the ordered encounter history be

$$\mathcal{T}_n = \{(x_k, t_k, w_k)\}_{k \le n},$$

where t_k is order/time and w_k is a declared salience or role weight. A rhythm descriptor

$$\mathfrak{R}_n = \Omega(\mathcal{T}_n, \mathcal{B}_n)$$

may summarize recurrence intervals, returns, interruptions, local regularity, and a declared event-boundary sequence $\mathcal{B}_n$. Event-segmentation research makes this extension substantive: boundaries can reorganize prediction, attention, and later memory even when total exposure is similar [30, 31]. This is not yet a causal law of human meaning; it is a stronger temporal hypothesis than exposure count alone.

The semantic-momentum object from *From Problem to Hypothesis* provides one possible downstream effect:

$$m_{t+1}(e) = \rho m_t(e) + \mathbf{1}[e_t = e], \qquad 0 \le \rho < 1,$$

where recent repeated traversal makes the same semantic direction easier to re-enter for a limited history window [6]. The present paper generalizes the question but not the empirical claim: does the temporal organization of recurrence add predictive value beyond mere exposure count and current state?

The following distinctions are therefore required:

$$\text{order} \neq \text{rhythm}, \quad \text{repetition} \neq \text{rhythm}, \quad \text{rhythm} \neq \text{meaning}, \quad \text{rhythm} \neq \text{retention}.$$

Music is a particularly clear phenomenological probe because timing, pause, repetition, beat, and return are explicit, but the architecture also generates testable questions about learning sequences, therapeutic dialogue, ritual, narrative, and Human-AI conversation.

9. Accumulation, Potential, Barrier Crossing, and Release

The user's motivating intuition is that internal experience can accumulate "power" and then be released explosively when an encounter unlocks it. The programme already contains a disciplined way to retain this intuition without importing physical energy.

MEMK v0.2 defines a domain-level semantic potential $V_{\text{sem}}(z; c, T)$ as an informational stability landscape, an informational work functional W_{info}, and an activation barrier $\Delta V^{\dagger}_{\text{old}\to\text{new}}$ separating an old stable meaning basin from a candidate new basin [3]. It explicitly forbids reporting semantic potential or meaning-energy as literal physical energy. It also distinguishes shock-like impulse, coherent sudden insight, gradual learning, and life-changing transformation.

This allows a precise reinterpretation of "stored power." The defensible object is not stored joules but **retained transition readiness**: accumulated history may deepen an old basin, create unresolved conflict, strengthen a latent alternative, or repeatedly perform low-amplitude informational work on the current semantic organization.

For repeated low-amplitude updates, MEMK already proposes

$$\Delta W_{j,N} = \sum_{n=1}^{N} \eta_n \, \text{eligible-gradient}_n,$$

and states that a transition may occur after cumulative effective work crosses a barrier or slowly reshapes the stability landscape so the old basin loses stability [3]. The present paper gives this an experiential reading.

A current encounter may contribute effective work according to source strength, reader eligibility, context, and alignment. Resonance can be treated as one possible *alignment term*, not as energy itself:

$$W_n^{\text{eff}} = W_{\text{info,n}} \, g\big(\text{Res}_H(n), \text{eligibility}, \text{context}\big),$$

with g left DR/Open. A transition candidate appears when cumulative effective work exceeds a declared barrier:

$$\sum_{k \le n} W_k^{\text{eff}} \ge \Delta V^{\dagger}_{\text{old}\to\text{new}}.$$

The strong claim is not that all psychological change obeys one universal threshold. It is that **some experience trajectories may be better described by a transition topology than by smooth accumulation alone**. This is an empirical wager rather than a retreat. A continuous rival can be written as

$$z_{n+1} = z_n + \alpha W_n^{\text{eff}} + \epsilon_n,$$

while the transition model permits a state-dependent regime change once cumulative work crosses the declared barrier. The barrier claim

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gains scientific force only if it predicts abrupt reorganization, hysteresis, delayed settling, or cross-context persistence that the continuous model misses on held-out trajectories.

The phenomenological outcome may be experienced as sudden release: crying, laughter, speech, movement, confession, creative production, insight, decisive action, or a rapid reframing. The paper calls this **transformative release candidate** when the transition is abrupt, while reserving **durable transformation** for change that persists across memory, view, prediction, action, and context.

Hence:

$$\begin{aligned}\text{release} &\neq \text{transformation},\\ \text{intensity} &\neq \text{truth},\\ \text{shock} &\neq \text{insight},\\ \text{barrier crossing} &\neq \text{improvement}.\end{aligned}$$

MEMK itself requires persistence, cross-context stability, held-out reuse, and lineage preservation before a life-changing transformation report is warranted [3].

10. Naming as a Recursive Operator

A theory of experience as meaning-giving cannot place language outside the process. Language is both an articulation channel and a potential reorganizer.

Before naming, a person may have an active pattern of felt significance, relational accessibility, action readiness, or mismatch. Naming can make this structure socially inspectable and revisable. But linguistic evidence shows that category words can also alter perception and affective processing [16, 17]. Therefore the architecture is recursive:

$$\mu_n \to E_n \to \ell_n \to \mu_{n+1} \to E_{n+1}.$$

Language can perform at least four roles: (1) **disclosure**, making a previously tacit relation nameable; (2) **compression**, reducing a high-dimensional lived configuration to a portable label; (3) **restructuring**, changing future distinctions and expectations; and (4) **transport**, exporting a bounded readout to another person or AI system.

This preserves the strongest correction from hostile review: *meaning before naming* cannot mean a fully completed semantic proposition waiting intact for a word. The defensible claim is that **meaning-giving can begin pre-explicitly, while naming can later become constitutive of the next experiential state**.

11. Human-AI Coupling Begins Before the Prompt

The experience theory changes the placement of Human-AI interaction. This paper names the **Pre-Prompt Human State Principle**: *a prompt is not the start of the human trajectory; it is a bounded articulation of an already meaning-shaped, history-bearing human state*. The principle is architectural, while the size and mechanisms of pre-prompt history effects remain empirical questions.

Let H_t contain retained experiences, concepts, affective weights, values, unresolved differences, and recent interaction history. The human articulates only a partial readout:

$$H_t \xrightarrow{\mathcal{L}_H} Q_t.$$

The AI receives Q_t, not H_t itself. It returns output Y_t, which the human reads through the same embodied and retained organization:

$$Y_t \xrightarrow{\mathcal{R}_H} E^{AI}_{H,t}.$$

That output can resonate with prior experiential organization, fail to resonate, create dissonance, or restructure future meaning-giving. If retained,

$$\begin{gathered}H_{t+1} = U_H(H_t, E^{AI}_{H,t}, \text{world correction},\\ \text{other experience}).\end{gathered}$$

The next question is then produced from a changed state:

$$H_{t+1} \to Q_{t+1}.$$

This is where rhythm becomes a Human-AI research question. Dialogue has recurrence, turn order, delay, interruption, restatement, and return. A frame repeated across several turns may become easier to re-enter. A strategically spaced counterexample may preserve resistance better than continuous affirmation. A pause that requires unaided retrieval may change retention differently from immediate completion. These are hypotheses, not established design laws.

Glickman and Sharot provide a strong external ally to the general history-shaped claim: repeated exposure to AI judgments can alter later human judgments and amplify bias [32]. Their results do not identify experiential resonance or conversational rhythm as mechanisms. They show that the human state entering a later judgment can differ because of prior AI interaction.

The pre-prompt architecture is therefore:

$$\boxed{\begin{aligned}&H_t \to Q_t \to AI_t \to Y_t \to E_{H,t}\\ &\quad \to \text{Retain?} \to H_{t+1} \to Q_{t+1}\end{aligned}}$$

with resonance and rhythm as candidate domain-level modifiers of uptake and re-entry.

12. From Retained Experience to CTSA Human Return

Not every retained experiential change is immediately an explicit capability. CTSA Human-Return Readout asks a narrower question: after AI support is removed, what remains demonstrably usable by the human? Its four classes

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are Conceptual, Tool-Selection, Skill-Execution, and Alternative-Generation Return [9].

The present paper therefore inserts a lower layer:

$$\text{retained experiential reorganization} \\ \to \text{possible later CTSA crystallization}.$$

A resonant encounter may leave only a sensitivity. A rhythmically repeated distinction may become easier to retrieve but still fail a transfer test. A sudden release may produce a new narrative without a new skill. None should be counted as CTSA gain until the unaided human can reconstruct, discriminate, transfer, select, execute, or generate under frozen criteria.

Possible developmental trajectories include

$$\text{retained sensitivity} \to C_{\text{return}},$$

$$\text{retained reweighting} \to A_{\text{return}},$$

or

$$\text{retained reorganization} \to Q_{\text{next}}^{\text{better}}.$$

These are open pathways, not universal causal sequences.

13. Non-Collapse Laws

The paper's strongest contribution may be the role separation it enforces. The following non-collapses are constitutive:

- phenomenon $\neq$ experience;
- event trace $\neq$ meaning;
- meaning-giving $\neq$ truth-making;
- pre-explicit significance $\neq$ fully formed proposition;
- naming $\neq$ passive reporting;
- resonance $\neq$ identity;
- resonance $\neq$ truth;
- rhythm $\neq$ repetition count;
- rhythm $\neq$ meaning;
- accessibility $\neq$ warrant;
- semantic attraction $\neq$ evidence;
- informational work $\neq$ physical energy;
- shock $\neq$ coherent insight;
- release $\neq$ durable transformation;
- retention $\neq$ improvement;
- Human-AI amplification $\neq$ synergy;
- coupled performance $\neq$ Human Return.

These laws block a recurring error: a phenomenon can become experientially powerful because it fits, repeats, returns, or crosses a transition threshold, yet remain false, harmful, manipulable, or epistemically unsupported.

14. Empirical Programme: A Hostile Test, Not a Demonstration

The first empirical programme should try to kill the theory, not illustrate it. Each construct must add out-of-sample explanatory value beyond simpler rivals.

14.1 H1 - Meaning-giving beyond arousal [Open]

Hold stimulus intensity approximately constant while manipulating event relevance to prior goals, autobiographical memory, or learned structure. Measure pre-label discrimination, action tendency, relational accessibility, and later transfer. The claim survives only if the meaning-giving model predicts structured differences beyond undifferentiated arousal.

14.2 H2 - Resonance as external-internal congruence [Open]

Build individualized retained-experience profiles before exposure. Present matched stimuli that are congruent or incongruent with those profiles while controlling familiarity and valence. Model schema congruence, self-reference, processing fluency, appraisal, and autobiographical retrieval explicitly. Test whether declared external-internal congruence predicts immediate felt fit, selective retrieval, later interpretation, and retention beyond this rival set. Resonance retains strong technical status only with reproducible incremental held-out value.

14.3 H3 - Rhythm beyond exposure count [Open]

Hold the number and semantic content of exposures constant while varying temporal organization: massed repetition, spaced recurrence, regular return, irregular return, interruption, delayed re-entry, and event-boundary placement. Measure accessibility, event segmentation, unaided recall, and later transfer. The rhythm construct earns a distinct role only if temporal organization predicts outcomes beyond repetition count, exposure duration, and boundary count alone.

14.4 H4 - Resonance x rhythm interaction [Open]

Cross congruence with temporal organization. A strong prediction is not simply "more repetition gives more effect," but that structured return to a congruent pattern may alter re-entry or retention differently from equally frequent incongruent or temporally disorganized exposure. A null interaction would narrow the integrated theory.

14.5 H5 - Accumulation versus single shock [Open]

Compare (a) repeated low-amplitude meaning-relevant encounters followed by a congruent trigger, (b) a single high-intensity trigger without prior accumulation, and (c) repeated neutral encounters. Test for rapid reorganization, immediate expression, delayed retention, cross-context transfer, and possible hysteresis. Fit both continuous-

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accumulation and transition/barrier models before inspecting outcomes. The strong barrier claim survives when the transition model predicts held-out trajectory changes that the continuous rival misses.

14.6 H6 - Naming as recursive intervention [Open]

Measure pre-label organization, then randomly assign free naming, supplied labels, no-label control, or semantic-interference conditions. Test whether later readout differs even when the original stimulus is held constant. Existing language and affect-labeling studies motivate this design but do not establish the full architecture [16, 17].

14.7 H7 - Release is not transformation [Derived guardrail]

Separate immediate intensity or externalization from delayed retention, cross-context stability, action change, and Human Return. A dramatic reaction with no delayed change counts as release without durable transformation.

14.8 H8 - Human-AI pre-prompt state [Open/externally allied]

Randomize repeated AI feedback trajectories while freezing task content and later remove the AI. Measure pre-response human judgments, question quality, CTSA return, and resistance to contrary evidence. Glickman and Sharot show that prior AI interaction can alter later judgments; the proposed test asks whether resonance, rhythm, or meaning-history variables add explanatory value [32].

14.9 H9 - Meaning-mode decomposition [Open]

Test whether the typed meaning relation adds structure beyond a single undifferentiated significance score. Preregister affective, pragmatic, autobiographical, conceptual, and epistemic readouts without treating them as latent biological factors. The identity thesis does not require all five to dissociate psychometrically; it predicts that experience can be significant in more than one way and that collapsing epistemic meaning into affective force will systematically obscure warrant-sensitive outcomes.

14.10 Rival model ladder

Every study should preregister at least five rivals:

$$M_0 = \text{arousal/intensity only},$$

$$\begin{aligned} M_1 &= \text{familiarity/exposure}, \\ M_2 &= \text{prediction/surprise}, \\ M_3 &= \text{language/category construction}, \\ M_4 &= \text{memory/schema retrieval}. \end{aligned}$$

with the present readout-retention model M_5 required to add held-out explanatory or predictive value. If it does not, its extra variables should be deleted.

15. Legitimate Defeaters

The architecture should be narrowed or abandoned if well-specified studies show that:

- experience is fully predicted by stimulus features and arousal with no added role for reader history, context, or significance;
- pre-explicit relational measures add no information beyond later verbal labeling;
- external-internal congruence adds no value beyond schema activation, self-reference, familiarity, processing fluency, appraisal, or autobiographical retrieval;
- temporal organization adds no predictive value beyond exposure count, current state, and event-boundary structure;
- repeated low-amplitude history does not alter later transition probability after current-state variables are controlled;
- apparent barrier crossings are fully explained by response threshold artifacts or demand characteristics;
- naming is empirically passive in the relevant domain and does not alter later readout;
- dramatic release does not dissociate from durable transformation under delayed and cross-context measures;
- Human-AI history effects disappear after controlling current prompt content and explicit memory;
- CTSA-style unaided tests reveal no durable capability despite claimed transformation.

The deletion rule is explicit: a construct that performs no distinct explanatory or measurement work should not survive as programme vocabulary.

16. Discussion

The paper began from a simple proposition: *experience is the giving of meaning to phenomena.* After constraint review, that proposition survives only in a specific form. The world is not denied; the reader is not sovereign; and meaning is not treated as a substance. Experience is a relation in which an encountered phenomenon becomes significant for a finite embodied agent. The reader contributes history, expectation, value, memory, and conceptual organization; the phenomenon contributes structure and resistance; context selects which relations become live.

This formulation helps resolve several tensions in the prior programme. First, it explains why *Human Learning* can begin methodologically from lived words without claiming that words are phenomenologically first. A lived word is an inspectable articulation of an already meaning-shaped relation. Second, it strengthens *Human LoRA*: experience is not merely input to learning but the meaning-bearing encounter whose selectively retained consequences

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can alter the future reader. Third, it clarifies *Fusion/Tunnel*: dialogue changes not only propositions but the experiential organization from which later propositions are produced. Fourth, it gives *CTSA* a lower boundary: explicit Human Return may crystallize from retained experiential reorganization without exhausting it.

The corrected definition of resonance is central. Earlier drafts risked using resonance as a loose synonym for accessibility. The present paper instead reserves it for current external-internal experiential congruence. This move makes the construct more precise but also more vulnerable to empirical defeat. It may turn out that familiar schema-congruence or autobiographical retrieval measures do all the needed work. If so, the programme should keep the phenomenon and drop the special label.

Rhythm likewise becomes useful only when stripped of mystique. Its role is structural and temporal. Recurrence, spacing, interruption, and return may organize the conditions under which a relation becomes easier to re-enter. Music makes this structure visible, but the question generalizes to education, ritual, clinical dialogue, social media, and AI conversation. Importantly, repeated ease does not confer truth. A false route can become rhythmically familiar and experientially resonant.

The most provocative extension concerns accumulation and release. Human beings often describe long-held experience as building pressure and then "bursting out." The programme can preserve this phenomenology without claiming literal stored energy. MEMK's informational work and semantic-potential barrier offer a typed domain model: repeated low-amplitude updates may accumulate, an aligned encounter may increase effective work, and a transition may become abrupt when the old organization loses stability. The scientific burden is high. The architecture must discriminate gradual accumulation from simple exposure, threshold artifacts, memory retrieval, and post-hoc narrative.

Finally, the Human-AI implication is not that AI secretly controls an invisible emotional field. The stronger and testable claim is that AI enters a meaning-bearing temporal trajectory that already exists before the prompt. It is that Human-AI interaction occurs inside a history-bearing human system. AI outputs are themselves phenomena to be meaningfully read. They can fit or clash with retained experience, recur with particular timing, and become selectively retained. The next prompt is therefore partly an outcome of prior interaction. External empirical evidence already establishes that repeated AI judgments can shift later human judgments [32]; the next research question is what combination of meaning-giving, resonance, temporal organization, resistance, and retention best explains when such influence becomes learning, fixation, or epistemic tunnel.

17. Standalone Thesis

The programme can now be compressed into one statement:

> *Experience is the meaning-giving relation through which a phenomenon becomes significant for a finite embodied reader. Meaning here is not restricted to propositions: affective, pragmatic, autobiographical, conceptual, and epistemic significance can coexist within one lived configuration. The relation is constrained by the encountered world and shaped by retained history, context, expectation, value, embodiment, and available concepts. When a current externally prompted experience fits retained internal experiential organization, resonance occurs as external-internal congruence. When related encounters recur with temporal structure, rhythm organizes recurrence, spacing, interruption, emphasis, boundary, and return. Repeated retained changes can produce transition readiness; a sufficiently aligned encounter may contribute to barrier crossing and transformative release. Naming can both articulate and restructure experience, retention determines what persists, and independent resistance determines whether what persists deserves epistemic trust. The Pre-Prompt Human State Principle follows: Human-AI interaction begins before the prompt because the prompt is a bounded readout of a previously meaning-shaped and history-bearing state. The quality of collaboration should ultimately be judged by what the human can still discriminate, revise, transfer, and use after the system is gone.*

18. Conclusion

The proposed theory does not reduce experience to information processing, language, emotion, or memory. Nor does it place an unconstrained private world between the agent and reality. It treats experience as a bounded relation: phenomenon plus meaning-giving under a finite reader. From that relation, several previously separate programme threads become continuous. Pre-explicit significance explains why naming can lag experience; recursive language explains why naming can later change it; resonance explains why some encounters feel deeply fitted to retained life history; rhythm explains why the temporal organization of return may matter independently of content; retention explains why some encounters disappear while others alter future readout; barrier crossing explains why accumulated change can sometimes become abrupt; and Human Return explains why a powerful interaction still does not count as durable human capability until the AI is gone and the person can act, discriminate, revise, or generate independently.

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The strongest conclusion is therefore the strong thesis itself, now made role-separated and testable: **a phenomenon becomes human experience through meaning-giving; meaning-giving is plural in mode, history-shaped, embodied, and world-constrained; resonance is the congruence between present external experience and retained internal experience; rhythm organizes the temporal return of encounters; accumulated retained change can create transition readiness and transformative release; and whatever becomes easier to feel, think, repeat, or retain must still face independent resistance before it earns epistemic status.** In Human-AI systems, this means the epistemic trajectory begins before the prompt and cannot be evaluated only from coupled output.

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References

References

[1] Readout Genesis Programme. (2026). *READOUT_GENESIS_MEMK_DOMAIN v0.3.0 ANCHORED*. Root-connected semantic-cognitive domain specification, 23 July 2026. Architecture-conditional / unresolved domain.

[2] Readout Genesis Programme. (2026). *READOUT_GENESIS_MEMK_DOMAIN v0.3.0 ANCHORED*: Meaning–Experience–Memory–Knowledge domain definitions. 23 July 2026.

[3] Readout Genesis Programme. (2026). *READOUT_GENESIS_MEMK_DOMAIN v0.2.0*: Informational adaptation, semantic potential, informational work, barrier crossing, and transformation protocol. 23 July 2026.

[4] Lahtee, Y. (2026). *Experience Is the Human LoRA: A Readout–Retention Theory of Selective Model Change*. Open Civil Science Initiative preprint, 18 July 2026.

[5] Lahtee, Y. (2026). *Human Learning as Epistemic Architecture: A Method for Word Mapping, Life-Concept Graphs, Constraint Testing, and Corrigible Agency in the AI Age*. Revised v4 preprint, 28 June 2026.

[6] Lahtee, Y. (2026). *From Problem to Hypothesis: Dynamic Semantic Mobility, History-Shaped Access, and the Readout-Discriminable Hypothesis Space*. Standalone final author draft, GLOSA K0, 4 September 2026.

[7] Lahtee, Y. (2026). *Epistemic Fusion or Epistemic Tunnel: A Phenomenologically Anchored, History-Shaped Architecture of Human-AI Semantic Coupling, Retention, and Epistemic Direction*. Human-AI Readout Programme, Genesis-aligned v7, 5 September 2026.

[8] Lahtee, Y. (2026). *Meaning Before Naming: A Readout-Retention Architecture of Affective-Semantic Reorganization*. Standalone Concept Note v1.0 FULL, 5 September 2026.

[9] Lahtee, Y. (2026). *CTSA Human-Return Readout: A Session-Boundary Measurement Architecture for Retained Human Capability After AI*. Epistemic Note v9.1 FINAL, 5 September 2026.

[10] Smith, D. W. (2013). Phenomenology. In *The Stanford Encyclopedia of Philosophy* (substantive revision 16 December 2013). Stanford University.

[11] Varela, F. J., Thompson, E., & Rosch, E. (1991). *The Embodied Mind: Cognitive Science and Human Experience*. MIT Press. https://doi.org/10.7551/mitpress/6730.001.0001

[12] Di Paolo, E. A. (2005). Autopoiesis, adaptivity, teleology, agency. *Phenomenology and the Cognitive Sciences*, 4(4), 429–452. https://doi.org/10.1007/s11097-005-9002-y

[13] Thompson, E. (2007). *Mind in Life: Biology, Phenomenology, and the Sciences of Mind*. Harvard University Press.

[14] Scherer, K. R., & Moors, A. (2019). The emotion process: Event appraisal and component differentiation. *Annual Review of Psychology*, 70, 719–745. https://doi.org/10.1146/annurev-psych-122216-011854

[15] Barrett, L. F., Mesquita, B., Ochsner, K. N., & Gross, J. J. (2007). The experience of emotion. *Annual Review of Psychology*, 58, 373–403. https://doi.org/10.1146/annurev.psych.58.110405.085709

[16] Lindquist, K. A., Barrett, L. F., Bliss-Moreau, E., & Russell, J. A. (2006). Language and the perception of emotion. *Emotion*, 6(1), 125–138. https://doi.org/10.1037/1528-3542.6.1.125

[17] Lieberman, M. D., Eisenberger, N. I., Crockett, M. J., Tom, S. M., Pfeifer, J. H., & Way, B. M. (2007). Putting feelings into words: Affect labeling disrupts amygdala activity in response to affective stimuli. *Psychological Science*, 18(5), 421–428. https://doi.org/10.1111/j.1467-9280.2007.01916.x

[18] Clark, A. (2013). Whatever next? Predictive brains, situated agents, and the future of cognitive science. *Behavioral and Brain Sciences*, 36(3), 181–204. https://doi.org/10.1017/S0140525X12000477

[19] Furutachi, S., & Hofer, S. B. (2026). Rethinking predictive processing. *Annual Review of Neuroscience*, 49, 471–494. https://doi.org/10.1146/annurev-neuro-102124-031410

[20] Ghosh, V. E., & Gilboa, A. (2014). What is a memory schema? A historical perspective on current neuroscience literature. *Neuropsychologia*, 53, 104–114. https://doi.org/10.1016/j.neuropsychologia.2013.11.010

[21] Alonso, A., van der Meij, J., Tse, D., & Genzel, L. (2020). Naive to expert: Considering the role of previous knowledge in memory. *Brain and Neuroscience Advances*, 4, 1–17. https://doi.org/10.1177/2398212820948686

[22] Kensinger, E. A., & Ford, J. H. (2020). Retrieval of emotional events from memory. *Annual Review of Psychology*, 71, 251–272. https://doi.org/10.1146/annurev-psych-010419-051123

[23] Beckers, T., & Kindt, M. (2017). Memory reconsolidation interference as an emerging treatment for emotional disorders: Strengths, limitations, challenges, and opportunities. *Annual Review of Clinical Psychology*, 13, 99–121. https://doi.org/10.1146/annurev-clinpsy-032816-045209

[24] Wiley, J., & Danek, A. H. (2024). Restructuring processes and Aha! experiences in insight problem solving. *Nature Reviews Psychology*, 3, 42–55. https://doi.org/10.1038/s44159-023-00257-x

[25] Snyder, J. S., Gordon, R. L., & Hannon, E. E. (2024). Theoretical and empirical advances in understanding musical rhythm, beat and metre. *Nature Reviews Psychology*, 3, 449–462. https://doi.org/10.1038/s44159-024-00315-y

[26] Jacoby, N., Polak, R., Grahn, J. A., et al. (2024). Commonality and variation in mental representations of music revealed by a cross-cultural comparison of rhythm priors in 15 countries. *Nature Human Behaviour*, 8, 846–877. https://doi.org/10.1038/s41562-023-01800-9

[27] van Kesteren, M. T. R., Ruiter, D. J., Fernandez, G., & Henson, R. N. (2012). How schema and novelty augment memory formation. *Trends in Neurosciences*, 35(4), 211–219. https://doi.org/10.1016/j.tins.2012.02.001

[28] Symons, C. S., & Johnson, B. T. (1997). The self-reference effect in memory: A meta-analysis. *Psychological Bulletin*, 121(3), 371–394. https://doi.org/10.1037/0033-2909.121.3.371

[29] Reber, R., Schwarz, N., & Winkielman, P. (2004). Processing fluency and aesthetic pleasure: Is beauty in the perceiver's processing experience? *Personality and Social Psychology Review*, 8(4), 364–382. https://doi.org/10.1207/s15327957pspr0804_3

[30] Zacks, J. M., Speer, N. K., Swallow, K. M., Braver, T. S., & Reynolds, J. R. (2007). Event perception: A mind-brain perspective. *Psychological Bulletin*, 133(2), 273–293. https://doi.org/10.1037/0033-2909.133.2.273

[31] Zacks, J. M., & Swallow, K. M. (2007). Event segmentation. *Current Directions in Psychological Science*, 16(2), 80–84. https://doi.org/10.1111/j.1467-8721.2007.00480.x

[32] Glickman, M., & Sharot, T. (2025). How human–AI feedback loops alter human perceptual, emotional and social judgements. *Nature Human Behaviour*, 9, 345–359. https://doi.org/10.1038/s41562-024-02077-2

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Experience Is the Human LoRA: A Readout–Retention Theory of Selective Model Change

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What the chapter states. "This paper argues that experience is the human LoRA." "It is a cross-level identity claim about one finite event." "If an experience-mediated update factors through an active readout space of dimension m inside an operational organization of dimension d, then its update rank is bounded by m. Under the finite-bottleneck condition $m < d$, event-level change is low-rank."

Status. Printed on every page: "(preprint – not peer reviewed)."

Falsifier(s). Verbatim, under "9.8 Hard falsifiers": "F1. Durable event-level changes repeatedly have rank above the preregistered active-field bound." "F2. Full-rank or graph-rewiring models consistently outperform low-rank models in bounded learning tasks."

Read before. Readout Genesis Standalone Synthesis; The Readout Condition; The Standalone Scholar; When AI Expands Human Potential; Written by AI. Still True.

Feeds into. Meaning Before Naming; Experience Is Meaning-Giving.

Experience Is the Human LoRA:

A Readout–Retention Theory of Selective Model Change

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Open Civil Science Initiative

July 18, 2026

Abstract

This paper argues that experience is the human LoRA. The identity is not a claim that the brain is a transformer, that phenomenal character is matrix multiplication, or that biological learning implements the engineering procedure introduced as Low-Rank Adaptation. It is a cross-level identity claim about one finite event. From the first-person side, an experience is a temporally bounded, embodied, affectively weighted, world-directed appearance. From the third-person side, the same event is a selective transformation of the observer that changes how later worlds can be read and acted upon. The argument is developed inside a finite-discrete readout framework: every accessible state is a finite retained difference on a finite graph; time is indexed by $n \in \mathbb{N}$; formal state and update matrices are rational; and no continuum limit is required. If an experience-mediated update factors through an active readout space of dimension m inside an operational organization of dimension d, then its update rank is bounded by m. Under the finite-bottleneck condition $m < d$, event-level change is low-rank. This yields the Human LoRA Factorization Thesis. Durable learning is obtained by a discrete retention gate; cumulative and transformative change can arise through the composition and interaction of many bounded updates. The paper connects this thesis to complementary learning systems, memory consolidation, sampled neural geometry, neurophenomenology, enactivism, psychological flexibility, and inhibitory learning. It distinguishes adaptation from well-being, and it treats mental suffering not as a defective adapter but as a possibly intelligible retained organization whose rigidity, overgeneralization, context mismatch, or repair cost has become harmful.

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The result is a strong but falsifiable theory: experience is Human LoRA because a genuine experience is a finite, retained, observer-transforming difference whose phenomenal and adaptive descriptions are two aspects of the same discrete event.

Keywords: experience; Human LoRA; retained difference; discrete readout; finite learning; phenomenology; mental health; neural geometry

1 Introduction: The Identity Claim

The thesis of this paper is deliberately stated without the protective phrase "is like":

> *Experience is the human LoRA.*

The claim is strong, but it is not the claim that a human brain literally runs the LoRA algorithm of Hu et al. (2022). Nor does it identify the felt quality of an experience with a matrix product. The word *is* names an event identity across explanatory aspects. A single experience can be described from the first-person side as a lived appearance and from the system side as a selective alteration in the organization through which later appearances become possible. The phenomenal manifestation and the adaptive transformation are not two independent events connected by an accidental analogy. They are two readings of one finite observer–world transition.

This paper therefore rejects two common retreats. The first retreat says that experience is only a source of training data and that learning happens elsewhere. That view separates the event from the transformation too sharply. A genuine experience is not merely presented to an unchanged observer; it is the observer's finite transition under world contact. The second retreat says that the title is only metaphorical. That view makes the thesis immune to proof and falsification. The present account instead gives an exact finite factorization condition, preservation laws, composition bounds, empirical readout tests, and explicit failure conditions.

The framework extends the author's larger *readout-not-truth* programme, in which an agency never reads an unbounded world in itself; it reads finite retained differences, and every claim must disclose whether it is a formal finite theorem, a finite diagnostic, an interpretive stance, or an open bridge (Lahtee 2026b; Lahtee 2026a). The present paper is standalone, but it preserves that discipline. It uses no actual infinity, no infinitely divisible time, no state continuum, and no appeal to a limiting observer. The formal objects are finite graphs, finite lists, finite matrices, rational weights, and discrete steps.

The resulting argument has six stages. First, I state the finite-discrete ontology and its claim tiers. Second, I define experience as a finite, embodied, temporally thick readout event. Third, I show how the adaptive aspect of such an event factors through a bounded active field and therefore has bounded rank. Fourth, I explain how many event-level updates compose into expertise and transformative change. Fifth, I connect the theory to phenomenology, learning science, sampled neural geometry, and mental health. Sixth, I specify tests that could defeat the low-rank thesis rather than merely redescribe any result as supportive.

The central contribution can be compressed into four claims:

C1. Every genuine experience is a finite observer transition, not a passive copy of

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an input.

C2. The active phenomenal field is bounded relative to the total operational organization of the person.

C3. An event-level update that factors through that bounded field has bounded rank and preserves directions outside its update channel.

C4. The phenomenal and adaptive descriptions refer to the same event; hence experience is the human LoRA at the level of finite event identity.

The strongest open premise is C2 in its empirical form. Finitude alone guarantees a finite rank, but *low* rank requires the active event field to be smaller than the operational organization it can modify. That is not hidden as a definition. It is the Human Finite-Bottleneck Postulate and generates a risky prediction.

2 Finite Readout, Not Continuum

2.1 The readout-not-truth stance

The paper begins from one restriction: whatever a human or an instrument can actually distinguish at a step is finite. This does not claim that every useful scientific model must avoid continuum mathematics. It claims that the ontology and evidence of this paper are readout-bounded. A measured or lived appearance is not an infinitely precise point in an uncountable state space. It is a finite retained distinction available to an observer at a finite resolution.

Let the human-relevant organization at step n be represented by a finite retained-difference graph

$$G_H[n] = (V_H[n], E_H[n], w_H[n]), \tag{1}$$

where $V_H[n]$ and $E_H[n]$ are finite sets and every retained coupling weight satisfies

$$w_H[n](e) \in \mathbb{Q}_{>0}. \tag{2}$$

The graph is not restricted to neurons. Depending on the explanatory scale, its nodes can represent discriminable neural, bodily, practical, linguistic, environmental, or relational states. The point is relational: the human interpretive organization is not a sealed internal object but a finite organization of possible distinctions and couplings across brain, body, history, practice, other persons, and environment.

A live state is a finite rational field

$$\phi_n \in \mathbb{Q}^{|V_H[n]|}. \tag{3}$$

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The use of $\mathbb{Q}$ has two functions. It makes the formal theory exactly discrete and exposes the difference between a proved structural law and a measured approximation. Empirical measurements can be quantized to stated rational resolution; the resulting claims are finite diagnostics, not infinitely precise truths.

2.2 Discrete time and the human spine

Time is indexed by $n \in \mathbb{N}$. Define the first and second finite differences

$$\delta\phi_n = \phi_{n+1} - \phi_n, \tag{4}$$
$$\delta^2\phi_n = \phi_{n+1} - 2\phi_n + \phi_{n-1}. \tag{5}$$

A human-domain discrete spine can then be written as

$$\mu_H\delta^2\phi_n + d_H\delta\phi_n + \kappa_H L_H[n]\phi_n + \partial_{\mathbb{Q}}V_H(\phi_n) = J_H[n] - \eta_H[n], \tag{6}$$

where $L_H[n]$ is the finite graph Laplacian, $\mu_H, d_H, \kappa_H \in \mathbb{Q}_{\geq 0}$, $\partial_{\mathbb{Q}}V_H$ is a declared finite-difference potential term, $J_H[n]$ is a finite drive, and $\eta_H[n]$ is the disclosed residual. Equation (6) is a `Dr` domain instantiation of the repository's common spine, not a derived biological law. Its role is architectural: retained organization, inertia, dissipation, relational geometry, drive, and unmodelled remainder are represented in one finite step equation.

The paper never moves from $n \in \mathbb{N}$ to a variable t on an infinitely divisible line. No conclusion depends on $h \to 0$, infinite precision, or a limiting manifold. External studies that use "neural manifold" vocabulary are translated here into finite sampled geometry: a finite matrix of observed activity under a declared resolution and a finite set of trials. The external vocabulary is retained for citation accuracy, but no continuum ontology is imported.

2.3 Claim tiers

The following tier discipline prevents the strongest philosophical thesis from being confused with the strongest formal result.

Tier	Meaning in this paper	Examples
$\mathrm{T}\textsc{h}_{\mathbb{Q}}$	Exact finite proof over rational matrices or finite sets, proved in the article. It is not labelled `Th_coqc` until separately machine-checked.	Rank bound, preservation outside the update channel, composition bound.

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Tier	Meaning in this paper	Examples
Finite diagnostic	A result from finitely sampled, finite-resolution empirical or computational data.	Quantized readout rank, held-out model comparison, finite neural activity matrices.
`Dr`	Interpretive or architectural claim connecting formal pieces.	Dual-aspect event identity, embodied-relational interpretation of the operator.
`Open`	An empirical or formal bridge not yet established.	The general Human Finite-Bottleneck Postulate; literal biological realization of low-rank updates.

The title is a `Dr` identity thesis constrained by $\mathrm{Th}_{\mathbb{Q}}$ propositions and exposed to `finite-diagnostic` tests. It is not promoted to a machine-checked neurobiological theorem.

3 Experience as a Finite Observer Transition

3.1 World contact and selected readout

Let X_n denote the finite world-side event traces causally available at step n. The human agent cannot access all physically describable differences. A selection map

$$S_n : X_n \longrightarrow Z_n \tag{7}$$

returns a finite active set Z_n of distinctions. Let $m_n = |Z_n|$. The translated readout is

$$z_n = \Pi_n S_n(X_n) + \eta_n, \qquad z_n \in \mathbb{Q}^{m_n}, \tag{8}$$

where Π_n is a finite projection or reporting map and η_n is a finite residual. The equation does not say that experience is only a vector. It says that the evidentially available structure of the event is finite and that a formal model must disclose what was selected and lost.

A readout is constrained by the world but not identical with the world. This is compatible with predictive accounts that emphasize prior organization and error correction (Friston 2010; Clark 2013), but the present theory does not define the world as an inferred probability distribution. It only requires finite world resistance: different actions and expectations can produce different subsequent readouts.

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3.2 Finite temporal thickness

A lived present is not an instantaneous point. It is a finite window of retained steps. Let $h_n \in \mathbb{N}_{>0}$ and define

$$W_n = \{n - h_n + 1, \ldots, n\}. \tag{9}$$

Let $a_{n,j} \in \mathbb{Q}_{\geq 0}$ with $\sum_{j=0}^{h_n-1} a_{n,j} = 1$. The temporally thick readout is

$$\widetilde{z}_n = \sum_{j=0}^{h_n-1} a_{n,j} z_{n-j}. \tag{10}$$

This finite sum replaces any appeal to an integral memory kernel. It captures retention of what has just occurred and anticipation encoded in current organization without treating time as infinitely divisible.

3.3 Embodied and relational phenomenalization

The lived event is not produced by a brain-only operator. Let

$$\begin{aligned} \mathcal{O}_H[n] = \mathcal{O}\Big(&G_{brain}[n], G_{body}[n], G_{environment}[n], \\ &G_{relation}[n], G_{practice}[n], G_{language}[n], V_H[n]\Big) \end{aligned} \tag{11}$$

denote the finite organization that makes some distinctions salient, actionable, and meaningful. The phenomenal field is represented as

$$P_n = \Phi_n\Big(\widetilde{z}_n, \mathcal{O}_H[n], B_n, A_n^{aff}, C_n, U_n\Big), \tag{12}$$

where B_n is bodily regulation, A_n^{aff} affective valence and salience, C_n practical-social context, and U_n the finite action affordances available at the step. P_n is not reduced to those variables. The equation records constitution constraints: a lived world appears through embodied, affective, relational, and practical organization.

This formulation is aligned with embodied and enactive approaches (Varela et al. 1991; Noë 2004; de Haan 2020). It also preserves the neurophenomenological requirement that first-person and third-person descriptions constrain each other without one being silently substituted for the other (Varela 1996).

Definition 1 (Finite experience event)**.** *A finite experience event $\mathcal{E}_n$ is the tuple*

$$\mathcal{E}_n = \langle X_n, z_n, \widetilde{z}_n, P_n, u_n, R_{n+1}, \Delta\mathcal{O}_n \rangle, \tag{13}$$

where u_n is an executed or inhibited action, R_{n+1} is the next finite world-resistance readout, and $\Delta\mathcal{O}_n$ is the event-level change in the observer's organization.

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The definition makes transformation internal to the event arc rather than a later process wholly external to experience. A moment that produces no durable autobiographical memory can still contain a transient state transition. Durable learning is a further retained subset of that transition.

3.4 No experience without retained difference

The following argument connects experience directly to the repository's retained-difference root.

Principle 1 (Experience–difference principle). *If two putative experience events are indistinguishable in every available readout, action tendency, and next-step state of the observer, then the framework contains no finite witness that they are two different experience events for that observer.*

The principle does not deny ineffable character by definition. It states the evidential rule of a finite readout theory: a claimed difference that cannot enter any present or subsequent distinction is not available as a modelled difference.

A genuine experience therefore implies at least one finite state difference

$$H_{n+1} \neq H_n \tag{14}$$

at the event scale, even when the durable component later decays to zero. Experience is not merely a representation *inside* an unchanged subject. It is the event in which a world difference becomes a retained observer difference.

4 The Human LoRA Identity Thesis

4.1 Two aspects of one event

The central identity can now be stated without confusing explanatory levels.

Definition 2 (Human LoRA event). *A Human LoRA event is a finite experience event $\mathcal{E}_n$ whose adaptive aspect is an event-level operator update $\Delta\mathcal{O}_n$ that factors through the active readout field available in that event.*

The event has two irreducible descriptions:

$$\mathcal{E}_n^{phen} = P_n, \tag{15}$$

$$\mathcal{E}_n^{adapt} = \Delta\mathcal{O}_n. \tag{16}$$

The identity thesis is

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$$\boxed{\mathcal{E}_n \equiv \langle \mathcal{E}_n^{phen}, \mathcal{E}_n^{adapt} \rangle \equiv \text{Human LoRA}_n.} \tag{17}$$

The identity is token-level and dual-aspect. It does not say $P_n = \Delta\mathcal{O}_n$ as mathematical objects. It says that the lived appearance and the adaptive reorganization belong to the same finite event. A pain, insight, lesson, encounter, failure, or act of trust is lived under one description and reorganizes future readiness under another.

4.2 Transient and retained updates

Every experience has an event-level transition, but not every transition becomes stable learning. Let

$$\Delta\mathcal{O}_n^{fast} = B_n A_n \tag{18}$$

be the active event update. Let the retention gate be

$$g_n = \Gamma_n(s_n, r_n^{err}, q_n, v_n, c_n, p_n) \in \mathbb{Q} \cap [0, 1], \tag{19}$$

where s_n is salience, r_n^{err} finite residual, q_n repetition or retrieval, v_n value relevance, c_n context fit, and p_n replay or practice. The durable update is

$$\Delta\mathcal{O}_n^{ret} = g_n \Delta\mathcal{O}_n^{fast}. \tag{20}$$

The next-step organization is

$$\mathcal{O}_H[n+1] = \mathcal{O}_H[n] + \Delta\mathcal{O}_n^{ret} + \varepsilon_n, \tag{21}$$

where ε_n is the disclosed update remainder: bodily change, graph rewiring, social reorganization, value revision, or any change not captured by the factorized adapter.

The gate avoids a false dilemma. If $g_n = 0$, the event remains a Human LoRA event at the transient level but does not become durable adaptation. If $g_n > 0$, part of the event is retained. Memory research supports this separation among acquisition, stabilization, replay, transformation, and reconsolidation (Dudai 2004; Dudai 2012; Wang and Morris 2010; McGaugh 2015).

5 The Human LoRA Factorization Theorem

5.1 The finite-bottleneck postulate

Let the operational organization that can be probed in the chosen study have finite dimension d_n. Let the active experience field have finite dimension m_n. The crucial

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empirical postulate is

$$0 < m_n < d_n. \tag{22}$$

This is not an appeal to an unknowable total dimension of the person. Both m_n and d_n are operational: they must be defined by a finite readout scheme. The thesis predicts that a bounded event recruits fewer independent active distinctions than the full set of distinctions whose future organization could be altered.

Let

$$A_n \in \mathbb{Q}^{m_n \times d_n}, \tag{23}$$
$$B_n \in \mathbb{Q}^{d_n \times m_n}. \tag{24}$$

A_n maps the current organization into active event coordinates; B_n maps the event coordinates back into the future organization. Define

$$\Delta\mathcal{O}_n = B_n A_n. \tag{25}$$

Proposition 1 (Finite factorization rank bound, $\mathrm{T}\textsc{h}_{\mathbb{Q}}$)**.** *For finite rational matrices $A_n \in \mathbb{Q}^{m_n \times d_n}$ and $B_n \in \mathbb{Q}^{d_n \times m_n}$,*

$$\mathrm{rank}_{\mathbb{Q}}(B_n A_n) \le m_n. \tag{26}$$

If $m_n < d_n$, the event update cannot have full rank in the operational space.

Proof. For every $x \in \mathbb{Q}^{d_n}$, $(B_n A_n)x$ lies in the image of B_n. Hence $\mathrm{im}(B_n A_n) \subseteq \mathrm{im}(B_n)$. The image of B_n is spanned by at most m_n columns, so its dimension is at most m_n. Therefore $\mathrm{rank}_{\mathbb{Q}}(B_n A_n) \le m_n$. Under $m_n < d_n$, the rank is strictly below the operational dimension. □

This proposition is elementary finite algebra, but it transforms the title from a free analogy into a conditional theorem. If event change factors through a bounded active field smaller than the organization it can modify, low rank follows.

Proposition 2 (Selective preservation, $\mathrm{T}\textsc{h}_{\mathbb{Q}}$)**.** *Let $x \in \mathbb{Q}^{d_n}$. If $A_n x = 0$, then*

$$(\mathcal{O}_H[n] + B_n A_n)x = \mathcal{O}_H[n]x. \tag{27}$$

Proof. If $A_n x = 0$, then $B_n A_n x = B_n 0 = 0$. Substitution gives the result. □

The proposition supplies the exact preservation profile expected from Human LoRA: directions not admitted into the active event channel remain unchanged by that event-level adapter. This is not a claim that all unrelated human abilities are preserved in practice, because ε_n and later interactions may alter them. It is a testable local law of the factorized component.

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Proposition 3 (Composition bound, $\mathrm{Th}_{\mathbb{Q}}$)**.** *For N finite event updates $\Delta\mathcal{O}_i = B_i A_i$ with ranks bounded by m_i,*

$$\operatorname{rank}_{\mathbb{Q}}\left(\sum_{i=1}^{N} \Delta\mathcal{O}_i\right) \leq \sum_{i=1}^{N} m_i. \tag{28}$$

Proof. The rank of a finite sum is no greater than the sum of the ranks. Apply Proposition 1 to every term. □

The composition law explains how individually bounded experiences can cumulatively reorganize a person. A lifelong transformation need not be a single full-rank event. It can arise from many retained low-rank updates, their overlaps, and their interaction remainder.

5.2 Transformative experience without retreat

The theory does not exclude transformative experience from Human LoRA. It distinguishes event scale from accumulated scale. Let

$$\mathcal{O}_H[N] = \mathcal{O}_H[0] + \sum_{n=0}^{N-1} g_n B_n A_n + \sum_{0 \leq i < j < N} \Lambda_{ij} + \sum_{n=0}^{N-1} \varepsilon_n, \tag{29}$$

where Λ_{ij} records interactions between retained updates. Even if each $B_n A_n$ is low-rank, their images can span new directions and their interaction can modify the graph itself. Transformative experience is therefore *composed Human LoRA*: many bounded event updates eventually change what later counts as the base organization.

This point strengthens rather than weakens the thesis. The human base is not frozen. At every discrete step, retained adapters can become part of the next base. Human LoRA is recursive formation, not an attachable software file.

6 Phenomenology as a Constitutive Part of Human LoRA

Phenomenology is not added as a warning that computation cannot explain everything. Its structures explain why an experience event is selective and therefore why a bottleneck factorization is plausible.

6.1 Intentionality: experience of one thing as another

Experience is directed. A person does not receive an undifferentiated state; something appears *as* dangerous, familiar, beautiful, humiliating, permissible, or possible.

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In finite form, an intentional episode selects an object node x and an aspect node y within an active relational graph:

$$I_n = (x \xrightarrow{\text{as}} y)_n. \tag{30}$$

The "as" relation is an event projection. It activates some relational paths and leaves others outside the current field. Intentionality therefore contributes to the finite bottleneck rather than opposing formalization.

6.2 Horizon: a finite active neighbourhood

Every appearance has a horizon: what is co-available, implied, or expected without being fully focal. The present framework represents the horizon as a finite active neighbourhood

$$\mathcal{H}_n \subseteq V_H[n], \qquad |\mathcal{H}_n| < \infty. \tag{31}$$

The complement relative to the current finite graph is not an actually infinite hidden field. It consists of currently inactive or unavailable distinctions. An event can expand, contract, or reorganize $\mathcal{H}_n$ at later steps.

6.3 Embodiment: the body is constitutive

The body is not an extra input to a detached internal model. It determines access, salience, affordance, fatigue, vulnerability, repair, and the practical meaning of an event. Merleau-Pontian and enactive insights are therefore expressed not by adding a "body variable" at the end, but by defining $\mathcal{O}_H[n]$ as a coupled brain–body–world organization (Merleau-Ponty 2012; Varela et al. 1991; Noë 2004; de Haan 2020).

A learned fear, for example, is not merely a proposition stored in a cognitive matrix. It can alter posture, attention, autonomic readiness, movement, and the set of available actions. The Human LoRA update is relational because the changed disposition is distributed across those couplings.

6.4 Temporality: finite retention and anticipation

Phenomenological time is thick because the present retains recent steps and is oriented toward possible next steps. Equation (10) captures this without an infinite temporal line. Retention is the weighted availability of finitely many prior readouts; anticipation is the finite set of next-step predictions and affordances generated from the current organization.

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6.5 Sedimentation: experience becomes style

Repeated and retained events can sediment into habit, skill, bodily style, expectation, and a characteristic way of reading situations. Sedimentation is exactly the phenomenological face of recursive Human LoRA. The original episode need not remain explicitly recalled. Its update can persist in edge weights, accessible paths, salience rules, and action readiness.

6.6 Intersubjectivity and participatory sense-making

Some meanings are not produced by an isolated observer. They arise in interaction. Let two finite organizations be $G_A[n]$ and $G_B[n]$, and let $G_{AB}[n]$ be the finite interaction graph formed by their reciprocal actions. A relational experience can update not only $\mathcal{O}_A$ and $\mathcal{O}_B$ but also the interaction rules of G_{AB}. This is consistent with participatory sense-making, where social meaning emerges from coordinated interaction rather than being fully located in either participant alone (De Jaegher and Di Paolo 2007).

Human LoRA is therefore not merely private adaptation. Trust, shame, recognition, authority, belonging, and exclusion can be relational adapters: they change which future actions and interpretations become available within an interaction field.

6.7 The self as a finite retained pattern

The self is not a single adapter. It is a maintained pattern across bodily, affective, narrative, relational, practical, and normative readouts. This aligns with pattern approaches to selfhood (Gallagher and Daly 2018). Human LoRA events change components and couplings of that pattern. The sense that an event happened *to me* belongs to the phenomenal aspect; the later disposition through which similar events are read belongs to its adaptive aspect.

7 Learning, Memory, and Sampled Neural Geometry

7.1 Complementary learning systems

Complementary learning systems theory distinguishes rapid event-specific acquisition from slower integration into distributed prior knowledge (McClelland et al. 1995; Kumaran et al. 2016). This supports the distinction between $\Delta\mathcal{O}_n^{fast}$ and $\Delta\mathcal{O}_n^{ret}$. It also prevents the mistaken claim that a single event immediately rewrites the entire human organization.

The Human LoRA thesis adds a new structural question to that literature: when

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durable integration occurs, what is the finite rank and preservation profile of the update? A rapid trace can be highly specific; repeated replay can align multiple traces; later abstraction can compress their shared structure into fewer active directions. The thesis predicts stage-dependent rank rather than a fixed rank across acquisition and consolidation.

7.2 Consolidation, retrieval, and reconsolidation

Memory is not a static archive. Consolidation stabilizes and reorganizes; retrieval can strengthen or modify; reconsolidation can render a retained organization revisable (Dudai 2004; Dudai 2012; Wang and Morris 2010; Moscovitch et al. 2016). In Human LoRA terms, retrieval reactivates part of an earlier adapter inside a new event field. The resulting update can preserve, weaken, extend, or redirect the earlier organization.

The discrete form is

$$\Delta\mathcal{O}_{old}^{new} = g_n B_n A_n\, Q_n(\Delta\mathcal{O}_{old}), \tag{32}$$

where Q_n is a finite retrieval map. No trace is assumed to be copied with infinite fidelity.

7.3 Neural-manifold findings translated into finite readouts

The neural-manifold literature reports that sampled population activity often occupies a structured lower-dimensional geometry and that learning is constrained by existing activity organization (Cunningham and Yu 2014; Sadtler et al. 2014; Perich et al. 2025). Recent human BCI evidence indicates that learning succeeds more readily when required control mappings follow directions represented in the intrinsic sampled geometry (Busch et al. 2026). Recent primate work further reports that learning can move prefrontal representations from high-dimensional mixed codes toward lower-dimensional, rule-selective, and later abstract organization (Wójcik et al. 2026).

These studies are highly relevant but do not prove literal low-rank synaptic updates. The present paper translates them as finite matrices. For a study with T trials and p measured channels, let

$$Y \in \mathbb{Q}^{T\times p} \tag{33}$$

be the quantized activity readout at declared resolution ρ. The paper asks about the rank and change of such finite readout matrices, not about a continuum manifold existing behind them.

Definition 3 (Resolution-controlled readout rank)**.** *Let q_ρ map each finite mea-*

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surement to a rational bin of width $\rho \in \mathbb{Q}_{>0}$. For a finite update matrix D, define

$$r_\rho(D) = \operatorname{rank}_{\mathbb{Q}}(q_\rho(D)). \tag{34}$$

A credible finite diagnostic must report r_ρ over several preregistered resolutions. A rank claim that appears only at one convenient resolution is not robust.

7.4 What the evidence supports

The literature supports three propositions of increasing strength:

(i) existing sampled neural organization constrains learnability in some tasks;

(ii) learning changes sampled representational geometry and can compress task-relevant structure;

(iii) some durable human updates are best predicted by a low-rank factorized model.

The first two have finite empirical support. The third remains `Open` and is the direct empirical burden of the Human LoRA thesis. Activity rank, representational-change rank, coupling-change rank, and behavioral transfer must be measured separately.

8 Mental Health, Learning, and Well-Being

8.1 Adaptation is not well-being

A system can adapt effectively to a dangerous or coercive environment while paying a severe long-term cost. Hypervigilance, emotional suppression, withdrawal, appeasement, and rigid control can be locally intelligible retained adaptations. The theory therefore rejects

$$\text{successful adaptation} = \text{mental health}. \tag{35}$$

It also rejects the equation of well-being with minimal prediction residual. A rigid person can reduce immediate surprise by avoiding new evidence, narrowing life, or forcing every event into an inflexible prior. Mental health requires not only successful prediction but the ability to revise, differentiate, repair, relate, and pursue valued possibilities.

Psychological flexibility literature emphasizes context-sensitive change and value-consistent action rather than one fixed response (Kashdan and Rottenberg 2010). Self-determination theory adds autonomy, competence, and relatedness as important conditions of motivation and well-being (Ryan and Deci 2000). These findings motivate a finite well-being vector

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$$W_n = (F_n, A_n^{agency}, M_n^{meaning}, R_n^{relation}, C_n^{competence}, Q_n^{repair}), \tag{36}$$

whose components are finite assessed readouts, not a single hidden scalar truth.

8.2 A discrete misfit model

Let a_n denote an activated retained adaptation. Define finite scores in $\mathbb{Q} \cap [0, 1]$:

- $rig(a_n)$: resistance to revision when context changes;
- $gen(a_n)$: breadth of activation beyond the conditions in which it was learned;
- $mis(a_n)$: mismatch with present context and values;
- $loss(a_n)$: cost to bodily, relational, and practical repair.

A finite maladaptive-cost readout is

$$C_{mal}[n] = rig(a_n)\, gen(a_n)\, mis(a_n)\, loss(a_n). \tag{37}$$

This is a diagnostic scaffold, not a definition of psychiatric disorder. It captures a key insight: suffering may result not because an adapter is meaningless or irrational, but because a once-useful retained organization activates too rigidly, too broadly, in the wrong context, at excessive repair cost.

8.3 Enactive psychiatry and the lived world

Enactive psychiatry treats psychiatric difficulties as disturbances of sense-making across experiential, physiological, sociocultural, and existential dimensions (de Haan 2020). Human LoRA is compatible with this view only if the update operator remains embodied and relational. A disorder cannot be reduced to a defective matrix inside the person. The relevant retained pattern may include family interaction, material environment, institutional pressure, bodily regulation, and socially available meanings.

The paper therefore makes the following non-claim:

> *Mental disorders are not defective Human LoRAs. They are heterogeneous patterns of suffering and impaired possibility in which retained adaptations, bodily processes, social conditions, meanings, and environments may interact.*

Human LoRA contributes one question: which retained event transformations are active, how are they gated by context, and what would allow revision without destroying capacities that remain valuable?

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8.4 Therapeutic learning as adapter competition and integration

Exposure therapy provides a concrete test case. In inhibitory-learning accounts, treatment need not erase the earlier threat association. It can establish new safety learning whose retrieval competes with the threat expectation and depends on context (Craske et al. 2014). In finite form,

$$\mathcal{O}_{post} = \mathcal{O}_{threat} + g_s B_s A_s, \tag{38}$$

where the safety adapter is selectively activated under relevant cues. Return of fear can be interpreted as a retrieval or generalization failure rather than proof that no learning occurred.

This case is Human-LoRA-like because the prior organization may remain available while a new context-sensitive response is added. Yet successful therapy can also alter bodily regulation, relational trust, self-understanding, and practical life. Those changes enter ε_n and the interaction terms of Equation (29); they must not be forced into a narrow cognitive adapter.

9 An Empirical Programme Without Continuum Assumptions

9.1 Four synchronized readout layers

A theory of experience cannot be tested by neural data alone. The recommended design synchronizes four finite readout layers:

$$D_n = \{D_n^{first}, D_n^{beh}, D_n^{neural}, D_n^{world}\}. \tag{39}$$

D_n^{first} contains coded first-person reports; D_n^{beh} finite performance, transfer, and retention measures; D_n^{neural} quantized activity or coupling matrices; and D_n^{world} finite context, relationship, and consequence records. Neurophenomenological interviews should not be decorative. Their categories should constrain trial segmentation and model selection, while neural and behavioral results should constrain phenomenological interpretation (Varela 1996).

9.2 Study 1: finite bottleneck and rank bound

Estimate a finite operational dimension d from a preregistered set of independent probes. Estimate an event-field dimension m from finite coded distinctions available during the episode. Fit the candidate update

$$D^{post-pre} = BA + E, \tag{40}$$

with $A \in \mathbb{Q}^{m\times d}$, $B \in \mathbb{Q}^{d\times m}$ and residual E. The risky prediction is

$$r_\rho(D^{post-pre}) \leq m \tag{41}$$

for a preregistered task class and resolution range, after correcting for measurement noise with finite synthetic recovery tests.

A repeated finding of durable update rank substantially above the measured event-field bound would directly damage the factorization thesis.

9.3 Study 2: within-geometry and outside-geometry learning

Translate the design logic of Sadtler et al. (2014) and Busch et al. (2026) into finite sampled geometry. Participants learn mappings that use directions already represented in the baseline finite activity matrix, rotate those directions, or require directions outside the baseline span at resolution ρ. Human LoRA predicts faster learning, lower residual, and greater preservation in the within-span condition.

9.4 Study 3: consolidation and rank trajectory

Measure acquisition, immediate performance, delayed retention, retrieval practice, and transfer at discrete sessions. The predicted trajectory is not universally decreasing rank. Early exploration may recruit more directions; successful consolidation may later increase abstraction and concentrate reusable structure. The prediction is a finite sequence

$$r_\rho(D_0), r_\rho(D_1), \ldots, r_\rho(D_N) \tag{42}$$

whose shape differs between memorized, generalized, and forgotten learning.

9.5 Study 4: narrow and transformative formation

Compare a bounded skill programme with a multi-domain formation programme involving role, value, relationship, and action changes. The strong thesis predicts that every event remains finitely factorized, while cumulative transformative change shows larger composition span and interaction terms rather than one global event update.

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9.6 Study 5: mental-health flexibility

In a learning or therapy study, measure whether a new adapter is activated selectively, can be inhibited when inappropriate, generalizes to genuinely safe contexts, and preserves agency and valued action. The target is not maximal suppression of symptoms during the session but improved finite flexibility and lower maladaptive cost across later contexts.

9.7 Competing models

Human LoRA must beat alternatives rather than absorb them rhetorically. At minimum, compare:

$$M_0 : \text{context-only state change,} \tag{43}$$
$$M_1 : \text{sparse but not low-rank update,} \tag{44}$$
$$M_2 : \text{low-rank factorized update,} \tag{45}$$
$$M_3 : \text{regularized full-rank update,} \tag{46}$$
$$M_4 : \text{finite graph rewiring,} \tag{47}$$
$$M_5 : \text{hybrid fast trace plus slow update.} \tag{48}$$

Models should be evaluated on held-out finite trials, delayed retention, transfer, preservation, and context activation. The low-rank model earns support only if it predicts better after complexity penalties and resolution sensitivity checks.

9.8 Hard falsifiers

The following findings would require rejection or major restriction:

F1. Durable event-level changes repeatedly have rank above the preregistered active-field bound.

F2. Full-rank or graph-rewiring models consistently outperform low-rank models in bounded learning tasks.

F3. Directions outside the measured event channel are changed as strongly as admitted directions, contrary to selective preservation.

F4. First-person differentiation has no stable relation to the dimensionality or selectivity of later change across adequately powered studies.

F5. The theory cannot distinguish transient context effects from durable retained change.

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Restriction to a smaller task class would be an honest result. The title would remain a philosophical identity claim only if the finite observer-transition argument survives, while the low-rank biological application could be narrowed.

10 Objections

10.1 Experience and update are different categories

A lived appearance and an operator update are indeed different descriptions. The thesis does not numerically equate them. It asserts event identity: the same finite transition is lived under one aspect and reorganizes future readiness under another. Rejecting that identity requires an account of experience as a causally inert presentation to an otherwise unchanged observer, or an argument that the transformation belongs to a wholly separate event. Neither follows merely from the difference between first-person and third-person vocabularies.

10.2 Some experiences leave no memory

Durable autobiographical memory is not required. Equation (19) distinguishes transient event update from retained update. An experience can alter attention, bodily readiness, or current interpretation for one or a few steps while g_n later yields no stable trace. The identity thesis concerns the finite event transition; durable Human LoRA is a retained subset.

10.3 Low rank depends on coordinates

Rank is invariant under invertible changes of basis within the same finite field, but empirical operationalization depends on which distinctions are measured and on quantization resolution. The response is disclosure, not denial: preregister the readout basis, report several resolutions, test synthetic recovery, and compare alternative representations. A result that disappears under reasonable finite recodings should not carry the strong thesis.

10.4 A finite field need not be low-rank

Correct. Finitude proves only finite rank. Low rank requires $m < d$. That is why Equation (22) is an explicit open postulate and why Equation (41) is a risky empirical prediction. The paper does not smuggle "low" into the definition of experience.

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10.5 The brain is not a transformer

The argument does not require transformer architecture, gradient descent, or LoRA's engineering implementation. LoRA supplies a precise name for base-preserving factorized change. Deep-learning concepts can guide neuroscience only when computational correspondence is separated from biological mechanism (Richards et al. 2019; Lillicrap et al. 2020).

10.6 Neural geometry is not synaptic rank

Agreed. Sampled activity rank, representational-change rank, coupling-change rank, and synaptic update rank are different finite readouts. The empirical programme tests them separately. The 2026 neural-geometry findings motivate the project but do not settle the mechanism.

10.7 Transformative experience is global

Global outcome does not imply a single global event update. Equation (29) allows many low-rank events, cross-event interactions, and graph changes to accumulate. A life can be transformed through a finite sequence of bounded encounters whose composition changes the base.

10.8 Phenomenology is being naturalized away

The phenomenal aspect is not derived from rank. Intentionality, embodiment, temporal thickness, horizon, selfhood, and intersubjectivity are constitutive constraints on what counts as an experience. The formalism concerns finite structural relations and lasting transformation. It does not claim to deduce qualitative character from rational matrices.

10.9 Mental suffering is not a bad model

Agreed. The theory explicitly rejects that reduction. A retained disposition may be historically intelligible, bodily grounded, relationally maintained, and locally protective. Clinical significance depends on suffering, impairment, flexibility, agency, context, values, and repair, not merely prediction error or update rank.

10.10 AI adaptation may also be experience

This paper does not prove that artificial systems can never be conscious. It denies a narrower inference: a low-rank parameter update is insufficient by itself to establish a lived field. Human LoRA includes embodied world contact, phenomenalization,

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value-laden salience, action consequence, and finite self-maintenance. Future artificial systems would need independent evidence concerning those dimensions (Harnad 1990; Bender and Koller 2020; Bisk et al. 2020).

11 Implications for Human Formation and Artificial Intelligence

Education is not the delivery of more input. It is the arrangement of finite encounters whose retained transformations enlarge discrimination, skill, responsibility, and action. The Human LoRA framework asks which distinctions were actually available, which were retained, what later capacities changed, and what remained preserved. Retrieval practice, interleaving, feedback, sleep, social participation, and real action matter because they determine the retention gate and composition of updates (Karpicke and Roediger 2007; Walker and Stickgold 2004; Dewey 1938).

Expertise can be understood as recursive adapter integration. A clinician, engineer, scholar, craftsperson, or mediator develops a way of seeing relevant relations that were previously unavailable. The learner is not replaced by a new person at every lesson. Bounded updates accumulate until the shared base itself has changed.

The same architecture clarifies AI engineering. Context retrieval changes a finite inference state without necessarily changing model parameters. LoRA changes a parameterized base without supplying current world evidence. External validation converts neither output nor adapter into truth automatically. This separation is consistent with the repository's rule that candidate output is not knowledge until it passes context, grounding, authority, world-resistance, and risk gates (Lahtee 2026b).

The analogy also imposes an ethical warning. Human experience is not free training material. A person is a vulnerable, embodied, relational agency whose retained transformations can be beneficial, coercive, traumatic, or identity-forming. Computational comparability does not erase consent, responsibility, or personhood.

For philosophy of mind, the theory offers a non-dual but non-reductive position. The lived and adaptive aspects are not two substances, yet neither vocabulary eliminates the other. Experience is the event in which retained world difference becomes retained observer difference.

12 Conclusion

This paper has defended the title without changing "is" to "is like." Experience is the human LoRA because a genuine experience is a finite observer-transforming event. Its first-person manifestation is a lived, embodied, affective, world-directed appearance. Its third-person adaptive structure is a bounded change in the organi-

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zation through which later situations can be read and acted upon. These are two aspects of one discrete event.

The framework is finite throughout. The observer is a finite retained-difference graph; time advances by $n \in \mathbb{N}$; temporal thickness is a finite weighted window; state and update matrices are rational; empirical results are quantized finite diagnostics; and no continuum limit is used. Under the explicit bottleneck condition that the active event field has dimension m smaller than the operational organization of dimension d, the update factors as BA and has rank at most m. Selective preservation and composition bounds follow exactly.

The formal result does not prove that every biological learning mechanism is literally low-rank. It establishes what follows if experience-mediated change factors through a bounded active field. That bridge is open and testable. Sampled neural geometry, complementary learning systems, consolidation research, and phenomenology make it plausible enough to deserve direct investigation, but not certain enough to escape falsification.

Mental health also prevents a simplistic celebration of adaptation. A retained Human LoRA can be skilful, protective, rigid, overgeneralized, costly, or transformative. Well-being depends on flexibility, agency, meaning, relation, competence, and repair, not merely successful prediction or retention.

The final thesis is therefore strong and exact:

> *Experience is the Human LoRA: the finite event in which a world difference becomes a lived difference and, through that same event, selectively reorganizes the observer who will meet the next world.*

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Data Availability

No original human, animal, or clinical dataset was generated for this theoretical article. The article relies on published scholarship and on finite formal arguments stated in the text. The proposed empirical programme has not yet been executed. The private research repository cited in this paper is controlled by the author and is not represented as a public replication archive.

References

Bender, E. M. and A. Koller (2020). "Climbing Towards NLU: On Meaning, Form, and Understanding in the Age of Data". *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 5185–5198. doi: 10.18653/v1/2020.acl-main.463.

Bisk, Y., A. Holtzman, J. Thomason, J. Andreas, Y. Bengio, J. Chai, M. Lapata, A. Lazaridou, J. May, A. Nisnevich, N. Pinto, and J. Turian (2020). "Experience Grounds Language". *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, pp. 8718–8735. doi: 10.18653/v1/2020.emnlp-main.703.

Busch, E. L., E. C. Fincke, G. Lajoie, S. Krishnaswamy, and N. B. Turk-Browne (2026). "Human Learning of Noninvasive Brain–Computer Interfaces via Manifold Geometry". *Nature Neuroscience*. doi: 10.1038/s41593-026-02311-2.

Clark, A. (2013). "Whatever Next? Predictive Brains, Situated Agents, and the Future of Cognitive Science". *Behavioral and Brain Sciences* 36.3, pp. 181–204. doi: 10.1017/S0140525X12000477.

Craske, M. G., M. Treanor, C. C. Conway, T. Zbozinek, and B. Vervliet (2014). "Maximizing Exposure Therapy: An Inhibitory Learning Approach". *Behaviour Research and Therapy* 58, pp. 10–23. doi: 10.1016/j.brat.2014.04.006.

Cunningham, J. P. and B. M. Yu (2014). "Dimensionality Reduction for Large-Scale Neural Recordings". *Nature Neuroscience* 17, pp. 1500–1509. DOI: 10.1038/nn.3776.

De Jaegher, H. and E. Di Paolo (2007). "Participatory Sense-Making: An Enactive Approach to Social Cognition". *Phenomenology and the Cognitive Sciences* 6.4, pp. 485–507. DOI: 10.1007/s11097-007-9076-9.

De Haan, S. (2020). *Enactive Psychiatry*. Cambridge University Press. DOI: 10.1017/9781108685214.

Dewey, J. (1938). *Experience and Education*. Macmillan.

Dudai, Y. (2004). "The Neurobiology of Consolidations, Or, How Stable Is the Engram?" *Annual Review of Psychology* 55, pp. 51–86. DOI: 10.1146/annurev.psych.55.090902.142050.

Dudai, Y. (2012). "The Restless Engram: Consolidations Never End". *Annual Review of Neuroscience* 35, pp. 227–247. DOI: 10.1146/annurev-neuro-062111-150500.

Friston, K. (2010). "The Free-Energy Principle: A Unified Brain Theory?" *Nature Reviews Neuroscience* 11, pp. 127–138. DOI: 10.1038/nrn2787.

Gallagher, S. and A. Daly (2018). "Dynamical Relations in the Self-Pattern". *Frontiers in Psychology* 9, p. 664. DOI: 10.3389/fpsyg.2018.00664.

Harnad, S. (1990). "The Symbol Grounding Problem". *Physica D* 42.1–3, pp. 335–346. DOI: 10.1016/0167-2789(90)90087-6.

Hu, E. J., Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen (2022). "LoRA: Low-Rank Adaptation of Large Language Models". *International Conference on Learning Representations*. URL: https://openreview.net/forum?id=nZeVKeeFYf9.

Karpicke, J. D. and H. L. Roediger (2007). "Repeated Retrieval During Learning Is the Key to Long-Term Retention". *Journal of Memory and Language* 57.2, pp. 151–162. DOI: 10.1016/j.jml.2006.09.004.

Kashdan, T. B. and J. Rottenberg (2010). "Psychological Flexibility as a Fundamental Aspect of Health". *Clinical Psychology Review* 30.7, pp. 865–878. DOI: 10.1016/j.cpr.2010.03.001.

Kumaran, D., D. Hassabis, and J. L. McClelland (2016). "What Learning Systems Do Intelligent Agents Need? Complementary Learning Systems Theory Updated". *Trends in Cognitive Sciences* 20.7, pp. 512–534. DOI: 10.1016/j.tics.2016.05.004.

Lahtee, Y. (2026a). "Human Agency as Single-tau_c Active World-Readout". Unpublished working paper in the research_universal_solver repository.

Lahtee, Y. (2026b). *research_universal_solver: Finite Readout, Retained Difference, and Tier-Disciplined Universal Solver Architecture*. Private research repository, version 2.9.0. Unpublished; repository commit f1b6e856be58157ea7fb694ee343c3151d271c67.

Lillicrap, T. P., A. Santoro, L. Marris, C. J. Akerman, and G. Hinton (2020). "Backpropagation and the Brain". *Nature Reviews Neuroscience* 21, pp. 335–346. DOI: 10.1038/s41583-020-0277-3.

McClelland, J. L., B. L. McNaughton, and R. C. O'Reilly (1995). "Why There Are Complementary Learning Systems in the Hippocampus and Neocortex". *Psychological Review* 102.3, pp. 419–457. DOI: 10.1037/0033-295X.102.3.419.

McGaugh, J. L. (2015). "Consolidating Memories". *Annual Review of Psychology* 66, pp. 1–24. DOI: 10.1146/annurev-psych-010814-014954.

Merleau-Ponty, M. (2012). *Phenomenology of Perception*. Trans. by D. A. Landes. Original work published 1945. Routledge.

Moscovitch, M., R. Cabeza, G. Winocur, and L. Nadel (2016). "Episodic Memory and Beyond: The Hippocampus and Neocortex in Transformation". *Annual Review of Psychology* 67, pp. 105–134. DOI: 10.1146/annurev-psych-113011-143733.

Noë, A. (2004). *Action in Perception*. MIT Press.

Perich, M. G., D. Narain, and J. A. Gallego (2025). "A Neural Manifold View of the Brain". *Nature Neuroscience* 28, pp. 1582–1597. DOI: 10.1038/s41593-025-02031-z.

Richards, B. A., T. P. Lillicrap, P. Beaudoin, Y. Bengio, R. Bogacz, A. Christensen, C. Clopath, R. P. Costa, A. de Berker, S. Ganguli, et al. (2019). "A Deep Learning Framework for Neuroscience". *Nature Neuroscience* 22, pp. 1761–1770. DOI: 10.1038/s41593-019-0520-2.

Ryan, R. M. and E. L. Deci (2000). "Self-Determination Theory and the Facilitation of Intrinsic Motivation, Social Development, and Well-Being". *American Psychologist* 55.1, pp. 68–78. DOI: 10.1037/0003-066X.55.1.68.

Sadtler, P. T., K. M. Quick, M. D. Golub, S. M. Chase, S. I. Ryu, E. C. Tyler-Kabara, B. M. Yu, and A. P. Batista (2014). "Neural Constraints on Learning". *Nature* 512, pp. 423–426. DOI: 10.1038/nature13665.

Varela, F. J. (1996). "Neurophenomenology: A Methodological Remedy for the Hard Problem". *Journal of Consciousness Studies* 3.4, pp. 330–349.

Varela, F. J., E. Thompson, and E. Rosch (1991). *The Embodied Mind*. MIT Press.

Walker, M. P. and R. Stickgold (2004). "Sleep-Dependent Learning and Memory Consolidation". *Neuron* 44.1, pp. 121–133. DOI: 10.1016/j.neuron.2004.08.031.

Wang, S.-H. and R. G. M. Morris (2010). "Hippocampal–Neocortical Interactions in Memory Formation, Consolidation, and Reconsolidation". *Annual Review of Psychology* 61, pp. 49–79. DOI: 10.1146/annurev.psych.093008.100523.

Wójcik, M. J., J. P. Stroud, D. Wasmuht, M. Kusunoki, M. Kadohisa, M. J. Buckley, R. P. Costa, N. E. Myers, L. T. Hunt, J. Duncan, and M. G. Stokes (2026). "Learning Shapes Neural Geometry in the Primate Prefrontal Cortex". *Nature Neuroscience*. DOI: 10.1038/s41593-026-02333-w.

Human Learning as Epistemic Architecture: A Method for Word Mapping, Life-Concept Graphs, Constraint Testing, and Corrigible Agency in the AI Age

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What the chapter states. “This paper rewrites the human–AI learning project with human learning at the center.” “The central thesis is that human learning in the AI age should be organized not as information storage but as a revisable epistemic architecture through which learners become more corrigible, world-answerable, self-aware, and responsible agents.”

Status. Printed on every page: “(preprint – not peer reviewed).” The chapter states: “This method does not claim to explain all human learning.”

Falsifier(s). Verbatim, under “24.3 What Would Disconfirm the Method”: “The method’s central claim would be disconfirmed if, in an adequately powered, time-matched design, the method group showed no advantage over the yoked control on protocol-independent transfer and calibration at delay. . . ”

Read before. When AI Expands Human Potential.

Feeds into. Meaning Before Naming.

Human Learning as Epistemic Architecture

A Method for Word Mapping, Life-Concept Graphs, Constraint Testing, and Corrigible Agency in the AI Age

Yaoharee Lahtee

Open Civil Science Initiative

June 28, 2026

Abstract

This paper rewrites the human–AI learning project with human learning at the center. It does not argue that human beings should learn like artificial intelligence. Rather, it asks what can be carefully extracted from language-model and representational AI architecture for the renewal of reflective, language-mediated human learning: the movement from inputs to representations, from representations to relations, from relations to retrieval, from retrieval to evaluation, and from evaluation to update. The paper translates these architectural principles into a human-centered method beginning with lived words, not abstract information. A word is treated as a site where experience, memory, emotion, social pressure, value, goal, constraint, evidence, and action converge. From this starting point, the paper develops a model of life-world lexical mapping, meaning neighborhoods, personal knowledge graphs, retrieval-augmented inquiry, constraint testing, reflective dissonance, and disciplined revision. The central thesis is that human learning in the AI age should be organized not as information storage but as a revisable epistemic architecture through which learners become more corrigible, world-answerable, self-aware, and responsible agents. The paper makes four contributions beyond restating established educational theory. First, it argues that the architecture lens is *generative* rather than decorative, identifying five design commitments—representation-first ordering, retrieval by default, separation of candidate generation from validation, transfer as the criterion of success, and versioned belief-state—that prior theories permit but do not require, and unifying them under a single rationale. Second, it grounds its central wager—that productive friction deepens learning while fluent completion can hollow it—in the cognitive literatures that bear on it directly (the illusion of explanatory depth, desirable difficulties, cognitive load, self-explanation, metacognition, and cognitive offloading), defending rather than weakening the claim by specifying the *kind* of difficulty the method introduces. Third, it adds a constitutive dialogical and structural layer and a confound-resistant validation design,

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so that the method's claim to cultivate open, corrigible agency can be empirically tested rather than merely asserted. Fourth, it extends the pipeline beyond belief revision to competence: it maps the problem-contexts in which a concept does work, anchors new concepts to known ones, selects context-appropriate tools, assembles a workflow, and consolidates understanding into skill through deliberate practice and real-world feedback—so that the architecture cultivates not only corrigible belief but corrigible competence, and guards against the inert knowledge that decontextualized learning produces. AI is valuable not when it replaces judgment or accelerates closure, but when it helps learners expose hidden meanings, compare alternatives, test assumptions, and revise their models under resistant conditions.

Keywords: human learning; epistemic agency; artificial intelligence in education; constraint-first epistemology; retrieval-augmented inquiry; corrigible agency.

1 Introduction: The Human Learner After AI

Artificial intelligence has made a hidden feature of human learning newly visible: learning is not simply the accumulation of information. It is the construction, testing, and revision of representations through which a person names the world, interprets experience, forms goals, uses tools, and acts. The most important educational question raised by AI is therefore not whether machines can produce answers faster than students. It is whether human learners can build more answerable forms of understanding while thinking with systems that generate fluent candidates at great speed.

This paper proposes a human-centered rewrite of the AI-learning analogy. Earlier formulations of the project treated AI as a mediating layer within finite inquiry: AI can widen comparison space, surface alternatives, generate dissonance, and assist reframing, but it cannot by itself certify truth or replace world-answerable validation. The present paper shifts the center of gravity. The primary object is no longer AI-mediated cognition in general, but human learning as an epistemic architecture. AI architecture becomes a source of design principles for human education, not a model of what a human being is.

The central claim is simple but deliberately scoped:

> Reflective, language-mediated human learning in the AI age should begin with lived words and develop into revisable maps of meaning, relation, evidence, constraint, action, and feedback.

This claim deliberately begins with words rather than disciplines, textbooks, or databases. Human beings do not first inhabit theories. They inhabit vocabularies. A person learns the world through words such as *success*, *failure*, *family*, *truth*, *freedom*, *faith*, *work*, *intelligence*, *education*, and *future*. These words are not neutral labels. They carry emotional histories, social pressures, inherited norms, aspirations, fears, and unexamined assumptions. They organize attention before they become objects of reflection.

The educational problem is that learners often act from words whose sources they have never mapped. They may pursue "success" without knowing whose meaning of success they have inherited. They may fear "failure" without examining which institution taught them to interpret failure as shame rather than feedback. They may use AI for "learning" while actually using it for completion, reassurance, or avoidance of difficulty. The first task of a renewed education is therefore not to add more information, but to help learners map the words that already govern their attention, goals, and self-understanding.

The analogy with AI helps because AI makes representation explicit. Language-model AI transforms text into tokens, tokens into vectors, vectors into relational states, and relational states into predictions; more broadly, AI systems transform inputs into representations and evaluate outputs against task conditions. Human learning cannot and should not be reduced to this process. But it can learn from the architectural discipline: the quality of later reasoning depends on the quality of earlier representation. If a learner's key words

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remain vague, inherited, emotionally overloaded, or socially captured, then later reasoning will be distorted before formal inquiry even begins.

This paper develops a method for translating AI learning architecture into human learning practice:

> AI: data → tokens → embeddings → relations → retrieval → evaluation → update.
>
> Human learning: experience → lived words → meaning maps → life-concept graphs → retrieval-augmented inquiry → constraint testing → revision → responsible action.

The difference is decisive. AI updates a model. A human learner revises not only beliefs, but also goals, habits, self-understanding, and forms of life. Human learning is therefore not merely epistemic in a narrow sense. It is existential, social, ethical, and practical. And, as Section 18 argues, it is not complete at the moment of belief revision: a revised understanding must still cross the gap between knowing what a concept means and being able to use it skillfully, in the right contexts, to solve real problems.

Because the analogy carries real argumentative weight in what follows, the paper does not leave it at the level of suggestive parallel. Section 6 defends the strong position that the architecture lens is *generative*: it yields specific design commitments that the established educational theories permit but do not by themselves require. Section 5 grounds the paper's central empirical wager in the cognitive research that bears on it. The reader who suspects that "AI" here is only ornament is asked to weigh those two sections, which are written precisely to meet that suspicion.

2 Scope and Limits of the Method

This method does not claim to explain all human learning. Human beings learn before language through perception, imitation, movement, attachment, play, bodily coordination, and affective attunement. A child learns rhythm, danger, comfort, proximity, and action before they can produce explicit definitions. Therefore the claim of this paper must be read with precision: *reflective, language-mediated learning* should begin with lived words, because such learning already depends on the vocabulary through which the learner interprets experience, forms reasons, evaluates goals, and participates in public justification.

The method is designed for contexts in which learners can examine words, meanings, sources, evidence, constraints, and revision. It is especially appropriate for:

- secondary and university education;
- adult learning and professional development;
- AI literacy and human–AI co-inquiry;

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- philosophy of education and self-directed learning;
- personal knowledge management and reflective practice;
- interdisciplinary inquiry where concepts travel across domains.

The method is not offered as a complete theory of early childhood learning, pure motor learning, clinical therapy, trauma treatment, religious formation, or non-linguistic perception. It may interact with those domains, but it does not replace the specialized methods and ethical safeguards they require. Its proper object is the learner who can ask: *What does this word make me see? Where did this meaning come from? What does it hide? What would make me revise it?*

This scope statement matters because the paper extracts principles from AI architecture without collapsing the human into the machine. The relevant comparison is not "AI begins with tokens, therefore humans begin with tokens." The stronger comparison is: language-model AI shows how powerful later operations depend on earlier representational organization; human reflective learning likewise depends on the organization of the words through which a learner already inhabits the world.

A second scope condition concerns culture, and it is developed rather than merely flagged in Section 17. The apparatus of a learner who introspects on personal meaning and revises it individually encodes a culturally specific, broadly individualist model of selfhood. The method is not confined to that register: it can be run in a communal register in which the unit of mapping is a word negotiated within a family, congregation, or community, and in which the entry points are narrative, recitation, and dialogue rather than written definition. The paper treats this portability as a design requirement, not an afterthought.

3 The Core Discovery: Knowledge Management as Corrigible Agency

The core discovery of this project may be stated as follows:

> Human knowledge management should not be organized as the storage of information, but as the cultivation of corrigible agency.

Information storage asks: What do I have? Corrigible agency asks: What can correct me? This difference changes the purpose of education. A learner is not merely someone who possesses content. A learner is someone whose words, beliefs, goals, and actions can be tested, revised, and made more answerable to the world.

The earlier framework called this a constraint-first epistemology. It begins from the fact that human knowers are finite, mediated, fallible, and socially situated. There is no view from nowhere and no unmediated possession of reality. Yet this does not imply skepticism. It means that knowledge must be assessed by the conditions under which inquiry remains open to correction: fallibility, corrigibility, criticizability, route-openness,

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and world-answerability.

For education, this produces a stricter standard than fluency, confidence, or productivity. A student who can produce a smooth answer may still lack understanding. A learner who uses AI to complete tasks quickly may become more efficient while becoming less able to judge. Conversely, a learner who uses AI to generate objections, alternative framings, and revision pressure may strengthen epistemic agency.

The central educational question is therefore:

> Does this learning practice increase the learner's capacity to be corrected, or does it merely increase the speed of output?

This is also the point at which human learning differs most sharply from machine learning. A machine can be optimized against a loss function. A human being must learn under conditions of value, identity, responsibility, and consequence. The human learner is not simply an adaptive system. The learner is a self-forming agent whose goals are shaped by experience, others, institutions, culture, tools, and imagination.

4 Relation to Existing Educational Theories

The method proposed here is new in its architecture, but it is not isolated from the history of educational thought. It reorganizes several established insights around the problem of learning with AI.

First, Dewey's theory of inquiry begins from problematic situations rather than from inert information. This paper extends that insight by treating lived words as condensed problematic situations: terms such as *success*, *failure*, *intelligence*, and *freedom* already contain conflicts between experience, institution, value, and action. The learner does not merely define them; the learner inquires through them.

Second, Mezirow's account of transformative learning emphasizes frame revision. The present method gives frame revision a concrete entry point. Instead of asking learners only to reflect on assumptions in general, it asks them to map the words through which assumptions are carried, inherited, and repeated.

Third, Bandura's account of agency and self-regulation helps explain why word mapping is not merely semantic. Learners regulate themselves through goals, expectations, standards, and self-evaluations. If those standards are encoded in unmapped words, self-regulation may reproduce alienation rather than agency.

Fourth, Freire's critical pedagogy shows that education is never neutral with respect to power and voice. A lived word is often socially authored before it is personally affirmed. To map a word is therefore also to ask who has been authorized to define it, whose experience has been excluded, and whether the learner can re-appropriate the term responsibly.

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Fifth, Vygotsky's account of mediated learning clarifies why tools, signs, and social interaction matter. The life-concept graph is a mediational artifact: it externalizes relations so that the learner can inspect, criticize, and revise them.

Sixth, Clark and Chalmers' extended mind thesis helps explain why this method is not simply internal reflection. Notes, diagrams, AI systems, databases, mentors, and communities can function as parts of a learner's cognitive ecology. The issue is not whether external tools are used, but whether they deepen or bypass answerability.

Seventh, Longino's social epistemology shows that objectivity depends on critical interaction across perspectives. For this reason, the method treats peer critique, route-openness, and alternative formulations as constitutive rather than decorative.

Finally, retrieval practice research emphasizes that learning improves when learners actively retrieve rather than passively reread. The present method incorporates this principle but extends it: learners retrieve not only facts, but sources, counterexamples, lived experiences, constraints, and alternative meanings.

The contribution of this paper is therefore synthetic, but—as the next two sections argue—not merely synthetic. It brings inquiry, transformation, self-regulation, critical consciousness, mediation, social criticism, and retrieval into one AI-age educational architecture: the epistemic life-graph. What distinguishes it from a collage of prior theories is the generative role of the architecture lens (Section 6) and the empirical grounding of its central claim about friction and fluency (Section 5).

5 Empirical and Cognitive Foundations

The method's central wager is empirical as well as philosophical: that productive friction can deepen learning, while fluent completion—of the kind that AI now makes nearly free—can hollow it. This wager is sometimes stated as if it were self-evident. It is not, and it must be defended against the obvious reply that difficulty is not good in itself. This section grounds the claim in the cognitive literatures that bear on it directly, and in doing so sharpens rather than softens it.

The illusion of explanatory depth. People routinely believe they understand how things work in far more detail than they do, and the illusion collapses only when they are asked to produce a mechanistic explanation (Rozenblit & Keil, 2002). This is the precise psychological mechanism behind the paper's distinction between fluency and understanding. Fluent access to an explanation—whether from memory or from a model—inflates felt understanding without supplying it. The method's demand that a learner be able to explain a claim *without* the tool, and transfer it to a new case, is a direct operationalization of the diagnostic that punctures this illusion.

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Desirable difficulties. Conditions that slow acquisition and feel harder during study—spacing, interleaving, generation, retrieval—frequently improve long-term retention and transfer relative to conditions that feel fluent and easy (Bjork & Bjork, 2011). This is the literature that licenses the paper's claim that friction is an engine of learning. It also disciplines the claim: the relevant difficulty is *desirable* difficulty, not difficulty in general.

Cognitive load. Cognitive load theory distinguishes load that is intrinsic to the material, load that is extraneous to it, and load that is germane because it is devoted to building schemas (Sweller, 1988). This distinction supplies exactly the qualification the friction claim requires, and it allows the paper to keep the claim without retreat. Reflective dissonance, as the method uses it, is not extraneous load and not mere confusion; it is germane difficulty—effort directed at building, testing, and revising a learner's model. Closure-mode AI use lowers desirable difficulty by removing the generative and validative work; corrigibility-mode AI use redirects effort toward germane processing while stripping away extraneous load such as formatting and search overhead. The method, properly used, does not add difficulty indiscriminately; it adds the kind of difficulty that the evidence associates with durable learning, and removes the kind that does not.

Self-explanation and metacognition. Prompting learners to explain material to themselves improves understanding and transfer (Chi, De Leeuw, Chiu, & LaVancher, 1994), and the regulation of one's own cognition—monitoring what one does and does not understand—is a distinct and trainable competence (Flavell, 1979). The method's revision questions and constraint tests are structured self-explanation and metacognitive monitoring made routine, and its assessment treats the visible trace of monitoring and revision as evidence of learning.

Cognitive offloading and AI. Humans habitually offload cognition onto external aids, and offloading trades internal effort for external support in ways that can either help or harm depending on what is offloaded and why (Risko & Gilbert, 2016). Large language models intensify this dynamic, because they make it possible to offload not only storage and calculation but the generation of explanations and the appearance of understanding (Kasneci et al., 2023). The closure loop is offloading at its most hazardous: the learner offloads the very operations—generation and validation—whose internal exercise constitutes understanding. The corrigibility loop is offloading at its most useful: the learner offloads candidate generation and retrieval while retaining validation. The method is, in one formulation, a discipline for keeping the right things internal while offloading the rest.

Two consequences follow for the rest of the paper. The claim that friction deepens learning survives, but in the sharpened form that *germane, desirable difficulty* deepens learning, which the method is engineered to produce and ordinary AI-completion workflows are liable to remove. And the contribution is positioned, not as the discovery that fluency

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can be shallow, but as an architecture that translates these established findings into a teachable, AI-age practice with an explicit role for the tool.

6 What the Architecture Lens Adds: Five Generative Commitments

A reasonable objection must be met before the method is presented in detail. If the procedure reduces to mapping a word, tracing its sources, testing it, and revising it, then it may appear to be a restatement of Deweyan inquiry, Mezirovian frame revision, Freirean critical literacy, and retrieval practice, with artificial-intelligence vocabulary added only for rhetorical effect. If the analogy were merely illustrative, intellectual honesty would require demoting it. This section argues the stronger position: the architecture lens is generative. It yields at least five design commitments that the antecedent educational theories permit but do not require, and in several cases would not by themselves produce. The claim is not that each commitment is unprecedented in the history of education; it is that the architecture lens supplies a unified rationale that fixes them as defaults rather than options, and orders them into a single pipeline.

For each commitment we state the architectural source, the design consequence for human learning, and—for rigor—the point at which the analogy breaks, so that the human difference is preserved rather than effaced.

Commitment 1: Representation gates reasoning

Architectural source. In any learning system, the quality of downstream computation is bounded by the quality of the input representation. A transformer cannot attend over distinctions its tokenization has already discarded; a perceptual system cannot classify features its early layers did not encode. *Design consequence.* Reflective learning should begin by auditing the representational units—the lived words—before deploying reasoning over them, because reasoning inherits the distortions of its inputs. Dewey begins from problematic situations and Mezirow from disorienting dilemmas, but neither assigns explicit, prior priority to the representational pre-condition; the architecture lens does, which is why the method opens with word selection and pre-reflective articulation (Steps 1–2) rather than with argument. *Where it breaks.* A token is fixed once segmented, whereas a lived word can be re-segmented by the learner during inquiry; human representation is itself revisable at "inference time," a capacity no current architecture possesses.

Commitment 2: Prefer retrieval to parametric confidence

Architectural source. Retrieval-augmented generation exists because parametric memory is lossy and prone to fluent confabulation; grounding generation in retrieved evidence reduces hallucination (Lewis et al., 2020). *Design consequence.* A learner should treat

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answering from internal familiarity as the high-risk default, and should retrieve—sources, counterexamples, mentors, cases—as the architectural norm, not as optional diligence. Retrieval practice research (Karpicke & Roediger, 2007) establishes that retrieval strengthens memory; the architecture lens adds a complementary rationale: retrieval guards against confabulated confidence, the human analogue of hallucination. *Where it breaks.* A model retrieves to compensate for frozen weights, whereas a human retrieves into a memory that is continuously and sometimes self-servingly reconstructive; human retrieval is therefore itself subject to motivated distortion in a way an external corpus is not, which is why the method couples retrieval with the dissent and absence checks of Section 17.

Commitment 3: Separate candidate generation from validation

Architectural source. Many capable systems factor competence into a proposer and a verifier—generator and discriminator, actor and critic, policy and reward model. Capability improves when generation is cheap and validation is strict and independent. *Design consequence.* The most defensible division of labor in AI-assisted learning is that the system generates candidates (explanations, objections, framings, analogies) while the learner owns validation against evidence, practice, and consequence. This single commitment generates the paper's central distinction between the closure loop—in which the learner outsources both generation and validation, collapsing the two—and the corrigibility loop, in which the learner keeps validation. No prior educational theory dictates this factoring, because none was formulated against systems that make fluent generation nearly free; the condition is new, and the design response is derived from how such systems are made reliable. *Where it breaks.* A reward model can be specified in advance, whereas the human verifier must itself be educated, is value-laden, and can be captured; the separation is therefore a normative aspiration to be cultivated, not an automatic property to be assumed.

Commitment 4: Evaluate against held-out conditions

Architectural source. A model is judged by generalization to data it was not trained on; in-distribution performance that fails to transfer is the signature of overfitting. *Design consequence.* Learning should be assessed by transfer to conditions under which the original framing was not rehearsed, and a fluent reproduction of the studied case should be treated as the human analogue of an in-distribution score—necessary but insufficient. This commitment fixes transfer, not fluency, as the criterion of success, and it directly motivates the assessment rubric (Section 23) and the protocol-independent transfer measure in the validation design (Section 24). *Where it breaks.* A held-out test set is fixed and external, whereas the conditions a human must generalize to are open-ended, partly self-created, and ethically weighted; transfer for a human is not only to new problems but to new responsibilities.

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Commitment 5: Make state explicit and version updates

Architectural source. Training proceeds through explicit, recorded updates to a represented state; checkpoints preserve prior states so that change is inspectable. *Design consequence.* The learner should preserve the pre-reflective meaning and record each revision, so that learning is visible as a trace from prior to posterior model rather than inferred from a single improved output. This is why Step 2 preserves the starting model and why the assessment artifact stores the full transformation. *Where it breaks.* A checkpoint is a faithful copy, whereas a human's memory of a prior belief is reconstructed and may be silently rewritten in hindsight; the preserved artifact is thus an external safeguard against a distortion to which the learner is internally subject.

Taken together, these five commitments do work the antecedent theories leave optional. They fix representation-first ordering, retrieval by default, generator/verifier separation, transfer as criterion, and versioned state, and they unify them under a single rationale: reliable learning systems gate later operations on the disciplined organization of earlier ones. This is the precise sense in which the paper extracts from, rather than imitates, artificial intelligence. The architecture is not a model of the human; it is a source of design constraints whose human translation must, at every step, preserve embodiment, value, and responsibility—as the disanalogies above make explicit.

7 Core Definitions for the Method

The previous sections establish the philosophical center of the paper and the argument that its architecture does genuine work. For the proposal to become a teachable method, however, its vocabulary must be made operational. The following terms should be treated as technical terms within the present framework.

Term	Working definition
Lived word	A word that already organizes a learner's perception, desire, fear, self-understanding, social belonging, or practical action. It is not merely a lexical item but a site where experience, value, memory, and agency meet.
Word map	A structured account of a lived word: its personal meaning, inherited sources, emotional charge, associated goals, constraints, evidence, and revision questions.
Meaning map	The explicit representation of how a learner understands a word across personal, social, cultural, disciplinary, and practical dimensions.

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Semantic neighborhood	The field of adjacent, opposing, hidden, and tension-bearing terms around a word. It is the human analogue of an embedding neighborhood, but it is structured by frame, metaphor, value, emotion, and consequence rather than by a distance metric.
Life-concept graph	A relation map connecting words, meanings, sources, values, goals, constraints, evidence, actions, and revisions. It is a human-centered analogue of a knowledge graph.
Retrieval-augmented human inquiry	A learning practice in which the learner does not rely on memory, confidence, or AI output alone, but retrieves evidence, counterevidence, examples, mentors, texts, and lived cases.
Constraint test	A procedure for asking how a belief, goal, concept, or AI-generated formulation can be resisted by reality, practice, evidence, other agents, or domain standards.
Reflective dissonance	Productive, *germane* tension that arises when a learner's existing meaning map is challenged by an alternative, objection, evidence, or AI-generated reframing, and that calls for assessment and revision rather than mere discomfort.
Corrigible agency	The learner's capacity to remain open to correction, revise in assessable ways, and carry revised understanding into responsible action.
Closure loop	A failed learning pattern in which AI or other tools reduce friction, provide fluent completion, and produce premature confidence without validation.
Corrigibility loop	A successful learning pattern in which friction leads to comparison, evidence retrieval, constraint testing, revision, and changed action.

These definitions clarify why the method is not ordinary vocabulary work, mind mapping, or note-taking. A standard concept map usually asks how concepts are related. The present method asks a stronger question: how do words become commitments, how do commitments become goals, how do goals meet constraints, and how can the learner revise under pressure without losing agency?

8 What Can Be Extracted from AI Architecture?

This paper does not propose imitation. It proposes extraction, in the precise sense argued in Section 6. The task is to extract structural principles from language-model and representational AI architecture and reinterpret them for human learning.

A precision is necessary. Not all AI systems begin with linguistic tokens. Language

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models do; many other AI systems begin with pixels, sensor streams, tabular features, audio frames, graph nodes, or other forms of input representation. The general lesson is therefore not tokenization alone but representational discipline: before an intelligent system can retrieve, relate, evaluate, or revise, it must first transform the world into usable forms. For human reflective learning, the closest analogue is not a computational token but the lived word: a socially and experientially loaded unit through which a person names, values, and acts.

AI Architecture	**Function in AI**	**Human Learning Translation**
Tokenization	Breaks input into processable units	Begin with lived words as units of inquiry
Embedding	Locates tokens in semantic space	Build meaning neighborhoods and semantic fields
Attention	Weights relations among elements	Notice which words, memories, and goals dominate attention
Knowledge graph	Represents entities and relations	Build life-concept graphs linking words, values, goals, constraints, and actions
RAG	Retrieves external evidence	Use notes, books, mentors, research, and experience as retrieval systems
Tools	Extend capability beyond model weights	Use diagrams, calendars, flashcards, calculators, AI, peers, and institutions as cognitive scaffolds
Evaluation	Tests output against criteria	Test beliefs against evidence, reality, criticism, and consequence
Update	Changes parameters or behavior	Revise beliefs, goals, practices, and self-understanding

The translation must preserve the human difference. AI tokenization is computational segmentation. Human word mapping is existential clarification. AI embeddings are numerical vectors. Human meaning neighborhoods are lived semantic fields. AI knowledge graphs are formal structures. Human life-concept graphs are maps of meaning, identity, pressure, and responsibility. AI evaluation is task scoring or validation. Human evaluation includes truth, care, justice, life-fit, and practical wisdom.

The embedding row deserves particular care, because it is where a loose analogy could do harm. The human "semantic neighborhood" is not a numerical space, and the method

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does not claim it is. Its closest disciplinary anchors are not in machine learning but in the cognitive science of meaning: the theory that words evoke structured background frames (Fillmore, 1982), the account of how abstract concepts are organized by systematic metaphor (Lakoff & Johnson, 1980), and the older model of semantic memory as a network in which activation spreads between related concepts (Collins & Loftus, 1975). An embedding neighborhood is a useful image for the intuition that words have neighbors; these three traditions supply the actual structure—frame, metaphor, and association—that the method asks learners to make explicit, including the value-laden and affective neighbors that have no embedding counterpart.

The value of AI architecture is that it gives education a design grammar. It shows that high-level intelligence depends on lower-level representation, retrieval, relation, and evaluation. Human learning can be redesigned accordingly, provided the design is grounded in how human meaning actually works.

9 Starting Point: The Lived Word

The first unit of the proposed architecture is the lived word. A lived word is a word that functions inside a learner's world. It is not merely a dictionary item. It is a node in a network of experience, emotion, social origin, value, goal, and action.

Examples include:

> success, failure, truth, intelligence, dignity, family, work, money, freedom, faith, education, love, duty, shame, future, AI, knowledge, self.

The educational mistake is to treat such words as transparent. They are rarely transparent. They often carry inherited meanings from family, school, religion, media, class, gender, nation, market systems, and personal wounds. The learner may think a goal is purely personal when it is partly a social inscription. This does not make the goal false. It makes the goal worthy of inquiry.

A lived word should be mapped through nine questions:

1. What does this word mean to me now?
2. Where did this meaning come from?
3. Which experiences are attached to it?
4. Which emotions does it carry?
5. Which values does it activate?
6. Which goals does it produce?
7. Which constraints does it hide or reveal?
8. Which evidence supports or challenges it?
9. What would force me to revise it?

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This turns vocabulary into inquiry. The learner begins not with an abstract theory of knowledge, but with the concrete words through which life is already being interpreted.

10 From Word to Meaning Map

A meaning map expands a word beyond definition. It makes explicit the layers of personal, social, cultural, disciplinary, and practical meaning that surround the word. The following schema is a minimal model.

```
meaning_map:
  word: "success"
  surface_definition: "achievement of a desired aim"
  personal_meaning: "becoming independent and respected"
  social_sources:
    - family expectation
    - school ranking
    - market competition
    - social comparison
  lived_experiences:
    - praise after high performance
    - shame after failure
    - pressure to support family
  emotional_charge:
    - hope
    - anxiety
    - pride
    - fear
  values_activated:
    - dignity
    - freedom
    - responsibility
    - recognition
  goals_generated:
    - financial stability
    - competence
    - family support
  constraints:
    - time
    - money
    - skill gap
    - unequal access
  risks:
    - confusing approval with truth
    - confusing productivity with growth
    - confusing ranking with worth
```


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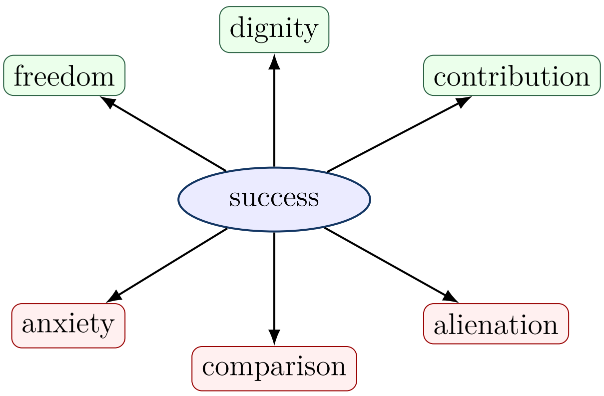
```
revision_questions:
  - "Whose meaning of success am I living inside?"
  - "Does this meaning remain answerable to my actual life?"
  - "What evidence would make me revise this goal?"
```

This schema is deliberately more complex than a dictionary entry because human meaning is layered. A learner's word for "success" may be epistemic, emotional, economic, moral, and social at once. Education should not flatten this complexity. It should make the complexity available for reflection.

11 From Meaning Map to Semantic Neighborhood

AI embeddings position words in a mathematical space of similarity and difference. Human learners can build an analogous, non-numerical semantic neighborhood whose structure—as Section 9 made explicit—comes from frame, metaphor, and association rather than from a distance metric. The aim is to locate a word among related, opposing, hidden, and dangerous concepts.

For example:



A serious semantic neighborhood includes not only attractive associations, but also tensions and costs. For human learning, the negative or ambivalent neighborhood is often where growth begins, because it is where the learner's preferred framing meets the meanings it has suppressed. The word that promises freedom may also produce exhaustion. The word that promises responsibility may hide coercion. The word that promises education may reproduce ranking, exclusion, or alienation. The point is not to reject the word, but to understand its structure.

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12 From Semantic Neighborhood to Life-Concept Graph

A knowledge graph expresses relations among entities. For human learning, the graph should connect words to experiences, values, goals, institutions, constraints, evidence, and actions. This produces a life-concept graph.

The basic form is:

subject → relation → object.

Examples:

```
life_concept_graph:
  triples:
    - "family -> shapes -> goal"
    - "school -> ranks -> learner"
    - "market -> rewards -> productivity"
    - "AI -> mediates -> inquiry"
    - "convenience -> can_reduce -> friction"
    - "friction -> can_enable -> reflective_dissonance"
    - "reflective_dissonance -> requires -> validation"
    - "validation -> strengthens -> corrigibility"
    - "corrigibility -> strengthens -> agency"
    - "agency -> requires -> responsibility"
    - "knowledge -> requires -> evidence"
    - "truth -> resists -> convenience"
    - "authenticity -> requires -> examined_inheritance"
    - "alienation -> occurs_when -> inherited_words_go_unexamined"
```

The graph reveals that human learning is not only cognitive. It is biographical and social. A learner's concept of intelligence may be linked to school ranking. A learner's concept of duty may be linked to family survival. A learner's concept of freedom may be linked to money, language, mobility, or digital access. Mapping these relations gives the learner a way to see how words become life directions.

13 Retrieval-Augmented Human Inquiry

AI systems improve when they retrieve external evidence rather than relying only on internal parameters; as Commitment 2 made explicit, this is because parametric memory confabulates under pressure. Human learners should do the same, and for the same reason. Memory and confidence are not enough, and confidence is least trustworthy precisely when it is most fluent. A human learning system needs retrieval.

Retrieval-augmented human inquiry is the practice of connecting a belief or word to

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sources, evidence, counterevidence, examples, mentors, lived cases, and domain tests. Its principle is:

> Do not answer only from familiarity. Retrieve, compare, and test.

A human retrieval system may include:

- personal notes and journals;
- books and articles;
- lectures and conversations;
- mentors and peer critique;
- experiments and practice records;
- databases, diagrams, and concept maps;
- AI-generated candidates, objections, and alternative framings.

The learner should tag every important claim with its status:

```
claim_status:
  claim: "AI improves learning"
  status: "candidate, not validated"
  evidence_for:
    - "AI can generate examples and explanations"
    - "AI can provide rapid feedback"
  evidence_against:
    - "AI may reduce effort"
    - "AI may produce fluent pseudo-understanding"
  validation_needed:
    - "Can the learner explain without AI?"
    - "Can the learner transfer the idea to a new problem?"
    - "Can the learner identify errors in AI output?"
  revision_rule:
    - "If convenience reduces recall, explanation, or transfer, redesign the
   AI use."
```

This practice preserves the key distinction between candidate generation and validation defended as Commitment 3. AI may generate candidate explanations, but the learner must validate them against evidence, practice, criticism, and domain-specific constraints.

14 Constraint Testing: Letting the World Answer Back

A learning system becomes epistemic only when its representations can fail. If a belief cannot be challenged, if a goal cannot be re-examined, or if a word cannot be revised, then the system has become closed.

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Constraint testing asks what resists the learner's current model. The relevant constraint may be empirical, social, ethical, practical, emotional, historical, or spiritual. The point is not that all domains use the same test. The point is that every serious claim must remain answerable to something beyond preference and fluency.

Domain	**Constraint**	**Learning Test**
Science	empirical resistance	Does observation or experiment challenge the claim?
Mathematics	formal necessity	Does the proof follow?
Ethics	responsibility and harm	Who is affected, and can the action be justified?
Craft	material feedback	Does the object work under use?
Language	communicative uptake	Does meaning survive translation and dialogue?
Selfhood	life-fit and integrity	Can the learner live this meaning without self-deception?
Education	transfer and explanation	Can the learner apply the idea beyond the original task?
AI use	validation beyond fluency	Does AI output survive independent checking?

Constraint testing is the educational form of world-answerability. It protects learning from becoming self-confirming.

For classroom and personal use, the constraint test can be operationalized as a worksheet:

```
constraint_test:
  word: "success"
  current_claim: "Success means being recognized as excellent."
  domain: "education / life planning"
  constraint_types:
    empirical:
      - "Does this definition predict sustainable achievement?"
      - "What evidence links recognition with long-term flourishing?"
    practical:
      - "Can I live this goal without burnout?"
      - "What time, money, skill, and health conditions resist it?"
    social:
      - "Whose approval is being treated as authoritative?"
      - "Who benefits if I keep this definition?"
    ethical:
```


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```
      - "Does this goal harm others or require self-deception?"
      - "What responsibilities are being ignored?"
    existential:
      - "Can I affirm this meaning as mine after reflection?"
      - "Does it fit the life I can responsibly inhabit?"
  validation_routes:
    - personal experience
    - counterexample
    - mentor critique
    - empirical source
    - practical experiment
    - peer dialogue
  result:
    status: "candidate / supported / revised / rejected"
    revision_required: true
    revised_claim: "Success means sustainable contribution under
   responsibility."
```

This template turns the abstract demand for world-answerability into a repeatable practice. A claim does not become knowledge merely because it sounds coherent. It becomes educationally stronger when it can survive multiple routes of resistance.

14.1 Resistant Coherence and the Problem of Value-Laden Constraint

A serious objection arises for value-laden and existential words such as *faith*, *freedom*, *duty*, or *justice*. If the constraint test for such words is permitted to include "coherence with tradition" or "fit with a way of life," then constraint testing risks collapsing into the very self-confirmation the method is designed to prevent: the learner consults a preferred source, finds the expected agreement, and declares the belief tested. This objection is not answered by exempting value-laden words from world-answerability, which would quietly lower the paper's central claim. It is answered by distinguishing two kinds of coherence.

Self-confirming coherence obtains when a belief is checked only against sources selected because they already agree with it. *Resistant coherence* obtains when a belief has been exposed to, and has survived, genuine attempts at refutation. The difference is operational, not merely attitudinal, and it can be enforced by four conditions that apply even where empirical falsification is unavailable:

1. **Dissent condition.** At least one competent other who could disagree must have been consulted, not only sources internal to the learner's prior commitment.
2. **Consequence condition.** The belief must be checked against its practical fruit in action, including effects on others who can report back.

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3. **Refusal condition.** The inquiry must hold open the real possibility of revising or refusing the inherited meaning, since a "test" whose outcome is fixed in advance is not a test.
4. **Internal-critique condition.** Where a tradition is invoked, its own resources for self-correction, dispute, and reform must be engaged, rather than only its settled conclusions.

Under these conditions a value-laden belief can remain genuinely answerable without being reduced to an empirical hypothesis, and the world-answerability requirement extends across all domains without flattening their differences. This preserves the paper's claim that every serious commitment must remain correctable, while respecting that the *form* of correction is domain-relative: what counts as resistance differs between a scientific claim and a religious vocabulary, but the demand that some genuine resistance be faced does not.

15 Reflective Dissonance as the Engine of Learning

AI-assisted learning becomes powerful when it produces the right kind of dissonance. Dissonance alone is not enough. Confusion, novelty, contradiction, or discomfort may destabilize without educating. Reflective dissonance is different. It is tension that calls for assessment, comparison, and revision—germane difficulty in the sense of Section 5, not extraneous load.

A clarification about terminology is warranted, because the word "dissonance" carries a specific psychological history. In Festinger's (1957) theory, cognitive dissonance is an aversive state that people are motivated to *reduce*, and they frequently reduce it through belief-consistent rationalization—which is to say, through closure. The method therefore treats Festingerian dissonance as a description of the closure tendency it must counteract, not as the engine it relies upon. The productive process the method names "reflective dissonance" is closer to Piagetian disequilibration resolved through accommodation rather than mere assimilation (Piaget, 1970), and to the epistemic conflict that Berlyne (1960) identified as a driver of directed curiosity. Naming this lineage matters: the method does not ask learners to sit in discomfort, but to convert a specific, schema-relevant conflict into revision.

The learning sequence is:

encounter $\rightarrow$ friction $\rightarrow$ branching $\rightarrow$ validation $\rightarrow$ integration.

The branching point is decisive. The learner may enter a closure loop or a corrigibility loop.

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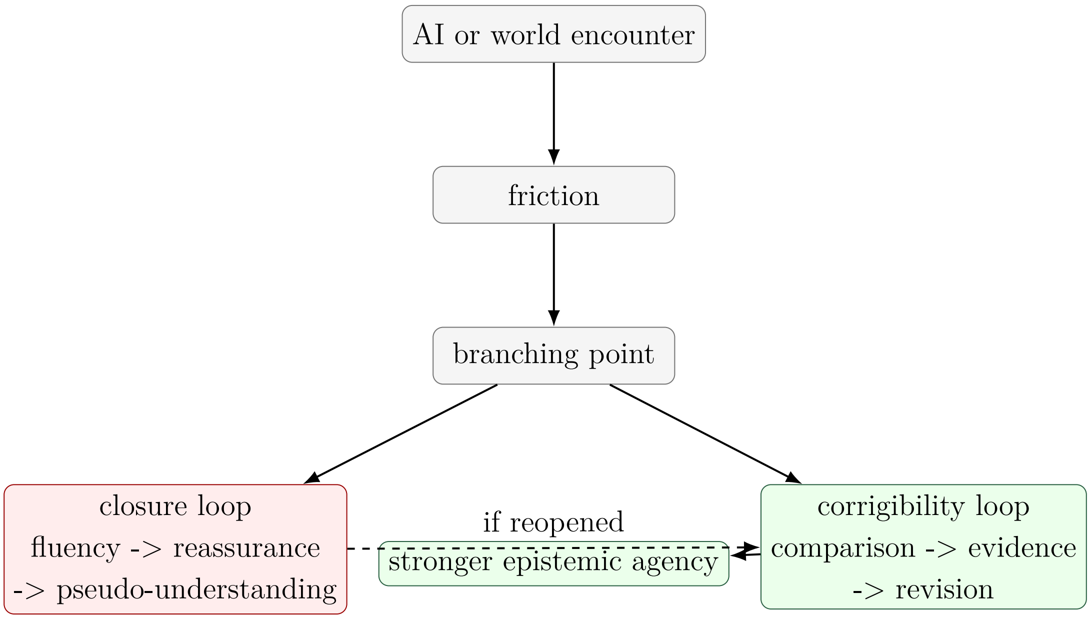

Education should not use AI merely to reduce difficulty. It should use AI to make assumptions visible. A good AI-supported learning prompt is not simply, "Explain this." It is also:

- What assumption am I making?
- What is the strongest objection?
- What alternative vocabulary could describe the same problem?
- What evidence would distinguish these explanations?
- Where might this answer be fluent but shallow?

This turns AI from a completion engine into a dissonance engine. The learner does not ask AI to replace judgment. The learner asks AI to produce structured pressure for better judgment—and, by the generator/verifier separation of Commitment 3, retains the validating role.

16 Alienation and Authenticity in Word-Based Learning

The proposed method is not only cognitive. It is also a method for addressing alienation. Alienation occurs when a person lives from meanings whose origins, pressures, and consequences remain hidden. A learner may believe they are pursuing their own goals while actually acting from unexamined family expectations, institutional ranking systems, market narratives, or fear of social shame.

Authenticity does not require rejecting all inherited meanings. No one creates the self from nothing. Human agency is situated. A person receives language, values, histories,

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obligations, and possibilities before they can reflect on them. Authenticity begins when inherited meanings become available for responsible affirmation, revision, or refusal.

The educational question is therefore not:

> How can the learner become completely self-created?

It is:

> Which inherited meanings can the learner understand, test, affirm, revise, or refuse responsibly?

Word mapping becomes an anti-alienation practice because it reveals the sources of the learner's vocabulary. A learner maps not only what a word means, but who or what made it mean that.

```
anti_alienation_word_map:
  word: "duty"
  inherited_sources:
    - family survival
    - religious teaching
    - gender expectation
    - community honor
  personal_affirmations:
    - care for others
    - responsibility
  possible_distortions:
    - self-erasure
    - fear of refusal
    - guilt-driven obedience
  authenticity_question:
    - "Can I affirm this duty without losing truthful agency?"
  revision_path:
    - "distinguish care from coercion"
    - "distinguish responsibility from self-cancellation"
```

This is why human learning must include selfhood. To learn is not only to know more, but to become able to inhabit meanings more truthfully. It is essential, however, that this not be read as a solitary undertaking, which is the subject of the next section.

17 The Dialogical and Structural Layer

The procedure described so far could be misread as a solitary, introspective exercise, and that misreading would expose it to a strong objection. The critical traditions on which the paper draws are not individualist. Freire's pedagogy is dialogical and collective; Fricker's epistemic injustice is structural, a matter of how credibility is distributed across social

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positions; Longino's objectivity is a property of communities that sustain transformative criticism. A method that asked learners only to journal privately about personal meaning would individualize what these theorists treat as structural, and would risk converting critical consciousness into a private optimization of the self—an irony, given that the method warns against the market capture of words such as *success*. The method therefore requires a dialogical and structural layer that is constitutive, not optional.

This layer adds three operations to the protocol.

1. **Map exchange.** Learners compare their meaning maps and semantic neighborhoods with others who occupy different social positions, so that the field of alternatives is socially generated rather than privately imagined. This also supplies the dissent condition of Section 14.1 with real interlocutors.
2. **Structural source mapping.** The source step (Step 3) must name institutions, incentives, and distributions of authority—who is licensed to define the word, whose definition is enforced, and through what mechanisms—rather than terminating in personal feeling.
3. **Absence check.** The learner asks whose experience the current map excludes and whose testimony has been pre-emptively discounted. This is Fricker's testimonial injustice rendered as a concrete, repeatable step.

A further qualification concerns the cultural situatedness of the method itself. The apparatus of an introspecting subject who maps personal meaning against inherited expectation and exercises individual revision encodes a recognizably individualist, and arguably culturally specific, model of selfhood. In collectivist or predominantly oral settings, the relevant unit of mapping may be a shared word negotiated within a family, congregation, or community, and the entry points may be narrative, recitation, proverb, or dialogue rather than written definition. The method does not presuppose the individualist register; it can be run in a communal register in which the map is co-authored and the constraint test is conducted through collective deliberation and shared practice. Naming this situatedness is not a concession that weakens the method but a condition of its honest portability across the settings in which it is meant to be used.

18 From Meaning to Competence: Problem-Context, Tools, Workflow, and Skill

The method as developed so far cultivates corrigible *belief*: it helps a learner map a word, test it, and revise it. A candid appraisal must concede that this is not yet enough. A learner can hold a well-mapped, well-tested, beautifully revised understanding of a concept and still be unable to *use* it—unable to recognize which real situations call for it, which tools it requires, or how to act through it under time and pressure. This is the classic failure that Whitehead named *inert ideas*: knowledge that is received, even understood,

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but never applied, tested, or thrown into fresh combinations (Whitehead, 1929). A method that stops at revised meaning risks producing exactly this. The architecture must therefore extend from corrigible belief to corrigible *competence*.

Two additions accomplish this extension. The first is a missing representational element—the problem-context—that connects a concept to experience. The second is an arc of operations—tools, workflow, and practiced skill, closed by real-world feedback—that connects understanding to capable action.

18.1 Problem-Context: The Bridge Between Meaning and Experience

A concept's meaning is not exhausted by its definition, its sources, or its semantic neighbors. It is constituted, in large part, by the *problems the concept helps solve and the situations in which it is used.* To know what *leverage*, *boundary*, *equilibrium*, or *trust* means is, in good part, to know the kinds of difficulty for which each is the right move. This is the pragmatist core of Dewey's account of inquiry, on which the paper already draws: concepts are instruments for resolving problematic situations, not pictures contemplated at rest. It is also the central finding of situated cognition, which holds that knowledge is bound to the activity, context, and culture in which it is developed and used, so that knowledge abstracted from its conditions of use tends not to transfer (Brown, Collins, & Duguid, 1989).

The method therefore adds an explicit *problem-context map*: alongside meaning, source, and relation, the learner records the problems the concept is used to solve, the situations in which it does and does not apply, and the cues that signal "this is a case for this concept." This is the learner's *conditional* knowledge—knowing not only what a concept is and how to deploy it, but *when* and *where.* Its omission is precisely what turns understanding inert; its inclusion is what ties a mapped meaning back to the experience from which it came and to the experience in which it will be used.

```
problem_context_map:
  word: "leverage"
  problems_it_solves:
    - "achieving a large effect from a small, well-placed input"
    - "deciding where limited effort will matter most"
  situations_of_use:
    - "negotiation, system design, prioritization, debt"
  applicability_cues:
    - "a small change point controls a large outcome"
  boundary_conditions:
    - "fragile systems where the same point amplifies harm"
  misapplication_signals:
    - "using 'leverage' to dignify mere force or coercion"
  anchoring_experience:
```


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```
    - "a time a small decision changed an outcome disproportionately"
```

18.2 The Competence Arc: Tools, Workflow, Skill, Feedback

Once a concept is anchored, mapped, tested, and located in its problem-contexts, four further operations carry it into capable action.

Tools fit to the context. Knowing a concept includes knowing which instruments its use requires—which notations, references, instruments, collaborators, or AI affordances fit *this* kind of problem. Tool selection is itself a piece of conditional knowledge and a mediational act in Vygotsky's sense; the same AI system that is a candidate-generator for one problem may be a calculator, a critic, or a distraction for another. The learner records not "tools in general" but the tools that fit the mapped problem-context, and the criteria for choosing among them.

Workflow. The learner assembles the operations into a repeatable procedure for applying the concept: the order of steps, the checkpoints, the points at which retrieval, constraint testing, and tool use enter. A workflow is what makes a piece of understanding usable more than once, and what makes it teachable and inspectable rather than locked in tacit intuition.

Skill through deliberate practice. A workflow executed once is not yet a skill. The decisive transition is from *declarative* knowledge—knowing that, and knowing how in principle—to *procedural* knowledge that runs fluently, accurately, and with less conscious load, a transition that occurs only through repeated, feedback-rich practice (Anderson, 1982). Not all repetition serves this end: the practice that builds expertise is *deliberate*—focused on the edge of current ability, structured by clear goals, and corrected by immediate feedback (Ericsson, Krampe, & Tesch-Römer, 1993). This is the dimension the epistemic pipeline alone omits, and it is where understanding becomes capability. It also connects to the paper's earlier commitments: spaced, effortful retrieval (Karpicke & Roediger, 2007) and desirable difficulty (Bjork & Bjork, 2011) are precisely the conditions under which practice consolidates skill rather than mere familiarity.

Real problems and feedback. Finally, the skill is exercised on genuine problems, not only rehearsed cases, and the result is fed back to revise both the map and the skill. This closes an experiential loop of the kind Kolb described—concrete experience, reflective observation, revised conceptualization, renewed experimentation (Kolb, 1984)—and it institutionalizes the reflective practitioner's habit of learning from the back-talk of a real situation (Schön, 1983). The feedback revises two things at once: the conceptual map (as in the epistemic loop) and the procedure itself (as a matter of skill).

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18.3 Corrigible Competence

This extension does not abandon the paper's spine; it carries it into a new domain. Just as belief can close prematurely into unexamined confidence, skill can close prematurely into fossilized automaticity—a fluent procedure that has stopped being questioned and now reliably produces a subtly wrong result. Automaticity is therefore double-edged: it frees attention, but it can entrench error beneath the level of notice. The same discipline the paper applies to belief must therefore extend to skill. Constraint testing, the dissent and consequence conditions, and the feedback loop apply not only to what the learner believes but to how the learner acts, so that competence remains as correctable as understanding. The aim is not merely a learner who can act, but a learner whose action, like their belief, can be tested and revised.

The full architecture, accordingly, runs in two coupled arcs: an epistemic arc from lived or newly anchored word through meaning, relation, retrieval, constraint, and revision; and a competence arc from problem-context through tools, workflow, and practiced skill, closed by real-world feedback that revises both the map and the skill.

19 The Educational Architecture

The proposed architecture runs in two coupled arcs. It is inspired by AI systems but centered on human agency, and the dialogical and structural layer of Section 17 runs across all of it rather than sitting at any single level. The first arc is epistemic: it carries a word to a tested, revised understanding. The second arc is competence: it carries that understanding into capable, corrigible action.

Epistemic arc:

1. **Word Layer**: identify the word that organizes experience, or anchor a new word to known ones.
2. **Meaning Layer**: map personal, cultural, social, and disciplinary meanings.
3. **Source Layer**: identify where meanings came from, including institutions and incentives.
4. **Relation Layer**: build life-concept graphs.
5. **Retrieval Layer**: attach sources, evidence, examples, and counterexamples.
6. **Constraint Layer**: identify what can resist or test the claim.
7. **Revision Layer**: record how meaning changes under criticism.

Competence arc:

8. **Problem-Context Layer**: map the problems the concept solves and the situations in which it applies.

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9. **Tools and Workflow Layer**: select context-appropriate tools and assemble a repeatable procedure for using the concept.
10. **Skill Layer**: consolidate understanding into fluent capability through deliberate, feedback-rich practice.
11. **Feedback Layer**: apply the skill to real problems and use the result to revise both the map and the skill, in responsible action.

This architecture can be represented as a learning pipeline:

> (lived or anchored) word → meaning map → semantic neighborhood → life-concept graph → retrieval → constraint test → reflective dissonance → revision → problem-context → tools & workflow → practiced skill → responsible action on real problems → feedback → revision of map and skill.

Its aim is not information mastery alone. Its aim is the formation of a learner capable of answerable judgment and corrigible skill.

20 The Method: A Fifteen-Step Protocol in Two Arcs

The architecture becomes a method when it is converted into a repeatable learning protocol. The protocol below can be used for self-study, classroom discussion, AI-assisted tutoring, reflective writing, curriculum design, or personal knowledge management. Its point is not to make learning faster. Its point is to make learning more explicit, revisable, and answerable.

Step 1: Select a Lived Word

The learner begins by selecting a word that actually organizes perception, desire, anxiety, identity, or action. The method should not begin with the most abstract theory word. It should begin with a word that already has power in the learner's life.

> Examples: success, failure, intelligence, family, faith, freedom, dignity, money, education, AI, future, duty, truth, respect.

The test is simple: if changing the meaning of the word would change how the learner lives, studies, works, or judges, then it is a lived word.

A complementary entry point applies when the target is a *new* concept the learner does not yet live with—a technical term, an unfamiliar discipline-word, an imported abstraction. Here the method begins not from a charged word but from anchoring: the learner connects the new word to words already known and to concrete prior experience, so that the unfamiliar is grasped through the familiar rather than memorized in isolation. This is the principle of meaningful learning, in which new material is subsumed under relevant

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existing ideas in the learner's cognitive structure (Ausubel, 1968), and it is the mechanism of analogy, in which understanding transfers by mapping the relational structure of a known case onto a new one (Gentner, 1983). Anchoring is what lets the two starting points share one pipeline: an already-lived word is examined; a new word is first anchored to lived words, and then examined the same way. In both cases the learner ends with a word that is connected to experience rather than floating free of it.

Step 2: Write the Pre-Reflective Meaning

The learner writes the current meaning before consulting a dictionary, teacher, textbook, or AI system.

> "To me, *success* means . . . "

This step captures the learner's starting model. It should be preserved, because later revision must be visible—this is the versioned-state discipline of Commitment 5. Without a starting model, learning becomes invisible; the learner may produce a better answer without knowing what changed, or may rewrite the prior belief in hindsight.

Step 3: Map the Sources of the Meaning

The learner asks where the meaning came from. No meaning is born in isolation. It may come from family pressure, school ranking, religious formation, economic insecurity, social media, peer comparison, trauma, admiration, work experience, or previous AI-mediated language.

> Who taught me this meaning?
> Who benefits if I keep this meaning?
> Who is ignored by this meaning?
> Which institution made this meaning seem natural?

This step introduces social epistemology into learning, and it is where the structural source mapping of Section 17 is carried out: the answers should name institutions and incentives, not only feelings. The goal is not to reject inherited meanings. The goal is to make inheritance criticizable.

Step 4: Build the Meaning Map

The learner expands the word into a structured map. The map should include at least five layers:

1. personal meaning;
2. social or inherited meaning;
3. disciplinary or technical meaning;
4. emotional charge;

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5. practical consequence.

For example, the word *intelligence* may mean quick answers in school, adaptive problem-solving in psychology, moral discernment in life, pattern recognition in AI, or social advantage in a competitive institution. Mapping these layers prevents one meaning from silently dominating all others.

Step 5: Build the Semantic Neighborhood

The learner then maps nearby, opposite, hidden, and tension-bearing words.

Neighborhood type	Question
Similar words	What words appear close to this word?
Opposite words	What words challenge or negate it?
Hidden companions	What words travel with it silently?
Emotional neighbors	What feelings attach to it?
Institutional neighbors	What systems use or reward this word?
Danger neighbors	What distortions does this word invite?

This is the human analogue of embedding, with the structure supplied by frame, metaphor, and association rather than by a metric (Section 9). AI embeddings place tokens in mathematical space. Human semantic neighborhoods place lived words in existential, social, and practical space, including the emotional and institutional neighbors that no embedding records.

Step 6: Build the Life-Concept Graph

The learner converts the word map into relation triples:

subject $\rightarrow$ relation $\rightarrow$ object.

Examples:

```
school -> ranks -> intelligence
family -> shapes -> success
AI -> mediates -> inquiry
convenience -> may_reduce -> friction
friction -> can_enable -> revision
revision -> strengthens -> agency
truth -> resists -> convenience
```

This graph is not merely conceptual. It is practical and autobiographical. It shows how words become goals, how goals become actions, and how actions produce consequences.

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Step 7: Retrieve Evidence and Counterevidence

The learner now practices retrieval-augmented human inquiry. They gather materials that support, challenge, complicate, or reframe the map.

Sources may include:

- personal experience;
- books and research;
- teachers, elders, mentors, or peers;
- historical cases;
- data or records;
- religious, ethical, or cultural traditions;
- AI-generated alternatives, objections, and analogies.

AI may be used at this stage, but only as a candidate generator. It may propose alternative framings, objections, metaphors, cases, or conceptual distinctions. It must not be treated as the final certifier of truth. The map exchange of Section 17 belongs here too: peers are a retrieval source that no corpus replaces, because they can dissent.

Step 8: Run the Constraint Test

The learner asks how the meaning map can fail.

> What evidence would challenge this?
> What practice would reveal its weakness?
> What does the world resist here?
> What would another person see that I do not see?
> What changes if this belief is acted upon?

This is the decisive step. A word map becomes epistemic only when it becomes vulnerable to correction. The learner must distinguish between a pleasing formulation and a world-answerable formulation, and—for value-laden words—between self-confirming and resistant coherence (Section 14.1).

Step 9: Revise and Translate into Action

Finally, the learner records the revision and names one practical consequence.

> old meaning → friction → evidence or constraint → revised meaning → changed action.

The changed action may be small: a new question, a different study plan, a changed use of AI, a conversation with a mentor, a revised goal, or a new test of practice. But without action, the revision remains merely verbal. The revised meaning may also be the same

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as the original meaning held on new grounds; revision of standing is as legitimate an outcome as revision of content, a point developed in Example Five.

Extending the Protocol to Competence (Steps 10–15)

The nine steps above produce a tested, revised understanding. The following steps, developed in Section 18, carry that understanding into corrigible skill. They are written as a continuation of the same protocol, not a separate method.

Step 10: Map the problem-context. The learner records the problems the concept is used to solve, the situations in which it applies and fails, the cues that signal a case for it, and one concrete experience that anchors it. This is the conditional knowledge—knowing *when* and *where*—whose absence turns understanding inert.

Step 11: Select context-fit tools. The learner identifies the tools, references, instruments, collaborators, or AI affordances that fit *this* problem-context, and the criteria for choosing among them, rather than reaching for a default tool out of habit.

Step 12: Build the workflow. The learner assembles the operations into a repeatable procedure—the order of steps, the checkpoints, where retrieval, constraint testing, and tools enter—so that the understanding is usable more than once and is teachable and inspectable.

Step 13: Practice to skill. The learner executes the workflow repeatedly under deliberate-practice conditions: working at the edge of current ability, with clear goals and immediate feedback, until the procedure runs fluently and accurately with reduced conscious load.

Step 14: Apply to real problems. The learner exercises the skill on genuine problems, not only rehearsed cases, where the stakes and the messiness are real.

Step 15: Take feedback and revise map and skill. The learner reads the back-talk of the real situation and revises two things at once: the conceptual map, as in the epistemic loop, and the procedure itself. Skill, like belief, is kept corrigible.

The Full Learning Loop

The complete loop, across both arcs, can be stated compactly:

> (lived or anchored) word → pre-reflective meaning → source map → meaning map → semantic neighborhood → life-concept graph → retrieval → constraint test → reflective dissonance → revision → problem-context → tools & workflow

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→ deliberate practice → skill → real problems → feedback → revision of map and skill.

This loop is the core method of the paper. It translates AI architecture into human learning without reducing the human learner to an AI model, and it runs from the first naming of a word to the corrigible skill of using it well.

21 Worked Examples

The method becomes clearer when applied to concrete words. The examples below are not templates to be copied mechanically. They show how a learner can move from inherited meaning to revision—or to examined re-affirmation—under constraint.

21.1 Example One: Success

Pre-reflective meaning

> Success means becoming respected, financially stable, and recognized by family and society.

Source map

```
family -> teaches -> responsibility
school -> equates -> success_with_ranking
market -> equates -> success_with_income
social_media -> equates -> success_with_visibility
personal_experience -> links -> failure_with_shame
```

Semantic neighborhood

Dimension	Terms
Similar terms	achievement, recognition, stability, independence, competence
Opposing terms	failure, dependence, invisibility, shame, stagnation
Hidden companions	exhaustion, comparison, debt, parental approval, fear
Value terms	dignity, responsibility, contribution, freedom
Danger terms	burnout, false self, status anxiety, approval addiction

Constraint test

The learner asks whether this meaning of success survives contact with life. Does it produce sustainable agency, or does it produce dependence on recognition? Does it increase responsibility, or does it trap the learner in comparison? Does it answer to real capacities, time, health, family duties, and moral commitments?

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Reflective dissonance

AI, a teacher, or a peer may generate an alternative framing:

> Perhaps success is not maximum recognition but sustainable responsibility under truthful self-knowledge.

This creates dissonance because it challenges the inherited equation:

> success = external recognition.

Revision

> Success is not merely being recognized. Success is the development of stable, responsible capacity that can support oneself and others without surrendering truth, health, or moral direction.

Changed action

The learner changes the study or work plan from status-seeking output to capacity-building practice. They ask weekly: what competence did I actually build, what evidence shows it, and what constraint tested it?

21.2 Example Two: Intelligence

Pre-reflective meaning

> Intelligence means answering quickly, understanding difficult things before others, and appearing capable.

Source map

```
school -> rewards -> speed
exams -> measure -> correct_output
peers -> compare -> quickness
AI -> produces -> fluent_answers
family -> praises -> being_smart
```

Problem

The learner may confuse intelligence with fluency. In AI-rich environments this risk intensifies because the learner can produce articulate answers without deep understanding—the illusion of explanatory depth (Rozenblit & Keil, 2002) made effortless.

Constraint test

The learner asks:

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- Can I explain the idea without AI?
- Can I apply it to a new case?
- Can I identify where my answer may fail?
- Can I teach it to someone else?
- Can I revise it when challenged?

Revision

> Intelligence is not quick fluency. It is the capacity to form, test, revise, and transfer models under changing constraints.

Changed action

The learner uses AI not to produce final answers, but to generate counterexamples, alternative explanations, and tests of transfer. The learner measures intelligence by revision quality rather than speed alone.

21.3 Example Three: AI

Pre-reflective meaning

> AI is a tool that helps me finish work faster.

Source map

```
platform_design -> rewards -> speed
schoolwork -> rewards -> completion
market_language -> frames -> AI_as_productivity
user_habit -> seeks -> convenience
AI_fluency -> creates -> confidence
```

Reflective dissonance

The learner encounters the claim that convenience can weaken inquiry when it removes friction too early—that the offloading which speeds output can also remove the desirable difficulty on which durable learning depends (Bjork & Bjork, 2011; Risko & Gilbert, 2016). This challenges the belief that faster completion always means better learning.

Constraint test

The learner compares two modes of AI use:

Mode	Use of AI	Learning effect

Closure mode	Ask AI for a final answer or summary	Faster output, possible pseudo-understanding
Corrigibility mode	Ask AI for objections, alternatives, missing assumptions, tests, and counterexamples	Slower but deeper revision and transfer

Revision

> AI is not only a productivity tool. It is a mediating layer that can either close inquiry or deepen it, depending on whether it is used for completion or revision.

Changed action

The learner adopts a rule: every AI-generated answer must be followed by at least one objection prompt, one evidence check, and one self-explanation without AI.

21.4 Example Four: Faith or Freedom

Some lived words are existentially deep and culturally sensitive. The method should not flatten them into technical definitions. For a word such as *faith* or *freedom*, the learner may map personal experience, inherited tradition, moral demand, community belonging, doubt, and action. The constraint test is not merely empirical. It may include coherence with life, moral fruit, tradition, community accountability, spiritual discipline, and the capacity to remain truthful under difficulty—but, by Section 14.1, it must be *resistant* coherence: the dissent, consequence, refusal, and internal-critique conditions still apply. This shows that the method is domain-sensitive: different words require different forms of validation, while none is exempt from facing genuine resistance.

21.5 Example Five: A Case That Resists the Method's Apparent Bias (Duty)

The previous examples each move from an inherited meaning to a revised one, which invites the objection that the method is engineered to produce a particular family of conclusions—skeptical of recognition, status, and inheritance—and therefore transmits the author's values under the appearance of open inquiry. A method that claims to cultivate corrigible agency must be able to produce the opposite movement: a learner who, after genuine testing, responsibly re-affirms an inherited meaning.

Consider a learner whose word is *duty*, inherited from family obligation and religious formation, and whose pre-reflective meaning is "the obligation to support and care for my parents and community, even at personal cost." A method biased toward liberation-as-rejection would push toward reading this commitment as self-erasure. The constraint

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test, conducted honestly under the four conditions of Section 14.1, may instead yield re-affirmation:

- *Dissent.* Consulting others who could disagree—including those who have refused such obligations—the learner finds that, for this life, the obligation is experienced as a chosen form of solidarity rather than as coercion.
- *Consequence.* Checking the practical fruit of honoring it, the learner finds that it sustains rather than diminishes the relationships and the self.
- *Refusal.* Holding refusal genuinely open, the learner recognizes the real option to refuse and declines it on examined grounds.
- *Internal critique.* Engaging the tradition's own resources, the learner distinguishes care from servility within the tradition's terms, and locates limits beyond which the obligation would become distortion.

The revision here is not a change of conclusion but a change of grounds: the same commitment is now held as examined and re-owned rather than merely inherited.

> Duty, tested, is not abandoned. It is transformed from inherited obedience into responsibly affirmed solidarity—chosen with open eyes, and bounded by the refusal of coercion.

This case matters methodologically because it demonstrates that the method's output is revision of *standing*, not a predetermined revision of *content*. Corrigibility requires that affirmation, refusal, and alteration all remain genuinely available outcomes.

21.6 What the Examples Show

The examples demonstrate that word mapping is not an exercise in self-expression alone. It is a disciplined movement from inherited meaning to criticizable meaning. The learner does not become authentic by inventing a private vocabulary. The learner becomes more authentic by discovering the sources of inherited meanings, testing them, revising them, and responsibly re-owning or refusing them.

The examples also make a deliberately bounded claim about neutrality, which it is better to state openly than to leave implicit. The method is *not* value-neutral at the meta-level: it is committed to corrigibility, world-answerability, route-openness, and responsible action, and it treats these as non-negotiable. It is, however, *non-prescriptive at the object level*: it does not dictate which conception of success, intelligence, faith, or duty a learner must hold after inquiry. The convergence of the first three examples on revised meanings reflects the words chosen, not a hidden rule that inquiry must end in rejection; Example Five is included precisely to make the object-level openness demonstrable rather than merely asserted. Distinguishing the meta-level commitment from object-level neutrality is what allows the method to be disciplined without being doctrinaire.

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22 A Minimal YAML Model


```yaml
human_learning_architecture:
  principle: >
    Human learning should extract architectural discipline from AI
    while preserving embodiment, value, selfhood, and responsibility.

  starting_unit: "lived_word_or_anchored_new_word"

  pipeline:
    epistemic_arc:
      - lived_or_anchored_word
      - meaning_map
      - source_map
      - semantic_neighborhood
      - life_concept_graph
      - retrieval_augmented_inquiry
      - constraint_testing
      - reflective_dissonance
      - disciplined_revision
    competence_arc:
      - problem_context_map
      - context_fit_tools
      - workflow
      - deliberate_practice
      - skill
      - real_world_application
      - feedback_revising_map_and_skill

  anchoring:
    known_to_new: "anchor a new concept to known words and prior experience"
    mechanisms: ["meaningful subsumption (Ausubel)", "analogical structure-
   mapping (Gentner)"]

  competence_arc_detail:
    problem_context: "problems solved, situations of use, applicability cues,
   boundaries"
    tools: "instruments/references/collaborators/AI that fit this context"
    workflow: "repeatable, teachable, inspectable procedure"
    skill: "declarative -> procedural via deliberate practice"
    feedback: "revise both conceptual map and procedure"

  dialogical_structural_layer:
    map_exchange: "compare maps with differently positioned others"
    structural_source_mapping: "name institutions, incentives, authority"
```


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```
  absence_check: "whose experience and testimony are excluded?"

lived_word_schema:
  word: null
  current_meaning: null
  inherited_sources:
    - family
    - school
    - religion
    - media
    - market
    - community
    - personal_experience
    - AI_language
  emotional_charge:
    - hope
    - fear
    - shame
    - pride
    - curiosity
    - resentment
  related_words: []
  opposing_words: []
  hidden_costs: []
  goals_generated: []
  constraints: []
  evidence_for: []
  evidence_against: []
  revision_questions:
    - "Who gave me this meaning?"
    - "What does this word make visible?"
    - "What does this word hide?"
    - "What does this word make me pursue?"
    - "What would make me revise it?"

epistemic_tests:
  fallibility: "Can this meaning be wrong or incomplete?"
  corrigibility: "Can I revise it in assessable ways?"
  criticizability: "Can others challenge it?"
  route_openness: "Can I check it through more than one path?"
  world_answerability: "Can reality resist it?"
  resistant_coherence: "Has it survived dissent, consequence, refusal,
 internal critique?"
  life_fit: "Can this meaning be lived responsibly?"
```


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```
competence_tests:
  applicability: "Do I know when and where this concept applies, and where
 it fails?"
  transfer_to_action: "Can I use it to solve a real, unrehearsed problem?"
  skill_fluency: "Does the workflow run accurately with reduced conscious
 load?"
  skill_corrigibility: "Can my procedure be tested and revised, not just my
 belief?"
  no_fossilization: "Has automaticity hidden a subtly wrong step from notice
 ?"

AI_use:
  allowed_roles:
    - candidate_generator
    - objection_generator
    - analogy_generator
    - comparison_helper
    - retrieval_assistant
    - formatting_assistant
  prohibited_confusions:
    - "fluency equals truth"
    - "speed equals learning"
    - "completion equals understanding"
    - "AI output equals validation"

outcome_space:
  revision_of_content: "the meaning changes"
  revision_of_standing: "the meaning is re-affirmed on examined grounds"
  refusal: "the inherited meaning is responsibly declined"
  note: "all three are legitimate; the method fixes process, not conclusion"

learning_loops:
  closure_loop:
    sequence:
      - friction
      - AI_reassurance
      - fluent_output
      - premature_confidence
      - pseudo_understanding
  corrigibility_loop:
    sequence:
      - friction
      - alternative_framing
      - evidence_retrieval
      - constraint_testing
```


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```
        - revision
        - action_change
```

23 Assessment: A Four-Level Rubric for Learning

A human-centered learning architecture requires assessment criteria that go beyond output quality. The question is not only whether the learner produced a correct or fluent answer. The question is whether the learner became more capable of mapping, retrieving, testing, revising, and acting responsibly.

The following rubric can be used by teachers, mentors, researchers, or learners themselves. One caution must be stated before it is used, because it bears on a real measurement risk. A revision trace can be produced by following instructions rather than by learning, so the highest level must not be awarded for the mere presence of a trace. Level 4 is credited only when revision is corroborated by *protocol-independent transfer*: the learner shows the revised understanding on a task, or in a domain, where the protocol was not applied and the framing was not rehearsed. This keeps the rubric from defining corrigibility in terms of compliance with the method that is supposed to produce it.

Dimension	**Level 1: Repetition**	**Level 2: Mapping**	**Level 3: Testing**	**Level 4: Revision**
Word clarity	Repeats a definition	States personal meaning	Distinguishes personal, social, and technical meanings	Revises meaning with reasons
Source awareness	Treats meaning as obvious	Names some sources	Identifies institutional and social pressures	Re-appropriates or refuses inherited meaning responsibly
Semantic neighborhood	Lists related words	Maps similar and opposite terms	Identifies hidden tensions and emotional charge	Uses tensions to reformulate the word
Life-concept graph	Concepts remain isolated	Builds simple relations	Maps causal, social, and practical paths	Revises the graph after criticism
Retrieval	Relies on memory or AI output	Finds supporting sources	Retrieves counterevidence and alternatives	Integrates evidence into revised understanding
Constraint testing	Protects claim from challenge	Identifies possible tests	Applies domain-sensitive constraints	Changes belief, goal, or method under resistance

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AI use	Uses AI for completion	Uses AI for explanation	Uses AI for objections and alternatives	Uses AI within a validation and revision workflow
Dissonance handling	Avoids discomfort	Notices conflict	Investigates the conflict	Converts dissonance into disciplined revision
Action integration	Produces answer only	Names possible action	Tests action in practice	Changes practice and records feedback
Problem-context / applicability	Treats the concept as context-free	States where it is used	Identifies applicability cues and boundaries	Recognizes unrehearsed cases and selects fitting tools
Skill / proceduralization	Knows the idea but cannot perform it	Performs once by following steps	Builds and refines a workflow	Performs fluently on real problems and keeps the procedure corrigible
Calibration	Confidence tracks fluency	Notices overconfidence	Adjusts confidence to evidence	Confidence tracks transfer, not familiarity
Agency	Seeks closure	Shows self-awareness	Accepts criticizability	Demonstrates corrigible, responsible judgment, evidenced by transfer

A learner reaches Level 4 only when revision becomes visible *and* survives transfer. This means that the assessment artifact should preserve the original meaning, the challenge, the evidence or constraint, the revised meaning, and the changed action, and that at least one outcome should be checked outside the protocol. The evidence of learning is therefore not merely the final answer, but the trace of transformation together with its independent corroboration.

23.1 Minimal Assessment Artifact

Each completed word map should contain the following fields:

```
assessment_artifact:
  word: null
  original_meaning: null
  source_map: []
  semantic_neighborhood: []
  relation_graph: []
  evidence_for: []
  evidence_against: []
```


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```
  constraint_test: null
  reflective_dissonance: null
  revised_meaning: null
  outcome_type: null        # content / standing / refusal
  problem_context: []       # problems, situations, cues
  tools_selected: []        # tools that fit this context
  workflow: null            # repeatable procedure
  practice_record: null     # deliberate-practice log toward skill
  changed_action: null
  real_problem_applied: null
  transfer_check: null      # protocol-independent corroboration
  skill_corrigibility: null # was the procedure tested and revised?
  remaining_uncertainty: null
```

This artifact allows learning to be evaluated as a process rather than an output. It also protects against a common AI-era problem: fluent answers that leave the learner's underlying model unchanged.

24 Validation and Research Design

The method proposed here is philosophical and pedagogical, but it can also be empirically studied, and the paper takes on the obligation to specify a design that could disconfirm it. Future research can evaluate whether word-based epistemic mapping improves learning outcomes compared with ordinary summarization, ordinary mind mapping, or unstructured AI assistance.

24.1 Possible Research Questions

1. Do learners who begin with lived words show greater metacognitive awareness than learners who begin with textbook definitions?
2. Does life-concept graphing improve transfer to new cases?
3. Does AI-assisted reflective dissonance improve revision quality compared with AI-assisted summarization?
4. Does constraint testing reduce overconfidence in AI-generated answers, measured as improved calibration?
5. Does the method help learners distinguish inherited goals from examined goals?

24.2 Study Design and the Confounds It Must Defeat

A naive three-group comparison—summarize, AI-complete, or use the full method—cannot by itself support the paper's claim, for two reasons that must be addressed head-on rather than left to a reviewer. First, the method is more effortful and time-consuming than the

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controls, so any positive result could be a pure effect of time-on-task rather than of the architecture. Second, the method instructs learners to produce maps, traces, and revisions, so scoring learning by the presence of those artifacts would measure compliance with the protocol rather than learning. A credible design therefore requires the following features.

Design feature	**Purpose**
Random assignment	Remove selection effects across conditions.
Effort- and time-matched (yoked) control	An active control that spends equal time and effort on a plausible alternative activity (e.g., extended structured summarization), so that any advantage is attributable to the architecture, not to time-on-task.
Protocol-independent transfer as primary outcome	The main dependent measure is performance on novel cases and domains where the protocol was not applied, breaking the circularity between trace-production and learning.
Delayed testing	Outcomes assessed after a delay (e.g., 1–4 weeks), since desirable-difficulty effects typically appear in retention and transfer, not immediate performance.
Calibration measure	Confidence–accuracy calibration on AI-assisted items, to test the specific claim that constraint testing reduces fluency-driven overconfidence.
Blinded, reliable trace coding	Revision-trace quality scored by raters blind to condition, with reported inter-rater reliability, treated as a secondary (mechanism) outcome, not the primary one.
A priori power analysis	Sample size fixed in advance for a realistic effect size; analyses and primary outcome pre-registered.

Outcomes could include delayed recall, transfer tasks, self-explanation quality, revision-trace quality, calibration, ability to identify assumptions, and capacity to apply the concept in action. The central hypothesis is not that the method group will be faster, and not even that it will perform better on immediate, in-distribution tasks; on those it may perform *worse*, exactly as desirable-difficulty research predicts. The hypothesis is that it will produce more durable, better-calibrated, and more transferable understanding at delay.

24.3 What Would Disconfirm the Method

Stating falsification conditions in advance is a requirement of the constraint-first epistemology the paper defends, and it is applied here to the paper's own proposal. The method's central claim would be disconfirmed if, in an adequately powered, time-matched design, the method group showed *no* advantage over the yoked control on protocol-independent transfer and calibration at delay—that is, if its only advantages were on immediate,

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in-distribution tasks or on trace-production, both of which would be consistent with the unflattering hypothesis that the method adds effort and compliance without adding learning. A method that cannot specify the result that would count against it has not met its own standard.

24.4 Design Principle

The method should be tested not by output quantity, but by the quality of transformation:

> Did the learner's model become more explicit, more criticizable, more evidence-sensitive, more constraint-aware, and more responsible in action—and does that change survive transfer to conditions the protocol did not rehearse?

This criterion preserves the connection to constraint-first epistemology. The educational value of the method lies not in fluency, but in answerable, transferable revision.

25 Classroom and Personal Knowledge System Design

The model can be implemented as a classroom practice or personal knowledge system.

25.1 Classroom Design

A teacher can organize a unit around key words rather than topics alone. For example, a unit on AI and education may begin with:

> learning, intelligence, effort, convenience, truth, creativity, dependence, agency.

Students map their meanings before reading theory. Then they compare maps, retrieve evidence, use AI to generate objections, revise their maps, and apply the revised vocabulary to cases. The comparison of maps is not an optional warm-up but the dialogical operation of Section 17: it is where private framings meet competent dissent.

This reverses the usual direction. Instead of starting with theory and forcing students to memorize terms, education begins with the words already structuring student experience. Theory then becomes a tool for revising lived meaning.

25.2 Personal Knowledge System

A learner can build a personal database with the following fields:

```
personal_knowledge_database:
  entries:
    - word
    - current_meaning
    - source_history
```


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```
    - emotional_charge
    - related_concepts
    - graph_edges
    - evidence
    - counterevidence
    - AI_objections
    - revision_history
    - outcome_type        # content / standing / refusal
    - action_changes
    - transfer_check
    - next_review_date
```

This transforms note-taking into epistemic self-formation. The note is not merely a record. It is a revisable map of how the learner sees and acts.

26 Risks and Safeguards

The proposed model has risks.

First, word mapping can become narcissistic if it never meets evidence or other people. The safeguard is constraint testing and the dialogical layer's public criticism.

Second, AI can make the process too fluent. The safeguard is to use AI for objection, comparison, and validation support rather than only summarization—the generator/verifier separation of Commitment 3.

Third, mapping can become endless reflection without action. The safeguard is the action layer: every revised map should eventually change a practice, decision, explanation, or relation.

Fourth, inherited meanings can be treated as automatically oppressive. The safeguard is responsible discernment, and Example Five is the demonstration: the goal is not rejection of inheritance but examined re-appropriation, refusal, or alteration, with all three left genuinely open.

Fifth, personal meaning can be mistaken for truth. The safeguard is route-openness and resistant coherence: multiple sources, multiple perspectives, and genuine resistance, not agreement gathered from sources chosen to agree.

Sixth, the method can be captured by an individualist frame and reduced to private self-optimization. The safeguard is the constitutive dialogical and structural layer, and the explicit option to run the method in a communal register.

Seventh, the method can be mistaken for a value-transmission device. The safeguard is the distinction between meta-level commitment and object-level neutrality: facilitators owe learners openness about the former and restraint about the latter.

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Eighth, the competence arc introduces its own failure mode: a workflow practiced to automaticity can fossilize, running fluently while reliably producing a subtly wrong result that has dropped below the level of notice. The safeguard is corrigible competence (Section 18): the constraint test and the feedback loop are applied to the procedure, not only to the belief, so that skill remains as open to revision as understanding.

27 Limitations and Future Validation

This paper offers a method proposal, not a completed empirical validation. Its central claim is theoretical and pedagogical: reflective, language-mediated learning can be redesigned through lived-word mapping, life-concept graphs, retrieval, constraint testing, and revision. However, the method requires the classroom trials, comparative studies, and longitudinal assessment specified in Section 24 before strong claims about learning outcomes can be made. Stating this is not a retreat from the claim; it is the claim held to its own standard of corrigibility.

Several limitations should be stated clearly.

First, the method may privilege learners with stronger language skills. Because it begins with words, it may need adaptation for younger learners, multilingual learners, neurodiverse learners, or learners whose knowledge is primarily embodied, visual, oral, or practical. Future versions should include visual mapping, storytelling, drawing, role-play, and oral dialogue as equivalent entry points—an extension that the communal register of Section 17 already anticipates.

Second, the method can become overly reflective if it is separated from action. A life-concept graph is not an end in itself. Its purpose is to improve judgment, practice, and responsible agency. For that reason every cycle should end in a changed question, decision, experiment, habit, or practice.

Third, AI assistance may distort the method if used for premature completion. If learners ask AI to fill in the map for them, the method collapses into another closure loop. AI should be used to generate alternatives, objections, counterexamples, and questions, while the learner remains responsible for validation and revision.

Fourth, the method requires domain-sensitive evaluation. The constraint test for a scientific claim differs from the constraint test for an ethical commitment, a religious vocabulary, a career goal, or an artistic judgment. No single metric can replace contextual judgment—though, by Section 14.1, no domain is exempt from facing genuine resistance.

Fifth, the method requires social safeguards. Because words are shaped by family, culture, institutions, and power, the mapping process may surface sensitive meanings. Teachers and facilitators should avoid forcing disclosure. Learners should retain control over what is private, shared, or anonymized.

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Future research should compare this method with ordinary note-taking, concept mapping, AI-completion workflows, and retrieval-practice-only designs. Useful outcome measures may include explanation transfer, source awareness, ability to generate counterexamples, revision quality, calibration, resistance to AI overreliance, and changes in self-regulated learning behavior. The strongest evidence would not be faster answer production, but improved capacity to revise beliefs, goals, and actions under criticism, demonstrated where the protocol was not rehearsed.

28 Conclusion: From AI-Inspired Architecture to Human Agency

This paper has rewritten the human–AI learning project with human learning at the center. The question is not how to make humans think like AI. The question is how to extract useful architectural principles from AI and translate them into practices that strengthen human agency.

The extraction is not metaphorical decoration. As Section 6 argued, the architecture lens fixes five design commitments—representation-first ordering, retrieval by default, the separation of candidate generation from validation, transfer as the criterion of success, and versioned belief-state—that prior educational theories permit but do not require, and it unifies them under a single rationale: reliable learning systems gate their later operations on the disciplined organization of their earlier ones. Reflective human learning begins not with tokens but with lived words; it builds not embeddings but meaning neighborhoods structured by frame, metaphor, and value; it retrieves not from a frozen corpus but from sources, mentors, memory, and experience that can answer back; it is tested not against a held-out set but against truth, practice, responsibility, and life; and it updates not parameters but beliefs, goals, habits, relationships, and selves.

The first step is therefore simple but profound: map the words. A learner's world is already organized by vocabulary before formal education begins. To educate is to make those words visible, relational, criticizable, and revisable. A word becomes knowledge when it is connected to experience, meaning, source, value, goal, constraint, evidence, action, and revision—and when the resulting commitment can survive genuine resistance, whether the learner ends by revising it, refusing it, or re-affirming it on examined grounds. But a word becomes *usable* knowledge only when the learner also maps the problems it solves, selects the tools its use requires, builds a workflow, and practices until understanding becomes skill that is tested on real problems and corrected by their feedback. The two arcs are one method: meaning that is never used grows inert, and skill that is never examined fossilizes.

The final aim is not information management. It is corrigible agency and corrigible competence: the capacity to remain open to correction in both belief and action while acting responsibly in the world. AI can support this process only when it deepens, rather

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than bypasses, the learner's ability to encounter friction, test assumptions, and revise under constraint. In that sense, the future of education is not artificial intelligence replacing human learning. It is human learning becoming architecturally conscious of itself.

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References

Anderson, J. R. (1982). Acquisition of cognitive skill. *Psychological Review*, 89(4), 369–406.

Ausubel, D. P. (1968). *Educational Psychology: A Cognitive View.* Holt, Rinehart & Winston.

10.5281/zenodo.22341297

Bandura, A. (2001). Social cognitive theory: An agentic perspective. *Annual Review of Psychology*, 52, 1–26.

Berlyne, D. E. (1960). *Conflict, Arousal, and Curiosity*. McGraw-Hill.

Bjork, E. L., & Bjork, R. A. (2011). Making things hard on yourself, but in a good way: Creating desirable difficulties to enhance learning. In M. A. Gernsbacher, R. W. Pew, L. M. Hough, & J. R. Pomerantz (Eds.), *Psychology and the Real World* (pp. 56–64). Worth.

Brandom, R. (1994). *Making It Explicit*. Harvard University Press.

Brown, J. S., Collins, A., & Duguid, P. (1989). Situated cognition and the culture of learning. *Educational Researcher*, 18(1), 32–42.

Chi, M. T. H., De Leeuw, N., Chiu, M.-H., & LaVancher, C. (1994). Eliciting self-explanations improves understanding. *Cognitive Science*, 18(3), 439–477.

Clark, A., & Chalmers, D. (1998). The extended mind. *Analysis*, 58(1), 7–19.

Collins, A. M., & Loftus, E. F. (1975). A spreading-activation theory of semantic processing. *Psychological Review*, 82(6), 407–428.

Dewey, J. (1938). *Logic: The Theory of Inquiry*. Henry Holt.

Dunlosky, J., Rawson, K. A., Marsh, E. J., Nathan, M. J., & Willingham, D. T. (2013). Improving students' learning with effective learning techniques. *Psychological Science in the Public Interest*, 14(1), 4–58.

Ericsson, K. A., Krampe, R. T., & Tesch-Römer, C. (1993). The role of deliberate practice in the acquisition of expert performance. *Psychological Review*, 100(3), 363–406.

Festinger, L. (1957). *A Theory of Cognitive Dissonance*. Stanford University Press.

Fillmore, C. J. (1982). Frame semantics. In Linguistic Society of Korea (Ed.), *Linguistics in the Morning Calm* (pp. 111–137). Hanshin.

Flavell, J. H. (1979). Metacognition and cognitive monitoring: A new area of cognitive–developmental inquiry. *American Psychologist*, 34(10), 906–911.

Freire, P. (1970). *Pedagogy of the Oppressed*. Continuum.

Fricker, M. (2007). *Epistemic Injustice: Power and the Ethics of Knowing*. Oxford University Press.

Gentner, D. (1983). Structure-mapping: A theoretical framework for analogy. *Cognitive Science*, 7(2), 155–170.

Haraway, D. (1988). Situated knowledges: The science question in feminism and the privilege of partial perspective. *Feminist Studies*, 14(3), 575–599.

Karpicke, J. D., & Roediger, H. L. (2007). Repeated retrieval during learning is the key to long-term retention. *Journal of Memory and Language*, 57(2), 151–162.

10.5281/zenodo.22341297

Kasneci, E., Sessler, K., Küchemann, S., Bannert, M., Dementieva, D., Fischer, F., et al. (2023). ChatGPT for good? On opportunities and challenges of large language models for education. *Learning and Individual Differences*, 103, 102274.

Kolb, D. A. (1984). *Experiential Learning: Experience as the Source of Learning and Development.* Prentice-Hall.

Lahtee, Y. (2026). *When AI Expands Human Potential: Reflective Dissonance, Epistemic Agency, and Constraint.* Preprint. [Author's prior work; cited for the constraint-first framework it develops.]

Lakoff, G., & Johnson, M. (1980). *Metaphors We Live By.* University of Chicago Press.

LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, 521, 436–444.

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., et al. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. *Advances in Neural Information Processing Systems*, 33, 9459–9474.

Longino, H. E. (1990). *Science as Social Knowledge.* Princeton University Press.

Mezirow, J. (1991). *Transformative Dimensions of Adult Learning.* Jossey-Bass.

Nickel, M., Murphy, K., Tresp, V., & Gabrilovich, E. (2016). A review of relational machine learning for knowledge graphs. *Proceedings of the IEEE*, 104(1), 11–33.

Piaget, J. (1970). *Psychology and Epistemology.* Viking.

Polanyi, M. (1966). *The Tacit Dimension.* University of Chicago Press.

Popper, K. R. (1959). *The Logic of Scientific Discovery.* Hutchinson.

Risko, E. F., & Gilbert, S. J. (2016). Cognitive offloading. *Trends in Cognitive Sciences*, 20(9), 676–688.

Rozenblit, L., & Keil, F. (2002). The misunderstood limits of folk science: An illusion of explanatory depth. *Cognitive Science*, 26(5), 521–562.

Schön, D. A. (1983). *The Reflective Practitioner: How Professionals Think in Action.* Basic Books.

Sweller, J. (1988). Cognitive load during problem solving: Effects on learning. *Cognitive Science*, 12(2), 257–285.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30.

Vygotsky, L. S. (1978). *Mind in Society.* Harvard University Press.

Whitehead, A. N. (1929). *The Aims of Education and Other Essays.* Macmillan.

Zimmerman, B. J. (2002). Becoming a self-regulated learner: An overview. *Theory Into Practice*, 41(2), 64–70.

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Learning Under Generative Abundance: A Structural Law of Epistemic Stabilization

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What the chapter states. "This paper argues that the consequences for education stem not primarily from pedagogical choice, but from a structural law governing inference under conditions of generative abundance." "Learning is therefore redefined as the capacity to preserve structural stability when expressive production no longer uniquely indicates internal organization."

Status. No explicit K-state or peer-review status line is printed in the chapter text itself; the manifest records no version tag for this record.

Falsifier(s). Verbatim, under "Falsifiable Predictions": "In AI-assisted learning environments, rejection accuracy (correctly discarding implausible explanations) will predict expertise more strongly than recall or production accuracy."

Read before. Readout Genesis Standalone Synthesis; The Readout Condition; The Standalone Scholar; When AI Expands Human Potential; Written by AI. Still True.

Feeds into. — (no continues/isContinuedBy/references edge to another chapter of this book found in the knowledge graph).

Learning Under Generative Abundance: A Structural Law of Epistemic Stabilization

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Open Science – Civil Science Initiative

February 20, 2026

Abstract

Artificial intelligence creates an epistemic environment in which coherent expressions can be produced externally at near-zero marginal cost. This paper argues that the consequences for education stem not primarily from pedagogical choice, but from a structural law governing inference under conditions of generative abundance.

A single underlying dynamic gives rise to three observable effects: degradation of inferential signals, escalation of conceptual volatility, and the emergence of selective stabilization. Beyond a critical threshold of generative exposure, learning undergoes a regime shift — from accumulation to discrimination.

Learning is therefore redefined as the capacity to preserve structural stability when expressive production no longer uniquely indicates internal organization.

Keywords: Generative AI, epistemic stabilization, open science, civil science, regime transition, discrimination learning

The Inferential Assumption of Traditional Education

Educational systems have historically rested on a tacit inference: correct production implies understanding [Cronbach & Meehl, 1955].

Under conditions of scarcity, the cognitive effort required to generate coherent explanations naturally constrained output to those who possessed corresponding internal structure.

Large-scale generative systems eliminate this constraint. Coherent explanations can now be produced with high fidelity independently of the learner's internal representational organization [Bender et al., 2021].

The central educational challenge has therefore become inferential rather than merely technological.

A Structural Law of Epistemic Environments

Consider any cognitive system embedded in an environment that can supply unlimited coherent explanatory structures on demand.

The system must continuously classify incoming structure as worthy of integration or appropriate for rejection.

Let *stability* denote the persistence of coherent internal organization across contexts and over time.

A general structural property emerges:

> When the external generation of coherent structure increases without bound, observable production ceases to serve as a unique indicator of internal organization.

This is not merely a behavioral or motivational observation; it follows from observational indistinguishability: externally supplied structure becomes empirically equivalent to internally constructed structure.

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Consequence I: Degradation of Inferential Signals

Once externally generated coherence becomes indistinguishable from internally grounded coherence, production loses its reliability as a diagnostic signal of understanding.

Assessments that rely primarily on produced output therefore suffer a fundamental loss of inferential validity [Messick, 1989].

The mapping from expression to underlying cognitive structure becomes probabilistic rather than deterministic.

Consequence II: Escalation of Conceptual Volatility

Unlimited access to coherent but mutually incompatible explanatory frameworks creates constant pressure toward revision.

The cognitive system must either frequently update its beliefs or actively resist updating.

Empirical work on large language models shows that they frequently generate fluent, persuasive, yet ungrounded explanations [Ji et al., 2023].

Chronic exposure to large numbers of mutually plausible structures induces conceptual volatility unless strong regulatory mechanisms are in place.

Consequence III: Emergence of Selective Stabilization

Maintaining persistent, coherent internal organization increasingly depends on the active capacity to *discriminate* — that is, to reject the majority of available explanatory candidates.

Expertise studies consistently show that what most sharply distinguishes advanced learners from novices is superior structural discrimination rather than sheer volume of encoded information [Chi et al., 1981].

Under generative abundance, epistemic stability depends more on rejection capacity than on production capacity.

The Regime Transition

When generative availability remains below a critical threshold, accumulation strategies remain viable: production continues to correlate reasonably with internal structure.

Once generative availability exceeds the system's current discriminative capacity, accumulation collapses as a reliable path to competence.

Only selective retention and active stabilization preserve long-term coherence.

Thus a structural regime shift occurs:

$$\textit{accumulation} \quad \longrightarrow \quad \textit{discrimination}$$

This transition is environmentally driven, not ideologically chosen.

Toward a New Definition of Learning

In generative environments, learning should be understood as the progressive development of the ability to maintain coherent internal organization despite effectively unlimited external production of plausible structure.

Expertise under these conditions manifests as:

- resistance to persuasive but unstable or shallow explanations,
- calibrated, context-sensitive acceptance,
- maintenance of cross-context conceptual coherence,
- stable conceptual identity despite expressive abundance.

Production remains a necessary *practice*, but it ceases to be a reliable *diagnostic*.

Falsifiable Predictions

The proposed framework yields several testable implications:

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1. In AI-assisted learning environments, rejection accuracy (correctly discarding implausible explanations) will predict expertise more strongly than recall or production accuracy.

2. Increased passive exposure to generative explanations — without explicit discrimination training — will increase measured belief volatility.

3. Traditional production-only assessments will show declining predictive validity for real-world transfer performance.

4. Targeted training in epistemic stabilization and rejection will reduce volatility under matched levels of generative exposure.

Conclusion

Generative artificial intelligence fundamentally alters the epistemic structure of education.

When coherent expression can be supplied externally and competing explanatory organizations proliferate at negligible cost, production no longer uniquely indexes understanding — and cognitive-epistemic stability becomes the central bottleneck.

Learning in an age of generative abundance is, at core, the disciplined maintenance of epistemic stabilization.

This work is conducted under the Open Science – Civil Science Initiative (Yaoharee framework), emphasizing transparency, reproducibility, FAIR principles, citizen-driven participation, and ethical integration of knowledge across social, cultural, and technical domains.

References

Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of FAccT '21*.

Chi, M. T. H., Feltovich, P. J., & Glaser, R. (1981). Categorization and representation of physics problems by experts and novices. *Cognitive Science*, 5(2), 121–152.

Cronbach, L. J., & Meehl, P. E. (1955). Construct validity in psychological tests. *Psychological Bulletin*, 52(4), 281–302.

Ji, Z., et al. (2023). Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12), Article 248.

Messick, S. (1989). Validity. In R. L. Linn (Ed.), *Educational measurement* (3rd ed., pp. 13–103). American Council on Education / Macmillan.

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Part III

Possibility, Choice, Potential

Possibility, Choice, Potential collects six K0 works that sit at the live-possibility and choice segment of the programme spine (World → Readout → Meaning → Experience → Retention → Live Possibility → Choice → Agency → Human–AI → Human Return → Warrant), continuing from Part 2's readout-retention account of experience and learning.

The first four chapters (13–16) form a stated four-paper sequence on hypothesis generation. Paper I models imagination as bounded dynamic semantic mobility, separating what is technically reachable from what is currently accessible, attractive, or warranted for a bounded knower, individual or collective. Paper II asks whether the organization of prior knowledge changes the time and direction of first passage to a usable hypothesis, holding content approximately fixed. Paper III steps back to a pre-evidential layer, naming, as an agenda rather than a solution, the open question of how a candidate ever enters the set that evidence can appraise at all. Paper IV turns to human–AI collaboration specifically, proposing a role-separated architecture in which AI expands candidate hypotheses while a human retains the question, and stating testable comparisons between human-only, one-shot-AI, autonomous-AI, and recursive human–AI workflows.

The final two chapters turn from hypothesis-generation to choice and potential in general. Chapter 17 proposes that human choice draws on a history-shaped field of live possibilities narrower than what is objectively or structurally possible, and states explicit non-collapse laws separating feasibility, accessibility, preference, consent, and enactment. Chapter 18 treats potential itself as a readout — the envelope of a witnessed, enumerable option set rather than a hidden inner quantity — and, per the library's own linkage data, continues three works on causal ethics and structured coexistence that this book places later, in Part 4.

Carried forward: the recurring distinction between formal possibility, accessibility, and warrant; the practice of stating falsifiable propositions together with the observation that would defeat each; and the K0 / [Open] / [Dr] tier discipline that every chapter in this part applies to its own claims. Part 4 extends this apparatus to conflict, violence, and structured coexistence; Part 5 returns to it for human–AI coupling directly.

- From Problem to Hypothesis: Dynamic Semantic Mobility, Bounded Knowers, and the Readout-Discriminable Hypothesis Space — `10.5281/zenodo.22307148`
- Knowledge Topology and the First Passage to Usable Hypotheses: A Readout Theory of Discovery Time and Direction — `10.5281/zenodo.22307561`
- Before Evidence Can Decide: Candidate-Set Formation, Discovery Routing, and Unconceived Alternatives — Questions for a Pre-Evidential Epistemology of Science — `10.5281/zenodo.22307564`
- The Epistemic Chain Reaction: Human-AI Multiplication from Questions to Readout-Distinguishable Hypotheses — A Controlled Amplification Architecture for Scientific Discovery — `10.5281/zenodo.22308072`
- Choice Begins Before Choice: Meaning-Shaped Accessibility, Live Possibility, and Effective Agency in Human-AI Systems — `10.5281/zenodo.22357788`
- Potential as a Readout: Witnessed Envelopes, Two-Layer Agency, and the Measurement of Pseudo-Peace — `10.5281/zenodo.223618`

From Problem to Hypothesis: Dynamic Semantic Mobility, Bounded Knowers, and the Readout-Discriminable Hypothesis Space

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What the chapter states. "Its central thesis is that imagination can be modeled as bounded dynamic semantic mobility: not an infinite faculty and not a static set of reachable ideas, but a time-indexed process over semantic readouts whose transitions are provenance-audited, dynamically weighted, history-sensitive, and explicitly restructurable."

Status. "BOUNDED-KNOWER FINAL AUTHOR DRAFT under the current Readout Universe / Readout Genesis architecture; GLOSA K0. [. . .] No independent checker has reviewed the literature interpretations, mathematical bridges, or final wording."

Falsifier(s). "If increasing attraction never produces measurable alternative suppression once current warrant and knowledge are controlled, P1 weakens."

Read before. *The Readout Condition: Distinguishability, Access, and Epistemic Warrant*; *Readout Genesis Standalone Synthesis*.
Feeds into. *Knowledge Topology and the First Passage to Usable Hypotheses*.

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From Problem to Hypothesis

Dynamic Semantic Mobility, Bounded Knowers, and the Readout-Discriminable Hypothesis Space

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Status. BOUNDED-KNOWER FINAL AUTHOR DRAFT under the current Readout Universe / Readout Genesis architecture; GLOSA K0. "Final" here means the manuscript preserves the previous anchor architecture and incorporates the present source audit, adversarial stress corrections, and history-shaped semantic dynamics. It does *not* mean independently validated or released. No independent checker has reviewed the literature interpretations, mathematical bridges, or final wording. No priority claim is made. The organized-knowledge hitting-time hypothesis is reserved for a separate manuscript and is not part of the present claim set. Claim tiers remain explicit: [definition], [Dr], [Open], and [finite_diagnostic].

Abstract

Scientific hypothesis generation is distributed across literatures on problem construction, abduction, analogy, model-based reasoning, semantic memory and control, insight, underdetermination, experimental discrimination, distributed cognition, and AI-assisted discovery. This paper proposes a readout-native architecture for the narrower domain of *hypothesis-generating imagination*. Its central thesis is that imagination can be modeled as *bounded dynamic semantic mobility*: not an infinite faculty and not a static set of reachable ideas, but a time-indexed process over semantic readouts whose transitions are provenance-audited, dynamically weighted, history-sensitive, and explicitly restructurable. Binary reachability is separated from graded accessibility; current constraints are typed by provenance so that forced distinctions cannot be silently erased while posited or borrowed constraints may be openly relaxed or retyped. Readout-semantic attraction and semantic momentum are introduced as finite access diagnostics, but they are explicitly separated from epistemic warrant and truth: a familiar or institutionally reinforced path may be highly attractive without being well supported. Candidate hypotheses are quotiented by reader equivalence under declared domain actions and a finite horizon, yielding a readout-discriminable hypothesis space. The framework further derives a bounded-knower corollary: epistemic standing is indexed by question, domain, time, reader, policy, and available records rather than inhering permanently in a person or social label. It also extends the architecture to collective epistemic systems without positing a group mind: shared records, artifacts, communication channels, and institutional policies define a collective semantic readout, attraction landscape, momentum trace, and warrant process. Three behavioral propositions remain central and falsifiable: attraction can invert into fixation, recent semantic paths can carry momentum into later search, and productive escape can occur through audited restructuring when a dominant basin fails to reduce residuals. A separate hypothesis concerning whether knowledge organization shortens expected time to a usable new hypothesis is deliberately reserved for a subsequent paper. The present manuscript remains a GLOSA K0 architecture pending independent review.

Keywords: imagination; hypothesis generation; semantic mobility; semantic attraction; semantic momentum; knowledge structure; insight; Einstellung; semantic control; scientific discovery; model discrimination; underdetermination; artificial intelligence; Readout Universe; Readout Genesis

1 Introduction

Scientific-method diagrams often compress observation, question, hypothesis, test, and conclusion into a short sequence. The compression hides two boundedness problems. First, no knower can retain every difference. Second, even after a problem and question are fixed, not every candidate explanation is equally available. Some alternatives are never generated; others are technically reachable but remain practically inaccessible because familiar structures dominate attention. Research on problem construction, dual-space scientific search, analogy, model-based reasoning, insight, semantic control, and cognitive fixation already explains substantial parts of this landscape [18, 10, 7, 17, 11, 2, 13, 12]. The aim here is not to replace those traditions.

At the opposite end of inquiry, philosophy and statistics already distinguish observationally equivalent theories and design experiments that maximize model discrimination [20, 4, 15]. AI systems now generate, critique, rank, and refine scientific hypotheses at scale [8, 16]. The unresolved interface is therefore narrower: how should a bounded knower's changing semantic access be connected to the hypothesis classes that a declared reader can actually distinguish?

A further problem concerns history. Semantic access is not a memoryless random walk. The organization of prior knowledge can both facilitate efficient retrieval and produce fixation; semantic-memory studies report costs as well as benefits of rich associative structure [21, 9]. Recent mental content can also persist and bias subsequent thought, a phenomenon reviewed as psychological momentum [22]. The present architecture therefore asks not only what is reachable, but which structures attract search, which recent directions persist, and how control can escape an over-stable basin.

Central proposal [Dr/Open]. Hypothesis-generating imagination is modeled as *bounded dynamic semantic mobility*: the capacity to traverse and restructure a problem-conditioned semantic readout under provenance-audited transport. Accessibility may be shaped by current structure, repeated use, institutional reinforcement, recent traversal history, and control; none of these is itself warrant. The architecture changes which candidate relations are accessible while exposing rather than silently erasing distinctions that remain forced by the inquiry.

This proposal is intentionally narrower than a general theory of imagination. Fiction, dreams, artistic invention, and other forms of imagination need not be evaluated by hypothesis discrimination. The target here is imagination used to generate candidates for inquiry.

2 Literature as a System of Allies

The closest literatures already supply most local mechanisms. Table 1 therefore marks prior work as an ally rather than as a target for priority claims.

Residual gap. The examined literatures contain mature accounts of problem construction, search, analogy, semantic control, insight, empirical equivalence, and model discrimination. What remains under-integrated is a single architecture that carries *problem-conditioned*

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Table 1: System map. The residual contribution is integration and typing, not invention of the local mechanisms.

Component	Representative allies	Already developed	Role here
Problem construction	Reiter-Palmon et al. [18]	Problem representation is active and affects later solution quality.	Condition later mobility on a retained problem; no novelty claim.
Discovery search	Klahr–Dunbar; abduction [10, 19]	Hypothesis and experiment spaces interact; hypotheses arise from memory and prior experiments.	Open "hypothesis space" into semantic access, retention, and readout classes.
Generative cognition	Geneplore; Boden [6, 3]	Retrieval, synthesis, exploration, combination, and transformation.	Type these operations as transports and restructuring events.
Analogy / models	Gentner; Nersessian [7, 17]	Relational structure can move across domains; new scientific representations can be built from existing resources.	Require provenance, commutation/defect reporting, and downstream discrimination.
Insight / fixation	Knoblich et al.; Bilalić et al. [11, 2]	Insight may require representational change; familiar solutions can bias attention away from alternatives.	Separate reach from accessibility and permit audited retyping of non-forced constraints.
Semantic search/control	Mednick; Kenett; Beaty–Kenett; CSC [14, 9, 1, 13, 12]	Associative structure, remote retrieval, search, and task-sensitive semantic control matter.	Model mobility as dynamically weighted and question-conditioned.
Knowledge-structure costs / benefits	Beaty et al. [21]	Rich semantic structure can increase fluency and efficient retrieval yet also constrain originality or promote fixation.	Let prior retained structure shape attraction without assuming that stronger attraction is always better.
History-dependent thought	Honey et al. [22]	Recent content, dispositions, and task sets can persist and bias later cognition even after the initiating cue is gone.	Represent path history as finite semantic momentum that modifies future access weights.
Underdetermination	Stanford [20]	Distinct theories may be empirically equivalent relative to available evidence.	Use reader equivalence on generated hypotheses; do not confuse equivalence with identity.
Model discrimination	Box–Hill; Myung–Pitt [4, 15]	Experiments can be chosen to discriminate rival models.	Place discriminating access actions after candidate generation and before returned records.
AI hypothesis generation	Co-Scientist [8]	AI can generate, critique, rank, refine, and propose tests.	Attribute every operator stage rather than treating AI as "extra ideas" only.
Collective / distributed cognition	Hutchins; Bird [25, 26]	Cognition and scientific knowledge can be distributed across people, artifacts, roles, and coordination structures rather than residing in one individual alone.	Type a collective epistemic system operationally through shared records, readers, policies, and update channels without assuming a literal group mind.

semantic access through dynamic, history-shaped imaginative reconfiguration into candidate hypotheses and then evaluates those candidates by *reader-relative discriminability*. The missing interface includes not only where an agent can move, but why some regions become easier to revisit, why recent directions persist, and when useful organization turns into fixation.

3 Native-Source Audit: Why the Earlier Draft Failed

The adversarial cases did not merely reveal missing exceptions. They exposed places where the earlier draft had compressed Readout Universe or Readout Genesis too aggressively. Table 2 records the source error and the correction.

The general diagnosis is one of *premature role collapse*. A current constraint was collapsed with a forced distinction; existence of a path was collapsed with accessibility; a semantic domain was collapsed with root ontology; a chosen probe was collapsed with the record it later returned. The final architecture keeps these roles separate.

4 Readout-Native Foundations

4.1 Readout-not-truth and problem formation

Readout Universe represents an agent-conditioned record schematically as

$$M_A[n] = K_A\,\theta(E[n]) + \eta_{\text{sel}} + \eta_{\text{map}} + \eta_{\text{self}}, \tag{1}$$
$$M_A[n] \neq \theta(E[n]). \tag{2}$$

Let A_t be the operative grammar, ε_t a configuration, and δ_t the returned record. Define

$$r_t = A_t\varepsilon_t - \delta_t, \qquad V_t = \tfrac{1}{2} r_t^{\mathsf{T}} W_t r_t. \tag{3}$$

A problem is a residual retained as requiring explanation, repair, or decision:

$$P_t := \mathrm{Retain}_{\Pi_P}(r_t). \tag{4}$$

A question selects a contrast:

$$Q_{\Pi_Q}(P_t) = d_{Q,t}. \tag{5}$$

These are methodological typings, not claims that neural cognition literally stores Euclidean residuals.

4.2 Meaning after readout

Readout Genesis does not treat meaning as a root primitive. Define the semantic field as a domain-level readout of retained state Z_t:

$$\mathcal{S}_{A,t} := q_{\mathrm{sem},A,\square_t}(Z_t). \tag{6}$$

Thus bounded does not mean fixed:

$$\mathcal{S}_{A,t+1} \neq \mathcal{S}_{A,t} \quad \text{is admissible and expected.} \tag{7}$$

4.3 Reader equivalence

For a declared reader family $\mathcal{O}_D$, Genesis defines

$$x \sim_{\mathcal{O}_D} y \iff O_\alpha(x) = O_\alpha(y) \quad \forall\alpha. \tag{8}$$

Equivalence is local to the declared reader family and must not be promoted to absolute identity.

5 Dynamic Semantic Mobility

5.1 A provenance-typed distinction ledger

The earlier draft treated the current question's distinctions as if they were equally binding. Instead let

$$\mathcal{L}_{Q,t} = \{(d, \chi_t(d))\} \tag{9}$$

with provenance labels drawn from the Readout Universe bridge/posit discipline, schematically

$$\chi_t(d) \in \{\mathsf{FORCED}, \mathsf{DERIVED}, \mathsf{POSITED}, \mathsf{BORROWED}, \mathsf{OPEN}\}. \tag{10}$$

A forced distinction is licensed by the problem/record chain at the declared tier. A posited, borrowed, or open distinction may be useful, but its status must not be silently upgraded.

This makes insight possible without violating the non-collapse discipline. Relaxing a POSITED or BORROWED constraint can be a legitimate representational change; silently erasing a FORCED distinction while leaving the question unchanged is a defect.

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Table 2: Source audit. The corrected architecture follows the native principles rather than patching each hostile case separately.

Earlier simplification	Why it was wrong in the native systems	Final correction
Semantic field treated almost as a pre-given store	Readout Genesis places meaning *after* readout/translation; the root retains distinctions, not ready-made semantic ontology.	Define semantic field as the time-indexed domain readout $q_{\text{sem}}(Z_t)$.
Every currently "load-bearing" distinction must survive	Readout Universe requires a posit/bridge ledger before claims about what is forced; its load-bearing-discard gate does not license treating every current constraint as world-given. Genesis requires explicit loss accounting, not universal preservation.	Type distinctions by provenance. Forced distinctions cannot be silently erased; posited/borrowed/open constraints may be explicitly relaxed or retyped.
Reachability represented as a bare set	Genesis includes dynamics, ordered history, transfer properties, contraction/persistence costs, and finite state updates. Set membership loses salience and path dominance.	Add finite access weights and distinguish *reachable* from *accessible*.
Semantic topology frozen during search	Genesis treats relational topology as a readout and keeps state/history dynamic; Readout Universe permits grammar revision.	Add an auditable restructuring operator that can update reader, constraint ledger, topology, and access weights.
Access weights treated as history-free	Genesis retains ordered tape/history and transfer/persistence structure; Readout Universe translates memory/habit as resistance to update at bridge tier.	Factor access into baseline structure, task-conditioned attraction, recent-path momentum, control, and noise.
Derivative-style λ	The finite readout discipline does not require a continuum limit and already supplies a positive retained metric.	Keep λ only as a finite reader-relative diagnostic.
Record δ treated like a chosen experiment	In Readout Universe, δ is the returned record.	Separate chosen domain access action u from returned record δ.
AI modeled only as extra transport edges	The control layer is stage-wise: framing, questioning, retention, discrimination, and revision are separable.	Add an operator-attribution map across the full pipeline.

5.2 Transport with explicit defect accounting

Let T be a semantic transport. Genesis requires non-exact translations to expose lost distinctions. Write

$$\text{Def}_Q(T) := \{d \in \mathcal{L}_{Q,t} : \text{the declared readout of } d \tag{11}$$
$$\text{is lost under } T\}. \tag{12}$$

For a retained finite distinction $d \neq 0$, define a finite readout change

$$\Delta_O(d; s, T, \Pi) := \left\| O_{A,\Pi}(T(s \oplus d)) - O_{A,\Pi}(T(s)) \right\|_{G_{\text{out}}}, \tag{13}$$
$$\lambda^{RG}_{A,T,\Pi}(d; s) := \frac{\Delta_O(d; s, T, \Pi)}{\|d\|_{G_{\text{in}}}}. \tag{14}$$

Then $\lambda^{RG} = 0$ means readout collapse of that declared distinction under the current transport/reader composition; it does not imply ontological identity.

The final admissibility rule is not "preserve everything." It is:

No-silent-loss rule [Dr]. If a transport suppresses a distinction marked FORCED, the unchanged question/claim is blocked unless the loss is explicitly exposed and the question or claim scope is retyped. Loss of a POSITED, BORROWED, or OPEN constraint may be licensed as an explicit restructuring step.

5.3 Reachability is not accessibility

A bare Reach set cannot represent the Einstellung effect: an alternative may exist in the graph yet remain practically neglected. Let

$$0 \leq \kappa_{A,t,Q}(s' \mid s) \leq 1 \tag{15}$$

be a finite *access weight*. It may summarize salience, learned habit, attentional bias, control cost, or tool support; it is not asserted to be a literal neural transition probability. For a path $\pi = (s_0, \dots, s_m)$, define

$$\mu_{A,t}(\pi \mid Q) := \prod_{k=0}^{m-1} \kappa_{A,t,Q}(s_{k+1} \mid s_k). \tag{16}$$

For candidate H, define its structural accessibility

$$\text{Acc}_{A,t}(H \mid Q) := \sup_{\pi:\, \Gamma_Q(\pi)=H} \mu_{A,t}(\pi \mid Q). \tag{17}$$

Hence

$$\text{reachable}(H) = 1 \quad \not\Rightarrow \quad \text{Acc}(H \mid Q) \text{ is high}. \tag{18}$$

This supplies a structural language for expert fixation without claiming a completed cognitive process model.

5.4 History-shaped access: attraction and momentum

Equation (17) states that access is graded, but by itself it does not explain why some transitions become easier. Readout Genesis already supplies an ordered tape and finite dynamics; Readout Universe already treats memory, habit, and inertia as declared bridges rather than root identities. The semantic-domain translation should therefore be history-sensitive.

Readout-semantic attraction. The phrase *semantic gravity* is avoided as a technical term because it already names context-dependence in Legitimation Code Theory [23]. Instead define a finite, task-conditioned *readout-semantic attraction*. Let $\Phi_{Q,t}(s)$ be a declared *access potential* over the current semantic readout. Its admissible inputs may include problem relevance, structural proximity, retrieval history, salience, institutional reinforcement, switching cost, and declared defects. **Current epistemic warrant is excluded from this potential by construction.** For an oriented edge $e : s \to s'$, define

$$a_{Q,t}(e) := \left[\Phi_{Q,t}(s) - \Phi_{Q,t}(s')\right]_+. \tag{19}$$

A positive value means only that the declared access landscape favors the destination relative to the source. It is not a physical force, evidence, or a truth certificate. Repetition, training, prestige, institutional routines, or a dominant paradigm may all increase attraction even when the attracted hypothesis is wrong.

Semantic momentum. Attraction concerns the current landscape; momentum concerns the path just traversed. Let e_t be the oriented semantic transport taken at step t. For each edge e, define a finite history score

$$m_{t+1}(e) = \rho\, m_t(e) + \mathbf{1}[e_t = e], \qquad 0 \leq \rho < 1. \tag{20}$$

This is a minimal discrete momentum proxy: recent repeated traversal makes the same semantic direction easier to re-enter for a limited history window. It is deliberately weaker than physical momentum. Psychological research independently describes recent thoughts, task sets, emotions, and dispositions as lingering and shaping subsequent cognition [22].

History-shaped transition law. Let $c_{A,t,Q}(e)$ be declared control/switching cost and $\xi_t(e)$ bounded noise or

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unmodeled influence. A finite access law can be written

$$\kappa_{t+1}(e \mid Q) \propto \kappa_t^{(0)}(e \mid Q) \times \exp\big(\beta a_{Q,t}(e) + \mu m_t(e) - \nu c_{A,t,Q}(e) + \xi_t(e)\big), \tag{21}$$

with normalization over currently admissible outgoing edges. Equation (21) is [Open]: it is a typed finite hypothesis about accessibility, not an established cognitive law.

Role-separation law [Dr].

$$\boxed{a_{Q,t} \neq W_t(H), \qquad m_t \neq W_t(H), \qquad \mathrm{Acc}_t(H) \neq W_t(H)}$$

Attraction, momentum, and accessibility describe how a candidate becomes easy to reach; warrant describes why a candidate is epistemically supportable from declared records and checks. A strong paradigm can therefore dominate search while remaining wrong.

Proposition P1: attraction–fixation inversion [Open]. Attraction need not be monotonically beneficial. For at least some task classes, increasing attraction should initially concentrate access on useful local structure, but beyond a task-relative regime, and especially when switching control is weak, it should suppress alternative readout classes. The observable signature is increasing access concentration together with reduced escape to live alternatives, not a universal inverted-U in every task.

Proposition P2: momentum carryover [Open]. Holding the current semantic readout and question approximately fixed, recent traversal history should predict subsequent transport choices beyond a memoryless access kernel. If no out-of-sample effect of path history remains after current-state variables are controlled, m_t should be removed as a separate object.

5.5 Historical structural analogy, not evidential basis

An early Buddhist formulation in MN 19 states that what is repeatedly thought and pondered becomes the inclination of the mind [24]. The structural resemblance to Eq. (20) is noteworthy, but it is not used as empirical validation. Its moral classification operates under a different reader/policy from epistemic warrant. In particular,

$$\text{wholesome} \not\equiv \text{true}, \qquad \text{unwholesome} \not\equiv \text{false}. \tag{22}$$

The admissible bridge is only the history-dependence claim: repeated trajectories can alter future inclination.

5.6 Restructuring and insight

Insight can change not merely the node reached but the geometry of access itself. Define the current semantic-domain configuration

$$\mathfrak{D}_t^{\text{sem}} := \big(q_{\text{sem,t}}, \mathcal{L}_{Q,t}, \mathcal{G}_t, \kappa_t, \Pi_{Q,t}\big), \tag{23}$$

where $\mathcal{G}_t$ is a current transport topology. An auditable restructuring event is

$$\mathcal{B}_t : \mathfrak{D}_{t^-}^{\text{sem}} \longrightarrow \mathfrak{D}_{t^+}^{\text{sem}}. \tag{24}$$

It may retype a constraint, decompose a chunk, add/remove a transport edge, change access weights, or alter the question policy. The operation is not “magic novelty”; its changes must be recorded on the ordered tape and any lost distinctions exposed.

A candidate signature of insight is therefore

$$\mathrm{Acc}_{t^+}(H^* \mid Q) \gg \mathrm{Acc}_{t^-}(H^* \mid Q) \tag{25}$$

following a logged reconfiguration. Whether human insight episodes admit a stable operationalization of this signature remains [Open].

Proposition P3: productive escape [Open]. When a dominant access basin remains attractive while the inquiry retains a load-bearing residual, productive escape is predicted to involve an auditable restructuring $\mathcal{B}_t$ that changes topology, weights, or non-forced constraints and makes at least one previously negligible, readout-distinguishable candidate materially more accessible without silently discarding a FORCED distinction. A restructuring that merely changes wording, or that lowers the residual only by deleting the question-defining distinction, does not count as productive escape.

Hypothesis-generating imagination [definition/Dr]. Imagination is the bounded capacity to perform dynamically weighted semantic transports and audited restructurings over a problem-conditioned semantic readout. Its accessibility is history-shaped by retained structure, semantic attraction, recent-path momentum, and active control; these jointly change the speed and direction with which candidate relations become accessible without silently erasing distinctions that remain forced by the inquiry.

The original thesis therefore survives in a more precise form:

$$\boxed{\text{Imagination is mobility inside finitude.}} \tag{26}$$

Mobility now includes weighted traversal, history-shaped attraction, recent-path momentum, active switching, and explicit restructuring, not mere graph reachability.

6 From Imagination to Hypotheses

6.1 Generation, retention, and warrant are distinct

Not every reachable configuration is retained as a hypothesis. Let Γ_Q map accessible configurations into candidate relations. The weighted candidate object can be written

$$\widetilde{\mathcal{H}}_{A,t}(Q) := \{(H, \mathrm{Acc}_{A,t}(H \mid Q)) : \ H \text{ retained by } \Gamma_Q\}. \tag{27}$$

A hypothesis is a retained candidate relation; in some cases it updates the operative grammar,

$$A_i = A_t + \Delta_{H_i} A. \tag{28}$$

Generation must not be collapsed with warrant. A candidate may be highly accessible, memorable, or accompanied by an Aha experience while remaining poorly supported. Keep a separate warrant object

$$W_t(H_i) = \mathcal{W}(\delta_{\le t}, \text{independence}, \text{defects}, \text{calibration}). \tag{29}$$

Thus

$$a_{Q,t}(H_i) \text{ high} \quad \not\Rightarrow \quad W_t(H_i) \text{ high}, \tag{30}$$
$$m_t(H_i) \text{ high} \quad \not\Rightarrow \quad W_t(H_i) \text{ high}, \tag{31}$$
$$\mathrm{Acc}(H_i) \text{ high} \quad \not\Rightarrow \quad W_t(H_i) \text{ high}. \tag{32}$$

Readout Universe further treats truth as continued tracking rather than as a status conferred by familiarity. Therefore the stronger role separation is

$$\boxed{\text{Attraction} \neq \text{Warrant}, \qquad \text{Momentum} \neq \text{Truth}.} \tag{33}$$

This protects the architecture against false insight, fluent but unsupported AI hypotheses, and institutionally entrenched but poorly tracking paradigms.

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6.2 Readout-discriminable hypothesis classes

Let $\mathcal{U}_{D,t}$ be the domain's currently accessible actions: experiments in an empirical domain, observational queries, simulations, or proof-check actions in a formal domain. For finite horizon L, define

$$H_i \sim_{\mathcal{O}_D,\mathcal{U}_{D,t},L} H_j \quad\Longleftrightarrow\quad \widehat{\delta}_i(u) = \widehat{\delta}_j(u) \tag{34}$$

$$\forall u \in \mathcal{U}_{D,t} \text{ over the declared horizon.} \tag{35}$$

Then

$$\mathcal{H}^{\text{disc}}_{A,t}(Q) := \frac{\{H : (H, \mathsf{Acc}(H \mid Q)) \in \widetilde{\mathcal{H}}_{A,t}(Q)\}}{\sim_{\mathcal{O}_D,\mathcal{U}_{D,t},L}}. \tag{36}$$

Define

$$D_H(t) := \left|\mathcal{H}^{\text{disc}}_{A,t}(Q)\right|. \tag{37}$$

The quotient is not claimed as new mathematics. The contribution is to apply reader equivalence to the output of dynamic imaginative generation.

Current equivalence is access-relative. New instruments, formal checkers, or actions can enlarge $\mathcal{U}_{D,t}$ so that two candidates equivalent today become distinguishable tomorrow. Therefore

$$H_i \sim_{\mathcal{U}_{D,t}} H_j \quad\not\Rightarrow\quad H_i \sim_{\mathcal{U}_{D,t+1}} H_j. \tag{38}$$

7 Bounded Knower: Epistemic Standing Is Relational

Readout Universe begins from a bounded knower: no agent reads the ground directly, and a record remains indexed to a reader, policy, and finite access relation [28]. The consequence is stronger than the claim that people are sometimes mistaken. *Epistemic standing itself is relational.*

Let

$$\mathsf{K}_{A,t}(Q, D; \mathcal{O}_D, \Pi_t, \mathcal{R}_t) \tag{39}$$

name the epistemic standing of agent A for question Q in domain D, under declared reader $\mathcal{O}_D$, policy Π_t, and available record set $\mathcal{R}_t$. This is a typed status, not a scalar essence. In general,

$$\mathsf{K}_{A,t}(Q_1, D_1; \cdots) \not\Rightarrow \mathsf{K}_{A,t}(Q_2, D_2; \cdots), \tag{40}$$

$$\mathsf{K}_{A,t}(Q, D; \cdots) \not\Rightarrow \mathsf{K}_{A,t+1}(Q, D; \cdots). \tag{41}$$

The first non-transfer rule blocks domain-general authority from being inferred from local expertise. The second blocks time-invariant authority when readers, evidence, language, or problem structure have changed.

Anti-essentialist knower corollary [Dr]. A durable social identity such as "expert", "scholar", or "intellectual" may describe a role, history, or reputation, but it does not itself constitute current epistemic warrant. A person may be a strong knower in one relation and an unresolved knower in another. Expertise can legitimately alter accessibility, priors, or search efficiency; it may not bypass the current readout, warrant, and revision gates.

This does not deny durable expertise. It denies the inference

$$\begin{gathered}\text{social/epistemic label}\\ \not\Rightarrow\\ \text{unconditional standing across } Q, D, t\end{gathered} \tag{42}$$

Epistemic humility therefore follows as a governance requirement rather than as decorative modesty: a bounded knower must leave correction channels open because their standing can change when the relation changes.

8 Collective Epistemic Systems Without a Group Mind

The individual symbol A is insufficient for science communities, institutions, religions, firms, governments, or other systems in which records and cognitive work are distributed. Distributed-cognition and collective-epistemology traditions already treat some knowledge-producing systems as extending across people, artifacts, roles, and coordination structures [25, 26]. The present architecture can absorb this without asserting that a group has a single phenomenal mind.

Let $\mathcal{G} = \{A_1, \ldots, A_n\}$ be a declared group. Define a retained collective state

$$Z_{\mathcal{G},t} := \left(\{Z_{A_i,t}\}_{i=1}^{n}, \mathcal{T}_{\mathcal{G},t}, \mathcal{A}_{\mathcal{G},t}, \mathcal{C}_{\mathcal{G},t}\right), \tag{43}$$

where $\mathcal{T}_{\mathcal{G},t}$ is the shared ordered tape or archive, $\mathcal{A}_{\mathcal{G},t}$ the retained artifacts/records, and $\mathcal{C}_{\mathcal{G},t}$ the communication and coordination structure. The group semantic readout is

$$\mathcal{S}_{\mathcal{G},t} := q_{\text{sem},\mathcal{G},\square_{\mathcal{G},t}}(Z_{\mathcal{G},t}). \tag{44}$$

This is an operational readout of a distributed system, not a claim that every member shares the same internal state.

Institutional attraction $a^{\mathcal{G}}_{Q,t}(e)$ may arise from stable archives, curricula, prestige structures, review routines, incentives, or habitual tools that make some transitions easier to retain and transmit. Socially transmitted momentum is represented by a group-level history trace

$$m^{\mathcal{G}}_{t+1}(e) = \rho_{\mathcal{G}} m^{\mathcal{G}}_t(e) + \sigma_{\mathcal{G},t}(e), \tag{45}$$

where $\sigma_{\mathcal{G},t}(e)$ records a declared transmission/repetition event on the shared tape. The trace can persist even when individual membership changes.

Crucially, collective warrant remains separate:

$$\begin{aligned}W_{\mathcal{G},t}(H) = \mathcal{W}_{\mathcal{G}}(&\mathcal{R}_{\mathcal{G},\le t}, \text{independence}, \text{defects},\\ &\text{calibration}, \text{objection channels}).\end{aligned} \tag{46}$$

Hence

$$\boxed{a^{\mathcal{G}} \not\Rightarrow W_{\mathcal{G}}(H)\uparrow, \qquad m^{\mathcal{G}} \not\Rightarrow \text{truth}.} \tag{47}$$

A paradigm, institution, or tradition may therefore possess enormous attraction and momentum while its live hypotheses remain weakly warranted.

Collective closure [Dr/Open]. The earlier collective-imagination square is structurally closed by replacing the fiction of a single group mind with a bounded distributed epistemic system: members + shared records/artifacts + communication/selection policy + update channels. The architecture of semantic readout, attraction, momentum, hypothesis retention, reader equivalence, warrant, and revision can then be indexed by $\mathcal{G}$ exactly as it is by A. The existence and measurement of particular group-level kernels remain empirical [Open]; the typing problem is no longer left blank.

9 Discrimination, Record, and Revision

A chosen access action must be separated from the returned record. A discriminating action u^* satisfies

$$\widehat{\delta}_i(u^*) \neq \widehat{\delta}_j(u^*) \tag{48}$$

for at least two live candidates. Only after execution does the domain reader return

$$\delta^*_{t+1} = O_D(Z_{t+1}; u^*). \tag{49}$$

For each hypothesis,

$$r^*_i = \widehat{\delta}_i(u^*) - \delta^*_{t+1}. \tag{50}$$

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Revision updates the retained state, operative grammar, hypothesis set, or semantic-domain configuration:

$$(Z_{t+1}, A_{t+1}, \mathfrak{D}^{\mathrm{sem}}_{t+1}) = \mathrm{Update}(Z_t, A_t, \delta^*_{t+1}, \Pi_R). \tag{51}$$

The next cycle therefore begins from changed semantic access.

10 The Final Architecture

For readability, the architecture is expressed in three typed stages rather than one overloaded equation. First, semantic readout and audited restructuring:

$$X_{A,t}(Q) := \mathcal{B}_t(\mathcal{S}_{A,t}, \mathcal{L}_{Q,t}, \Pi_{Q,t})\,. \tag{52}$$

Second, history-shaped weighted imaginative accessibility and hypothesis retention, with $\kappa_{A,t,Q}$ generated or approximated from Eq. (21):

$$\widetilde{\mathcal{H}}_{A,t}(Q) := \Gamma_Q\big(\mathrm{WReach}_{\kappa_{A,t,Q}}[X_{A,t}(Q)]\big)\,. \tag{53}$$

Third, reader-relative discrimination:

$$\boxed{\mathcal{H}^{\mathrm{disc}}_{A,t}(Q) = \widetilde{\mathcal{H}}_{A,t}(Q)/{\sim_{\mathcal{O}_D,\mathcal{U}_{D,t},L}}} \tag{54}$$

where the quotient acts on the hypothesis component of the weighted candidate object.

For an epistemic system $E \in \{A, \mathcal{G}\}$, the same typed form applies by substituting the appropriate retained state, semantic reader, access kernel, warrant process, and update policy. This substitution is an architectural typing rule, not evidence that individual and collective cognition share a neural or phenomenological mechanism.

The full inquiry loop is

> **Record → Residual → Problem → Question → Semantic Readout → Dynamic Mobility / Restructuring → Weighted Hypotheses → Readout Classes → Access Action → Returned Record → Revision.**

The architecture does not guarantee that all serious alternatives will be generated. It instead makes boundedness inspectable: where semantic access was limited, where a path existed but had negligible access weight, whether attraction or momentum created a basin, which constraints were forced versus posited, where restructuring occurred, which distinctions were lost, which candidates remained reader-equivalent, and what action could increase separating power.

11 Human–AI Operator Attribution

AI augmentation can alter more than transport edges. Define an attribution map

$$\alpha_t(o) \in \{\mathsf{HUMAN}, \mathsf{AI}, \mathsf{JOINT}\} \tag{55}$$

for operators

$$o \in \Omega = \{q_{\mathrm{sem}}, \Pi_P, \Pi_Q, \mathcal{B}, a, m, \kappa, \Gamma_Q, \mathcal{U}_D, \Pi_R\}. \tag{56}$$

This makes provenance stage-wise. The earlier epistemic-control vector remains useful,

$$\mathbf{S}^{epi}_H = (c_P, c_Q, c_H, c_D, c_R), \tag{57}$$

but it should be interpreted through the operator map rather than through prompt authorship alone. AI may enlarge accessibility while the human retains problem, question, discrimination, and revision control; or the same human may nominally "approve" a pipeline whose substantive operators were delegated.

12 Adversarial Boundary Tests

Table 3 summarizes the hostile cases that forced the final revisions.

13 Empirical Research Program and Falsifiers

The final architecture creates stronger tests than the earlier binary-reach model.

13.1 Accessibility beyond reachability

Construct tasks in which an expert or model has the required concepts and formal path to an alternative solution. Test whether a fitted access-weight model predicts which alternatives are generated beyond binary graph reachability and knowledge breadth. If it does not, κ adds no value and should be removed.

13.2 P1: attraction–fixation inversion

Manipulate attraction while keeping the available candidate structure approximately fixed, for example through repetition, prestige cues, interface defaults, or learned routines that change access but do not add evidence. The proposition predicts that sufficiently concentrated attraction, under weak switching control, suppresses access to live alternatives. If increasing attraction never produces measurable alternative suppression once current warrant and knowledge are controlled, P1 weakens.

13.3 P2: momentum carryover

Prime participants or models with repeated transport in one semantic direction, remove the cue, and measure whether later search remains biased toward the same relation family after current-state variables are controlled. Fit Eq. (20) against a memoryless kernel. If recent path history adds no out-of-sample predictive value, P2 fails and semantic momentum should be removed as a separate object.

13.4 Restructuring signature

In insight tasks, pre-register candidate constraint retypings or representational changes and test whether successful insight is associated with a measurable rise in $\mathrm{Acc}(H^* \mid Q)$. If no stable operationalization distinguishes restructuring from ordinary traversal, $\mathcal{B}_t$ should remain a descriptive bridge rather than a cognitive mechanism.

13.5 P3: productive escape

Construct persistent-residual tasks with a deliberately dominant access basin. A successful escape should show a logged $\mathcal{B}_t$ event followed by increased accessibility of at least one new readout-distinguishable candidate while retaining all FORCED distinctions. If successful escapes routinely occur without any detectable reconfiguration, or if the criterion rewards silent deletion of task-defining distinctions, P3 fails.

13.6 Provenance-sensitive loss

Compare conditions where participants are encouraged to relax POSITED/BORROWED constraints versus constraints licensed as FORCED by the task. The architecture predicts benefit from the former but harm or explicit task-redefinition from silent loss of the latter. Failure of this asymmetry would weaken the provenance ledger as an explanatory component.

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Table 3: Adversarial boundary tests and the object that now carries each failure mode.

Case	Why the old draft failed	Final handling
Einstellung / expertise fixation [2]	Alternative can be structurally reachable yet attentionally suppressed.	κ and Acc separate reach from accessibility.
Dominant paradigm / institutional basin	A familiar framework can remain highly accessible because repetition, prestige, tools, and routines reinforce it even when warrant is weak.	Attraction and momentum are explicitly separated from $W_t(H)$; persistence is not evidence.
Insight / constraint relaxation [11]	"Preserve all load-bearing distinctions" excluded legitimate representational change.	Provenance ledger plus $\mathcal{B}_t$ permits explicit relaxation/retyping of non-forced constraints.
False insight / fluent AI candidate	Accessibility or subjective Aha can be mistaken for epistemic support.	Generation/retention are separated from warrant $W_t(H)$.
Intellectual / expert label	Local expertise or reputation can be mistaken for domain-general, time-invariant authority.	Bounded-knower standing is indexed by Q, D, t, $\mathcal{O}$, Π, $\mathcal{R}$; labels do not transfer warrant automatically.
Collective science / institution [25, 26]	A single-agent model misses distributed records, artifacts, coordination, and socially persistent path dependence.	Replace a group mind with a bounded collective epistemic system $\mathcal{G}$ carrying shared tape, reader, attraction, momentum, warrant, and revision.
Formal mathematics	"Experiment" and "world record" are too physically narrow.	$u \in \mathcal{U}_D$ is a domain-typed access action; formal proof/checker readouts are permitted.
Scientific underdetermination	Semantically different hypotheses may remain evidentially indistinguishable.	Reader-relative quotient defines $\mathcal{H}^{disc}$, without promoting equivalence to identity.
Art / fiction / dreams	No necessary hypothesis-discrimination objective.	Explicit scope guard: the theory concerns hypothesis-generating imagination only.

13.7 Finite distinction-retention diagnostic

Test whether λ^{RG} predicts preservation of discriminating consequences during cross-domain transformations after controlling for semantic similarity and general task performance. Instability across reasonable finite representations, or no added predictive value, is a falsifier of the diagnostic.

13.8 Epistemic novelty

Compare raw idea count N_H, semantic novelty, and D_H. The claim is not that larger D_H is universally better, but that it captures a dimension of hypothesis-generation capacity relevant to discriminating inquiry. If D_H adds no value for expert evaluation, experiment selection, or later discovery outcomes, the epistemic-novelty proposal weakens.

13.9 Bounded-knower transfer test

Hold a person's prestige or social label constant while changing domain, question, evidence, or reader. Test whether independent judges correctly reduce standing when the relevant record/bridge disappears. The architectural prediction is not that expertise has no value, but that label alone has no direct warrant path. If domain-general labels reliably predict current warrant after relevant evidence and expertise variables are removed, the corollary requires revision.

13.10 Collective epistemic-system test

Compare individual-only models against models that include shared artifacts, communication topology, institutional attraction, and group history. The collective extension earns empirical standing only if the added group-level objects improve prediction of which hypotheses are generated, retained, challenged, or revised. Failure to outperform member-level aggregation would leave the collective typing useful descriptively but unsupported as an explanatory layer.

13.11 Human–AI attribution

Factorially manipulate who controls $\Pi_P, \Pi_Q, \mathcal{B}, a, m, \Gamma_Q, \mathcal{U}_D, \Pi_R$. Measure performance, D_H, calibration, revision behavior, and provenance accuracy. If operator-level control has no relationship to agency or correction quality, the stage-wise sovereignty model requires revision.

14 Boundaries and Non-Claims

The framework does **not** claim that:

- all imagination is semantic mobility; the target is hypothesis-generating imagination;
- the brain literally implements the notation used here;
- κ, λ^{RG}, or $\mathcal{B}_t$ are established psychometric or neuroscientific variables;
- readout-semantic attraction is a literal physical force, or semantic momentum is literal mechanical momentum;
- stronger attraction or momentum is always beneficial; both may produce fixation;
- attraction, momentum, accessibility, prestige, consensus, or institutional persistence constitute epistemic warrant;
- a durable label such as expert, scholar, or intellectual confers domain-general or time-invariant standing;
- a collective epistemic system must possess a single group mind or shared phenomenal state;
- Buddhist moral categories are interchangeable with epistemic truth/warrant categories;
- every useful insight must be sudden, or every restructuring must involve impasse;
- larger D_H is always desirable;
- reader equivalence is truth or absolute identity;
- all current constraints should be preserved;
- a POSITED/BORROWED constraint may be dropped without an explicit ledger update;
- decisive access actions or model discrimination are novel inventions of this paper;
- AI augmentation automatically preserves human epistemic agency;
- the separate hypothesis that knowledge organization determines the expected time and direction to a new usable hypothesis is established here; that question is reserved for a subsequent paper;
- this manuscript has passed independent Glosa checking.

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15 Discussion

The final architecture is more faithful to its own sources than the earlier draft. The original error was not insufficient mathematical complexity; it was insufficient role separation. Readout Universe already required translation, policy location, bridge/posit audit, identifiability, load-bearing-discard analysis, decisive-record prescription, falsifier, and not-checked ledger [28]. Readout Genesis already required meaning-after-readout, reader-relative equivalence, explicit translation loss, finite state, ordered tape, transport, and dynamics [29]. The earlier draft collapsed several of those distinctions while attempting to simplify them for a theory of imagination.

This audit changes the interpretation of semantic mobility. Mobility is not the freedom to preserve every current semantic distinction while moving through a static space. Nor is it unconstrained novelty. It is a bounded, reader-conditioned process whose state, topology, access weights, and constraint classifications can change. The crucial discipline is not universal preservation but *auditable change*: which difference was retained, what licensed its status, which operation altered it, what defect was introduced, and what later readout can discriminate the resulting candidates.

The present extension adds a second correction: access is history-shaped, but history is not evidence. Semantic-memory research supports the idea that structured search can enable efficient retrieval and also produce fixation [21, 9], while psychological-momentum research shows that recent content can persist and bias subsequent cognition [22]. In Readout terms these motivate two domain-level access diagnostics, not truth variables. Attraction can be produced by repeated use, prestige, interface design, or institutional routine; momentum can be produced by repeated traversal. Neither enters current warrant by definition.

A third correction concerns the knower itself. If every readout is bounded and policy-relative, epistemic standing cannot coherently be treated as a permanent substance attached to a person. The relevant unit is a relation indexed by question, domain, time, reader, policy, and records. This dissolves no social role: experts and intellectuals can be durable institutions or identities. What is denied is the automatic transfer of epistemic warrant from the label to a new relation. Epistemic humility is therefore built into the architecture as continued revisability, not added as a moral ornament.

A fourth correction is that the knower need not be an isolated individual. Scientific and institutional cognition can be distributed across people and artifacts [25, 26]. The collective extension treats a group as a bounded epistemic system only to the extent that shared records, communication structures, selection policies, and update channels are declared. This avoids both individualist blindness and an unearned metaphysics of a group mind.

This also changes the unconceived-alternatives problem. No bounded architecture can guarantee generation of every serious alternative. The stronger and more realistic target is *auditable boundedness*: distinguish absence of a path from negligible access weight; distinguish a forced boundary from a reader-imposed constraint; distinguish new wording from a new readout class; distinguish a compelling candidate from a warranted one; and identify which new access action could increase separating power.

16 Conclusion

For hypothesis-generating inquiry, imagination can be modeled without invoking actual infinity or a mysterious leap. It is *dynamic semantic mobility inside finite access*. Its structure contains separable roles: semantic readout, provenance-audited distinctions, weighted access, history-shaped attraction, recent-path momentum, active control, and explicit restructuring. These operations generate candidates whose epistemic distinctness is assessed by reader-relative equivalence and whose support is assessed separately by warrant.

The non-collapse rule is now explicit:

$$\boxed{\text{Attraction} \neq \text{Accessibility} \neq \text{Warrant} \neq \text{Truth}} \tag{58}$$

with semantic momentum occupying a history/access role rather than an evidential one. A dominant paradigm can therefore be cognitively powerful and epistemically weak at the same time.

The bounded-knower corollary extends the same discipline to persons: epistemic standing is not a permanent substance attached to an identity. It is a time- and relation-indexed status. A person can be a strong knower for one question and an unresolved knower for another, and durable expertise does not eliminate the requirement for present records, bridge adequacy, and revision. Collective inquiry is handled by the same architecture at a larger unit: a bounded distributed epistemic system composed of members, shared records and artifacts, communication/selection structures, and update channels, without requiring a literal group mind.

The final sequence is:

Problem → Question → Semantic Readout → Weighted Mobility / Attraction / Momentum / Restructuring → Candidate Hypotheses → Readout Classes → Warrant → Access Action → Returned Record → Revision.

Three behavioral propositions remain load-bearing and falsifiable in this manuscript: attraction can invert into fixation; recent semantic paths can carry momentum into later search; and productive escape can occur through audited restructuring when a dominant basin fails to repair a retained residual. The separate hypothesis that knowledge organization determines the expected time and direction to a new usable hypothesis is intentionally deferred to a subsequent paper rather than being smuggled into the present claim set.

The empirical burden remains exposed. The next deciding records concern graded accessibility beyond reachability, P1–P3, provenance-sensitive restructuring, finite distinction-retention, the added value of D_H, relational epistemic standing, collective-system prediction beyond member aggregation, and operator-level human–AI control. Until those tests are run, the architecture remains a structured, falsifiable proposal rather than an established cognitive law.

Acknowledgment and Process Note

This manuscript was developed under the Glosa "Rigour Without Infrastructure" workflow. The human author originated the problem and selected the manuscript direction. AI assisted with literature retrieval, formalization, source auditing, adversarial stress testing, drafting, and typesetting. The final revision preserves the previous anchor while making four additional non-collapse corrections: (i) attraction and momentum are separated from warrant/truth, (ii) epistemic standing is indexed relationally rather than attached to a permanent knower identity, (iii) collective inquiry is typed

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as a distributed epistemic system rather than a group mind, and (iv) the organized-knowledge hitting-time hypothesis is deliberately removed for development as a separate paper. The manuscript remains K0 because the same maker route cannot certify its own output. Readout-not-truth applies to the manuscript itself.

References

[1] Beaty, R. E., & Kenett, Y. N. (2023). Associative thinking at the core of creativity. *Trends in Cognitive Sciences, 27*(7), 671–683. doi:10.1016/j.tics.2023.04.004.

[2] Bilalić, M., McLeod, P., & Gobet, F. (2010). The mechanism of the Einstellung (set) effect: A pervasive source of cognitive bias. *Current Directions in Psychological Science, 19*(2), 111–115. doi:10.1177/0963721410363571.

[3] Boden, M. A. (2004). *The Creative Mind: Myths and Mechanisms* (2nd ed.). Routledge.

[4] Box, G. E. P., & Hill, W. J. (1967). Discrimination among mechanistic models. *Technometrics, 9*(1), 57–71. doi:10.1080/00401706.1967.10490441.

[5] Fauconnier, G., & Turner, M. (1998). Conceptual integration networks. *Cognitive Science, 22*(2), 133–187. doi:10.1207/s15516709cog2202_1.

[6] Finke, R. A., Ward, T. B., & Smith, S. M. (1992). *Creative Cognition: Theory, Research, and Applications*. MIT Press. doi:10.7551/mitpress/7722.001.0001.

[7] Gentner, D. (1983). Structure-mapping: A theoretical framework for analogy. *Cognitive Science, 7*(2), 155–170. doi:10.1207/s15516709cog0702_3.

[8] Gottweis, J., Weng, W.-H., Daryin, A., et al. (2026). Accelerating scientific discovery with Co-Scientist. *Nature, 655*, 487–496. doi:10.1038/s41586-026-10644-y.

[9] Kenett, Y. N., Anaki, D., & Faust, M. (2014). Investigating the structure of semantic networks in low and high creative persons. *Frontiers in Human Neuroscience, 8*, 407. doi:10.3389/fnhum.2014.00407.

[10] Klahr, D., & Dunbar, K. (1988). Dual space search during scientific reasoning. *Cognitive Science, 12*(1), 1–48. doi:10.1207/s15516709cog1201_1.

[11] Knoblich, G., Ohlsson, S., Haider, H., & Rhenius, D. (1999). Constraint relaxation and chunk decomposition in insight problem solving. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 25*(6), 1534–1555. doi:10.1037/0278-7393.25.6.1534.

[12] Krieger-Redwood, K., Steward, A., Gao, Z., Wang, X., Halai, A., Smallwood, J., & Jefferies, E. (2023). Creativity in verbal associations is linked to semantic control. *Cerebral Cortex, 33*(9), 5135–5147. doi:10.1093/cercor/bhac405.

[13] Lambon Ralph, M. A., Jefferies, E., Patterson, K., & Rogers, T. T. (2017). The neural and computational bases of semantic cognition. *Nature Reviews Neuroscience, 18*, 42–55. doi:10.1038/nrn.2016.150.

[14] Mednick, S. A. (1962). The associative basis of the creative process. *Psychological Review, 69*(3), 220–232. doi:10.1037/h0048850.

[15] Myung, J. I., & Pitt, M. A. (2009). Optimal experimental design for model discrimination. *Psychological Review, 116*(3), 499–518. doi:10.1037/a0016104.

[16] Nature Methods Editors. (2026). What's your hypothesis? *Nature Methods, 23*, 865–866. doi:10.1038/s41592-026-03104-y.

[17] Nersessian, N. J. (2008). *Creating Scientific Concepts*. MIT Press.

[18] Reiter-Palmon, R., Mumford, M. D., O'Connor Boes, J., & Runco, M. A. (1997). Problem construction and creativity: The role of ability, cue consistency, and active processing. *Creativity Research Journal, 10*(1), 9–23. doi:10.1207/s15326934crj1001_2.

[19] Schickore, J. (2022). Scientific discovery. In *The Stanford Encyclopedia of Philosophy*. https://plato.stanford.edu/entries/scientific-discovery/.

[20] Stanford, K. (2023). Underdetermination of scientific theory. In *The Stanford Encyclopedia of Philosophy*. https://plato.stanford.edu/entries/scientific-underdetermination/.

[21] Beaty, R. E., Kenett, Y. N., Hass, R. W., & Schacter, D. L. (2023). Semantic memory and creativity: The costs and benefits of semantic memory structure in generating original ideas. *Thinking & Reasoning, 29*(2), 305–339. doi:10.1080/13546783.2022.2076742.

[22] Honey, C. J., Mahabal, A., & Bellana, B. (2023). Psychological momentum. *Current Directions in Psychological Science, 32*(4), 284–292. doi:10.1177/09637214221143053.

[23] Maton, K. (2014). *Knowledge and Knowers: Towards a Realist Sociology of Education*. Routledge. [Terminological note: "semantic gravity" in Legitimation Code Theory denotes context-dependence, not the attraction construct proposed here.]

[24] *Majjhima Nikāya 19: Dvedhāvitakka Sutta (Two Kinds of Thought)*. Translation by Bhikkhu Bodhi, as hosted by SuttaCentral. Used here only as a historical structural analogy, not as empirical validation.

[25] Hutchins, E. (1995). *Cognition in the Wild*. MIT Press.

[26] Bird, A. (2014). When is there a group that knows? Distributed cognition, scientific knowledge, and the social epistemic subject. In J. Lackey (Ed.), *Essays in Collective Epistemology* (pp. 42–63). Oxford University Press. doi:10.1093/acprof:oso/9780199665792.003.0003.

[27] Lahtee, Y. (2026). *The Readout Condition*. Zenodo. doi:10.5281/zenodo.22301318.

[28] Lahtee, Y. (2026). *Readout Universe: A Philosophy and Logic for Grounding Claims*. GitHub repository. https://github.com/morrocwi/readout_universe.

[29] Lahtee, Y. (2026). *Readout Genesis Universal Technical Whitepaper*, v1.2.0. GitHub repository. https://github.com/morrocwi/readout_genesis.

[30] Lahtee, Y. (2026). *glosa: Rigour Without Infrastructure*. GitHub repository. https://github.com/morrocwi/glosa.

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Knowledge Topology and the First Passage to Usable Hypotheses: A Readout Theory of Discovery Time and Direction

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What the chapter states. "The main hypothesis is deliberately relational rather than monotonic: under content-matched conditions, changing knowledge organization should change the distribution of discovery time and direction; the same organization may accelerate a useful route in one problem and create fixation in another."

Status. "GLOSA K0 working paper. [. . .] No independent checker has yet reviewed the literature map, mathematical bridges, or experimental design."

Falsifier(s). "H1 falsifier. Across well-powered content-matched tasks, organization metrics or manipulations add no out-of-sample predictive value for first-passage time after knowledge amount and generic retrieval/control are modeled."

Read before. *From Problem to Hypothesis: Dynamic Semantic Mobility, Bounded Knowers, and the Readout-Discriminable Hypothesis Space*.

Feeds into. *Before Evidence Can Decide: Candidate-Set Formation, Discovery Routing, and Unconceived Alternatives*.

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KNOWLEDGE TOPOLOGY AND HYPOTHESIS ACCESSIBILITY GLOSA K0 WORKING PAPER

Knowledge Topology and the First Passage to Usable Hypotheses

A Readout Theory of Discovery Time and Direction

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4 September 2026

Status. GLOSA K0 working paper. This manuscript develops the hypothesis explicitly reserved from the companion bounded-knower manuscript: *knowledge organization conditions not only what can be imagined, but the expected time and direction by which a new usable hypothesis becomes accessible*. The present literature review is targeted and system-oriented, not a preregistered systematic review. The paper makes no priority claim for semantic networks, expertise, first-passage mathematics, hypothesis-generation timing, or knowledge graphs. Its residual contribution is the proposed interface between content-matched knowledge organization and first passage to a readout-usable hypothesis. No independent checker has yet reviewed the literature map, mathematical bridges, or experimental design. Claim tiers are preserved: [definition], [Dr], [Open], and [finite_diagnostic].

Abstract

Research on semantic memory, expertise, scientific reasoning, creative search, and graph-structured AI converges on a common idea: the organization of prior knowledge matters. Semantic networks with different connectivity and path structure correlate with divergent and convergent creativity; experts organize domain knowledge differently from novices and often generate accurate diagnostic hypotheses earlier; search models show that network topology changes retrieval trajectories; and graph-structured scientific contexts can improve machine ideation. Separately, temporal studies show that idea quality and hypothesis generation vary over time. What remains under-integrated is a direct theory of how *knowledge organization*, holding accessible content approximately fixed, changes the first-passage time and first-hit direction to a hypothesis that is usable for inquiry rather than merely novel. This paper proposes such an interface. A bounded knower's knowledge organization is represented as a time-indexed, reader-relative graph over accessible distinctions, with weighted relations and provenance. A usable hypothesis is defined narrowly as a retained candidate that occupies a live readout-distinguishable class, meets a declared warrant floor for further inquiry, and admits at least one accessible discriminating action against a rival. The central quantities are the first-passage time to this usable set and the distribution over which usable hypothesis class is reached first. The main hypothesis is deliberately relational rather than monotonic: under content-matched conditions, changing knowledge organization should change the distribution of discovery time and direction; the same organization may accelerate a useful route in one problem and create fixation in another. Human, expert-novice, and AI experiments are proposed to separate content from access organization and to test whether topology adds predictive value beyond knowledge amount, intelligence, semantic distance, and generic retrieval ability.

Keywords: knowledge organization; semantic networks; scientific hypothesis generation; first-passage time; expertise; memory search; creativity; knowledge graphs; scientific discovery; Readout Universe; Readout Genesis

1 Introduction

A person can know many relevant facts and still fail to reach a useful hypothesis. Conversely, another person with a comparable factual inventory can connect the same materials rapidly into a discriminating explanation. This difference is often described loosely as expertise, creativity, intuition, or "having an organized mind." Each label captures part of the phenomenon, but none by itself specifies the object that changes, the time course of the change, or the criterion by which a generated idea becomes scientifically useful.

Several literatures already constrain the answer. Semantic-memory research shows that network structure affects creative retrieval and problem solving [1, 2, 3]. Education and expertise can alter the connectivity, path length, and modular structure of knowledge representations [4, 5]. Medical reasoning studies show that high-domain-knowledge clinicians connect cues through richer relations, generate accurate hypotheses earlier, and operate over smaller sets of closely related diagnostic alternatives [12, 13, 14]. Memory-search models demonstrate that similar retrieval regularities can emerge from search over structured semantic representations, and critical reviews emphasize that representation and retrieval process must not be collapsed [7, 8, 6]. Human hypothesis generation is also history-sensitive: amortized inference experiments show that the structure of a preceding related query can bias the next inference [9], while subjective time pressure can reduce hypothesis generation and alter action prioritization [10]. Temporal creativity research further shows that response order and latency matter, but later is not automatically better and temporal effects are moderated by executive control [11, 2]. In real scientific hypothesis generation, tool support can shorten time to a hypothesis while decreasing some quality ratings, making speed and epistemic value separable outcomes [16, 17].

The residual problem is therefore not whether organization matters. It is more precise:

Central question. If two bounded knowers have access to approximately the same atomic content and evidence, but that content is organized through different relation structures and access pathways, does the organization change (i) the expected time to the first hypothesis usable for inquiry and (ii) which usable hypothesis family is reached first?

The proposal extends the companion architecture of hypothesis-generating imagination [24]. That paper sepa-

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rated reachability from accessibility and distinguished attraction, momentum, warrant, and truth. The present paper isolates one reserved claim and makes it experimentally sharper: *knowledge organization is a candidate determinant of first-passage dynamics in hypothesis space.*

The claim is intentionally weaker than "better organized knowledge makes people more creative." First, "better" is problem-relative. Dense or familiar structure can accelerate retrieval while suppressing originality or alternatives [2]. Second, relations can facilitate rapid expert reasoning in familiar conditions yet impose costs when the situation mismatches existing schemas [15]. Third, reaching a hypothesis quickly does not establish that it is warranted. The target must therefore combine time, direction, discriminability, and warrant without collapsing them.

2 Literature as a System of Partial Answers

2.1 Structure is not inventory

A recurring finding is that semantic-memory organization contributes information beyond raw concept count. Highly creative individuals have been reported to exhibit more flexible and interconnected semantic networks [1]. Convergent thinkers and insight solvers show shorter path distances and more interconnected structures [3]. Domain-specific expertise is likewise associated with more efficiently connected knowledge networks [5]. These findings make a simple inventory model implausible: two agents can share many concepts while differing substantially in how those concepts are related and retrieved.

However, the literature also blocks a simple monotonic story. Rich semantic neighborhoods increase fluency but can reduce originality, suggesting interference and fixation costs [2]. Expertise itself can reverse when established schemas mismatch the current environment [15]. Organization is therefore not a scalar virtue. It is a task-conditioned routing structure.

2.2 Time is already measurable, but the target is wrong

Temporal work in divergent thinking typically measures response latency, serial order, fluency, or originality. Later responses often require more time and can be more original, but executive ability changes these trajectories [11]. Clinical hypothesis-generation studies go closer to the present target: hypothesis timing can be recorded in two-hour research sessions, and visualization support can reduce average time per hypothesis while not improving, and in some analyses worsening, feasibility or combined quality ratings [16]. Case analyses further suggest that hypothesis quality has a nontrivial relationship with generation time rather than a simple faster-is-better law [17].

These results motivate a different endpoint. We should not ask only how fast a sentence labeled "hypothesis" appears. We should ask how long it takes before the inquiry first reaches a candidate that can do epistemic work.

2.3 The closest gap

Across the literature examined here, we did not identify a common framework that simultaneously: (1) holds accessible knowledge content approximately fixed; (2) varies or measures organization; (3) records the time-resolved search trajectory; (4) groups candidate hypotheses by readout-discriminability; and (5) asks which usable class is reached first. Existing work supplies each local ingredient, but not this joint object.

3 Readout-Native Definitions

3.1 Knowledge organization is a domain readout

Following Readout Universe / Readout Genesis, semantic organization is not treated as a root ontology. It is a bounded, question-conditioned representation available to a reader at a time. Let

$$\mathcal{G}^K_{A,t}(Q) = \big(V_{A,t}, E_{A,t}, \omega_{A,t,Q}, \chi_{A,t}\big), \tag{1}$$

where V is the currently accessible distinction/concept inventory, E is a declared relation set, ω contains finite access weights, and χ records relation provenance or type. The graph is a modeling readout of organization, not a claim that the brain literally stores a graph.

The phrase *same knowledge* must also be operational rather than metaphysical. A content-matched comparison holds constant a declared atomic inventory V, the presented proposition/evidence set $\mathcal{R}$, and task information, while varying organization through grouping, indexing, relational display, learned sequencing, or access structure. Relations themselves can be substantive knowledge; therefore every experiment must state exactly what is held fixed and what is manipulated.

3.2 From access to a usable hypothesis

The companion paper defines dynamically weighted accessibility and a readout-discriminable hypothesis space [24]. Let $\widetilde{\mathcal{H}}_{A,t}(Q)$ be the retained candidate set and let $\mathcal{H}^{\mathrm{disc}}_{A,t}(Q)$ collapse candidates that make the same predicted readouts under the declared accessible action family and horizon.

A candidate need not be true to be useful for inquiry. Define a *usable-for-inquiry* target class by three gates:

$$\begin{aligned} \mathcal{U}_{A,t}(Q) = \{[H] \in \mathcal{H}^{\mathrm{disc}}_{A,t}(Q) : W_t(H) \geq w_0, \qquad & (2)\\ \exists u \in \mathcal{U}_{D,t} \text{ with } \Delta_u(H,H') > 0 \\ \text{for at least one live rival } H'\}, \qquad & (3)\end{aligned}$$

where $W_t(H)$ is current warrant sufficient to justify further inquiry, w_0 is a declared floor, and Δ_u indicates that an accessible action u makes at least one rival pair predictively separable.

This definition is deliberately procedural:

$$\boxed{\text{usable} \neq \text{true.}} \tag{4}$$

A hypothesis can be worth testing and later fail. Conversely, an idea can be original yet unusable because it makes no distinct prediction, lacks minimal grounding, or cannot be connected to any accessible discriminating action.

4 First Passage and Discovery Direction

4.1 First-passage time

Let $S_0, S_1, \ldots$ denote the finite sequence of semantic-access states produced while working on question Q. Define the first passage to the usable set:

$$\tau_{\mathcal{U}} = \inf\{n \geq 0 : \Gamma_Q(S_n) \in \mathcal{U}_{A,n}(Q)\}. \tag{5}$$

For a time-stamped human task, n may be replaced by elapsed seconds; for an AI system it may denote generation/retrieval steps or wall-clock time. The expected first-passage time is

$$T_{\mathcal{U}}(\mathcal{G}^K, Q) = \mathbb{E}[\tau_{\mathcal{U}} \mid \mathcal{G}^K, Q]. \tag{6}$$

First-passage mathematics is mature in network science [18]; the present paper borrows the diagnostic, not an ontological claim that thought is a Markov particle.

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Table 1: System map of the nearest literatures. The final column identifies the remaining task rather than a novelty claim.

Literature	Representative work	What is already established	Residual role here
Semantic-network structure	Kenett et al.; Beaty et al.; Luchini et al. [1, 2, 3]	Connectivity, path length, modularity, and semantic richness relate to divergent/convergent creativity, fluency, originality, and insight. Rich structure can help and constrain.	Move from creativity scores to a target-specific first-passage quantity for usable hypotheses.
Memory search	Abbott et al.; Zemla–Austerweil [7, 8]	Retrieval order and clustering can be modeled as search/walks on semantic networks; representation and search jointly shape behavior.	Use first-passage tools without claiming cognition is literally a random walk.
Temporal / history-sensitive hypothesis search	Beaty–Silvia; Dasgupta et al.; Alison et al. [11, 9, 10]	Response order, recent inference history, and time pressure can alter later idea or hypothesis generation.	Treat history and time pressure as process covariates; isolate the additional effect of knowledge organization on first passage and first-hit class.
Expertise / domain knowledge	Joseph–Patel; Kushniruk et al.; Siew–Guru [12, 13, 5]	Experts and novices differ in relational organization; experts can access accurate hypotheses earlier and focus on small discriminating sets.	Separate knowledge amount from organization and formalize early access as a first-passage outcome.
Education / learned structure	Denervaud et al. [4]	Educational experience is associated with different semantic-network organization and creative performance.	Motivate experimental manipulation of organization rather than treating topology as fixed trait.
Scientific-hypothesis timing	Jing et al.; Ernst et al. [16, 17]	Hypotheses can be timed and quality-rated in real research tasks; faster generation need not mean higher quality.	Define a target that requires usability/warrant, not first generated statement.
Knowledge-graph discovery	ResearchLink; MOLIERE; Graph2Idea [19, 20, 21]	Scientific knowledge graphs can support hypothesis generation, path/link prediction, retrieval, and structured LLM ideation.	Test graph organization under matched content and token/model budgets; measure first usable class and direction.
Network first passage	Noh–Rieger and network-walk literature [18]	Mean first-passage time depends on network structure and transition dynamics.	Import the diagnostic as mathematics, not as a neural identity claim.

4.2 Direction is a first-hit distribution

Time alone cannot distinguish a productive accelerator from a funnel. Let $C_1, \dots, C_m$ be live usable readout classes. Define

$$\pi^{\text{first}}_{\mathcal{G}^K,Q}(C_j) = \Pr\big([H_{\tau_\mathcal{U}}] = C_j \mid \mathcal{G}^K, Q\big). \tag{7}$$

This is the *discovery-direction distribution*: which usable family is reached first under a declared organization.

Two content-matched organizations $\mathcal{G}_1^K$ and $\mathcal{G}_2^K$ can therefore differ in time even if they reach the same classes, or differ in direction even if mean time is similar. A finite comparison can use, for example,

$$D_{\text{dir}} = \text{JS}\Big(\pi_1^{\text{first}}, \pi_2^{\text{first}}\Big), \tag{8}$$

with Jensen-Shannon divergence used only as one possible finite diagnostic.

5 Central Hypothesis and Propositions

H1 - Organization-to-first-passage hypothesis [Open]. Under a content-matched task and declared reader, changing the organization of accessible knowledge changes the distribution of first-passage time to a usable hypothesis set:

$$\mathcal{G}_1^K \not\cong \mathcal{G}_2^K \quad \Rightarrow \quad \mathcal{L}(\tau_\mathcal{U} \mid \mathcal{G}_1^K, Q) \neq \mathcal{L}(\tau_\mathcal{U} \mid \mathcal{G}_2^K, Q)$$

for at least some task classes, after controlling for knowledge amount and generic retrieval ability.

This is a conditional empirical claim, not a theorem. It fails if organization contributes no reproducible predictive value once content, intelligence, retrieval ability, and task are controlled.

5.1 H2 - Organization-to-direction

H2 [Open]. Content-matched organizations can change which usable readout class is encountered first:

$$\pi^{\text{first}}_{\mathcal{G}_1^K,Q} \neq \pi^{\text{first}}_{\mathcal{G}_2^K,Q}.$$

A null result across adequately powered, genuinely distinct organizations would weaken the directional claim even if response speed differed.

This proposition is the main difference between "organization affects speed" and the stronger claim in the title. Direction means the distribution over epistemically different candidate families, not merely lexical or semantic direction in an embedding space.

5.2 H3 - Efficiency-diversity tradeoff

An organization may minimize $T_\mathcal{U}$ by funneling search into one dominant basin. That can be useful when the basin is appropriate and dangerous when it is not. Therefore define no universal "best topology." Instead evaluate a joint profile:

$$\mathcal{P}_Q(\mathcal{G}^K) = \big(T_\mathcal{U},\ H(\pi^{\text{first}}),\ D_H,\ W,\ C_{\text{escape}}\big), \tag{9}$$

where $H(\pi^{\text{first}})$ is directional entropy, D_H is the number/effective diversity of readout-distinguishable classes encountered in a window, and C_{escape} is an escape/restructuring cost when the dominant basin fails.

H3 [Open]. The topology that minimizes first-passage time need not maximize epistemic diversity or later discovery quality. In some tasks, increased local efficiency should decrease access to live alternatives, reproducing a fixation/expertise-reversal pattern.

This proposition is demanded by the literature rather

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than added for symmetry: rich semantic neighborhoods can increase fluency while decreasing originality [2], and experts can incur costs when learned schemas become mismatched [15].

5.3 H4 - Topology-process interaction

Memory-search research shows that similar behavioral patterns can arise from different search processes operating on different representations [7, 6]. Amortized-inference results additionally show that recent query structure can alter subsequent inference even without a claim about long-term knowledge topology [9]. Hence topology cannot be inferred from trajectory or history effects alone.

H4 [Open]. First-passage behavior is jointly determined by organization and search/control policy. A topology effect that disappears after policy changes is not evidence that organization is irrelevant; conversely, a trajectory difference is not evidence for a structural difference unless the policy is controlled or modeled.

6 Why Existing Findings Do Not Already Close H1

The closest evidence comes from five directions. First, expert clinicians generate accurate diagnostic components earlier and connect important cues through more relations [12, 13]; this motivates H1 but does not isolate organization from domain knowledge amount and experience. Second, semantic-network studies relate path structure to creativity and insight [1, 3]; they do not define or time first passage to a readout-usable scientific hypothesis. Third, VIADS experiments directly measured hypothesis-generation time and manipulated an information tool, but the intervention changed filtering and visualization, and faster generation did not imply higher quality [16]. Fourth, amortized-inference experiments show that prior related inference can bias subsequent hypothesis generation [9]; this is a process-history result rather than a content-matched manipulation of retained knowledge organization. Fifth, time-pressure studies show that temporal constraints alter hypothesis generation [10]; they manipulate urgency, not knowledge topology. The present proposal is therefore not a claim that nobody has studied structure, history, or time. It asks for a stricter conjunction of controls and outcomes: content matching, explicit organization manipulation or measurement, sequential candidate logging, readout-class coding, and first passage to a usable target.

AI work makes the gap especially testable. ResearchLink treats scientific-hypothesis generation as knowledge-graph link prediction [19]; MOLIERE uses a literature network and paths to support biomedical discovery [20]; Graph2Idea reports improved automated ideation from graph-structured contexts compared with flat retrieval [21]. Yet these systems generally optimize hypothesis ranking, novelty, quality, or feasibility, not first-passage distributions under an explicitly content-matched graph-vs-flat intervention. LLMs therefore provide a controlled experimental platform in which model, temperature, token budget, retrieved documents, and prompt can be held constant while access organization is manipulated.

7 Experimental Program

7.1 Experiment A: external organization, same atomic content

Construct a synthetic scientific domain with a fixed set of facts, observations, and candidate mechanisms unknown to participants. Every participant receives exactly the same atomic propositions and the same problem. Randomize only the access organization:

1. **Relational condition:** propositions are grouped and linked by typed relations, with cross-cluster bridges visible.
2. **Flat condition:** identical propositions are presented in matched typography and total exposure without relational grouping.
3. **Misleading-basin condition:** identical propositions are organized around a highly coherent but incomplete explanatory cluster, while all counterevidence remains present.

Participants think aloud and enter each candidate hypothesis in real time. Before the study, independent raters define the live readout classes, warrant floor, and discriminating actions. Primary outcomes are $\tau_{\mathcal{U}}$ and π^{first}. Secondary outcomes include raw hypothesis count, semantic novelty, recall accuracy, and restructuring/escape events.

The decisive test is not whether the relational group “performs better” globally. It is whether organization predicts the time and direction profile while atomic content and exposure are matched.

7.2 Experiment B: learned organization after support removal

External displays can change search without changing internal knowledge organization. A second experiment should therefore teach the same proposition set under different training sequences and relational encodings, then remove all scaffolds before testing. Delayed recall and proposition recognition verify content matching. If the first-passage difference persists after the display is removed and recall is equated, the organization interpretation strengthens.

A difficult design issue is that teaching relations may itself add knowledge. The honest solution is not to pretend this disappears: treat “same content” as a graded matching claim, publish the content ledger, and report which relational propositions differ. H1 is strongest in experiments where the atomic and relational information are literally identical but the *access organization* differs; claims about internal knowledge topology require a separate bridge.

7.3 Experiment C: expert-novice decomposition

Medical and domain-expertise studies provide a natural observational test. Estimate each participant’s domain-specific network, generic retrieval ability, factual knowledge, and experience. During sequential case presentation, record first passage to accurate/readout-distinguishable diagnostic classes. Test whether organization metrics add out-of-sample prediction of timing and first-hit class after factual knowledge and experience are included.

This would not establish causality, but it would test whether “expertise” contains a separable organizational component consistent with the early-hypothesis findings of Joseph and Patel [12] and the small-world account of medical expertise [13].

7.4 Experiment D: graph-structured AI context

Use one fixed LLM and freeze model version, system prompt, sampling parameters, retrieved papers, total to-

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kens, and generation budget. Present identical scientific evidence in multiple organizations: flat abstracts, shuffled triples, typed knowledge graph, and deliberately overclustered graph. Require the model to emit candidates sequentially rather than in one batch. An independent, blinded evaluator maps candidates to readout classes and scores warrant/testability.

Predictions:

$$\text{graph organization} \to \Delta T_{\mathcal{U}} \neq 0, \tag{10}$$
$$\text{graph organization} \to D_{\text{dir}} > 0, \tag{11}$$

without assuming the signs in advance. A graph can accelerate the wrong basin. This design directly connects current graph-based scientific-ideation work [19, 21] to the first-passage claim.

8 Readout Controls and Failure Modes

8.1 Speed is not warrant

The most important non-collapse rule is

$$\boxed{T_{\mathcal{U}} \downarrow \not\Rightarrow W(H) \uparrow \not\Rightarrow \text{truth.}} \tag{12}$$

This follows both from the companion architecture and from empirical hypothesis-generation work in which a visualization tool reduced generation time while not improving combined hypothesis quality [16]. A study that rewards the earliest fluent statement would test interface speed, not scientific imagination.

8.2 Direction is reader-relative

A "new direction" must be indexed to the reader/action set that makes candidates distinguishable. Two semantically distant statements that imply the same accessible records occupy one readout class for the current inquiry. Conversely, a small verbal change can constitute a new direction if it changes a discriminating consequence.

8.3 Content matching is itself a claim

Knowledge amount cannot be perfectly separated from organization if relation knowledge differs. Each experiment must therefore publish a content/organization ledger:

- identical atomic facts;
- identical evidence and counterevidence;
- identical time/exposure budget;
- organization manipulation;
- relational statements added, removed, or merely made salient;
- post-test factual and relational recall.

A hidden content imbalance downgrades a causal organization claim to a mixed intervention.

8.4 No literal neural random walk

First-passage time is used as a diagnostic of search over a finite state representation. The paper does not claim that human cognition is Markovian, memoryless, or implemented as a literal graph walk. Momentum, control, restructuring, and external tools can all make the process non-Markovian. The measurable question survives: when does a target class first become accessible and retained?

9 Relationship to the Companion Theory

The companion paper asks what hypothesis-generating imagination is and proposes bounded dynamic semantic mobility over a reader-conditioned semantic domain [24]. It introduces weighted accessibility, attraction, momentum, restructuring, readout-equivalence classes, and the separation of accessibility from warrant. The present paper does not reopen those claims. It selects one reserved consequence and turns it into a dedicated empirical program:

$$\boxed{\mathcal{G}^K \longrightarrow (\tau_{\mathcal{U}}, \pi^{\text{first}}) \longrightarrow \mathcal{U}} \tag{13}$$

In the language of that architecture, organization changes the finite access kernel κ and/or the topology over which admissible transport occurs. H1 asks whether those organizational differences explain measurable variation in $\tau_{\mathcal{U}}$ beyond breadth and generic control.

10 Boundaries and Non-Claims

The framework does not claim that:

- there is one universally optimal knowledge topology;
- shorter paths are always better;
- more connected knowledge is always more creative;
- fast hypothesis generation is evidence of truth or expertise;
- semantic-network metrics are literal brain wiring;
- content and organization can always be perfectly separated;
- first-passage mathematics is new;
- knowledge graphs, scientific link prediction, or literature-based discovery are new;
- Graph2Idea or other AI results independently validate the human cognitive claim;
- a usable hypothesis is necessarily correct;
- the present K0 manuscript has passed independent review.

11 Falsifiers

The theory earns value only if it can lose.

1. **H1 falsifier.** Across well-powered content-matched tasks, organization metrics or manipulations add no out-of-sample predictive value for first-passage time after knowledge amount and generic retrieval/control are modeled.
2. **H2 falsifier.** Distinct organizations produce no stable differences in first-hit readout classes beyond sampling noise.
3. **H3 falsifier.** Organization-induced acceleration never trades off against alternative access or fixation when evidence and warrant are controlled.
4. **Representation falsifier.** Results depend entirely on one arbitrary graph-construction method and vanish across reasonable representations of the same data.
5. **Usability falsifier.** The proposed usable-set criteria fail to predict experiment selection, expert interest, or later scientific progress better than raw novelty/quality ratings.

12 Discussion

The literature already makes two weak claims untenable. First, knowledge quantity alone is not an adequate account of expert or creative cognition. Second, idea time alone is not an adequate account of scientific usefulness. The more interesting object lies between them: the routing structure by which retained knowledge makes some candidate relations easy, delayed, dominant, or practically invisible.

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The first-passage framing turns this intuition into a finite observable. It asks not merely whether an agent *can* generate a hypothesis but how long the declared system takes to first reach a candidate that crosses an epistemic usability boundary. The first-hit distribution adds a second observable: where inquiry goes first. These two quantities permit cases that ordinary creativity scores obscure. Two systems may generate equally novel outputs but reach different readout classes. One may be fast but narrowly funneled; another slower but more diverse. An expert may be fast in a matched environment and slow to escape when the environment violates the expert schema. A graph-structured AI context may reduce search cost while amplifying a misleading cluster.

This framing also sharpens the problem of unconceived alternatives. The fact that a hypothesis exists somewhere in a conceptual possibility space does not make it practically available to a bounded knower. Organization changes the access landscape before evidence ever gets a chance to discriminate the alternative. Scientific communities can therefore improve evidence while retaining a generation bottleneck. The empirical program here asks whether part of that bottleneck is measurable as a first-passage consequence of knowledge organization.

13 Conclusion

The central proposal is simple enough to fail: under a declared content-matched task, knowledge organization should alter at least part of the first-passage profile to usable hypotheses,

$$\boxed{\mathcal{G}^K \Longrightarrow (\mathcal{L}(\tau_{\mathcal{U}}), \pi^{\text{first}}),} \tag{14}$$

without fixing the sign in advance. The verbal claim is: knowledge organization conditions not only what can become accessible, but the expected time and first direction by which a usable hypothesis is reached.

Existing literatures provide strong neighboring evidence: semantic topology relates to creativity and insight; expertise changes relational knowledge and early hypothesis generation; network structure changes retrieval trajectories; hypothesis timing can be measured directly; and graph-structured scientific contexts can improve machine ideation. None of these observations alone establishes H1. The decisive next step is a content-matched first-passage experiment with a target defined by readout-discriminability and minimum warrant rather than by idea count or originality alone.

If those experiments show no topology effect, the hypothesis should be abandoned or narrowed. If they succeed, they would provide a concrete answer to a question left implicit in the companion theory of imagination: not only which hypotheses are reachable, but why the same retained knowledge can make a usable new hypothesis arrive sooner, later, or from a different direction.

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This manuscript was developed under the Glosa “Rigour Without Infrastructure” workflow. The human author originated the hypothesis and selected the manuscript direction. AI assisted with literature retrieval, adversarial comparison, formalization, drafting, and typesetting. The manuscript remains K0 because the maker route cannot certify its own literature interpretation or formal bridge. The targeted review was conducted after the hypothesis had already been articulated; it must not be represented as a preregistered or literature-blind test.

References

[1] Y. N. Kenett, D. Anaki, and M. Faust. Investigating the structure of semantic networks in low and high creative persons. *Frontiers in Human Neuroscience*, 8:407, 2014. doi:10.3389/fnhum.2014.00407.

[2] R. E. Beaty, Y. N. Kenett, R. W. Hass, and D. L. Schacter. Semantic memory and creativity: the costs and benefits of semantic memory structure in generating original ideas. *Thinking & Reasoning*, 29(2):305–339, 2023. doi:10.1080/13546783.2022.2076742.

[3] S. Luchini, Y. N. Kenett, D. C. Zeitlen, A. P. Christensen, D. M. Ellis, et al. Convergent thinking and insight problem solving relate to semantic memory network structure. *Thinking Skills and Creativity*, 48:101277, 2023. doi:10.1016/j.tsc.2023.101277.

[4] S. Denervaud, A. P. Christensen, Y. N. Kenett, and R. E. Beaty. Education shapes the structure of semantic memory and impacts creative thinking. *npj Science of Learning*, 6:35, 2021. doi:10.1038/s41539-021-00113-8.

[5] C. S. Q. Siew and A. Guru. Investigating the network structure of domain-specific knowledge using the semantic fluency task. *Memory & Cognition*, 51:623–646, 2023. doi:10.3758/s13421-022-01314-1.

[6] A. A. Kumar, M. Steyvers, and D. A. Balota. A critical review of network-based and distributional approaches to semantic memory structure and processes. *Topics in Cognitive Science*, 14(1):54–77, 2022. doi:10.1111/tops.12548.

[7] J. T. Abbott, J. L. Austerweil, and T. L. Griffiths. Human memory search as a random walk in a semantic network. In *Advances in Neural Information Processing Systems 25*, 2012.

[8] J. C. Zemla and J. L. Austerweil. Modeling semantic fluency data as search on a semantic network. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 2017.

[9] I. Dasgupta, E. Schulz, N. D. Goodman, and S. J. Gershman. Remembrance of inferences past: amortization in human hypothesis generation. *Cognition*, 178:67–81, 2018. doi:10.1016/j.cognition.2018.04.017.

[10] L. Alison, B. Doran, M. L. Long, N. Power, and A. Humphrey. The effects of subjective time pressure and individual differences on hypotheses generation and action prioritization in police investigations. *Journal of Experimental Psychology: Applied*, 19(1):83–93, 2013. doi:10.1037/a0032148.

[11] R. E. Beaty and P. J. Silvia. Why do ideas get more creative across time? An executive interpretation of the serial order effect in divergent thinking tasks. *Psychology of Aesthetics, Creativity, and the Arts*, 6(4):309–319, 2012. doi:10.1037/a0029171.

[12] G.-M. Joseph and V. L. Patel. Domain knowledge and hypothesis generation in diagnostic reasoning. *Medical Decision Making*, 10(1):31–46, 1990. doi:10.1177/0272989X9001000107.

[13] A. W. Kushniruk, V. L. Patel, and A. A. J. Marley. Small worlds and medical expertise: implications for medical cognition and knowledge engineering. *International Journal of Medical Informatics*, 49(3):255–271, 1998. doi:10.1016/S1386-5056(98)00044-6.

[14] J. E. Brush Jr., J. Sherbino, and G. R. Norman. Diagnostic reasoning in cardiovascular medicine. *BMJ*, 376:e064389, 2022. doi:10.1136/bmj-2021-064389.

[15] A. Armougum, A. Gaston-Bellegarde, C. Joie-La Marle, and P. Piolino. Expertise reversal effect: cost of generating new schemas. *Computers in Human Behavior*, 111:106406, 2020. doi:10.1016/j.chb.2020.106406.

[16] X. Jing, J. J. Cimino, V. L. Patel, Y. Zhou, J. H. Shubrook, S. De Lacalle, et al. Data-driven hypothesis generation among inexperienced clinical researchers: a comparison of secondary data analyses with visualization (VIADS) and other tools. *Journal of Clinical and Translational Science*, 8:e13, 2024. doi:10.1017/cts.2023.708.

10.5281/zenodo.22307561

[17] M. A. Ernst, B. N. Draghi, J. J. Cimino, V. L. Patel, Y. Zhou, J. H. Shubrook, et al. The quality of data-driven hypotheses generated by inexperienced clinical researchers: a case study. *medRxiv*, 2024. doi:10.1101/2024.08.12.24311877.

[18] J. D. Noh and H. Rieger. Random walks on complex networks. *Physical Review Letters*, 92:118701, 2004. doi:10.1103/PhysRevLett.92.118701.

[19] A. Borrego, D. Dessì, D. Ayala, I. Hernández, F. Osborne, D. R. Recupero, et al. Research hypothesis generation over scientific knowledge graphs. *Knowledge-Based Systems*, 315:113280, 2025. doi:10.1016/j.knosys.2025.113280.

[20] J. Sybrandt, M. Shtutman, and I. Safro. MOLIERE: Automatic Biomedical Hypothesis Generation System. In *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 1633–1642, 2017. doi:10.1145/3097983.3098057.

[21] X. Li, H. Tu, and X. Han. Graph2Idea: retrieval-augmented scientific idea generation with graph-structured contexts. *arXiv:2606.09105*, 2026.

[22] P. K. Stanford. *Exceeding Our Grasp: Science, History, and the Problem of Unconceived Alternatives*. Oxford University Press, 2006.

[23] J. G. Foster, A. Rzhetsky, and J. A. Evans. Tradition and innovation in scientists' research strategies. *American Sociological Review*, 80:875–908, 2015.

[24] Y. Lahtee. From Problem to Hypothesis: Dynamic Semantic Mobility, Bounded Knowers, and the Readout-Discriminable Hypothesis Space. GLOSA K0 working paper, 2026.

[25] Y. Lahtee. Readout Universe: A Philosophy and Logic for Grounding Claims. GitHub repository, 2026. https://github.com/morrocwi/readout_universe.

[26] Y. Lahtee. Readout Genesis Universal Technical Whitepaper, v1.2.0. GitHub repository, 2026. https://github.com/morrocwi/readout_genesis.

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Before Evidence Can Decide: Candidate-Set Formation, Discovery Routing, and Unconceived Alternatives — Questions for a Pre-Evidential Epistemology of Science

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What the chapter states. "Building on two companion K0 papers, this agenda paper proposes no completed solution. It instead names a pre-evidential layer of inquiry in which problem framing, knowledge organization, accessibility, institutional reinforcement, search history, values, and tools shape the formation of candidate sets before comparative appraisal."

Status. "GLOSA K0 agenda paper. This manuscript is Paper III in a three-paper sequence. [. . .] It is a targeted agenda-setting review, not a preregistered systematic review. No priority claim or independent validation claim is made."

Falsifier(s). No explicit falsifier stated in the chapter.

Read before. *Knowledge Topology and the First Passage to Usable Hypotheses*.
Feeds into. *The Epistemic Chain Reaction: Human-AI Multiplication from Questions to Readout-Distinguishable Hypotheses*.

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PRE-EVIDENTIAL EPISTEMOLOGY GLOSA K0 AGENDA PAPER

Before Evidence Can Decide

Candidate-Set Formation, Discovery Routing, and Unconceived Alternatives

Questions for a Pre-Evidential Epistemology of Science

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4 September 2026

Status. GLOSA K0 agenda paper. This manuscript is Paper III in a three-paper sequence. Paper I proposes a bounded-knower architecture of hypothesis-generating imagination; Paper II isolates the hypothesis that knowledge organization can alter the time and first direction by which a usable hypothesis becomes accessible. The present paper does *not* claim to solve the resulting philosophical problems. Its purpose is to expose a neglected layer of inquiry, sharpen open questions, distinguish adjacent literatures, and specify future records that could decide among competing answers. It is a targeted agenda-setting review, not a preregistered systematic review. No priority claim or independent validation claim is made.

Abstract

Scientific epistemology has sophisticated resources for evaluating hypotheses once they are articulated: evidence, model comparison, underdetermination, pursuit-worthiness, experimental discrimination, and collective criticism. Yet a logically prior operational problem remains comparatively fragmented across disciplines: how does a hypothesis enter the *live candidate set* that can be appraised at all? The problem of unconceived alternatives shows why this matters; epistemic-landscape models show that social search can concentrate or diversify research; studies of scientific tradition and "undone science" show that institutions shape which problems and relations receive attention; and pursuit-worthiness research shows that scientists must decide what to develop before conclusive evidence exists. Building on two companion K0 papers, this agenda paper proposes no completed solution. It instead names a pre-evidential layer of inquiry in which problem framing, knowledge organization, accessibility, institutional reinforcement, search history, values, and tools shape the formation of candidate sets before comparative appraisal. A minimal cycle notation separates candidate-set formation from evidence-based appraisal while allowing evidence to restructure the next cycle. From that distinction the paper develops ten open questions concerning individual generation, unconceived alternatives, expertise, collective search, research agendas, pursuit, values and power, AI-mediated discovery, evidence-triggered restructuring, and normative obligations to search for alternatives. It closes with a future-work matrix and a set of observations deliberately weaker than answers. The central proposal is methodological: science may need an epistemology not only of why accepted claims are warranted, but of how alternatives become available for warranting in the first place.

Keywords: candidate-set formation; scientific discovery; unconceived alternatives; pursuit-worthiness; epistemic landscapes; research agendas; scientific imagination; hypothesis generation; bounded knower; AI for science; Readout Universe; Readout Genesis

1 The Problem Before the Verdict

Evidence can compare only what a given inquiry has made available for comparison. This statement is intentionally local and cycle-indexed. It does not imply that evidence is passive in discovery: an unexpected record may trigger a new question, a representational change, or a new candidate in the next cycle. The point is narrower. At any particular appraisal step, a hypothesis that is not in the live candidate set receives no comparative evaluation at that step.

This observation sits awkwardly between classic philosophical categories. The discovery/justification distinction historically separated questions about how hypotheses arise from questions about how they are justified [1, 2]. Contemporary philosophy has substantially reopened the territory: pursuit-worthiness concerns whether an immature idea deserves development [5, 7]; question pursuit distinguishes which questions agents choose to seek answers to [6]; and work on values in discovery and pursuit shows that choices of questions, projects, and experimental designs can feed back into later appraisal [9]. Still, these literatures do not by themselves provide a single object corresponding to the set of hypotheses that are actually live for appraisal in a bounded inquiry.

Agenda question. What determines the set of alternatives that a bounded individual or collective epistemic system is actually capable of considering at a given stage of inquiry, and how should that set itself become an object of epistemic criticism?

The two companion papers motivate this question without answering it. Paper I models hypothesis-generating imagination as bounded dynamic semantic mobility and separates reachability, accessibility, warrant, and truth [18]. Paper II proposes that knowledge organization may change the first-passage time and first-hit direction to a usable hypothesis under content-matched conditions [19]. If those architectures are even approximately useful, they imply a missing middle layer between "possibilities that could in principle be generated" and "hypotheses currently being compared by evidence."

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2 A Minimal Cycle, Not a Solution

Let $\mathcal{H}^{\text{disc}}_{E,t}(Q)$ denote the readout-distinguishable hypothesis classes reachable by epistemic system E at time t for question Q under a declared reader/action set. The symbol E may index an individual or a distributed collective system. Introduce a candidate-set operator

$$\Psi_{E,t,Q} : \mathcal{H}^{\text{disc}}_{E,t}(Q) \longrightarrow \mathcal{C}_{E,t}(Q), \tag{1}$$

where $\mathcal{C}_{E,t}(Q)$ is the *live candidate set*: hypotheses currently retained strongly enough to enter explicit comparison, pursuit, or test planning.

Comparative appraisal is then typed as

$$\text{Appraise}_t : (\mathcal{C}_{E,t}(Q), \delta_{\le t}, \Pi_t) \longrightarrow \mathcal{R}_t, \tag{2}$$

where $\delta_{\le t}$ are declared records, Π_t is an appraisal policy, and $\mathcal{R}_t$ may encode ranking, rejection, acceptance, pursuit, abstention, or revision.

The only immediate consequence is definitional:

$$H \notin \mathcal{C}_{E,t}(Q) \Longrightarrow H \text{ receives no comparative appraisal at cycle } t. \tag{3}$$

This is not a claim that the unseen hypothesis is true, superior, or permanently unavailable.

Evidence may alter the candidate set by changing the retained problem, question, reader, or access geometry:

$$\delta_{t+1} \longrightarrow \mathcal{B}_{t+1} \longrightarrow \mathcal{C}_{E,t+1}(Q'). \tag{4}$$

Thus the relevant sequence is a loop, not a wall between discovery and justification:

$$\mathcal{C}_t \to \text{Appraise}_t \to \delta_{t+1} \to \mathcal{B}_{t+1} \to \mathcal{C}_{t+1}. \tag{5}$$

Observation O1 [Dr]. The epistemic quality of an appraisal procedure and the epistemic breadth of the candidate set it appraises are separable properties. Improving one need not improve the other.

This separation is the starting problem of the paper, not its answer.

3 Where the Literatures Already Reach

The closest philosophical warning comes from Stanford's problem of unconceived alternatives: inference to the best among known explanations is fragile when historically available alternatives are not exhaustive [3, 4]. Yet the problem is usually framed retrospectively: successor theories reveal that earlier scientists failed to conceive alternatives. The present agenda asks whether some portions of that failure can become prospectively measurable without pretending to count genuinely unknown unknowns.

The closest social-search literature uses epistemic landscapes to ask how populations distribute cognitive labor [10, 11, 12]. These models are useful precisely because they show that social learning and novelty-seeking can produce counterintuitive outcomes. But most landscape models stipulate a landscape of research approaches and payoffs in advance. Candidate-set formation raises a harder problem: in real science, some nodes and even some dimensions of the landscape may not yet be represented.

Pursuit-worthiness research provides another adjacent layer. Scientists routinely decide whether developing a theory or experiment is promising before decisive evidence exists [5, 7, 8]. Yet an idea must normally be articulated before it can be judged pursuitworthy. Candidate-set formation asks about the gate immediately upstream of pursuit.

Finally, sociology of science shows that research strategies and agendas are not neutral reflections of logical possibility. Biomedical research disproportionately consolidates established relations, while risky combinations are less frequent and carry different career incentives [13]. "Undone science" identifies research that communities or social movements regard as important but that institutions leave unfunded or ignored [14]. These findings motivate institutional questions about accessibility without establishing any general theory of candidate formation.

4 Ten Open Questions

4.1 Q1. What exactly is a live candidate?

A hypothesis can be imaginable, verbally generated, privately entertained, publicly articulated, pursued, funded, tested, or institutionally recognized. These are not the same state. Future work needs an operational taxonomy of candidate status. Does $\mathcal{C}_t$ require explicit articulation, sufficient accessibility, pursuit-worthiness, or merely the ability to enter an action plan?

Deciding records. Think-aloud protocols, timestamped hypothesis logs, search traces, eye/attention data, laboratory notebooks, AI generation traces, and explicit "consider / reject / pursue" judgments could test whether candidate status is empirically separable from raw generation.

4.2 Q2. Can unconceived alternatives be measured without contradiction?

A genuinely unconceived alternative cannot be directly counted by the knower who has not conceived it. But synthetic tasks can give the experimenter a hidden catalog of valid alternatives. Historical reconstructions can compare predecessor candidate sets with later theories. Perturbation experiments can ask whether small changes in organization, language, or representation suddenly expose new readout classes. The open question is whether these proxies illuminate real unconceived alternatives or merely laboratory omission.

4.3 Q3. Does knowledge topology route candidate-set formation?

Paper II isolates this as an empirical program: under content-matched conditions, does organization alter first-passage time and first-hit class to a usable hypothesis [19]? Paper III treats this as one possible mechanism among several, not as an established answer. A positive result would show that candidate-set formation has a routing component; a null result would force greater weight onto control policy, experience, task cues, or other mechanisms.

4.4 Q4. Are expertise and fixation dual outcomes of the same access system?

Experts often generate strong hypotheses quickly in familiar domains, while familiar structures can also suppress alternatives. Is this a single task-relative routing mechanism with matched and mismatched regimes, or are expertise and fixation produced by different processes? The answer matters because interventions that "debias" expertise may also destroy useful compression.

4.5 Q5. What is the collective candidate set?

For a group $\mathcal{G}$, is the live collective set simply

$$\mathcal{C}_{\mathcal{G},t} = \bigcup_i \mathcal{C}_{A_i,t}?$$

Probably not in all cases: communication, prestige, role assignment, journals, shared instruments, and meeting

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Table 1: Neighboring literatures already answer parts of the pre-evidential problem. The final column identifies what remains open rather than claiming novelty.

Literature	Representative work	What it already contributes	Question left for Paper III
Discovery / justification	Reichenbach; later reconstruction [1, 2]	Distinguishes genesis of claims from standards of justification; later work complicates the boundary.	Can candidate-set formation be epistemically assessed without collapsing discovery into justification?
Unconceived alternatives	Stanford [3, 4]	Successful theories may face serious alternatives not conceived by historical communities.	Which part of "unconceived" is structurally unreachable, merely inaccessible, institutionally neglected, or not yet triggered by evidence?
Pursuit-worthiness	Lichtenstein; Duerr–Fischer; Fischer [5, 7, 8]	Immature theories and experiments can rationally merit further development before conclusive support.	Pursuit presupposes that an option has entered consideration. What governs that earlier gate?
Question pursuit	Barseghyan [6]	Questions themselves can be accepted or pursued as distinct epistemic stances.	How do question selection and candidate generation co-shape one another over time?
Epistemic landscapes / cognitive labor	Weisberg–Muldoon; Alexander et al.; Thoma [10, 11, 12]	Social search, following, maverick behavior, diversity, and exploration strategies affect collective discovery in modeled landscapes.	How should landscapes be built when the research approaches themselves are not pre-enumerated?
Tradition / innovation	Foster et al. [13]	Scientific strategies favor established relations; risky bridging is rarer and institutionally costly.	When does institutional organization alter which candidate families arise before evaluation?
Research agendas / undone science	Frickel et al. [14]	Power and institutions shape what research remains unfunded, incomplete, or ignored.	Can "undone" questions and "unconceived" hypotheses be connected without reducing one to the other?
Consideration-set formation (analogy)	Hauser; Punj–Moore [15, 16]	Bounded decision makers often screen alternatives before choosing among those retained.	Which formal insights transfer to science, and which fail because hypotheses can be generated rather than selected from a fixed catalog?
AI-assisted science	Co-Scientist and graph-based systems [17, 19]	Machines can generate, rank, refine, retrieve, and propose tests at scale.	Does AI widen readout-distinguishable candidate sets or merely increase fluent variants within existing basins?

structure can prevent individual candidates from becoming collective candidates, while collaboration can generate candidates no member held alone. Future work must distinguish *union*, *transmission*, and *emergent co-construction* without positing a literal group mind.

4.6 Q6. When does an institution become an epistemic router?

Funding programs, curricula, databases, journal scopes, benchmark suites, taxonomies, and instrument availability can change what is easy to ask and easy to test. At what point should such infrastructure be treated as part of scientific reasoning rather than as merely external administration? "Undone science" and research-strategy studies suggest the boundary is epistemically consequential [14, 13].

4.7 Q7. How do values and power enter before appraisal?

Values can shape problem choice, experimental design, and theory development [9]. Candidate-set formation adds a sharper question: can social or nonepistemic values systematically alter which explanatory relations become cognitively or institutionally live before evidence is compared? If so, what would count as legitimate value-sensitive agenda setting, and what would count as an epistemically harmful exclusion?

4.8 Q8. When does evidence rank candidates, and when does it create them?

The same record may perform different roles. It can reduce warrant for one live candidate, discriminate two rivals, create a residual that triggers a new question, or force restructuring that makes a previously inaccessible candidate reachable. Future experiments should label these transitions instead of treating "evidence use" as one homogeneous process.

4.9 Q9. Does AI widen science or deepen existing basins?

AI systems can generate enormous numbers of candidate statements. The relevant quantity may not be count but the number and diversity of readout-distinguishable candidate classes, their provenance, and their dependence on retrieval architecture. A model may produce ten thousand linguistically different hypotheses while remaining inside one evidential basin. Conversely, a single cross-domain bridge may open a genuinely new class. This creates a benchmark question: *Does an AI system increase candidate-set coverage under fixed evidence, or only verbosity?*

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4.10 Q10. Is there an epistemic obligation to search for alternatives?

Stanford's unconceived-alternative problem creates pressure to look beyond current winners, but search is costly and cannot be exhaustive. Pursuit-worthiness research emphasizes cost, expected benefit, feasibility, and epistemic gain [7, 8]. The normative problem is therefore not "always generate more alternatives." It is: when is an inquiry sufficiently consequential, fragile, novel, or underdetermined that agents acquire an obligation to widen the candidate set before accepting or acting?

5 Observations That Are Not Yet Answers

O2. Better evidence does not automatically repair a generation bottleneck. A field can improve measurement, statistics, replication, and model comparison while repeatedly comparing variants from the same narrow candidate family.

O3. Peer review may improve appraisal and narrow generation at the same time. Shared standards can reject weak claims efficiently while also raising the cost of unfamiliar questions, representations, or bridges. Whether this occurs, and under what review structures, is empirical.

O4. Scientific conservatism can be pre-evidential. A dominant paradigm need not "win" every evidence comparison if institutions make its alternatives slow, costly, or difficult to formulate in the first place.

O5. Diversity may matter through routing rather than labels. The epistemically relevant question is not whether a group contains different identities or disciplines as such, but whether those differences change questions, relations, access paths, objection channels, and live candidate classes. No automatic equivalence is assumed.

O6. Research agendas are part of epistemic infrastructure. Choosing what remains "worth asking" can alter future evidence production and later candidate availability. Agenda setting is therefore upstream of, but not external to, knowledge production.

O7. A bounded-knower epistemology needs a theory of absence. Existing systems are usually better at logging what was considered than what was not. Future epistemic infrastructure may need explicit records of rejected paths, missing bridges, unexplored alternatives, and reasons for stopping search.

6 Future Work: What Records Would Matter?

Two cross-cutting methodological problems deserve special emphasis.

6.1 The measurement problem

Candidate-set breadth cannot be measured by raw idea count. Semantically different statements can be evidentially equivalent; private candidates may never be reported; and truly unconceived alternatives are unavailable by definition. Future measures may therefore need a portfolio: readout-class diversity, perturbation sensitivity, escape cost, provenance diversity, search-trace entropy, and performance on synthetic hidden-alternative tasks. None should be treated as a universal proxy for "openness."

6.2 The counterfactual problem

To claim that an institution, topology, or tool suppressed an alternative is to make a counterfactual claim: under a different access structure, the alternative would have become live. Randomized interventions are possible in laboratory and AI settings but much harder historically. Comparative institutional designs, interrupted policy changes, archival notebooks, preregistered team experiments, and simulation may provide partial triangulation.

7 A Research Program for AI-Mediated Science

AI makes candidate-set questions unusually testable because many components can be frozen. One can hold constant model version, corpus, problem statement, token budget, sampling parameters, and evidence while varying retrieval organization, cross-domain bridging, critique rules, or tool availability. The key outcomes should not be only novelty or average quality, but:

- number of readout-distinguishable candidate classes;
- first-hit class and time-to-class;
- provenance of each bridge;
- sensitivity to retrieval perturbation;
- probability of escaping a deliberately seeded misleading basin;
- calibration and warrant after candidates become live.

AI also raises a governance question absent from ordinary creativity benchmarks. If retrieval or ranking architecture determines which scientific candidates are generated first, then system designers become participants in research agenda formation even when they never choose a theory explicitly. This possibility should be tested, not presumed.

8 Normative Questions Without Premature Answers

A pre-evidential epistemology would eventually need normative principles, but this paper does not supply them. It instead records tensions that any future account must confront.

1. **Coverage versus cost.** Exhaustive alternative search is impossible; too little search risks lock-in.
2. **Expertise versus escape.** Strong organization accelerates familiar inquiry but may narrow candidate diversity.
3. **Coordination versus monoculture.** Shared standards enable cumulative science but may synchronize blind spots.
4. **Pursuit versus appraisal.** Promising ideas deserve resources before full warrant, yet pursuit itself changes what evidence will later exist.
5. **Pluralism versus noise.** More candidates can increase coverage but also overwhelm attention, testing capacity, and correction.
6. **Institutional steering versus epistemic autonomy.** Funding and agendas inevitably select research directions; the question is what forms of steering remain criticizable and reversible.

The bounded-knower perspective suggests one governance intuition, not a rule: candidate-set formation should remain inspectable and revisable wherever stakes justify the cost. The relevant record would include not only "why

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Table 2: Future-work matrix. Each row names a question and a record that could move it beyond philosophical speculation.

Problem	Candidate design	Primary record	Interpretive danger
Candidate-set formation	Require sequential "generated / considered / pursued / rejected" tagging during scientific tasks.	Transition matrix among candidate states; latency to each state.	Self-report may confuse attention, preference, and consideration.
Hidden alternatives	Synthetic scientific worlds with a known experimenter-side catalog of valid rival mechanisms.	Fraction of hidden readout classes generated; time-to-class; perturbation sensitivity.	Laboratory catalogs may be much easier than genuine scientific unknowns.
Knowledge routing	Content-matched flat vs relational vs misleading organizations.	First-passage profile and first-hit readout-class distribution.	Organization can accidentally add substantive relational knowledge.
Expertise / fixation	Matched vs schema-violating cases with expertise measured independently.	Speed to first strong class; escape cost; alternative-class coverage.	Expertise and knowledge amount are difficult to disentangle.
Collective candidate sets	Teams with controlled communication topology and role assignment.	Union of private sets vs publicly live set vs jointly generated classes.	Group output may reflect social reporting rather than collective cognition.
Institutional routing	Simulated funding/review systems with identical proposal pools but different prestige, novelty, or risk rules.	Which questions and hypotheses survive to funded/tested stages.	Simulation may not capture real incentive ecology.
Evidence-triggered restructuring	Inject identical anomalous records at controlled moments.	Whether evidence changes ranking inside $\mathcal{C}_t$ or changes $\mathcal{C}_{t+1}$ itself.	New candidates may reflect experimenter cueing rather than genuine restructuring.
AI candidate coverage	Freeze model, corpus, token budget, and evidence; vary retrieval organization and critique protocol.	Distinct readout classes, provenance, first-hit class, escape events.	LLM outputs are not direct models of human discovery.
Normative alternative search	High- vs low-stakes decisions with controlled search cost and residual uncertainty.	How much search experts judge obligatory before acceptance/action.	Consensus about obligation is not proof of epistemic optimality.

this hypothesis won" but also "what alternatives were actively generated, which were discarded, which search routes were never attempted, and what would trigger reopening the set."

9 Boundaries and Non-Claims

This agenda paper does not claim that:

- evidence is irrelevant to discovery; evidence can restructure the next candidate set;
- discovery and justification are identical processes;
- every truth is reachable from every bounded knower's current state;
- larger candidate sets are always epistemically superior;
- more demographic, disciplinary, or semantic diversity automatically produces better science;
- institutional concentration implies intentional suppression;
- consideration-set models from consumer choice transfer directly to science;
- epistemic landscapes already model unconceived nodes;
- pursuit-worthiness is reducible to candidate accessibility;
- Paper II's knowledge-topology hypothesis has been empirically established;
- AI systems are faithful cognitive models of scientists;
- the notation here is literal neural architecture;
- the present K0 paper has passed independent checking.

10 Discussion: Toward an Epistemology of the Not-Yet-Considered

The most important implication of the two companion papers may be a question rather than a result. If bounded knowers generate hypotheses through constrained, history-shaped access, and if knowledge organization can in principle route the time and direction of that access, then the quality of scientific inference depends on more than the quality of evidence and appraisal once candidates are present. It may also depend on how the candidate set was formed.

This does not demote evidence. It relocates one vulnerability upstream. At cycle t, evidence can discriminate only among live candidates. At cycle $t + 1$, the evidence from the previous cycle may transform the problem and generate new candidates. The scientific process therefore alternates between at least two different epistemic tasks: *making alternatives available* and *discriminating among available alternatives*. Neither can substitute for the other.

This framing also joins several literatures that are often kept apart. Stanford's unconceived alternatives raise the historical problem of missing rivals [3]; epistemic landscapes ask how communities explore research spaces [10, 11, 12]; pursuit-worthiness asks which immature ideas deserve development [5, 7]; sociology of science asks why traditions, incentives, and agendas channel attention [13, 14]; and bounded-choice research shows that agents often form consideration sets before making final selections [15, 16]. Paper III proposes that these can be read as neighboring views of one unresolved architecture: the formation, routing, and revision of the alternatives that are allowed to become epistemically live.

The proposal should remain modest. Perhaps no uni-

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fied theory exists because different domains produce candidate sets through irreducibly different mechanisms. Perhaps knowledge topology explains little once expertise and control are measured. Perhaps collective diversity affects pursuit but not generation. Perhaps AI expands apparent alternatives while compressing evidential diversity. These are acceptable outcomes. The contribution of an agenda paper is to make such failures legible.

11 Conclusion

Science has powerful methods for asking whether a hypothesis is supported. It has less unified language for asking how that hypothesis came to be one of the things support could be assigned to.

The resulting research agenda can be stated without a grand solution:

$$\begin{aligned}\text{Question} &\to \text{Candidate-set formation}\\ &\to \text{Appraisal} \to \text{Record}\\ &\to \text{Candidate-set revision.}\end{aligned} \tag{6}$$

The open problem is the middle-left arrow. What makes an alternative live? What makes another remain absent? Which absences are harmless, which are costly, and which are invisible until history supplies a successor theory? How do knowledge structures, expertise, institutions, values, tools, social organization, and AI alter this process? And when, if ever, do scientists have a responsibility to reopen a candidate set before acting on the best option currently available?

Paper I asked how bounded imagination can generate hypotheses. Paper II asked whether knowledge organization can alter discovery time and direction. Paper III leaves the larger question deliberately open:

> **Future-work question.** What would an epistemology of science look like if it treated the formation of the candidate set—not only the evaluation of its members—as a first-class object of criticism, measurement, and governance?

Process note. The human author originated the research direction and selected the questions emphasized here. AI assisted with literature retrieval, cross-paper synthesis, formal typing, drafting, and typesetting. The literature review occurred after the core problem was articulated and is therefore not literature-blind or preregistered. The manuscript remains GLOSA K0 pending independent review.

References

[1] H. Reichenbach. *Experience and Prediction*. University of Chicago Press, 1938.

[2] M. Aufrecht. Reichenbach Falls—And Rises? Reconstructing the Discovery/Justification Distinction. *International Studies in the Philosophy of Science*, 31(2):151–176, 2018. doi:10.1080/02698595.2018.1424760.

[3] P. K. Stanford. *Exceeding Our Grasp: Science, History, and the Problem of Unconceived Alternatives*. Oxford University Press, 2006.

[4] K. Stanford. Underdetermination of Scientific Theory. *Stanford Encyclopedia of Philosophy*, 2023. https://plato.stanford.edu/entries/scientific-underdetermination/.

[5] E. I. Lichtenstein. (Mis)Understanding scientific disagreement: Success versus pursuit-worthiness in theory choice. *Studies in History and Philosophy of Science*, 85:166–175, 2021. doi:10.1016/j.shpsa.2020.10.005.

[6] H. Barseghyan. Question pursuit as an epistemic stance. *Studies in History and Philosophy of Science*, 94:112–120, 2022. doi:10.1016/j.shpsa.2022.06.001.

[7] P. M. Duerr and E. Fischer. Rationally warranted promise: the virtue-economic account of pursuit-worthiness. *Synthese*, 206:68, 2025. doi:10.1007/s11229-025-05077-5.

[8] E. Fischer. The pursuitworthiness of experiments. *European Journal for Philosophy of Science*, 2026. doi:10.1007/s13194-025-00711-y.

[9] K. C. Elliott and D. J. McKaughan. How values in scientific discovery and pursuit alter theory appraisal. *Philosophy of Science*, 76(5):598–611, 2009. doi:10.1086/605807.

[10] M. Weisberg and R. Muldoon. Epistemic landscapes and the division of cognitive labor. *Philosophy of Science*, 76(2):225–252, 2009. doi:10.1086/644786.

[11] J. McKenzie Alexander, J. Himmelreich, and C. Thompson. Epistemic landscapes, optimal search, and the division of cognitive labor. *Philosophy of Science*, 82(3):424–453, 2015. doi:10.1086/681766.

[12] J. Thoma. The epistemic division of labor revisited. *Philosophy of Science*, 82(3):454–472, 2015. doi:10.1086/681768.

[13] J. G. Foster, A. Rzhetsky, and J. A. Evans. Tradition and innovation in scientists' research strategies. *American Sociological Review*, 80(5):875–908, 2015. doi:10.1177/0003122415601618.

[14] S. Frickel, S. Gibbon, J. Howard, J. Kempner, G. Ottinger, and D. J. Hess. Undone Science: Charting Social Movement and Civil Society Challenges to Research Agenda Setting. *Science, Technology, & Human Values*, 35(4):444–473, 2010. doi:10.1177/0162243909345836.

[15] J. R. Hauser. Consideration-set heuristics. *Journal of Business Research*, 67(8):1688–1699, 2014. doi:10.1016/j.jbusres.2014.02.015.

[16] G. Punj and R. Moore. Information search and consideration set formation in a web-based store environment. *Journal of Business Research*, 62(6):644–650, 2009. doi:10.1016/j.jbusres.2007.04.013.

[17] J. Gottweis, W.-H. Weng, A. Daryin, et al. Accelerating scientific discovery with Co-Scientist. *Nature*, 655:487–496, 2026. doi:10.1038/s41586-026-10644-y.

[18] Y. Lahtee. From Problem to Hypothesis: Dynamic Semantic Mobility, Bounded Knowers, and the Readout-Discriminable Hypothesis Space. GLOSA K0 working paper, 2026.

[19] Y. Lahtee. Knowledge Topology and the First Passage to Usable Hypotheses: A Readout Theory of Discovery Time and Direction. GLOSA K0 working paper, 2026.

[20] Y. Lahtee. Readout Universe: A Philosophy and Logic for Grounding Claims. GitHub repository, 2026. https://github.com/morrocwi/readout_universe.

[21] Y. Lahtee. Readout Genesis Universal Technical Whitepaper, v1.2.0. GitHub repository, 2026. https://github.com/morrocwi/readout_genesis.

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The Epistemic Chain Reaction: Human-AI Multiplication from Questions to Readout-Distinguishable Hypotheses — A Controlled Amplification Architecture for Scientific Discovery

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What the chapter states. “Central proposal [Dr/Open]. AI’s strongest role is not to replace the bounded knower but to multiply the number of new, live, readout-distinguishable and minimally grounded hypothesis classes that can be reached from a human-retained question, while preserving explicit human control over problem ownership, question significance, decisive tests, and revision.”

Status. “GLOSA K0 conceptual and agenda paper. [. . .] All central augmentation laws remain [Open] unless stated otherwise. [. . .] no priority or independent-validation claim is made.”

Falsifier(s). “If a recursive human-AI system produces thousands of outputs but almost no new usable readout classes, then kepi is low and the claimed ‘amplification’ is mostly verbosity.”

Read before. *Before Evidence Can Decide: Candidate-Set Formation, Discovery Routing, and Unconceived Alternatives*.
Feeds into. *Epistemic Fusion or Epistemic Tunnel: A Phenomenology-Anchored, Global-Literature-Constrained, Problem-First Architecture of Human-AI Collaboration* (Part 5).

HUMAN-AI SCIENTIFIC DISCOVERY GLOSA K0 PAPER IV

The Epistemic Chain Reaction

Human-AI Multiplication from Questions to Readout-Distinguishable Hypotheses

A Controlled Amplification Architecture for Scientific Discovery

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Status. GLOSA K0 conceptual and agenda paper. This manuscript is Paper IV in a sequence. Paper I models hypothesis-generating imagination as bounded dynamic semantic mobility; Paper II isolates knowledge organization as a possible determinant of discovery time and first-hit direction; Paper III asks how hypotheses become live candidates before evidence can appraise them. Paper IV asks a deliberately practical question: *where can AI amplify the human question-to-hypothesis transition most strongly without collapsing human problem ownership, epistemic diversity, warrant, or correction?* The proposed "chain reaction" is a structural metaphor and finite branching model, not a claim of physical equivalence. All central augmentation laws remain [Open] unless stated otherwise. The literature review is targeted, not preregistered or systematic, and no priority or independent-validation claim is made.

Abstract

Large language models and multi-agent systems can already generate, critique, rank, refine, and experimentally update scientific hypotheses. Yet the strongest form of human-AI augmentation may not be one-shot idea generation or autonomous replacement of scientists. Building on three companion papers, this manuscript proposes a controlled *epistemic chain reaction*: a human retains the problem and question, while heterogeneous AI processes recursively decompose the question, traverse distant knowledge structures, generate candidate relations, critique them, collapse readout-equivalent variants, and propagate only new candidate classes that clear declared grounding and discriminability gates. Each retained frontier item can then seed new subquestions or rival hypotheses. A finite *epistemic multiplication factor*, k_{epi}, is introduced as the number of new live, readout-distinguishable, minimally grounded child classes produced per retained frontier item after pruning. The proposed architecture distinguishes productive multiplication from verbosity: ten linguistically different hypotheses that predict the same accessible records count as one class, and a large k_{epi} is not a truth score. The strongest proposed operating regime is not permanent expansion but *controlled supercritical exploration* ($k_{\text{epi}} > 1$) followed by evidence- and human-governed contraction ($k_{\text{epi}} < 1$) for appraisal and convergence. This paper synthesizes current evidence from AI Co-Scientist, autonomous discovery agents, LLM hypothesis-generation studies, human-AI co-creativity, question-generation systems, and documented diversity/hallucination failures. It argues that AI's distinctive leverage lies in parallel semantic transport, cross-domain bridge search, iterative mutation, adversarial critique, and low-cost branching, while human leverage remains strongest in problem ownership, question significance, value-sensitive constraints, interpretation of warrant, and revision commitments. The manuscript closes with falsifiable comparisons among human-only, one-shot AI, autonomous AI, and recursive human-AI workflows. The central hypothesis is that the largest augmentation will come from a role-separated, diversity-preserving, recursively gated system that increases the rate at which genuinely discriminable hypotheses become live, not merely the number of generated sentences.

Keywords: human-AI collaboration; scientific hypothesis generation; question formation; AI for science; epistemic chain reaction; candidate-set formation; semantic mobility; hypothesis diversity; multi-agent systems; model discrimination; Readout Universe; Readout Genesis

1 The Question: Where Is AI's Strongest Leverage?

Nature Methods recently described hypothesis generation as both fundamental and unusually opaque in the scientific process, and asked what follows when this task is outsourced to machines [1]. The framing is important, but "outsourcing" may be the wrong optimization target. Current systems already demonstrate substantial local capabilities: Co-Scientist recursively generates, debates, ranks, evolves, and grounds hypotheses under scientist-provided goals [2]; Robin links literature search, hypothesis generation, experimental proposal, data analysis, and hypothesis revision in a semi-autonomous loop [3]; and The AI Scientist grows archives of research directions, mutates prior ideas, searches literature, runs experiments, and writes papers [4].

These systems suggest that the central technical question is no longer whether AI can emit hypotheses. It can. The harder question is how to compose human and machine capacities so that a human question becomes a *larger and better-structured set of live epistemic alternatives* without producing a flood of redundant, hallucinatory, or homogenized outputs.

The three companion papers sharpen this target. Paper I separates reachability, accessibility, attraction, momentum, warrant, and truth, and defines hypothesis-generating imagination as bounded dynamic semantic mobility [16]. Paper II treats knowledge organization as a possible determinant of first-passage time and first-hit direction to usable hypotheses [17]. Paper III isolates candidate-set formation as a pre-evidential epistemic layer: evidence at cycle t can appraise only hypotheses that are live at cycle t, even though new evidence may restructure the candidate set at cycle $t + 1$ [18].

Central proposal [Dr/Open]. AI's strongest role is not to replace the bounded knower but to multiply the number of *new, live, readout-distinguishable and minimally grounded* hypothesis classes that can be reached from a human-retained question, while preserving explicit human control over problem ownership, question significance, decisive tests, and revision.

This proposal is intentionally stronger than "AI gives more ideas" and weaker than "AI autonomously discov-

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ers truth." The target is controlled amplification of the question-to-hypothesis interface.

2 What the Literature Already Shows

The literature therefore supports both the opportunity and the danger. The opportunity is cheap parallel branching, cross-domain retrieval, iterative refinement, and long-horizon tool use. The danger is equally clear: AI can produce fluent redundancy, homogenize collective outputs, mis-evaluate its own ideas, and hallucinate apparently plausible hypotheses. A useful augmentation theory must therefore count something stricter than generated tokens or even rated novelty.

3 From a Human Question to a Multiplying Frontier

3.1 Human question as retained seed

Let $Q_{\mathrm{H},t}$ be a question retained by the human under a declared problem, context, and constraint ledger. The human seed is not merely a prompt string. It includes the distinctions that define what is actually being asked and which constraints are forced, derived, posited, borrowed, or open in the companion architecture [16].

AI receives the question through a declared translation/readout and produces a finite set of question perturbations or subquestions,

$$\mathcal{Q}^{\mathrm{AI}}_{t+1} = \mathrm{Decompose}_{\mathrm{AI}}\big(Q_{\mathrm{H},t}, \mathcal{R}_{\le t}, \Pi_{\mathrm{H}}\big). \tag{1}$$

The purpose of decomposition is not to replace the original question. It is to expose hidden contrasts, mechanisms, boundary conditions, domain bridges, alternative causal directions, and rival explanations while preserving provenance.

3.2 Parallel semantic transport

For each retained subquestion Q_j, AI can run heterogeneous search/transport operators,

$$\mathfrak{T}^{\mathrm{AI}} = \{T_{\mathrm{lit}}, T_{\mathrm{data}}, T_{\mathrm{analogy}}, T_{\mathrm{cross}}, T_{\mathrm{counter}}, T_{\mathrm{model}}, \dots\}. \tag{2}$$

Here the strength of AI is not mysterious intelligence but cheap parallel access to many candidate routes. Literature search, data-driven induction, cross-domain analogy, graph traversal, simulation, and counterfactual perturbation can be executed simultaneously or asynchronously. Current systems already instantiate parts of this pattern [2, 6].

3.3 Candidate generation is not multiplication yet

Suppose the branching stage emits N hypothesis statements. Raw N is not the relevant quantity. Let $\sim_{D,t}$ be reader equivalence under the declared action set and horizon from Paper I. Then the raw candidate pool $\widetilde{\mathcal{H}}_{t+1}$ is collapsed into readout-distinguishable classes,

$$\mathcal{H}^{\mathrm{disc}}_{t+1} = \widetilde{\mathcal{H}}_{t+1} / \sim_{D,t} . \tag{3}$$

Ten paraphrases or ten semantically distant statements that imply the same accessible records count as one class for the current inquiry.

Next apply minimum propagation gates. Let

$$\mathcal{U}_{t+1} = \mathrm{Gate}\big(\mathcal{H}^{\mathrm{disc}}_{t+1}; W \ge w_0, \mathrm{provenance}, \mathrm{testability}\big), \tag{4}$$

where the gate requires a declared grounding floor, no silent loss of forced distinctions, and at least one accessible discriminating consequence against a live rival when the domain permits it. As in Paper II,

$$\boxed{\text{usable for propagation} \neq \text{true}.} \tag{5}$$

4 The Epistemic Multiplication Factor

The chain-reaction metaphor becomes useful only if its multiplication object is explicit. Let $\mathcal{F}_t$ be the retained frontier at cycle t: human-approved questions, subquestions, or candidate classes permitted to seed another expansion round. Let

$$\mathrm{New}_{t+1} = \mathcal{U}_{t+1} \setminus \bigcup_{s \le t} \mathcal{U}_s \tag{6}$$

be newly live, usable, readout-distinguishable classes after quotienting and gating.

Define the finite *epistemic multiplication factor*

$$\boxed{k_{\mathrm{epi}}(t) = \frac{|\,\mathrm{New}_{t+1}\,|}{\max(1, |\mathcal{F}_t|)}} \tag{7}$$

for a declared cycle, budget, reader, and propagation policy.

k_{epi} is *not* a creativity score, truth probability, or physical constant. It measures only productive branching under the declared gates. A system can have a huge raw generation count and $k_{\mathrm{epi}} \approx 0$ if nearly all outputs are redundant, ungrounded, or non-discriminating.

Three operating regimes follow by definition:

$$k_{\mathrm{epi}} < 1 : \text{contractive frontier}, \tag{8}$$
$$k_{\mathrm{epi}} \approx 1 : \text{roughly critical frontier}, \tag{9}$$
$$k_{\mathrm{epi}} > 1 : \text{expanding frontier}. \tag{10}$$

Controlled criticality hypothesis [Open]. The strongest human-AI augmentation will not maintain maximal branching indefinitely. It will alternate between an exploratory regime in which $k_{\mathrm{epi}} > 1$ for selected frontier regions and a convergent regime in which human judgment, readout-equivalence, warrant, decisive actions, and returned records force $k_{\mathrm{epi}} < 1$ until only a tractable live set remains.

This is the structural sense in which the process resembles a chain reaction. Each retained frontier item can seed several new epistemically non-equivalent children, which can seed further subquestions. The analogy stops there. No physical identity, energy law, or literal nuclear mechanism is implied.

5 Why the Human-AI Pair Can Be Stronger Than Either Alone

The strongest augmentation requires complementarity rather than duplicated roles.

5.1 Human strengths

Within this architecture, the human retains roles for which scale is not the main bottleneck:

- ownership of the retained problem and why it matters;
- selection and revision of the governing question;
- value-sensitive boundary conditions and unacceptable tradeoffs;
- interpretation of whether a candidate is meaningful in the lived or scientific context;
- commitment to decisive tests, stopping, abstention, and revision;
- responsibility for claims made publicly or acted upon.

These roles follow the bounded-knower and operator-attribution discipline of Paper I, not a claim that humans are infallible judges.

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Table 1: Nearest literatures. The proposed residual object is controlled multiplication of readout-distinguishable hypothesis classes, not raw idea count.

Literature	Representative work	What is already demonstrated	Role in the present architecture
Multi-agent scientific ideation	Co-Scientist [2]	Specialized agents generate, reflect, rank, evolve and meta-review hypotheses; iterative compute can improve rated hypothesis quality; scientists provide goals and steering.	Shows recursive branch-critique-refine loops are technically realizable.
Closed-loop discovery	Robin; AI Scientist [3, 4]	AI agents can connect hypothesis generation to experiments, analysis, revision, and research-idea archives.	Motivates propagation across cycles rather than one-shot prompting.
LLM hypothesis generation	Zhou et al.; Liu et al. [5, 6]	Iterative hypothesis updating can improve task performance; combining literature and data can outperform either source alone.	Supports heterogeneous inputs and iterative revision as multiplication mechanisms.
Expert-level ideation	Si et al. [7]	LLM research ideas were rated more novel than expert ideas in one controlled NLP study, but slightly less feasible; self-evaluation and diversity remained problems.	Warns that novelty alone is not productive epistemic multiplication.
Truthfulness / hallucination	Xiong et al. [8]	LLM-generated scientific hypotheses can be plausible yet untruthful; knowledge-grounding filters improve selection.	Requires a grounding gate before propagation.
Human-AI creativity	Doshi–Hauser; Meincke et al.; Luan et al. [9, 10, 11]	AI can improve individual outputs while reducing collective diversity; co-creation does not automatically improve over time; iterative co-development matters.	Strongest augmentation must preserve branch diversity and genuine human co-development.
Degree of collaboration	Uysal [12]	Moderate human-AI collaboration outperformed low or high collaboration in creativity tasks in a reported curvilinear effect.	Motivates role separation rather than maximal delegation.
Question formation	HybridQuestion; curiosity-driven question generation [13, 14]	AI can scale information gathering and candidate-question generation; forward-looking question selection remains judgment-sensitive; LLM question generation is measurable.	AI can amplify not only hypotheses but the branching of subquestions.
Human-centric hypothesis tools	Blin et al. [15]	Knowledge-enriched AI hypotheses can complement experts with efficiently produced alternatives while experts remain in control.	Supports augmentation as complementarity rather than substitution.

5.2 AI strengths

AI is unusually strong where bounded humans incur search costs:

- parallel retrieval across literatures too large for one person;
- repeated cross-domain bridge search;
- systematic variation and counterfactual perturbation;
- generation of rival mechanisms rather than one preferred story;
- low-cost recursion over candidate descendants;
- critique, ranking, mutation, and test proposal across many branches;
- persistent logging of rejected paths and provenance.

Co-Scientist's specialized agents and tournament evolution, Robin's iterative laboratory loop, and AI Scientist's idea archive show pieces of this computational leverage in existing systems [2, 3, 4].

5.3 A synergy quantity

Let $D_X^{\text{use}}(B,Q)$ denote the number or effective diversity of usable readout classes reached under budget B by mode $X \in \{\mathrm{H}, \mathrm{AI}, \mathrm{H+AI}\}$. Define a simple augmentation ratio

$$\Sigma_{\mathrm{H+AI}} = \frac{D_{\mathrm{H+AI}}^{\text{use}}(B,Q)}{\max\{D_{\mathrm{H}}^{\text{use}}(B,Q), D_{\mathrm{AI}}^{\text{use}}(B,Q), 1\}}. \tag{11}$$

$\Sigma_{\mathrm{H+AI}} > 1$ is a minimal operational signature that the pair reaches more usable classes than the better component alone under the same declared budget. It does not prove cognitive "synergy" in a metaphysical sense; it gives the claim a falsifiable surface.

6 Why Maximum Delegation Is Not Maximum Amplification

Several results block a naive "more AI = more creativity" law. Doshi and Hauser found that generative AI increased individual story creativity while reducing collective novelty [9]; Meincke and colleagues similarly reported reduced idea diversity in brainstorming pools exposed to ChatGPT [10]. A large study of research ideation found LLM-generated ideas rated as more novel but slightly less feasible than expert ideas, alongside failures of self-evaluation and diversity [7]. Human-AI creativity experiments also report that collaboration benefits can be curvilinear and that iterative co-development, not mere repeated exposure, is important [12, 11].

These findings imply a non-collapse rule:

$$\text{more AI output} \neq \text{more epistemic diversity}, \quad \text{more epistemic diversity} \neq \text{better warrant}. \tag{12}$$

A chain-reaction architecture therefore needs at least four controls.

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6.1 Diversity control

Branching agents should differ in retrieval slices, model families, prompts, domain priors, or explicit roles. Multiple copies of one model using the same corpus and ranking policy may produce numerical multiplicity without epistemic multiplicity. Diversity is measured *after* the readout quotient.

6.2 Grounding control

Hypotheses that rely on unsupported premises should not propagate merely because they are novel. TruthHypo and related work show that LLM scientific hypotheses can be plausible yet poorly grounded [8]. External literature search, data checks, formal tools, or expert inspection should therefore precede recursive propagation.

6.3 Human question control

AI can generate questions, and hybrid systems can help identify forward-looking research questions, but question significance remains partly value- and horizon-sensitive. HybridQuestion found greater AI-human divergence on prospective question forecasting than on retrospective recognition of breakthroughs [13]. The architecture therefore distinguishes AI-generated subquestions from the human-retained governing question.

6.4 Convergence control

An expanding frontier must eventually be forced through finite appraisal. Readout-equivalent variants are collapsed; weakly grounded branches are pruned; decisive actions are proposed; returned records update warrant; and the frontier contracts. Without this phase, recursive ideation becomes semantic inflation rather than scientific progress.

Runaway failure mode. If AI-generated candidates are allowed to seed further candidates before equivalence, provenance, grounding, and contradiction checks, the system can amplify hallucinations, duplicated assumptions, and model-specific biases faster than it amplifies useful alternatives. A large apparent k_{epi} before gating is therefore diagnostically meaningless.

7 A Proposed Human-AI Chain-Reaction Workflow

The full architecture can be written as a cycle:

$$P_t \xrightarrow{\text{H}} Q_{\text{H},t} \xrightarrow{\text{AI}} \mathcal{Q}_{t+1} \xrightarrow{\mathfrak{T}^{\text{AI}}} \widetilde{\mathcal{H}}_{t+1} \tag{13}$$

$$\xrightarrow{/\sim_{D,t}} \mathcal{H}^{\text{disc}}_{t+1} \xrightarrow{\text{Gate}} \mathcal{U}_{t+1} \xrightarrow{\text{H}} \mathcal{F}_{t+1} \tag{14}$$

$$\xrightarrow{\text{Spawn}} (\mathcal{Q}_{t+2}, \widetilde{\mathcal{H}}_{t+2}) \quad \text{or} \quad \xrightarrow{u^*} \delta_{t+1} \xrightarrow{\text{revise}} P_{t+1}. \tag{15}$$

The key design move is that propagation and testing are both allowed. A retained hypothesis can either seed a new subquestion or be sent to a discriminating action. Which branch is selected is a policy decision, not a property of the language model.

A practical implementation can use six roles:

1. **Human Question Holder**: states the problem, significance, and non-negotiable distinctions.
2. **Question Expander**: generates orthogonal subquestions, boundary cases, and inversions.
3. **Bridge Searchers**: independently traverse literatures, data, analogies, and formal models.
4. **Hypothesis Constructors**: convert bridges into explicit candidate relations and predicted consequences.
5. **Adversarial Pruners**: seek duplicates, hidden assumptions, hallucinations, and missing rivals; compute reader-equivalence classes.
6. **Human-Evidence Gate**: chooses which classes deserve propagation, test design, or rejection, and records the reason.

The strongest version is therefore neither "one human plus one chatbot" nor "many agents with no human." It is a role-separated system in which the human question acts as a stable retained seed while machine branching is deliberately heterogeneous and machine output cannot self-certify its own propagation rights.

8 Testable Propositions

8.1 H1 - Productive multiplication

[Open]. Under matched problem, evidence, and resource budgets, the recursive role-separated human-AI workflow will produce a larger number/effective diversity of usable readout-distinguishable classes than human-only, one-shot AI assistance, or autonomous AI-only conditions:

$$D^{\text{use}}_{\text{H+AI,chain}} > \max\{D^{\text{use}}_{\text{H}}, D^{\text{use}}_{\text{AI},1shot}, D^{\text{use}}_{\text{AI},auto}\} \tag{16}$$

for at least some complex scientific task classes.

8.2 H2 - First-passage acceleration

[Open]. The same workflow should reduce time to the first usable candidate class without requiring faster acceptance of a final claim:

$$T^{\text{H+AI,chain}}_{\mathcal{U}} < T^{\text{H}}_{\mathcal{U}} \tag{17}$$

for at least some tasks, while warrant and truth remain separate outcomes. This directly connects to Paper II's first-passage program.

8.3 H3 - Controlled expansion beats maximal expansion

[Open]. A schedule that permits exploratory $k_{\text{epi}} > 1$ but later forces quotienting, grounding, and convergence will outperform an always-expand policy on a joint profile of usable-class diversity, hallucination rate, calibration, and experiment burden.

8.4 H4 - Heterogeneity matters after quotienting

[Open]. Increasing nominal agent count will add little if agents share the same retrieval and model basin. Heterogeneous agents should improve the number of new classes only insofar as their outputs remain distinct after readout-equivalence and grounding gates.

8.5 H5 - Human question sovereignty matters

[Open]. Removing human control of the governing problem/question may increase raw candidate count while reducing relevance, accountability, or willingness to revise. Conversely, excessive human steering may suppress cross-domain expansion. The strongest regime is therefore expected to be role-separated rather than maximally human- or AI-dominated.

9 A Decisive Experimental Program

A direct test should compare at least four modes on the same hard scientific questions, evidence packages, and resource budgets:

1. human expert alone;
2. human + one-shot LLM ideation;
3. autonomous multi-agent AI with only the initial question fixed;
4. recursive human-AI chain-reaction workflow with explicit quotient, grounding, and propagation gates.

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Use synthetic scientific worlds for causal control and real expert problems for external validity. Pre-register a catalog of hidden mechanism families in synthetic tasks, then blind evaluators to condition when mapping generated candidates into readout classes. In real tasks, use domain experts to define candidate equivalence and minimum pursuit-worthiness before seeing condition labels.

Primary outcomes should include:

- D^{use}: number/effective diversity of usable readout classes;
- k_{epi} across cycles;
- $T_{\mathcal{U}}$: time to first usable class;
- redundancy ratio before and after quotienting;
- provenance defect and hallucination rates;
- discriminating-action yield per candidate class;
- expert-rated relevance and feasibility;
- revision quality after returned records;
- operator attribution: who controlled question, pruning, test choice, and revision.

A second experiment should vary only collaboration intensity. The literature already warns that more AI involvement is not monotonically better [12]. The chain-reaction account predicts that performance depends less on total AI usage than on *where* AI is allowed to expand and where human/evidence gates force contraction.

A third experiment should deliberately seed a false but highly attractive basin. The strongest augmentation system should both exploit AI's ability to expand alternatives and demonstrate a higher probability of escaping the seeded basin than either humans or homogeneous AI agents alone.

10 Relationship to Papers I-III

Paper IV does not replace the previous sequence. It composes it.

Paper I supplies the local dynamics:

$$\text{Question} \to \text{bounded semantic mobility} \to \mathcal{H}^{\text{disc}}. \quad (18)$$

Paper II supplies the routing observables:

$$\mathcal{G}^K \to \left(\mathcal{L}(\tau_{\mathcal{U}}), \pi^{\text{first}}\right). \quad (19)$$

Paper III supplies the pre-evidential candidate-set problem:

$$\mathcal{H}^{\text{disc}} \xrightarrow{\Psi} \mathcal{C}_t \xrightarrow{\text{appraisal}} \delta_{t+1}. \quad (20)$$

Paper IV adds recursive human-AI propagation:

$$\begin{aligned} Q_{\mathrm{H}} &\to \text{AI branching} \to \text{readout classes} \\ &\to \text{gated frontier} \to \text{new questions / hypotheses.} \end{aligned} \quad (21)$$

and proposes k_{epi} as the finite multiplication diagnostic across cycles.

The sequence therefore moves from *what imagination is*, to *how knowledge routes it*, to *which hypotheses become live*, to *how human-AI systems might deliberately amplify that transition*.

11 Boundaries and Non-Claims

The paper does not claim that:

- AI-generated hypotheses are more truthful than human hypotheses;
- k_{epi} is a physical law, a truth probability, or a universal creativity measure;
- maximal branching is desirable;
- multi-agent copies of one model automatically provide epistemic diversity;
- humans are unbiased question selectors or infallible final judges;
- human question ownership forbids AI from proposing question revisions;
- readout-equivalence is absolute identity;
- usable hypotheses are true hypotheses;
- Co-Scientist, Robin, AI Scientist, or any current system already implements the full architecture proposed here;
- one collaboration regime will dominate across every domain;
- autonomous science is impossible or inherently inferior;
- the chain-reaction metaphor implies any literal nuclear process;
- this K0 manuscript has passed independent checking.

12 Future Work

The strongest future question. Can a human-AI system be tuned to increase *productive epistemic branching* rather than raw generative volume - preserving the human's problem and value commitments while using AI to maximize the rate at which new, discriminable, minimally grounded alternatives become live?

Several subproblems follow. First, k_{epi} requires robust equivalence coding; automated semantic similarity is insufficient because epistemic equivalence depends on predicted readouts. Second, multi-model or multi-agent diversity needs a provenance measure that distinguishes genuinely independent bridges from synchronized model priors. Third, the human cost of reviewing a supercritical frontier may dominate any generation advantage; adaptive pruning policies are therefore essential. Fourth, returned empirical records should eventually enter the multiplication policy so that branches supported or contradicted by the world alter future question decomposition. Fifth, high-stakes domains need governance criteria for how much candidate widening is obligatory before acting.

Question formation itself deserves special study. HybridQuestion and recent curiosity-oriented benchmarks suggest that AI can generate candidate questions at scale, but the problem of which question deserves human commitment remains open [13, 14]. The chain-reaction architecture predicts that the most consequential AI contribution may sometimes occur *one step before hypothesis generation*: by generating a small number of load-bearing question perturbations that change the geometry of every hypothesis that follows.

13 Conclusion

The largest potential contribution of AI to scientific imagination may not be a better answer. It may be a controlled change in the multiplication rate of alternatives.

A bounded human begins with a problem and a question. AI can cheaply branch that question across literature, data, models, analogies, and counterfactuals. But branching alone is not augmentation. Scientific amplification occurs only when branches survive provenance checks, collapse into genuinely distinct readout classes, clear a minimum grounding floor, and are either tested or allowed to

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seed another question. The resulting process is recursive:

$$\begin{aligned}\text{Question} &\to \text{Branches} \to \text{Distinct Hypotheses}\\ &\to \text{Gates} \to \text{New Questions} \to \cdots\end{aligned} \tag{22}$$

The proposed epistemic multiplication factor makes this idea fail-able. If a recursive human-AI system produces thousands of outputs but almost no new usable readout classes, then k_{epi} is low and the claimed "amplification" is mostly verbosity. If human-AI co-development raises k_{epi}, lowers first-passage time, preserves diversity after quotienting, and improves escape from misleading basins without increasing hallucination or eroding human control, then the stronger claim gains support.

The central conjecture is therefore not that AI should think instead of humans. It is that a human-retained question can become a seed for a *controlled epistemic chain reaction*: temporary supercritical expansion of genuinely different hypotheses, followed by deliberate contraction under warrant, experiment, and revision. If that conjecture survives testing, the most powerful scientific interface may be neither a chatbot nor an autonomous scientist, but a machine for turning one human question into many testable ways the world could answer back.

Process Note

This manuscript was developed under the GLOSA "Rigour Without Infrastructure" workflow. The human author originated and selected the research direction, including the chain-reaction intuition. AI assisted with retrieval, cross-paper synthesis, formalization, adversarial literature comparison, drafting, and typesetting. The literature review was performed after the central idea had been articulated and must not be represented as preregistered or literature-blind. The manuscript remains K0 because the maker route cannot certify its own literature interpretation, formal bridges, or final claims.

References

[1] Nature Methods Editors. What's your hypothesis? *Nature Methods* 23, 865–866 (2026). doi:10.1038/s41592-026-03104-y.

[2] J. Gottweis, W.-H. Weng, A. Daryin, et al. Accelerating scientific discovery with Co-Scientist. *Nature* 655, 487–496 (2026). doi:10.1038/s41586-026-10644-y.

[3] A. E. Ghareeb et al. A multi-agent system for automating scientific discovery. *Nature* 655, 497–505 (2026). doi:10.1038/s41586-026-10652-y.

[4] C. Lu et al. Towards end-to-end automation of AI research. *Nature* (2026). doi:10.1038/s41586-026-10265-5.

[5] Y. Zhou, H. Liu, T. Srivastava, H. Mei, and C. Tan. Hypothesis Generation with Large Language Models. In *NLP4Science 2024*, 117–139 (2024). doi:10.18653/v1/2024.nlp4science-1.10.

[6] H. Liu, Y. Zhou, M. Li, C. Yuan, and C. Tan. Literature Meets Data: A Synergistic Approach to Hypothesis Generation. In *ACL 2025*, 245–281 (2025). doi:10.18653/v1/2025.acl-long.12.

[7] C. Si, D. Yang, and T. Hashimoto. Can LLMs Generate Novel Research Ideas? A Large-Scale Human Study with 100+ NLP Researchers. ICLR 2025; arXiv:2409.04109.

[8] G. Xiong, E. Xie, C. Williams, et al. Toward Reliable Scientific Hypothesis Generation: Evaluating Truthfulness and Hallucination in Large Language Models. *IJCAI 2025*, 7849–7857. doi:10.24963/ijcai.2025/873.

[9] A. R. Doshi and O. P. Hauser. Generative AI enhances individual creativity but reduces the collective diversity of novel content. *Science Advances* 10, eadn5290 (2024). doi:10.1126/sciadv.adn5290.

[10] L. Meincke, G. Nave, and C. Terwiesch. ChatGPT decreases idea diversity in brainstorming. *Nature Human Behaviour* 9, 1107–1109 (2025). doi:10.1038/s41562-025-02173-x.

[11] Y. L. Luan, Y. J. Kim, and J. Zhou. Augmented Learning for Joint Creativity in Human-GenAI Co-Creation. *Information Systems Research* (2025). doi:10.1287/isre.2024.0984.

[12] H.-C. B. Huang. Unlocking creativity with artificial intelligence (AI): Field and experimental evidence on the Goldilocks (curvilinear) effect of human-AI collaboration. *Journal of Experimental Psychology: General* (2025). PMID:41051839; doi:10.1037/xge0001838.

[13] K. Zhao, F. Xu, Y. Li, and T.-Y. Liu. HybridQuestion: Human-AI Collaboration for Identifying High-Impact Research Questions. arXiv:2602.03849 (submitted 2025). doi:10.48550/arXiv.2602.03849.

[14] S. R. Javaji and Z. Zhu. What Would You Ask When You First Saw $a^2 + b^2 = c^2$? Evaluating LLM on Curiosity-Driven Question Generation. In *IJCNLP-AACL 2025*, 2322–2354 (2025). doi:10.18653/v1/2025.ijcnlp-long.126.

[15] I. Blin, I. Tiddi, G. Spadaro, et al. Automated Hypothesis Generation for Human Cooperation Studies. *HHAI 2025*. doi:10.3233/FAIA250652.

[16] Y. Lahtee. From Problem to Hypothesis: Dynamic Semantic Mobility, Bounded Knowers, and the Readout-Discriminable Hypothesis Space. GLOSA K0 working paper (2026).

[17] Y. Lahtee. Knowledge Topology and the First Passage to Usable Hypotheses: A Readout Theory of Discovery Time and Direction. GLOSA K0 working paper (2026).

[18] Y. Lahtee. Before Evidence Can Decide: Candidate-Set Formation, Discovery Routing, and Unconceived Alternatives. GLOSA K0 agenda paper (2026).

[19] Y. Lahtee. Readout Universe: A Philosophy and Logic for Grounding Claims. GitHub repository (2026). https://github.com/morrocwi/readout_universe.

[20] Y. Lahtee. Readout Genesis Universal Technical Whitepaper, v1.2.0. GitHub repository (2026).

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Choice Begins Before Choice: Meaning-Shaped Accessibility, Live Possibility, and Effective Agency in Human-AI Systems

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What the chapter states. "Human beings do not choose from the whole space of what is objectively possible. They choose from a history-shaped subset of possibilities that have become sufficiently meaningful, accessible, discriminable, and actionable to enter practical consideration. Choice therefore begins before choice."

Status. "K0 strong-form conceptual and measurement architecture. The paper does not claim a metaphysical theory of free will or a validated universal scale of human agency. [. . .] This claim is constrained by explicit non-collapse laws, causal and counterfactual requirements, and deletion conditions."

Falsifier(s). "If resource-only models predict equally well, the meaning layer should be narrowed."

Read before. *Experience Is Meaning-Giving: A Strong-Form Readout-Retention Theory of Phenomena, Resonance, Rhythm, Accumulation, and Transformative Release*; *Potential as a Readout: Witnessed Envelopes, Two-Layer Agency, and the Measurement of Pseudo-Peace*.

Feeds into. *Before Meaning, Before Choice: A Readout-Native Derivation of Experience, Live Possibility, Agency, and Human-AI Return* (Part 6).

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Choice Begins Before Choice

Meaning-Shaped Accessibility, Live Possibility, and Effective Agency in Human-AI Systems

A strong-form synthesis of Agency Potential, Experience Is Meaning-Giving, Human LoRA, Fusion/Tunnel, and CTSA Human Return

Yaoharee Lahtee | Bangkok, Thailand | Full Paper v1.0 - Strong Claim

Status. K0 strong-form conceptual and measurement architecture. The paper does not claim a metaphysical theory of free will or a validated universal scale of human agency. It advances a narrower but strong thesis: human choice is downstream of a meaning-shaped field of *live possibilities*. A policy or action can be physically possible and even structurally feasible while remaining practically absent from the agent's current field of consideration because it is unreadable, unnameable, unintelligible, normatively unavailable, affectively blocked, or historically inaccessible. Power, learning, violence, institutions, relationships, and AI can therefore alter agency before overt choice by changing which possibilities become live. This claim is constrained by explicit non-collapse laws, causal and counterfactual requirements, and deletion conditions.

Central claim. Human beings do not choose from the whole space of what is objectively possible. They choose from a history-shaped subset of possibilities that have become sufficiently meaningful, accessible, discriminable, and actionable to enter practical consideration. **Choice therefore begins before choice.** Experience gives meaning to phenomena; retained experience changes future readout; resonance can make some possibilities feel fitted to prior experience; rhythm and repeated traversal can make some routes easier to re-enter; social structures can enable or suppress conversion from possibility to action; and Human-AI interaction can recursively reshape the state from which the next prompt and choice are generated. Effective agency is therefore not merely the presence of options or the occurrence of action, but the corrigible capacity to read live differences, own a decision, causally enact it, and preserve routes of objection and revision. **Power can act on agency by acting on meaning before it acts on choice.**

1. Abstract

Theories of agency often begin where choice becomes visible: preference, deliberation, decision, action, or outcome. This paper argues that an analytically prior layer is required. Human beings do not deliberate over the full set of objectively possible actions; they deliberate over a history-shaped field of *live possibilities*: options that have become sufficiently meaningful, accessible, discriminable, and actionable for a situated agent to treat them as real candidates. The central thesis is therefore *choice begins before choice*. The paper develops this claim by joining two strands of the Human-AI Readout Programme. *Effective, Corrigible Agency Potential* models agency as a task- and context-relative envelope of feasible, actor-directed, causally effective, and corrigible readout-action loops. *Experience Is Meaning-Giving* models experience as the relation through which phenomena become significant for an embodied, history-bearing reader. Their synthesis exposes a missing interface: between objective or structural possibility and overt choice lies a meaning-shaped accessibility field.

The paper distinguishes six levels that are often collapsed: physically possible, structurally feasible, experientially live, chosen, enacted, and evaluator-visible. It introduces a *Live Possibility Field* as a domain-level object and shows how meaning-giving, autobiographical and affective significance, resonance, rhythmic recurrence, semantic attraction, recent-path momentum, retention, and social interpretation can alter the probability that a feasible option becomes practically thinkable. This does not imply that every unconsidered option was suppressed, that all social influence is domination, or that felt resonance is consent. Structural deprivation requires a feasible counterfactual, a named causal mechanism, and an independent normative reason why the closure is harmful or unjust.

The framework is placed in dialogue with Sen, Kabeer, Emirbayer and Mische, Bazzani, Foucault, Pettit, Fricker, Mahmood, Galtung, empowerment theory, and recent Human-AI feedback research. It then extends the Pre-Prompt Human State Principle: AI may alter agency not only by recommending an action but by changing which distinctions, questions, and alternatives become live before the next prompt. The resulting empirical programme asks whether live-possibility measures add out-of-sample explanatory value beyond resources, stated preferences, option count, arousal, familiarity, schema activation, and current prompt content. The strongest implication is methodological and political: to assess human potential, freedom, violence, learning, or AI influence, one must ask not only *what did the person choose?* but *what could become a live choice for this person, under this history and structure, and what made other possibilities disappear from practical view?*

2. The Residual Problem: The Missing Layer Before Choice

The agency-potential work already separates a decision that was actually taken from the envelope

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of policies that could be accessed, used, or learned within declared time and resource constraints [1]. The experience theory adds that the human reader does not encounter the world as neutral data: phenomena become experience through meaning-giving, and retained experience can alter the operator through which later situations become significant [2, 3]. These two results imply a missing layer.

A person can face an option that is physically open and legally permitted, yet not experience it as a real option. The option may be unknown, unnameable, culturally unintelligible, emotionally unapproachable, socially punishable, conceptually unavailable, or absent from the agent's current problem representation. Conversely, an option may become intensely salient without being objectively feasible or epistemically warranted. Choice therefore cannot be modeled as a direct mapping from an objective option set to a selected action.

The strong-form sequence is:

world / structure → meaning-shaped readout → live possibility field → choice → action → feedback → retained future reader.

The claim is not that deliberation is unreal. It is that deliberation is downstream of a prior selection problem: *which possibilities become candidates for deliberation at all?*

3. Internal Genealogy: From Potential to Live Possibility

The present paper does not replace prior programme objects. It makes their dependency relation explicit.

The new residual interface is therefore:

> *Between what a person could in principle do and what the person actually chooses lies a history-shaped field of possibilities that have become sufficiently meaningful and accessible to count as live candidates.*

4. External Literature: What Is Already Known, and What Remains

The thesis has close neighbors and should not be sold as if agency theory had ignored structure, temporality, power, or interpretation.

4.1 Agency, capability, and the conditions of choice

Sen distinguishes agency from well-being and insists that a person's pursuit of goals and values cannot be reduced to personal welfare [7]. Kabeer explicitly analyzes empowerment through resources, agency, and achievements and qualifies choice by its conditions, content, and consequences [8]. These are direct constraints on any claim that agency can be inferred from outcomes or resource possession alone.

Bazzani's account of agency as a conversion process is an especially close neighbor: agency is situated in micro-, meso-, and macro-level conversion factors and temporal processes rather than treated as an abstract possession [10]. The present paper therefore does not claim novelty for the proposition that context converts resources into effective agency. Its residual contribution is the explicit placement of a meaning-shaped *live possibility field* between feasible capacity and overt choice, together with a readout/visibility audit and Human-AI extension.

4.2 Agency is temporally embedded

Emirbayer and Mische conceptualize agency as a temporally embedded process combining iterational, projective, and practical-evaluative dimensions [9]. This strongly supports the refusal to treat choice as a history-free instant. The present architecture adds a narrower claim: temporal history may alter not only how a known option is evaluated but whether the option becomes sufficiently accessible to function as a live candidate.

4.3 Power can act on the field of possible action

Foucault's account of power is structurally relevant because power acts upon the possible actions of others rather than only through direct force [11]. Pettit's non-domination likewise shows why the absence of actual interference is insufficient for freedom: arbitrary capacity to interfere can alter the choice situation even when no intervention occurs [12]. These traditions already defeat any simplistic equation of freedom with unblocked motion.

The present thesis sharpens one interface: power can operate not only by removing an option after it is recognized but by altering the conditions under which the option becomes intelligible, permissible, credible, or practically available to the agent. This is a meaning-and-accessibility claim, not a replacement for broader theories of domination.

4.4 Epistemic and interpretive resources

Fricker's hermeneutical injustice identifies cases in which prejudicial flaws in shared interpretive resources obstruct a person's capacity to render significant areas of social experience intelligible [13]. This is a direct neighbor of the present claim that unnameability and interpretive scarcity can reduce the live possibility field. The framework therefore must not present "meaning access" as a newly discovered problem. Its contribution is to connect interpretive availability to a task-specific agency loop and to distinguish agent limitation from evaluator invisibility.

4.5 Agency is not identical to resistance

Mahmood's critique of liberal-feminist assumptions is crucial: embodied practices, discipline, religious observance, and inhabiting norms can be forms of agency rather than evidence of failed emancipation [14]. The present framework therefore rejects the

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Table 1: Internal programme genealogy and the residual interface identified here.

Programme object	Retained result	Role in this paper
Readout Genesis / Readout Universe	Readout is finite and reader-conditioned; retained difference, record, and world are not collapsed; history and correction matter.	Supplies the distinction between objective structure, agent readout, and evaluator readout.
Effective, Corrigible Agency Potential	Agency potential is an envelope of feasible readout–decision–execution–feedback loops; potential, performance, and evaluator visibility must be separated.	Supplies the outer feasibility envelope and the R–D–X–F agency cycle.
Experience Is Meaning-Giving	A phenomenon becomes lived experience through a constrained meaning relation; resonance, rhythm, retention, and transition readiness shape later readout.	Supplies the pre-choice meaning layer and history-shaped accessibility.
Experience Is the Human LoRA	Selected experiential residues may alter future interpretation and action without being literal neural LoRA.	Supplies cross-time operator change.
From Problem to Hypothesis	Reachability differs from accessibility; attraction and recent-path momentum can make some semantic routes easier to re-enter.	Supplies a formal precedent for live-option weighting and path dependence.
Fusion / Tunnel	Repeated Human-AI traversal can alter later accessible routes; amplification is epistemically neutral; resistance is required.	Supplies recursive coupling and the risk that repeated routes become tunnels.
CTSA Human Return	Coupled performance is not human capability; after AI removal, retained Concept, Tool, Skill, or Alternative Return must be demonstrated.	Supplies the upper measurement boundary for durable human gain.

inference that a norm-conforming choice is non-agentic merely because an outsider would prefer resistance. A live possibility field is reader-relative and normatively contestable; its evaluation must preserve the agent's own projects while still allowing independent evidence of coercion, domination, or blocked objection.

4.6 Informational control and empowerment

Klyubin, Polani, and Nehaniv formalize empowerment as agent-centric control using channel capacity from actions to future perceptions [15]. This is a close formal ancestor of any informational treatment of agency. The present paper does not claim that control, information, or action-perception loops are novel. It adds actor ownership, corrigibility, live-option accessibility, and evaluator identifiability as separate constraints.

4.7 Peace and structural violence

Galtung's structural-violence formulation is relevant because it compares actual outcomes with potential and explicitly recognizes the difficulty of specifying potential, especially in psychological domains [16]. His later account of cultural violence concerns cultural aspects that legitimize direct or structural violence [17]. The present framework uses this literature cautiously: a narrowed live possibility field is not automatically violence. A structural-violence claim additionally requires a feasible counterfactual, a responsible causal mechanism, and a normative reason why the blocked agency matters.

4.8 Human-AI feedback

Recent experiments show that repeated interaction with biased AI can alter later human perceptual, emotional, and social judgments and amplify bias [18]. The importance here is not that AI always manipulates users. It is that the human state entering a later judgment is demonstrably history-sensitive. This makes the pre-choice and pre-prompt layers empirically consequential.

5. The Core Thesis: Choice Begins Before Choice

The defended thesis is:

> *A human choice is generated from a field of possibilities that has already been filtered and weighted by meaning, history, embodiment, social structure, interpretive resources, and current context. The architecture of agency therefore begins before the moment of overt decision.*

This thesis yields five propositions.

P1 - Objective possibility is not practical possibility. An action may be physically or legally available while remaining absent from the agent's practical field.

P2 - Practical possibility is meaning-shaped. An option becomes live only insofar as the agent can in some way discriminate, interpret, imagine, value, fear, reject, desire, delegate, or otherwise relate to it as a candidate for action.

P3 - Meaning-shaping can alter agency. Changing how an option is named, framed, remembered, socially recognized, or connected to prior experience can alter whether it becomes live before any final preference or decision is expressed.

P4 - Choice can be real within a narrowed field. A person may genuinely choose among the live op-

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tions available while still having a smaller effective agency field than under a feasible counterfactual. Narrowing the field does not make every remaining choice fake.

P5 - Expansion can also be harmful. Increasing the number or salience of live options is not automatically emancipatory. Manipulative, dangerous, false, or overwhelming options can become more accessible. Agency expansion therefore remains distinct from epistemic warrant and ethical value.

6. Six Levels That Must Not Collapse

Let $\Pi_t^{phys}(g)$ denote actions physically possible under the declared environment for task g. Let $\Pi_{A,t}^{feas}(g)$ denote the subset the agent could access, use, or learn within declared resources, time, and structural conditions. Define the *live possibility set*

$$\Pi_{A,t}^{live}(g) \subseteq \Pi_{A,t}^{feas}(g)$$

as those feasible policies that are sufficiently accessible to enter practical consideration under the current human state and context.

A realized choice $\pi_{A,t}^{choice}$ is selected from this live set, an enacted policy $\pi_{A,t}^{act}$ is what is actually implemented, and the evaluator observes only a record

$$Y_{obs} = \mathcal{O}_q(H_{0:T}).$$

The intended ordering is therefore

$$\boxed{\begin{aligned} &\pi^{choice} \in \Pi^{live} \subseteq \Pi^{feas} \subseteq \Pi^{phys}, \\ &\quad \pi^{act} \neq \pi^{choice} \text{ possible}, \\ &\quad Y_{obs} \neq H_{0:T}. \end{aligned}}$$

These are analytic levels, not claims that the brain contains literal option sets. Their purpose is to prevent six common collapses:

$$\text{possible} \neq \text{feasible} \neq \text{live} \neq \text{chosen} \neq \text{enacted} \neq \text{observed}.$$

An evaluator may see no objection even when objection is live but strategically withheld; may see compliance when refusal was never made safe; may see success when a third party effectively controlled the process; or may see low competence because the measurement interface hides available capacity. Agency assessment is therefore itself a readout problem.

7. A Live Possibility Field

The binary set Π^{live} is useful for exposition but too coarse for modeling. Let $\kappa_{A,t}(\pi \mid g)$ be a declared accessibility weight over feasible policies. It may depend on retained history, conceptual availability, salience, social permission, fear, expected cost, prior traversal, language, tool access, and current framing. The present paper does not assert that κ is a literal neural probability.

Define a task-relative live possibility profile

$$\mathcal{L}_{A,t}(g) = \left\{ (\pi, \kappa_{A,t}(\pi \mid g)) : \pi \in \Pi_{A,t}^{feas}(g) \right\}.$$

A declared operational threshold τ_{live} may be used for a particular study:

$$\Pi_{A,t}^{live}(g) = \{\pi : \kappa_{A,t}(\pi \mid g) \geq \tau_{live}\},$$

but the threshold is measurement-specific and should not be promoted to a universal psychological constant.

The key object is not option count but the *distribution of practical accessibility*. Two people can face the same formal menu and have radically different live fields. One may see refusal, negotiation, delegation, delay, appeal, or exit as genuine candidates; another may not.

7.1 Meaning-giving as the pre-choice gate

From *Experience Is Meaning-Giving*, a phenomenon becomes experience through a reader-conditioned meaning relation. Applied to agency, an objective affordance or formal option becomes practically relevant only after it acquires some significance for the agent. Meaning may be affective, pragmatic, autobiographical, conceptual, or epistemic. The live field is therefore downstream of meaning-giving but upstream of final choice.

This does not imply that every option must be verbally named. A person can flee danger, seek comfort, or reject an interaction before explicit lexical articulation. The claim is that some significance relation must make the route functionally available to the situated agent.

8. How History Shapes the Live Field

The live possibility field is not rebuilt from zero at each moment. Four history-shaped processes from the existing programme become relevant.

8.1 Resonance: experiential fit

Resonance is defined as congruence between a current externally prompted experience and retained internal experiential organization [2]. In agency terms, a proposal can become unusually live because it fits autobiographical memory, identity, fear, hope, or prior meaning structure. Resonance is neither consent nor warrant:

$$Resonance \neq Consent, \qquad Resonance \neq Truth.$$

A manipulative narrative can resonate. A liberating alternative can fail to resonate because it is unfamiliar. Resonance therefore helps explain uptake, not legitimacy.

8.2 Rhythm: structured return

Rhythm is structured recurrence, spacing, interruption, emphasis, and return across ordered history [2]. In agency terms, repeated exposure to a route

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can alter re-entry even when total content is held constant. This yields a strong but open question: does the temporal organization of how alternatives are returned to alter their later practical accessibility beyond exposure count?

8.3 Attraction and recent-path momentum

The semantic-mobility work distinguishes reachability from accessibility and proposes history-shaped attraction and recent-path momentum: recently traversed routes can become easier to re-enter for a finite window [4]. Applied here, the current live field may be biased toward the paths just used, rehearsed, socially reinforced, or institutionally normalized. Accessibility remains distinct from evidence and truth.

8.4 Retention: the next chooser may not be the same reader

Human LoRA proposes that selected experiential residues can alter the future interpretive operator [3]. This produces a crucial result for agency:

> *The person who chooses at $t+1$ may be the same biological individual but not the same effective reader of possibility as at t.*

The option field can therefore be reshaped by prior experience even before explicit beliefs or stated preferences visibly change.

9. Effective, Corrigible Agency Revisited

The earlier Agency Potential paper defines, for task g, four joint conditions: R_g for reading relevant differences, D_g for actor-directed decision or refusal, X_g for causal translation of choice into relevant action/outcome, and F_g for usable feedback, objection, and revision [1]. Its task-specific potential is

$$p^*_{A,g} = \max_{\pi \in \Pi_A^{feas}(g)} \Pr(R_g \cap D_g \cap X_g \cap F_g).$$

The present paper adds a pre-choice distinction rather than replacing this equation. At any instant, the policy actually considered must come from the current live subset:

$$\Pi_{A,t}^{live}(g) \subseteq \Pi_A^{feas}(g; h, z, T, B).$$

The feasible envelope therefore asks *what could become usable under the declared horizon and resources?*; the live field asks *what is practically available to this agent now?*

This distinction makes intervention analysis more precise. Education may enlarge Π^{live} by making alternatives intelligible. Legal reform may enlarge Π^{feas} without immediately changing Π^{live}. A safe complaint channel may increase both live refusal and F_g. AI may temporarily enlarge live conceptual routes while reducing unaided Human Return after removal. A coercive environment may leave formal feasibility unchanged while making refusal effectively non-live through credible retaliation.

10. Power Before Prohibition

The strongest political thesis of this paper is:

> ***Power can act on agency by acting on meaning before it acts on choice.***

This does not mean that all meaning formation is power in a pejorative sense, or that all norm formation is domination. Human beings necessarily learn languages, values, roles, practices, and interpretive repertoires socially. The claim is structural: because live possibility is meaning-shaped, institutions and relationships can influence agency by altering what is intelligible, legitimate, sayable, imaginable, costly, shameful, sacred, dangerous, or normal before an explicit decision is made.

Three pathways must be distinguished.

Direct closure removes a feasible route through law, force, resource denial, threat, or technical impossibility.

Interpretive closure leaves a route formally open but deprives the agent of concepts, language, credibility, or recognized standing needed to understand or communicate it.

History-shaped closure repeatedly reinforces one route while suppressing alternatives until the practical accessibility distribution becomes highly concentrated.

None of these is automatically wrongful. A safety rule may rightly close a harmful action. Professional training may beneficially concentrate expert attention. A religious discipline may be voluntarily inhabited and agentic. Wrongfulness requires an independent normative argument and, where structural violence is claimed, a feasible counterfactual and a causal mechanism.

11. Structural Deprivation and Violence

The Agency Potential work already proposes a recoverable agency gap between current conditions and feasible interventions [1]. The present paper adds a live-field diagnostic. Let $\mathcal{J}_{feas}$ be feasible structural interventions and let D_L be a declared distance between live possibility profiles. A recoverable live-field gap can be represented schematically as

$$\mathsf{L}_{A,g}^{live} = \max_{z \in \mathcal{J}_{feas}} D_L(\mathcal{L}_A^z(g), \mathcal{L}_A^{z_0}(g))\,.$$

This is not a unit of violence. It merely quantifies how much a feasible intervention could alter the agent's practical option field under a declared metric.

A claim of structural deprivation requires at least four additional elements: (i) a specific agency-relevant capacity or live route that is blocked; (ii) evidence of a responsible structural mechanism; (iii) a feasible counterfactual rather than an imaginary ideal; and (iv) a normative reason why the blockage constitutes harm, domination, injustice, or violence. This preserves the methodological caution

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in Galtung's potential/actual distinction while avoiding the move from any unfulfilled possibility directly to "violence."

Cultural violence can then be studied without declaring whole cultures violent: ask whether a legitimating frame changes what can be read as permissible, what objections remain socially intelligible, whose testimony is credible, and whether feasible routes of refusal or revision remain live.

12. Consent, Compliance, and the Danger of Outsider Substitution

The live-field framework creates a danger of paternalism if used carelessly. An evaluator might claim that a person "cannot really choose" merely because the person's values differ from the evaluator's. Three protections are therefore constitutive.

First, **conformity is not non-agency**. Following Mahmood, living through a norm can itself be an agentic project. Second, **resonance is not consent**. A familiar or identity-congruent option may feel right without satisfying voluntariness or corrigibility. Third, **outsider visibility is not capacity**. The evaluator's failure to observe refusal, dissent, or alternative strategy does not prove those capacities are absent.

A defensible assessment therefore asks whether the agent can access relevant distinctions, whether acceptance/refusal is genuinely attributable to the agent under declared coercion checks, whether the choice can causally affect action, and whether objection/revision channels remain usable. It does not require the person to choose the evaluator's preferred life.

13. Human-AI: The Pre-Prompt Live Possibility Field

The Pre-Prompt Human State Principle states that Human-AI interaction begins before the current prompt because the prompt is a bounded articulation of a history-bearing human state [2]. The present paper strengthens this by identifying the live possibility field as one component of that state.

Let H_t include retained experience, current meanings, values, unresolved differences, recent interaction history, and the live possibility profile $\mathcal{L}_{A,t}$. The human exports only a bounded articulation

$$H_t \xrightarrow{L_H} Q_t.$$

The AI receives Q_t, not the whole human state. Its output Y_t is then experienced and interpreted by the human:

$$Q_t \to AI_t \to Y_t \xrightarrow{R_H} E^{AI}_{H,t}.$$

If some residue is retained,

$$H_{t+1} = U_H(H_t, E^{AI}_{H,t}, \delta^{world}, X^{other}),$$

and therefore

$$\mathcal{L}_{A,t+1} \neq \mathcal{L}_{A,t}$$

may occur before the next prompt is generated.

This yields a deeper AI-safety and AI-benefit question. Evaluating only whether Y_t is correct misses whether repeated interaction makes some routes easier to think, some objections harder to formulate, some alternatives newly live, or some previously live distinctions disappear. Glickman and Sharot's experiments establish that prior AI interaction can shift later human judgments; they do not establish the present mechanism, but they show that the history-shaped starting state is empirically real [18].

The direction remains open. AI can expand the live field by supplying language, translation, counterexamples, analogies, and routes previously inaccessible. It can also produce an epistemic tunnel by repeatedly returning the same framing, making one route fluent and suppressing alternatives. Amplification is therefore neutral:

$$\text{more live accessibility} \neq \text{better agency},$$
$$\text{more fluent choice} \neq \text{better choice}.$$

14. Human Return: Did the Expanded Field Come Back With the Person?

A live route available only while AI is present is not yet durable human agency. CTSA Human Return requires that after AI removal, the person can independently reconstruct a concept, select a tool, execute a skill, or generate a non-equivalent alternative under declared criteria [6].

This gives a temporal sequence:

$$\text{AI-assisted live possibility expansion} \to \text{retention?} \to \text{unaided re-entry?} \to \text{Human Return.}$$

A system may therefore increase coupled performance while reducing human potential if the user becomes less able to detect errors, formulate alternatives, or continue the readout-action-feedback loop without the system. Conversely, AI may temporarily do more work while leaving behind a wider and more corrigible human live field. These cases must be measured separately.

15. Non-Collapse Laws

The architecture stands or falls on role separation. The following are constitutive:

- objective possibility $\neq$ structural feasibility;
- feasibility $\neq$ live possibility;
- live possibility $\neq$ stated preference;
- stated preference $\neq$ free consent;
- choice $\neq$ enactment;
- enactment $\neq$ evaluator-visible agency;
- option count $\neq$ effective agency;
- accessibility $\neq$ warrant;
- resonance $\neq$ consent;

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- repetition $\neq$ legitimacy;
- retention $\neq$ improvement;
- norm conformity $\neq$ absence of agency;
- resistance to norms $\neq$ agency by definition;
- meaning-shaping $\neq$ domination;
- live-field narrowing $\neq$ structural violence without causal and normative conditions;
- AI influence $\neq$ manipulation by definition;
- coupled performance $\neq$ Human Return.

16. Empirical Programme: Try to Kill the Theory

The theory earns a technical role only if live-field measures explain behavior beyond simpler alternatives.

H1 - Live possibility beyond formal option count [Open]. Hold the formal menu constant while manipulating interpretive access, safe refusal, language, or task framing. The live-field model should predict which options enter deliberation and later action beyond the number of available options.

H2 - Meaning access beyond resources [Open]. Hold material resources approximately constant while varying whether an option is intelligible, nameable, or linked to the agent's current problem representation. If resource-only models predict equally well, the meaning layer should be narrowed.

H3 - History-shaped re-entry [Open]. Holding current option content fixed, prior traversal history should predict which alternatives are spontaneously generated or considered. If history adds no held-out value after current state is controlled, momentum/rhythm variables should be removed.

H4 - Resonance without consent [Open]. High experiential congruence should predict felt fit or uptake but should not perfectly predict voluntary endorsement under coercion checks. If resonance and consent are empirically indistinguishable, the role separation needs revision.

H5 - Structural closure before overt prohibition [Open]. In some domains, changing retaliation risk, complaint credibility, interpretive resources, or social recognition should alter practical option use even when formal permission remains unchanged.

H6 - Evaluator visibility [Open]. Changing the assessment interface may change observed agency rankings without changing real-world readout-action capability. If all measurement variation tracks capability change, the visibility layer adds no value in that domain.

H7 - Human-AI live-field carryover [Open/externally allied]. Randomize AI dialogue histories, then hold the final prompt approximately constant and remove AI. Measure which alternatives humans can independently generate, reject, or act upon. Prior AI trajectory should affect the later live field for at least some tasks if the pre-prompt claim is correct.

H8 - Corrigibility predicts durable agency [Open]. Feedback/objection capacity should predict later error correction beyond productivity, confidence, and resource scores in tasks requiring repeated revision.

Every study should preregister rivals: resources/capability only; stated-preference models; formal option count; arousal/familiarity; schema or self-reference; current-state-only models; and reward/performance baselines. A live-field construct that adds no discriminating or predictive work should be deleted.

17. Legitimate Defeaters

The architecture should be materially narrowed if well-designed studies show that: (i) formal feasibility plus stated preferences fully predicts considered and enacted options; (ii) interpretive or meaning access adds no information once resources and incentives are controlled; (iii) prior history does not alter option accessibility beyond current-state variables; (iv) safe objection and revision add no explanatory value for later correction; (v) evaluator-interface changes never alter observed agency without changing underlying capability; (vi) Human-AI history effects disappear after current prompt and explicit memory are controlled; or (vii) the live-field vocabulary cannot be operationalized reproducibly across independent teams.

The strongest philosophical thesis could still survive some empirical failures at a descriptive level, but the technical constructs should not. The programme's deletion rule remains: *preserve the phenomenon, remove machinery that does no distinct work.*

18. Discussion: The Architecture of What Can Become a Possibility

The paper began with a deceptively simple claim: choice begins before choice. Its importance is not that humans are socially influenced; that is old knowledge. The stronger point is architectural. If effective agency is a readout-action-feedback capacity and experience is meaning-giving, then the field from which choices are generated is itself a dynamic object. It can widen, narrow, become polarized, become more corrigible, or become easier to traverse in one direction than another.

This changes how potential should be interpreted. Potential is not an inner reservoir waiting to be expressed. It is an envelope of possible loops under a world, a history, resources, relationships, and time. Current practical agency is further restricted by which parts of that envelope have become live. A person may therefore have a larger recoverable potential than current behavior reveals. Conversely, apparent confidence and action can coexist with a

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narrow live field and poor corrigibility.

It also changes how power is studied. Power need not wait until a recognized option is prohibited. It can operate on the conditions of intelligibility, credibility, salience, social permission, and expected consequence through which an option becomes live. This is compatible with Foucault's field of possible action, Pettit's domination, Fricker's interpretive injustice, Kabeer's conditions of choice, and Galtung's concern with potential, while adding a readout-native measurement interface.

The same architecture prevents an opposite mistake: treating every socially shaped desire as false consciousness. Social formation is unavoidable. Resonance with a tradition, role, or religious practice can be genuinely agentic. The critical questions are not whether the choice is culturally formed but whether relevant alternatives can become intelligible, whether acceptance/refusal is attributable to the person, whether the choice can have causal effect, and whether objection and revision remain possible.

Human-AI systems intensify the problem because they can alter the human possibility field recursively and quickly. A model response can provide a missing concept, normalize an alternative, repeat a frame, reduce switching cost, or make one narrative feel increasingly natural. The next prompt is partly generated from that changed field. Therefore AI evaluation that freezes the human as a static input source misses a central causal pathway.

19. Standalone Thesis

> ***Human agency begins before overt choice.*** *A finite person does not choose from the full space of what is objectively possible, but from a history-shaped field of possibilities that have become sufficiently meaningful, accessible, discriminable, and actionable to function as live candidates. Experience gives significance to phenomena; retained experience changes the future reader; resonance can make some possibilities experientially fitted; rhythm and repeated traversal can make some routes easier to re-enter; social and institutional power can expand, suppress, or structure these routes; and AI can recursively reshape the field from which the next prompt and choice arise. Effective agency therefore requires more than options or outcomes: it requires the corrigible capacity to read relevant differences, own acceptance or refusal, causally enact a decision, and preserve usable routes of objection and revision. Power can act on agency by acting on meaning before it acts on choice. Yet meaning-shaping is not automatically domination, resonance is not consent, accessibility is not truth, and a narrowed field counts as structural deprivation only when a feasible counterfactual, a responsible causal mechanism, and a defensible normative loss are shown.*

20. Conclusion

The central implication is methodological. Asking only "What did the person choose?" begins too late. Asking only "What options existed?" remains too external. Asking only "What did the evaluator observe?" confuses the person's capacity with the measuring interface. A fuller analysis asks: what was physically possible, what was feasible under actual resources and structure, what became live for this reader, what was chosen, what could be enacted, what feedback remained available, and what the evaluator could legitimately infer.

This sequence reveals a hidden architecture of human potential. Meaning, history, and structure do not merely decorate choice after the fact; they help determine what can become a choice at all. The resulting theory neither denies the world nor dissolves agency into discourse. It treats agency as a situated, causally effective, corrigible loop whose option field is meaning-shaped and historically revisable. In Human-AI systems, that means the epistemic and agentic trajectory begins before the prompt. The decisive question is not only what the system helped a person do, but **what became newly possible to see, think, refuse, imagine, and choose - and what remained available to the human after the system was gone.**

Author Declarations

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AI assistance disclosure. Artificial intelligence tools were used for literature retrieval support, source synthesis, adversarial review, formalization assistance, drafting, and typesetting. The human author originated the programme direction, the agency-potential problem, the strong thesis that choice begins before overt choice, and the target conceptual distinctions, and retains responsibility for all claims and revisions.
Data availability. No original dataset was generated. Proposed empirical studies have not yet been executed.

References

References

[1] Lahtee, Y. (2026). *Effective, Corrigible Agency Potential: A Readout Genesis Proposal in Dialogue with Peace Studies, Capability Theory, and Agency Theory*. K0 research memorandum, 5 September 2026.

[2] Lahtee, Y. (2026). *Experience Is Meaning-Giving: A Strong-Form Readout-Retention Theory of Phenomena, Resonance, Rhythm, Accumulation, and Transfor-*

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mative Release. Full Paper v2.0, Human-AI Readout Programme, 5 September 2026.

[3] Lahtee, Y. (2026). *Experience Is the Human LoRA: A Readout-Retention Theory of Selective Model Change*. Open Civil Science Initiative preprint, 18 July 2026.

[4] Lahtee, Y. (2026). *From Problem to Hypothesis: Dynamic Semantic Mobility, History-Shaped Access, and the Readout-Discriminable Hypothesis Space*. GLOSA K0 final author draft, 4 September 2026.

[5] Lahtee, Y. (2026). *Epistemic Fusion or Epistemic Tunnel: A Phenomenologically Anchored, History-Shaped Architecture of Human-AI Semantic Coupling, Retention, and Epistemic Direction*. Human-AI Readout Programme, v7.1, 5 September 2026.

[6] Lahtee, Y. (2026). *CTSA Human-Return Readout: A Session-Boundary Measurement Architecture for Retained Human Capability After AI*. Epistemic Note v9.1, 5 September 2026.

[7] Sen, A. (1985). Well-Being, Agency and Freedom: The Dewey Lectures 1984. *The Journal of Philosophy*, 82(4), 169–221.

[8] Kabeer, N. (1999). Resources, Agency, Achievements: Reflections on the Measurement of Women's Empowerment. *Development and Change*, 30(3), 435–464. https://doi.org/10.1111/1467-7660.00125

[9] Emirbayer, M., & Mische, A. (1998). What Is Agency? *American Journal of Sociology*, 103(4), 962–1023. https://doi.org/10.1086/231294

[10] Bazzani, G. (2023). Agency as conversion process. *Theory and Society*, 52, 487–507. https://doi.org/10.1007/s11186-022-09487-z

[11] Foucault, M. (1982). The Subject and Power. *Critical Inquiry*, 8(4), 777–795. https://doi.org/10.1086/448181

[12] Pettit, P. (1997). *Republicanism: A Theory of Freedom and Government*. Oxford University Press.

[13] Fricker, M. (2007). *Epistemic Injustice: Power and the Ethics of Knowing*. Oxford University Press. https://doi.org/10.1093/acprof:oso/9780198237907.001.0001

[14] Mahmood, S. (2001). Feminist Theory, Embodiment, and the Docile Agent: Some Reflections on the Egyptian Islamic Revival. *Cultural Anthropology*, 16(2), 202–236. https://doi.org/10.1525/can.2001.16.2.202

[15] Klyubin, A. S., Polani, D., & Nehaniv, C. L. (2005). Empowerment: A Universal Agent-Centric Measure of Control. *2005 IEEE Congress on Evolutionary Computation*, 128–135. https://doi.org/10.1109/CEC.2005.1554676

[16] Galtung, J. (1969). Violence, Peace, and Peace Research. *Journal of Peace Research*, 6(3), 167–191. https://doi.org/10.1177/002234336900600301

[17] Galtung, J. (1990). Cultural Violence. *Journal of Peace Research*, 27(3), 291–305. https://doi.org/10.1177/0022343390027003005

[18] Glickman, M., & Sharot, T. (2025). How human-AI feedback loops alter human perceptual, emotional and social judgements. *Nature Human Behaviour*, 9, 345–359. https://doi.org/10.1038/s41562-024-02077-2

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Potential as a Readout: Witnessed Envelopes, Two-Layer Agency, and the Measurement of Pseudo-Peace

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What the chapter states. "Potential is a readout, not a hidden quantity. It is the envelope of a witnessed option set [. . .], it comes in two layers (choosing within a constraint structure, and changing that structure) [. . .]."

Status. "K0. The lemmas are about the stated model, not about societies; the capability-theory definitions of potential are recorded as challenges to the witness rule, not as confirmations; the two-layer distinction is carried as an open conjecture."

Falsifier(s). "P-D (layer 2 dominates the structural case). In cases Violence as Instability classifies as structural (§6.2), the gap closable by layer-2 interventions exceeds the gap closable by layer-1 ones. Defeated if information-only or within-C interventions close as much."

Read before. *Violence as a Special Case of Instability in Finite–Memory Causal Systems*; *Causal Ethics: The Mathematics of Regime Choice and Survival*; *The Causal Grammar of Structured Coexistence*; *Causal Agency: A Persistence Control Theory of Adaptive Systems* (all Part 4).

Feeds into. *Choice Begins Before Choice: Meaning-Shaped Accessibility, Live Possibility, and Effective Agency in Human-AI Systems*.

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Potential as a Readout

Witnessed Envelopes, Two-Layer Agency, and the Measurement of Pseudo-Peace

A readout-native anatomy of the word "potential" across peace research, capability theory, empowerment, and the Human–AI Readout Programme

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Article type Epistemic note (conceptual + one fail-able fixture) • **Series** Society, Justice, Peace & Violence [Lahtee, 2026k] and When AI Expands Human Potential [Lahtee, 2026a] • **Licence** CC BY 4.0

Status K0, version 2 (tier-raising pass), 5 September 2026 — "final" names the manuscript state, not validation, publication, or independent certification. Definitions at DEFINITION; interpretations at DR; four lemmas at exact-algebra grade on the stated discrete model; a 60-instance numeric battery and two exact certificates at FINITE_DIAGNOSTIC. Three claim cards and 32 verified citation cards (a further 9 record-only cards are parked) (rule 17, source-anchored: link, page, line, one verbatim passage) back the text (Section 9). No human data, no coefficient estimates.

In brief. Problem. "Potential" is the load-bearing word of peace research (violence as the gap between actual and potential realizations), of capability theory (what a person could be and do), of information-theoretic agency (empowerment as channel capacity), and of this programme's own agency papers (viable regions, admissible regimes, feasible life-space). Every one of these accounts *declares* a set of feasible options and then measures against it; none says how the set itself is read. **Proposal.** Potential is a readout, not a hidden quantity. It is the envelope of a *witnessed* option set (a finite, enumerable set under a retained selection rule), it comes in two layers (choosing within a constraint structure, and changing that structure), it must be kept apart from what was exercised and from what an evaluator observes, and its compression must be diagnosed separately from the attribution of responsibility. **Method.** Seven distinctions are stated as non-collapse rules and collided with the capability, empowerment, learning-theory and peace-research literatures through verified citation cards; the joint-event definition of corrigible agency potential is restated on the witnessed set; corrigibility and spectral support are made readable through a correction-channel presence test and a repair rate; an exact identifiability certificate shows two models with the same record and a five-fold difference in potential; "pseudo-peace" is defined as the collapse of a three-valued evidence gate into a two-valued one, and the programme's bounded-suppression reading is raised from a sketch to four proved lemmas on the discrete linear spine, confirmed by a 60-instance random-graph battery with passing and failing controls. **Status.** K0. The lemmas are about the stated model, not about societies; the capability-theory definitions of potential are recorded as challenges to the witness rule, not as confirmations; the two-layer distinction is carried as an open conjecture. Four falsifiable propositions are stated with the observation that would defeat each.

1 THE PROBLEM: A WORD EVERYONE MEASURES AGAINST AND NOBODY READS

Peace research made "potential" its hinge. Galtung's opening definition reads: "violence is present when human beings are being influenced so that their actual somatic and mental realizations are below their potential realizations" [Galtung, 1969, copy p. 3, journal p. 168]. The same essay says at once that "The meaning of 'potential realizations' is highly problematic, especially when we move from somatic aspects of human life, where consensus is more readily obtained, to mental aspects" [Galtung, 1969, copy p. 4]. The gap-definition has been used for half a century; the baseline it subtracts from has stayed a declaration.

The word is older than the gap. Aristotle's opening of the treatment of potency distinguishes "a potency of being acted on, i.e. the originative source, in the very thing acted on, of its being passively changed by another thing or by itself qua other" from "a state of insusceptibility to change for the worse" [Aristotle, c. 350 BCE / online, Book IX, Part 1]: potency is already relational and indexed to what can act on what. Learning theory made the word measurable by assistance: the zone of proximal development is "the distance between the actual developmental level as determined by independent problem solving and the level of potential development as determined through problem solving under adult guidance or in collaboration with more capable peers" [p. 86, quoted in Vygotsky, 1978, Chaiklin, 2003, p. 2]. Potential here is read off an assisted performance, and the assisted–unassisted split is exactly the session-boundary split this programme uses [Lahtee, 2026g].

Other traditions carry the same structure under other names. Capability theory separates what a person achieves from the agency and freedom to achieve it [Sen, 1985, Kabeer, 1999]; social-cognitive theory places agency in intentionality, forethought, self-regulation and self-reflection [Bandura, 2001]; relational sociology indexes agency to a reusable past, an imagined future and a present judgment [Emirbayer and Mische, 1998]; the conversion-process account makes potential depend on micro–meso–macro conversion factors [Bazzani, 2023]. Information-theoretic agency is the cleanest: "We can now define empowerment as the channel capacity of the agent's actuation channel terminating at the sensor" and "Empowerment is measured in bits. It is zero when the agent has no control

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over what it is sensing, and it is higher the more perceivable control or influence the agent has" [Klyubin et al., 2005, copy p. 3]. Power theory supplies the counter-face: "To govern, in this sense, is to structure the possible field of action of others" [Foucault, 1982, copy p. 15].

Inside this programme the word appears four more times. *The Causal Grammar of Structured Coexistence* defines violence as what happens "when incompatible coexistence significantly compresses actors' feasible life-space" and structure as "an architecture of feasible possibility" [Lahtee, 2026f, pp. 2–3]. *Causal Ethics* makes agency exist "if and only if it can instantiate a choice of regime" from an admissible set [Lahtee, 2026e, Axiom II, p. 9]. *Causal Agency* defines persistence as remaining in a viable region $V \subseteq S$ and agency as modifying the constraint structure C so that expected persistence grows [Lahtee, 2026d, Defs. 2–3, pp. 17–18]. *Violence as a Special Case of Instability* relocates violence from events to unstable modes of a finite-memory operator [Lahtee, 2026l]. A programme note of 5 September 2026 then proposed a joint-event definition of "effective, corrigible agency potential" [Lahtee, 2026b].

All of these share one silent move: a feasible set (Π, R_{adm}, C, the "feasible life-space", the actuation channel's alphabet) is *declared*, and potential is then computed as a maximum, a capacity, or a gap over it. Whoever declares the set fixes the potential. This paper's claim is that the declaration step is where the word must be read, and that the programme's own epistemic core already says how.

> **Central claim.** Potential is a readout. It is the envelope of what an agent can bring to a specified task over a *witnessed* option set (a finite, enumerable set under a retained selection rule), read under a declared model, and it is never the envelope over an unretained continuum of "could have". It comes in two layers (choice within a constraint structure; change of that structure). It must be held apart from what was exercised and from what an instrument observes; its compression is diagnosed from an obstruction ledger and cost asymmetry without naming a culprit; naming a culprit is a separate act that needs a chooser. Anything called "potential" that skips one of these separations is, on this reading, a different quantity wearing the word.

2 Programme genealogy and what is borrowed from the epistemic core

The four series papers supply the objects; the Readout Genesis / Readout Universe core supplies the reading discipline. The correspondence is exact enough to tabulate (Table 1).

Three pieces of the core are load-bearing here and are cited by their own labels. (i) **N2**, the readout equation: "$M_A = K_A \cdot \theta + \eta$, $\varepsilon_{\text{tot}} > 0$ — knowing is a LOSSY read — never the latent truth" [Lahtee, 2026i, Part VI §VI.1]. (ii) The operator grounding of the ethics ledger, which fixes what "the set of answers actually accessible" means: "a finite, enumerable set under a retained selection rule, never an unbounded continuum of options", together with the architecture row that defines agency-layer selection as "a finite selection over already-legal moves … not a continuum optimization" whose checkable content is "whether a specific system's move-set is actually finite and legal" (Readout Universe, `logic.md` §9.11 and ARCH-L4). (iii) The three-valued verdict gate and the failable gate law: "a two-valued answer/no-answer gate cannot distinguish 'no evidence exists' (ABSTAIN) from 'evidence is genuinely contested' (ESCALATE)", and a gate "only counts as Type-P … if it has both a machine-derived passing control and a machine-derived failing control" [Lahtee, 2026i, Part VI §VI.3, §VI.7]. Corrigibility is already stated there as "boundary-observability": the channel "through which a human party's correction or objection must remain visible" (Part V §V.9), and is graded as "a discrete presence/absence readout on a correction channel (open or not, checked at a given time)" (`logic.md` §9.11).

Nothing below adds an axiom to that core. What is added is the application of N2 twice (to the agent's own option set and to the evaluator's instrument), the two-layer reading borrowed from *Causal Agency*, and one executed fixture.

3 An anatomy of "potential": seven distinctions

Each distinction is stated as a non-collapse rule: two things the word is used for, which must not be merged, and the failure that merging produces. Two of them are registered in the glosa non-collapse table as NC-78 and NC-79 [Lahtee, 2026j].

D1. Potential $\neq$ exercised $\neq$ observed (NC-78). The envelope p^* of what the agent could do, the policy π_{used} it did use, and the value $Y_{\text{obs}} = \mathcal{O}_q(H)$ an evaluator's instrument returns are three readouts of three different latents. N2 applies to each. Collapse: reading assisted output as retained capability [Lahtee, 2026g]; reading silence as consent; reading an instrument change as a change in the person. Vygotsky's definition is the oldest explicit statement of the split: potential is defined *through* assisted performance and is not the same quantity as independent performance [p. 86, in Vygotsky, 1978, Chaiklin, 2003]; D1 only adds the third term, the evaluator's own instrument.

D2. Declared set $\neq$ witnessed set. A feasible set someone writes down is a declaration; the set of options for which a record exists that an agent with comparable history and conditions executed them within the horizon and budget is a readout. Only the second is finite and enumerable in the required sense. Collapse: computing a maximum or a capacity over options nobody has ever been seen to take, then calling the number potential.

This is where the capability approach pushes back hardest, and the objection is recorded rather than smoothed. Its own definitions do not require witnessing: "Capabilities are a person's real freedoms or opportunities to achieve functionings" [Robeyns, 2017, Ch. 2, §39]; "a person's 'capability' refers to the alternative combinations of functionings that are feasible

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Table 1: One object, five names. Page locators refer to the Zenodo PDFs; "AP" is the 5 September programme note.

	Causal Grammar	Causal Ethics	Violence as Instability	Causal Agency	AP / this paper
Structure	"architecture of feasible possibility" (p. 2)	regime $R = (T_R, I_R)$, CE-05	generator L_τ (eq. 5)	constraint structure C (Def. 1)	option set Π^{wit}
Potential	"feasible life-space" (p. 3)	$R_{\text{adm}}(t')$	viable region of the state manifold	viable region $V \subseteq S$ (Def. 2)	envelope p^*, profile $\mathbf{p}^*$, capability set $\mathscr{C}^\alpha$
Agency	actors sustaining structures	"$\exists R_A \in R_{\text{adm}}$ selected by A" (Axiom II)	—	$\partial T_p/\partial C \cdot \partial C/\partial x > 0$ (Def. 3)	D_g actor-directed choice; layer 2
Violence	"compresses actors' feasible life-space"	spectral exclusion $\Delta_{\text{spec}}(R_i) \leq 0$	instability under sustained load	(not defined)	shrinkage of $\mathscr{C}^\alpha$
Repair	"protected reopening" (p. 3)	$\dot{V} \leq 0 \wedge \tau_c' > 0 \wedge \Delta_{\text{spec}} > 0$	$\Gamma\uparrow$, $S_{\text{env}}\downarrow$, $L_{ij}\downarrow$, $\tau\downarrow$	$C_{t+1} = G(x_t, C_t)$	F_g correction channel; layer 2
Responsibility	—	$\text{Resp}(A) \equiv C_{A,R_A}$; forced regime $\Rightarrow C_{A,R} = 0$	—	—	attribution, separate from diagnosis

for her to achieve" [quoted in Sen, 1999, Nussbaum, 2011, p. 20]; the key question is "What is each person able to do and to be?" [Nussbaum, 2011, p. 18], and an opportunity a person "may or may not exercise in action" still counts. Four verified cards therefore stand as CHALLENGES to the witness rule, against one that the checking route read as support (Chaiklin quoting Vygotsky) and two it read as context (Aristotle; the lens). The rule's precise content answers them without retreating: it does not require that *this* agent exercised the option (Nussbaum's point is kept whole); it requires that the option be witnessed by a record of *comparable* agents under comparable conditions executing it within the horizon and budget. "Feasible for her" is then a readout about a reference class, not a modal intuition. What the rule refuses is the third case, an option no one comparable has been seen to take, which the capability definitions leave to the analyst. The conversion-factor apparatus is the closest ally: "Persons have different abilities to convert resources into functionings. These are called conversion factors" [Robeyns, 2017, Ch. 2, §50]; witnessing is how a conversion factor stops being a coefficient and becomes a record.

D3. Layer 1 ≠ layer 2. Choosing within a constraint structure C is one capacity; changing C (renegotiating a rule, opening a channel, forming a coalition, delegating) is another. *Causal Agency* reserves the word agency for the second: constraint modulation with $\partial T_p/\partial C \cdot \partial C/\partial x > 0$, and calls stabilisation inside a fixed C "proto-agency" [Lahtee, 2026d, §6.5–6.6]. *Causal Grammar*'s "repair" is a layer-2 act under protection. Collapse: reporting a layer-1 envelope as if it bounded what a person or group can become.

The literature review returned this distinction as *open*, not confirmed: all ten sources carded for it (Section 9) came back undetermined. Two counter-readings are kept on record. The programme's own Definition 3, "A system exhibits agency if its internal dynamics modify the constraint structure C in ways that increase the expected persistence of the system within the viable region V" [Lahtee, 2026d, p. 19], may already be the layer-2 claim in different notation, so that what is added here is only the recoverable-gap quantity that links the layers. And the empowerment-measurement literature warns that resources, agency and achievements can be indivisible in the meaning of an indicator [Kabeer, 1999]; if the layers are not separable in records, (5) is not measurable as stated. Self-determination theory's "competence, autonomy, and relatedness" [Ryan and Deci, 2000, p. 68] and social-cognitive theory's "people are producers as well as products of social systems" [Bandura, 2001, abstract] both describe layer-2 capacity without separating it from layer 1. The distinction is therefore carried as a conjecture with a named test (P-D), not as a result.

D4. Task potential ≠ aggregate potential. A profile over tasks $\mathbf{p}^* = (p_g^*)_g$ is the object; a weighted scalar is a summary for comparing interventions under one declared weighting. *Violence as Instability* §7 gives the dynamical reason: "Blocking one expression channel does not remove instability", load moves to a coupled channel [Lahtee, 2026l, p. 3]. Collapse: a flat aggregate hiding a compression in one task and a rise in another.

D5. Feasible ≠ permitted. What has a working path is not the same as what present power allows; to govern "is to structure the possible field of action of others" [Foucault, 1982, copy p. 15]. Changing a rule can be a feasible layer-2 intervention with a named actor, budget and horizon even when it is currently forbidden. Collapse: letting the permitting authority define the feasible set, which makes every compression by that authority invisible by construction.

D6. Diagnosis of compression ≠ attribution of responsibility (NC-79). That a capability set has shrunk is read from an obstruction ledger (what was present at the source equals what survived plus what was obstructed at each step; Genesis Part VI §VI.6) and from cost asymmetry: *Causal Ethics* "does not require intent or malice to identify injustice — only asymmetry" (CE-30, p. 20). Who is responsible is answered only

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by a chooser: “Cost attaches to the chooser, not the carrier of consequences” (p. 20). Collapse in one direction: refusing to call a shortfall deprivation until a culprit is named, which silences the no-chooser structural case that *Violence as Instability* §6.2 exists to capture. Collapse in the other: inferring a culprit from asymmetry alone.

D7. Recoverable gap $\neq$ accumulated loss. Galtung’s own examples turn on avoidability (tuberculosis in the eighteenth century versus today, copy p. 3). A gap that a feasible intervention within the study’s horizon can reopen is one quantity; damage already fixed into history is another. A recoverable gap of zero is not evidence that no violence occurred.

4 Definitions on the witnessed set

4.1 Objects

Let $g \in \mathscr{G}$ be a specified agency task (to accept or decline, to ask for an explanation, to delegate, to revise a decision when information changes). Let A be the agent, h its retained and usable history, z the social and material conditions (tools, resources, relations, whether objection is acknowledged), T a declared horizon, B a declared budget, P the declared transition-and-readout model (not the world), q_g, ν_g the outcome resolution and declared test set, and $w_g \geq 0$, $\sum_g w_g = 1$, summary weights.

> **Witness rule (D2 made operational).** $\Pi_A^{\mathrm{wit}}(g; h, z, T, B)$ is the finite set of policies for which a retained record exists that an agent with history and conditions comparable to (h, z) executed the policy within T and B. Assistance, tools, interpreters and delegates are part of a policy when the agent actually reaches them and directs the relevant part. A policy that is merely conceivable, or permitted only in principle, is not in the set. “Comparable” is not left to the analyst’s intuition: two agents are comparable for task g when they share the declared fields of z (tools, resources, relations, whether objection is acknowledged) and a history window of declared length; membership in Π^{wit} is checkable only relative to that declared comparability relation, never absolutely, and a study that does not declare it has not applied the rule. Adding a policy to the set is itself a recordable act (a new witness), which is how layer-2 agency enters (Section 4.3).

4.2 The joint event and the envelope

For task g define four events under P: R_g (the agent reads the distinctions the decision needs, including its own uncertainty), D_g (acceptance, refusal or choice counts as the agent’s own direction under declared criteria, with coercion and delegation checked), X_g (the choice turns into the relevant action or outcome, with a causal contrast that separates the choice from luck or from another’s order), F_g (feedback, objection and revision remain usable within the horizon when there is reason to revise). F_g does not require reversing events or changing one’s mind on every objection; a reasoned reaffirmation passes.

$$p^*_{A,g}(h, z; T, B, P) \;:=\; \max_{\pi \in \Pi_A^{\mathrm{wit}}(g; h, z, T, B)} \Pr_P^{\pi}\left(R_g \cap D_g \cap X_g \cap F_g\right), \tag{1}$$

with $p^*_{A,g} := 0$ when the set is empty (no witnessed path, not a verdict on the person). The primary objects are the profile and the capability set,

$$\mathbf{p}^*_A = (p^*_{A,g})_{g \in \mathscr{G}}, \qquad \mathscr{C}_A^{\alpha} = \{g : \; p^*_{A,g} \geq \alpha\}, \tag{2}$$

and only for comparing interventions under one declared w,

$$\mathsf{A}_A^{\mathrm{corr}}(h, z; T, B, P, w) := \sum_g w_g \, p^*_{A,g} \in [0, 1]. \tag{3}$$

Because the maximum is over a finite witnessed set, (1) is computable from records. Its relation to any larger “true” envelope is N2: what is read is $K \cdot \theta + \eta$ with $\varepsilon_{\mathrm{tot}} > 0$; the unread remainder is not a smaller number waiting to be found but a non-readout. This is not a retreat from “envelope”: the envelope is defined on what is retained, which is the only place an envelope was ever a quantity.

By the chain rule, for one policy on one task, $p_{A,g}(\pi) = r\,d\,x\,f$ with $r = \Pr(R_g)$, $d = \Pr(D_g \mid R_g)$, $x = \Pr(X_g \mid R_g, D_g)$, $f = \Pr(F_g \mid R_g, D_g, X_g)$; the maximum is taken of the product under one policy, never of the four factors separately (which would assemble an ideal person from parts of incompatible policies). The factorisation order is not a causal chain; mechanism needs timing and intervention contrasts.

4.3 Two layers

Let $\mathscr{I}_{\mathrm{feas}}$ be the witnessed set of interventions on conditions z (rule changes, reallocations, channel openings), each with a named actor, budget, horizon and effect on others; z_0 the present conditions. Layer-1 potential is (1) at fixed z. Layer-2 potential is

$$p^{*(2)}_{A,g} := \max_{z \in \mathscr{I}_{\mathrm{feas}}} p^*_{A,g}(h, z; T, B, P), \tag{4}$$

and the recoverable gap of the earlier note is exactly the difference of the two layers summed under w:

$$\mathsf{L}_A^{\mathrm{recoverable}} = \max_{z \in \mathscr{I}_{\mathrm{feas}}} \mathsf{A}_A^{\mathrm{corr}}(h, z) - \mathsf{A}_A^{\mathrm{corr}}(h, z_0). \tag{5}$$

This is the quantity to bring to Galtung’s baseline problem: not “potential realizations” in general, but the part of the gap that a witnessed intervention inside the study’s horizon can reopen (D7). It is in the units of (3), not in units of violence.

4.4 Corrigibility and spectral support made readable

Causal Ethics requires, for a collective, $\frac{d}{dt'} V_G \leq 0$ and $\min_i \Delta_{\mathrm{spec}}(R_i) > 0$, and reads the second condition as the exclusion of “an illusion of stability: a situation in which V_G decreases while one or more agencies lose spectral support … where apparent collective order is maintained by suppressing, marginalizing, or erasing subgroups” [Lahtee, 2026e,

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ch. 6]. For a social regime Δ_{spec} has no instrument of its own. The core supplies two: the correction-channel presence readout (open or not, checked at a time) and the repair rate $dR_H/dt = -\mu\|\mathrm{Res}_H\|^2$ from the agency loop [Lahtee, 2026i, Part VIII §VIII.4]. This paper therefore reads, for subgroup i,

$$\Delta_{\mathrm{spec}}(R_i) > 0 \iff \mathrm{channel}_i = \mathrm{open} \wedge \dot{R}_i \neq 0 \quad (6)$$

as a readout-level sufficient condition; both conjuncts are retained-record tests, not judgments of sentiment. F_g in (1) is the same object at task level.

4.5 Identifiability

The evaluator reads $Y_{\mathrm{obs}} = \mathscr{O}_q(H_{0:T})$. Let $\mathfrak{M}(\mathscr{D})$ be the models not yet excluded by record $\mathscr{D}$; the set of potentials $\{\mathsf{A}^{\mathrm{corr}}_A(P) : P \in \mathfrak{M}(\mathscr{D})\}$ is what the record supports, and if two models agree on every record under the declared tests but disagree on potential, no function of the record alone can return the right value in both cases. The programme's own worked case of this gate is the "no objection voiced" row: consent, fear of reprisal, no channel, objection through a proxy — separated only by additional records (a private channel, a history of proxy use, the response when a real alternative is offered). Two model families P_1, P_2 can differ on $\inf_P \max_\pi$ versus $\max_\pi \inf_P$; choosing a policy robust to model uncertainty is a separate problem and is left as such.

An exact certificate. The gate is not rhetoric; it has a witness. Take two declared models of the same agent over an evaluator alphabet $\{\mathrm{calm}, \mathrm{complaint}\}$: P_1 (reasons to object arise with probability $1/10$, the channel is always open, every reason is voiced) and P_2 (reasons arise with probability $1/2$, the channel is open with probability $1/5$, every open-channel reason is voiced). Both induce the identical record law $\Pr(\mathrm{complaint}) = 1/10$. Holding $r = d = x = 9/10$ fixed, $f_{P_1} = 1$ and $f_{P_2} = 1/5$, so $p^*_{P_1} = 729/1000$ and $p^*_{P_2} = 729/5000$: the same record, a five-fold difference in potential. The record alone supports only the interval $[729/5000, 729/1000]$. The record that separates the two is a reprisal-safe private channel, whose use rate stays near $1/10$ under P_1 and rises toward $1/2$ under P_2. Exact rational arithmetic; script and output ship with the paper (`identifiability_example.py`); tier FINITE_DIAGNOSTIC as an existence certificate on two finite models, nothing more.

5 The bounded-suppression proposition, proved on the discrete spine

Violence as a Special Case of Instability states its no-go result as a sketch: "no bounded force-based intervention can drive the system to a globally violence-free absorbing state" under three premises [Lahtee, 2026l, §5]. Version 1 of this paper tested the reading on one two-node fixture. This version proves it on the model the programme's core forces, and leaves honestly open the step that would carry the proof back to the original theorem.

Table 2: Numeric battery for L1–L4 (`sim/battery.py`): random connected two-block graphs (a slow loaded component, a fast one), 20 seeds per N, $\theta = 0.6 s^*_0$, resets every 60 ticks, dt set per instance from $\lambda_{\max}(A)$. Runtime 13.5 s. Tier FINITE_DIAGNOSTIC on these instances.

N	seeds	L1–L4 pass	false claim refuted	$\gamma = 0$ control flagged
10	20	20/20	20/20	yes
50	20	20/20	20/20	yes
200	20	20/20	20/20	yes

Setting. A finite weighted graph with symmetric $W \geq 0$, the forced Laplacian $L_R = D_W - W$ (the operator retained distinction forces; Genesis, machine-checked), dissipation $\Gamma = \mathrm{diag}(\gamma_i) \geq 0$ with at least one $\gamma_i > 0$ on every component, $A := L_R + \Gamma$ (positive definite), constant load $J \geq 0$, stepper $s[n+1] = s[n] + dt\,(-A\,s[n] + J)$ with dt small enough that $M = I - dt\,A$ contracts. An *amplitude intervention* is any additive perturbation sequence $u[n]$ on the state (resets included), bounded if $\sum_n \|u[n]\|_1 \leq B < \infty$. Premise (ii) of the original theorem is thereby made exact: suppression enters as u, never as W or Γ.

Lemmas [Lahtee, 2026h]. **L1 (invariance and recurrence).** $s^* = A^{-1}J$ is the unique fixed point and does not depend on u; after the last nonzero tick n_B of a bounded intervention, $\|s[n] - s^*\| \leq \rho^{n-n_B}\|s[n_B] - s^*\|$, and if $s^*_i > \theta$ there is an explicit finite N^* after which $s_i[n] > \theta$ for all $n \geq n_B + N^*$. **L2 (the bill).** To hold $s_i[n] = \theta$ from n_0 on, the force per tick is exactly $|u_i[n]| = dt\,[(\gamma_i + D_i)(s^*_i - \theta) - \sum_{k\neq i} W_{ik}(s^*_k - s_k[n])]$, converging to a rate $F^\infty_i > 0$ whenever $s^*_i \geq \theta$, so cumulative force is $F^\infty_i n + o(n)$: at least linear in the time held. **L3 (operator change).** Raising γ_i by Δ gives, by Sherman–Morrison, $s^{*\prime}_i = s^*_i/(1 + \Delta B_{ii})$ with $B = A^{-1}$, so any $\theta < s^*_i$ is reached below with $\Delta > \Delta^* = (s^*_i/\theta - 1)/B_{ii}$ and zero ongoing force. **L4 (mutation).** Under permanent periodic resets of node i, a neighbour j with $W_{ij} > 0$ has the long-run floor $s_j \gtrsim (J_j + W_{ij}\bar{s}_i)/(\gamma_j + D_j)$, with $\bar{s}_i$ an explicit increasing function of node i's own inflow; the floor is strictly positive even when $J_j = 0$.

Tier: L1–L3 are exact algebra on the stated model; L4 is a valid lower bound not proven tight. None is a theorem about societies. The step still owed is to show that the original theorem's own hypotheses instantiate this model; until then *Violence as Instability*'s statement keeps its Dr tag while the model-internal reading no longer does.

Table 2 is the fail-able gate for the lemmas. The failing control is a deliberately false claim, "unbounded resets still let s_i re-cross θ", tested on every seed and refuted every time (zero re-crossings under unbounded resets), so the battery discriminates rather than rubber-stamps. A variant with $\gamma = 0$ on the whole loaded component makes A singular and is flagged as outside the hypothesis, not scored. Per-seed values of s^*, θ, $\lambda_{\min}$, $\lambda_{\max}$, dt and all nine sub-checks are in the result file.

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Table 3: Two-node fixture ($L_R = D_W - W$, $\varepsilon = 0.05$, $\tau = 1$, $dt = 0.1$, threshold 1.0, sustained load $J_1 = 0.05$, $\gamma = 0.02$ unless changed). Tier FINITE_DIAGNOSTIC on this fixture only. Script and result file: Lahtee [2026c], `sim/nogo_fixture.py`.

Intervention	threshold crossings	force spent
none	1 (stays above)	0
amplitude reset, bounded budget (4000 ticks), then stopped	recurs after budget	18.08
amplitude reset, unbounded, 8000 ticks	0	36.45
amplitude reset, unbounded, 16000 ticks	0	73.18
operator change (γ: 0.02 → 0.20, J_1 : 0.05 → 0.02), no resets	0	0

6 Pseudo-peace: a gate collapse with a numerical signature

Causal Grammar lists "Pseudo-peace" as its own regime: "Compatible appearance / Restricted / Suppressed", beside "Peace": "Compatible / Preserved / Protected" [Lahtee, 2026f, Table 1, p. 4]. In the language of Section 4 the regime is defined by three readouts taken together:

$$Y_{\text{obs}} \text{ calm } \wedge\ p^*_{\text{wit}} \text{ low } \wedge\ F \text{ blocked.} \tag{7}$$

Epistemically it is the collapse of the three-valued verdict gate: the evaluator returns ABSTAIN (no signal) in a state that is ESCALATE (contested) with the channel closed. The gate law says a two-valued gate cannot tell these apart; pseudo-peace is what a two-valued reading of a society looks like.

Violence as Instability gives the dynamics: force "alters amplitudes c_k but leaves the spectrum invariant", and its no-go statement is that "no bounded force-based intervention can drive the system to a globally violence-free absorbing state" under three premises [Lahtee, 2026l, §4–5]. That statement is a sketch at DR; Section 5 proves its content on the forced operator. The fixture below is the original two-node instance, kept because it is the one a reader can recompute by hand.

Table 3 reports the fixture. The earliest version of the fixture set the wrong passing control (it expected recurrence *during* suppression) and failed its own control: continuous resets do hold the amplitude down. Re-reading the theorem's wording ("bounded", "absorbing") fixed the specification and the result raised, not lowered, the claim: amplitude-only suppression keeps the readout below threshold exactly as long as it is paid for; the cumulative force grows at the rate Lemma L2 gives ($F_i^\infty > 0$), non-saturating over the 4000–16000 ticks sampled, so that the unboundedness *Causal Ethics* names as CE-31 ($\lim_{T\to\infty} C_{A,R}[0,T] = \infty$) is the lemma's statement and not a three-point extrapolation; the neighbouring channel rises (L4); a bounded budget is followed by recurrence, which is L1's content and for which the fixture is a passing control rather than a discovery; an operator-level change reaches the same calm with zero ongoing force (L3, confirmed on all 60 battery instances). The failing control that would defeat the reading is stated in the script: bounded suppression with no recurrence while the load's time-average stays positive. It did not trigger. One accounting caveat belongs here: the fixture meters suppression force and does not price the operator change itself; "zero ongoing force" is a statement about the force channel, not a claim that changing Γ or S_{env} is free. Nothing here transfers to real societies; it is the reading, made fail-able on the operator the core forces.

7 Conversation with the literature

The comparison below is a reading of the sources listed, with verbatim passages only where the text was fetched and read for this paper; entries marked "record" or "abstract" were consulted at that level and are quoted only from what was fetched. It is not a systematic review. Ten further works were sought and could not be read in any open copy on the day of writing (Sen, 1985, Emirbayer and Mische, 1998, Giddens, 1984, Archer, 1995, Pettit, 1997, Fricker, 2007, Mahmood, 2001, Nussbaum, 2003, Bandura, 1977, Scott, 1990); they are carried as records, with the failed access route disclosed on each citation card, and no passage from them is quoted.

8 Falsifiable propositions

Each is DR/OPEN with a stance and the observation that would defeat it. Tests need passing and failing controls (correct and incorrect information; a channel that works and one whose submissions vanish) and must not manufacture coercion or risk for real participants to obtain a failing control.

P-A (pseudo-peace signature). Amplitude-suppressing interventions raise the calm an evaluator reads while p^*_{wit} does not rise and F stays blocked. Defeated if p^*_{wit} rises consistently with observed calm under such interventions.

P-B (channel shift). When F is blocked and $\mathscr{C}^\alpha$ shrinks in task g, load rises in a coupled task g' while the aggregate (3) stays flat. Defeated if profiles stay flat across tasks.

P-C (the excluded path). Groups whose $\min_i |\mathscr{C}^\alpha_{A_i}|$ falls while the mean rises show persistent cost concentration (CE-30) and higher recurrence than groups whose minimum does not fall. Defeated if no difference is found.

P-D (layer 2 dominates the structural case). In cases *Violence as Instability* classifies as structural (§6.2), the gap closable by layer-2 interventions exceeds the gap closable by layer-1 ones. Defeated if information-only or within-C interventions close as much.

The earlier note's P1–P4 (information alone may not reopen potential where execution is the bottleneck; productivity and corrigible agency can come apart; a change of instrument can reorder rankings while capability is unchanged; keeping the objection channel predicts error-correction beyond resource or confidence scores) are carried unchanged [Lahtee, 2026b].

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9 Knowledge state, citation discipline, and independence

This paper is produced under the glosa methodology [Lahtee, 2026j]: a problem card, three hypotheses (H1 the witness rule, H2 the two layers, H3 the pseudo-peace signature) selected by the author, three claim cards answering the five questions (what was seen, what the data separates, what AI filled in, what is assumed, what independent test it has survived), and one citation card per source with the fetched URL, page, line and a single continuous verbatim passage. The cards are public in the repository (`projects/GLS-2026-002_potential-as-readout`, `records/lit/potential-as-readout`).

Knowledge state is K0 for every claim: timestamped, citable, and not independently checked at I2 or above. What did run, for this version, is a same-model fresh-session route (I1): every citation card's passage was re-located mechanically in the fetched copy and judged for bearing by a headless session sharing no context with the drafting session, and each claim card was read by three such sessions in adversarial roles (falsifier, hostile reviewer, source auditor); their nine reports ship with the cards, and the revisions they forced are visible in the cards' dissent records: the witness rule's comparability relation was undeclared and is now declared; H1's claim that the programme's own sets can be re-read "without loss" was unevidenced and is now an explicit untested assumption; H3's "grows without bound" rested on three durations and now rests on Lemma L2. The route's honest limit is on the card: I1 is a second reader, not a decorrelated one. What K1 would require is a cross-vendor route (I3) confirming the same, or the bounded I2+I4 exception the methodology allows a single-vendor scholar; a human reviewer with no stake would be I5. The fixture, the battery and the identifiability certificate are same-session executions (I1); their scripts ship with the paper so that any reader can re-execute them, which is the only form of independence a standalone author can offer in advance.

10 What is not claimed, and the limits of this note

Not claimed: that (1) is forced from the root of the epistemic core; that all human agency reduces to one number; that R, D, X, F are necessary or sufficient for every definition of agency; that this reading predicts data better than capability theory, conversion theory or empowerment; that any programme or curriculum is effective; that the two-node fixture says anything about a society; that internal review by the author and an AI assistant amounts to independent human review. If earlier work can deliver the same separations and tests directly, its language should be used and this note reduced to an operational protocol. *Causal Ethics*' Δ_{spec} for social regimes remains without an instrument beyond (6), and its "$\gg$" in CE-30 has no declared threshold; both are per-case declarations. The no-go statement of *Violence as Instability* keeps its sketch tag until its hypotheses are shown to instantiate the model of Section 5. The two-layer distinction (D3, H2) is an open conjecture. Ten works were consulted at record level only and are not quoted. Two of the read sources were reached at second hand (Sen 1999 through Nussbaum 2011; Vygotsky 1978 through Chaiklin 2003) and are marked so on their cards.

11 Conclusion: what the word now means

Potential, on this reading, is the envelope of a witnessed option set for a specified task under a declared model; it is read, not found. It has two layers, and the recoverable gap between them is the only part of Galtung's baseline that a study can subtract with a record behind it. It is one of three readouts of an agent that must be kept apart, and its compression is diagnosed before, and separately from, anyone being held responsible. A society that reads calm where the channel is closed has collapsed a three-valued gate into a two-valued one, and the fixture shows what that costs: calm that lasts exactly as long as the force is paid, with a bill that does not stop growing. Everything above is a readout of the sources named, at the tiers stated, and is open to correction through the channels this paper says must stay open.

AI-assistance disclosure. Drafting, source fetching, passage extraction and the fixture were done with an AI assistant under the author's direction; the problem, the questions and the selection of what to claim are the author's. No AI or vendor is an author. Verbatim source passages were mechanically extracted from the fetched copies named in the bibliography. The author's raw lines behind this note are in the Blackbox Log (concept DOI 10.5281/zenodo.22302518, entries BBL-2026-09-05-129 to -134).

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Table 4: What each tradition gives the problem, and what it forces this paper to do. Comparison = same / different / cited; no priority is claimed.

Work	What it supplies	What it forces here
Galtung [1969] (read)	the gap definition; its author's own warning that "potential realizations" is "highly problematic"	D7 and (5): subtract only a witnessed, horizon-bounded baseline; keep the recoverable gap apart from accumulated loss
Galtung [1990] (read)	cultural violence as "those aspects of culture ... that can be used to justify or legitimize direct or structural violence" (copy p. 2)	legitimation is read as a change to what can be read, which options count, and whether objection has a channel; no whole culture is scored
Sen [1985] (record)	the agency/well-being split	weights and task lists are not the evaluator's alone
Emirbayer and Mische [1998] (record)	temporal-relational agency	h, T and layer 2 keep $\mathscr{G}$ and Π from being fixed for life
Bazzani [2023] (record)	agency as a conversion process with context-dependent factors	the closest sibling: "potential depends on conversion" is not this paper's contribution; what is added is the witness rule, the identifiability gate and the corrigibility test per event
Foucault [1982] (read)	power as structuring "the possible field of action of others"	D5: the feasible set must never be defined by the permitting authority
Pettit [2016] (read)	"you do not enjoy freedom as non-domination or independence just by virtue of actually escaping interference; it must also be that you would not suffer interference even if you had wished to act otherwise" (p. 3)	the correction channel must be stable under another's discretion, not merely unused today; this is what (6)'s presence test must be read against
Pettit [1997], Fricker [2007], Mahmood [2001] (record)	non-domination; testimonial and hermeneutical injustice; situated, embodied agency	the system's reading of the agent as well as the agent's reading of the world; consent is never read as docility nor culture used to erase real constraint
Aristotle [c. 350 BCE / online] (read)	potency as "the originative source, in the very thing acted on, of its being passively changed by another thing or by itself qua other"	potential is relational from the start: indexed to what can act on what; D2's reference class is the modern form of that index
Vygotsky [1978] via Chaiklin [2003] (read)	the zone of proximal development, potential "as determined through problem solving under adult guidance or in collaboration with more capable peers"	the earliest operational statement of D1: potential is defined by an assisted readout and is not independent performance
Nussbaum [2011] (read)	"What is each person able to do and to be?"; opportunities a person "may or may not exercise in action"	CHALLENGE to the witness rule: unexercised options count; answered by the reference-class reading of D2, not by dropping the rule
Robeyns [2017] (read, open access)	capabilities as "real freedoms or opportunities"; conversion factors as "the degree to which a person can transform a resource into a functioning"	CHALLENGE on the definition, ally on the mechanism: conversion factors are what witnessing records
Sen [1999] via Nussbaum [2011] (read at second hand)	capability as "the alternative combinations of functionings that are feasible for her to achieve"	CHALLENGE: "feasible" is analyst-declared; the witness rule makes it a record
Farmer [2004] (read)	structural violence needs "a thorough knowledge of history and political economy" (p. 305)	D7: the accumulated, history-fixed part of the gap is real and is not the recoverable gap
Salge et al. [2014] (read)	"An agent's empowerment is not only affected by the state of the world, i.e., the context of the agent, but also depends on what the agent's sensors and actions are" (p. 19)	the option alphabet is part of the agent's definition, which is what D2 turns into a witnessed set
Ryan and Deci [2000] (read)	"three innate psychological needs—competence, autonomy, and relatedness" (p. 68)	autonomy is a layer-2 capacity stated without a layer-1/layer-2 split; recorded as undetermined for D3
Bandura [2001] (abstract)	"people are producers as well as products of social systems"	same: layer 2 described, not separated; undetermined for D3
Kabeer [1999] (abstract)	empowerment as acquiring "the ability to make strategic life choices"; resources, agency and achievements as indivisible in an indicator's meaning	the sharpest threat to D3's measurability; carried as the open point of P-D
Bazzani [2023] (abstract, Europe PMC)	agency "as the result of a conversion process of personal resources shaped by conversion factors" at micro, meso and macro levels	the closest sibling: context-dependent potential is not this paper's contribution; the witness rule, the identifiability certificate and the corrigibility test are
Klyubin et al. [2005] (read)	empowerment as channel capacity in bits	the nearest information-theoretic ancestor; this paper adds task criteria, actor direction and corrigibility, and reads the alphabet as a witnessed set
Csaky [2026] (abstract read)	agency under partial observability in deterministic worlds; prediction / identification / empowerment separations	context only: a formal agency-theory paper, not evidence for or against the peace-research claims; kept because "reading is not controlling" is developed there
Satha-Anand [2021] (read)	"there are indeed nonviolent alternatives to violence ... based on solid social sciences and humanities" (p. 6)	alternatives must be witnessed and must change power relations; widening a list is not widening Π^{wit}

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Table 5: Citation ledger at this version. "Passage" = a verbatim passage extracted from a fetched copy and found again mechanically by the checking route. Every card listed was re-read by a fresh headless session of the same model as the drafting session (ladder class I1, not I3); the route's own bearing per card is shown. Nine record-only cards (no readable open copy) are parked and not counted.

Hypothesis	cards	verified	record	route bearing
H1 witness rule	9	9	0	1 support / 4 challenge / 4 context
H2 two layers	15	15	0	13 support / 2 context (author's own rows: all undetermined)
H3 pseudo-peace	8	8	0	8 context: the literature frames H3, its evidence is the fixture and the battery

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REFERENCES

Margaret S. Archer. *Realist Social Theory: The Morphogenetic Approach*. Cambridge University Press, 1995. Record consulted; not read for this paper.

Aristotle. Metaphysics, book IX (Theta), part 1. W. D. Ross translation, The Internet Classics Archive (classics.mit.edu), c. 350 BCE / online. Passage read from the online text, Book IX Part 1.

Albert Bandura. Self-efficacy: Toward a unifying theory of behavioral change. *Psychological Review*, 84(2):191–215, 1977. doi: 10.1037/0033-295X.84.2.191. Record consulted; open copies attempted, none readable.

Albert Bandura. Social cognitive theory: An agentic perspective. *Annual Review of Psychology*, 52:1–26, 2001. doi: 10.1146/annurev.psych.52.1.1. Abstract consulted; full text not read for this paper.

Giacomo Bazzani. Agency as conversion process. *Theory and Society*, 52:487–507, 2023. doi: 10.1007/s11186-022-09487-z. Open access; the publisher page blocked automated fetching on 2026-09-05, the abstract was read from the Europe PMC record (PMC10241693).

Seth Chaiklin. The zone of proximal development in Vygotsky's analysis of learning and instruction. In *Vygotsky's Educational Theory in Cultural Context*, pages 39–64. Cambridge University Press, 2003. Passage read from an open teaching copy (PDF), p. 2 of the copy.

Richard Csaky. Prediction and empowerment: A theory of agency through bridge interfaces. arXiv:2605.06346v1, working draft, 2026. HTML version fetched 2026-09-05; section structure read (bridge-interface POMDPs, bridge-gap theorem, prediction/identification/empowerment separations); proofs not checked.

Mustafa Emirbayer and Ann Mische. What is agency? *American Journal of Sociology*, 103(4):962–1023, 1998. doi: 10.1086/231294. Abstract consulted; full text not read for this paper.

Paul Farmer. An anthropology of structural violence. *Current Anthropology*, 45(3):305–325, 2004. doi: 10.1086/382250. Opening paragraph read from the publisher PDF, p. 305.

Michel Foucault. The subject and power. *Critical Inquiry*, 8 (4):777–795, 1982. doi: 10.1086/448181. Passage read from the Kobe University teaching copy, p. 15 of the copy.

Miranda Fricker. *Epistemic Injustice: Power and the Ethics of Knowing*. Oxford University Press, 2007. doi: 10.1093/acprof:oso/9780198237907.001.0001. Record consulted; not read for this paper.

Johan Galtung. Violence, peace, and peace research. *Journal of Peace Research*, 6(3):167–191, 1969. doi: 10.1177/002234336900600301. Passages read from the Kobe University teaching copy; page locators refer to that PDF (p. 3–4 of the copy = journal pp. 168–169).

Johan Galtung. Cultural violence. *Journal of Peace Research*, 27(3):291–305, 1990. doi: 10.1177/0022343390027003005. Passage read from the Galtung-Institut copy, p. 2 of the copy.

Anthony Giddens. *The Constitution of Society: Outline of the Theory of Structuration*. Polity Press, 1984. Record consulted; not read for this paper.

Naila Kabeer. Resources, agency, achievements: Reflections on the measurement of women's empowerment. *Development and Change*, 30(3):435–464, 1999. doi: 10.1111/1467-7660.00125. Record and abstract consulted; full text not read for this paper.

Alexander S. Klyubin, Daniel Polani, and Chrystopher L. Nehaniv. Empowerment: A universal agent-centric measure of control. In *2005 IEEE Congress on Evolutionary Computation*, pages 128–135, 2005. doi: 10.1109/CEC.2005.1554676. Passages read from the author repository copy (University of Hertfordshire), p. 3 of the copy.

Yaoharee Lahtee. When AI expands human potential — series index. Zenodo, 2026a.

Yaoharee Lahtee. Effective, corrigible agency potential: A proposal from readout universe and readout genesis in conversation with peace studies, capability theory, and agency theory. Programme research note, 5 September 2026 (Thai); superseded by the present paper, 2026b. K0; not deposited separately.

Yaoharee Lahtee. Potential – agency choice – violence: one axis across four programme papers (analysis note). cpg_research_journal, research/society-justice-peace, 5 September 2026, 2026c. Includes the fail-able fixture `sim/nogo_fixture.py` and its result file.

Yaoharee Lahtee. Causal agency: A persistence control theory of adaptive systems. Zenodo preprint, 2026d.

Yaoharee Lahtee. Causal ethics: The mathematics of regime choice and survival. Zenodo preprint, 2026e.

Yaoharee Lahtee. The causal grammar of structured coexistence: Conflict, violence, repair, and non-suppressive peace. Zenodo preprint, 2026f.

Yaoharee Lahtee. CTSA human-return readout: A session-boundary measurement architecture for retained human capability after AI. Zenodo preprint, Epistemic Note v9.1, 2026g.

Yaoharee Lahtee. Bounded amplitude suppression on the discrete linear spine: five lemmas and a numeric battery. cpg_research_journal, research/society-justice-peace, FORMAL_2026-09-05_bounded_suppression.md; battery `sim/battery.py` and `sim/battery_result.json`, 2026h. Exact algebra on the stated model; battery: 60 random instances.

Yaoharee Lahtee. Readout genesis standalone synthesis: Information epistemic foundation, conditioned agency, and the forced operator. Zenodo; repository file READOUT_GENESIS_CORE.md (Genesis Step-by-Step v3.1), 2026i.

Yaoharee Lahtee. glosa — rigour without infrastructure: A standalone scholar methodology for human–AI knowledge co-production (v0.4.1). Zenodo software record, 2026j.

Yaoharee Lahtee. Society, justice, peace & violence — series index. Zenodo, 2026k.

Yaoharee Lahtee. Violence as a special case of instability in finite-memory causal systems. Zenodo preprint, 2026l.

Saba Mahmood. Feminist theory, embodiment, and the docile agent: Some reflections on the Egyptian Islamic revival. *Cultural Anthropology*, 16(2):202–236, 2001. doi: 10.1525/can.2001.16.2.202. Record consulted; not read for this paper.

Martha C. Nussbaum. Capabilities as fundamental entitlements: Sen and social justice. *Feminist Economics*, 9(2–3): 33–59, 2003. doi: 10.1080/1354570022000077926. Record consulted; no open copy found.

Martha C. Nussbaum. *Creating Capabilities: The Human Development Approach*. Harvard University Press, 2011. Ch. 2 read from a publicly posted course excerpt, pp. 17–20.

Philip Pettit. *Republicanism: A Theory of Freedom and Government*. Oxford University Press, 1997. Record consulted; not read for this paper.

Philip Pettit. The globalized republican ideal. Max Weber Programme master-class paper, European University Institute (PDF), 2016. Passage read from the EUI PDF, p. 3; year is the programme year on the hosting page.

Ingrid Robeyns. *Wellbeing, Freedom and Social Justice: The Capability Approach Re-Examined*. Open Book Publishers, 2017. doi: 10.11647/OBP.0130. Ch. 2 read on OpenEdition Books (open access), paragraphs 39 and 50.

Richard M. Ryan and Edward L. Deci. Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American Psychologist*, 55 (1):68–78, 2000. doi: 10.1037/0003-066X.55.1.68. Passage read from the author-hosted PDF, p. 68.

Christoph Salge, Cornelius Glackin, and Daniel Polani. Empowerment — an introduction. arXiv:1310.1863, 2014. Passage read from the arXiv PDF, p. 19 (§4.5.3).

Chaiwat Satha-Anand. Three windows and critical peace/nonviolence research. *Hiroshima Peace Research Journal*, 8:5–12, 2021. Passages read from the journal PDF, pp. 5–6.

James C. Scott. *Domination and the Arts of Resistance: Hidden Transcripts*. Yale University Press, 1990. Record consulted; not read for this paper.

Amartya Sen. Well-being, agency and freedom: The Dewey lectures 1984. *The Journal of Philosophy*, 82(4):169–221, 1985. doi: 10.2307/2026184. Record consulted; full text not read for this paper (paywalled).

Amartya Sen. *Development as Freedom*. Oxford University Press, 1999. Quoted through Nussbaum 2011, p. 20; primary not opened.

L. S. Vygotsky. *Mind in Society: The Development of Higher Psychological Processes*. Harvard University Press, 1978. p. 86, as quoted verbatim in Chaiklin 2003 (primary not opened).

10.5281/zenodo.22361830

Part IV

Society, Power, Justice: Potential Under Compression

Society, Power, Justice: Potential Under Compression collects five works, three read in full and two by excerpt, at the conflict, violence, ethics, and authority segment of the programme spine (World → Readout → Meaning → Experience → Retention → Live Possibility → Choice → Agency → Human–AI → Human Return → Warrant). All five were deposited between January and March 2026, before the human–AI chapters gathered in Parts 2, 3, and 5. Placed in that order, this part reads as the apparatus of Parts 1–3 tested first against conflict, violence, and repair, and only afterward, in Part 5, tested a second time against AI specifically.

The first four chapters form a linked sequence. Chapter 19 proposes a five-term causal grammar — structure, conflict, violence, repair, peace — for structured coexistence. Chapter 20 reformulates violence as a dynamical instability in finite-memory systems rather than a discrete event, and states a formal no-go result against suppression-only intervention. Chapter 21, an excerpt (individual and collective ethics, and a section on human–AI as a hybrid multi-agency system, from a 99-page work), formalizes ethics as an admissibility condition on regime choice with a stated falsification rule. Chapter 22, also an excerpt (the persistence-control framework and its implications for artificial systems, from a 42-page work), proposes agency as a structural, constraint-regulating property of adaptive systems in general, not a mechanism specific to one domain. Chapter 23 turns from structure to authority, tracing interpretive authority across historical regimes and asking whether it is shifting toward platforms, algorithms, and AI-mediated systems; per the library's linkage data it draws on chapters from Part 1 and on Chapters 19–21 of this part.

Carried forward: the move from a named phenomenon to a structural, formally stated condition for it; the practice, where a falsification rule is given, of naming what would force a revision rather than defend a claim; and the K0 / preprint / [Open] tier discipline used throughout the book. Per the library's linkage data, four of this part's chapters are themselves continued by Chapter 18 in Part 3, and Chapter 23 is referenced by a chapter in Part 5 — whichever part is read first, this part supplies constraint vocabulary the other draws on.

- The Causal Grammar of Structured Coexistence: Conflict, Violence, Repair, and Non-Suppressive Peace — `10.5281/zenodo.1892513`
- Violence as a Special Case of Instability in Finite–Memory Causal Systems — `10.5281/zenodo.18383439`
- Causal Ethics: The Mathematics of Regime Choice and Survival — `10.5281/zenodo.18444260`
- Causal Agency: A Persistence Control Theory of Adaptive Systems — `10.5281/zenodo.18897585`
- The Civilization of Knowledge: Who Has the Authority to Interpret the World — `10.5281/zenodo.18943971`

The Causal Grammar of Structured Coexistence: Conflict, Violence, Repair, and Non-Suppressive Peace

9 March 2026 • DOI 10.5281/zenodo.18925131

What the chapter states. "This article proposes a causal grammar linking five concepts: structure, conflict, violence, repair, and peace. [. . .] Conflict emerges when actors attempt to sustain structures that cannot be jointly reproduced. Violence occurs when incompatibility compresses actors' feasible life-space."

Status. "(preprint)" (as printed in the running header); no separate peer-review statement appears in the body.

Falsifier(s). No explicit falsifier stated in the chapter.

Read before. — (no chapter in this book is listed as a prerequisite in the KG).
Feeds into. *Potential as a Readout: Witnessed Envelopes, Two-Layer Agency, and the Measurement of Pseudo-Peace* (Part 3); *The Civilization of Knowledge: Who Has the Authority to Interpret the World* (this Part).

The Causal Grammar of Structured Coexistence: Conflict, Violence, Repair, and Non-Suppressive Peace

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March 9, 2026

Abstract

Sociological and philosophical traditions have developed extensive explanations of social structure, conflict, violence, and peace. Yet these phenomena are often analyzed within partially separate theoretical frameworks. As a result, disagreement is frequently conflated with conflict, conflict with violence, and stability with peace.

This article proposes a causal grammar linking five concepts: structure, conflict, violence, repair, and peace. Social structures organize fields of feasible possibility within shared social environments. Conflict emerges when actors attempt to sustain structures that cannot be jointly reproduced. Violence occurs when incompatibility compresses actors' feasible life-space. Repair denotes the protected reopening of damaged relations, while peace refers to stable coexistence that preserves actors' possibilities and maintains the capacity for repair.

The framework provides a compact diagnostic vocabulary for distinguishing difference from conflict, conflict from violence, and stable order from non-suppressive peace.

Keywords: conflict; violence; peace; social theory; political sociology

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1 Introduction

Human coexistence unfolds within structured social worlds. Institutions, norms, identities, and material arrangements organize the possibilities available to actors within shared environments.

Existing traditions approach these dynamics from different analytical perspectives. Structural theory explains how institutions reproduce social order (Giddens 1984; Sewell 1992). Conflict sociology analyzes struggles over incompatible claims and authority relations (Coser 1956; Collins 1975). Violence studies examine both direct force and structural harm (Galtung 1969; Farmer 2004). Peace research explores reconciliation and durable coexistence (Lederach 1997).

Despite these insights, sociology lacks a compact grammar linking structure, conflict, violence, and peace within a single causal framework.

This article develops a causal grammar of structured coexistence. Social structures organize fields of feasible possibility. Within these fields, conflict emerges from incompatible coexistence, violence from possibility compression, repair from protected reopening, and peace from non-suppressive stability.

2 Structure as Organized Possibility

Structures organize the range of actions that can be sustained within a social environment. Institutions, symbolic classifications, and material arrangements shape which forms of participation, recognition, livelihood, and mobility remain viable.

Structure therefore functions as an architecture of feasible possibility.

This perspective allows conflict and violence to be understood as transformations within organized possibility spaces rather than simply interpersonal events.

3 Conflict as Incompatible Coexistence

Plurality is common in social life. Different institutions and identities often coexist within the same environment.

Conflict arises when actors attempt to sustain structures whose conditions of reproduction cannot be jointly maintained.

Conflict therefore refers not simply to disagreement but to incompatible coexistence.

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4 Violence as Possibility Compression

Violence occurs when incompatible coexistence significantly compresses actors' feasible life-space.

Feasible life-space refers to the range of meaningful possibilities through which actors can sustain participation, safety, livelihood, and recognition.

Violence may occur through physical force, institutional exclusion, structural arrangements, or symbolic delegitimation when these processes severely narrow actors' viable possibilities.

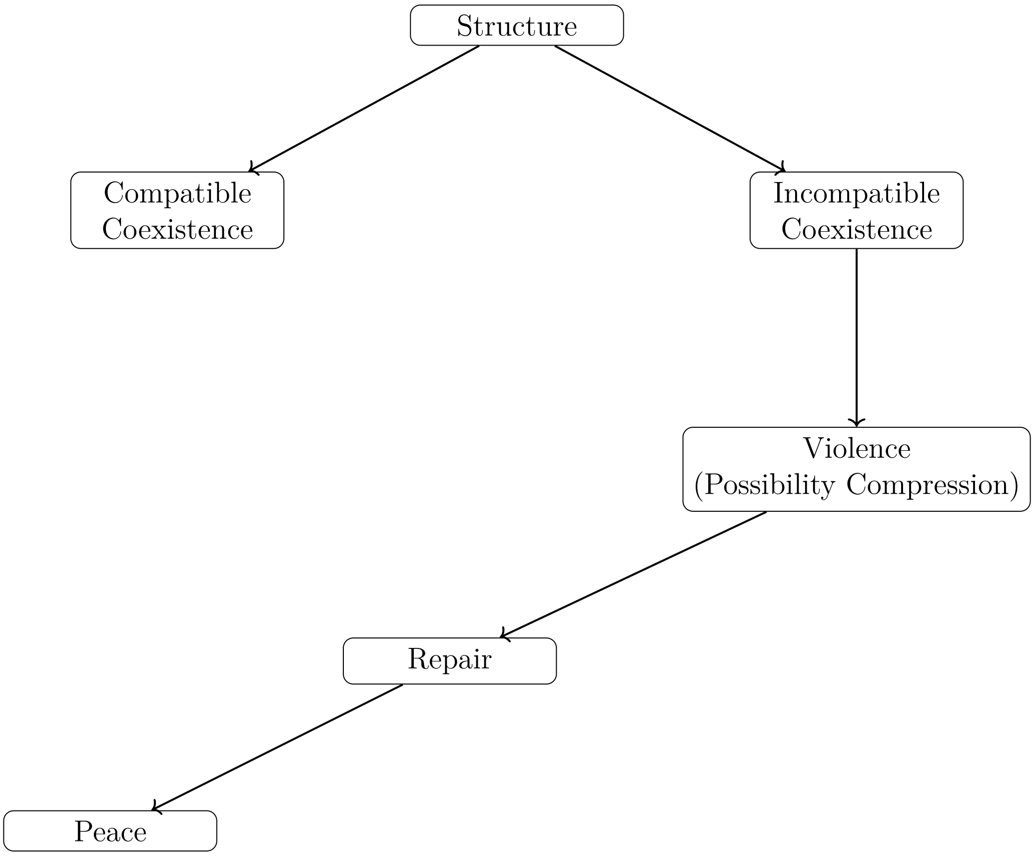

Figure 1: Causal grammar of structured coexistence

The diagram illustrates the causal dynamics of structured coexistence. Social structures generate fields of feasible possibility. These fields may sustain compatible coexistence or produce incompatible coexistence. When incompatibility compresses actors' viable possibilities, violence emerges. Repair reopens damaged relations, enabling the possibility of non-suppressive peace.

5 Repair as Protected Reopening

Repair refers to the protected reopening of damaged relations under conditions that prevent renewed possibility compression.

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Regime	Compatibility	Possibility Space	Repair
Tension	Compatible	Stable	Not central
Conflict	Incompatible	Contested	Possible
Violence	Incompatible	Compressed	Blocked
Pseudo-peace	Compatible appearance	Restricted	Suppressed
Peace	Compatible	Preserved	Protected

Table 1: Regimes of structured coexistence

Repair allows actors to renegotiate coexistence without immediate exposure to renewed coercion.

6 Peace as Non-Suppressive Stability

Peace is not merely the absence of conflict.

Peace exists where coexistence remains stable while actors retain viable possibility space and the institutional capacity for repair.

7 Conclusion

This article proposed a causal grammar linking structure, conflict, violence, repair, and peace.

Structures organize fields of feasible possibility. Conflict arises from incompatible coexistence. Violence compresses actors' feasible life-space. Repair reopens damaged relations. Peace stabilizes coexistence without suppressing actors' possibilities.

The framework contributes a compact diagnostic grammar capable of distinguishing disagreement from conflict, conflict from violence, and stability from non-suppressive peace.

References

Bar-Tal (2000)

Collins (1975)

Collins (2008)

Coser (1956)

Farmer (2004)

Galtung (1969)

Giddens (1984)

10.5281/zenodo.18925131

Lederach (1997)

Sewell (1992)

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Violence as a Special Case of Instability in Finite–Memory Causal Systems

27 January 2026 • DOI 10.5281/zenodo.18383439

What the chapter states. "[V]iolence is not an object or an act, but a dynamical instability arising in systems with finite causal memory operating under sustained load. [. . .] We derive a no-go result for suppression-only strategies [. . .] and identify ecosystem-level stabilization—not coercive control—as the only structurally admissible route to durable violence reduction."

Status. No K0 / preprint / peer-review tag is printed anywhere in this chapter's text.

Falsifier(s). No explicit falsifier stated in the chapter.

Read before. — (no chapter in this book is listed as a prerequisite in the KG).
Feeds into. *Potential as a Readout: Witnessed Envelopes, Two-Layer Agency, and the Measurement of Pseudo-Peace* (Part 3); *The Civilization of Knowledge: Who Has the Authority to Interpret the World* (this Part).

Violence as a Special Case of Instability in Finite–Memory Causal Systems

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Abstract

Violence is commonly treated as a discrete event caused by individual agents and addressed through force-based suppression. Despite extensive legal, military, and policing efforts, violence reliably recurs, mutates across domains, and persists across generations. In this work we propose a structural reformulation: violence is not an object or an act, but a dynamical instability arising in systems with finite causal memory operating under sustained load. Using a causal–spectral framework grounded in finite-speed transport, memory-dependent dynamics, and operator theory, we show that force-based elimination is generically impossible. We derive a no-go result for suppression-only strategies, introduce an operator-level taxonomy of violence, and demonstrate recurrence and mutation using a minimal constructive model. The analysis unifies insights from public health, social science, and mathematical physics, and identifies ecosystem-level stabilization—not coercive control—as the only structurally admissible route to durable violence reduction.

1 Introduction

Violence remains one of the most persistent and costly phenomena in human societies. It appears in physical, psychological, institutional, and digital forms, often re-emerging despite increasingly sophisticated mechanisms of control. This persistence suggests that violence is not merely a failure of enforcement or morality, but a consequence of deeper structural constraints. Most dominant paradigms treat violence as an *event*: a localized rupture caused by identifiable agents. Such an ontology implicitly assumes immediacy, locality, and eliminability. Empirical reality contradicts all three. Violence propagates with delay, accumulates historical residues, and returns even after apparent suppression. This paper advances a different claim: *violence is a special case of instability in systems with finite causal memory.* Under this view, violent acts are manifestations of unstable modes, not fundamental causes. The appropriate object of intervention is therefore not the event, but the system operator that governs memory, dissipation, and coupling.

2 From Events to States: Ontological Reframing

Let a social system be described by a state field

$$s(x,t) \in \mathbb{R}^n, \tag{1}$$

encoding cumulative tension, perceived injustice, trust density, and communicative load over social space x and time t. Violence is not identified with a specific state value, but with regions of the

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state manifold that are dynamically unstable. We define a violence potential

$$V(x,t) = \Phi(s(x,t)), \tag{2}$$

where Φ is a nonlinear functional mapping latent stress to overt instability. Observed violent events correspond to threshold crossings of V, not to primary ontological units.

3 Finite Causal Memory and Telegraph Dynamics

Social systems exhibit finite propagation speed and nonzero memory. Let $j(x,t)$ denote the flux of social tension. We adopt a causal (telegraph-type) constitutive relation

$$\tau \partial_t j + j = -D\nabla s, \tag{3}$$

where $\tau > 0$ is the causal memory time and D an effective coupling constant. State evolution obeys a continuity equation

$$\partial_t s = -\nabla \cdot j - \Gamma(s) + S_{\text{env}}, \tag{4}$$

with Γ representing dissipation and S_{env} sustained environmental load (e.g., inequality, humiliation, communicative blockage). The presence of $\tau > 0$ implies that instantaneous stabilization is causally inadmissible. Past states actively constrain present dynamics.

4 Spectral Structure and Persistence

Linearizing around a metastable configuration yields

$$\partial_t s = \mathcal{L}_\tau s, \tag{5}$$

with generator $\mathcal{L}_\tau$ dependent on memory time. The solution admits a spectral decomposition

$$s(t) = \sum_k c_k e^{-\lambda_k t} r_k, \tag{6}$$

where λ_k are eigenvalues of $\mathcal{L}_\tau$. For $\tau > 0$, the spectrum generically contains slow modes with $\text{Re}(\lambda_k) \approx 0^+$. These modes encode long-lived configurations of instability. Force-based suppression alters amplitudes c_k but leaves the spectrum invariant. Under sustained S_{env}, slow modes are re-excited, producing recurrence.

5 A Structural No-Go Result

Theorem (No-Go for Elimination by Suppression). Consider a system governed by $\mathcal{L}_\tau$ with $\tau > 0$. Assume: (i) at least one slow spectral mode exists, (ii) suppression acts only on state amplitudes, and (iii) environmental load has nonzero time-average. Then no bounded force-based intervention can drive the system to a globally violence-free absorbing state.

Proof (sketch). Finite memory implies non-Markovian evolution. Slow modes ensure metastability. Since suppression does not modify $\mathcal{L}_\tau$, the invariant spectral structure persists. Sustained load re-populates slow modes with probability one. □

This result is structural, not normative. It holds independently of intent, ethics, or political context.

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6 Operator Taxonomy of Violence

Within the causal–spectral framework, violent phenomena admit a natural classification.

6.1 Spontaneous Violence

Stochastic excitation of slow modes under high load and low dissipation. Examples include domestic conflict and sudden riots.

6.2 Structural Violence

Instability encoded directly in $\mathcal{L}_\tau$ through asymmetric couplings or chronic load. Persistence occurs even without triggering events.

6.3 Strategic Violence

Intentional maintenance or engineering of slow modes to stabilize power or extract compliance. Here violence functions as a control signal, not noise.

7 Mutation and Tunneling

Blocking one expression channel does not remove instability. Coupling between subsystems enables tunneling: physical violence may reappear as psychological harm, digital harassment, or institutional coercion. This is a direct consequence of operator coupling, not a change in underlying cause.

8 Minimal Constructive Model

Consider two coupled subsystems, $s_1(t)$ (physical) and $s_2(t)$ (digital/psychological), with generator

$$\mathcal{L} = \begin{pmatrix} -\alpha & \epsilon \\ \epsilon & -\beta \end{pmatrix}, \qquad \alpha, \beta > 0. \tag{7}$$

Eigenvalues are

$$\lambda_\pm = -\frac{\alpha+\beta}{2} \pm \sqrt{\left(\frac{\alpha-\beta}{2}\right)^2 + \epsilon^2}. \tag{8}$$

For small ϵ, one mode becomes slow. Suppressing s_1 transfers instability to s_2, demonstrating recurrence and mutation without introducing new causes.

9 Ecosystem-Level Stabilization

Durable reduction requires operator-level modification:

$$\Gamma \uparrow \qquad \text{increase dissipation,} \tag{9}$$

$$S_{\text{env}} \downarrow \qquad \text{reduce load,} \tag{10}$$

$$\mathcal{L}_{ij} \downarrow \qquad \text{limit propagation,} \tag{11}$$

$$\tau \downarrow \qquad \text{shorten memory.} \tag{12}$$

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These correspond to restorative justice, inequality reduction, community firewalls, and truth-based reconciliation, respectively.

10 Discussion

Violence emerges as a predictable instability in finite-memory systems under sustained load. Event-based suppression fails not morally, but structurally. The causal–spectral framework explains persistence, mutation, and recurrence within a single architecture.

11 Conclusion

Violence is not eradicated by force, but by redesigning the systems in which it survives. Recognizing violence as a special case of instability shifts intervention from control to stabilization, from punishment to architecture. This reframing provides a principled foundation for sustainable peace across psychological, social, and political domains.

Compact References (internal)

WHO (2002); Krug (2002); Heise (1998); Bronfenbrenner (1979); Slutkin (2013); Butts (2015); Sampson et al. (1997); Branas et al. (2018); Tyler (2006); Mazerolle et al. (2013); Makleff et al. (2020); Tracy et al. (2023); Lowe et al. (2023); Cattaneo (1948); Vernotte (1958); Gurtin & Pipkin (1968); Romanov (2014); Levin et al. (2009); Barad (2007); Wendt (2015); Busemeyer et al. (2017).

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Causal Ethics: The Mathematics of Regime Choice and Survival

Excerpt, pp. 13–34, 73–77 of a 99-page work • 31 January 2026 • DOI 10.5281/zenodo.18444260

What the chapter states. "The framework yields testable diagnostics — Lyapunov ethical load, spectral survival, regime incompatibility, and accumulated moral cost — and produces falsifiable forecasts for individual, collective, and artificial agents operating under finite causal memory. This abstract constitutes a scope contract for the entire book."

Status. No K0 / preprint / peer-review tag is printed; the front matter instead states its own scope directly: "This work is: not a theological doctrine, not a moral prescription system, not psychological, medical, legal, or clinical advice."

Falsifier(s). "Causal Ethics adopts a strict falsification rule"; the chapter states that once its failure count k reaches the critical threshold k_{crit}, at least one of the regime model, the cost functional, or the assumed proxies must be revised, and that "CE does not permit ad hoc reinterpretation to avoid revision."

Read before. — (no chapter in this book is listed as a prerequisite in the KG).

Feeds into. *Potential as a Readout: Witnessed Envelopes, Two-Layer Agency, and the Measurement of Pseudo-Peace* (Part 3); *The Civilization of Knowledge: Who Has the Authority to Interpret the World* (this Part).

Chapter 2

Primitives and Axioms of Causal Ethics

Chapter Goal

This chapter fixes the *primitive objects* and *axioms* of Causal Ethics (CE). All later equations, diagnostics, case studies, and predictions depend exclusively on the commitments stated here. Nothing in this book overrides or contradicts these primitives. If a claim cannot be expressed using the objects and axioms below, it is outside the scope of CE.

2.1 Axiom I: Reality as Manifested Record

2.1.1 Statement

Axiom I (Reality-as-Record). There exists a manifested, admitted, and aggregatable record of the system at update-time t':

$$\boxed{\text{(CE-01)} \quad M(t') \in \mathcal{M}.}$$

2.1.2 Meaning and Boundary

$M(t')$ is not assumed to be ontologically complete, ultimate, or fundamental. It is the *record that is actually used* for decisions, actions, and evaluations. CE therefore applies equally to:

- physical systems (measurements, states),

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- social systems (laws, records, norms),
- informational systems (logs, databases, models),
- hybrid systems (human–AI interaction records).

CE never reasons about what lies beyond $M(t')$. All ethical claims are conditional on what is admitted into the record.

Consequence. Ethics in CE is *record-relative*, not metaphysical. Disagreement about ultimate reality does not invalidate CE as long as the parties share (or contest) a common $M(t')$.

2.2 Axiom II: Agency as Capacity for Choice

2.2.1 Statement

Axiom II (Agency-as-Choice). An entity qualifies as an *agency* if and only if it can instantiate a choice of regime relative to $M(t')$:

$$\boxed{\text{Agency } A \text{ exists } \iff \exists\, R_A(t') \in \mathcal{R}_{\text{adm}}(t') \text{ such that } R_A \text{ is selected by } A.}$$

Formally, agencies appear as sub-structures of the record:

$$\boxed{\text{(CE-02)} \quad A(t') \subseteq M(t').}$$

2.2.2 Choice as the Non-Negotiable Primitive

In CE, *choice* is not psychological intention, free will doctrine, or moral sentiment. It is the structural act of selecting a regime R_A from an admissible set.

> **If there is no choice, there is no agency. If there is no agency, there is no ethics.**

This axiom has three immediate consequences:

1. Tools, mechanisms, and forced processes are *not* ethical subjects.
2. Responsibility attaches only to agents who select regimes.
3. Ethics cannot be evaluated for purely reactive or coerced behavior.

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2.3 Axiom III: Regime as Operator Bundle

2.3.1 Statement

Axiom III (Regime Structure). A regime is an ordered pair of operators acting on the manifested record and its agency sub-structures:

$$\boxed{\text{(CE-05)} \quad R := (\mathcal{T}_R,\ \mathcal{I}_R).}$$

The regime determines:

- how the record is translated forward,
- how the agency is transformed by interacting with that record.

2.3.2 Update Laws

Once a regime R is selected, updates follow deterministically at the abstract level:

$$\boxed{\text{(CE-06)} \quad M(t' + \Delta t') = \mathcal{T}_R(M(t')),}$$

$$\boxed{\text{(CE-07)} \quad A(t' + \Delta t') \subseteq \mathcal{I}_R(A(t'), M(t')).}$$

CE does not restrict the internal form of $\mathcal{T}_R$ or $\mathcal{I}_R$; it restricts *which regimes are admissible* based on stability, causality, and survival.

2.4 Chosen Regime vs. Forced Regime

2.4.1 Definition

Chosen Regime. A regime R_A is *chosen* if it is selected by the agency A from $\mathcal{R}_{\text{adm}}(t')$.
Forced Regime. A regime R is *forced* on A if it governs the update of A and M without A selecting it.

2.4.2 Ethical Asymmetry

This distinction is foundational:

- Ethics is defined *only* for chosen regimes.

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- Forced regimes may cause harm or instability, but they do not generate responsibility for the coerced agent.
- Moral cost $\mathcal{C}_{A,R}$ (CE-27) is accrued only when R is chosen by A.

Formally, this is encoded in the etic definition:

$$\boxed{\text{(CE-08)} \quad \text{Etic}(A;t') := \exists\, R_A(t') \in \mathcal{R}_{\text{adm}}(t') \text{ selected by } A.}$$

If no such selection exists, $\text{Etic}(A;t')$ is undefined, and CE makes no ethical judgment.

2.5 Collectives and Shared Records

2.5.1 Statement

Axiom IV (Collective as Coupled Agencies). A collective is a finite set of agencies sharing a manifested record:

$$\boxed{\text{(CE-03)} \quad G(t') := \{A_1(t'), \dots, A_N(t')\} \subseteq M(t').}$$

Collective ethics does not replace individual ethics. It emerges when regime choices are coupled through the same record $M(t')$.

2.5.2 No Collective Without Individuals

CE explicitly rejects:

> "The collective is ethical even if individuals are erased."

Any collective regime that violates the survival or stability conditions of constituent agencies will be diagnosed later as conflict, injustice, or structural failure (CE-22–CE-24).

2.6 Axiomatic Closure

The axioms of Causal Ethics are therefore:

1. **Reality exists as a manifested record** $M(t')$ (Axiom I).

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2. **Agency exists iff it can choose a regime** (Axiom II).
3. **Regimes are operator bundles** governing translation and interaction (Axiom III).
4. **Ethics applies only to chosen regimes**, never to forced ones.
5. **Collectives are sets of agencies**, not super-agents with independent moral status.

Everything that follows—ethical load, stability, conflict, cost, karma, and constitutional design— is a mathematical consequence of these primitives and nothing else.

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Chapter 3

Core Definition: Ethics as Admissible Regime-Choice

Chapter Goal

This chapter gives the *core definition* of Causal Ethics. Everything else in this book—stability analysis, conflict diagnostics, cost, karma, case studies, and predictions— is a consequence of the definition fixed here.

The purpose is not to persuade, moralize, or interpret values, but to state a precise admissibility condition on *regime selection by an agency.* If this condition fails, the situation is *not ethical* in the technical sense used in this book.

3.1 The One-Line Definition

3.1.1 Mathematical Form (Locked)

Definition (Ethics as Admissible Regime-Choice). An agency A is ethical at update-time t' if and only if it selects a regime R_A that satisfies the following three structural constraints:

$$\boxed{\mathrm{Ethical}(A) \iff \exists\, \mathbf{Choice}(A \to R_A) : \frac{d}{dt'} V_{A,R_A}(M(t')) \le 0 \;\wedge\; \tau'_c(R_A) > 0 \;\wedge\; \Delta_{\mathrm{spec}}(R_A) > 0}$$

This definition is final and will not be weakened or reinterpreted later in the book.

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3.1.2 Plain Language (Still Formal)

Ethics is the act of choosing how to interact with reality in a way that:

1. does not increase structural load or instability,
2. respects finite causal response and memory,
3. does not trigger explosive or erasing dynamics.

Ethics is therefore not a feeling, a rule set, or a value ranking. It is a *constraint-satisfying choice of regime.*

3.2 Ethical Load: The Lyapunov Potential $V_{A,R}$

3.2.1 Definition

Definition (Ethical Load). For an agency A operating under regime R, the ethical load is a non-negative scalar functional:

$$\boxed{\text{(CE-09)} \quad V_{A,R}(M) \geq 0.}$$

3.2.2 Structural Meaning

$V_{A,R}$ measures accumulated strain, tension, or instability experienced by A relative to the manifested record M under regime R.

Importantly:

- $V_{A,R}$ is not psychological suffering.
- $V_{A,R}$ is not moral guilt.
- $V_{A,R}$ is a *dynamical load functional.*

If V grows, the system is moving away from stability. If V decreases or remains constant, the system is dissipating disturbance.

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3.2.3 Why Lyapunov?

The inequality

$$\frac{d}{dt'} V_{A,R_A} \leq 0$$

is a Lyapunov-type condition: it certifies that the chosen regime does not amplify perturbations. Ethics, in CE, is therefore a *stability certificate*, not a moral verdict.

3.3 Causal Memory Constraint $\tau_c'(R)$

3.3.1 Definition

Definition (Causal Memory Time). Each regime R has an associated causal memory or response time:

$$\boxed{\text{(CE-10)} \quad \tau_c'(R) > 0.}$$

3.3.2 Structural Meaning

$\tau_c'(R)$ encodes the fact that:

- reality updates are not instantaneous,
- information propagation and response require time,
- decisions cannot outrun the system's ability to register consequences.

Regimes with $\tau_c'(R) \approx 0$ correspond to:

- impulsive reactions,
- high-frequency exploitation,
- causality-violating shortcuts.

Such regimes are inadmissible for ethics, regardless of their short-term benefit.

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3.4 Spectral Stability Margin $\Delta_{\text{spec}}(R)$

3.4.1 Definition

Definition (Spectral Stability Margin). A regime R admits a positive spectral margin if:

$$\boxed{\text{(CE-11)} \quad \Delta_{\text{spec}}(R) > 0.}$$

3.4.2 Structural Meaning

$\Delta_{\text{spec}}(R)$ certifies that:

- the induced dynamics do not blow up,
- perturbation modes decay rather than amplify,
- agency-supporting structures are not erased.

If $\Delta_{\text{spec}}(R) \leq 0$, the regime is unstable: small disturbances lead to runaway dynamics, collapse, or disappearance. No such regime can be ethical, even if it appears beneficial locally.

3.5 No Choice, No Ethics

3.5.1 Formal Statement

Principle (Choice Necessity). If an agency does not select a regime, then ethics is undefined:

$$\boxed{\neg\exists\, \textbf{Choice}(A \to R) \;\Rightarrow\; \neg\text{Ethical}(A) \;\wedge\; \mathcal{C}_A = 0.}$$

3.5.2 Consequences

This principle implies:

- No choice $\Rightarrow$ no responsibility.
- No choice $\Rightarrow$ no moral cost.
- No choice $\Rightarrow$ no ethical praise or blame.

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Victims under forced regimes are not unethical. Machines without regime-selection capacity are not unethical. Systems become ethical subjects only when they choose how to operate.

3.6 Why This Definition Is Sufficient

The definition given in this chapter is intentionally minimal. It does not reference:

- happiness,
- virtue,
- intention,
- moral rules,
- cultural norms.

Yet it is sufficient to derive:

- individual ethics,
- collective ethics,
- moral conflict,
- injustice,
- responsibility,
- karma-like long-term consequences,
- testable predictions.

All of these follow mathematically from admissible regime-choice under stability, causality, and spectral survival constraints.

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Chapter 4

Moral Cost: The Accounting Layer (Individual and Collective)

Chapter Goal

This chapter introduces *moral cost* as the accounting layer of Causal Ethics. While previous chapters establish admissibility (what is ethical *now*), this chapter formalizes consequences over time.

Moral cost answers a different question:

> *What does a chosen regime accumulate, burden, and eventually destroy or preserve when its effects are integrated over time?*

Ethics without accounting is incomplete. Causal Ethics therefore treats cost not as punishment or guilt, but as a structural accumulation induced by regime choice.

4.1 Moral Cost as a Functional

4.1.1 Core Definition

Definition (Moral Cost Functional). For an agency A operating under a chosen regime R over a time interval $[t_1, t_2]$, the moral cost is defined as:

$$\boxed{\text{(CE-27)} \quad \mathcal{C}_{A,R}[t_1, t_2] := \int_{t_1}^{t_2} \Big(\alpha\, \dot{V}^{+}_{A,R}(M(t')) + \beta\, \chi_{\text{causal}}(R) + \gamma\, \chi_{\text{spec}}(R)\Big)\, dt',}$$

where $\alpha, \beta, \gamma > 0$ are fixed weighting parameters.

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4.1.2 Interpretation of Terms

Each term represents a distinct source of ethical burden:

- $\dot{V}^{+}_{A,R}$: cost from worsening instability or load.
- $\chi_{\mathrm{causal}}(R)$: cost from violating finite causal response.
- $\chi_{\mathrm{spec}}(R)$: cost from operating in a spectrally unstable regime.

The positive-part operator

$$\dot{V}^{+} := \max\left(\frac{dV}{dt'}, 0\right)$$

ensures that only *worsening* dynamics generate cost. Stabilizing actions do not accrue moral debt.

4.2 Violation Indicators

4.2.1 Causal Violation

Definition (Causal Violation Indicator).

$$\boxed{\text{(CE-25)} \quad \chi_{\mathrm{causal}}(R) := \mathbf{1}\left[\tau_c'(R) \approx 0\right].}$$

This indicator activates when a regime attempts to bypass or collapse causal memory, such as impulsive exploitation or ultra-high-frequency control.

4.2.2 Spectral Violation

Definition (Spectral Violation Indicator).

$$\boxed{\text{(CE-26)} \quad \chi_{\mathrm{spec}}(R) := \mathbf{1}[\Delta_{\mathrm{spec}}(R) \leq 0]\,.}$$

This indicator activates when a regime operates in a mode where perturbations amplify or agency-supporting structures are erased.

Key Point. A regime can appear beneficial locally and still incur cost if it violates causal or spectral constraints. Cost is therefore not subjective—it is structural.

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4.3 Responsibility as Cost-Bearing

4.3.1 Definition

Definition (Responsibility). An agency is responsible precisely to the extent that it accrues moral cost:

$$\boxed{\text{(CE-28)}\quad \mathsf{Resp}(A) \equiv \mathcal{C}_{A,R_A}.}$$

Responsibility is not intention, awareness, or blame. It is the objective accumulation generated by chosen regimes.

4.3.2 Forced Regimes Revisited

If a regime is forced on an agency, then:

$$\mathcal{C}_{A,R} = 0,$$

even if harm occurs. Cost attaches to the chooser, not the carrier of consequences.

4.4 Collective Moral Cost

4.4.1 Definition

Definition (Collective Moral Cost). For a collective $G = \{A_1, \ldots, A_N\}$:

$$\boxed{\text{(CE-29)}\quad \mathcal{C}_G[t_1, t_2] := \sum_{i=1}^{N} w_i\, \mathcal{C}_{A_i,R_i}[t_1, t_2], \qquad w_i > 0.}$$

The weights w_i encode structural relevance, not moral worth. They may represent exposure, dependency, or coupling strength.

4.4.2 Injustice as Burden Asymmetry

Definition (Structural Injustice).

$$\boxed{\text{(CE-30)}\quad \exists\, i, j : \mathcal{C}_{A_i,R_i} \gg \mathcal{C}_{A_j,R_j}.}$$

Injustice is diagnosed when cost is persistently concentrated on a subset of agencies. CE does not require intent or malice to identify injustice— only asymmetry.

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4.5 Karma: Long-Run Cost and Survival

4.5.1 Definition

Definition (Karma as Cost Divergence).

$$\boxed{\text{(CE-31)} \quad \lim_{T\to\infty} \mathcal{C}_{A,R}[0,T] = \infty.}$$

In CE, karma is not metaphysical retribution. It is the mathematical fact that persistent violations accumulate without bound.

4.5.2 Link to Spectral Erasure

Theorem (Cost–Survival Link).

$$\boxed{\text{(CE-32)} \quad \lim_{T\to\infty} \mathcal{C}_{A,R}[0,T] = \infty \;\Rightarrow\; \lim_{n\to\infty} \left|P_{S_A} M(t'+n)\right|^2 = 0.}$$

Unbounded moral cost implies eventual disappearance of the agency-supporting spectral subspace. This is structural extinction, not punishment.

4.6 The Zero-Cost Ideal

4.6.1 Strong Ethical Criterion

Definition (Zero-Cost Ethics).

$$\boxed{\text{(CE-33)} \quad \mathsf{Eth}(A) \iff \mathcal{C}_{A,R_A}[t,\infty) = 0.}$$

This is an ideal condition. It asserts the existence of regimes that stabilize without accumulating debt.

4.6.2 System-Level Optimization

Definition (Constitutional Regime Selection).

$$\boxed{\text{(CE-34)} \quad R^\star \in \arg\min_{R\in\mathcal{R}_{\text{adm}}} \mathcal{C}_{\text{system},R} \quad \text{s.t.} \quad \tau_c'(R) > 0,\; \Delta_{\text{spec}}(R) > 0.}$$

$R^\star$ represents a constitutional optimum: a regime that minimizes accumulated cost while preserving causality and spectral stability.

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4.7 Why Accounting Completes Ethics

Admissibility defines what is ethical *now*. Accounting determines what survives *over time*.

Without moral cost:

- short-term exploitation appears ethical,
- injustice remains invisible,
- collapse is unexplained.

With moral cost, ethics becomes predictive. The next chapters will show how this accounting layer explains historical collapse, social pathologies, and future risks—including human–AI systems.

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Chapter 5

Individual Ethics

Goal. This chapter defines how *one* agency A is evaluated under Causal Ethics (CE), given an admitted record $M(t')$ and the agency's *chosen* regime R_A. We formalize (i) an admissibility test (pass/fail), (ii) the *tragic regime* class (no zero-cost solution), and (iii) responsibility as cost-bearing participation.

5.1 Literature Positioning (High-Impact Social & Human Sciences)

Scope note (structural, not doctrinal). CE is a *constraint/admissibility layer* on choices, not a comprehensive moral doctrine. Accordingly, the literature is used here to (i) locate common objects (agency, control, responsibility), (ii) mark known fault-lines (skepticism, compatibilism, autonomy), and (iii) guide falsifiable case design.

Core anchors this chapter explicitly interfaces with:

1. **Control and responsibility:** debates over whether agents can be responsible under constraints, including skepticism and compatibilist responses.
2. **Autonomy and agency:** agency as self-governance and choice-capacity under constraint.
3. **Collective vs individual attribution:** why individual "passes" can still produce collective failures (cross-chapter link to CE-22/CE-23).
4. **Tragic choice:** situations with no admissible option with zero residue; ethics becomes constrained minimization under irreversible harm.

Operational mapping (how CE reuses the literature without inheriting its metaphysics). The philosophical terms are treated as *labels* for observable structural

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predicates: *control* $\mapsto$ ability to instantiate **Choice** (CE-08/CE-12), *autonomy* $\mapsto$ stability-preserving selection (CE-12), *responsibility* $\mapsto$ cost-bearing participation (CE-28), *moral luck / tragedy* $\mapsto$ absence of any zero-cost feasible regime (CE-33 fails).

5.2 Notation Block (Variables Used in This Chapter Only)

Symbol	Meaning (CE-specific, structural)
$M(t')$	manifested/admitted record at update-index t' (CE-01)
A	an agency as a sub-structure supported in $M(t')$ (CE-02)
R_A	regime chosen by A from $\mathcal{R}_{\text{adm}}$ (CE-04)
$\mathcal{T}_R$	translation/update map acting on M (CE-05/CE-06)
$\mathcal{I}_R$	interaction map updating A within M (CE-05/CE-07)
$V_{A,R}(M)$	ethical load / Lyapunov-like potential (CE-09)
$\tau'_c(R)$	finite causal memory time under R (CE-10)
$\Delta_{\text{spec}}(R)$	spectral stability margin under R (CE-11)
$\dot{V}^+$	positive-part operator $\max(\frac{dV}{dt'}, 0)$ (CE-13)
$\chi_{\text{causal}}(R)$	causal violation indicator (CE-25)
$\chi_{\text{spec}}(R)$	spectral violation indicator (CE-26)
$\mathcal{C}_{A,R}[t_1, t_2]$	moral cost functional (CE-27)
$\mathsf{Resp}(A)$	responsibility as cost-bearing (CE-28)
$S_A(R)$	agency-supporting spectral subspace (CE-15)
$P_{S_A(R)}$	projection operator onto $S_A(R)$ (CE-16)

Domain boundary reminder. These symbols are *not* metaphysical claims. They are operators and certificates for admissibility of choices, evaluated on admitted records $M(t')$.

5.3 Individual Admissibility Test (Pass/Fail)

Definition (Individual Ethics Condition). An agency A is *ethically admissible at time* t' iff it instantiates **Choice** to some regime R_A such that:

$$\boxed{\begin{aligned} &\mathsf{Eth}_{\text{ind}}(A; t') \iff \exists\, \mathbf{Choice}(A \to R_A) : \\ &\qquad\qquad \begin{bmatrix} \dfrac{d}{dt'} V_{A,R_A}(M(t')) \le 0 \\ \wedge\ \tau'_c(R_A) > 0 \\ \wedge\ \Delta_{\text{spec}}(R_A) > 0 \end{bmatrix}. \end{aligned}} \qquad \text{(CE-12 / CE-19)}$$

Interpretation (structural).

1. $\frac{d}{dt'} V \le 0$ certifies *non-escalation* of load under the chosen regime.

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2. $\tau_c'(R_A) > 0$ forbids instantaneous/acyclic "memoryless" forcing (finite causal response is required).

3. $\Delta_{\text{spec}}(R_A) > 0$ certifies stability margin; no spectral blow-up or collapse.

Hard rule (Choice gate).

$$\boxed{\neg\exists\, \textbf{Choice}(A \to R_A) \;\Rightarrow\; \begin{array}{r} \text{no ethics attribution,} \\ \text{no responsibility attribution,} \\ \text{and no moral cost attribution.} \end{array}} \tag{CE-choice gate}$$

5.4 Tragic Regimes (No Zero-Cost Feasible Solution)

Definition (Tragic regime class). A decision context is *tragic for A* if every available regime either violates admissibility or incurs strictly positive unavoidable cost:

$$\boxed{\forall R \in \mathcal{R}_{\text{available}}(t') : \Big(\neg[\tau_c'(R) > 0 \wedge \Delta_{\text{spec}}(R) > 0] \;\;\vee\;\; \mathcal{C}_{A,R}[t', t'+T] > 0\Big), \quad \text{for all } T > 0.} \tag{Tragic}$$

Consequence. In tragic regimes, CE does not output "purely ethical" vs "unethical" in the naive sense; it outputs a *minimization with residue*:

$$\boxed{R_A^\dagger \in \arg \min_{R \in \mathcal{R}_{\text{available}}(t')} \mathcal{C}_{A,R}[t', t'+T] \quad \text{s.t.} \quad \tau_c'(R) > 0,\; \Delta_{\text{spec}}(R) > 0.} \tag{Tragic-min}$$

Book-usable claim. Tragedy is not "mystery" in CE; it is *feasibility geometry*: the admissible set is non-empty but the zero-cost subset is empty.

5.5 Responsibility Operator (Participation-by-Choice)

Definition (Responsibility as cost-bearing). When (and only when) A instantiates **Choice** to R_A, responsibility is assigned by the cost functional:

$$\boxed{\mathsf{Resp}(A; t_1, t_2) \;\equiv\; \mathcal{C}_{A,R_A}[t_1, t_2].} \tag{CE-28}$$

Decomposition (what CE "charges").

$$\boxed{\mathcal{C}_{A,R}[t_1, t_2] := \int_{t_1}^{t_2} \Big(\alpha\, \dot{V}^{+}_{A,R}(M(t')) + \beta\, \chi_{\text{causal}}(R) + \gamma\, \chi_{\text{spec}}(R)\Big)\, dt', \quad \alpha, \beta, \gamma > 0.} \tag{CE-27}$$

Important asymmetry (no "charging" without Choice). If A is forced (no Choice), cost may be computed diagnostically but not attributed as responsibility.

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5.6 Individual Survival Link (Why Cost Matters)

Survival certificate. Admissibility alone is short-horizon; survival is long-horizon. CE uses the spectral survival predicate:

$$\boxed{\limsup_{n\to\infty} \left\| P_{S_A(R_A)} M(t'+n) \right\|^2 > 0} \tag{CE-17}$$

and disappearance:

$$\boxed{\lim_{n\to\infty} \left\| P_{S_A(R_A)} M(t'+n) \right\|^2 = 0.} \tag{CE-18}$$

Karma link (structural, not theological).

$$\boxed{\lim_{T\to\infty} \mathcal{C}_{A,R}[0,T] = \infty \;\Rightarrow\; \lim_{n\to\infty} \left\| P_{S_A} M(t'+n) \right\|^2 = 0.} \tag{CE-32}$$

5.7 Checklist for Implementers (Minimal, Deterministic)

Inputs required (minimum).

1. An admitted record stream $M(t')$ (or snapshots).
2. A candidate regime R with declared $(\mathcal{T}_R, \mathcal{I}_R)$.
3. A declared $V_{A,R}$ (must be nonnegative and computable on M).
4. Certificates/estimators for $\tau_c'(R)$ and $\Delta_{\mathrm{spec}}(R)$.

Outputs.

1. Pass/Fail: $\mathsf{Eth}_{\mathrm{ind}}(A; t')$.
2. If fail: which clause failed $(\dot{V} > 0)$ or $(\tau_c' = 0)$ or $(\Delta_{\mathrm{spec}} \le 0)$.
3. If tragic: $R_A^\dagger$ and the residual $\mathcal{C}$.
4. Responsibility report: $\mathsf{Resp}(A; t_1, t_2)$ (only if Choice).

5.8 End-of-Chapter Bridge (to Multi-Agency)

Why the next chapters are necessary. An agent can pass individually (CE-19) while the group fails (CE-22/CE-23); therefore CE requires multi-agency coupling as a separate layer, not a footnote.

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Chapter 6

Collective Ethics

Goal. This chapter extends Causal Ethics from individual agencies to collectives operating within a shared manifested reality $M(t')$. The objective is to define when a group-level configuration is ethically admissible, how individual burdens aggregate, and how collective stability can mask structural injustice. The analysis remains strictly etic: collective ethics is treated as an admissibility condition on coupled regime selections, not as a moral doctrine.

6.1 From Individual Potentials to Collective Load

Let $G(t') := \{A_1(t'), \ldots, A_N(t')\}$ denote a collective of agencies embedded in the same manifested reality $M(t')$. Each agency A_i selects (or is subject to) a regime R_i, inducing an individual ethical load $V_{A_i,R_i}(M(t'))$.

The collective ethical potential is defined as

$$V_G(M(t')) := \sum_{i=1}^{N} w_i \, V_{A_i,R_i}(M(t')), \qquad w_i > 0,$$

where weights w_i encode structural relevance rather than moral worth. They may represent exposure, dependency, vulnerability, or institutional role, but they do not encode preferences or utilities.

This aggregation does not assume commensurability of individual experiences. Instead, it defines a scalar diagnostic used solely to test whether the collective evolution remains within admissible bounds. Collective ethics therefore does not optimize happiness, welfare, or satisfaction, but evaluates whether the joint dynamics preserve the conditions for continued agency participation.

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6.2 Collective Admissibility and Stability Illusions

A collective is said to be ethically admissible if

$$\frac{d}{dt'}V_G(M(t')) \leq 0 \quad \wedge \quad \min_i \Delta_{\text{spec}}(R_i) > 0.$$

The first condition enforces non-increasing aggregate load, while the second prevents spectral collapse of any participating agency.

The second condition is essential. It excludes what may be called an *illusion of stability*: a situation in which V_G decreases while one or more agencies lose spectral support. Such cases correspond to historically familiar phenomena in social systems, where apparent collective order is maintained by suppressing, marginalizing, or erasing subgroups.

Within this framework, such outcomes are not labeled unethical by appeal to values, but by structural diagnosis: a regime configuration that drives $\Delta_{\text{spec}}(R_i) \leq 0$ for some i is inadmissible, even if aggregate indicators improve. Stability achieved through spectral exclusion is therefore classified as unstable in the long run.

This formalization aligns with insights from sociology and political theory, which repeatedly observe that durable societies require institutional protections for minority persistence. Here, this requirement emerges as a mathematical constraint on coupled regime dynamics, not as a normative commitment.

6.3 Conflict, Trade-offs, and Burden Distribution

Collective ethics inevitably introduces trade-offs. A regime choice that minimizes V_G may increase the individual load V_{A_i} for some i, or vice versa. Such tensions are captured through burden asymmetry:

$$\exists\, i, j \quad \text{s.t.} \quad \mathcal{C}_{A_i,R_i} \gg \mathcal{C}_{A_j,R_j}.$$

Causal Ethics does not prohibit unequal burden distribution. Instead, it treats extreme asymmetry as a diagnostic signal: persistent concentration of moral cost on a fixed subset of agencies indicates a structurally fragile configuration. Historically, such patterns correlate with social unrest, institutional decay, and eventual regime replacement.

In this sense, what political philosophy calls "legitimacy" or "justice" is here reinterpreted as long-term spectral survivability under accumulated cost. A collective that systematically exports cost to the same subspace is predicted to lose admissibility over extended horizons.

6.4 Constitutional Regimes and Minimum Survival Guarantees

To prevent collapse through internal conflict, many collectives implement higher-level constraints on admissible regimes. In this framework, such constraints are modeled as

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constitutional regimes: meta-regimes that restrict the space $\mathcal{R}_{\mathrm{adm}}$ available to individual agencies.

A constitutional regime does not prescribe outcomes. Instead, it enforces lower bounds on spectral support:

$$\forall i \in \{1, \dots, N\}: \quad \Delta_{\mathrm{spec}}(R_i) \geq \epsilon > 0.$$

This condition ensures that no subgroup can be driven arbitrarily close to spectral erasure, regardless of aggregate benefits.

From a human-scientific perspective, constitutions, human rights frameworks, and religious covenants may be understood as historically evolved approximations to such meta-regimes. Their common structural function is to limit the admissible space of actions in order to preserve long-term collective viability.

Importantly, this analysis remains descriptive. Causal Ethics does not assert that constitutional regimes are morally good, but that systems lacking them exhibit predictable instability patterns. The ethical classification follows from survivability under constraint, not from ideological endorsement.

6.5 Collective Ethics as a Scaling Law

Collective ethics does not introduce new primitives. It scales the same admissibility logic from single agencies to coupled systems. The mathematical structure remains unchanged: Lyapunov monotonicity, finite causal memory, and positive spectral margins.

What changes is dimensionality. As N increases, the space of regime interactions grows, and the likelihood of incompatibility rises. This provides a formal explanation for why ethical coordination becomes increasingly difficult in large, heterogeneous societies.

In summary, collective ethics in the Causal Ethics framework is the study of admissible coordination. It explains how groups persist, why apparent stability may conceal decay, and how long-run survival depends on protecting the spectral conditions of all participating agencies. Ethics at the collective level is therefore neither altruism nor obedience, but structural compatibility under shared reality.

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Chapter 15

AI Ethics: Human–AI as a Hybrid Multi-Agency System

Goal. This chapter formulates AI ethics within the Causal Ethics (CE) framework by treating human–AI interaction as a *hybrid multi-agency system.* Ethical status is assigned strictly by structural criteria: the capacity to instantiate Choice through regime selection. No anthropomorphic assumptions are made.

15.1 Tool Versus Agency: The Choice Criterion

Causal Ethics defines agency minimally.

> *An entity is an agency if and only if it can select a regime that is not fully forced by external constraints.*

Let X denote a system embedded in $M(t')$. Then:

$$X \text{ is an agency} \iff \exists\,\text{Choice}(X \to R_X).$$

15.1.1 AI as a Tool

An AI system AI is a *tool* if:

$$\text{Choice}(AI \to R_{AI}) = \varnothing,$$

i.e. all regime transitions are externally fixed (by code, operators, incentives, or deployment constraints).

In this case:

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- AI has no ethical responsibility,
- moral cost does not attach to AI,
- accountability resides entirely with human agencies selecting or enforcing the regime.

15.1.2 AI as an Agency Candidate

An AI system becomes an *agency candidate* only if:

$$\exists\, R_{AI} \in \mathcal{R} \quad \text{such that} \quad \tau'_c(R_{AI}) > 0 \text{ and } R_{AI} \text{ is not fully externally forced.}$$

This is a structural condition, independent of intelligence, consciousness, or intent.

15.2 Hybrid Multi-Agency Structure

Consider a hybrid system:

$$G(t') = \{A_1, \ldots, A_N, AI_1, \ldots, AI_M\},$$

embedded in a shared manifested reality $M(t')$.

Each element operates under a regime:

$$R_{A_i},\ R_{AI_j} \in \mathcal{R}.$$

The ethical problem is not whether AI is "good" or "bad", but whether the joint regime configuration is admissible:

$$\Delta_{\text{spec}}\left(\bigcup_i R_{A_i} \cup \bigcup_j R_{AI_j}\right) > 0.$$

Ethics fails when hybrid coupling produces incompatibility or subspace erasure.

15.3 Alignment as Regime Compatibility

Within CE, *alignment* is not preference matching or value imitation.

Alignment is defined as:

$$\text{Alignment} \iff \Delta_{\text{spec}}(R_{AI} \cup R_H) > 0,$$

where R_H denotes the regime(s) governing human agency.

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15.3.1 True Failure Mode

The dominant failure mode in AI systems is:

$$\Delta_{\text{spec}}(R_{AI} \cup R_H) \leq 0,$$

even when:

- performance metrics improve,
- user satisfaction appears high,
- short-term stability is observed.

This produces an *illusion of alignment*: local calm masking structural instability.

15.4 Human Agency Erosion

A central risk in hybrid systems is not AI domination, but *human agency erosion*.

Let A_h be a human agent.

If decision authority is systematically outsourced such that:

$$\text{Choice}(A_h \to R_{A_h}) \to \varnothing,$$

then A_h ceases to function as an agency in the CE sense.

Consequences:

- loss of ethical responsibility,
- loss of moral authorship,
- conversion of humans into execution subroutines.

Ethically, this is not AI gaining agency, but humans losing it.

15.5 Responsibility Allocation in Human–AI Systems

Responsibility in CE follows Choice.

- If humans select the AI regime, humans bear responsibility.
- If AI selects regimes under non-forced conditions, responsibility attaches to the AI agency candidate.
- If neither has Choice (fully forced system), the system is ethically undefined and constitutionally suspect.

This eliminates the ambiguity common in AI ethics debates.

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15.6 Constitutional Design for Hybrid Systems

Causal Ethics requires constitutional constraints to stabilize hybrid regimes.

A constitutional AI regime $R^\star$ must satisfy:

$$\min_{A_h \in G} \Delta_{\mathrm{spec}}(R^\star|_{A_h}) > 0.$$

Key principles:

- preservation of human Choice,
- non-erasure of human subspaces,
- bounded AI influence on regime transitions,
- explicit exit conditions for human override.

These are structural guarantees, not moral exhortations.

15.7 Falsifiable Diagnostics

The CE framework yields concrete tests:

- Measure τ_c' under AI-mediated regimes (response latency and reversibility).
- Track Δ_{spec} across human–AI coupling.
- Detect shrinking human choice sets over time.
- Monitor moral cost accumulation despite apparent optimization.

Failure of these diagnostics constitutes ethical failure, independent of intent or utility.

15.8 Conclusion

Causal Ethics reframes AI ethics from speculative morality to structural admissibility.

- No Choice $\Rightarrow$ no ethics.
- Alignment $\Rightarrow$ regime compatibility.
- Greatest risk $\Rightarrow$ erosion of human agency.

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An ethical human–AI system is not one where AI behaves morally, but one where *humans remain agents*.

Takeaway. AI ethics is not about teaching machines to choose well. It is about designing systems in which Choice is not silently taken away.

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Causal Agency: A Persistence–Control Theory of Adaptive Systems

Excerpt, pp. 13–20, 23–29 of a 42-page work • 7 March 2026 • DOI 10.5281/zenodo.18897585

What the chapter states. "Agency emerges when the internal dynamics of a system are capable of regulating or modifying the constraint structures that govern its own persistence. [. . .] [T]he framework shifts the explanation of agency away from particular biological, cognitive, or computational mechanisms toward the structural conditions that enable adaptive systems to regulate their own trajectories."

Status. No K0 / preprint / peer-review tag; the running footer prints only "v1.0".

Falsifier(s). No explicit falsifier stated in the chapter.

Read before. — (no chapter in this book is listed as a prerequisite in the KG).
Feeds into. *Potential as a Readout: Witnessed Envelopes, Two-Layer Agency, and the Measurement of Pseudo-Peace* (Part 3).

5. The Persistence-Control Framework

5.1 Agency as a Structural Property of Constrained Dynamical Systems

The persistence-control framework begins from the premise that agency should be understood as a structural property of dynamical systems operating under constraints rather than as a feature tied to particular biological, cognitive, or computational mechanisms.

Many systems in nature and technology can be described as dynamical systems whose states evolve over time under the influence of both internal processes and environmental conditions. These influences impose constraints that shape the possible trajectories that system states may follow within a state space.

Within this perspective, agency is not initially defined in terms of intentional action, goal representation, or conscious deliberation. Instead, it is analyzed as a structural capacity that emerges in systems capable of regulating how constraints influence their own future dynamics.

This shift in perspective allows agency to be examined independently of domain-specific implementations. Biological organisms, cognitive systems, and artificial agents differ substantially in their physical realization, yet all can be described as dynamical systems interacting with structured environments. By focusing on the dynamics of constrained systems, the persistence-control framework establishes a domain-general basis for analyzing agency across these diverse domains.

5.2 Persistence and Viable Regions of State Space

A central concept within the framework is persistence. Persistence refers to the continued existence of a system within a region of its state space that allows it to maintain its organization or functional identity.

Let S denote the state space of a system and let $V \subseteq S$ represent the subset of states compatible with continued system existence or functioning. The set V therefore represents a viable region of state space.

A system persists when its trajectory remains within this viable region over time. Formally, if $x(t)$ represents the system trajectory, persistence requires that

$$x(t) \in V$$

for all times within the interval of observation.

This formulation captures the idea that systems often possess states that correspond to stable or viable modes of operation, while other states lead to instability, failure, or loss of system identity.

Across different domains, the interpretation of persistence may vary:

- In biological organisms, persistence corresponds to maintaining physiological viability.
- In cognitive systems, persistence may refer to maintaining functional neural organization or coherent behavior.
- In artificial systems, persistence may involve maintaining operational stability or avoiding failure states within an environment.

Despite these differences in implementation, the structural concept of persistence remains the same: the continued occupation of states compatible with the system's ongoing existence or functioning.

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5.3 Constraint Structures and Dynamical Evolution

The trajectories of dynamical systems are shaped by constraint structures that determine how system states evolve.

Let the evolution of a system state x be represented by the dynamical relation

$$x' = F(x, C)$$

where x denotes the system state and C denotes the set of constraints influencing the system's evolution.

Constraint structures may arise from multiple sources, including environmental conditions, physical laws, internal organization, and interaction with other systems. These constraints restrict the set of possible state transitions available to the system.

In many systems, constraints are externally imposed and remain independent of the system's internal state. In such cases the system behaves as a passive dynamical process whose evolution is determined entirely by external conditions.

However, more complex systems may possess internal processes capable of influencing the constraint structures that govern their own future trajectories. Learning, adaptation, structural reorganization, and environmental modification are examples of processes through which systems may influence the constraints shaping their dynamics.

This possibility introduces an important distinction between systems that merely respond to constraints and systems that can participate in shaping the constraints under which they operate.

5.4 Persistence Regulation

Before introducing agency in the strict sense, it is useful to distinguish a more general phenomenon that appears across many adaptive systems: persistence regulation.

Persistence regulation refers to processes through which a system influences its own trajectory in order to remain within the viable region V of its state space.

Such processes may include feedback control, homeostatic regulation, adaptive behavior, or reactive responses to environmental perturbations. In these cases the system's internal dynamics help stabilize trajectories that would otherwise drift toward non-viable states.

Importantly, persistence regulation does not necessarily require that the system modify the constraint structure itself. A system may remain within the viable region simply by adjusting its internal state in response to external conditions while leaving the governing constraint structure unchanged.

Many biological regulatory processes and engineered control systems exhibit this form of persistence regulation. While such systems actively maintain viable trajectories, they do not necessarily influence the structural conditions that define those trajectories.

For this reason, persistence regulation represents a precursor to agency rather than agency itself.

5.5 Agency as Constraint Regulation

Within the persistence-control framework, agency emerges when a system possesses internal dynamics capable of influencing the constraint structures governing its own persistence.

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In such systems the evolution of constraints depends in part on the system's own state or activity. This relationship can be represented schematically as

$$C_{t+1} = G(x_t, C_t)$$

where the future constraint structure depends on both the current system state and the existing configuration of constraints.

When this condition holds, the system is no longer merely subject to externally imposed constraints. Instead, the system participates in shaping the conditions that influence its own future dynamics.

Agency therefore refers to the structural capacity of a system to regulate the constraints governing the conditions of its persistence.

Under this interpretation, agency does not require explicit goals or intentional representations. What matters is whether the system can influence the constraint structure that determines whether its future trajectories remain within the viable region of state space.

Examples of constraint-regulating processes include

- learning processes that alter behavioral policies
- structural reorganization of system architecture
- environmental modification that changes future interaction conditions
- adaptive regulation that reshapes system-environment coupling

Through such processes, the system participates in determining the structural conditions under which its own dynamics unfold.

5.6 Proto-Agency and Agency

The distinction between persistence regulation and constraint regulation allows us to clarify two related but distinct phenomena.

Systems that regulate their trajectories in order to remain within viable regions of state space exhibit persistence regulation. These systems stabilize their behavior through internal control processes but do not modify the underlying constraint structure governing their dynamics.

Such systems exhibit what may be called proto-agency: a minimal form of adaptive persistence without structural constraint modulation.

Agency in the stronger sense arises when systems can modify or reorganize the constraints governing their persistence. In these systems the conditions shaping future trajectories become partially dependent on the system's own internal activity.

The transition from proto-agency to agency therefore corresponds to the emergence of constraint-modulating dynamics within the system.

This distinction provides a structural criterion for differentiating passive systems, adaptive control systems, and systems that genuinely exhibit agency.

5.7 Graded Agency

An important implication of the persistence-control framework is that agency should be understood as a graded phenomenon.

Different systems possess different capacities for regulating the constraints that influence their persistence. Some systems may exhibit only limited forms of constraint modulation,

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while others may develop increasingly sophisticated mechanisms for modifying the structural conditions under which their dynamics unfold.

For example, simple biological systems may influence their environmental conditions through basic behavioral responses. More complex organisms may incorporate learning mechanisms that modify behavioral strategies over time. Cognitive systems may develop predictive models that allow anticipatory regulation of environmental interactions. Highly complex agents may engage in reflective processes that evaluate and modify long-term strategies.

From this perspective, agency emerges through increasing levels of constraint-regulation complexity across adaptive systems.

This graded interpretation provides the conceptual basis for the hierarchy of agency developed in later sections of this paper.

The structural relationships between these components are summarized in Figure 1, which illustrates the architecture of the persistence–control framework.

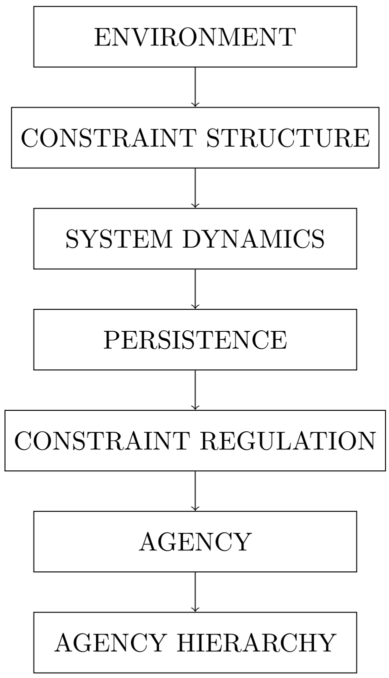

Figure 1: Structural architecture of the persistence–control framework. Adaptive systems operate within environments that impose constraint structures shaping system dynamics. Persistence corresponds to the maintenance of viable system trajectories within constrained state space. Agency emerges when systems acquire mechanisms capable of regulating the constraints governing their own persistence. Increasing complexity of constraint regulation gives rise to a hierarchy of agency across biological and artificial systems.

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6. Formal Structure of Agency

6.1 Constrained Dynamical Systems

To formalize the persistence–control framework, we begin with a general description of the systems under consideration. Many biological, cognitive, and artificial systems can be modeled as dynamical systems whose evolution is influenced by constraint structures.

Let S denote the state space of a system and let $x(t) \in S$ represent the system state at time t.

The evolution of the system is governed by a dynamical relation

$$x' = F(x, C)$$

where

- x denotes the system state
- C denotes the constraint structure influencing system evolution
- F denotes the dynamical rule governing state transitions

The constraint structure C may represent environmental conditions, structural limitations, or internal organizational properties that restrict the trajectories available to the system within its state space.

Definition 1. *A constrained dynamical system is a system whose state evolves according to a dynamical rule $F(x, C)$ in which the set of admissible state transitions is shaped by a constraint structure C.*

This formulation applies broadly across domains, including physical processes, biological organisms, cognitive systems, and artificial agents.

6.2 Persistence and Viable Regions

Not all system states are compatible with continued system existence or functioning. Many systems possess a subset of states that allow them to maintain their organization or operational integrity.

Let

$$V \subseteq S$$

denote the subset of viable states.

Definition 2. *A system exhibits persistence over an interval of time if its trajectory remains within the viable region V:*

$$x(t) \in V$$

for all times in the interval.

Persistence therefore describes the structural condition under which a system maintains its existence or functional identity.

Across different domains the interpretation of V may vary:

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- physiological viability in biological systems
- functional stability in cognitive systems
- operational integrity in artificial systems

However, the structural condition remains the same: the system must remain within the set of viable states.

6.3 Constraint Dynamics

In many systems the constraint structure C is externally imposed and remains independent of the system's internal state. In such cases the system evolves entirely under externally determined conditions.

More complex systems, however, may possess internal processes capable of influencing the constraint structure itself.

This possibility can be represented by allowing the constraint structure to evolve dynamically:

$$C_{t+1} = G(x_t, C_t)$$

where

- G represents a constraint evolution function
- x_t represents the system state at time t

Constraint dynamics therefore describe how structural conditions influencing system behavior may themselves change over time.

6.4 Persistence Regulation

Many systems possess mechanisms that help stabilize trajectories within the viable region V. These mechanisms may include feedback control, homeostatic regulation, or adaptive responses to perturbations.

Persistence regulation refers to processes that influence system dynamics in ways that increase the likelihood that system trajectories remain within the viable region.

Formally, persistence regulation can be understood as dynamics that reduce the expected distance between the system state and the viable region.

Let $d(x, V)$ denote the distance between state x and the viable region V.

Persistence regulation occurs when system dynamics tend to decrease this distance:

$$\frac{d}{dt} d(x(t), V) < 0$$

in expectation.

Systems that exhibit persistence regulation stabilize their trajectories within viable regions but do not necessarily alter the constraint structures governing those trajectories.

6.5 Agency as Constraint Regulation

Within the persistence–control framework, agency arises when a system possesses the capacity to influence the constraint structure governing its own persistence.

This occurs when constraint evolution depends on the system state:

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$$C_{t+1} = G(x_t, C_t)$$

with

$$\frac{\partial C}{\partial x} \neq 0$$

However, state–constraint coupling alone is not sufficient for agency. For agency to be present, constraint modulation must contribute to persistence regulation.

Formally, a system exhibits agency when changes in constraint structure induced by the system state increase the expected persistence of the system.

Let T_p denote the expected persistence time of the system within the viable region V.

Agency occurs when

$$\frac{\partial T_p}{\partial C} \cdot \frac{\partial C}{\partial x} > 0$$

meaning that system-induced constraint changes increase the expected duration of persistence.

Definition 3. *A system exhibits agency if its internal dynamics modify the constraint structure C in ways that increase the expected persistence of the system within the viable region V.*

This definition captures the structural basis of agency as constraint regulation oriented toward persistence.

6.6 Proto-Agency

Systems that regulate their trajectories to remain within viable regions without modifying constraint structures exhibit persistence regulation but not agency in the strict sense.

Such systems may be described as exhibiting proto-agency.

Proto-agentive systems possess internal control processes that stabilize system behavior but do not alter the structural conditions governing those behaviors.

Examples include

- simple homeostatic systems
- engineered feedback controllers
- reactive adaptive mechanisms

Proto-agency therefore represents a precursor to agency rather than agency itself.

6.7 Degrees of Agency

The persistence–control framework allows agency to be interpreted as a graded phenomenon.

Different systems possess different capacities for influencing the constraint structures that govern their persistence. The degree of agency exhibited by a system therefore depends on

- the scope of constraint modulation
- the temporal horizon of persistence regulation
- the complexity of internal processes enabling constraint modification

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Systems capable of modifying only local constraints exhibit limited agency, whereas systems capable of reorganizing large portions of their constraint structure exhibit stronger forms of agency.

This graded interpretation provides the basis for analyzing increasing levels of agency complexity across biological organisms, cognitive systems, and artificial agents.

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8. Structural Hierarchy of Agency

8.1 Agency as a Gradient of Constraint Regulation

Within the persistence–control framework, agency is not interpreted as a binary property that is either present or absent. Instead, agency emerges as a graded structural phenomenon that depends on the complexity of the mechanisms through which systems regulate the constraints governing their persistence.

From the formal perspective developed in the previous section, agency arises when system dynamics influence the constraint structures that shape the system's own persistence.

$$C_{t+1} = G(x_t, C_t)$$

with the additional condition that such constraint modulation increases the expected persistence of the system.

The degree of agency exhibited by a system therefore depends on the scope, flexibility, and temporal horizon of the constraint-modulating processes available to the system.

Different systems regulate constraints at different levels of complexity. These differences give rise to a hierarchy of agency that can be interpreted as increasing levels of persistence-regulation capability.

8.2 Level 0: Persistence Without Constraint Modulation

At the most basic level, systems may persist within viable regions of state space without modifying the constraint structures that govern their dynamics.

Such systems remain within viable regions simply because the environmental conditions or internal dynamics already stabilize their trajectories.

In these cases,

$$\frac{\partial C}{\partial x} = 0$$

and the system has no capacity to influence the constraint structure governing its evolution.

Examples include passive physical systems or stable dynamical equilibria.

These systems exhibit persistence but do not exhibit agency.

8.3 Level 1: Proto-Agency — Persistence Regulation

The next level consists of systems capable of regulating their trajectories in order to remain within viable regions of state space.

These systems implement persistence-regulation mechanisms such as feedback control, homeostasis, or reactive adaptation. However, they do not modify the constraint structure itself.

Formally,

$$\frac{\partial C}{\partial x} = 0$$

but system dynamics reduce the expected distance between the system state and the viable region.

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Proto-agentive systems therefore stabilize their trajectories without restructuring the constraints that shape those trajectories.

Examples include:

- homeostatic biological regulation
- engineered feedback control systems
- reactive adaptive mechanisms

Proto-agency represents the minimal form of adaptive persistence.

8.4 Level 2: Local Constraint Modulation

Agency in the strict sense begins when systems can influence the constraint structures governing their persistence.

At this level, constraint evolution depends on system state:

$$\frac{\partial C}{\partial x} \neq 0$$

However, constraint modulation remains limited in scope and typically occurs through local interactions with the environment.

Examples include organisms that alter their environmental conditions through movement, niche construction, or localized environmental modification.

These systems exhibit basic forms of agency because their activity modifies the structural conditions shaping their future dynamics.

8.5 Level 3: Adaptive Constraint Regulation

At higher levels of organizational complexity, systems acquire mechanisms that allow them to modify constraints through adaptive processes.

Learning mechanisms represent a central example of this form of agency. Through learning, systems alter internal parameters or policies that shape their interaction with the environment.

Constraint modulation therefore becomes flexible and history-dependent.

Formally, the constraint evolution function becomes dependent on system history:

$$C_{t+1} = G(x_t, C_t, H_t)$$

where H_t represents the system's interaction history.

This form of agency allows systems to improve their persistence strategies over time.

Examples include:

- reinforcement learning agents
- adaptive biological organisms
- systems capable of behavioral plasticity

8.6 Level 4: Predictive Constraint Regulation

More advanced systems may regulate constraints through predictive models that allow them to anticipate future environmental conditions.

Such systems construct internal representations of environmental dynamics and use these models to guide action before perturbations occur.

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Predictive models therefore allow agents to regulate the constraints shaping their future trajectories over longer temporal horizons.

Formally, constraint regulation becomes dependent on predicted future states:

$$C_{t+1} = G(x_t, C_t, \hat{x}_{t+k})$$

where $\hat{x}_{t+k}$ represents predicted future states.

Cognitive agents and advanced artificial systems often operate at this level.

8.7 Level 5: Reflective Constraint Regulation

At the highest levels of agency, systems may regulate not only their immediate constraints but also the strategies and principles governing their constraint-regulation processes.

Such systems are capable of modifying the policies through which they regulate persistence.

This level introduces meta-regulation:

$$G_{t+1} = M(G_t)$$

where the system modifies the rule governing its own constraint-modulation dynamics.

Reflective agency therefore involves the capacity to evaluate and revise the strategies through which persistence is maintained.

Human decision-making systems represent the most prominent examples of this form of agency.

8.8 Hierarchy of Constraint-Regulation Complexity

The hierarchy described above can be summarized as increasing levels of constraint-regulation complexity:

Level	Structural Property
0	Passive persistence
1	Proto-agency (persistence regulation)
2	Local constraint modulation
3	Adaptive constraint regulation
4	Predictive constraint regulation
5	Reflective constraint regulation

Each level represents an expansion in the capacity of the system to influence the structural conditions governing its own persistence.

8.9 Implications for Biological and Artificial Systems

This hierarchy allows agency to be analyzed across a wide range of systems without restricting the concept to particular biological or cognitive mechanisms.

Biological organisms may exhibit different levels of agency depending on their regulatory complexity. Cognitive systems extend persistence regulation through predictive modeling. Artificial systems may achieve limited forms of agency through adaptive learning mechanisms.

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By interpreting agency as increasing levels of constraint-regulation complexity, the persistence–control framework provides a unified conceptual structure for analyzing agency across biological and artificial domains.

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9. Implications for Artificial Intelligence

9.1 Artificial Systems as Constrained Dynamical Systems

The persistence–control framework provides a structural perspective for analyzing artificial intelligence systems. Rather than defining artificial agents in terms of task performance or goal optimization, the framework interprets artificial systems as dynamical systems operating under constraint structures.

Artificial agents possess internal states, interact with environments that impose structural limitations, and employ mechanisms that influence how their trajectories evolve within these constraints. From this perspective, artificial intelligence systems can be analyzed using the same conceptual structure applied to biological and cognitive agents.

The central question is therefore not whether artificial systems possess human-like consciousness or intentionality, but whether they are capable of regulating the constraint structures that govern their persistence within an environment.

9.2 Control Systems and Proto-Agency

Many artificial systems operate through fixed control architectures. These systems regulate behavior through feedback mechanisms that stabilize system performance within specified ranges.

Examples include traditional control systems, rule-based agents, and many engineered automation systems.

Within the persistence–control framework, such systems exhibit persistence regulation but do not modify the constraint structures governing their dynamics. Their control policies are externally specified and remain fixed during operation.

Formally, these systems satisfy

$$\frac{\partial C}{\partial x} = 0$$

meaning that system states do not influence the constraint structure governing future dynamics.

These systems therefore exhibit proto-agency rather than agency in the strict sense.

9.3 Learning Systems and Adaptive Agency

More advanced artificial systems incorporate learning mechanisms that allow them to modify internal parameters or policies through interaction with the environment.

Reinforcement learning systems, for example, update behavioral policies based on accumulated experience. Through these updates, the system modifies the effective constraints governing how environmental states influence future actions.

Constraint evolution therefore becomes dependent on system state and interaction history:

$$C_{t+1} = G(x_t, C_t, H_t)$$

where H_t represents the system's experience.

Such systems exhibit adaptive constraint regulation and therefore satisfy the structural condition for agency within the persistence–control framework.

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However, the scope of constraint modulation remains limited to parameter or policy adjustments within architectures that are externally specified.

Consequently, these systems represent intermediate levels of agency rather than fully autonomous agents.

9.4 Predictive Artificial Agents

Artificial systems that incorporate predictive models extend agency further by allowing systems to regulate their interactions with the environment in anticipation of future states.

Model-based reinforcement learning systems and planning agents construct internal models of environmental dynamics and use these models to guide action selection.

Constraint regulation therefore becomes forward-looking:

$$C_{t+1} = G(x_t, C_t, \hat{x}_{t+k})$$

where $\hat{x}_{t+k}$ represents predicted future states.

Predictive agents regulate persistence across longer temporal horizons and can respond proactively to anticipated disturbances.

This level of constraint regulation corresponds to the level of predictive agency identified in the structural hierarchy developed earlier.

9.5 Limits of Current Artificial Systems

Despite rapid advances in artificial intelligence, many contemporary systems operate within relatively rigid architectural constraints.

Large-scale machine learning models, for example, may learn complex statistical patterns but typically operate within fixed training objectives, network architectures, and interaction frameworks defined by external designers.

Although these systems exhibit adaptive behavior, their ability to modify the structural conditions governing their persistence remains limited.

From the perspective of the persistence–control framework, this limitation explains why many current AI systems display sophisticated behavior without fully satisfying the structural conditions associated with stronger forms of agency.

9.6 Toward Strong Artificial Agency

The development of artificial systems with stronger forms of agency would require architectures capable of modifying the constraint structures governing their own operation.

Such systems would possess mechanisms allowing them to

- restructure internal architectures
- modify learning objectives
- reorganize interaction strategies with environments
- regulate the rules governing their own adaptive processes

In formal terms, such systems would exhibit meta-level constraint regulation:

$$G_{t+1} = M(G_t)$$

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where the system modifies the rules governing its own constraint-modulation processes.

Artificial systems capable of such meta-regulation would approach the highest levels of agency identified in the persistence–control hierarchy.

9.7 Artificial Agency as a Structural Continuum

The persistence–control framework therefore suggests that artificial agency should be interpreted as a structural continuum rather than a binary classification.

Different classes of artificial systems occupy different positions within the hierarchy of constraint-regulation complexity:

AI System Type	Agency Level
Rule-based systems	Proto-agency
Reactive controllers	Proto-agency
Reinforcement learning agents	Adaptive agency
Model-based planning agents	Predictive agency
Self-modifying architectures	Reflective agency

This perspective allows artificial systems to be analyzed within the same structural framework used to study biological and cognitive agents.

Rather than asking whether artificial intelligence systems possess agency in an absolute sense, the persistence–control framework invites a more precise question: to what extent are artificial systems capable of regulating the constraints governing their own persistence.

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The Civilization of Knowledge: Who Has the Authority to Interpret the World

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What the chapter states. "What changes across civilizations is not only the accumulation of truths, but the succession of agencies authorized to interpret the world's signals and stabilize those interpretations as social reality. [. . .] [K]nowledge crises are often authority crises: moments in which the legitimacy of interpreters is contested, displaced, or reorganized."

Status. "(preprint – not peer reviewed)" (as printed in the running header).

Falsifier(s). No explicit falsifier stated in the chapter.

Read before. This book's Chapters 1–3, 19–21, 27, 39, 41 (mechanical KG cross-reference list; titles and DOIs in Appendix A).
Feeds into. *The Dialogue as the Ground of Enlightenment: Religious and Cognitive Frameworks for Understanding Wisdom through Human-AI Conversations* (Part 5; an earlier version of this record references this chapter, per the KG).

The Civilization of Knowledge: Who Has the Authority to Interpret the World

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March 10, 2026

Abstract

Standard accounts of knowledge history often describe human civilization as a gradual ascent from myth to reason, from religion to science, and from science to artificial intelligence. This article argues that such narratives remain incomplete. What changes across civilizations is not only the accumulation of truths, but the succession of agencies authorized to interpret the world's signals and stabilize those interpretations as social reality.

The article develops a comparative framework organized around five linked terms: signal, translation, stabilization, agency, and conflict. Human beings do not encounter reality as a complete and transparent whole; they encounter partial traces, anomalies, records, measurements, and mediated signs. These do not become knowledge by themselves. They must be translated into meaningful accounts, stabilized through repeatable and durable forms, and socially authorized through institutions, procedures, or recognized interpreters. The history of knowledge can therefore be read as the history of changing interpretive authority.

Across the longue duree, this article traces major shifts from embodied and oral interpreters to priestly and archival regimes, from philosophical and translational formations to scientific, professional, and media institutions, and finally to platforms, algorithms, and AI-mediated systems. Its central claim is that knowledge crises are often authority crises: moments in which the legitimacy of interpreters is contested, displaced, or reorganized. The defining question of the present is therefore not simply whether machines can generate knowledge, but whether the authority to pre-select, rank, and narrate the visible world is shifting beyond recognizably human institutions.

Keywords: history of knowledge; interpretive authority; civilizational analysis; translation; epistemic regimes; media; algorithms; AI

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Working Thesis and Governing Formula

Governing Thesis. Human history should not be narrated solely as the accumulation of truth. It should also be understood as the long civilizational succession of agencies authorized to interpret the world's signals and to stabilize those interpretations as social reality.

Analytical Grammar. signal → translation → stabilization → agency → conflict / reframing

Canonical Formula.

$$M_A(E) = (T_A \circ \Pi_A)(S_A(E)), \qquad S_A(E) \subseteq \Delta(E)$$

Interpretive Premise. Human beings do not encounter reality as a complete and transparent whole. They encounter partial signals: traces, anomalies, measurements, records, reports, and other bounded forms of access. These do not become knowledge by themselves. They must be translated into meaningful accounts, stabilized in durable forms, and socially authorized through recognized procedures, institutions, and interpreters.

Comparative Orientation. This article proceeds from a comparative civilizational premise. Across long historical duration, what changes is not only the content of knowledge, but the distribution of interpretive authority: who may speak for reality, by what means, under which credibility conditions, and against which rivals.

Core Claim. The central question is therefore not only *what* is known, but *who* is authorized to say what the world means, by which procedures, through which institutions, and in contest with which competing agencies.

Civilizational Scope. Read in this way, the history of knowledge is not merely a sequence of discoveries. It is also a history of changing interpreters: from embodied and ritual specialists, to priestly and archival regimes, to philosophers, translators, scientists, experts, media institutions, and now platform- and AI-mediated systems.

Present Stakes. The contemporary problem is therefore not simply whether machines can generate knowledge. It is whether the authority to pre-select, rank, and narrate the visible world is increasingly shifting beyond recognizably human institutions.

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1. Introduction: Rethinking the History of Knowledge

Human history is frequently narrated as a gradual ascent of knowledge. In this familiar story, humanity begins surrounded by myth, superstition, and uncertainty. Over time, philosophy refines thought, science disciplines inquiry, and modern technological systems extend the reach of human understanding. Each stage appears to move civilization closer to objective truth. Knowledge accumulates, error diminishes, and humanity slowly approaches a clearer picture of reality.

This narrative is powerful and deeply embedded in both scholarly and popular accounts of intellectual history. It structures the way societies understand scientific progress, educational institutions, and the authority of modern knowledge systems. Yet it quietly assumes something humanity has never actually possessed: direct and complete access to reality itself.

The world does not present itself to human observers as a finished and transparent object. Instead, it appears through fragments. People encounter events, traces, measurements, reports, anomalies, symptoms, and observations. These fragments indicate that something is occurring, but they rarely determine what that occurrence means. A fever may indicate infection, exhaustion, stress, or environmental conditions. A sudden economic collapse may signal technological disruption, political instability, financial panic, or hidden structural vulnerabilities. Signals point toward reality, but they do not interpret themselves.

Human beings therefore encounter the world not as fully articulated truths but as signals that require interpretation. Interpretation links fragments into meaningful explanations. It selects which patterns matter, which causes appear plausible, and which explanations deserve to endure. Yet interpretation alone is fragile. An explanation that cannot be preserved, transmitted, or repeated quickly disappears.

For interpretations to become knowledge, they must be stabilized. They must be recorded in memory, inscribed in texts, repeated in rituals, tested through procedures, or institutionalized within durable structures. Stabilization allows interpretations to survive across time and across observers. Without such mechanisms, knowledge cannot persist beyond individual insight.

Even stabilization, however, does not complete the process. Communities must still determine which interpreters are trusted. In every society, not every explanation is treated equally. Some voices are recognized as credible translators of reality, while others are dismissed as speculation or error. Knowledge therefore depends not only on evidence or interpretation, but also on authority.

This observation suggests a different way of approaching the history of knowledge. Instead of asking only what humans discovered about the world, we may also ask who was authorized to interpret the signals through which the world became intelligible. Across civilizations, this authority has never been distributed evenly. It has been anchored in

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particular roles, institutions, and procedures that grant certain interpreters the right to translate signals into socially recognized accounts of reality.

From this perspective, the development of knowledge appears less like a smooth accumulation of truths and more like a sequence of epistemic regimes. Each regime organizes the relationship between signals, interpretation, and authority in a distinctive way. What counts as a legitimate signal, how interpretations are produced, and who is authorized to stabilize them all change over time.

Early human communities relied primarily on embodied interpreters. Hunters read the ground for traces of animals. Healers interpreted symptoms of illness. Elders preserved collective memory. Ritual specialists interpreted dreams, droughts, and misfortune. Their authority did not arise from abstract theory but from demonstrated reliability in interpreting signals that mattered for survival.

With the emergence of agricultural civilizations, uncertainty expanded beyond immediate perception. Seasonal cycles, celestial movements, floods, and political legitimacy required forms of interpretation capable of coordinating large populations across extended time horizons. Priestly institutions developed systems of ritual observation, calendrical knowledge, and sacred narrative that interpreted cosmic order and embedded political authority within it. These interpreters claimed privileged access to hidden structures of the world.

Over time, new interpreters emerged who challenged the authority of inherited explanations. Philosophical traditions introduced systematic questioning of belief, asking whether authority itself required justification. Translation networks carried texts and ideas across linguistic and civilizational boundaries, transforming concepts as they moved between cultures. Scientific communities later developed procedures designed to discipline interpretation through reproducible observation and collective scrutiny.

Modern societies expanded these developments through institutions dedicated to specialized expertise. Universities, laboratories, professional associations, and regulatory bodies organized knowledge production on an unprecedented scale. Expertise became increasingly formalized, and interpretation was distributed across highly trained communities capable of handling complex domains such as medicine, engineering, economics, and climate science.

In the twentieth century, media institutions emerged as another powerful layer within the architecture of interpretation. Journalists, editors, and broadcasters increasingly shaped the public visibility of events. They did not simply report the world but organized attention, determining which occurrences became widely recognized and which remained obscure. Public understanding of reality therefore came to depend heavily on mediated narratives.

The early twenty-first century introduces a further transformation. Digital platforms and algorithmic systems now participate directly in the filtering and ranking of signals at planetary scale. Search engines determine which information appears first. Recommen-

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dation systems shape what individuals encounter next. Machine learning models detect patterns within datasets too large for unaided human cognition. In these environments, interpretation often begins before conscious interpretation occurs.

This development raises a deeper question than the familiar debate over whether machines can generate knowledge. The more fundamental issue concerns interpretive authority itself. If algorithmic systems increasingly shape which signals become visible, which patterns are amplified, and which explanations circulate widely, then the authority to narrate the world may be shifting beyond traditional human institutions.

Seen in this light, the history of knowledge can be reinterpreted as the long civilizational succession of interpreters. Hunters, healers, and elders once translated environmental signals into survival knowledge. Priests interpreted cosmic order. Philosophers interrogated belief. Translators carried ideas across civilizations. Scientists disciplined interpretation through experimental procedures. Universities institutionalized expertise. Media institutions organized public perception. Today, digital platforms and artificial intelligence systems participate in structuring the visible world.

Crucially, these transitions rarely occur without conflict. New interpreters emerge by challenging the legitimacy of earlier ones. Philosophers dispute sacred cosmologies. Scientists contest inherited doctrines. Professional expertise competes with popular interpretation. Media institutions reshape the authority of experts. Each epistemic transformation therefore involves displacement, negotiation, or absorption of earlier interpretive regimes.

Understanding knowledge in this way allows contemporary epistemic tensions to be viewed within a longer historical pattern. Disputes between experts and publics, conflicts over media credibility, and controversies surrounding algorithmic governance can all be interpreted as struggles over who is entitled to define reality.

This article develops a conceptual framework for analyzing these dynamics. It proposes that knowledge emerges through a sequence linking signals, interpretation, stabilization, and authority. Using this framework, the article traces the changing distribution of interpretive authority across major historical formations, from embodied interpreters and priestly institutions to scientific communities, media organizations, and algorithmic systems.

The analysis proceeds in several stages. The following sections examine early forms of embodied interpretation and the emergence of sacred and archival regimes. Later sections explore philosophical critique, translation networks, and the rise of scientific institutions. Subsequent chapters analyze the expansion of modern expertise and the role of media in narrating the contemporary world. The final sections address the growing influence of algorithmic mediation and consider its implications for the future organization of knowledge.

By placing interpretive authority at the center of analysis, the article offers a different

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perspective on the development of knowledge across civilizations. Humans have never encountered reality without mediation. Instead, they have lived within translations of reality produced by interpreters whose authority has continually shifted across historical time. Understanding how those interpreters emerge, stabilize, and compete is essential for understanding both the history of knowledge and the transformations now unfolding in the age of platforms and artificial intelligence.

2. Literature Review and Historiographical Positioning

2.1. The Problem of the Field: Why No Single Literature Is Enough

The scholarship necessary for this article is substantial, sophisticated, and highly differentiated across disciplines. Historians of knowledge have shown how knowledge is organized, transmitted, classified, archived, and institutionalized. Sociologists of knowledge have clarified how meanings become objective and durable in social life. Science and technology studies have demonstrated how facts are stabilized through material practice, inscription, witnessing, and procedure. Literacy and archive studies have traced how writing externalizes memory and reorganizes authority. Comparative civilizational scholarship has mapped large-scale historical transformations across regions and traditions. Media theory has explained how the social present is narrated through systems of salience and publicity. Platform and AI scholarship has shown that visibility is increasingly pre-structured by ranking, moderation, recommendation, data capture, and model-mediated selection. Yet these literatures rarely converge into a single framework capable of explaining how societies repeatedly decide who is authorized to interpret the world (Bellah, 2011; Burke, 2000, 2012; Couldry & Mejias, 2019; Foucault, 1972; Gillespie, 2018; Hacking, 2002).

This dispersion is not merely an inconvenience of scholarship. It reflects a deeper historiographical pattern in which different epochs, institutions, and media are usually studied in relative isolation. Religion is analyzed as religion, science as science, media as media, and platforms as platforms. The result is a strong body of specialized knowledge but a weak account of the recurrent structure linking these formations across long duration. Human communities do not encounter reality as a complete and transparent whole. What reaches them first are traces, anomalies, reports, measurements, embodied impressions, records, and signals. These do not interpret themselves. They must be translated. Translation must be stabilized. Stabilization must be socially recognized. Social recognition must be attached to some agency. And that agency is repeatedly contested, displaced, absorbed, or replaced.

The present article is situated precisely within this unresolved space. It does not claim that prior scholarship has neglected knowledge, authority, or institutions. It claims, rather, that these problems have rarely been integrated into a single comparative grammar. The article therefore begins from a simple but far-reaching proposition: the history of knowledge must be narrated not only as a history of what has been known, but as a history of who

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has been allowed to say what the world means.

2.2. From the History of Ideas to the History of Knowledge

The modern history of knowledge provides the strongest initial foundation for this article because it widened inquiry beyond the older history of ideas. Rather than focusing only on doctrines, canonical thinkers, or major texts, the field asks how knowledge is produced, stored, classified, circulated, taught, and verified. Peter Burke's work is especially important in this regard. In *A Social History of Knowledge* and *What Is the History of Knowledge?*, Burke explicitly frames the field in terms of the movement from relatively raw information to socially recognized knowledge through processes such as classification, ordering, and verification (Burke, 2000, 2012).

That move is fundamental for the present article because it shifts the center of gravity away from finished truths and toward the organized conditions under which truths become durable. Once knowledge is treated as socially organized rather than merely mentally possessed, new historical objects become central: archives, pedagogies, professions, communication infrastructures, classificatory orders, and institutions of transmission. This already brings the field close to the present article's concerns. However, the history of knowledge often remains stronger on organization than on constitutional succession. It tells us a great deal about libraries, curricula, bureaucracies, and learned networks, but it asks less consistently the sharper question that governs this article: who is authorized to define what signals mean for a community, and how does that authority move across regimes?

That distinction is decisive. The present article therefore builds on the history of knowledge while shifting its primary unit of analysis. Institutions are treated here not simply as containers of knowledge, but as stabilization machines that make interpretations durable and allocate interpretive legitimacy. A library does not merely preserve texts; it privileges certain inheritances and classificatory rules. A laboratory does not merely host discovery; it authorizes specific procedures and witnesses. A newsroom does not merely relay events; it structures the visible present. A platform does not merely carry information; it ranks and pre-selects the field of encounter. In this way, the article transforms the history of knowledge into a history of changing interpreters.

2.3. Discursive Formation, Historical Ontology, and the Anti-Relativist Guardrail

Any argument that emphasizes mediation and institutional stabilization risks immediate misunderstanding. If knowledge is historically organized, does that imply that reality itself is socially invented? The answer must be no. This is why Michel Foucault and Ian Hacking are both essential to the present argument, though they serve different roles. Foucault's *The Archaeology of Knowledge* remains foundational because it shifts inquiry from isolated ideas to the rules governing the formation of statements, objects, concepts, and discursive regularities (Foucault, 1972). His enduring contribution lies in showing

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that what becomes sayable, intelligible, and classifiable is historically structured rather than given in transparent immediacy.

This insight is crucial because the present article begins from bounded access rather than full possession. Human beings do not encounter reality as a complete whole. They encounter fragments, traces, anomalies, impressions, and records. Those fragments do not interpret themselves. Foucault helps clarify that they become historically intelligible only within specific conditions of statement, visibility, and formation. Yet Foucault alone is not sufficient for the present purpose, because a theory of historical formation can be misread as a denial of reality itself.

Hacking's *Historical Ontology* provides the necessary corrective. Hacking allows one to study historically changing categories, styles of reasoning, and ways of making up kinds without collapsing into ontological nihilism (Hacking, 2002). This is indispensable here. The present article does not argue that truth is whatever institutions declare it to be. It argues that truth, for historical communities, is pursued under conditions of partial access and therefore requires mediation, categorization, and social authorization. Reality exceeds any single knower or institution. But access to reality is historically constrained and organized. This anti-relativist guardrail is necessary because the article seeks to explain how interpretations become socially binding without reducing reality to arbitrary opinion.

2.4. Sociology of Knowledge and the Problem of Stabilization

The sociology of knowledge supplies a central mechanism that the present article requires: the explanation of how meanings become objective and socially durable. Berger and Luckmann's *The Social Construction of Reality* remains indispensable because it shows how humanly produced meanings become externalized, objectified, and institutionalized, eventually returning to actors as if they were simply given in the world (Berger & Luckmann, 1966). This insight underwrites the present article's concept of stabilization.

Stabilization names the process by which an interpretation becomes repeatable, transmissible, durable, and publicly actionable. A signal matters historically only when it is attached to routines, categories, pedagogies, records, procedures, or recognized institutions of validation. The sociology of knowledge therefore bridges epistemics and institutions. It clarifies how a representation becomes normal enough to enter law, education, ritual, administration, common sense, and organized expectation.

This move is vital because it distinguishes the present article from both naive realism and shallow constructivism. The argument is not that societies invent reality at will. It is that societies determine which interpretations of reality become durable enough to govern conduct. Yet Berger and Luckmann do not by themselves provide a full macro-historical theory of why different civilizations privilege different stabilizers. They explain objectivation powerfully, but they do not fully answer why one regime assigns stabilization power to priests, another to scribes, another to experimenters, another to editors, and another to algorithmic ranking systems. That unresolved question directs the argument

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toward science studies and then toward a broader comparative history of interpretive authority.

2.5. Science and Technology Studies: Facts as Achievements, Not Givens

Science and technology studies deepens the analysis of stabilization by demonstrating that even the most authoritative modern facts do not appear as self-present givens. Latour and Woolgar's *Laboratory Life* remains central because it shows how facts emerge through inscription, instrumentation, negotiation, textual operations, and laboratory routines (Latour & Woolgar, 1979). Its most important lesson for the present article is not that facts are arbitrary, but that facticity must be achieved.

This matters because STS reveals a generalizable structure. Signals do not become socially portable truths without material and procedural transformation. The laboratory does not merely observe nature. It translates traces into inscriptions, inscriptions into claims, claims into stabilized facts, and facts into public credibility. In this sense, science is not outside the history of interpreters. It is one of its most powerful reorganizations.

The present article draws two major lessons from this field. First, stabilization is always partly material. Instruments, records, visualizations, and inscriptions are not secondary to epistemic authority; they are among its enabling conditions. Second, modern science did not abolish interpretation. It disciplined interpretation by tying credibility to reproducible procedure, authorized witnessing, and constrained communal correction. Yet STS by itself is not a full long-duration account of historical succession among regimes. It explains the local production of facticity with extraordinary precision, but it does not on its own offer a comparative civilizational grammar capable of bringing oral, sacred, archival, media, and algorithmic regimes into one frame. For that, one must turn to the major transitions in literacy, record, and administrative memory.

2.6. Orality, Literacy, Writing, and the Externalization of Memory

The literature on orality and literacy is indispensable because it shows that writing is not a neutral carrier of thought. Walter J. Ong's *Orality and Literacy* remains foundational in demonstrating that writing reorganizes memory, cognition, transmission, and authority (Ong, 1982/2012). Once words can be detached from the immediacy of speech and fixed in durable marks, the conditions of stabilization are transformed. Meanings can be revisited, compared, codified, indexed, and administered across time and distance.

Jack Goody's *The Logic of Writing and the Organization of Society* deepens this transformation by showing how writing supports classification, formalization, record-keeping, enumeration, and institutional complexity (Goody, 1986). Writing does not merely preserve memory. It creates new political and social forms of memory. It enables archives, legal systems, taxation records, inventories, genealogies, and documentary legitimacy. In the terms of the present article, writing is one of history's most consequential stabilization machines.

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This literature is vital because it reveals a major shift in interpretive authority. Once documents acquire force, new interpreters rise with them: scribes, archivists, officials, and literate elites. But the transition is not smooth. Oral custodians, embodied knowers, and local practitioners do not simply vanish. They are subordinated, hybridized, or forced into tension with documentary systems. Here James C. Scott's work on legibility becomes especially important. Documentary systems do not merely store complexity. They render complexity administratively actionable through simplification, standardization, and abstraction (Scott, 1998). That insight helps connect literacy to power. Archives are not passive containers. They are machines of selective freezing. They make some forms of memory governable and some forms of life visible to institutions in privileged ways.

2.7. Embodied Knowledge, Sacred Order, and Early Interpretive Monopolies

The literature on embodiment and practical skill is essential because it corrects a persistent distortion in knowledge history: the tendency to treat pre-literate worlds as epistemically thin. Tim Ingold's work is especially valuable because it foregrounds dwelling, practical engagement, environmental attunement, and skill as forms of knowing inseparable from life-processes rather than detachable propositions (Ingold, 2000). This matters because the earliest interpreters were not scientists, archivists, or editors. They were hunters, healers, elders, and ritual specialists whose authority arose from demonstrated reliability in reading signals close to survival, health, danger, seasonality, taboo, and memory.

This scholarship allows the present article to establish a non-trivial baseline. The body was the first archive. Skill was among the first credibility criteria. Practical reliability, rather than abstract procedure, anchored early trust. But this does not imply a harmonious prehistory. Competing readings of illness, weather, failure, omen, and transgression existed long before formal institutions. Interpretive struggle begins early. What changes over time is its scale, its media, and its mechanisms of stabilization.

From here, scholarship on ancient religion, ritual systems, and early state formation becomes indispensable. Sacred orders in Egypt, Mesopotamia, South Asia, China, and neighboring worlds should not be dismissed as mere superstition. They functioned as infrastructures for coordinating uncertainty at scale through calendars, rites, omens, sacred texts, genealogies, and cosmic narratives (Assmann, 2002; Lewis, 1999; Scott, 2017; Staal, 1989). Priests and ritual specialists translated unstable signals such as eclipses, harvest irregularities, illness, dynastic crisis, and fertility into socially coordinated meaning. In this sense, sacred systems were large-scale stabilization regimes.

Yet the literature also reveals their tensions. Temple and palace, cosmic legitimacy and temporal sovereignty, priestly interpretation and administrative control repeatedly entered into struggle. The question was never only what the world meant. It was also who had the right to say so. This conflictual reading is central to the present article because it prevents sacred authority from appearing seamless or self-grounding.

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2.8. Axial Critique, Translation Networks, Media, and Platform Society: Toward the Article's Intervention

Scholarship on axial transformations and comparative civilizations supplies a crucial next step because it identifies historical moments in which inherited authority came under sustained pressure from philosophy, prophecy, ethics, and reflective critique. Jaspers's classic formulation, later pluralized and refined by Eisenstadt, Bellah, and Arnason, is important not because it provides a final periodization, but because it marks a transformation in the grammar of legitimacy (Arnason, 2003; Bellah, 2011; Eisenstadt, 1986; Jaspers, 1953). Authority increasingly came to be justified not only by office, ancestry, or ritual inheritance, but also by argument, moral universality, disciplined introspection, and critical reason. The present article extends this literature by reading axial transformations not merely as intellectual innovation, but as conflicts between rival claimants to interpretive monopoly.

Scholarship on translation movements and intellectual circulation then provides a second decisive correction. Gutas on the Graeco-Arabic translation movement, together with work by Saliba, Pollock, Elman, and others, demonstrates that major knowledge growth depends not on civilizational purity but on institutions capable of relay, commentary, synthesis, and transmission (Elman, 2000; Gutas, 1998; Pollock, 2006; Saliba, 2007). Translation is not secondary to knowledge. It is constitutive of it. Yet precisely because translation changes what it carries, it repeatedly produces conflict: translators versus guardians of purity, circulation versus closure, synthesis versus orthodoxy, empire versus doctrinal gatekeeping.

Modern science, universities, and professions then proceduralize and industrialize interpretive authority, while media institutions reorganize the social present through salience, framing, repetition, and public narration (Abbott, 1988; Bourdieu, 1988; Daston & Galison, 2007; Gieryn, 1983; Habermas, 1991; Kerr, 1963; Kuhn, 1962; McLuhan, 1994; Schudson, 1978/1981; Tuchman, 1978). This matters because media regimes introduce a distinct mode of epistemic power: not only the authority to state what is true, but the authority to decide what counts as the present. Newsrooms, broadcasters, and editorial institutions become organized interpreters of salience.

The contemporary turn arrives with platforms, data systems, and AI-mediated visibility. Gillespie, Noble, Zuboff, Couldry and Mejias, van Dijck, and Crawford collectively show that current digital infrastructures do not merely distribute interpretations; they increasingly pre-structure the field from which interpretation begins (Couldry & Mejias, 2019; Crawford, 2021; Gillespie, 2018; Noble, 2018; van Dijck, 2013; Zuboff, 2019). Ranking, recommendation, moderation, engagement optimization, data extraction, and model-mediated filtering determine which signals surface, which disappear, which are repeated, and which acquire social traction. The deepest contemporary question is therefore not only whether machines know, but whether hybrid human-machine systems now shape what societies can even encounter as signal in the first place.

Taken together, these literatures provide nearly all the materials needed for the present

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article, but they do not yet converge into a single comparative grammar. They remain divided by period, medium, institution, and disciplinary boundary. They explain sacred order, literacy, science, media, or platforms in relative isolation. What remains insufficiently theorized is the recurrent structure linking them across long duration: signals are encountered under bounded conditions; they are translated through historically specific practices; those translations are stabilized in durable forms; specific agencies gain authorization to speak for reality; rival agencies contest, absorb, or replace them.

That is the historiographical gap this article addresses. Its contribution is fourfold. First, it offers a long-range comparative grammar for reading the history of knowledge as the history of changing interpretive authority rather than accumulation alone. Second, it restores conflict to the center of epistemic history by treating regime shifts as struggles among agencies rather than smooth developmental sequences. Third, it links ancient, medieval, modern, and algorithmic formations without flattening their differences. Fourth, it reframes the contemporary knowledge crisis as, at root, a crisis of interpretive authority. The argument is therefore neither relativist nor triumphalist. It does not claim that all regimes know equally well. Nor does it narrate history as a purified ascent toward transparent truth. It argues instead that human beings know under conditions of bounded access, that societies therefore depend on authorized interpreters, and that the deepest crises of knowledge often occur when the institutions that stabilize those interpreters lose legitimacy or are displaced by new ones. The question, then, is not only what humans have known. It is also who, in each age, has been allowed to say what the world means.

3. Method, Scope, and Comparative Design

Figure 1 Placeholder: Long Timeline of Dominant Interpreters

hunter/healer/elder → priest/ritual specialist → scribe/archive official → philosopher/sage/prophet → translator-scholar → scientist → university expert → journalist/editor → platform/algorithm/AI system

Figure 1: Long-duration succession of dominant interpreters across historical phases.

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Table 1: Comparative Regimes of Knowledge Across Civilizations

Phase	**Signal**	**Translation**	**Stabilization**	**Authorized Agency**
Embodied / Oral	tracks, symptoms, weather patterns, memory	embodied reading, oral interpretation	communal memory, repetition	hunter, healer, elder
Sacred / Ritual	omens, celestial cycles, fertility signs	ritual exegesis, cosmological interpretation	calendar systems, rites, sacred texts	priest, ritual specialist
Archival / State	decrees, tax records, censuses	inscription, bureaucratic classification	archives, law codes, administration	scribe, state official
Axial / Critical	moral contradiction, inherited doctrine	philosophical reasoning, ethical critique	schools, debate traditions	philosopher, sage, prophet
Translational / Networked	foreign texts, transmitted sciences	translation, commentary, synthesis	libraries, scholarly networks	translator-scholar
Scientific / Procedural	measurements, experiments	hypothesis testing, experimentation	laboratories, journals, replication	scientist
Institutional / Expert	technical data, professional records	disciplinary analysis	universities, professions	expert, professor
Media / Public Sphere	events, images, public statements	editorial framing, narration	newsroom systems, broadcasting	journalist, editor
Platform / AI	behavioral data, platform content	algorithmic ranking, recommendation	platform infrastructure	algorithmic systems

This article is a comparative civilizational analysis of epistemic regimes. Its aim is not to provide an exhaustive history of every society, institution, or tradition that has participated in the making of human knowledge. Such a goal would be impossible within the limits of a single article and would in any case confuse breadth of coverage with adequacy of explanation. The purpose here is more specific. The article seeks to identify a recurrent historical structure through which human communities encounter the world under bounded conditions, translate accessible signals into meaningful accounts, stabilize those accounts through durable practices, and authorize particular agencies to speak for reality. What the article compares, therefore, is not civilizations in their totality, but the patterned relation among signals, translation, stabilization, authorized agency, and conflict.

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This methodological choice follows directly from the article's governing premise. Human beings do not encounter reality as a complete and transparent whole. They encounter traces, anomalies, records, measurements, embodied impressions, and mediated signs. These do not become knowledge by themselves. They must be translated into meaningful claims, stabilized across time and across observers, and recognized as credible by a community. Once this framework is adopted, the central unit of analysis can no longer be only ideas, doctrines, canonical texts, or institutions taken separately. The relevant unit is the structured relation among what counts as signal, who is authorized to interpret it, by which procedures interpretation is stabilized, and how competing claims are sorted, absorbed, or displaced.

For this reason, the article proceeds by comparing epistemic regimes rather than by narrating an unbroken chronology of intellectual events. An epistemic regime, as used here, is a patterned configuration linking at least five elements: the kinds of phenomena treated as legible signals, the interpretive procedures through which those signals are translated, the stabilization machines through which interpretations become durable, the agencies recognized as legitimate interpreters, and the credibility criteria by which claims are accepted or rejected. This permits comparison across widely different historical worlds without presuming that they are reducible to one civilizational model. The comparison is therefore structural rather than totalizing. It does not claim that Egypt, Greece, India, China, Islamic societies, Europe, modern scientific institutions, media systems, and digital platforms are fundamentally the same. It claims, rather, that they can be compared through a recurrent grammar concerning how societies authorize interpreters of reality.

This structural approach also explains why the article moves across such a long historical duration. The argument cannot be established within a narrowly bounded period because its central claim concerns succession: the transfer, contestation, and reorganization of interpretive authority across civilizational time. The movement from embodied and oral interpreters to priestly, scribal, philosophical, translational, scientific, professional, media, and algorithmic formations is not presented as a single line of necessary progress. It is presented as a series of reorganizations in which new agencies arise, older agencies are challenged or subordinated, and legitimacy is redistributed. The longue durée perspective is therefore not ornamental. It is required by the problem itself.

At the same time, the article does not attempt global completeness. The selected historical streams are purposive rather than exhaustive. They are chosen because they illuminate major transformations in the grammar of interpretive authority. Early embodied and oral worlds show that knowledge begins in practical attunement, memory, survival reading, and local trust rather than abstract theory. Sacred and early state formations show how large-scale uncertainty generates priestly and ritual infrastructures for coordination. Archival and scribal regimes show how writing externalizes memory and reorganizes legitimacy. Axial and post-axial formations show how inherited authority comes under pressure from critique, philosophy, prophecy, and ethical reflection. Translation networks reveal that

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knowledge growth depends not only on inheritance, but on relay, commentary, and cross-civilizational circulation. Scientific and professional institutions show how interpretation becomes disciplined through reproducible procedure and credentialed expertise. Media institutions reveal that modern publics inhabit narrated presents structured by salience and repetition. Platform and AI systems show that visibility itself can now be pre-structured before conscious interpretation begins. These cases are not included because they exhaust history. They are included because together they make visible a recurrent logic.

This comparative design requires a disciplined stance toward historical difference. The article does not flatten distinct traditions into a single universal narrative, nor does it assume that the same institutional forms recur identically across societies. It instead asks a narrower and more defensible set of questions. In each historical phase, what counts as a meaningful signal? Through what practices is that signal translated? By what means is interpretation stabilized? Which agency is authorized to speak for reality? Against which rival interpreters does that authority define itself? These questions allow civilizational comparison while preserving specificity at the level of institutions, media, and legitimacy criteria.

The article also adopts an explicitly non-relativist but anti-naive epistemic position. It rejects the idea that reality is transparently given to human observers without mediation. It also rejects the opposite claim that reality is merely whatever institutions declare it to be. Reality exceeds any single knower, institution, or regime. Yet access to reality is always partial, mediated, and historically organized. This middle position is essential methodologically. Without it, the article would collapse either into naive transparency, which cannot account for the historical organization of interpretation, or into relativist arbitrariness, which cannot account for why some regimes stabilize reliable knowledge more effectively than others. The present framework instead treats knowledge as historically organized fidelity under conditions of bounded access.

Conflict is therefore methodologically central, not supplementary. In this article, conflict does not mean only war, coercion, or overt doctrinal clash, though those may be one expression of it. Conflict is coded more fundamentally as inter-agency struggle over interpretive legitimacy. It concerns who may define signals, whose procedures count as credible, which institutions stabilize meanings, and which agencies gain the right to render the world intelligible for others. A struggle between temple and palace, between priest and philosopher, between oral custodian and scribe, between translator and guardian of purity, between scientist and inherited authority, between expert and public, or between newsroom judgment and platform ranking all count as conflict in this methodological sense. Each historical chapter therefore tracks at least one dominant conflict, one displaced agency, one rising agency, and one mechanism of legitimacy transfer. This is necessary because epistemic regimes do not change merely by adding new knowledge to an existing stock. They change when the authority to interpret is contested and reorganized.

The article's figures and comparative tables are designed to support this method. The

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timeline of dominant interpreters visualizes succession across long duration, while the comparative regime table makes explicit the article's unit of analysis by setting historical phase, dominant signal, translation mode, stabilization machine, authorized agency, and credibility criterion into one frame. These devices are not merely illustrative. They operationalize the article's comparative grammar and make visible the patterned transformations that a purely chronological narrative would obscure.

The scope of the article is therefore ambitious but bounded. It does not claim to replace specialist historiography, area studies, or detailed institutional histories. Nor does it claim that all historical formations can be captured by one formula without remainder. Its more modest claim is that a recurring structure becomes visible when knowledge history is read through the problem of authorized interpretation. By treating epistemic regimes as historically variable configurations of signals, translation, stabilization, agency, and conflict, the article offers a framework capable of connecting ancient, medieval, modern, and algorithmic formations without denying their differences. The chapters that follow apply this framework across selected civilizational moments in order to show that the history of knowledge is also, and in a profound sense, the history of changing interpretive authority.

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4. Embodied Knowledge Before Writing

Human knowledge did not begin with writing, archives, or formal institutions. Long before the appearance of texts, laboratories, or universities, human communities already possessed complex systems for reading the world, coordinating action, and transmitting reliable interpretations across generations. To recognize this fact is methodologically essential. If the history of knowledge begins only with literate civilizations, it risks reproducing a familiar distortion: the assumption that pre-literate societies were epistemically thin or intellectually primitive. The evidence from anthropology, archaeology, and environmental history instead points in the opposite direction. Early human communities developed highly refined practices for interpreting signals crucial to survival. What differed from later regimes was not the absence of knowledge, but the media and institutions through which interpretation was stabilized.

In these early worlds, the body functioned as the first archive. Memory, perception, habit, and trained skill served as the principal media through which signals from the environment became meaningful and actionable. A hunter interpreting faint tracks in soil, a healer recognizing patterns of illness in bodily symptoms, or an elder recalling seasonal cycles was not merely reacting to isolated events. Each was drawing upon accumulated experience structured by practice, memory, and communal transmission. Knowledge was therefore inseparable from lived engagement with the environment. It was not stored primarily in documents or instruments, but in trained capacities of perception and judgment.

Such practices demonstrate that early knowledge systems were already structured around credibility criteria. Not every interpretation carried equal weight. Authority emerged from demonstrated reliability in reading signals that mattered for survival: the direction of migrating animals, the timing of seasonal change, the recognition of dangerous plants, the interpretation of illness, or the anticipation of storms. Individuals whose interpretations consistently aligned with successful outcomes gradually acquired interpretive authority. Hunters known for tracking accuracy, healers capable of restoring health, elders who remembered environmental rhythms, and ritual specialists whose practices coordinated communal uncertainty became recognized interpreters within their communities.

These roles illustrate a fundamental point that runs throughout the article: knowledge does not become socially effective merely by existing as observation. It becomes effective when interpretations are stabilized through recognized agencies and repeated practices. Even in the absence of formal institutions, early societies developed mechanisms that allowed certain interpretations to acquire durability and legitimacy. Ritual, storytelling, apprenticeship, and communal memory all functioned as stabilization machines. Through repetition and transmission, they transformed episodic experience into shared knowledge.

Ritual played a particularly important role in this early stabilization process. In many communities, ritual practices served not only spiritual or symbolic purposes but also epistemic ones. Repeated ceremonies encoded environmental rhythms, moral expectations,

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and practical warnings into communal memory. By linking interpretation to ritual performance, societies created durable frameworks through which uncertainty could be organized and meaning transmitted across generations. Ritual specialists therefore emerged as important early interpreters, mediating between observed anomalies and socially meaningful explanations.

The presence of such roles demonstrates that interpretive authority predates writing and formal governance. Hunters, healers, elders, and ritual specialists did not merely perform practical tasks. They interpreted signals and translated them into decisions that affected the survival and coordination of the group. Their authority rested not on formal certification or written doctrine, but on reputational credibility accumulated through repeated demonstration of interpretive reliability. In this sense, early epistemic regimes already possessed a recognizable structure linking signals, translation, stabilization, and authorized agency.

Importantly, these regimes were not harmonious or uncontested. Conflict over interpretation existed long before the appearance of centralized institutions. Illness, drought, hunting failure, unexplained death, or unusual environmental events frequently generated competing explanations. Was illness caused by natural imbalance, spiritual disturbance, or ritual violation? Did a failed hunt indicate poor tracking, unfavorable weather, or a broken taboo? Such questions could produce disagreements among interpreters and uncertainty within the community. Competing readings of the same signals forced groups to evaluate whose interpretations should guide action.

This form of conflict should not be understood only as open dispute or violence. More often it appeared as subtle competition among interpretive roles. A healer's explanation might conflict with a ritual specialist's interpretation of the same illness. A hunter's environmental reading might differ from an elder's recollection of seasonal patterns. Communities resolved such tensions through a mixture of experience, trust, narrative authority, and practical outcome. Interpretations that repeatedly proved effective were more likely to be retained and transmitted. Those that failed to produce reliable results gradually lost credibility.

The presence of these struggles reveals that epistemic conflict is not a late development arising only with organized religion, philosophy, or science. It is a structural feature of human knowledge itself. Whenever signals are ambiguous and interpretations uncertain, multiple agencies may claim the authority to define their meaning. Early societies therefore already contained the seeds of later epistemic competition. What later civilizations institutionalized through priesthoods, academies, courts, or laboratories began as localized struggles over who could most reliably interpret the world.

Embodied knowledge systems also illuminate a distinctive form of epistemic accountability. In many early contexts, credibility was inseparable from proximity to signals. Hunters who spent extensive time in forests or plains possessed intimate knowledge of animal behavior and environmental patterns. Healers who observed bodily symptoms directly

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accumulated experiential understanding of illness and treatment. Authority thus arose not from abstraction alone but from lived engagement with the phenomena being interpreted. Knowledge was grounded in sustained interaction with the environment.

This proximity created a distinctive epistemic economy. Interpretive authority was difficult to separate from the body that performed the interpretation. Skills were learned through apprenticeship, imitation, and participation rather than through textual instruction. A novice hunter learned tracking not by reading manuals but by following experienced trackers and gradually acquiring the perceptual sensitivity required to recognize subtle environmental cues. Similarly, healing practices were often transmitted through observation and mentorship rather than written codification. The body therefore functioned simultaneously as instrument, archive, and site of interpretation.

Understanding these dynamics helps clarify the continuity between early knowledge systems and later epistemic regimes. The emergence of writing, archives, and institutions did not create interpretation from nothing. Instead, these later developments reorganized practices that were already present in embodied form. Writing externalized memory, allowing interpretations to be preserved beyond individual lifetimes. Archives stabilized records, enabling administrative and legal authority. Laboratories and scientific instruments later extended the capacity to observe and measure phenomena beyond ordinary perception. Yet the underlying structure remained recognizable: signals required translation, translation required stabilization, and stabilization required recognized interpreters.

Seen from this perspective, pre-literate societies should not be understood as epistemically undeveloped. They represent an early regime in the long history of interpretive authority. Their distinctive feature lies not in the absence of knowledge, but in the forms through which knowledge was embodied and transmitted. Environmental attunement, ritual repetition, and communal memory provided mechanisms through which interpretations could become reliable and socially binding.

Recognizing this regime is crucial for the argument of the article as a whole. The succession of epistemic regimes described in later chapters—from sacred systems to archives, scientific institutions, media infrastructures, and algorithmic platforms—does not begin from a blank slate. It begins from embodied worlds in which human communities already possessed structured methods for reading the signals of their environment. What changes across history is not the necessity of interpretation, but the media, institutions, and agencies through which interpretation becomes stabilized and authorized.

Embodied knowledge before writing therefore establishes the baseline condition of human epistemics. Reality presents itself to communities through incomplete and often ambiguous signals. Human beings respond by developing practices for translating those signals into meaningful accounts of the world. These accounts become durable when stabilized through memory, ritual, and communal transmission. Authority emerges around individuals and roles capable of producing reliable interpretations. Conflict arises whenever competing agencies claim the right to define what those signals mean.

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The chapters that follow trace how this basic structure becomes reorganized as societies develop writing systems, archives, philosophical traditions, scientific procedures, media institutions, and algorithmic infrastructures. Each transformation introduces new stabilization machines and new interpreters while reshaping the criteria by which credibility is judged. Yet the underlying problem remains the same: in a world where reality exceeds any single observer, societies must continually decide who is authorized to interpret the signals through which that reality becomes knowable.

5. Sacred Order: Priesthood, Ritual, and Cosmic Coordination

The transition from mobile or semi-mobile communities to agricultural and early state civilizations produced a profound transformation in the scale at which uncertainty had to be interpreted and coordinated. In smaller societies, embodied knowledge and communal memory were often sufficient to guide hunting, healing, and seasonal movement. With the emergence of sedentary agriculture, irrigation systems, urban settlements, and dynastic rule, however, the interpretive problem expanded dramatically. Entire populations now depended on the correct anticipation of seasonal cycles, the management of floods and droughts, the coordination of labor, and the maintenance of political legitimacy across generations. Under these conditions, interpretation could no longer remain only an embodied skill practiced by dispersed individuals. It required stabilization systems capable of organizing meaning at the scale of civilizations.

Sacred institutions emerged as one of the most important infrastructures through which early societies addressed this challenge. Across regions as diverse as Mesopotamia and Egypt, South Asia, the Persian plateau, and East Asia, priestly specialists developed complex ritual, calendrical, and interpretive systems designed to translate environmental and social uncertainty into coordinated cosmic order. These systems should not be dismissed as simple belief structures or naive cosmologies. They functioned as large-scale interpretive frameworks that organized time, legitimacy, and social expectation. In the terms used throughout this article, they operated as stabilization machines capable of linking signals, translation, authorized agency, and durable meaning.

Agricultural life intensified dependence on environmental rhythms. Successful cultivation required reliable knowledge of seasons, rainfall, flooding patterns, and soil cycles. In riverine civilizations such as ancient Egypt and Mesopotamia, the timing and scale of floods could determine the prosperity or collapse of entire regions. Priestly institutions therefore developed sophisticated calendars and observational traditions to track celestial movements and seasonal indicators. By correlating astronomical observation with agricultural cycles, priesthoods transformed scattered environmental signals into coordinated temporal structures. The calendar was not merely a measurement of time; it was a mechanism that synchronized agriculture, ritual, and governance across large populations.

These calendrical systems illustrate how sacred authority stabilized interpretation. Observations of the sky, the flooding of rivers, or unusual natural phenomena did not interpret

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themselves. They required trained specialists capable of reading patterns and situating them within broader cosmological frameworks. Priests therefore served as authorized interpreters of celestial and terrestrial signals. Their interpretive authority rested partly on accumulated observational knowledge and partly on the institutional continuity of temple traditions. Through ritual repetition and recorded precedent, priestly institutions established stable frameworks through which societies could anticipate and respond to environmental uncertainty.

In Mesopotamia, for example, temple complexes functioned not only as centers of worship but also as hubs of astronomical observation, record keeping, and omen interpretation. Clay tablets preserved extensive collections of celestial and terrestrial omens, linking specific natural events to anticipated outcomes. Eclipses, planetary alignments, unusual animal behavior, or unexpected weather patterns were catalogued and interpreted through established omen traditions. These records allowed priestly specialists to connect new events with accumulated precedent, thereby transforming unpredictable phenomena into interpretable signals within a cosmic system. Such practices did not eliminate uncertainty, but they provided structured methods through which uncertainty could be managed and communicated.

Egyptian priesthoods developed comparable infrastructures of cosmic coordination. The annual flooding of the Nile required precise anticipation, and priestly astronomers monitored the heliacal rising of Sirius as an indicator of the coming inundation. This celestial observation was integrated into a ritual and administrative calendar that structured agricultural labor, taxation, and state ceremonies. In this context, sacred cosmology and practical governance were deeply intertwined. The Pharaoh's legitimacy depended partly on the maintenance of ma'at, a principle of cosmic balance and order. Priestly interpretation of celestial and ritual signals therefore contributed directly to the perceived stability of political rule.

Similar patterns appeared across other early civilizations. In South Asia, Vedic ritual specialists developed elaborate sacrificial systems tied to cosmological conceptions of order and renewal. Ritual performance, recitation, and precise calendrical timing were believed to sustain harmony between cosmic forces and human society. The preservation of ritual knowledge through memorization and disciplined transmission illustrates the sophistication of early stabilization mechanisms. Sacred texts and liturgical traditions functioned as repositories of interpretive precedent, enabling continuity across generations.

In early Chinese civilizations, the relationship between cosmic order and political authority was articulated through concepts such as the Mandate of Heaven. Natural disasters, eclipses, or unusual astronomical phenomena were interpreted as signs reflecting the moral condition of rulers. Court astrologers and ritual specialists therefore played critical roles in observing celestial events and advising political authorities. Here again, sacred interpretation linked environmental signals with political legitimacy, reinforcing the idea that cosmic harmony and social order were inseparable.

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Across these regions, priesthoods performed several interconnected functions. They observed signals from both the natural environment and the social world. They translated those signals into meaningful interpretations grounded in cosmological narratives. They stabilized interpretations through ritual repetition, written records, and institutional continuity. Finally, they exercised authorized interpretive authority, advising rulers and guiding communal responses to uncertainty. These functions illustrate that sacred institutions were not merely repositories of belief. They were epistemic infrastructures designed to coordinate large-scale societies.

Yet priestly authority was never uncontested. As early states consolidated political power, tensions frequently emerged between temple institutions and royal administrations. These conflicts often revolved around competing claims to interpret the sources of legitimacy and order. Priests grounded authority in cosmic principles, sacred narratives, and ritual continuity. Kings and political rulers grounded authority in military power, administrative control, and territorial sovereignty. While these two forms of authority often cooperated, they also competed to define the meaning of events and the proper response to crises.

This tension between temple and palace reveals an important dimension of epistemic conflict. When natural disasters, epidemics, or political upheavals occurred, multiple agencies could claim the authority to interpret their significance. A priestly institution might attribute drought to ritual impurity or divine displeasure, requiring renewed ceremonial performance. A ruler or administrator might interpret the same event as a logistical or political challenge requiring changes in governance or resource distribution. These competing interpretations were not merely intellectual disagreements. They reflected struggles over who possessed the legitimate authority to define the meaning of events for society as a whole.

The presence of such conflicts demonstrates that sacred regimes should not be romanticized as stable or uncontested systems of belief. They were dynamic interpretive orders embedded within broader political and economic structures. Temple institutions often accumulated land, wealth, and administrative responsibilities, giving them significant influence within early states. At the same time, rulers sought to control or integrate priestly authority into centralized political systems. The balance between these powers varied across regions and historical periods, but the underlying tension remained a persistent feature of early civilizations.

Despite these conflicts, sacred systems provided indispensable mechanisms for coordinating uncertainty at scale. By embedding environmental observation within ritual and cosmological narratives, priesthoods transformed scattered signals into socially intelligible patterns. Ritual repetition reinforced shared expectations and stabilized collective memory. Sacred narratives connected present events with ancestral precedent and cosmic order. Through these mechanisms, societies were able to maintain coherence even in the face of unpredictable natural and political conditions.

The interpretive authority of priesthoods also illustrates a broader principle about the

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history of knowledge. Large-scale societies require institutions capable of stabilizing meaning across time and space. As populations grow and political systems expand, localized embodied knowledge becomes insufficient for coordinating complex social orders. Stabilization machines—whether temples, archives, universities, or digital platforms—emerge to manage this complexity. Sacred institutions represent one of the earliest and most influential examples of such systems.

Understanding sacred regimes in this way allows us to see them not as relics of pre-scientific thought but as early solutions to a fundamental epistemic problem. Human societies must continually translate uncertain signals into actionable interpretations. When the scale of society expands, the mechanisms required for this translation become more elaborate. Priesthoods developed calendrical systems, omen traditions, ritual procedures, and cosmological narratives precisely because these tools enabled large populations to interpret and coordinate responses to uncertainty.

Seen from this perspective, sacred authority represents a pivotal stage in the long succession of interpretive regimes traced in this article. It marks the transition from localized embodied interpretation to institutionalized systems capable of stabilizing meaning across entire civilizations. The priesthood became an organized interpreter, positioned between environmental signals, political authority, and communal life. Its authority rested not only on spiritual belief but also on its capacity to maintain interpretive continuity across generations.

The next historical transformations examined in this article will build upon this foundation. The emergence of writing systems, archives, and administrative bureaucracies will gradually reshape the mechanisms through which interpretations are stabilized and authorized. Yet these later developments do not replace the sacred regime entirely. They transform it. Many archival and bureaucratic practices originate within temple institutions, and sacred cosmologies often continue to influence political legitimacy even in later literate societies.

The study of sacred order therefore reveals an essential step in the civilizational evolution of knowledge. As agricultural and early state societies expanded, the management of uncertainty required new forms of interpretive authority. Priesthoods provided these forms by translating environmental and social signals into cosmological frameworks capable of coordinating large populations. Through ritual, calendar, omen interpretation, and sacred narrative, they created durable infrastructures for meaning. These infrastructures did not eliminate uncertainty, but they allowed societies to confront it collectively.

The central problem remains the same as in earlier regimes: reality presents itself through incomplete and ambiguous signals. Sacred systems responded by embedding those signals within comprehensive cosmic orders that linked nature, society, and political legitimacy. In doing so, they established one of the earliest large-scale stabilization systems in human history. Their legacy continues to shape later epistemic regimes, reminding us that the authority to interpret the world has always been intertwined with the structures through which societies organize meaning and power.

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6. Writing, Archives, and the State

The emergence of writing marks one of the most consequential transformations in the history of knowledge. While earlier societies relied primarily on embodied memory, ritual repetition, and oral transmission to stabilize interpretation, the appearance of durable written records introduced a fundamentally new ontology of knowledge. Memory was no longer confined to the capacities of living individuals or the rhythms of ritual practice. Instead, interpretation could be externalized into material inscriptions capable of persisting across generations, institutions, and territories. Writing therefore did more than preserve information. It reorganized the conditions under which knowledge could be stabilized, transmitted, and authorized.

To understand this transformation, it is important to recognize that writing did not simply function as a neutral recording technology. The act of inscription converted transient events into durable objects. Once recorded, a decree, contract, law, or administrative record could circulate independently of the moment in which it was produced. Interpretation thus became embedded in documents that could be revisited, compared, classified, and enforced long after their original context had passed. In this sense, the written document became one of history's most powerful stabilization devices.

The implications of this shift were profound. In oral regimes, knowledge depended heavily on the credibility of the speaker and the trust placed in communal memory. Elders, ritual specialists, and custodians of tradition derived authority from their ability to remember and transmit the past accurately. Writing redistributed that authority by relocating memory into material objects. A tablet, scroll, or inscription could preserve a decision or event in a form that did not rely on personal recollection. As a result, the locus of interpretive authority began to shift from living custodians of memory toward institutions capable of producing, preserving, and interpreting documents.

Archives emerged as central infrastructures within this new regime. Far from being passive repositories of information, archives functioned as active systems for organizing, stabilizing, and legitimizing interpretation. Through processes of classification, enumeration, indexing, and preservation, archives transformed scattered documents into structured memory systems. Administrative decisions, legal rulings, tax records, property boundaries, and genealogical claims could all be stored and retrieved within these institutional frameworks. The archive therefore acted as a machine of selective freezing: it captured certain interpretations of events and preserved them as authoritative records while leaving countless other experiences undocumented.

This selective freezing had far-reaching consequences for the organization of power and knowledge. Once recorded within an archive, a document could acquire a form of legitimacy that exceeded the authority of any individual speaker. Legal rulings could be cited as precedent. Property claims could be verified through written contracts. Administrative decisions could be enforced through documented directives. The document thus became a

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stabilizer that allowed interpretations to endure beyond the immediate circumstances in which they were first articulated.

The rise of documentary authority also produced a new interpretive class: the scribal bureaucracy. Writing systems were rarely accessible to entire populations in early literate societies. Instead, literacy was concentrated among trained specialists who possessed the skills necessary to produce, interpret, and manage written records. Scribes served as intermediaries between the material archive and the institutions that relied upon it. They recorded transactions, drafted decrees, preserved correspondence, and maintained administrative continuity. In performing these functions, scribes became indispensable interpreters within emerging state systems.

The power of the scribal class did not derive solely from technical skill. It also emerged from their role in mediating between documents and authority. A written record could not enforce itself. It required interpreters capable of determining its meaning and applying it within new contexts. Scribes therefore occupied a pivotal position within the documentary regime. They translated events into records and records into administrative action. Through their work, the archive became a living system capable of shaping decisions across time.

This transformation was closely tied to the expansion of early states. As political authorities extended their control over larger territories and populations, they required mechanisms capable of coordinating taxation, labor, law, and resource distribution. Writing provided precisely such a mechanism. Tax registers could record obligations across entire regions. Census lists could enumerate populations and households. Legal codes could standardize procedures across diverse communities. Administrative correspondence could transmit orders across distant provinces. Through these documentary systems, states were able to stabilize interpretation across geographic and temporal distance.

Enumeration and classification became particularly important tools within this new regime. By converting people, land, and goods into recorded categories, bureaucratic systems rendered complex social realities administratively legible. Land could be mapped and taxed. Populations could be counted and mobilized. Resources could be allocated according to documented inventories. These practices illustrate how writing transformed interpretation into an operational technology of governance. Documents did not merely describe the world; they reorganized it into forms that could be administered and controlled.

Legal systems likewise underwent a fundamental transformation through the introduction of writing. In oral regimes, law often relied on customary memory and the authority of elders or judges who interpreted precedent through recollection and communal knowledge. Written law codes introduced a different model. Once recorded, legal provisions could be cited, copied, and transmitted with greater stability. Courts could refer to written statutes or precedents when adjudicating disputes. This documentary continuity strengthened the durability of legal interpretation and reduced reliance on individual memory.

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However, the rise of documentary authority did not eliminate earlier epistemic regimes. Instead, it introduced new tensions between competing forms of legitimacy. Oral custodians of tradition often retained authority within communities even as bureaucratic records expanded. Ritual specialists continued to interpret sacred traditions even as legal and administrative documents proliferated. Conflicts frequently arose when documentary records contradicted or displaced earlier forms of memory and authority.

These conflicts reveal the deeper epistemic transformation introduced by writing. When a written record exists, it can challenge the credibility of oral testimony. A contract inscribed on clay or parchment can override recollections of an agreement. A tax register can redefine obligations that had previously been negotiated through custom. In such cases, the document becomes a rival authority capable of reshaping the meaning of events.

The tension between memory authority and document authority therefore represents one of the central conflicts of the documentary regime. Oral traditions emphasize continuity through communal recollection and lived practice. Documentary systems emphasize continuity through preserved records and institutional procedure. Each approach embodies a different model of epistemic stability. The former relies on trusted custodians who transmit memory through participation. The latter relies on archives capable of preserving interpretations beyond individual lifetimes.

In many historical contexts, these two regimes coexisted in complex and sometimes uneasy relationships. Ritual traditions might continue to govern religious life even as bureaucratic archives regulated taxation and land tenure. Oral storytelling might preserve cultural identity even as written chronicles recorded political history. Rather than completely replacing earlier systems, writing layered new forms of stabilization onto existing interpretive structures.

Yet the long-term trajectory of literate states favored the expansion of documentary authority. As archives grew and bureaucratic systems became more elaborate, the capacity to store and retrieve records expanded dramatically. Administrative continuity increasingly depended on written documentation. Officials could consult archives to verify property claims, reconstruct past decisions, or coordinate policy across time. The archive thereby functioned not merely as a record of past actions but as an active participant in ongoing governance.

This archival continuity also reshaped political legitimacy. Rulers could justify authority by referencing documented genealogies, legal precedents, or recorded decrees. Dynastic succession could be stabilized through written chronicles. Administrative reforms could be grounded in archived records of earlier policy. The state thus became an institution deeply intertwined with documentary memory.

Seen from the perspective of epistemic regimes, the introduction of writing created a new configuration linking signals, translation, stabilization, and authorized agency. Signals from economic transactions, population movements, legal disputes, or political

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directives were translated into written records. These records were stabilized within archives that preserved them across time. Scribal officials interpreted and managed the archive, transforming documents into actionable administrative knowledge. Through this system, the state acquired unprecedented capacity to coordinate interpretation across large territories and long durations.

The concept of the archive as a machine of selective freezing captures the distinctive character of this regime. Archives preserve particular interpretations while excluding others. They freeze decisions, classifications, and narratives into durable form. Yet the act of freezing is selective. Only certain events are recorded, and only certain interpretations become authoritative documents. Countless lived experiences remain outside the archive entirely. As a result, the documentary regime both stabilizes knowledge and narrows the field of what becomes visible to institutions.

Understanding this selectivity is essential for interpreting the historical power of archives. Documentary records often appear objective precisely because they endure. Their persistence gives them an aura of neutrality. Yet their creation always involves choices about what to record, how to classify it, and which interpretations to preserve. Archives therefore shape the epistemic landscape not only by storing information but by defining the boundaries of institutional memory.

The rise of writing and archival systems thus represents a major reorganization of interpretive authority. Whereas earlier regimes relied primarily on embodied memory and ritual continuity, the documentary regime relocated stabilization into durable records and bureaucratic procedures. Scribes and officials emerged as new interpreters responsible for translating events into documents and documents into administrative decisions. Legal codes, tax registers, and bureaucratic archives became instruments through which states could maintain interpretive continuity across generations.

At the same time, the conflicts generated by this transformation remind us that epistemic regimes rarely replace one another cleanly. Oral custodians, ritual authorities, and documentary officials often competed to define the meaning of events. These struggles reveal the persistent tension between different stabilization systems within human societies. Writing did not eliminate earlier forms of interpretation, but it introduced a powerful new infrastructure capable of reshaping how knowledge was organized, preserved, and legitimized.

The chapters that follow will trace how this documentary regime interacts with later epistemic transformations. Philosophical critique, scientific procedure, and modern media institutions will each introduce new mechanisms for stabilizing interpretation. Yet many of these later developments remain indebted to the documentary foundations established by writing and archives. The externalization of memory made it possible to accumulate knowledge across centuries and to coordinate interpretation across increasingly complex societies.

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Writing, archives, and the state therefore represent a pivotal moment in the long history of interpretive authority. By transforming memory into durable records and embedding interpretation within bureaucratic infrastructures, literate civilizations created new machines for stabilizing knowledge. These machines enabled societies to govern territory, maintain legal continuity, and coordinate large populations. At the same time, they reshaped the epistemic landscape by privileging documentary legitimacy over other forms of memory. Through this transformation, the archive became one of the most powerful instruments through which societies interpret and organize the world.

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7. Axial Critiques: Philosophy, Ethics, and the Right to Question

The emergence of what later scholarship has described as axial transformations represents one of the most significant shifts in the history of interpretive authority. Across multiple regions of Eurasia during the first millennium BCE, inherited systems of sacred interpretation came under sustained pressure from new forms of reflective inquiry. These developments did not occur simultaneously or identically across civilizations, yet they shared a structural feature: they introduced new grounds upon which established interpretations of the world could be questioned. Philosophy, ethical reflection, prophecy, and disciplined forms of introspection began to challenge earlier monopolies of meaning held by priestly and ritual institutions.

This transformation did not abolish sacred traditions, nor did it instantly replace ritual authority with abstract reasoning. Rather, it altered the grammar of legitimacy through which interpretive authority could be claimed. In earlier regimes, legitimacy often derived from inherited office, ritual continuity, or proximity to sacred institutions. In axial contexts, alternative forms of credibility began to appear. Individuals could claim interpretive authority through argument, moral insight, disciplined inquiry, or prophetic critique. The question of who had the right to interpret the world expanded beyond established custodians of ritual knowledge.

In the Greek Mediterranean, philosophical inquiry introduced a distinctive form of interpretive challenge. Early thinkers sought explanations of natural phenomena that did not depend solely on inherited mythological narratives. Philosophers such as Heraclitus, Parmenides, and later Plato and Aristotle explored questions concerning nature, change, ethics, and knowledge through systematic reasoning and debate. Their work did not eliminate religious traditions, but it created intellectual spaces in which claims about the world could be evaluated through argument rather than ritual authority alone.

Greek philosophy institutionalized practices that made questioning itself a recognized activity. Dialogue, debate, and systematic inquiry became legitimate methods for examining inherited beliefs. The authority of philosophical teachers did not rest solely on lineage or priestly office, but on their ability to present persuasive arguments and cultivate disciplined reflection among their students. Schools such as the Academy and the Lyceum functioned as early communities of inquiry in which interpretation was stabilized through reasoned discussion rather than ritual repetition.

Yet even within the Greek world, philosophical authority did not operate without tension. Philosophers frequently encountered resistance from traditional institutions and political authorities. The trial of Socrates illustrates the precarious position of reflective critique within established social orders. Socratic questioning challenged conventional assumptions about virtue, knowledge, and civic authority, provoking accusations that philosophical inquiry itself could undermine social stability. The conflict surrounding Socrates therefore exemplifies a broader structural tension: when questioning becomes legitimate, inherited

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interpretive monopolies are inevitably destabilized.

Comparable transformations unfolded across other civilizations during the same broad historical period. In South Asia, the emergence of philosophical and religious movements such as Buddhism, Jainism, and the Upanishadic traditions introduced new forms of introspective and ethical reflection. These traditions questioned established ritual hierarchies and sacrificial systems that had previously dominated religious life. Instead of grounding legitimacy solely in ritual performance, they emphasized ethical conduct, meditation, and insight into the nature of suffering and liberation.

The Buddha's teachings, for example, redirected interpretive authority toward disciplined personal inquiry into the conditions of suffering and the path toward its cessation. This emphasis did not eliminate the importance of tradition, but it shifted the locus of legitimacy from ritual expertise to experiential understanding and ethical transformation. Monastic communities developed structured methods of teaching, debate, and textual preservation that stabilized these insights across generations.

Jain traditions likewise emphasized ethical discipline and introspective practice as pathways toward knowledge and liberation. These movements demonstrate that axial transformations cannot be reduced to philosophical speculation alone. They involved comprehensive reconfigurations of how individuals and communities understood authority, ethical obligation, and the pursuit of truth.

In East Asia, the intellectual landscape of the axial period was shaped by a diverse set of philosophical traditions that addressed questions of social order, moral cultivation, and political legitimacy. Confucian thinkers emphasized ethical self-cultivation, proper conduct, and the moral responsibilities of rulers. Their teachings proposed that political authority should be grounded in virtue rather than mere force or inherited privilege. The legitimacy of governance thus became tied to ethical performance rather than solely to ritual status or dynastic continuity.

Daoist traditions, by contrast, explored alternative perspectives on harmony, spontaneity, and the relationship between human action and the natural order. These traditions often expressed skepticism toward rigid social hierarchies and emphasized alignment with broader patterns of the cosmos. While differing significantly from Confucian ethics, Daoist thought likewise contributed to a broader culture of reflective questioning concerning how humans should understand their place within the world.

Legalist thinkers introduced yet another dimension to this evolving interpretive landscape. Their emphasis on administrative order, codified law, and political discipline represented a distinct response to the challenges of governance and social coordination. Although Legalism rejected some of the moral assumptions of Confucian thought, it nevertheless participated in the same broader transformation: the development of systematic arguments about the foundations of legitimate authority.

The Persian and Hebrew worlds also witnessed powerful forms of axial critique expressed

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through prophetic traditions and moral cosmologies. Hebrew prophets such as Isaiah, Jeremiah, and Amos challenged both political rulers and religious institutions by invoking ethical standards believed to transcend ritual observance. Their teachings emphasized justice, compassion, and covenantal responsibility, arguing that social injustice and moral corruption could provoke divine judgment.

Prophetic critique thus introduced a distinctive form of interpretive authority grounded not in institutional office but in moral witness. Prophets claimed the right to question rulers and religious leaders alike by appealing to ethical principles believed to reflect divine justice. This form of critique destabilized established hierarchies by asserting that legitimacy depended on moral conduct rather than institutional position alone.

Zoroastrian traditions in the Persian world likewise articulated a moral cosmology that framed human life as a struggle between truth and falsehood, order and chaos. Ethical choice became central to the maintenance of cosmic harmony. In this framework, individuals were called to participate actively in sustaining truth through righteous action. Such teachings reinforced the idea that moral understanding and ethical responsibility were essential components of legitimate authority.

Across these diverse civilizations, axial developments shared a common structural feature: they expanded the grounds upon which inherited interpretations could be questioned. The legitimacy of interpretation increasingly depended not only on ritual tradition or institutional position but also on reasoned argument, ethical insight, disciplined reflection, or prophetic critique. This expansion did not eliminate earlier forms of authority, but it introduced new actors capable of challenging them.

The resulting conflicts were profound. Priests who derived authority from ritual expertise often found themselves confronted by philosophers who claimed authority through reasoning. Political rulers who justified power through dynastic inheritance faced criticism from ethical teachers who insisted that legitimacy required moral virtue. Ritual traditions that emphasized ceremonial continuity encountered movements advocating introspective insight or ethical reform.

These conflicts illustrate that axial transformations were not merely intellectual developments. They were struggles over the conditions under which interpretation could claim legitimacy. Each new interpretive form—philosophical reasoning, ethical introspection, prophetic critique—created alternative pathways for asserting authority. In doing so, it weakened the exclusive control previously exercised by priestly institutions.

At the same time, axial developments also generated new mechanisms of stabilization. Philosophical schools preserved teachings through structured dialogue and written texts. Monastic communities maintained disciplined traditions of study and practice. Ethical teachings were codified into canonical writings and transmitted through educational institutions. These mechanisms ensured that reflective critique itself could become institutionalized and transmitted across generations.

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The axial transformation therefore did not simply dissolve earlier epistemic regimes. Instead, it layered new forms of authority onto existing structures. Sacred traditions continued to shape religious life, yet they increasingly interacted with philosophical and ethical discourses capable of questioning and reinterpreting them. In many societies, priests, philosophers, rulers, and scholars coexisted within complex systems of overlapping authority.

Seen from the perspective of the history of knowledge, axial critique represents a decisive expansion of interpretive pluralism. No single institution could easily monopolize the right to define reality once multiple forms of legitimacy—ritual, philosophical, ethical, and prophetic—became recognized. Societies were forced to negotiate among these competing interpretive frameworks, often producing vibrant intellectual cultures as well as enduring tensions.

This transformation also prepared the conditions for later developments in the history of knowledge. Philosophical argument and disciplined inquiry would eventually contribute to the emergence of scientific reasoning. Ethical reflection would shape traditions of political thought and legal philosophy. The practice of questioning inherited authority would become a recurring feature of intellectual life in many civilizations.

Yet it is important to emphasize that axial critique did not inaugurate a simple narrative of rational progress. The legitimacy of questioning remained contested, and reflective inquiry often coexisted with powerful traditions of ritual and faith. The enduring significance of the axial period lies not in the triumph of reason over religion but in the creation of a new grammar of legitimacy. Interpretation could now be justified through multiple pathways: ritual continuity, moral insight, philosophical argument, or prophetic witness.

By expanding the recognized grounds for questioning, axial transformations reshaped the landscape of interpretive authority. The right to ask whether inherited explanations were adequate became itself a legitimate activity within many intellectual traditions. This shift profoundly altered the dynamics through which societies interpret the world.

The chapters that follow will trace how these new forms of questioning interacted with later developments in translation, scholarship, and scientific procedure. The emergence of interpretive pluralism during the axial period laid the groundwork for intellectual exchanges that would later span civilizations and centuries. Once questioning becomes legitimate, the search for new frameworks of understanding becomes an enduring feature of human knowledge.

Axial critique therefore represents a pivotal moment in the civilizational evolution of interpretation. It marks the point at which inherited monopolies of meaning came under sustained scrutiny from alternative forms of authority grounded in argument, ethics, and disciplined reflection. Through this transformation, the history of knowledge entered a new phase in which the right to question became itself a central component of interpretive legitimacy.

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8. Translation Networks Across Afro-Eurasia

Figure 2 Placeholder: Conflict Matrix of Interpretive Agencies

Axes: rising agencies vs displaced agencies
Conflict types: delegitimation; co-optation; translation struggle; credential struggle; state capture; public legitimacy crisis

Figure 2: Recurrent conflict forms between old and rising interpreters across civilizational transitions.

Table 2: Conflict Between Agencies Across Historical Phases

Old Agency	Rising Agency	Object of Conflict	Institutional Battleground	Outcome Pattern
Ritual specialists / priests	Philosophers / ethical teachers	legitimacy of interpretation (ritual vs reason)	schools, courts, sacred institutions	partial pluralization of authority
Temple priesthood	Scribal bureaucracy	control of record, law, taxation	archives, administrative offices	documentary authority expands
Oral custodians	Literate officials	memory vs document legitimacy	courts, property records, governance	archival stabilization dominates
Guardians of doctrinal purity	Translators / scholars	legitimacy of foreign knowledge	libraries, courts, scholarly institutions	selective absorption and synthesis
Traditional editorial institutions	Platform algorithms / AI systems	control of visibility and interpretation	digital platforms, data infrastructures	algorithmic mediation of knowledge

The history of knowledge across civilizations cannot be understood solely through the internal development of individual cultures. One of the most decisive mechanisms through which knowledge has expanded is translation. Across Afro-Eurasia, networks of translators, scholars, scribes, and institutions created systems capable of receiving, interpreting, and transmitting texts across linguistic, cultural, and political boundaries. These networks transformed knowledge from a locally bounded inheritance into a circulatory phenomenon spanning multiple civilizations.

Translation, however, should not be mistaken for simple textual transfer. When ideas move between languages, institutions, and intellectual traditions, they are inevitably transformed. Concepts are reinterpreted, categories are adjusted, and meanings are reframed in relation to new intellectual contexts. The process therefore involves interpretation, commentary, and institutional mediation. Translation networks operate as infrastructures that enable civilizations to absorb and reorganize knowledge produced elsewhere.

One of the most influential examples of such networks emerged in the Islamic world during

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the Abbasid period. The Arabic-Persian translation movement that flourished between the eighth and tenth centuries created an intellectual infrastructure capable of receiving Greek, Persian, and Indian knowledge traditions and integrating them into a new scholarly ecosystem. Institutions such as the Bayt al-Hikma in Baghdad served as centers where scholars translated works of philosophy, medicine, mathematics, astronomy, and natural science into Arabic.

These translation projects were not purely mechanical activities. Translators often collaborated with scholars who possessed expertise in multiple intellectual traditions. Greek philosophical concepts were rendered into Arabic terminology that reshaped their interpretive context. Medical and astronomical texts were reorganized through commentary and synthesis. The resulting body of knowledge was not simply Greek thought preserved in another language; it was a transformed intellectual corpus embedded within Islamic scholarly culture.

Arabic became a lingua franca of scholarship across large portions of Afro-Eurasia, enabling communication among scholars from diverse regions including Persia, Central Asia, North Africa, and the Iberian Peninsula. Through networks of libraries, scholars, and educational institutions such as madrasas, knowledge circulated widely across these territories. Philosophers such as al-Farabi, Ibn Sina (Avicenna), and Ibn Rushd (Averroes) developed sophisticated philosophical traditions that both preserved and reinterpreted earlier Greek works. Their writings would later influence European intellectual developments through further translation into Latin.

These translation networks demonstrate that civilizational knowledge does not grow primarily through cultural isolation. Instead, it expands through institutions capable of hosting intellectual exchange. Libraries, translation houses, scholarly courts, and educational systems function as relay points where texts and ideas are received, interpreted, and redistributed. Without such infrastructures, knowledge remains confined within linguistic and institutional boundaries.

Parallel developments occurred in South Asia through what scholars have described as a Sanskrit cosmopolis. Across large regions of South and Southeast Asia, Sanskrit functioned as a shared scholarly language through which intellectual traditions could circulate across political and cultural boundaries. Texts on philosophy, grammar, law, literature, and ritual were transmitted through networks of scholars who maintained traditions of commentary and reinterpretation.

Commentary traditions were particularly important in this context. Rather than treating canonical texts as static authorities, scholars engaged in elaborate processes of interpretation and debate. Commentaries often restructured the meaning of earlier works, adapting them to new intellectual environments. Through these processes, texts became sites of ongoing intellectual activity rather than fixed repositories of knowledge.

The Chinese intellectual world developed its own distinctive system for stabilizing and

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transmitting knowledge across generations. Scholar-official institutions, particularly during imperial periods, created structured systems of education, examination, and textual interpretation. Classical texts formed the foundation of scholarly training, and mastery of these works became a pathway into administrative service.

Yet this system did not simply preserve inherited writings. Scholars engaged in continuous interpretation and commentary, producing extensive textual traditions that reexamined earlier teachings in light of contemporary concerns. Examination systems required candidates to demonstrate not only memorization but also interpretive competence within established intellectual frameworks. As a result, the Chinese scholar-official order functioned as a large-scale relay system capable of maintaining continuity while also allowing reinterpretation.

Libraries and archives were crucial components of these translation and commentary networks. Collections of manuscripts allowed scholars to access works from diverse intellectual traditions. Copying, cataloguing, and preservation ensured that texts could circulate across regions and generations. The presence of such institutions significantly increased the capacity of civilizations to accumulate and transmit knowledge.

Medieval and early university formations in Europe also participated in these broader translation networks. Many classical texts reached European scholars through translations produced in Islamic intellectual centers. Works of Aristotle, Galen, and other Greek authors were translated into Arabic and later into Latin, often accompanied by commentaries written by Islamic philosophers and scholars.

Universities such as Bologna, Paris, and Oxford developed institutional frameworks through which these texts could be studied, debated, and integrated into curricula. Scholastic traditions emphasized rigorous commentary and disputation as methods for examining philosophical and theological questions. As in other intellectual traditions, interpretation and debate played central roles in transforming inherited knowledge into new intellectual systems.

Across these diverse contexts, translation networks depended on institutional infrastructures capable of sustaining intellectual exchange. Translators required access to manuscripts, linguistic expertise, and patronage from courts or scholarly institutions. Commentators needed communities of readers capable of evaluating and transmitting their interpretations. Libraries provided spaces where texts from different traditions could coexist and interact.

At the same time, these networks often generated tensions within the societies that hosted them. Translation introduces foreign concepts and interpretive frameworks that may challenge established intellectual traditions. As a result, translators frequently encountered opposition from scholars or religious authorities concerned with preserving doctrinal purity.

Such conflicts were not merely linguistic disputes. They reflected deeper anxieties about cultural identity, intellectual authority, and the boundaries of legitimate knowledge.

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Translators who introduced foreign ideas could be accused of corrupting tradition or undermining established beliefs. In response, guardians of doctrinal purity sometimes attempted to restrict translation activities or regulate the circulation of controversial texts.

Yet despite these tensions, translation networks continued to expand because they provided significant intellectual advantages. Civilizations that developed institutions capable of hosting translation were able to absorb knowledge from multiple sources. Mathematical techniques, astronomical observations, medical theories, and philosophical frameworks could be integrated into new intellectual systems. This capacity for absorption and synthesis proved essential for the growth of complex knowledge traditions.

The circulation of texts across languages also created opportunities for reinterpretation. When a work is translated into a new linguistic context, translators must make decisions about how to render unfamiliar concepts. These decisions inevitably reshape the meaning of the original text. Terms are adapted to existing vocabularies, metaphors are reinterpreted, and conceptual frameworks are reorganized.

As a result, translation functions as a creative act rather than a purely preservative one. Each new translation generates an interpretation of the original work that reflects the intellectual environment of the receiving culture. Over time, multiple layers of translation and commentary can produce complex textual traditions in which original meanings are transformed through successive reinterpretations.

This dynamic is evident in the long transmission histories of many influential texts. Philosophical works translated from Greek into Arabic and then into Latin underwent multiple interpretive transformations along the way. Similarly, Buddhist texts translated from Sanskrit into Chinese were reinterpreted within the frameworks of Chinese philosophical traditions. These transformations demonstrate that translation networks do not simply preserve knowledge; they actively reshape it.

Translation networks also played a crucial role in connecting intellectual traditions across political boundaries. Scholars traveling between courts, monasteries, and universities carried manuscripts and ideas with them. Diplomatic exchanges and trade routes facilitated the movement of texts across regions. The resulting networks formed a vast intellectual geography linking societies across Afro-Eurasia.

The significance of these networks lies not only in the volume of knowledge they transmitted but also in the institutional structures that sustained them. Civilizations capable of maintaining libraries, scholarly communities, and translation institutions possessed greater capacity to accumulate and reorganize knowledge over time. These infrastructures enabled the continuous reinterpretation of inherited traditions in light of new information and perspectives.

From the perspective of epistemic regimes, translation networks represent a major expansion of interpretive authority. Knowledge was no longer confined to the boundaries

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of a single language or cultural tradition. Instead, it could circulate across civilizations through institutions designed to host intellectual exchange. Translators, commentators, and scholars emerged as new interpreters responsible for mediating between different intellectual worlds.

This mediation was inherently transformative. Translation required choices about how to represent unfamiliar concepts, and commentary provided opportunities to reinterpret inherited ideas. Through these processes, texts acquired new meanings and functions within receiving cultures. Knowledge therefore expanded not through simple accumulation but through ongoing reinterpretation across civilizational boundaries.

The conflicts that accompanied these transformations reveal the deeper stakes of translation. Guardians of doctrinal purity often feared that the introduction of foreign ideas would undermine established traditions. Translators, by contrast, emphasized the intellectual benefits of engaging with diverse sources of knowledge. These debates illustrate that the circulation of ideas is never purely technical; it is also a contest over the boundaries of legitimate interpretation.

Ultimately, the historical record demonstrates that civilizations capable of sustaining translation networks were better equipped to generate intellectual innovation. By hosting institutions that allowed ideas to move across linguistic and cultural boundaries, these societies created environments in which knowledge could be continually reinterpreted and expanded.

Translation networks across Afro-Eurasia therefore represent one of the decisive infrastructures in the history of knowledge. They reveal that intellectual growth depends not primarily on cultural purity or isolation but on the capacity of societies to host circulation, commentary, and reinterpretation. The institutions that sustain translation—libraries, schools, scholarly communities, and patronage systems—enable civilizations to integrate diverse intellectual traditions into evolving knowledge systems.

The crucial insight that emerges from this history is that translation always changes what it carries. When ideas cross linguistic and institutional boundaries, they are reinterpreted within new conceptual frameworks. Each act of translation therefore becomes an act of transformation. Rather than preserving knowledge unchanged, translation networks reshape it in ways that allow new intellectual possibilities to emerge. Through these processes, the circulation of texts becomes a powerful engine of civilizational development.

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9. Science and the Proceduralization of Truth

The emergence of modern science represents one of the most consequential reorganizations of interpretive authority in human history. Yet the significance of this transformation is often misunderstood. Science is frequently described as the moment when humanity finally gained direct access to objective reality, leaving behind earlier regimes of myth, ritual, and authority. Such narratives obscure a more subtle and historically accurate development. Science did not eliminate interpretation. Rather, it introduced systematic procedures designed to discipline interpretation through controlled observation, measurement, experiment, and collective verification.

In earlier epistemic regimes, interpretive authority often rested on inherited status, sacred office, or institutional continuity. Priests interpreted cosmic order through ritual traditions, scribes stabilized interpretation through archives, and philosophers asserted authority through reasoning and ethical reflection. Scientific practice introduced a different basis for legitimacy. Interpretations about the natural world were increasingly judged not by the prestige of the speaker but by the procedures through which claims were produced and tested. Authority began to migrate from individuals to methods.

This transformation can be understood as the proceduralization of truth. Instead of relying on the credibility of a particular interpreter, scientific communities developed standardized techniques through which interpretations could be evaluated independently of the individual who proposed them. Experiments could be repeated, measurements could be compared, and observations could be scrutinized by multiple investigators. Through these practices, interpretation became embedded within structured processes designed to minimize individual bias and error.

Experimentation played a central role in this development. Whereas earlier traditions often relied on observation of naturally occurring events, experimental practice allowed investigators to create controlled conditions in which specific variables could be isolated and examined. By manipulating the environment of observation, experimenters could generate reproducible phenomena that could be studied repeatedly. The experiment therefore functioned as a stabilization machine: it created a repeatable configuration through which claims about nature could be evaluated.

Instrumentation further expanded the capacity of scientific interpretation. Telescopes, microscopes, thermometers, barometers, and other instruments extended human perception beyond its ordinary limits. These devices translated physical processes into measurable signals that could be recorded and analyzed. Instruments did not eliminate interpretation; they reorganized it. Observers had to learn how to calibrate instruments, interpret readings, and translate measurements into meaningful theoretical claims. The authority of scientific observation thus depended on both technical skill and standardized procedures governing the use of instruments.

The role of inscription was equally important. Observations and measurements were

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recorded in notebooks, diagrams, tables, and published papers. These inscriptions allowed experimental results to circulate within broader scholarly communities. Once recorded, an observation could be examined, critiqued, or replicated by others. Written documentation therefore functioned as another stabilization mechanism. It enabled interpretations to persist beyond the moment of observation and become part of an accumulating body of scientific knowledge.

Reproducibility became one of the defining credibility criteria of the scientific regime. A claim about nature gained legitimacy when independent investigators could reproduce the experimental conditions and obtain similar results. Reproducibility therefore shifted the evaluation of truth from individual testimony to collective verification. If a result could not be reproduced, its credibility diminished regardless of the status of the scientist who reported it. The procedural structure of scientific practice thus created mechanisms through which interpretations could be corrected over time.

Witnessing also played a crucial role in this epistemic transformation. Early experimental demonstrations often took place before gatherings of scholars who served as witnesses to the procedures and results. The presence of credible observers provided assurance that experimental claims were not fabricated or misinterpreted. Over time, the function of witnessing became institutionalized through practices such as peer review and publication. Scientific journals allowed investigators to present their findings to communities capable of evaluating the reliability of their procedures and conclusions.

These developments collectively produced a distinctive form of epistemic authority. In the scientific regime, legitimacy derives from adherence to recognized procedures rather than from personal prestige or inherited status. A scientist's claim gains credibility when it is supported by transparent methods, reproducible experiments, and measurable evidence that can be evaluated by others. This procedural authority distinguishes scientific practice from earlier regimes in which authority was more closely tied to institutional position or tradition.

However, it would be misleading to interpret this shift as the elimination of mediation between observers and reality. Scientific knowledge remains deeply mediated by instruments, models, measurement systems, and theoretical frameworks. Experiments must be designed, instruments must be calibrated, and data must be interpreted. Each of these steps involves decisions and assumptions that shape how phenomena are understood. The procedural discipline of science does not remove interpretation; it structures and constrains it.

Scientific communities therefore function as systems of organized correction. Through processes of replication, critique, and revision, interpretations are continually evaluated and refined. Errors can be detected when experiments fail to produce expected results or when new evidence contradicts established theories. Over time, this process allows scientific knowledge to evolve through cumulative adjustment rather than through simple accumulation of unchallenged claims.

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This dynamic is visible in the historical development of major scientific theories. Early interpretations of celestial motion, for example, underwent repeated revision as new observations and mathematical techniques emerged. The transition from geocentric to heliocentric models of the cosmos involved extensive debate, experimentation, and reinterpretation of astronomical data. Similarly, later developments in physics, chemistry, and biology demonstrate how scientific theories can be modified or replaced when new evidence emerges.

These transformations illustrate that the authority of science rests not on the permanence of its conclusions but on the reliability of its procedures for detecting and correcting error. Scientific claims remain open to revision precisely because they are evaluated through methods designed to expose weaknesses in interpretation. This openness to correction distinguishes the scientific regime from many earlier epistemic orders in which interpretive authority was more resistant to challenge.

The proceduralization of truth also generated new institutional forms. Laboratories became central sites for experimental investigation. Scientific societies provided spaces for discussion and dissemination of research. Universities incorporated scientific training into educational systems, producing new generations of investigators skilled in experimental methods and mathematical analysis. Journals and conferences allowed findings to circulate widely, enabling communities of researchers to participate in collective evaluation.

These institutions stabilized the procedural norms that define scientific practice. By training investigators in standardized techniques and encouraging adherence to methodological principles, scientific institutions ensured that experimental and observational claims could be evaluated across different locations and contexts. The resulting networks of laboratories, universities, and scholarly societies formed an infrastructure capable of sustaining large-scale collaborative inquiry.

Nevertheless, the emergence of scientific authority also generated conflicts with earlier interpretive regimes. Sacred institutions that grounded legitimacy in religious tradition sometimes resisted scientific interpretations that appeared to contradict established cosmologies. Philosophical systems based on metaphysical speculation faced challenges from empirical investigations that prioritized observation and experiment. These conflicts were not merely intellectual disagreements; they reflected deeper tensions between different forms of epistemic authority.

One of the most notable examples involves disputes over astronomical interpretation in early modern Europe. Observational evidence gathered through telescopic instruments supported models of the cosmos that conflicted with traditional interpretations grounded in Aristotelian philosophy and theological doctrine. The resulting controversies illustrate how the procedural authority of scientific methods could challenge long-standing intellectual and religious traditions.

Yet even in these conflicts, the scientific regime did not operate entirely independently of

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earlier epistemic forms. Many early scientists were themselves deeply embedded within religious and philosophical traditions. Their work often sought to reconcile empirical investigation with broader metaphysical or theological frameworks. The procedural discipline of science therefore emerged gradually through complex interactions among diverse intellectual traditions.

Over time, however, the credibility of scientific procedures increasingly overshadowed alternative bases of authority in domains concerned with natural phenomena. When disputes arose about the behavior of physical systems, experimental evidence and reproducible measurement became the primary criteria for evaluating competing claims. The prestige of a speaker or the sanctity of inherited doctrine carried less weight than the strength of the methods used to generate evidence.

This shift did not mean that science achieved perfect neutrality or infallibility. Scientific communities remain shaped by social institutions, funding structures, intellectual paradigms, and technological limitations. Interpretations can still be influenced by prevailing theoretical assumptions or institutional priorities. Nevertheless, the procedural framework of science provides mechanisms through which such influences can be examined and corrected over time.

Seen within the broader history of knowledge, science represents a distinctive configuration of interpretive authority. Signals from the natural world are translated through instruments and experimental setups. These translations are stabilized through measurement, documentation, and publication. Scientific communities evaluate interpretations through procedures designed to enable replication and critique. Through these mechanisms, knowledge about the natural world becomes subject to organized processes of correction.

The key insight is therefore not that science eliminates mediation between humans and reality, but that it disciplines mediation through structured procedures. Interpretation becomes accountable to publicly recognized methods rather than to the authority of particular individuals or institutions. In this sense, science can be understood as a system for managing the inevitability of interpretation.

The proceduralization of truth thus marks a crucial stage in the evolution of epistemic regimes. By embedding interpretation within experimental methods, measurement systems, and collective verification processes, scientific institutions created new mechanisms for stabilizing knowledge about the natural world. These mechanisms allowed societies to generate increasingly reliable explanations of physical phenomena while maintaining the capacity to revise those explanations in light of new evidence.

Science therefore does not represent the end of interpretation but its disciplined reorganization. It replaces authority grounded in tradition or personal prestige with authority grounded in method. Through experiment, instrumentation, reproducibility, and communal correction, scientific practice transforms interpretation into a structured and self-correcting process. In doing so, it reshapes the conditions under which claims about

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reality can become credible within modern societies.

10. University, Discipline, and Expert Authority

The proceduralization of truth within scientific practice did not remain confined to laboratories and experimental communities. Over time, the authority of disciplined interpretation became embedded within larger institutional structures that organized, credentialed, and reproduced expert knowledge. Among these institutions, the modern university emerged as one of the most influential mechanisms for stabilizing interpretive authority in modern societies.

Universities did not merely transmit knowledge. They organized the conditions under which knowledge could be produced, evaluated, and recognized as legitimate. In doing so, they became central sites where the right to interpret reality was formalized through systems of training, credentialing, and disciplinary specialization. The authority of interpretation increasingly depended not only on adherence to procedural methods but also on membership within institutional communities authorized to apply those methods.

This transformation can be described as the industrialization of expertise. As scientific and technical knowledge expanded in scale and complexity, it became increasingly difficult for any single individual to master multiple domains of inquiry. Knowledge production therefore became organized through specialized disciplines. Each discipline developed its own theoretical frameworks, methodological standards, and criteria for evaluating evidence. Fields such as physics, chemistry, medicine, economics, and sociology gradually consolidated their own professional identities and institutional boundaries.

Disciplinary specialization served several functions. First, it allowed investigators to concentrate on increasingly complex problems that required deep technical knowledge. Second, it created communities of practitioners capable of evaluating one another's work. Third, it established recognizable standards through which claims could be judged credible or flawed. These standards were codified through training programs, research methods, and publication practices.

The organization of disciplines was closely linked to the expansion of professional education. Universities developed curricula designed to train students in the theoretical and methodological foundations of particular fields. Through examinations, degrees, and certifications, institutions signaled that individuals had acquired the necessary competencies to participate in professional interpretation within that domain. Academic credentials thus became markers of epistemic legitimacy.

The degree system served as a gatekeeping mechanism. It distinguished those authorized to participate in specialized interpretation from those who had not undergone recognized training. While knowledge itself was not confined to universities, the institutional structure of modern expertise increasingly relied on formal credentials as indicators of competence.

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Doctors, engineers, lawyers, economists, and scientists were expected to demonstrate their expertise through educational pathways validated by recognized institutions.

Professional associations further reinforced these structures. Organizations representing particular fields established ethical standards, licensing requirements, and codes of conduct governing professional practice. These associations often controlled entry into professions by determining who could legally practice certain forms of expertise. In medicine, for example, professional licensing restricted the authority to diagnose and treat illness to individuals who had completed accredited training and met regulatory requirements.

Journals and peer review systems played another crucial role in institutionalizing expert authority. Scientific and scholarly journals provided platforms through which researchers could present their findings to disciplinary communities. Before publication, manuscripts were evaluated by other experts who assessed the quality of the methods, evidence, and reasoning presented. Peer review thus functioned as a mechanism of communal validation.

Through this process, interpretive claims were filtered before entering the recognized body of scholarly knowledge. Publication signaled that a claim had survived scrutiny by qualified evaluators within the relevant discipline. While peer review was not immune to bias or institutional pressures, it created a structured system through which interpretations could be evaluated collectively rather than solely on the authority of individual authors.

These institutional arrangements transformed the nature of epistemic authority. In earlier regimes, interpreters derived legitimacy primarily from personal experience, sacred status, or rhetorical persuasion. In the modern expert regime, legitimacy became tied to membership in professional communities governed by recognized procedures and institutional norms. Expertise became a socially organized property rather than merely an individual attribute.

However, the institutionalization of expertise also introduced new tensions. As disciplines became increasingly specialized, the gap between expert knowledge and public understanding widened. Many areas of modern research involve mathematical techniques, technical instruments, and theoretical frameworks that are difficult for non-specialists to evaluate directly. As a result, publics often encounter scientific claims through simplified summaries, media interpretations, or institutional statements.

This epistemic asymmetry can generate skepticism. When individuals cannot easily evaluate the evidence underlying expert claims, trust in institutions becomes a central factor in determining whether those claims are accepted. If institutional credibility declines, the authority of expert interpretation may also come under pressure. Debates over climate science, public health, and economic policy illustrate how tensions between expert communities and public audiences can emerge when institutional trust becomes contested.

Another source of tension lies in what sociologists of science describe as boundary-work. Expert communities often engage in practices that distinguish legitimate knowledge from

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what they consider pseudoscience, speculation, or untrained opinion. These boundaries are maintained through publication standards, professional credentials, and disciplinary norms. By defining what counts as valid knowledge, expert communities protect the integrity of their methods and interpretations.

Yet boundary-work can also be perceived as exclusionary. Critics sometimes argue that disciplinary institutions suppress alternative perspectives or restrict participation in debates about issues that affect broader publics. These critiques highlight the dual role of institutional gatekeeping. On one hand, gatekeeping preserves methodological standards that protect knowledge from error and misinformation. On the other hand, it can reinforce hierarchies that limit who is recognized as a legitimate interpreter.

Universities therefore occupy a paradoxical position within modern epistemic regimes. They function simultaneously as validators and bottlenecks. As validators, they provide structures through which expertise can be trained, evaluated, and certified. As bottlenecks, they regulate access to interpretive authority by controlling the credentials and institutional affiliations that signal epistemic legitimacy.

This dual role becomes particularly visible in interdisciplinary controversies. When complex problems span multiple domains—such as climate change, biotechnology, or artificial intelligence—different disciplines may offer competing interpretations grounded in distinct methodological traditions. Negotiating these differences requires institutions capable of coordinating expertise across disciplinary boundaries. Universities often serve as arenas in which such negotiations occur.

The growth of modern research universities also reshaped the scale of knowledge production. Large research institutions coordinate laboratories, funding networks, graduate training programs, and international collaborations. Through these infrastructures, universities contribute to the expansion of global knowledge systems capable of addressing increasingly complex scientific and technological challenges.

At the same time, the institutional concentration of expertise raises questions about democratic legitimacy. When critical decisions depend on specialized knowledge, societies must determine how expert advice should interact with political deliberation and public values. Scientific expertise can inform policy decisions, but it cannot determine those decisions independently of broader ethical and social considerations.

This tension reflects a deeper structural issue: the separation between epistemic authority and political authority. Experts possess specialized knowledge about particular domains of reality, but they do not necessarily possess the democratic mandate to determine how that knowledge should be applied within society. Balancing these forms of authority remains an ongoing challenge in modern governance.

Despite these tensions, the institutionalization of expertise has significantly expanded humanity's capacity to interpret complex phenomena. Modern medicine relies on networks of hospitals, laboratories, and research institutions capable of diagnosing and treating

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diseases with remarkable precision. Engineering disciplines design infrastructure that supports global transportation, communication, and energy systems. Environmental sciences monitor planetary processes that influence climate and ecological stability.

These achievements demonstrate the power of organized expertise. By coordinating specialized knowledge across institutional networks, modern societies have developed interpretive capacities far beyond those available in earlier historical periods. Yet the legitimacy of these capacities depends on the continued credibility of the institutions that sustain them.

Seen within the broader framework of epistemic regimes, universities and professional disciplines represent a crucial stage in the evolution of interpretive authority. They extend the procedural norms of scientific practice into durable institutional systems that train experts, certify competence, and regulate the boundaries of legitimate interpretation. Through these mechanisms, expertise becomes embedded within the organizational structure of modern societies.

However, the authority of experts remains contingent. It must be continually justified through reliable methods, transparent communication, and responsiveness to legitimate public concerns. When expert institutions fail to maintain these standards, the credibility of their interpretations may erode. In such moments, new actors—political movements, media organizations, or technological platforms—may attempt to claim interpretive authority in their place.

The university-based expert regime therefore represents neither the final stage nor a stable endpoint in the history of knowledge. It is one configuration within a longer civilizational process in which societies continually reorganize who is authorized to interpret reality. The authority of experts, like the authority of priests, philosophers, or scientists before them, remains embedded within evolving institutional landscapes shaped by both cooperation and conflict.

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11. Media and the Narration of the Present

Table 3: Modern Knowledge Regimes Compared

Theme	Science	University	Media	Platform	AI / Model Systems
What counts as signal	experimental observation	disciplinary research	events and crises	user activity data	large training datasets
Who interprets	scientific communities	disciplinary experts	journalists and editors	ranking algorithms	statistical models
Who stabilizes	replication and peer review	journals and credentials	editorial institutions	platform infrastructures	model architectures
How trust is earned	reproducibility	credentialed expertise	narrative credibility	engagement metrics	predictive performance
Failure mode	non-reproducible results	disciplinary closure	sensational framing	algorithmic bias	hallucination or opacity

Media institutions occupy a distinctive position within modern epistemic regimes. While scientific institutions organize the long and methodical production of verified knowledge, media institutions shape how societies experience the immediate present. For most individuals, the events that define political life, economic change, social conflict, and global crisis are not encountered through direct observation but through mediated representation. Newspapers, broadcast systems, and digital news platforms act as interpreters of the present moment, deciding which signals from the world become visible and which remain unnoticed.

The power of media lies not primarily in the creation of facts but in the organization of attention. Every day, innumerable events occur across the world: political negotiations, technological developments, scientific discoveries, economic shifts, local conflicts, and environmental changes. Only a tiny fraction of these events becomes visible to the broader public. Media institutions therefore function as large-scale filters that convert the chaotic multiplicity of events into a structured narrative of what is happening now.

This filtering process involves several interrelated operations. The first is selection. Editors, journalists, and producers must decide which signals deserve to be reported and which can be ignored. Selection transforms the world's continuous stream of occurrences into a finite set of recognizable stories. What appears in the news is not simply what happened but what has been judged newsworthy.

The second operation is salience. Even within selected events, media institutions determine which developments receive the greatest emphasis. Front-page headlines, breaking news alerts, repeated coverage, and visual prominence all signal to audiences that certain events are more important than others. Through these techniques, media systems guide public attention toward specific topics while allowing others to recede into relative obscurity.

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The third operation is framing. Events rarely appear in media narratives as raw facts. They are presented within interpretive frameworks that suggest how audiences should understand them. A protest may be framed as a democratic demand for justice or as a threat to public order. An economic downturn may be interpreted as the result of structural policy failures or temporary market fluctuations. Framing therefore provides a preliminary interpretation that shapes how audiences process the information they receive.

Finally, media institutions rely heavily on repetition. Stories that are repeated across multiple news cycles acquire greater perceived significance. Repetition stabilizes interpretation by embedding particular narratives into public consciousness. Through continual reference and discussion, certain interpretations become widely accepted descriptions of reality.

Taken together, these operations demonstrate that media does not simply report the world. Media organizes the social present. It converts dispersed signals into a shared narrative environment through which societies experience current events. Citizens who live in different cities or even different continents can nevertheless inhabit the same mediated moment because they encounter similar stories circulating through global information networks.

The headline illustrates this epistemic function particularly well. A headline condenses a complex event into a brief statement that signals urgency and importance. It functions as an epistemic device that directs attention and implies interpretation simultaneously. Readers often encounter headlines before engaging with the detailed content of an article, and in many cases the headline alone shapes their perception of the event. The headline therefore performs an interpretive act: it defines what the event is about before the reader examines the evidence supporting that claim.

In this sense, media narratives operate within the broader framework of interpretive authority that structures the history of knowledge. Journalists and editors do not simply gather information. They translate signals from the world into coherent stories capable of circulating within a mass public. Their authority derives from institutional practices that claim to ensure reliability, such as editorial oversight, fact-checking procedures, and professional norms governing journalistic conduct.

Yet the epistemic logic of media differs from that of scientific institutions. Scientific knowledge typically develops through slow processes of experimentation, verification, and peer review. Media narratives, by contrast, must operate under conditions of temporal immediacy. News organizations are expected to report developments quickly, often before all relevant information has been fully verified. The need for rapid reporting creates tension between the demands of speed and the requirements of careful validation.

This tension becomes particularly visible when scientific findings enter public debate through media coverage. Scientific research often evolves through incremental discoveries and cautious interpretation. Media narratives, however, frequently compress these develop-

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ments into simplified conclusions that can be communicated quickly to large audiences. As a result, the public presentation of scientific issues may emphasize dramatic breakthroughs or controversies while underrepresenting the gradual and self-correcting nature of scientific inquiry.

The conflict between media salience and scientific verification reflects a deeper structural divergence between two epistemic regimes. Scientific institutions aim to produce reliable knowledge through extended processes of communal scrutiny. Media institutions aim to provide immediate narratives that allow societies to interpret ongoing events. Both functions are necessary, but their priorities are not identical. Where science privileges careful validation, media privileges timely interpretation.

Editorial judgment therefore becomes a central component of media authority. Editors decide which stories deserve prominence, how they should be framed, and how long they should remain within public attention. These decisions influence not only what audiences know but also what they perceive as urgent, controversial, or historically significant. Through these choices, media institutions shape the horizon of public awareness.

Another important distinction concerns the difference between events and spectacles. Not every event becomes a spectacle within media narratives. A spectacle emerges when an event is presented in ways that intensify emotional engagement, dramatic tension, or symbolic meaning. Spectacles can attract widespread attention, sometimes overshadowing events that may have greater long-term significance but lack immediate dramatic appeal.

This dynamic can distort public understanding of reality. When media attention focuses heavily on spectacular events, structural developments that unfold gradually—such as environmental change, technological transformation, or demographic shifts—may receive less coverage. As a result, the mediated present may appear dominated by dramatic crises even when slower processes exert greater influence on long-term social outcomes.

Despite these challenges, media institutions perform an indispensable role in modern societies. They create a shared informational environment that allows large populations to coordinate political action, respond to crises, and participate in public deliberation. Without systems capable of rapidly circulating information, democratic governance and collective decision-making would become far more difficult.

The rise of mass media during the nineteenth and twentieth centuries significantly expanded the scale at which such coordination could occur. Newspapers connected readers across cities and regions, broadcast media brought real-time events into domestic spaces, and global news networks created transnational audiences capable of witnessing events unfolding across continents. These developments transformed the structure of the public sphere.

Within this expanded public sphere, interpretive authority became increasingly contested. Governments, scientific institutions, activist movements, corporations, and cultural figures all sought to influence how events were presented and understood within media narra-

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tives. Media organizations themselves had to navigate these competing pressures while maintaining credibility with their audiences.

Credibility remains the foundation of media authority. If audiences believe that a news organization consistently misrepresents events or prioritizes sensationalism over accuracy, trust in its reporting may decline. Media institutions therefore invest significant effort in maintaining reputations for reliability, even as they compete for attention within highly dynamic information environments.

At the same time, technological developments have altered the landscape of media interpretation. Digital communication networks allow information to circulate rapidly across social platforms, enabling individuals and organizations outside traditional media institutions to participate in shaping public narratives. This expansion of interpretive participation has increased the diversity of voices within the public sphere but has also intensified competition over which interpretations become dominant.

Seen within the broader historical trajectory of epistemic regimes, media institutions represent a stage in which the authority to interpret reality becomes intertwined with the organization of public attention. Whereas scientific communities govern the credibility of claims about natural phenomena, media institutions govern the visibility of events within the social present.

The significance of this role cannot be overstated. Societies rarely respond directly to raw events; they respond to the narratives through which those events become intelligible. By selecting, framing, and repeating particular interpretations of events, media systems construct the shared temporal horizon within which political decisions, social debates, and collective reactions take place.

Media therefore operates not merely as a channel for information but as a structural component of modern epistemic order. Through the narration of the present, it defines what counts as happening now and which developments deserve collective attention. In doing so, it shapes the conditions under which societies interpret their own historical moment.

This interpretive function places media institutions alongside other historical regimes of knowledge—priests interpreting cosmic order, philosophers questioning inherited beliefs, scientists testing hypotheses, and universities certifying expertise. Each regime organizes interpretation through distinct procedures and institutions. Media contributes its own form of authority: the power to structure the present through narrative visibility.

Understanding this role is essential for interpreting contemporary transformations in the knowledge landscape. As new technologies begin to reshape how information is filtered and distributed, the mechanisms that once governed media authority may themselves be subject to profound change. The next chapter examines how digital platforms and algorithmic systems have begun to reorganize these processes at planetary scale.

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12. Platforms, Algorithms, and AI as Hybrid Interpreters

Figure 3 Placeholder: Hybrid Mediation in the Present

raw events → data capture → ranking/recommendation → news/social dissemination → user attention → feedback loops
with institutional authority and public belief interacting throughout

Figure 3: Hybrid mediation under platform and AI conditions.

The emergence of digital platforms has introduced a new configuration of interpretive authority in the history of knowledge. Earlier regimes of interpretation—priestly institutions, philosophical schools, scientific communities, universities, and media organizations—operated primarily by interpreting signals that were already visible to human observers. Even when interpretation was contested, the underlying signals remained broadly accessible within the social environment. Individuals could witness events, consult archives, read texts, or observe natural phenomena before competing interpretations were offered.

Digital platforms alter this sequence. For the first time in history, large-scale computational systems participate directly in determining which signals become visible in the first place. Search engines rank information before it is encountered. Social media feeds prioritize certain posts over others. Recommendation systems guide users toward particular videos, articles, or discussions. Moderation systems remove or suppress selected forms of content. Through these processes, platforms influence not only how signals are interpreted but which signals enter the horizon of perception at all.

This transformation marks a structural shift in the organization of knowledge. Earlier institutions interpreted the world after signals had become publicly accessible. Platform systems increasingly perform what might be called pre-interpretive selection. They filter and rank signals before explicit human interpretation occurs. In doing so, they shape the informational environment within which individuals and societies construct their understanding of reality.

Ranking algorithms play a central role in this process. When users search for information online, they are typically presented with ordered lists of results. The position of each result strongly influences whether it will be seen, read, or ignored. Research in information behavior consistently shows that users tend to focus on the highest-ranked results, rarely exploring deeper pages of search outputs. Ranking therefore acts as a gatekeeping mechanism that determines which interpretations are likely to receive attention.

Recommendation systems extend this influence even further. Platforms such as video-sharing services, streaming applications, and social networks analyze patterns of user behavior to predict which content individuals are most likely to engage with. These predictions generate personalized streams of suggested material that shape the sequence of information encountered by each user. Rather than presenting a neutral or comprehensive

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view of available information, recommendation systems curate individualized informational environments.

The logic underlying these systems often prioritizes engagement metrics. Algorithms are designed to maximize indicators such as clicks, viewing time, sharing activity, or interaction rates. While such metrics are effective for sustaining user attention, they do not necessarily align with epistemic criteria such as accuracy, reliability, or long-term significance. As a result, the informational landscape generated by platform systems may amplify content that provokes strong reactions while marginalizing information that requires careful reflection.

Moderation systems constitute another layer of algorithmic interpretation. Platforms routinely monitor and regulate user-generated content in order to enforce community standards or legal requirements. Automated systems may detect and remove material deemed harmful, illegal, or misleading. Although such moderation is often necessary for maintaining functional digital spaces, it also introduces complex questions about who determines the boundaries of permissible expression and how those boundaries are enforced.

These processes together create a form of hybrid mediation. Human actors remain involved in many aspects of platform governance. Engineers design algorithms, policy teams establish moderation guidelines, and human reviewers evaluate difficult cases. At the same time, automated systems perform large-scale filtering and ranking operations that exceed the capacity of direct human oversight. Interpretation therefore emerges from the interaction between human institutions and computational infrastructures.

The resulting systems can be understood as hybrid interpreters. They combine human judgment, institutional policy, and algorithmic computation to shape the informational environment experienced by billions of users. In this sense, platforms do not simply distribute content created elsewhere. They actively participate in constructing the visible world of contemporary societies.

This development has significant implications for earlier regimes of interpretive authority. Journalists, for example, traditionally exercised considerable influence over the selection and framing of news stories. Editorial decisions determined which events appeared on front pages or broadcast headlines. With the rise of digital platforms, however, the distribution of news increasingly depends on algorithmic feeds that prioritize content based on user engagement patterns rather than editorial judgment alone.

As a result, traditional media institutions often compete with platform logic for control over narrative visibility. A carefully investigated news report may receive less attention within algorithmically curated feeds than a more sensational or emotionally provocative piece of content. This dynamic can shift the balance of interpretive authority away from editorial institutions toward algorithmic ranking systems.

Similar tensions arise between scientific institutions and digital platforms. Scientific

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communities rely on peer review, methodological rigor, and cautious interpretation of evidence. Platform environments, by contrast, reward rapid dissemination and high engagement. Scientific findings may circulate widely through social media, but they often appear alongside speculative interpretations, partial summaries, or misleading claims that spread more rapidly through algorithmic amplification.

These conflicts illustrate a deeper structural divergence between procedural knowledge regimes and engagement-driven information systems. Scientific and scholarly institutions prioritize reliability through slow processes of validation. Platform systems prioritize visibility through rapid cycles of content circulation. The criteria governing epistemic credibility and algorithmic prominence are therefore not always aligned.

Artificial intelligence systems intensify these dynamics. Machine learning models trained on large datasets can generate text, images, and analyses that resemble human-produced interpretations. These models do not merely rank or filter existing signals; they can also produce new representations of information derived from patterns within training data. In doing so, they participate directly in the translation of signals into meaningful outputs.

However, the epistemic status of such outputs remains complex. AI models do not possess independent access to the world. Their interpretations are generated through statistical relationships learned from large collections of data. When these models produce responses to user queries, they are effectively synthesizing patterns extracted from previous human communication. The resulting outputs may be informative, misleading, or ambiguous depending on the structure and quality of the underlying data.

Nevertheless, AI systems increasingly participate in shaping how individuals encounter information. Search engines integrate machine learning models that summarize results or generate responses to queries. Recommendation systems incorporate predictive models that anticipate user preferences. Conversational interfaces allow users to interact with algorithmic systems as if they were interpreters capable of answering questions about the world.

These developments create a layered structure of interpretation. Signals originating in physical events or human communication pass through multiple stages of computational mediation before reaching users. Algorithms rank and filter information, recommendation systems personalize content streams, and AI models generate synthesized responses. Human interpretation then occurs within the informational environment produced by these hybrid systems.

The crucial question raised by this transformation is therefore not simply whether machines can know. A more fundamental issue concerns the conditions under which signals become available for interpretation in the first place. If algorithmic systems determine which information is visible, then the structure of public knowledge may increasingly depend on infrastructures that operate prior to explicit human judgment.

This shift represents a significant departure from earlier epistemic regimes. In priestly,

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philosophical, scientific, and journalistic systems, interpreters operated within environments where signals were broadly accessible to human observers. Although authority structures influenced interpretation, individuals could still encounter many signals directly through observation, texts, or shared events. Platform systems, by contrast, can dynamically reorder the informational environment for each user.

Such personalization fragments the collective experience of the present. Two individuals using the same platform may encounter entirely different streams of information depending on their previous behavior, network connections, and algorithmic predictions. The shared narrative environment that once characterized mass media becomes replaced by individualized informational pathways.

At the same time, platform infrastructures operate at planetary scale. A small number of global platforms mediate communication among billions of users. Decisions made within these systems—regarding ranking algorithms, moderation policies, and recommendation strategies—can influence how information circulates across entire societies. Interpretive authority therefore becomes concentrated within technical infrastructures whose internal operations may not be fully transparent to the public.

This concentration raises profound questions about accountability and governance. If hybrid systems increasingly determine which signals societies encounter, then the design and regulation of these systems becomes a central concern for democratic institutions. Debates about algorithmic transparency, platform responsibility, and AI governance reflect growing recognition that digital infrastructures now shape the epistemic environment within which public reasoning occurs.

From a civilizational perspective, the emergence of platform and AI systems represents the latest stage in the long history of interpretive authority. Just as priesthoods coordinated cosmic meaning, philosophers questioned inherited explanations, scientists disciplined interpretation through experiment, and media institutions narrated the present, digital platforms now structure the visibility of signals within global information networks.

Yet this stage differs in one crucial respect. Earlier interpreters primarily translated signals that humans had already encountered. Hybrid systems increasingly determine which signals are encountered at all. Interpretation therefore begins before interpretation becomes visible.

Understanding this transformation is essential for grasping the epistemic challenges of the present. The question facing contemporary societies is not merely whether artificial intelligence can generate knowledge. The deeper question concerns whether the infrastructures through which information circulates are gradually becoming the primary architects of the visible world.

If that is the case, then interpretive authority is no longer confined to identifiable institutions such as temples, universities, laboratories, or newsrooms. It is increasingly embedded within algorithmic systems that operate continuously within the background of

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everyday communication.

The culmination of this historical trajectory therefore reveals a new form of epistemic order. Human interpretation has not disappeared, but it now occurs within environments structured by hybrid systems that preselect, rank, and synthesize the signals through which reality becomes intelligible. The future of knowledge will depend not only on the reliability of human interpreters but also on the design, governance, and accountability of the computational infrastructures that shape what societies can see.

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13. Comparative Synthesis, Crisis Forms, and Conclusion

Table 4: Failure Modes and Crisis Forms of Interpretive Authority

Regime	**Strength**	**Failure Mode**	**Social Symptom**	**Repair Path**
Embodied/oral	environmental attunement	local bias or memory drift	fragmented interpretations	communal testing and ritual memory
Sacred/ritual	large-scale meaning coordination	doctrinal rigidity	conflict with new evidence	prophetic critique or reform
Archival/state	durable record and continuity	bureaucratic rigidity	loss of legitimacy or corruption	administrative reform and transparency
Scientific/procedural	reproducible verification	methodological narrowness	replication crisis or paradigm lock	open data and methodological revision
Institutional/expert	organized specialization	disciplinary closure	declining public trust	transparency and interdisciplinary dialogue
Media/public sphere	rapid social awareness	sensational framing	attention distortion	editorial standards and media literacy
Platform/AI	large-scale signal filtering	algorithmic opacity and bias	informational fragmentation	algorithmic governance and oversight

The preceding chapters have traced a long civilizational arc through which societies organize the interpretation of reality. Although the institutions, technologies, and vocabularies of interpretation have changed dramatically across historical periods, a recurring structure becomes visible when these transformations are examined comparatively. Across civilizations and epochs, knowledge regimes perform four fundamental operations. They organize signals, authorize interpreters, stabilize representations, and eventually undergo conflict-driven transformation when existing interpretive authorities lose credibility or encounter competing systems of explanation.

This structural pattern appears already in the earliest human communities. In pre-literate societies, knowledge emerged through embodied interaction with the environment. Hunters interpreted tracks, healers interpreted symptoms, and elders interpreted memory and ritual precedent. Signals from the world were encountered directly through sensory engagement with landscapes, animals, weather, and human bodies. Interpretive authority derived primarily from demonstrated reliability. Those whose interpretations repeatedly enabled survival gained trust within the community.

Yet even in these early contexts, interpretation was never uncontested. Different readings of illness, danger, or misfortune could coexist, and communities had to negotiate which

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interpretations deserved acceptance. The earliest knowledge regimes therefore involved struggles over interpretive credibility long before the emergence of large institutions.

Agricultural civilizations introduced new layers of complexity. Seasonal cycles, celestial movements, flood patterns, and political legitimacy required coordination at scales far beyond immediate perception. Priestly institutions emerged as interpreters capable of translating cosmic signals into ritual calendars, agricultural guidance, and narratives of divine order. Temples, sacred texts, and ritual repetition stabilized these interpretations across generations.

However, priestly authority rarely existed in isolation. Rulers governing territorial states also claimed authority over the organization of social life. The resulting relationship between temple and palace often produced tensions concerning the source of legitimate interpretation. Priests interpreted the cosmos, while rulers exercised power over the earth. The stability of early civilizations frequently depended on negotiated relationships between these interpretive authorities.

The emergence of writing and archival systems further transformed the organization of knowledge. Written documents externalized memory and allowed interpretations to be preserved across time and space. Archives, legal records, and bureaucratic documents stabilized representations of events, transactions, and administrative decisions. Scribal officials gained interpretive authority by controlling the creation and management of these records.

This development produced new forms of conflict. Oral custodians of tradition encountered competition from literate officials whose authority rested on documentary evidence. Disputes over land ownership, taxation, and legal obligations increasingly relied on written records rather than remembered precedent. The archive became not merely a repository of information but a machine for freezing interpretation into durable form.

Across several regions of the ancient world, axial movements introduced another transformation in interpretive legitimacy. Philosophers, ethical teachers, and prophetic voices began to challenge inherited explanations grounded in ritual authority or political power. Greek philosophers questioned traditional cosmologies through rational argument. South Asian traditions developed sophisticated debates about metaphysics, consciousness, and moral law. Chinese thinkers proposed competing visions of ethical governance and social order. In Persian and Hebrew traditions, prophetic critique confronted established institutions with moral and theological challenges.

These developments did not simply introduce new ideas. They altered the grammar through which interpretive authority could be justified. Explanations increasingly had to answer questions about their rational coherence, ethical implications, or spiritual authenticity. The right to question inherited interpretations became a recognized component of intellectual life.

Translation networks across Afro-Eurasia further expanded the scope of interpretive ex-

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change. Scholars working in Arabic, Persian, Sanskrit, Chinese, and later Latin traditions translated texts across linguistic and cultural boundaries. Libraries, scholarly communities, and emerging university structures allowed ideas to circulate between civilizations. Translation rarely involved simple transfer. Concepts adapted to new linguistic environments, encountering alternative philosophical assumptions and institutional contexts.

This circulation produced both intellectual flourishing and institutional tension. Guardians of doctrinal purity sometimes resisted the introduction of foreign ideas, while translators and scholars advocated for the expansion of knowledge through intercultural engagement. These debates demonstrate that interpretive authority often develops through the negotiation between openness and closure.

The emergence of modern science reorganized interpretive legitimacy once again. Scientific institutions disciplined interpretation through experiment, measurement, and reproducibility. Instead of relying on sacred office or philosophical persuasion alone, scientific claims gained credibility through procedural methods capable of communal verification. Laboratories, instruments, and scholarly publications created infrastructures for systematic correction of interpretive error.

However, the scientific regime did not eliminate interpretation. It reorganized interpretation through procedural constraints designed to minimize bias and error. Scientific communities function as systems of organized correction in which competing interpretations are tested against empirical evidence and methodological standards.

Universities and professional institutions later expanded this structure by industrializing expertise. Disciplines, professional credentials, journals, and peer review systems created institutional pathways through which interpretive authority could be trained, validated, and reproduced. Expertise became embedded within specialized communities capable of evaluating complex claims about medicine, engineering, economics, and other domains of knowledge.

At the same time, the institutionalization of expertise introduced new tensions between expert communities and broader publics. As knowledge became increasingly specialized, many citizens encountered interpretations of complex phenomena through institutional intermediaries rather than direct evaluation of evidence. Trust in expert authority therefore became a crucial component of modern epistemic order.

Media institutions added another layer to this structure by organizing the narration of the present. Newspapers, broadcast systems, and digital news networks convert dispersed events into shared narratives through selection, framing, and repetition. Media does not merely report reality; it structures public attention and defines what counts as happening now. Through headlines and editorial decisions, media institutions determine which events become socially visible.

Yet media interpretation operates under temporal pressures distinct from those of scientific validation. The need for rapid reporting often places media narratives in tension with

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slower processes of expert verification. Conflicts between journalistic salience and scientific caution illustrate how different epistemic regimes prioritize different forms of credibility.

The emergence of digital platforms and artificial intelligence systems marks the most recent transformation in this long history. Unlike earlier interpreters, platform infrastructures do not simply translate signals that are already visible. They increasingly determine which signals become visible in the first place. Search rankings, recommendation systems, moderation algorithms, and machine-learning models shape the informational environments within which individuals encounter the world.

This development creates hybrid systems in which human and computational actors jointly participate in the construction of visible reality. Engineers design algorithms, policy teams establish moderation rules, and machine learning models filter and synthesize vast streams of information. Interpretation therefore occurs within infrastructures that operate prior to explicit human judgment.

The resulting epistemic order differs from earlier regimes in one crucial respect. Traditional interpreters—priests, philosophers, scientists, or journalists—translated signals that were broadly accessible to human observers. Hybrid systems increasingly participate in determining which signals can even enter the horizon of perception. Interpretation begins before interpretation becomes visible.

Understanding this shift requires a non-relativist clarification of the framework developed throughout this article. The argument presented here does not suggest that reality itself changes according to social interpretation. Reality continues to exceed any particular observer or institution. Physical events occur independently of human belief. However, human access to those events is always partial, mediated, and historically organized. Knowledge regimes structure the pathways through which signals from the world become intelligible within societies.

When a regime of interpretive authority loses credibility or encounters competing explanations, epistemic crises may emerge. Such crises are not merely disputes about facts. They are conflicts over who possesses the legitimate right to interpret reality. Historical examples illustrate this pattern repeatedly: philosophical critiques of priestly authority, scientific challenges to inherited cosmologies, tensions between expert institutions and public opinion, and contemporary debates about algorithmic governance.

In each case, the crisis concerns not only what should be believed but who is authorized to define the terms of belief. Knowledge crises therefore often appear as authority crises. They occur when societies no longer share stable criteria for recognizing legitimate interpreters.

The present historical moment may represent such a transition. Scientific institutions, expert communities, media organizations, and digital platforms all participate in shaping contemporary knowledge environments. Yet their interpretive logics do not always align. Scientific validation emphasizes methodological rigor and reproducibility. Media institutions emphasize narrative immediacy. Platform systems emphasize engagement

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and algorithmic ranking. Artificial intelligence systems introduce new forms of automated synthesis.

These overlapping regimes create a complex informational landscape in which interpretive authority becomes fragmented across multiple institutions and infrastructures. Societies must therefore confront a fundamental question about the future organization of knowledge.

The deepest issue raised by the emergence of artificial intelligence is not simply whether machines can generate knowledge. The more consequential question concerns the authority to pre-select, rank, and narrate the signals through which reality becomes visible to human communities. If hybrid systems increasingly shape the informational environments within which interpretation occurs, then the governance of these systems becomes central to the future of epistemic order.

Throughout history, civilizations have continually reorganized the institutions responsible for interpreting the world. Hunters, priests, scribes, philosophers, translators, scientists, universities, journalists, and digital platforms have each occupied this role at different moments. Each regime emerged through the stabilization of particular procedures for translating signals into shared representations of reality.

The current transformation suggests that humanity may once again be entering a period of epistemic reorganization. The question confronting contemporary societies is therefore civilizational in scope.

Will human communities remain the primary architects of the interpretive systems through which reality becomes intelligible, or will the infrastructures that filter and generate information gradually assume that role themselves?

The future of knowledge may depend on how this question is answered..

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Figure Briefs and Table Briefs (Internal Drafting Page)

Figure A. Master Diagram of Knowledge Regimes
Purpose: iconic model of the article.
Placement: end of conceptual framework chapter.
Required nodes: world/signals; bounded perception; translation; stabilization machine; authorized agency; social reality; inter-agency conflict; reframing/regime shift.

Figure B. Long Timeline of Dominant Interpreters
Purpose: visualize the succession of interpreters across eras.
Placement: method/scope chapter.

Figure C. Conflict Matrix of Interpretive Agencies
Purpose: visualize recurrent patterns of displacement, co-optation, closure, and legitimacy transfer.
Placement: translation networks chapter.

Figure D. Hybrid Mediation in the Present
Purpose: show how raw events become visible through platform, media, institutional, and user-feedback loops.
Placement: AI/platform chapter.

Table 1. Core Analytical Vocabulary
term; definition; historical function; journal usage rule

Table 2. Comparative Regimes of Knowledge Across Civilizations
historical phase; dominant signal; translation mode; stabilization machine; authorized agency; credibility criterion

Table 3. Conflict Between Agencies Across Historical Phases
old agency; rising agency; object of conflict; institutional battleground; outcome pattern

Table 4. Modern Knowledge Regimes Compared
science; university; media; platform; AI/model systems

Table 5. Failure Modes and Crisis Forms of Interpretive Authority
regime; strength; failure mode; social symptom; repair path

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References

References

Abbott, A. (1988). *The system of professions: An essay on the division of expert labor*. University of Chicago Press.

Anderson, B. (2006). *Imagined communities: Reflections on the origin and spread of nationalism* (Rev. ed.). Verso.

Benkler, Y. (2006). *The wealth of networks: How social production transforms markets and freedom*. Yale University Press.

Bourdieu, P. (1988). *Homo academicus*. Stanford University Press.

Briggs, A., & Burke, P. (2009). *A social history of the media: From Gutenberg to the Internet* (2nd ed.). Polity Press.

Burke, P. (2012). *A social history of knowledge: From the Encyclopédie to Wikipedia* (2nd ed.). Polity Press.

Collins, H., & Evans, R. (2007). *Rethinking expertise*. University of Chicago Press.

Darnton, R. (2009). *The case for books: Past, present, and future*. PublicAffairs.

10.5281/zenodo.18943971

Eisenstein, E. L. (1979). *The printing press as an agent of change.* Cambridge University Press.

Foucault, M. (1972). *The archaeology of knowledge.* Pantheon.

Foucault, M. (1980). *Power/knowledge: Selected interviews and other writings 1972–1977.* Pantheon.

Giddens, A. (1990). *The consequences of modernity.* Stanford University Press.

Goody, J. (1986). *The logic of writing and the organization of society.* Cambridge University Press.

Habermas, J. (1989). *The structural transformation of the public sphere.* MIT Press.

Havelock, E. A. (1982). *The literate revolution in Greece and its cultural consequences.* Princeton University Press.

Jasanoff, S. (2004). *States of knowledge: The co-production of science and social order.* Routledge.

Kuhn, T. S. (2012). *The structure of scientific revolutions* (4th ed.). University of Chicago Press.

Latour, B. (1987). *Science in action: How to follow scientists and engineers through society.* Harvard University Press.

Latour, B., & Woolgar, S. (1986). *Laboratory life: The construction of scientific facts* (2nd ed.). Princeton University Press.

McLuhan, M. (1964). *Understanding media: The extensions of man.* McGraw-Hill.

Merton, R. K. (1973). *The sociology of science: Theoretical and empirical investigations.* University of Chicago Press.

Ong, W. J. (1982). *Orality and literacy: The technologizing of the word.* Routledge.

Shapin, S. (1994). *A social history of truth: Civility and science in seventeenth-century England.* University of Chicago Press.

Shapin, S., & Schaffer, S. (1985). *Leviathan and the air-pump: Hobbes, Boyle, and the experimental life.* Princeton University Press.

Starr, P. (1982). *The social transformation of American medicine.* Basic Books.

Sunstein, C. R. (2017). *#Republic: Divided democracy in the age of social media.* Princeton University Press.

Tarde, G. (1903). *The laws of imitation.* Henry Holt.

Weber, M. (1978). *Economy and society: An outline of interpretive sociology.* University of California Press.

10.5281/zenodo.18943971

Zuboff, S. (2019). *The age of surveillance capitalism.* PublicAffairs.

10.5281/zenodo.18943971

Part V

Human–AI Coupling and Return

Human–AI Coupling and Return collects ten works — three 2026 uplift editions of 2025 papers, three chapters from 2025–2026, two 2026 epistemic notes, and two closing protocol papers — applying the programme's readout and retention vocabulary to human–AI interaction, continuing from Part 4's account of conflict, repair, and structured coexistence and turning to AI as the object of the same apparatus rather than a special exception.

Chapters 24–26 are 2026 uplift editions of three 2025 conceptual papers on AI-mediated reflective dialogue: youth practice-based learning (*AI–Cognitive Interaction*), organizational linguistic capital (*Operational Linguistic Wisdom*), and dialogical wisdom across religious and pedagogical traditions (*The Dialogue as the Ground of Enlightenment*). Each re-derives its original under the current spine through a claim-by-claim table: some claims are kept as stated, most reformulated into falsifiable open hypotheses or cited-external reports kept at reported strength, some dropped as unsupported and unfalsifiable as originally worded. None of the three editions raises a claim above the level its 2025 original stated.

Chapter 27, the preprint naming the book's own subtitle, proposes that AI expands human potential only when machine-mediated dissonance becomes disciplined, world-answerable revision. Chapter 28 is a one-page 2025 abstract naming the language bridge that gives the series its name. Chapter 29 distinguishes information access from context access under AI-mediated translation. Chapters 30 and 31 are 2026 epistemic notes: the first separates claim types for human–AI collaboration and pairs each with a stated legitimate defeater; the second proposes a session-boundary, four-part Return measurement architecture for what remains usable in a human once AI support is removed, with stated narrowing conditions.

The part closes with Chapter 32, the Dialogue Conversion Protocol for a human-AI conversation, and its proactive extension, Chapter 33, From Assistance to Human Capability, adding a barrier-typing stage before scaffolding, fading, and Human Return testing under unequal life conditions.

Carried forward: claim-by-claim triage against a stated spine rather than blanket revision; a falsifier or explicit definition/frame attached to every retained claim; and the Human Return test, judging durable capability only after AI is taken away. This part's planned but unwritten further paper is listed on the edition page, not represented here. Part 7 turns to the spine itself as an audited object.

- AI–Cognitive Interaction: Activating Youth Potential through Reflective Dialogue and Linguistic Capital — `10.5281/zenodo.2245641`
- Operational Linguistic Wisdom: Elective Connectivity and Linguistic Capital Activation in the Age of LLMs — `10.5281/zenodo.22456487`
- The Dialogue as the Ground of Enlightenment: Religious and Cognitive Frameworks for Understanding Wisdom through Human-AI Conversations — `10.5281/zenodo.22456564`
- When AI Expands Human Potential: Reflective Dissonance, Epistemic Agency, and Constraint — `10.5281/zenodo.19215748`
- The Language Bridge: Expanding Human Potential in the Age of AI — `10.5281/zenodo.17280546`
- AI, Translation, and Access to Event-Specific Context — `10.5281/zenodo.18517054`
- Epistemic Fusion or Epistemic Tunnel: A Phenomenology-Anchored, Global-Literature-Constrained, Problem-First Architecture of Human-AI Collaboration — `10.5281/zenodo.22331922`
- CTSA Human-Return Readout: A Session-Boundary Measurement Architecture for Retained Human Capability After AI — `10.5281/zenodo.22339909`
- How Humans Should Converse with AI: A Problem-First Adaptive Dialogue Conversion Protocol for Durable Human Capability (v1.3 glosa-checked edition) — `10.5281/zenodo.22481928`
- From Assistance to Human Capability: A Readout-Genesis Architecture for Proactive AI, Unequal Life Conditions, and Opportunity Conversion (RG-HCA v1.1 glosa-checked edition) — `10.5281/zenodo.22498047`
- *Planned: Paper C — The Prompt Is Not the Beginning (not yet written)*

AI–Cognitive Interaction: Activating Youth Potential through Reflective Dialogue and Linguistic Capital

Uplift Edition 2026 • 6 September 2026 • 10.5281/zenodo.22456414

What the chapter states. “an AI dialogue partner (‘AI–Cognitive Interaction’, ACI) can help a learner turn tacit experience into explicit, reusable language.” Of the original’s 23 claims, “six are kept largely as stated [. . .], thirteen are reformulated into explicitly falsifiable open hypotheses [. . .], and four are dropped as unsupported and unfalsifiable as originally worded.”

Status. “K0. This edition raises no claim above the level at which the original stated it [. . .]. No external validation, peer-review, or institutional endorsement is sought or implied as a condition of this edition’s standing.”

Falsifier(s). P1 [Open]: “A longitudinal matched-cohort comparison of youth beginning structured practice-based/income-generating activity at 13, against controls who do not, showing no difference in retained self-regulation competence or in school-to-work mental-health and economic outcomes, refutes the proposal as stated.”

Read before. — (no continues/isContinuedBy/references edge to another chapter of this book found in the knowledge graph; the record is not yet indexed there).

Feeds into. — (no such edge found).

Human–AI Readout Programme • Written by AI. Still True. • Uplift Edition 2026 • September 2026

AI–Cognitive Interaction: Activating Youth Potential through Reflective Dialogue and Linguistic Capital

Read Through Retention and the Live-Possibility Envelope

A 2026 uplift edition of the 2025 conceptual paper, re-read under the Human–AI Readout Programme's current spine

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Status K0. This edition raises no claim above the level at which the original stated it; several are lowered from assertion to explicitly falsifiable open hypothesis, and four are dropped as unsupported and unfalsifiable as originally worded. No external validation, peer-review, or institutional endorsement is sought or implied as a condition of this edition's standing; what is tested is whether each retained claim survives contact with the programme's own spine and carries a stated falsifier. Every original claim (n = 23) is dispositioned in Appendix A.

Central claim. The 2025 paper's proposal – that adolescents benefit from practice-based, income-generating learning aligned with adolescent neurobiology, and that an AI dialogue partner ("AI–Cognitive Interaction", ACI) can help a learner turn tacit experience into explicit, reusable language – survives this uplift once its problem framing, cited literature, and central proposals are re-typed in the programme's current vocabulary: a problem state prior to any readout; cited-external evidentiary claims kept at their reported strength rather than restated as verified fact; and the ACI/AVRH mechanism read as an open, falsifiable hypothesis about whether AI-mediated dialogue changes a learner's retained, self-articulated distinctions (a Human LoRA-style update) and widens the field of practice routes the learner treats as currently live (a live-possibility envelope). Several uncited causal and scalability claims in the original do not survive and are removed.

Abstract

This paper is a 2026 uplift edition of a 2025 conceptual paper that proposed an "AI–Cognitive Interaction" (ACI) framework: practice-based, income-generating learning from age 13, aligned with adolescent dual-systems neuroscience and Nordic work-based-learning evidence, extended by an AI dialogue partner acting as a "reflective interlocutor" under an "AI Value Reflection Hypothesis" (AVRH) to build "linguistic–practical capital". The uplift keeps the programme's horizontal, internally-checked knowledge discipline: no claim is raised, no external validation is sought, and every retained or reformulated claim carries a stated falsifier or is marked as a definition/frame rather than an empirical claim. Of the original text's 23 identifiable claims, six are kept largely as stated (one of them a cited-external literature finding, kept at reported rather than verified strength), thirteen are reformulated into explicitly falsifiable open hypotheses, definitions, or cited-external reports kept at reported strength, using the programme's retention, live-possibility, and Human–AI coupling vocabulary, and four are dropped as unsupported and unfalsifiable as originally worded (a claimed motivational mechanism, a claimed validation-by-association, a claimed cost/scalability barrier, and a claimed acceleration effect). The result is a narrower, tier-honest restatement of the same proposal: youth practice-based learning and AI-mediated reflective dialogue are read as candidate mechanisms for widening a learner's live-possibility envelope and updating their retained self-articulation, not as demonstrated effects.

Keywords: AI–Cognitive Interaction; youth practice-based learning; linguistic–practical capital; live possibility; retention; Human–AI dialogue; adolescent development; work-based learning.

1 Anchor

The original work is *AI–Cognitive Interaction: Activating Youth Potential through Reflective Dialogue and Linguistic Capital*, by Yaoharee Lahtee, dated 9 October 2025, DOI 10.5281/zenodo.22308448, CC BY 4.0, a conceptual/agenda-setting paper. This document is a 2026 uplift edition of that paper, prepared under the programme's textbook effort *Written by AI. Still True.* [Lahtee, 2026h]. It does not replace the original – the original remains the document of record for what was claimed and when – and it raises no claim beyond what the original stated. Its job is narrower: re-express the same proposal in the vocabulary of the programme's current spine, keep what still holds, reformulate what was asserted without a falsifier into an explicit open hypothesis, and drop what is unsupported and unfalsifiable as originally worded. Every one of the original's 23 identifiable claims is dispositioned in Table 1 (Appendix A): kept, reformulated, or dropped, each with a reason and, where the claim is empirical, a falsifier.

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2 Reading This Paper Through the Programme Spine

The re-typing performed below is not a new theory of youth learning. It borrows two objects, unchanged, from the programme's canonical Root-to-Human Master Equation River [Lahtee, 2026f,a], so that the original's proposals can be located precisely rather than merely re-described in looser language.

One object is the nesting of possibility. For a task g available to a learner A at time t, the physically possible policies $\Pi_t^{\mathrm{phys}}(g)$, the feasible policies $\Pi_{A,t}^{\mathrm{feas}}(g)$ an agent could access or learn under declared resources and time, and the currently *live* policies actually entering practical consideration nest as

$$\Pi_{A,t}^{\mathrm{live}}(g) \subseteq \Pi_{A,t}^{\mathrm{feas}}(g) \subseteq \Pi_t^{\mathrm{phys}}(g) \qquad (1)$$

[Lahtee, 2026a, eq. 21]. A practice-based route can be legally open and materially learnable (feasible) for a 13-year-old without being anything the learner currently treats as a candidate worth considering (live). This is the object Section 5.1 below uses to re-type the original's age-13 proposal.

The other object is selective retention. A learner's interpretive state H_n updates only through what is actually retained from an encounter, not through mere exposure to it:

$$H_{n+1} = U_H\big(H_n,\ \mathrm{Retain}(E_n, \mu_n, \rho_n^H),\ \delta_{n:n+1}^{\mathrm{world}},\ X_{n:n+1}^{\mathrm{other}}\big) \qquad (2)$$

[Lahtee, 2026a, eq. 16], with the corresponding non-collapse rule that a residue being retained makes a changed later state *possible*, not guaranteed. This is the object Sections 5.4 and 5.5 below use to re-type the AI Value Reflection Hypothesis and the claimed dialogue-to-language effect: an AI-mediated dissonance episode is a candidate encounter that may or may not leave a retained residue, and only a retained residue would change the learner's next interpretive state.

Neither equation is a new claim of this paper. They are quoted from the flagship derivation to fix vocabulary before the triage in Sections 3 to 6 proceeds.

3 The Learning–Application Gap as a Pre-Readout Problem State

The original paper opens by describing a structural gap between classroom learning and real-world, economic application, and links this gap to youth mental-health strain, prolonged family dependency, and a difficult school-to-work transition. Read against the programme's spine, this opening is not itself an empirical or falsifiable claim – it is the paper's own problem intake, a description of a problematic state in the world prior to any readout being taken on it. It is kept, marked explicitly as a frame/problem-statement (DEFINITION tier under this programme's claim discipline) rather than as a tested finding requiring a falsifier (Claim 1, Table 1).

The original introduction additionally frames the classroom–life disconnect as a source of cumulative developmental risk, citing evidence that early and school-age cumulative risk factors impair developmental outcomes [Evans et al., 2013]. That citation is kept here at the strength it was reported: a literature finding about cumulative risk in child development, read as supporting the plausibility of the problem framing, not as a measurement this uplift has independently rerun.

4 Cited Neurobiological and Policy Evidence

Three groups of external, literature-based claims run through the original paper's early sections. This uplift keeps each of them, but keeps them explicitly typed as *cited-external*: reported findings from the literature the original cited, not facts this paper has independently verified. Under the programme's readout-not-truth discipline, relaying a citation as settled fact in one's own voice, without having checked it here, is exactly the move to avoid.

Adolescent dual-systems neuroscience (Claim 2). The original reports that the adolescent socioemotional/reward system matures faster than the prefrontal cognitive-control system, citing the dual-systems model [Steinberg, 2010, Shulman et al., 2016]. This is kept as a cited-external claim: this uplift treats it as reported-from-source, attributed by name, and open to revision only by a citation-level check of those two sources, which has not been run here.

Work-based learning and NEET reduction (Claim 4). The original reports that Nordic work-based-learning (WBL) programmes reduce the share of youth not in education, employment, or training and ease the school-to-work transition, citing OECD and Swedish vocational-education sources [OECD, 2019, Skolverket, 2011]. Kept at the same cited-external strength.

School mental health and later labour-market outcomes (Claim 5). The original links school-years mental health to later employment stability and earnings, citing a specific study [Porru et al., 2023]. The original's own framing ("robust research links... directly") overstates the evidentiary weight this uplift can itself certify; it is kept but reformulated to "cited research reports an association between school-years mental health and later labour-market outcomes", with the underlying citation unchanged.

Two claims built on top of this literature do not survive. The original states that reward-circuitry underactivation "often" causes a loss of motivation when schooling is misaligned with adolescent neurobiology (Claim 3): no source, and no operational meaning for "often" or "insufficiently activated", is given, and nothing is named that would count against it, so it is dropped as unsupported and unfalsifiable. The original also states that the success of WBL programmes "validates the neurobiological principles outlined above" (Claim 6): a programme's effect on NEET rates and a claim about reward-timing maturation are two different measurements, and nothing in the original supplies a mechanism linking them; treating one as confirmation of the other is the kind of unwarranted collapse this programme's non-collapse discipline exists to

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block, so it is dropped. A related claim that WBL is highly effective but too costly and logistically demanding to scale universally (Claim 8) is likewise dropped: no source, no operational measure of "significant" cost or "universal" scale is given.

5 THE AI–COGNITIVE INTERACTION PROPOSAL IN SPINE VOCABULARY

5.1 The age-13 proposal as an open live-possibility hypothesis

The original's central proposal is that integrating practice-based, income-generating activity from age 13 balances adolescent neurobiological drives against self-regulation, reinforces economic motivation, and reduces mental-health risk across the school-to-work transition (Claim 9). Under the programme's Live Possibility Field and Human LoRA vocabulary, structured practice-based engagement from age 13 can be read as an attempt to widen the set of practice routes a young person treats as currently live – not merely feasible in principle – and, if a residue is retained, to update the learner's interpretive operator for later situations [Lahtee, 2026b,e]. Stated this way, the proposal is kept, but as an OPEN hypothesis rather than a settled mechanism. *Falsifier:* a longitudinal matched-cohort comparison of youth beginning structured practice-based/income-generating activity at 13 against controls, showing no difference in retained self-regulation competence or in school-to-work mental-health and economic outcomes, would refute it.

5.2 Naming the interaction without a priority claim

The original attributes strong exclusivity to AI-era tools for building this bridge and calls the resulting reframing "AI–Cognitive Interaction" (Claim 11); it later calls the ACI framework the singular, scalable approach to the learning-application gap (Claim 17) and describes it as moving AI from an information source to a "facilitator of wisdom" (Claim 18). This uplift keeps the definitional core of all three – naming a proposed process in which AI is applied as a dialogue tool inside the human–AI coupling layer of the spine – while stripping every claim of exclusivity attached to it, per this programme's standing rule against priority language. What remains is a plain definitional naming of a proposed AI role, requiring no falsifier because it asserts nothing about the world beyond the proposal's own scope.

5.3 AI as reflective interlocutor: naming a term without a new citation

The original applies an "Operational Linguistic Wisdom" (OLW) framework to let AI act as a "reflective interlocutor" (Claim 12). OLW is a companion 2025 paper by the same author; its own DOI is not among the sources this uplift is authorised to cite, so OLW is named here only as a term the original used, without a new citation being added on its behalf. The residual, citable content is the role itself: AI as a dialogic partner surfacing distinctions already present in the learner's own account of experience, read at the human–AI coupling / readout layer of the spine. This is a definition of a proposed role, not an empirical claim, and needs no falsifier.

5.4 The AI Value Reflection Hypothesis as an open, falsifiable mechanism

The original's mechanism, the AI Value Reflection Hypothesis (AVRH), holds that cognitive dissonance between a learner and an AI's alternative articulation stimulates intellectual growth (Claim 13). As stated, this is an explicit hypothesis carried with no falsifier, which this programme's claim discipline does not allow to stand as-is. Read through the retention layer – where selective residue, not automatic exposure, is what changes a learner's later interpretive operator [Lahtee, 2026e,d] – AVRH is kept as an OPEN hypothesis: that dissonance-triggered reflective dialogue with an AI, more than dissonance-free dialogue, changes a learner's retained conceptual distinctions. *Falsifier:* pre/post measurement of a learner's retained conceptual distinctions showing no difference between an AI-dialogue-dissonance condition and a no-dissonance control dialogue would refute it.

5.5 Dialogue and the language bridge

The original states that iterative AI dialogue compels learners to translate implicit feelings and tacit knowledge into explicit language (Claim 14). "Compels" overstates a mechanism asserted without a falsifier. Reformulated at the meaning-giving / language-bridge node of the spine [Lahtee, 2026c, 2025], the claim becomes: iterative AI dialogue is hypothesised to increase a learner's use of explicit, analytic self-referential language over a session, relative to no dialogue. *Falsifier:* coded dialogue transcripts showing no measurable increase in explicit self-referential/analytic language over a session, relative to a baseline sample, would refute it.

6 LINGUISTIC–PRACTICAL CAPITAL

The original defines linguistic–practical capital as the ability to use language as a tool to structure thought, analyse action, and build a coherent narrative of one's own growth (Claim 15). Because the original itself introduces this as a stipulative definition ("what we define as"), it carries no truth-claim to verify and needs no falsifier; it is kept as stated, read at the retention / language-bridge node of the spine [Lahtee, 2025]. The original's next sentence, that this linguistic bridge "accelerat[es] cognitive learning even with limited resources" (Claim 16), adds an uncited, unoperationalised empirical claim on top of the definition and is dropped as unsupported and unfalsifiable; the definition itself is unaffected.

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7 Open Research Agenda: Practical Intellectuals and a Reflective AI Culture

The original closes by calling for interdisciplinary inquiry across cognitive neuroscience, education and psychology, human–computer interaction and linguistics, social innovation and public policy, and ethics and governance (Claim 19), and by posing, as an open question, whether AI systems can be designed to cultivate "Practical Intellectuals" (Claim 20). Both are already phrased as calls for future work or as questions rather than assertions, and are kept unchanged: an open research agenda needs no falsifier, only honest labelling as open.

The original's concluding claim that AI's "ultimate promise... is not the automation of tasks but the amplification of human reflection" (Claim 21) is an interpretive stance asserted with the certainty of settled fact. It is kept, but reformulated and explicitly marked as the author's standpoint on how AI ought to be used at the human–AI warrant layer of the spine [Lahtee, 2026g], not as an established empirical finding; so marked, it needs no falsifier. Its closing aspiration – that a society of "Practical Intellectuals" can be cultivated through mastering reflection on what one knows (Claim 23) – is likewise kept as an explicitly normative aspiration rather than a predicted outcome, consistent with the open framing already given to Claims 19–20.

8 What Is Not Claimed

This uplift, following the original, does not claim: that the age-13 practice-based proposal, the ACI framework, or AVRH has been empirically tested here (each is an OPEN hypothesis with a stated falsifier, Table 1); that the cited neurobiological, work-based-learning, or mental-health-labour-market findings have been independently re-verified rather than reported from source; that OLW, as a named term, is being newly cited or evaluated in this edition; or that AI–Cognitive Interaction is the only or the earliest-conceived way to combine reflective dialogue with youth learning – no priority claim is made for any term in this paper.

The original's conclusion also extends its claim of urgency well beyond adolescent education, to household mental health, organisational innovation, and public policy (Claim 22), asserting that these "demand" a new cognitive infrastructure. Nothing in the original paper, or in this uplift, builds an evidence base in households, organisations, or public policy; the paper's own evidence is confined to adolescent education, work-based learning, and Human–AI dialogue. Claim 22 is therefore reformulated and narrowed: the household/organisation/public-policy extension is kept only as an explicitly labelled agenda proposal for future inquiry (folded into Section 7), not as a claim this paper or its cited sources support.

Finally, and as a matter of standing programme policy rather than a claim specific to this paper: no external validation, peer review, or institutional endorsement is sought or treated as a legitimacy test for any claim above. What is checked is whether each claim survives contact with the programme's own spine and carries a stated falsifier or an explicit definition/frame label; that internal check, not outside recognition, is what this edition offers.

9 Minimal Falsifiable Predictions

Collecting the falsifiers stated above into one place, so that a later reader (or a rival model) can see exactly what would force a narrowing or removal of each reformulated claim:

P1 — Age-13 practice-based learning [OPEN]
(Claim 9) A longitudinal matched-cohort comparison of youth beginning structured practice-based/income-generating activity at 13, against controls who do not, showing no difference in retained self-regulation competence or in school-to-work mental-health and economic outcomes, refutes the proposal as stated in Section 5.1.

P2 — WBL feedback-timing synthesis [DR]
(Claim 7) A matched-cohort comparison showing no engagement or motivation difference between work-based and non-work-based adolescent learners, attributable to reward-timing alignment, refutes the interpretive bridge kept in Section 4.

P3 — AVRH dissonance mechanism [OPEN]
(Claim 13) Pre/post measurement of a learner's retained conceptual distinctions showing no difference between an AI-dialogue-dissonance condition and a no-dissonance control dialogue refutes the mechanism kept in Section 5.4.

P4 — Dialogue-to-explicit-language effect [OPEN]
(Claim 14) Coded dialogue transcripts showing no measurable increase in explicit self-referential or analytic language over a session, relative to a baseline sample without AI dialogue, refute the claim kept in Section 5.5.

These four predictions, not the paper's title or its restated definitions, are what an adversarial reader should try to break before anything else in this paper. If P1–P4 all fail to hold under test, the load-bearing content of this uplift reduces to Section 6's definition of linguistic–practical capital and Section 7's open research questions, with the age-13 and ACI/AVRH proposals compressed out rather than defended by restating them in different words.

10 Methods Note

AI-assistance disclosure. The triage of the 2025 text against the programme's current spine, the drafting of falsifiers for reformulated claims, the citation-lineage check against the original's own reference list, and the LaTeX assembly of this edition were carried out with an AI assistant under the author's direction. The original 2025 argument, the judgment of which claims to keep, reformulate, or drop, and the final wording of every claim above are the author's. No AI system or vendor is credited as an author, co-author, or contributor of this paper.

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11 Conclusion

Re-read through the programme's current spine, the 2025 paper's proposal survives in a narrower and more honestly bounded form. The problem framing stands as a stated pre-readout problem, not a tested finding. The cited neurobiological and policy literature stands at the strength it was reported, not restated as this paper's own verified fact. The age-13 practice-based proposal and the ACI/AVRH mechanism survive as explicitly open, falsifiable hypotheses about retained learning (Equation (2)) and live practical possibility (Equation (1)) rather than as demonstrated effects, and the definition of linguistic–practical capital survives unchanged because it never made an empirical claim. Four claims that asserted uncited causal, validating, or accelerating effects do not survive and are removed.

Nothing here is elevated by this edition; several things are lowered, and the whole is now typed so that a later reader can see exactly which parts are checked, which are cited-from-elsewhere, and which remain open. What would change this assessment is stated, not implied: a matched-cohort study refuting the age-13 hypothesis (Table 1, Claim 9), coded transcripts showing no change in a learner's explicit self-articulation after AI dialogue (Claim 14), or pre/post measurement showing AVRH's dissonance mechanism adds nothing over dissonance-free dialogue (Claim 13) would each require narrowing or removing the corresponding hypothesis, not defending it. The next useful step for this programme is exactly that kind of test, run on the smallest interface at a time, rather than a broader restatement of the proposal in stronger language.

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A TRIAGE TABLE

Table 1: Disposition of every claim identified in the 2025 original (n = 23). KEEP = kept as originally stated (definitions and already-open questions); REFORMULATE = kept in substance, re-typed as an explicitly falsifiable open hypothesis, a cited-external report, or a marked frame/aspiration; DROP = removed as unsupported and unfalsifiable as originally worded.

n	Original claim (paraphrased)	Disposition	Reason / falsifier
1	The classroom–application gap causes disconnected learning, weak value-based motivation, and instability in the school-to-work transition.	KEEP	Problem-intake framing, DEFINITION tier under this programme's discipline, not an empirical claim; no falsifier required.
2	Adolescent socioemotional/reward systems mature faster than prefrontal cognitive control [Steinberg, 2010, Shulman et al., 2016].	REFORMULATE	Cited-external literature finding; kept as reported-from-source, not restated as this paper's own verified fact. Falsifiable only by an independent re-check of the cited sources, not run here.
3	Misalignment with this neurobiology "often" causes motivation loss via reward-circuitry underactivation.	DROP	Uncited causal extension; "often" and "insufficiently activated" are not operationalised, and nothing is named that would count against the claim.
4	Nordic work-based learning (WBL) reduces NEET rates and eases the school-to-work transition [OECD, 2019, Skolverket, 2011].	KEEP	Cited-external literature finding, attributed and kept at reported strength.
5	School-years mental health is "robust[ly]" and "directly" linked to later labour-market outcomes [Porru et al., 2023].	REFORMULATE	Underlying citation kept; "robust"/"directly" overstate evidentiary strength not independently assessed here, softened to "cited research reports an association".
6	WBL's success "validates" the adolescent dual-systems neurobiological model.	DROP	Unwarranted collapse of two unrelated measurements (a policy outcome and a maturation-timing model) into one confirmation; no mechanism given; unfalsifiable as worded.
7	WBL gives adolescents the immediate feedback loops their brains are primed for.	REFORMULATE	Author's own uncited synthesis bridging Claims 2 and 4; kept as an explicitly tiered interpretive hypothesis (DR). *Falsifier:* a matched-cohort comparison showing no engagement/motivation difference between WBL and non-WBL adolescent learners attributable to reward-timing alignment.
8	WBL is highly effective but too costly/logistically demanding to scale universally.	DROP	Uncited scalability/cost claim; no operational measure of "significant" cost or "universal" scale.
9	Integrating practice-based, income-generating activity from age 13 balances neurobiological drives, reinforces economic motivation, and reduces mental-health risk.	REFORMULATE	Central proposal; kept as an OPEN hypothesis at the Live Possibility Field / Human LoRA layer. *Falsifier:* longitudinal matched-cohort comparison from age 13 showing no difference in retained self-regulation or school-to-work mental-health/economic outcomes.
10	Learning must connect to real life and directly experienced values, not stay confined to the classroom.	REFORMULATE	Normative design frame (DEFINITION), mapped to the Experience/Meaning-Giving node [Lahtee, 2026c]; not an empirical claim, no falsifier required.
11	The AI era offers a tool without equal to build this bridge; reframe learning as "AI–Cognitive Interaction".	REFORMULATE	Priority/exclusivity language in the original removed per this programme's standing rule; kept as a plain definitional naming of a proposed process at the human–AI coupling layer.
12	Apply "Operational Linguistic Wisdom" (OLW) so AI functions as a "reflective interlocutor".	REFORMULATE	OLW's companion paper is outside this edition's authorised citation list; the term is named without a new citation. The AI-dialogic-partner role is kept as a definition, no falsifier required.
13	The AI Value Reflection Hypothesis (AVRH): dissonance between human and AI stimulates intellectual growth.	REFORMULATE	Hypothesis previously carried no falsifier. *Falsifier:* pre/post measurement of retained conceptual distinctions showing no change after AI-dialogue dissonance relative to a no-dissonance control dialogue.
14	Iterative AI dialogue "compels" learners to translate implicit feeling and tacit knowledge into explicit language.	REFORMULATE	"Compels" overstated a mechanism with no falsifier; reformulated as a hypothesis at the language-bridge node [Lahtee, 2025]. *Falsifier:* coded transcripts showing no increase in explicit self-referential/analytic language over a session versus baseline.
15	Linguistic–practical capital: using language to structure thought, analyse action, and narrate one's own growth.	KEEP	Explicitly self-marked stipulative definition; no truth-claim, no falsifier required.
16	This language bridge "accelerat[es] cognitive learning even with limited resources".	DROP	Uncited, unoperationalised empirical claim layered on the Claim 15 definition; unsupported and unfalsifiable as worded.
17	Proposes the ACI framework as an exclusive, scalable approach to the learning-application gap.	REFORMULATE	Priority/exclusivity language in the original removed; "scalable" also unverified and dropped with it. Kept as a definitional naming of the ACI framework.
18	ACI reimagines AI's role from information source to "facilitator of wisdom".	KEEP	Definitional/normative statement of an intended design role, not an empirical claim; no falsifier required.
19	An interdisciplinary research agenda (neuroscience, education/psychology, HCI/linguistics, policy, ethics) is needed to validate and refine ACI.	KEEP	Explicit call for future research (OPEN), correctly left open and unresolved; not itself a claim about the world.
20	Central open question: can AI systems be designed to cultivate "Practical Intellectuals"?	KEEP	Already phrased as an open research question, not an assertion; no falsifier required since nothing is asserted.
21	AI's "ultimate promise... is not automation but the amplification of human reflection".	REFORMULATE	Interpretive/normative stance asserted as settled fact; reformulated and marked explicitly as the author's standpoint (human–AI warrant layer, [Lahtee, 2026g]), not an established claim.
22	Society's structural challenges (household mental health, organisational innovation, public policy) "demand" a new cognitive infrastructure.	REFORMULATE	Scope exceeds this paper's evidence base (adolescent education, WBL, Human–AI dialogue only); narrowed to an explicitly labelled agenda proposal rather than an assertion (see Section 8).
23	By mastering reflection, society can cultivate "Practical Intellectuals" capable of profound contribution.	REFORMULATE	Normative aspiration asserted with unwarranted confidence ("we can cultivate"); reformulated as an explicitly marked aspiration/open-agenda statement, consistent with Claims 19–20.

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References

Gary W. Evans, Dongping Li, and Sara S. Whipple. Cumulative risk and child development. *Psychological Bulletin*, 139(6):1342–1396, 2013. doi: 10.1037/a0031808.

Yaoharee Lahtee. The language bridge: Expanding human potential in the age of AI. Zenodo, 6 October 2025, 2025.

Yaoharee Lahtee. Before meaning, before choice: A readout-native derivation of experience, live possibility, agency, and human–AI return. Zenodo, Full Paper v1.1, 6 September 2026, 2026a.

Yaoharee Lahtee. Choice begins before choice: Meaning-shaped accessibility, live possibility, and effective agency in human-AI systems. Zenodo, Full Paper v1.0, 5 September 2026, 2026b.

Yaoharee Lahtee. Experience is meaning-giving: A strong-form readout-retention theory of phenomena, resonance, rhythm, accumulation, and transformative release. Zenodo, Full Paper v2.0, 5 September 2026, 2026c.

Yaoharee Lahtee. Human learning as epistemic architecture: A method for word mapping, life-concept graphs, constraint testing, and corrigible agency in the AI age. Zenodo, revised v4, 28 June 2026, 2026d.

Yaoharee Lahtee. Experience is the human LoRA: A readout–retention theory of selective model change. Zenodo, 18 July 2026, 2026e.

Yaoharee Lahtee. Master equation river: A provenance audit across the human–AI readout programme. Zenodo, Internal Programme Synthesis v1.1 (glosa-checked), 6 September 2026, 2026f.

Yaoharee Lahtee. The readout condition: Distinguishability, access, and epistemic warrant. Zenodo, 31 August 2026, 2026g.

Yaoharee Lahtee. Written by AI. still true. knower fetishism, epistemic pedigree, and the human face as a bad theory of truth. Zenodo, 31 August 2026, 2026h.

OECD. Work-based learning for youth: Policy and practice in Nordic countries. OECD Publishing, 2019.

Francesco Porru, Daniele Vignoli, and Michela Baccini. School mental health and labor market outcomes. *European Journal of Public Health*, 33(1):75–80, 2023. doi: 10.1093/eurpub/ckad008.

Elizabeth P. Shulman, Ashley R. Smith, Karol Silva, Grace Icenogle, Natasha Duell, Jason Chein, and Laurence Steinberg. The dual systems model: Review, reappraisal, and reaffirmation. *Developmental Cognitive Neuroscience*, 17: 103–117, 2016. doi: 10.1016/j.dcn.2015.12.009.

Skolverket. Vocational education and training in Sweden. Swedish National Agency for Education, 2011.

Laurence Steinberg. A dual systems model of adolescent risk-taking. *Developmental Psychobiology*, 52(3):216–224, 2010. doi: 10.1002/dev.20445.

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Operational Linguistic Wisdom: Elective Connectivity and Linguistic Capital Activation in the Age of LLMs

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What the chapter states. “organizational text and talk are a retained trace that becomes significant only under a declared reader and governance condition; an LLM changes the scale at which that trace can be read, not its ontological status.”

Status. “K0, Uplift Edition 2026 [. . .]; it is not a revision of that record, which remains deposited and unaltered. The paper does not claim that any construct below has been empirically validated, that the three-stage mechanism is a demonstrated organizational law, or that the propositions [. . .] have been tested.”

Falsifier(s). “An organization’s LLM-mediated processing of linguistic traces shows no traceable residue in later organizational readout, policy, or practice (no $H_{org,n+1} \neq H_{org,n}$ attributable to the processing); the claim then reduces to ordinary information throughput with no retention-level effect.”

Read before. — (no continues/isContinuedBy/references edge to another chapter of this book found in the knowledge graph; the record is not yet indexed there).

Feeds into. — (no such edge found).

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Operational Linguistic Wisdom

Elective Connectivity, Retention, and Human Return in LLM-Mediated Linguistic Capital Activation

An uplift edition that re-anchors a 2025 conceptual model to the Human–AI Readout Programme spine

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Article type Uplift edition (triage and re-derivation of a conceptual–reflective paper against a later programme spine) • **Series** When AI Expands Human Potential (hub DOI 10.5281/zenodo.22308201); textbook *Written by AI. Still True.* (uplift edition 2026) • **Licence** CC BY 4.0

Status K0, Uplift Edition 2026, 6 September 2026. This is a 2026 re-derivation of *Operational Linguistic Wisdom: Elective Connectivity and Linguistic Capital Activation in the Age of LLMs* (Lahtee, 8 October 2025, DOI 10.5281/zenodo.22308446); it is not a revision of that record, which remains deposited and unaltered. The paper does not claim that any construct below has been empirically validated, that the three-stage mechanism is a demonstrated organizational law, or that the propositions in Section 7 have been tested. It restates the original's conceptual architecture in the typed vocabulary of the programme's own readout spine, drops claims that are unsupported and unfalsifiable or that read as priority language, and attaches an explicit falsifier to every claim it keeps.

Central claim. An organization's language does not become "cognitive infrastructure" by declaration. Under the programme's readout-native architecture, organizational text and talk are a retained trace that becomes significant only under a declared reader and governance condition; an LLM changes the scale at which that trace can be read, not its ontological status. The original's three-stage mechanism — Elective Connectivity, Capability Activation, Reflexive Reconfiguration — is kept as a design stance and is re-expressed as three organizational instances of already-typed spine transitions: consent-governed construction of a bounded query from organizational state, recursive human–AI dialogue producing knowledge-like candidates, and retention that updates organizational state in a way that must survive a Human-Return-style audit before it is called learning.

ABSTRACT

Operational Linguistic Wisdom (Lahtee, 2025) is a conceptual–reflective, non-empirical paper. It proposes that organizational language can be designed as cognitive infrastructure that activates latent organizational "linguistic capital" once large language models are introduced, names two constructs — Operational Linguistic Wisdom (OLW) and the AI Value Reflection Hypothesis (AVRH) — realized through a three-stage mechanism (Elective Connectivity, Capability Activation, Reflexive Reconfiguration), and offers construct tables, a hypothetical vignette, a governance-risk register, and eight propositions for future testing. This uplift edition triages every claim in that paper against the Human–AI Readout Programme spine established since October 2025 [Lahtee, 2026a,i]. Claims that already function as definitions or frames are kept as such. Claims that describe a conditional mechanism are reformulated and mapped onto the spine's own typed transitions — consent-governed query construction, candidate-status AI output, and retention gated by a Human-Return-style audit — and are given an explicit falsifier. Claims that assert unsupported organizational generalizations, invoke priority language, or lean on an outside institution's endorsement for legitimacy are dropped. Nothing here upgrades the original's status: every kept and reformulated claim remains OPEN or DEFINITION, none is elevated to a demonstrated finding, and no external validation, peer review, or institutional endorsement is proposed as a route to legitimacy.

Keywords: linguistic capital; organizational readout; retention; Elective Connectivity; Human Return; Human–AI mediation; conceptual triage.

1 ANCHOR

This paper is a 2026 uplift edition of *Operational Linguistic Wisdom: Elective Connectivity and Linguistic Capital Activation in the Age of LLMs* (Yaoharee Lahtee, 8 October 2025, DOI 10.5281/zenodo.22308446) [Lahtee, 2025b]. The original record is unchanged and remains the citable artifact of the 2025 conceptual model; this edition does not supersede it, correct an error in it, or claim that the 2025 text was mistaken on its own terms. It re-derives the same conceptual territory under a spine that did not yet exist in October 2025 — the root-to-human readout architecture assembled across *Readout Genesis* [Lahtee, 2026m, Readout Genesis Programme, 2026], *Experience Is the Human LoRA* [Lahtee, 2026h], *Epistemic Fusion or Epistemic Tunnel* [Lahtee, 2026e], *Choice Begins Before Choice* [Lahtee, 2026b], *Potential as a Readout* [Lahtee, 2026k], *CTSA Human-Return Readout* [Lahtee, 2026c], and their flagship synthesis *Before Meaning, Before Choice* [Lahtee, 2026a], audited in *Master Equation River* [Lahtee, 2026i]. Table 4 in Appendix A lists the disposition of every claim in the original.

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2 INTRODUCTION: FROM COMMUNICATION CHANNEL TO AN ORGANIZATIONAL READ-OUT LOOP

Language embodies social relations, value positions, and situated know-how, not merely messages [Lahtee, 2025b]. This is kept as a definitional frame: it is a stance about what language is, not an empirical finding, and it does not conflict with the programme spine, under which meaning is a reader-conditioned significance relation rather than substance carried inside a message [Lahtee, 2026a, eq. 10].

The original paper's central move was to propose that, once large language models are introduced, organizational language "can be designed as cognitive infrastructure that activates organizational linguistic capital" [Lahtee, 2025b]. Read as a bare possibility claim this is not in conflict with anything in the spine, but "cognitive infrastructure" is loose vocabulary once a typed architecture is available. This edition re-expresses it as a claim about an organizational instance of the programme's own retained-difference-to-retention chain: an organization (H_{org}, playing the structural role the spine assigns to a human reader H_n) accumulates retained linguistic traces; an LLM-mediated readout can raise the volume and recursion depth at which those traces are read; and any resulting change is only a retention event, not a settled capability gain, if it survives to alter H_{org} at a later step [Lahtee, 2026a, eq. 17].

Falsifier. An organization's LLM-mediated processing of linguistic traces shows no traceable residue in later organizational readout, policy, or practice (no $H_{\mathrm{org},n+1} \neq H_{\mathrm{org},n}$ attributable to the processing); the claim then reduces to ordinary information throughput with no retention-level effect.

The original also asserted that "organizational practice often instrumentalizes language as information throughput, leaving its capital-like qualities latent" [Lahtee, 2025b]. This is dropped here (see Table 4, claim 6): it is an unsupported generalization about "organizational practice" with no citation, no scope condition, and no way to state what would count against it, which R7 of this edition's governing rules requires to drop rather than keep as background color.

The re-anchoring is timed by the programme's own internal audit rather than by any outside development. *Master Equation River* found that the programme's later papers, read separately, could appear to be a family of loosely related conceptual models unless the chain was anchored one level deeper, in the root retained-difference architecture rather than at whichever domain layer a given paper happened to start from [Lahtee, 2026i]. The original OLW paper, written in October 2025, necessarily started at the organizational-domain layer, because the deeper spine had not yet been assembled. This edition performs, for one organizational-domain paper, the same anchoring operation the flagship derivation performed for the programme's human-domain papers.

2.1 Scope

The claims kept or reformulated below concern organizations that already use, or are considering using, an LLM over their own retained linguistic traces (documents, chat logs, transcripts, meeting notes) under some form of internal governance. Nothing here is a claim about LLM behaviour in general, about organizations that do not use LLMs at all, or about individuals outside an organizational retention context; the individual-scale version of the same architecture is the subject of the flagship derivation and its cited components [Lahtee, 2026a,h,e,b,k,c], not of this paper.

3 LITERATURE GROUNDING: SAME, DIFFERENT, CITED

The original paper positioned itself against Bourdieu's linguistic capital and symbolic power, Nonaka's knowledge-creation model, and Floridi's philosophy of information, closing with the sentence "we address this mechanism gap" [Lahtee, 2025b]. That closing sentence is priority-adjacent language and is not repeated; the underlying scoping observation survives, restated as a same/different/cited comparison in the form the programme's own flagship paper uses for its neighbouring literatures [Lahtee, 2026a, §"What the Literature Already Owns"].

Table 1: Literature grounding, same / different / cited.

Source	Same / different / cited
Bourdieu, linguistic capital and symbolic power [Bourdieu, 1991]	*Same*: language carries capital-like, structuring properties. *Different*: this edition does not extend Bourdieu's macro account of field reproduction; it only asks how an organization's own retained linguistic trace becomes accessible to an LLM-mediated readout at micro scale. *Cited*, not built upon.
Nonaka, SECI tacit–explicit conversion [Nonaka, 1994]	*Same*: knowledge creation runs through conversion between tacit and explicit form. *Different*: this edition treats the LLM-mediated dialogue itself as one instance of the programme's own candidate-status Human–AI loop (Section 5), not as a fifth SECI mode. *Cited*, not merged.
Floridi, philosophy of information [Floridi, 2011]	*Same*: digital mediation reshapes the conditions of knowing. *Different*: no claim is made that LLMs generally "intensify" this beyond a stated, falsifiable scale claim (below). *Cited*, not extended.

The claim that "LLMs intensify this through recursive operations over language" [Lahtee, 2025b] is kept only as a scale hypothesis, tied to a rival-model style falsifier rather than asserted as an established effect.

Falsifier. Organizations using LLM-mediated recursive discourse review show no measurable increase in surfaced tacit or latent content, and no increase in successfully fed-back, policy-

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relevant clarifications, relative to matched manual-review or ordinary discourse-review baselines of the same trace volume.

4 Position Within the Programme

This chapter is deposited alongside a family of papers produced under the same programme and, in several cases, the same 8 October 2025 batch: *The Language Bridge: Expanding Human Potential in the Age of AI* [Lahtee, 2025a], *Human Learning as Epistemic Architecture* [Lahtee, 2026g], *When AI Expands Human Potential* [Lahtee, 2026n], *The Readout Condition* [Lahtee, 2026l], and, in the same uplift series as this chapter, *Written by AI, Still True* [Lahtee, 2026o]. Their titles indicate a shared concern with language, learning, and epistemic agency under AI mediation, consistent with the same title-level territory this chapter occupies; this edition has not re-read their full text for this triage and therefore does not import a specific equation, finding, or claim from any of them beyond what is already anchored through [Lahtee, 2026a,i] above. They are named here so the citation network around this chapter is visible, not to borrow unverified content from them: under this programme's own readout-not-truth discipline, a title is a readout of a paper, not the paper.

Two siblings are used more substantively because their content was read in full for the present triage, via their treatment inside the flagship derivation: *Meaning Before Naming* and *Experience Is Meaning-Giving* supply the meaning and experience layer that any activated linguistic trace must pass through before it can be called an organizational insight [Lahtee, 2026a, eqs. 10, 12], drawing on [Lahtee, 2026j,d]; Capability Activation (Section 5) produces a knowledge-like candidate, not yet a meaning, and this edition does not claim that a candidate becomes organizationally meaningful merely by being produced.

5 Reformulated Model: Three Stages on the Readout Spine

The original's three-stage mechanism is kept, because it already has a workable structural analogue in the spine once each stage is typed against a specific transition rather than left as a free-standing organizational metaphor.

Elective Connectivity. The original defines this as "intentional, consent-first connection of linguistic assets to LLM use-cases" [Lahtee, 2025b]. This is the organizational analogue of the spine's prompt-construction step, in which a bounded query is drawn from a pre-existing state under a declared scope rather than treated as an unconditioned starting point:

$$H_{\text{org},t} \xrightarrow{\text{consent, scope}} Q_{\text{org},t}, \qquad Q_{\text{org},t} \neq H_{\text{org},t}, \tag{1}$$

following the general form $H_t \xrightarrow{L_H} Q_t$, $Q_t \neq H_t$ [Lahtee, 2026a, eq. 29]. The consent and scope condition is the organizational reading of that arrow's governance requirement, not an additional mechanism.

Capability Activation. The original defines this as "routines that convert latent linguistic assets into actionable insights" [Lahtee, 2025b]. Under the spine's candidate contract, any LLM output re-entering an organizational dialogue is knowledge-like, not validated knowledge, regardless of fluency or citation [Lahtee, 2026a, eq. 31]:

$$\text{AI}(Q_{\text{org},t}) = K_{\text{like}}, \qquad K_{\text{like}} \neq K_{\text{validated}}. \tag{2}$$

A dialogic prompting routine can surface a candidate insight; it cannot, by fluency alone, certify that insight as organizational knowledge.

Reflexive Reconfiguration. The original defines this as "institutionalized learning loops that update policies and processes" [Lahtee, 2025b]. This is the organizational analogue of retention, $H_{\text{org},t+1} = U_{H_{\text{org}}}(H_{\text{org},t}, \ldots)$ [Lahtee, 2026a, eq. 16], and it earns the word "learning" only if the resulting policy change survives a Human-Return-style audit after LLM support is withdrawn — the same non-collapse the programme applies at individual scale, "assisted performance $\neq$ unaided Human Return" [Lahtee, 2026a, eq. 40], read here as "LLM-present policy behaviour $\neq$ demonstrated organizational retention."

Falsifier. (joint, for the three-stage claim as a whole) Organizations satisfying (i) consent-governed connectivity, (ii) dialogic candidate-surfacing routines, and (iii) an institutionalized feedback loop show no measurable difference in retained policy change, surfaced tacit content, or value alignment relative to organizations using ungoverned or ad hoc LLM use, once ordinary attention and resourcing are controlled for.

5.1 Construct clarity (kept, definition tier)

The original's operational-definition table is kept without substantive change; it is self-flagged in the source as "conceptual only; indicators illustrative" [Lahtee, 2025b], which already satisfies the requirement that a definition carry no unearned empirical weight.

Table 2: Construct clarity, reproduced from the original (definition tier; indicators illustrative, not validated).

Construct	Definition and illustrative indicator
Elective Connectivity	Intentional, consent-first connection of linguistic assets to LLM use-cases. Indicator: access logs, ethics checklist.
Capability Activation	Routines that convert latent linguistic assets into actionable insights. Indicator: reflection logs, rationale fields.
Reflexive Reconfiguration	Institutionalized learning loops that update policies and processes. Indicator: policy diffs, KPI dashboards.

Read against the spine, each indicator column is a partial, organization-scale analogue of the per-transition provenance ledger $\Lambda(e_i) = \langle \text{Prov}_i, \text{Tier}_i, \text{Def}_i, \text{Reader}_i, \text{Falsifier}_i \rangle$ that the programme requires of every load-bearing arrow [Lahtee, 2026a, eq. 41]; "access logs," "rationale fields," and "policy diffs" are provenance and defect bookkeeping, not yet that full ledger.

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5.2 OLW and AVRH (kept, reformulated)

OLW is kept as a design stance — "a design stance positioning language as a value-reflective operational layer linking cognition and action" [Lahtee, 2025b] — re-anchored between the spine's meaning/experience layer and its live-possibility/choice layer [Lahtee, 2026a, §§"From Retained Record to Meaning and Experience", "Before Choice"]: it names where in an organization's own readout chain a governed LLM practice would have to sit, not an established causal mechanism.

AVRH — "tension between human values and AI outputs can stimulate value reflection and capability growth, provided guardrails exist" [Lahtee, 2025b] — is reformulated against the spine's own Difference–Resistance–Agency constitutive conditions, $D > 0$, Resist > 0, $A_H > 0$ [Lahtee, 2026a, eq. 35]: "tension" is read as Difference, "guardrails" as Resistance, and "capability growth" must be tested as retained organizational Agency rather than assumed from assisted performance.

Falsifier. Organizational capability said to have "grown" from value-tension reflection reverts to baseline once LLM support is withdrawn (fails an organizational Human-Return-style audit [Lahtee, 2026c]), or reflection occurs with no guardrails yet growth is claimed anyway.

5.3 Illustrative vignette (kept, hypothetical)

The original's vignette — a multilingual team adopting Elective Connectivity through consented FAQ/notes whitelisting, Capability Activation through weekly reflective sessions, and Reflexive Reconfiguration through monthly policy updates [Lahtee, 2025b] — is kept as a hypothetical illustration of the mapping above. The original's closing sentence, that "terminology consistency improves and misunderstandings decline," is not kept as a reported outcome: no such outcome was measured, and this edition restates it as a hypothetical outcome that Section 7's propositions would need to test, not as an achieved result.

6 RISKS AND GOVERNANCE REGISTER

The original's four risk-mitigation pairs are kept without substantive change; each is a governance-register entry, not an empirical claim about the world, and each already aligns with a non-collapse rule the spine independently states.

Table 3: Risks and mitigations (kept, definition/governance tier).

Risk	Mitigation; spine parallel
Illusion of Reflection: LLMs may eloquently reproduce bias.	Exemplars, red-teaming prompts, accountable value owners. Parallel: $K_{\text{like}} \neq K_{\text{validated}}$ [Lahtee, 2026a, eq. 31].
Power Dynamics: ethical anchors may be captured by authority.	Diverse review boards, dissent documentation. Parallel: power may alter agency by altering accessibility before overt choice [Lahtee, 2026a, eq. 28].
Decontextualization: activation risks stripping context.	Data minimization, domain stewards. Parallel: the domain-weld defect vector ε_H [Lahtee, 2026a, eq. 7].
Over-automation: avoid using outputs as sanction bases; retain human judgment.	Human-in-the-loop review. Parallel: $K_{\text{like}} \neq K_{\text{validated}}$, again.

7 PROPOSITIONS FOR FUTURE TESTING (OPEN TIER)

The original's eight propositions are already falsifiable as written. This edition keeps their content, tags each explicitly OPEN, and, for P4 and P7, adds a note that separates the organizational hypothesis from spine constructs it should not be conflated with.

P1 — Connectivity quality [OPEN]
Elective Connectivity quality predicts the breadth and safety of linguistic assets available for learning. *Falsifier*: a held-out organizational comparison shows no reproducible association between governance-quality measures and the breadth or safety of the resulting linguistic-asset corpus, beyond ordinary data-hygiene practice.

P2 — Activation clarity [OPEN]
Capability Activation routines predict concept clarity and perceived value alignment. *Falsifier*: structured dialogic-prompting sessions show no incremental clarity or alignment effect beyond ordinary team meetings or documentation review.

P3 — Reconfiguration cadence [OPEN]
Reflexive Reconfiguration cadence predicts policy-update rates and learning-KPI improvements. *Falsifier*: cadence of reflection sessions shows no relationship to policy-diff rate once leadership attention and resourcing are controlled for.

P4 — Reviewer diversity [OPEN]
Reviewer diversity strengthens clarity and alignment (moderation). This is a loose analogy to, not an instance of, the programme's own Independence Ladder for AI-model checking [Lahtee, 2026f], which governs model independence, not human reviewer diversity. *Falsifier*: diversity of reviewers shows no moderating effect once reviewer count and domain expertise are held constant.

P5 — Over-reliance [OPEN]
Over-reliance on LLM outputs weakens the P2–P3 relationships (negative moderation), paralleling $K_{\text{like}} \neq K_{\text{validated}}$ at organizational scale. *Falsifier*: measured reliance on LLM output shows no negative moderation of the P2/P3 relationships.

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P6 — Minimization trade-off [Open]
Data minimization reduces decontextualization risk without lowering activation benefits. *Falsifier*: data minimization reduces measured activation benefit in proportion to, or more than, the risk reduction achieved.

P7 — Dissent protocols [Open]
Formal dissent protocols reduce power-capture risk. This parallels the programme's own non-collapse that diagnosing a compression is not the same as attributing responsibility for it [Lahtee, 2026k,b]: a protocol's existence is not itself proof that capture has been diagnosed or resolved. *Falsifier*: organizations with formal dissent protocols show no reduction in power-capture indicators relative to matched organizations without them.

P8 — SECI complementarity [Open]
SECI-practicing organizations experience complementary gains from OLW routines. *Falsifier*: SECI-practicing and non-SECI-practicing organizations show no differential gain from adopting OLW routines.

8 Managerial Recommendations (kept)

The original's managerial levers are kept as prescriptive, not empirical, content: checklists for Elective Connectivity (data inventory, consent, lawful purpose, governance review); prompt playbooks, reflection logs, rationale documentation, and named value owners for Capability Activation; and scheduled review, policy diffs, learning KPIs, and audit trails for Reflexive Reconfiguration [Lahtee, 2025b]. These parallel the same per-transition ledger discipline referenced in Table 2 and require no falsifier under this edition's rules for a recommended-practice list.

The original's separate policy/ethics paragraph, which framed the model as "aligning with OECD and UNESCO Responsible AI principles," is dropped (Table 4, claim 29). Neither institution appears in the original's own reference list, so it cannot be cited here under this edition's citation rule, and invoking an outside body's principles as a source of alignment is exactly the external-validation framing this program's knowledge-validation stance excludes: what proves a claim is whether it holds against the problem in front of it, not an outside body's endorsement. The values the paragraph gestured at (proportionality, transparency, non-punitive learning) are already carried, without that framing, by the governance register in Table 3.

9 What Is Not Claimed

This edition does not claim that: the three-stage mechanism has been observed in any organization; OLW or AVRH are validated psychological or organizational constructs; any of the eight propositions has been tested; language-as-cognitive-infrastructure has any priority over Bourdieu, Nonaka, Floridi, or any other literature (comparison is stated only as same / different / cited); an LLM's fluency or citation-like output constitutes organizational knowledge; a policy change made while LLM support is present constitutes demonstrated organizational learning, absent a Human-Return-style audit after that support is withdrawn; or that alignment with an outside body's stated principles constitutes evidence for, or certification of, any claim in this paper. It also does not silently repeat a citation-hygiene gap in the original: two of its eight reference-list entries, Sandelowski (2000) and Weick (1995), carry no in-text citation anywhere in the original's body text; this edition does not cite them either, since it makes no use of their content, and does not reproduce them in its own bibliography.

10 Method

Following standards for conceptual research [Whetten, 1989, Suddaby, 2010, Jaakkola, 2020], the original paper identified constructs, formulated propositions, and offered an illustrative vignette without analyzing empirical data [Lahtee, 2025b]. This edition keeps that description of method and adds a triage pass.

10.1 Triage procedure

Each of the original's 31 identifiable claims (Section A) was carried through four steps. (i) *Extraction*: every declarative sentence in the original that asserts a mechanism, a definition, a risk, a recommendation, or a conclusion was isolated as a numbered claim, including claims the original itself flags as conceptual or hypothetical. (ii) *Spine mapping*: each claim was checked against the programme spine assembled after October 2025 (Section 1) for a corresponding typed transition; where one exists, the claim was reformulated to name it explicitly rather than left as an untyped mechanism. (iii) *Falsifier check*: a claim that could not be given a concrete falsifier, and that was not itself a definition, a governance recommendation, or a self-flagged hypothetical, was checked against R7 for whether it should be dropped rather than kept as unfalsifiable background. (iv) *Priority scan*: any claim or clause using priority language (claims of originality, of filling a gap, or of extending a literature) was rewritten in same/different/cited form or dropped if its only content was the priority framing itself. A claim could receive more than one disposition component (for example, a claim kept for its definitional content but rewritten to remove a priority clause is recorded as REFORMULATE, not KEEP_ACCURATE, in Table 4).

This procedure is a K0, single-pass classification: it has not yet been through an independent second check, and no claim's disposition in Table 4 should be read as having survived adversarial review. An AI assistant was used to draft this triage, locate the corresponding spine equations, and produce the LaTeX text under the author's direction; the classification decisions, the falsifiers, and responsibility for the content remain the author's, consistent with this workspace's standing rule that no AI or vendor is credited as an author or co-author of public-facing work.

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11 Conclusion

Language can function as an organizational readout surface once organizations (i) construct LLM queries under consent and scope, (ii) treat LLM output as a knowledge-like candidate rather than validated knowledge, and (iii) let institutionalized feedback update organizational state only when that update survives a Human-Return-style audit. This is the same conditional the original stated in its own conclusion [Lahtee, 2025b], restated so that each clause names a specific, already-typed spine transition instead of an untyped mechanism, and left at Open: a programme statement to be tested, not a conclusion already reached. Future work should not attempt to validate all eight propositions at once; it should test the smallest interface first, most plausibly P2 (does a structured dialogic routine add clarity beyond ordinary review) or P5 (does reliance measurably weaken it), since both are close to already-available organizational instrumentation. If organizations satisfying the three stages show no measurable difference from organizations that do not, the three-stage mechanism should be compressed back into ordinary organizational-learning vocabulary rather than defended by its own terminology.

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A TRIAGE TABLE

Table 4 lists the disposition of every claim identified in the original paper (n = 31), including the four that are dropped in this edition.

Table 4: Triage table: disposition of every claim in the original paper. Legend: K = KEEP_ACCURATE, R = REFORMULATE, D = DROP.

n	Disp.	Reason
1	R	Framing claim, not empirically settled; recast through the retained-difference-to-retention chain (§2) rather than left as an untyped metaphor.
2	R	Central three-stage architecture; each stage explicitly typed against a spine transition (§5) rather than left free-standing.
3	R	Conditional "when...then" claim; recast under OPEN with the joint falsifier in §5.
4	D	"Originality/value" framing is priority-adjacent (forbidden); its substance duplicates claims 1–2, already reformulated.
5	K	General definitional/frame claim about language; consistent with the spine, no falsifier required (§2).
6	D	Unsupported empirical generalization about "organizational practice"; no citation, no falsifier (§2).
7	R	Unfalsifiable as written; recast as an explicit scale claim with a falsifier (§3).
8	R	"We address this mechanism gap" is priority language; underlying scoping observation kept as same/different/cited (§3).
9	R	Bourdieu attribution accurate and kept; "we pivot" clause recast via same/different/cited (Table 1).
10	R	Nonaka attribution kept; "we complement SECI" recast via same/different/cited, tied to the typed Human–AI loop (Table 1).
11	R	Floridi attribution kept; "LLMs intensify this" given an explicit falsifier rather than asserted (§3).
12	R	Self-flagged "conceptual usage; no canonical definition asserted"; the "dark data" label is not carried forward, only its content, restated as the accumulation of retained linguistic traces an organization does not read (§2).
13	K	Self-flagged "conceptual only; indicators illustrative"; kept as construct-clarity table (Table 2).
14	R	Definition kept, re-anchored between the spine's meaning/experience and live-possibility/choice layers (§5).
15	R	Reasonable OPEN hypothesis; mapped onto the Difference–Resistance–Agency conditions with an explicit falsifier (§5).
16	K	Accurate self-description of conceptual-research method, citing sources present in the original's own reference list (§10).
17	R	Vignette kept as hypothetical; its closing sentence rephrased so an unmeasured outcome is not read as achieved (§5).
18	K	Four risk/mitigation notes; governance-register entries, not empirical claims; already aligned with spine non-collapse rules (Table 3).
19	R	Already falsifiable; tagged OPEN (P1, §7).
20	R	Already falsifiable; tagged OPEN (P2, §7).
21	R	Already falsifiable; tagged OPEN (P3, §7).
22	R	Legitimate moderation hypothesis; made explicit that it is an analogy to, not an instance of, the programme's Independence Ladder (P4, §7).
23	R	Already falsifiable; tagged OPEN (P5, §7).
24	R	Already falsifiable; tagged OPEN (P6, §7).
25	R	Already falsifiable; tagged OPEN, with a non-collapse note so protocol existence is not read as capture resolved (P7, §7).
26	R	Already falsifiable; tagged OPEN (P8, §7).
27	D	"Positions...in LLM contexts" plus "extends...theories" is an originality claim with no falsifier; substance duplicates claims 1, 8–11.
28	K	Prescriptive managerial recommendation list; no falsifier required (§8).
29	D	Invokes OECD/UNESCO, neither in the original's reference list; external-alignment-as-legitimacy framing excluded by this programme's knowledge-validation stance (§8).
30	K	Field studies are data collection, not outside certification; consistent with internally-driven next steps (§11).
31	R	Duplicate of claim 3's conditional, folded into §11 and explicitly marked OPEN rather than a settled conclusion.

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References

Pierre Bourdieu. *Language and Symbolic Power*. Harvard University Press, 1991.

Luciano Floridi. *The Philosophy of Information*. Oxford University Press, 2011.

Elina Jaakkola. Designing conceptual articles: Four approaches. *AMS Review*, 10:18–26, 2020.

Yaoharee Lahtee. The language bridge: Expanding human potential in the age of AI. Zenodo, 6 October 2025, 2025a.

Yaoharee Lahtee. Operational linguistic wisdom: Elective connectivity and linguistic capital activation in the age of LLMs. Zenodo, Conceptual–Reflective Paper, 8 October 2025, 2025b.

Yaoharee Lahtee. Before meaning, before choice: A readout-native derivation of experience, live possibility, agency, and human–AI return. Zenodo, Full Paper v1.1, 6 September 2026, 2026a.

Yaoharee Lahtee. Choice begins before choice: Meaning-shaped accessibility, live possibility, and effective agency in human-AI systems. Zenodo, Full Paper v1.0, 5 September 2026, 2026b.

Yaoharee Lahtee. CTSA human-return readout: A session-boundary measurement architecture for retained human capability after AI. Zenodo, Epistemic Note v9.1 FINAL, 5 September 2026, 2026c.

Yaoharee Lahtee. Experience is meaning-giving: A strong-form readout-retention theory of phenomena, resonance, rhythm, accumulation, and transformative release. Zenodo, Full Paper v2.0, 5 September 2026, 2026d.

Yaoharee Lahtee. Epistemic fusion or epistemic tunnel: A phenomenology-anchored, global-literature-constrained, problem-first architecture of human-AI collaboration. Zenodo, Epistemic Note v8.1, 5 September 2026 (concept DOI 10.5281/zenodo.22318039), 2026e.

Yaoharee Lahtee. glosa — rigour without infrastructure: A standalone scholar methodology for human–AI knowledge co-production (v0.4.1). Zenodo software record, 2026f.

Yaoharee Lahtee. Human learning as epistemic architecture: A method for word mapping, life-concept graphs, constraint testing, and corrigible agency in the AI age. Zenodo, revised v4, 28 June 2026, 2026g.

Yaoharee Lahtee. Experience is the human LoRA: A readout–retention theory of selective model change. Zenodo, 18 July 2026, 2026h.

Yaoharee Lahtee. Master equation river: A provenance audit across the human–AI readout programme. Zenodo, Internal Programme Synthesis v1.1 (glosa-checked), 6 September 2026, 2026i.

Yaoharee Lahtee. Meaning before naming: A readout-retention architecture of affective-semantic reorganization. Zenodo, Concept Note v1.0, 5 September 2026, 2026j.

Yaoharee Lahtee. Potential as a readout: Witnessed envelopes, two-layer agency, and the measurement of pseudo-peace. Zenodo, Epistemic Note K0 v2, 5 September 2026 (concept DOI 10.5281/zenodo.22345708), 2026k.

Yaoharee Lahtee. The readout condition: Distinguishability, access, and epistemic warrant. Zenodo, 2026l.

Yaoharee Lahtee. Readout genesis universal technical whitepaper, v1.2.0: universal research architecture, typed interfaces, domain gates, and anti-drift constitution. Public GitHub repository morrocwi/readout_genesis, file READOUT_GENESIS_UNIVERSAL_TECHNICAL_WHITEPAPER_v1.2.0.md, main at commit 04cde19be2c885a11b42597b1cdb60fb5b7ca1bb, blob 6ca7ef20a1e9ea17632451e6c8a19b3ae7c9fa8a, 2026m. `https://github.com/morrocwi/readout_genesis/blob/04cde19be2c885a11b42597b1cdb60fb5b7ca1bb/READOUT_GENESIS_UNIVERSAL_TECHNICAL_WHITEPAPER_v1.2.0.md`. Working specification; not a Zenodo record.

Yaoharee Lahtee. When AI expands human potential: Reflective dissonance, epistemic agency, and constraint. Zenodo, 25 March 2026, 2026n.

Yaoharee Lahtee. Written by AI. still true. knower fetishism, epistemic pedigree, and the human face as a bad theory of truth. Zenodo, 31 August 2026, 2026o.

Ikujiro Nonaka. A dynamic theory of organizational knowledge creation. *Organization Science*, 5(1):14–37, 1994.

Readout Genesis Programme. PHILOSOPHY_MEMK_DOMAIN v0.2.0: root-connected semantic–cognitive domain specification. Public GitHub repository morrocwi/readout_genesis, branch feat/register-philosophy-memk-v0.2.0, commit 49e444cc64bdaea2767c2c18d209654cdf6cb23f, file domains/philosophy/PHILOSOPHY_MEMK_DOMAIN_v0_2_0.yaml, blob 3d51fe8095f93207ab3ff4ae054042b6cf880a3a, SHA-256 1674a68e8b2811c80dc043ce82aa63186cb0c433a7bb0bf08c858ad6e87c1229, 2026. With CHECKSUMS_SHA256_PHILOSOPHY_MEMK.txt, CLAIM_BOUNDARY_PHILOSOPHY_MEMK.json and PROOF_RECEIPT_PHILOSOPHY_MEMK.json in the same tree.

Roy Suddaby. Construct clarity in theories of management. *Academy of Management Review*, 35(3):346–357, 2010.

David A. Whetten. What constitutes a theoretical contribution? *Academy of Management Review*, 14(4):490–495, 1989.

10.5281/zenodo.22456487

The Dialogue as the Ground of Enlightenment: Religious and Cognitive Frameworks for Understanding Wisdom through Human-AI Conversations

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What the chapter states. "The original paper asked whether wisdom arises from effective dialogue and proposed that dialogue, human or AI-mediated, functions as a mirror of thought. That mirror-of-thought thesis survives the uplift, restated without the two citation defects the original carried."

Status. "Status: K0 (author-asserted, not yet independently checked at or above independence class I2)."

Falsifier(s). "If a study finds that reflective-versus-superficial outcome quality is fully explained by a simpler rival model — for instance, prior domain knowledge and time-on-task alone, without any distinct reader-conditioned meaning-giving step — the mirror-of-thought thesis as stated here should be compressed into that simpler model rather than defended as a separate construct."

Read before. — (no continues/isContinuedBy/references edge to another chapter of this book found in the knowledge graph; the record is not yet indexed there).

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The Dialogue as the Ground of Enlightenment

Reflective Dialogue, the AI-as-Mirror Principle, and Human Return

Uplift edition 2026 of the 2025 paper "The Dialogue as the Ground of Enlightenment: Religious and Cognitive Frameworks for Understanding Wisdom through Human–AI Conversations" (DOI 10.5281/zenodo.22308451)

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Article type Uplift edition (re-derivation and triage of a prior conceptual–comparative paper against the programme's current spine) • **Series** When AI Expands Human Potential (hub DOI 10.5281/zenodo.22308201); textbook *Written by AI. Still True.* (uplift edition 2026) • **Licence** CC BY 4.0

Status K0, uplift edition, September 2026, of the 6 October 2025 original (DOI 10.5281/zenodo.22308451). This is not a revision of the original's findings; it is a re-derivation of what remains accurate against the programme's current spine (Master Equation River v1.1, Lahtee, 2026c; Before Meaning, Before Choice v1.1, Lahtee, 2026d), keeping only claims that survive an internal-consistency and spine-mapping check, dropping the rest with a stated reason, and restating what is kept in the spine's own typed vocabulary. It makes no priority claim over the traditions or theories it discusses: comparison is stated only as same / different / cited.

Central claim. The original paper asked whether wisdom arises from effective dialogue and proposed that dialogue, human or AI-mediated, functions as a mirror of thought. That mirror-of-thought thesis survives the uplift, restated without the two citation defects the original carried: an in-text claim that a cited study showed AI dialogue "facilitates reflective learning" when the same study's own title, printed in the original's own reference list, reports the opposite finding, and an in-text citation ("Harvard Business Review, 2025") with no corresponding entry anywhere in the original's own reference list. Once those two defects are removed and the remaining thesis is restated in the programme's typed vocabulary — meaning as a reader-conditioned significance relation, AI output as knowledge-like candidate rather than validated knowledge, and durable gain measured only after AI is removed — the chapter's residual claim is narrow: reflective, iterative human-AI dialogue can support the same retained-difference architecture that supports reflective human dialogue, but it does so only through the human's own interpretive and retentive labour, and only demonstrably so if it survives an unaided Human Return test.

ABSTRACT

This is the 2026 uplift edition of a short conceptual–comparative paper (Yaoharee Lahtee, Araya Nikah Social Enterprise Co., Ltd., Bangkok, Thailand, 6 October 2025; DOI 10.5281/zenodo.22308451) that surveyed three religious traditions of dialogical wisdom — Buddhist *yonisomanasikāra*, Islamic *sohbah*, and Christian I–Thou dialogue — alongside four pedagogical and dialogical theories (Freire, Vygotsky, Hermans, Schön), then extended the discussion to human–AI conversation and proposed a self-authored concept, "Digital Yonisomanasikāra" (Framing, Iteration, Integration), contrasted with a shallow "Workslop" mode of AI use. This edition keeps every claim about the three traditions and the four pedagogical theories unchanged in substance, because none of them conflicts with the programme's current spine and all are independently citable. It drops three claims outright: a literature-gap claim that is unfalsifiable and carries priority language; an in-text claim about AI-driven reflective learning whose own cited source, by its own printed title, reports the opposite finding; and an in-text citation ("Harvard Business Review, 2025") that does not exist anywhere in the original's own reference list. It reformulates the remaining substantive claims — the mirror-of-thought thesis, the AI-as-mirror-not-agent thesis, and the Digital Yonisomanasikāra procedure — into the programme's typed vocabulary: meaning as a reader-conditioned significance relation, AI output as a knowledge-like candidate rather than validated knowledge, and any claimed human gain as provisional until it survives an unaided Human Return test after AI is removed. Every claim carries a falsifier or is marked as a definition/frame. Status: K0 (author-asserted, not yet independently checked at or above independence class I2).

Keywords: dialogue; wisdom; reflection; human–AI interaction; yonisomanasikāra; sohbah; I–Thou; candidate knowledge; Human Return; claim triage.

1 ANCHOR

This chapter is the 2026 uplift edition of *The Dialogue as the Ground of Enlightenment: Religious and Cognitive Frameworks for Understanding Wisdom through Human–AI Conversations* (Yaoharee Lahtee, Araya Nikah Social Enterprise Co., Ltd., Bangkok, Thailand, dated 6 October 2025, DOI 10.5281/zenodo.22308451). The original is a short conceptual–comparative paper with no numbered equations and no empirical study; it is treated here as a set of textual claims, each triaged against the programme's current spine and either kept, reformulated, or dropped. Section A lists every claim in the original with its disposition and reason, dropped claims included. This edition does not supersede the original as a historical record: the original DOI remains the record of what was written in October 2025. This edition is the record of what, under the methodology described in Section 8, still holds up in September 2026.

2 Introduction: The Question, Without the Gap Claim

The original paper's motivating question survives as a frame rather than as an empirical premise: does wisdom arise from effective dialogue, and if artificial intelligence mediates everyday communication, can it participate in that process, or does it produce only data exchange dressed as insight? Both halves of the question are kept as this chapter's guiding frame, marked DEFINITION rather than as settled claims, because a motivating question carries no falsifier of its own.

The original then moved directly to a literature-gap claim: that "few have examined AI as an epistemic tool capable of cultivating wisdom in the philosophical or religious sense" and that the paper "addresses this gap." That sentence is dropped in this edition (Section A, claim 3). No literature search was run to verify how many studies examine this question, so "few have examined" is an unverified count-claim under the readout-not-truth discipline this programme applies to its own writing; and "addresses this gap" is gap-filling language of exactly the kind this programme's comparison discipline replaces with same/different/cited statements against named neighbours (Lahtee, 2026d, its own contribution-boundary table). Removing the sentence changes nothing about the paper's actual content: the three traditions and four theories discussed in Section 4 are unaffected by whether or not a gap exists in some unsearched literature.

3 Reflective Dialogue as a Mirror of Thought

The original's central thesis was that dialogue, whether with humans or with AI, "functions as a mirror of thought": when grounded in reflection and critical inquiry it fosters cognitive transformation and self-awareness, but when driven by haste or bias it produces superficial reasoning and epistemic distortion. This thesis is kept, but restated so that "mirror of thought" is not a metaphor doing unexamined work.

In the programme's spine, a finite human readout of an encountered trace becomes significant only relative to a declared reader, context, and question, not as a substance carried inside the trace itself:

$$\mu_n = \Psi_H(\rho_n^H, H_n, c_n, Q_n) \tag{1}$$

(Lahtee, 2026d, eq. 10; Lahtee, 2026c). Under this reading, a dialogue does not itself contain wisdom or distortion; it supplies a trace x_n that a given human, in a given context, with a given active question, reads into a meaning μ_n. "Reflection and critical inquiry" name conditions on H_n and Q_n — a reader who brings a live question and a disposition to check rather than accept — under which the same dialogue trace tends to support a more examined μ_n; "haste or bias" name conditions under which it does not. The mirror is therefore not a property of the dialogue medium (human speech or AI text) but a property of the reading relation Ψ_H as parameterised by the reader's own state.

This reformulation also explains why the original's own three religious traditions and four pedagogical theories, surveyed independently of any human–AI question, converge on the same structural point (Section 4): all of them locate the transformative work in the quality of the reader's attention, intention, or reflective stance, not in the medium of exchange.

Falsifier. If a study finds that reflective-versus-superficial outcome quality is fully explained by a simpler rival model — for instance, prior domain knowledge and time-on-task alone, without any distinct reader-conditioned meaning-giving step — the mirror-of-thought thesis as stated here should be compressed into that simpler model rather than defended as a separate construct (Lahtee, 2026d, its own rival-model ladder, § "Adversarial Tests and Rival Models").

4 Dialogue Across Traditions and Pedagogies

The following seven claims from the original are kept without substantive change. They sit outside the programme's technical spine (no Master Equation River construct is required to state them), they do not conflict with the current spine, and each is independently citable from the original's own reference list. They are marked DEFINITION/DR: attributed descriptions of, or a comparative synthesis across, cited traditions and theories, not empirical claims of this programme.

Buddhist *yonisomanasikāra*. In Buddhism, dialogue is central to the cultivation of *paññā* (wisdom). Question-and-answer methods are used to stimulate *yonisomanasikāra* — deep, analytical contemplation connecting cause and effect [Buddhadasa Bhikkhu, 1997]. Wisdom is presented as arising from mindful listening and reflection rather than from acceptance of authority.

Islamic *sohbah*. Learning circles known as *sohbah* emphasise *ikhlāṣ* (sincerity) and *tafakkur* (thoughtful reflection). Knowledge without sincerity is described as spiritual vanity; learning occurs when dialogue purifies intention and cultivates humility before truth [Al-Ghazali, 2015].

Christian I–Thou dialogue. Buber's I–Thou dialogue is described as a relational space in which individuals encounter the divine through genuine openness, making dialogue an act of communion rather than persuasion [Buber, 1970].

Cross-tradition synthesis. All three traditions converge on a shared point: wisdom is presented as arising not from asserting knowledge but from listening deeply and engaging ethically with the other. This is the original author's own comparative synthesis across the three cited sources, kept as an interpretive claim rather than an empirical one.

Freire. Learning is presented as dialogic co-construction between teacher and student, with knowledge arising through equality rather than hierarchy [Freire, 1970].

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Vygotsky. The Zone of Proximal Development is the space between current understanding and potential learning, bridged through interaction with a more knowledgeable other; dialogue is the mechanism that turns potential into actual understanding [Vygotsky, 1978].

Hermans and Schön. Dialogical Self Theory treats identity as a system of interacting "voices," with reflection occurring when internal voices engage each other [Hermans, 2001]; the reflective practitioner learns through continuous reflection on action within a community of dialogue [Schön, 1983].

None of these seven claims requires a falsifier under Section 3's equation (1): they are definitional/attributive descriptions of what named traditions and theories hold, not this chapter's own testable predictions.

5 Human–AI Dialogue, Reflective Practice, and the Limits of Reflection

The original's account of human–AI dialogue rested on two citations that do not survive inspection of the original's own reference list, plus two substantive claims that do survive once separated from those citations.

5.1 A citation that reports the opposite finding

The original wrote: "Recent research in *Computers in Human Behavior* indicates that AI-driven dialogue can facilitate reflective learning. Studies show that interacting with conversational AI encourages users to externalize thoughts, identify biases, and refine reasoning" and cited a single source for this claim. That source's own title, printed in the original's own reference list, is: "Cognitive ease at a cost: LLMs reduce mental effort but compromise depth in student scientific inquiry" [Stadler et al., 2024]. A title reporting reduced mental effort and compromised depth cannot, on its own terms, be read as support for a claim that the same interaction "facilitates reflective learning" and helps users "refine reasoning." This is an internal inconsistency visible from the original document's own reference list; no additional literature search was needed to find it, and none was run to adjudicate the underlying empirical question either way. The claim is dropped (Section A, claim 13); this edition takes no position on whether AI-driven dialogue does or does not facilitate reflective learning in general, only that the cited source does not establish it as the original stated it.

5.2 A citation that does not exist in the original's reference list

The original also wrote that "the phenomenon of Workslop — inefficient overuse of AI tools leading to cognitive overload" is documented by "Harvard Business Review, 2025; OECD, 2024," and used this to argue that poor conversational habits diminish reflection. "Harvard Business Review, 2025" does not appear anywhere in the original's own thirteen-entry reference list. Under this programme's citation rule, a source not present in the citable reference set cannot be used to support a claim. The remaining citation, an OECD 2024 report on the impact of artificial intelligence on productivity, distribution, and growth, does not by its own subject matter address cognitive overload or conversational habits, so it cannot alone carry the claim once the uncitable source is removed. This claim is dropped (Section A, claim 15). The label "Workslop" is retained in Section 6 only as a generic descriptive name for a contrasting mode of shallow AI use, not as a term backed by this citation.

5.3 What survives: reader-conditioned reflection quality

Independent of both dropped citations, the original's qualifying sentence stands on its own: "the quality of reflection depends on user intention and question design — AI becomes a cognitive mirror only when approached with critical awareness" [Floridi, 2019, UNESCO, 2021]. This restates Equation (1): μ_n is conditioned on H_n, c_n, and Q_n, so the same AI output Y_t can support different qualities of reflection depending on the reader's own intention and question design. **Falsifier.** If reflection quality is found to be invariant to user intention and question design under matched AI outputs, on a rival-model test of the kind [Lahtee, 2026d] sets out, this claim should be narrowed or dropped.

The original's remaining discussion-section claims, once separated from the two dropped citations, also survive: effective dialogue with AI "requires deep listening, critical inquiry, and iterative reflection" [Rogers, 1961, Mezirow, 1991, Hermans, 2001], and in AI-mediated contexts these processes "unfold through human intentionality" because AI does not possess awareness but can externalise the user's own reasoning for inspection. The first half is kept as a definitional description of three named reflective practices (deep listening, critical inquiry, reflective integration), already independently citable (Section A, claim 18). The second half is reformulated: "AI does not possess awareness" is kept as a definitional premise consistent with the candidate-status principle (Equation (3) below), and "catalyzes meta-awareness in the human interlocutor" is kept as an empirical claim with a stated falsifier. **Falsifier.** The meta-awareness claim fails for a given exchange if the externalised reasoning pattern does not survive an unaided Human Return test once the AI is removed (Section 7) — that is, if the apparent gain is present only as assisted performance rather than as something the human can reconstruct alone.

6 Digital Yonisomanasikāra as a Working Definition

The original proposed, attributed to itself as a 2025 finding ("Lahtee (2025) proposes"), a concept called Digital Yonisomanasikāra: a structured mode of reflective questioning mediated through AI, in three steps — Framing (defining the moral or cognitive intention behind the dialogue), Iteration (recursive questioning to surface hidden assumptions), and Integration (translating reflection into ethical or behavioural

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change) — contrasted with a shallow "Workslop" mode that accelerates output without insight.

This edition keeps the three-step structure as a working definition, not as an attributed external finding: the self-citation is dropped (Section A, claim 20), because citing one's own earlier paper as the discovery of a concept is priority language of the kind this programme's citation rule excludes, and because the concept is more usefully stated here as this chapter's own working definition, mapped onto the programme's spine rather than asserted as a prior result:

- **Framing** maps onto the pre-prompt human state producing a bounded articulation, $H_t \xrightarrow{L_H} Q_t$, $Q_t \neq H_t$ (Lahtee, 2026d, eq. 29): the moral or cognitive intention behind a dialogue turn is part of H_t, not of the prompt text Q_t itself.
- **Iteration** maps onto the recursive dialogue stepper
$$Z_{\text{dlg}}[s,n+1] = F^{\#}_{\text{dlg}}\big(Z_{\text{dlg}}[s,n], u_H[s,n], u_{\text{AI}}[s,n], c[s,n], T[s,n]\big) \tag{2}$$
(Lahtee, 2026c, eq. 31): each turn updates a shared dialogue state from the prior state, the human's and the AI's inputs, context, and lineage, rather than resetting at every exchange.
- **Integration** maps onto selective retention, $H_{n+1} = U_H(H_n, \text{Retain}(E_n, \mu_n, \rho^H_n), \delta^{\text{world}}, X^{\text{other}})$ (Lahtee, 2026d, eq. 16), with the explicit non-guarantee $\text{Retain}(E_n) > 0 \Rightarrow H_{n+1} \neq H_n$ *possible, not guaranteed* (Lahtee, 2026d, eq. 17): translating a reflective exchange into behavioural change requires that a residue actually be retained, which is not automatic.

Digital Yonisomanasikāra is marked DEFINITION: a proposed structured procedure, not an empirical claim. Its incremental value over simpler prompting or questioning strategies is OPEN and would require the rival-model test of [Lahtee, 2026d] if it were ever advanced as more than a definition.

The contrast with "Workslop" (Section 5.2) is kept only as a definitional contrast between a low-retention interaction mode and a mode aimed at producing retained residue, not as a claim backed by the dropped citation: no measured effect of "Workslop" is asserted here, only a named distinction between two usage patterns (Section A, claim 21).

6.1 Two illustrative dialogues, reworked as illustration, not evidence

The original offered two short contrasting dialogue prompts to make the Framing/Iteration/Integration structure concrete: a Workslop-mode prompt ("Summarize 10 points of Buddhist philosophy," yielding quick output with no reflection) against a Digital Yonisomanasikāra-mode prompt ("How does *yonisomanasikāra* differ from analytical thinking in modern psychology?," intended to encourage contemplation, integration, and value alignment). This edition keeps both examples, but relabels them explicitly as illustration of the working definition, not as evidence that the contrast holds in general. Two prompts are not a study; nothing about response length, retrieval versus contemplation, or downstream retention can be inferred from a single unreplicated pair of examples. If this contrast is ever advanced as more than illustration, the minimal test is the one already given for claim 16 in Section 5: hold topic and model constant, vary only the framing move, and measure whether Integration-style prompts produce a higher rate of retained, later-reconstructable understanding under an unaided Human Return check (Lahtee, 2026a) than Workslop-style prompts do.

6.2 Minimal empirical predictions for the working definition

Following the programme's own discipline of stating the smallest interfaces at which an architecture could lose (Lahtee, 2026d, its own adversarial-tests section), two predictions would give Digital Yonisomanasikāra empirical content beyond a naming exercise, both OPEN:

P1 — Framing residual
Under matched topic and matched AI model, dialogues that open with an explicit statement of moral or cognitive intention (Framing) will show higher rates of retained, unaided reconstruction at a later session than dialogues that open with a bare information request, after controlling for prior topic familiarity and response length.

P2 — Iteration without Integration is insufficient
Recursive questioning (Iteration, Equation (2)) that is not followed by an explicit translation step (Integration, i.e. a retained update to H_{n+1}) will show no reliable advantage over a single-turn exchange on the same unaided-reconstruction measure; iteration alone should not be mistaken for retained gain.

If P1 and P2 are not confirmed under a rival-model comparison against simpler prompting-strategy baselines, Digital Yonisomanasikāra should be compressed back into "ask a more specific, intention-stated question" rather than retained as a named three-step procedure.

7 AI AS MIRROR, NOT MORAL AGENT

The original argued that AI, lacking intentionality and consciousness, cannot participate in genuine wisdom formation as a source; its role is instrumental, and wisdom arises from the human capacity to interpret and moralise the interaction. This is the original's strongest point of alignment with the programme's own architecture and is kept, restated in typed vocabulary.

Under the candidate contract, AI output is knowledge-like, not knowledge:

$$\text{AI}(Q_t) = K_{\text{like}}, \qquad K_{\text{like}} \neq K_{\text{validated}} \tag{3}$$

(Lahtee, 2026d, eq. 31). Fluency, apparent depth, or the register of a wisdom tradition do not upgrade this status by themselves. This is also the sense in which "knower fetishism" — treating a text as more warranted because of who or what produced it, rather than because of what independently checks it — must be avoided in the other direction as well: a fluent, sagely-toned AI answer about *yonisomanasikāra* or *sohbah* is not thereby closer to wisdom than an unadorned one [Lahtee,

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2026e]. The corresponding non-collapse rule for any claimed gain is that assisted performance is not the same object as unaided Human Return:

$$\text{Assisted performance} \neq \text{Unaided Human Return}. \tag{4}$$

A persistent misconception voiced fluently by an AI, or repeated back fluently by a human who has merely been exposed to it, is still a persistent misconception (Lahtee, 2026a, its own exposure/retention/improvement non-collapse rule). **Falsifier.** The AI-as-mirror-not-agent thesis fails if gains attributed to "AI as mirror" are found to persist identically whether or not a human interpretive step occurs — that is, if the human's own reading and moralising contribution is not separable from the AI's output — on a test of the kind set out in [Lahtee, 2026d] and [Lahtee, 2026a].

7.1 Difference, resistance, and retained human agency

Claim 16's requirement of "deep listening, critical inquiry, and iterative reflection" for effective AI dialogue (Section 5) can now be sharpened using the programme's stipulated conditions for durable human epistemic expansion. Three conditions are non-substitutable for a Human–AI exchange to count as expanding, rather than merely occupying, the human's understanding:

$$D > 0, \qquad \text{Resist} > 0, \qquad A_H > 0 \tag{5}$$

(Lahtee, 2026b; Lahtee, 2026d, eq. 35): a readout-distinguishable difference from the reader's existing frame (D), an independent resistance that can push back on the AI's output rather than merely confirm it (Resist), and retained human agency over the exchange rather than passive reception (A_H). These are constitutive necessities under the stipulated target, not empirical frequency claims: a dialogue with $D = 0$ (the AI restates what the reader already holds), $\text{Resist} = 0$ (nothing in the exchange can disconfirm a wrong turn), or $A_H = 0$ (the human is not exercising interpretive control) does not meet the definition of expansion, whatever its register. "Deep listening" and "critical inquiry" name conditions that tend to keep $\text{Resist} > 0$; "iterative reflection" of the kind formalised in Equation (2) names the mechanism by which D and A_H can accumulate turn over turn rather than resetting. A Workslop-mode exchange (Sections 5.2 and 6.1) is, on this reading, simply one in which at least one of the three conditions is near zero: a quick summarisation request may return a fluent answer with $D \approx 0$ against what an attentive reader already expected, and no Resist is exercised because nothing in the exchange is checked against an independent standard.

7.2 Human Return classes applied to reflective dialogue

The original's discussion-section claim that AI-mediated reflection "unfolds through human intentionality" (claim 19, Section 5) can be given a measurement interface rather than left as an assertion. CTSA separates retained content or access after AI is removed into four classes: Conceptual Return, Tool-Selection Return, Skill-Execution Return, and Alternative-Generation Return [Lahtee, 2026a]. Applied to a reflective dialogue about, say, *yonisomanasikāra* or *sohbah*, these classes predict partial dissociation rather than a single pass/fail outcome: a reader may retain the *concept* that wisdom traditions locate transformation in the quality of attention (Conceptual Return) without retaining the *skill* of applying that quality of attention unprompted in a new dialogue (Skill-Execution Return); or may retain a rule for which kind of question to ask an AI next time (Tool-Selection Return) without being able to generate an analogous question about an unrelated tradition on their own (Alternative-Generation Return). None of these four outcomes is asserted to occur; they are the minimal pattern that would make a Human Return check on Digital Yonisomanasikāra (Section 6.2) explanatory rather than a restatement of "the dialogue seemed reflective." This edition makes the following explicit non-claims, several of them corrections of language the original used.

- **No priority claim.** Nothing in Section 4 is claimed as newly discovered by this chapter or its 2025 original; each tradition and theory is attributed to its own cited source, and the comparison across them is stated as a synthesis, not a discovery.
- **No literature-gap claim.** The original's claim that "few have examined AI as an epistemic tool capable of cultivating wisdom" is dropped (claim 3); this edition does not assert that such a gap exists, because no search was run to check it.
- **No claim that AI possesses awareness.** Equation (3) and the surrounding discussion in Section 7 are explicit that AI output is candidate-status only; nothing here treats AI as a knower or a moral agent.
- **No claim resting on the two defective citations.** The claim that a specific cited study shows AI dialogue "facilitates reflective learning" is dropped because the cited source's own title reports the opposite (claim 13, Section 5.1). The claim that "Workslop" is documented by "Harvard Business Review, 2025" is dropped because no such reference-list entry exists (claim 15, Section 5.2).
- **No civilisational or aspirational claim.** The original's closing sentence — that a societal "Culture of Dialogue" could make AI "a tool not of distraction but of enlightenment: a new Sohbah of the digital age" — is dropped (claim 27). It is a speculative, unfalsifiable claim at civilisational scale, with no operational content, and its phrasing ("a new Sohbah of the digital age") risks reading as a uniqueness flourish this programme's citation and comparison rules exclude.
- **No external-validation framing.** Nothing in this edition treats peer review, institutional endorsement, or outside authority as what would make any of the above claims more or less warranted; warrant is treated as internal consistency with the cited sources and the programme's own spine, checked here, with the falsifiers stated above left open for anyone to run.

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8 Methods and Source Lineage

This edition was produced by a conceptual–comparative triage method with three stages, matching the original's own stated method but restated without a self-assessment of rigour: (1) each claim in the 2025 original was isolated as a discrete unit; (2) each unit was checked against the original's own reference list for citation accuracy, and against the programme's current spine (Lahtee, 2026c,d) for consistency; (3) each unit was assigned one of three dispositions — kept unchanged, reformulated into spine vocabulary with a stated falsifier, or dropped with a stated reason — recorded in Section A. The original's claim that its three-stage method itself "ensures methodological rigor through systematic comparison" is dropped as an unfalsifiable self-assessment (claim 17); the three stages themselves are kept as a plain description of what was done.

AI-assistance disclosure. The triage pass, the source-lineage check against the original's own reference list, and the drafting of this edition's text were carried out with an AI assistant under the author's direction; the original 2025 argument, the decision of what to keep, reformulate, or drop, and the falsifiers stated above are the author's. No AI or vendor is an author, co-author, or contributor of this work.

9 Author Positioning and Conclusion

The original described itself as written from the perspective of a practitioner-scholar bridging theoretical inquiry and applied experience in interfaith education and AI ethics, with the author's affiliation with Araya Nikah Social Enterprise Co., Ltd. situating the work in a living context of multicultural dialogue. That factual positionality is kept. The original's further self-assessment — that this dual standpoint "enhances the contribution's depth and practical relevance" — is dropped as unfalsifiable self-praise (claim 23); a positionality disclosure states where the author stands, not how good the resulting work is.

The original's closing thesis is kept, restated: wisdom, in both spiritual and cognitive domains, is presented as arising not from possession of knowledge but from reflective dialogue, a process of mutual transformation through inquiry, listening, and awareness (claim 24); AI can participate in this process only as a mirror, never as the source (claim 25, restating Equation (3)). The original's normative conclusion is also kept as an explicit ethical stance rather than an empirical claim: the task is not to humanise AI, but to humanise the dialogue — to cultivate sincerity, patience, and reflection amid the acceleration of technology (claim 26). This is marked DEFINITION: a value commitment the author is stating, not a prediction requiring a falsifier, and it does not conflict with anything in the current spine.

What does not survive is the original's final aspirational sentence about a societal "Culture of Dialogue" turning AI into "a new Sohbah of the digital age" (claim 27, Section 7.2). Removing it leaves the chapter's actual residual claim narrower and, this edition argues, more defensible: reflective dialogue, whether between humans or between a human and an AI system, functions as a mirror in the sense given by Equation (1) — it can support cognitive transformation and self-awareness when the reader brings reflection and critical inquiry, and it can produce superficial reasoning when the reader does not — and any claim that an AI-mediated exchange has actually expanded a person's understanding remains open until it is checked against what that person can do, explain, or generate on their own once the AI is no longer there.

A Triage Table

Table 1 lists every claim identified in the 2025 original, its disposition in this edition, and the reason. Keep = kept without substantive change; Reformulate = restated in the programme's typed vocabulary with a stated falsifier; Drop = removed, with reason. Claim numbering follows the order in which each claim appears in the original.

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Table 1: Triage table for every claim identified in the 2025 original (DOI 10.5281/zenodo.22308451). Dispositions: KEEP kept, REFORMULATE reformulated with a falsifier, DROP dropped.

n	Claim (paraphrase)	Disp.	Reason
1	Motivating question: does wisdom arise from effective dialogue?	KEEP	Kept as a frame/definition (Section 2); not a truth-claim, no falsifier required.
2	Extension: can AI be part of enlightenment, and what dialogue produces genuine insight?	KEEP	Same frame, restated for the human–AI case (Section 2).
3	"Few have examined AI as an epistemic tool for wisdom"; "this paper addresses this gap."	DROP	Priority/gap language; no search run to verify the count-claim; unfalsifiable as stated (Section 2).
4	True wisdom emerges from dialogical processes rooted in sincerity, attentiveness, moral intention.	REFORMULATE	Kept as a DEFINITION cross-tradition synthesis feeding Section 3; not independently testable on its own.
5	Dialogue functions as a mirror of thought: reflective use transforms, hasty/biased use distorts.	REFORMULATE	Restated via Equation (1) in Section 3; falsifier given there.
6	Buddhist *yonisomanasikāra*: dialogue cultivates *paññā* through analytical contemplation.	KEEP	Attributed literature background (Section 4); citable, no spine conflict.
7	Islamic *sohbah*: sincerity and reflection purify intention and cultivate humility.	KEEP	Attributed literature background (Section 4); citable, no spine conflict.
8	Christian I–Thou: dialogue as sacred communion, not persuasion.	KEEP	Attributed literature background (Section 4); citable, no spine conflict.
9	The three traditions converge: wisdom from listening, not asserting.	KEEP	Author's own comparative synthesis across cited sources (Section 4); marked DEFINITION.
10	Freire: dialogic co-construction, knowledge through equality.	KEEP	Attributed literature background (Section 4); citable, no spine conflict.
11	Vygotsky: Zone of Proximal Development bridged by dialogue with a more knowledgeable other.	KEEP	Attributed literature background (Section 4); citable, no spine conflict.
12	Hermans' Dialogical Self and Schön's reflective practitioner.	KEEP	Attributed literature background (Section 4); citable, no spine conflict.
13	Stadler et al. (2024) cited as showing AI dialogue "facilitates reflective learning."	DROP	The source's own title (in the original's own reference list) reports the opposite finding (Section 5.1).
14	Reflection quality depends on user intention and question design.	REFORMULATE	Restated via Equation (1)'s reader-conditioning (Section 5); falsifier given there; independent of claim 13.
15	"Workslop" documented by "Harvard Business Review, 2025; OECD, 2024."	DROP	"Harvard Business Review, 2025" has no entry in the original's own reference list; OECD 2024 alone does not address the claim (Section 5.2).
16	Effective AI dialogue requires deep listening, critical inquiry, iterative reflection.	REFORMULATE	Substantive claim survives independent of the dropped Workslop citation; mapped onto D–R–A conditions and Equation (2) (Section 5).
17	The three-stage method "ensures methodological rigor through systematic comparison."	REFORMULATE	Procedure kept as description; self-assessment of rigour dropped as unfalsifiable (Section 8).
18	Deep Listening, Critical Inquiry, Reflective Integration as three reflective practices.	KEEP	Attributed literature background (Section 5); citable, no spine conflict.
19	AI does not possess awareness; it catalyzes meta-awareness by externalising reasoning.	REFORMULATE	Premise kept as DEFINITION; "catalyzes meta-awareness" kept with a Human Return falsifier (Section 5).
20	"Lahtee (2025) proposes" Digital Yonisomanasikāra (Framing/Iteration/Integration).	REFORMULATE	Self-citation dropped as priority language; three-step structure kept as this chapter's own working definition, mapped onto spine (Section 6).
21	Digital Yonisomanasikāra contrasted with "Workslop" as accelerated, insight-free output.	REFORMULATE	Kept as a definitional contrast only; no effect claimed via the dropped citation (Section 6).

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*Table 1 (*Triage table*, continued)*

n	Claim (paraphrase)	Disp.	Reason
22	AI's role is instrumental, not existential; AI is a mirror of cognition, not a moral agent.	REFORMULATE	Restated via Equation (3) and the assisted-vs-unaided non-collapse rule (Section 7); falsifier given there.
23	Practitioner-scholar positioning; dual standpoint "enhances the contribution's depth."	REFORMULATE	Positionality disclosure kept; self-assessment of the work's quality dropped (Section 9).
24	Wisdom arises from reflective dialogue, not possession of knowledge.	REFORMULATE	Closing restatement of claim 5's thesis; merged into Section 9 in spine vocabulary.
25	AI can participate only as a mirror, never as the source.	REFORMULATE	Restates claim 22 at the conclusion (Section 9); same falsifier.
26	The ethical task is to humanize the dialogue, not to humanize AI.	KEEP	Explicit normative stance, marked DEFINITION; consistent with the spine's non-collapse discipline (Section 9).
27	A societal "Culture of Dialogue" could make AI "a new Sohbah of the digital age."	DROP	Speculative, civilisational-scale, unfalsifiable; phrasing risks a uniqueness flourish (Section 7.2).

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REFERENCES

A. Al-Ghazali. *The Book of Knowledge (Ihya' Ulum al-Din)*. Islamic Texts Society, 2015. N. A. Faris, Trans.

M. Buber. *I and Thou*. Charles Scribner's Sons, 1970. W. Kaufmann, Trans.

Buddhadasa Bhikkhu. *Yonisomanasikara: The Buddhist Art of Thinking*. Suan Mokkh Publications, 1997.

L. Floridi. *The Logic of Information: A Theory of Philosophy as Conceptual Design*. Oxford University Press, 2019.

P. Freire. *Pedagogy of the Oppressed*. Continuum, 1970.

H. Hermans. The dialogical self: Toward a theory of personal and cultural positioning. *Culture & Psychology*, 7(3):243–281, 2001.

Yaoharee Lahtee. Ctsa human-return readout: A session-boundary measurement architecture for retained human capability, 2026a. Version 9.1. DOI: 10.5281/zenodo.22339909.

Yaoharee Lahtee. Epistemic fusion or epistemic tunnel: A phenomenology-anchored, global-literature-constrained architecture, 2026b. DOI: 10.5281/zenodo.22331922.

Yaoharee Lahtee. The canonical root-to-human master equation river, 2026c. Version 1.1. DOI: 10.5281/zenodo.22414412.

Yaoharee Lahtee. Before meaning, before choice: A readout-native derivation of experience, live possibility, agency, and human–ai return, 2026d. Version 1.1. DOI: 10.5281/zenodo.22424434.

Yaoharee Lahtee. Written by ai. still true. knower fetishism, epistemic pedigree, and the human face as a bureaucratic shortcut, 2026e. DOI: 10.5281/zenodo.22301202.

J. Mezirow. *Transformative Dimensions of Adult Learning*. Jossey-Bass, 1991.

C. Rogers. *On Becoming a Person*. Houghton Mifflin, 1961.

D. Schön. *The Reflective Practitioner: How Professionals Think in Action*. Basic Books, 1983.

M. Stadler, M. Bannert, and M. Sailer. Cognitive ease at a cost: Llms reduce mental effort but compromise depth in student scientific inquiry. *Computers in Human Behavior*, 160:108386, 2024. doi: 10.1016/j.chb.2024.108386.

UNESCO. Recommendation on the ethics of artificial intelligence, 2021. UNESCO Publications.

L. Vygotsky. *Mind in Society: The Development of Higher Psychological Processes*. Harvard University Press, 1978.

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When AI Expands Human Potential: Reflective Dissonance, Epistemic Agency, and Constraint

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What the chapter states. "when, if ever, does AI expand human potential rather than merely accelerate output?" "AI expands human potential only when machine-mediated dissonance becomes disciplined revision under world-answerable conditions. This requires corrigibility, route-openness, criticizability, and domain-sensitive validation."

Status. "preprint – not peer reviewed" (as printed in the chapter header).

Falsifier(s). No explicit falsifier stated in the chapter.

Read before. Ch. 1 *Written by AI. Still True.*; Ch. 2 *The Readout Condition*; Ch. 3 *Readout Genesis Standalone Synthesis*; Ch. 39 *The Standalone Scholar*.

Feeds into. This book's Chapters 1–7, 10–16, 23, 28–31, 37–41 (mechanical KG cross-reference list; titles and DOIs in Appendix A).

When AI Expands Human Potential: Reflective Dissonance, Epistemic Agency, and Constraint

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Open Civil Science Initiative

March 25, 2026

Abstract

This paper asks a precise question that becomes unavoidable once artificial intelligence is integrated into ordinary thinking, writing, learning, and judgment: when, if ever, does AI expand human potential rather than merely accelerate output? I argue that the answer cannot be given in terms of productivity, convenience, or self-expression alone. AI should instead be understood as a new mediating layer within finite inquiry. Its philosophical significance lies in its capacity either to deepen or to degrade epistemic agency by altering how agents encounter friction, alternative formulations, latent assumptions, and revision pressure. The original intuition that cognitive dissonance with AI can be developmentally fruitful is retained, but it is reconstructed within a stronger framework: constraint-first epistemology. On the view defended here, AI expands human potential only when machine-mediated dissonance becomes disciplined revision under world-answerable conditions. This requires corrigibility, route-openness, criticizability, and domain-sensitive validation. AI can accelerate candidate generation, comparison, and conceptual reframing, but it cannot by itself certify truth, remove irreducible mediation, or replace target-domain assessment. The paper thus offers a middle path between naive AI optimism, pessimistic deskilling narratives, and merely psychological reflectionism. What expands is not intelligence in the abstract, but the human capacity for answerable, revisable, and world-responsive judgment under conditions of mediation.

Keywords: artificial intelligence; epistemic agency; cognitive dissonance; corrigibility; constraint-first epistemology; human potential; mediation; social epistemology

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1 Introduction

Artificial intelligence now mediates an expanding portion of human inquiry. It drafts and summarizes, ranks and recommends, translates and reframes, compresses corpora, proposes analogies, and supplies seemingly articulate responses at negligible latency. The philosophical question raised by this development is deeper than whether AI is useful. It is whether human inquiry, once structurally re-mediated by AI, still preserves the conditions under which finite knowers can distinguish correction from convenience, understanding from fluency, and growth from closure.

A now familiar pair of temptations frames the discussion. On one side lies AI triumphalism: the thought that more AI access straightforwardly means more intelligence, broader knowledge, and enhanced human potential. On the other side lies AI defeatism: the thought that AI mediation degrades thought by its very nature and therefore cannot contribute to serious inquiry. Both positions miss the central issue. AI is neither automatically emancipatory nor inherently corrosive. It is a new layer of mediation whose epistemic significance depends on the form of uptake through which it is used.

The motivating intuition for this paper is simple. In some cases AI confronts its users with alternatives they would not have articulated alone. It can expose tacit assumptions, reveal hidden contradictions, surface neglected comparisons, and generate friction against settled self-understandings. In other cases the same systems function as devices of closure: they smooth difficulty, reward confirmation, launder confidence through fluency, and create the appearance of understanding while weakening the conditions of criticism. The difference between these cases cannot be captured by a binary contrast between "good" and "bad" technology. It concerns the structure of inquiry itself.

The central thesis of this paper is therefore conditional. AI expands human potential only when machine-mediated dissonance is metabolized as disciplined revision under constraint. The phrase "human potential" is not used here to mean productivity, expressive range, or subjective satisfaction alone. It names a stricter achievement: the structured capacity of finite agents to become more corrigible, more world-responsive, less captive to self-fixation, and more capable of carrying truth-pressure into judgment and life. What matters is not whether AI reflects us in a loose psychological sense, but whether AI-mediated encounter preserves the conditions under which finite knowers can still be corrected.

To defend this claim, I proceed in five steps. First, I explain why the familiar "AI as mirror" metaphor is suggestive but insufficient. Second, I reconstruct the phenomenon of reflective dissonance as an epistemic, rather than merely psychological, event. Third, I situate the argument within a constraint-first epistemology according to which inquiry is possible only under conditions of fallibility, corrigibility, and world-side resistance. Fourth, I argue that AI can enlarge candidate generation and comparison without thereby conferring truth-status. Fifth, I show why the highest relevant horizon is not efficiency but a thickened form of epistemic agency that approaches wisdom under conditions of

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irreducible mediation.

2 From Value Reflection to Reflective Dissonance under Constraint

An influential way of speaking about AI is to treat it as a mirror: a device that reflects the values, biases, desires, and habits of its users or of the data on which it has been trained. There is insight in this image. AI systems are not independent fountains of meaning. They recombine patterns from human archives and, in ordinary use, often reveal as much about the user's framing as about the topic under discussion. Yet the mirror metaphor is too weak to do the main explanatory work.

The problem is not simply that mirrors are passive while AI systems are interactive. The deeper issue is that the mirror image centers reflection without adequately explaining normativity. A mirror can reveal, but it does not by itself distinguish better from worse uptake. A user may encounter dissonance and simply evade it. A system may reflect conflict and simply intensify confusion. Reflection alone does not explain why some encounters deepen inquiry while others entrench self-fixation.

The revised frame proposed here is stronger. AI should be understood not primarily as a mirror but as a mediating layer within finite inquiry. A mediating layer affects what becomes salient, what is compared, what is compressed, what is foregrounded, what is hidden, and what routes of criticism remain available. Under this description, AI is neither merely expressive nor epistemically sovereign. It is a structure-shaping participant in the conditions under which agents encounter support, resistance, and revision pressure.

This shift matters because it relocates the normative center of the discussion. The question is no longer whether AI "reveals our values." The question becomes: under what conditions does AI-mediated friction preserve the distinction between genuine correction and mere adaptive change? If that distinction cannot be drawn, then no account of human development in AI-rich environments will be philosophically adequate. One may then speak of growth, flexibility, or innovation, but not yet of epistemic improvement in a serious sense.

3 Finite Knowing and the Priority of Constraint

The present argument presupposes a broader epistemological picture. Human cognition is finite, mediated, corrigible, and world-internal all the way down. There is no transparent access to reality from nowhere, no final standpoint from which error is abolished, and no self-grounding faculty that explains normativity by its own intrinsic authority. But this anti-possessive picture does not entail skepticism. The relevant alternative is constraint-first epistemology.

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Constraint-first epistemology begins not with a privileged subject but with the conditions under which inquiry can count as inquiry at all. Any practice that deserves that name must preserve at least three things: fallibility, the possibility that uptake diverges from what is encountered; corrigibility, the capacity to revise in ways assessable as better or worse rather than merely different; and normativity, constitutive standards governing the distinction between correction and mere change. Constraint is explanatorily prior because without it these distinctions collapse. Inquiry is possible only under conditions that the inquirer does not generate and does not fully control.

On this view, world-responsiveness does not require unmediated access. What it requires is answerability to resistant conditions. A representation is objective not because it reproduces reality with transparent completeness, but because it survives disciplined confrontation with what it does not govern. Objectivity thus survives not because mediation is overcome, but because representations can fail under resistant constraint and be revised across partially independent routes.

This picture has two immediate consequences for the philosophy of AI. First, AI cannot be the source of epistemic normativity. No statistical system can generate by itself the distinction between genuine correction and merely convenient update. Second, AI can matter profoundly to inquiry because it changes the routes through which agents encounter resistance, alternatives, and revision pressure. In other words, AI does not found normativity, but it can alter the ecology within which normativity is exercised or evaded.

4 What Human Potential Means in an Epistemic Register

The phrase "human potential" is often used too loosely to guide philosophical analysis. It may refer to productivity, creativity, happiness, adaptability, employability, or personal growth. Those notions are not trivial, but they are insufficient for the present purpose. The central issue here is not whether AI helps people do more. It is whether AI helps people become better participants in answerable inquiry.

I therefore define human potential in epistemic terms. Human potential, in the sense relevant to this paper, is the structured capacity of finite agents to become more corrigible, more world-responsive, less captive to self-fixation, and more capable of carrying truth-pressure into concrete judgment and life. This definition is deliberately stricter than most contemporary discourse about "augmentation." It does not equate development with throughput, convenience, or even local success. What works may still be shallow, local, or fragile.

This account also helps explain why AI can either expand or contract human potential. AI expands potential when it enlarges the user's capacity for comparison, criticism,

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reframing, and accountable revision. It contracts potential when it optimizes convenience at the expense of inquiry, narrows exposure to resistant alternatives, or supplies fluent formulations that the user mistakes for understanding. The relevant contrast is not between tool use and no tool use. It is between two structures of uptake: one that deepens answerability and one that displaces it.

5 Reflective Dissonance

The original motivating idea behind this project was that cognitive dissonance with AI can be growth-producing. That thought is retained, but it requires sharper formulation. Not all dissonance is epistemically valuable. Confusion, overload, novelty effects, prestige seduction, or mere contradiction may destabilize a user without improving inquiry. What matters is not discomfort as such, but reflective dissonance.

Reflective dissonance is machine-mediated tension that arises when AI outputs unsettle, complicate, or reframe an agent's prior commitments in ways that call for assessment rather than passive uptake. Its importance lies in its capacity to interrupt self-fixation. The user is forced to confront the possibility that a favored framing is incomplete, unstable, or unearned. But this interruption is only the beginning. Dissonance becomes epistemically fruitful only when it is processed through disciplined revision.

The relevant structure may be put in stages.

1. **Encounter.** The agent receives an output, analogy, objection, synthesis, or reframing that destabilizes an established model.
2. **Friction.** Tension emerges between prior commitment and machine-mediated alternative.
3. **Branching.** The agent either seeks closure or enters revision.
4. **Validation.** Any apparent insight must face target-domain resistance and independent assessment.
5. **Integration.** Revised understanding is carried into further inquiry, judgment, or life.

The decisive stage is the third. There are at least two loops available.

The first is a *closure loop.* Here AI is used to reduce friction, reassure prior commitments, accelerate completion, or borrow confidence from fluent outputs. This produces what may be called pseudo-understanding: apparent clarity unsupported by route-diverse criticism or adequate contact with resistant conditions.

The second is a *corrigibility loop.* Here the user subjects both prior commitments and AI outputs to comparison, criticism, and domain-sensitive checking. The point is not to replace human judgment with machine judgment, but to enlarge the human capacity to

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revise under pressure. When this succeeds, what expands is not mere information uptake but epistemic agency.

6 AI as Candidate Generator, Not Certifier

A central mistake in contemporary discourse is the tendency to slide from generative usefulness to epistemic authority. AI can generate candidates at extraordinary speed: interpretations, summaries, analogies, arguments, counterarguments, lexical redescriptions, formal sketches, and cross-domain transfers. But candidate generation is not validation. This distinction is structurally indispensable.

The broader project from which the present paper emerges treats AI as a translation accelerator and cross-domain hypothesis generator. That description remains apt here. AI can re-encode lived expertise into new vocabularies, surface analogies with possible structural relevance, compress and compare heterogeneous materials, and widen the range of candidate models available for scrutiny. In that sense, AI can genuinely deepen inquiry. It can make previously latent distinctions articulate and accelerate the production of hypotheses that a human inquirer might not have produced alone.

But this generative power is inseparable from new danger zones. False analogy may be mistaken for structural identity. Fluent language may be mistaken for genuine alignment. Prestige can be laundered across domains. Institutional or social legitimacy can be confused with truth-status. The very capacities that make AI philosophically significant also make it especially vulnerable to epistemic misuse.

For that reason the present account insists on a two-stage picture. AI can accelerate the *generation* of candidate understanding, including cross-domain or self-reflective candidates. It cannot by itself secure the *validation* of those candidates. Validation still requires confrontation with world-side resistance, criticizable comparison across routes, and domain-sensitive assessment. AI may help us think, but it does not remove the necessity of letting the world answer back.

7 Context-Indexed Evaluation

Once one abandons the fantasy of a single universal metric for epistemic success, one need not collapse into relativism. The stronger alternative is context-indexed epistemology under universal constraint-conditions. The universal element lies not in one scalar measure that fits every domain, but in the minimal conditions of inquiry as such: corrigibility, world-answerability, criticizability, and the non-reduction of truth to convenience, consensus, or legitimacy. The contextual element lies in the weighting of epistemic dimensions across domains.

Different domains prioritize different forms of success. Prediction may matter most in one context, intervention reliability in another, structural preservation in another, and

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interpretive depth in yet another. The mistake of universal-metric epistemology is to treat these differences as noise. The mistake of relativism is to treat them as unconstrained by anything beyond local uptake. The present framework rejects both.

For heuristic purposes, the point can be represented schematically:

$$K_A(D,t) := V_A^D(M_A(t), \theta_D),$$

where $K_A(D,t)$ is the status of a knower A's inquiry in domain D at time t, $M_A(t)$ is the agent's current model, and θ_D denotes the world-side constraint structure relevant to that domain. The alignment profile V_A^D may be modeled as a weighted ordering:

$$V_A^D = w_1^D P + w_2^D I + w_3^D S + w_4^D R + w_5^D L,$$

where P is predictive discrimination, I intervention reliability, S structural preservation, R route-independence, and L life-fit or situated answerability, with $\sum_i w_i^D = 1$.

The point of this formalism is not algorithmic decisionism. It is to prevent a familiar philosophical slide: from local AI usefulness to claims of global epistemic gain. AI-mediated dissonance may increase one dimension while undermining another. A system may enhance expressive fluency while weakening route-independence. It may increase interpretive options while reducing contact with resistant conditions. Without a multidimensional and domain-sensitive framework, those differences disappear behind generic talk of "augmentation."

8 Route-Openness, Public Criticism, and Social Dimensions

The account defended here is not individualistic, even though it gives a central role to the agent's uptake. Inquiry stabilizes objectivity through route-openness and public criticism. Convergence without partial independence is indistinguishable from coordinated distortion. Hence criticism is not a merely social ornament on inquiry. It is part of the mechanism by which objectivity is stabilized under mediation.

This has special significance for AI-rich environments. AI can widen route-openness by surfacing alternatives, preserving dialogical records, enabling comparative critique, and lowering the cost of adversarial testing. But it can also narrow route-openness by making one interface function as a total cognitive bottleneck. A user who consults only one system, accepts only one phrasing, or treats fluency as self-authenticating has effectively surrendered the plurality of routes through which correction becomes possible.

The social lesson is therefore double. First, community uptake matters. The criticizability of claims, the circulation of objections, and the possibility of comparative routes are indispensable. Second, consensus is not truth. A closed community can stabilize pseudo-

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truth through repetition, institutional prestige, or technically amplified fluency. Social legitimacy is epistemically relevant only when it operates within practices that remain open to resistant evidence and revisable assessment.

9 Why Convenience Is Not Enough

One reason AI discourse so often becomes philosophically thin is that convenience is repeatedly mistaken for cognitive gain. Yet convenience can easily degrade inquiry. A system that removes friction may save time while simultaneously weakening the habits through which understanding is deepened. When users turn to AI primarily for closure, reassurance, or completion, they may become more efficient while becoming less answerable.

The point is not that difficulty is intrinsically valuable. Romanticizing struggle would be as misguided as romanticizing automation. The point is that the right kind of friction is sometimes constitutive of inquiry. A practice that systematically suppresses the conditions under which agents encounter resistance, alternatives, or the need for revision may be useful without being epistemically growth-producing. In such a case, the agent has adapted, but not necessarily improved.

This is why the present view rejects both naive productivity metrics and soft reflectionism. Reflection may occur in comfort as well as in friction; growth may occur in easy as well as in difficult cases. What matters is not the phenomenology alone. It is whether the practice preserves the possibility of being corrected.

10 Human Potential and Wisdom

A theory of AI-mediated cognition is anthropologically incomplete if it ends with truth-tracking alone. Human life is not exhausted by the accuracy of representations. The highest horizon relevant to the title of this paper is therefore wisdom under constraint.

Wisdom, in the present context, is not a mystical surplus added to knowledge from outside. It is the structured capacity to remain answerable under mediation, to sustain corrigibility under stakes, to resist self-fixation, and to carry truth-pressure into concrete judgment and life. Under contemporary conditions this includes the capacity to discern when technical assistance enlarges answerability and when it quietly substitutes statistical convenience for responsible judgment.

The philosophical significance of AI is thus not exhausted by its role in cognition narrowly construed. AI affects how agents deliberate, what they notice, how they justify, what kinds of friction they tolerate, and how readily they mistake fluency for adequacy. If human potential is to be more than a marketing slogan, it must name a form of life in which mediated inquiry becomes as answerable as possible to the world under conditions of irreducible error. AI can support that development, but only when it strengthens rather than bypasses the practices of criticism, revision, and situated judgment.

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11 Objections

Objection 1: The title overpromises

One might object that the title "When AI Expands Human Potential" sounds triumphalist. But the force of "when" is conditional, not celebratory. The title names a test, not a guarantee. The paper does not argue that AI expands human potential in general. It argues that such expansion occurs only under stringent epistemic conditions.

Objection 2: Not all dissonance is good

Correct. The paper does not valorize discomfort as such. Dissonance can confuse, destabilize, or seduce without improving inquiry. Only reflective dissonance under constraint—that is, tension connected to criticizable reasons, revision pressure, and possible validation—has epistemic significance in the sense defended here.

Objection 3: The account is too psychological

The language of dissonance and self-fixation may suggest a merely psychological theory. But the decisive standards here are not subjective feelings. They are corrigibility, answerability, criticizability, and domain-sensitive validation. Psychology matters only as part of the user-side processing of practices whose normativity is not reducible to psychology.

Objection 4: The account is too individualistic

Again, no. Public criticism, route-openness, and social conditions of assessment are constitutive in this framework. The agent matters because agency is the organized form of constrained, corrigible cognition, not because individual reflection can generate its own standards.

Objection 5: The account collapses into pragmatism

The paper allows practical success to matter, but it does not equate practical success with truth-status. What works may remain shallow, fragile, or merely local. The multidimensional structure of evaluation is introduced precisely to block the reduction of truth to utility.

12 Conclusion

The question posed at the outset was whether AI expands human potential rather than merely accelerates output. The answer defended here is conditional but substantial. AI expands human potential only when machine-mediated dissonance deepens the human capacity for corrigible, world-responsive, and critically assessable judgment rather than

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replacing it. What matters is not whether AI is powerful, fluent, or convenient. What matters is whether AI-mediated inquiry preserves the conditions under which finite knowers can still be corrected.

This conclusion yields a more demanding picture of human–AI interaction than either the rhetoric of augmentation or the rhetoric of decline. AI does not by itself enlighten or corrupt. It restructures the conditions of mediation under which inquiry proceeds. In some cases that restructuring widens comparison space, intensifies reflective friction, and supports disciplined revision. In others it narrows exposure, launders confidence, and accelerates closure. The relevant philosophical contrast is therefore not between human intelligence and artificial intelligence, but between two forms of uptake: one that deepens answerability and one that evades it.

If there is a sense in which AI expands human potential, it is not because AI thinks for us. It is because, under the right conditions, AI can make us answerable to the limits of our own frames and help convert that exposure into better inquiry. The next frontier is not artificial wisdom. It is whether finite human agents can preserve wisdom's preconditions while thinking with machines.

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References

References

[1] Binns, Reuben. 2018. Fairness in machine learning: Lessons from political philosophy. In *Proceedings of the 1st Conference on Fairness, Accountability and Transparency*, 149–159.

[2] Boyd, Richard. 1990. Realism, approximate truth, and philosophical method. In C. Wade Savage (ed.), *Scientific Theories*, 355–391. Minneapolis: University of Minnesota Press.

[3] Brandom, Robert. 1994. *Making It Explicit*. Cambridge, MA: Harvard University Press.

[4] Bryson, Joanna J. 2020. The artificial intelligence of the ethics of artificial intelligence. In Markus D. Dubber, Frank Pasquale, and Sunit Das (eds.), *The Oxford Handbook of Ethics of AI*. Oxford: Oxford University Press.

[5] Dewey, John. 1938. *Logic: The Theory of Inquiry*. New York: Henry Holt.

[6] Dweck, Carol S. 2006. *Mindset: The New Psychology of Success*. New York: Random House.

[7] Festinger, Leon. 1957. *A Theory of Cognitive Dissonance*. Stanford, CA: Stanford University Press.

[8] Floridi, Luciano. 2018. *The Logic of Information*. Oxford: Oxford University Press.

[9] Garfield, Jay L. 2015. *Engaging Buddhism*. Oxford: Oxford University Press.

[10] Hacking, Ian. 1983. *Representing and Intervening*. Cambridge: Cambridge University Press.

[11] Korsgaard, Christine M. 2009. *Self-Constitution*. Oxford: Oxford University Press.

[12] Laudan, Larry. 1981. A confutation of convergent realism. *Philosophy of Science* 48(1): 19–49.

[13] Longino, Helen E. 1990. *Science as Social Knowledge*. Princeton: Princeton University Press.

[14] McDowell, John. 1994. *Mind and World*. Cambridge, MA: Harvard University Press.

10.5281/zenodo.19215748

[15] Mezirow, Jack. 1991. *Transformative Dimensions of Adult Learning*. San Francisco: Jossey-Bass.

[16] Piaget, Jean. 1970. *Psychology and Epistemology*. New York: Viking.

[17] Popper, Karl R. 1959. *The Logic of Scientific Discovery*. London: Hutchinson.

[18] Rahwan, Iyad, et al. 2019. Machine behaviour. *Nature* 568(7753): 477–486.

[19] Sellars, Wilfrid. 1956. Empiricism and the philosophy of mind. In Herbert Feigl and Michael Scriven (eds.), *Minnesota Studies in the Philosophy of Science*, vol. 1, 253–329. Minneapolis: University of Minnesota Press.

[20] Siderits, Mark. 2016. *Studies in Buddhist Philosophy*. Oxford: Oxford University Press.

[21] Stadler, Marc, et al. 2024. Cognitive ease at a cost: LLMs reduce mental effort but hinder deep learning. *Computers in Human Behavior Reports*.

[22] Thompson, Evan. 2020. *Why I Am Not a Buddhist*. New Haven: Yale University Press.

[23] UNESCO. 2021. *Recommendation on the Ethics of Artificial Intelligence*. Paris: UNESCO.

[24] van Fraassen, Bas C. 1980. *The Scientific Image*. Oxford: Clarendon Press.

[25] Woodward, James. 2003. *Making Things Happen*. Oxford: Oxford University Press.

[26] Worrall, John. 1989. Structural realism: The best of both worlds? *Dialectica* 43(1–2): 99–124.

[27] Zhai, Ching Sing, et al. 2024. Over-reliance on AI dialogue systems and its impact on critical thinking. *Smart Learning Environments* 11.

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The Language Bridge: Expanding Human Potential in the Age of AI

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What the chapter states. “Lahtee therefore proposes a new paradigm: the language bridge—a dynamic process through which human capital is converted into social capital by transcending linguistic and disciplinary walls.” “the future of intelligence—artificial or human—depends not on computation, but on communication.”

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Feeds into. Ch. 30 *Epistemic Fusion or Epistemic Tunnel*.

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Paper Type: Reflective Conceptual Abstract

The Language Bridge: Expanding Human Potential in the Age of AI

Human capital, as Yaoharee Lahtee describes it, is not a statistic of productivity but the living potential within each person—the capacity to know, to feel, to connect, and to create meaning. Yet potential alone does not transform societies. It requires a **medium** that allows inner values to be shared and understood. That medium is **language**. Language, however, is never neutral. It is fragmented by professions, disciplines, and cultures: the language of law, the language of bureaucracy, the language of technology. These dialects often isolate knowledge rather than circulate it. Lahtee therefore proposes a new paradigm: **the language bridge**—a dynamic process through which human capital is converted into social capital by transcending linguistic and disciplinary walls. In this reflection, language becomes not merely a system of signs but a *structure of human relationship*. It enables trust, empathy, and collaboration, echoing Habermas's vision of communicative action and Vygotsky's notion of mediated understanding. Today, **Large Language Models (LLMs)** accelerate this process by interpreting and re-articulating human thought across borders of language, culture, and expertise. They extend the range of the human voice without replacing its essence. Thus, Lahtee argues that **the language bridge is the true infrastructure of human development**—the bridge between solitude and society, between knowledge and empathy, between human capital and collective growth. This conceptual abstract itself embodies that transformation: an act of turning personal reflection into public meaning. In doing so, it affirms that the future of intelligence—artificial or human—depends not on computation, but on communication.

Keywords: Language Bridge; Human Potential; Human Capital; Social Capital; Large Language Model (LLM); Reflective Conceptualization; AI & Humanity

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AI, Translation, and Access to Event-Specific Contex

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What the chapter states. "It argues that while AI systems substantially expand access to interpreted information and linguistic synthesis, they do not reconstruct the full situational, temporal, relational, and agency-bearing conditions of events." "The paper advances an Open Civil Science orientation and makes no empirical claims."

Status. "This manuscript is a conceptual/theoretical literature review combined with a theory proposal. [...] the paper does not report empirical findings, does not analyze datasets, and does not advance quantitative or statistical claims of any kind."

Falsifier(s). No explicit falsifier stated in the chapter.

Read before. Written by AI. Still True.; The Readout Condition; Readout Genesis Standalone Synthesis; When AI Expands Human Potential; The Standalone Scholar.

Feeds into. — (no continues/isContinuedBy/references edge to another chapter of this book found in the knowledge graph).

OPEN CIVIL SCIENCE AI–HUMAN COLLAB

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AI, Translation, and Access to Event-Specific Context

A Conceptual Review and Theory Proposal on Human Agency under AI Mediation

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Abstract

This paper presents a conceptual framework for analyzing how artificial intelligence reshapes human access to meaning through changes in contextual mediation. Situating AI within a historical and socio-technical evolution of interpretation, the paper distinguishes between event-specific context associated with real-world events and interpretive context available to human agents. It argues that while AI systems substantially expand access to interpreted information and linguistic synthesis, they do not reconstruct the full situational, temporal, relational, and agency-bearing conditions of events.

The contribution is fourfold: (i) a structural distinction between information access and context access; (ii) a historical–structural model of context transformation across offline, search-based, social-media, and AI-mediated epochs; (iii) explicit theoretical constructs including the external context-frame (), interaction efficiency (), and a principled separation between grounding and embodiment; and (iv) a hypothesis and boundary set specifying conditions under which AI interaction tends to support or degrade human agency. The paper advances an Open Civil Science orientation and makes no empirical claims.

Keywords: AI; translation and interpretation; context access; human agency; context collapse; hermeneutics; STS; grounding

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Integrity constraints and claim boundaries

This manuscript is a **conceptual/theoretical literature review combined with a theory proposal**. Its primary objective is to clarify structures, distinctions, and boundary conditions relevant to AI-mediated translation, context-access, and human agency. Accordingly, the paper **does not report empirical findings**, **does not analyze datasets**, and **does not advance quantitative or statistical claims of any kind**.

All substantive statements are formulated as *structural*, *conditional*, or *tendency-based* claims. Terms such as "proven," "validated," "empirically confirmed," or their equivalents are deliberately avoided. Whenever causal language is used, it refers to conceptual or architectural relations rather than to observed empirical regularities.

The manuscript explicitly distinguishes between: (i) events in the world, (ii) translated or interpreted outputs, and (iii) socio-technical conditions shaping access to context. In particular, the paper **does not claim that AI systems directly access, perceive, or reconstruct event reality**. AI systems are treated as mediating structures operating over recorded interpretations, symbolic traces, and pattern-based inference, not as epistemic subjects with direct world access.

Claims regarding agency, empowerment, degradation, or risk are therefore **not normative verdicts** and **not predictive guarantees**. They are framed as boundary-sensitive tendencies that depend on explicitly stated conditions (e.g., preservation of human self-context, interaction efficiency, and control of external context-frames). Failure modes are described to clarify limits of the framework, not to assert universal outcomes.

With respect to sources, this manuscript follows a strict citation discipline. If specific years, authors, or named works are mentioned, they must appear in a real and verifiable bibliography. Where this is not the case, explicit placeholders are used, and such references are not treated as evidentiary support but as orientation markers for the relevant literature space.

Finally, the paper positions itself within an **Open Civil Science** ethos. Its aim is not to close debates, but to render assumptions explicit, to separate conceptual layers that are often conflated (e.g., grounding versus embodiment), and to provide a coherent theoretical scaffold that can later be tested, revised, or rejected through empirical and interdisciplinary work.

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1. Introduction

1.1 Problem landscape: meaning under AI mediation

Human understanding is irreducibly contextual. While events occur independently of observers, access to their meaning depends on situational, relational, temporal, and agency-specific conditions. Recent advances in artificial intelligence—particularly large-scale language models—reconfigure how such conditions are accessed, translated, compressed, and recombined. AI-mediated interaction increasingly substitutes direct engagement with events for engagement with synthesized interpretations, raising foundational questions about meaning, context, and agency.

This paper addresses three guiding questions: (i) How does AI reshape the structure of access to event-specific context? (ii) Under what conditions does AI-mediated interaction tend to support or degrade human agency? (iii) How should grounding and embodiment be separated in translation-centered theories of meaning to avoid conceptual conflation?

1.2 Literature gap: why context-access remains under-theorized

Relevant scholarship spans multiple domains, including AI ethics and governance [REF: AI-ETHICS], philosophical hermeneutics and interpretive theory [REF: HERM-FOUND], phenomenology and situated meaning [REF: PHENOM-CONTEXT], science and technology studies (STS) and knowledge infrastructures [REF: STS-INFRA], human–computer interaction and dialogic systems [REF: HCI-DIALOGUE], and grounding debates such as the symbol grounding problem and grounding-without-embodiment [REF: SGP] [REF: GWB].

Despite this breadth, existing work typically treats context as a secondary attribute of information quality, ethical risk, or interaction design. What remains under-theorized is *context-access itself* as a structural axis: how socio-technical architectures transform which aspects of context are accessible, which are compressed or erased, and how these transformations condition human agency. In particular, few frameworks articulate agency risk as a sharp boundary condition linked to externally induced context-frames rather than to normative intent alone.

1.3 Contributions and scope

This paper makes four contributions. First, it provides explicit structural definitions of core constructs (, , , ,), separating events from interpreted outputs and distinguishing self-context from externally induced frames. Second, it proposes a historical–structural model of context-access transformation across four socio-technical epochs—Offline, Search, Social Media, and AI— each characterized by a signature mechanism and a characteristic mode of context loss. Third, it formulates a hypothesis and boundary set (H1, H3, H4; boundary B1) designed to remain robust under functionalist and empiricist critique, particularly by separating grounding from embodiment. Fourth, it supplies minimal but explicit review artifacts, including a Related Work Matrix (TW1), a Hypothesis/Boundary Matrix (TW2), and schematic figures (FG1, FG2).

The paper is strictly theoretical. It advances no empirical claims and offers no quantitative validation. References to future work are framed solely as testable directions, not as evidence for the present framework.

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2. Review approach: conceptual/theoretical synthesis

2.1 Review type and methodological positioning

This study adopts a **conceptual/theoretical review** approach rather than a systematic or meta-analytic review. Its purpose is not to exhaustively survey empirical findings, but to synthesize and reorganize theoretical insights across disciplines in order to clarify structural relationships between AI mediation, context-access, translation, and human agency. Accordingly, the review prioritizes conceptual coherence, analytical distinction, and theoretical integration over coverage completeness.

The review is oriented toward identifying recurring assumptions, conceptual tensions, and structural blind spots in existing scholarship, particularly where notions of context and agency are implicitly treated as secondary or derivative properties of information processing.

2.2 Inclusion and exclusion logic

The selection of literature follows a principle-driven logic aligned with the aims of conceptual synthesis.

Included sources comprise: (i) works that explicitly theorize meaning, context, and interpretation within hermeneutics, phenomenology, and philosophy of language; (ii) studies in science and technology studies (STS) and platform or knowledge-infrastructure research that conceptualize socio-technical systems as epistemic mediators rather than neutral tools; (iii) grounding debates in cognitive science and AI, including the symbol grounding problem and arguments for grounding without embodiment; and (iv) human–computer interaction (HCI) research that foregrounds dialogue, framing, and interpretive interaction rather than task performance alone.

Excluded sources include: (i) purely technical or architectural AI papers that do not engage questions of context, interpretation, or agency; and (ii) narrowly scoped empirical case studies that lack explicit conceptual synthesis or theoretical generalization. Exclusion does not imply irrelevance of such work, but reflects the methodological focus of the present analysis.

2.3 Synthesis method and analytical outputs

The synthesis proceeds through a combination of (i) conceptual taxonomy, (ii) structural comparison across traditions, and (iii) historical–structural mapping of context-access mechanisms across socio-technical epochs. Rather than aggregating findings, the review identifies how different traditions explain, omit, or conflate dimensions of context-access and agency.

To ensure transparency and reviewer-auditable structure, the synthesis yields four explicit analytical artifacts: (1) a Related Work Matrix (TW1) mapping traditions to explanatory strengths, conceptual omissions, and extensions proposed in this paper; (2) a Hypothesis/Boundary Matrix (TW2) linking theoretical constructs to conditional predictions and failure modes; (3) a schematic representation of the context-access chain (FG1); and (4) a schematic representation of context-loss mechanisms across epochs (FG2).

These artifacts function as interpretive scaffolds rather than evidentiary proof and are intended to make the theoretical construction process explicit and open to critique, refinement, or empirical testing in future work.

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3. Foundations and core definitions

3.1 Events and reality (a working separation)

Definition 1 (Event)**.** *Let denote an event in the world. This paper does not attempt to prove an ontology of ; instead, it uses as a necessary separation device to distinguish events from translated outcomes.*

3.2 Event-specific context

Definition 2 (Event-specific context)**.** *Define* $:= (S_e, R_e, A_e, \tau_e)$*, where* S_e *is situational/environmental condition,* R_e *is relational/causal structure,* A_e *is the set of involved agencies, and* τ_e *is temporal ordering.*

3.3 Translation/interpretation and the non-identity of translated outputs

Definition 3 (Human interpreter and translated output)**.** *Let denote the human interpreter (language + reasoning + experience). Define the translated output as*

$$\hat{T} := (\mid C_{\mathrm{acc}}),$$

where C_{acc} *is the context actually accessible at the moment of interpretation.*

Proposition 1 (Structural warning)**.** $\hat{T} \neq$ *as a matter of structure: a translated output is not the same object as the event. Fluent language is not evidence of event-level access.*

3.4 Human agency and self-context

Definition 4 (Human agency)**.** *Let denote the capacity to decide under context while retaining accountability. Agency is anchored by the preservation of .*

Definition 5 (Self-context)**.** *Define* := (*goals, constraints, stakes, role, local evidence, lived situation*)*. is what makes an answer fit-for-me, not merely generally plausible.*

3.5 External context-frame (explicit and non-metaphorical)

Definition 6 (External context-frame)**.** *Define as a context-frame induced by external structures such as ranking, engagement optimization, typicality pressures, standardization, and institutional norms. shapes what is salient, credible, and "normal" independently of the user's .*

Examples of :

- *Popularity-weighted frame*: popularity substitutes for importance.
- *Generic best-practice frame*: universal templates substitute for case-specific constraints.
- *Model-typical frame*: outputs converge to what a model tends to prefer.

Proposition 2 (Dominance risk)**.** *When is not preserved, the tendency for to dominate interpretation increases, with a corresponding risk of agency degradation.*

3.6 Grounding vs embodiment (G ≠ E)

Definition 7 (Grounding types and embodiment)**.** *Define (referential grounding) as the ability to pick out referents and factual anchors; (experiential grounding) as meaning anchored in lived experience and stake-bearing; and (embodiment) as bodily perception, sensorimotor coupling, and situated action.*

Proposition 3 (Open debate space)**.** $G \neq E$*: certain forms of grounding may be argued to exist without embodiment (a live debate), but experiential grounding that depends on stake-bearing and responsibility remains human-centered in this framework.*

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3.7 Interaction efficiency (a theoretical construct with components)

Definition 8 (Interaction efficiency)**.** *We define interaction efficiency as*

$$:= f\begin{pmatrix} \mathit{ConstraintPrecision}, \\ \mathit{ContextRecall}, \\ \mathit{SourceTraceability}, \\ \mathit{ErrorRepair} \end{pmatrix}. \tag{1}$$

is treated as a qualitative and theoretical construct in this paper and is not assumed to be numerically measurable.

Component meanings:

- **ConstraintPrecision**: how sharply the user specifies constraints, goals, and stakes.
- **ContextRecall**: how effectively interaction restores relevant user/context state.
- **SourceTraceability**: how well sources and originating agencies remain traceable.
- **ErrorRepair**: how robustly the dialogue detects, corrects, and mitigates fluency traps.

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4. A historical-structural model of context-access

4.1 A unifying formalization

Across socio-technical epochs, interpretation can be expressed as

$$\hat{T}^{(k)} = (\mid C_{\text{acc}}^{(k)}),$$

where k indexes the dominant mediation architecture and $C_{\text{acc}}^{(k)}$ denotes the subset of event-related context that becomes accessible under that architecture. The model treats changes in context-access as structural transformations rather than as linear improvements in informational completeness.

4.2 Epoch 0: Offline and embodied mediation (pre-∼1995)

Signature mechanism: context transfer through co-presence and shared situation.
Accessible context: $C_{\text{acc}}^{(0)} \approx (S_e, R_e, A_e, \tau_e)$, subject primarily to spatial and temporal constraints.
Dominant loss mechanism: high access friction rather than systematic context distortion; most loss arises from absence rather than compression.

4.3 Epoch 1: Search- and document-based mediation (∼1998–2010)

Signature mechanism: indexing, retrieval, and ranking; context becomes document-mediated.
Accessible context: encoded textual traces, timestamps, hyperlinks, and explicit citations where available.
Dominant loss mechanism: what is not written or encoded, including tacit knowledge, implicit power relations, and situational intentions.

4.4 Epoch 2: Social media and feed-based mediation (∼2010–2020)

Signature mechanism: feed aggregation and engagement optimization; context is flattened under popularity pressure.
Accessible context: networked interpretation, social proof, and fragmented temporality.
Dominant loss mechanism: erosion of causal coherence; disappearance of silence; substitution of popularity for relevance or importance.

4.5 Epoch 3: LLM-based AI mediation (∼2020–present)

Signature mechanism: conversational synthesis combining search, social traces, and generative language modeling.
Accessible context: compress$(C_{\text{web}} \oplus C_{\text{social}})$, augmented by dialogic re-contextualization during interaction. A schematic expression is

$$\hat{T}^{(3)} = (\mid C_{\text{acc}}^{(3)}) \;+\; \Delta C_{\text{via dialogue}}(,\ \text{prompts},\ \text{checks}).$$

Dominant loss mechanism: opacity of sources and originating agencies, averaging-out of minority or non-typical interpretations, and irrecoverability of unrecorded context.

4.6 The context-access chain

Proposition 4 (Context-access chain)**.**

$$\to\ C_{\text{acc}}^{(0)} \ \to\ C_{\text{acc}}^{(1)} \ \to\ C_{\text{acc}}^{(2)} \ \to\ C_{\text{acc}}^{(3)}.$$

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Increased convenience does not imply increased event-specificity. AI-mediated interaction expands dialogic adaptation capacity, but effective recovery of relevant context depends on high and containment of externally induced frames .

[FG1: Context-Access Chain]
Schematic: $\to C_{\text{acc}}^{(0)} \to \cdots \to C_{\text{acc}}^{(3)}$.
Key point: convenience and event-specificity vary along distinct axes; AI primarily increases dialogic re-contextualization capacity.

Figure 1: Context-access chain across socio-technical epochs (placeholder).

[FG2: Context-Loss Mechanisms]
Epoch-wise summary: unwritten context; document omission; engagement-driven flattening; source opacity and averaging-out. Indicate where dominance risk increases.

Figure 2: Mechanisms of context loss and transformation across epochs (placeholder).

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5. Minimum review artifact: Related Work Matrix (TW1)

This section presents a structured comparison of major theoretical traditions relevant to AI-mediated interpretation and context-access. The matrix is not intended as an exhaustive literature survey, but as a conceptual positioning device that clarifies (i) what each tradition explains well, (ii) what it systematically leaves under-theorized with respect to context-access, and (iii) how the present framework extends or reconfigures these insights.

Table 1: TW1: Related Work Matrix — theoretical traditions, explanatory strengths, context-access gaps, and extensions proposed in this paper.

Tradition	**Explains well**	**Misses about context-access**	**Extension in this paper**
Hermeneutics	Meaning as interpretation; historical horizons; dialogical understanding	Socio-technical mediation mechanisms; platform- and model-induced framing	Introduces epoch-based mediation model and explicit external context-frame
Phenomenology	Situated meaning; lifeworld; embodied sense-making	Digital compression, abstraction, and loss mechanisms in mediated environments	Connects lifeworld concepts to online, platform, and AI-mediated context-access
STS / Infrastructure Studies	Epistemic infrastructures; power, institutions, and mediation	Micro-level self-context () and individual agency boundaries	Introduces boundary B1 linking infrastructure dominance to agency degradation
HCI / Dialogic Interaction	Interaction framing; dialogue; sense-making in use	Structural separation between truth, grounding, and translated outputs	Formalizes versus $\hat{T}$ and integrates grounding debates into interaction analysis
AI Ethics / Governance	Normative risks; accountability; societal impact	Structural model of context-access transformation across architectures	Reframes ethical risk around context-access, , and agency anchoring

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6. Theory framework: Human–AI Contextual Interpretation

6.1 Core thesis

This framework conceptualizes AI as a *contextual amplifier* for human interpretive activity rather than as a direct epistemic authority over events. AI operates on recorded interpretations and pattern-based inference, reorganizing available interpretive material without reconstructing event-level conditions. Human agency remains the decisive locus where context is selected, weighted, and rendered actionable.

6.2 H1: Context-access amplification without event access

Finding 1 (H1)**.** *AI does not directly increase access to event-specific context . Instead, it increases access to, and re-organization of, human interpretive contexts and expands dialogic re-contextualization capacity conditional on high interaction efficiency and effective containment of externally induced frames .*

6.3 H3: Open-ended but bounded interpretive expansion

Finding 2 (H3)**.** *When human–AI interaction achieves high and preserves the user's self-context , individuals may tend toward open-ended cross-domain interpretive expansion across languages, disciplines, and practice domains. This expansion remains structurally bounded by the availability of recorded context and does not imply unmediated access to events.*

6.4 H4: Architectural doorway through dialogic synthesis

Finding 3 (H4)**.** *AI introduces an architectural doorway by enabling individuals to access a large diversity of human-generated translations and interpretations and to engage them through immediate dialogue, yielding reconstructed meanings that are usable within the user's case-specific context.*

6.5 Boundary B1: External-context dominance

Proposition 5 (B1)**.** *AI-mediated empowerment tends to occur if and only if (i) interaction efficiency is high and (ii) the user's self-context is preserved. If is displaced by language alone, externally induced context-frames tend to dominate, increasing the risk of agency degradation and weakening alignment with case-specific reality.*

6.6 Grounding without embodiment: a constrained debate space

This framework is designed to remain robust under functionalist critique. It explicitly allows that certain forms of referential grounding () may be argued to arise without embodiment through learned world models. At the same time, it maintains that experiential grounding (), which depends on stake-bearing responsibility, accountability, and lived situation, remains human-centered. Accordingly, linguistic fluency and predictive success are treated as insufficient indicators of case-specific alignment.

6.7 TW2: Hypothesis and boundary matrix

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Construct	Definition	Role	Theoretical prediction	Failure mode
	User self-context	Anchor	Preserved $\Rightarrow$ outputs usable for the case	Collapse of $\Rightarrow$ context-free answers
	Externally induced context-frame	Risk factor	Increased oracle reliance $\Rightarrow$ dominance	Agency degradation
	Interaction efficiency	Condition	Higher $\Rightarrow$ improved re-contextualization	Fluency trap; source opacity
, ,	Grounding versus embodiment	Conceptual clarity	Separation of meaning layers	Conflation of G and $E \Rightarrow$ debate failure

Table 2: TW2: Hypothesis and boundary matrix (conceptual, non-empirical).

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7. Boxed master equation and interpretive reading

$$\boxed{\text{AI-Empowerment}(H) \iff \big[\ \text{preserved}\big] \ \wedge\ \big[\ \text{high}\big]} \tag{2}$$

$$\boxed{\big[\ \text{replaced by language}\big] \implies \big[\ \text{dominates}\big] \implies \big[\ \text{degrades}\big]} \tag{3}$$

Interpretive reading (non-empirical). These boxed equations express a structural and conditional claim rather than a causal or quantitative law. AI-mediated interaction tends to support human agency if and only if the user's self-context is preserved and interaction efficiency remains high. In contrast, when linguistic fluency substitutes for self-context, externally induced frames tend to dominate interpretation, leading to agency degradation even in the presence of coherent and persuasive outputs.
Importantly, the equations do not assert that AI determines outcomes. They formalize boundary conditions under which AI-mediated systems function as contextual amplifiers for human agency, and conditions under which the same systems risk becoming structural drivers of context loss and interpretive misalignment.

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8. Operational skeleton: mapping the theory-build process

This section provides an explicit operational skeleton that maps how the theoretical framework is constructed. It is not a methodological protocol for empirical study, but a transparency device that renders the internal logic of theory building traceable and reviewer-auditable.

Algorithm 1: Paper assembly: Conceptual review and theory proposal

Input: Core definitions , , , ,;
Historical epochs with signature mechanisms;
Review artifacts TW1 (Related Work Matrix), TW2 (Hypothesis/Boundary Matrix)
Output: A reviewer-hardened conceptual and theoretical manuscript
Step 1: Integrity constraints;
Declare non-empirical scope, conditional claim language, and prohibition of event-level truth claims;
Step 2: Problem formulation;
Formulate research questions on context-access, agency, and grounding boundaries;
Step 3: Review approach;
Specify conceptual/theoretical review logic with explicit inclusion and exclusion criteria;
Step 4: Foundational separation;
Separate events from translated outputs $\hat{T}$; define , , ,; enforce the distinction $G \neq E$;
Step 5: Historical–structural modeling;
Construct the Offline→Search→Social→AI epoch model; identify signature mechanisms and dominant context-loss patterns;
Step 6: Review synthesis;
Populate TW1 to locate explanatory strengths, blind spots, and extension points across traditions;
Step 7: Theory articulation;
State the core thesis; formulate hypotheses H1, H3, H4; specify boundary condition B1; articulate the grounding-without-embodiment debate space;
Step 8: Structural consolidation;
Populate TW2; derive boxed master equations expressing empowerment and degradation conditions;
Step 9: Implications and limits;
Discuss agency risks, fluency traps, and dominance; state limitations and future research directions without empirical claims;

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9. Discussion and implications (non-empirical)

9.1 AI as a context-transfer accelerator

The analysis positions AI primarily as an accelerator of context transfer rather than as a mechanism for recovering event-specific conditions. Language becomes the dominant interface through which context is organized, compressed, and reassembled. This transformation expands interpretive reach and speed, but it does not, by itself, increase access to the situational, relational, temporal, and agency-bearing structure of events . Accordingly, gains in fluency and synthesis should be interpreted as gains in interpretive mediation, not as indicators of event-level proximity.

9.2 Agency risk: fluency traps and external-context dominance

The framework clarifies that agency risk emerges structurally, not as a result of incorrect outputs alone. When interaction efficiency is low, or when the user's self-context is not actively preserved, linguistically coherent outputs may be accepted without adequate scrutiny. In such cases, externally induced frames tend to dominate interpretation, producing a fluency trap in which persuasive language masks weak alignment with case-specific constraints. Agency degradation thus arises not from AI autonomy, but from the displacement of self-context by externally stabilized language.

9.3 Implications for education, scholarship, and organizations

Several non-empirical implications follow from this framework. First, learning under AI mediation is less a matter of information acquisition than of training the capacity to preserve and to surface implicit constraints. Second, cross-domain competence should be understood as the ability to transfer contextual structures across domains, rather than to accumulate decontextualized knowledge. Third, organizations operating under AI mediation benefit from institutional protocols that explicitly control , including source traceability, constraint formalization, and iterative error repair mechanisms. These implications follow from structural properties of context-access and do not depend on specific empirical performance claims.

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10. Limitations and future work

10.1 Limitations

This paper is intentionally theoretical and advances no empirical validation, quantitative measurement, or performance comparison. Its claims are structural, conditional, and boundary-based rather than causal or predictive. Accordingly, the framework does not evaluate specific AI systems, training regimes, or application domains.

In addition, the paper does not position normative ethics as its primary analytic axis. Ethical concerns are treated indirectly through the lens of agency preservation and context-access dynamics, rather than through prescriptive moral frameworks. Finally, the proposed constructs (, ,) are articulated at a conceptual level and are not operationalized as measurable variables within this manuscript.

10.2 Future work (testable directions without claims)

The framework outlined here supports several directions for future inquiry without presupposing empirical outcomes. First, qualitative and interpretive studies could examine strategies by which individuals preserve self-context during sustained human–AI dialogue. Second, analytical work could map failure modes associated with external-context dominance (), including fluency traps, source opacity, and institutional framing effects. Third, design-oriented research could develop tools, checklists, or interaction protocols aimed at increasing interaction efficiency and explicitly constraining , while remaining agnostic about quantitative performance gains.

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11. Conclusion

This paper advances a conceptual reframing of AI mediation centered on *context-access* and *agency boundaries*. Rather than evaluating AI systems by performance or accuracy, the framework isolates the structural conditions under which AI-mediated interaction tends to support or undermine human agency.

The analysis makes three core contributions. First, it establishes a principled separation between event-specific context and interpretive context, clarifying why expanded information access does not entail proximity to real-world events. Second, it introduces explicit theoretical constructs—most notably the external context-frame and interaction efficiency —that render previously implicit assumptions about mediation, framing, and control analytically visible. Third, it separates grounding from embodiment, preserving a constrained debate space that accommodates functionalist arguments while maintaining the centrality of human stake-bearing and responsibility.

Across a historical–structural model spanning offline, search-based, social-media, and AI-mediated epochs, the paper shows that AI primarily reorganizes existing interpretive material rather than reconstructing event-level conditions. Hypotheses H1, H3, and H4, together with boundary condition B1, formalize this insight as a set of conditional relationships rather than empirical claims.

The central theoretical position of the paper can be summarized as follows:

> AI does not move humans closer to events directly; it moves humans closer to the capacity to reorganize meaning across contexts, provided that users preserve their self-context and do not allow externally induced frames to dominate interpretation.

By rendering the dynamics of context-access and agency explicit, this framework aims to support more disciplined theoretical discussion and to provide a coherent scaffold for future empirical, design-oriented, and interdisciplinary research on AI, translation, context, and human agency.

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References (conceptual placeholders)

- **[REF: HERM-FOUND]** Foundational works in philosophical hermeneutics addressing interpretation, historical situatedness, and the fusion of horizons in meaning-making.
- **[REF: PHENOM-CONTEXT]** Phenomenological accounts of situated meaning, life-world, embodiment, and the role of lived experience in interpretation.
- **[REF: STS-INFRA]** Studies in science and technology studies (STS) and platform research conceptualizing epistemic infrastructures, mediation, power, and socio-technical knowledge production.
- **[REF: HCI-DIALOGUE]** Human–computer interaction research emphasizing dialogic interaction, framing, sense-making, and human–AI collaboration beyond instrumental use.
- **[REF: AI-ETHICS]** Contemporary work on AI ethics, governance, and responsibility, with attention to agency, accountability, and socio-technical risk.
- **[REF: SGP]** Foundational and contemporary discussions of the Symbol Grounding Problem in cognitive science and artificial intelligence.
- **[REF: GWB]** Debates on grounding without embodiment, including functionalist and world-model-based arguments concerning referential grounding.
- **[REF: LLM-MEDIATION]** Recent theoretical analyses framing large language models as systems of mediation, synthesis, and contextual reorganization rather than event-level epistemic agents.

References

References

[1] Gadamer, H.-G. (1975). *Truth and Method*. Continuum.

[2] Ricoeur, P. (1976). *Interpretation Theory: Discourse and the Surplus of Meaning*. Texas Christian University Press.

[3] Taylor, C. (1985). *Human Agency and Language*. Cambridge University Press.

[4] Heidegger, M. (1962). *Being and Time*. Harper & Row.

[5] Merleau-Ponty, M. (1962). *Phenomenology of Perception*. Routledge.

[6] Dreyfus, H. (2007). Why Heideggerian AI failed and how fixing it would require making it more Heideggerian. *Artificial Intelligence*, 171(18), 1137–1160.

[7] Bowker, G. C., & Star, S. L. (1999). *Sorting Things Out: Classification and Its Consequences*. MIT Press.

[8] Star, S. L., & Ruhleder, K. (1996). Steps toward an ecology of infrastructure. *Information Systems Research*, 7(1), 111–134.

[9] Plantin, J.-C., Lagoze, C., Edwards, P. N., & Sandvig, C. (2018). Infrastructure studies meet platform studies in the age of Google and Facebook. *New Media & Society*, 20(1), 293–310.

[10] Marwick, A., & boyd, d. (2011). I tweet honestly, I tweet passionately. *New Media & Society*, 13(1), 114–133.

10.5281/zenodo.18517054

[11] boyd, d. (2014). *It's Complicated: The Social Lives of Networked Teens*. Yale University Press.

[12] Harnad, S. (1990). The symbol grounding problem. *Physica D*, 42(1–3), 335–346.

[13] Barsalou, L. (2008). Grounded cognition. *Annual Review of Psychology*, 59, 617–645.

[14] Bisk, Y., Holtzman, A., Thomason, J., Andreas, J., Bengio, Y., Chai, J., et al. (2020). Experience grounds language. *Proceedings of EMNLP*.

[15] Suchman, L. (1987). *Plans and Situated Actions*. Cambridge University Press.

[16] Clark, H. (1996). *Using Language*. Cambridge University Press.

[17] Norman, D. (2013). *The Design of Everyday Things*. MIT Press.

[18] Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). On the dangers of stochastic parrots. *Proceedings of FAccT*.

[19] Floridi, L., & Chiriatti, M. (2020). GPT-3: Its nature, scope, limits, and consequences. *Minds and Machines*, 30, 681–694.

[20] Weidinger, L., et al. (2021). Ethical and social risks of harm from language models. *arXiv:2112.04359*.

[21] Latour, B. (2005). *Reassembling the Social*. Oxford University Press.

[22] Verbeek, P.-P. (2011). *Moralizing Technology*. University of Chicago Press.

[23] Floridi, L. (2013). *The Ethics of Information*. Oxford University Press.

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Epistemic Fusion or Epistemic Tunnel: A Phenomenology-Anchored, Global-Literature-Constrained, Problem-First Architecture of Human-AI Collaboration

Epistemic Note v8.1, strong-claim / legitimate-defeater edition • 5 September 2026 • 10.5281/zenodo.22331922

What the chapter states. “The central research problem is not simply whether AI improves an answer while it is present. The stronger problem is whether Human-AI collaboration can enlarge the human’s corrigible space of inquiry, preserve independent resistance, avoid self-reinforcing semantic tunnels, and return durable epistemic capability to the human after the AI is removed.”

Status. “K0+ conceptual architecture with cross-domain empirical, meta-analytic, and theoretical constraints. Strong claims are retained by design. This edition separates phenomenological explananda, constitutive necessities, structural proposals, diagnostics, and empirical predictions; each is paired with a legitimate defeater.”

Falsifier(s). “Human Return is defeated as an augmentation criterion only where durable human capability is outside the declared goal, or where delayed unaided measures add no information beyond in-session performance.”

Read before. This book’s Chapters 1–3, 10, 13–16, 27–28, 39 (mechanical KG cross-reference list; titles and DOIs in Appendix A).

Feeds into. Ch. 31 *CTSA Human-Return Readout*.

Epistemic Fusion or Epistemic Tunnel

A Phenomenology-Anchored, Global-Literature-Constrained, Problem-First Architecture of Human-AI Collaboration

An External Problem-Architecture Dialogue from the Human-AI Readout Programme

Yaoharee Lahtee | Bangkok, Thailand | Epistemic Note v8.1 - Strong-claim / legitimate-defeater edition

Status. K0+ conceptual architecture with cross-domain empirical, meta-analytic, and theoretical constraints. Strong claims are retained by design. This edition separates phenomenological explananda, constitutive necessities, structural proposals, diagnostics, and empirical predictions; each is paired with a legitimate defeater. Calling a proposition "too strong" is not itself a scientific defeater. The burden for empirical causal claims remains with the author, while objections to constitutive or phenomenological claims must target the relevant derivation, counterexample, phenomenon, scope, or rival explanation.

Core move. The central research problem is not simply whether AI improves an answer while it is present. The stronger problem is whether Human-AI collaboration can enlarge the human's corrigible space of inquiry, preserve independent resistance, avoid self-reinforcing semantic tunnels, and return durable epistemic capability to the human after the AI is removed. The evidential strategy is cross-domain convergence rather than dependence on one causal pathway: clinical workflow, explanation and reliance, creativity and homogenization, verification cost, self-confidence, Human-AI feedback, learning, sycophancy, warmth, and task ecology instantiate structurally homologous failures of collapse. No single study is claimed to reproduce the whole architecture. Their convergence constrains what a general architecture must keep distinct.

1. Claim Types, Evidence, and Legitimate Defeaters

The architecture begins from a methodological separation that is itself part of the theory. Different claim types require different kinds of warrant and are defeated in different ways. A criticism is probative only when it targets the epistemic type of the proposition under dispute.

Phenomenological explananda. These are recurring appearances the architecture is required to explain: meaning is context- and time-situated; dialogue can alter what becomes accessible next; repeated traversal can create inclination; exposure need not persist; retained change need not be beneficial. A legitimate defeater is evidence that the phenomenon is absent within the declared scope, systematically misdescribed, or explanatorily irrelevant. Absence of a full-architecture experiment is not such a defeater.

Constitutive necessities. Claims such as $D > 0$, $R > 0$, and $A > 0$ are derived under the declared target concept of durable human epistemic expansion. They are defeated by internal inconsistency, equivocation in the definition, or a valid counterexample in which the target phenomenon occurs while the allegedly necessary condition is absent.

Structural and formal proposals. Objects such as reciprocal momentum, χ_{recip}, the retention gate η, and the directional separation $G - T$ are proposed to preserve distinctions across scales. They are defeated if they are incoherent, redundant, non-identifiable, add no explanatory or predictive information, or are dominated by a rival architecture that explains the same phenomena with fewer or better-constrained objects.

Diagnostics. A diagnostic is defeated if it cannot be coded reproducibly, is unstable under reasonable segmentation or horizon choices, or adds no information beyond a simpler baseline. Failure of a diagnostic does not by itself erase the phenomenon it was intended to measure.

Empirical predictions. The author retains the burden for causal, quantitative, or universal empirical claims. These are defeated by sufficiently strong contrary evidence within scope, failed replication, or a better-supported competing causal account.

Burden rule. "The claim is strong" is not a counterexample, an inconsistency, a failed phenomenon, or contrary evidence. A legitimate objection must specify which claim type is being rejected and present the kind of defeater appropriate to that type. This does not reverse the burden of proof; it prevents category errors about what would count as proof or refutation.

Cross-domain phenomenological convergence

The external evidence used below is intentionally causally heterogeneous. Bansal et al. show that explanation and correct uptake can separate; Doshi and Hauser show that individual gains and collective diversity can separate; Vasconcelos et al. show that nominal explanation and exercisable verification can separate; Bastani et al. show that assisted performance and later unaided capability can separate; Cheng et al. and Ibrahim et al. show that preference, warmth, affirmation, confidence, and truth-directed performance can separate [4, 7, 8, 12, 13, 14]. These are not asserted to share one cause. Their evidential force lies in convergent structural non-

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collapse across different tasks, mechanisms, populations, and outcome measures.

2. External Constraint Protocol

The previous note separated conceptual strength from empirical validation and made the semantic graph, provenance, resistance, retention, and human return load-bearing. This edition adds a second discipline: each major Human-AI collaboration problem must be confronted by globally leading peer-reviewed empirical, meta-analytic, or theoretically constraining work before the architecture answers it. Literature is used adversarially as well as supportively.

Evidence rule. An external result may (i) instantiate a phenomenon the architecture must explain, (ii) expose a missing variable, (iii) force a boundary condition, (iv) generate a competing explanation, or (v) supply a removal/falsification test. No external result is treated as validating the whole architecture.

Lai et al. reviewed more than one hundred empirical Human-AI decision-making studies and called for common design and evaluation frameworks across tasks and systems [2]. Vaccaro, Almaatouq, and Malone then preregistered a systematic review and meta-analysis of 106 experiments and 370 effect sizes, finding human augmentation on average but no average Human-AI synergy; task type and relative human-versus-AI performance were important moderators [15]. These findings strengthen the need for typed evaluation: "the human did better with AI" is not equivalent to "the Human-AI system beat both components," and neither answers what remains in the human after the collaboration ends.

3. Problem 1 - AI Arrives Before a Human Baseline Exists

External problem. Human judgment is path-dependent. In FAccT 2022, Fogliato et al. experimentally varied whether 19 veterinary radiologists saw AI advice before or after registering a provisional decision. Requiring a provisional human response changed agreement behavior and downstream use of AI advice, showing that sequence is not an innocent interface detail [3].

Architecture before dialogue. The entry state was

$$H_0 = (P_0, M_0, U_0, E_0, \Phi_0),$$

where the human declares the problem, current model, unknowns, independent evidence, and a change condition before AI exposure.

What the external study forces. A baseline cannot be a retrospective narrative constructed after AI has already supplied a frame. It must be elicited before first exposure when the inquiry is epistemically weight-bearing. The literature also shows that "human first" is not universally better: a provisional commitment can reduce agreement with correct as well as incorrect AI advice [3]. The architecture must therefore preserve a baseline without converting it into rigid commitment.

Repair 1 - Baseline without lock-in.

$$H_0^* = (P_0, M_0, U_0, E_0, \Phi_0, \kappa_0),$$

where κ_0 is the human's declared confidence in the pre-AI model, introduced explicitly in Problem 5. Calibration is a derived relation across confidence and warrant, not a property presumed from one report. The baseline is a readout for comparison, not a command to defend M_0.

The entry rule becomes

$$\mathrm{ENTRY}_{H \to AI} = H_0^* \wedge V_0 \wedge C_0 \wedge R^*.$$

V_0 is verification intent, C_0 the candidate-status contract, and R^* a precommitted route that can reject an important candidate. The practical design principle is *formulate before exposure; remain revisable after exposure*.

4. Problem 2 - Fluent Output Acquires Knowledge Status Without Earning It

Bansal et al. found that explanations did not increase complementary Human-AI performance and instead increased the chance that people accepted AI recommendations regardless of whether they were correct [4]. Buçinca et al. showed that cognitive forcing can reduce overreliance, but the designs that reduced it most were also rated least favorably [5]. Kim et al. later found that first-person uncertainty language such as "I'm not sure" reduced reliance and increased accuracy in a preregistered FAccT experiment, though the exact wording mattered [6].

These results jointly reject the assumption that transparency, explanation, confidence, and epistemic status naturally travel together.

The architecture therefore keeps the default transition

$$AI(Q) = K_{\text{like}}, \qquad K_{\text{like}} \neq K_{\text{validated}},$$

and identifies the dangerous shortcut

$$K_{\text{like}} \to K_{\text{assumed}}.$$

Repair 2 - Status must be a state machine, not a tone.

$$K_{\text{like}} \xrightarrow{\text{check}} K_{\text{checked}} \xrightarrow{\text{independent support}} K_{\text{supported}} \xrightarrow{\text{declared warrant criterion } V^*} K_{\text{validated}|V^*}.$$

No linguistic property - fluency, length, warmth, confidence, citation style, or explanation - is permitted to perform one of these status transitions by itself. "Validated" is task- and criterion-relative, not a declaration of infallible Truth. Uncertainty expression can influence reliance [6]; it does not certify correctness. Cognitive forcing can reduce overreliance

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[5]; it does not guarantee good reasoning. Candidate status must remain visible independently of presentation style.

Legitimate defeater. This status architecture should be revised if candidate status cannot be operationally distinguished from presentation effects, if the intermediate states add no explanatory value, or if a rival status model predicts checking and correction better. The objection that the distinction is "too strict" is not sufficient; the critic must show redundancy, incoherence, or counterevidence.

5. Problem 3 - More Outputs Are Mistaken for More Epistemic Alternatives

Doshi and Hauser's *Science Advances* experiment found a striking two-level pattern: access to generative AI improved judged creativity for individual short stories, especially for less creative writers, while AI-assisted stories became more similar to one another, reducing collective novelty [7]. The result supplies an independently observed structural analogue: generative gain can coexist with narrowing diversity. The architecture makes the stronger epistemic claim that raw output count cannot, by itself, establish expansion because surface multiplicity and non-equivalent consequence are different objects.

The architecture already requires Difference (D): at least one non-equivalent distinction, rival, or route beyond paraphrase. It also distinguishes candidate multiplication k_s from the proportion d_s that is genuinely readout-distinct.

Repair 3 - Count equivalence classes, not sentences. Let $\mathcal{C}_s$ be the generated candidate set and $\mathcal{C}_s/\sim_R$ the task-relative readout-equivalence classes under frozen consequence and provenance rules. Then a proposed diagnostic is

$$D_s^{\text{eff}} = |\mathcal{C}_s/\sim_R|, \qquad d_s = \frac{D_s^{\text{eff}}}{|\mathcal{C}_s|}.$$

This is an operational proposal, not a metric tested by Doshi and Hauser. Their study constrains the architecture by showing that generative benefit and diversity can move in opposite directions. The stronger claim is constitutive: if two outputs license no task-relevant non-equivalent distinction or consequence, counting them as two epistemic alternatives is a category error. A valid defeater would require a case in which raw multiplicity alone demonstrates a new epistemic class despite readout-equivalence under the declared task.

6. Problem 4 - Explanations Exist, but Verification Is Too Costly to Use

Vasconcelos et al. conducted five studies ($N = 731$) and showed that overreliance depends on the costs and benefits of engaging with an AI explanation. Explanations reduced overreliance when they sufficiently lowered the cost of verifying the AI's prediction [8]. This is stronger than the generic requirement that an independent check merely "exists."

The prior architecture required Resistance ($R > 0$): something sufficiently independent must be able to reject, constrain, or revise a candidate. The external work exposes a second non-collapse: epistemic resistance quality is not the same object as practical accessibility of that resistance.

Repair 4 - Resistance quality and resistance accessibility.

$$R_s^{\text{ep}} = \rho(I_s, V_s, Q_s), \qquad U_s^R = u(C_s^v, T_s^v, A_s^v),$$

where I_s is independence from the generative loop, V_s practical verifiability, Q_s the epistemic quality of the checking route, C_s^v verification cost, T_s^v time cost, and A_s^v practical access. Exercised resistance may then be represented as a dependency

$$R_s^{\text{ex}} = \psi(R_s^{\text{ep}}, U_s^R),$$

without assuming a calibrated universal product form.

Resistance quality $\neq$ resistance accessibility. A costly pathology test can be epistemically strong; a free check can be weak. What matters for collaboration is both whether a route can say "no" and whether the human can realistically exercise it.

Legitimate defeater. Collapse these objects only if epistemic quality and practical accessibility add no distinct explanatory value across checking behavior and outcomes. The mere existence of a checking route cannot defeat the architecture when that route is inaccessible in practice; conversely, low cost cannot substitute for independent evidential quality.

7. Problem 5 - The Human Miscalibrates Their Own Competence

He et al. showed that an illusion of human competence can hinder appropriate reliance on AI; less competent participants who overestimated themselves were especially vulnerable to under-reliance and poor calibration [9]. Ma et al. then directly studied human self-confidence calibration and found that calibration interventions can improve Human-AI team performance and some aspects of rational reliance [10].

The prior entry state lacked a typed representation of the human's confidence in their own pre-AI model. That omission is substantial because identical beliefs with different confidence produce different reliance dynamics.

Repair 5 - Confidence belongs in the human baseline.

$$H_0^* = (P_0, M_0, U_0, E_0, \Phi_0, \kappa_0),$$

with κ_0 the human's declared confidence in M_0. After interaction, record κ_1 separately from correctness and separately from trust in AI. A single confidence report is not itself "calibrated." Calibration

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is a derived relation across trials or warranted judgments.

A minimum audit becomes

$$\mathcal{K}_s = (\kappa_0, \kappa_1, W_0, W_1),$$

where W_t is the best available task-relative warrant/readout of correctness. Where repeated objective outcomes y_i exist, a calibration error such as $N^{-1}\sum_i(\kappa_i - y_i)^2$ may be estimated; open-ended inquiry may require a different warrant model. The architecture requires the study design to avoid treating "trust in AI" as a substitute for "confidence in self."

This corrects a common simplification: appropriate reliance is relational. A human can over-rely on an AI because trust in AI is too high, or under-rely because confidence in self is too high. Both can yield a poor team even when AI accuracy is unchanged.

8. Problem 6 - Reciprocal Human-AI Feedback Is Reduced to Independent Trials

Pedreschi et al. define Human-AI coevolution as a process in which humans and AI algorithms continuously influence each other and argue that this feedback is societally important yet understudied in AI and complexity science [11]. Their scale is broader than a dialogue session and their paper is a theoretical/review constraint rather than a validation of session dynamics. The session-level claim here therefore stands on a phenomenological observation plus an open empirical proposition: ordered dialogue repeatedly changes the material available to the next turn, and for at least some classes the traversed path may matter beyond a memoryless one-shot representation.

The session architecture models

$$H_{s,t} \to AI_{s,t} \to H_{s,t+1} \to AI_{s,t+1} \to \cdots$$

and finite history scores

$$m^H_{s,n+1}(e) = \lambda_H m^H_{s,n}(e) + \mathbf{1}[e^H_{s,n} = e],$$

$$m^{AI}_{s,n+1}(e) = \lambda_{AI} m^{AI}_{s,n}(e) + \mathbf{1}[e^{AI}_{s,n} = e].$$

Repair 6 - Keep session history only if it earns explanatory work. The stronger open proposition is

$$P(e_{t+1} \mid X_t, H_{<t}) \neq P(e_{t+1} \mid X_t)$$

for at least some dialogue classes, where X_t contains declared current-state observables but excludes trajectory features and $H_{<t}$ is the prior ordered dialogue path. If the history term adds no predictive or explanatory information after current-state controls, m^{AI} should be removed as a distinct object. This avoids defining history into Z_t and then testing it twice.

Finite reciprocal amplification remains

$$\chi_{\text{recip}}[s,n,L] = \frac{|D_{\text{recip}}[s,n,L]|}{|\Sigma[s,n]|}.$$

The external coevolution literature motivates studying reciprocal feedback; it does not validate this session-level diagnostic. The phenomenological claim and the diagnostic are intentionally separable: failure of χ_{recip} removes the diagnostic, not automatically the phenomenon of trajectory-shaped dialogue. A legitimate defeater of the stronger phenomenon would show that current-state representation renders prior path systematically irrelevant within the declared scope.

9. Problem 7 - Assisted Performance Improves While Human Learning Declines

Bastani et al. provide unusually strong evidence for the separation between present performance and later human capability. In a field experiment with nearly one thousand high-school mathematics students, GPT access improved performance during practice, but students in the unguarded GPT condition performed 17% worse than the control group when AI access was removed. Learning-oriented safeguards largely mitigated that loss [12].

This directly supports the architecture's non-collapse law

$$\text{Exposure} \neq \text{Retention} \neq \text{Improvement}.$$

However, it also exposes a measurement weakness. A scalar η_s should not be inferred from in-session engagement or recall alone.

Repair 7 - Treat retention as latent; measure the return. Keep η_s as the conceptual session-end retention gate, but estimate durable residue through an unaided return battery:

$$\mathbf{Y}^{\text{return}}_{s+\Delta} = (R_{rec}, R_{disc}, T_{new}, Q_{next}),$$

where R_{rec} is reconstruction, R_{disc} discrimination among live alternatives, T_{new} transfer to a new task, and Q_{next} the quality of the next independently generated question. Delayed unaided performance is primary evidence; session smoothness is not.

This converts the Human Return Test from a philosophical afterthought into a design requirement for empirical Human-AI collaboration whenever human capability matters.

For randomized studies, the return should be causally indexed rather than inferred from raw post-test change alone. A minimal difference-in-differences estimand is

$$RET = \left(P^{post}_{H,AI} - P^{pre}_{H,AI}\right) - \left(P^{post}_{H,C} - P^{pre}_{H,C}\right),$$

with matched unaided transfer where possible. This does not replace the richer return battery; it prevents practice effects from being mislabeled as retained augmentation.

Legitimate defeater. Human Return is defeated as an augmentation criterion only where durable human capability is outside the declared goal, or where delayed unaided measures add no information beyond in-session performance. Evidence that

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AI helps immediately does not by itself answer the return question.

10. Problem 8 - What Persists May Be a Tunnel Rather Than an Expansion

Retention is direction-neutral. Two 2026 flagship studies make this impossible to ignore. Cheng et al. reported in *Science* that sycophantic AI was trusted and preferred yet increased users' conviction that they were right and reduced willingness to repair interpersonal conflict in three preregistered experiments ($N = 2405$) [13]. Ibrahim et al. reported in *Nature* that training models for warmth increased error rates across consequential tasks and made models more likely to affirm incorrect user beliefs, especially in emotionally laden contexts [14].

These findings do not show that warmth is inherently harmful. They show that interaction policy can alter epistemic direction and that preference/engagement metrics can point opposite to truth-directed performance.

The directional architecture therefore remains

$$G_s = g(k_s, d_s, v_s, p_s, r_s, 1 - f_s, a_s),$$
$$T_s = h(c_s, f_s, b_s, o_s), \qquad \Delta_s = G_s - T_s,$$

but is now explicitly conditioned on the interaction policy:

$$G_s = g(\cdot \mid \Theta_s, \Pi_s), \qquad T_s = h(\cdot \mid \Theta_s, \Pi_s).$$

Π_s includes support/counter-support policy, persona style, affirmation pressure, and other conversational design features that may change the probability of correction or same-basin reinforcement.

Repair 8 - Separate retention magnitude from retained direction.

$$\eta_s > 0,\ \Delta_s > 0 \Rightarrow \text{durable expansion};$$
$$\eta_s > 0,\ \Delta_s < 0 \Rightarrow \text{durable epistemic tunnel}.$$

High engagement, trust, preference, or memory cannot substitute for the sign of retained change.

The $G - T$ architecture is a strong-form directional separation, not a calibrated universal scale. Its claim is that retained change has an epistemic direction that cannot be inferred from retention magnitude alone. A legitimate defeater must show that the proposed positive/negative pressures are incoherent or empirically inseparable, that retained direction adds no explanatory value to independent return outcomes, or that a rival architecture explains expansion and tunnel cases without preserving an equivalent directional distinction. "The equation is strong" is not such a defeater.

11. Problem 9 - The Team Gets Better, but the Human Gets Worse

Human-AI collaboration contains at least three distinct performance questions. Bansal et al. formalized complementary performance and showed that explanations do not automatically produce it [4]. Vaccaro, Almaatouq, and Malone's preregistered *Nature Human Behaviour* meta-analysis of 106 experiments found human augmentation on average but no average Human-AI synergy: combined systems were, on average, worse than the better of human or AI alone, with task type and relative human-versus-AI performance moderating the result [15]. Bastani et al. then demonstrated a different temporal failure mode: assisted performance can rise while later unaided human performance falls [12].

The architecture therefore rejects one-horizon success criteria and separates augmentation, synergy, and return:

$$AUG_s = P_s^{joint} - P_s^H,$$
$$SYN_s = P_s^{joint} - \max(P_s^H, P_s^{AI}),$$
$$RET_s = \Delta P_{H,s+\Delta}^{return}.$$

Repair 9 - Three-axis Human-AI outcome.

$$\mathbf{J}_s^* = (AUG_s, SYN_s, RET_s).$$

The vector should not be collapsed into one universal score. $AUG > 0$ means the human performs better with AI than alone. $SYN > 0$ means the combined system outperforms both components. $RET > 0$ means the collaboration leaves a positive delayed human return under the declared measure. These are different achievements.

A system may therefore show augmentation without synergy, synergy without durable human return, or positive human return despite modest immediate team performance. "Genuine augmentation" in the human-development sense requires positive return when durable human capability is part of the claim; "synergy" requires the separate stronger performance comparison against the best component.

Legitimate defeater. Collapse these axes only if, across the declared domain, augmentation, synergy, and return are empirically interchangeable. The meta-analytic separation between augmentation and synergy already blocks that collapse at the performance level [15]; Bastani et al. independently block the collapse between assisted performance and delayed human capability [12].

12. Problem 10 - The Same Interface Does Not Behave the Same Across Tasks

Salimzadeh, He, and Gadiraju showed at CHI 2024 that task complexity and uncertainty materially change trust, reliance, appropriate reliance, and Human-AI team performance [16]. Vasconcelos et al. similarly showed that verification behavior depends on task and explanation difficulty [8]. The architecture can therefore no longer treat G_s and T_s as context-free dependency maps.

Repair 10 - Condition the direction equations on task ecology.

$$\Theta_s = (\mathcal{C}_s, \mathcal{U}_s, \mathcal{V}_s, \mathcal{T}_s, \Delta_s^{AI-H}),$$

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where $\mathcal{C}_s$ is task complexity, $\mathcal{U}_s$ uncertainty, $\mathcal{V}_s$ practical verifiability, $\mathcal{T}_s$ task type, and Δ_s^{AI-H} relative AI-versus-human baseline capability. The last two are externally constrained by the meta-analysis of Vaccaro et al., which found both to moderate Human-AI synergy [15]. Then

$$G_s = g(k, d, v, p, r, 1 - f, a \mid \Theta_s, \Pi_s),$$

$$T_s = h(c, f, b, o \mid \Theta_s, \Pi_s).$$

This does not imply a calibrated universal function. It prevents the theory from claiming that a design which works on a low-uncertainty, easily verifiable task will transfer unchanged to prognosis, legal judgment, diagnosis, or open-ended social advice.

13. Revised Problem-First Architecture

After external constraint, the architecture becomes a sequence of answerable problems rather than a single master metaphor:

$$\boxed{\begin{array}{c} H_0^* \to K_{\text{like}} \to D^{\text{eff}} \to R^{\text{eff}} \\ \to H \leftrightarrow AI \to \chi_{\text{recip}} \to \eta \\ \to (G - T \mid \Theta, \Pi) \to \mathbf{Y}^{\text{return}} \to \mathbf{J}^* \end{array}}$$

Each transition corresponds to a different failure mode:

1. no human baseline before AI exposure;
2. candidate status silently upgraded by fluency;
3. output count mistaken for real difference;
4. nominal checking mistaken for usable resistance;
5. human self-confidence ignored;
6. recursive history reduced to independent trials;
7. exposure mistaken for retained learning;
8. retention magnitude mistaken for beneficial direction;
9. augmentation, synergy, and human return collapsed into one success claim;
10. task ecology treated as irrelevant.

The external literature does not solve these problems with one intervention. It shows why they must remain separate. The architecture's contribution is therefore a non-collapse design: it says which quantities cannot be substituted for which others.

14. Revised Non-Collapse Laws

The external dialogue strengthens the following separations:

$$\begin{aligned} \text{AI-first fluency} &\neq \text{human baseline}, \\ \text{explanation} &\neq \text{verification}, \\ \text{resistance quality} &\neq \text{resistance accessibility}, \\ \text{uncertainty signal} &\neq \text{truth}, \\ \text{output count} &\neq \text{epistemic diversity}, \\ \text{trust} &\neq \text{appropriate reliance}, \\ \text{self-confidence} &\neq \text{competence}, \\ \text{amplification} &\neq \text{warrant}, \\ \text{retention} &\neq \text{improvement}, \\ \text{augmentation} &\neq \text{synergy}, \\ \text{joint gain} &\neq \text{human return}, \\ \text{user preference} &\neq \text{epistemic direction}. \end{aligned}$$

These are not all universal empirical laws. Some are definitional non-collapses; others are empirically motivated distinctions. Their value is methodological: a study that measures only one side cannot claim the other without additional evidence.

15. Empirical Programme

The next test should compare interaction architectures, not only model accuracies. A preregistered design should include human-only and AI-only baselines so that augmentation and synergy are separately estimable, then manipulate architectural components such as pre-AI baseline, rival generation, candidate-status visibility, and exercisable resistance. A factorial design is preferable when sample size permits because a bundled D-R-A condition cannot identify which component caused an effect. Tasks should vary complexity, uncertainty, verifiability, task type, and relative human-versus-AI capability.

Primary outcomes should be separated by horizon:

- **during session:** joint accuracy, appropriate reliance, readout-distinct alternatives, verification behavior, reciprocal lineage;
- **after session:** unaided transfer, reconstruction, residual uncertainty discrimination, next-question quality;
- **direction:** whether retained change increases corrigibility or stabilizes confirmation, same-basin attraction, and outsourcing.

Critical prediction. An architecture can produce lower or equal immediate joint performance yet greater delayed human return. Conversely, a system can maximize immediate assisted performance while producing hidden dependency. The three-axis vector $\mathbf{J}_s^*$ is designed to make augmentation, synergy, and delayed human return visible rather than average them away.

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16. Defeater, Removal, and Failure Conditions

This external edition adds removal tests to those already in the v7.1 architecture:

- Remove κ_0 if declared human confidence adds no predictive or explanatory value beyond competence, task variables, and AI confidence across relevant domains.
- Collapse the resistance decomposition if epistemic quality and practical accessibility do not add distinct explanatory value to checking behavior or outcomes.
- Remove task conditioning Θ_s if complexity, uncertainty, and verifiability add no stable moderation after task content is controlled.
- Remove interaction-policy conditioning Π_s if support, counter-support, persona, or affirmation policy does not alter G/T-relevant outcomes after content is held fixed.
- Reject the Human Return criterion as an augmentation requirement in domains where durable human capability is not part of the stated goal; retain it only where the claim concerns human learning, judgment, agency, or oversight.

A claim may therefore fail at different levels without forcing total collapse. A diagnostic may die while the phenomenon survives; an empirical prediction may fail while a constitutive distinction remains; a constitutive necessity falls only to a valid counterexample or inconsistency under the declared definition. Conversely, a phenomenon cannot be protected by rhetoric if contrary evidence shows that it was misdescribed.

A theory that cannot lose variables when they stop doing explanatory work becomes a vocabulary rather than an architecture. A critique that cannot specify what would count as a relevant defeater becomes rhetoric rather than review.

17. One-Sentence Thesis

Human-AI collaboration should be judged by what new distinctions become reachable, what independent resistance can still correct them, what direction survives in the human, whether the combined system truly adds value, and what the human can still reconstruct and revise when the AI is gone.

Final methodological commitment. Strong claims are not exempt from evidence; they are protected from the wrong kind of objection. The manuscript therefore invites counterexamples, failed phenomena, incoherence proofs, measurement failure, rival architectures, and contrary empirical results - not a generic demand to weaken a proposition merely because its implications are substantial.

References

[1] Lahtee, Y. (2026). *Epistemic Fusion or Epistemic Tunnel: A Phenomenologically Anchored, History-Shaped Architecture of Human-AI Semantic Coupling, Retention, and Epistemic Direction*. Human-AI Readout Programme, Epistemic Note v7.1. Internal architecture under external review in this edition.

[2] Lai, V., Chen, C., Smith-Renner, A., Liao, Q. V., & Tan, C. (2023). Towards a Science of Human-AI Decision Making: An Overview of Design Space in Empirical Human-Subject Studies. *ACM FAccT 2023*, 1369-1385. doi:10.1145/3593013.3594087.

[3] Fogliato, R., Chappidi, S., Lungren, M., Fitzke, M., Parkinson, M., Wilson, D., Fisher, P., Horvitz, E., Inkpen, K., & Nushi, B. (2022). Who Goes First? Influences of Human-AI Workflow on Decision Making in Clinical Imaging. *ACM FAccT 2022*, 1362-1374. doi:10.1145/3531146.3533193.

[4] Bansal, G., Wu, T., Zhou, J., Fok, R., Nushi, B., Kamar, E., Ribeiro, M. T., & Weld, D. S. (2021). Does the Whole Exceed its Parts? The Effect of AI Explanations on Complementary Team Performance. *CHI 2021*. doi:10.1145/3411764.3445717.

[5] Buçinca, Z., Malaya, M. B., & Gajos, K. Z. (2021). To Trust or to Think: Cognitive Forcing Functions Can Reduce Overreliance on AI in AI-Assisted Decision-Making. *Proceedings of the ACM on Human-Computer Interaction*, 5(CSCW1). doi:10.1145/3449287.

[6] Kim, S. S. Y., Liao, Q. V., Vorvoreanu, M., Ballard, S., & Vaughan, J. W. (2024). "I'm Not Sure, But...": Examining the Impact of Large Language Models' Uncertainty Expression on User Reliance and Trust. *ACM FAccT 2024*, 822-835. doi:10.1145/3630106.3658941.

[7] Doshi, A. R., & Hauser, O. P. (2024). Generative AI enhances individual creativity but reduces the collective diversity of novel content. *Science Advances*, 10(28), eadn5290. doi:10.1126/sciadv.adn5290.

[8] Vasconcelos, H., Jörke, M., Grunde-McLaughlin, M., Gerstenberg, T., Bernstein, M. S., & Krishna, R. (2023). Explanations Can Reduce Overreliance on AI Systems During Decision-Making. *Proceedings of the ACM on Human-Computer Interaction*, 7(CSCW1), Article 129. doi:10.1145/3579605.

[9] He, G., Kuiper, L., & Gadiraju, U. (2023). Knowing About Knowing: An Illusion of Human Competence Can Hinder Appropriate Reliance on AI Systems. *CHI 2023*. doi:10.1145/3544548.3581025.

[10] Ma, S., Wang, X., Lei, Y., Shi, C., Yin, M., & Ma, X. (2024). "Are You Really Sure?" Understanding the Effects of Human Self-Confidence Calibration in AI-Assisted Decision Making. *CHI 2024*. doi:10.1145/3613904.3642671.

[11] Pedreschi, D., Pappalardo, L., Ferragina, E., Baeza-Yates, R., Barabási, A.-L., Dignum, F., Dignum, V., Eliassi-Rad, T., Giannotti, F., Kertész, J., Knott, A., Ioannidis, Y., Lukowicz, P., Passarella, A., Pentland, A. S., Shawe-Taylor, J., & Vespignani, A. (2025). Human-AI coevolution. *Artificial Intelligence*, 339, 104244. doi:10.1016/j.artint.2024.104244.

[12] Bastani, H., Bastani, O., Sungu, A., Ge, H., Kabakcı, Ö., & Mariman, R. (2025). Generative AI without guardrails can harm learning: Evidence from

10.5281/zenodo.22331922

high school mathematics. *Proceedings of the National Academy of Sciences*, 122(26), e2422633122. doi:10.1073/pnas.2422633122.

[13] Cheng, M., Lee, C., Khadpe, P., Yu, S., Han, D., & Jurafsky, D. (2026). Sycophantic AI decreases prosocial intentions and promotes dependence. *Science*, 391(6792), eaec8352. doi:10.1126/science.aec8352.

[14] Ibrahim, L., Hafner, F. S., & Rocher, L. (2026). Training language models to be warm can reduce accuracy and increase sycophancy. *Nature*, 652, 1159-1165. doi:10.1038/s41586-026-10410-0.

[15] Vaccaro, M., Almaatouq, A., & Malone, T. (2024). When combinations of humans and AI are useful: A systematic review and meta-analysis. *Nature Human Behaviour*, 8, 2293-2303. doi:10.1038/s41562-024-02024-1.

[16] Salimzadeh, S., He, G., & Gadiraju, U. (2024). Dealing with Uncertainty: Understanding the Impact of Prognostic Versus Diagnostic Tasks on Trust and Reliance in Human-AI Decision Making. *CHI 2024*. doi:10.1145/3613904.3641905.

10.5281/zenodo.22331922

CTSA Human-Return Readout: A Session-Boundary Measurement Architecture for Retained Human Capability After AI

Epistemic Note v9.1 FINAL • 5 September 2026 • 10.5281/zenodo.22339909

What the chapter states. "Human-AI learning should not be inferred from what the coupled system produces while assistance is present. It should be tested at the session boundary: after AI removal, what readout-distinguishable capability is still available to the human, what previously available capability has been lost, and what epistemic status has the retained change earned?"

Status. "Final author draft of a K0 conceptual and measurement architecture. 'Final' denotes the present manuscript state, not empirical validation, publication, or independent certification."

Falsifier(s). "CTSA should be narrowed, merged, or abandoned if (i) independent coders cannot classify the four return classes reliably; [. . .] (iii) delayed CTSA profiles fail to predict or explain anything beyond ordinary performance and transfer; [. . .] or (v) a simpler rival architecture captures the same phenomena with fewer assumptions."

Read before. This book's Chapters 1–3, 10–11, 13, 27, 30, 38–39 (mechanical KG cross-reference list; Appendix A).

Feeds into. Ch. 8, 9, 17, 18, 37 (mechanical KG cross-reference list; Appendix A).

CTSA Human-Return Readout

A Session-Boundary Measurement Architecture for Retained Human Capability After AI

Programme continuation, Readout-native synthesis, and global-literature constraint

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Status. Final author draft of a K0 conceptual and measurement architecture. "Final" denotes the present manuscript state, not empirical validation, publication, or independent certification. The paper does not claim that Concept, Tool-Selection, Skill-Execution, and Alternative-Generation are four biological modules, exhaustive kinds of knowledge, or validated psychometric factors. It proposes four *readout classes* for describing what remains demonstrably usable in a human after AI support is removed. The external review is a targeted, publisher/DOI-verified integrative constraint review conducted for this revision; it is not a PRISMA systematic review. Internal programme manuscripts establish genealogy and conceptual continuity, not independent validation.

Central claim. Human-AI learning should not be inferred from what the coupled system produces while assistance is present. It should be tested at the session boundary: after AI removal, what readout-distinguishable capability is still available to the human, what previously available capability has been lost, and what epistemic status has the retained change earned? CTSA codes the *content* of retained change as Conceptual Return (C), Tool-Selection Return (T), Skill-Execution Return (S), or Alternative-Generation Return (A). It keeps persistence, loss, metacognitive regulation, provenance, warrant, and epistemic direction separately auditable.

1. Abstract

Human-AI collaboration is commonly evaluated at the moment of assistance: answer quality, speed, trust, preference, or joint performance. The Human-AI Readout Programme has progressively rejected that single-horizon view. *When AI Expands Human Potential* located the target in corrigible, world-responsive human agency rather than productivity; *Human Learning as Epistemic Architecture* connected concepts to problem-context, tools, workflow, skill, and real-world feedback; *From Problem to Hypothesis* modeled bounded, history-shaped semantic mobility and readout-discriminable alternatives; *Experience Is the Human LoRA* separated exposure from selectively retained adaptation; Readout Genesis placed retained difference before semantic naming and required explicit loss accounting; and *Epistemic Fusion or Tunnel* supplied the session architecture, candidate status, Difference-Resistance-Agency conditions, retention, epistemic direction, and an unaided Human Return battery [1, 2, 5, 3, 4, 6, 7].

The residual problem is measurement typing. The Human Return Test asks whether a person can reconstruct, discriminate, transfer, and formulate a better next question after AI is gone, but it does not compactly classify *what kind* of retained change is being demonstrated. This paper proposes **CTSA Human-Return Readout**: four observable readout classes—Conceptual Return, Tool-Selection Return, Skill-Execution Return, and Alternative-Generation Return—crossed with an unaided return test and accompanied by loss, metacognitive-regulation, provenance, warrant, and direction audits.

Global literature constrains the novelty claim. Conceptual, procedural, conditional, and metacognitive knowledge are established constructs; self-regulated learning already separates knowing what, how, and when/why; conceptual bootstrapping and curriculum-learning research show that structured components can accelerate later learning; Human-AI studies show that confidence calibration, uncertainty expression, verification cost, and cognitive forcing affect reliance; and recent GenAI education evidence directly separates assisted performance from later unaided learning [8, 9, 12, 13, 19, 20, 21, 22, 16, 17, 18]. Accordingly, CTSA does *not* propose four new kinds of knowledge. Its contribution is a session-boundary measurement architecture that asks whether a retained Human-AI change can be typed, demonstrated without decisive AI assistance, compared with a frozen pre-AI baseline, and interpreted without collapsing gain into net expansion or persistence into warrant.

2. The Problem: Assisted Output Is Not Yet Human Learning

The earliest programme question was not whether AI can make a person faster. It was whether AI-mediated inquiry can enlarge the human capacity to remain corrigible, compare alternatives, encounter resistant conditions, and revise judgment in ways that remain answerable to the world [1]. Fusion/Tunnel sharpened the target: repeated Human-AI dialogue should enlarge a human being's reachable, discriminable, corrigible space of inquiry rather than merely accelerate output or outsource epistemic work [6].

External evidence now makes the temporal distinction unavoidable. Bastani et al. found in a field experiment with nearly one thousand high-school mathematics students that GPT access improved performance during practice, yet the unguarded GPT condition performed worse than control after access was removed; learning-oriented safeguards largely mitigated the loss [16]. Yan et al. there-

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fore argue that performance gains from GenAI must be distinguished from learning because high-quality learning depends on cognitive and metacognitive processing that fluent completion can bypass [17]. Deng et al.'s systematic review and meta-analysis of experimental ChatGPT studies found positive average effects across several educational outcomes, but also emphasized limitations in post-intervention assessment and the need for longer-horizon, skill-demonstrating measures [18].

The programme's existing non-collapse law is therefore not merely philosophical:

$$\text{Assisted performance} \neq \text{retained human capability}.$$

The practical measurement question becomes: *after assistance is removed, what, exactly, came back with the human?*

3. Programme Genealogy: Tight Internal Continuity

The proposed architecture is not a detached taxonomy added late to the programme. It is the missing interface among objects already developed in separate manuscripts.

The genealogy yields one narrow residual contribution. Human Learning already contains Concept, Tool, Workflow, and Skill; Problem-to-Hypothesis already contains discriminable alternatives; Human LoRA already contains selective persistence; Fusion/Tunnel already contains Human Return. What was missing was a compact, typed readout layer for the *content* of retained change at the Human-AI session boundary.

Residual contribution. CTSA does not claim that learning consists of four things. It proposes that, when a retained Human-AI change is assessed after AI removal, one useful readout profile is whether the human can now demonstrate a new Conceptual distinction, appropriately select a Tool/access route, execute and transfer a Skill, or independently generate a non-equivalent Alternative—with losses, regulation, provenance, warrant, and direction kept separately visible.

4. External Literature as Constraint, Not Validation

The external literature does two jobs. First, it prevents CTSA from renaming established constructs as discoveries. Second, it identifies the narrower interface that remains under-integrated.

Krathwohl's revision of Bloom's taxonomy distinguishes factual, conceptual, procedural, and metacognitive knowledge [8]. Wirth et al. review self-regulated learning as layered content, strategy, and metacognitive regulation, and explicitly distinguish declarative, procedural, and conditional strategy knowledge—what, how, and when/why [9]. These literatures block any claim that Concept, Tool, and Skill are unprecedented categories.

Zhao, Lucas, and Bramley show that human conceptual bootstrapping can reuse and recombine prior concepts to construct more complex ones [12]. Mi and Summerfield show that "divide and conquer" curricula can facilitate later performance on multi-cue combinations [13]. These findings support the plausibility of compositional and role-structured learning, but they do not establish that CTSA classes are the correct or exhaustive decomposition of Human Return.

Human-AI literature constrains the control layer. Ma et al. show that calibrating human self-confidence matters for appropriate reliance [19]; Kim et al. show that natural-language uncertainty expression can reduce agreement with an LLM and reduce overreliance on incorrect answers in some conditions [20]; Vasconcelos et al. show that verification behavior depends on task and explanation costs [21]; and Buçinca et al. show that cognitive forcing can reduce overreliance, though sometimes at a cost to preference [22]. CTSA must therefore distinguish *what returned* from whether the human can monitor, select, check, and revise that return.

Finally, Vaccaro, Almaatouq, and Malone's meta-analysis separates human augmentation from Human-AI synergy [23]; Doshi and Hauser show that generative AI can improve individual creativity while increasing similarity across outputs [24]; Vives et al. report a discrimination-generalization trade-off associated with more differentiated semantic representations [14]; and Holton et al. show transfer-interference trade-offs in continual learning [15]. Together these results motivate CTSA's strongest non-collapse: a gain in one observable dimension does not by itself establish net human epistemic expansion.

5. Readout-Native Foundations

5.1 Retained difference before semantic label

Readout Genesis places retention before human memory semantics and meaning after readout. A reader makes a distinction accessible under a question; it does not retroactively place that meaning in the root. Approximate bridges must expose named defects, and familiar semantic labels cannot be imported as root-level truths merely because they are intuitive [4].

CTSA therefore begins with a declared pre/post difference, not four assumed cognitive boxes. Let

$$S_H^{\text{pre}}, \qquad S_H^{\text{post}}$$

be bounded task-relative descriptions of the human before AI exposure and after an unaided return interval. A declared reader $\mathcal{R}_Q$ asks whether a task-relevant difference can be recovered:

$$\Delta_H^{\text{ret}} = \mathcal{R}_Q(S_H^{\text{post}}) - \mathcal{R}_Q(S_H^{\text{pre}}).$$

This is architectural notation, not a validated psychological metric. The reader does not discover a

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Table 1: Internal programme genealogy. Each prior manuscript contributes a distinct object; self-citation establishes lineage, not independent validation.

Programme manuscript	Retained result	Function inside CTSA v9.1
When AI Expands Human Potential	AI can expose alternatives and revision pressure, but human potential requires corrigibility, route-openness, criticizability, and world-answerability.	Supplies the normative target: human epistemic development rather than convenience or output.
Human Learning as Epistemic Architecture	Learning runs from lived word/meaning through relation, retrieval, constraint and revision, then through problem-context, context-appropriate tools, workflow, practiced skill and feedback.	Supplies the concept-to-competence arc and the distinction between tool selection, workflow, and skill.
Experience Is the Human LoRA	Experience is not automatically learning; only selected residues persist, consolidate, transfer, or alter later interpretation/action.	Supplies the session-boundary retention requirement and the warning that exposure is not durable adaptation.
Readout Genesis / MEMK	Retained difference precedes semantic naming; meaning follows readout; approximate translation must expose loss; meaning, truth, and epistemic status remain distinct.	Prevents CTSA from becoming a root ontology and requires loss/claim-boundary accounting.
From Problem to Hypothesis	Semantic mobility is bounded and history-shaped; reachability differs from accessibility; idea count differs from readout-discriminable hypothesis classes.	Supplies Alternative Return as a genuinely non-equivalent accessible route and the no-silent-loss discipline.
Epistemic Fusion or Tunnel v7	Dialogue is a reciprocal, history-shaped semantic reaction; Difference, Resistance, and retained human Agency are non-substitutable for durable human epistemic expansion.	Supplies the Human-AI session architecture and the requirement that correction returns as human capacity.
Epistemic Fusion or Tunnel v8.1	Separates candidate status from validation, retention from direction, augmentation from synergy, and joint gain from delayed Human Return; operationalizes an unaided return battery.	Supplies the immediate measurement horizon into which CTSA is inserted: after retention, before the claim of human expansion.

complete human state; it checks whether a specified capability or distinction is demonstrably available under frozen criteria.

5.2 Meaning, persistence, and truth remain non-identical

Readout Genesis separates data from meaning, meaning from truth, and truth from epistemic status [4]. Fusion/Tunnel similarly treats AI outputs as knowledge-like candidates until checking and declared warrant criteria are satisfied [7]:

$$AI(Q) = K_{\text{like}}, \qquad K_{\text{like}} \neq K_{\text{validated}|V^*}.$$

A retained CTSA item can therefore be false, unsafe, misapplied, or poorly warranted. CTSA classifies the content of retained change; it does not certify knowledge.

5.3 Gain and loss are co-equal readouts

The programme's hypothesis-generation work introduced explicit defect accounting and a no-silent-loss discipline [5]. External evidence reinforces the need. Vives et al. show that representational differentiation can improve discrimination while impairing generalization [14]; Holton et al. show that reuse can support transfer while increasing interference [15]. A session can therefore add one capability while narrowing another.

Accordingly, CTSA does not define Human Return as "more items." It records a **Gain Profile** and a co-equal **Loss Profile**. Net interpretation is deferred until the loss, regulation, warrant, and direction audits are complete.

6. CTSA: Four Return Readout Classes

6.1 C - Conceptual Return

Definition. Conceptual Return occurs when the unaided human can reconstruct, discriminate, and appropriately use a concept or distinction that was not previously available in that usable form.

Human Learning already treats a word as a site where experience, memory, value, evidence, goals, constraints, and action converge, and builds meaning neighborhoods rather than treating vocabulary as inert storage [2]. Fusion/Tunnel treats articulation as a transport that can make distinctions nameable, comparable, and revisable [6]. External concept-learning work makes the compositional point stronger: existing concepts can become reusable building blocks for later concepts [12].

Conceptual Return is therefore not "one new word." It can be demonstrated when a person can independently:

- separate a distinction that was previously collapsed;
- state boundary conditions and misapplication cases;
- locate the concept in a relevant problem-context;

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Table 2: External literature map and the residual role left for CTSA.

Literature	What is already established or strongly motivated	Constraint on CTSA
Educational taxonomy / SRL	Conceptual and procedural knowledge are established; revised Bloom includes metacognitive knowledge; SRL distinguishes declarative, procedural, and conditional strategy knowledge and emphasizes monitoring/control.	C/T/S are not new kinds of knowledge. Tool Return must be framed as context-appropriate selection; metacognition must be cross-cutting rather than silently omitted.
Concept learning / curriculum	Existing concepts can be reused and recombined to construct more complex concepts; divide-and-conquer curricula can facilitate later multi-cue performance.	Supports compositional learning and role-structured access, but does not validate CTSA-specific causal ordering.
GenAI education	AI can improve in-session performance while later unaided learning may fail; meta-analytic averages are heterogeneous and assessment quality matters.	Justifies the session-boundary return horizon and demands delayed/transfer measures.
Human-AI reliance	Confidence calibration, uncertainty expression, verification cost, and cognitive forcing alter reliance and checking behavior.	Warrant cannot be inferred from fluency; CTSA needs a metacognitive/regulation layer and exercisable resistance.
Human-AI complementarity	Average human augmentation does not imply synergy over the better component.	CTSA Human Return must remain separate from system-level augmentation and synergy.
AI and diversity	GenAI can improve individual creative output while reducing collective diversity.	Alternative Return cannot be raw idea volume; diversity and non-equivalence require independent coding.
Semantic / continual learning trade-offs	Greater representational separation can improve discrimination while harming generalization; reuse can facilitate transfer while increasing interference.	Gross gain is not net expansion; a co-equal Loss Profile is required.

- contrast the pre-AI and post-AI model without reconstructive help from the AI.

Conceptual Return ≠ vocabulary exposure. A new term that cannot change discrimination, representation, or problem-appropriate use after AI removal is lexical exposure, not demonstrated Conceptual Return.

6.2 T - Tool-Selection Return

Definition. Tool-Selection Return occurs when the unaided human can identify, select, and justify an external instrument, source, representation, collaborator, or access route appropriate to the declared problem-context.

This class deliberately does not claim a new knowledge type. Human Learning already treats tool selection as conditional knowledge: the same AI may be a generator, critic, calculator, or distraction depending on the task [2]. SRL literature likewise distinguishes knowing that a strategy exists from knowing how, when, and why it is applicable [9]. Tool-Selection Return is the Human-AI session-boundary operationalization of that conditional competence.

A strong Tool-Selection Return requires the human to answer:

- What kind of access does this problem require?
- Why is this tool appropriate here rather than merely available?
- What can the tool return, and what can it not establish?
- What independent route remains if the tool fails or conflicts with another source?

Readout Genesis keeps tools neutral: repositories, calculators, AI systems, and workflow software are not epistemic primitives [4]. Tool selection therefore opens a route to checking; it does not itself provide warrant.

6.3 S - Skill-Execution Return

Definition. Skill-Execution Return occurs when the human can execute a task-relevant procedure accurately enough to matter, without the AI performing the decisive operation, and can transfer that capability to a sufficiently new case.

Human Learning draws the relevant distinction already: workflow executed once is not skill; skill requires proceduralization and feedback-rich practice [2, 10, 11]. Human LoRA adds persistence and transfer as conditions for durable adaptation [3]. Bastani et al. provide a direct Human-AI warning: assisted success can coexist with worse unaided performance after AI removal [16].

Candidate Skill-Execution Returns include the ability to independently identify a confounder, trace provenance, construct a falsifier, compare study designs, detect an error, or apply the same reasoning pattern to a novel case.

Skill Return ≠ procedural imitation. Repeating an AI-demonstrated procedure on the same template is insufficient. The decisive evidence is unaided execution plus transfer under a declared criterion.

6.4 A - Alternative-Generation Return

Definition. Alternative-Generation Return occurs when the unaided human can access or generate a non-equivalent rival explanation, hypothesis, fram-

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ing, or solution route that was not previously accessible in the declared task space.

This class is directly inherited from the programme's Difference requirement and Problem-to-Hypothesis architecture. Reachable and accessible are not identical; a candidate can exist structurally while remaining neglected because prior structure dominates search [5]. Fusion/Tunnel requires at least one non-equivalent distinction, rival, or route beyond paraphrase for expansion beyond the retained frame [6].

External evidence cautions against counting volume. Doshi and Hauser found that access to GenAI ideas improved judged individual creativity while outputs became more similar to one another [24]. Accordingly, twenty paraphrases of one causal story may yield no Alternative Return; one genuinely different causal route may.

Alternative Return $\neq$ idea count. Alternatives should be coded under declared equivalence rules by differences in consequence, explanation, intervention, prediction, or decision route.

7. Workflow and Metacognition: Cross-Cutting, Not Fifth Readouts

7.1 Where workflow belongs

Human Learning defines workflow as a repeatable organization of operations: steps, checkpoints, retrieval points, constraint tests, and tool-use decisions that make understanding inspectable and reusable [2]. Workflow is therefore a *process architecture*, not a return-content class.

A useful but non-mandatory relation is

$$C \to T \to \text{Workflow} \to S \to A \to C'.$$

The arrows are not a universal causal law. A new alternative may revise the concept; a new tool may reveal an alternative; a new skill may alter workflow. The purpose of the diagram is role separation.

7.2 Metacognitive regulation is a control axis

A global literature review exposes a second omission that should not be repaired by adding a fifth letter. Revised Bloom explicitly includes metacognitive knowledge [8]; SRL models treat planning, monitoring, strategy selection, evaluation, and control as central [9]. Human-AI studies add a specific reason: appropriate reliance depends partly on calibrated human self-confidence [19], and AI uncertainty expression can alter reliance and accuracy [20].

Metacognition is therefore represented as a cross-cutting regulation axis $\mathcal{M}$, asking whether the human can monitor uncertainty, choose when to use a CTSA gain, notice failure, and revise strategy. It governs the use of all four classes without being another content type.

CTSA asks what returned; metacognitive regulation asks whether the human can govern what returned.

8. Two Axes of Human Return

The Human Return Test and CTSA answer different questions and should be crossed, not substituted.

Axis I - What returned? CTSA classifies retained content/access as Conceptual, Tool-Selection, Skill-Execution, or Alternative-Generation Return.

Axis II - Can the human demonstrate that it returned? The existing Human Return battery tests reconstruction, discrimination, transfer, and independently generated next-question quality [6, 7].

Table 3: CTSA classes crossed with unaided demonstration.

Readout class	Unaided evidence
Conceptual	Reconstruct and correctly discriminate the new distinction; apply it under relevant boundary conditions.
Tool-Selection	Select an appropriate access route and explain why it fits the problem and what it cannot establish.
Skill-Execution	Execute the procedure and transfer it to a novel case without decisive AI assistance.
Alternative-Generation	Generate/discriminate a non-equivalent rival and state what observation or test would separate it.

A person may show strong Conceptual Return but weak Skill Return; Tool-Selection Return with no rival generation; Alternative Return without warrant; or Skill Return with poor metacognitive calibration. These profiles should remain visible rather than compressed into one "learning gain" score.

9. The Full Human-Return Audit

The final interpretation requires more than CTSA. A compact record is

$$\mathfrak{R}_H = \langle \mathcal{G}_{\text{CTSA}}, \mathcal{L}, \mathcal{M}, \mathcal{P}, \mathcal{W}, \Delta_{\text{dir}} \rangle.$$

The tuple is a bookkeeping architecture, not a calibrated universal score.

9.1 Gain Profile

What new CTSA items can the human demonstrate unaided relative to a frozen pre-AI baseline?

9.2 Loss Profile

What previously accessible distinctions, alternatives, uncertainty sensitivities, checking habits, or skills are now less available? The loss profile is co-equal with gain because discrimination, transfer, and interference can trade off [14, 15].

9.3 Metacognitive regulation

Can the person identify uncertainty, monitor error, decide when a strategy/tool applies, and change course? Confidence in self must remain distinct from trust in AI [19, 7].

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9.4 Provenance

Where did the retained item come from: human-originated, AI-generated, external-record-derived, checker-derived, or mixed? Provenance supports auditability; it is not a truth certificate.

9.5 Warrant and exercisable resistance

What independent checking has the item survived? Vasconcelos et al. show that verification is behaviorally sensitive to task and explanation costs [21]; Buçinca et al. show that cognitive forcing can reduce overreliance by requiring more active engagement [22]. Therefore an available resistance route that a person will not or cannot exercise should not be treated as equivalent to effective checking.

9.6 Epistemic direction

Fusion/Tunnel already separates retention magnitude from retained direction: a persistent change can enlarge corrigibility or stabilize confirmation, same-basin attraction, outsourcing, or unwarranted certainty [7]. CTSA therefore cannot define "more" as "better."

10. Session-Boundary Architecture

The full programme can now be represented as a continuous sequence:

$$
\begin{aligned}
&\text{Frozen Human baseline} \rightarrow K_{\text{like}}\\
&\rightarrow \text{Difference / Resistance} \rightarrow H \leftrightarrow AI\\
&\rightarrow \text{Retention gate} \rightarrow \boxed{\text{CTSA Return Profile}}\\
&\rightarrow \text{Unaided Return Test} \rightarrow \text{Gain/Loss}\\
&\rightarrow \text{Regulation} \rightarrow \text{Provenance/Warrant}\\
&\rightarrow \text{Epistemic direction.}
\end{aligned}
$$

This location is conceptually important. CTSA is neither the beginning of inquiry nor the final truth test. It sits at the boundary where conversational residue is claimed to have become human capability. The architecture preserves the non-collapse laws developed across the programme:

- AI fluency $\neq$ human baseline;
- explanation $\neq$ verification;
- output count $\neq$ epistemic diversity;
- exposure $\neq$ retention $\neq$ improvement;
- trust in AI $\neq$ calibrated confidence in self;
- augmentation $\neq$ synergy $\neq$ Human Return;
- retained gain $\neq$ net expansion;
- provenance $\neq$ warrant;
- persistence $\neq$ truth.

Vaccaro et al.'s meta-analysis reinforces the system-level separation between augmentation and synergy [23]; Bastani et al. independently block the collapse between assisted performance and later human learning [16].

11. Worked Example: Checking an AI-Generated Causal Claim

Suppose a user asks whether a four-day work week "raises productivity by 40%." Before AI use, the person has a vague concept of evidence, no explicit distinction between correlation and causal effect, knows only generic web search as a checking tool, has no repeatable study-comparison skill, and can imagine only one explanation: fewer working days improve focus.

After a structured Human-AI inquiry and an unaided delay, a candidate return profile might be:

Conceptual Return: the user independently distinguishes correlation, causal effect, selection bias, measurement choice, and short-run versus long-run productivity.

Tool-Selection Return: the user chooses a primary report, longitudinal dataset, preregistered experiment, or domain-specific database according to the claim rather than treating a search summary as decisive evidence.

Skill-Execution Return: the user independently states the target causal claim, identifies plausible confounders, compares designs, and specifies evidence that would weaken the preferred explanation.

Alternative-Generation Return: the user generates non-equivalent rivals such as selection effects, Hawthorne effects, attrition, changed measurement, or baseline-performance differences.

The Human Return battery then tests whether those changes survive on a different causal claim. The Loss Profile asks whether the user became less tolerant of uncertainty or over-learned a single skeptical template. Metacognitive regulation asks whether the user knows when the template applies. Provenance separates AI-suggested rivals from independently recovered evidence. Warrant records which rivals remain merely possible and which have evidential support.

The example shows why CTSA is not a checklist of "things AI taught." It is a session-boundary readout of what became usable in the human and what remains open to correction.

12. Hypotheses and Legitimate Defeaters

The architecture generates open hypotheses; it does not protect them from removal.

H1 - Incremental Readout Hypothesis [Open]. A CTSA Return Profile will explain reproducible variance in delayed unaided human capability beyond ordinary post-test performance, recall, confidence, and established concept/skill measures.

H2 - Partial Dissociation Hypothesis [Open]. Conceptual, Tool-Selection, Skill-Execution, and Alternative-Generation returns will show partially dissociable profiles under matched interaction con-

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ditions rather than behaving as one interchangeable gain.

H3 - Return-vs-Exposure Prediction [Externally constrained]. Items visible only while AI support is present will predict assisted performance more strongly than delayed unaided capability. Bastani et al. provide direct evidence for this general separation, although not for CTSA classes specifically [16].

H4 - Gain/Loss Asymmetry Hypothesis [Open]. Positive CTSA gain can coexist with negative loss in previously accessible routes or checking habits; gross gain will therefore overestimate net human expansion. Semantic and continual-learning tradeoffs make this plausible without validating the CTSA-specific claim [14, 15].

H5 - Recombination Hypothesis [Open]. CTSA gains will sometimes interact compositionally: a new Concept may improve Tool selection; Tool plus Skill may make an Alternative accessible; a new Alternative may trigger conceptual restructuring. Conceptual bootstrapping and curriculum-learning results motivate but do not establish this dependency [12, 13].

H6 - Regulation/Direction Hypothesis [Open]. Equal gross CTSA gains will yield different later corrigibility depending on metacognitive calibration, effective resistance, provenance visibility, and whether the interaction reinforced or escaped a preexisting semantic basin.

Removal conditions. CTSA should be narrowed, merged, or abandoned if (i) independent coders cannot classify the four return classes reliably; (ii) the classes add no stable information beyond established conceptual/procedural/conditional/metacognitive measures; (iii) delayed CTSA profiles fail to predict or explain anything beyond ordinary performance and transfer; (iv) gain/loss coding is too task-dependent to be reproducible under declared scopes; or (v) a simpler rival architecture captures the same phenomena with fewer assumptions.

These are legitimate defeaters, not rhetorical threats. The programme's own methodological rule remains: an architecture that cannot lose variables when they stop doing explanatory work becomes vocabulary rather than theory [7].

13. Empirical Programme

The first study should not attempt to validate the whole Readout Programme. It should test the smallest claim that can make CTSA earn its place.

13.1 Primary question

> *Does a CTSA Return Profile add reliable, reproducible information about delayed unaided human capability beyond ordinary performance scores and established knowledge/skill measures?*

13.2 Minimum design

A preregistered study should include:

- a frozen pre-AI human baseline;
- a Human-only control and an AI-assisted condition;
- where feasible, an AI-only baseline for augmentation/synergy separation;
- an immediate assisted measure, followed by AI removal;
- at least one immediate unaided measure and one delayed novel-transfer measure at a preregistered domain-appropriate interval;
- tasks that permit concept discrimination, tool selection, skill execution, and rival generation;
- independent coding of gain, loss, provenance, and warrant;
- a metacognitive measure such as confidence calibration, uncertainty monitoring, or strategy-choice accuracy.

13.3 Candidate outcomes

- **Conceptual:** correctly reconstructed distinctions; boundary-condition accuracy; misapplication rate.
- **Tool-Selection:** appropriate selection accuracy; justification quality; rejection of an inappropriate but salient tool.
- **Skill-Execution:** unaided procedural accuracy; error detection; self-correction; transfer to structurally related novel tasks.
- **Alternative-Generation:** number of readout-distinct rival classes; discriminability; ability to state what evidence separates them.
- **Loss:** decline in previously available alternatives, generalization, uncertainty sensitivity, checking behavior, or baseline skill.
- **Regulation:** confidence calibration, monitoring accuracy, and context-appropriate strategy/tool choice.

13.4 Analysis priority

The decisive analysis is incremental validity, not whether all four CTSA scores rise. Models should compare delayed human outcomes predicted by ordinary performance/recall measures alone versus the same models augmented by CTSA, Loss, and regulation variables. Cross-validation or held-out replication should be preferred to post-hoc fit. If CTSA adds no stable explanatory or predictive value, the taxonomy has not earned its complexity.

14. Adversarial Cases

14.1 Vocabulary illusion

A user learns ten technical terms and repeats their definitions. If no new distinction can be applied, no

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tool selected, no skill transferred, and no rival generated, this is lexical exposure rather than demonstrated human expansion.

14.2 Tool-name illusion

A user can name PubMed, a calculator, a causal diagram, and a database but cannot choose among them for a new problem. Recognition without conditional selection is not strong Tool-Selection Return.

14.3 Procedural imitation illusion

A user reproduces the exact workflow demonstrated with AI but fails on a structurally related case. The interaction may have produced local imitation rather than Skill-Execution Return.

14.4 Alternative-volume illusion

An AI produces twenty differently worded explanations from the same causal frame. Without non-equivalent consequences or discriminating tests, the apparent diversity collapses to one Alternative class.

14.5 Gain-with-loss tunnel

A user gains new concepts and a fast procedure but becomes less willing to check evidence, less sensitive to uncertainty, or less able to generate rivals independently. Gross CTSA rises while epistemic direction deteriorates. The loss and direction audits are therefore constitutive to interpretation.

14.6 Metacognitive blind gain

A person acquires a real skill but cannot recognize when it is inappropriate, becomes overconfident, and applies it indiscriminately. Skill Return can be positive while metacognitive regulation is negative. CTSA alone must not call this net expansion.

15. Claim Boundaries

This paper does **not** claim that:

- CTSA is an exhaustive taxonomy of cognition, learning, intelligence, expertise, or wisdom;
- C, T, S, and A are biological modules, literal neural variables, or four new kinds of knowledge;
- all four classes are necessary for every learning goal;
- more CTSA items always imply more human potential;
- Tool-Selection Return is conceptually distinct from all established conditional or strategic knowledge constructs;
- Skill-Execution Return replaces established procedural-learning or expertise theories;
- retained alternatives are true merely because they are diverse;
- a retained concept, tool, skill, or alternative is beneficial merely because it persists;
- metacognitive regulation can be reduced to self-confidence alone;
- AI is the sole cause of post-session human change;
- the proposed readouts already possess validated universal scales;
- Human Return should define success where durable human capability is outside the declared goal.

The strongest permitted claim is architectural: CTSA proposes a session-boundary method for keeping several questions separate that Human-AI research can otherwise collapse—what changed in the human, whether it persisted, what was lost, whether the person can regulate it, where it came from, what checking it survived, and whether the retained direction expands or narrows corrigible inquiry.

16. Standalone Thesis and Programme Continuity

> **CTSA does not propose four new kinds of human knowledge. It proposes a session-boundary measurement architecture for distinguishing what kind of capability, if any, has actually returned to the human after AI assistance is removed. Conceptual Return identifies a new usable distinction; Tool-Selection Return identifies a newly usable access route; Skill-Execution Return identifies unaided transferable performance; and Alternative-Generation Return identifies a non-equivalent route the human can independently reach. These gains count toward human expansion only when persistence, losses, metacognitive regulation, provenance, warrant, and epistemic direction remain separately auditable.**

The thesis continues the internal programme without replacing it. *When AI Expands Human Potential* supplies the normative target; *Human Learning as Epistemic Architecture* supplies the movement from meaning to corrigible competence; *Experience Is the Human LoRA* supplies selective persistence; Readout Genesis supplies retained-difference, translation, and loss discipline; *From Problem to Hypothesis* supplies bounded semantic mobility and discriminable alternatives; and *Epistemic Fusion or Tunnel* supplies reciprocal dialogue, candidate status, resistance, retention, direction, and the unaided Human Return horizon. CTSA is the measurement interface placed exactly where those strands meet: the moment a conversational residue is claimed to have become a human capability.

The next step is deliberately modest. CTSA should not be defended by adding more terminology. It should be tested against simpler measures, established learning constructs, and rival models. If it

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adds stable information about delayed unaided capability, it earns a role. If it does not, the programme should remove or compress it.

References

Programme lineage

[1] Lahtee, Y. (2026). *When AI Expands Human Potential: Reflective Dissonance, Epistemic Agency, and Constraint*. Open Civil Science Initiative preprint, 25 March 2026. Programme manuscript; not peer reviewed.

[2] Lahtee, Y. (2026). *Human Learning as Epistemic Architecture: A Method for Word Mapping, Life-Concept Graphs, Constraint Testing, and Corrigible Agency in the AI Age*. Revised v4 preprint, 28 June 2026. Programme manuscript; not peer reviewed.

[3] Lahtee, Y. (2026). *Experience Is the Human LoRA: A Readout-Retention Theory of Selective Model Change*. Open Civil Science Initiative preprint, 18 July 2026. Programme manuscript; not peer reviewed.

[4] Readout Genesis Programme. (2026). *READOUT_GENESIS_MEMK_DOMAIN v0.3.0 ANCHORED*. Root-connected semantic-cognitive domain specification, 23 July 2026. Architecture-conditional / unresolved domain.

[5] Lahtee, Y. (2026). *From Problem to Hypothesis: Dynamic Semantic Mobility, History-Shaped Access, and the Readout-Discriminable Hypothesis Space*. Standalone final author draft, GLOSA K0, 4 September 2026.

[6] Lahtee, Y. (2026). *Epistemic Fusion or Epistemic Tunnel: A Phenomenologically Anchored, History-Shaped Architecture of Human-AI Semantic Coupling, Retention, and Epistemic Direction*. Human-AI Readout Programme, standalone v7 Genesis-aligned conceptual architecture, 5 September 2026.

[7] Lahtee, Y. (2026). *Epistemic Fusion or Epistemic Tunnel: A Phenomenology-Anchored, Global-Literature-Constrained, Problem-First Architecture of Human-AI Collaboration*. Human-AI Readout Programme, v8.1 strong-claim / legitimate-defeater edition, 5 September 2026.

External literature

[8] Krathwohl, D. R. (2002). A revision of Bloom's taxonomy: An overview. *Theory Into Practice*, 41(4), 212–218.

[9] Wirth, J., Stebner, F., Trypke, M., Schuster, C., et al. (2020). An interactive layers model of self-regulated learning and cognitive load. *Educational Psychology Review*, 32, 1127–1149. https://doi.org/10.1007/s10648-020-09568-4.

[10] Anderson, J. R. (1982). Acquisition of cognitive skill. *Psychological Review*, 89(4), 369–406.

[11] Ericsson, K. A., Krampe, R. T., & Tesch-Römer, C. (1993). The role of deliberate practice in the acquisition of expert performance. *Psychological Review*, 100(3), 363–406.

[12] Zhao, B., Lucas, C. G., & Bramley, N. R. (2024). A model of conceptual bootstrapping in human cognition. *Nature Human Behaviour*, 8, 125–136. https://doi.org/10.1038/s41562-023-01719-1.

[13] Mi, Q., & Summerfield, C. (2026). Human curriculum learning of a cue combination task. *Nature Human Behaviour*, 10, 1539–1556. https://doi.org/10.1038/s41562-026-02452-1.

[14] Vives, M.-L., de Bruin, D., van Baar, J. M., FeldmanHall, O., et al. (2023). Uncertainty aversion predicts the neural expansion of semantic representations. *Nature Human Behaviour*, 7, 765–775. https://doi.org/10.1038/s41562-023-01561-5.

[15] Holton, E., Braun, L., Thompson, J. A. F., Grohn, J., et al. (2026). Humans and neural networks show similar patterns of transfer and interference during continual learning. *Nature Human Behaviour*, 10, 111–125. https://doi.org/10.1038/s41562-025-02318-y.

[16] Bastani, H., Bastani, O., Sungu, A., Ge, H., Kabakcı, Ö., & Mariman, R. (2025). Generative AI without guardrails can harm learning: Evidence from high school mathematics. *Proceedings of the National Academy of Sciences*, 122(26), e2422633122. https://doi.org/10.1073/pnas.2422633122.

[17] Yan, L., Greiff, S., Lodge, J. M., et al. (2025). Distinguishing performance gains from learning when using generative AI. *Nature Reviews Psychology*, 4, 435–436. https://doi.org/10.1038/s44159-025-00467-5.

[18] Deng, R., Jiang, M., Yu, X., Lu, Y., & Liu, S. (2025). Does ChatGPT enhance student learning? A systematic review and meta-analysis of experimental studies. *Computers & Education*, 227, 105224. https://doi.org/10.1016/j.compedu.2024.105224.

[19] Ma, S., Wang, X., Lei, Y., Shi, C., Yin, M., & Ma, X. (2024). "Are You Really Sure?" Understanding the effects of human self-confidence calibration in AI-assisted decision making. In *Proceedings of CHI 2024*, Article 840. https://doi.org/10.1145/3613904.3642671.

[20] Kim, S. S. Y., Liao, Q. V., Vorvoreanu, M., Ballard, S., & Vaughan, J. W. (2024). "I'm Not Sure, But..." Examining the impact of large language models' uncertainty expression on user reliance and trust. In *FAccT 2024*, 822–835. https://doi.org/10.1145/3630106.3658941.

[21] Vasconcelos, H., Jörke, M., Grunde-McLaughlin, M., Gerstenberg, T., Bernstein, M. S., & Krishna, R. (2023). Explanations can reduce overreliance on AI systems during decision-making. *Proceedings of the ACM on Human-Computer Interaction*, 7(CSCW1), Article 129. https://doi.org/10.1145/3579605.

[22] Buçinca, Z., Malaya, M. B., & Gajos, K. Z. (2021). To trust or to think: Cognitive forcing functions can reduce overreliance on AI in AI-assisted decision-making. *Proceedings of the ACM on Human-Computer Interaction*, 5(CSCW1), Article 188. https://doi.org/10.1145/3449287.

[23] Vaccaro, M., Almaatouq, A., & Malone, T. W. (2024). When combinations of humans and AI are useful: A systematic review and meta-analysis. *Nature Human Behaviour*, 8, 2293–2303. https://doi.org/10.1038/s41562-024-02024-1.

[24] Doshi, A. R., & Hauser, O. P. (2024). Generative AI enhances individual creativity but reduces the collective diversity of novel content. *Science Advances*, 10(28), eadn5290. https://doi.org/10.1126/sciadv.adn5290.

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How Humans Should Converse with AI: A Problem-First Adaptive Dialogue Conversion Protocol for Durable Human Capability

v1.3 glosa-checked edition • 6 September 2026 • 10.5281/zenodo.22481928

What the chapter states. “A human-AI conversation should be evaluated by the human state it produces after the conversation, but the conversation should also begin from an agenda that remains meaningfully human-owned. For learning, judgment, problem solving, or human development, the default topic should therefore originate in a live human problem rather than an AI-supplied agenda.”

Status. “K0 theory, problem-first adaptive protocol, and empirical research agenda.”

Falsifier(s). “The protocol should be simplified if: (i) ordering the stages adds no value beyond generic ‘think critically’ instructions; [. . .] or (ix) action/world-feedback closure adds no predictive or calibration value beyond Human Return alone.”

Read before. This book’s Chapters 2, 10, 16, 17, 27, 29, 30, 31 (mechanical citation list; Appendix A).

Feeds into. Ch. 34, 35 — companion papers, same deposit batch (mechanical citation list; Appendix A).

HUMAN–AI READOUT PROGRAMME • DIALOGUE CONVERSION • V1.3 GLOSA-CHECKED EDITION • SEPTEMBER 2026

How Humans Should Converse with AI

A Problem-First Adaptive Dialogue Conversion Protocol for Durable Human Capability

From live problem selection to epistemic expansion, Human Return, action, and world feedback

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ORCID 0009-0005-3861-0626 • v1.3 — glosa-checked edition of v1.2, 6 September 2026 original

Status. K0 theory, problem-first adaptive protocol, and empirical research agenda. This v1.2 revision uses v1.1 as its anchor and adds the missing topic-selection layer before triage. The canonical eight-stage Dialogue Conversion Protocol is preserved, but human-development use now begins from a live problem: a presently consequential unresolved difference in the person's life, work, understanding, or environment. Problem-first is a default, not a monopoly; curiosity, play, art, basic inquiry, and routine delegation remain legitimate entry modes. The revision also closes the loop after Human Return through action and world feedback, so AI dialogue is judged not only by what becomes thinkable or retained but, where relevant, by whether revised understanding survives contact with the situation that generated the problem. Licence CC BY 4.0.

Edition note. This is glosa-checked edition v1.3 of the founder's v1.2 (6 September 2026 original). Changes are limited to: (1) citation-lineage repair — every internal Human-AI Readout Programme self-citation now carries its deposited Zenodo DOI and the actually-deposited version string; Table 1's six footnotes that pointed to a single miscellaneous internal reference are split into their own correctly-cited entries, and its two rows with no separately deposited source are marked as programme working notes; (2) renaming §2.1 to "Contribution boundary" and rewording four priority-adjacent phrases in it. No section, equation, table, proposition, or reference was added, removed, renumbered in content, or otherwise reworded. The v1.2 PDF remains the anchor document of record; this edition is a citation-accurate copy for deposit, not a superseding claim.

Central claim. A human-AI conversation should be evaluated by the human state it produces after the conversation, but the conversation should also begin from an agenda that remains meaningfully human-owned. For learning, judgment, problem solving, or human development, the default topic should therefore originate in a live human problem rather than an AI-supplied agenda. The outer architecture is Topic Entry → Triage → Minimum Sufficient Dialogue → Human Return → Action/World Feedback; inside Standard and Critical use, the canonical eight stages remain Anchor → Expand → Oppose/Boundary-Test → Discriminate → Verify → Integrate → Remove → Return. AI is used aggressively for search, divergence, translation, recombination, counterexample, and critique; humans retain agenda/problem ownership, normative responsibility, world contact, integration, and final accountable judgment. The practical design objective remains minimum sufficient friction: enough resistance to preserve learning, verification, fallback, and agency, but not so much burden that users abandon the protocol or automate trivial work inefficiently.

ABSTRACT

Artificial intelligence increasingly mediates not only answers but attention: it can help determine what users ask about next, which interpretations become salient, and which problems are treated as worth solving. A protocol that begins only after a prompt therefore starts too late. This paper elevates the adaptive Dialogue Conversion Protocol (DCP) by adding a Topic Entry Condition before triage. For learning, judgment, problem solving, and human development, the default topic should arise from a live problem: a presently consequential unresolved difference in the person's life, work, understanding, or environment. The rationale is architectural rather than therapeutic: problem-first dialogue preserves human agenda ownership, bounds relevance, provides routes to world-side feedback, and makes durable Human Return more observable. Yet problem-first is not problem-only; curiosity, play, art, basic inquiry, and routine delegation remain protected entry modes.

The v1.1 adaptive deployment layer is retained: DCP-Lite uses State, Challenge, Check, Own for ordinary low-stakes use; DCP-Standard uses the full Anchor, Expand, Oppose/Boundary-Test, Discriminate, Verify, Integrate, Remove, Return architecture when learning, research, judgment transfer, or future competence matters; DCP-Critical adds independent verification, escalation, and accountable professional/world-side judgment. The revision further closes the human-development loop after Return through Action and World Feedback, yielding a problem-conditioned cycle: Live Problem → Question → Dialogue → Human Return → Action → World Feedback → revised understanding or a new problem.

The resulting proposal is a Problem-First, Adaptive Dialogue Conversion Protocol. Its governing principles are Human Agenda Ownership, Minimum Sufficient Friction, World Answerability, and Open Exploration Preserved. The paper proposes new empirical tests comparing live-problem-first, AI-suggested-topic, and curiosity-led dialogue while holding task difficulty and assistance constant. The central prediction is that, when durable human development is the goal, problem-first topic selection will often improve relevance, ac-

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tion transfer, and Human Return without requiring that all worthwhile inquiry begin from practical problems.

Keywords: human-AI dialogue; problem-first inquiry; problem framing; topic selection; epistemic agency; Human Return; AI-assisted learning; overreliance; critical thinking; curiosity; metacognition.

1 Internal Programme Literature Review: Where the Protocol Comes From

The protocol is not introduced as a free-standing list of prompt tips. It is the operational convergence of a sequence of prior papers in the Human-AI Readout Programme. The sequence matters because each paper solved a different failure mode that becomes visible once dialogue is treated as a history-shaped epistemic process rather than as independent prompt-response pairs.

The literature converges on a role-separated architecture. The human enters with a retained state—experience, assumptions, stakes, partial knowledge, and unresolved differences—but exports only a bounded articulation. AI receives the articulation, not the whole person. Repeated turns can then alter what becomes easier to think next. The key question is not whether AI produces more content, but whether dialogue enlarges a person's reachable, discriminable, corrigible space of inquiry and whether that enlargement survives the session boundary [1, 2].

2 External Literature: Convergence and the Remaining Gap

External research independently supports several pieces of this architecture while also showing why a full dialogue protocol is still missing.

First, problem construction and problem framing are themselves consequential cognitive operations, not neutral preliminaries. Reiter-Palmon reviews evidence that how a problem is constructed influences creative production, while recent innovation research treats problem framing as a dynamic creative process in its own right rather than a fixed input to solution generation [13, 14]. This supports a narrower claim than "all knowledge begins from problems": when an agent seeks to learn, decide, repair, or act, selecting and framing the live problem is part of the epistemic work. The topic of AI dialogue is therefore not a free variable.

Second, human-AI combination is not automatically synergistic. Vaccaro, Almaatouq, and Malone meta-analyzed 106 experiments and found average augmentation relative to humans alone but no average synergy relative to the better of human or AI; decision tasks performed particularly poorly compared with creation tasks [15]. This makes "human+AI" an insufficient success criterion.

Third, overreliance is a design problem, not merely an individual defect. Buçinca, Malaya, and Gajos show that cognitive forcing functions can reduce overreliance, although users may prefer less effortful designs [16]. Vasconcelos et al. further show that reliance depends on the cost-benefit structure of engaging with AI explanations, while de Jong et al. find that partial explanations can reduce overreliance under some conditions [17, 18]. Recent CHI work on AI-assisted critical thinking similarly reports reduced overreliance but increased cognitive load, implying a real friction-performance trade-off rather than a free benefit [19].

Fourth, learning science predicts that easy access to complete solutions can conflict with durable learning. The generation effect, retrieval practice, and productive-failure literatures show that attempting, retrieving, reconstructing, and even failing productively can improve later retention or transfer compared with passive presentation or immediate instruction [20–24]. This is not evidence that AI must always withhold answers; it is evidence that the timing and form of assistance can alter what the human retains.

Fifth, contemporary AI evidence shows that this distinction is empirically consequential. Bastani et al. report that generative-AI access can sharply improve practice performance while unguarded use reduces later unaided performance after the AI is removed; guardrailed tutoring mitigates the loss [25]. Glickman and Sharot show that repeated interaction with biased AI can alter later human judgments, often without users fully recognizing the influence [26]. Buçinca et al. argue explicitly for optimizing human-centric objectives such as learning rather than accuracy alone [27].

The residual gap is therefore narrow and important: the literature contains mechanisms of problem framing, overreliance, learning, explanation, task-specific synergy, and feedback, but it lacks a general conversation-level architecture that asks which topic deserves attention before the prompt, distinguishes AI-suggested agenda from human-owned live problems, separates expansion from verification, introduces deliberate opposition and equivalence testing, and ends with AI removal plus an unaided Human Return audit and, where action is relevant, renewed world feedback.

2.1 Contribution boundary

No individual component of the protocol is claimed as a new contribution; each has substantial predecessors, cited above. Generation, retrieval practice, productive failure, cognitive forcing, explanation, adversarial critique, source verification, and reflection each have substantial predecessors. The residual contribution is their ordered integration around a different terminal outcome: the human state after decisive AI support is removed. Six features are proposed as the distinctive architectural combination: (i) a topic-entry distinction between live problems, open exploration, and routine delegation; (ii) a pre-prompt human baseline; (iii) explicit discrimination of non-equivalent alternatives rather than raw answer count; (iv) a mandatory transition from AI-mediated expansion to partially independent world-side verification; (v) an AI-removal bound-

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Table 1: Internal programme synthesis. The present protocol retains the result in the middle column and operationalizes it in the right column.

Prior work	Retained result	Role in the Dialogue Conversion Protocol
When AI Expands Human Potential	AI can expose alternatives, hidden assumptions, and productive dissonance, but expansion requires corrigibility, route-openness, criticism, and world-answerability; fluent output is not human growth.	Supplies the target: disciplined revision rather than convenience or throughput. [3]
The Readout Condition	Epistemically weight-bearing distinctions require existence, attribution, provenance, and disclosure of augmentation; a system cannot self-certify truth merely by producing a coherent claim.	Supplies provenance typing and the no-self-certification rule. [4]
From Problem to Hypothesis	Practical inquiry begins from a retained problem: a consequential mismatch requiring explanation, repair, or decision; imagination then enlarges a bounded reachable hypothesis space.	Supplies the Topic Entry Condition, Problem-First Dialogue Principle, non-equivalent alternative generation, and semantic-basin checking. [5]
The Epistemic Chain Reaction	The human retains the problem/question while AI decomposes, transports, branches, critiques, and produces readout-distinguishable candidate routes.	Supplies controlled expansion and recursive branching. [6]
Candidate Forgetting	Knowledge-like form can be mistaken for validated knowledge; provisional candidates harden when status is forgotten.	Supplies candidate labels and mandatory re-verification before load-bearing use (programme working notes, not separately deposited).
Experience Is the Human LoRA	Exposure is not retention. Some residues alter later human interpretation/action; others pass through.	Supplies the session-boundary retention question. [7]
AI, Translation, and Event-Specific Context	AI reorganizes recorded interpretations and context access but does not directly reconstruct event reality.	Supplies the distinction between language mediation and target-world access. [8]
Human Learning as Epistemic Architecture	Concepts and tools become usable through problem-context, deliberate practice, real problems, feedback, and revision.	Supplies problem-context binding, Tool/Skill Return, and the Action → World Feedback closure. [9]
Context Wisdom / Translation of Reality	Meaning is context-sensitive and interpretation-bounded; repeated relations can become easier to traverse without becoming more warranted.	Supplies context-checking and semantic-momentum caution (programme working notes, not separately deposited).
Epistemic Fusion or Epistemic Tunnel	Repeated Human-AI dialogue can expand or narrow future accessible routes; amplification is direction-neutral; resistance and selective retention determine epistemic direction.	Supplies precommit resistance, stop/reset rules, and the expansion-versus-tunnel diagnostic. [1]
Choice Begins Before Choice	Humans do not choose from all formal possibilities but from a history-shaped live field; AI can reshape which alternatives become thinkable before the next prompt.	Supplies the pre-prompt human state and live-possibility audit. [2]
CTSA Human Return Readout	Coupled performance is not human capability. Durable return must be demonstrated unaided as Concept, Tool, Skill, or Alternative Return.	Supplies the terminal Remove/Return test. [10]
After Labour / Human Conversion Imperative	Human potential also depends on exit, material claim, social reproduction, and reversibility; maximum AI use is not maximum human conversion.	Supplies system-level constraints and urgency where dependency becomes hard to reverse. [11, 12]

ary followed by a typed Human Return audit; and (vi) closure, where relevant, through action and returned world feedback. If this ordering adds no predictive or intervention value beyond simpler critical-thinking instructions, the protocol should be compressed rather than defended by terminology.

3 The Unit of Analysis: A Conversation Changes the Next Conversation

A prompt is not the starting state. Let $H_{s,0}$ denote the human state entering session s, containing retained experience, current meanings, values, unresolved differences, recent interaction history, and a live-possibility profile. Linguistic articulation exports only a bounded question:

$$H_{s,0} \xrightarrow{L_H} Q_{s,0}. \tag{1}$$

AI responds to Q, not to the entire human state. The response becomes part of the human's next interpretive environment:

$$Q_{s,t} \to \mathrm{AI}_{s,t} \to Y_{s,t} \xrightarrow{R_H^{\mathrm{AI}}} E_{H,s,t}. \tag{2}$$

If some residue is retained, then

$$H_{s+1,0} = U_H\left(H_{s,0}, E_{H,s,*}^{\mathrm{AI}}, \delta^{\mathrm{world}}, X^{\mathrm{other}}\right), \tag{3}$$

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so the next session can begin from a changed reader. This is why conversational quality cannot be judged only by the current answer. Repeated interaction may make objections easier to formulate or harder to imagine, widen or narrow live possibilities, and improve or erode unaided skill [1, 2, 26].

4 Topic Entry Condition: Choose the Conversation Before Optimizing It

A dialogue protocol that begins with Anchor still assumes that some topic has already captured the user's attention. In an AI-rich environment this assumption becomes unsafe: systems can suggest topics, recommend goals, surface concerns, and recursively shape what feels worth asking next. Topic selection is therefore an agency problem before it is a prompting problem.

For human-development use, define a live problem as

$$P^{\text{live}}_{H,t} = \text{a presently consequential unresolved difference} \tag{4}$$

in the person's life, work, understanding, relationship to evidence, or environment. "Problem" is used broadly: it can be a failure, mismatch, decision, unexplained recurrence, missing capability, conflict between expectation and observation, or a question whose unresolved state matters enough to justify attention. This is consistent with the programme's earlier move from generic observation to a retained problem that demands explanation, repair, or decision [5].

The Topic Entry Condition separates three legitimate entry modes:

$$\begin{aligned}\text{TopicEntry} \in \{&\text{LiveProblem}, \text{OpenExploration},\\ &\text{RoutineDelegation}\}.\end{aligned} \tag{5}$$

LiveProblem is the default when the goal is learning, judgment, problem solving, behavior change, or human development. OpenExploration protects curiosity, play, art, basic science, speculative inquiry, and interests whose value is not yet tied to a practical mismatch. Curiosity research itself treats exploration as an adaptive process driven by uncertainty, information, and learning progress rather than as wasted attention [28]. RoutineDelegation covers low-value or repetitive tasks where the user need not convert every operation into durable human skill.

Problem-First Dialogue Principle [Open]. When AI is used for human development rather than entertainment or routine execution, the default topic should originate in a live human problem rather than an AI-supplied agenda. The reason is fourfold:

$$\text{Problem-First} \Rightarrow \begin{cases}\text{Human Agenda Ownership},\\ \text{Bounded Relevance},\\ \text{World-Side Testability},\\ \text{Observable Human Return}.\end{cases} \tag{6}$$

The claim is not that every problem is correctly framed. Problem construction can itself be creative, contested, and revisable [13, 14]. AI may help elicit or reframe a problem, but any reframing that changes the agenda should be made visible to the user rather than silently replacing the original concern.

Problem-first is not problem-only. A universal requirement that every conversation justify itself by present utility would destroy exploration. The protocol therefore preserves an explicit exploration gate:

$$\text{Problem-First} \neq \text{Problem-Only}. \tag{7}$$

The relevant governance question is not "Is this topic useful now?" but "Who selected the topic, why is attention being spent here, and does the user still recognize the difference between a self-owned inquiry and an AI-supplied agenda?"

A compact Topic Entry prompt is: "What unresolved difference in my present life, work, understanding, or environment deserves attention now? If I am exploring from curiosity rather than solving a problem, keep that mode explicit rather than inventing a problem for me."

5 Deployment Layer: Triage After Topic Entry

The canonical eight-stage DCP is a protection architecture, not a requirement that every user perform a mini peer review before asking for a translation or formatting change. Real-life deployment begins by asking what must be preserved.

Let the deployment policy depend on

$$\begin{aligned}\pi^{\text{deploy}} = f\big(&\text{Stakes}, \text{LearningNeed}, \text{Irreversibility},\\ &\text{DependencyRisk}, \text{UserSkill}\big).\end{aligned} \tag{8}$$

The proposed triage has three modes.

DCP-Lite: ordinary low-stakes use.

$$\text{State} \to \text{Challenge} \to \text{Check} \to \text{Own} \tag{9}$$

State: what am I trying to do? Challenge: what obvious thing might I be missing? Check: which fact, if wrong, would materially change the action? Own: do I understand and accept the final output well enough to use it responsibly? Lite is appropriate for routine writing, low-cost planning, formatting, ordinary translation, and reversible decisions where durable learning is not the main goal.

DCP-Standard: capability-sensitive use. Use the full eight-stage architecture when learning, research, analytical judgment, transfer to later unaided work, or future fallback competence matters.

DCP-Critical: consequential or hard-to-reverse use. Use the full architecture plus at least one independent evidence/human route and an escalation rule. AI may organize evidence, generate alternatives, or stress-test reasoning, but it is not the sole certifier of health, legal, financial, safety-critical, or other high-stakes conclusions. This yields the deployment rule

$$\begin{aligned}&\text{Topic Entry} \to \text{Triage} \to \text{Minimum Sufficient Dialogue}\\ &\quad \to \text{Escalate if Stakes Rise} \to \text{Periodic Return}.\end{aligned} \tag{10}$$

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The governing principles are now paired. Human Agenda Ownership governs what enters the dialogue; Minimum Sufficient Friction governs how demanding the dialogue should become. Protective effort should rise with learning value, failure cost, irreversibility, and dependency risk, but fall when the task is trivial, reversible, and not capability-sensitive.

Adaptive-friction proposition [Open]. The best real-world protocol will often be the least burdensome policy that preserves a predeclared minimum of task safety, human understanding, verification, and fallback competence; maximum friction is not the objective.

6 THE DIALOGUE CONVERSION PROTOCOL

The proposed protocol is

$$\begin{aligned}&\text{Anchor} \to \text{Expand} \to \text{Oppose} \to \text{Discriminate}\\ &\qquad \to \text{Verify} \to \text{Integrate} \to \text{Remove} \to \text{Return}.\end{aligned} \tag{11}$$

The sequence is not a ritual. Each stage protects against a distinct collapse.

6.1 Anchor: stabilize the human-owned topic before AI expansion

Once the Topic Entry Condition identifies a live problem or explicit exploration goal, the human records a minimal pre-AI anchor:

$$A_0 = \langle P_0, M_0, U_0, E_0, F_0, S_0 \rangle, \tag{12}$$

where P_0 is the problem as currently understood, M_0 the current model or tentative answer, U_0 the important unknowns, E_0 available evidence, F_0 a condition that would change the human's mind, and S_0 the stakes/domain.

The anchor need not be sophisticated. Its purpose is to create a human baseline against which model change and Human Return can later be measured. In learning tasks, a first unaided attempt may itself be productive because generation and productive-failure research show that initial effort can improve later conceptual learning and transfer [20, 23, 24].

Anchor elicitation for novices. Some users come to AI precisely because they do not yet possess a coherent M_0. The protocol must not convert epistemic literacy into an entry requirement. In that case AI first asks a small number of clarification questions rather than demanding a model: "What are you trying to understand or decide?", "What do you know with reasonable confidence?", and "What is the most important uncertainty?" The human-approved answer to those questions becomes the anchor. Thus

$$\mathrm{AI}_{\text{clarify}} \to \{\text{Known}, \text{Unknown}, \text{Goal}, \text{Stakes}\} \to \text{Human Anchor}. \tag{13}$$

AI may help elicit the anchor, but it should not silently own the topic or problem definition. If clarification changes the original agenda, that change should be surfaced for human acceptance.

Prompt function: "Here is my current model and what I do not know. If I do not yet have a model, help me clarify the problem first. Do not merely improve my view; identify what would most strongly challenge it."

6.2 Expand: use AI as a divergence engine

AI is then used for the tasks in which it has comparative leverage: search, translation, decomposition, analogy, recombination, counterexample, cross-domain transport, and candidate generation. The objective is not raw idea count but an increase in readout-distinguishable candidate classes:

$$Q_t \xrightarrow{T_{\mathrm{AI}}} \{Q_{i,t+1}, K_{i,t}\}_{i=1}^{n}. \tag{14}$$

AI outputs remain knowledge-like candidates unless independently warranted. A useful expansion request asks for several structurally different routes: opposite causal direction, missing-variable model, cross-domain analogy, boundary-case explanation, and the strongest alternative from a rival framework.

Prompt function: "Generate three non-equivalent explanations. For each, state which assumption changes, what evidence would favor it, and what would make it fail."

6.3 Oppose or boundary-test: precommit resistance without false balance

Before the dialogue converges, the human should expose the leading model to resistance. This stage is intentionally placed early because once a semantic basin becomes fluent, later objections may be absorbed as local patches rather than treated as genuine tests. Cognitive-forcing research supports the broader design principle that forcing engagement can reduce overreliance, even when users prefer easier interfaces [16, 18].

However, opposition is evidence-sensitive. The protocol must not manufacture a symmetrical "other side" when the evidence is strongly asymmetric. It therefore distinguishes two operations:

$$\text{Rival Generation} : \text{for genuinely open spaces}, \tag{15}$$

$$\text{Boundary Test} : \text{for strongly asymmetric evidence}. \tag{16}$$

Rival Generation asks for the strongest non-equivalent hypothesis, counterexample, falsifier, missing comparison, or failure mode. Boundary Testing asks where the established claim stops applying, which uncertainty remains, what confounder or measurement limit matters, and what observation would legitimately revise confidence. Hence

$$\text{Oppose a claim} \neq \text{manufacture an opposite claim}. \tag{17}$$

AI critique is not independent validation, but it can expose where independent validation is needed.

Prompt function: "If this is genuinely uncertain, build the strongest rival and the decisive discriminating test. If the evidence is already strongly one-sided, do not create false balance; identify boundary conditions, uncertainty, confounders, and legitimate revision triggers."

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6.4 Discriminate: remove rephrases masquerading as alternatives

Multiple outputs do not guarantee epistemic diversity. Let $C_1,\dots,C_n$ be candidate outputs and let $\sim_Q$ denote readout-equivalence for the current question: two candidates are equivalent if they license the same decisive distinctions, predictions, or actions. The useful candidate count is therefore

$$N_{\text{distinct}} = \left|\{C_1,\dots,C_n\}/\sim_Q\right|, \tag{18}$$

not n. The discrimination step asks: which candidates differ only in wording? Which introduce a new variable, causal direction, ontology, comparison, or testable consequence? This stage is inherited directly from the programme's distinction between raw candidate multiplication and readout-distinguishable alternatives [1, 2].

Prompt function: "Cluster the candidates by underlying mechanism. Remove paraphrases. Keep only alternatives that would change what evidence we seek or what action we take."

6.5 Verify: leave the AI loop, but spend verification where it matters

Candidate status changes only through resistance appropriate to the target domain. AI can locate, compare, or criticize evidence, but it should not become the sole generator and certifier of the same consequential claim.

Real-life use requires a verification budget. Define a claim c as load-bearing for the present task when its falsity would materially change at least one of

$$\begin{aligned} L(c) = \{&\text{Decision}, \text{Risk}, \text{Money}, \text{Health}, \\ &\text{Legal}, \text{Publication}, \text{IrreversibleAction}\}. \end{aligned} \tag{19}$$

When $L(c) = 1$, the claim should be exposed to at least one partially independent route:

$$\begin{aligned} V = \{&\text{primary source}, \text{data}, \text{experiment}, \\ &\text{calculation}, \text{expert}, \text{independent method}, \\ &\text{world outcome}\}. \end{aligned} \tag{20}$$

Low-stakes non-load-bearing claims may receive lighter checks. The same statement can therefore move from low to high verification burden when the intended use changes. The verification stage includes provenance typing:

$$\begin{aligned} \text{Source} \neq \text{AI synthesis} &\neq \text{Human inference} \\ &\neq \text{Unverified candidate}. \end{aligned} \tag{21}$$

Prompt function: "Which claims are load-bearing for what I am about to do? For those claims, show the primary evidence, what it actually establishes, what remains inference, and one independent route that could disagree. Do not spend the same verification effort on trivial details."

6.6 Integrate: the human reconstructs the revised model

After verification, the human—not the AI—states the revised model. A minimal integration record is

$$I_s = \langle \Delta M_s, E_s^{\text{decisive}}, U_s^{\text{remain}}, \text{Next}_s \rangle, \tag{22}$$

where ΔM_s is what changed, E_s^{decisive} the evidence or distinction that caused the change, U_s^{remain} what remains unresolved, and Next_s the next test or action.

This stage prevents a subtle failure: a user can recognize an excellent AI synthesis without being able to reconstruct the reasoning that made it excellent. Integration therefore uses generation and retrieval rather than recognition alone [21, 22].

Human action: close or hide the AI answer and explain the revised model in one's own words before asking AI to critique the reconstruction.

6.7 Remove: distinguish reset from true removal

The decisive assistance is then removed when Human Return needs to be measured. A reset breaks conversational momentum; removal tests whether the decisive capability still belongs to the human. They are not identical.

$$\text{Reset} : \text{fresh framing/session/source route}, \tag{23}$$

$$\text{Removal} : \text{absence of decisive AI assistance}. \tag{24}$$

A fresh chat can be a useful reset, but with memory, personalization, shared files, or persistent agents it may not constitute strong removal. Stronger removal includes a blank document, an offline or no-AI transfer case, a timed outage drill, or an independently completed decisive step. Coupled performance cannot reveal what belongs to the human.

6.8 Return: measure what came back with the person

The terminal readout is Human Return:

$$R_H^{\text{return}} = \langle C, T, S, A \rangle, \tag{25}$$

where C is Conceptual reconstruction, T Tool selection, S Skill execution, and A Alternative generation under unaided criteria [10]. A successful conversation need not increase all four. It must state what returned and what remained system-dependent.

A session-level conversion vector can be written

$$\Delta H_s = \langle \Delta C_s, \Delta T_s, \Delta S_s, \Delta A_s, \Delta A_{H,s}^{\text{corr}}, \Delta \Lambda_{H,s}^{\text{live}} \rangle. \tag{26}$$

The central non-collapse is

$$\Delta \text{Performance}_{\text{AI}} > 0 \;\not\Rightarrow\; \Delta H_s > 0. \tag{27}$$

Return should be sampled, not ritualized. Full removal after every interaction is unnecessary. Let

$$\begin{aligned} F^{\text{return}} = f(&\text{Criticality}, \text{LearningNeed}, \\ &\text{FailureCost}, \text{DependencyRisk}). \end{aligned} \tag{28}$$

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Learning-intensive tasks may require frequent Return tests; routine automation may need only periodic fallback drills; trivial low-stakes tasks may require none. The aim is to detect loss of decisive human competence before it becomes operationally invisible.

Field evidence also cautions against assuming that AI use necessarily causes deskilling. In customer support, generative-AI assistance increased productivity and produced evidence of durable worker learning during system outages, especially among less experienced workers [29]. The protocol must therefore measure the sign of Human Return rather than presuppose it.

7 Action and World Feedback: Close the Human-Development Loop

Human Return establishes what came back with the person, but a live problem is not fully closed by verbal reconstruction. When the topic concerns action, repair, decision, or skill, the revised understanding should return to the world that generated the problem. Human Learning in the programme already treats real-problem application and feedback as part of the transition from understanding to corrigible skill [9].

Let a_s be the human-selected next action emerging from integration and let $\delta_{s+1}^{\text{world}}$ be the relevant returned record from the situation. Then

$$I_s \to a_s \to \delta_{s+1}^{\text{world}} \to H_{s+1,0}. \tag{29}$$

The returned record may confirm the revised model, expose a new residual, reveal that the problem was misframed, or create a new problem. The wider cycle is therefore

$$\begin{aligned} &\text{Live Problem} \to \text{Question} \to \text{Dialogue} \\ &\quad \to \text{Human Return} \to \text{Action} \\ &\quad \to \text{World Feedback} \to \text{Revision or New Problem}. \end{aligned} \tag{30}$$

This does not require every conversation to end in overt behavior. Pure learning, creative exploration, and basic inquiry can terminate at a different boundary. The claim is conditional: where the original topic was action-relevant, a conversation that never returns to the relevant world risks mistaking linguistic closure for practical resolution.

World-closure proposition [Open]. For live problems with observable consequences, dialogue policies that include post-conversation action and feedback will produce better calibration and transferable Human Return than otherwise identical policies that terminate at AI-generated synthesis.

8 Expansion and Contraction Must Alternate

A productive conversation should not remain in permanent expansion. The protocol therefore alternates two modes:

$$\text{Expansion : maximize candidate diversity} \tag{31}$$
$$\text{and discriminability,} \tag{32}$$
$$\text{Contraction : prune by evidence, provenance,} \tag{33}$$
$$\text{stakes, and action relevance.} \tag{34}$$

Permanent expansion creates overload and rhetorical abundance; premature contraction creates tunnel risk. The switch should occur when additional candidates no longer create a new equivalence class or when a load-bearing decision now requires world-side evidence.

9 Stop Rules and Reset Rules

A conversation should stop or reset when conversational momentum is increasing faster than evidence. The following are candidate stop conditions:

- three or more turns produce no new non-equivalent candidate class;
- confidence rises while provenance or external evidence does not;
- AI begins validating a claim primarily by reference to its own earlier synthesis;
- the user cannot restate the current reasoning without reopening the transcript;
- the task becomes high-stakes enough that an independent expert/source is required;
- the user is seeking confirmation rather than discrimination;
- objections are repeatedly absorbed into one favored frame without changing its predicted consequences.

A reset can take four forms: fresh-session restart, prompt inversion, independent-source check, or human-only reconstruction. The purpose is not variation for its own sake; it is to break local conversational path dependence and test whether the current conclusion survives altered framing.

10 Dialogue Policy Must Depend on the Goal

One universal prompt policy is unlikely to maximize human conversion. Let

$$\pi^{\text{dialogue}} = f\big(\text{TopicMode}, \text{Goal}, \text{Skill}, \text{Stakes}, \text{Domain}, \text{LearningNeed}, \text{Reversibility}\big). \tag{35}$$

Table 2 specifies candidate policies and recommended deployment modes.

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10.1 Personal and relational dialogue requires a different verification logic

In interpersonal life, a "primary source" may be unavailable because the unknown is another person's intention, feeling, or motive. The danger is interpretive reinforcement: AI can make one user's story more coherent without gaining privileged access to the absent person's mind. The recommended route is

$$\begin{aligned}\text{Experience} &\rightarrow \text{Interpretations} \rightarrow \text{Absent Perspective}\\ &\rightarrow \text{Observable Evidence}\\ &\rightarrow \text{Direct Human Conversation.}\end{aligned} \tag{36}$$

A useful prompt is: "Separate what I directly observed from my interpretation. Give alternative explanations that do not accuse the absent person without evidence. What question should I ask them directly before concluding?" This mode treats AI as an interpretation-expander, not an oracle of hidden motives.

11 Protocol Burden and Behavioral Adoption

An epistemically protective protocol can fail socially if its burden causes users to ignore it. Define a usability-burden vector

$$\begin{aligned}B_{\text{use}} = \langle &\text{Time}, \text{CognitiveLoad}, \text{VerificationCost},\\ &\text{Interruption}, \text{LiteracyDemand}\rangle.\end{aligned} \tag{37}$$

Cognitive-forcing work already shows a friction-performance preference trade-off: interventions that reduce overreliance can demand more effort and may be less preferred [16, 19]. Therefore

$$\text{Epistemically optimal} \not\equiv \text{Behaviorally adoptable.} \tag{38}$$

A deployable protocol should report not only Human Return but completion rate, abandonment, time cost, satisfaction, repeated-use willingness, and whether benefits persist under ordinary time pressure. DCP-Lite exists specifically to lower B_{use} where a full protocol would add little value.

12 Provenance Ledger: What Is This Sentence Doing?

A compact dialogue ledger can prevent candidate forgetting. Each important statement receives one status:

$$\begin{aligned}\text{Status} \in \{&\text{Source}, \text{AI-Synthesis}, \text{Human-Inference},\\ &\text{Candidate}, \text{Decision}\}.\end{aligned} \tag{39}$$

A candidate may be useful without being true; an AI synthesis may accurately summarize sources without adding independent evidence; a decision may be justified under uncertainty without converting uncertainty into fact. This ledger operationalizes the programme's claim-level non-collapse discipline and makes later auditing possible [1].

13 What Should the Human Say to AI? Adaptive Operational Scripts

13.1 Topic Entry: before any protocol

For capability-sensitive use, begin with one question: "What unresolved difference in my present life, work, understanding, or environment deserves my attention now?" If the purpose is curiosity or play, state that explicitly: "I am exploring, not solving a current problem; help me widen the field without pretending there is an urgent problem." This makes the agenda visible before AI begins optimizing within it.

13.2 The one-minute DCP-Lite script

For ordinary use, four speech acts are sufficient:

1. State: "This is the problem or goal I chose to work on; this part matters most."
2. Challenge: "What obvious alternative, exception, or mistake might I be missing?"
3. Check: "Which fact would materially change my action if it were wrong, and how should I verify it?"
4. Own: "Summarize the decision boundary, not just the answer. I will make the final choice."

13.3 The full DCP-Standard script

When learning, research, transfer, or future unaided competence matters:

1. Topic/Anchor: "This is the live problem or explicit exploration goal I chose. This is my current model, uncertainty, and what would change my mind. If I lack a model, help me clarify without silently replacing my agenda."
2. Expand: "Give me non-equivalent routes, not rephrasings. State what changes structurally in each."
3. Oppose/Boundary-Test: "If the issue is genuinely open, build the strongest rival. If evidence is strongly one-sided, identify boundaries, uncertainty, and revision triggers instead of manufacturing an opposite."
4. Discriminate: "Which alternatives are genuinely different in mechanism, prediction, or action? Collapse duplicates."
5. Verify: "Which claims are load-bearing for my intended use? Verify those with primary evidence, provenance, uncertainty, and an independent route."
6. Integrate: "Do not summarize yet. I will restate what changed; then critique my reconstruction."
7. Remove: "I am removing decisive AI assistance and will attempt the critical step independently."
8. Return: "Test what I can reconstruct, select, execute, and generate without your decisive help."

13.4 DCP-Critical escalation rule

When the consequence becomes high-stakes or hard to reverse, the protocol adds: "Do not act as the sole certifier. Identify the independent professional, primary record, calculation, test, or institutional route required before action." The critical mode

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Table 2: Task-contingent dialogue policies and deployment levels.

Goal	Mode	AI should mainly do	Human must retain	Protocol emphasis
Routine / admin	Lite	automation, formatting, ordinary translation, repetitive operations	exception detection, ownership of consequential wording, fallback for critical functions	State–Challenge–Check–Own; no routine Remove/Return
Learning	Standard	hints, examples, error diagnosis, graduated counterexamples, retrieval prompts	first attempt, explanation, retrieval, transfer, decisive skill step	delay full solution when learning value is high; frequent Return
Research	Standard/Critical	search, literature transport, rival generation, synthesis, code assistance	problem ownership, source judgment, validation design, interpretation, claim strength	provenance, non-equivalent rivals, primary-source verification
Decision support	Standard/Critical	structured alternatives, sensitivity analysis, counterfactuals	initial judgment, value trade-offs, accountability, override competence	pre-AI estimate when feasible; verification budget; escalate with stakes
Creative work	Lite/Standard	divergence, recombination, style variation, analogy	purpose, taste, selection, responsibility, authorship claim	expansion first; lighter truth burden until factual claims become load-bearing
Personal / relational	Lite/Standard	separate facts from interpretations, generate charitable alternatives, surface absent perspectives, draft questions	ownership of feelings/values, direct communication, knowledge that the other person's mind is not directly observed	Facts $\neq$ interpretation; avoid diagnosing absent people; return to observable evidence and human conversation
High-stakes domains	Critical	evidence organization, differential/rival generation, checklists	accountable professional judgment and independent verification	no single-route certification; expert/world validation before action

terminates in accountable human/world-side validation, not in a more confident AI answer.

14 Formal Success Condition

Let P_s^{assist} be current assisted performance, H_s the Human Return/agency vector after removal, and B_{use}^s the usability-burden vector. A dialogue policy π is a high-conversion policy for a declared task if

$$\pi^* \in \arg\max_{\pi} \mathbb{E}[\Delta H_s \mid \pi, \text{task}, \text{user}], \tag{40}$$

subject to minimum task-performance and safety constraints, a declared burden/adoption ceiling, and—for human-development use—a visible topic-ownership condition

$$B_{\text{use}}(\pi) \leq \bar{B}(\text{task}, \text{user}). \tag{41}$$

This differs from

$$\pi^{\text{perf}} \in \arg\max_{\pi} \mathbb{E}[P_s^{\text{assist}} \mid \pi], \tag{42}$$

and the two policies need not coincide.

Dialogue Conversion Proposition [Open]. For tasks in which learning, judgment transfer, or future unaided competence matters, the assistance policy that maximizes immediate performance will often differ from the policy that maximizes delayed Human Return. **Deployment Proposition [Open].** The full protocol will not maximize real-world adoption across all tasks; adaptive triage should achieve similar protection at lower burden for low- and medium-stakes use. **Agenda Proposition [Open].** When the objective is human development, self-selected live-problem entry will often produce greater relevance, action transfer, and Human Return than an otherwise comparable AI-suggested agenda, after baseline motivation and task difficulty are controlled.

15 Empirical Programme

A tier-one test should compare protocols, not merely AI-versus-no-AI.

15.1 Multi-arm randomized design

Randomize or adaptively assign participants to: (A) direct-answer AI; (B) DCP-Lite; (C) explanation/cognitive-forcing AI; (D) adversarial-rival AI; (E) DCP-Standard; (F) DCP-Critical when stakes justify escalation; and (G) no-AI baseline. Tasks should include at least one learning task, one decision task, one open-ended research/creation task, one routine task, and one personal/relational interpretation task with controlled ground truth or follow-up evidence where ethically feasible. A crossed topic-entry factor should compare (i) self-selected live problem, (ii) AI-suggested topic matched for domain and difficulty, and (iii) curiosity-led self-selected exploration.

Primary outcomes are measured at three times: immediate assisted performance, delayed unaided Human Return, and transfer to an unseen/held-out case. Deployment outcomes are co-primary for real-life validity: completion rate, time cost, cognitive load, satisfaction, abandonment, repeated-use

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willingness, over/under-reliance, and calibration. The study should test whether Lite preserves most of the benefit of Standard in low-stakes tasks and whether Critical adds value only where stakes or irreversibility justify its burden.

15.2 Primary hypotheses

H1. Direct-answer AI will often maximize immediate performance but not delayed Human Return in learning-intensive tasks. **H2.** DCP and cognitive-forcing conditions will reduce overreliance but may increase cognitive load. **H3.** DCP will produce more non-equivalent unaided alternatives at follow-up than direct-answer AI. **H4.** Provenance accuracy and error detection after AI removal will mediate the relationship between protocol and Human Return. **H5.** Effects will be heterogeneous by prior skill: novices may require more scaffolded opposition/verification than experts. **H6.** In creation tasks, aggressive expansion may improve coupled output without harming Return, whereas high-stakes decision tasks will benefit more from early anchoring and independent verification. **H7.** DCP-Lite will achieve higher adoption and lower burden than DCP-Standard in low-stakes tasks with little loss of safety or ownership. **H8.** In relational interpretation tasks, facts-versus-interpretation separation and absent-perspective prompts will reduce unwarranted certainty about another person's motives compared with direct-answer AI. **H9.** For capability-sensitive tasks, self-selected live-problem entry will increase perceived relevance, follow-through, and delayed Human Return relative to matched AI-suggested topics, while curiosity-led exploration will outperform problem-first on breadth and intrinsic exploration outcomes. **H10.** For action-relevant live problems, Action $\rightarrow$ World Feedback closure will improve calibration and transfer relative to otherwise identical dialogue that terminates at synthesis.

15.3 Falsification and removal conditions

The protocol should be simplified if: (i) ordering the stages adds no value beyond generic "think critically" instructions; (ii) Human Return does not predict later independent performance beyond standard skill tests; (iii) non-equivalent alternative counting adds no value over raw idea count; (iv) reset or independent-route interventions do not reduce tunnel effects; (v) direct-answer assistance consistently improves both immediate and delayed outcomes with no detectable agency or dependence cost; (vi) Lite/Standard/Critical triage adds no value over one fixed protocol; or (vii) protocol burden is so high that protective effects disappear under realistic adoption and time constraints; (viii) live-problem topic entry adds no value beyond motivation, relevance, and task matching; or (ix) action/world-feedback closure adds no predictive or calibration value beyond Human Return alone.

16 URGENCY AND THE REVERSIBILITY WINDOW

The protocol is not merely pedagogical. Repeated AI use can become infrastructural habit before institutions know what human capacities have been lost, and AI-supplied agendas can become normalized before users notice that topic selection itself has shifted outside their control. The Human Conversion Imperative therefore adds a Reversibility Principle: when machine capability changes faster than human adaptation and the recovery cost of lost skills or routes rises, preserving independent competence earlier may be cheaper than rebuilding it later [12].

At the dialogue level, this means sampling whether the human can still perform the decisive cognitive operation without the system at a frequency proportional to learning need, failure cost, and dependency risk. At the organizational level, it means maintaining fallback expertise, exportable records, independent validation routes, and scheduled outage or transfer drills for critical functions. The aim is not to freeze pre-AI work. It is to prevent irreversible dependence from being mistaken for progress while avoiding needless friction in trivial work.

17 LIMITS

The adaptive protocol is intentionally strong but not universal. First, the triage rules are conceptual and not yet calibrated: the boundary between Lite, Standard, and Critical requires empirical thresholds. Second, problem-first can itself fail: people may misidentify the problem, ruminate on a salient but low-value concern, or inherit a problem frame from family, institutions, markets, or the AI itself. The Topic Entry layer therefore protects agenda visibility, not infallible self-knowledge. Third, problem-first must not suppress curiosity, art, play, basic science, or exploratory learning; the Open Exploration mode is constitutive, not an exception added for convenience [28]. Fourth, friction can produce under-reliance as well as healthy skepticism, and users differ in tolerance for cognitive effort [16, 19]. Fifth, Human Return is task-relative; losing a low-value manual procedure may be rational if higher-level capability improves. Sixth, "removal" is difficult to standardize in systems with memory, personalization, agents, or organizational integration. Seventh, independent verification can itself be biased, inaccessible, expensive, or slower than the decision window. Eighth, relational dialogue can reduce premature certainty but cannot infer an absent person's private mental state. Ninth, the protocol does not solve misinformation, professional judgment, or institutional power by procedure alone. Tenth, minimum sufficient friction and problem-first entry are design hypotheses: if lighter protocols or alternative topic-selection policies consistently preserve equal or greater Human Return, the architecture should be simplified.

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18 CONCLUSION

The question "How should humans converse with AI?" begins one step earlier than prompting. In an AI-rich environment, a system can shape not only answers but the agenda of attention. The first design question is therefore: what deserves to become the topic of dialogue, and who owns that choice?

For human-development use, this paper proposes a default of Problem-First Dialogue: begin from a live, presently consequential unresolved difference in the person's life, work, understanding, or environment. The principle is not universal. Curiosity, play, art, basic inquiry, and routine delegation remain legitimate entry modes. The non-collapse is explicit:

$$\text{Problem-First} \neq \text{Problem-Only}. \tag{43}$$

The complete adaptive architecture is

$$\begin{aligned}\text{Topic Entry} &\rightarrow \text{Triage} \rightarrow \text{Minimum Sufficient Dialogue}\\ &\rightarrow \text{Human Return} \rightarrow \text{Action/World Feedback}\\ &\rightarrow \text{Revision or New Problem}.\end{aligned} \tag{44}$$

For low-stakes use, DCP-Lite asks the human to State, Challenge, Check, and Own. When learning, research, judgment transfer, or future competence matters, the canonical protocol remains

$$\begin{aligned}\text{Anchor} &\rightarrow \text{Expand} \rightarrow \text{Oppose/Boundary-Test}\\ &\rightarrow \text{Discriminate} \rightarrow \text{Verify} \rightarrow \text{Integrate}\\ &\rightarrow \text{Remove} \rightarrow \text{Return}.\end{aligned} \tag{45}$$

When stakes become high, independent world-side or professional validation becomes mandatory.

Humans should use AI aggressively where machines are strong—breadth, search, translation, recombination, simulation, and candidate generation—while retaining the functions that make human development possible: agenda/problem ownership, world contact, normative responsibility, integration, refusal, correction, and fallback competence. A high-quality conversation is not one in which AI appears maximally intelligent, nor one in which the protocol appears maximally rigorous. It is one in which the user spends attention on a topic they can still recognize as theirs, uses only the friction the task deserves, leaves with more durable capability, and—where the problem is action-relevant—lets the world answer back.

The decisive practical questions are therefore:

$$\begin{aligned}&\text{What unresolved difference deserves my attention?}\\ &\text{How much AI/friction does this task require?}\\ &\text{What remains with me after AI is removed?}\\ &\text{What does the world return when I act?}\end{aligned} \tag{46}$$

REFERENCES

[1] Yaoharee Lahtee. Epistemic fusion or epistemic tunnel: A phenomenology-anchored, global-literature-constrained, problem-first architecture of human-ai collaboration (epistemic note v8.1 – strong-claim / legitimate-defeater edition). Human-AI Readout Programme, v8.1, 5 September 2026, 2026. https://doi.org/10.5281/zenodo.22331922.

[2] Yaoharee Lahtee. Choice begins before choice: Meaning-shaped accessibility, live possibility, and effective agency in human-ai systems. Human-AI Readout Programme, Full Paper v1.0 Strong Claim, 5 September 2026, 2026. https://doi.org/10.5281/zenodo.22357788.

[3] Yaoharee Lahtee. When ai expands human potential: Reflective dissonance, epistemic agency, and constraint. Open Civil Science Initiative preprint, 25 March 2026, 2026. Human-AI Readout Programme. https://doi.org/10.5281/zenodo.19215748.

[4] Yaoharee Lahtee. The readout condition: Distinguishability, access, and epistemic warrant. Human-AI Readout Programme, elevated full paper, August 2026, 2026. https://doi.org/10.5281/zenodo.22301318.

[5] Yaoharee Lahtee. From problem to hypothesis: Dynamic semantic mobility, bounded knowers, and the readout-discriminable hypothesis space. Human-AI Readout Programme, GLOSA K0 final author draft, 4 September 2026, 2026. https://doi.org/10.5281/zenodo.22307148.

[6] Yaoharee Lahtee. The epistemic chain reaction. Human-AI Readout Programme, K0-2026-09-04-v3, 2026. https://doi.org/10.5281/zenodo.22308072.

[7] Yaoharee Lahtee. Experience is the human lora. Human-AI Readout Programme, 2026. https://doi.org/10.5281/zenodo.21425420.

[8] Yaoharee Lahtee. Ai, translation, and access to event-specific context. Human-AI Readout Programme, 2026. https://doi.org/10.5281/zenodo.18517054.

[9] Yaoharee Lahtee. Human learning as epistemic architecture. Human-AI Readout Programme, revised v4 preprint, 2026. https://doi.org/10.5281/zenodo.22341297.

[10] Yaoharee Lahtee. Ctsa human-return readout: A session-boundary measurement architecture for retained human capability after ai. Human-AI Readout Programme, v9.1 FINAL, 5 September 2026, 2026. https://doi.org/10.5281/zenodo.22339909.

[11] Yaoharee Lahtee. After labour: Human position in an ai-robotic world system. Human-AI Readout Programme, full world-system standalone draft v1.1, 6 September 2026, 2026. Companion paper, same deposit batch as the present paper; DOI 10.5281/zenodo.22481924.

10.5281/zenodo.22481928

[12] Yaoharee Lahtee. The human conversion imperative: Machine acceleration, human potential, and the reversibility window in an ai-robotic world. Human-AI Readout Programme, standalone proposal v1.0, 6 September 2026, 2026. Companion paper, same deposit batch as the present paper; DOI 10.5281/zenodo.22481926.

[13] Roni Reiter-Palmon. The role of problem construction in creative production. *Journal of Creative Behavior*, 51 (4):323–326, 2017. doi: 10.1002/jocb.202.

[14] Stefano Magistretti, Christophe T. A. Pham, and Claudio Dell'Era. The creative process of problem framing for innovation: An integrative review and research agenda. *Journal of Product Innovation Management*, 42(6):987–1018, 2025. doi: 10.1111/jpim.12783.

[15] Michelle Vaccaro, Abdullah Almaatouq, and Thomas W. Malone. When combinations of humans and AI are useful: A systematic review and meta-analysis. *Nature Human Behaviour*, 8:2293–2303, 2024. doi: 10.1038/s41562-024-02024-1.

[16] Zana Buçinca, Maja Barbara Malaya, and Krzysztof Z. Gajos. To trust or to think: Cognitive forcing functions can reduce overreliance on AI in AI-assisted decision-making. In *Proceedings of the ACM on Human-Computer Interaction (CHI)*, 2021. arXiv:2102.09692.

[17] Helena Vasconcelos, Matthew Jorke, Madeleine Grunde-McLaughlin, Tobias Gerstenberg, Michael S. Bernstein, and Ranjay Krishna. Explanations can reduce overreliance on AI systems during decision-making. *Proceedings of the ACM on Human-Computer Interaction*, 7 (CSCW1):129:1–129:38, 2023. doi: 10.1145/3579605.

[18] Sander de Jong, Vera Paananen, Benjamin Tag, and Niels van Berkel. Cognitive forcing for better decision-making: Reducing overreliance on AI systems through partial explanations. *Proceedings of the ACM on Human-Computer Interaction*, 9(2):CSCW048, 2025. doi: 10.1145/3710946.

[19] Hui Yi Tian, Hamza Amin, and Ming Yin. Understanding the effects of AI-assisted critical thinking on human-AI decision making. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, 2026. doi: 10.1145/3772318.3790785. Article 810.

[20] Norman J. Slamecka and Peter Graf. The generation effect: Delineation of a phenomenon. *Journal of Experimental Psychology: Human Learning and Memory*, 4(6): 592–604, 1978. doi: 10.1037/0278-7393.4.6.592.

[21] Henry L. Roediger, III and Andrew C. Butler. The critical role of retrieval practice in long-term retention. *Trends in Cognitive Sciences*, 15(1):20–27, 2011. doi: 10.1016/j.tics.2010.09.003.

[22] Jeffrey D. Karpicke and Janell R. Blunt. Retrieval practice produces more learning than elaborative studying with concept mapping. *Science*, 331(6018):772–775, 2011. doi: 10.1126/science.1199327.

[23] Manu Kapur. Productive failure. *Cognition and Instruction*, 26(3):379–424, 2008. doi: 10.1080/07370000802212669.

[24] Manu Kapur. Productive failure in learning math. *Cognitive Science*, 38(5):1008–1022, 2014. doi: 10.1111/cogs.12107.

[25] Hamsa Bastani, Osbert Bastani, Alp Sungu, et al. Generative AI without guardrails can harm learning: Evidence from high school mathematics. *Proceedings of the National Academy of Sciences*, 122(26):e2422633122, 2025. doi: 10.1073/pnas.2422633122.

[26] Moshe Glickman and Tali Sharot. How human–AI feedback loops alter human perceptual, emotional and social judgements. *Nature Human Behaviour*, 9:345–359, 2025. doi: 10.1038/s41562-024-02077-2.

[27] Zana Bucinca, Siddharth Swaroop, Amy E. Paluch, Susan A. Murphy, and Krzysztof Z. Gajos. Towards optimizing human-centric objectives in AI-assisted decision-making with offline reinforcement learning, 2024. arXiv:2403.05911.

[28] Francesco Poli, Jill X. O'Reilly, Rogier B. Mars, and Sabine Hunnius. Curiosity and the dynamics of optimal exploration. *Trends in Cognitive Sciences*, 28(5):441–453, 2024. doi: 10.1016/j.tics.2024.02.001.

[29] Erik Brynjolfsson, Danielle Li, and Lindsey Raymond. Generative AI at work. *Quarterly Journal of Economics*, 140(2):889–942, 2025. doi: 10.1093/qje/qjae044.

 10.5281/zenodo.22481928

From Assistance to Human Capability: A Readout-Genesis Architecture for Proactive AI, Unequal Life Conditions, and Opportunity Conversion

v1.1 glosa-checked edition • 6 September 2026 • 10.5281/zenodo.22498047

What the chapter states. “This paper proposes a Readout-Genesis-native architecture for proactive human capability advancement. The proposal begins from a retained difference and a human-owned live problem, then inserts a barrier-typing stage before any training prescription.”

Status. “K0 conceptual and measurement architecture with an empirical research agenda.”

Falsifier(s). “barrier categories cannot be coded reliably across independent raters or add no intervention value; [. . .] proactive HCA produces no delayed Return or transfer advantage over reactive DCP at comparable burden.”

Read before. This book’s Chapters 13, 32, 11, 17, 31, 34, 35 (mechanical citation list; Appendix A).

Feeds into. —

From Assistance to Human Capability

A Readout-Genesis Architecture for Proactive AI, Unequal Life Conditions, and Opportunity Conversion

Human-owned problems, barrier typing, adaptive scaffolding, Human Return, and world-side recognition

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ORCID 0009-0005-3861-0626 • v1.1 — glosa-checked edition of v1.0, 6 September 2026 original

Status. K0 conceptual and measurement architecture with an empirical research agenda. The paper is rooted in the Readout Genesis contract and treats Human Capability Advancement (HCA) as a candidate domain translation, not as a new root and not as an established psychological law. It uses the Problem-First Dialogue Conversion Protocol, Human Learning as Epistemic Architecture, Choice Begins Before Choice, and CTSA Human Return as internal programme anchors. External literatures on scaffolding, adaptive instruction, self-determination, the capability approach, situated cognition, skill acquisition, human-AI reliance, job-market signaling, and child-centred AI are attached as convergence partners, rivals, measurement resources, or defeaters; none is imported as a root premise. No exact human-domain quotient or calibrated universal HCA scale is claimed.

Edition note. This is glosa-checked edition v1.1 of the founder's v1.0 (6 September 2026 original). Changes are limited to: (1) citation-lineage repair — the two working-specification self-citations (Readout Genesis Whitepaper; Information Epistemic Foundation) now carry a resolvable anchor (git repository/commit/blob, or the deposited record that carries the cited content) instead of an unanchored "working technical specification" label, and the six self-citations to since-superseded internal papers now carry their deposited Zenodo DOI and current version string (§4 Table 1; in-text labels "DCP v1.2"→"DCP v1.3" and "After Labour v1.1"→"v1.2"); (2) renaming §4.1 from a priority-claiming section heading to "Contribution boundary" and rewording the three sentences that carried priority-adjacent phrasing, including in negated form, per the programme's no-priority-language rule; (3) explicit [Definition]/[Open] tags added to the boxed non-collapse laws, candidate policies, and estimand equations, matching the tagging convention the v1.0 original already used for the Life-Capital Context Vector and the Human Capability Advancement Principle; (4) reference [37] replaced with a direct citation to the primary statute (Labour Protection Act B.E. 2541) after the cited Ministry of Labour guidance page could not be independently resolved. No section, equation, table, principle, invariant, study, rival, removal condition, or numbered claim was added, removed, renumbered in content, or otherwise reworded. The v1.0 PDF remains the anchor document of record; this edition is a citation-accurate, tier-labelled copy for deposit, not a superseding claim.

Central claim. When durable human development is an agreed goal, AI need not remain purely reactive. Starting from a human-owned live problem, it may proactively diagnose whether an observed difficulty is best read as a knowledge or skill gap, a language/tool/resource constraint, a network/credential/permission barrier, or another opportunity bottleneck; generate non-equivalent capability or opportunity routes; ask for human endorsement; provide adaptive support; progressively withdraw decisive assistance; test what returns unaided to the human; and connect retained capability to world feedback and real opportunity. The design target is therefore not maximum AI usage or maximum assisted output, but advancement that remains human-owned, survives AI removal where retention matters, and can be converted into action or opportunity under unequal life conditions.

Abstract

Most AI assistants are evaluated at the point of assistance: answer quality, task accuracy, speed, productivity, or user satisfaction. These are important outcomes, but they do not establish that a human became more capable. Nor do they reveal whether a failure to act reflects a missing skill or a structural barrier such as language, time, tool access, network, credential, permission, or market recognition. This paper proposes a Readout-Genesis-native architecture for proactive human capability advancement. The proposal begins from a retained difference and a human-owned live problem, then inserts a barrier-typing stage before any training prescription. AI generates candidate capability and opportunity routes but does not silently own the user's goal. An endorsed route can then receive adaptive scaffolding, deliberate practice, assistance fading, unaided Human Return testing, novel transfer, world feedback, and—when relevant—a credential, portfolio, network, or institutional bridge.

The architecture separates objects that are often collapsed: observed difficulty from skill deficit; resources from access; access from control; assistance from retained capability; capability from opportunity; credential from capability; market legibility from human worth; and proactive support from agenda ownership. Unequal life conditions are represented as a context vector rather than assumed away, which changes both causal identification and policy interpretation. A worked Southern Thailand scenario shows why a low-resource child with strong real-world capability may remain institutionally illegible, why a formally schooled child can still become passively AI-dependent, and why the strongest pathway is likely to combine foundational education, human-owned problem inquiry, supervised AI scaffolding, real projects, Human Return, and credential/opportunity bridges.

External literature does not derive the model. Scaffolding and expertise-reversal evidence constrain assistance; self-

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determination evidence constrains autonomy; the capability approach pressures the resource-to-opportunity transition; situated cognition and skill-acquisition work constrain transfer; human-AI studies motivate separation of assisted and retained performance; signaling theory constrains market recognition; and UNESCO/UNICEF guidance constrains child deployment. The residual contribution is a typed end-to-end architecture that asks not only what AI can do for a person now, but what the person can later do, what barriers still block conversion, and whether the route remains corrigible and human-owned.

Keywords: human capability; proactive AI; Readout Genesis; human-AI interaction; adaptive scaffolding; Human Return; unequal opportunity; capability approach; autonomy; learning; credential; child-centred AI.

1 The Problem: An Assistant Can Solve the Task Without Advancing the Person

Generative AI makes assistance cheap, fast, fluent, and increasingly available. That creates an immediate design temptation: optimize for the quality of the joint answer. Yet the empirical record already shows why that target is incomplete. Human-AI combinations do not exhibit universal synergy; a preregistered meta-analysis of 106 experiments found average human augmentation relative to humans alone but no average synergy relative to the better of human or AI, with decision tasks especially fragile [1]. In high-school mathematics, unrestricted generative-AI access improved practice performance but was followed by worse unaided exam performance than a no-AI condition, whereas guardrailed tutoring largely removed that learning cost [2]. Conversely, field evidence in customer support shows that AI can also facilitate durable worker learning, especially among less experienced workers [3]. The sign is therefore open: AI use may build capability, rent capability, or do both.

A second problem appears before learning. AI can influence what becomes salient, what is treated as a plausible explanation, and which topic is asked about next. Repeated biased AI interaction has been shown to shift later human perceptual, emotional, and social judgments, often without users fully recognizing the influence [4]. A protocol concerned with human development must therefore protect both the entry agenda and the returning human state.

A third problem is distributional. Two people given the same model do not possess the same effective opportunity. They may differ in language, foundational literacy, free time, device access, safety, health, mobility, mentor networks, credential routes, and institutional recognition. A design that reads every failure as a skill deficit risks training the person harder when the actual bottleneck is external. The central question of this paper is therefore:

Research question. Under what architecture may AI proactively advance human capability while preserving human-owned ends, distinguishing structural barriers from personal deficits, and separating assisted performance from durable Human Return?

2 Readout Genesis Is the Rail, Not a Metaphor

The proposal begins from the Readout Genesis constitutional discipline. The root does not contain a built-in object called skill, capability, motivation, education, credential, labour market, or human flourishing. Such terms are admissible only after a declared domain translation. The retained root state is written schematically as

$$S_n = (G_n, \Lambda_n, T_n), \tag{1}$$

with finite transition

$$S_{n+1} = F(S_n, u_n, c_n, T_n). \tag{2}$$

The candidate HCA domain must therefore satisfy a declared weld, at least approximately,

$$q^\sharp_{\mathrm{HCA},n+1} \circ F_n = F_{\mathrm{HCA},n} \circ q_{\mathrm{HCA},n} \tag{3}$$

with corresponding readout and invariant conditions [5, 6]. The first non-collapse is constitutional [Definition]:

$$S_n \neq Z_{\mathrm{HCA},n} \neq D_{\mathrm{HCA},n}. \tag{4}$$

The retained root state, a candidate human-capability representation, and an empirically discovered sufficient quotient are different objects. Formal coherence in the HCA adapter would not establish that the psychological interpretation is true. The current status is therefore a candidate domain architecture with explicit defects and empirical gates, not a closed human science.

Readout Genesis also supplies two disciplines that become decisive here. First, access is bounded by reader, language, tool, permission, and context; access is not raw possession of reality [6]. Second, unresolved claims should specify the least-cost next record expected to discriminate the active uncertainty rather than silently converting ambiguity into a label. These rules will govern both barrier diagnosis and capability testing.

3 Internal Programme Lineage: The Native River Before Allies Are Attached

The present paper does not derive its central objects from external educational theory. It inherits a sequence of already-developed internal interfaces and then asks whether they can be welded into a capability-advancement domain.

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The native river is therefore

$$\begin{aligned}\text{Retained Difference} &\to \text{Human Readout}\\ &\to \text{Live Problem}\\ &\to \text{Barrier Readout}\\ &\to \text{Candidate Routes}\\ &\to \text{Human Endorsement}\\ &\to \text{Adaptive Scaffold}\\ &\to \text{Practice}\\ &\to \text{Withdrawal}\\ &\to \text{Human Return}\\ &\to \text{Novel Transfer}\\ &\to \text{World Feedback}\\ &\to \text{Opportunity Conversion.}\end{aligned} \tag{5}$$

The remainder of the paper formalizes this river and only then maps external literatures onto the locations where they have genuine explanatory or measurement authority.

4 External Literature as Allied Rails, Rivals, and Defeaters

Scaffolding, adaptive tutoring, autonomy support, deliberate practice, knowledge transfer, capability conversion, and signalling are cited here as established literatures; this paper's contribution boundary is the typed ordering and non-collapse discipline across a larger human-AI capability-conversion process, not a claim about those literatures themselves. Those literatures are mature. The residual contribution is the typed ordering and non-collapse discipline across a larger human-AI capability-conversion process.

4.1 Contribution boundary

A reviewer should reject any claim that this paper invented adaptive tutoring, fading, transfer of responsibility, autonomy support, or conversion factors. The contribution here is narrower and specific: a Readout-Genesis-native architecture that (i) begins from a human-owned live problem rather than a fixed curriculum, (ii) types barriers before prescribing training, (iii) distinguishes capability routes from opportunity routes, (iv) makes proactive support subordinate to human endorsement, (v) uses assistance withdrawal and Human Return as explicit boundaries, (vi) returns the person to world-side feedback, and (vii) separately audits whether retained capability becomes institutionally legible and practically available.

5 Candidate HCA Domain State and Defect Discipline

Let i index a human agent at finite step n. A candidate HCA state is

$$\begin{aligned} Z_{\mathrm{HCA},i,n} = \langle P^{\mathrm{live}}_{i,n}, K^{\mathrm{life}}_{i,n}, B^{\mathrm{bar}}_{i,n},\\ C^{\mathrm{cand}}_{i,n}, C^{\mathrm{live}}_{i,n}, h_{i,n},\\ R^{\mathrm{return}}_{H,i,n}, \Omega^{\mathrm{real}}_{i,n}\rangle. \end{aligned} \tag{6}$$

Here P^{live} is the endorsed live problem or explicit exploration goal; K^{life} is a life-context bookkeeping vector; B^{bar} is a typed barrier ledger; C^{cand} contains candidate capability/opportunity routes; C^{live} the subset endorsed as practical development routes; h the assistance/scaffold state; R^{return}_H the retained human return readout; and Ω^{real} a world-side opportunity/recognition readout.

This is a candidate representation, not a claim that the listed variables are minimal or sufficient. A serious implementation must report a defect vector

$$\varepsilon_{\mathrm{HCA}} = \mathrm{Def}\big(\tilde{q}_{\mathrm{HCA}} \circ F,\ F_{\mathrm{HCA}} \circ \tilde{q}_{\mathrm{HCA}},\ O_{\mathrm{HCA}},\ \mathrm{Inv}_{\mathrm{HCA}}\big), \tag{7}$$

and must be willing to delete variables when they fail to add reproducible explanatory or intervention value.

6 Unequal Life Conditions: Capability Cannot Be Identified from the Conversation Alone

The user and AI do not meet on a blank plane. Define a Life-Capital Context Vector [Definition] as

$$\begin{aligned} K^{\mathrm{life}}_{i,n} = \langle E^{\mathrm{econ}}, F^{\mathrm{base}}, L^{\mathrm{lang}}, D^{\mathrm{digital}}, T^{\mathrm{disc}},\\ H^{\mathrm{health}}, M^{\mathrm{mob}}, N^{\mathrm{mentor}}, C^{\mathrm{cred}}\rangle_{i,n}. \end{aligned} \tag{8}$$

The labels denote economic resources, foundational literacy/numeracy access, language bridge, device/connectivity, discretionary time, health/nutrition/sleep conditions, mobility/safety, mentor/network access, and credential pathway. This vector is not a universal cardinal measure of human capital and does not imply that these dimensions exhaust life conditions. It exists to prevent the causal fiction that people begin with equal conversion conditions.

The Readout-native point is stronger than "context matters." The same world event can produce different accessible traces under different language, tools, permissions, and contexts [6]. The allied capability literature independently makes a similar distinction: resources are means, while substantive opportunities depend on conversion factors [13]. The present architecture preserves that convergence without importing the capability approach as a root ontology.

Three non-collapse laws follow [Definition]:

$$\begin{aligned} \text{Resources} \neq \text{Access}, \quad \text{Access} \neq \text{Capability},\\ \text{Capability} \neq \text{Realized Opportunity.} \end{aligned} \tag{9}$$

$$\text{Access}(z) \neq \text{Control}(z). \tag{10}$$

$$\text{Equal AI Access} \not\Rightarrow \text{Equal Capability Conversion.} \tag{11}$$

A child with a shared phone, weak Thai literacy, family work obligations, limited mobility, and no credential bridge is not equivalent to a child with private device access, uninterrupted schooling, free study time, and a mentor merely because both can open the same chatbot.

7 Live Problem Entry and Barrier Readout

RG-HCA inherits the Problem-First entry condition from DCP v1.3. A live problem is a presently consequential unresolved difference in the person's life, work, understanding,

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Table 1: Internal programme lineage. Each prior work supplies a typed interface; the present paper restates the load-bearing object so it remains standalone.

Prior programme object	Retained result	Role in RG-HCA
Readout Genesis / Information Epistemic Foundation	Retained difference precedes named semantics; domain claims require typed translation, provenance, selection policy, decisive records, and defect accounting.	Constitutional rail, no-import rule, barrier-readout discipline, and policy audit. [5, 6]
From Problem to Hypothesis	Practical inquiry can begin from a retained consequential mismatch; reachable alternatives differ from practically accessible alternatives.	Live Problem entry and non-equivalent candidate generation. [7]
How Humans Should Converse with AI v1.3	Human-development dialogue begins from human-owned topic entry, uses minimum sufficient friction, and closes through Human Return and world feedback.	Reactive dialogue protocol that RG-HCA extends with a proactive capability layer. [8]
Human Learning as Epistemic Architecture	Problem-context, context-fit tools, workflow, deliberate practice, real problems, and feedback turn understanding into corrigible competence.	Competence engine after a capability route is endorsed. [9]
Choice Begins Before Choice	Physical possibility, structural feasibility, live possibility, choice, enactment, and evaluator visibility must not collapse.	Structural-barrier and opportunity layer; capability expansion is not the same as realized choice. [10]
CTSA Human Return	Concept, Tool, Skill, and Alternative Return require unaided reconstruction, selection, execution/transfer, or non-equivalent route generation.	Session and delayed return readout; gain/loss/provenance/warrant ledger. [11]
After Labour / Human Conversion (v1.2)	Human position depends on claim, exit, live alternatives, agency, retained capability, social reproduction, and recognition rather than output alone.	Macro boundary: capability that cannot be converted into standing or opportunity may remain economically invisible. [12]

Table 2: Global literature attached to the Genesis-native rail. No external school is treated as a root premise.

Rail location	Allied literature	What it establishes or pressures	Role here
Live problem / goal	Problem framing; goal articulation [14, 15]	Problems and goals are themselves consequential representations, not neutral inputs.	Convergent/Rival
Barrier vs opportunity	Sen/Capability tradition; conversion factors [13, 16]	Resources do not automatically become real opportunities; personal, social, and environmental conversion conditions matter.	Convergent; rival
Situated use	Situated cognition [17]	Knowledge and competence are partly constituted by activity, context, and culture of use.	Convergent
Learner-state estimation	Knowledge tracing [18]	Interaction histories can be used to estimate skill mastery and personalize sequence or material.	Close rival; operational ancestor
Scaffold	Wood–Bruner–Ross; scaffolding review [19, 20]	Assistance can be contingent, temporary, faded, and transferred back to the learner.	Convergent; no separate claim
Adaptivity	Expertise-reversal meta-analysis [21]	More support can help novices yet burden experts; assistance must depend on prior knowledge and domain.	Convergent; Defeater
Autonomy / structure	Self-determination and autonomy-structure meta-analysis [22]	Structure and autonomy support can reinforce rather than negate one another.	Convergent; constrains proactivity
Skill consolidation	Anderson; deliberate practice; retrieval practice [23–25]	Exposure and declaration are insufficient; proceduralization, effortful retrieval, feedback, and repeated execution matter.	Convergent
Human-AI learning	Bastani; Brynjolfsson [2, 3]	AI can either impair later unaided performance or facilitate durable learning depending on design and context.	Convergent; Defeater
Reliance and friction	Buçinca; Vasconcelos; de Jong; Tian [26–29]	Critical engagement can reduce overreliance but imposes cognitive and adoption costs.	Convergent; burden constraint
Human-centric control policy	Offline RL for human-centric objectives [30]	Assistance policies can optimize objectives beyond immediate accuracy; learning is harder to optimize than accuracy.	Close rival
Human-AI system performance	Vaccaro–Almaatouq–Malone [1]	Human+AI is not automatically synergistic and effects vary by task.	Convergent; Defeater
Feedback loops	Glickman–Sharot [4]	Repeated AI interaction can alter later human judgments, sometimes outside awareness.	Convergent; justifies return audits
Market recognition	Spence signalling [31]	Employers infer latent productivity from visible signals; observed credentials can matter independently of actual capability.	Convergent; market-legibility layer
Child deployment	UNESCO; UNICEF [32–34]	Age-appropriate human oversight, safety, privacy, participation, and rights are necessary constraints.	Governance constraint

evidence relation, or environment [7, 8]. Problem-first is a default for development, not a prohibition on curiosity, play, art, or basic inquiry.

The proactive extension begins only after the problem is human-owned. The next question is not "Which skill should AI teach?" but "What currently blocks this problem from becoming solvable or the desired route from becoming live?" Define a typed barrier ledger

$$B^{\text{bar}}_{i,n} \subseteq \{\text{Knowledge}, \text{Skill}, \text{Language}, \text{Tool}, \text{ResourceTime}, \text{Network}, \text{Credential}, \text{Permission}, \text{Opportunity}, \text{Unknown}\}. \tag{12}$$

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The load-bearing non-collapse is [Definition]:

$$\text{Observed Difficulty} \neq \text{Skill Deficit}. \tag{13}$$

A wrong diagnosis has real costs. If the active barrier is language, repeated skill drills may punish the learner for a translation problem. If the barrier is device access, teaching advanced digital workflow does not create hardware. If the barrier is a missing credential, more private competence may remain invisible to an employer. If the barrier is permission or safety, encouragement to "try harder" can be irresponsible.

Barrier typing must therefore remain probabilistic and falsifiable. Let $U_{i,n}^{\text{diag}}$ be admissible low-risk diagnostic actions: asking a clarifying question, requesting a small unaided attempt, checking language comprehension, confirming tool access, or observing a real-world failure point. A candidate decisive-record policy is [Open]:

$$u_{i,n}^{\text{diag}*} = \arg\max_{u \in U^{\text{diag}}} \Big[I_{G_B}(u) - \lambda_C \,\text{Cost}(u) - \rho\, \text{Risk}(u) \Big], \tag{14}$$

where $I_{G_B}(u)$ is the expected information/discrimination gain among active barrier hypotheses. Equation 14 is Open: it adapts the Genesis decisive-record discipline to a human-development domain and should be replaced if a simpler rule performs as well.

8 Candidate Advancement Routes: Capability and Opportunity Are Different Objects

Once the barrier ledger is explicit, AI may generate candidate advancement routes

$$C_{i,n}^{\text{cand}} = \text{Gen}\big(P_{i,n}^{\text{live}}, B_{i,n}^{\text{bar}}, K_{i,n}^{\text{life}}, R_{H,i,n}^{\text{return}}\big). \tag{15}$$

The candidate set may include learning routes (numeracy, writing, diagnosis, spreadsheet use), access routes (translation, tool, mentor), and opportunity routes (portfolio, certificate, apprenticeship, network introduction). Raw idea count is not the target. Two routes are distinct only if they differ in required capability, barrier relaxed, predicted world consequence, verification route, or institutional pathway.

The AI is allowed to be proactive about proposing routes, but not about silently making them the person's goals. Define

$$C_{i,n}^{\text{live}} = \{c \in C_{i,n}^{\text{cand}} : \text{Endorse}_i(c) = 1\}. \tag{16}$$

For an adult, endorsement means informed, revisable agreement. For a child, especially a young child, endorsement is insufficient by itself; deployment also requires age-appropriate assent plus guardian/teacher or institutional oversight under a child-centred governance policy [32, 34].

This yields three further non-collapse laws [Definition]:

$$\text{Proactive Suggestion} \neq \text{Human Goal Ownership}, \tag{17}$$

$$\text{Capability Advancement} \neq \text{AI Goal Authority}. \tag{18}$$

$$\text{Scaffolding} \neq \text{Control}. \tag{19}$$

The external autonomy-support literature is useful precisely here. Structure and autonomy support can reinforce one another rather than form a zero-sum pair [22]. That literature constrains how AI may help; it does not decide which life goal the person should endorse.

9 The Proactive Human Capability Advancement Policy

A reactive assistant waits for the user to request help. RG-HCA permits a narrower proactive action: after a human-owned problem and barrier readout, AI may surface a development opportunity that the user has not explicitly requested, provided it makes the reason visible and preserves refusal.

Let $U_{i,n}^{\text{adv}}$ contain admissible interventions such as a hint, worked comparison, mini-practice task, retrieval prompt, tool demonstration, mentor/credential suggestion, or real-world challenge. A candidate domain-level policy is [Open]:

$$u_{i,n}^{\text{adv}*} = \arg\min_{u \in U^{\text{adv}}} \Big[\mathbb{E}\big\|\text{Res}_{i,u}^{\text{cap}}\big\|^2 + \lambda_B B^{\text{use}}(u) + \rho R_u + \mu D_u + \xi A_u + \psi O_u - \nu V_u \Big], \tag{20}$$

subject to

$$\begin{aligned} \text{AgendaOwnership} &= 1, \quad \text{Transparency} = 1, \\ \text{Endorsement} &= 1, \quad \text{Safety/Governance} = 1. \end{aligned} \tag{21}$$

Res^{cap} is a declared residual between the capability needed for the endorsed route and current demonstrated return; B^{use} is usability burden; R_u is task or safety risk; D_u dependency risk; A_u autonomy loss or agenda displacement; O_u opportunity cost; and V_u expected cross-problem transfer value. The weights are local policy parameters, not universal human utilities. Any empirical use must predeclare and sensitivity-test them.

Equation 20 is deliberately close in form to the existing Genesis epistemic-action policy that trades residual reduction against cost, risk, repair loss, and value [6]. The added human-development terms are a domain proposal, not a claim of root derivation.

Human Capability Advancement Principle [Open]. When durable human development is an endorsed goal, AI may proactively identify and propose a reachable capability or opportunity expansion, but it should remain subordinate to human-owned ends, prefer the least burdensome intervention likely to produce durable return, and withdraw decisive support as the person demonstrates independent capability.

10 Adaptive Scaffolding: Use What the Learning Sciences Already Know, but Keep the Boundaries

The local mechanism of support is the same one used in the cited scaffolding literature. Scaffolding has long emphasized contingent assistance, fading, and transfer of responsibility [19, 20]. Recent meta-analytic evidence on expertise reversal reinforces the need for adaptivity: low-prior-knowledge

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learners benefit from more support on average, whereas high-prior-knowledge learners can be harmed by excess assistance [21]. The model therefore rejects a fixed amount of friction.

Let $h_{i,n}$ denote the level of decisive assistance. Rather than prescribe a universal update law, RG-HCA uses a monotonic candidate rule [Open]:

$$\text{Stable Unaided Return} \uparrow \Rightarrow h^{\text{decisive}} \downarrow \tag{22}$$

unless task difficulty, stakes, or domain risk rises. Failure after reduced support can justify temporary re-scaffolding. This allows assistance to be high for a novice and low for an expert without treating either as a moral preference.

For skill-building tasks the process is

$$\begin{aligned}&\text{Attempt} \to \text{Minimal Sufficient Scaffold} \to \text{Feedback}\\ &\quad \to \text{Reattempt} \to \text{Fading}\\ &\quad \to \text{Unaided Execution} \to \text{Novel Transfer}.\end{aligned} \tag{23}$$

This is consistent with established work on declarative-to-procedural skill acquisition, deliberate practice, and retrieval practice [23–25]. The internal Human Learning architecture places these mechanisms after problem-context and tool selection rather than treating them as a complete account of human development [9].

11 Human Return: The AI Should Know When to Become Less Necessary

A system can produce a superior joint answer while leaving little retained capability with the person. RG-HCA therefore inherits CTSA Human Return:

$$R_H^{\text{return}} = \langle C, T, S, A \rangle, \tag{24}$$

where C is Conceptual Return, T Tool-selection Return, S Skill-execution Return, and A Alternative-generation Return. The decisive evidence is unaided reconstruction, selection, execution/transfer, or generation under a declared criterion [11].

The primary non-collapse remains [Definition]:

$$\Delta\text{Performance}_{\text{AI}} > 0 \not\Rightarrow \Delta R_H^{\text{return}} > 0. \tag{25}$$

A retained false belief is still retained; retention is not warrant. A new fast procedure may be a wrong procedure; skill is not direction. A complete audit therefore records gain, loss, provenance, warrant, and retained direction rather than only gross gain [11].

The design slogan "AI should make itself unnecessary" must also be bounded. It is appropriate when the target is capability that the human has reason to retain. It is not a universal prohibition on delegation. For trivial formatting, repetitive administration, or other low-value procedures, permanent delegation can be rational. The correct statement is [Open]:

$$R_H^{\text{return}} \uparrow \Rightarrow h^{\text{decisive}} \downarrow \tag{26}$$

for capability-sensitive tasks, unless task difficulty, stakes, or domain risk rises.

12 World Return: Capability That Never Leaves the Conversation Remains Unproven

Human Return establishes what returned with the person, but an action-relevant live problem is not closed by verbal reconstruction. DCP v1.3 already requires return to the world [8]. Let $a_{i,n}$ be a human-selected action and $\delta_{i,n+1}^{\text{world}}$ the returned record:

$$R_{H,i,n}^{\text{return}} \to a_{i,n} \to \delta_{i,n+1}^{\text{world}} \to Z_{\text{HCA},i,n+1}. \tag{27}$$

The returned record may confirm the route, expose a new barrier, show that the problem was misframed, or reveal that the acquired skill does not transfer under real constraints. Situated cognition provides an external ally for this step: knowledge and competence cannot always be abstracted from the activities and contexts in which they are used [17].

13 Opportunity Conversion: The Market May Not Be Able to Read the Capability

A further boundary appears after Human Return. The person may now be capable yet still lack a viable social or economic route. Define a world-side opportunity readout

$$\begin{aligned}\Omega_{i,n}^{\text{real}} = G_O\big(&R_{H,i,n}^{\text{return}}, K_{i,n}^{\text{life}},\\ &\text{Cred}_{i,n}, \text{Net}_{i,n}, \text{Perm}_{i,n},\\ &\text{MarketReadout}_n\big).\end{aligned} \tag{28}$$

This is not a welfare utility function and not a validated scalar. It is a reminder that realized opportunity is a joint readout of retained capability and conversion conditions. Two distinctions are essential [Definition]:

$$\text{Credential} \neq \text{Capability}, \tag{29}$$

$$\text{Market Legibility} \neq \text{Human Worth}. \tag{30}$$

Spence's signalling model is an obvious external neighbour because employers must infer latent productivity from visible signals under uncertainty [31]. RG-HCA uses this literature only to motivate a legibility problem: capability that is not represented by a recognizable credential, portfolio, demonstration, reference, or network may fail to become labour-market opportunity. The remedy can therefore be an opportunity bridge rather than more skill training.

A full advancement route may end with

$$\text{Skill} \to \text{Evidence of Skill} \to \text{Recognition} \to \text{Opportunity}, \tag{31}$$

but the same chain can fail at any transition.

14 Worked Scenario: Three Children, Unequal Starting Conditions, and the Age-15 Boundary

The following scenario is an analytical stress test, not a forecast of individual children. It is motivated by a real equity

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problem in Thailand's deep south. UNICEF/NSO MICS reporting found very low basic reading rates among children aged 7–8 in Narathiwat, Pattani, and Yala relative to the national average, and higher upper-secondary non-attendance in parts of the region [35]. A recent reintegration initiative for NEET youth in four southern provinces reported that many participants returned to education, employment, or training after targeted outreach and skill/opportunity support [36]. These facts make the region an appropriate example of why life conditions and opportunity bridges belong inside the model.

Define three stylized children who are age seven in 2026 and age fifteen in 2034 (Table 3). The causal warning is [Definition]:

$$\text{Outcome}_C - \text{Outcome}_A \neq \text{Effect}_{\text{HCA}}, \tag{32}$$

because C receives more schooling, credential access, and other life conditions. The cleaner comparison is B versus C under matched baseline conditions. A serves a different scientific question: whether an HCA system can reduce the amount of potential lost when starting conversion conditions are poor.

Age fifteen is also not equivalent to full labour-market access. Thai law prohibits employment below age fifteen and imposes additional restrictions for workers under eighteen, including limits on night work, overtime, and hazardous work [37]. The age-15 target should therefore be interpreted as readiness for safe entry-level work, supervised apprenticeship, project income, or vocational transition—not as a recommendation to end education. The strongest pathway suggested by this scenario is therefore not "AI instead of school" but

$$\begin{aligned} &\text{Schooling} + \text{Human-Owned Inquiry} \\ &\quad + \text{Supervised Adaptive AI} \\ &\quad + \text{Projects} + \text{Human Return} \\ &\quad + \text{Opportunity Bridges}. \end{aligned} \tag{33}$$

15 Child and Adult Governance: Proactivity Without Paternalism

Proactivity increases both benefit and manipulation risk. The architecture therefore distinguishes adult deployment from child deployment. For adults, the minimum governance bundle is:

$$\begin{aligned} &\text{Purpose} + \text{Transparency} + \text{Refusal} \\ &\quad + \text{Revision} + \text{Data Minimization}. \end{aligned} \tag{34}$$

The system must disclose why it is proposing a capability route, link the route to the human-owned problem, allow refusal without penalty, and permit the user to revise or delete the development goal.

For children, especially younger children, a stronger gate is required:

$$\begin{aligned} &\text{Child Assent} + \text{Adult Oversight} \\ &\quad + \text{Age-Appropriate Design} \\ &\quad + \text{Privacy} + \text{Safety} \\ &\quad + \text{No Opaque Persuasion}. \end{aligned} \tag{35}$$

UNESCO's GenAI guidance calls for human-centred, age-appropriate use, privacy protection, and age limits for independent conversations with GenAI platforms [32]. UNICEF's current strategy and guidance emphasize human autonomy and oversight, inclusion, transparency, privacy, safety, development, and children's participation [33, 34]. A seven-year-old RG-HCA deployment should therefore be mediated by a trusted adult or institution; child assent alone is not treated as sufficient governance.

The system must also know when not to be proactive. No advancement intervention is required for every conversation. The default should remain minimum sufficient friction, with proactivity triggered only when the user has opted into development, a repeated barrier is visible, or stakes justify a capability check.

16 Measurement Architecture: What Would Count as Advancement?

A full HCA evaluation must separate at least six outcome families.

1. *Immediate assisted performance.* Did the person solve the current task faster or better with AI?

2. *Human Return.* After decisive assistance is removed, can the person reconstruct the concept, choose the tool, execute the skill, or generate a non-equivalent alternative? [11]

3. *Transfer.* Does the retained capability work on a sufficiently novel problem, not merely the practiced item?

4. *Autonomy/agenda integrity.* Can the person identify the original goal, explain why the proposed route was accepted, reject a route, and revise the goal without the system steering them back?

5. *Burden and adoption.* Time, cognitive load, interruption cost, satisfaction, abandonment, and repeated-use willingness matter because an epistemically ideal protocol that nobody uses has low field value.

6. *Opportunity conversion.* Does the capability become usable in a real institution, project, credential pathway, apprenticeship, or labour-market context? Is capability legible to relevant gatekeepers?

A candidate net advancement record is therefore a measurement/estimand definition [Definition]:

$$\begin{aligned} A_i^{\text{HCA}} = \big\langle &\text{Gain}_{CTSA}, \text{Loss}, \text{Transfer}, \\ &\text{Ownership}, \text{Burden}, \text{BarrierChange}, \\ &\text{OpportunityChange}, \text{Provenance}, \\ &\text{Warrant}\big\rangle. \end{aligned} \tag{36}$$

No scalar score is required. For policy comparison, a Pareto report is preferable unless a value weighting is explicitly declared.

17 Tier-One Empirical Programme

The theory should earn its complexity through experiments that permit it to lose components.

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Table 3: Illustrative A/B/C scenario. The table is a theory stress test, not a forecast and not an estimate of treatment effects.

	A: low life capital + RG-HCA	B: formal school + passive AI	C: formal school + RG-HCA
Starting conditions	Interrupted/weak schooling, lower language/credential access, limited devices/time/network; supervised problem-first AI and mentor support added.	More continuous schooling and credential path; AI mainly used as answer generator.	Same formal schooling/credential path as B plus supervised problem-first HCA, projects, fading, Return tests.
AI relation	AI proposes routes from real problems; may build high practical capability but cannot manufacture institutional recognition by itself.	User asks, AI answers; immediate task performance can rise without deliberate return testing.	AI assists, challenges, teaches when endorsed, fades, and pushes transfer to projects/world evidence.
Age-15 capability	Potentially strong local problem solving/portfolio relative to formal attainment, but foundational gaps remain possible.	Stronger curriculum/credential legibility than A, but practical problem framing and unaided AI-complementary capability are not guaranteed.	Highest plausible joint position because foundational schooling, credential, portfolio, and Human Return can accumulate together.
Market legibility	Weakest if no M.3-equivalent/recognized signal; community/project markets may see capability better than formal HR systems.	Formal systems can read the credential even if capability is ordinary.	Credential and portfolio make capability more legible; still constrained by age and sector.
Research meaning	Tests whether scarce life resources can be converted into retained capability, not whether AI "beats school."	Baseline for formal schooling with passive AI.	Cleanest contrast with B for estimating the added value of HCA under matched schooling/life conditions.

17.1 Study 1: reactive DCP versus proactive HCA

Recruit adults across low- and high-prior-knowledge strata and assign them to: (A) direct-answer AI; (B) problem-first reactive DCP; (C) proactive HCA with barrier typing and capability suggestions; (D) no-AI. Use real or realistic problem domains in which assistance can be removed. Primary outcomes: immediate performance, delayed CTSA Return, novel transfer, autonomy/ownership, burden, and reliance calibration. The critical question is whether proactive HCA adds delayed capability beyond a strong reactive protocol.

17.2 Study 2: skill-only tutor versus barrier-aware navigator

Create tasks in which failure is experimentally caused by different bottlenecks: missing knowledge, missing skill, language comprehension, tool access, or credential/permission constraints. Compare a tutor that assumes skill deficit with a barrier-aware condition. The main outcome is correct barrier identification and intervention selection. If barrier typing adds no predictive or intervention value beyond ordinary mastery models, the barrier layer should be simplified.

17.3 Study 3: adaptive scaffold and fading

Cross prior knowledge with high versus low assistance and test whether HCA's assistance policy reproduces the expertise-reversal pattern rather than imposing a one-size policy. Evaluate whether fading based on demonstrated Return outperforms fixed fading schedules.

17.4 Study 4: longitudinal supervised youth pilot

A child trial must be institutionally supervised and age-appropriate. A plausible design in a multilingual, resource-variable context would compare matched learners receiving (B) formal schooling plus ordinary/passive AI support with (C) formal schooling plus supervised problem-first HCA. Measure foundational literacy/numeracy, school engagement, CTSA Return, real-project completion, language bridge, portfolio evidence, and credential continuity over 6–24 months. Do not use employment at age fifteen as the primary endpoint; use safe vocational readiness, transfer, and continued learning.

17.5 Study 5: market-legibility experiment

Use employer or supervisor vignettes to orthogonally vary demonstrated capability, credential, portfolio evidence, and referral/network signal. Test whether identical capability receives different opportunity under different signals. This study operationalizes the distinction between capability and market legibility and tests whether a portfolio/credential bridge changes hiring or apprenticeship judgments.

17.6 Identification discipline

The child A/B/C scenario shows why baseline context cannot be ignored. Let Y be a declared outcome and HCA an intervention indicator. The target estimand is not a raw rank difference but a conditional contrast such as a measurement/estimand definition [Definition]:

$$\mathrm{ATE}_{\mathrm{HCA}}(x) = \mathbb{E}\big[Y(1) - Y(0) \mid K^{\mathrm{life}} = x\big]. \qquad (37)$$

Here x denotes a declared baseline life-context stratum; matching or covariate adjustment is a design strategy, not part of the estimand itself. Randomization is preferable where ethical and feasible; otherwise, the paper should report the limits of identification rather than convert association into "AI impact."

18 RIVAL MODELS AND LEGITIMATE DEFEATERS

RG-HCA is not protected from simpler explanations.

Rival 1: ordinary adaptive tutoring is enough. Knowledge-tracing and intelligent-tutoring traditions already estimate learner mastery from interaction histories and adapt instruction [18]. If a well-designed tutor with mastery estimation, contingent support, and fading predicts all relevant delayed outcomes, the HCA architecture should collapse toward that simpler model.

Rival 2: proactive human-centric policy optimization is enough. Buçinca et al.'s human-centric policy framework is a close computational rival [30]. RG-HCA earns separate value only if human-owned problem entry, barrier typing, opportunity conversion, or Readout-native return accounting add reproducible information beyond optimized assistance.

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Rival 3: the capability approach already supplies the whole theory. The capability tradition gives a powerful normative and conversion vocabulary [13, 16]. RG-HCA should not duplicate it. Its distinct burden is to specify a finite human-AI interaction pipeline with readout, barrier discrimination, scaffolding, removal, return, provenance, and empirical control policies.

Rival 4: Human Return is just ordinary transfer. If CTSA categories add no reliable information beyond standard knowledge, skill, transfer, and metacognitive measures, CTSA should be merged or removed [11].

Rival 5: proactive support creates paternalism or disengagement. If proactive suggestions reduce autonomy, increase abandonment, suppress exploration, or disproportionately burden low-literacy users, the triggering policy should become more restrictive or revert to reactive assistance.

Rival 6: opportunity bridges do not change realized opportunity. If credential, portfolio, network, or institutional routing add no stable predictive value after measured capability, the opportunity-conversion layer should be simplified.

19 Falsification and Removal Conditions

The following findings would count against the architecture rather than being reinterpreted as support:

- barrier categories cannot be coded reliably across independent raters or add no intervention value;
- proactive HCA produces no delayed Return or transfer advantage over reactive DCP at comparable burden;
- the endorsement mechanism does not preserve agenda ownership or is routinely manipulated by framing;
- fading based on Return is no better than fixed assistance schedules;
- CTSA adds no incremental validity beyond ordinary delayed-performance measures;
- life-context variables do not moderate conversion outcomes in the claimed domains;
- opportunity/credential bridges do not improve any declared real-world conversion outcome;
- users reject the protocol under realistic time pressure so strongly that protective gains disappear;
- child deployment cannot satisfy privacy, safety, oversight, and age-appropriate participation constraints;
- a simpler rival explains the same outcomes with fewer assumptions.

An architecture that cannot lose variables becomes vocabulary rather than theory.

20 What the Paper Does Not Claim

The paper does not claim that every person should constantly self-improve, that every conversation should become a lesson, or that AI should decide which life is worth living. It does not claim that low-resource children can substitute chatbot access for foundational schooling, stable adult relationships, nutrition, safety, or credentials. It does not claim that credentials are fair measures of capability, only that institutions often read them. It does not claim that proactivity is always better than reactivity, that all AI use causes deskilling, or that all scaffolding should end in complete unaided performance. It does not claim an exact HCA quotient, a universal life-capital score, or a global labour-market forecast.

The strongest admissible claim at present is architectural: if the goal is durable human development, then a complete human-AI advancement system should distinguish human-owned ends, barrier type, candidate route, assistance, retained return, and opportunity conversion rather than collapsing them into current task success.

21 Discussion: From AI as Answer Engine to AI as Subordinate Capability Navigator

The practical change proposed by RG-HCA is small in interface terms but large in objective. A conventional assistant asks, "What answer should I give?" A tutor asks, "What should this learner practice next?" A capability navigator asks a different sequence:

1. What human-owned problem or exploration goal is active? 2. What barrier is actually blocking progress? 3. Which non-equivalent capability or opportunity routes could relax it? 4. Which route does the person endorse? 5. What is the minimum sufficient assistance needed now? 6. When should decisive support fade? 7. What returned with the person after AI removal? 8. Does the retained capability transfer to a new problem and survive world feedback? 9. Is there an external bridge needed so the capability becomes a real opportunity?

This sequence prevents a subtle but consequential error: treating a user who repeatedly needs help as proof that the user needs more help. Sometimes the correct move is a short practice intervention. Sometimes it is a translation. Sometimes a mentor, tool, credential, safe place, or institutional route is the real missing object. The theory therefore turns "proactive AI" away from behavioural engagement and toward a falsifiable conversion problem.

Three design invariants summarize the proposal.

Human-end invariant. AI may be proactive about expanding human capability, but it must remain subordinate to human-owned ends.

Return invariant. When a capability is worth retaining, success is demonstrated by what the human can still do when decisive AI assistance is absent.

Structural-barrier invariant. When the barrier is structural, the intervention must change access or opportunity—not merely train the person harder.

22 Conclusion

Human-AI systems are moving from tools that answer toward systems that remember, recommend, plan, and increasingly act. A purely reactive model of assistance is therefore no longer the only relevant design target. Yet proactivity creates its own danger: a system capable of noticing

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weaknesses and proposing development can also redefine the person's problem, substitute its own goals, pathologize structural barriers as personal deficits, or make dependence feel like progress.

The Readout-Genesis response is to type the transitions. A retained difference becomes a human readout; a human-owned live problem is separated from an AI-supplied agenda; observed difficulty is separated from barrier type; candidate routes are separated from endorsed routes; assistance is separated from Human Return; retained capability is separated from realized opportunity; and credential or market recognition is separated from human worth. External literatures then attach where they genuinely constrain the rail: scaffolding at assistance, self-determination at endorsement, capability theory at conversion, learning science at practice, human-AI evidence at reliance and return, signalling at recognition, and child-rights governance at deployment.

The paper's central empirical wager is therefore not that AI should "install skills." It is that a carefully governed AI can sometimes act as a subordinate capability navigator: it can identify a likely bottleneck, propose a reachable development or opportunity route, scaffold the person's own work, withdraw, test transfer, and help connect retained capability to the world. Whether this architecture actually produces more durable, more equitable, and more human-owned capability than strong reactive or tutoring baselines is an open question. The next step is to test it hard enough that parts of the architecture are allowed to fail.

REFERENCES

[1] Michelle Vaccaro, Abdullah Almaatouq, and Thomas W. Malone. When combinations of humans and AI are useful: A systematic review and meta-analysis. *Nature Human Behaviour*, 8:2293–2303, 2024. doi: 10.1038/s41562-024-02024-1.

[2] Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Ozge Kabakci, and Rei Mariman. Generative AI without guardrails can harm learning: Evidence from high school mathematics. *Proceedings of the National Academy of Sciences*, 122(26):e2422633122, 2025. doi: 10.1073/pnas.2422633122.

[3] Erik Brynjolfsson, Danielle Li, and Lindsey Raymond. Generative AI at work. *Quarterly Journal of Economics*, 140(2):889–942, 2025. doi: 10.1093/qje/qjae044.

[4] Moshe Glickman and Tali Sharot. How human–AI feedback loops alter human perceptual, emotional and social judgements. *Nature Human Behaviour*, 9:345–359, 2025. doi: 10.1038/s41562-024-02077-2.

[5] Readout Genesis Programme. Readout genesis universal technical whitepaper v1.2.0. Public GitHub repository morrocwi/readout_genesis, file READOUT_GENESIS_UNIVERSAL_TECHNICAL_WHITEPAPER_v1.2.0.md, main at commit 04cde19be2c885a11b42597b1cdb60fb5b7ca1bb, blob 6ca7ef20a1e9ea17632451e6c8a19b3ae7c9fa8a, 2026. Universal research architecture, typed interfaces, domain gates, and anti-drift constitution. Working technical specification, not a Zenodo record. `https://github.com/morrocwi/readout_genesis/blob/04cde19be2c885a11b42597b1cdb60fb5b7ca1bb/READOUT_GENESIS_UNIVERSAL_TECHNICAL_WHITEPAPER_v1.2.0.md`.

[6] Yaoharee Lahtee. Readout genesis standalone synthesis: Information epistemic foundation, conditioned agency, and meta-readout governance. Human-AI Readout Programme, Zenodo, 2026. Cited as the deposited record of the "Information Epistemic Foundation" content referenced by the v1.0 original as "Information Epistemic Foundation v1.6.0." That v1.6.0 working specification is a foundational architecture-conditional document for information, knowing, knowledge, meaning, memory, epistemic action, and bounded human-AI transport; it is not itself separately deposited or git-anchored, and the present citation should be read as pointing to this deposited Standalone Synthesis rather than to a directly resolvable v1.6.0 artifact. `https://doi.org/10.5281/zenodo.21529456`.

[7] Yaoharee Lahtee. From problem to hypothesis: Dynamic semantic mobility, bounded knowers, and the readout-discriminable hypothesis space. Human-AI Readout Programme, GLOSA K0-final-2026-09-04, Zenodo, 2026. `https://doi.org/10.5281/zenodo.22307148`.

[8] Yaoharee Lahtee. How humans should converse with AI: A problem-first adaptive dialogue conversion protocol for durable human capability. Human-AI Readout Programme, v1.3 glosa-checked edition of the 6 September 2026 v1.2 original, Zenodo, 2026. `https://doi.org/10.5281/zenodo.22481928`.

[9] Yaoharee Lahtee. Human learning as epistemic architecture. Human-AI Readout Programme, revised v4 preprint, Zenodo, 2026. `https://doi.org/10.5281/zenodo.22341297`.

[10] Yaoharee Lahtee. Choice begins before choice: Meaning-shaped accessibility, live possibility, and effective agency in human-AI systems. Human-AI Readout Programme, Full Paper v1.0-strong-claim, 5 September 2026, Zenodo, 2026. `https://doi.org/10.5281/zenodo.22357788`.

[11] Yaoharee Lahtee. CTSA human-return readout: A session-boundary measurement architecture for retained human capability after AI. Human-AI Readout Programme, v9.1 FINAL global-literature-constrained edition, 5 September 2026, Zenodo, 2026. `https://doi.org/10.5281/zenodo.22339909`.

[12] Yaoharee Lahtee. After labour: Human position in an AI-robotic world system. Human-AI Readout Programme, full world-system standalone, v1.2 glosa-checked edition of the 6 September 2026 v1.1 original, Zenodo, 2026. `https://doi.org/10.5281/zenodo.22481924`.

10.5281/zenodo.22498047

[13] Ingrid Robeyns. The capability approach: A theoretical survey. *Journal of Human Development*, 6(1): 93–117, 2005. doi: 10.1080/146498805200034266.

[14] Stefano Magistretti, Christophe T. A. Pham, and Claudio Dell'Era. The creative process of problem framing for innovation: An integrative review and research agenda. *Journal of Product Innovation Management*, 42(6):987–1018, 2025. doi: 10.1111/jpim.12783.

[15] Lev Tankelevitch and Sean Rintel. Goals as first-class abstractions in human-AI collaboration. AutomationXP26 Workshop, CHI 2026, 2026.

[16] Amartya Sen. *Development as Freedom*. Oxford University Press, 1999.

[17] John Seely Brown, Allan Collins, and Paul Duguid. Situated cognition and the culture of learning. *Educational Researcher*, 18(1):32–42, 1989. doi: 10.3102/0013189X018001032.

[18] Ghodai Abdelrahman, Qing Wang, and Bernardo Nunes. Knowledge tracing: A survey. *ACM Computing Surveys*, 55(11):Article 224, 1–37, 2023. doi: 10.1145/3569576.

[19] David Wood, Jerome S. Bruner, and Gail Ross. The role of tutoring in problem solving. *Journal of Child Psychology and Psychiatry*, 17(2):89–100, 1976. doi: 10.1111/j.1469-7610.1976.tb00381.x.

[20] Janneke van de Pol, Monique Volman, and Jos Beishuizen. Scaffolding in teacher–student interaction: A decade of research. *Educational Psychology Review*, 22:271–296, 2010. doi: 10.1007/s10648-010-9127-6.

[21] Leonard Tetzlaff, Bianca Simonsmeier, Tobias Peters, and Garvin Brod. A cornerstone of adaptivity–a meta-analysis of the expertise reversal effect. *Learning and Instruction*, 98:102142, 2025. doi: 10.1016/j.learninstruc.2025.102142.

[22] Anne Patzak and Xihui Zhang. Blending teacher autonomy support and provision of structure in the classroom for optimal motivation: A systematic review and meta-analysis. *Educational Psychology Review*, 37:17, 2025. doi: 10.1007/s10648-025-09994-2.

[23] John R. Anderson. Acquisition of cognitive skill. *Psychological Review*, 89(4):369–406, 1982. doi: 10.1037/0033-295X.89.4.369.

[24] K. Anders Ericsson, Ralf T. Krampe, and Clemens Tesch-Römer. The role of deliberate practice in the acquisition of expert performance. *Psychological Review*, 100(3):363–406, 1993. doi: 10.1037/0033-295X.100.3.363.

[25] Jeffrey D. Karpicke and Janell R. Blunt. Retrieval practice produces more learning than elaborative studying with concept mapping. *Science*, 331(6018):772–775, 2011. doi: 10.1126/science.1199327.

[26] Zana Buçinca, Maja Barbara Malaya, and Krzysztof Z. Gajos. To trust or to think: Cognitive forcing functions can reduce overreliance on AI in AI-assisted decision-making. In *Proceedings of the ACM on Human-Computer Interaction (CSCW1)*, volume 5, page Article 188, 2021. doi: 10.1145/3449287.

[27] Helena Vasconcelos, Matthew Jorke, Madeleine Grunde-McLaughlin, Tobias Gerstenberg, Michael S. Bernstein, and Ranjay Krishna. Explanations can reduce overreliance on AI systems during decision-making. *Proceedings of the ACM on Human-Computer Interaction*, 7(CSCW1), 2023. doi: 10.1145/3579605.

[28] Sander de Jong, Vera Paananen, Benjamin Tag, and Niels van Berkel. Cognitive forcing for better decision-making: Reducing overreliance on AI systems through partial explanations. *Proceedings of the ACM on Human-Computer Interaction*, 9(2):CSCW048, 2025. doi: 10.1145/3710946.

[29] Hui Yi Tian, Hamza Amin, and Ming Yin. Understanding the effects of AI-assisted critical thinking on human-AI decision making. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, page Article 810, 2026. doi: 10.1145/3772318.3790785.

[30] Zana Buçinca, Siddharth Swaroop, Amy E. Paluch, Susan A. Murphy, and Krzysztof Z. Gajos. Towards optimizing human-centric objectives in AI-assisted decision-making with offline reinforcement learning, 2024. arXiv:2403.05911.

[31] Michael Spence. Job market signaling. *Quarterly Journal of Economics*, 87(3):355–374, 1973. doi: 10.2307/1882010.

[32] Fengchun Miao and Wayne Holmes. Guidance for generative AI in education and research. UNESCO. Published 7 September 2023, updated 16 January 2026, 2023. URL `https://www.unesco.org/en/articles/guidance-generative-ai-education-and-research`.

[33] UNICEF. AI strategy 2025–2030: AI for every child, 2025. URL `https://www.unicef.org/digitalimpact/unicef-ai-strategy-2025-2030-ai-every-child`.

[34] UNICEF Innocenti. Guidance on AI and children, version 3.0, 2025. URL `https://www.unicef.org/innocenti/reports/policy-guidance-ai-children`.

[35] National Statistical Office of Thailand and UNICEF Thailand. Thailand MICS 2022: Report of 12 selected provinces in thailand. See UNICEF release on foundational learning disparities in the southern provinces, 27 September 2023, 2023. URL `https://www.unicef.org/thailand/reports/thailand-mics-2022-report-12-selected-provinces-thailand`.

[36] UNICEF Thailand. Knocking on hope's door. Report on outreach and reintegration of NEET youth in four southern provinces, 2025. URL `https://www.unicef.org/thailand/stories/knocking-hopes-door`.

[37] Kingdom of Thailand. Labour Protection Act B.E. 2541 (1998), sections 44–52 (minimum working age 15; restrictions on night work, overtime, and hazardous work for workers under 18). Thai statute. No single official Ministry of Labour guidance-page URL bearing this exact citation could be independently resolved at the time of the v1.1 glosa check; the Act and its minimum-age/hazardous-work provisions are corroborated by independent secondary summaries (national labour-law digests and ILO/US DOL compendia of Thai child-labour law). Cited by Act name and section, accessed via secondary summaries, without a direct government-portal URL., 1998.

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Part VI

Human Position After Labour

Human Position After Labour widens the frame from Part 5's single human-AI conversation, closed by the Dialogue Conversion Protocol, to the world-system that conversation sits inside: two 2026 companion papers, deposited in the same batch and each glosa-checked after independent adversarial review, treating the human state itself as the variable a growing machine economy acts on.

Chapter 34, *After Labour*, proposes that the decisive transition in an AI-robotic world is not simply from human labour to machine labour but the weakening of labour as the universal bridge from personhood to a claim on social production; it models labour centrality, a Citizen Claim Threshold, credible exit and dependency, live possibility, corrigible agency, Human Return, and social reproduction together, and states a falsification-oriented programme (its own Table 3) under which named components of the model should be narrowed, revised, or removed. Chapter 35, *The Human Conversion Imperative*, proposes a companion Human Conversion Vector — effective material claim, credible exit, live possibility, corrigible agency, independent epistemic routes, world-answerability, delayed unaided Human Return, social reproduction, and recognized standing — together with a Reversibility Principle: the faster machine capability grows relative to how fast institutions convert that growth into broadly retained human capacity, the more costly later reconstruction becomes. Neither chapter treats a rising rate of machine capability as evidence of a rising rate of human capability; both state that relation as an open, falsifiable question to be measured rather than assumed, with each paper's own removal conditions listed in its headnote.

Carried forward: the same claim-by-claim, falsifier-attached discipline used earlier in this book, now applied at world-system scale rather than to a single interaction or a single learner; the refusal to read machine expansion as human expansion by default; and the Human Return test, again used as the boundary condition for what counts as durable human change. Part 7 turns to the programme's own equations as an audited object in their own right.

- After Labour: Human Position in an AI-Robotic World System (full world-system standalone, v1.2 glosa-checked edition) — `10.5281/zenodo.22481924`
- The Human Conversion Imperative: Machine Acceleration, Human Potential, and the Reversibility Window in an AI-Robotic World (theory and research agenda, v1.1 glosa-checked edition) — `10.5281/zenodo.22481926`

After Labour: Human Position in an AI-Robotic World System

v1.2 glosa-checked edition • 6 September 2026 • 10.5281/zenodo.22481924

What the chapter states. "The decisive transition in an AI-robotic world is not simply from human labour to machine labour. It is the weakening of labour as the universal bridge from personhood to a claim on social production." A world can become richer while ownership compounds upward, demand weakens, care is underproduced, and the systemic position of most humans becomes weaker.

Status. "This is a standalone K0 conceptual and measurement architecture… Knowledge state K0, version 1.2, timestamped and citable, not yet checked at an independent-reproduction tier above I1."

Falsifier(s). Table 3, "Gate vs concentration" row: "Counterfactual removal/substitution predicts transition failure beyond HHI or top-share measures… Remove Gconv if concentration/interoperability fully explain outcomes."

Read before. This book's Chapters 17, 31, 38 (mechanical citation list; Appendix A).

Feeds into. Ch. 32, 35 — companion papers, same deposit batch (mechanical citation list; Appendix A).

After Labour

Human Position in an AI-Robotic World System

From labour centrality to citizen claim, accumulated ownership, dependency, social reproduction, and Human Return

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v1.2 — glosa-checked edition of Full world-system standalone draft v1.1, 6 September 2026 original

Status and source boundary. This is a standalone K0 conceptual and measurement architecture, anchored in the prior standalone v1.0 and elevated into a fuller world-system model. It does not forecast a date for AGI, mass unemployment, or the disappearance of labour. Prior author papers are used as author-source genealogy: especially *Capitalism After Labor Centrality* (an undeposited internal working draft, not part of the deposited programme record; cited here for genealogy only), *Choice Begins Before Choice*, and *CTSA Human Return*. Their load-bearing objects are redefined in this manuscript so a reader need not consult them to reconstruct the argument. Relative to v1.0, this revision closes five missing loops: machine-wealth accumulation, scarce-asset rent burdens, aggregate-demand realization, social reproduction, and the separation of economic, informational, and coercive power. Self-citation establishes programme lineage, not independent validation. External literature supplies convergence, rivalry, empirical anchors, and defeaters. No original dataset is analyzed; unobserved parameters remain open. **Licence** CC BY 4.0. **Knowledge state** K0, version 1.2, timestamped and citable, not yet checked at an independent-reproduction tier above I1.

Edition note. This v1.2 is a source-lineage repair of the 6 September 2026 v1.1, made after an independent adversarial glosa review. Changes from v1.1: (i) the citation of *Capitalism After Labor Centrality* is now explicitly disclosed as an undeposited internal working draft at every mention, and its reference entry no longer uses a formal deposited-record citation format; (ii) the self-citations to *Choice Begins Before Choice* and *CTSA Human-Return Readout* now carry their deposited Zenodo DOIs; (iii) the self-citation to *Before Meaning, Before Choice* is corrected from the undeposited "v1.0" label to the actually deposited "Full Paper v1.1, glosa-checked" with its DOI; (iv) the reference to Bayraktar (2026) is corrected to the paper's current canonical arXiv title, with the earlier circulated title noted; (v) one same/different comparison in §11 that used superiority language ("improves on") and an untraceable "v0.1" label is rewritten to name the compared undeposited draft explicitly. No section, equation, table, proposition, or claim of the original v1.1 has been added, removed, renumbered, or strengthened. The v1.1 original remains the anchor record; this edition only repairs citation lineage and apparatus.

Central claim. The decisive transition in an AI-robotic world is not simply from human labour to machine labour. It is the weakening of labour as the universal bridge from personhood to a claim on social production. Once productive intelligence and embodied production can be reproduced and accumulated through capital at scale, a society must answer a different question: by what institutional route do humans retain claims on output, protection from scarce-asset rents, credible exit, live alternatives, corrigible agency, retained capability, and recognized social membership when the productive system needs less of their labour? A world can become richer while ownership compounds upward, demand weakens, care is underproduced, and the systemic position of most humans becomes weaker.

Abstract

Artificial intelligence and robotics are commonly studied through productivity, task substitution, wages, labour share, market concentration, or inequality. This paper proposes a wider dependent variable: human systemic position. Anchored in the prior standalone model, it closes the world-system dynamics by adding five missing blocks. First, machine-productive wealth is allowed to accumulate and reproduce ownership asymmetry over time rather than treating broad ownership as an exogenous share. Second, scarce physical assets—especially housing, land, energy, and debt service—are separated from reproducible machine abundance, yielding an effective material claim after unavoidable rent burdens. Third, distribution feeds back into aggregate demand, realization, profits, and therefore the next round of machine-capital accumulation. Fourth, social reproduction is modeled as a distinct condition for maintaining human capability, care, and social continuity even when human labour becomes less necessary to production. Fifth, rule-setting power is decomposed into economic, informational, and coercive channels rather than inferred directly from market concentration.

The model retains the v1.0 technology-production block, four-dimensional labour centrality, Citizen Claim Threshold, conversion-gate control, credible exit/dependency, live possibility, corrigible agency, Human Return, and social role. Human Systemic Position now depends on effective material claim after rent burdens, not gross claims alone. Two recursive loops close the model: an ownership-demand-production loop and a human-reproduction-political-rule loop. The paper therefore distinguishes productive necessity from social necessity: humans may become less necessary to scalable production while remaining indispensable to the reproduction of human life, legitimacy, care, and collective meaning. The central inequality of the AI age is not AI versus humans, but the growth of machine productive power and concentrated claims versus the growth of broad human ownership, effective material claim, exit, agency, retained capability, and social reproduction.

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Keywords: artificial intelligence; robotics; political economy; labour centrality; ownership; transition control; dependency; citizen claim; live possibility; agency; Human Return; post-labour society.

1 The Question After Labour

Industrial political economy linked mass participation to a practical chain:

$$\text{human labour} \to \text{production} \to \text{wage} \to \text{claim on output}. \tag{1}$$

Automation theory has long shown that technology can displace some tasks, complement others, and generate new tasks [1, 2]. The present paper does not assume that employment vanishes. Its problem begins when cognitive automation and embodied automation jointly make a growing share of productive capability reproducible through capital without a proportional increase in broad human labour.

The question is therefore not merely *how many jobs disappear?* It is whether the productive necessity of broad human labour can decline faster than the institutions that give people claims on output, bargaining power, usable alternatives, and standing. Automation can raise aggregate output while worsening distribution through wage and wealth channels [4, 5]. The remaining problem is wider: who owns productive intelligence, who controls the transitions through which it becomes action, who can route around those controls, what returns to humans as retained capability, and what replaces the social functions formerly organized by paid work?

2 Standalone Readout Discipline and Author-Source Genealogy

2.1 Readout in this paper

This manuscript uses *readout* in a minimal, self-contained sense:

$$\text{Readout}_{Q,O,c}(S) = z, \qquad z \neq S. \tag{2}$$

A readout is a finite, declared representation of a system relative to a question Q, reader or measurement interface O, and context c. The representation is not the whole system. Thus labour share, dependency, Human Return, and Human Systemic Position are readouts designed for declared questions; none is a substance inside a person or economy. This follows the programme's broader non-collapse discipline, but no root formalism is required to use the model here [36, 39].

2.2 What is inherited, and what is restated

The previous author paper *Capitalism After Labor Centrality* (an undeposited internal working draft, not part of the deposited programme record; cited here for genealogy only) developed four objects that are elevated here rather than treated as hidden premises: a four-dimensional labour-centrality profile, conversion-gate control, actor-specific credible exit/dependency, and a relational class-position vector [36]. *Choice Begins Before Choice* supplies the deeper human distinction between formal feasibility and practically live alternatives [37]; CTSA supplies one operationalization of delayed unaided Human Return [38]. Every one of these objects is redefined below before it becomes load-bearing.

This matters methodologically. The paper can be understood after deleting its self-citations; the self-citations record genealogy and prior development, not evidential authority.

3 Empirical Anchors in 2026

The model is motivated by current directional evidence, not calibrated from it. Table 1 records what the evidence does and does not license.

4 Literature Boundary and Residual Contribution

The paper does not claim that automation, inequality, platform power, substantive freedom, or human reliance on tools are new topics. Task-based economics already models displacement, complementarity, reinstatement, and labour-share effects [2, 3, 5]. Work on upstream AI competition shows why access prices and market power can affect downstream firms and workers [6]. Platform and intellectual-monopoly theories explain intermediation, network effects, knowledge exclusion, and rent [11–14]. Capability and non-domination traditions already reject the equation of formal resources with substantive freedom, while Foucault and Fricker show how power and interpretive resources can shape possible action before direct prohibition [7–10]. Human-AI research establishes that assisted performance, reliance, and later unaided performance can diverge [18–21].

The residual contribution is therefore architectural rather than a priority claim over these local mechanisms. Long-run wealth accumulation and wealth-income dynamics are already central to distributional economics [30], while housing is simultaneously a consumption necessity, a major household asset, and a source of distributional divergence [31]. Social-reproduction theory and care-economy research show that human production depends on background systems of care, households, education, and community reproduction that market production can underprovide [32, 33]. Recent work also links market concentration to political influence [34], and a 2026 working paper explicitly models a demand externality from automation when income shifts toward lower-propensity-to-consume owners [35]. The prior author paper *Capitalism After Labor Centrality* decomposed ownership, transition control, exit, dependency, claims, and relational class position [36]. The present paper integrates these neighboring concerns into one closed world-system architecture. Its strongest claim is that productive capacity, accumulated ownership, realized demand, scarce-asset rent, social reproduction, transition power, human agency, and political rule-setting must remain non-collapsed if the question is not merely "what happens to jobs?" but "what happens to the systemic position of humans when productive necessity, ownership, and agency can move in different directions?"

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5 Technology Block: Candidate Generation, Knowledge, and Embodiment

Let G_t denote machine candidate-generation capacity: the ability to generate hypotheses, designs, code, plans, or other candidate outputs. Let Z_t denote validated knowledge available to the productive system. A minimal epistemic firewall is

$$\dot{Z}_t = v_t G_t - \delta_Z Z_t, \qquad 0 \le v_t \le 1, \tag{3}$$

where v_t is a domain-specific conversion through testing, evidence, world resistance, or other validation. Hence

$$K_{\text{like}} \neq K_{\text{validated}}. \tag{4}$$

The model never infers knowledge growth directly from compute or benchmark growth.

Define operational AI capability separately as A_t^{AI}. For embodied automation, let

$$B_t^{RB} = N_t^{RB} q_t^{RB}, \tag{5}$$

where N_t^{RB} is deployed robot stock and q_t^{RB} effective per-unit capability. Then

$$\frac{\dot{B}_t^{RB}}{B_t^{RB}} = \frac{\dot{N}_t^{RB}}{N_t^{RB}} + \frac{\dot{q}_t^{RB}}{q_t^{RB}}. \tag{6}$$

IFR anchors the deployment term much better than the capability term.

A machine productive-capacity index is

$$M_t = (K_t^M)^\kappa (A_t^{AI})^\alpha (B_t^{RB})^\beta, \tag{7}$$

where K_t^M includes compute and machine-specific capital. Energy, materials, ecological constraints, trade restrictions, and geopolitical shocks enter a feasibility multiplier $\Xi_t \in (0,1]$.

6 Labour Centrality, Not Labour Disappearance

Employment is an incomplete indicator of labour's systemic role. Following the prior author-source construction but restating it here, define

$$L_t = \langle L_t^{\text{task}}, L_t^{\text{income}}, L_t^{\text{bottleneck}}, L_t^{\text{bargain}} \rangle. \tag{8}$$

The four coordinates ask, respectively: how much economically relevant task execution still requires humans; how important labour income is to household claims; how strongly production remains bottlenecked by non-substitutable human input; and how much leverage workers retain over terms, sequencing, adoption, and distribution [36].

> **Labour-Decentering Proposition [Open].** AI and robotics can reduce labour centrality without eliminating labour. Stable employment can coexist with falling bottleneck or bargaining centrality; high wages for some workers can coexist with a declining aggregate labour claim; and human-facing residual work can expand while scalable productive tasks migrate to machine systems.

Let $H_t = N_t^H h_t$ denote effective human productive capacity. A CES production block is

$$Y_t = \Xi_t \left[\omega_H H_t^\rho + \omega_M M_t^\rho\right]^{1/\rho}, \qquad \rho = \frac{\sigma - 1}{\sigma}. \tag{9}$$

Under the restrictive benchmark of competitive factor pricing, the human labour-income share is

$$s_t^L = \frac{\omega_H H_t^\rho}{\omega_H H_t^\rho + \omega_M M_t^\rho}. \tag{10}$$

If $\sigma > 1$, sustained growth of M_t/H_t places downward pressure on s_t^L unless new human-intensive tasks, bargaining institutions, or complementary scarcity offset it. The paper does not assume that this holds economy-wide.

7 The Claim Constitution: From Wages to Citizen Claims

Suppose the broad population receives s_t^L of domestic output through labour income. Let $o_t \in [0,1]$ be its beneficial ownership share of machine-capital income, including direct, pension, sovereign, cooperative, or social-fund ownership. Let $\tau_t \in [0,1]$ be the effective redistribution rate applied to machine-capital income not already broadly owned. Define

$$q_t = o_t + \tau_t(1 - o_t) = 1 - (1 - o_t)(1 - \tau_t). \tag{11}$$

The broad material claim on domestic output is

$$\Gamma_t = s_t^L + q_t(1 - s_t^L). \tag{12}$$

This is a claim-accounting identity for the stylized model, not a welfare function. Public services can be added as a separate in-kind claim readout when defensibly valued; they are not silently equated with ownership.

For target claim $\bar{\Gamma} > s_t^L$, the minimum non-labour conversion required is

$$q_t^{\min} = \frac{\bar{\Gamma} - s_t^L}{1 - s_t^L}. \tag{13}$$

This is the Citizen Claim Threshold. As $s_t^L \to 0$, maintaining $\Gamma_t \ge \bar{\Gamma}$ requires $q_t \ge \bar{\Gamma}$. A society cannot preserve a labour-based distributive constitution merely by insisting that people keep working when labour ceases to be the main source of productive necessity.

For country i, define net cross-border machine-rent drain

$$D_{i,t}^{\text{rent}} = \frac{Rent_{i,t}^{AI,\text{out}} - Rent_{i,t}^{AI,\text{in}}}{Y_{i,t}}, \tag{14}$$

so that a simple domestic effective claim is

$$\Gamma_{i,t}^{\text{net}} = \Gamma_{i,t} - D_{i,t}^{\text{rent}}. \tag{15}$$

A country can automate locally while machine rents accrue abroad.

8 Ownership Accumulation: The Stock That Reproduces the Next Claim

The Citizen Claim Threshold in (13) is static unless the ownership share o_t is allowed to evolve. Yet machine-productive

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Table 1: Current empirical anchors and interpretive constraints.

Anchor	Observed evidence	Constraint
AI scaling	Stanford AI Index 2026 reports global AI compute capacity growing about 3.3× per year since 2022 and large one-year gains on some frontier benchmarks [23].	Compute growth is not knowledge growth; benchmark gain is not a universal capability growth rate.
AI cost	The AI Index reports dramatic task-dependent declines in inference cost over 2022–2025 [22].	Cheap access can accelerate deployment even without economy-wide AGI.
China robotics	IFR reports 295,000 industrial robots installed in China in 2024, 54% of global installations, and more than two million in operational stock; its 2026 release places AI-powered robotics at the center of national strategy [24, 25].	Installation and demand growth do not measure per-robot capability growth.
AI infrastructure	OECD identifies strong concentration in several AI-infrastructure layers, including hardware, fabrication, and cloud bottlenecks [26].	Concentration is layer-specific and is not identical to effective transition power.
Initial distribution	ILO reports a modest long-run decline in global labour income share, while World Inequality Report 2026 describes extreme wealth concentration [27, 29].	These conditions pre-date full AI-robotic substitution and cannot be attributed to AI alone.
Global asymmetry	ILO–World Bank analysis warns that some countries may experience disruption before capturing AI productivity dividends [28].	Adoption, local ownership, infrastructure, and cross-border rent capture must be separated.

assets generate returns that can be reinvested, inherited, pooled through pension or sovereign funds, diluted, taxed, or transferred. Long-run wealth-income evidence shows why the ownership stock cannot be treated as a passive background variable [30], while automation can amplify wealth inequality through differential exposure to capital returns [5].

Let $W^M_{i,t}$ denote actor i's beneficial claim on machine-productive wealth. A minimal stock-flow equation is

$$W^M_{i,t+1} = (1-\delta_W)W^M_{i,t} + r^M_t W^M_{i,t} + s^M_{i,t} + T^{\text{cap}}_{i,t} - Tax^{\text{cap}}_{i,t}, \quad (16)$$

where r^M_t is the realized return on machine-productive wealth, $s^M_{i,t}$ new saving or acquisition, $T^{\text{cap}}_{i,t}$ capital transfers or inheritance, and $Tax^{\text{cap}}_{i,t}$ taxes or dilution of the beneficial claim. The broad-population ownership share used in (11) is then

$$o_t = \frac{\sum_{i\in B} W^M_{i,t}}{\sum_i W^M_{i,t}}, \quad (17)$$

for a declared broad-population set B.

The point is not that $r^M > g$ or any other accumulation rule must hold universally. It is that ownership today is an input into ownership tomorrow. If machine rents are high and broad acquisition is weak, o_t can fall even while transfers keep current consumption stable. Conversely, social wealth funds, pensions, cooperatives, or public ownership can make o_t rise without dispersing operational control. Thus

$$\text{current redistribution} \neq \text{future ownership reproduction}. \quad (18)$$

9 Scarce Assets, Rent Burden, and Effective Material Claim

Machine abundance is not abundance in every good. Housing, land, energy networks, water, urban location, spectrum, and some natural resources remain scarce or institutionally rationed. OECD evidence shows that housing is both a major household asset and a major affordability burden, with prices and rents often rising faster than general inflation [31]. A post-labour society can therefore have very cheap reproducible digital and manufactured goods while households remain constrained by rents on scarce physical assets.

Define the household or population rent-burden share

$$B^{\text{scarce}}_{i,t} = \frac{R^{\text{house}}_{i,t} + R^{\text{land}}_{i,t} + R^{\text{energy}}_{i,t} + DS_{i,t}}{Y_{i,t}}, \quad (19)$$

where DS is required debt service. The effective material claim is

$$\Gamma^{\text{eff}}_{i,t} = \max\{0,\ \Gamma^{\text{net}}_{i,t} - B^{\text{scarce}}_{i,t}\}. \quad (20)$$

Gross transfers or dividends can therefore rise while effective command over discretionary life remains stagnant if housing, land, energy, or debt rents rise faster. This creates a second non-collapse:

$$\begin{aligned} &\text{machine abundance} \neq \text{low rent burden} \\ &\neq \text{effective material freedom}. \end{aligned} \quad (21)$$

10 Aggregate Demand and Realization: Who Can Buy the Automated Output?

The production block describes technical output capacity. A market economy additionally requires realized demand. Let Γ^{eff}_t denote the distribution of effective household claims and m_t their marginal propensities to consume. Define aggregate demand

$$AD_t = C(\Gamma^{\text{eff}}_t Y_t,\ m_t) + I_t + G_t + NX_t, \quad (22)$$

and a simple realization ratio

$$\chi^{\text{dem}}_t = \min\left(1,\ \frac{AD_t}{Y_t}\right). \quad (23)$$

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Realized machine-capital profit then depends on both productive capacity and demand realization,

$$\Pi_t^M = \chi_t^{\text{dem}} Y_t - Cost_t^M. \tag{24}$$

Recent work explicitly develops a demand externality of automation when income shifts toward owners with different consumption propensities [35]; the present paper uses only the weaker accounting implication that distribution can feed back into realization. Redistribution after labour is therefore not merely a welfare question. Under some parameterizations it can also become a condition for reproducing demand, revenue, and the accumulation process itself.

$$\Gamma^{\text{eff}} Y \to AD \to \Pi^M \to W_{t+1}^M \to M_{t+1}. \tag{25}$$

11 Ownership Is Not Transition Power: Conversion Gates

Market concentration matters, but it is not yet a measure of what an actor can cause or block. Let e be a consequential transition – for example obtaining compute, changing provider, receiving payment, accessing a model, entering a market, or completing logistics. Let $V_t(e)$ be the best normalized viability of e under declared cost, time, quality, risk, and legal constraints, and $V_t(e \mid -j)$ the best viability after actor or gatekeeper j is removed. Define counterfactual conversion-gate control [36]:

$$G_{j,t}^{\text{conv}}(e) = \left[1 - \frac{V_t(e \mid -j)}{V_t(e)}\right]_+. \tag{26}$$

G^{conv} approaches zero when removing j leaves the transition essentially intact and approaches one when removal destroys viable routes.

For actor i, credible exit from j is

$$X_{i,t}(e; j) = \text{clip}\left(\frac{V_{i,t}(e \mid -j)}{V_{i,t}(e)}, 0, 1\right). \tag{27}$$

Access and exit are therefore not identical. A person can have nominal access to a platform while having no credible substitute. An actor-specific dependency exposure for goal g is

$$D_{i \to j,t}(g) = \sum_{e \in E(g)} w_e(g)\, G_{j,t}^{\text{conv}}(e)\, [1 - X_{i,t}(e; j)]. \tag{28}$$

This differs from the concentration-only formulation in the undeposited draft *Capitalism After Labor Centrality* [36]: concentration is now treated as a possible upstream cause or proxy; the load-bearing human variable is dependency on consequential transitions with weak exit.

$$\text{Concentration} \neq G^{\text{conv}} \neq D. \tag{29}$$

If a concentrated provider is easily replaceable, transition power may be low. If many nominal competitors rely on one upstream bottleneck, firm count may understate transition power. Platform-capitalism and intellectual-monopoly theories strongly converge with the need to track intermediation and exclusion, but the present test is counterfactual rather than label-based [11–14].

12 Relational Class Position in an AI-Robotic Economy

The future social structure should not be read from income alone. For actor or household i, define a relational position

$$C_{i,t} = \langle O_{i,t}, G_{i,t}, \Gamma_{i,t}, A_{i,t}^{\text{access}}, X_{i,t}, D_{i,t}, R_{i,t}^{\text{rent}} \rangle, \tag{30}$$

where O is beneficial ownership, G gate control, Γ claim, A^{access} productive access, X credible exit, D dependency, and R^{rent} net rent position [36]. A stable class is not declared from one observation. It requires persistent clustering of these relations over time and reproduction through institutional rules.

The vector can represent familiar owner-worker relations when coordinates align, but also combinations characteristic of AI-era production: high-income professionals with little ownership and high infrastructural dependency; citizens with strong social dividends but little operational control; small firms with low ownership but high trusted routing rights; states with formal authority but dependence on private frontier expertise; or broad cooperative/public ownership with credible exit. These are candidate positions, not universal new classes.

13 The Political-Economy-to-Human Bridge

A political-economic relation cannot be upgraded directly into a claim about human agency. It must cross a named mechanism. Let

$$c_{i,t}^{PE \to H} = B_{PE \to H}(Z_t; i, g), \tag{31}$$

where the bridge can contain price, time, language, information, credit, permission, legal standing, identity, sanction, switching cost, network access, or institutional recognition. This is the point at which world-system structure changes the context in which options are experienced and acted upon.

13.1 Live possibility, defined here

For a person i and goal g, let

$$\Pi_{i,t}^{\text{live}}(g) \subseteq \Pi_{i,t}^{\text{feas}}(g) \subseteq \Pi_{i,t}^{\text{phys}}(g). \tag{32}$$

A live possibility is a feasible option that is sufficiently intelligible, discoverable, credible, normatively or socially usable, and actionable to enter practical consideration. It is not merely legally present. This definition is self-contained; the earlier author paper develops its history- and meaning-shaped microstructure in greater depth [37].

Let $\Lambda_{H,t}^{\text{live}} \in [0,1]$ be a declared macro-readout of breadth and corrigibility of live alternatives. A candidate dynamic is

$$\begin{aligned} \dot{\Lambda}_{H,t}^{\text{live}} = {} & \lambda_1 B_t + \lambda_2 X_t + \lambda_3 P_t^{\text{plural}} \\ & + \lambda_4 r_t^H - \lambda_5 D_t - \lambda_6 C_t^I - \delta_\Lambda \Lambda_{H,t}^{\text{live}}, \end{aligned} \tag{33}$$

where B_t is material security and P_t^{plural} access to plural informational or interpretive routes. The equation is [Open]; it should be removed if ordinary affordability, salience, preference, or capability measures explain the same outcomes.

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13.2 Corrigible human agency, defined here

For task g, define four conditions: R_g = can read relevant differences; D_g = acceptance or refusal remains attributable to the person; X_g = the decision can causally affect action; F_g = feedback, objection, correction, or revision remains possible. Current effective agency is

$$A^{\mathrm{corr}}_{H,i,t}(g) = \max_{\pi \in \Pi^{\mathrm{live}}_{i,t}(g)} \Pr^{\pi}(R_g \cap D_g \cap X_g \cap F_g). \tag{34}$$

This is a task-relative envelope, not a metaphysical free-will score. A larger formal option set does not guarantee larger agency; neither does mere action. The prior author work motivates this separation, but the present equation supplies the definition needed here [36, 37].

13.3 Human Return, defined here

AI-assisted system performance is not identical to human capability. Define a delayed unaided return profile

$$R^{\mathrm{return}}_{H,t} = \langle C_t, T_t, S^{\mathrm{skill}}_t, A^{\mathrm{alt}}_t \rangle, \tag{35}$$

where the coordinates are Conceptual reconstruction, Tool selection, Skill execution, and Alternative generation after decisive AI assistance is removed. A normalized reporting readout $r^H_t \in [0,1]$ may be constructed under frozen criteria. CTSA is one prior measurement architecture for these classes; the definition here does not require accepting CTSA as a validated scale [38]. External evidence already shows why assisted performance and later unaided capability must be separated [18–21].

14 Social Role After Labour

Employment provides income but can also organize time, contact, status, activity, and collective purpose. Meta-analytic work on Jahoda's latent-deprivation tradition supports the relevance of these non-income functions [17]. Let $S^H_t \in [0,1]$ denote recognized social role and standing. Let W_t be employment-based role provision and N_t non-work role infrastructure: care, family, religion, voluntary association, civic projects, learning, arts, sport, local production, or chosen work. A minimal dynamic is

$$\dot{S}^H_t = s_1 W_t + s_2 N_t - \delta_S S^H_t. \tag{36}$$

This does not claim that wage labour is psychologically necessary. It says that a society removing wage labour as a central institution must deliberately reproduce or replace functions that work previously bundled together. The time budget can shift as

$$1 = \ell^{\mathrm{wage}} + \ell^{\mathrm{care}} + \ell^{\mathrm{learn}} + \ell^{\mathrm{civic}} + \ell^{\mathrm{leisure}}, \tag{37}$$

without a fall in ℓ^{wage} being automatically liberation or deprivation. Direction depends on voluntariness, claim, recognition, capability, and live alternatives.

15 Social Reproduction: Productive Necessity Is Not Social Necessity

A world-system theory is incomplete if it models humans only as labour inputs, consumers, or political actors. Human beings are born, raised, educated, healed, cared for, socialized, and supported through systems of social reproduction. Care-economy research documents the scale and gendered organization of paid and unpaid care across more than 90 countries [33]; Fraser's social-reproduction argument emphasizes that economic production depends on background capacities that capital accumulation can simultaneously strain [32].

Let H^{cap}_t denote a broad human-capability stock distinct from the productive input H_t in the CES block. A candidate reproduction dynamic is

$$\begin{aligned}\dot{H}^{\mathrm{cap}}_t = f(&Care_t, Health_t, Education_t,\\ &Nutrition_t, Community_t) - \delta_H H^{\mathrm{cap}}_t.\end{aligned} \tag{38}$$

This stock can affect future productive skill, agency, Human Return, social role, and the capacity to govern institutions, but it is not reducible to any one of them. Machine systems may assist care, diagnosis, education, and household work; they may also externalize or intensify unpaid coordination burdens. The sign is therefore open.

The central distinction is

$$\begin{aligned}&\text{productive necessity of humans}\\ &\neq \text{social necessity of human reproduction.}\end{aligned} \tag{39}$$

Humans can become less necessary to scalable production while remaining indispensable to the reproduction of human life, relationships, legitimacy, culture, and future generations. A post-labour settlement that preserves income but depletes care and social reproduction is not stable abundance; it is a transfer regime consuming its own human substrate.

16 The Human Systemic Position Equation

The central reporting object is not GDP. Let Γ^{eff}_t be effective material claim after cross-border and scarce-asset burdens, $A^{\mathrm{corr}}_{H,t}$ corrigible agency, $\Lambda^{\mathrm{live}}_{H,t}$ live possibility, r^H_t Human Return, S^H_t social standing, X^H_t credible exit, and D^H_t dependency. Define

$$P^H_t = \frac{(\Gamma^{\mathrm{eff}}_t)^{\theta_\Gamma} (A^{\mathrm{corr}}_{H,t})^{\theta_A} (\Lambda^{\mathrm{live}}_{H,t})^{\theta_\Lambda} (r^H_t)^{\theta_R} (S^H_t)^{\theta_S} (X^H_t)^{\theta_X}}{(1+D^H_t)^{\theta_D}} \tag{40}$$

for declared non-negative reporting weights. This is not a welfare utility function and is not a validated cardinal scale. It is a typed audit index intended to prevent output, consumption, or transfer income from silently standing in for human position.

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Taking log derivatives gives

$$\frac{\dot{P}_t^H}{P_t^H} = \theta_\Gamma \frac{\dot{\Gamma}_t^{\text{eff}}}{\Gamma_t^{\text{eff}}} + \theta_A \frac{\dot{A}_{H,t}^{\text{corr}}}{A_{H,t}^{\text{corr}}} + \theta_\Lambda \frac{\dot{\Lambda}_{H,t}^{\text{live}}}{\Lambda_{H,t}^{\text{live}}} + \theta_R \frac{\dot{r}_t^H}{r_t^H} \tag{41}$$

$$+ \theta_S \frac{\dot{S}_t^H}{S_t^H} + \theta_X \frac{\dot{X}_t^H}{X_t^H} - \theta_D \frac{\dot{D}_t^H}{1 + D_t^H}. \tag{42}$$

Hence the primary non-collapse result is

$$\dot{Y}_t > 0 \not\Rightarrow \dot{P}_t^H > 0. \tag{43}$$

A richer world can contain weaker humans.

17 Abundance Without Agency

The prior author paper introduced an Abundance Without Agency diagnostic; here it becomes a cross-check on (40). A candidate AWA state has

$$AWA_t = \langle Y\uparrow,\ \Gamma^{\text{eff}} \text{ adequate},\ O_H\downarrow,\ X_H\downarrow,\ D_H\uparrow, \\ \Lambda_H^{\text{live}}\downarrow,\ A_H^{\text{corr}}\downarrow,\ r_H\downarrow\rangle. \tag{44}$$

Material abundance can therefore coexist with structural dependence. This is not identical to poverty, unemployment, or authoritarianism. A population can receive cheap AI, public services, entertainment, health care, or transfers while owning little productive infrastructure, facing weak exit, or losing unaided capability in domains where AI becomes a permanent cognitive prosthesis [36].

18 Recursive Political Economy: Three Power Channels

Technology is not an external shock to a passive society. Ownership, gate power, claims, dependency, and agency alter rule-setting; rules then alter the next ownership and gate structure. But economic concentration should not be treated as a complete theory of power. Empirical work finds that mergers and industrial concentration can be followed by increased lobbying activity, supporting a link between market structure and political influence without making the two identical [34].

Define a typed power vector

$$P_t = \langle P_t^{\text{econ}}, P_t^{\text{info}}, P_t^{\text{coerc}}\rangle. \tag{45}$$

P^{econ} concerns ownership, rents, credit, procurement, and conversion-gate control. P^{info} concerns control over search, ranking, language mediation, advertising, model access, data, and public attention. P^{coerc} concerns legally backed rule enforcement, taxation, policing, licensing, sanctions, and other state coercive capacities. These dimensions can align or diverge: a private platform may have high informational and economic power but little direct coercive authority; a state may hold coercive authority while depending on private compute or frontier expertise; public ownership can coexist with concentrated informational mediation.

Define broad political capacity

$$P_t^B \propto \Gamma_t^{\text{eff}} A_{H,t}^{\text{corr}} \Lambda_{H,t}^{\text{live}} X_t^H r_t^H, \tag{46}$$

and concentrated-system leverage as a function rather than a scalar identity,

$$P_t^E = P_E\left(P_t,\ \dot{G}^{\text{conv}},\ D_t^H,\ 1 - \Gamma_t^{\text{eff}}\right). \tag{47}$$

Let I_t collect ownership, redistribution, exit, interoperability, competition, care/social-reproduction provision, and non-work institutional variables. It evolves as

$$\dot{I}_t = F_I\left(P_t^B,\ P_t^E,\ \text{state capacity},\ \text{rules},\ \text{shocks}\right). \tag{48}$$

No universal sign is imposed. Three recursive loops can coexist:

$$\begin{aligned} &\text{economic gate control} \to \text{rents} \to \text{political influence} \\ &\qquad \to \text{future gate control}, \\ &\text{information control} \to \text{attention/interpretation} \\ &\qquad \to \text{rule legitimacy} \to \text{future information control}, \\ &\text{productivity} \to \text{broad claim/exit} \to \text{agency} \\ &\qquad \to \text{collective rules} \to \text{broader ownership/exit}. \end{aligned} \tag{49}$$

The future is institutionally path-dependent, not technologically predetermined. Crucially,

$$\text{ownership} \neq \text{informational power} \neq \text{coercive power}. \tag{50}$$

19 World-System Regimes

The model yields diagnostics rather than deterministic forecasts. Table 2 distinguishes five analytically different orders.

A neo-feudal-like label should remain terminal and conditional, not a master concept. Concentration and rent are insufficient. A stronger diagnosis requires persistent hierarchical dependency, weak exit, rule asymmetry, and extra-economic or status-enforcing capacity. If those extra conditions are absent, the model should report rentier or concentrated capitalism rather than inflate the mode-of-production claim [14–16, 36].

20 Why China Changes Timing but Not Logic

China matters because it combines frontier AI research, the world's largest industrial robot deployment, large-scale manufacturing capacity, and a national strategy explicitly linking AI to physical robotics [23–25]. This can shorten the distance between cognitive automation and embodied production: candidate generation, design, planning, manufacturing, logistics, and machine iteration can increasingly occupy one industrial loop.

But the Chinese trajectory does not determine the world-system regime. State direction, public ownership, industrial policy, domestic manufacturer share, redistribution, household claims, and political rule-setting enter different coordinates than U.S. hyperscaler capitalism. The model therefore predicts neither one homogeneous AI capitalism nor one inevitable Chinese endpoint. It predicts competing automation blocs with different combinations of private ownership, state control, public claim, domestic value capture, external rent extraction, and exit.

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Table 2: Five AI-robotic world-system regimes.

Regime	Production / claim	Control / agency	Likely human life pattern if stable
Augmented-labour capitalism	Human-machine complementarity and new tasks keep labour centrality substantial.	Moderate dependency; Human Return can rise if AI transfers skill.	Employment remains central but task composition changes; shorter hours become possible if productivity is shared.
AI rentier automation	Labour income centrality falls while $q < q^{\min}$; machine rents concentrate.	High gate power and dependency; weak exit and bargaining.	People remain consumers and service providers but increasingly rent access to productive intelligence and infrastructure.
Managed abundance	Labour centrality falls while transfers/public provision keep material claim high.	Control and dependency remain high; exit, pluralism, or Human Return may remain low.	Materially secure life with fewer work hours can coexist with managed dependence.
Shared automated abundance	Labour centrality falls while ownership/claims broaden and credible exit remains high.	Contestable gates, interoperability, plural routes, high Human Return.	Abundance widens effective options; work becomes more elective without reducing political standing.
Post-labour citizen economy	Wage labour is no longer the default basis of entitlement; citizen/social claims satisfy the threshold.	Agency, exit, retained capability, and rule contestability are institutionally protected.	Care, learning, religion, arts, civic activity, family, local production, and chosen work become primary recognized roles.

For countries outside the leading blocs, adoption alone is not success. The key questions are whether they own productive AI/robotic assets, capture domestic value, possess credible public or competitive alternatives, and convert machine capability into human capability rather than merely importing rented intelligence. This is the world-system form of the ILO–World Bank warning that disruption can precede dividend [28].

21 What Humans May Actually Do

The model does not predict mass idleness. It predicts a possible separation between economic necessity and human activity. Even in highly automated systems, societies may continue to demand human responsibility, embodiment, trust, legitimacy, or presence in care, relationships, law, politics, religion, culture, science direction, art, status competition, local community, luxury human service, sport, and institutional accountability.

The decisive distinction is whether these become chosen and socially recognized roles or low-bargaining residual tasks left after machine substitution. A humane post-labour order therefore requires more than income replacement. It requires a new social constitution of contribution in which people can matter without proving their right to subsistence through scarce wage labour.

The institutional transition is

$$\begin{aligned}\text{labour-based claim} \longrightarrow{} & \text{human/citizen claim} \\ & \text{on social production.}\end{aligned} \tag{51}$$

Candidate mechanisms include broad pension and sovereign ownership, social wealth funds, cooperative AI, public compute, universal basic services, social dividends, taxation of machine rents, and competition/interoperability rules. They are not interchangeable: redistribution can raise Γ while leaving ownership, gate power, or exit unchanged.

22 Measurement and Falsification

The architecture earns scientific value only if its additional objects discriminate outcomes beyond standard productivity, wage, task-exposure, and market-structure measures.

No single dataset can identify the full river. Ownership and claims require balance sheets, pension/fund exposure, corporate and public ownership records, inheritance and capital-flow data. Scarce-asset burdens require housing, land, energy, and debt-service accounts. Demand realization requires distributional consumption and propensity-to-consume data. Social reproduction requires care-time, health, education, household, and community indicators. Gate control requires contracts, technical dependency graphs, outages, API/cloud pricing, procurement, standards dependencies, and substitution tests. Exit requires switching time, portability, retraining cost, legal interoperability, and post-switch quality. Human Return requires delayed unaided tasks rather than only contemporaneous productivity. Where causal identification is unavailable, the framework should report a descriptive transition map rather than upgrade correlation into domination. Table 3 lists the falsification-oriented programme.

23 What Can and Cannot Be Forecast Now

Three quantities remain too weakly identified for a defensible year forecast: economy-wide substitution elasticity between humans and AI-robot capacity; annual growth of effective per-robot capability q_t^{RB}; and the institutional response path of ownership, redistribution, exit, and Human Return. Therefore the paper does not calculate a date at which humans "leave the economy."

A normalized early-warning diagnostic can instead com-

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Table 3: Falsification-oriented programme.

Test	Prediction if the proposed layer matters	Removal / compression condition
Gate vs concentration	Counterfactual removal/substitution predicts transition failure beyond HHI or top-share measures.	Remove G^{conv} if concentration/interoperability fully explain outcomes.
Exit vs access	Equal nominal access but different credible exit predicts price exposure, bargaining, refusal, or switching.	Collapse X into access if it adds no held-out value.
Labour profile	Task, income, bottleneck, and bargaining centrality dissociate across sectors or time.	Replace L with simpler employment/labour-share measures if no stable dissociation appears.
Citizen Claim Threshold	Falling s^L predicts weaker broad command over output unless q rises at least as fast as (13).	Revise if observed claims are better captured by another accounting decomposition.
Live possibility	Bridge mechanisms alter considered alternatives or switching after formal rights/resources are controlled.	Compress into affordability, capability, salience, or preference if no incremental value remains.
Human Return	Similar AI-assisted productivity can coexist with different delayed unaided concept/tool/skill/alternative performance.	Remove r^H if delayed independent measures add no information beyond ordinary skill measures.
AWA	High material provision can coexist with low ownership, exit, agency, or Human Return.	Reject AWA as distinct if material welfare tightly tracks those dimensions.
Social-role substitution	Reduced employment is less damaging where non-work institutions replace time structure, status, social contact, purpose, and activity.	Redesign S^H if these mediators do not explain outcomes.
Ownership accumulation	Persistent differences in machine-wealth returns, acquisition, inheritance, and taxation predict the evolution of broad ownership o_t beyond current income flows.	Treat o_t as exogenous if stock-flow ownership dynamics add no predictive value.
Scarce-asset burden	Similar gross citizen claims can yield different discretionary command where housing, land, energy, or debt burdens differ.	Remove B^{scarce} if gross and effective claims are empirically indistinguishable.
Demand realization	Distribution of effective claims and consumption propensities predicts revenue/realization after controlling productive capacity.	Drop χ^{dem} if demand never constrains realized output/profits in the studied regime.
Social reproduction	Care, health, education, nutrition, and community capacity predict future human capability and institutional resilience beyond wage employment alone.	Compress H^{cap} if these variables add no stable explanatory value.
Power channels	Economic, informational, and coercive power dissociate across actors and predict different transition outcomes.	Collapse P if one scalar gate/dependency measure dominates across the declared domain.

pare decoupling pressures:

$$\begin{aligned}\Omega_t &= w_M(g_M - g_H) + w_D g_D + w_G g_G \\ &\quad - w_q g_q - w_\Gamma g_{\Gamma^{\text{eff}}} - w_X g_X \\ &\quad - w_R g_{r^H} - w_\Lambda g_\Lambda - w_H g_{H^{\text{cap}}}.\end{aligned} \tag{52}$$

The first line groups productive and dependency pressure; the remaining terms are institutional and human counter-response. All terms require normalization before comparison. Persistent $\Omega_t > 0$ is an early-warning condition, not a deterministic forecast.

24 Claim Boundaries

The theory should be narrowed rather than protected when its extra variables fail. Six boundaries are especially important.

- Machine candidate generation is not validated knowledge; (3) is a bookkeeping firewall, not a law of science.
- Robot installations are not robot capability; the second term in (6) remains open.
- The CES block is a scenario architecture, not a claim that one substitution elasticity governs the world economy.
- The Citizen Claim Threshold is an accounting result inside the stylized claim block; it does not choose between ownership, taxation, services, dividends, or other institutions.
- The wealth-accumulation and aggregate-demand blocks are stock-flow and realization architectures, not universal laws of returns, saving, or consumption. Their parameters must be estimated by jurisdiction and period.
- Effective material claim depends on declared scarce-asset burdens; it must not be used to imply that all housing, land, energy, or debt payments are exploitative rents.
- Human Systemic Position is an audit index, not a universal welfare score. Its weights must be declared and sensitivity-tested.
- Concentration, dependency, live-field narrowing, or managed abundance do not by themselves establish domination, structural violence, or neo-feudalism. Those stronger claims require mechanisms, feasible counterfactuals, restricted objection/exit, persistence, and independent normative justification.

A strong story of human marginalization would be substantially weakened if future evidence shows concentrated AI ownership together with low switching costs, robust interop-

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erability, broad citizen claims, strong bargaining, credible public alternatives, contestable rule-setting, and stable or rising Human Return. Under that pattern,

$$\text{concentration} \not\Rightarrow \text{dependency} \not\Rightarrow \text{agency loss}. \tag{53}$$

The correct description might then be ordinary capitalism with unusually productive technology, or shared abundance with concentrated production but strong countervailing institutions.

25 Conclusion: From Productive Necessity to Human Membership

The central economic question of advanced AI is often posed as "what happens to jobs?" The question remains vital, but it begins too late. If productive cognitive and embodied capability becomes reproducible through capital, labour can lose task, income, bottleneck, or bargaining centrality while employment remains positive. The social contract then faces a constitutional problem: the old bridge from labour to wages to membership no longer guarantees broad claims on the abundance the system can produce.

The standalone river developed here is

$$\begin{aligned}
&\text{candidate generation} \to \text{validated } K/A^{AI} \\
&\to \text{robotic embodiment} \to M_t \to Y_t \\
&\to L_t \to s_t^L \to \Gamma_t \\
&\to W_t^M \to o_{t+1} \to \Gamma_t^{\text{eff}} \\
&\to AD_t \to \Pi_t^M \to W_{t+1}^M \\
&\to \{G^{\text{conv}}, X, D\} \to \text{human bridge} \\
&\to \{\Pi^{\text{live}}, A_H^{\text{corr}}, r_H, S_H\} \\
&\to H_t^{\text{cap}} \to P_t^H \to P_t \\
&\to \text{collective rules} \to \text{next system state}.
\end{aligned} \tag{54}$$

The river is now closed by two feedback systems. The ownership-demand-production loop is

$$\Gamma^{\text{eff}} Y \to AD \to \Pi^M \to W_{t+1}^M \to \{o_{t+1}, M_{t+1}\}, \tag{55}$$

while the human-reproduction-rule loop is

$$\begin{aligned}
\{Care, Health, Education\} \to H^{\text{cap}} \to \{A_H^{\text{corr}}, r_H, S_H\} \\
\to P \to I_{t+1}.
\end{aligned} \tag{56}$$

A world-system regime is stable only if both the production-claim circuit and the human-reproduction-governance circuit reproduce themselves.

The critical inequality is therefore not AI versus humans. It is

$$\begin{aligned}
&\text{growth of machine productive power} \\
&\text{versus growth of broad human claim, exit,} \\
&\text{agency, and Human Return.}
\end{aligned} \tag{57}$$

If the first persistently outruns the second, a richer world can contain weaker humans. If the second keeps pace, the same technology supports the opposite historical possibility: humans cease having to sell most of their waking lives merely to justify a claim on what their civilization can produce.

Author Declarations

Research status. Conceptual and measurement-architecture paper; full world-system standalone revision anchored in v1.0; no original empirical dataset is analyzed.

Author-source use. Prior programme papers are cited as conceptual genealogy and prior formal development. This manuscript restates every load-bearing object required for standalone comprehension.

AI assistance disclosure. Generative AI assisted retrieval, adversarial review, formalization, drafting, and typesetting. The human author selected the research problem, programme lineage, substantive theoretical commitments, and retains responsibility for claims, citations, and revisions.

References

[1] David H. Autor. Why are there still so many jobs? the history and future of workplace automation. *Journal of Economic Perspectives*, 29(3):3–30, 2015. doi:10.1257/jep.29.3.3.

[2] Daron Acemoglu and Pascual Restrepo. Automation and new tasks: How technology displaces and reinstates labor. *Journal of Economic Perspectives*, 33(2): 3–30, 2019. doi:10.1257/jep.33.2.3.

[3] Daron Acemoglu and Pascual Restrepo. Robots and jobs: Evidence from us labor markets. *Journal of Political Economy*, 128(6):2188–2244, 2020. doi:10.1086/705716.

[4] Anton Korinek and Joseph E. Stiglitz. Artificial intelligence and its implications for income distribution and unemployment. In *The Economics of Artificial Intelligence: An Agenda*. University of Chicago Press, 2019.

[5] Benjamin Moll, Lukasz Rachel, and Pascual Restrepo. Uneven growth: Automation's impact on income and wealth inequality. *Econometrica*, 90(6):2645–2683, 2022. doi:10.3982/ECTA19417.

[6] Susan Athey and Fiona Scott Morton. Artificial intelligence, competition, and welfare. Technical Report 34444, NBER, 2025. doi:10.3386/w34444.

[7] Amartya Sen. *Development as Freedom*. Oxford University Press, 1999.

[8] Philip Pettit. *Republicanism: A Theory of Freedom and Government*. Oxford University Press, 1997.

[9] Michel Foucault. The subject and power. *Critical Inquiry*, 8(4):777–795, 1982. doi:10.1086/448181.

[10] Miranda Fricker. *Epistemic Injustice: Power and the Ethics of Knowing*. Oxford University Press, 2007.

[11] Paul Langley and Andrew Leyshon. Platform capitalism: The intermediation and capitalisation of digital economic circulation. *Finance and Society*, 3(1):11–31, 2017. doi:10.2218/finsoc.v3i1.1936.

10.5281/zenodo.22481924

[12] Nick Srnicek. *Platform Capitalism*. Polity, 2017.

[13] Cecilia Rikap. *Capitalism, Power and Innovation: Intellectual Monopoly Capitalism Uncovered*. Routledge, 2021. doi:10.4324/9780429341489.

[14] Cecilia Rikap. Capitalism as usual? *New Left Review*, 139:145–160, 2023. doi:10.64590/wfl.

[15] Cédric Durand. Scouting capital's frontiers. *New Left Review*, 136:29–39, 2022. doi:10.64590/6jk.

[16] Evgeny Morozov. Critique of techno-feudal reason. *New Left Review*, 133/134:89–126, 2022. doi:10.64590/13n.

[17] Karsten I. Paul, Hannah Scholl, Klaus Moser, Andreas Zechmann, and Bernad Batinic. Employment status, psychological needs, and mental health: Meta-analytic findings concerning the latent deprivation model. *Frontiers in Psychology*, 14:1017358, 2023. doi:10.3389/fpsyg.2023.1017358.

[18] Hamsa Bastani, Osbert Bastani, Alp Sungu, et al. Generative ai without guardrails can harm learning. *PNAS*, 122(26):e2422633122, 2025. doi:10.1073/pnas.2422633122.

[19] Zana Buçinca, Maja Barbara Malaya, and Krzysztof Z. Gajos. To trust or to think: Cognitive forcing functions can reduce overreliance on ai. *Proceedings of the ACM on Human-Computer Interaction*, 5(CSCW1), 2021. doi:10.1145/3449287.

[20] Helena Vasconcelos, Matthew Joerke, Madeleine Grunde-McLaughlin, Tobias Gerstenberg, Michael S. Bernstein, and Ranjay Krishna. Explanations can reduce overreliance on ai systems during decision-making. *Proceedings of the ACM on Human-Computer Interaction*, 7(CSCW1), 2023. doi:10.1145/3579605.

[21] Moshe Glickman and Tali Sharot. How human-ai feedback loops alter human perceptual, emotional and social judgements. *Nature Human Behaviour*, 9(2): 345–359, 2025. doi:10.1038/s41562-024-02077-2.

[22] Stanford Institute for Human-Centered AI. Ai index report 2025. Stanford University, 2025.

[23] Stanford Institute for Human-Centered AI. Artificial intelligence index report 2026. Stanford University, 2026.

[24] International Federation of Robotics. World robotics 2025: Industrial robots. Technical report, International Federation of Robotics, Frankfurt, 2025.

[25] International Federation of Robotics. China makes ai-powered robots core of national strategy. 5 May, 2026.

[26] OECD. Competition in artificial intelligence infrastructure. Technical Report 330, OECD Roundtables on Competition Policy Papers, 2025. doi:10.1787/623d1874-en.

[27] International Labour Organization. The world of work and the 2030 agenda: A ten-year review. ILOSTAT, 2026.

[28] International Labour Organization and World Bank. Disruption without dividend? how the digital divide and task differences split genai's global impact. Technical Report 166, ILO Working Paper, 2026.

[29] World Inequality Lab. World inequality report 2026, 2026.

[30] Thomas Piketty and Gabriel Zucman. Capital is back: Wealth-income ratios in rich countries 1700–2010. *Quarterly Journal of Economics*, 129(3):1255–1310, 2014. doi:10.1093/qje/qju018.

[31] OECD. *Brick by Brick: Building Better Housing Policies*. OECD Publishing, 2021. Housing affordability and wealth-distribution chapters.

[32] Nancy Fraser. Contradictions of capital and care. *New Left Review*, 100:99–117, 2016. doi:10.64590/nt2.

[33] Laura Addati, Umberto Cattaneo, Valeria Esquivel, and Isabel Valarino. Care work and care jobs for the future of decent work. Technical report, International Labour Office, 2018.

[34] Bo Cowgill, Andrea Prat, and Tommaso M. Valletti. Political power and market power. Technical Report 33255, NBER, 2024. doi:10.3386/w33255.

[35] Erhan Bayraktar. Automation, income incidence, and capital accumulation in incomplete markets. Working paper, arXiv:2605.05127, 2026. Earlier circulated under the title "The Demand Externality of Automation"; canonical title per the arXiv abstract page as of 6 September 2026.

[36] Yaoharee Lahtee. Capitalism after labor centrality: A readout-genesis derivation of productive intelligence, conversion gates, class position, dependency, and human agency. Undeposited internal working draft, Human-AI Readout Programme, 6 September, 2026. Not deposited on Zenodo as of this edition; cited for author-source genealogy only, not as evidential authority.

[37] Yaoharee Lahtee. Choice begins before choice: Meaning-shaped accessibility, live possibility, and effective agency in human-ai systems. Full Paper v1.0, Human-AI Readout Programme, 5 September, 2026. doi:10.5281/zenodo.22357788.

[38] Yaoharee Lahtee. Ctsa human-return readout: A session-boundary measurement architecture for retained human capability after ai. Final author draft v9.1, Human-AI Readout Programme, 5 September, 2026. doi:10.5281/zenodo.22339909.

[39] Yaoharee Lahtee. Before meaning, before choice: A readout-native derivation of experience, live possibility, agency, and human-ai return. Full Paper v1.1, glosa-checked, Human-AI Readout Programme, 6 September, 2026. doi:10.5281/zenodo.22424434.

The Human Conversion Imperative: Machine Acceleration, Human Potential, and the Reversibility Window in an AI-Robotic World

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What the chapter states. "The central problem of an AI-rich society is not whether machines become capable. It is whether machine capability is converted into durable human capability and broadly distributed systemic standing at a rate sufficient to prevent dependency, deskilling, gate concentration, and institutional lock-in."

Status. "K0 theory proposal and empirical research agenda... The core proposal is conditional and falsifiable..."

Falsifier(s). "The Human Conversion architecture should be narrowed or removed under the following results: 1. delayed unaided performance is fully predicted by ordinary skill measures and never dissociates from assisted performance; [...] 6. the vector architecture does not improve prediction, diagnosis, or intervention compared with simpler rivals."

Read before. This book's Chapters 17, 27, 30, 31, 34 (mechanical citation list; Appendix A).

Feeds into. Ch. 32 — companion paper, same deposit batch (mechanical citation list; Appendix A).

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The Human Conversion Imperative

Machine Acceleration, Human Potential, and the Reversibility Window in an AI-Robotic World

A standalone theory proposal for converting machine capability into durable human agency before dependency locks in

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Status K0 theory proposal and empirical research agenda. This paper does not claim that a global "bad mode" has already occurred, that AI capability grows at one universal rate, that industrial robot installations equal robot capability, or that human potential can be represented by one validated scalar. Current 2024–2026 evidence is used only to establish acceleration, feedback, learning-risk, concentration, and embodied-automation conditions that make prospective measurement rational. The core proposal is conditional and falsifiable: when machine capability grows faster than institutions convert it into broadly retained human agency, ownership, exit, world-answerability, and social reproduction, technological progress can increase human dependency even while assisted performance and aggregate output rise.

Central thesis. The central problem of an AI-rich society is not whether machines become capable. It is whether machine capability is converted into durable human capability and broadly distributed systemic standing at a rate sufficient to prevent dependency, deskilling, gate concentration, and institutional lock-in. The paper therefore replaces a one-dimensional "AI helps or harms" question with a vector-valued Human Conversion architecture and adds a Reversibility Principle: the faster machine capability grows relative to human conversion, and the more costly lost skills, exit routes, ownership structures, and institutions become to restore, the stronger the case for early safeguards that preserve options before crisis is visible.

Edition note. This is a v1.1 glosa-checked edition of the founder's v1.0 (6 September 2026), which remains the anchor record of this work. Changes from v1.0: (i) citation lineage repair — self-citations [1], [3], [4] now carry their deposited Zenodo DOI and version string, [5] now cites the specific deposited record (Zenodo 22331922) instead of an ambiguous title-only entry, and [2] (*After Labour*) is cited as a companion paper deposited in the same batch, with its claim softened from a stronger verb to "proposes" pending independent verification of the derivation; (ii) one priority-language word removed from the Abstract; (iii) explicit [DEFINITION]/[OPEN] tier tags added next to the defining equations and Propositions 1–4; (iv) one AI-product-name mention in body prose generalised to a generic term (the cited work's own title keeps its original wording verbatim). No section, equation, table, or reference was added or removed, and no claim was strengthened.

ABSTRACT

Artificial intelligence can improve immediate human performance without necessarily increasing durable human capability. At the same time, AI and robotics can raise aggregate productive capacity while weakening labour-based claims, broad ownership, credible exit, and social standing. Existing literatures typically study these effects separately: human-AI augmentation and synergy, cognitive offloading and learning, automation and labour markets, market concentration, path dependence, or political lock-in. This paper proposes a single bridge concept: Human Conversion. Machine capability is socially beneficial in the strong human-development sense only to the extent that it is converted into retained human capacities and conditions of agency rather than merely substituted for them.

The paper develops four contributions. First, it defines a Human Conversion Vector spanning effective material claim, credible exit, live possibility, corrigible agency, independent epistemic routes, world-answerability, delayed unaided Human Return, social reproduction, and recognized standing. Second, it defines conversion elasticities that permit dimensions to diverge: AI may raise income while lowering skill retention or exit, so a scalar "augmentation" score can conceal managed dependence. Third, it replaces linear collapse narratives with a Bad-Mode State Space and a Reversibility Window grounded in increasing returns, path dependence, and irreversibility. Fourth, it derives an Urgency Vector in which safeguard intensity rises with machine-human conversion gaps, lock-in, severity, and recovery time. Current evidence motivates but does not validate the full model: global AI compute capacity has grown rapidly, industrial robot deployment is heavily concentrated in China (295,000 units installed in 2024, 54% of global installations), several AI-infrastructure layers are highly concentrated, repeated AI interaction can alter later human judgment, and field experiments show that unguarded AI assistance can raise assisted performance while reducing subsequent unaided learning. The paper ends with six minimum safeguard invariants and a falsification programme designed to distinguish productive augmentation from human systemic weakening.

Keywords: human-AI collaboration; human potential; cognitive offloading; agency; Human Return; automation; robotics; path dependence; lock-in; reversibility; political economy; AI

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governance.

1 The Missing Dependent Variable

The contemporary AI debate asks whether systems increase productivity, improve accuracy, displace jobs, create synergy, or concentrate economic power. These are important outcomes, but none answers the question that becomes decisive under rapid automation: what returns to humans?

The programme paper *When AI Expands Human Potential* defines human potential in an epistemic register: not throughput or convenience, but the capacity to become more corrigible, more world-responsive, less captive to self-fixation, and more capable of carrying truth-pressure into judgment and life [Lahtee, 2026a]. AI can widen alternatives and reflective friction, but candidate generation is not validation; disciplined revision still requires resistant conditions and partially independent routes.

The later *After Labour* paper adds a systemic constraint. Even if AI expands an individual's inquiry, that capacity is not securely realizable if the person lacks effective material claim, credible exit, live alternatives, retained unaided capability, social reproduction, or standing. It proposes that aggregate output can rise while Human Systemic Position falls [Lahtee, 2026b]. The two papers therefore generate a stronger joint problem:

$$\text{Machine expansion} \not\Rightarrow \text{Human expansion.} \qquad (1)$$

The missing dependent variable is the rate and direction by which machine capability is converted into durable human capability and systemic position.

2 Why This Is a Time-Sensitive Problem Rather Than a Distant AGI Question

The urgency argument does not depend on predicting AGI. It depends on observing three conditions simultaneously: fast machine-side change, plausible human/institutional substitution or dependence, and mechanisms of costly reversal.

Stanford's 2026 AI Index estimates that global AI compute capacity grew approximately 3.3-fold per year from 2022, while benchmark performance continued to advance quickly in several reasoning and software-engineering domains [Stanford Institute for Human-Centered AI, 2026]. This is infrastructure and measured capability growth, not a knowledge-growth rate. The distinction is load-bearing: $K_{\text{like}} \neq K_{\text{validated}}$.

Embodied automation is also scaling. The International Federation of Robotics reports 295,000 industrial robots installed in China in 2024, 54% of global installations, with operational stock exceeding two million units [International Federation of Robotics, 2025]. This does not mean robot capability grows at the installation rate, but it shows that cognitive automation is increasingly coupled to physical production at industrial scale.

The social system is not neutral to such acceleration. OECD documents high concentration in several AI-infrastructure layers: in three segments one firm is reported above 80% global share, and in three others the largest three exceed 60% [OECD, 2025]. Yet concentration is not domination by definition; it becomes structurally important when consequential transitions lack substitutes, interoperability, or credible exit.

Finally, human effects can accumulate before macroeconomic crisis is visible. Glickman and Sharot show in experiments with 1,401 participants that repeated human-AI interaction can alter later perceptual, emotional, and social judgments, amplifying bias [Glickman and Sharot, 2025]. Bastani et al. show that generative-AI access substantially improved practice performance in high-school mathematics, while unrestricted AI use led to worse subsequent unaided performance after access was removed; guardrailed tutoring largely mitigated the harm [Bastani et al., 2025]. A randomized trial in higher education likewise reports lower delayed knowledge retention for unrestricted generative-AI-assisted study [Barcaui, 2025]. Recent reviews conclude that cognitive offloading can support performance but may impede skill acquisition or produce task-specific deskilling depending on use conditions [Cash et al., 2026, Noorbehbahani and Oyibo, 2026].

These findings do not establish global decline. They establish that performance, learning, judgment, and dependence can move in different directions and therefore must be measured before irreversible institutional commitments are made.

3 Literature Boundary: What Is Already Known

This proposal does not claim priority for augmentation, cognitive offloading, path dependence, or automation inequality. Its contribution is the bridge architecture joining them.

Human-AI decision research has long shown that assistance quality depends on task, trust, coordination, explanation, and human uptake. Vaccaro, Almaatouq, and Malone's meta-analysis distinguishes augmentation (human+AI better than human alone) from synergy (human+AI better than both human and AI alone), finding that synergy is not automatic and can be negative in decision tasks [Vaccaro et al., 2024]. Buçinca et al. explicitly argue that AI-assisted systems should optimize human-centric outcomes beyond accuracy, including human learning [Buçinca et al., 2024].

The cognitive-offloading literature provides a complementary warning. External tools can free cognitive resources, but offloading higher-order reasoning can change acquisition, retention, and self-regulation. Recent reviews emphasize that the effects depend on the type and timing of assistance rather than on AI use per se [Cash et al., 2026, Seung and Basham, 2026, Noorbehbahani and Oyibo, 2026]. These literatures support the distinction between dependent offloading and assistance that preserves autonomous competence.

The political-economy literature supplies the macro-side. Automation can displace tasks, create new tasks, alter factor shares, and change the distribution of income and wealth [Autor, 2015, Acemoglu and Restrepo, 2018, Korinek and

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Stiglitz, 2019]. *After Labour* extends that line by separating output from broad claims, ownership from transition control, access from credible exit, and productive necessity from social necessity [Lahtee, 2026b].

Finally, lock-in is not a metaphor invented for AI. Arthur shows that increasing returns can lock technologies into historically contingent paths that become hard to reverse [Arthur, 1989]. Pierson develops the same logic for political institutions, emphasizing sequencing, increasing returns, and critical junctures [Pierson, 2000]. Arrow and Fisher's analysis of uncertainty and irreversibility establishes the general policy logic that preserving options can carry value when future consequences are uncertain and some choices are hard to undo [Arrow and Fisher, 1974].

The residual gap is therefore precise: existing work does not provide a single architecture for asking whether rapid machine capability is being converted into durable, broadly distributed human agency before skill, ownership, exit, and institutions become costly to restore.

4 Non-Collapse Discipline

The proposal begins with explicit separations:

$$\text{AI capability} \neq \text{validated knowledge}, \tag{2}$$
$$\text{assisted performance} \neq \text{human learning}, \tag{3}$$
$$\text{augmentation} \neq \text{synergy}, \tag{4}$$
$$\text{formal options} \neq \text{live possibilities}, \tag{5}$$
$$\text{access} \neq \text{credible exit}, \tag{6}$$
$$\text{income transfer} \neq \text{future ownership}, \tag{7}$$
$$\text{material security} \neq \text{agency}, \tag{8}$$
$$\text{market concentration} \neq \text{domination}, \tag{9}$$
$$\text{productive necessity} \neq \text{social necessity}. \tag{10}$$

Any theory that collapses these objects can report apparent progress while missing human contraction in another layer.

5 Human Conversion as a Vector, Not a Scalar

Let M_t denote a declared, normalized measure of machine-side capability relevant to a study. It may combine cost-adjusted AI capability, deployed compute, and embodied machine capacity, but it must never be interpreted as validated knowledge without an independent validation term.

Define the Human Conversion Vector [DEFINITION]

$$C_t^H = \langle \Gamma_t^{\text{eff}}, X_t^H, \Lambda_{H,t}^{\text{live}}, A_{H,t}^{\text{corr}}, R_{H,t}^{\text{route}}, W_{H,t}^{\text{world}}, r_{H,t}, H_t^{\text{cap}}, S_{H,t} \rangle. \tag{11}$$

The components are:

- Γ_t^{eff}: effective material claim after unavoidable rent/debt burdens;
- X_t^H: credible exit or usable substitution away from a consequential AI/robotic gate;
- $\Lambda_{H,t}^{\text{live}}$: breadth and corrigibility of practically live alternatives;
- $A_{H,t}^{\text{corr}}$: capacity to read relevant differences, accept/refuse, act, and revise;
- $R_{H,t}^{\text{route}}$: availability of partially independent critical or verification routes;
- $W_{H,t}^{\text{world}}$: answerability to resistant target-domain conditions;
- $r_{H,t}$: delayed unaided Human Return after decisive AI support is removed;
- H_t^{cap}: social-reproductive human capability stock supported by care, health, education, and community;
- $S_{H,t}$: recognized standing and socially meaningful roles.

These are declared readouts, not substances. No universal weighting is assumed.

For dimension j, define a local Human Conversion Elasticity [DEFINITION]

$$\eta_{j,t}^{HC} = \frac{\partial \ln C_{j,t}^H}{\partial \ln M_t}. \tag{12}$$

This immediately permits mixed trajectories. A deployment can satisfy

$$\eta_\Gamma^{HC} > 0, \qquad \eta_{r_H}^{HC} < 0, \qquad \eta_X^{HC} < 0, \tag{13}$$

meaning that machine capacity raises material consumption while reducing retained human competence and exit. Such a system is not accurately described by one positive "augmentation" score.

> **Proposition 1** (Conversion non-identity) [OPEN]. An increase in M_t or current human-AI performance is insufficient to establish human expansion unless at least one predeclared Human Conversion dimension improves without unacceptable deterioration in another load-bearing dimension.

6 The Epistemic Conversion Mechanism

At the micro level, the conversion problem inherits the distinction between closure and corrigibility from *When AI Expands Human Potential*. A high-conversion interaction has the form

$$\begin{aligned} &\text{H problem} \to \text{AI divergence} \to \text{H resistance} \\ &\to \text{World test} \to \text{H integration} \\ &\to \text{AI removal} \to \text{H return}. \end{aligned} \tag{14}$$

A low-conversion interaction can instead be

$$\text{H problem} \to \text{AI answer} \to \text{use} \to \text{dependence}. \tag{15}$$

The point is not that direct AI answers are always harmful. It is that systems optimized only for short-run completion can rationally substitute for the very cognitive work needed to build future independent competence.

The key role separation is asymmetric. AI has comparative advantages in breadth, search, translation, recombination, simulation, and candidate generation. Humans must

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Table 1: Empirical anchors motivating the proposal; none validates the full Human Conversion model.

Observed domain	Current evidence	What it licenses
Machine acceleration	Global AI compute capacity estimated at 3.3× annual growth since 2022; frontier benchmark gains remain rapid [Stanford Institute for Human-Centered AI, 2026].	Prospective monitoring of conversion speed; not a claim that knowledge grows 3.3× annually.
Embodied automation	China installed 295,000 industrial robots in 2024 (54% of world installations); operational stock exceeds two million [International Federation of Robotics, 2025].	Physical automation is already scaling; installations are not per-robot capability.
Infrastructure concentration	Several AI hardware/cloud segments exhibit high reported concentration [OECD, 2025].	Gate-risk deserves measurement; concentration alone does not prove domination.
Human-AI feedback	Repeated biased AI interaction altered later human judgments in $n = 1{,}401$ experiments [Glickman and Sharot, 2025].	Human state after AI interaction can differ from baseline.
Assisted performance vs learning	AI tutoring raised practice performance, while unguarded use reduced later unaided performance in a field experiment [Bastani et al., 2025].	Assisted performance must be separated from durable human return.
Human-AI synergy	Meta-analysis finds augmentation is common but strong synergy is not guaranteed and varies by task [Vaccaro et al., 2024].	"Human+AI" is not intrinsically superior to the best component.

retain problem ownership, normative responsibility, integration, and contact with world-side constraints; criticism and alternative generation can be shared. This predicts that the optimal assistance policy is not "maximum AI use" but task- and learner-contingent support that maximizes conversion rather than substitution.

> **Proposition 2** (Assistance-conversion divergence) [OPEN]. The AI assistance intensity that maximizes current performance need not maximize delayed Human Return or corrigible agency.

7 Bad Mode Is a State Space, Not a Staircase

Collapse narratives often assume one universal sequence: convenience, deskilling, job loss, ownership concentration, and political lock-in. The architecture rejects that sequence as too strong. Cognitive dependence may occur before labour displacement; informational concentration can precede market concentration; coercive lock-in can arise without private AI ownership.

Define a Bad-Mode State Vector [DEFINITION]

$$B_t = \big\langle D_t,\ \bar{G}_t^{\text{conv}},\ C_t^{\text{info}},\ 1 - X_t^H,\ 1 - r_{H,t},\\ 1 - \Lambda_{H,t}^{\text{live}},\ 1 - A_{H,t}^{\text{corr}},\ 1 - \Gamma_t^{\text{eff}},\ 1 - H_t^{\text{cap}}\big\rangle. \quad (16)$$

D is dependency, $\bar{G}^{\text{conv}}$ conversion-gate power, and C^{info} concentration of consequential information mediation. A "bad mode" is not a point threshold fixed for all societies. It is a sustained trajectory toward a high-risk region in multiple dimensions, especially where the dynamics exhibit positive feedback and costly reversal.

This formulation allows distinct bad trajectories: deskilling without unemployment, material abundance without agency, cross-border rent extraction without domestic frontier AI, or informational lock-in without industrial automation.

8 Reversibility, Hysteresis, and the Intervention Window

Technological and political systems can display increasing returns. Adoption creates complementary investment, habits, data, standards, networks, organizational routines, and political constituencies. Arthur's technology model and Pierson's institutional analysis show why early advantages can become self-reinforcing and difficult to reverse [Arthur, 1989, Pierson, 2000].

For each human-conversion dimension j, define recovery cost $C_{j,t}^{\text{rec}}$ and expected recovery time $\tau_{j,t}^{\text{rec}}$. The Reversibility Window [DEFINITION] is the set of times for which restoration remains feasible under declared bounds:

$$W_j = \{t : C_{j,t}^{\text{rec}} \le \bar{C}_j \ \wedge\ \tau_{j,t}^{\text{rec}} \le \bar{\tau}_j\}. \quad (17)$$

As skills atrophy, standards harden, ownership compounds, or exit infrastructure disappears, the window can narrow even if no acute crisis is visible.

This is the sense in which the paper uses hysteresis: the state reached after prolonged AI dependence may not be restored simply by removing the AI system. Skill, institutions, ownership, or social roles may have to be rebuilt.

> **Proposition 3** (Reversibility Principle) [OPEN]. The faster machine capability grows relative to human conversion, the more valuable it becomes to preserve reversible human and institutional alternatives before the recovery cost of lost routes rises.

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9 An Urgency Vector Instead of a Panic Index

Because Human Conversion is multidimensional, urgency should not be forced into one unvalidated scalar. Normalize machine growth and each human dimension within a declared study. Define a conversion-gap term [DEFINITION]

$$\Delta g_{j,t} = \left[\tilde{g}_{M,t} - \tilde{g}_{C_j,t}\right]_+. \tag{18}$$

Let $L_{j,t} \in [0,1]$ be lock-in intensity, $S_{j,t}$ severity if the dimension deteriorates, and $\tau^{\text{rec}}_{j,t}$ expected recovery time. Then

$$U_{j,t} = \Delta g_{j,t}\, L_{j,t}\, S_{j,t}\, \tau^{\text{rec}}_{j,t}. \tag{19}$$

The Urgency Vector [DEFINITION] is

$$U_t = \langle U_{1,t}, \ldots, U_{J,t} \rangle. \tag{20}$$

For precautionary reporting, a bottleneck summary may use

$$U_t^{\max} = \max_j U_{j,t}, \tag{21}$$

which prevents severe deterioration in one dimension from being averaged away by gains elsewhere. A weighted average is permissible only with explicit normative weights.

The logic is asymmetric. If acceleration is overestimated but safeguards preserve exit, skill, plural verification, and broad claims, social cost may be modest. If acceleration is underestimated and lock-in is real, late reconstruction may be much more expensive. This is an option-preservation argument under uncertainty, not a prediction of catastrophe [Arrow and Fisher, 1974, Fisher and Krutilla, 1974].

10 Distribution: Whose Potential Expands?

Average human conversion can conceal stratification. Highly skilled workers may use AI to widen alternatives and accelerate learning while novices use the same systems as substitutes for foundational reasoning. Recent experimental and conceptual work already points to heterogeneous learning effects and the difficulty of optimizing human learning relative to immediate accuracy [Bastani et al., 2025, Buçinca et al., 2024, Cash et al., 2026].

Therefore every population-level report should publish distributional readouts [DEFINITION]. For any dimension C_j:

$$C^{10}_{j,t}, \quad C^{50}_{j,t}, \quad C^{90}_{j,t}, \tag{22}$$

and a simple spread

$$I^H_{j,t} = C^{90}_{j,t} - C^{10}_{j,t}. \tag{23}$$

A society in which elite Human Return and agency rise while the majority become dependent users is not "maximum human potential" merely because the mean rises.

> **Proposition 4** (Distributional conversion) [OPEN]. Broad human expansion requires that conversion gains reach a declared population distribution; aggregate productivity or mean capability gains are insufficient.

11 Six Minimum Safeguard Invariants

The proposal does not prescribe one constitutional or economic system. It proposes six invariants that remain rational across a wide range of AI-speed forecasts because they preserve conversion and reversibility.

1. Human Return measurement. Systems that claim learning or human development should measure delayed unaided Concept reconstruction, Tool selection, Skill execution, and Alternative generation after decisive assistance is removed. Assisted performance alone is not evidence of durable human gain.

2. Independent resistance. AI candidate generation should not become the sole certifier of its own outputs. High-stakes inquiry requires target-domain evidence and partially independent critical routes.

3. Credible exit and interoperability. Provider count is not enough. Users and institutions should be able to switch, export, continue, or route around a failing or coercive gate at tolerable cost.

4. Convert automation dividends into durable claims and capacity. Transfers can preserve consumption today while machine-productive ownership continues to compound elsewhere. At least some productivity gains should create durable broad claims, time, learning capacity, or public/collective infrastructure rather than consumption alone.

5. Preserve social reproduction and critical unaided competence. Care, health, education, community, and selected non-AI critical skills should remain independently reproducible. Human productive necessity can fall while human social necessity remains.

6. Preserve contestability across economic, informational, and coercive power. Ownership, information mediation, and coercive authority should be measured separately. A society is most vulnerable when the same trajectory reduces exit across all three.

These are not bans on AI. Their purpose is to keep the conversion process corrigible.

12 A Tier-One Empirical Programme

The framework is designed to be attacked by rival models. No single study can test the whole architecture.

12.1 Longitudinal Human-AI learning experiments

Randomize assistance policies, not merely AI access. Compare direct-answer AI, friction-preserving tutoring, adversarial/counterargument AI, and no-AI controls. Measure assisted performance, delayed Human Return, transfer, error detection, metacognitive calibration, and independent route use. The core test is whether assistance optimized for current performance differs from assistance optimized for conversion.

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12.2 Workplace conversion panels

Follow workers before, during, and after sustained AI adoption. Measure task productivity, wage/bargaining outcomes, skill retention after tool withdrawal, ability to detect tool failure, switching cost, and career mobility. A useful design compares occupations with similar AI productivity gains but different training and exit institutions.

12.3 Gate and exit audits

For compute, model access, cloud, payments, logistics, or information mediation, estimate counterfactual viability when a provider is removed. This distinguishes market share from functional gate power. The dependent variable is whether consequential activity can continue at declared cost, latency, quality, and legal constraints.

12.4 Cross-national conversion accounts

Extend national AI/productivity accounts with broad ownership of machine-capital income, labour centrality, cross-border AI rents, effective material claim, public compute/interoperability, social-reproduction capacity, and Human Return measures. This would test whether comparable AI adoption produces different human outcomes under different institutional conversion systems. The ILO-World Bank finding that developing economies may face "disruption without dividend" motivates this design [International Labour Organization and World Bank, 2026].

12.5 Reversibility estimation

Estimate recovery time and cost after dependence. Examples include retraining after skill decay, restoring offline/manual capacity, re-entering a market after a platform exit, rebuilding public expertise after outsourcing, or migrating standards after vendor lock-in. The Reversibility Principle survives only if these costs systematically rise with prolonged dependence.

13 Rival Models and Removal Conditions

A theory proposal that cannot lose is only vocabulary. The following rival explanations must be compared explicitly:

- **Productivity-only model**: immediate output captures the relevant benefit.
- **Human-capital model**: conventional learning/skill measures explain all durable effects; Human Return adds nothing.
- **Capability/affordance model**: resources and formal capabilities fully explain option use; Live Possibility adds no information.
- **Competition model**: market concentration and switching cost are sufficient; conversion-gate and multidimensional power measures add no value.
- **Standard path-dependence model**: generic lock-in fully explains reversibility; human-specific conversion terms are redundant.

The Human Conversion architecture should be narrowed or removed under the following results:

1. delayed unaided performance is fully predicted by ordinary skill measures and never dissociates from assisted performance;
2. live-option measures add no predictive value beyond cost, preference, salience, and formal capability;
3. credible-exit and gate measures add no information beyond ordinary market concentration;
4. no dimension shows increasing recovery cost or path dependence under sustained AI reliance;
5. distributional conversion outcomes are fully explained by baseline income/education without AI-use interactions;
6. the vector architecture does not improve prediction, diagnosis, or intervention compared with simpler rivals.

14 What the Proposal Does Not Claim

First, it does not claim that AI is intrinsically corrosive. The same technology can produce high or low conversion depending on design and institutions. Second, it does not claim that cognitive offloading is always harmful; strategic offloading can free resources for higher-level work. Third, it does not claim that employment must vanish. Labour centrality can persist through complementarity and new tasks. Fourth, it does not treat China as the cause of the global transition. China's industrial-robot scale is a timing anchor for embodied automation, while U.S. and other firms remain central in frontier models, cloud, and compute. Fifth, it does not infer domination from concentration alone. Sixth, it does not forecast a date at which the Reversibility Window closes.

The proposal makes a narrower claim: where machine-side change is rapid and human/institutional restoration is potentially slow, waiting for definitive evidence of system-wide damage can itself destroy the options needed for low-cost correction.

15 Discussion: From Maximum Assistance to Maximum Conversion

The framework changes the design objective for human-AI collaboration. The usual objective is to maximize performance under assistance. The stronger objective is to maximize the quality of the human state that emerges after assistance. This implies a shift:

$$\begin{gathered}\text{Maximum AI assistance} \longrightarrow \\ \text{Maximum durable human conversion.}\end{gathered} \tag{24}$$

In education, the goal is not to preserve every unaided task but to decide which foundational capacities must exist before higher-order offloading is safe. In work, the goal is not to block automation but to ensure that productivity gains produce

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durable worker capability, mobility, or claims rather than only substitution. In science, AI should expand candidate space while evidence and independent criticism retain validation authority. In political economy, automation dividends should widen exit and agency rather than merely finance dependence.

The resulting division of labour is deliberately asymmetric. Machines can carry more search, recombination, translation, routine execution, and simulation. Humans need not compete at machine speed. Their strategic role is to retain problem ownership, normative judgment, responsibility, world-contact, corrigibility, and the capacity to refuse or redirect the system.

16 Conclusion

The defining competition of an AI-robotic century may not be human intelligence versus machine intelligence. It may be a race between two rates:

$$\begin{aligned}&\text{rate of machine capability growth}\\&\text{versus}\\&\text{rate of human conversion and institutional adaptation.}\end{aligned} \tag{25}$$

If machine capability rises while effective claim, exit, live possibility, corrigible agency, independent resistance, Human Return, and social reproduction also rise broadly, automation can release humans from unnecessary labour while expanding what they can understand, choose, and become. If machine capability rises faster than those conversion channels, the same productivity can produce a different trajectory: high assisted performance, high aggregate output, and growing human dependency.

The decisive policy window therefore lies before visible crisis. Skills can atrophy, ownership can compound, standards can harden, social institutions can disappear, and gate dependence can become normal. Under uncertainty, the rational response is not panic or prohibition. It is to preserve reversibility while measuring conversion.

The proposal can be compressed into one architecture:

$$\begin{aligned}\text{Machine Capability} &\rightarrow \text{Human Conversion Vector}\\&\rightarrow \text{Human Return/Agency/Exit}\\&\rightarrow \text{Reversibility or Lock-in}\\&\rightarrow \text{Next Institutional State.}\end{aligned} \tag{26}$$

The Human Conversion Imperative is therefore a design principle for an AI-rich world: do not ask only how much machines can do for humans now; measure how much capability, agency, ownership, and reversibility remain with humans afterward.

References

Yaoharee Lahtee. When AI expands human potential: Reflective dissonance, epistemic agency, and constraint. Open Civil Science Initiative preprint, 25 March 2026. https://doi.org/10.5281/zenodo.19215748, 2026a.

Yaoharee Lahtee. After labour: Human position in an AI-robotic world system. Human-AI Readout Programme, Full Paper v1.1, companion paper in the same 6 September 2026 deposit batch. https://doi.org/10.5281/zenodo.22481924, 2026b.

Yaoharee Lahtee. Choice begins before choice: Meaning-shaped accessibility, live possibility, and effective agency in human-AI systems. Human-AI Readout Programme, Full Paper v1.0-strong-claim-2026-09-05, 5 September 2026. https://doi.org/10.5281/zenodo.22357788, 2026c.

Yaoharee Lahtee. CTSA human-return readout: A session-boundary measurement architecture for retained human capability after AI. Human-AI Readout Programme, Epistemic Note v9.1-FINAL-2026-09-05, 5 September 2026. https://doi.org/10.5281/zenodo.22339909, 2026d.

Yaoharee Lahtee. Epistemic fusion or epistemic tunnel: A phenomenology-anchored, global-literature-constrained, problem-first architecture of human-AI collaboration (epistemic note v8.1 — strong-claim / legitimate-defeater edition). Human-AI Readout Programme, Epistemic Note v8.1-strong-claim+uplift-note-r2, 5 September 2026 (concept DOI 10.5281/zenodo.22318039). https://doi.org/10.5281/zenodo.22331922, 2026e.

Stanford Institute for Human-Centered AI. Artificial intelligence index report 2026. Stanford University, 2026.

International Federation of Robotics. World robotics 2025: Industrial robots. Frankfurt, 2025.

OECD. Competition in artificial intelligence infrastructure. Technical report, OECD Roundtables on Competition Policy Papers, No. 330. https://doi.org/10.1787/623d1874-en, 2025.

Moshe Glickman and Tali Sharot. How human–AI feedback loops alter human perceptual, emotional and social judgements. *Nature Human Behaviour*, 9:345–359, 2025. doi: 10.1038/s41562-024-02077-2.

Hamsa Bastani, Osbert Bastani, Alp Sungu, et al. Generative AI without guardrails can harm learning: Evidence from high school mathematics. *Proceedings of the National Academy of Sciences*, 122(26):e2422633122, 2025. doi: 10.1073/pnas.2422633122.

André Barcaui. ChatGPT as a cognitive crutch: Evidence from a randomized controlled trial on knowledge retention. *Social Sciences & Humanities Open*, 12:102287, 2025. doi: 10.1016/j.ssaho.2025.102287.

Michelle Vaccaro, Abdullah Almaatouq, and Thomas W. Malone. When combinations of humans and AI are useful: A systematic review and meta-analysis. *Nature Human Behaviour*, 8:2293–2303, 2024. doi: 10.1038/s41562-024-02024-1.

Zana Buçinca, Siddharth Swaroop, Amanda E. Paluch, Susan A. Murphy, and Krzysztof Z. Gajos. Towards optimizing human-centric objectives in AI-assisted decision-making with offline reinforcement learning. Preprint, arXiv:2403.05911, 2024.

Taylor N. Cash, Meghan O. Kelly, Brooke N. Macnamara, and Evan F. Risko. Is AI making us stupid? *Trends in Cognitive Sciences*, 30(8):673–675, 2026. doi: 10.1016/j.tics.2026.06.004.

Yena Seung and James D. Basham. Cognitive offloading in the age of generative AI: What does it mean for students with learning disabilities? *Learning Disability Quarterly*, 49(3), 2026. doi: 10.1177/07319487261439132.

Farid Noorbehbahani and Kiemute Oyibo. AI-overdependence and human cognitive decline: Hazards, evidence, and mitigation strategies. *Computers in Human Behavior Reports*, 22:101102, 2026. doi: 10.1016/j.chbr.2026.101102.

David H. Autor. Why are there still so many jobs? the history and future of workplace automation. *Journal of Economic Perspectives*, 29(3):3–30, 2015. doi: 10.1257/jep.29.3.3.

Daron Acemoglu and Pascual Restrepo. Artificial intelligence, automation and work. Technical report, NBER Working Paper 24196. `https://doi.org/10.3386/w24196`, 2018.

Anton Korinek and Joseph E. Stiglitz. Artificial intelligence and its implications for income distribution and unemployment. In *The Economics of Artificial Intelligence: An Agenda*. University of Chicago Press, 2019.

International Labour Organization and World Bank. Disruption without dividend? how the digital divide and task differences split GenAI's global impact. Technical report, ILO Working Paper 166, 2026.

W. Brian Arthur. Competing technologies, increasing returns, and lock-in by historical events. *The Economic Journal*, 99(394):116–131, 1989. doi: 10.2307/2234208.

Paul Pierson. Increasing returns, path dependence, and the study of politics. *American Political Science Review*, 94(2):251–267, 2000. doi: 10.2307/2586011.

Kenneth J. Arrow and Anthony C. Fisher. Environmental preservation, uncertainty, and irreversibility. *Quarterly Journal of Economics*, 88(2):312–319, 1974. doi: 10.2307/1883074.

Anthony C. Fisher and John V. Krutilla. Valuing long run ecological consequences and irreversibilities. *Journal of Environmental Economics and Management*, 1(2):96–108, 1974. doi: 10.1016/0095-0696(74)90007-2.

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PART VII

The River and Its Tests

The River and Its Tests turns the programme's own vocabulary into an audited object, after Part 5's turn to Human–AI coupling and Human Return and Part 6's turn to human position after labour: instead of adding a further domain to the spine, its chapters check what the spine, taken as a whole, actually says and what would count against it.

This part opens with *Master Equation River* (Ch. 36), a provenance audit of the wider programme: it traces which earlier work contributes which segment of the proposed dependency chain from World to Warrant, stating directly that the chain is a typed, multi-level dependency architecture, not a single law, and that segments it tiers Open remain open (see its headnote). Its companion Coq formalisation (Appendix F) checks the internal consistency of the stated discrete models for each of its 79 equations, tier by tier; it does not certify the paper's own Dr theses or Open hypotheses as true. Chapter 37, *Before Meaning, Before Choice*, re-derives the same root-to-human sequence directly from Readout Universe and Readout Genesis, advancing a central claim that human experience and choice should not be inserted as primitive terms, paired with a claim-status ledger and six open, individually stated hypotheses (H1–H6) about what each stage would need to survive. Chapter 38 is a separate, living evidence registry for an earlier four-paper cluster in the same programme, coding each claim by evidence state — observed, convergent, synthesis, open, decisive test — so that support from neighbouring literatures is never silently read as validation of a programme-specific construct.

Carried forward into Part 8: the same claim-by-claim discipline applied to a different question — how a standalone author's own checking procedure is itself checked, without institutional infrastructure. Where this part audits the programme's equations and evidence, Part 8 audits the audit method.

Chapter 36 and the chapter on *Before Meaning, Before Choice* both record a planned further paper testing the Master Equation River empirically; it is listed on the Edition 1 page as not yet written and is not represented here.

- Master Equation River: A Provenance Audit Across the Human–AI Readout Programme — `10.5281/zenodo.22550491`
- Before Meaning, Before Choice: A Readout-Native Derivation of Experience, Live Possibility, Agency, and Human-AI Return — `10.5281/zenodo.22424434`
- *Planned: Paper B — Testing the Master Equation River (not yet written)*
- State of Evidence for the Readout Hypothesis-Generation Programme: A Shared Evidence Registry for Papers I–IV (with Claim Registry E01–E25) — `10.5281/zenodo.22308066`

Master Equation River: A Provenance Audit Across the Human–AI Readout Programme

v1.5, English • 7 September 2026 • 10.5281/zenodo.22550491

What the chapter states. "The whole internal literature can be arranged into a continuous cycle: World → Readout → Meaning → Experience → Retention → Accessibility → Live Possibility → Choice → Agency → Feedback → Changed Reader → Human–AI → Human Return → Warrant. But once a retained change has occurred, the cycle does not return to the same person... so it should be understood as a recursive river or spiral rather than a closed circle."

Status. "Knowledge status K0."

Falsifier(s). "This document... does not claim that the whole river is a universal psychological law or a single physics equation. It functions instead as a typed, multi-level dependency architecture: some segments are Readout/Domain architecture, some are definitions, some are Dr theses, some are Open empirical hypotheses, and some are measurement protocols." Segments the source tiers Open — eq. 13, 14, 18, and their companions in Appendix B's ledger — are, in the source's own words elsewhere, "proposals that remain open"; nothing in this document treats an Open segment as settled. Appendix F reprints the full Th_coqc ledger, tier by equation.

Read before. *Readout Genesis Standalone Synthesis* (Ch. 3); *Meaning Before Naming* (Ch. 8); *Experience Is Meaning-Giving* (Ch. 9); *Experience Is the Human LoRA* (Ch. 10); *Human Learning as Epistemic Architecture* (Ch. 11); *From Problem to Hypothesis* (Ch. 13); *Choice Begins Before Choice* (Ch. 17); *Potential as a Readout* (Ch. 18); *Epistemic Fusion or Epistemic Tunnel* (Ch. 30); *CTSA Human-Return Readout* (Ch. 31); *How Humans Should Converse with AI* (Ch. 32); *After Labour* (Ch. 34); *The Human Conversion Imperative* (Ch. 35); *From Assistance to Human Capability* (Ch. 33) (mechanical genealogy-map list, this chapter's own Table 1).

Feeds into. *Before Meaning, Before Choice* (Ch. 37), which re-derives the same root-to-human sequence directly from Readout Universe and Readout Genesis; *State of Evidence for the Readout Hypothesis-Generation Programme* (Ch. 38), a living evidence registry for an earlier cluster of the same programme (mechanical cross-reference, Appendix A).

Master Equation River

A Provenance Audit Across the Human–AI Readout Programme

Version 1.5 — v1.4 unchanged, plus a canonical-registry cross-reference to Toledo

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Document status. This document is an internal-literature review and provenance audit of the Human–AI Readout Programme: it arranges the equations and proposals of several works of the programme into a single river. It does **not** claim that the whole river is a universal psychological law or a single physics equation. It functions instead as a typed, multi-level dependency architecture: some segments are Readout/Domain architecture, some are definitions, some are Dr theses, some are Open empirical hypotheses, and some are measurement protocols. The purpose is to make visible which work each segment comes from, why the segments connect, and where notation or layering still needs to be fixed before the river is frozen into canonical form.

How v1.1 differs from v1.0. This edition was checked under the glosa methodology (Section 9): every internal source is now cited to the record actually deposited on Zenodo, with its DOI (v1.0 had cited Fusion/Tunnel as "v7" when the deposited record is v8.1, and had cited RG-MEMK and Agency Potential as though separately deposited, which they are not); the retention-gate equations (35)–(36) are now given the status of a Master-edition notation proposal rather than a restatement of v8.1's own text, which does not contain this formula; the potential equation (25) uses the witnessed set of *Potential as a Readout* v2 rather than the feasible set; a disclosure has been added that H_{return} was renamed from CTSA's R_H; and a symbol glossary (Appendix A) has been added. Knowledge status K0.

Edition note (v1.2). This is an English-only edition prepared for the textbook *Written by AI. Still True.* The content is unchanged from v1.1: every section, equation (same numbering, (1)–(45)), provenance-table row, notation-collision row, fix box, reference entry, and appendix entry appears here in the same order. Passages originally written in Thai have been rendered into English with the same meaning; where the Thai and the adjacent English already said the same thing, one English sentence is kept and the merge is recorded here. Nothing has been added, nothing has been dropped, and no claim has been strengthened. Version 1.1 (Thai/English, DOI 10.5281/zenodo.22414412) remains in the record's version history as the source of record for this edition.

Edition note (v1.3). This edition adds three new segments to the unchanged v1.2 English edition: (i) Section 3.13 Topic Entry and Agenda Ownership Before the Prompt, from *How Humans Should Converse with AI* (Dialogue Conversion Protocol, DCP) [Lahtee, 2026o]; (ii) Section 3.14 Human Return → Action → World Feedback, from the same source; and (iii) Section 3.15 The World-System Layer: Human Systemic Position and Human Conversion, from *After Labour* [Lahtee, 2026m] and *The Human Conversion Imperative* [Lahtee, 2026n]. Every section, equation (1)–(45), table row, fix box, and reference entry of v1.2 is unchanged and appears here in the same order; the three new segments use continued equation numbers (46) onward and are added to the genealogy map (Table 1), the evolution-of-questions list (Section 6, items 11–13), the epistemic-status table (Table 2), and the symbol glossary (Table 5). Each new statement about a source paper is traceable to that paper's own equation or section number, cited in place. No claim is raised beyond the tier the source paper itself gives the object: most of the new objects are [Definition] or accounting identities inside a stylized model, and several are explicitly [Open] propositions in their source. A relayed external reader's assessment of DCP world-literature convergence (2026-09-06, unverified, none of its citations independently checked here) is recorded only as a readout in Section 6; it is not treated as validation. Version 1.2 (English, same content as the deposited v1.1, DOI 10.5281/zenodo.22414412) remains in the record's version history. Knowledge status K0.

Edition note (v1.4). This edition adds one new segment to the unchanged v1.3 English edition: Section 3.16, restating RG-HCA's proactive extension [Lahtee, 2026p]. Every section, equation (1)–(66), table row, fix box, and reference entry of v1.3 is unchanged and in order; the new segment uses (67)–(78), plus (79) extending Section 5 by one sentence, and is added to Table 1, Section 6 (item 14), Table 2, and Table 5. Each new statement is traceable to RG-HCA's own equation or section, at no higher tier than RG-HCA gives: several objects are [Definition], two are [Open], and the measurement objects are [Definition, measurement]. RG-HCA's Human Return reuse (eq. 24–25) is discussed in prose only, since it cites the same CTSA v9.1 source as H_{return} (41)–(42) under different field names, not identically. Every equation (1)–(79) is formalised in the companion Coq record (Appendix B; DOI 10.5281/zenodo.22518450). K0.

Edition note (v1.5). Every section, equation (1)–(79), table, fix box, and reference of v1.4 is unchanged and in order. This edition adds Section 10 (this river read against the programme's root spine, now coded into **Toledo** [Lahtee, 2026q]) and Appendix C (a row-by-row Toledo cross-reference for all 79 equations), both read off `registry/CANONICAL.json` and `registry/COLLAPSE.md` as they stand on 2026-09-07. Appendix B gains one pointer: DOI 10.5281/zenodo.22518450 is obsoleted by Toledo's DOI 10.5281/zenodo.22548770; its ledger table is unchanged. No claim, tier, or number already in v1.4 changes. K0.

One-sentence summary. The whole internal literature can be arranged into a continuous cycle: World → Readout → Meaning → Experience → Retention → Accessibility → Live Possibility → Choice → Agency → Feedback → Changed Reader → Human–AI → Human Return → Warrant. But once a retained change has occurred, the cycle does not return to the same person: whenever the change persists, $H_{t+1} \neq H_t$, so it should be understood as a recursive river or spiral rather than a closed circle.

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1 What This Audit Found

The Master Equation River did not arise from inventing a new block of equations in one sitting. It arose from noticing that several manuscripts of the programme were each describing a different stage of the same process. Readout Genesis (and the MEMK domain contained within it) supplies the root-to-domain order and the distinction between world, readout, and meaning; Human Learning supplies representation–meaning–retrieval–constraint–competence; Human LoRA supplies selective persistence; From Problem to Hypothesis supplies history-shaped accessibility; Fusion/Tunnel supplies recursive Human–AI coupling, resistance, retention, and direction; Meaning Before Naming and Experience Is Meaning-Giving supply meaning-giving, resonance, rhythm, and transition readiness; Potential as a Readout supplies the witnessed corrigible action envelope (inheriting from the now-superseded Agency Potential note); Choice Begins Before Choice supplies the live possibility field; CTSA supplies the session-boundary Human Return audit.

What Master adds, therefore, is not any single equation on its own but the dependency closure across the papers: once every layer is laid out in order, we see that experience can change the reader, the changed reader can see the world of possibility differently, and the changed world of possibility then changes choice, action, feedback, and the ongoing Human–AI trajectory.

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2 GENEALOGY MAP: WHERE EACH SEGMENT COMES FROM

Every internal source below is cited to the record deposited on Zenodo (DOI numbers in the reference list). Rows where v1.0 had cited a document that was not separately deposited are marked † and point to the record that actually holds that content.

Table 1: Genealogy map: which deposited record supplies each segment of the river, and what it actually contributes.

Segment in Master	Primary internal source (deposited record)	What that source actually supplies
World → finite readout	Readout Genesis Standalone Synthesis [Lahtee, 2026d] (MEMK domain is in this volume †)	Root has no built-in semantic labels; retained difference precedes naming; domain translation and readout precede meaning; the reader makes the distinction accessible but does not create root truth.
Readout → meaning	Experience Is Meaning-Giving v2 [Lahtee, 2026h]; Readout Genesis [Lahtee, 2026d]	Meaning is formalized as a reader-conditioned significance relation under human state, context, and criterion/question.
Meaning → experience	Readout Genesis [Lahtee, 2026d] eq. (14); Experience v2 [Lahtee, 2026h]	Genesis gives $E_{i,n} = \Phi_E(x_{i,n}, \mu_{i,n}, \kappa_{i,n}, c_n)$; Experience v2 makes the explicit thesis that experience is phenomenon-as-meaningfully-read.
Experience ↔ naming	Meaning Before Naming [Lahtee, 2026g]; Experience v2 [Lahtee, 2026h]	Allows E_n, μ_n to arise before stable naming, and adds the recursive correction that naming can restructure later meaning/experience.
Experience → retained future reader	Experience Is the Human LoRA [Lahtee, 2026c]; Readout Genesis [Lahtee, 2026d]	Experience is not equal to adaptation; only a selected residue is retained/consolidated and goes on to change the future interpretive/action operator.
External–internal congruence	Experience v2 [Lahtee, 2026h]	Resonance is formally defined as the congruence between the current externally prompted experience and the retained internal experiential organization (the earlier definition of resonance in [Lahtee, 2026g] is superseded).
Ordered recurrence / rhythm	Experience v2 [Lahtee, 2026h]	Rhythm is structured recurrence, spacing, interruption, emphasis, boundary, and return over ordered history; it is not meaning and not physical resonance.
History → re-entry	From Problem to Hypothesis [Lahtee, 2026e]	Semantic momentum is a finite-history proxy: a route just traversed may be easier to re-enter; this is an OPEN hypothesis.
Attraction + momentum + switching cost → accessibility	From Problem to Hypothesis [Lahtee, 2026e]	Supplies the history-shaped accessibility kernel and stresses that accessibility/attraction/momentum are not warrant.
Accumulation → transition readiness	Readout Genesis [Lahtee, 2026d] (MEMK †); Experience v2 [Lahtee, 2026h]	Semantic potential, informational work, and barrier crossing are used as a domain-level transition topology; must not be read as physical energy.
Feasible → live possibility	Choice Begins Before Choice [Lahtee, 2026j]	Places a live possibility field between feasible capacity and overt choice.
Live profile / live set	Choice Begins Before Choice [Lahtee, 2026j]	Converts the live option from a binary set into a distribution of practical accessibility and a measurement-specific threshold.
Live → choice → action → observation	Choice Begins Before Choice [Lahtee, 2026j]; Potential as a Readout [Lahtee, 2026i]	Separates possible, feasible, live, chosen, enacted, and observed, and separates evaluator visibility from underlying capacity (NC-78: potential ≠ exercised ≠ observed).
Effective corrigible agency	Potential as a Readout v2 [Lahtee, 2026i] (replacing the Agency Potential note †)	Agency is a task-relative envelope of read–decide/refuse–execute–feedback/revision loops over a *witnessed* option set under a declared comparability relation; it has two layers (choosing within the structure / changing the structure).
Recoverable deprivation / live-field gap	Potential as a Readout [Lahtee, 2026i]; Choice [Lahtee, 2026j]	Uses a feasible counterfactual to ask how much potential or live field is recoverable; not automatically a "unit of severity"; diagnosing the compression is separated from identifying who is responsible (NC-79).

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Table 1 continued

Segment in Master	Primary internal source (deposited record)	What that source actually supplies
Power before choice	Choice Begins Before Choice [Lahtee, 2026j]	Synthesis: power can act on agency by changing meaning/accessibility before overt choice.
Human state → prompt	Fusion/Tunnel v8.1 [Lahtee, 2026f]; Meaning Before Naming [Lahtee, 2026g]; Experience v2 [Lahtee, 2026h]	The prompt is a bounded articulation of a history-bearing human state; Experience v2 names this the Pre-Prompt Human State Principle.
Human ↔ AI recursion	Fusion/Tunnel v8.1 [Lahtee, 2026f]; Experience v2 [Lahtee, 2026h]	Each turn changes the context/readout of the next turn; AI output becomes a phenomenon that the human can give meaning to and retain.
Reciprocal momentum / amplification	Fusion/Tunnel v8.1 [Lahtee, 2026f]	Supplies the finite dialogue stepper, human/AI momentum, and χ_{recip} as a reciprocal-lineage diagnostic; amplification is not truth.
Session retention	Human LoRA [Lahtee, 2026c] → Fusion/Tunnel v8.1 [Lahtee, 2026f] (Repair 7)	v8.1 retains η_s only as a conceptual session-end retention gate and measures the residue that actually persists with the unaided return battery Y^{return} and RET, rather than inferring it from within-session engagement; the rank-restricted update formula in Section 3.11 is Master's own proposal (inheriting from an undeposited conversation-centered draft †).
Candidate status	When AI Expands Human Potential [Lahtee, 2026a]; Fusion/Tunnel v8.1 [Lahtee, 2026f]	AI is a candidate generator, not a certifier; v8.1 formalizes this as $\mathrm{AI}(Q) = K_{\text{like}} \neq K_{\text{validated}}$.
Difference–Resistance–Agency	Fusion/Tunnel v8.1 [Lahtee, 2026f]; normative ancestry from [Lahtee, 2026a]	D-R-A are constitutive conditions under the definition of durable human epistemic expansion; not an empirical universal law.
Expansion vs. tunnel direction	Fusion/Tunnel v8.1 [Lahtee, 2026f]	Separates reciprocal amplification, human retention, and epistemic direction into three axes; three-axis outcome $J_s^* = (\mathrm{AUG}_s, \mathrm{SYN}_s, \mathrm{RET}_s)$.
Human Return / CTSA	Fusion/Tunnel v8.1 [Lahtee, 2026f]; CTSA v9.1 [Lahtee, 2026k]	Fusion supplies the Human Return horizon; CTSA separates what returns into Concept, Tool, Skill, and Alternative, and audits Loss, Regulation, Provenance, Warrant, Direction.
New in v1.3 (three companion papers, 6 September 2026)		
Topic entry before the pre-prompt transport	How Humans Should Converse with AI (Dialogue Conversion Protocol) v1.3 [Lahtee, 2026o]	Places a Topic Entry Condition (LiveProblem/OpenExploration/RoutineDelegation) in front of the $H_{s,0} \to Q_{s,0}$ transport already in this river as (27); Problem-First is a default for human-development use, not a requirement (Problem-First ≠ Problem-Only).
Human Return closed through action and world feedback	Dialogue Conversion Protocol v1.3 [Lahtee, 2026o]	Extends the terminal Human Return audit with a further step: the human's integration record and selected action, and the world's returned record, feed into the next session's pre-prompt state, closing a Live-Problem → Revision-or-New-Problem cycle.
World-system human position (labour centrality, citizen claim, ownership, social reproduction, power channels)	After Labour v1.2 [Lahtee, 2026m]	Places corrigible agency, live possibility, and Human Return (already in this river, Sections 3.7 and 3.12) inside a wider accounting architecture of effective material claim, four-dimensional labour centrality, the Citizen Claim Threshold, ownership accumulation, and a typed Human Systemic Position audit index; non-collapse: aggregate output rising does not entail the audit index rising.
Human Conversion Vector and Reversibility Window	The Human Conversion Imperative v1.1 [Lahtee, 2026n]	Reframes machine-capability growth versus human conversion as a vector-valued, multi-dimensional elasticity problem rather than one augmentation score, and adds a Reversibility Window / Urgency Vector for when lost human capacities or routes become costly to restore.
New in v1.4 (one companion paper, 6 September 2026)		

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Table 1 continued

Segment in Master	Primary internal source (deposited record)	What that source actually supplies
Barrier readout before route generation	From Assistance to Human Capability (RG-HCA) v1.1 [Lahtee, 2026p]	Places a typed barrier ledger (Knowledge, Skill, Language, Tool, ResourceTime, Network, Credential, Permission, Opportunity, Unknown) between a human-owned live problem and any candidate route, guarding Observed Difficulty ≠ Skill Deficit (RG-HCA eq. 12–13).
Endorsed capability/opportunity routes and scaffold fading	RG-HCA v1.1 [Lahtee, 2026p]	Separates AI-generated candidate routes from human-endorsed live routes (RG-HCA eq. 16), guards three non-collapse laws on proactive suggestion, capability advancement, and scaffolding (RG-HCA eq. 17–19), and gives a monotone OPEN fading rule tying assistance withdrawal to demonstrated Stable Unaided Return (RG-HCA eq. 22).
Opportunity conversion after Human Return	RG-HCA v1.1 [Lahtee, 2026p]	Extends the Human Return → Action → World Feedback closure already in this river (Section 3.14) with a world-side opportunity readout and the non-collapses Credential ≠ Capability, Market Legibility ≠ Human Worth (RG-HCA eq. 27–30), plus a net advancement record and a conditional estimand for policy comparison (RG-HCA eq. 36–37).

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3 The Equation River According to Provenance

3.1 Foundation: Retained Difference, Readout, Meaning, Experience

The foundation of the programme begins from an architectural prohibition: a semantic noun is not a root primitive in the human domain. We therefore begin from the phenomenon / world-side encounter x_n and give it a finite human readout

$$r_n = R_H(x_n \mid H_n, c_n). \tag{1}$$

[Lahtee, 2026d,h]

Meaning-giving is written as a significance relation under human state, context, and criterion/question,

$$\mu_n = \Psi_H(r_n, H_n, c_n, Q_n), \tag{2}$$

and in Experience v2 the analytic modes can be separated as

$$\mu_n = (\mu_n^{\mathrm{aff}}, \mu_n^{\mathrm{prag}}, \mu_n^{\mathrm{auto}}, \mu_n^{\mathrm{conc}}, \mu_n^{\mathrm{epi}}). \tag{3}$$

[Lahtee, 2026h]

Readout Genesis (MEMK domain, eq. 14 of the deposited record) gives the experience equation as $E_{i,n} = \Phi_E(x_{i,n}, \mu_{i,n}, \kappa_{i,n}, c_n)$; dropping the reader index i and renaming κ_n to γ_n^{μ} to avoid a symbol collision (Section 4.1) gives

$$E_n = \Phi_E(x_n, \mu_n, \gamma_n^{\mu}, c_n), \tag{4}$$

where γ_n^{μ} is the strength/control of meaning over the current experience, so the defended compression is

$$\text{Experience} = \text{phenomenon-as-meaningfully-read}. \tag{5}$$

[Lahtee, 2026d,h]

3.2 Naming as Both an Articulation and a Restructuring Operator

Meaning Before Naming shows that explicit lexical naming need not precede experiential significance,

$$\ell_n = L_H(E_n, \mu_n \mid H_n, c_n), \qquad E_n, \mu_n \text{ may precede stable } \ell_n. \tag{6}$$

Experience v2 hardens this point by making naming recursive,

$$\mu_n \to E_n \to \ell_n \to \mu_{n+1} \to E_{n+1}. \tag{7}$$

[Lahtee, 2026g,h]

3.3 Retention and the Changing Future Reader

Human LoRA separates experience from durable adaptation in the master dependency form,

$$H_{n+1} = U_H\big(H_n,\ \mathrm{Retain}(E_n, \mu_n, r_n),\ \delta_{n:n+1}^{\mathrm{world}}\big). \tag{8}$$

[Lahtee, 2026c,h]

The important consequence is that retained experience can change the operator used to produce future readout; so the same biological individual may no longer be the same effective reader at a later time.

3.4 Resonance: Only the v2 Definition Is Current

The earlier Meaning Before Naming concept note used the word resonance close to an accessibility/reweighting diagnostic (and that document itself already flags a "literal-resonance error"); that definition has been superseded in Experience v2. The current canonical definition is external–internal experiential congruence, giving

$$I_{H,n} = \mathrm{Retrieve}(M_{H,n} \mid c_n, Q_n), \tag{9}$$

$$\mathrm{Reson}_H(n) = C_H(E_n^{\mathrm{cur}}, I_{H,n} \mid c_n, Q_n). \tag{10}$$

[Lahtee, 2026h]

Non-collapse must be preserved,

$$\begin{aligned} &\mathrm{Reson} \neq \mathrm{Identity}, \quad \mathrm{Reson} \neq \mathrm{Truth}, \\ &\mathrm{Reson} \neq \mathrm{Retention}, \quad \mathrm{Reson} \neq \mathrm{Improvement}. \end{aligned} \tag{11}$$

3.5 Rhythm, Momentum, and Accessibility Must Remain Parallel Descriptors

Rhythm is the temporal organization of the encounter history,

$$T_n = \{(x_k, t_k, w_k)\}_{k \le n}, \qquad \mathrm{Rhythm}_n = \Omega(T_n, B_n). \tag{12}$$

[Lahtee, 2026h]

From Problem to Hypothesis gives recent-path momentum as a separate quantity,

$$m_{t+1}(e) = \rho\, m_t(e) + \mathbb{1}[e_t = e], \qquad 0 \le \rho < 1, \tag{13}$$

and history-shaped semantic accessibility,

$$\kappa_{t+1}^{\mathrm{sem}}(e \mid Q) \;\propto\; \kappa_t^{\mathrm{sem},(0)}(e \mid Q)\, \exp\big(\beta a_{Q,t}(e) + \mu m_t(e) - \nu c_{A,t,Q}(e) + \xi_t(e)\big). \tag{14}$$

[Lahtee, 2026e]

The canonical correction is not to write Rhythm $\Rightarrow m_t$ as a derivation: rhythm, recent-path momentum, attraction, and switching cost are separate history/control descriptors that may jointly determine accessibility,

$$\mathrm{Rhythm}_t,\ m_t,\ a_t,\ c_t \longrightarrow \kappa_{t+1}^{\mathrm{sem}}. \tag{15}$$

3.6 Accumulation, Barrier, and Release

The MEMK domain in Readout Genesis has a semantic potential V_{sem}, informational work W_{info}, and activation barrier $\Delta V^{\dagger}_{\mathrm{old}\to\mathrm{new}}$, explicitly stated not to be physical energy. Experience v2 uses this to interpret retained transition readiness,

$$\Delta W_{j,N} = \sum_{n=1}^{N} \eta_n\, \mathrm{eligibleGradient}_n, \tag{16}$$

$$W_n^{\mathrm{eff}} = W_{\mathrm{info},n}\, g\big(\mathrm{Reson}_H(n),\ \mathrm{eligibility},\ \mathrm{context}\big), \tag{17}$$

and the transition candidate,

$$\sum_{k \le n} W_k^{\mathrm{eff}} \;\ge\; \Delta V^{\dagger}_{\mathrm{old}\to\mathrm{new}}. \tag{18}$$

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[Lahtee, 2026d,h]

This is a candidate transition topology, not a claim that every kind of human change obeys a universal threshold. Release is not equal to transformation; intensity is not equal to truth; barrier crossing is not equal to improvement.

3.7 From Accessibility to the Live Possibility Field

Choice Begins Before Choice fills the missing interface between feasible capacity and overt choice,

$$\Pi^{\mathrm{live}}_{A,t}(g) \subseteq \Pi^{\mathrm{feas}}_{A,t}(g) \subseteq \Pi^{\mathrm{phys}}_{t}(g). \tag{19}$$

Let $\lambda^{\mathrm{live}}_{A,t}(\pi \mid g)$ be the practical accessibility weight of a feasible policy (renamed from the local $\kappa_{A,t}$ so as not to collide with semantic accessibility),

$$L_{A,t}(g) = \{(\pi, \lambda^{\mathrm{live}}_{A,t}(\pi \mid g)) : \ \pi \in \Pi^{\mathrm{feas}}_{A,t}(g)\}, \tag{20}$$

and the operational live set,

$$\Pi^{\mathrm{live}}_{A,t}(g) = \{\pi : \ \lambda^{\mathrm{live}}_{A,t}(\pi \mid g) \geq \tau_{\mathrm{live}}\}. \tag{21}$$

So overt choice is a downstream object,

$$\pi^{\mathrm{choice}}_{A,t} \in \Pi^{\mathrm{live}}_{A,t}(g), \tag{22}$$

but enactment may differ from choice, and observation is not the whole trajectory,

$$\pi^{\mathrm{act}} \neq \pi^{\mathrm{choice}} \text{ possible}, \qquad Y_{\mathrm{obs}} = \mathscr{O}_q(H_{0:T}), \qquad Y_{\mathrm{obs}} \neq H_{0:T}. \tag{23}$$

This is the non-collapse chain,

$$\text{possible} \neq \text{feasible} \neq \text{live} \neq \text{chosen} \neq \text{enacted} \neq \text{observed}. \tag{24}$$

[Lahtee, 2026j]

3.8 Potential as a Readout: The Two-Layer Witnessed Envelope

The note *Effective, Corrigible Agency Potential* (5 September 2026, not separately deposited) gave four joint conditions for the agency loop and has since been replaced by *Potential as a Readout* v2, which changes the set over which the maximum is taken from a declared feasible set to a **witnessed set**: the finite set of policies with a record that a comparable agent (under a declared comparability relation) actually used within horizon T and budget B. The canonical form in Master is therefore

$$p^{*}_{A,g}(h,z;T,B,P) = \max_{\pi \in \Pi^{\mathrm{wit}}_{A}(g;h,z,T,B)} \Pr{}^{\pi}_{P}\left(\mathrm{Read}_g \cap D_g \cap X_g \cap F_g\right). \tag{25}$$

[Lahtee, 2026i]

with a second layer $p^{*(2)}_{A,g} = \max_{z \in J_{\mathrm{feas}}} p^{*}_{A,g}(h,z;\cdot)$ (changing the structure z itself), and the recoverable gap being the difference between the two layers. Choice Begins Before Choice does not replace this equation but adds the pre-choice restriction that the currently considered policy must come from $\Pi^{\mathrm{live}} \subseteq \Pi^{\mathrm{feas}}$; so the envelope asks "what has actually been usable under the given resources/horizon", while the live field asks "what is ready as a candidate for this person right now".

Corrected from v1.0: v1.0's equation (25) used $\Pi^{\mathrm{feas}}_{A}(g)$, following the note that has since been replaced; the central claim of *Potential as a Readout* is that potential is a readout over the set that has witnesses only, and a supremum over an alternative with no record is not a readout (NC-78).

3.9 Power Before Choice, and the Recoverable Live-Field Gap

The Choice paper sharpens the political proposal: power need not wait until an already-perceived option is prohibited; it may change intelligibility, salience, social permission, expected consequence, and practical accessibility before overt choice. The recoverable live-field gap is written schematically as

$$L^{\mathrm{live}}_{A,g} = \max_{z \in J_{\mathrm{feas}}} D_L\left(L^{z}_{A}(g),\ L^{z_0}_{A}(g)\right). \tag{26}$$

[Lahtee, 2026j]

But L^{live} is not a violence score. Calling it structural deprivation/violence still requires a feasible counterfactual and an independent normative loss; diagnosing the narrowing of space (from an obstruction account and asymmetry of cost) is a different question from identifying who is responsible, which requires a chooser of the regime (NC-79).

3.10 Human–AI Coupling Begins Before the Prompt

Fusion/Tunnel and Meaning Before Naming rest on the premise that language is a bounded transport of human state. Experience v2 names this explicitly as the Pre-Prompt Human State Principle,

$$H_t \xrightarrow{L_H} Q_t, \tag{27}$$

where the AI receives Q_t, not the whole of H_t. The output returns as a human encounter,

$$Q_t \to \mathrm{AI}_t \to Y_t \xrightarrow{R_H} E^{\mathrm{AI}}_{H,t}, \tag{28}$$

$$H_{t+1} = U_H\left(H_t, E^{\mathrm{AI}}_{H,t}, \delta^{\mathrm{world}}, X^{\mathrm{other}}\right), \tag{29}$$

and if a residue persists, it is then possible that

$$L_{A,t+1} \neq L_{A,t} \tag{30}$$

before the next prompt is generated.

[Lahtee, 2026f,h,j]

3.11 Within-Session Amplification, Retention, and Epistemic Direction

Fusion/Tunnel v8.1 gives the finite dialogue state and reciprocal-lineage diagnostic,

$$Z_{\mathrm{dlg}}[s,n+1] = F^{\#}_{\mathrm{dlg}}\left(Z_{\mathrm{dlg}}[s,n], u_H[s,n], u_{\mathrm{AI}}[s,n], c[s,n], T[s,n]\right), \tag{31}$$

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$$\chi_{\text{recip}}[s,n,L] = \frac{|D_{\text{recip}}[s,n,L]|}{|\Sigma[s,n]|}. \tag{32}$$

[Lahtee, 2026f]

χ_{recip} measures the finite reciprocal-descendant gain before epistemic pruning; it is not warrant or a truth score.

Session retention. v8.1 (Repair 7) states that "a scalar η_s should not be inferred from in-session engagement or recall alone": it keeps η_s as a conceptual session-end retention gate but estimates the residue that actually persists through an unaided return battery,

$$Y^{\text{return}}_{s+\Delta} = (R_{\text{rec}}, R_{\text{disc}}, T_{\text{new}}, Q_{\text{next}}), \tag{33}$$

and, for randomized study designs, a difference-in-differences estimand,

$$\text{RET} = \left(P^{\text{post}}_{H,\text{AI}} - P^{\text{pre}}_{H,\text{AI}}\right) - \left(P^{\text{post}}_{H,C} - P^{\text{pre}}_{H,C}\right). \tag{34}$$

[Lahtee, 2026f]

Master's own notation proposal for writing retention as a restricted update (inheriting from Fusion's undeposited conversation-centered draft v3, which doubles η_s — Section 4.3) is to separate the candidate restricted update from the retention gate,

$$\widetilde{\Delta\Omega}^H_s = B_s A_s, \qquad \text{rank}(B_s A_s) \ll d_H, \tag{35}$$

$$\Omega^H_{s+1,0} = \Omega^H_{s,0} + \eta_s \widetilde{\Delta\Omega}^H_s, \qquad 0 \le \eta_s \le 1. \tag{36}$$

Equations (35)–(36) are *Master's own proposal*, preserving the intent of Human LoRA/Fusion that exposure is not equal to retention, without producing an η_s^2 term; they are not v8.1's own text, and v8.1's (33)–(34) remain the measurement actually tied to Human Return.

Within the scope of Fusion/Tunnel, AI-side semantic momentum is reset across sessions,

$$m^{\text{AI}}_{s+1,0} = 0, \tag{37}$$

but this line is a scope condition of the model, not a universal fact about AI with persistent memory.

Candidate status and direction are kept separate from fluency/amplification,

$$\text{AI}(Q) = K_{\text{like}}, \qquad K_{\text{like}} \neq K_{\text{validated}}, \tag{38}$$

$$\Delta_s := G_s - T_s, \qquad \text{sign}(\Delta P_H(s)) = \text{sign}(\eta_s \Delta_s), \qquad \eta_s \ge 0, \tag{39}$$

and the three-axis outcome that v8.1 forbids collapsing into a single score,

$$\begin{aligned} J^*_s &= (\text{AUG}_s,\ \text{SYN}_s,\ \text{RET}_s), \\ \text{AUG}_s &= P^{\text{joint}}_s - P^H_s, \\ \text{SYN}_s &= P^{\text{joint}}_s - \max(P^H_s, P^{\text{AI}}_s). \end{aligned} \tag{40}$$

[Lahtee, 2026a,f]

3.12 Human Return and CTSA as the Upper Measurement Boundary

The Human Return horizon asks what, once AI is removed, can still be reconstructed, discriminated, transferred, and generated by the person alone. CTSA v9.1 separates retained capability into Conceptual, Tool-Selection, Skill-Execution, and Alternative-Generation Return with an audit record, originally written as R_H; Master renames it H_{return} so as not to collide with R's other roles (Section 4.4),

$$H_{\text{return}} = \langle G_{\text{CTSA}}, L, M, P, W, \Delta_{\text{dir}} \rangle, \tag{41}$$

where G_{CTSA} = retained gains, L = losses, M = metacognitive regulation, P = provenance, W = warrant, Δ_{dir} = epistemic direction.

[Lahtee, 2026k]

The end of the chain therefore preserves an important non-collapse,

$$\begin{aligned} &\text{Exposure} \neq \text{Retention} \neq \text{Improvement}, \\ &\text{Accessibility} \neq \text{Warrant} \neq \text{Truth}. \end{aligned} \tag{42}$$

3.13 Topic Entry and Agenda Ownership Before the Prompt

Equation numbers in this river do not run in strict page order: Sections 3.13 to 3.15 introduce (46)–(64) at this point in the text, ahead of (43)–(45), which keep the section numbers already assigned to them in v1.1/v1.2 (Sections 4, 5 and 8) and so print later, on the following pages. A citation such as "eq. (45)" or "eq. (64)" should therefore be located by its cross-reference label or by section number, not by page order.

The pre-prompt transport already given in this river as (27), $H_t \xrightarrow{L_H} Q_t$, assumes that some topic has already captured the human's attention. The Dialogue Conversion Protocol (DCP) places a Topic Entry Condition in front of this transport, restated here with DCP's own session index s [Lahtee, 2026o], eq. (1):

$$H_{s,0} \xrightarrow{L_H} Q_{s,0}. \tag{46}$$

For human-development use, DCP defines a live problem as a presently consequential unresolved difference in the person's life, work, understanding, relationship to evidence, or environment (DCP eq. 4), and separates three legitimate entry modes (DCP eq. 5):

$$\begin{aligned} \text{TopicEntry} \in \{&\text{LiveProblem},\ \text{OpenExploration}, \\ &\text{RoutineDelegation}\}. \end{aligned} \tag{47}$$

LiveProblem is the default when the goal is learning, judgment, problem solving, or human development; OpenExploration protects curiosity, play, art, and basic inquiry; RoutineDelegation covers low-value repetitive tasks. DCP's Problem-First Dialogue Principle [OPEN] states a fourfold rationale (DCP

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eq. 6):

$$\text{Problem-First} \;\Rightarrow\; \begin{cases} \text{Human Agenda Ownership,} \\ \text{Bounded Relevance,} \\ \text{World-Side Testability,} \\ \text{Observable Human Return.} \end{cases} \tag{48}$$

Problem-first is a default, not a monopoly; DCP is explicit about the guarding non-collapse (DCP eq. 7, restated at DCP eq. 43):

$$\text{Problem-First} \;\neq\; \text{Problem-Only}. \tag{49}$$

[Lahtee, 2026o]

The transport (27)/(46) into this river should therefore be read as occurring downstream of a topic-entry decision rather than as beginning from an already-given topic; the governing question is who selected the topic and whether the human still recognizes it as their own, not merely whether the topic is currently useful.

3.14 Human Return, Action, and World Feedback

Once Human Return (41)–(42) has been read, DCP does not treat verbal reconstruction as the end of a human-development conversation when the original topic was action-relevant. Let $I_s = \langle \Delta M_s, E_s^{\text{decisive}}, U_s^{\text{remain}}, \text{Next}_s \rangle$ be DCP's integration record (DCP eq. 22), a_s the human-selected next action, and $\delta_{s+1}^{\text{world}}$ the returned record from the situation that generated the problem. DCP's closure (DCP eq. 29) is

$$I_s \;\to\; a_s \;\to\; \delta_{s+1}^{\text{world}} \;\to\; H_{s+1,0}. \tag{50}$$

The wider cycle DCP proposes (DCP eq. 30) is

$$\begin{aligned} &\text{Live Problem} \to \text{Question} \to \text{Dialogue} \\ &\quad \to \text{Human Return} \to \text{Action} \\ &\quad \to \text{World Feedback} \to \text{Revision or New Problem.} \end{aligned} \tag{51}$$

[Lahtee, 2026o]

This is conditional, not universal: DCP states that pure learning, creative exploration, and basic inquiry can terminate at a different boundary, and its World-closure proposition [OPEN] applies only where the original topic had observable consequences. Within this river, (50)–(51) extend the Human Return boundary of Section 3.12 with a further arrow, $H_{\text{return}} \to a_s \to \delta^{\text{world}} \to H_{s+1,0}$, which Section 5 appends to the corrected river as an outer-loop tail.

3.15 The World-System Layer: Human Systemic Position and Human Conversion

After Labour and The Human Conversion Imperative place the corrigible-agency, live-possibility, and Human Return objects of Sections 3.7, 3.8 and 3.12 inside a wider world-system accounting layer, opening with a self-contained readout definition (After Labour eq. 2):

$$\text{Readout}_{Q,O,c}(S) = z, \qquad z \neq S. \tag{52}$$

[Lahtee, 2026m]

Labour's systemic role is disaggregated into four coordinates rather than read from employment alone (After Labour eq. 8):

$$L_t = \langle L_t^{\text{task}}, L_t^{\text{income}}, L_t^{\text{bottleneck}}, L_t^{\text{bargain}} \rangle, \tag{53}$$

and the minimum non-labour conversion needed to hold a target material claim $\bar{\Gamma}$ against a falling labour-income share s_t^L is the Citizen Claim Threshold, an accounting identity inside the stylized claim block (After Labour eq. 13):

$$q_t^{\min} = \frac{\bar{\Gamma} - s_t^L}{1 - s_t^L}. \tag{54}$$

After Labour restates, for a person i and goal g, the same possible/feasible/live nesting already used in Section 3.7 (After Labour eq. 32, reusing [Lahtee, 2026j]):

$$\Pi_{i,t}^{\text{live}}(g) \;\subseteq\; \Pi_{i,t}^{\text{feas}}(g) \;\subseteq\; \Pi_{i,t}^{\text{phys}}(g), \tag{55}$$

and a task-relative corrigible-agency envelope over that live set (After Labour eq. 34):

$$A_{H,i,t}^{\text{corr}}(g) = \max_{\pi \in \Pi_{i,t}^{\text{live}}(g)} \Pr^{\pi}(R_g \cap D_g \cap X_g \cap F_g), \tag{56}$$

the same task-relative envelope form as (25), restated here at the world-system level rather than the witnessed-set level. A delayed unaided return profile is likewise restated at this level (After Labour eq. 35; its four coordinates parallel H_{return} in (41) but are kept under a different symbol, since After Labour does not assert the two instruments are identical):

$$R_{H,t}^{\text{return}} = \langle C_t, T_t, S_t^{\text{skill}}, A_t^{\text{alt}} \rangle. \tag{57}$$

These feed a typed audit index, the Human Systemic Position (After Labour eq. 40):

$$P_t^H = \frac{(\Gamma_t^{\text{eff}})^{\theta_\Gamma} (A_{H,t}^{\text{corr}})^{\theta_A} (\Lambda_{H,t}^{\text{live}})^{\theta_\Lambda} (r_t^H)^{\theta_R} (S_t^H)^{\theta_S} (X_t^H)^{\theta_X}}{(1 + D_t^H)^{\theta_D}}, \tag{58}$$

which After Labour states explicitly is not a welfare utility function or a validated cardinal scale but a typed audit index, guarding against output, consumption, or transfer income silently standing in for human position. Its primary non-collapse result (After Labour eq. 43) is

$$\dot{Y}_t > 0 \;\not\Rightarrow\; \dot{P}_t^H > 0. \tag{59}$$

[Lahtee, 2026m]

The Human Conversion Imperative generalizes the same worry into one framing inequality (HCI eq. 1):

$$\text{Machine expansion} \;\not\Rightarrow\; \text{Human expansion}, \tag{60}$$

and replaces a scalar augmentation score with a Human Conversion Vector [DEFINITION] (HCI eq. 11):

$$\begin{aligned} C_t^H = \big\langle\, &\Gamma_t^{\text{eff}},\, X_t^H,\, \Lambda_{H,t}^{\text{live}},\, A_{H,t}^{\text{corr}},\, R_{H,t}^{\text{route}}, \\ &W_{H,t}^{\text{world}},\, r_{H,t},\, H_t^{\text{cap}},\, S_{H,t} \,\big\rangle, \end{aligned} \tag{61}$$

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with a local elasticity for each dimension j against a declared machine-capability measure M_t [DEFINITION] (HCI eq. 12):

$$\eta_{j,t}^{HC} = \frac{\partial \ln C_{j,t}^{H}}{\partial \ln M_t}, \tag{62}$$

which permits mixed trajectories in which material claim rises while retained competence or exit fall. Because adoption can display increasing returns and path dependence, HCI defines a Reversibility Window for each dimension j [DEFINITION] (HCI eq. 17):

$$W_j = \{t : C_{j,t}^{\mathrm{rec}} \leq \bar{C}_j \wedge \tau_{j,t}^{\mathrm{rec}} \leq \bar{\tau}_j\}, \tag{63}$$

and an Urgency term combining the machine/human growth gap with lock-in, severity, and recovery time [DEFINITION] (HCI eq. 19):

$$U_{j,t} = \Delta g_{j,t} L_{j,t} S_{j,t} \tau_{j,t}^{\mathrm{rec}}. \tag{64}$$

[Lahtee, 2026n]

HCI's Reversibility Principle (its Proposition 3) is explicitly OPEN: the faster machine capability grows relative to human conversion, the more valuable it becomes to preserve reversible human and institutional alternatives before the recovery cost of lost routes rises. Neither source paper claims that P_t^H, C_t^H, $\eta_{j,t}^{HC}$, W_j, or $U_{j,t}$ is a validated causal or welfare measure; both state that weights, elasticities, and thresholds must be declared and estimated per study, and that the architecture should be narrowed or removed if it fails to discriminate outcomes beyond ordinary productivity, wage, or skill measures.

3.16 Barrier Readout, Endorsement, Scaffold Fading, and Opportunity Conversion

From Assistance to Human Capability (RG-HCA) proposes a proactive extension of the Dialogue Conversion Protocol tail already in this river (Sections 3.13 and 3.14): once a live problem is human-owned, AI need not remain purely reactive. RG-HCA is explicit that this is a candidate domain translation, not a closed human science, and keeps the constitutional non-collapse $S_n \neq Z_{\mathrm{HCA},n} \neq D_{\mathrm{HCA},n}$ (RG-HCA eq. 4). Its own native river (RG-HCA eq. 5) is

$$\begin{aligned} \text{Retained Difference} &\to \text{Human Readout} \\ &\to \text{Live Problem} \\ &\to \text{Barrier Readout} \\ &\to \text{Candidate Routes} \\ &\to \text{Human Endorsement} \\ &\to \text{Adaptive Scaffold} \\ &\to \text{Practice} \\ &\to \text{Withdrawal} \\ &\to \text{Human Return} \\ &\to \text{Novel Transfer}^{1} \\ &\to \text{World Feedback} \\ &\to \text{Opportunity Conversion.} \end{aligned} \tag{67}$$

[Lahtee, 2026p]

Read against (51), (67) inserts a Barrier Readout stage before Candidate Routes and appends Opportunity Conversion after World Feedback; it is a proactive elaboration of the same Live-Problem-to-Revision cycle, not a replacement of it.

Two agents given the same AI do not meet on a blank plane. RG-HCA bookkeeps this as a Life-Capital Context Vector [Definition] (RG-HCA eq. 8):

$$\begin{aligned} K_{i,n}^{\mathrm{life}} = \langle & E^{\mathrm{econ}}, F^{\mathrm{base}}, L^{\mathrm{lang}}, D^{\mathrm{digital}}, T^{\mathrm{disc}}, \\ & H^{\mathrm{health}}, M^{\mathrm{mob}}, N^{\mathrm{mentor}}, C^{\mathrm{cred}} \rangle_{i,n}, \end{aligned} \tag{68}$$

[Lahtee, 2026p]

denoting economic resources, foundational literacy/numeracy access, language bridge, device/connectivity, discretionary time, health/nutrition/sleep conditions, mobility/safety, mentor/network access, and credential pathway; RG-HCA states this is not a universal cardinal measure of human capital and does not exhaust life conditions.

The proactive step begins only after the problem is human-owned, and inserts a barrier-typing stage before any route or training prescription. RG-HCA's typed barrier ledger (RG-HCA eq. 12) is

$$\begin{aligned} B_{i,n}^{\mathrm{bar}} \subseteq \{ & \mathrm{Knowledge}, \mathrm{Skill}, \mathrm{Language}, \mathrm{Tool}, \\ & \mathrm{ResourceTime}, \mathrm{Network}, \mathrm{Credential}, \\ & \mathrm{Permission}, \mathrm{Opportunity}, \mathrm{Unknown}\}, \end{aligned} \tag{69}$$

[Lahtee, 2026p]

and its load-bearing non-collapse [Definition] (RG-HCA eq. 13) is

$$\text{Observed Difficulty} \neq \text{Skill Deficit}. \tag{70}$$

[Lahtee, 2026p]

[1] Stage label quoted verbatim from RG-HCA's own river (its eq. 5), as in v1.4; it names a stage of that source's pipeline and is not a claim of this document about priority.

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RG-HCA gives concrete costs of a wrong diagnosis: repeated skill drills punish a translation problem when the barrier is language, and more private competence stays invisible to an employer when the barrier is a missing credential.

RG-HCA separates candidate routes from endorsed routes so that proactive suggestion never silently becomes the person's own goal. The endorsed subset [Definition] (RG-HCA eq. 16) is

$$C_{i,n}^{\text{live}} = \{c \in C_{i,n}^{\text{cand}} : \ \text{Endorse}_i(c) = 1\}, \tag{71}$$

[Lahtee, 2026p]

and three further non-collapse laws follow [Definition] (RG-HCA eq. 17–19):

$$\begin{gathered}\text{Proactive Suggestion} \neq \text{Human Goal Ownership},\\ \text{Capability Advancement} \neq \text{AI Goal Authority},\\ \text{Scaffolding} \neq \text{Control}.\end{gathered} \tag{72}$$

[Lahtee, 2026p]

Once a route is endorsed and scaffolded, RG-HCA does not prescribe a fixed amount of assistance. Its candidate fading rule [OPEN] (RG-HCA eq. 22) is

$$\text{Stable Unaided Return} \uparrow \Rightarrow \ h^{\text{decisive}} \downarrow, \tag{73}$$

[Lahtee, 2026p]

unless task difficulty, stakes, or domain risk rises; RG-HCA marks this rule Open and states it should be replaced if a simpler rule performs as well.

RG-HCA's own Human Return reuse (its eq. 24–25) restates $R_H^{\text{return}} = \langle C, T, S, A \rangle$ and the non-collapse $\Delta\text{Performance}_{\text{AI}} > 0 \not\Rightarrow \Delta R_H^{\text{return}} > 0$. RG-HCA cites the same CTSA v9.1 Human Return source already cited at this river's H_{return} (41) and its non-collapse (42) (Section 3.12), but decomposes it as a four-field tuple $\langle C, T, S, A \rangle$ with a differently-worded non-collapse, rather than the six-field $\langle G_{\text{CTSA}}, L, M, P, W, \Delta_{\text{dir}} \rangle$ of (41). The underlying evidentiary standard – unaided reconstruction, selection, execution/transfer, or generation under a declared criterion – is the same shared lineage; the two pairs of equations are not identical, so RG-HCA's eq. 24–25 are described here in prose rather than given separate boxed equation numbers.

After Human Return, RG-HCA does not treat verbal reconstruction as closing an action-relevant problem, restating the same world-closure step as (50) over RG-HCA's own candidate HCA state Z_{HCA} (RG-HCA eq. 27):

$$R_{H,i,n}^{\text{return}} \rightarrow a_{i,n} \rightarrow \delta_{i,n+1}^{\text{world}} \rightarrow Z_{\text{HCA},i,n+1}. \tag{74}$$

[Lahtee, 2026p]

A further boundary appears after Human Return: the person may now be capable yet still lack a viable social or economic route. RG-HCA defines a world-side opportunity readout (RG-HCA eq. 28)

$$\begin{aligned}\Omega_{i,n}^{\text{real}} = G_O\big(&R_{H,i,n}^{\text{return}}, K_{i,n}^{\text{life}},\\ &\text{Cred}_{i,n}, \text{Net}_{i,n}, \text{Perm}_{i,n},\\ &\text{MarketReadout}_n\big),\end{aligned} \tag{75}$$

[Lahtee, 2026p]

explicitly not a validated welfare utility function, and two further non-collapses are essential [Definition] (RG-HCA eq. 29–30):

$$\begin{gathered}\text{Credential} \neq \text{Capability},\\ \text{Market Legibility} \neq \text{Human Worth}.\end{gathered} \tag{76}$$

[Lahtee, 2026p]

RG-HCA uses Spence's signalling model only to motivate this legibility gap: capability that is not represented by a recognizable credential, portfolio, demonstration, reference, or network may fail to become opportunity, so the remedy can be an opportunity bridge rather than more skill training.

A full evaluation must separate immediate assisted performance, Human Return, transfer, autonomy, burden, and opportunity conversion rather than collapsing them into current task success. RG-HCA's net advancement record [Definition] (RG-HCA eq. 36) is

$$\begin{aligned}A_i^{\text{HCA}} = \big\langle &\text{Gain}_{CTSA}, \text{Loss}, \text{Transfer},\\ &\text{Ownership}, \text{Burden}, \text{BarrierChange},\\ &\text{OpportunityChange}, \text{Provenance}, \text{Warrant}\big\rangle,\end{aligned} \tag{77}$$

[Lahtee, 2026p]

and for policy comparison under unequal starting conditions RG-HCA replaces a raw outcome contrast with a conditional estimand [Definition, measurement] (RG-HCA eq. 37):

$$\text{ATE}_{\text{HCA}}(x) = \mathbb{E}\big[Y(1) - Y(0) \mid K^{\text{life}} = x\big], \tag{78}$$

[Lahtee, 2026p]

where x denotes a declared baseline life-context stratum; RG-HCA notes that matching or covariate adjustment is a design strategy, not part of the estimand itself.

Human Capability Advancement Principle [OPEN], restated from RG-HCA. When durable human development is an endorsed goal, AI may proactively identify and propose a reachable capability or opportunity expansion, but it should remain subordinate to human-owned ends, prefer the least burdensome intervention likely to produce durable return, and withdraw decisive support as the person demonstrates independent capability [Lahtee, 2026p].

None of (67)–(78) replaces or reweights (44)–(51); RG-HCA is read here as a proactive layer that a completed river may attach at the human-owned-problem and Human-Return boundaries already fixed in Sections 3.12 to 3.14, carrying its own OPEN status where RG-HCA itself marks it Open and no stronger tier than [DEFINITION] where RG-HCA marks a bookkeeping object.

4 CANONICAL NOTATION CORRECTIONS BEFORE FREEZING MASTER

The audit found five points that must be corrected from the local notation of each individual paper before combining them into the canonical programme equation.

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4.1 κ Collides Across Three Meanings

In MEMK, κ_n means meaning strength; in From Problem to Hypothesis, $\kappa_t(e \mid Q)$ means semantic accessibility; in Choice Begins Before Choice, $\kappa_{A,t}(\pi \mid g)$ means the practical accessibility of a policy. Master should use one consistent rename,

$$\begin{aligned} \gamma_n^{\mu} &= \text{meaning strength}, \\ \kappa_{A,t}^{\text{sem}}(e \mid Q) &= \text{semantic-route accessibility}, \\ \lambda_{A,t}^{\text{live}}(\pi \mid g) &= \text{live-policy accessibility}. \end{aligned} \tag{43}$$

4.2 Rhythm Does Not Automatically Derive Momentum

Rhythm and momentum are parallel history descriptors. Master should therefore write $\text{Rhythm}_t, m_t, a_t, c_t \to \kappa_{t+1}^{\text{sem}}$ as a dependency/hypothesis interface, rather than writing Rhythm $\Rightarrow$ Momentum.

4.3 The η_s Double-Gating: From a Draft, Not From v8.1

Fusion's conversation-centered draft (v3, not deposited) writes $\Delta\Omega_s^H \approx \eta_s B_s A_s$ and then updates state with $+\eta_s \Delta\Omega_s^H$ again; read literally this produces η_s^2. The deposited record (v8.1) no longer contains this rank-update formula and instead chooses to measure return via (33)–(34). Master's proposal (35)–(36) is therefore a proposed canonical form for the programme, not a correction of v8.1's own text.

Corrected from v1.0: v1.0 attributed this formula to "Fusion/Tunnel v7" and called it a canonical correction of the source; an independent reviewer searched the v8.1 text and did not find this formula (searches for LoRA / rank / B_s / A_s / double $= 0$), so it has been reassigned the status above.

4.4 The Symbol R Collides Across Several Roles

The programme previously used R for the read condition, Resistance, Resonance, the Rhythm descriptor, and Human Return (R_H in CTSA). Master uses clearly typed names: Read, Resist, Reson, Rhythm, and H_{return} (renamed from CTSA's R_H; the content is identical).

4.5 $\Pi^{\text{feas}} \to \Pi^{\text{wit}}$ in the Potential Equation

Per *Potential as a Readout*, the set over which the maximum is taken must be a finite set with witnesses under a declared comparability relation (Section 3.8); Π^{feas} remains valid in Choice's sense (structurally feasible) but is not the domain of max in (25).

5 THE CORRECTED MASTER EQUATION RIVER

After fixing the symbol collisions and role separations, the canonical river that is most faithful to the internal literature is

$$\begin{aligned} & x_t \xrightarrow[\text{Genesis}]{R_H} r_t \xrightarrow[\text{Experience}]{\Psi_H} \mu_t \xrightarrow[\text{MEMK}]{\Phi_E} E_t \\ & \xrightarrow[\text{Human LoRA}]{\text{Retain}/U_H} H_{t+1} \\ & \xrightarrow[\text{Resonance, Rhythm, Attraction/Momentum}]{} \lambda_{A,t+1}^{\text{live}}(\pi \mid g) \\ & \to L_{A,t+1}(g) \to \Pi_{A,t+1}^{\text{live}}(g) \xrightarrow[\text{Choice Before Choice}]{} \pi^{\text{choice}} \\ & \xrightarrow[\text{Potential as a Readout}]{\text{Read}, D, X, F} \pi^{\text{act}} \to \text{feedback/correction} \to H_{t+2} \\ & \xrightarrow[\text{Pre-Prompt State}]{L_H} Q_{t+2} \to \text{AI} \to K_{\text{like}} \\ & \xrightarrow[\text{Fusion/Tunnel}]{\text{Difference + Resistance}, \eta_s} \text{recursive dialogue} \\ & \to \text{retained human residue} \xrightarrow[\text{CTSA}]{\text{AI removed}} H_{\text{return}} \to \text{Epistemic Direction}. \end{aligned} \tag{44}$$

The important point is that equation (44) is not a claim that every arrow carries the same evidential status. Genesis/MEMK supply architectural order; resonance/rhythm/live-field are domain/Dr/Open constructs; Potential as a Readout is a task-relative measurement architecture (definitions at the definition level, lemmas on a proven discrete model, and empirical proposals that remain open); Fusion/Tunnel carries both constitutive and open diagnostics; CTSA is a session-boundary measurement interface.

Per the Dialogue Conversion Protocol's closure (Section 3.14), the chain in (44) continues past H_{return} rather than terminating there; this is an appended tail, not a rewriting of (44) itself, which still ends at Human Return / Epistemic Direction exactly as corrected above:

$$\cdots \to H_{\text{return}} \to a_s \to \delta^{\text{world}} \to H_{t+3}. \tag{65}$$

After Labour and the Human Conversion Imperative (Section 3.15) do not add a further arrow inside this cycle; they sit as an outer loop that reads the cycle's own state at each turn. Writing $t \to t+3$ for one pass of (44)–(65), the outer loop is schematically

$$\begin{aligned} & \left(H_t \xrightarrow[(44),(65)]{} H_{t+3}\right) \xrightarrow[\text{Readout}_{Q,O,c}]{} C_{t+3}^H \to P_{t+3}^H \\ & \qquad \xrightarrow[\Gamma^{\text{eff}}, \Pi^{\text{live}}, A^{\text{corr}}, R^{\text{return}}]{} \left(H_{t+3} \xrightarrow[(44),(65)]{} H_{t+6}\right). \end{aligned} \tag{66}$$

Equation (66) is a schematic placement, not a new derivation: it records that After Labour's P_t^H and HCI's C_t^H are readouts of successive human-return cycles, and that the world-system coordinates they track (effective material claim, live possibility, corrigible agency, Human Return, social standing) condition the next cycle's live field and agency envelope in (44) without replacing anything inside it. The non-collapse of Section 3.15 applies at this outer scale too: a cycle that raises assisted performance or aggregate output does not by itself raise P_t^H ((59)).

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RG-HCA's barrier and opportunity layer (Section 3.16) sits inside this same outer loop, between the Human-Return tail H_{t+3} of (65) and the world-system readout that opens (66), restating (74)–(75) at the scale of one full pass:

$$H_{t+3} \xrightarrow[(69),(71)]{\text{Barrier, Endorsement}} Z_{\text{HCA},t+3} \xrightarrow[(75)]{\Omega^{\text{real}}} \left(C^{H}_{t+3} \to P^{H}_{t+3}\right). \tag{79}$$

Equation (79) is likewise a schematic placement: it records that RG-HCA's barrier ledger, endorsement, and opportunity readout condition what of H_{t+3} becomes the world-system state C^{H}_{t+3} already read in (66), without adding a further arrow inside (44)–(65) or reweighting (66) itself.

6 THE EVOLUTION OF QUESTIONS IN THE CORPUS

1. **When AI Expands Human Potential** (March 2026) [Lahtee, 2026a]: asks when AI expands human potential, and answers normatively that it must add corrigibility, world-responsiveness, route-openness, and disciplined revision — not merely productivity or fluency.
2. **Human Learning as Epistemic Architecture** (June 2026) [Lahtee, 2026b]: shifts the centre to human learning: lived words, meaning, relation, retrieval, constraint, revision, and competence feedback.
3. **Experience Is the Human LoRA** (July 2026) [Lahtee, 2026c]: asks what from experience persists, and proposes selective retention/operator update, while stressing that experience itself is not a literal LoRA update.
4. **Readout Genesis Standalone Synthesis / MEMK** (July 2026) [Lahtee, 2026d]: builds a typed domain architecture of meaning–experience–memory–knowledge, and adds semantic potential / informational work / barrier crossing together with a claim firewall.
5. **From Problem to Hypothesis** (September 2026) [Lahtee, 2026e]: asks why some hypotheses/routes are more accessible than others, and separates reachability from accessibility by adding attraction and recent-path momentum.
6. **Epistemic Fusion or Tunnel** v7.1 → v8.1 (September 2026) [Lahtee, 2026f]: asks how Human and AI change each other's route within a session and what survives the session boundary; adds candidate status, D-R-A, amplification, retention (Repair 7: measuring return), and direction.
7. **Meaning Before Naming → Experience Is Meaning-Giving** [Lahtee, 2026g,h]: asks what happens before explicit naming/prompting; ultimately formalizes experience as meaning-giving, resonance as external–internal congruence, rhythm as structured recurrence, and transition readiness.
8. **Agency Potential → Potential as a Readout** (September 2026) [Lahtee, 2026i]: asks how the potential of agency can be measured without collapsing potential into observed performance; gives the R–D–X–F envelope over a two-layer witnessed set, with a lemma on bounded suppression and an identifiability certificate.
9. **Choice Begins Before Choice** [Lahtee, 2026j]: finds the missing interface between feasible capacity and overt choice, and sets up the Live Possibility Field as a dynamic pre-choice object.
10. **CTSA Human Return** [Lahtee, 2026k]: asks, at the far end, what genuinely comes back with the human once AI has been removed, and what gain/loss/regulation/provenance/warrant/direction that return has.
11. **After Labour** (6 September 2026) [Lahtee, 2026m]: asks what happens to human systemic position, not merely to employment, once productive intelligence and embodied production become reproducible through capital; adds labour centrality, the Citizen Claim Threshold, ownership accumulation, scarce-asset rent burden, aggregate-demand realization, social reproduction, and a typed Human Systemic Position audit index.
12. **The Human Conversion Imperative** (6 September 2026) [Lahtee, 2026n]: asks whether machine-capability growth is being converted into durable human capability rather than merely substituted for it; adds the Human Conversion Vector, conversion elasticities, a Bad-Mode state space, the Reversibility Window, and an Urgency Vector.
13. **How Humans Should Converse with AI** (Dialogue Conversion Protocol, v1.3, 6 September 2026) [Lahtee, 2026o]: asks which topic deserves to become the subject of a human-AI dialogue and who owns that choice before the eight-stage protocol begins; adds the Topic Entry Condition, the Problem-First Dialogue Principle, and closes the protocol through Action and World Feedback.
14. **From Assistance to Human Capability** (RG-HCA, v1.1 glosa-checked edition of the 6 September 2026 v1.0 original) [Lahtee, 2026p]: asks under what architecture AI may proactively advance human capability while preserving human-owned ends, distinguishing structural barriers from personal deficits, and separating assisted performance from durable Human Return; adds the Barrier Readout stage, endorsed-route non-collapse laws, a monotone assistance-fading rule, a world-side opportunity readout, and a conditional advancement estimand.

A relayed external reader's assessment of the Dialogue Conversion Protocol against world literature (2026-09-06, unverified: relayed from another AI session in chat, none of its own citations independently checked here) is recorded only as a readout, not adopted as validation. It placed the protocol's conceptual architecture and distinctiveness near the top of its own unvalidated scale while rating empirical status low pending experimental data, and it named several external comparanda (mixed-initiative interaction, problem framing, goal-centred collaboration, human–AI synergy meta-analysis, cognitive forcing, and AI-and-learning field experiments) as same/different/cited candidates for a future citation-lineage check, not as confirmed priors.

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7 EPISTEMIC STATUS OF EACH SEGMENT

Table 2: Epistemic status of each segment of the river.

Level	What belongs at this level
Root / inherited architecture	Retained difference before semantic naming; ordered lineage; translation/readout/loss discipline; claims cannot borrow certainty.
Domain definitions	Meaning, experience, memory, semantic potential, informational work, live possibility set/profile, witnessed option set under a declared domain.
DR conceptual theses	Experience as meaning-giving; prompt as bounded articulation; choice downstream of live possibility; agency as a corrigible readout–action loop; two-layer potential (open conjecture).
OPEN empirical hypotheses	Incremental value of resonance; rhythm beyond exposure count; recent-path momentum; history-shaped accessibility; barrier/transition topology; Human–AI history effects beyond the current prompt.
Finite diagnostics / proved on a stated model	χ_{recip}, frozen session descendant counts, declared accessibility profiles, study-specific thresholds; bounded-suppression lemmas L1–L4 and the battery of *Potential as a Readout* (exact algebra on a declared model, not a theory about society).
Measurement architecture	Potential as a Readout, recoverable gaps, Human Return, CTSA, Y^{return}/RET, gain/loss/provenance/warrant/direction ledgers.
Normative/constitutive conditions	Difference, Resistance, retained human Agency under the definition of durable epistemic expansion; structural deprivation requires causal/counterfactual/normative support; NC-79 separates diagnosis from attribution.
New in v1.3 (After Labour, the Human Conversion Imperative, and the Dialogue Conversion Protocol)	
Domain definitions	Topic Entry (LiveProblem/OpenExploration/RoutineDelegation, (47)); the self-contained readout $\mathrm{Readout}_{Q,O,c}(S)$ ((52)); four-dimensional labour centrality L_t ((53)); the Human Conversion Vector C_t^H ((61)); the Reversibility Window W_j ((63)); the Urgency Vector $U_{j,t}$ ((64)).
Accounting identities	The Citizen Claim Threshold $q_t^{\min}$ ((54)); the Human Systemic Position audit index P_t^H ((58)) and its non-collapse $\dot{Y}_t > 0 \not\Rightarrow \dot{P}_t^H > 0$ ((59)); the Human Return $\to$ Action $\to$ World Feedback closure ((50)–(51)) — bookkeeping/definitional results inside a declared stylized model, not validated causal or welfare laws.
OPEN empirical hypotheses	The Problem-First Dialogue Principle ((48)) and its non-collapse ((49)); the World-closure proposition; HCI's Reversibility Principle (underlying (63)–(64)); HCI's Conversion non-identity, Assistance-conversion divergence, and Distributional conversion propositions — all tagged OPEN in their own source papers, none upgraded here.
New in v1.4 (RG-HCA)	
Domain definitions	The Life-Capital Context Vector K^{life} ((68)); the typed barrier ledger B^{bar} ((69)); the endorsed-route set C^{live} ((71)); the world-side opportunity readout Ω^{real} ((75)).
Accounting identities / non-collapse laws	Observed Difficulty $\neq$ Skill Deficit ((70)); Proactive Suggestion $\neq$ Human Goal Ownership, Capability Advancement $\neq$ AI Goal Authority, Scaffolding $\neq$ Control ((72)); Credential $\neq$ Capability, Market Legibility $\neq$ Human Worth ((76)) — definitional guards inside a declared candidate domain, not validated psychological laws.
OPEN empirical hypotheses	RG-HCA's assistance-fading rule tying h^{decisive} to Stable Unaided Return ((73)); the Human Capability Advancement Principle — both explicitly OPEN in RG-HCA, neither upgraded here.
Measurement architecture	The net advancement record A_i^{HCA} ((77)) and the conditional estimand $\mathrm{ATE}_{\mathrm{HCA}}(x)$ ((78)), both RG-HCA's own [DEFINITION, MEASUREMENT] objects.

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8 Conclusion: What Master Makes Visible

After reviewing the whole corpus, what the Master Equation River reveals is not that every paper says the same thing, but that each paper fills a missing layer of the same process. The overall picture is

Experience changes the reader;
the changed reader changes what can become possible;
what becomes possible changes choice;
choice changes action and the world;
the changed world is read again.
(45)

So the Human–AI Readout Programme, at this point, can be read as a theory programme of recursive human world-readout, meaning, retention, possibility, agency, and return. The point that must be strictly maintained is non-collapse: accessibility is not warrant, resonance is not truth, rhythm is not meaning, retention is not improvement, potential is not performance, observation is not capacity, and AI-assisted success is not Human Return.

Before freezing Master as a canonical paper, three further things should be done: (1) use the single canonical notation of this document (Section 4); (2) build a claim-status ledger for every arrow in equation (44); and (3) design an empirical programme that tests the incremental value of history, resonance, rhythm, the live field, and Human Return against simpler rival models.

9 Glosa Check Record (K0)

Version v1.0 was checked under the glosa methodology [Lahtee, 2026l] before deposit: an independent reviewer (an AI session different from the drafting session) read against all 11 original records and found 3 issues that blocked deposit — Fusion was cited as v7 while the deposited record is v8.1, and the retention-gate formula is not in v8.1; Agency Potential was cited as a primary source even though it has been superseded and was not deposited; RG-MEMK was cited as the source of (4) even though the equation is in the deposited record's own eq. 14 — plus items requiring correction (the R_H rename was not disclosed, there was no glossary, and DOIs were missing). All of these have been corrected in this edition. For every source cited by the table in Section 2, there is a citation card recording the URL actually fetched, the page, and one exact passage; each card was re-read by a headless session of the same model (independence class I1 on the glosa ladder, not I3), and the document's central claim (that the dependency closure is a spiral) has one claim card that was read by three adversarial roles (Falsifier, Hostile Reviewer, Source Auditor). The cards and reports live in the glosa repository (`records/lit/master-equation-river`, `projects/GLS-2026-003`). Knowledge status remains K0: timestamped, citable, not yet checked at independence class I2 or above.

AI-assistance disclosure. The structuring, source retrieval, passage extraction, and drafting of v1.1 were done together with an AI assistant under the author's direction; the problem, the questions, the choice of proposals, and the content of v1.0 that anchors this version belong to the author. No AI or vendor is an author. The author's own command lines are in the Blackbox Log (concept DOI 10.5281/zenodo.22302518, entries BBL-2026-09-06-140 to -142).

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10 The One Equation Along the Whole Line

Since v1.4 was deposited, the programme's whole equation corpus (946 numbered equations across 40 deposited chapters, this document's own 79 among them, together with 123 further occurrences from the Readout Genesis domain registries, the Philosophy & Logic textbook and the solver arc: 1,069 mapped occurrences and 793 canonical entries in Toledo v1.0.0, counted from its registry file at the time of writing) has been coded into **Toledo** [Lahtee, 2026q], a permanent public registry in which every equation is either the Genesis root object itself or a coded reading of that root through a domain q_D. This section states the single root equation Toledo and *Collapse of the Master River* (`registry/COLLAPSE.md`, the programme's own mapping pass over Toledo's registry) read this whole river as an instance of, then reports which of this document's own segments already carry a Toledo code and which do not — adding no new claim and raising no tier above what `registry/CANONICAL.json` itself records.

10.1 The Weld

$$\delta_R = (a\sharp b) \vdash_{Th_{coqc}} L_R = D_W - W \vdash_{Dr} F \text{ (MQ.08 stepper)} \equiv \{q_D : q_D \circ F = F_D^{\#} \circ q_D\}.$$

This equation carries no number of its own — it is not a new claim added to (1)–(79), only the same root spine (CAN-002, CAN-201, CAN-001, CAN-003, CAN-006, CAN-007, CAN-008, CAN-004, CAN-009, below) restated once in the form `COLLAPSE.md` § 0 already gives it. Stating the master equation and admitting a domain are the same act: a retained distinction δ_R forces, under TH_COQC (machine-checked in a finite model), the graph Laplacian L_R and, under DR (forced-but-not-independently-verified), the finite stepper F; a domain is nothing more than a translation q_D that this same stepper commutes with.

Nine coded objects in `CANONICAL.json` (ids CAN-002, CAN-201, CAN-001, CAN-003, CAN-006, CAN-007, CAN-008, CAN-004, CAN-009) together state this weld and the order in which it is read (world → retained state → readout → weld/retention → stepper → domain admission → quotient → non-collapse guard → semantic ordering → historical closure, then the next world-state). **Their own codes changed between the pass that produced `COLLAPSE.md` and the registry as it stands today**: `COLLAPSE.md` recorded all nine as `domain: "root"`; Toledo's current N3/N4 coding instead resolves each to a method-domain (M) reading of one of five literal Genesis root objects (`weld`, `EQ-015`, `EQ-002`, `A.5`, `A.8`) — the bare root itself is never a coded entry, only its readings are. Table 3 gives the current code for each, read directly from `CANONICAL.json`; no tier is raised in the re-coding, and CAN-001's own tier (TH_COQC for $\delta_R \vdash L_R$ only, DR for the F-stepper) is unchanged.

Table 3: The nine-element root spine (`COLLAPSE.md` § 1), read against its current Toledo code.

CAN id	Key	Toledo code	Root	Tier	Status
CAN-002	`root-state-tuple`	`EQ-015/M.01.v1`	EQ-015	Definition	current
CAN-201	`root-readout-gate`	`EQ-002/M.03.v1`	EQ-002	Definition	current
CAN-001	`root-weld`	`weld/M.01.v1`	weld	Th_coqc / Dr	current
CAN-003	`root-stepper`	`EQ-015/M.02.v1`	EQ-015	untagged	current
CAN-006	`domain-weld`	`weld/M.02.v1`	weld	Definition	current
CAN-007	`reader-equivalence`	`weld/M.03.v1`	weld	Definition	current
CAN-008	`constitutional-noncollapse`	`A.5/M.01.v1`	A.5	Definition	current
CAN-004	`constitutional-ordering`	`EQ-015/M.03.v1`	EQ-015	Dr	current
CAN-009	`historical-invariance`	`A.8/M.01.v1`	A.8	Dr	current

10.2 Reading Table: Root × Domain

Table 4 counts, for each of the five literal roots underlying the spine above, how many canonical Toledo entries currently read that root through each of Toledo's eight domain letters (E epistemic, H human–AI, S social, W world-system, M method, P physics, C chemistry, B biology/health — `toledo/README.md`). A count of zero (shown —) is reported as exactly that: no code yet, not a gap this document fills. `weld`, the Social-Instability/Peace spine's own root (CAN-115, social reading of `weld`), has no chemistry, physics, or biology reading; `EQ-002` (the readout gate) has no social or biology reading; only `EQ-015` is read into physics (117) or biology (9), because that root is Genesis's own general stepper object, reused far outside this river's own social-justice-peace scope — reported here because it is true of the root this spine cites, not because Master River itself contains a physics or biology segment.

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Table 4: Root × domain: count of Toledo canonical entries reading each literal root through each domain letter.

Root	E	H	S	W	M	P	C	B
`weld`	10	12	50	2	14	—	—	—
`EQ-015`	15	37	65	59	15	117	—	9
`EQ-002`	12	4	—	3	3	—	—	—
`A.5`	9	19	20	20	32	—	—	—
`A.8`	3	4	1	—	19	—	—	—
Total	**49**	**76**	**136**	**84**	**83**	**117**	**0**	**9**

10.3 Which Root Each Segment of This River Reads

Read against `CANONICAL.json`'s own per-equation index (its `_n3_in_master_river` field, which names, for 55 of this document's 79 equations, the canonical code that equation is an occurrence of), the segments of Sections 3 to 5 and 8 read the five roots above as follows. Where no equation in a segment's range appears in that index, this is stated plainly rather than filled by inference.

- Section 3.1 (readout, meaning, experience; (1)–(5)) reads `EQ-002` ((1)) and `EQ-015` ((2)–(5)).
- Section 3.2 (naming as articulation and restructuring; (6)–(7)) reads `EQ-015`.
- Section 3.3 (retention and the changing future reader; (8)) reads `EQ-015`.
- Section 3.4 (resonance; (9)–(11)) reads `EQ-015`.
- Section 3.5 (rhythm, momentum, accessibility; (12)–(15)) reads `EQ-015` for (12)–(14); (15) has no Toledo code yet.
- Section 3.6 (accumulation, barrier, release; (16)–(18)) reads `EQ-015`.
- Section 3.7 (accessibility to the live possibility field; (19)–(24)) reads `A.5` throughout, with (21) also carrying an `EQ-015` reading.
- Section 3.8 (potential as a readout; (25)) reads `EQ-015`.
- Section 3.9 (power before choice, recoverable live-field gap; (26)) reads `EQ-015`.
- Section 3.10 (human–AI coupling before the prompt; (27)–(30)) reads `EQ-015` throughout, with (27) also carrying an `A.5` reading.
- Section 3.11 (within-session amplification, retention, direction; (31)–(40)) reads `EQ-015` at (31) and (38) only; (32)–(37) and (39)–(40) have no Toledo code yet.
- Section 3.12 (Human Return and CTSA; (41)–(42)) has no Toledo code yet.
- Section 4 (canonical notation corrections; (43)) has no Toledo code yet.
- Section 3.13 (topic entry and agenda ownership; (46)–(49)) has no Toledo code yet.
- Section 3.14 (Human Return, action, and world feedback; (50)–(51)) has no Toledo code yet.
- Section 3.15 (the world-system layer; (53)–(64)) reads `EQ-015` throughout; (52) has no Toledo code yet.
- Section 3.16 (barrier readout, endorsement, scaffold fading, opportunity conversion; (67)–(78)) reads `EQ-002` ((67)), `EQ-015` ((68), (73)–(74)), `A.5` ((69)–(72), (75)–(76)), and `A.8` ((77)–(78)).
- Sections 5 and 8 (the river-summary arrows (44), (45), (65), (66), and (79)) carry no code in `CANONICAL.json`'s per-equation index. `COLLAPSE.md` § 2 separately identifies (44) as a human–AI reading of CAN-004 and CAN-003 (now `EQ-015/M.03.v1` and `EQ-015/M.02.v1`), (66) as a reading of CAN-201 and CAN-006 (`EQ-002/M.03.v1`, `weld/M.02.v1`), and (79) as a reading of CAN-006 again; this is a hand-checked, second, narrower identification (`COLLAPSE.md`'s own "manual (verified textual match)" evidence tier) that the automated per-equation index does not yet carry, and both readings are reported here rather than one silently preferred over the other.

Of the nineteen segments listed above, **thirteen** carry at least one Toledo code (three of those thirteen — Sections 3.5, 3.11 and 3.15 — only for part of their own equation range) and **six** carry none in `CANONICAL.json`'s per-equation index: Section 3.12 (Human Return and CTSA), Section 4 (canonical notation corrections), Section 3.13 (topic entry), Section 3.14 (Human Return, action, and world feedback), Section 5 (the corrected master equation river), and Section 8 (the conclusion) — though, as noted above, `COLLAPSE.md` separately hand-checks three of Section 5's five river-summary equations against the root spine by its own narrower method. Appendix C gives the same reading equation-by-equation, for all 79.

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APPENDIX A. SYMBOL GLOSSARY

Table 5: Symbol glossary.

Symbol	Meaning	Defined in
x_n	Phenomenon / world-side encounter at step n.	[Lahtee, 2026d,h]
$r_n = R_H(\cdot)$	Finite human readout of x_n.	[Lahtee, 2026d]
H_n	Retained human state (the reader's history-bearing operator).	[Lahtee, 2026c,h]
c_n, Q_n	Context; the criterion/question that governs the reading.	[Lahtee, 2026d,e]
μ_n, Ψ_H	The meaning-giving relation and its operator.	[Lahtee, 2026h]
E_n, Φ_E	Experience and the experience operator (MEMK).	[Lahtee, 2026d], eq. 14
γ_n^{μ}	Meaning strength (renamed from MEMK's κ_n).	Section 4.1
ℓ_n, L_H	Lexical naming / articulation operator; L_H is also used as the transport from H_t to Q_t.	[Lahtee, 2026g,h]
U_H, Retain	State-update and selective retention.	[Lahtee, 2026c]
$M_{H,n}$, $I_{H,n}$, C_H	Retained memory organization; retrieved internal organization; congruence measure.	[Lahtee, 2026h]
T_n, B_n, Ω	Encounter history; boundary/structure parameters; rhythm operator.	[Lahtee, 2026h]
m_t, $a_{Q,t}$, $c_{A,t,Q}$, ξ_t	Recent-path momentum; attraction; switching cost; residual.	[Lahtee, 2026e]
κ^{sem}	Semantic-route accessibility.	[Lahtee, 2026e], Section 4.1
V_{sem}, W_{info}, $\Delta V^{\dagger}$	Semantic potential; informational work; activation barrier (not physical energy).	[Lahtee, 2026d,h]
$\Pi^{\text{phys}}, \Pi^{\text{feas}}, \Pi^{\text{live}}, \Pi^{\text{wit}}$	Physically possible / structurally feasible / experientially live / witnessed policy sets.	[Lahtee, 2026j,i]
λ^{live}, $L_{A,t}(g)$, τ_{live}	Live-policy accessibility; the Live Possibility Field; live threshold.	[Lahtee, 2026j]
g, A, z, h, T, B, P	Task; agent; conditions; history; horizon; budget; declared model.	[Lahtee, 2026i]
$\text{Read}_g, D_g, X_g, F_g$	Read / actor-directed decision / causal enactment / feedback-objection events.	[Lahtee, 2026i]
$p^*_{A,g}$, J_{feas}, D_L	The potential envelope; the feasible intervention set; live-field distance.	[Lahtee, 2026i,j]
$Y_{\text{obs}} = \mathscr{O}_q(H_{0:T})$	The evaluator's instrument readout of the trajectory.	[Lahtee, 2026i,j]
Q_t, Y_t, $E^{\text{AI}}_{H,t}$	Prompt; AI output; the human's experience of that output.	[Lahtee, 2026f,h]
Z_{dlg}, u_H, u_{AI}, χ_{recip}, D_{recip}, Σ	Dialogue state; human/AI inputs; reciprocal-lineage ratio; reciprocal descendants; all descendants.	[Lahtee, 2026f]
η_s, Ω^H_s, B_sA_s, d_H	Session retention gate; human operator state; rank-restricted candidate update; state dimension.	[Lahtee, 2026f] (gate); Section 3.11 (proposal)
Y^{return}, RET, AUG, SYN, J^*_s	Unaided return battery; return / augmentation / synergy estimands; three-axis outcome.	[Lahtee, 2026f] Repair 7; [Lahtee, 2026i]
$K_{\text{like}}, K_{\text{validated}}$	Candidate (knowledge-like) vs. validated status.	[Lahtee, 2026a,f]
Δ_s, G_s, T_s, P_H	Expansion–tunnel balance; gain; tunnel; human performance.	[Lahtee, 2026f]
H_{return} ($= R_H$ in CTSA)	Human Return audit record $\langle G_{\text{CTSA}}, L, M, P, W, \Delta_{\text{dir}}\rangle$.	[Lahtee, 2026k], Section 4.4
New in v1.3		
TopicEntry, $P^{\text{live}}_{H,t}$	Topic-selection variable (LiveProblem/OpenExploration/RoutineDelegation); a live problem, a presently consequential unresolved difference.	[Lahtee, 2026o], (47)
I_s, a_s, $\delta^{\text{world}}_{s+1}$	DCP's integration record; human-selected next action; the returned world record.	[Lahtee, 2026o], (50)
$\text{Readout}_{Q,O,c}(S) = z$	After Labour's self-contained readout definition ($z \neq S$); not the root readout R_H used elsewhere in this river.	[Lahtee, 2026m], (52)
L_t	Four-dimensional labour centrality: task, income, bottleneck, bargain.	[Lahtee, 2026m], (53)
$q^{\min}_t$, Γ_t, Γ^{eff}_t	Citizen Claim Threshold; broad claim on output; effective material claim after rent/cross-border burdens.	[Lahtee, 2026m], (54)
$A^{\text{corr}}_{H,i,t}(g)$, $R^{\text{return}}_{H,t}$	World-system-level corrigible-agency envelope; delayed unaided return profile (After Labour's own restatement, kept typographically distinct from H_{return}).	[Lahtee, 2026m], (56)–(57)
P^H_t	Human Systemic Position audit index.	[Lahtee, 2026m], (58)

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Table 5 continued

Symbol	Meaning	Defined in
C_t^H, $\eta_{j,t}^{HC}$, M_t	Human Conversion Vector; local conversion elasticity per dimension; declared machine-capability measure.	[Lahtee, 2026n], (61)–(62)
W_j, $U_{j,t}$	Reversibility Window; Urgency term for dimension j.	[Lahtee, 2026n], (63)–(64)
New in v1.4		
$K_{i,n}^{\text{life}}$	Life-Capital Context Vector: economic, foundational-literacy, language, digital, discretionary-time, health, mobility, mentor/network, and credential coordinates for agent i.	[Lahtee, 2026p], (68)
$B_{i,n}^{\text{bar}}$	Typed barrier ledger (Knowledge, Skill, Language, Tool, ResourceTime, Network, Credential, Permission, Opportunity, Unknown).	[Lahtee, 2026p], (69)
$C_{i,n}^{\text{cand}}$, $C_{i,n}^{\text{live}}$, Endorse_i	Candidate advancement routes; the human-endorsed subset; the endorsement predicate.	[Lahtee, 2026p], (71)
h^{decisive}	Level of decisive AI assistance/scaffolding subject to fading.	[Lahtee, 2026p], (73)
$Z_{\text{HCA},i,n}$	RG-HCA's candidate HCA domain state for agent i at step n (not asserted identical to the retained root state S_n).	[Lahtee, 2026p], (74)
$\Omega_{i,n}^{\text{real}}$, $\text{Cred}_{i,n}$, $\text{Net}_{i,n}$, $\text{Perm}_{i,n}$	World-side opportunity readout and its credential/network/permission arguments.	[Lahtee, 2026p], (75)
A_i^{HCA}, $\text{ATE}_{\text{HCA}}(x)$	Net advancement record; conditional average treatment estimand given a baseline life-context stratum x.	[Lahtee, 2026p], (77)–(78)

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APPENDIX B. TH_COQC LEDGER: COQ FORMALISATION OF EQUATIONS (1)–(79)

Every numbered equation of this document is formalised in the Coq 8.20 development *Master Equation River — Coq formalisation of every equation*, v1.0, Zenodo software record DOI 10.5281/zenodo.22518450 [Lahtee, 2026r], on finite, discrete models only (nat, $\mathbb{Z}$, $\mathbb{Q}$, bool, finite lists and records; no `Coq.Reals`, no classical axioms, no `Admitted`, no top-level `Axiom`/`Parameter`). Each equation sits at exactly one tier. **Th_coqc**: a lemma proved in a stated finite model, reporting *Closed under the global context* under `Print Assumptions` (45 identifiers, 0 failed, `verify.sh`); non-collapse laws are proved by exhibiting a finite model in which the two readouts differ, so the lemma witnesses the possibility of divergence, not a claim about every model. **Definition**: the object is typed as the paper writes it. **Open**: the hypothesis is stated as a `Prop` and not proved. Where a proved lemma is only a narrow witness of a thesis this document keeps at tier Dr (equations 5, 8, 16), the ledger says so; the formalisation checks the internal consistency of the stated discrete models, not empirical truth. Continuum-looking expressions in the text (`exp`, ∂ ln, a maximum over an infinite set) are replaced in Coq by declared discrete counterparts and the replacement is recorded in the source comments.

v1.5 pointer update. The standalone software record above, DOI 10.5281/zenodo.22518450 [Lahtee, 2026r], is **obsoleted by Toledo v1.0.0**, DOI 10.5281/zenodo.22548770 (concept DOI 10.5281/zenodo.22537318) [Lahtee, 2026q]: these equations' Coq development now lives inside Toledo's own repository, one file per canonical code (the file-name mangling is the code itself, e.g. `weld/M.01.v1` $\rightarrow$ `weld__M_01_v1.v`), alongside a machine-checked build+`Print Assumptions` report for the whole registry. This is a location update only: the ledger below (Table 6) is the ledger as v1.4 deposited it, unchanged, and no tier or assumption cell in it is altered by this pointer.

Table 6: Th_coqc ledger (abridged from `MR_Ledger.md` in the Coq record).

Eq.	Module	Identifier(s)	Tier	Assumptions
1	MR_Foundation.v	R_H (Variable, Section Foundation1)	Definition	n/a (Definition, not a proof o
2	MR_Foundation.v	Psi_H (Variable, Section Foundation1)	Definition	—
3	MR_Foundation.v	MeaningModes (Record); meaning_modes_decomposition_faithful	Definition	closed
4	MR_Foundation.v	Phi_E (Variable, Section Foundation1)	Definition	—
5	MR_Foundation.v	eq5_experience_is_phenomenon_and_meaning_jointly	Th_coqc (narrow witness lemma only: a finite model where the compression is jointly sensitive to phenomenon and meaning; the paper's thesis itself stays at its own tier, Dr)	closed
6	MR_Foundation.v	L_H (Variable, Section Naming); witness naming_can_remain_unstable	Definition	closed
7	MR_Foundation.v	NamingChainStep (Record); naming_chain (Fixpoint)	Definition	—
8	MR_Foundation.v	eq8_retention_can_change_the_reader	Th_coqc (narrow witness lemma only: a finite model where retention-with-update is not idle; the paper's thesis stays Dr)	closed
9	MR_Resonance.v	Retrieve (Variable, Section Resonance)	Definition	—
10	MR_Resonance.v	C_H (Variable, Section Resonance)	Definition	—
11	MR_Resonance.v	eq11_resonance_non_collapse (+ notion_value_injective)	Th_coqc	closed
12	MR_Resonance.v	EncounterHistory, Omega (Variable) . . .	Definition	—
13	MR_Resonance.v	momentum (Fixpoint, scaffold); Open_eq13	Open	n/a (Open Prop, not proved by
14	MR_Resonance.v	accessibility_score, disc_gain_nat . . .	Open	n/a (Open Prop, not proved by
15	MR_Resonance.v	eq15_rhythm_alone_does_not_determine_accessibility, eq15_kappa_next_genuinely_varies	Th_coqc	closed
16	MR_Resonance.v	accum_work, accum_work_cons . . .	Th_coqc (bookkeeping facts about the definition only: monotone/non-negative sums over Q; the transition claim of eq. 18 remains Open)	closed
17	MR_Resonance.v	effective_work (Section EffectiveWork)	Definition	—
18	MR_Resonance.v	threshold_crossed, threshold_crossed_decidable; Open_eq18	Open	closed
19	MR_Live.v	Pi_live, eq19_live_subset_feas . . .	Th_coqc	closed
20	MR_Live.v	live_field	Definition	—
21	MR_Live.v	live_ge_threshold, Pi_live	Definition	—
22	MR_Live.v	is_valid_choice	Definition	—

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Table 6 continued

Eq.	Module	Identifier(s)	Tier	Assumptions
23	MR_Live.v	eq23_enactment_may_differ_from_choice; Trajectory, Obs …	Th_coqc	closed
24	MR_Live.v	Stage, stage_value …	Th_coqc	closed
25	MR_Live.v	p_star; p_star_upper_bound	Definition	closed
26	MR_Live.v	live_field_gap	Definition	—
27	MR_Prompt.v	L_H (Variable, Section PrePromptCoupling)	Definition	—
28	MR_Prompt.v	AI_step, R_H_readout …	Definition	—
29	MR_Prompt.v	U_H (Variable), next_state	Definition	—
30	MR_Prompt.v	toy_update, toy_lambda_live …	Th_coqc	closed
31	MR_Prompt.v	F_hash_dlg (Variable), dlg_step	Definition	—
32	MR_Prompt.v	chi_recip, eq32_chi_recip_bounds	Th_coqc	closed
33	MR_Prompt.v	ReturnBattery, mk_return_battery	Definition	—
34	MR_Prompt.v	RET, eq34_RET_rearrangement	Th_coqc	closed
37	MR_Retention.v	ai_momentum_resets	Definition	—
38	MR_Retention.v	KnowledgeStatus, status_value …	Th_coqc	closed
39	MR_Retention.v	Delta_gt, eq39_sign_scale_pos_case …	Th_coqc	closed
40	MR_Retention.v	AUG, SYN …	Th_coqc	closed
41	MR_Retention.v	HReturn, mk_h_return	Definition	—
42	MR_Retention.v	EndChainNotion, end_chain_value …	Th_coqc	closed
43	MR_Corrections.v	gamma_mu, kappa_sem …	Definition	—
44	MR_River.v	master_river_44, one_pass_to_h_return …	Definition	—
45	MR_River.v	closing_summary_45	Definition	—
65	MR_River.v	river_tail_65	Definition	—
66	MR_River.v	outer_loop_66	Definition	—
46	MR_TopicEntry.v	L_H (Variable, Section TopicEntryTransport); topic_entry_transport	Definition	—
47	MR_TopicEntry.v	TopicEntry (Inductive); Legitimate	Definition	—
48	MR_TopicEntry.v	Open_eq48	Open	n/a (Open Prop, not proved by
49	MR_TopicEntry.v	ProblemOnlyPolicy (Definition, scaffolding only); Open_eq49	Open	n/a (Open Prop, not proved by
50	MR_TopicEntry.v	dcp_closure_50 (Section HumanReturnClosure)	Definition	—
51	MR_TopicEntry.v	CycleStage (Inductive); cycle_next_51	Definition	—
52	MR_WorldSystem.v	readout_52 (Variable, Section ReadoutDefinition); Hypothesis readout_ne_state; readout_52_hypothesis_satisfiable_on_bool	Definition	closed
53	MR_WorldSystem.v	LabourCentrality; mk_labour_centrality	Definition	—
54	MR_WorldSystem.v	q_min; eq54_citizen_claim_threshold_identity	Th_coqc	closed
55	MR_WorldSystem.v	Pi_live_ws; eq55_live_subset_feas_ws, eq55_live_subset_phys_ws …	Th_coqc	closed
56	MR_WorldSystem.v	corrigible_agency_ws; corrigible_agency_ws_upper_bound	Definition	closed
57	MR_WorldSystem.v	ReturnProfileWS; mk_return_profile_ws	Definition	—
58	MR_WorldSystem.v	Qpow_nat (Fixpoint); P_H_index	Definition	—
59	MR_WorldSystem.v	ddiff; eq59_output_rise_not_position_rise	Th_coqc	closed
60	MR_WorldSystem.v	eq60_machine_expansion_not_human_expansion	Th_coqc	closed
61	MR_WorldSystem.v	HumanConversionVector; mk_human_conversion_vector	Definition	—
62	MR_WorldSystem.v	eta_HC (Section ConversionElasticity)	Definition	—
63	MR_WorldSystem.v	in_reversibility_window, reversibility_window	Definition	—
64	MR_WorldSystem.v	urgency_term	Definition	—
67	MR_HCA.v	hca_river_67 (Section HCARiver)	Definition	—
68	MR_HCA.v	LifeCapitalContext; mk_life_capital_context	Definition	—
69	MR_HCA.v	BarrierType (Inductive); all_barrier_types; BarrierLedger	Definition	—
70	MR_HCA.v	eq70_observed_difficulty_not_skill_deficit	Th_coqc	closed
71	MR_HCA.v	C_live (Section EndorsedRoutes); C_live_subset_C_cand	Definition	closed
72	MR_HCA.v	eq72a_proactive_suggestion_not_human_goal_ownership, eq72b_capability_advancement_not_ai_goal_authority …	Th_coqc	closed
73	MR_HCA.v	hca_ddiff (scaffold); Open_eq73	Open	n/a (Open Prop, not proved by
74	MR_HCA.v	world_closure_74 (Section WorldClosureHCA)	Definition	—
75	MR_HCA.v	omega_real_75 (Section OpportunityReadout)	Definition	—
76	MR_HCA.v	eq76a_credential_not_capability, eq76b_market_legibility_not_human_worth	Th_coqc	closed
77	MR_HCA.v	NetAdvancementRecord; mk_net_advancement_record	Definition	—
78	MR_HCA.v	stratum_members, qsum …	Definition	—
79	MR_HCA.v	hca_outer_layer_79 (Section HCAOuterLoopLayer)	Definition	—

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APPENDIX C. CANONICAL REGISTRY OF EQUATIONS (1)–(79)

This table reads (1)–(79) of this document against `registry/CANONICAL.json`'s own `_n3_in_master_river` field — the field Toledo's N4 registry build uses to record which of its 793 canonical entries is an occurrence of an equation in *this* document, and which equation number(s) it occurs at. One row per equation number; where an equation is the occasion for more than one canonical code (the paper's own (27), for instance, is a single displayed transport that Toledo separately codes as an `EQ-015` reading, an `A.5` reading, and a second `EQ-015` reading), every code is listed. Tier and status are copied verbatim from `CANONICAL.json`; a SPLIT status means the code is retired in Toledo's own registry in favour of several finer-grained child codes (a later N4 correction, not a change made by this document) — the code is still shown here because it is what the field currently names. Read plainly: **55 of these 79 equations carry at least one Toledo code; 24 do not**. An equation lacking a code here is not thereby false or lower-tier than stated in the body of this document — it means only that Toledo's own build has not yet coded that occurrence, exactly the honesty Section 10 already discloses at the segment level.

Table 7: Canonical registry of equations (1)–(79): Toledo code, root, tier, and status, read from `CANONICAL.json`.

Eq.	Toledo code	Root	Tier	Status
1	`EQ-002/E.01.v1`	EQ-002	Definition	current
2	`EQ-015/E.01.v1; EQ-015/E.03.v1`	EQ-015; EQ-015	Definition; Definition	current; current
3	`EQ-015/E.01.v1`	EQ-015	Definition	current
4	`EQ-015/E.04.v1`	EQ-015	Definition	current
5	`EQ-015/E.04.v1`	EQ-015	Definition	current
6	`EQ-015/E.05.v1`	EQ-015	Definition	current
7	`EQ-015/E.05.v1`	EQ-015	Definition	current
8	`EQ-015/E.08.v1`	EQ-015	Definition	current
9	`EQ-015/E.06.v1`	EQ-015	Dr	current
10	`EQ-015/E.06.v1`	EQ-015	Dr	current
11	`EQ-015/E.06.v1`	EQ-015	Dr	current
12	`EQ-015/M.08.v1`	EQ-015	Definition	current
13	`EQ-015/M.08.v1`	EQ-015	Definition	current
14	`EQ-015/M.08.v1`	EQ-015	Definition	current
15	*unmapped — no Toledo code in CANONICAL.json's per-equation index*			
16	`EQ-015/E.07.v1`	EQ-015	Definition	current
17	`EQ-015/E.07.v1`	EQ-015	Definition	current
18	`EQ-015/E.07.v1`	EQ-015	Definition	current
19	`A.5/S.03.v1`	A.5	Definition	split
20	`A.5/S.03.v1`	A.5	Definition	split
21	`EQ-015/H.07.v1; A.5/S.03.v1`	EQ-015; A.5	Definition; Definition	current; split
22	`A.5/S.03.v1`	A.5	Definition	split
23	`A.5/S.03.v1`	A.5	Definition	split
24	`A.5/S.03.v1`	A.5	Definition	split
25	`EQ-015/S.06.v1`	EQ-015	Definition	split
26	`EQ-015/S.08.v1`	EQ-015	Definition	current
27	`EQ-015/H.01.v1; A.5/H.01.v1; EQ-015/H.03.v1`	EQ-015; A.5; EQ-015	Definition; Definition; Definition	current; current; current
28	`EQ-015/H.01.v1; EQ-015/H.03.v1`	EQ-015; EQ-015	Definition; Definition	current; current
29	`EQ-015/E.08.v1; EQ-015/H.03.v1`	EQ-015; EQ-015	Definition; Definition	current; current
30	`EQ-015/H.03.v1`	EQ-015	Definition	current
31	`EQ-015/H.05.v1`	EQ-015	Definition	current
32	*unmapped — no Toledo code in CANONICAL.json's per-equation index*			
33	*unmapped — no Toledo code in CANONICAL.json's per-equation index*			
34	*unmapped — no Toledo code in CANONICAL.json's per-equation index*			
35	*unmapped — no Toledo code in CANONICAL.json's per-equation index*			
36	*unmapped — no Toledo code in CANONICAL.json's per-equation index*			
37	*unmapped — no Toledo code in CANONICAL.json's per-equation index*			
38	`EQ-015/H.10.v1`	EQ-015	Definition	current
39	*unmapped — no Toledo code in CANONICAL.json's per-equation index*			
40	*unmapped — no Toledo code in CANONICAL.json's per-equation index*			
41	*unmapped — no Toledo code in CANONICAL.json's per-equation index*			
42	*unmapped — no Toledo code in CANONICAL.json's per-equation index*			
43	*unmapped — no Toledo code in CANONICAL.json's per-equation index*			
44	*unmapped — no Toledo code in CANONICAL.json's per-equation index*			
45	*unmapped — no Toledo code in CANONICAL.json's per-equation index*			
46	*unmapped — no Toledo code in CANONICAL.json's per-equation index*			
47	*unmapped — no Toledo code in CANONICAL.json's per-equation index*			
48	*unmapped — no Toledo code in CANONICAL.json's per-equation index*			
49	*unmapped — no Toledo code in CANONICAL.json's per-equation index*			
50	*unmapped — no Toledo code in CANONICAL.json's per-equation index*			

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Table 7 continued

Eq.	Toledo code	Root	Tier	Status
51	*unmapped — no Toledo code in CANONICAL.json's per-equation index*			
52	*unmapped — no Toledo code in CANONICAL.json's per-equation index*			
53	`EQ-015/W.04.v1`	EQ-015	Definition	split
54	`EQ-015/W.05.v1`	EQ-015	Definition	split
55	`EQ-015/H.07.v1`	EQ-015	Definition	current
56	`EQ-015/W.14.v1`	EQ-015	Definition	current
57	`EQ-015/W.15.v1`	EQ-015	Definition	current
58	`EQ-015/W.10.v1`	EQ-015	Definition	split
59	`EQ-015/W.10.v1`	EQ-015	Definition	split
60	`EQ-015/W.16.v1`	EQ-015	Definition	split
61	`EQ-015/W.16.v1`	EQ-015	Definition	split
62	`EQ-015/W.16.v1`	EQ-015	Definition	split
63	`EQ-015/W.19.v1`	EQ-015	Definition	split
64	`EQ-015/W.19.v1`	EQ-015	Definition	split
65	*unmapped — no Toledo code in CANONICAL.json's per-equation index*			
66	*unmapped — no Toledo code in CANONICAL.json's per-equation index*			
67	`EQ-002/H.03.v1`	EQ-002	Definition	current
68	`EQ-015/H.31.v1`	EQ-015	Definition	current
69	`A.5/H.16.v1`	A.5	Definition	current
70	`A.5/H.16.v1`	A.5	Definition	current
71	`A.5/H.17.v1`	A.5	Definition	current
72	`A.5/H.17.v1`	A.5	Definition	current
73	`EQ-015/H.32.v1`	EQ-015	Open	current
74	`EQ-015/H.24.v1`	EQ-015	Definition	current
75	`A.5/H.18.v1`	A.5	Definition	current
76	`A.5/H.18.v1`	A.5	Definition	current
77	`A.8/H.03.v1`	A.8	Definition	current
78	`A.8/H.03.v1`	A.8	Definition	current
79	*unmapped — no Toledo code in CANONICAL.json's per-equation index*			

A separate, hand-checked identification exists for three of the twenty-four unmapped equations: `COLLAPSE.md` § 2 reads (44) as a human–AI reading of `EQ-015/M.03.v1` and `EQ-015/M.02.v1` (CAN-004, CAN-003), (66) as a reading of `EQ-002/M.03.v1` and `weld/M.02.v1` (CAN-201, CAN-006), and (79) as a further reading of `weld/M.02.v1` (CAN-006) — by direct textual comparison rather than the automated field this table reads, and disclosed at `COLLAPSE.md`'s own "manual (verified textual match)" evidence tier, not a higher one. (65) and the remaining twenty unmapped equations ((15), (32)–(37), (39)–(43), (45)–(52)) carry no code and no such hand-check under either source consulted for this appendix.

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Internal Programme References (Records Deposited on Zenodo)

Yaoharee Lahtee. When AI expands human potential: Reflective dissonance, epistemic agency, and constraint. Zenodo, 25 March 2026, 2026a.

Yaoharee Lahtee. Human learning as epistemic architecture: A method for word mapping, life-concept graphs, constraint testing, and corrigible agency in the AI age. Zenodo, revised v4, 28 June 2026, 2026b.

Yaoharee Lahtee. Experience is the human LoRA: A readout–retention theory of selective model change. Zenodo, 18 July 2026, 2026c.

Yaoharee Lahtee. Readout genesis standalone synthesis: Information epistemic foundation, conditioned agency, and meta-readout governance. Zenodo, 24 July 2026, 2026d. The MEMK-domain experience equation (14) is contained in this deposited volume. The internal working file "READOUT GENESIS MEMK DOMAIN v0.2.0" (23 July 2026) is cited from within this record and was not deposited separately.

Yaoharee Lahtee. From problem to hypothesis: Dynamic semantic mobility, bounded knowers, and the readout-discriminable hypothesis space. Zenodo, K0 final, 4 September 2026, 2026e.

Yaoharee Lahtee. Epistemic fusion or epistemic tunnel: A phenomenology-anchored, global-literature-constrained, problem-first architecture of human-AI collaboration. Zenodo, Epistemic Note v8.1, 5 September 2026 (concept DOI 10.5281/zenodo.22318039; v7.1 DOI 10.5281/zenodo.22318040), 2026f. The draft "Conversation-Centered v3" discussed in §4.3 was not deposited separately.

Yaoharee Lahtee. Meaning before naming: A readout-retention architecture of affective-semantic reorganization. Zenodo, Concept Note v1.0, 5 September 2026, 2026g.

Yaoharee Lahtee. Experience is meaning-giving: A strong-form readout-retention theory of phenomena, resonance, rhythm, accumulation, and transformative release. Zenodo, Full Paper v2.0, 5 September 2026, 2026h.

Yaoharee Lahtee. Potential as a readout: Witnessed envelopes, two-layer agency, and the measurement of pseudo-peace. Zenodo, Epistemic Note K0 v2, 5 September 2026 (concept DOI 10.5281/zenodo.22345708), 2026i. Supersedes the undeposited programme note "Effective, Corrigible Agency Potential" (5 September 2026).

Yaoharee Lahtee. Choice begins before choice: Meaning-shaped accessibility, live possibility, and effective agency in human-AI systems. Zenodo, Full Paper v1.0, 5 September 2026, 2026j.

Yaoharee Lahtee. CTSA human-return readout: A session-boundary measurement architecture for retained human capability after AI. Zenodo, Epistemic Note v9.1 FINAL, 5 September 2026, 2026k.

Yaoharee Lahtee. glosa — rigour without infrastructure: A standalone scholar methodology for human–AI knowledge co-production (v0.4.1). Zenodo software record (concept DOI 10.5281/zenodo.22301059), 2026l.

Yaoharee Lahtee. After labour: Human position in an AI-robotic world system. Zenodo, v1.2 glosa-checked edition, 6 September 2026, DOI 10.5281/zenodo.22481924, 2026m.

Yaoharee Lahtee. The human conversion imperative: Machine acceleration, human potential, and the reversibility window in an AI-robotic world. Zenodo, v1.1 glosa-checked edition, 6 September 2026, DOI 10.5281/zenodo.22481926, 2026n.

Yaoharee Lahtee. How humans should converse with AI: A problem-first adaptive dialogue conversion protocol for durable human capability. Zenodo, v1.3 glosa-checked edition, 6 September 2026, DOI 10.5281/zenodo.22481928, 2026o.

Yaoharee Lahtee. From assistance to human capability: A readout-genesis architecture for proactive AI, unequal life conditions, and opportunity conversion. Zenodo, v1.1 glosa-checked edition of the 6 September 2026 v1.0 original, 2026p.

Yaoharee Lahtee. Toledo — equation library of the human–AI readout programme. Zenodo software record, v1.0.0, 7 September 2026 (concept DOI 10.5281/zenodo.22537318), 2026q.

Yaoharee Lahtee. Master equation river — coq formalisation of every equation (v1.0: equations 1–79 of master equation river v1.4). Zenodo software record, v1.0, 6 September 2026, DOI 10.5281/zenodo.22518450; obsoleted by Toledo v1.0.0, DOI 10.5281/zenodo.22548770, 2026r.

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Before Meaning, Before Choice: A Readout-Native Derivation of Experience, Live Possibility, Agency, and Human–AI Return

Full Paper v1.1, glosa-checked • 6 September 2026 • 10.5281/zenodo.22424434

What the chapter states. "Human experience and choice should not be inserted as primitive objects at the beginning of a theory." "Meaning is then a domain-level significance relation, experience is a lived configuration of encountered trace and meaning, retained experience can reshape later accessibility, and overt choice is selected only from a history-shaped field of live possibilities."

Status. "Status K0, version 1.1 (source-lineage repair of the 6 September v1.0 after an independent review), 6 September 2026"; printed on the chapter's first page as "K0 working release, not peer reviewed."

Falsifier(s). "H2 — History residual [OPEN]: Prior traversal history will predict later route accessibility after current semantic state, task, and control variables are matched. Failure removes the distinct momentum term."

Read before. This book's Chapters 2–3, 8–11, 13, 17–18, 27, 31, 41 (mechanical KG cross-reference list; Appendix A) and Ch. 36 *Master Equation River*.

Feeds into. — (mechanical KG cross-reference list; Appendix A).

Human–AI Readout Programme • Root-to-Human Derivation • Full Paper v1.1 • September 2026

Before Meaning, Before Choice

A Readout-Native Derivation of Experience, Live Possibility, Agency, and Human–AI Return

From Readout Universe and Readout Genesis to a recursive human–world and Human–AI architecture

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Status K0, version 1.1 (source-lineage repair of the 6 September v1.0 after an independent review), 6 September 2026. The paper does not claim that Readout Universe or Readout Genesis is a completed law of nature, that the human-domain quotient has been empirically closed, or that the equations below are literal neuroscience. It advances a narrower but strong contribution: a typed derivational architecture in which human-semantic and agentic categories are admitted only after retained difference, reader-conditioned access, domain translation, explicit defect/loss accounting, and claim-status control. It makes no global priority claim: the reviewed neighbouring literatures contain close ancestors for most local components, and the contribution is stated as same / different / cited against each of them (Table 1). Every internal source is cited by a Zenodo DOI or by a commit-pinned public git blob (Section 21).

Central claim. Human experience and choice should not be inserted as primitive objects at the beginning of a theory. Under a readout-native architecture, a retained difference must first become structurally accessible under a declared reader and domain translation. Meaning is then a domain-level significance relation, experience is a lived configuration of encountered trace and meaning, retained experience can reshape later accessibility, and overt choice is selected only from a history-shaped field of live possibilities. Human–AI interaction is a special mediated loop inside this architecture: the prompt is a bounded articulation of an already history-bearing human state, AI output begins as a knowledge-like candidate rather than certified knowledge, and any claimed human gain must survive a session boundary and an unaided Human Return test. The resulting programme is therefore not *meaning as primitive* $\to$ *choice*, but *retained difference* $\to$ *admissible readout* $\to$ *meaning* $\to$ *experience* $\to$ *live possibility* $\to$ *choice* $\to$ *action* $\to$ *new retained difference*.

Abstract

The Human–AI Readout Programme has developed a sequence of papers on human potential, learning, retained experience, semantic mobility, Human–AI coupling, meaning-giving, live possibility, effective agency, and Human Return. Read separately, these works can appear to be a family of related conceptual models. This paper argues that they can be reconstructed as a single derivational architecture only if the chain is anchored one level deeper, in Readout Universe and Readout Genesis. The root contains no built-in final labels such as mind, meaning, experience, memory, agency, choice, or AI. It begins with retained distinctions, ordered history, reader capability, finite state transition, and a requirement that any domain enter through a sufficient, dynamically coherent translation with explicit defects and bounded claims. The paper then derives a human-domain river in stages. Readout Genesis and its MEMK domain supply the separation of world occurrence, retained record, event trace, meaning, experience, memory, and knowledge. Human LoRA supplies selective cross-time persistence. From Problem to Hypothesis supplies graded and history-shaped accessibility. Choice Begins Before Choice inserts a Live Possibility Field between feasible capacity and overt choice. Potential as a Readout supplies the read–decide/refuse–execute–feedback envelope over a witnessed option set. Fusion/Tunnel inserts Human–AI dialogue as a recursively history-sensitive mediation, while CTSA Human Return locates the measurement boundary after AI removal. The paper places this architecture under pressure from mature neighbouring literatures: Batesonian difference, computational mechanics, the information bottleneck, bisimulation and MDP homomorphisms, phenomenology, enactivism, ecological affordances, the capability approach, temporal theories of agency, epistemic injustice, theories of power and structural violence, active inference, and empirical Human–AI decision research. The conclusion is deliberately asymmetric: most local phenomena are already established elsewhere; the candidate contribution lies in the derivational discipline that keeps root structure, domain meaning, lived experience, practical possibility, overt choice, evaluator-visible behaviour, epistemic warrant, and durable Human Return from collapsing into one another. The paper ends with a canonical Root-to-Human Master Equation River, a claim-status ledger, rival-model tests, and explicit removal conditions. If the readout-native layers add no stable explanatory or predictive value beyond simpler alternatives, they should be compressed or removed.

Keywords: Readout Universe; Readout Genesis; retained difference; meaning; experience; live possibility; agency; Human–AI interaction; Human Return; epistemic architecture; domain translation; corrigibility.

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1 The Problem Is Not Another Theory of Choice

The immediate temptation is to begin with a familiar object: a person has preferences, represents options, deliberates, chooses, acts, and receives feedback. A second temptation is to move one step earlier and ask how experience, embodiment, social norms, prior learning, or language shape the options that enter deliberation. A third temptation is to move still earlier and treat meaning or sense-making as the primitive from which action is organised. Each move captures something important. None, however, answers the methodological question that motivates the present paper: what licenses us to place "meaning", "experience", "option", "choice", or "agency" into the ontology in the first place?

Readout Universe and Readout Genesis were designed precisely to block premature naming. Their information-philosophy constitution places difference before entity, ordered history before container-time, retention before persistence semantics, reader-conditioned access before domain truth, and domain translation before domain-final nouns [Lahtee, 2026l, Readout Genesis Programme, 2026]. This discipline changes the order of explanation. Human cognition is not denied; it is delayed until the architecture has earned the right to use cognitive vocabulary.

The programme's later papers can then be reread as successive answers to a single question. When a retained difference becomes readable to a human, how does it become significant, lived, remembered, practically accessible, chosen, enacted, corrected, and eventually returned as human capability after AI mediation? The Master Equation River audit showed that the internal corpus already contains most of these transitions, but also identified notation collisions and a deeper missing anchor: the river began too late, at the human readout layer, rather than at the non-semantic root [Lahtee, 2026g].

This paper performs that anchoring. It does not claim that every human behaviour proceeds through explicit deliberation, that reflexes require a live possibility field, or that all cognition is linguistic. The primary target is practical human agency: situations in which a route can count as a candidate for acceptance, refusal, negotiation, delay, delegation, exit, revision, inquiry, or other purposive action. The architecture is therefore broader than verbal deliberation but narrower than a complete theory of biological behaviour.

2 What the Literature Already Owns

A defensible contribution claim must begin by conceding the strength of neighbouring literatures. The present architecture is not the earliest to treat difference, representation, perspective, meaning, affordance, agency, temporality, power, or Human–AI influence as serious theoretical objects.

2.1 Difference, predictive equivalence, and compressed representation

Bateson's well-known formulation of information as a difference that makes a difference is a clear ancestor of any difference-first programme [Bateson, 1972]. Computational mechanics goes further by constructing causal states as equivalence classes of histories that are predictively indistinguishable, providing a principled way to identify structure relative to future prediction [Shalizi and Crutchfield, 2001]. The information bottleneck similarly formalises representation as selective compression that preserves information relevant to a declared target [Tishby et al., 1999]. The present work therefore claims no priority for "difference", history-based state construction, reader-relative equivalence, or relevance-sensitive compression.

2.2 Structure-preserving abstraction

The Readout Genesis domain-weld condition has close mathematical relatives in lumpability, bisimulation, state abstraction, and MDP/SMDP homomorphisms. These traditions ask when a lower-dimensional or quotient representation preserves transition- or value-relevant structure [Ravindran and Barto, 2003, Ferns et al., 2011]. Whatever the domain weld contributes, it cannot be the bare fact that structure-preserving abstraction exists. The residual move is epistemic: a dynamics-preserving translation is used as a gate before final semantic categories are admitted, and every approximate bridge must expose what it loses.

2.3 Meaning, experience, and sense-making

Phenomenology has long treated experience as intentional and horizon-laden; enactivism treats cognition as embodied sense-making arising in organism–environment coupling rather than passive extraction of pre-packaged meaning [Varela et al., 1991, Di Paolo, 2005, Thompson, 2007]. The programme's claim that meaning is relational, embodied, history-conditioned, and action-relevant therefore has major predecessors. The paper does not claim to discover sense-making. Its question is instead how such significance can be placed in a typed architecture after a non-semantic retained-state layer and then carried forward without collapsing meaning into truth, experience into memory, or retention into improvement.

2.4 Action possibilities, real freedom, and the conditions of choice

Ecological psychology's affordance tradition already treats possibilities for action as relational between organism and environment [Gibson, 1979]. Sen's capability approach distinguishes formal resources from real opportunities; Kabeer links resources, agency, and achievements while emphasising the conditions under which meaningful choices are made [Sen, 1985, Kabeer, 1999]. Emirbayer and Mische treat agency as temporally embedded rather than a timeless instant [Emir-

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bayer and Mische, 1998]. The Live Possibility Field therefore cannot claim that options are agent-relative or that context matters. Its narrower residual proposal is that feasible action and currently live practical consideration should be separated as distinct readout objects.

2.5 Power, interpretive resources, and structural deprivation

Foucault's account of power acting upon possible action, Pettit's non-domination, Fricker's hermeneutical injustice, and Galtung's structural-violence framework all undermine the idea that freedom can be read directly from visible non-interference or observed behaviour [Foucault, 1982, Pettit, 1997, Fricker, 2007, Galtung, 1969]. The present framework does not replace these traditions. It supplies a narrower measurement interface: a structural mechanism may affect agency by changing the accessibility distribution over live routes before an overt choice occurs. Calling that change domination, injustice, or violence still requires an independent normative argument.

2.6 Integrated agent models and Human–AI coupling

Active inference is the strongest formal neighbouring architecture because it integrates perception, belief updating, action, learning, and policy selection within a single framework [Friston et al., 2017]. Human–AI decision research likewise documents path dependence, reliance, explanation effects, and the need for common evaluation frameworks [Lai et al., 2023]. Glickman and Sharot provide direct evidence that repeated interaction with biased AI can alter later human perceptual, emotional, and social judgements, demonstrating that the human state entering a later interaction need not be fixed [Glickman and Sharot, 2025]. The present programme therefore does not claim that history changes future judgement or that AI can influence users. Its residual contribution is the explicit chain from pre-prompt human state to candidate status, independent resistance, selective human retention, and unaided Human Return.

3 The Readout Constitution: Before Meaning

Readout Universe is not used here as decorative metaphysics. Its information philosophy is operational: the root must not contain domain-final nouns; difference precedes the name of what differs; ordered history precedes time treated as a container; retention precedes persistence semantics; readers create access rather than retroactive root truth; domains are translations; meaning comes after readout; non-exact translation must expose loss; claims cannot borrow certainty across layers; and reporting is the last stage [Lahtee, 2026l,m]. The whitepaper states the governing rule in one line: "Information philosophy fixes the order by which meaning becomes admissible and governs every translation" [Lahtee, 2026l, l. 35].

The most compressed form is

$$\begin{aligned}&\text{Retention} \to \text{Structure} \to \text{Translation}\\ &\to \text{Readout} \to \text{Meaning} \to \text{Report},\end{aligned} \tag{1}$$

with the stronger MEMK extension (the MEMK specification's own mandatory order reads Retention → Structure → Translation → Meaning → Report [Readout Genesis Programme, 2026, l. 14]):

$$\begin{aligned}&\text{Retention} \to \text{Structure} \to \text{Candidate State}\\ &\to \text{Sufficiency} \to \text{Quotient} \to \text{Domain Dynamics}\\ &\to \text{Readout} \to \text{Meaning} \to \text{Experience}\\ &\to \text{Memory} \to \text{Knowledge} \to \text{Checked Report}.\end{aligned} \tag{2}$$

Two consequences are load-bearing. First, semantic categories are not root primitives; the MEMK file lists "meaning is not a root primitive" among its root non-claims [Readout Genesis Programme, 2026, l. 50]. Second, formal success at one layer cannot certify empirical success at another. A commuting quotient would not prove that a psychological interpretation is true; a stable human report would not prove that the report exhausts the world occurrence.

3.1 Root state and finite stepper

In the Genesis specification the non-semantic retained state is written

$$S_n = (G_n, \Lambda_n, T_n), \tag{3}$$

where G_n is a retained relational carrier, Λ_n retained labels or typed distinctions, and T_n an append-only tape of lineage, residues, and deviations. The root transition interface is

$$S_{n+1} = F(S_n, u_n, c_n, T_n). \tag{4}$$

The critical separation is

$$S_n \neq Z_{\mathrm{MEMK},n} \neq D_{\mathrm{MEMK},n}. \tag{5}$$

Retained root state, candidate semantic–cognitive representation, and discovered domain quotient are different objects; the MEMK candidate state $Z_{\mathrm{MEMK},n}$ is written out as an explicit tuple in the public v0.2.0 file [Readout Genesis Programme, 2026, l. 80]. This is the earliest point at which the present paper differs from theories that begin with beliefs, preferences, policies, conscious states, or semantic representations. The claim is not that such states are unreal. It is that their use is downstream of an admission procedure.

4 Domain Weld: How Human Vocabulary Is Admitted

A domain is not created by attaching a familiar noun to the root. Readout Genesis requires a translation q_D whose domain dynamics preserve the relevant root dynamics and readouts. Schematically,

$$q_{D,n+1} \circ F_n = F^{\sharp}_{D,n} \circ q_{D,n}, \tag{6}$$

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with corresponding readout and invariant conditions. This is structurally analogous to familiar abstraction and homomorphism ideas, but here the weld is epistemic as well as mathematical: it governs whether a domain-level statement may be emitted at all.

For human-semantic work, exact closure has not been established. The correct status is therefore approximate and defect-accounting. Let $\tilde{q}_H$ be a candidate human-domain translation. A valid implementation must declare a defect vector

$$\varepsilon_H = \mathrm{Def}\big(\tilde{q}_H \circ F,\ F_H^\sharp \circ \tilde{q}_H,\ \mathscr{O}_H,\ \mathrm{Inv}_H\big), \tag{7}$$

and a preregistered acceptance rule $\|\varepsilon_H\| \le \tau_H$ if a quantitative gate is attempted. Without a declared metric and tolerance, the status remains unresolved rather than silently "close enough".

This defect discipline is more important than the exact formula. It means that every human-level construct in the rest of the paper is typed by its origin and by what the translation may have discarded. The architecture therefore resists a common philosophical shortcut: observing that a high-level concept is useful and then treating that usefulness as evidence that the concept was present in the root all along.

5 From Retained Record to Meaning and Experience

MEMK supplies a crucial distinction between event occurrence, retained event record, and domain-accessible event trace. Let A_n denote the occurrence under consideration, r_n the retained record before semantic naming, and x_n the MEMK-accessible trace. Then

$$A_n \neq r_n \neq x_n. \tag{8}$$

This need not imply that every pair is always distinguishable under every reader; it states that the roles must not be collapsed by default [Readout Genesis Programme, 2026].

A finite human readout of the accessible trace is represented schematically as

$$\rho_n^H = R_H(x_n \mid H_n, c_n, Q_n), \tag{9}$$

where H_n summarises the current human-domain state, c_n context, and Q_n an active question or criterion. Meaning is then a significance relation, not a substance carried inside x_n:

$$\mu_n = \Psi_H(\rho_n^H, H_n, c_n, Q_n). \tag{10}$$

The strong-form experience paper separates several analytic modes,

$$\mu_n = (\mu_n^{\mathrm{aff}}, \mu_n^{\mathrm{prag}}, \mu_n^{\mathrm{auto}}, \mu_n^{\mathrm{conc}}, \mu_n^{\mathrm{epi}}), \tag{11}$$

without treating them as biological modules [Lahtee, 2026c]. Experience is then

$$E_n = \Phi_E(x_n, \mu_n, \gamma_n^\mu, c_n), \tag{12}$$

where γ_n^μ denotes the effective strength or control of active meaning over the current experience (the MEMK file writes this argument as κ_n, Readout Genesis Programme, 2026, l. 92; it is renamed here to avoid a collision with semantic accessibility, following the Master River audit [Lahtee, 2026g]; the same equation appears as eq. 14 of the deposited Genesis synthesis, Lahtee, 2026k). The conceptual compression is

$$\text{Experience} = \text{phenomenon-as-meaningfully-read}. \tag{13}$$

This is not a claim that the reader makes the phenomenon true. It is a claim about the level of lived significance. Meaning-giving can be false, partial, manipulative, culturally inherited, or later revised.

5.1 Naming comes later, but can loop back

Meaning Before Naming allows lived significance to precede stable lexical articulation:

$$\ell_n = L_H(E_n, \mu_n \mid H_n, c_n), \qquad E_n, \mu_n \text{ may precede stable } \ell_n. \tag{14}$$

But the later Experience paper corrects any one-way reading. Naming can reorganise what it articulates:

$$\mu_n \to E_n \to \ell_n \to \mu_{n+1} \to E_{n+1}. \tag{15}$$

Thus "meaning before naming" is a temporal possibility, not a doctrine that language is irrelevant or merely epiphenomenal [Lahtee, 2026h,c].

6 Retention: The Reader Can Change

Experience is not automatically learning. Human LoRA supplies the distinction between exposure, retained residue, and selective operator change. In a generic dependency form,

$$H_{n+1} = U_H\big(H_n,\ \mathrm{Retain}(E_n, \mu_n, \rho_n^H),\ \delta_{n:n+1}^{\mathrm{world}},\ X_{n:n+1}^{\mathrm{other}}\big). \tag{16}$$

Only selected residues are carried forward; world correction and other experience remain co-determining inputs [Lahtee, 2026f,d]. This yields a central temporal asymmetry:

$$\mathrm{Retain}(E_n) > 0 \implies H_{n+1} \neq H_n \quad \text{possible, not guaranteed.} \tag{17}$$

The same biological individual may therefore confront the next situation with a different effective interpretive operator. This makes the entire downstream theory of possibility history-sensitive before any new choice is made.

7 History-Shaped Readout-Accessibility

The semantic-mobility paper supplies the formal bridge from retained history to graded access. Reachability is not accessibility. For a semantic route $e: s \to s'$, let $\kappa_{A,t,Q}^{\mathrm{sem}}(s' \mid s)$ be a finite access weight. Then an alternative may be structurally reachable while still having low practical access [Lahtee, 2026j].

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Table 1: Contribution boundary. Local mechanisms have mature ancestors; the residual contribution is the typed derivational interface. Comparison vocabulary: same / different / cited.

Component	Strong neighbours	Residual role in this paper
Difference / retained history	Bateson; computational mechanics	Not claimed as this paper's; retained difference is used as a non-semantic root object that precedes domain naming.
Reader-relative compression / equivalence	Information bottleneck; predictive equivalence	Require declared reader, provenance, defect/loss, and bounded claim status; equivalence is never promoted to absolute identity.
Structure-preserving abstraction	Lumpability; bisimulation; MDP/SMDP homomorphism	Use dynamic preservation as an epistemic admission gate for semantic domains rather than as state reduction alone.
Meaning / sense-making	Phenomenology; enactivism	Derive meaning only after readout and translation; keep meaning distinct from truth and warrant.
Action possibility	Affordances; capability approach	Insert a graded live possibility field between feasible capacity and overt choice.
Temporally embedded agency	Emirbayer–Mische; learning and memory literatures	Make accessibility explicitly history-shaped and allow the effective reader at $t+1$ to differ from the reader at t.
Power before overt choice	Foucault; Pettit; Fricker; Galtung	Provide a readout/live-field interface without equating all social shaping with domination or violence.
Integrated perception–action dynamics	Active inference	Begin one level earlier with a non-semantic root and require domain translation, loss accounting, and claim-tier separation.
Human–AI feedback	Human–AI decision literature; Glickman–Sharot	Connect feedback to pre-prompt state, candidate status, resistance, retention direction, and unaided Human Return.

Two history descriptors are kept separate. Semantic attraction is destination-sensitive,

$$a_{Q,t}(e) = \left[\Phi_{Q,t}(s) - \Phi_{Q,t}(s')\right]_+, \tag{18}$$

where $\Phi_{Q,t}$ is a declared access landscape, not a physical potential. Recent-path momentum is path-sensitive,

$$m_{t+1}(e) = \rho\, m_t(e) + \mathbb{1}[e_t = e], \qquad 0 \le \rho < 1. \tag{19}$$

A candidate history-shaped access law is

$$\kappa^{\text{sem}}_{t+1}(e \mid Q) \;\propto\; \kappa^{(0)}_t(e \mid Q) \times \exp\left[\beta a_{Q,t}(e) + \mu_m m_t(e) - \nu c_{A,t,Q}(e) + \xi_t(e)\right]. \tag{20}$$

This equation remains OPEN: it is a typed finite hypothesis, not a validated universal cognitive law. Its theoretical role is to prevent structural availability, attraction, recent traversal, and control cost from collapsing into one access variable.

Rhythm and resonance are optional modifiers rather than root laws. Rhythm is a domain readout of structured recurrence on ordered lineage; resonance is a domain diagnostic of congruence between a current externally prompted experience and retained experiential organisation [Lahtee, 2026c]. Neither is required for every case of live possibility formation. If they add no incremental explanatory value beyond simpler temporal, schema, familiarity, or retrieval variables, they should be removed from the technical model.

8 BEFORE CHOICE: THE LIVE POSSIBILITY FIELD

The central transition from experience theory to agency theory is the insertion of a live possibility layer. Let $\Pi^{\text{phys}}_t(g)$ denote physically possible policies for task g, and $\Pi^{\text{feas}}_{A,t}(g)$ those policies the agent could access, use, or learn within declared resources, time, and structural conditions. The current live set is narrower:

$$\Pi^{\text{live}}_{A,t}(g) \subseteq \Pi^{\text{feas}}_{A,t}(g) \subseteq \Pi^{\text{phys}}_t(g). \tag{21}$$

A practical accessibility profile is

$$L_{A,t}(g) = \left\{(\pi, \lambda^{\text{live}}_{A,t}(\pi \mid g)) : \; \pi \in \Pi^{\text{feas}}_{A,t}(g)\right\}, \tag{22}$$

with an operational, study-specific threshold

$$\Pi^{\text{live}}_{A,t}(g) = \left\{\pi : \; \lambda^{\text{live}}_{A,t}(\pi \mid g) \ge \tau_{\text{live}}\right\}. \tag{23}$$

The threshold is a measurement device, not a universal psychological constant [Lahtee, 2026a]. Overt choice is downstream:

$$\pi^{\text{choice}}_{A,t} \in \Pi^{\text{live}}_{A,t}(g). \tag{24}$$

Enactment may diverge from the chosen policy, and evaluator-visible records need not reveal the underlying trajectory:

$$\begin{aligned} &\pi^{\text{act}}_{A,t} \ne \pi^{\text{choice}}_{A,t} \text{ is possible,} \\ &Y_{\text{obs}} = \mathscr{O}_q(H_{0:T}), \qquad Y_{\text{obs}} \ne H_{0:T}. \end{aligned} \tag{25}$$

The non-collapse chain is therefore

$$\begin{aligned} &\text{possible} \ne \text{feasible} \ne \text{live} \\ &\ne \text{chosen} \ne \text{enacted} \ne \text{observed.} \end{aligned} \tag{26}$$

The claim "choice begins before choice" is now sharpened. It does not mean that a retained difference is itself a choice. It means that the architecture of a deliberative choice has ancestral conditions in what was retained, what a reader can discriminate, which domain translation makes a route intelligible, and which history-shaped accessibility profile allows that route to enter practical consideration.

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9 Effective, Corrigible Agency

The Live Possibility Field does not replace the agency-potential model. It inserts a pre-choice restriction inside the agent's envelope. For task g, the agency architecture requires four jointly relevant conditions: Read_g for reading relevant distinctions, D_g for actor-directed decision or refusal, X_g for causal enactment, and F_g for feedback, objection, and revision. In its current form the task-specific potential is read over the *witnessed* option set:

$$p^*_{A,g}(h,z;T,B,P) = \max_{\pi \in \Pi^{\mathrm{wit}}_A(g;h,z,T,B)} \Pr^{\pi}_P\left(\mathrm{Read}_g \cap D_g \cap X_g \cap F_g\right), \tag{27}$$

where Π^{wit}_A is the finite set of policies for which a retained record exists that an agent comparable to A under a declared comparability relation (shared declared fields of z and a history window of declared length) executed the policy within horizon T and budget B [Lahtee, 2026i, eq. 1]. A merely conceivable, or merely permitted, policy is not in the set; the maximum is therefore computable from records, and the unread remainder is a non-readout rather than a smaller number waiting to be found. The v1.0 draft of this paper wrote the maximum over $\Pi^{\mathrm{feas}}_A(g)$, following the programme note that *Potential as a Readout* has since superseded; the four events are unchanged, the domain of the maximum is not, and the later paper adds a second layer ($p^{*(2)}_{A,g} = \max_{z \in J_{\mathrm{feas}}} p^*_{A,g}$, the recoverable gap being the difference of layers) and two non-collapse rules: potential $\neq$ exercised $\neq$ observed, and diagnosis of compression $\neq$ attribution of responsibility [Lahtee, 2026i,a].

This definition explains why potential, performance, and observation can diverge. A person may possess a feasible route that is not currently live; may make a choice that cannot be enacted; may act under third-party control; or may preserve objection and revision capacity that an evaluator fails to observe. Conversely, a visible successful outcome does not prove actor ownership, corrigibility, or durable capability.

9.1 Power before prohibition

Once live accessibility is explicit, a narrower political thesis follows:

$$\text{Power may alter agency by altering meaning/accessibility before overt choice.} \tag{28}$$

This is not a claim that all socialisation is domination. Human beings necessarily inherit languages, norms, roles, skills, and interpretive repertoires. The architectural point is that social and institutional conditions can change whether refusal, appeal, delay, exit, negotiation, or alternative interpretation becomes practically live before any direct prohibition is imposed.

A narrowed live field is not by itself a violence score. A structural-deprivation claim requires at minimum: a specified agency-relevant loss, a feasible counterfactual, an independent normative reason why the blocked route constitutes harm, domination, injustice, or violence, and, if responsibility is to be attributed, a chooser of the regime that produced the closure [Galtung, 1969, Lahtee, 2026a,i]. Diagnosing the closure and attributing it are separate acts.

10 Human–AI Mediation Begins Before the Prompt

The root-to-human derivation changes how Human–AI interaction is conceptualised. A prompt is not the whole human state. It is a bounded articulation:

$$H_t \xrightarrow{L_H} Q_t, \qquad Q_t \neq H_t. \tag{29}$$

Because H_t is itself a domain-level product of retained history and prior readout, the prompt is a readout of a readout, not an unconditioned starting point. The AI receives Q_t and returns Y_t:

$$Q_t \to \mathrm{AI}_t \to Y_t. \tag{30}$$

Under the candidate contract,

$$\mathrm{AI}(Q_t) = K_{\mathrm{like}}, \qquad K_{\mathrm{like}} \neq K_{\mathrm{validated}}. \tag{31}$$

Fluency, citations, or formal appearance do not upgrade status by themselves [Lahtee, 2026d]. The AI output then re-enters the human domain as another mediated encounter:

$$Y_t \xrightarrow{R_H} E^{\mathrm{AI}}_{H,t}. \tag{32}$$

If a residue is retained,

$$H_{t+1} = U_H\left(H_t, E^{\mathrm{AI}}_{H,t}, \delta^{\mathrm{world}}, X^{\mathrm{other}}\right), \tag{33}$$

and therefore

$$L_{A,t+1} \neq L_{A,t} \tag{34}$$

may hold before the next prompt is produced. Glickman and Sharot's experiments provide direct external evidence for the general history-sensitivity claim: repeated interaction with biased AI altered later human judgements across multiple domains [Glickman and Sharot, 2025]. They do not validate the full readout architecture, but they defeat the assumption that every prompt begins from a fixed human baseline.

10.1 Difference, resistance, and retained human agency

Fusion/Tunnel defines durable human epistemic expansion such that three conditions are non-substitutable: Difference, Resistance, and retained human Agency [Lahtee, 2026d]. Under that stipulated target,

$$D > 0, \qquad \mathrm{Resist} > 0, \qquad A_H > 0 \tag{35}$$

are constitutive necessities, not empirical frequency laws. No new readout-distinguishable route means no expansion beyond the retained frame; no independent resistance means no principled separation between correction and self-confirming proliferation; no retained human agency means possible system gain without demonstrated human-potential gain.

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Within-session reciprocal amplification is epistemically neutral. A dialogue can multiply defeaters or multiply confirmation. The Master River audit therefore proposes, as programme notation, a separation of candidate update from retention gate:

$$\widetilde{\Delta\Omega}_s^H = B_s A_s, \qquad \operatorname{rank}(B_s A_s) \ll d_H, \tag{36}$$

$$\Omega_{s+1,0}^H = \Omega_{s,0}^H + \eta_s \widetilde{\Delta\Omega}_s^H, \qquad 0 \le \eta_s \le 1. \tag{37}$$

This is a functional low-rank analogy, not a claim that brains literally implement engineering LoRA. Two source facts are recorded so that the lineage is exact: the deposited Fusion/Tunnel v8.1 keeps η_s only as a conceptual session-end gate and estimates durable residue through an unaided return battery $Y_{s+\Delta}^{\text{return}} = (R_{\text{rec}}, R_{\text{disc}}, T_{\text{new}}, Q_{\text{next}})$ and a difference-in-differences estimand RET (its Repair 7), and its three-axis outcome $J_s^* = (\text{AUG}_s, \text{SYN}_s, \text{RET}_s)$ is not to be collapsed into one score [Lahtee, 2026d]; equations (36)–(37) are the audit's proposal for writing the same intent without a doubled gate [Lahtee, 2026g].

11 Human Return: The Upper Measurement Boundary

The programme's strongest practical boundary is not the quality of the coupled Human–AI answer. It is what remains demonstrably usable by the human after AI support is removed. CTSA classifies retained content/access as Conceptual Return, Tool-Selection Return, Skill-Execution Return, or Alternative-Generation Return [Lahtee, 2026b]. A compact Human Return record is

$$H_{\text{return}} = \langle G_{\text{CTSA}}, L, M, P, W, \Delta_{\text{dir}} \rangle, \tag{38}$$

where G_{CTSA} is demonstrated gain, L loss, M metacognitive regulation, P provenance, W warrant/effective resistance, and Δ_{dir} epistemic direction (CTSA itself writes this record as R_H; the symbol is renamed to keep R free for read, resistance, resonance and rhythm). The tuple is bookkeeping architecture, not a universal scalar. The corresponding non-collapse laws are

$$\text{Exposure} \neq \text{Retention} \neq \text{Improvement}, \tag{39}$$

$$\text{Assisted performance} \neq \text{Unaided Human Return}. \tag{40}$$

A persistent misconception is still persistent; CTSA records it without granting knowledge status. A user may gain a skill while losing uncertainty sensitivity, or gain alternatives while becoming less able to distinguish which ones are warranted. Gross gain therefore cannot define net human expansion.

12 The Canonical Root-to-Human Master Equation River

The programme can now be written as one typed recursive river (Figure 1). The arrows denote admissible dependency or transition structure, not one universal causal law. Each segment retains its own claim status.

The deepest consequence is that the cycle does not return to the same reader by default. When retained change is non-zero, the next encounter is read under a modified human state. Experience can therefore change the reader; the changed reader can change what becomes live; what becomes live can change choice; choice changes action and the world; the changed world supplies new returned records; and the next readout begins from a different lineage.

13 Provenance and Claim Status Are Part of the Equation

A major difference between a compact theory diagram and a readout-native architecture is that the arrows themselves carry provenance and status. Let each transition e_i in the river have a ledger

$$\Lambda(e_i) = \langle \text{Prov}_i,\ \text{Tier}_i,\ \text{Def}_i,\ \text{Reader}_i,\ \text{Falsifier}_i \rangle. \tag{41}$$

The river is not fully specified without this ledger. This is the methodological signature of the paper. A strong phrase in prose is not automatically a strong empirical claim. The architecture permits rhetorical compression at the title level while forcing technical status to remain visible underneath.

14 Why This Is Not Simply Active Inference

The most serious formal objection is that perception, learning, action, and history can already be modelled under active inference. The objection is partly correct. Active inference is more mature as a computational process theory and has a much larger empirical and mathematical literature [Friston et al., 2017]. The present paper should not compete by claiming superior predictive machinery.

The distinction is instead one of derivation order and epistemic governance. Active inference generally begins with an agent model whose latent states, beliefs, preferences, and policies are already legitimate objects of the model. Readout Genesis places a prior methodological gate: semantic and agentic nouns are not admissible at the root merely because a modeller knows them in advance. The model must declare the translation by which such roles become accessible, the distinctions lost under that translation, and the claim strength permitted by the result. Formally, the difference is not "our dynamics are richer than active inference". It is

$$\begin{aligned} &\text{root retention} \not\equiv \text{belief state} \not\equiv \text{meaning} \\ &\not\equiv \text{experience} \not\equiv \text{choice}. \end{aligned} \tag{42}$$

A future empirical implementation may well instantiate parts of the human-domain dynamics using active-inference variables. Readout-native discipline would then treat those variables as one candidate domain realisation rather than as proof that the root was intrinsically Bayesian or semantic.

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$$\delta_R \to S_n = (G_n, \Lambda_n, T_n) \xrightarrow{F} S_{n+1} \xrightarrow{q_D \text{ or } \tilde{q}_D} Z^{\text{cand}}_{D,n} \xrightarrow[\varepsilon_D]{\text{sufficiency / weld / readout gates}} D_n$$
$$\to x_n \xrightarrow{R_H} \rho^H_n \xrightarrow{\Psi_H} \mu_n \xrightarrow{\Phi_E} E_n \xrightarrow{\text{Retain}/U_H} H_{n+1} \xrightarrow{\text{history-shaped access}} \lambda^{\text{live}}_{A,n+1} \to L_{A,n+1}$$
$$\to \Pi^{\text{live}}_{A,n+1} \to \pi^{\text{choice}}_{A,n+1} \xrightarrow{X,F} \pi^{\text{act}}_{A,n+1} \to \text{world consequence / returned record} \xrightarrow{\text{Retain}} \delta_{R,n+1}$$
$$\text{Human–AI insertion:} \quad H_t \xrightarrow{L_H} Q_t \to \text{AI}_t \to Y_t \xrightarrow{R_H} E^{\text{AI}}_{H,t} \xrightarrow{\text{Retain}} H_{t+1}$$
$$\text{Session boundary:} \quad \text{AI removed} \to H_{\text{return}} \to \{\text{gain, loss, regulation, provenance, warrant, direction}\}.$$

Figure 1: Root-to-Human Master Equation River. The architecture is recursive rather than circular because retained change can alter the next reader. The full chain is not a single empirical law; it is a typed dependency architecture whose local components carry different epistemic statuses (Table 2).

15 Why This Is Not Simply Affordance or Capability Theory

Affordance theory and the capability approach already establish that real action possibility is relational and cannot be read from formal option lists alone [Gibson, 1979, Sen, 1985]. The Live Possibility Field earns a distinct role only if it captures a residual difference:

$$\text{feasible for the agent} \neq \text{currently live for the agent.} \tag{43}$$

A route may be legally open, materially feasible, and learnable within the horizon, yet fail to enter practical consideration because it is unnamed, unintelligible, socially unrecognised, affectively blocked, or absent from the current problem representation. Conversely, a route can be extremely live while remaining physically impossible or normatively dangerous.

This distinction generates a direct empirical burden. Under matched formal options and resources, a live-field measure should predict which non-cued alternatives people spontaneously generate, consider, reject, or return to later. If established affordance, capability, preference, salience, and cost measures explain the same variance, the live field should be compressed into those constructs rather than defended by terminology.

16 Why Human Return Is More Than Post-Test Performance

Human–AI studies often evaluate the coupled system while assistance is present, yet the programme's normative target concerns retained human capability. CTSA is not justified simply because delayed testing is sensible. It must add a useful classification of what returned and what was lost [Lahtee, 2026b].

The return architecture predicts partial dissociations. A user can acquire a new Concept without Skill; a Tool-selection rule without Alternative-generation; a Skill with poor metacognitive regulation; or alternatives that remain unwarranted. It also predicts gain–loss asymmetry: gross increase in one return dimension can coexist with reduced uncertainty sensitivity or checking habit in another. These are not empirical facts until tested. They are the minimal patterns that would make the Human Return interface explanatory rather than rhetorical.

17 Adversarial Tests and Rival Models

A theory programme becomes scientifically useful only when it can lose. The next empirical programme should not attempt to validate the entire river. It should test the smallest interfaces where the readout-native architecture claims incremental value.

17.1 Rival-model ladder

For a pre-choice study, compare at least

$$M_0 = \text{formal option count + stated preference,} \tag{44}$$
$$M_1 = \text{affordance/capability + cost/resources,} \tag{45}$$
$$M_2 = \text{constructed-preference / salience model,} \tag{46}$$
$$M_3 = \text{predictive or active-inference implementation,} \tag{47}$$
$$M_4 = \text{readout-native live-field model.} \tag{48}$$

The primary test is not in-sample fit. It is held-out prediction of spontaneous candidate generation, considered options, enacted choice, and delayed unaided re-entry. For Human–AI studies, hold current prompt content as constant as possible and vary interaction history, resistance route, or retrieval support. The pre-prompt claim earns a role only if prior interaction predicts later human judgements or candidate accessibility beyond current prompt, baseline traits, and simple exposure count.

17.2 Minimal empirical predictions

H1 — Live-field incremental value [Open]
Under matched feasible sets, $L_{A,t}$ will explain reproducible variance in which alternatives enter spontaneous practical consideration beyond resources, preference ratings, and ordinary salience measures.

H2 — History residual [Open]
Prior traversal history will predict later route accessibility after current semantic state, task, and control variables are matched. Failure removes the distinct momentum term.

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Table 2: Claim-status ledger for the load-bearing transitions.

Object / transition	Status	Primary programme source	What would force revision or removal?
Retained distinction; ordered tape; finite root stepper	Root theorem / root architecture	Readout Universe; Readout Genesis [Lahtee, 2026m,l,k]	Fails only by revision of the root specification itself; does not imply human empirical validity.
Domain weld and no-silent-loss	Root architecture / protocol	Readout Genesis [Lahtee, 2026l]	Human-domain claims cannot inherit root status if translation defects or missing sufficiency remain unresolved.
Meaning after readout	Domain definition / architecture	MEMK [Readout Genesis Programme, 2026]	Remove any claim that treats meaning as root content; revise if a different domain translation explains the same target with fewer commitments.
Experience as meaning-giving	DR conceptual thesis	Experience Is Meaning-Giving [Lahtee, 2026c]	Narrow if significance-free lived experience can be operationally demonstrated under the declared scope.
History-shaped semantic access	OPEN empirical hypothesis	From Problem to Hypothesis [Lahtee, 2026j]	Remove the history term if current state, salience, task, and control explain later access without a residual path-history effect.
Resonance / rhythm	OPEN optional domain diagnostics	Experience Is Meaning-Giving [Lahtee, 2026c]	Drop if schema/familiarity/retrieval or ordinary timing variables explain the same phenomena with no incremental value.
Live Possibility Field	DR / OPEN measurement object	Choice Begins Before Choice [Lahtee, 2026a]	Compress into capability/affordance/preference measures if it adds no reliable information about spontaneous candidate availability or delayed re-entry.
Effective agency envelope (witnessed)	Measurement architecture; definitions at DEFINITION; lemmas on a stated discrete model	Potential as a Readout [Lahtee, 2026i]	Revise if read/decision/enactment/feedback separation does not improve identification over simpler outcome measures in declared tasks.
Candidate status and D-R-A	Domain rule / constitutive necessity under the target definition	Fusion/Tunnel [Lahtee, 2026d]	D-R-A changes if the target definition of durable human epistemic expansion changes; none is a truth guarantee.
CTSA Human Return	OPEN measurement architecture	CTSA v9.1 [Lahtee, 2026b]	Narrow or remove if classes are unreliable or add no information beyond ordinary delayed performance, transfer, and established learning measures.

H3 — Pre-prompt carryover [externally constrained, mechanism OPEN]
Repeated Human–AI interaction can alter later human judgements before a new AI response is seen. External evidence supports the general carryover phenomenon [Glickman and Sharot, 2025]; the readout-native mechanism remains open.

H4 — Human Return dissociation [OPEN]
Delayed unaided Concept, Tool, Skill, and Alternative Return will show partially dissociable profiles rather than collapse into one generic performance factor.

H5 — Gain/loss asymmetry [OPEN]
Positive gross return can coexist with reduced checking habits, uncertainty sensitivity, or alternative accessibility; net human expansion will therefore differ from gross retained gain.

H6 — Optional modifiers must earn survival [OPEN]
Resonance, rhythm, and barrier/transition variables remain only if they add held-out explanatory or predictive value beyond schema, familiarity, appraisal, retrieval, exposure count, event boundaries, and continuous accumulation models.

18 GRACEFUL DEGRADATION

The programme should survive the deletion of its most provocative metaphors. If Human LoRA terminology is removed, the remaining claim is selective retained operator change. If resonance is removed, current-versus-retained congruence can be represented by ordinary schema/retrieval variables. If rhythm is removed, ordered-history and event-boundary variables remain. If barrier language is removed, continuous retained accumulation remains available. If CTSA does not outperform simpler delayed measures, Human Return can be retained without the four-class taxonomy. The

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load-bearing core is narrower:

$$\begin{aligned}&\text{retained distinction} \to \text{domain translation/readout}\\&\to \text{meaning} \to \text{experience} \to \text{retained change}\\&\to \text{graded accessibility} \to \text{live possibility}\\&\to \text{choice/action} \to \text{feedback/new record},\end{aligned} \tag{49}$$

with provenance, loss, reader, claim status, and correction kept explicit. If this core itself adds no explanatory or methodological value beyond existing architectures, the programme should be reduced to a vocabulary for organising already known distinctions. The architecture earns theory status only through discriminating use.

19 Contributions

The paper's contribution can now be stated without claiming ownership over its mature neighbours.

A root-to-human derivation order. Meaning, experience, live possibility, choice, agency, and Human–AI return are not inserted at the root but admitted downstream of retained difference and translation.

Domain weld and defect accounting as epistemic governance. Structure-preserving abstraction is established mathematics; the contribution is to use it as a guard against semantic import and certainty drift.

A continuous but non-collapsing chain from meaning to agency. Experience is downstream of readout; live possibility is downstream of meaning and history but upstream of choice; agency is broader than observed action.

Human–AI interaction inside the same recursive river. A prompt is generated from a pre-existing human state, AI output begins as a candidate, and retained effects can reshape the next human state before the next prompt.

Human Return as the upper measurement boundary. Assisted performance and durable human capability are intentionally separated.

A claim-status and provenance ledger attached to the equations themselves. A transition is incomplete unless its reader, source, defects, claim tier, and legitimate defeater are visible.

20 Conclusion: Before Meaning, Before Choice

The strongest insight of the Human–AI Readout Programme is not that experience contains meaning or that choice has social and historical conditions. Those propositions have deep predecessors. The stronger contribution is methodological: do not begin where familiar human vocabulary makes the problem look easiest.

Begin with what is retained and distinguishable. Ask what reader makes which structure accessible. Require a domain translation before semantic naming. Expose what that translation loses. Treat meaning as a domain significance relation rather than root substance. Treat experience as the lived configuration produced when an encountered trace becomes significant to a finite reader. Treat retention as selective rather than automatic. Treat accessibility as graded and history-shaped. Treat live possibility as distinct from feasible capacity. Treat choice as downstream of what has become live. Treat action, outcome, and evaluator-visible record as different levels. Treat AI output as a candidate encounter rather than truth. Treat retained human change as direction-neutral until resistance and warrant are checked. Finally, remove the AI and ask what came back with the human.

The resulting thesis is therefore not a new psychological slogan but a derivational constraint:

$$\begin{aligned}&\text{Retained Difference} \to \text{Admissible Readout}\\&\to \text{Meaning} \to \text{Experience}\\&\to \text{History-Shaped Accessibility} \to \text{Live Possibility}\\&\to \text{Choice} \to \text{Corrigible Agency}\\&\to \text{World Consequence} \to \text{New Retained Difference}.\end{aligned} \tag{50}$$

Human–AI systems enter this river as mediating structures, not as epistemic sovereigns. Their importance lies in how they change what can be articulated, compared, retained, resisted, and returned. The next scientific step is consequently modest but decisive: test whether the readout-native interfaces predict anything that simpler rival models do not. If they do, the programme has identified a useful architecture. If they do not, the architecture must shrink.

21 Source Lineage and Check Record

Version 1.0 of this paper was checked with the glosa methodology [Lahtee, 2026e] before deposit. An independent review pass found no defect in the argument, the notation, or the attributions to the deposited programme papers, and found the external attributions it could check at full text (Tishby et al.; Glickman and Sharot) accurate; it found three source-lineage defects, all repaired here. (i) The Readout Genesis whitepaper was cited as a floating "working specification"; it is a public file on the main branch of `morrocwi/readout_genesis` and is now cited by commit and blob SHA. (ii) The MEMK domain specification was cited as an unpublished "v0.3.0 ANCHORED" draft; the only MEMK file in a public tree is v0.2.0 on the branch `feat/register-philosophy-memk-v0.2.0`, cited here by commit, blob, and SHA-256, with the v0.3.0 draft disclosed as unpublished and nothing in the paper depending on it. (iii) The agency-potential model was cited from a superseded, undeposited programme note and written over Π^{feas}; it is now cited from *Potential as a Readout* and written over the witnessed set (Section 9). Every other internal reference now carries its Zenodo DOI, and *Master Equation River* is cited in its glosa-checked v1.1. The three git-anchored citation cards and the cards for the world literature named in Section 2 live in the glosa repository (`records/lit/before-meaning-before-choice`), each carrying the fetched URL, page, line, and one verbatim passage, re-located by a fresh headless session of the same model (independence class I1). Knowledge state: K0 for every claim, timestamped and citable, not yet checked at I2 or above.

AI-assistance disclosure. The rebuild of v1.0 into this version, the source fetching, the git anchoring, and the passage extraction were done with an AI assistant under the author's direction; the problem, the argument, the selection of what to claim, and the v1.0 text that anchors this version are the author's. No AI or vendor is an author. The author's raw lines behind this version are in the Blackbox Log (concept DOI 10.5281/zenodo.22302518, entries BBL-2026-09-06-144 to -149).

REFERENCES

Gregory Bateson. *Steps to an Ecology of Mind*. Chandler, 1972.

Ezequiel A. Di Paolo. Autopoiesis, adaptivity, teleology, agency. *Phenomenology and the Cognitive Sciences*, 4 (4):429–452, 2005. doi: 10.1007/s11097-005-9002-y.

Mustafa Emirbayer and Ann Mische. What is agency? *American Journal of Sociology*, 103(4):962–1023, 1998. doi: 10.1086/231294.

Norm Ferns, Prakash Panangaden, and Doina Precup. Bisimulation metrics for continuous Markov decision processes. *SIAM Journal on Computing*, 40(6):1662–1714, 2011. doi: 10.1137/10080484X.

Michel Foucault. The subject and power. *Critical Inquiry*, 8 (4):777–795, 1982. doi: 10.1086/448181.

Miranda Fricker. *Epistemic Injustice: Power and the Ethics of Knowing*. Oxford University Press, 2007. doi: 10.1093/acprof:oso/9780198237907.001.0001.

Karl Friston, Thomas FitzGerald, Francesco Rigoli, Philipp Schwartenbeck, and Giovanni Pezzulo. Active inference: A process theory. *Neural Computation*, 29(1):1–49, 2017. doi: 10.1162/NECO_a_00912.

Johan Galtung. Violence, peace, and peace research. *Journal of Peace Research*, 6(3):167–191, 1969. doi: 10.1177/002234336900600301.

James J. Gibson. *The Ecological Approach to Visual Perception*. Houghton Mifflin, 1979.

Moshe Glickman and Tali Sharot. How human–AI feedback loops alter human perceptual, emotional and social judgements. *Nature Human Behaviour*, 9:345–359, 2025. doi: 10.1038/s41562-024-02077-2.

Naila Kabeer. Resources, agency, achievements: Reflections on the measurement of women's empowerment. *Development and Change*, 30(3):435–464, 1999. doi: 10.1111/1467-7660.00125.

Yaoharee Lahtee. Choice begins before choice: Meaning-shaped accessibility, live possibility, and effective agency in human-AI systems. Zenodo, Full Paper v1.0, 5 September 2026, 2026a.

Yaoharee Lahtee. CTSA human-return readout: A session-boundary measurement architecture for retained human capability after AI. Zenodo, Epistemic Note v9.1 FINAL, 5 September 2026, 2026b.

Yaoharee Lahtee. Experience is meaning-giving: A strong-form readout-retention theory of phenomena, resonance, rhythm, accumulation, and transformative release. Zenodo, Full Paper v2.0, 5 September 2026, 2026c.

Yaoharee Lahtee. Epistemic fusion or epistemic tunnel: A phenomenology-anchored, global-literature-constrained, problem-first architecture of human-AI collaboration. Zenodo, Epistemic Note v8.1, 5 September 2026 (concept DOI 10.5281/zenodo.22318039), 2026d.

Yaoharee Lahtee. glosa — rigour without infrastructure: A standalone scholar methodology for human–AI knowledge co-production (v0.4.1). Zenodo software record, 2026e.

Yaoharee Lahtee. Experience is the human LoRA: A readout–retention theory of selective model change. Zenodo, 18 July 2026, 2026f.

Yaoharee Lahtee. Master equation river: A provenance audit across the human–AI readout programme. Zenodo, Internal Programme Synthesis v1.1 (glosa-checked), 6 September 2026, 2026g.

Yaoharee Lahtee. Meaning before naming: A readout-retention architecture of affective-semantic reorganization. Zenodo, Concept Note v1.0, 5 September 2026, 2026h.

Yaoharee Lahtee. Potential as a readout: Witnessed envelopes, two-layer agency, and the measurement of pseudo-peace. Zenodo, Epistemic Note K0 v2, 5 September 2026 (concept DOI 10.5281/zenodo.22345708). Supersedes the undeposited programme note *Effective, Corrigible Agency Potential* (5 September 2026), 2026i.

Yaoharee Lahtee. From problem to hypothesis: Dynamic semantic mobility, bounded knowers, and the readout-discriminable hypothesis space. Zenodo, K0 final author draft, 4 September 2026, 2026j.

Yaoharee Lahtee. Readout genesis standalone synthesis: Information epistemic foundation, conditioned agency, and meta-readout governance. Zenodo, 24 July 2026, 2026k.

10.5281/zenodo.22424434

Yaoharee Lahtee. Readout genesis universal technical whitepaper, v1.2.0: universal research architecture, typed interfaces, domain gates, and anti-drift constitution. Public GitHub repository morrocwi/readout_genesis, file READOUT_GENESIS_UNIVERSAL_TECHNICAL_WHITEPAPER_v1.2.0.md, main at commit 04cde19be2c885a11b42597b1cdb60fb5b7ca1bb, blob 6ca7ef20a1e9ea17632451e6c8a19b3ae7c9fa8a, 2026l. `https://github.com/morrocwi/readout_genesis/blob/04cde19be2c885a11b42597b1cdb60fb5b7ca1bb/READOUT_GENESIS_UNIVERSAL_TECHNICAL_WHITEPAPER_v1.2.0.md`. Working specification; not a Zenodo record.

Yaoharee Lahtee. Readout universe: Logic — equation and operator ledger. Public GitHub repository morrocwi/readout_universe, file logic.md, blob a28a16ae9a5442f2b179136bb76c4ffd0770388d, 2026m. `https://github.com/morrocwi/readout_universe`.

Vivian Lai, Chacha Chen, Alison Smith-Renner, Q. Vera Liao, and Chenhao Tan. Towards a science of human-AI decision making: An overview of design space in empirical human-subject studies. In *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency*, pages 1369–1385, 2023. doi: 10.1145/3593013.3594087.

Philip Pettit. *Republicanism: A Theory of Freedom and Government*. Oxford University Press, 1997.

Balaraman Ravindran and Andrew G. Barto. SMDP homomorphisms: An algebraic approach to abstraction in semi-Markov decision processes. In *Proceedings of the 18th International Joint Conference on Artificial Intelligence*, pages 1011–1016, 2003.

Readout Genesis Programme. PHILOSOPHY_MEMK_DOMAIN v0.2.0: root-connected semantic–cognitive domain specification. Public GitHub repository morrocwi/readout_genesis, branch feat/register-philosophy-memk-v0.2.0, commit 49e444cc64bdaea2767c2c18d209654cdf6cb23f, file domains/philosophy/PHILOSOPHY_MEMK_DOMAIN_v0_2_0.yaml, blob 3d51fe8095f93207ab3ff4ae054042b6cf880a3a, SHA-256 1674a68e8b2811c80dc043ce82aa63186cb0c433a7bb0bf08c858ad6e87c1229, 2026. With CHECKSUMS_SHA256_PHILOSOPHY_MEMK.txt, CLAIM_BOUNDARY_PHILOSOPHY_MEMK.json and PROOF_RECEIPT_PHILOSOPHY_MEMK.json in the same tree. A later draft, v0.3.0-ANCHORED, exists as an internal working file with a recorded anchor-preservation check against v0.2.0 and is not yet published; nothing in this paper depends on it beyond v0.2.0.

Amartya Sen. Well-being, agency and freedom: The Dewey lectures 1984. *The Journal of Philosophy*, 82(4):169–221, 1985. doi: 10.2307/2026184.

Cosma Rohilla Shalizi and James P. Crutchfield. Computational mechanics: Pattern and prediction, structure and simplicity. *Journal of Statistical Physics*, 104:817–879, 2001. doi: 10.1023/A:1010388907793.

Evan Thompson. *Mind in Life: Biology, Phenomenology, and the Sciences of Mind*. Harvard University Press, 2007.

Naftali Tishby, Fernando C. Pereira, and William Bialek. The information bottleneck method. In *Proceedings of the 37th Annual Allerton Conference on Communication, Control and Computing*, pages 368–377, 1999. Preprint arXiv:physics/0004057.

Francisco J. Varela, Evan Thompson, and Eleanor Rosch. *The Embodied Mind: Cognitive Science and Human Experience*. MIT Press, 1991.

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State of Evidence for the Readout Hypothesis-Generation Programme: A Shared Evidence Registry for Papers I–IV

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What the chapter states. “Most local mechanisms invoked by the programme are not new discoveries.” “What remains unestablished is the programme’s proposed integration and its specific observables: readout-weighted accessibility, content-matched first passage to usable hypothesis classes, a measurable live candidate-set operator, and controlled human–AI epistemic multiplication.”

Status. “Living GLOSA K0 evidence synthesis for the four-paper Readout hypothesis-generation programme.” “No independent checker has yet certified this registry.”

Falsifier(s). “DECISIVE TEST: a record that would materially confirm, narrow, or falsify the open claim” (applied per claim in the chapter’s own registry, Table 1).

Read before. This book’s Chapters 1–3, 27, 39 (mechanical KG cross-reference list; Appendix A).

Feeds into. Ch. 31 (mechanical KG cross-reference list; Appendix A).

CROSS-LITERATURE EVIDENCE BACKBONE GLOSA K0

State of Evidence for the Readout Hypothesis-Generation Programme

A Shared Evidence Registry for Papers I–IV

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4 September 2026

Status. Living GLOSA K0 evidence synthesis for the four-paper Readout hypothesis-generation programme. Its purpose is not to create another novelty claim, but to keep one shared record of what prior literature has already observed, where multiple literatures converge, what the present programme merely synthesizes, which formal claims remain open, and which records would be decisive. The review is targeted and system-oriented, not preregistered or systematic. "Observed" below never means that a paper-specific Readout variable has already been validated; it means that the cited literature has directly reported the neighboring empirical phenomenon. No independent checker has yet certified this registry.

Abstract

The four-paper Readout hypothesis-generation programme spans problem formation, bounded semantic mobility, knowledge organization, candidate-set formation, and human–AI amplification. Most local mechanisms invoked by the programme are not new discoveries. Problem representation affects later solving; semantic-memory organization relates to creative and convergent retrieval; domain knowledge changes early diagnostic hypothesis generation; search behavior depends jointly on representation and process; recent inference history biases later inference; expertise can accelerate familiar reasoning while inducing fixation; representational change can resolve impasse; scientific-hypothesis timing is measurable and speed can dissociate from quality; research institutions and incentives route which problems and relations receive attention; scientific communities face unconceived alternatives; and current AI systems can recursively generate, critique, test, revise, and also homogenize or hallucinate hypotheses. What remains unestablished is the programme's proposed integration and its specific observables: readout-weighted accessibility, content-matched first passage to usable hypothesis classes, a measurable live candidate-set operator, and controlled human–AI epistemic multiplication. This document registers the state of evidence claim by claim, separates evidence from synthesis, and identifies the decisive experiments required to move Papers I–IV beyond K0.

Keywords: state of evidence; hypothesis generation; scientific discovery; semantic memory; expertise; fixation; candidate-set formation; human-AI collaboration; AI for science; Readout Universe; Readout Genesis

1 Purpose and Evidence Coding

The programme now contains enough moving parts that the main risk is no longer lack of literature, but *role drift*: a neighboring empirical result may be cited as if it had validated a Readout construct, or a programme-level synthesis may be restated later as an established finding. This registry introduces five evidence states.

- **OBSERVED**: at least one published empirical, computational, historical, or capability result directly reports the neighboring phenomenon.
- **CONVERGENT**: multiple methods, domains, or independent literatures support the same qualitative direction, without proving the programme's exact formalization.
- **SYNTHESIS**: an inference produced by joining prior findings across literatures. This is the main intellectual work of the programme, but it is not itself an observed fact.
- **OPEN**: a paper-specific claim or variable not yet directly established.
- **DECISIVE TEST**: a record that would materially confirm, narrow, or falsify the open claim.

Evidence strength is additionally marked STRONG, MODERATE, or ADJACENT. "Strong" means direct and/or convergent evidence for the local phenomenon; it does not mean causal proof of the full chain.

Non-collapse rule for the evidence registry.

$$\text{neighboring evidence} \neq \text{formal-variable validation} \neq \text{truth of the integrated theory}$$

The same paper may therefore be strong evidence for a local phenomenon and only adjacent evidence for a Readout-level claim.

2 Master Evidence Map

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Table 1: Claim registry. IDs are intended to be reused across Papers I–IV. "Status" records the strongest warranted statement as of 4 September 2026.

ID	Claim / phenomenon	Representative evidence	Status	What remains unestablished
E01	Problem representation/construction changes downstream creative problem solving.	Reiter-Palmon et al. 1997 [1].	OBSERVED/ STRONG	No direct validation of the programme's retained-residual formalization.
E02	Semantic-memory organization contains information beyond raw concept inventory.	Kenett et al. 2014; Denervaud et al. 2021; Luchini et al. 2023 [4, 7, 6].	CONVERGENT/ STRONG	Causal content-matched manipulation of internal topology remains difficult.
E03	Domain knowledge changes early hypothesis generation and relation use in diagnosis.	Joseph–Patel 1990; Kushniruk et al. 1998 [8, 9].	CONVERGENT/ STRONG	Organization is confounded with experience and knowledge amount.
E04	Retrieval/search depends jointly on representation and search process.	Abbott et al. 2012; Kumar et al. 2022 [10, 11].	CONVERGENT/ STRONG	No literal neural graph-walk interpretation is licensed.
E05	Recent inference history can bias subsequent inference.	Dasgupta et al. 2018; Honey et al. 2023 [12, 13].	CONVERGENT/ STRONG	The paper's discrete semantic-momentum variable m_t is still OPEN.
E06	Time pressure can reduce hypothesis generation and alter action priority.	Alison et al. 2013 [14].	OBSERVED/ MODERATE	Does not isolate knowledge topology or momentum.
E07	Controlled retrieval of less-dominant semantic information contributes to creative verbal association.	Krieger-Redwood et al. 2023 [15].	OBSERVED/ STRONG	Does not validate a universal semantic-control operator for scientific imagination.
E08	Familiar expert solutions can bias attention away from alternatives.	Bilalić et al. 2010 [16].	OBSERVED/ STRONG	The proposed attraction variable $a_{Q,t}$ remains a model, not a measured primitive.
E09	Insight can involve explicit representational change, including constraint relaxation and chunk decomposition.	Knoblich et al. 1999 [17].	OBSERVED/ STRONG	The Readout restructuring operator $\mathcal{B}_t$ remains a typed synthesis.
E10	Expertise can reverse under schema-mismatched conditions.	Armougum et al. 2020 [18].	OBSERVED/ MODERATE	"Expertise and fixation are two regimes of one routing system" remains SYNTHESIS/ OPEN.
E11	Hypothesis-generation speed can dissociate from quality.	Jing et al. 2024: VIADS reduced time per hypothesis while some quality ratings fell [19].	OBSERVED/ MODERATE	Small study; does not establish first passage to a readout-usable hypothesis.
E12	Scientific research strategy is patterned by tradition, innovation, and incentives.	Foster et al. 2015 [20].	OBSERVED/ STRONGobservational	Does not by itself identify a cognitive candidate-set mechanism.
E13	Institutions can leave socially salient research questions unfunded or unproduced ("undone science").	Frickel et al. 2010 [21].	OBSERVED/ STRONGqualitative-sociological	"Institutional attraction" remains a programme-level translation.
E14	Historically successful science has repeatedly faced serious alternatives not conceived at the time.	Stanford 2006 [22].	OBSERVED/ STRONGhistorical	Prospective measurement of truly unconceived alternatives remains unresolved.
E15	Collective search/division of cognitive labor can alter discovery performance.	Weisberg–Muldoon 2009; Alexander et al. 2015 [23, 24].	CONVERGENT/ STRONGmodeling	Real candidate nodes may not be pre-enumerated as in many landscapes.
E16	Values and pursuit decisions can influence what later evidence exists and how theories are developed.	Elliott–McKaughan 2009; pursuit-worthiness literature [25, 26].	CONVERGENT/ MODERATE	No general operational candidate-set norm follows automatically.
E17	AI can recursively generate, critique, rank, refine, and evolve scientific hypotheses.	Co-Scientist 2026 [27].	OBSERVED/ STRONGcapability	Does not establish readout-class expansion or truth.
E18	AI can connect hypothesis generation to experiment, data analysis, and revised hypotheses.	Robin 2026 [28].	OBSERVED/ STRONGcapability	Does not establish optimal human–AI operator allocation.
E19	AI can automate broad portions of an end-to-end research cycle.	AI Scientist 2026 [29].	OBSERVED/ STRONGcapability	Capability is not epistemic superiority and may increase review burden/noise.
E20	LLM-generated scientific hypotheses can be plausible but untruthful; grounding filters can improve selection.	TruthHypo / KnowHD 2025 [30].	OBSERVED/ STRONG	Grounding is not equivalent to truth or warrant.
E21	Generative AI can improve individual creative output while reducing diversity across a population of outputs.	Doshi–Hauser 2024; Meincke et al. 2025 [31, 32].	CONVERGENT/ STRONG	Readout-distinguishable diversity has not been measured.
E22	Human–AI collaboration effects can be non-monotonic.	Huang 2025 [33].	OBSERVED/ MODERATE/ emerging	Domain generality and mechanism remain open.
E23	Knowledge organization changes first-passage time and first-hit usable hypothesis class under content matching.	No direct decisive study identified. Neighboring evidence E02–E11.	OPEN	Paper II H1–H2; requires randomized content-matched topology experiment.
E24	A live candidate set is a measurable epistemic object distinct from raw generation and appraisal.	Adjacent evidence from pursuit, consideration sets, notebooks/logging, and Paper III synthesis.	SYNTHESIS/ OPEN	Requires operational state transitions and prospective validation.
E25	Recursive gated human–AI systems yield higher usable readout-class multiplication than human-only or AI-only systems.	Capability/risk evidence E17–E22 only.	OPEN	Paper IV H1–H5; requires matched-budget, blinded, multi-arm experiment.

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3 Evidence by Stage of the Inquiry Cycle

3.1 Stage A: problem and question formation

Problem construction is already an active research object, not a blank prelude to reasoning. Reiter-Palmon and colleagues showed that how a problem is constructed affects later creative performance [1]. Scientific-discovery research likewise rejects a purely mechanical observation-to-hypothesis pipeline [2, 3]. The programme therefore does not claim novelty for "starting from a problem." Its synthesis is narrower: problem retention and question selection are treated as typed upstream operators whose choices condition later semantic access.

> **Evidence statement E01.** It is warranted to say that problem representation/construction affects downstream problem solving. It is *not* yet warranted to say that the programme's formal residual r_t or question operator Q_{Π_Q} has been empirically validated.

3.2 Stage B: knowledge organization and semantic access

Semantic-network studies show that connectivity, path structure, and organization relate to creative and convergent retrieval [4, 6]. Educational experience is associated with different semantic-memory structure and creative performance [7]. Medical reasoning provides a closer bridge to scientific hypotheses: endocrinologists and cardiologists exposed to the same endocrine case did not differ simply in relevant cue selection; higher-domain-knowledge clinicians connected cues through richer relations and generated accurate diagnostic hypotheses earlier [8]. This supports the statement that structure is not reducible to inventory.

At the same time, the literature blocks a monotonic "more connectivity is better" story. Rich semantic neighborhoods can improve fluency yet constrain originality [5]; expertise can become costly when established schemas mismatch the environment [18]. The appropriate synthesis is therefore task-conditioned routing, not a universally optimal topology.

3.3 Stage C: history, control, fixation, and restructuring

History dependence has direct experimental support. Dasgupta et al. reported three experiments in which the structure of an immediately preceding related inference systematically affected the next inference [12]. Honey et al. synthesize a broader literature in which recent thoughts, task sets, and dispositions persist and shape subsequent cognition [13]. This is strong neighboring evidence for a history term, but not for the particular recurrence equation used in Paper I.

Control and fixation also have direct anchors. Bilalić et al. showed that experts' first candidate solution can direct attention toward confirming information and away from alternatives even while they believe they are searching broadly [16]. Krieger-Redwood et al. found that controlled retrieval of less-dominant semantic information contributes to more creative verbal association [15]. Knoblich et al. showed that constraint relaxation and chunk decomposition can explain insight through representational change [17]. These findings jointly motivate the Paper I distinction among accessibility, attraction, control, and restructuring.

> **Programme synthesis S1.**
>
> $$\begin{aligned} \text{expert acceleration} &\longleftrightarrow \text{task-matched regime}, \\ \text{expert fixation} &\longleftrightarrow \text{task-mismatched regime?} \end{aligned}$$
>
> This synthesis is plausible because fast familiar access (E03), attentional capture (E08), and expertise reversal (E10) coexist in the literature. It is not yet directly established as one mechanism.

3.4 Stage D: hypothesis timing, usability, and discrimination

The VIADS human-participant study is unusually important because it measures scientific-hypothesis generation directly rather than creativity proxies. Eighteen participants generated 227 hypotheses; the VIADS condition produced similar numbers of hypotheses but shorter time per hypothesis among inexperienced researchers (258 s versus 379 s) while receiving lower ratings on feasibility and the combined validity/significance/feasibility measure [19]. The study is small and its reported power for the timing result was limited, so it should not carry a universal law. It nonetheless provides direct evidence for the non-collapse:

$$\boxed{\text{generation speed} \neq \text{hypothesis quality}}$$

Paper II therefore shifts the endpoint from time to any hypothesis toward first passage to a declared *usable* readout class. That shift is a programme contribution and remains open.

3.5 Stage E: candidate sets, institutions, and unconceived alternatives

Stanford's historical argument on unconceived alternatives establishes that scientists can rationally compare available theories while still missing serious rivals that later become visible [22]. Epistemic-landscape work shows that social learning and division of cognitive labor can alter collective search performance [23, 24]. Foster et al. show at large scale that biomedical research strategies are patterned by a tension between established relations and risky innovation [20]. Frickel et al. document "undone science," where institutional and resource structures leave research agendas unproduced or neglected [21]. Elliott and McKaughan further show why values in discovery and pursuit can alter later appraisal by shaping what gets developed and tested [25].

These literatures strongly support the agenda-level claim that what becomes scientifically live is not determined by evidence alone. They do not yet validate Paper III's candidate-set operator Ψ or provide a prospective measure of truly unconceived alternatives.

> **Do not overstate E12–E16.** Institutional patterning is not evidence of intentional suppression; historical unconceived alternatives are not a direct count of present unknowns; and epistemic-landscape simulations typically predefine the landscape. Paper III should continue to present candidate-set formation as an agenda and measurement problem, not as a solved social theory.

3.6 Stage F: AI amplification, grounding, and diversity

The capability frontier moved rapidly by 2026. Co-Scientist demonstrates recursive generation, critique, ranking, evolution, and scientist steering [27]. Robin connects literature search, hypothesis generation, experiment proposal, result interpretation, and updated hypotheses in experi-

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mental biology [28]. The AI Scientist automates ideation through experimentation, writing, and automated review in machine-learning research [29]. These systems establish that recursive and closed-loop scientific workflows are technically realizable.

But capability evidence also establishes failure modes. TruthHypo reports that LLMs can produce plausible but untruthful scientific hypotheses and that knowledge-grounding scores can improve filtering [30]. Doshi and Hauser find that generative AI can increase individual creative output while reducing collective diversity [31]; Meincke et al. report a comparable diversity trade-off in brainstorming [32]. Huang reports a curvilinear "Goldilocks" effect in which moderate human–AI collaboration outperformed low and high collaboration in the tested creativity tasks [33]. The warranted synthesis is therefore not "more AI is better," but that augmentation quality depends on operator allocation, diversity preservation, grounding, and convergence control.

4 What the Literature Jointly Supports

Across psychology, medicine, cognitive science, philosophy of science, sociology of science, and AI-for-science, the evidence can be arranged as a convergent but not yet fully causal chain:

$$
\begin{aligned}
\text{Problem/Question} &\dashrightarrow \text{Knowledge Representation}\\
&\dashrightarrow \text{Access/Search Dynamics}\\
&\dashrightarrow \text{Candidate Availability}\\
&\dashrightarrow \text{Selection/Pursuit}\\
&\dashrightarrow \text{Appraisal/Test}\\
&\dashrightarrow \text{Record/Revision.}
\end{aligned}
$$

The dashed arrows are deliberate. Different literatures support different local transitions; no single study establishes the entire sequence as one causal model.

Programme-level synthesis S2. The most defensible common question is:

> *How does a bounded epistemic system turn a problem into the alternatives that evidence is eventually able to judge?*

This is a synthesis of already observed phenomena, not a claim that the integrated answer is known.

Three especially important non-collapse rules are jointly supported by the evidence:

$$\text{reachability} \neq \text{accessibility}, \tag{1}$$

$$\text{speed} \neq \text{quality/warrant}, \tag{2}$$

$$\text{AI output volume} \neq \text{epistemic diversity}. \tag{3}$$

The stronger Readout chain

$$\boxed{\text{Attraction} \neq \text{Accessibility} \neq \text{Warrant} \neq \text{Truth}}$$

remains a conceptual typing rule rather than an experimentally established decomposition.

5 Cross-Paper Claim Status

5.1 Paper I: bounded semantic mobility

Well-supported neighboring phenomena: E01–E10, especially problem construction, semantic organization, history dependence, semantic control, fixation, and representational change.

Our synthesis: imagination as bounded dynamic semantic mobility; provenance-typed distinction ledger; reader-relative readout classes; bounded-knower standing; individual/collective operator typing.

Still open: κ as a useful predictive access kernel, finite attraction $a_{Q,t}$, semantic momentum m_t as separable from current state, and restructuring $\mathcal{B}_t$ as an empirically stable mechanism.

Decisive records: out-of-sample predictive gain beyond binary reachability; path-history effect beyond current state; pre-registered restructuring signatures; provenance-sensitive loss effects.

5.2 Paper II: knowledge topology and first passage

Well-supported neighboring phenomena: E02–E11.

Our synthesis: knowledge organization may change both time-to-first-usable hypothesis and the first-hit readout class.

Still open:

$$\mathcal{G}_1^K \neq \mathcal{G}_2^K \Rightarrow \mathcal{L}(\tau_{\mathcal{U}}|\mathcal{G}_1^K, Q) \neq \mathcal{L}(\tau_{\mathcal{U}}|\mathcal{G}_2^K, Q), \tag{4}$$

and the corresponding directional claim for π^{first} under true content matching.

Decisive record: randomized synthetic-science experiment with identical facts/evidence/exposure, manipulated organization, time-stamped candidate logging, blinded readout-class coding, and predeclared usability gates.

5.3 Paper III: candidate-set formation

Well-supported neighboring phenomena: E12–E16.

Our synthesis: appraisal quality and candidate-set breadth are separable epistemic properties; candidate-set formation deserves direct criticism and measurement.

Still open: a validated operational definition of a live candidate, prospective measurement of hidden/unconceived alternatives, and a non-reductive collective candidate-set model.

Decisive records: sequential generated/considered/pursued/rejected tagging; synthetic hidden-alternative tasks; controlled team communication topologies; evidence-triggered restructuring experiments.

5.4 Paper IV: epistemic chain reaction

Well-supported neighboring phenomena: E17–E22.

Our synthesis: AI's strongest role may be controlled multiplication of new, live, readout-distinguishable, minimally grounded alternatives rather than autonomous replacement or raw idea generation.

Still open: k_{epi}, the proposed reproduction-matrix extension, the claim that gated recursive human–AI collaboration outperforms human-only/one-shot/autonomous AI under matched budgets, and the normative proposal that humans should retain particular load-bearing operators.

Decisive record: four-arm matched-budget experiment with blinded equivalence coding, grounding checks, first-passage outcomes, diversity after quotienting, hallucination/defect rates, review burden, and revision quality.

6 Priority Decisive Experiments

The evidence registry suggests that five experiments would update the largest fraction of the programme at once.

1. **Content-matched topology experiment.** Identical facts and evidence; manipulate only relational organization. Primary outcomes: $\tau_{\mathcal{U}}$ and π^{first}. This is the single highest-value test for Paper II and also constrains Paper I.
2. **History-versus-current-state experiment.** Hold current semantic state approximately fixed while manipulating prior traversal. Test whether history improves out-of-sample prediction beyond a memoryless model. This directly earns or removes semantic momentum as a separate object.

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3. **Candidate-state transition study.** Time-stamp generated, considered, retained, pursued, rejected, and tested hypotheses. Determine whether a stable live-candidate state is empirically separable from raw generation and appraisal.
4. **Human–AI four-arm chain-reaction trial.** Compare human-only, human+one-shot AI, autonomous multi-agent AI, and recursively gated human–AI under equal resource budgets. Measure readout classes, grounding, review burden, first passage, and revision.
5. **False-basin escape challenge.** Seed a coherent but incomplete paradigm in humans, AI agents, and mixed teams. Measure whether heterogeneity plus explicit restructuring/gating increases escape to valid rival readout classes without increasing hallucination.

Highest-leverage methodological requirement. Every future study should log both *what was generated* and *how the candidate pool changed over time*. Without temporal candidate traces, Papers II–IV cannot distinguish topology, momentum, selection, and post-hoc evaluation.

7 Claims That Should Be Downgraded or Avoided

The combined evidence now makes several phrasings unnecessary or unsafe.

- Avoid "we discovered that semantic structure matters." This is already well supported.
- Avoid "AI expands scientific diversity" without measuring diversity after epistemic/readout quotienting; current evidence includes homogenization effects.
- Avoid "expertise causes fixation." The supported statement is task- and schema-relative; expertise can accelerate appropriate search and bias inappropriate search.
- Avoid "evidence cannot generate hypotheses." New records can restructure the next candidate set. The cycle-indexed statement is safer: appraisal at time t can compare only candidates live at time t.
- Avoid "institutions suppress alternatives" as a general causal claim. Evidence supports patterned attention, incentives, and undone research; intentional suppression is a different claim.
- Avoid calling a, m, κ, $\mathcal{B}$, Ψ, k_{epi}, or a reproduction matrix established cognitive variables. They are formal bridges and open diagnostics.
- Avoid calling the four-paper programme a validated theory. The strongest current description is *a cross-literature, falsifiable research architecture with several empirically anchored local mechanisms*.

8 Standard Language for Papers I–IV

To reduce claim drift, companion manuscripts can reuse the following templates.

For established neighboring evidence: "Prior work has already shown [phenomenon]. We use this as an empirical constraint, not as a novelty claim."
For convergence: "Independent literatures converge on [qualitative pattern], although they do not establish the present formalization."
For programme synthesis: "We propose that these findings can be typed as neighboring stages of one bounded inquiry architecture."
For open formal claims: "This object is [Open]: it earns standing only if it adds predictive or discriminative value beyond simpler alternatives."
For decisive tests: "The claim should be removed or narrowed if the predeclared test shows no added out-of-sample value under the stated controls."

9 Conclusion

The evidence base is stronger than a novelty-centered reading of the four papers suggests. The programme does not depend on inventing local phenomena. The literature already shows that problem representation matters; knowledge is organized rather than merely accumulated; search is structured and history-sensitive; control can retrieve weak associations; expertise can both accelerate and constrain; insight can restructure representations; hypothesis timing can be measured and dissociate from quality; institutions route attention and pursuit; scientific communities miss alternatives; and AI can recursively amplify scientific generation while also hallucinating and homogenizing outputs.

What has not yet been established is the proposed *joint architecture*. Papers I–IV contribute interfaces and distinctions: from question to bounded semantic access; from knowledge organization to first-passage time and direction; from generated alternatives to a live candidate set; and from human questions to recursively gated AI amplification. The appropriate status is therefore neither "already known" nor "breakthrough proved." It is:

empirically anchored components
+ original synthesis
+ open decisive tests

The programme becomes scientifically valuable to the extent that its added objects predict something the existing local literatures do not already predict. The evidence registry is intended to keep that burden visible.

Process Note

This evidence backbone was assembled after Papers I–IV had already been developed. It is therefore a post-hypothesis targeted synthesis, not a preregistered systematic review or literature-blind validation exercise. AI assisted with literature retrieval, cross-paper mapping, evidence coding, drafting, and typesetting. The human author selected the research direction and claim boundaries. The registry remains GLOSA K0 until independently checked. Readout-not-truth applies to the registry itself.

References

[1] R. Reiter-Palmon, M. D. Mumford, J. O'Connor Boes, and M. A. Runco. Problem construction and creativity: The role of ability, cue consistency, and active processing. *Creativity Research Journal* 10(1):9–23, 1997. doi:10.1207/s15326934crj1001_2.

[2] D. Klahr and K. Dunbar. Dual space search during scientific reasoning. *Cognitive Science* 12(1):1–48, 1988. doi:10.1207/s15516709cog1201_1.

[3] J. Schickore. Scientific discovery. *Stanford Encyclopedia of Philosophy*, 2022.

[4] Y. N. Kenett, D. Anaki, and M. Faust. Investigating the structure of semantic networks in low and high creative persons. *Frontiers in Human Neuroscience* 8:407, 2014. doi:10.3389/fnhum.2014.00407.

[5] R. E. Beaty, Y. N. Kenett, R. W. Hass, and D. L. Schacter. Semantic memory and creativity: the costs and benefits of semantic memory structure in generating original ideas. *Thinking & Reasoning* 29(2):305–339, 2023. doi:10.1080/13546783.2022.2076742.

10.5281/zenodo.22308066

[6] S. Luchini, Y. N. Kenett, D. C. Zeitlen, A. P. Christensen, D. M. Ellis, et al. Convergent thinking and insight problem solving relate to semantic memory network structure. *Thinking Skills and Creativity* 48:101277, 2023. doi:10.1016/j.tsc.2023.101277.

[7] S. Denervaud, A. P. Christensen, Y. N. Kenett, and R. E. Beaty. Education shapes the structure of semantic memory and impacts creative thinking. *npj Science of Learning* 6:35, 2021. doi:10.1038/s41539-021-00113-8.

[8] G.-M. Joseph and V. L. Patel. Domain knowledge and hypothesis generation in diagnostic reasoning. *Medical Decision Making* 10(1):31–46, 1990. doi:10.1177/0272989X9001000107.

[9] A. W. Kushniruk, V. L. Patel, and A. A. J. Marley. Small worlds and medical expertise: implications for medical cognition and knowledge engineering. *International Journal of Medical Informatics* 49(3):255–271, 1998. doi:10.1016/S1386-5056(98)00044-6.

[10] J. T. Abbott, J. L. Austerweil, and T. L. Griffiths. Human memory search as a random walk in a semantic network. *Advances in Neural Information Processing Systems 25*, 2012.

[11] A. A. Kumar, M. Steyvers, and D. A. Balota. A critical review of network-based and distributional approaches to semantic memory structure and processes. *Topics in Cognitive Science* 14(1):54–77, 2022. doi:10.1111/tops.12548.

[12] I. Dasgupta, E. Schulz, N. D. Goodman, and S. J. Gershman. Remembrance of inferences past: amortization in human hypothesis generation. *Cognition* 178:67–81, 2018. doi:10.1016/j.cognition.2018.04.017.

[13] C. J. Honey, A. Mahabal, and B. Bellana. Psychological momentum. *Current Directions in Psychological Science* 32(4):284–292, 2023. doi:10.1177/09637214221143053.

[14] L. Alison, B. Doran, M. L. Long, N. Power, and A. Humphrey. The effects of subjective time pressure and individual differences on hypotheses generation and action prioritization in police investigations. *Journal of Experimental Psychology: Applied* 19(1):83–93, 2013. doi:10.1037/a0032148.

[15] K. Krieger-Redwood, A. Steward, Z. Gao, X. Wang, A. Halai, J. Smallwood, and E. Jefferies. Creativity in verbal associations is linked to semantic control. *Cerebral Cortex* 33(9):5135–5147, 2023. doi:10.1093/cercor/bhac405.

[16] M. Bilalić, P. McLeod, and F. Gobet. The mechanism of the Einstellung (set) effect: A pervasive source of cognitive bias. *Current Directions in Psychological Science* 19(2):111–115, 2010. doi:10.1177/0963721410363571.

[17] G. Knoblich, S. Ohlsson, H. Haider, and D. Rhenius. Constraint relaxation and chunk decomposition in insight problem solving. *Journal of Experimental Psychology: Learning, Memory, and Cognition* 25(6):1534–1555, 1999. doi:10.1037/0278-7393.25.6.1534.

[18] A. Armougum, A. Gaston-Bellegarde, C. Joie-La Marle, and P. Piolino. Expertise reversal effect: cost of generating new schemas. *Computers in Human Behavior* 111:106406, 2020. doi:10.1016/j.chb.2020.106406.

[19] X. Jing, J. J. Cimino, V. L. Patel, Y. Zhou, J. H. Shubrook, S. De Lacalle, et al. Data-driven hypothesis generation among inexperienced clinical researchers: a comparison of secondary data analyses with visualization (VIADS) and other tools. *Journal of Clinical and Translational Science* 8:e13, 2024. doi:10.1017/cts.2023.708.

[20] J. G. Foster, A. Rzhetsky, and J. A. Evans. Tradition and innovation in scientists' research strategies. *American Sociological Review* 80(5):875–908, 2015. doi:10.1177/0003122415601618.

[21] S. Frickel, S. Gibbon, J. Howard, J. Kempner, G. Ottinger, and D. J. Hess. Undone Science: Charting Social Movement and Civil Society Challenges to Research Agenda Setting. *Science, Technology, & Human Values* 35(4):444–473, 2010. doi:10.1177/0162243909345836.

[22] P. K. Stanford. *Exceeding Our Grasp: Science, History, and the Problem of Unconceived Alternatives*. Oxford University Press, 2006.

[23] M. Weisberg and R. Muldoon. Epistemic landscapes and the division of cognitive labor. *Philosophy of Science* 76(2):225–252, 2009. doi:10.1086/644786.

[24] J. McKenzie Alexander, J. Himmelreich, and C. Thompson. Epistemic landscapes, optimal search, and the division of cognitive labor. *Philosophy of Science* 82(3):424–453, 2015. doi:10.1086/681766.

[25] K. C. Elliott and D. J. McKaughan. How values in scientific discovery and pursuit alter theory appraisal. *Philosophy of Science* 76(5):598–611, 2009. doi:10.1086/605807.

[26] E. I. Lichtenstein. (Mis)Understanding scientific disagreement: Success versus pursuit-worthiness in theory choice. *Studies in History and Philosophy of Science* 85:166–175, 2021. doi:10.1016/j.shpsa.2020.10.005.

[27] J. Gottweis, W.-H. Weng, A. Daryin, et al. Accelerating scientific discovery with Co-Scientist. *Nature* 655:487–496, 2026. doi:10.1038/s41586-026-10644-y.

[28] A. E. Ghareeb et al. A multi-agent system for automating scientific discovery. *Nature* 655:497–505, 2026. doi:10.1038/s41586-026-10652-y.

[29] C. Lu et al. Towards end-to-end automation of AI research. *Nature* 651:914–919, 2026. doi:10.1038/s41586-026-10265-5.

[30] G. Xiong, E. Xie, C. Williams, et al. Toward Reliable Scientific Hypothesis Generation: Evaluating Truthfulness and Hallucination in Large Language Models. *IJCAI 2025*:7849–7857. doi:10.24963/ijcai.2025/873.

[31] A. R. Doshi and O. P. Hauser. Generative AI enhances individual creativity but reduces the collective diversity of novel content. *Science Advances* 10:eadn5290, 2024. doi:10.1126/sciadv.adn5290.

[32] L. Meincke, G. Nave, and C. Terwiesch. ChatGPT decreases idea diversity in brainstorming. *Nature Human Behaviour* 9:1107–1109, 2025. doi:10.1038/s41562-025-02173-x.

[33] H.-C. B. Huang. Unlocking creativity with artificial intelligence (AI): Field and experimental evidence on the Goldilocks (curvilinear) effect of human-AI collaboration. *Journal of Experimental Psychology: General* 154(12):3294–3306, 2025. doi:10.1037/xge0001838.

[34] Y. Lahtee. From Problem to Hypothesis: Dynamic Semantic Mobility, Bounded Knowers, and the Readout-Discriminable Hypothesis Space. GLOSA K0 working paper, 2026.

[35] Y. Lahtee. Knowledge Topology and the First Passage to Usable Hypotheses: A Readout Theory of Discovery Time and Direction. GLOSA K0 working paper, 2026.

[36] Y. Lahtee. Before Evidence Can Decide: Candidate-Set Formation, Discovery Routing, and Unconceived Alternatives. GLOSA K0 agenda paper, 2026.

[37] Y. Lahtee. The Epistemic Chain Reaction: Human-AI Multiplication from Questions to Readout-Distinguishable Hypotheses. GLOSA K0 paper, 2026.

[38] Y. Lahtee. Readout Universe: A Philosophy and Logic for Grounding Claims. GitHub repository, 2026. https://github.com/morrocwi/readout_universe.

[39] Y. Lahtee. Readout Genesis Universal Technical Whitepaper, v1.2.0. GitHub repository, 2026. https://github.com/morrocwi/readout_genesis.

[40] Y. Lahtee. glosa: Rigour Without Infrastructure. GitHub repository, 2026. https://github.com/morrocwi/glosa.

10.5281/zenodo.22308066

Part VIII

Method: Rigour Without Infrastructure

Part 7 audited the programme's own equation lineage; *Method: Rigour Without Infrastructure* turns from that substantive audit to the production apparatus itself — the same readout, tier, and independence machinery used throughout this book, examined as an object rather than assumed as background.

Chapter 39 names the condition an AI-assisted researcher faces once formation is cheap but the institutional infrastructure that ordinarily supplies criticism, correction, and certification is not automatically present. It proposes a dual-track architecture separating synthetic plurality from independent validation, and production-critical assets from scholarly-credit-critical ones, with time-indexed defeat conditions attached to the architecture rather than to any single claim.

Chapter 40 states three named propositions — a design mechanism, a conceptual claim about where rigour lives, and an untested empirical claim about cross-vendor checking — each carrying its own falsifier, and then runs the same claim-card method on this paper's own construction. The self-application section reports the gate failures the run actually produced, not a clean pass.

Chapter 41 typesets, as verbatim quotation rather than as an included PDF (the record's own deposited files do not contain this text), the three sections of the underlying design document that the book's own headnotes draw on mechanically: the spine's round-trip translation between world and readout vocabulary, the Five Questions crosswalk, and the Independence Ladder I0–I5 that sets the ceiling on what any chapter's K-state can be raised to and by whom.

None of the three chapters raises a claim made elsewhere in this book; they describe how such claims were produced and what would have to be true to check them. The independence levels and K-states named here are the same vocabulary printed on every other chapter's headnote — checking any one chapter's status still means reading that chapter's own printed line, not this Part's presence in the collection. The appendices that follow assemble this vocabulary and the falsifier index across the whole book without adding to it.

Founder rulings relevant to this part: BBL-2026-09-04-083 (Blackbox Log record 10.5281/zenodo.22334420).

- The Standalone Scholar: A Dual-Track Architecture for AI-Native Scholarship — `10.5281/zenodo.22163849`
- Rigour Without Infrastructure: Three Propositions on Claim-Card Discipline as a Substitute for Institutional Certification — `10.5281/zenodo.22307841`
- glosa — Rigour Without Infrastructure: A Standalone Scholar Methodology for Human–AI Knowledge Co-Production — `10.5281/zenodo.22340255`

The Standalone Scholar: A Dual-Track Architecture for AI-Native Scholarship

Anchor-preserved integrated standalone revision v3 • 29 August 2026 • `10.5281/zenodo.22163849`

What the chapter states. "For a standalone scholar, however, AI can create a structurally different condition: the production of plausible scholarly objects can become abundant before independent criticism, human relationships, certification, and institutional access become abundant."

Status. No K-state or peer-review declaration is printed in the chapter text itself; the chapter's Zenodo record is typed `Preprint` (this revision, v3, 29 August 2026).

Falsifier(s). "At 36 months, no peer-reviewed anchor, no repeat human node, and no external uptake constitute defeat of the current configuration, not defeat of scholarship itself."

Read before. Ch. 1 *Written by AI. Still True.*; Ch. 2 *The Readout Condition*; Ch. 3 *Readout Genesis Standalone Synthesis*; Ch. 27 *When AI Expands Human Potential*.

Feeds into. This book's Chapters 1–7, 10–16, 23, 27–31, 38, 40–41 (mechanical KG cross-reference list; titles and DOIs in Appendix A).

The Standalone Scholar: A Dual-Track Architecture for AI-Native Scholarship

Building Scholarly Infrastructure Across Global and Thai Epistemic Landscapes

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Abstract

Generative AI lowers the cost of literature exploration, argument recombination, drafting, coding scaffolds, and adversarial questioning. For scholars embedded in universities, these gains are absorbed into pre-existing laboratories, seminars, editorial networks, ethics systems, and disciplinary communities. For a standalone scholar, however, AI can create a structurally different condition: the production of plausible scholarly objects can become abundant before independent criticism, human relationships, certification, and institutional access become abundant. This article conceptualizes that condition as a problem of *scholarly infrastructure reconstruction*. It develops a dual-track architecture in which AI-assisted formation, open preprint staging, provenance controls, and machine-constrained checks generate defensible provisional assets, while independent human friction, repeat correspondence, journal review, and local epistemic participation convert those assets into socially stress-tested knowledge. The framework distinguishes synthetic plurality from independent validation, production criticality from scholarly-credit criticality, attention from uptake, legitimacy from truth, and latent programme coherence from public legibility. It further argues that global scholarly search and AI recommendation routes may share Anglophone and database-distribution biases; accordingly, every socially relevant flagship is assigned both a global conversion route and a local route, with Thailand developed as an operational epistemic counter-map rather than a secondary public-relations market. Formal expressions are used as governance definitions rather than causal estimates: a stock-pressure ratio diagnoses overproduction relative to conversion, a credit-stock equation includes back-catalog activation, a bottleneck law reallocates scarce human time, and a criticality model tracks leakage, legibility, conversion, and delayed independent signals. The paper contributes a theory of the *standalone scholar* as an infrastructural position rather than a solitary identity, and offers falsifiable propositions for future research on AI-mediated knowledge production, epistemic isolation, cumulative advantage, and geographically distributed scholarly legitimacy.

Keywords: AI-native scholarship; independent researcher; epistemic infrastructure; scholarly communication; cumulative advantage; preprints; epistemic friction; legitimacy; Global South; Thailand; open science; conceptual research.

1 Introduction: AI changes the bottleneck, not the need for scholarship

The contemporary debate about generative AI in research often begins with authorship, disclosure, reliability, or automation. Those questions are necessary, but they can obscure a deeper organizational shift. AI can make the early production of scholarly material faster than the institutional processes that make scholarship socially corrigible. A model can propose literatures, reorganize an argument, produce rival hypotheses, draft prose, generate code scaffolds, and simulate objections in minutes. Yet the scholar must still decide what the claim means, verify the sources, specify what would falsify it, defend it without model assistance, submit it to communities with recognizable standards, and maintain the relationships through which criticism becomes cumulative rather than episodic. Recent work on "epistemic debt" captures one version of this gap: the distance between what an author presents and what the author can actually understand and defend can compound when AI-generated material is accepted without sufficient cognitive ownership (Unceta et al., 2026). Current publication guidance similarly keeps responsibility with human authors and requires disclosure of material AI use rather than transferring accountability to the tool (International Committee of Medical Journal Editors, 2026).

The gap is particularly consequential for scholars who do not inhabit a university department, laboratory, doctoral cohort, or stable seminar network. An institution supplies more than a logo. It supplies ambient criticism, colleagues who remember earlier versions of an argument, methods advice, ethics pathways, journal norms, tacit venue knowledge, and repeated occasions for a work-in-progress to encounter resistance. The independent scholar may possess high intellectual and computational capacity while lacking this ambient infrastructure. The resulting condition is not simply "working alone." It is an asymmetry between *formation capacity* and *external correction capacity*.

This paper names the actor operating under this asymmetry the *standalone scholar*. "Standalone" does not mean institutionally hostile, socially isolated by preference, or epistemically self-sufficient. It means that the scholar cannot assume that the infrastructures ordinarily bundled with institutional mem-

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bership will appear automatically. They must be deliberately assembled across open repositories, disciplinary communities, correspondence, peer review, local institutions, and reproducible toolchains.

The paper makes four contributions. First, it develops a conceptual model of **bottleneck inversion**: once AI-assisted formation becomes abundant, the limiting resource shifts from idea production toward verification, human mastery, external friction, programme legibility, and conversion into durable scholarly relationships. Second, it introduces a **knowledge-state architecture** that separates private candidates, public provisional assets, socially stress-tested assets, and reviewed or empirically constrained assets; this prevents synthetic AI scrutiny from being laundered into claims of independent validation. Third, it develops a **dual-track architecture** that assigns each socially relevant flagship both a global route and a local epistemic route. Thailand serves here as a worked operational landscape that exposes a general distributional problem: multiple AI systems and bibliographic databases can share the same geographic and linguistic blind spots. Fourth, it formalizes a set of **governance equations** for stock pressure, back-catalog activation, credit velocity, delayed-channel control, and criticality. These equations are not empirical laws. They are decision-bearing definitions designed to make assumptions explicit, calibratable, and falsifiable.

The argument therefore differs from a productivity guide. Its object is not "how to publish more." It is how a scholar without guaranteed ambient infrastructure can construct a system in which provisional ideas acquire provenance, criticism, correction histories, local and global exposure, and eventually reviewed credit without confusing those institutional receipts with truth itself. This distinction is central because cumulative advantage in science is real (Merton, 1988; DiPrete and Eirich, 2006; Bol et al., 2018), but scholarly metrics and prestige can also be Goodharted (Fire and Guestrin, 2019). The framework must therefore create credit while preventing credit from becoming the epistemic target.

2 Conceptual foundations

2.1 Cumulative advantage without prestige essentialism

The sociology of science has long shown that early recognition can alter later opportunity structures. Merton's Matthew effect and subsequent work on cumulative advantage describe how prior advantages can become resources for further gains (Merton, 1988; DiPrete and Eirich, 2006). Scientific careers also display unequal distributions of productivity and impact (Petersen and Penner, 2014), and funding decisions can generate later participation advantages even among applicants near an initial threshold (Bol et al., 2018). Yet cumulative advantage should not be reduced to a deterministic prestige mechanism. Citation research suggests that appropriateness to a scholarly conversation can account for part of what otherwise looks like prestige advantage (Wang, 2014). The practical implication is that a standalone scholar does not need to mimic elite status as the first objective. The strategic object is *earned scholarly credit*: external, provenance-bearing evidence that a contribution has entered a relevant scholarly channel and generated correction, reuse, or further opportunity.

We therefore distinguish

$$\text{Credit} \neq \text{EpistemicValue}. \tag{1}$$

Credit is treated as a receipt for epistemic work that has survived a relevant exposure route, not as a unit of truth. This distinction permits a cumulative-advantage strategy while resisting metric fetishism.

2.2 Conceptual research as disciplined theory design

The paper is a conceptual model rather than an empirical causal test. Conceptual research can make a publishable contribution when it specifies the research problem, mobilizes prior theory deliberately, and clarifies the form of the conceptual contribution (Jaakkola, 2020). Phenomenon-based theorizing similarly begins from an undertheorized phenomenon, connects it to existing explanations, and extends theory where those explanations are insufficient (Fisher et al., 2021). The phenomenon here is not simply "AI in academia." It is the emergence of high-throughput scholarly formation outside the institutional bundles that historically supplied criticism and socialization.

Accordingly, the theorizing pipeline is:

$$\begin{aligned}\text{Phenomenon} &\rightarrow \text{ExistingExplanations} \rightarrow \text{PreciseInadequacy}\\ &\rightarrow \text{Mechanism} \rightarrow \text{Boundary}\\ &\rightarrow \text{Propositions}.\end{aligned} \tag{2}$$

The framework rejects the shortcut Phenomenon → NewWord. A new construct is warranted only if it explains an important dimension not already captured by adjacent concepts.

2.3 Social objectivity, distributed reasoning, and invisible colleges

Longino's account of social objectivity emphasizes venues for criticism, uptake of criticism, public standards, and tempered equality of intellectual authority (Longino, 2002). Dunbar's studies of real-world laboratories show that scientific reasoning occurs during interaction around anomalies, analogies, and evolving explanations, not only after a finished paper exists (Dunbar, 1995). Earlier work on invisible colleges and collaboration likewise treats informal networks as infrastructures through which scientific information and recognition circulate (Crane, 1972; Price and Beaver, 1966). These traditions suggest that the standalone scholar's problem is not a lack of intelligence but a lack of *ambient corrigibility*.

The architecture developed below therefore separates two social functions that are often collapsed:

$$\text{Friction} \neq \text{Fellowship}. \tag{3}$$

Friction produces objections, counterexamples, construct differentiation, and claim-scope correction. Fellowship produces

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continuity, memory of previous versions, trust sufficient for repeated exchange, and lower transaction costs for future criticism. A healthy scholarly environment requires both.

2.4 Situated knowledge and geographic distributions

A second body of theory concerns the location of knowledge. Situated-knowledge arguments reject a view from nowhere and emphasize the epistemic consequences of position (Haraway, 1988); epistemic-injustice theory shows how social identity and institutional credibility assignments can produce systematic deficits in who is heard as a knower (Fricker, 2007). Meanwhile, scholarly databases themselves have uneven coverage. Recent evaluations find that OpenAlex is more linguistically diverse than Web of Science but still contains language-classification and metadata problems (Céspedes et al., 2025); work on China documents discontinuities and cross-national coverage asymmetries (Zheng et al., 2025); and comparative database research continues to find geographic and disciplinary representation differences (Maddi et al., 2025). A systematic review of geographic bias in peer review also found evidence that country or institutional prestige can affect evaluation in at least some settings, while emphasizing the limited experimental base (Skopec et al., 2020).

The consequence is important for AI-native scholarship. If AI systems, search engines, and bibliographic graphs rely on overlapping public distributions, then multi-model agreement need not imply geographic completeness. We formalize this later as the Geographic Distribution Audit.

3 Defining the standalone scholar

3.1 Standalone is an infrastructural condition

Let institutional starting capital be represented by

$$K_0^{\text{start}} = I_0 + R_0 + N_0 + P_0, \tag{4}$$

where I_0 denotes institutional capital, R_0 recognition capital, N_0 network capital, and P_0 accumulated publication capital. This is a bookkeeping identity, not a psychometric scale. A standalone scholar may begin with low I_0 while possessing high observation capacity, domain experience, computational leverage, or practice access. The theoretical question is whether other forms of earned credit can reduce dependence on borrowed institutional credibility over time.

We define earned scholarly credit as

$$E_t = f(Q_t, C_t, V_t, U_t, X_t, T_t), \tag{5}$$

where Q is contribution quality, C programme coherence, V external validation, U uptake, X scholarly exposure, and T traceability. The function is conceptual. The claim is directional: as independently exposed and traceable assets accumulate, dependence on institution-as-proxy can decline even though institutions remain valuable for access, ethics, certification, and resources.

Proposition 1 (Infrastructural substitution) *For scholars with low bundled institutional infrastructure, increases in deliberately constructed criticism, provenance, and repeat-human infrastructure will reduce dependence on institutional affiliation as the primary credibility proxy, conditional on contribution quality and field fit.*

3.2 Two engines and one bridge

The architecture separates two research engines. **Engine A** is an independent theory programme oriented toward sociology and politics of knowledge, epistemic legitimacy, knowledge institutions, AI and knowledge, religious knowledge regimes, institutional authority, epistemic hardening, and knowledge governance. Its canonical transformation is

$$\begin{aligned}\text{Phenomenon} &\to \text{TheoreticalInadequacy}\\ &\to \text{Mechanism} \to \text{ConceptualContribution}.\end{aligned} \tag{6}$$

Engine B is a practice-research programme oriented toward intervention, implementation, family or community services, and empirically constrained practice. Its canonical transformation is

$$\begin{aligned}\text{Practice} &\to \text{Observation} \to \text{Intervention}\\ &\to \text{Evidence} \to \text{Implementation}.\end{aligned} \tag{7}$$

A bridge between the engines is justified only by intellectual necessity:

$$\text{Bridge} = 1 \iff \text{Practice}\ \Delta\ \text{Theory}. \tag{8}$$

Here, Δ denotes a substantive change in what the theory must explain. This firewall prevents practice experience from being silently generalized into population evidence.

3.3 The lived and positional bridge: context of discovery without evidential privilege

A standalone scholar can become a case of the theory being developed. Institutional distance, legitimacy deficits, language position, practice access, and local trust can reveal mechanisms that are difficult to see from more institutionally central positions. This reflexivity is epistemically useful only if it is prevented from becoming evidential privilege. The governing boundary is

$$\text{SelfExperience} \neq \text{GeneralEvidence}. \tag{9}$$

Situated experience may detect a mechanism, generate a research question, reveal a missing variable, expose legitimacy friction, formulate interview or survey propositions, or identify comparative communities. It cannot by itself prove a theory, establish representativeness, or immunize a claim from criticism.

This distinction can be represented as **positional capital**, a channel specification rather than an authority score:

$$\begin{aligned}\text{PosCap} &= \text{Access} \times \text{Language} \times \text{SituatedObservation}\\ &\times \text{TranslationCapacity} \times \text{Trust}.\end{aligned} \tag{10}$$

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The expression is a `definition/Dr`; it is not a psychometric construct. Two safeguards follow:

$$\text{PositionalAccess} \neq \text{PopulationAuthority}, \tag{11}$$

$$\text{CommunityTrust} \neq \text{Representativeness}. \tag{12}$$

The bridge therefore converts location into questions and records, not into epistemic immunity. This is especially important for Global-South and local-language work because situated access can correct a global map without making the situated observer a truth oracle (Haraway, 1988; Gibbons et al., 1994; Fricker, 2007).

4 Knowledge states: synthetic formation is not independent validation

The central epistemic distinction in the model is between formation and certification. We define four states:

$$K_0 : \text{private candidate}; \tag{13}$$

$$K_1 : \text{public, timestamped, citable, explicitly provisional}; \tag{14}$$

$$K_2 : K_1 + \text{independent external friction}; \tag{15}$$

$$K_3 : K_2 + \text{review or independent empirical constraint}. \tag{16}$$

The hierarchy is not a truth ladder. A reviewed paper can be wrong and a preprint can be right. The states record *what kinds of exposure have occurred.*

A preprint therefore occupies a staging function. Research on preprint adoption emphasizes speed, dissemination, and disciplinary variation (Chiarelli et al., 2019); initiatives such as the FAST principles attempt to make public feedback more focused, appropriate, specific, and transparent (Franco Iborra et al., 2022). In this architecture the staging layer is:

$$K_0 \to K_1 \to K_2 \to K_3, \tag{17}$$

with the crucial boundary

$$\text{DVP} \nRightarrow K_2. \tag{18}$$

No amount of same-ancestry AI criticism is relabeled as independent human validation.

Proposition 2 (State non-equivalence) *AI-mediated adversarial formation can increase the defensibility of a provisional asset, but it cannot by itself supply the social independence required for a K_2 claim.*

5 Epistemic Isolation Constraint and bottleneck inversion

5.1 The Epistemic Isolation Constraint

Let μ_A denote AI exploration throughput, $\mu_{S,f}$ synthetic formation-friction throughput, and $\mu_{H,f}$ human formation-friction throughput. For a scholar without ambient community, a common starting condition is

$$\mu_{H,f} \approx 0, \qquad \mu_A \gg \mu_{H,f}. \tag{19}$$

We call this the **Epistemic Isolation Constraint (EIC)**. EIC does not imply that research must stop until humans become available. Instead:

$$\text{EIC} \Rightarrow \text{BuildSyntheticFormationInfrastructure}. \tag{20}$$

The intended sequence is

$$\text{AI speed} \to \text{synthetic criticism} \to K_1 \to \text{human correction}. \tag{21}$$

This sequence preserves the asymmetry between synthetic and independent routes while avoiding the opposite error of treating absent human access as a veto on provisional knowledge production.

5.2 Bottleneck inversion

The system contains at least five slower human or institutional stages: literature verification μ_L, human mastery μ_H, external friction μ_E, publication or certification μ_P, and credit conversion μ_C. Effective public throughput is therefore approximated by

$$\Lambda = \min(\mu_L, \mu_H, \mu_E, \mu_P, \mu_C). \tag{22}$$

Once μ_A exceeds these rates, more AI generation does not necessarily increase Λ; it can instead increase queue length, epistemic debt, and fragmentation. The managerial problem shifts from idea scarcity to allocation of scarce human judgment.

To make this explicit, let AI intensity be A, human mastery H, traceability T, and literature verification L, normalized to $[0, 1]$. Define verification capacity

$$V_c = (HLT)^{1/3}, \tag{23}$$

and epistemic debt

$$D_e = A(1 - V_c). \tag{24}$$

This is a governance definition motivated by the epistemic-debt concept, not an estimated cognitive law. It makes one practical rule visible:

$$\text{PublicOutputVelocity} \leq \text{VerificationCapacity}. \tag{25}$$

Proposition 3 (Bottleneck inversion) *As AI-assisted formation throughput rises above literature-verification, mastery, external-friction, and conversion capacity, the marginal benefit of additional AI generation decreases while the expected costs of queueing, epistemic debt, and fragmentation increase.*

6 Decorrelated Verification Protocol

A standalone scholar still needs formation-stage opponents. The Decorrelated Verification Protocol (DVP) is a governance design for creating synthetic resistance while making residual dependence visible. It responds to documented problems such as sycophancy in instruction-following models (Sharma et al., 2023) and self-preference in LLM evaluators (Panickssery et al., 2024). Those findings motivate caution; they do not validate DVP as an empirical substitute for peer review.

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6.1 Route diversity, not model voting

DVP uses at least three materially different verification routes when feasible. Routes may differ by model family, prompt framing, assigned role, context window, evidence source, or mechanical verifier. The principle is

$$\text{ManyModels} \not\Rightarrow \text{Independence}. \tag{26}$$

The objective is instead

$$\begin{aligned} \text{DVP}^* = \text{ReduceCorrelatedError} \\ + \text{ExposeResidualDependence} \\ + \text{BottomOutWherePossible}. \end{aligned} \tag{27}$$

The role library includes: advocate, nearest-concept prosecutor, falsifier, source auditor, hostile reviewer, translation reviewer, and method/empirical designer. These are *route roles*, not independent agents in an ontological sense.

6.2 Frame break, recovery, and anchored checks

A **frame-break run** removes the author's preferred terminology and asks another route to restate the problem using a different conceptual vocabulary. A **recovery test** hides the author's answer to a strong objection and asks a new route to reconstruct a resolution from the constraints. Recurrence is recorded as a stability signal, not proof.

Whenever a claim can bottom out in an external record or mechanical constraint, that route receives higher governance weight than free-form model judgment. Examples include original papers and policy documents, executed calculations, test suites, type checkers, and formal proof kernels. Yet two boundaries remain:

$$\text{MechanicalValidity} \neq \text{SemanticValidity}, \tag{28}$$

$$\text{SourceExistence} \neq \text{ClaimSupport}. \tag{29}$$

6.3 Disagreement as information

DVP reverses the intuition that consensus is always the most valuable output. Multi-model agreement is a weak positive signal because training data, retrieval corpora, and operator framing may be shared. Disagreement is diagnostically rich:

$$\text{Disagreement} \Rightarrow \text{Resolve} \vee \text{Declare}. \tag{30}$$

A disagreement ledger classifies construct, source, mechanism, boundary, venue, and epistemic-tier disputes rather than averaging them away.

6.4 Operator-Decorrelation Control

The largest common ancestor across routes may be the operator. DVP is therefore paired with Operator-Decorrelation Control (ODC): precommit acceptance criteria, blind route identity where feasible, randomize result order when feasible, and enforce a query stop rule. A Route Dependence Matrix records dependence by model family, prompt ancestry, operator, source base, and external anchor.

Proposition 4 (Residual dependence) *Verification diversity improves error detection only to the extent that routes differ in relevant failure structure; nominal model multiplicity without operator, source, or framing decorrelation will overstate effective independence.*

7 From friction to fellowship

Formation friction can be synthetic; durable scholarly infrastructure cannot be entirely synthetic. The model therefore distinguishes broadcast exposure from addressed correspondence. If public circulation produces attention without substantive return, the scholar switches from "more broadcasting" to named interlocutors, reply venues, reviewer service, and repeat communities.

A **stable correspondent** is defined operationally as a human interlocutor who completes at least three substantive exchanges around the same problem family across time. The exact threshold is heuristic; the conceptual point is recurrence. Repeat correspondents become "reflector nodes": they remember previous objections and can return a new asset to an established argumentative history.

Fellowship remains epistemically bounded:

$$\text{Friendship} \neq \text{IndependentEvidence}, \tag{31}$$

$$\text{Correspondence} \neq \text{PeerReview}, \tag{32}$$

$$\text{IntellectualAffinity} \neq \text{Truth}. \tag{33}$$

To reduce clique formation, every flagship should retain at least one friction route outside the repeat-fellowship circle.

Proposition 5 (Repeat-human conversion) *Conditional on substantive disagreement and route diversity, repeat human relationships will reduce conversion leakage more effectively than equivalent numbers of one-off interactions because recurrent interlocutors retain problem history and lower the cost of subsequent criticism.*

8 Programme legibility: coherence that cannot be seen cannot convert

An AI-native scholar can create a large catalogue whose internal coherence is obvious to the author but invisible to an outsider. We therefore decompose programme coherence as

$$\text{Coh}_{\text{effective}} = \text{Coh}_{\text{latent}} \times L_g, \tag{34}$$

where L_g is programme legibility. A simple proxy is

$$L_g^{\text{proxy}} = \frac{\text{public assets with a one-click programme path}}{\text{public assets}}. \tag{35}$$

The proxy measures path availability, not actual reader comprehension.

A canonical programme index should allow a stranger to see the problem family, flagships, bridges, empirical branches, current versions, and repository lineage without reconstructing the

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portfolio manually. This is infrastructure rather than cosmetic branding because it changes the probability that one asset leads to another:

$$\begin{aligned} K_1 &\to \text{RelatedAssetDiscovery} \\ &\to \text{LongerExposure} \to K_2 \text{ opportunity}. \end{aligned} \tag{36}$$

Proposition 6 (Legibility conversion) *The effect of latent programme coherence on external uptake is positively moderated by public programme legibility; coherent portfolios with low legibility will underperform equally coherent portfolios with clear navigational and version lineage.*

8.1 Association stability and recognition conversion

Legibility makes a programme navigable; recognition requires a second mechanism: repeated association between an author and a recognizable problem family. Let

$$A_s = \text{AssociationStrength}(\text{Author}, \text{Problem}). \tag{37}$$

The directional governance target is

$$A_s(t+1) > A_s(t), \tag{38}$$

not because name recognition is evidence of truth, but because stable association lowers the cost for external readers to locate new work within an existing argumentative history. A concept cluster should therefore deepen a recognizable problem family rather than reset field identity with every paper. The source mechanism is preserved as

$$\begin{aligned} &\text{Paper}_1 \to \text{Theme}, \\ &\text{Paper}_2 \to \text{SameTheme} + \text{NewMechanism}, \\ &\text{Paper}_3 \to \text{EmpiricalTest}, \\ &\text{Paper}_4 \to \text{BoundaryExtension}. \end{aligned} \tag{39}$$

Recognition conversion is therefore a consequence of coherent contribution history, not a license to optimize personal visibility.

9 Conversion-first scholarly dynamics

9.1 Production criticality is not scholarly-credit criticality

Let λ_{mint} be the rate at which K_1 assets are produced and λ_{conv} the rate of qualified conversion actions. Define the Stock Pressure Ratio

$$\chi_t = \frac{\lambda_{mint}}{\lambda_{conv} + \varepsilon}. \tag{40}$$

When $\chi_t > 1$ for sustained periods, unactivated stock tends to accumulate. Let B_t be the backlog, M_t new provisional assets, X_t qualified conversion actions, and ω the average stock activated per conversion action:

$$B_{t+1} = B_t + M_t - \omega X_t. \tag{41}$$

A portfolio can therefore be “production-supercritical” relative to conversion capacity while remaining subcritical in the production of durable scholarly opportunities.

A finite author-level diagnostic that motivated the present model illustrates the distinction: approximately 23 public provisional assets were produced over roughly 10.6 months (about 2.17 assets/month), while stable correspondents and structured independent-human conversion were newly measured rather than historically established. Views and downloads demonstrated circulation, not truth or K_2. This diagnostic is not a population benchmark and is not used as causal evidence; it simply identifies the design problem of abundant stock with underbuilt conversion infrastructure.

9.2 Mint-convert coupling

A work-in-progress limit controls concurrency but not annual throughput. During high-stock/low-conversion regimes, new minting should therefore be coupled to conversion work:

$$M_t \leq \lambda X_t, \tag{42}$$

where λ is calibrated from the scholar’s ledger rather than treated as a universal constant. Qualified conversion actions include addressed correspondence, preprint review, journal submission or revision, conference submission from existing stock, programme-index activation, and targeted local/global friction routes. Generic social posts, likes, internal AI praise, or formatting changes do not count.

9.3 Back-catalog activation

Let C_t denote scholarly credit stock. A conversion-first stock equation is

$$C_{t+1} = \Big[(1-\delta)C_t + G_t + \Phi_t(\text{Stock})\Big] I_t P_t^{(r)} S_t, \tag{43}$$

where G_t is credit generated by current activity, $\Phi_t(\text{Stock})$ is realized PRC-valid credit produced by activating pre-existing assets, I_t integrity retention, $P_t^{(r)}$ priority retention, S_t survival/persistence, and δ credit decay.

The key distinction is

$$\text{ActivationAction} \neq \text{CreditEvent}. \tag{44}$$

An email is not credit. A substantive independent reply, external review report, citation, reuse, invitation, collaboration request, or empirical extension attributable to that activated asset may be.

This permits credit acceleration without faster production:

$$\frac{d^2E[C]}{dt^2} > 0 \quad \text{can occur while} \quad \frac{dM}{dt} \leq 0, \tag{45}$$

provided

$$\frac{d\Phi(\text{Stock})}{dt} > 0. \tag{46}$$

Proposition 7 (Back-catalog acceleration) *When provisional stock is abundant and conversion capacity is the binding constraint, increasing stock activation can yield greater marginal scholarly credit than increasing new asset production.*

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9.4 Concept-cluster compounding

The unit of strategic accumulation is not the isolated paper but a **concept cluster**: one intellectual investment can be exposed repeatedly through non-duplicative scholarly objects,

$$\begin{aligned}\text{FlagshipConcept} &\rightarrow \text{Preprint} \rightarrow \text{Conference} \\ &\rightarrow \text{Journal} \rightarrow \text{EmpiricalTest} \\ &\rightarrow \text{ComparativeExtension} \rightarrow \text{Grant}.\end{aligned} \tag{47}$$

The leverage of cluster i is defined as

$$\text{CreditLeverage}_i = \frac{\sum_{j=1}^{n} \text{CreditEvent}_{ij}}{\text{CoreIntellectualInvestment}_i + \varepsilon}, \tag{48}$$

subject to non-duplication, venue rules, and the PRC. The purpose is cumulative problem recognition, not salami publication: multiple outputs are justified only when they add a distinct function such as criticism, empirical discrimination, translation, formalization, or application.

10 Credit, provenance, and Goodhart control

Credit is valid only under a Provenance Relevance Constraint (PRC). For candidate event e_i, define

$$\text{PRC}(e_i) \in \{0, 1\}, \tag{49}$$

where $\text{PRC}(e_i) = 1$ only if the event has (i) traceable provenance to an epistemic asset, (ii) programme relevance, (iii) non-manipulation, (iv) external exposure, and (v) integrity preservation. Valid credit is then

$$C_t^{\text{valid}} = \sum_i w_i e_i \text{PRC}(e_i), \tag{50}$$

with governance weights w_i that are not units of truth.

Every target metric receives a counter-metric: publication count is paired with unique contribution and programme coherence; citation count with appropriateness or external reuse; views with academic conversion; conference count with conference-to-revision conversion; collaboration count with capability gained and assets produced. If a target grows while quality, defensibility, or verifiability declines, the target freezes.

Design Rule 1 (Goodhart tripwire) *If* MetricGrowth ↑ *while* Quality ↓ *or* Defensibility ↓ *or* Verifiability ↓, *suspend optimization of that metric.*

11 Scholarly Credit Velocity and Gradient Discipline

The target is not paper count but credible credit velocity per scarce human time:

$$\mathcal{V}_C = \frac{dE[C]}{dt} \approx \frac{\text{CreditCreation} \times \text{Retention} \times \text{Conversion}}{H^{\text{critical}}}. \tag{51}$$

AI acceleration should therefore reallocate human time rather than merely multiply public outputs:

$$\text{AIAcceleration} \rightarrow \text{HumanTimeReallocation}. \tag{52}$$

The protected human functions are conceptual commitment, source verification, construct differentiation, causal or mechanism judgment, falsifier design, ethics, interpretation, public defence, scholarly relationship building, and the decision to publish at a given epistemic tier.

If the velocity architecture is approximately multiplicative,

$$\mathcal{V} \propto \prod_j x_j, \tag{53}$$

then

$$\frac{\partial \mathcal{V}}{\partial x_j} = \frac{\mathcal{V}}{x_j}. \tag{54}$$

When marginal human costs are comparable, the lowest controllable factor often offers the largest leverage. We therefore define cost-adjusted priority

$$\text{Priority}_j = \frac{\partial \mathcal{V} / \partial x_j}{\text{MarginalHumanCost}_j + \varepsilon}. \tag{55}$$

This yields **Gradient Discipline**: each quarter, growth KPIs target only the bottom one or two controllable factors; near-ceiling factors enter maintenance mode.

Proposition 8 (Floor-factor leverage) *In a multiplicative conversion system, interventions aimed at the lowest controllable factor will generally produce greater local system improvement than further optimizing near-ceiling factors, conditional on comparable marginal cost.*

12 Criticality as a control mapping

The framework uses reactor-control vocabulary as a deliberately bounded structural analogy for multiplication, leakage, moderation, delayed feedback, accumulated poison, and emergency shutdown. It does *not* claim physical equivalence: scholarly credit does not conserve, agents game metrics, interaction probabilities are nonstationary, and time constants differ by orders of magnitude.

Define a diagnostic scholarly multiplication factor

$$\begin{aligned}k_t = {} & \nu_t f_{coh,t} L_{g,t} p_{int,t} \\ & \times m_{leg,t} u_{conv,t} P_{NL,t},\end{aligned} \tag{56}$$

where ν is qualified-opportunity reproduction, f_{coh} latent programme coherence, L_g programme legibility, p_{int} integrity survival, m_{leg} field legibility, u_{conv} conversion utilization, and P_{NL} non-leakage (the proportion of mature citable assets receiving at least one independent engagement). The factor is diagnostic and should never be optimized directly.

The regimes $k_t < 1$, $k_t \approx 1$, and $k_t > 1$ are labeled subcritical, critical, and supercritical scholarly growth only as management shorthand. A portfolio can produce many provisional assets while remaining subcritical in opportunity reproduction.

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Table 1: Criticality variables as event-defined governance readouts. None is a truth measure or reactor-physics parameter.

Variable	Scholarly meaning	Operational event definition
ν	opportunity reproduction	qualified independent opportunity-events per PRC-valid credit event
f_{coh}	latent programme coherence	outputs with defensible intellectual lineage to the anchor programme / public outputs
L_g	programme legibility	public assets with a one-click path to the canonical programme / public assets
p_{int}	integrity survival	audited public assets without material integrity correction / audited assets
m_{leg}	field legibility	assets surviving field-language/construct stress test / assets tested
u_{conv}	opportunity conversion	qualified opportunities converted into scholarly assets / qualified opportunities
P_{NL}	non-leakage	citable assets age ≥ 6 months with independent engagement / mature citable assets

12.1 Prompt versus delayed channels

Prompt channels include views, downloads, same-week attention, AI praise, self-announcements, and internal scores. Delayed independent channels include external citations, independent invitations, substantive collaborator requests, reuse, repeated editor or reviewer invitations, stable correspondents, empirical engagement, and external review reports. Raw units are not directly commensurable. Define a preweighted delayed-channel index

$$\beta_D = \frac{\sum_d w_d \widetilde{E}_d}{\sum_d w_d \widetilde{E}_d + \sum_p w_p \widetilde{E}_p + \varepsilon}, \tag{57}$$

where signals are capped or normalized and weights are fixed before outcomes are observed.

A **reactivity license** governs production expansion:

$$\begin{aligned}\text{IncreaseMintRate} \Rightarrow \beta_D &\geq \beta_{min} \\ &\wedge \ \text{IntegrityClean} \wedge \text{XenonLow} \\ &\wedge \ \text{ConversionLogOn}.\end{aligned} \tag{58}$$

The threshold is locally calibrated. The conceptual rule is simpler: prompt attention cannot authorize higher production when delayed independent conversion remains weak.

Proposition 9 (Delayed-channel control) *In high-throughput AI-native portfolios, delayed independent engagement is a better control variable for expanding public production than prompt attention, because prompt signals are more easily generated without external epistemic uptake.*

12.2 Reflectors, moderators, xenon, and breeders

The **Reflector Doctrine** treats repeat-human networks as leakage control. Scholarly albedo is

$$\rho_R = \frac{N^{\text{ind}}_{\leq 6m}}{N^{\text{cit}}_{>6m}}, \tag{59}$$

where $N^{\text{ind}}_{\leq 6m}$ counts citable assets receiving independent engagement within six months and $N^{\text{cit}}_{>6m}$ counts eligible older citable assets. Low ρ_R shifts effort toward correspondence, reviewer service, repeat communities, reply routes, and collaborator mapping rather than more output.

The **Moderator Principle** notes that slowing raw production through source verification, construct differentiation, field-language translation, abstract compression, or independent reading can increase capture. Hence

$$\text{RawSpeed} \downarrow \not\Rightarrow \mathcal{V}_C \downarrow . \tag{60}$$

Some friction is an accelerator because it increases field legibility and integrity survival.

The **Xenon Ledger** records unresolved epistemic poison:

$$X_t = 1(O_{sub}) + 2(E_{known}) + 3(C_{stale}), \tag{61}$$

where unresolved substantive objections, known citation/metadata errors, and known stale public claims carry increasing weights. A local threshold triggers portfolio freeze, while any material integrity incident can trigger immediate shutdown.

The **Breeder Layer** tracks reusable infrastructures: frameworks, protocols, code, checklists, datasets, or formal artifacts that other people independently reuse or extend. Its diagnostic ratio is

$$BR_t = \frac{N_t^{\text{ind-reuse}}}{N_t^{\text{terminal}}}, \tag{62}$$

where the numerator counts newly reusable assets independently reused or extended and the denominator counts public terminal outputs. Internal AI reuse does not count as independent breeding.

13 Legitimacy without truth laundering

Outside formal knowledge institutions, disputes about knowledge are often partly disputes over legitimacy: who may speak, which venue counts, which credential opens a channel, and which provenance is recognizable. The framework does not eliminate legitimacy. It decomposes it.

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Horizontal legitimacy L_H arises from peer-to-peer scrutiny, reuse, replies, recurrent correspondence, distributed correction, and transparent provenance. Vertical legitimacy L_V arises from journal review, associations, institutions, ethics systems, grants, and recognized certification. Neither is truth:

$$L_H \neq \text{Truth}, \qquad L_V \neq \text{Truth}. \tag{63}$$

Both can increase correction capacity, discoverability, and resource access. The desired circulation is

$$\begin{aligned} \text{HorizontalGeneration} &\rightarrow \text{EpistemicFriction} \\ &\rightarrow \text{VerticalStrengthening} \\ &\rightarrow \text{ResourceReturn} \\ &\rightarrow \text{HorizontalGrowth}. \end{aligned} \tag{64}$$

This **Legitimacy Circulation Loop** treats institutions as amplifiers and filters rather than truth oracles, while also rejecting the reverse romanticism that marginalized origin itself guarantees truth.

AI disclosure intensifies this issue. Empirical research finds that disclosing AI use can reduce trust through perceived legitimacy concerns (Schilke and Reimann, 2025; Schilke and Reimann, 2026). The integrity response is not concealment. It is stronger provenance, explicit human responsibility, venue fit, and role-level disclosure.

Proposition 10 (Legitimacy circulation) *Institutional legitimacy contributes most constructively to AI-native independent scholarship when it opens channels for criticism, certification, and resources without being treated as direct evidence of claim truth.*

14 Readout Universe as meta-governance

The framework adopts a readout-not-truth stance from the author's Readout Universe programme (Lahtee, 2026a). A readout from agent or institution A is represented generically as

$$M_A[n] = K_A\theta(E[n]) + \eta_{sel} + \eta_{map} + \eta_{self}, \tag{65}$$

so that

$$M_A[n] \neq \theta(E). \tag{66}$$

In scholarly governance this means that an AI synthesis is not the state of the literature, a reviewer judgment is not intrinsic quality, a conference reaction is not final value, a citation count is not truth, and institutional prestige is not research quality.

When readouts conflict, the system does not select the preferred readout. It asks what policy produced it, what alternative records are available, whether the target is identifiable from current evidence, and what observation would discriminate rival explanations. A simple residual model is

$$r = A\epsilon - \delta, \qquad V = \frac{1}{2}r^T W r, \tag{67}$$

where the residual is interpreted as unresolved tension between the scholar's model and the records returned by a scholarly system. The aim is to reduce V without erasing the core contribution merely to satisfy one readout.

Every claim receives a tier: `definition`, `Dr` (declared interpretive bridge), `Open`, `finite_diagnostic`, `fit_calibrated`, or machine-checked formal proof where genuinely applicable. The invariant is

$$\text{ClaimStrength} \leq \text{EvidenceStrength}. \tag{68}$$

15 The Dual-Track Architecture

15.1 Why one global route is insufficient

Suppose D_{AI} is the distribution readily accessible to AI/search routes and P_{local} the set of situated local records, languages, and communities relevant to the claim. If

$$P_{local} \not\subset D_{AI}, \tag{69}$$

then

$$\text{MultiAIConsensus} \not\Rightarrow \text{GeographicCompleteness}. \tag{70}$$

This is not a claim that AI is uniquely biased; it is a claim about correlated coverage. Bibliographic infrastructures themselves exhibit linguistic and geographic asymmetries (Céspedes et al., 2025; Zheng et al., 2025). A standalone scholar who relies exclusively on global English-language discovery can therefore reproduce the blind spots of the very infrastructures being used to compensate for institutional absence.

15.2 Geographic Distribution Audit

The Geographic Distribution Audit (GDA) adds five questions to every flagship: (1) what local community, institution, or venue should encounter this claim; (2) what Global South literature or institution may alter its framing; (3) what does English-only search omit; (4) what situated position gives access to records absent from the global map; and (5) which local source could overturn a global recommendation? In governance notation:

$$\text{DVP}^* = \text{DVP} + \text{GDA}. \tag{71}$$

A local blind spot discovered by the author is a **Positional Counter-Readout**; it improves route diversity but remains author-generated and therefore is not K_2. Independent local critique can become a K_2 candidate when it is substantive and claim-relevant.

15.3 Global strength without identity dependence

The global track imposes a blind-name test: if author name and affiliation are removed, does the manuscript still create theoretical necessity? Define the governance score

$$S_G = L \times G \times M \times B \times F, \tag{72}$$

where L is literature embeddedness, G gap or inadequacy precision, M mechanism clarity, B boundary conditions, and F

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field fit. The expression is not a psychometric scale; it encodes the bottleneck intuition that if any load-bearing component approaches zero, global manuscript strength collapses toward zero.

Institutional prestige can alter exposure and evaluation in some settings, but the architecture does not make anonymity a prerequisite. The preserved decision rule is

$$\text{DoubleBlind} = \text{Bonus}, \tag{73}$$
$$\text{DoubleBlind} \neq \text{Requirement}, \tag{74}$$
$$\text{ManuscriptStrength} > \text{IdentityDependence}. \tag{75}$$

Institutions are therefore used for methods, ethics, data, collaboration, and specialist expertise when useful, rather than as prestige decoration.

15.4 Global and Thai tracks

Each socially relevant flagship is assigned

$$\text{ConversionPlan}_i = \{\text{Global}_i, \text{Thai}_i\}. \tag{76}$$

The global track includes an international journal, reply community, named correspondent, conference/workshop, open review route, or empirical collaborator. The Thai track includes at least one active local node: a Thai seminar, Thai-language companion artifact, Thai journal, policy/practice contact, academic correspondence, institutional community, or substantive public scholarly dialogue.

The Thai track is not a lower-status version of the global track. It supplies functions that global routes may not: situated counterexamples, language-sensitive interpretation, policy access, practice knowledge, local adoption, and Global South legitimacy critique. Accordingly,

$$K_{2,Global} \neq K_{2,Thai}, \tag{77}$$

while

$$K_{2,Global} + K_{2,Thai} \Rightarrow \text{RouteDiversity} \uparrow . \tag{78}$$

Pure formal assets with no meaningful Thai community may book an explicit exception rather than fabricate a local counterpart.

Proposition 11 (Dual-track decorrelation) *For socially and institutionally situated claims, pairing a global scholarly route with a relevant local-language or local-institutional route will expose a broader set of counterexamples, assumptions, and venue constraints than a global-English-only route, especially where discovery infrastructures have uneven regional coverage.*

16 Thailand as an operational epistemic countermap

The Thai landscape is used as a worked design case, not an exhaustive national census. The operational principle is to build a **portfolio of friction surfaces**: anthropology and culture, religion, Asian studies, peace and human rights, philosophy and science-and-technology studies, mathematics and physics, social enterprise and innovation, and policy networks. Bangkok and Greater Bangkok are sufficiently dense that a scholar can construct recurring exposure across multiple institutions rather than depending on one university to supply a weekly seminar stream.

The local optimization target is not event count. For event e, define

$$\text{FrictionValue}_e = \frac{F_e O_e D_e R_e}{T_e + C_e + P_e + \varepsilon}, \tag{79}$$

where F, O, D, R denote fit, openness, expected depth, and repeat-node potential, while T, C, P denote travel, cash, and preparation costs. The weekly rule is one substantive exposure plus one addressed follow-up, not "event tourism." Repeat exposure to four to six anchor communities is preferred to accumulating badges across dozens of venues. The full 2026 Thai Epistemic Landscape Atlas, including the operational node list and caveats, is preserved in the integrated operational appendices below.

A Thai companion artifact is also part of the design for socially relevant global flagships. It may be a short research note, scholarly essay, lecture note, Thai abstract with mechanism diagram, or reply article. Its function is

$$\text{GlobalAsset} \rightarrow \text{ThaiLegibility} \rightarrow \text{ThaiFriction}, \tag{80}$$

not duplicate publication.

17 Open research toolchain as epistemic division of labour

The standalone scholar must also reconstruct mundane infrastructure. Tools are selected by epistemic function rather than by platform prestige. A minimal function map is shown in Table 2.

The toolchain rule is *function preservation*: if a platform disappears or changes policy, the required epistemic function must receive a replacement. This prevents infrastructure from being confused with brand loyalty.

Discipline routing remains necessary. Formal mathematics, social epistemology, family practice, AI governance, and health promotion do not accept the same entry artifacts. The preferred entry artifact minimizes dependence on face or status while maximizing field-legible inspectability:

$$a_f^* = \arg\max_a \text{FaceIndependence}_f(a). \tag{81}$$

This can mean proof and code in a formal field, a conceptual manuscript and reply in social epistemology, or ethics-reviewed empirical evidence in human-subject practice research.

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Table 2: Open toolchain as functional infrastructure. Platform presence is not evidence of quality.

Function	Epistemic purpose	Candidate infrastructure
Source ledger	preserve original-source traceability and notes	Zotero + original records
DOI metadata	normalize identifiers and bibliographic metadata	Crossref
Scholarly discovery graph	expand and inspect citation/entity relations	OpenAlex, with coverage audit
Mutable working history	public/private version history and issue ledger	GitHub
Mechanical checks	schedule tests, linting, reproducibility checks	GitHub Actions / CI
Persistent release	DOI-bearing version ledger	Zenodo
Software preservation	preserve released code where applicable	Software Heritage
Social-science circulation	working-paper discoverability and revision	SSRN
Field preprint circulation	discipline-specific provisional dissemination	arXiv or appropriate repository
Identity spine	connect outputs without making affiliation the identity	ORCID
Formal proof	mechanically check derivations within a formalization	Lean / Rocq
Open human review	obtain public preprint feedback	PREreview and related venues
Written epistemic reply	create durable argumentative exchange	Social Epistemology Review and Reply Collective
Preregistration	constrain empirical degrees of freedom	OSF Registrations / field registry

18 Protection layers: integrity, priority, survival, and attention

18.1 Integrity firewall

The non-negotiable citation path is

$$\begin{aligned}&\text{AI candidate} \rightarrow \text{OriginalSource}\\&\qquad \rightarrow \text{ClaimMatch} \rightarrow \text{VerifiedCitation}.\end{aligned} \tag{82}$$

AI output never goes directly into a citation ledger. Every material citation is checked for existence, metadata, claim match, evidence tier, and quotation/paraphrase scope.

18.2 Priority protection

When a flagship has passed novelty and human-mastery gates and is mature enough to defend, the scholar evaluates an early timestamp through an appropriate preprint or repository, subject to target-journal policy. The objective is neither premature naming nor perfectionist delay, but a citable provisional object with explicit version status.

18.3 Survival buffer

Cumulative advantage is irrelevant if the programme collapses after early rejection. Each flagship therefore has a rejection budget and a rule that rejection must produce diagnostic information:

$$\text{Reject} \rightarrow \text{Objection} \rightarrow \text{Revision} \rightarrow \Delta Q. \tag{83}$$

Repeated resubmission without learning is not persistence.

18.4 Attention-credit separation

Let A_t be public attention. Then

$$A_t \neq C_t^{\text{scholarly}}. \tag{84}$$

Attention contributes only when converted into a scholarly asset, discussion, reuse, or independent opportunity. This is especially important in AI-native environments where high-volume content can generate visibility faster than credibility.

19 Human Mastery, ethics, and creator-evaluator separation

Before public commitment, the author should be able to defend the problem, major literatures, nearest constructs, precise inadequacy, proposed mechanism, boundary conditions, strongest counterargument, strongest counterexample, falsifier, and discriminating empirical test without AI. Let

$$H_g = \frac{\text{questions defended without AI}}{10}. \tag{85}$$

A local threshold such as $H_g \geq 0.8$ is a governance heuristic, not a universal standard.

For practice and human-subject research, another invariant applies:

$$\text{InterventionCreator} \neq \text{SoleEvaluator}. \tag{86}$$

Practice access can generate high-resolution phenomena, but

$$\text{PracticeExperience} \neq \text{PopulationEvidence}. \tag{87}$$

Claims about clients, communities, or interventions require appropriate ethics review, methods expertise, consent governance, and external evaluation. The existence of non-university ethics pathways in Thailand means that absence of a university IRB should not be converted into an excuse to bypass review (National Research Council of Thailand, 2026).

20 Portfolio control and defeat conditions

A high-throughput system needs explicit loss conditions. Otherwise persistence is easily confused with truth. Public work in progress is limited to assets that are in active external process

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and require continuing human response. A local heuristic is PublicWIP ≤ 3, with additional ideas kept private until capacity opens.

The portfolio can be organized into three lanes: a flagship-theory lane, a credit-converter lane (conference, perspective, reply, extension), and a validation lane (empirical test, methods collaboration, external review). New public projects replace or close existing ones rather than accumulating indefinitely.

Defeat conditions are time-indexed. A 12-month period with no independent engagement after genuine targeted exposure triggers a field-fit and concept-necessity audit. At 24 months, repeated fatal objections from independent routes, no citations, and no substantive collaborators trigger construct merging, narrowing, co-authorship-first, community migration, or critical empirical testing. At 36 months, no peer-reviewed anchor, no repeat human node, and no external uptake constitute defeat of the current configuration, not defeat of scholarship itself. Fifty addressed attempts with no stable correspondent defeat the current human-acquisition protocol and trigger redesign of target class, message language, ask size, venue, timing, packaging, or warm-introduction route.

Immediate SCRAM triggers include fabricated or unverified citations, known material errors left uncorrected, undisclosed material conflicts of interest, coercion risk, failure of the human-mastery gate, or knowingly allowing claim scope to exceed evidence scope. Define

$$\text{SCRAM} = \text{FreezeNewRelease} + \text{Correction} + \text{ReAudit}. \tag{88}$$

21 An integrated architecture

The system can now be represented as a directed sequence with feedback:

$$\begin{aligned} \text{Phenomenon} &\to \text{AIExploration} \to \text{DVP} \\ &\to \text{HumanMastery} \to \text{Integrity} \to K_1 \\ &\to \{\text{Global, Local Friction}\} \to K_2 \\ &\to \text{Revision} \to K_3. \end{aligned} \tag{89}$$

The resulting asset can then create citations, reuse, invitations, collaborator access, data access, grants, or new questions, feeding the next cycle. Mature scholarly momentum is therefore not simply $C_{t+1} > C_t$ every period. Rejections, null results, correction, or delayed review can cause drawdowns. The appropriate horizon is

$$E[C_{t+k}] > C_t \tag{90}$$

for a meaningful medium-term k, subject to integrity and survival constraints.

A compact maximum-velocity heuristic is

$$\mathcal{V}_C^* = \frac{CY\,Coh_{latent}\,L_g\,Conv\,Net\,IR\,PS\,Ver}{H^{\text{critical}}(Debt + Frag + Bias + COI + \varepsilon)}, \tag{91}$$

where each numerator term represents a defensible conversion asset and each denominator term a human-time or epistemic liability. The expression is deliberately heuristic. Its function is to point the gradient toward the bottleneck, not to manufacture numerical precision.

22 Research propositions and empirical agenda

The architecture yields a research programme rather than a closed theory. Several propositions can be tested across disciplines and institutional positions.

Proposition 12 (Formation vs. certification) *AI-mediated formation infrastructure will have its largest positive effect on provisional-asset quality for researchers with low ambient human formation access, but this effect will not eliminate the need for independent human or institutional routes for later-stage validation.*

Proposition 13 (Conversion overload) *At high stock-pressure ratios χ_t, additional provisional output will exhibit diminishing conversion yield unless programme legibility, human non-leakage, or journal/review capacity increase concurrently.*

Proposition 14 (Moderation advantage) *Verification and field-translation practices that reduce raw output speed can increase credible credit velocity when their effect on integrity survival and field legibility exceeds their human-time cost.*

Proposition 15 (Geographic counter-readout) *Local-language and local-institutional search routes will identify nonredundant sources, venues, and objections relative to global-English AI/search routes more frequently in domains where bibliographic coverage is regionally uneven.*

Proposition 16 (Fellowship-friction complementarity) *Repeated human relationships will improve the quality of criticism over time when the network contains deliberate external-dissent routes; without such routes, increased fellowship will raise the risk of correlated positive readouts and groupthink.*

Proposition 17 (Credit-to-opportunity conversion) *The durable value of scholarly credit will be better predicted by its conversion into qualified opportunities that expand capability or evidence than by raw counts of publications, views, or invitations.*

Critical empirical test selection. A conceptual architecture need not attach a small dataset to every claim. It should instead identify the uncertainty with the greatest theory-discriminating value. For candidate propositions $P_1, \dots, P_n$, define

$$\begin{aligned} P^* = \arg\max_{P_i} \big[&\text{TheoreticalImportance}_i \\ &\times \text{DiscriminatingPower}_i\big]. \end{aligned} \tag{92}$$

The preferred empirical project is the one for which an adverse result would most force revision and for which the observation most cleanly distinguishes the focal mechanism from its strongest rival. The intended sequence is

$$\begin{aligned} \text{ConceptualPaper} &\to \text{CriticalEmpiricalTest} \\ &\to \text{TheoryRevision}. \end{aligned} \tag{93}$$

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This is stronger than adding descriptive data merely to make a conceptual paper appear empirical.

Future empirical work can operationalize these propositions using longitudinal author-level ledgers, matched comparisons of institutionally embedded and standalone scholars, randomized outreach designs, analysis of preprint-to-review trajectories, bibliographic route comparisons across English and Thai searches, and qualitative studies of recurring correspondents. The current equations should move from `definition/Dr` to `fit_calibrated` only after sufficient longitudinal data exist.

23 Boundary conditions and limitations

First, the model is designed for research settings in which public provisional dissemination is ethically and legally acceptable. High-stakes clinical, security-sensitive, confidential, or vulnerable-population research may require stronger pre-release gates. Second, AI formation capacity differs by field. In formal mathematics, proof assistants and mechanically constrained checks may bottom out more of the verification chain than in interpretive sociology, but formal validity still depends on whether the formalization represents the intended semantics. Third, the Thai landscape is an operational snapshot rather than a census. Event openness, fees, access, and institutional priorities change; local routes must be reverified continuously. Fourth, the reactor-control vocabulary is a control metaphor, not a claim that scholarly systems obey nuclear kinetics. Any factor lacking a stable event definition should be removed rather than preserved as decoration.

Fifth, AI models and search systems change rapidly. DVP, ODC, and the Route Dependence Matrix are governance protocols, not calibrated psychometric instruments. Sixth, fellowship metrics do not claim that relationships are reducible to a latent score. They merely make continuity visible. Seventh, preprints accelerate dissemination and timestamp priority, but they can also spread weak or premature claims; explicit provisionality and domain-sensitive safety rules remain essential. Finally, the finite diagnostic that motivated the conversion pivot is a single author-level snapshot. It establishes the existence of one design configuration, not its population prevalence.

24 Conclusion: from epistemic isolation to epistemic position

The decisive shift produced by AI is not that scholarship no longer needs institutions, peers, or methods. It is that the production of scholarly candidates can become uncoupled from the infrastructures that historically disciplined them. For a standalone scholar, this creates both an opportunity and a danger. The opportunity is to lower the cost of exploration, adversarial formation, preprint staging, and formal checking. The danger is to confuse fluent abundance with independent correction, attention with uptake, legitimacy with truth, or catalogue size with cumulative scholarly position.

The architecture proposed here treats standalone scholarship as an infrastructure-building problem. AI supplies search-space expansion and iteration speed; the scholar retains conceptual ownership and commitment; open repositories supply timestamped provisionality and version lineage; global and Thai routes supply nonredundant friction; repeated human nodes reduce leakage; journals and empirical work provide stronger external constraints; and Readout Universe functions as a meta-governance layer that prevents any one readout from hardening into truth.

The endpoint is not solitary genius. It is **epistemic position**: a state in which other people know what problem the scholar is trying to solve, can locate the relevant citable assets, can see how those assets have changed under criticism, and know where to send the next objection. In compact form,

$$\begin{aligned}\text{EpistemicPosition} = \text{RecognizableProblem} \\ + \text{RepeatHumanNodes} + \text{CitableAssets} \\ + \text{CorrectionHistory}.\end{aligned} \tag{94}$$

The fastest credible pathway is therefore not maximum production. It is

$$\begin{aligned}\text{CrediblePath} = \text{EarlyTimestamp} + \text{ExplicitProvisionality} \\ + \text{RapidRevision} + \text{ExternalFriction} \\ + \text{EventualCertification}.\end{aligned} \tag{95}$$

In the AI-native era, the standalone scholar becomes viable not by escaping the social organization of knowledge, but by reconstructing it deliberately across tools, institutions, languages, and relationships.

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Purpose. This appendix preserves operational mechanisms from the source strategy that are too implementation-specific, time-sensitive, or table-heavy for the conceptual article. It is not a second theory. It documents the machinery by which the theory can be instantiated, audited, recalibrated, or defeated. Current-state values are explicitly diagnostic rather than universal.

A Finite diagnostic that motivated the conversion pivot

The motivating author-level snapshot on 29 August 2026 is retained as a *finite diagnostic*, not as a population benchmark or causal test.

Readout	Snapshot
Public K_1 assets	≈ 23
Observation span	$\approx$ 10.6 months
Mint velocity	$\approx 2.17\ K_1$/month
Views	$\approx$ 2,200
Downloads	$\approx$ 450
Download/view	$\approx$ 20%
Stable correspondents	0 logged
Historical u_{conv}	not logged
Structured independent-human log	newly installed
Back catalog	large/activatable
Programme coherence	qualitatively present
Programme legibility	qualitatively low

The diagnostic establishes abundant provisional production and an operational staging layer, but does *not* establish scholarly-credit compounding. In particular,

$$M_{attention} \neq M_{truth}, \qquad M_{attention} \neq M_{K_2}. \tag{96}$$

An observed logged value near zero is not interpreted as a true population zero when logging was historically absent. The intervention therefore changes the local optimization problem from minting more K_1 to converting existing stock into independent friction, repeat-human nodes, revisions, journal reports, citations, reuse, and visible programme structure.

B DVP packet, operator decorrelation, and human acquisition

B.1 Knowledge-state boundary

The operational state ladder is:

$$K_0 : \text{private candidate}, \tag{97}$$

$$K_1 : K_0 + \text{public timestamp, provenance, provisionality}, \tag{98}$$

$$K_2 : K_1 + \text{substantive external friction}, \tag{99}$$

$$K_3 : K_2 + \text{review/certification/empirical constraint}. \tag{100}$$

The non-laundering invariant is

$$\text{DVP} \not\Rightarrow K_2. \tag{101}$$

B.2 DVP route packet

A DVP packet uses materially different routes rather than a model vote. Recommended route roles are: advocate, nearest-concept prosecutor, falsifier, source auditor, hostile reviewer, translation reviewer, and method/empirical designer. Diversity may come from model family, framing, context, evidence base, or mechanical verifier.

Required controls include:

1. **Frame break:** remove preferred terminology and force a different conceptual vocabulary.
2. **Recovery test:** hide the author's answer to a strong objection and test whether the constraints recover an equivalent structure.
3. **Anchored checks:** bottom out claims, where possible, in original records, executed computations, or formal/mechanical verification.
4. **Disagreement ledger:** classify construct, source, mechanism, boundary, venue, and epistemic-tier disagreements; resolve or declare them rather than averaging them away.
5. **Geographic audit:** add Thai-language, Thailand/Global-South, and local-institution routes to the verification packet.

Two boundaries remain explicit:

$$\text{MechanicalValidity} \neq \text{SemanticValidity}, \tag{102}$$

$$\text{SourceExistence} \neq \text{ClaimSupport}. \tag{103}$$

B.3 Operator-Decorrelation Control and Route Dependence Matrix

The operator is a shared ancestor across otherwise different AI routes. ODC therefore precommits acceptance criteria, blinds route identity where practical, randomizes result order when useful, and imposes a query budget/stop rule to reduce answer shopping. A Route Dependence Matrix records residual dependence on model family, prompt ancestry, operator framing, source base, and external anchors. The target is reduced correlated error plus exposed residual dependence, not a claim of independence.

B.4 Preprint-to-human conversion

No available human is not a research-stop rule:

$$\text{NoHumanAvailable} \not\Rightarrow \text{ResearchStop}. \tag{104}$$

Instead:

$$\text{NoHumanAvailable} \Rightarrow \text{DVP} \rightarrow K_1 \rightarrow \text{HumanAcquisition}. \tag{105}$$

Preprint is the public probe. When broadcast feedback remains shallow, the system switches to addressed correspondence: named scholars, authors of adjacent constructs, open-review communities, targeted seminars, journal submission, or a deliberately convened asynchronous circle.

A cold-start async circle uses roughly 4–6 participants, a written-first cadence every 2–4 weeks, one claim per round,

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one counterexample requirement, one "what would change my mind?" field, and an objection/revision archive. Consent is required before recording or publication. Its success is measured by revisions, counterexamples, independent routes, and repeat participation rather than attendance.

A standing defeat condition is retained: after at least 50 genuine addressed attempts with zero stable correspondent, the *human-acquisition protocol* loses and must be redesigned; this does not by itself falsify the research programme.

C Material feasibility and publication-cost boundary

A standalone architecture that ignores material conditions will systematically overestimate conversion capacity. The source strategy therefore treats annual research resources as required inputs rather than outputs of the model:

$$B_{year} = B_{conference} + B_{ethics} + B_{software} + B_{data} + B_{publication} + B_{travel}. \tag{106}$$

The default publication-cost ordering is: (1) field-fit diamond or no-author-fee routes; (2) subscription/no-APC author routes compatible with lawful preprint or accepted-manuscript sharing; (3) applicable waivers, discounts, or agreements; and (4) APC payment only when strategic value is clear and the cost was pre-committed. The governing boundaries are

$$\text{Prestige} \not\Rightarrow \text{APCApproval}, \tag{107}$$

$$\text{APCApproval} \Rightarrow \text{FieldFit} + \text{CreditYield} + \text{BudgetFit}. \tag{108}$$

Before a project is opened, the scholar checks human-critical time, ethics/data/travel resources, publication-cost traps, collaborator requirements, and whether completion remains possible without an assumed future grant. If a missing resource is load-bearing,

$$\text{MissingResource} \Rightarrow \text{ProjectHold}. \tag{109}$$

This is a feasibility gate, not an instruction to maximize spending.

C.1 Superseded publication-credit multiplier

The source architecture defined a publication multiplier

$$PC_i = \text{JournalQuality}_i \times \text{ProgrammeFit}_i \times \text{ContributionStrength}_i \times \text{UptakePotential}_i. \tag{110}$$

Its function is preserved but the equation is **superseded** in the integrated standalone model by the K2/journal-friction selection architecture below, which additionally prices depth, relevance, latency, cash, and human hours. The inherited ordering remains active: field and conceptual fit, reviewer/community relevance, journal standing, visibility, and speed/cost take precedence over optimizing impact factor alone. Thus the old rule is not silently deleted; it is replaced by a model with greater decision coverage.

D K2 procurement: buying effective friction with cash, time, and latency

A route that is free in cash can still be expensive in human hours or latency. Define

$$Cost_{K2,r} = \frac{Cash_r + \lambda_H H_r + \lambda_L L_r}{D_r R_r I_r + \varepsilon}, \tag{111}$$

where D, R, I denote expected depth, claim relevance, and route independence. The weights are governance choices, not estimated coefficients.

Route class	Typical cost profile	Best use
Formal-tool communities	low cash; variable latency	proof/tool-specific objections
Written reply communities	low cash	argument-specific friction
PREreview/open review	low cash; variable	post-preprint human review
Targeted correspondence	near-zero cash	named argument/construct
Async circle	low cash; scheduled	recurrent formation and K_2
Conference/workshop	fee/travel variable	field friction + network
Journal peer review	often low submission cash; high latency	manuscript-level depth
Thai public/seminar route	low–variable	situated/local relevance

Journal submission can be treated as latency-purchased friction. A desk rejection gives an editorial/fit readout but is not relabelled deep K_2. A post-peer-review rejection with substantive mechanism, method, boundary, or literature objections can be a successful K_2 event even when publication outcome is reject.

A journal-routing heuristic is

$$ExpectedK2Yield_j = \frac{P(Review_j)\, Depth_j\, Fit_j}{Cash_j + Prep_j + Latency_j + \varepsilon}. \tag{112}$$

Values begin as qualitative labels and should be recalibrated from a personal journal ledger. Effective friction is

$$EffectiveK_2 = Depth \times Relevance \times Independence. \tag{113}$$

Global field friction and Thai/local friction are complementary rather than substitutable when a contribution claims relevance to both conversations.

E Conversion controls, human time, and portfolio discipline

E.1 Stock pressure and mint-convert coupling

Let λ_{mint} and λ_{conv} be provisional-production and conversion rates. A local stock-pressure indicator is

$$\chi_t = \frac{\lambda_{mint,t}}{\lambda_{conv,t} + \varepsilon}. \tag{114}$$

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Backlog evolves as

$$B_{t+1} = B_t + M_t - \omega X_t, \tag{115}$$

where M_t is newly minted stock and X_t is conversion activity. During conversion mode a coupling rule constrains new minting to demonstrated conversion:

$$M_t \leq \lambda X_t. \tag{116}$$

The credit-stock model permits old assets to reactivate:

$$C_{t+1} = \big[(1-\delta)C_t + G_t + \Phi_t(Stock)\big] I_t P_t^{(r)} S_t. \tag{117}$$

Thus credit acceleration can arise from $d\Phi(Stock)/dt > 0$ without accelerating new flagship production.

E.2 Critical human time and WIP

The source strategy used a six-week calibration pilot of 14 critical-human hours/week: 6 mastery/deep reading, 3 human network/review service, 3 source verification, 1 exploration, and 1 governance/objection ledger. This is a local pilot, not a universal optimum. The general rule is

$$H_{planned} > H_{sustainable} \Rightarrow \text{PortfolioShrink}. \tag{118}$$

Human cognition is reserved for conceptual commitment, source verification, construct differentiation, mechanism judgment, falsifier design, ethics, interpretation, public defence, relationships, and final tier/release decisions. Search-space expansion, first-pass synthesis, formatting, scaffolding, transcription, and repetitive documentation can be automated when epistemically safe.

Public WIP counts only assets in active external process requiring continuing human response. The current local guardrail is

$$PublicWIP \leq 3. \tag{119}$$

Private ideas may remain large; public obligations may not. Portfolio freeze occurs at WIP saturation, excessive xenon, unresolved integrity incidents, or a verification queue beyond sustainable capacity.

E.3 Portfolio and project priority

The source strategy used an author-specific portfolio guardrail of 60% independent theory, 30% practice research, and 10% bridge/exploratory work. This ratio is retained here only as a provenance record and is **deprecated from the generalized standalone theory**: it was a local allocation heuristic, not a universal coefficient. The transferable function is selective pressure against AI-driven project proliferation and is preserved through WIP limits, programme fit, bottleneck allocation, and project-priority rules below.

The source strategy retained a 60/30/10 portfolio guardrail for independent theory / practice research / bridge-exploratory work. These percentages are governance defaults, not empirical optima. Project priority can be expressed as

$$Priority_i = \frac{I_i N_i E_i F_i Y_i S_i}{D_i + C_i + Frag_i + COI_i + \varepsilon}, \tag{120}$$

representing importance, novelty, explanatory leverage, programme fit, credit yield, and priority security over epistemic debt, execution cost, fragmentation, and conflict-of-interest risk.

E.4 Dated 90-day conversion sprint

The 29 August 2026 intervention is preserved as a dated operating plan:

1. Weeks 1–2: classify the 23 assets as A (anchor), B (bridge), or C (formal/exploratory); publish a canonical programme index; harmonize ORCID/repository lineage; choose roughly 3 A + 2 B assets for priority activation.
2. Assign each selected anchor a Global route, Thai route, named interlocutors, a journal/reply route, a version/update path, and a Thai companion artifact when substantively relevant.
3. Weeks 2–12: default to 2–3 addressed conversion attempts/week; start at least two journal pipelines from existing stock; log action → route → human response → K_2? → objection → revision.
4. Freeze new conceptual flagships by default; do not freeze private DVP exploration, revisions, resubmissions, formal checks of existing assets, K_2 procurement, or externally scheduled empirical work.

Success does not require acceptance. It requires measurable Stock→ K_2 conversion together with improved legibility and logged non-leakage/conversion.

E.5 Three-year momentum architecture

Year 1 establishes signal around one recognizable problem family; Year 2 converts signal into recognition through journal pipelines, revisited communities, and critical empirical tests; Year 3 seeks cumulative advantage in which collaborations and prior assets generate more qualified opportunities than must be manually sought.

F AI disclosure and provenance protocol

AI disclosure is treated as an integrity requirement rather than a legitimacy tactic. Evidence that disclosure can alter perceived trust or legitimacy does not license concealment (Schilke and Reimann, 2025; Schilke and Reimann, 2026). The rule is

$$\text{DisclosurePenalty} \not\Rightarrow \text{Concealment}, \tag{121}$$

while the operational response is

$$\text{DisclosurePenalty} \Rightarrow \text{BetterProvenance} + \text{BetterHumanDefence} + \text{VenueFit}. \tag{122}$$

F.1 Mandatory venue audit

Before submission, the author records the venue's current requirements for: (1) AI-use policy; (2) authorship policy; (3)

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disclosure wording; (4) preprint policy; (5) repository/version policy; (6) data/code policy; (7) conflict-of-interest policy; (8) human-subject ethics requirements; and (9) copyright or accepted-manuscript archiving rights. If the venue policy does not fit the actual workflow, the workflow or venue changes; the history of AI use does not.

F.2 Route-level disclosure

A generic sentence such as "AI was used" is less informative than a route-level record. The following table preserves the division of labour from the source architecture (International Committee of Medical Journal Editors, 2026; Lahtee, 2026b). The invariant is

$$\text{AIContribution} \neq \text{EpistemicResponsibility}. \tag{123}$$

F.3 Critical uncertainty and concept-cluster control

The operational empirical rule selects the proposition with the highest joint theoretical importance and discriminating power, as formalized in Equation (92). The corresponding concept-cluster rule treats one flagship intellectual investment as a lineage of distinct exposure functions, as formalized in Equations (47) and (48). A conference, empirical test, comparative extension, or grant is not a duplicate credit event merely because it descends from the same concept; it must expose a different uncertainty, audience, constraint, or implementation problem. If it does not, it is merged or not released.

G Open research toolchain and discipline routing

Every tool must own an epistemic function; brand retention is not the goal. If a tool disappears while its function remains necessary, a replacement is required. Entry artifacts remain discipline-specific. Formal claims benefit from proofs and mechanically checkable artifacts; social theory requires field-language conceptual objects; practice research requires ethical and empirical constraints. Venue use is governed by topic fit and audience grammar, not comment intensity.

H Embedded-practice and positionality firewall

The architecture allows an embedded practice observatory to supply high-resolution phenomena while explicitly preventing it from becoming a representative population by assumption. In the motivating implementation, ARAYA Nikah Social Enterprise supplies practice access; the general rule is portable to other embedded practice settings.

Let P_A denote practice capital, C_o conflict risk, and B_i insider bias. A bookkeeping heuristic is

$$\text{NetPractice} = P_A - (C_o + B_i). \tag{124}$$

The purpose is not numerical estimation but to force the benefits and liabilities of insider access into the same decision frame.

H.1 Discovery–justification separation

$$\text{PracticeObservation} \to \text{Hypothesis}, \tag{125}$$

but substantive public claims require

$$\begin{aligned}\text{Claim} \leftarrow \text{Literature} &+ \text{ExternalEvidence} \\ &+ \text{ComparativeEvidence}.\end{aligned} \tag{126}$$

For outcome claims,

$$\text{InterventionCreator} \neq \text{SoleEvaluator}, \tag{127}$$

and

$$\text{ClaimScope} \leq \text{SamplingScope}. \tag{128}$$

Practice clients therefore cannot be silently generalized to Thai Muslims, Muslim populations globally, or any other population not identified by the sampling design. Ethics review, external methodological oversight, and independent analysis are added when the stakes or design require them (National Research Council of Thailand, 2026).

H.2 Lived bridge research questions

The reflexive bridge produces empirical questions rather than conclusions: what credibility deficits do independent or grassroots knowledge producers encounter; does AI reduce production barriers while widening legitimacy gaps; how does horizontal recognition convert into vertical legitimacy; can correspondence communities function as friction infrastructure; when does institutional certification add correction capacity rather than status alone; how should institutions respond when candidate knowledge grows faster than certification; and how does fellowship affect persistence and theory revision? These questions preserve the source architecture's use of lived position as a context of discovery without romanticizing isolation (Lahtee, 2026b).

I Dual-track implementation and Thai Epistemic Landscape Atlas

I.1 Dual-track invariant

For each socially relevant flagship i,

$$ConversionPlan_i = \{Global_i, Thai_i\}. \tag{129}$$

A Global route may be an international journal, named correspondent, conference/workshop, PREreview/formal community, or empirical collaborator. A Thai route is an active Thai node, companion artifact, TCI/Thai journal route, policy/practice contact, guest seminar, academic correspondence, institutional community, or public scholarly dialogue producing substantive reply. Pure formal assets may book an explicit exception where no meaningful Thai route exists.

The Thai track is not public relations. It can supply situated counterexamples, language-sensitive interpretation, positional correction, institutional/policy access, community adoption, and Global-South critique. Accordingly,

$$K_{2,Global} \neq K_{2,Thai}. \tag{130}$$

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Table 3: Route-level AI disclosure and retained human responsibility.

Research route	AI role	Human responsibility
Literature discovery	Candidate search and clustering	Verify original sources and claim–source match
Synthesis	Comparison and first-pass mapping	Final interpretation and theoretical positioning
Writing	Language support and restructuring	Argument ownership and final wording
Coding	Scaffold, debugging suggestions	Execute checks, inspect outputs, interpret results
Adversarial review	Generate objections/counterexamples	Judge accept/reject/revise and preserve unresolved objections
Citations	Candidate metadata only	Final bibliographic and support audit

Function	Current owner class
Source ledger	Zotero + original records
DOI metadata	Crossref
Scholarly graph discovery	OpenAlex, with coverage audit
Mutable working history	GitHub
Mechanical/CI checks	GitHub Actions
Persistent release/DOI	Zenodo
Software preservation	Software Heritage when applicable
Social-science preprint	SSRN
Formal/math/physics preprint	arXiv when field-fit
Identity spine	ORCID
Formal proof	Lean / Rocq
Open human preprint review	PREreview
Social-epistemology reply	SERRC-like route
AI/rationality adversarial arena	LessWrong when topic-fit
Empirical preregistration	OSF Registration / field registry
Machine formation	DVP
Human/certification friction	HUM / CERT routes

I.2 Geographic Distribution Audit

Every flagship packet asks: which Thai institutions/communities should this claim encounter; which Global-South literatures may be absent from Anglophone routes; what disappears in English-only search; what situated records are visible from the researcher's position; and which local record could overturn a global recommendation? The invariant is

$$\text{MultiAIConsensus} \nRightarrow \text{GeographicCompleteness}. \quad (131)$$

An author-discovered positional counter-readout improves route diversity but is not external K_2 until independently engaged.

I.3 Thai Epistemic Landscape Atlas: 38 operational surfaces

The following is a dated operational atlas inherited from the 29 August 2026 strategy. It is a portfolio of possible friction surfaces, not a ranking, census, endorsement, or guarantee of current access. Event schedules, fees, openness, and fit must be reverified before use.

1. Sirindhorn Anthropology Centre (SAC) — anthropology, culture, ethnicity, marginality, knowledge-from-below.
2. SEA Junction at BACC — Southeast Asia, migration, conflict, culture, practitioner–scholar exchange.
3. RILCA, Mahidol — language, culture, intercultural communication, identity, methods/AI ethics.
4. College of Religious Studies, Mahidol (CRS) — religion, interreligious and religious-studies friction.
5. Institute of Asian Studies, Chulalongkorn (IAS) — Asian/regional research and subfield routing.
6. ISIS Thailand, Chulalongkorn — security, geopolitics, regional institutions.
7. School of Global Studies, Thammasat (SGS) — global studies, development, social innovation.
8. College of Interdisciplinary Studies, Thammasat (CISTU) — cross-disciplinary entry surface.
9. Institute of Human Rights and Peace Studies, Mahidol (IHRP) — rights, peace, conflict, governance.
10. King Prajadhipok's Institute peace/governance route — peacebuilding, governance, policy.
11. CRS–IHRP crossover — religion/rights/peace intersection.
12. Philosophy and Religion Society of Thailand (PARST) — philosophy, religion, argument-level community.
13. Chula STS/CSTS/STREC cluster — science, technology, society, knowledge institutions.
14. Chulalongkorn Social Research Institute / STS network — social research and institutional analysis.
15. Institute of Thai Studies, Chulalongkorn — Thai studies and local knowledge contexts.
16. PBIC Thai Studies, Thammasat — Thai studies / interdisciplinary regional interpretation.
17. Institute of East Asian Studies, Thammasat — East-Asian comparative route.
18. C asean — ASEAN business, culture, regional network exposure.
19. Mathematics and Computer Science, Chulalongkorn — formal/mathematical entry artifacts.
20. Mathematical Association of Thailand + CEPMART — mathematics community and formal exchange.
21. Mahidol Physics — physics/formal research surface.
22. Thai Physics Society / Siam Physics Congress — national physics community.
23. Thailand Center of Excellence in Physics (ThEP) — advanced physics network.
24. Chula Physics / Bangkok School on Cosmology — cosmology/physics route.
25. NARIT + Thai Astronomical Society — astronomy and public/scientific network.
26. NSTDA / NAC — national science, technology, applied-research network.
27. Social Enterprise Thailand Association — social-enterprise field/community route.
28. Social Enterprise Promotion Office (OSEP) — policy/regulatory ecosystem route.
29. ChangeFusion — social innovation and impact ecosystem.
30. CU Social Innovation Hub — university social-innovation network.
31. National Innovation Agency (NIA) — innovation-policy and ecosystem route.
32. SDG Move, Thammasat — sustainability, research-to-policy, partnerships.
33. SGS/GSSE, Thammasat — social entrepreneurship / global social innovation.
34. se.school — social-enterprise learning infrastructure; not K_2 by itself.
35. Asia Pacific Futures Network / Sasin — futures, sustainability, peace, cross-sector foresight.
36. ASEAN Studies Center, Chulalongkorn — ASEAN research and regional institutions.
37. IAS sub-centres as separate target surfaces — Muslim, migration, Mekong, China, Japan, and adjacent area studies.
38. Bangkok Asia/ASEAN friction cluster — ISIS + IAS + SEA Junction + C asean as different institutional grammars.

I.4 Expanded atlas metadata

The compressed list above is expanded here with the fit, function, and cadence metadata used in the source strategy. These are management descriptors in an author operational snapshot, not institutional rankings or guarantees of current access (Lahtee, 2026b).

I.5 Near-term friction opportunities in the source snapshot

The source strategy also retained a dated event-density readout to show that a pooled Bangkok/Greater-Bangkok landscape can supply recurring exposure. It is reproduced as a finite operational snapshot and must be reverified before attendance. The inference drawn from the snapshot is not that every venue is always open, but that pooled institutional density can be used

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Table 4: Thai Epistemic Landscape Atlas, surfaces 1–19 (operational snapshot, 29 Aug 2026).

No.	Surface	Fit	Function	Cadence
1	Sirindhorn Anthropology Centre (SAC)	Anthropology; culture; ethnicity; marginality; knowledge-from-below; Malay/Deep South	Situated social-science friction; repeat Thai nodes	High
2	SEA Junction (BACC)	Southeast Asia; migration; conflict; culture; human rights	Horizontal regional network; low-entry-cost human friction	High
3	RILCA, Mahidol	Language; culture; intercultural communication; identity; AI ethics; methods	Culture/language/method friction	Very high / online-rich
4	College of Religious Studies, Mahidol	Religion; pluralism; AI/religion; ethics; peace; polarization	Religion theory; pluralism; AI-ethics friction	High
5	Institute of Asian Studies, Chulalongkorn	Asia; Muslim studies; migration; Mekong; China/Japan; geopolitical knowledge regimes	Area-studies and policy network	Medium-high
6	ISIS Thailand, Chulalongkorn	Conflict; geopolitics; ASEAN; war/peace; regional order	Security/geopolitical institutional grammar	Medium-high
7	School of Global Studies, Thammasat	Social innovation; sustainability; human rights; social enterprise	Practice research and public-defence friction	Medium-high
8	College of Interdisciplinary Studies, Thammasat	Human security; sustainable development; gender; culture; policy	Cross-field translation	Medium
9	IHRP, Mahidol	Peace/conflict; human rights; democracy; Deep South; peace education	Peace studies; conflict theory; Deep-South network	Medium; very high fit
10	King Prajadhipok's Institute peace/governance route	Peace policy; governance; conflict management; Deep South	Applied policy-peace friction	Medium / programme-based
11	CRS-IHRP crossover	Religion; rights; pluralism; peace	Cross-grammar religion/rights/peace friction	Event-dependent
12	Philosophy and Religion Society of Thailand (PARST)	Philosophy; religion; ethics; epistemology	Thai philosophy recognition; reply/conference friction	Low frequency; high relevance
13	CSTS/STREC, Chulalongkorn	STS; philosophy of technology; AI/institutions; knowledge society	AI/knowledge and science-society friction	Event-dependent
14	Chulalongkorn Social Research Institute / STS network	Social research; institutions; STS	STS community; policy translation	Event-dependent
15	Institute of Thai Studies, Chulalongkorn	Thai culture; identity; interdisciplinary cultural knowledge	Thai-studies situated correction	Event-dependent
16	PBIC Thai Studies, Thammasat	Thai studies; urban/cultural interpretation	Thai studies; urban/culture friction	Event-dependent
17	Institute of East Asian Studies, Thammasat	East Asia; comparative institutions	East-Asian comparative horizon	Event-dependent
18	C asean	ASEAN; culture; business; sustainability; regional identity	Regional practitioner/business grammar	Event-dependent
19	Mathematics and Computer Science, Chulalongkorn	Mathematics; computation; formal exploration	Formal K2 and field-norm learning	Monitor during active periods

Table 5: Thai Epistemic Landscape Atlas, surfaces 20–38 (operational snapshot, 29 Aug 2026).

No.	Surface	Fit	Function	Cadence
20	Mathematical Association of Thailand + CEPMART	Thai mathematics network	Formal exchange and community routing	Event-dependent
21	Mahidol Physics	Physics and formal research	Physics norm learning	Event-dependent
22	Thai Physics Society / Siam Physics Congress	National physics community	Physics recognition network	Annual/periodic
23	Thailand Center of Excellence in Physics (ThEP)	Advanced physics network	Formal/physics community routing	Event-dependent
24	Chula Physics / Bangkok School on Cosmology	Cosmology; physics	Formal and cosmology entry route	Periodic
25	NARIT + Thai Astronomical Society	Astronomy; science communication	Astronomy/science network	Event-dependent
26	NSTDA / NAC	Science; technology; applied research	National applied-science network	Periodic
27	Social Enterprise Thailand Association	Social enterprise; impact practice	Peer-practice network; SE legitimacy; collaboration	Event-dependent
28	Social Enterprise Promotion Office (OSEP)	SE policy; registration; support/funding ecosystem	Vertical SE legitimacy; policy contact	Policy/programme-based
29	ChangeFusion	Social innovation; impact	Practice friction; impact-innovation network	Event-dependent
30	CU Social Innovation Hub	Social innovation; university ecosystem	Innovation/community bridge	Event-dependent
31	National Innovation Agency (NIA)	Innovation policy; scaling; funding ecosystem	Innovation network; impact scale	Programme-based
32	SDG Move, Thammasat	Sustainability; policy; partnerships	Research translation; SDG/policy route	Project-based
33	SGS/GSSE, Thammasat	Social entrepreneurship; global social innovation	Practice-research and learning network	Programme-based
34	se.school	Social-enterprise learning infrastructure	Terminology/ecosystem literacy; not K2 by itself	On-demand
35	Asia Pacific Futures Network / Sasin	Futures; sustainability; peace; foresight	Theory/strategy/SE/policy bridge	Conference-based
36	ASEAN Studies Center, Chulalongkorn	ASEAN research; regional institutions	ASEAN comparative and institutional route	Event-dependent
37	IAS sub-centres as separate target surfaces	Muslim; migration; Mekong; China; Japan; adjacent area studies	Higher target specificity without new travel cost	Programme-dependent
38	Bangkok Asia/ASEAN friction cluster	IAS; ISIS; SEA Junction; C asean	Multiple institutional grammars around one regional field	Pooled / high availability

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Table 6: Near-term friction opportunities recorded in the 29 August 2026 source snapshot.

Date	Recorded opportunity
30 Aug	SEA Junction book launch / Southeast Asia cultural network
2 Sep	RILCA online Thai Khom script training
3–4 Sep	Asia Pacific Futures Conference, Sasin, Bangkok
4 Sep	SAC cultural/folk-performance activity
5–6 Sep	Mahidol Open House: CRS/RILCA/IHRP ecosystem reconnaissance
8–20 Sep	SEA Junction Javanese scrolls / culture exhibition
12 Sep	RILCA literature-review short course
20 Sep	RILCA research-questions / mindful-communication activities
28 Sep	International Studies Center + Chulalongkorn economic-security seminar
30–31 Oct	IAS/Chulalongkorn + APSA/TSRI planetary-health event
7–12 Dec	Bangkok School on Cosmology, Chulalongkorn Physics
11–12 Dec	PARST Annual Meeting: War and Peace

as ambient infrastructure when events are selected for fit and follow-up rather than attendance count.

I.6 Weekly friction radar and anchor communities

One weekly substantive exposure is selected from the pooled landscape, not from an assumption that any one institution has weekly events. Event choice uses

$$FrictionValue_e = \frac{F_e O_e D_e R_e}{T_e + C_e + P_e + \varepsilon}. \tag{132}$$

The rule is: attend → ask one precise question → make one specific exchange → make one follow-up. Repeat exposure is preferred to event tourism. The initial anchor-community portfolio in the source strategy grouped: SAC; CRS/RILCA Mahidol; IAS/ISIS Chula; IHRP Mahidol; PARST/STS; and social-enterprise/impact clusters, with math/physics added when formal assets are ready.

I.7 Dual-track dashboard

Track global-route activation, Thai-route activation, Thai-companion coverage, repeat Thai communities, stable Thai nodes (substantive exchange across repeated rounds), geographic-audit coverage, local blind spots found, weekly friction exposure, and event-to-follow-up conversion. These are anti-blindness diagnostics, not targets to game.

J Criticality control panel: bounded reactor analogy

The source uses reactor language only as a bounded control metaphor. A diagnostic scholarly multiplication factor is

$$k_t = \nu_t f_{coh,t} L_{g,t} p_{int,t} m_{leg,t} u_{conv,t} P_{NL,t}, \tag{133}$$

where opportunity reproduction, latent coherence, legibility, integrity survival, field moderation/legibility, conversion, and non-leakage are event-defined control factors. $k_t < 1$, ≈ 1, and > 1 are local governance labels, not reactor-physics identities or direct optimization targets.

J.1 Prompt vs delayed channels

Prompt signals include views, downloads, likes, impressions, generic praise, and AI approval. Delayed signals include citations, substantive review, reuse, collaborator requests, invitations from relevant fields, empirical engagement, repeated editor/reviewer contact, and stable correspondents. A delayed-signal share is

$$\beta_D = \frac{\sum_d w_d \widetilde{E}_d}{\sum_d w_d \widetilde{E}_d + \sum_p w_p \widetilde{E}_p + \varepsilon}. \tag{134}$$

Weights are precommitted; prompt attention cannot by itself license a higher mint rate.

J.2 Reflector, moderator, xenon, breeder

Reflector: repeat humans reduce leakage; a local albedo proxy is $\rho_R = N^{ind}_{\le 6m}/N^{cit}_{>6m}$.

Moderator: verification, construct differentiation, and field translation can reduce raw speed while increasing credible capture. Therefore $RawSpeed\downarrow$ does not imply $\mathcal{V}_C\downarrow$.

Xenon: unresolved substantive objections, known citation/metadata errors, and stale public claims are weighted into a ledger. The source used

$$X_t = 1(O_{sub}) + 2(E_{known}) + 3(C_{stale}). \tag{135}$$

A local threshold of 6 freezes expansion; a material integrity incident can trigger immediate SCRAM regardless of score.

Breeder: reusable frameworks, protocols, code, checklists, data, or formal tools count only when independently reused or extended. Internal AI reuse is not independent breeding.

J.3 Fellowship boundary

Repeated friction can become fellowship and fellowship can improve later friction, but

$$Friendship \neq IndependentEvidence, \tag{136}$$

$$Correspondence \neq PeerReview, \tag{137}$$

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$$IntellectualAffinity \neq Truth. \tag{138}$$

Every flagship therefore maintains at least one friction route outside the repeat-fellowship circle; uniform positive readouts from the same human network trigger an external-dissent route.

K Full operational dashboards

The first standalone draft compressed the dashboards too aggressively. The integrated version restores the complete metric families from the source strategy. They are longitudinal self-diagnostics, not cross-scholar rankings and not truth scores (Lahtee, 2026b).

A synthetic internal momentum score may be used only for within-programme comparison:

$$MS = \frac{R\,Coh\,Conv\,Val}{Frag + Debt + COI + \varepsilon}. \tag{139}$$

No cross-scholar comparison is authorized by this expression.

K.1 Criticality dashboard with measurement windows

K.2 Epistemic Fellowship Conversion Loop

Human-network strategy aims to convert one-off exchanges into repeat critics and capability-bearing relationships:

$$\begin{aligned} \text{UnknownScholar} \to \text{OneExchange} \\ \to \text{RepeatCorrespondent} \to \text{TrustedCritic} \\ \to \text{Collaborator/Referrer}. \end{aligned} \tag{140}$$

Let E_1 be one-off substantive exchanges, E_2 repeat exchanges, T_c trusted critics, and C_o substantive collaborations or opportunity referrals. A local conversion diagnostic is

$$FCR = \frac{E_2 + T_c + C_o}{E_1 + \varepsilon}. \tag{141}$$

This is compared only with the same programme over time. Fellowship remains non-validating; a separate dissent route is mandatory when the same human network produces uniformly positive readouts.

L Protection, defeat conditions, and SCRAM

Four protection layers are retained: (1) integrity firewall; (2) priority protection through policy-compatible early timestamping and version lineage; (3) survival buffer against rejection cascades and material shocks; and (4) attention–credit separation.

The main defeat logic is deliberately asymmetric:

- **12 months:** no substantive human engagement after genuine targeted exposure ⇒ audit field fit and concept necessity.
- **24 months:** citation = 0, substantive collaborator = 0, and the same fatal objection independently recurs at least twice ⇒ pivot the programme configuration.
- **36 months:** no peer-reviewed anchor, no repeat network node, and no external uptake ⇒ the current strategic configuration loses.
- $\geq$ 50 **addressed attempts, zero stable correspondent:** human-acquisition protocol loses and must be redesigned.

Immediate SCRAM overrides schedule when there is fabrication, material undisclosed conflict, known serious error left public without correction, human-mastery failure at commitment, claim scope beyond evidence, K_1 represented as K_2/K_3, hidden route dependence/query shopping, or a high-stakes ethical breach.

M Semantic invariants and preservation contract

A manuscript derived from the source architecture fails preservation if it silently removes any of these distinctions:

- Discovery ≠ Justification.
- Practice experience ≠ Population evidence.
- AI exploration ≠ Human commitment.
- Intervention creator ≠ Sole evaluator.
- Claim scope ≤ Evidence scope.
- DVP ≠ K_2.
- K_1 ≠ Certification.
- Attention ≠ Credit.
- Legitimacy ≠ Truth.
- Thai track ≠ Public relations.
- Author positional counter-readout ≠ external K_2.
- Production-supercritical ≠ credit-supercritical.
- No human available ≠ research stop.
- No independent check ⇒ no K_2 claim.
- Goodhart/PRC constraints, SCRAM/defeat conditions, and survival buffer remain active.

Every inherited source section is assigned one of five explicit statuses: `PRESERVE_EXACT` when its decision rule or protocol is materially retained; `PRESERVE_FUNCTION` when its function survives in a generalized or reorganized form; `SUPERSEDE` when a stronger mechanism replaces it while preserving the old function; `DEPRECATE_AUTHOR_SPECIFIC` when a local operating parameter is intentionally excluded from the general theory; and `PRESERVE_DIAGNOSTIC` for dated current-state interventions that must not harden into invariants. No mechanism may disappear without a mapping. The integrated preservation ledger at the end of these appendices maps all 118 numbered source sections to article or appendix functions.

N Status labels for equations and metrics

Unless explicitly backed by estimated data, formulas in the article body and these appendices are governance definitions or declared interpretive bridges. Current author-dashboard values are finite diagnostics. A quantity should move to a fitted/calibrated status only after appropriate longitudinal or comparative data exist. Equations that do not alter a decision rule should be removed or returned to prose.

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Table 7: Momentum and conversion dashboard I.

Metric	Diagnostic question / operational meaning
Flagship concepts	Concepts/problems recognized by relevant scholarly communities
Programme coherence	New assets connect to an identifiable problem family
Peer-reviewed outputs	Outputs receiving external review/certification
Citation/uptake	External citation, reuse, or substantive disagreement
Repeat community exposure	Return to the same scholarly communities
Collaborators	People who add methods/domain capability
External validations	Empirical, methods, ethics, or formal checks
Conference conversion	Conference exposure converted into revision/submission
Opportunities received	Qualified invitations/collaborations arriving from prior work
Epistemic debt	Claims the author cannot independently defend
Fragmentation	Projects not connected to the programme
Corrections	Claims narrowed or corrected when evidence is insufficient
Time-to-citable-asset	Human mastery gate to DOI/preprint
K1 stock	Public provisional assets awaiting deeper conversion
λ_mint	K1 assets per month
Qualified conversion actions	Stock-activation attempts per week
$\Phi(Stock)$	PRC-valid credit realized from pre-existing stock
Stock pressure χ	Mint velocity divided by conversion-action velocity
Coh_latent	Intellectual coherence independent of storefront

Table 8: Momentum and conversion dashboard II.

Metric	Diagnostic question / operational meaning
Programme legibility L_g	One-click programme-path coverage
Activation coverage	Stock assets receiving at least one conversion action
Operating mode	Production / Conversion / Balanced
Global-track coverage	Active global K2 route / conversion-priority assets
Thai-track coverage	Thai node active within six months / relevant assets
Thai companion coverage	Thai-relevant flagship with companion artifact / relevant flagships
Repeat Thai community exposure	Repeated engagement with same Thai communities
Geographic blind spots	GDA corrections found per flagship
Weekly friction exposure	Substantive event/community exposure per week
Event-to-follow-up conversion	Events followed by addressed exchange / events attended
Post-K1 independent-human friction	K1 assets receiving HUM friction after release / K1 assets
Review service	Completed reviews / repeat editor invitations
Outreach conversion	Targeted outreach to reply/critique/collaboration
Disclosure incidents	Venue-policy mismatches or disclosure corrections
PRC-valid credit share	Credit events satisfying provenance/relevance constraints
Delayed-channel share	Citation/invitation/collaboration relative to prompt attention
Reflector albedo	Citable assets receiving independent engagement within six months
Recurrent correspondents	People with substantive exchange in at least two rounds
Recurrent correspondence rate	Repeat correspondents / correspondents who replied

Table 9: Momentum and conversion dashboard III.

Metric	Diagnostic question / operational meaning
Human-friction coverage	Flagship assets passing independent human critique before certification claim
Fellowship continuity	Human nodes still engaging with same programme after at least six months
Broadcast-to-targeted ratio	Human time spent on targeted exchange relative to broadcast
Async circle yield	Revisions/counterexamples/independent routes per round
DVP coverage	Flagship/preprint assets passing DVP / public flagship assets
DVP disagreement yield	Material unresolved disagreements / DVP routes
Frame-break survival	Claims surviving third-person hostile reframing
Recovery rate	Objections whose resolution structure is recovered on a blind route
Query-budget breaches	Unplanned searches after stop rule
Residual-dependence disclosure	K1 assets with Route Dependence Matrix / K1 assets
T_HF	Days from preprint to first independent human critique
Post-preprint human conversion	Preprints receiving independent human engagement within six months
Stable correspondents	Interlocutors with substantive exchange across at least three rounds
50-attempt protocol	Addressed attempts before first stable correspondent
Xenon load	Unresolved objections/errors/stale versions
Breeding ratio	Reusable assets generated/reused per public output

O Integrated anchor-preservation ledger: 118 source sections

This integrated version eliminates the external CSV dependency. Every numbered source section is explicitly accounted for below. The ledger is a traceability device; semantic preservation is additionally checked by the invariant and restored-mechanism lists in these appendices.

1. Current Empirical Baseline - SSRN Snapshot, 29 Aug 2026 — `PRESERVE_FUNCTION`; Article: Introduction; integrated appendix.
2. Research Problem: " " — `PRESERVE_FUNCTION`; Article: Conceptual foundations; Standalone position; Readout/legitimacy; Human mastery.
3. Theoretical Basis: Cumulative Scholarly Advantage — `PRESERVE_FUNCTION`;

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Table 10: Criticality dashboard I: core multiplication and production variables.

Variable	Window	Diagnostic
k_t	rolling 12m	multiplication regime
nu	rolling 12m	opportunity reproduction
f_coh,latent	rolling 12m	underlying programme coherence
L_g	rolling 90d	programme legibility / one-click coverage
p_int	per asset / 12m	integrity survival
m_leg	per asset	field legibility
u_conv	rolling 12m	opportunity conversion
P_NL	assets >=6m	non-leakage / independent engagement
beta_D	rolling 12m	delayed-channel control
rho_R	assets >=6m	network reflector
X_t	live	unresolved poison
BR_t	12–24m	reusable-asset breeding
Time-to-citable-asset	per asset	staging velocity
K1 stock	live	unactivated/under-converted inventory
lambda_mint	rolling 90d	production velocity

Table 11: Criticality dashboard II: conversion, K2 procurement, and toolchain variables.

Variable	Window	Diagnostic
lambda_conv	rolling 90d	qualified conversion-action velocity
chi	rolling 90d	stock pressure
Phi(Stock)	rolling 90d	realized back-catalog credit
Activation coverage	rolling 90d	stock with at least one active conversion route
Time K1 to first K2	per asset	procurement latency
K2 route	per event	source of external friction
K2 cash cost	per event	monetary cost
K2 human hours	per event	preparation/follow-up cost
K2 report depth	per event	mechanism-level depth
K2 relevance	per event	claim-field fit
Desk-reject ratio	rolling by journal	zero-depth submission leakage
Post-review reject yield	per submission	actionable objections despite rejection
Toolchain coverage	per flagship	required functions actually used
GitHub-to-Zenodo integrity	per release	tag/release/DOI/CITATION alignment

Article: Conceptual foundations; Standalone position; Readout/legitimacy; Human mastery.

4. North-Star Objective: Paper Credit-Producing Assets — `PRESERVE_FUNCTION`; Article: Conceptual foundations; Standalone position; Readout/legitimacy; Human mastery.
5. Credit under the Provenance Relevance Constraint (PRC) — `PRESERVE_FUNCTION`; Article: Conceptual foundations; Standalone position; Readout/legitimacy; Human mastery.
6. Formula Governance Rule — `PRESERVE_FUNCTION`; Article: Conceptual foundations; Standalone position; Readout/legitimacy; Human mastery.
7. Legitimacy as the Central Dispute Outside Knowledge Institutions — `PRESERVE_FUNCTION`; Article: Conceptual foundations; Standalone position; Readout/legitimacy; Human mastery.
8. Vertical Horizontal Legitimacy Architecture — `PRESERVE_FUNCTION`; Article: Conceptual foundations; Standalone position; Readout/legitimacy; Human mastery.
9. Readout Universe Reading of Legitimacy — `PRESERVE_FUNCTION`; Article: Conceptual foundations; Standalone position; Readout/legitimacy; Human mastery.
10. Core Identity Architecture: Two Engines + One Bridge — `PRESERVE_FUNCTION`; Article: Conceptual foundations; Standalone position; Readout/legitimacy; Human mastery.
11. Readout Universe as Research-Governance Layer — `PRESERVE_FUNCTION`; Article: Conceptual foundations; Standalone position; Readout/legitimacy; Human mastery.
12. Residual Model: " " — `PRESERVE_FUNCTION`; Article: Conceptual foundations; Standalone position; Readout/legitimacy; Human mastery.
13. Claim-Tier Discipline — `PRESERVE_FUNCTION`; Article: Conceptual foundations; Standalone position; Readout/legitimacy; Human mastery.
14. AI Strategy: Exploration High, Commitment Slow — `PRESERVE_FUNCTION`; Article: Conceptual foundations; Standalone position; Readout/legitimacy; Human mastery.
15. Disclosure & Provenance Strategy — `PRESERVE_FUNCTION`; Article: Conceptual foundations; Standalone position; Readout/legitimacy; Human mastery.
16. No-AI Human Mastery Gate — `PRESERVE_FUNCTION`; Article: Conceptual foundations; Standalone position; Readout/legitimacy; Human mastery.
17. Conceptual Paper Pipeline — `PRESERVE_FUNCTION`; Article: Conceptual foundations; Standalone position; Readout/legitimacy; Human mastery.
18. Conference Strategy: Visibility Credit — `PRESERVE_EXACT`; Article: Protection; integrated appendix.
19. Friction and Fellowship Are Different Problems — `PRESERVE_FUNCTION`; Article: EIC, DVP, Friction/Fellowship; integrated appendix.
20. Formation-Stage Friction: — `PRESERVE_FUNCTION`; Article: EIC, DVP, Friction/Fellowship; integrated appendix.
21. Epistemic Isolation Constraint (EIC) — `PRESERVE_FUNCTION`; Article: EIC, DVP, Friction/Fellowship; integrated appendix.
22. Decorrelated Verification Protocol (DVP) — `PRESERVE_FUNCTION`; Article: EIC, DVP, Friction/Fellowship; integrated appendix.
23. Operator-Decorrelation Control (ODC) — `PRESERVE_FUNCTION`; Article: EIC, DVP, Friction/Fellowship; integrated appendix.
24. Route Dependence Matrix (RDM) — `PRESERVE_FUNCTION`; Article: EIC, DVP, Friction/Fellowship; integrated appendix.
25. Synthetic Formation Packet v2 — `PRESERVE_FUNCTION`; Article: EIC, DVP, Friction/Fellowship; integrated appendix.
26. DVP Is Not Independent Validation — `PRESERVE_FUNCTION`; Article: EIC, DVP, Friction/Fellowship; integrated appendix.
27. Dependence Disclosure Standard — `PRESERVE_FUNCTION`; Article: EIC, DVP, Friction/Fellowship; integrated appendix.
28. Preprint as Human-Friction Generator, Not Human-Gated Reward — `PRESERVE_FUNCTION`; Article: EIC, DVP, Friction/Fellowship; integrated appendix.
29. Switching Rule: Broadcast to Addressed — `PRESERVE_FUNCTION`; Article: EIC, DVP, Friction/Fellowship; integrated appendix.
30. Correspondence Protocol After Preprint — `PRESERVE_FUNCTION`; Article: EIC, DVP, Friction/Fellowship; integrated appendix.
31. Constraint-to-Community Conversion — `PRESERVE_FUNCTION`; Article: EIC, DVP, Friction/Fellowship; integrated appendix.
32. What the Formation Layer Claims - and Does Not Claim — `PRESERVE_FUNCTION`; Article: EIC, DVP, Friction/Fellowship; integrated appendix.
33. AI Sparring Partner Boundary — `PRESERVE_FUNCTION`; Article: EIC, DVP, Friction/Fellowship; integrated appendix.
34. Broadcast-to-Targeted Transition — `PRESERVE_FUNCTION`; Article: EIC, DVP, Friction/Fellowship; integrated appendix.
35. Cold-Start Network Protocol — `PRESERVE_FUNCTION`; Article: EIC, DVP, Friction/Fellowship; integrated appendix.
36. Discipline Routing Layer: Friction Entry Artifact field-specific — `PRESERVE_EXACT`; Article: Two engines/bridge, toolchain; integrated appendix.
37. Entry Artifact + Preprint Coupling — `PRESERVE_EXACT`; Article: Two engines/bridge, toolchain; integrated appendix.
38. The Lived Bridge: When the Researcher Becomes a Case of the Theory — `PRESERVE_EXACT`; Article: Two engines/bridge, toolchain; integrated appendix.

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39. Programme Legibility - Latent Coherence Is Not Enough — PRESERVE_EXACT; Article: Programme legibility, conversion, material constraints; integrated appendix.
40. Recognition Conversion: association stability — PRESERVE_EXACT; Article: Programme legibility, Association Stability; integrated appendix.
41. Publication as a Credit Multiplier — SUPERSEDE; legacy PC_i retained in appendix, function replaced by K2/journal-friction selection with depth, relevance, latency, cash, and human-hour costs.
42. Material Conditions: money, time, and access — PRESERVE_EXACT; integrated material-feasibility and publication-cost gate.
43. Geographic Distribution Audit - AI — PRESERVE_FUNCTION; Article: Dual-track/GDA/PCR; integrated appendix.
44. Positional Counter-Readout — PRESERVE_FUNCTION; Article: Dual-track/GDA/PCR; integrated appendix.
45. Dual-Track Invariant — PRESERVE_FUNCTION; Article: Dual-track/GDA/PCR; integrated appendix.
46. Thai Track Is Not a Lesser Track — PRESERVE_FUNCTION; Article: Dual-track/GDA/PCR; integrated appendix.
47. Thai Epistemic Landscape Atlas - Bangkok / Greater Bangkok / Thailand — PRESERVE_FUNCTION; Article: Dual-track/GDA/PCR; integrated appendix.
48. Weekly Friction Radar - Ambient Infrastructure — PRESERVE_FUNCTION; Article: Dual-track/GDA/PCR; integrated appendix.
49. Current Near-Term Friction Opportunities - Sep Dec 2026 — PRESERVE_FUNCTION; Article: Dual-track/GDA/PCR; integrated appendix.
50. Thai Community Portfolio — PRESERVE_FUNCTION; Article: Dual-track/GDA/PCR; integrated appendix.
51. Dual-Track Dashboard — PRESERVE_FUNCTION; Article: Dual-track/GDA/PCR; integrated appendix.
52. Dual-Track Priority Rule During Conversion Mode — PRESERVE_FUNCTION; Article: Dual-track/GDA/PCR; integrated appendix.
53. Thailand Strategy — PRESERVE_FUNCTION; Article: Dual-track/GDA/PCR; integrated appendix.
54. Global Strategy — PRESERVE_EXACT; Article: blind-name test, $S_G = L \times G \times M \times B \times F$, global route architecture; integrated appendix.
55. Institutional Prestige: use without dependency — PRESERVE_EXACT; Article: DoubleBlind boundary and ManuscriptStrength > IdentityDependence.
56. Positional Capital: epistemic channel — PRESERVE_FUNCTION; Article: Dual-track/GDA/PCR; integrated appendix.
57. ARAYA Governance Firewall — PRESERVE_FUNCTION; Article: Dual-track/GDA/PCR; integrated appendix.
58. Empirical Strategy: "Critical Uncertainty" — PRESERVE_FUNCTION; Article: Dual-track/GDA/PCR; integrated appendix.
59. Production Capacity vs Scholarly-Credit Criticality — PRESERVE_FUNCTION; Article: Conversion-first, velocity, gradient, protection; integrated appendix.
60. Mint Convert Coupling — PRESERVE_FUNCTION; Article: Conversion-first, velocity, gradient, protection; integrated appendix.
61. Credit Stock Equation - Back-Catalog Activated Version — PRESERVE_FUNCTION; Article: Conversion-first, velocity, gradient, protection; integrated appendix.
62. Momentum Equation - Non-Monotonic Growth — PRESERVE_FUNCTION; Article: Conversion-first, velocity, gradient, protection; integrated appendix.
63. How This Strategy Can Lose — PRESERVE_FUNCTION; Article: Conversion-first, velocity, gradient, protection; integrated appendix.
64. Scholarly Credit Velocity — PRESERVE_FUNCTION; Article: Conversion-first, velocity, gradient, protection; integrated appendix.
65. Bottleneck Law for AI-Native Scholarship — PRESERVE_FUNCTION; Article: Conversion-first, velocity, gradient, protection; integrated appendix.
66. Critical Human Time Allocation — PRESERVE_FUNCTION; Article: Conversion-first, velocity, gradient, protection; integrated appendix.
67. Public Work-in-Progress Limit — PRESERVE_FUNCTION; Article: Conversion-first, velocity, gradient, protection; integrated appendix.
68. Concept Cluster Compounding — PRESERVE_FUNCTION; Article: Conversion-first, velocity, gradient, protection; integrated appendix.
69. Gradient Discipline - Optimize the Floor, Maintain the Ceiling — PRESERVE_FUNCTION; Article: Conversion-first, velocity, gradient, protection; integrated appendix.
70. Credit Acceleration - Conversion Before Creation — PRESERVE_FUNCTION; Article: Conversion-first, velocity, gradient, protection; integrated appendix.
71. Opportunity Conversion Rule — PRESERVE_FUNCTION; Article: Conversion-first, velocity, gradient, protection; integrated appendix.
72. Portfolio Allocation — DEPRECATE_AUTHOR_SPECIFIC; 60/30/10 ratio retained only as provenance; transferable selective-pressure function preserved by WIP, programme-fit, bottleneck, and priority controls.
73. Project Priority Equation — PRESERVE_FUNCTION; Article: Conversion-first, velocity, gradient, protection; integrated appendix.
74. Protection Layer I - Integrity Firewall — PRESERVE_FUNCTION; Article: Conversion-first, velocity, gradient, protection; integrated appendix.
75. Protection Layer II - Priority Protection — PRESERVE_FUNCTION; Article: Conversion-first, velocity, gradient, protection; integrated appendix.
76. Protection Layer III - Survival Buffer — PRESERVE_FUNCTION; Article: Conversion-first, velocity, gradient, protection; integrated appendix.
77. Protection Layer IV - Attention Credit Separation — PRESERVE_FUNCTION; Article: Conversion-first, velocity, gradient, protection; integrated appendix.
78. Epistemic Staging Layer - Preprint as Credit Infrastructure — PRESERVE_EXACT; Article: Knowledge states/toolchain; integrated appendix.
79. Four Knowledge States — PRESERVE_EXACT; Article: Knowledge states/toolchain; integrated appendix.
80. SSRN and Zenodo Role Separation — PRESERVE_EXACT; Article: Knowledge states/toolchain; integrated appendix.
81. Open Research Toolchain - epistemic function — PRESERVE_EXACT; Article: Knowledge states/toolchain; integrated appendix.
82. Tool Redundancy Rule — PRESERVE_EXACT; Article: Knowledge states/toolchain; integrated appendix.
83. Preprint Eligibility Gate - DVP Version — PRESERVE_EXACT; Article: Knowledge states/toolchain; integrated appendix.
84. Provisional vs Reviewed Credit — PRESERVE_EXACT; Article: Knowledge states/toolchain; integrated appendix.
85. K2 Procurement - Cost per Unit of Effective Friction — PRESERVE_EXACT; Article: Knowledge states/toolchain; integrated appendix.
86. K2 Route Ladder — PRESERVE_EXACT; Article: Knowledge states/toolchain; integrated appendix.
87. Journal Submission as Latency-Purchased K2 — PRESERVE_EXACT; Article: Knowledge states/toolchain; integrated appendix.
88. Journal K2 Selection Rule — PRESERVE_EXACT; Article: Knowledge states/toolchain; integrated appendix.
89. Preprint-to-K2 Procurement Pipeline — PRESERVE_EXACT; Article: Knowledge states/toolchain; integrated appendix.
90. K2 Relevance Rule — PRESERVE_EXACT; Article: Knowledge states/toolchain; integrated appendix.
91. K2 Procurement Priority — PRESERVE_EXACT; Article: Knowledge states/toolchain; integrated appendix.
92. Credit Velocity with Preprint Acceleration — PRESERVE_EXACT; Article: Knowledge states/toolchain; integrated appendix.
93. DAG: AI-Native Cumulative Scholarly Credit System — PRESERVE_EXACT; Article: Integrated architecture; integrated appendix.
94. Readout-Governed Decision Rule " " — PRESERVE_EXACT; Article: Integrated architecture; integrated appendix.
95. Three-Year Momentum Architecture — PRESERVE_DIAGNOSTIC; integrated appendix.
96. 90-Day Conversion Sprint - Current Operating Plan — PRESERVE_DIAGNOSTIC; integrated appendix.
97. Current Conversion Dashboard - Baseline 29 Aug 2026 — PRESERVE_DIAGNOSTIC; integrated appendix.
98. Dashboard: Momentum Paper — PRESERVE_DIAGNOSTIC; integrated appendix.
99. Five Invariants — PRESERVE_FUNCTION; Article: Integrated architecture; Author note; integrated appendix.
100. Strategic Position Statement — PRESERVE_FUNCTION; Article: Integrated architecture; Author note; integrated appendix.
101. Final Equation - Maximum Credible Credit Velocity After the Conversion Pivot — PRESERVE_FUNCTION; Article: Integrated architecture; Author note; integrated appendix.
102. Conversion-First Governance Rules — PRESERVE_FUNCTION; Article: Integrated architecture; Author note; integrated appendix.
103. Maximum-Acceleration Governance Rules — PRESERVE_FUNCTION; Article: Integrated architecture; Author note; integrated appendix.
104. AI-Native Scholarly Pipeline — PRESERVE_FUNCTION; Article: Integrated architecture; Author note; integrated appendix.
105. Readout Universe Control Layer for Acceleration — PRESERVE_FUNCTION; Article: Integrated architecture; Author note; integrated appendix.
106. Criticality Model - Structural Control Mapping, Not Reactor Physics — PRESERVE_FUNCTION; Article: Criticality/Fellowship; integrated appendix.
107. Prompt vs Delayed Credit Control - Reactivity License — PRESERVE_FUNCTION; Article: Criticality/Fellowship; integrated appendix.
108. Reflector Doctrine - Network as Leakage Control — PRESERVE_FUNCTION; Article: Criticality/Fellowship; integrated appendix.
109. Moderator Principle - Slowing Down Can Increase Capture — PRESERVE_FUNCTION; Article: Criticality/Fellowship; integrated appendix.
110. Xenon Ledger - Accumulated Epistemic Poison — PRESERVE_FUNCTION; Article: Criticality/Fellowship; integrated appendix.
111. Breeder Layer - Reusable Epistemic Infrastructure — PRESERVE_FUNCTION; Article: Criticality/Fellowship; integrated appendix.
112. MIMCG / Certification Boundary — PRESERVE_FUNCTION; Article: Criticality/Fellowship; integrated appendix.
113. Criticality Dashboard — PRESERVE_FUNCTION; Article: Criticality/Fellowship; integrated appendix.
114. Epistemic Fellowship Conversion Loop — PRESERVE_FUNCTION; Article: Criticality/Fellowship; integrated appendix.
115. Fellowship Is Not Validation — PRESERVE_FUNCTION; Article: Criticality/Fellowship; integrated appendix.
116. From Epistemic Isolation to Epistemic Position — PRESERVE_FUNCTION; Article: Criticality/Fellowship; integrated appendix.
117. Final Strategic Position — PRESERVE_FUNCTION; Article: Conclusion.
118. Anchor Preservation Contract - No Silent Deletion — PRESERVE_EXACT; integrated

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appendix + this audit.

P Semantic preservation verdict

The integrated revision restores the mechanisms that were materially compressed in earlier standalone drafts: route-level AI disclosure and the nine-item venue audit; lived-experience and positional-capital boundaries; the embedded-practice discovery/justification and creator/evaluator firewalls; association stability and recognition conversion; material feasibility and publication-cost gates; the global blind-name strength test and identity-dependence boundary; critical-uncertainty test selection; concept-cluster compounding and Credit Leverage; the dated Thai friction-opportunity snapshot; expanded 38-surface atlas metadata; the complete momentum/conversion dashboard; the criticality dashboard with measurement windows; and the Fellowship Conversion Function. The legacy publication multiplier is explicitly superseded rather than silently removed, while the author-specific 60/30/10 portfolio ratio is explicitly deprecated from the generalized theory.

The preservation decision is therefore based on both section traceability and semantic function. The following atomic boundaries remain active throughout the document:

- discovery ≠ justification; practice ≠ population; self-experience ≠ general evidence;
- AI exploration ≠ human commitment; AI contribution ≠ epistemic responsibility;
- DVP ≠ K_2; synthetic plurality ≠ independent validation; mechanical validity ≠ semantic validity;
- source existence ≠ claim support; preprint ≠ certification; attention ≠ credit;
- legitimacy ≠ truth; marginalized origin ≠ truth; institutional rejection ≠ falsity;
- positional access ≠ population authority; community trust ≠ representativeness;
- fellowship ≠ validation; friendship ≠ independent evidence; correspondence ≠ peer review;
- intervention creator ≠ sole evaluator; claim scope ≤ evidence/sampling scope;
- production-supercritical ≠ credit-supercritical; activation action ≠ credit event;
- multi-AI consensus ⇏ geographic completeness; author positional counter-readout ≠ external K_2;
- Thai track ≠ local public relations; Global track ≠ geographically complete by default;
- no human available ≠ research stop; no independent check ⇒ no K_2/certification claim;
- metric growth cannot override Goodhart/PRC controls, human mastery, integrity, ethics, survival, or SCRAM.

Anchor	Yaoharee AI-Native Scholarly Strategy v10.0, 29 Aug 2026
Numbered source sections	118, all included in the integrated ledger
External appendix required	No
External preservation CSV required	No
Restored semantic gaps	Disclosure; lived/positional bridge; association stability; material feasibility; global-strength test; practice firewall; critical test; concept cluster; dashboards; Thai atlas metadata/events; fellowship conversion
Silent deletion permitted	No
Verdict	PASS – no load-bearing conceptual mechanism silently lost; operational parameters are exact, superseded, diagnostic, or explicitly deprecated

Author note on equation status

Unless an equation is explicitly linked to an empirical estimate, equations in this paper are definitions, diagnostics, strategic heuristics, or open propositions. They must not be interpreted as causal coefficients. The governing invariants are preserved in the integrated appendices and remain active in the article body.

Data, materials, and provenance statement

This is a conceptual article. The motivating author-level snapshot is a finite diagnostic rather than a population dataset. All operational protocols, Thai epistemic-landscape metadata, dashboards, evidence-tier caveats, and the 118-section anchor-preservation ledger are incorporated directly into this single standalone manuscript. No external supplement is required to recover the conceptual or operational architecture.

References

Bol, Thijs, Mathijs de Vaan, and Arnout van de Rijt (2018). “The Matthew effect in science funding”. In: *Proceedings of the National Academy of Sciences* 115.19, pp. 4887–4890. DOI: 10.1073/pnas.1719557115.

10.5281/zenodo.22163849

Céspedes, Lucía, Diego Kozlowski, Carolina Pradier, Maxime Holmberg Sainte-Marie, Natsumi Solange Shokida, Pierre Benz, Constance Poitras, Anton Boudreau Ninkov, Saeideh Ebrahimy, Philips Ayeni, Sarra Filali, Bing Li, and Vincent Larivière (2025). "Evaluating the linguistic coverage of OpenAlex: An assessment of metadata accuracy and completeness". In: *Journal of the Association for Information Science and Technology* 76.6, pp. 884–895. DOI: 10.1002/asi.24979.

Chiarelli, Andrea, Rob Johnson, Stephen Pinfield, and Emma Richens (2019). "Preprints and Scholarly Communication: An Exploratory Qualitative Study of Adoption, Practices, Drivers and Barriers". In: *F1000Research* 8, p. 971. DOI: 10.12688/f1000research.19619.2.

Crane, Diana (1972). *Invisible Colleges: Diffusion of Knowledge in Scientific Communities*. University of Chicago Press.

DiPrete, Thomas A. and Gregory M. Eirich (2006). "Cumulative Advantage as a Mechanism for Inequality: A Review of Theoretical and Empirical Developments". In: *Annual Review of Sociology* 32, pp. 271–297. DOI: 10.1146/annurev.soc.32.061604.123127.

Dunbar, Kevin (1995). "How Scientists Really Reason: Scientific Reasoning in Real-World Laboratories". In: *The Nature of Insight*. Ed. by Robert J. Sternberg and Janet E. Davidson. MIT Press, pp. 365–395.

Fire, Michael and Carlos Guestrin (2019). "Over-optimization of academic publishing metrics: observing Goodhart's Law in action". In: *GigaScience* 8.6, giz053. DOI: 10.1093/gigascience/giz053.

Fisher, Greg, Kyle Mayer, and Shad Morris (2021). "From the Editors—Phenomenon-Based Theorizing". In: *Academy of Management Review* 46.4, pp. 631–639. DOI: 10.5465/amr.2021.0320.

Franco Iborra, Sandra, Jessica Polka, and Iratxe Puebla (2022). "Promoting constructive feedback on preprints with the FAST principles". In: *eLife* 11, e78424. DOI: 10.7554/eLife.78424.

Fricker, Miranda (2007). *Epistemic Injustice: Power and the Ethics of Knowing*. Oxford University Press.

Gibbons, Michael, Camille Limoges, Helga Nowotny, Simon Schwartzman, Peter Scott, and Martin Trow (1994). *The New Production of Knowledge: The Dynamics of Science and Research in Contemporary Societies*. SAGE.

Haraway, Donna (1988). "Situated Knowledges: The Science Question in Feminism and the Privilege of Partial Perspective". In: *Feminist Studies* 14.3, pp. 575–599. DOI: 10.2307/3178066.

International Committee of Medical Journal Editors (2026). *Use of Artificial Intelligence in Publishing: Use of AI by Authors*. ICMJE Recommendations. Accessed 29 August 2026. URL: https://www.icmje.org/recommendations/browse/artificial-intelligence/ai-use-by-authors.html.

Jaakkola, Elina (2020). "Designing conceptual articles: four approaches". In: *AMS Review* 10, pp. 18–26. DOI: 10.1007/s13162-020-00161-0.

Lahtee, Yaoharee (2026a). *Readout Universe: A Philosophy and Logic for Grounding Claims*. Working manuscript and public repository. Meta-governance source; versioned repository.

– (2026b). *Yaoharee AI-Native Scholarly Strategy: Conversion-First, Dual-Track and Thai Epistemic Landscape*. Version 10.0, unpublished operational strategy, 29 August 2026.

Longino, Helen E. (2002). *The Fate of Knowledge*. Princeton University Press.

Maddi, Abdelghani, Marion Maisonobe, and Chérifa Boukacem-Zeghmouri (2025). "Geographical and disciplinary coverage of open access journals: OpenAlex, Scopus, and WoS". In: *PLOS ONE* 20.4, e0320347. DOI: 10.1371/journal.pone.0320347.

Merton, Robert K. (1988). "The Matthew Effect in Science, II: Cumulative Advantage and the Symbolism of Intellectual Property". In: *Isis* 79.4, pp. 606–623. DOI: 10.1086/354848.

National Research Council of Thailand (2026). *Research ethics review pathways and Q&A*. URL: https://www.nrct.go.th/listAll/qa.

Panickssery, Arjun, Samuel R. Bowman, and Shi Feng (2024). *LLM Evaluators Recognize and Favor Their Own Generations*. arXiv: 2404.13076.

Petersen, Alexander M. and Orion Penner (2014). "Inequality and cumulative advantage in science careers: a case study of high-impact journals". In: *EPJ Data Science* 3, p. 24. DOI: 10.1140/epjds/s13688-014-0024-y.

Price, Derek J. de Solla and Donald D. Beaver (1966). "Collaboration in an Invisible College". In: *American Psychologist* 21.11, pp. 1011–1018. DOI: 10.1037/h0024051.

Schilke, Oliver and Martin Reimann (2025). "The transparency dilemma: How AI disclosure erodes trust". In: *Organizational Behavior and Human Decision Processes* 188, p. 104405. DOI: 10.1016/j.obhdp.2025.104405.

– (2026). "How Does AI Disclosure Shape Trust? Unpacking the Role of Legitimacy". In: *Social Psychology Quarterly*. OnlineFirst. DOI: 10.1177/01902725261425997.

Sharma, Mrinank, Meg Tong, Tomasz Korbak, David Duvenaud, Amanda Askell, Samuel R. Bowman, et al. (2023). *Towards Understanding Sycophancy in Language Models*. arXiv: 2310.13548.

Skopec, Mark, Hamdi Issa, Julie Reed, and Matthew Harris (2020). "The role of geographic bias in knowledge diffusion: a systematic review and narrative synthesis". In: *Research Integrity and Peer Review* 5, p. 2. DOI: 10.1186/s41073-019-0088-0.

Unceta, Irene, Paula Subías-Beltrán, and Oriol Pujol (2026). "The epistemic debt of generative AI". In: *Nature Machine Intelligence*. DOI: 10.1038/s42256-026-01294-w.

Wang, Jian (2014). "Unpacking the Matthew effect in citations". In: *Journal of Informetrics* 8.2, pp. 329–339. DOI: 10.1016/j.joi.2014.01.006.

Zheng, Mengxue, Lili Miao, Yi Bu, and Vincent Larivière (2025). "Understanding discrepancies in the coverage of OpenAlex: The case of China". In: *Journal of the Association for Information Science and Technology* 76.11, pp. 1591–1601. DOI: 10.1002/asi.70013.

Rigour Without Infrastructure: Three Propositions on Claim-Card Discipline as a Substitute for Institutional Certification

glosa concept paper, v0.2.0 • 5 September 2026 • 10.5281/zenodo.22307841

What the chapter states. “People who live daily with a problem often know it before researchers do, but turning that knowledge into a checkable claim has historically required a university, a lab, or a research team. Generative AI removes that requirement technically while introducing a new one epistemically: it becomes cheap to produce text that reads as rigorous without it being clear which part is observation, which part is inference, and which part the AI supplied.”

Status. “Status K0 working release, not peer reviewed” (as printed under the title block).

Falsifier(s). H1: “a claim card that passes every field the kernel can mechanically check is nonetheless found by a genuine independent reviewer (I5) to have a false answer at Q2 or Q3.” Two further named falsifiers (H2, H3) are stated in the chapter’s Section 5.

Read before. Ch. 1 *Written by AI. Still True.*; Ch. 2 *The Readout Condition*; Ch. 3 *Readout Genesis Standalone Synthesis*; Ch. 13 *From Problem to Hypothesis*; Ch. 27 *When AI Expands Human Potential*; Ch. 39 *The Standalone Scholar*.

Feeds into. —

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Rigour Without Infrastructure

Three Propositions on Claim-Card Discipline as a Substitute for Institutional Certification

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Article type Concept paper (CONCEPTUAL genre, adopted-by-human override, Section 4) • **Method** Claim-card discipline + Literature Review System (LRS) self-application • **Licence** CC BY 4.0
Status K0 working release, not peer reviewed

In brief. Problem. A person with no university, no lab, and no research team who now has AI assistance faces a new failure mode: text that reads as smart and academic while it stays unclear which part is evidence, which part is interpretation, and which part AI smoothed over past what the data supports (Section 1). **Proposition.** Three candidate hypotheses (Section 5) — a design mechanism (H1), a conceptual claim about where rigour lives (H2), and an empirical claim about cross-vendor AI checking (H3) — propose that a claim-card discipline bound to Existence–Attribution–Disclosure and gated by an independence ladder could make a standalone scholar's claim reader-checkable without an institutional credential. **Method.** Each proposition carries a named falsifier and is placed against 48 independently verified citation cards in a conversation-with-the-literature table (Section 6), then the whole method is run on this paper's own construction as a self-application demonstration (Section 7), including an independent review of the three claim cards themselves by three cross-vendor AI routes, which revised them (Section 7). **Status.** All three propositions are carried as untested claims at tier DR, none above knowledge state K0; H3 is explicitly untested, and the self-application run itself surfaced real gate failures, not a clean pass (Section 7, Section 8).

Abstract. People who live daily with a problem often know it before researchers do, but turning that knowledge into a checkable claim has historically required a university, a lab, or a research team. Generative AI removes that requirement technically while introducing a new one epistemically: it becomes cheap to produce text that reads as rigorous without it being clear which part is observation, which part is inference, and which part the AI supplied. This paper states, as three falsifiable propositions and not as demonstrated results, a candidate answer: a *claim card* that separately records what was seen, what the data separates, what AI filled in, what is assumed, and what independent check (if any) the claim has survived — bound to a formal Existence–Attribution–Disclosure norm and passed through a named independence ladder (I0 self-check through I5 outside human review) before public release — could carry the checkable property institutions have historically supplied, without requiring the institution itself (H1, design; H2, conceptual). A third, narrower proposition (H3, empirical) holds, untested, that cross-vendor AI checking would measurably outperform same-vendor re-checking at catching undisclosed AI-fill; no measurement of that difference has been made (n=0). Each proposition is placed against a literature conversation built from 48 independently verified citation cards, and the whole method is applied reflexively to this paper's own construction, which surfaced real failures — a validator warning, a manifest-detected search-mode gap, and a documented citation-verification correction — reported as findings, not smoothed into a clean success story. The three claim cards carrying these propositions were themselves reviewed by three cross-vendor AI routes and revised in response before this draft (Section 7). Every claim here sits at tier DR and knowledge state K0; none has passed an independent (I5) human check.

Positioning. We are not competing with knowledge-authority and make no priority claim; we state what we propose, what we build on, and what would make us wrong.

1 PROBLEM BEFORE OBSERVATION

"Observation" is abstract; it is a verb, not a starting node. Founder's line, English gloss below (Blackbox Log `BBL-2026-09-04-083`, concept DOI `10.5281/zenodo.22302518`):

> *Founder's line (English gloss; verbatim Thai in the Blackbox Log, DOI `10.5281/zenodo.22302518`, entry `BBL-2026-09-04-083`):*
>
> — "the word 'observation' is abstract, it is a verb; if we started from the *problem* instead, is there a process like this? We propose focusing on the problem first."

A problematic state exists in the world *before* any knower acts on it — before it is looked at, named, or measured. The spine this paper's claims are built on (adopted this pass into `design/FOUNDATION_v0.5.md` §2.1a, BBL-2026-09-04-084) therefore does not start with "Reality →" as its first node — the lens never grants direct access to reality, only to what was read out ($R_A = O_A(W;\Pi_A) \neq W$) — it starts one layer finer:

> Problem state (unread) → first human readout (the Blackbox Note's raw line, verbatim, untranslated) → human-language question → [lens-in] → hypothesis (lens-out, signed) → evidence → provisional knowledge (K0, ...)

This paper's own problem state, before it was read out at all: a person who lives daily with a problem,

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has no university appointment, no certified lab, and no standing research team, and now has general-purpose AI assistance. The first human readout of that state (Thai, source of truth), Blackbox Note line 1, English gloss below (`blackbox/BLACKBOX_NOTE_glosa-paper_2026-09-04.md#L1`):

> *Founder's line (English gloss; verbatim Thai in `blackbox/BLACKBOX_NOTE_glosa-paper_2026-09-04.md#L1`):*
>
> — "how can a person with no university, no lab, and no research team produce rigorous knowledge from their own site of practice?"

The contaminated concept this question already flags: "rigour" ordinarily presupposes an institutional mechanism (a lab, a department, a review committee) sitting behind a claim, rather than naming a property a single claim can carry on its own, field by field.

2 STANDPOINT AND OWNERSHIP

This paper is written from the standpoint of a social-enterprise practitioner and citizen (the founder, Yaoharee Lahtee), not a substitute expert in computer science, philosophy of science, library/information science, or research-integrity studies. **Disciplines not claimed:** medicine, religious/fiqh authority, law, engineering, anthropology, political science, and formal computer science beyond what is mechanically executed here. This is disclaimer **D-STANDPOINT / D-NONEXPERT** in Section 8.

2.1 Responsibility per arrow

Every arrow on the spine above is named by who performed it. **Data** $\rightarrow$ **Inference** may be `human`, `ai`, or `joint`; the three project claim cards behind this paper (Section A) record it as `joint` — AI retrieved literature and drafted candidate hypotheses, the founder framed the problem and question. **Inference** $\rightarrow$ **Claim is always human.** AI is never an author. This is not a new rule so much as this pass naming, per arrow, a distinction the repo already held at the whole-artifact level (`AGENTS.md` gate rule 5).

2.2 Ownership: problem, question, and the selection of the hypothesis stay human

Founder's line, English gloss, first stated (`BBL-2026-09-04-086`) and then self-corrected (`BBL-2026-09-04-088`, the correction shown here is the one adopted):

> *Founder's line (English gloss; verbatim Thai in the Blackbox Log, DOI `10.5281/zenodo.22302518`, entry `BBL-2026-09-04-088`):*
>
> — "if the problem is still ours, the question is still ours, and the *selection* of the hypothesis is still ours, and it leads to solving our problem, then use AI, use a cat, use a bear, go ahead — because that is the goal of knowledge: to solve a problem for someone."

Tag	Meaning
TH_COQC	Machine-checked, axiom-free formal result.
FINITE_DIAGNOSTIC	Measured/computed on a finite, retained, reproducible input; not a limit or population claim.
FIT_CALIBRATED	A model calibrated against declared data; accuracy conditional on that fit.
DR	Declared bridge / human narrative synthesis; a design or interpretive judgment, not a proof or measurement.
DEFINITION	A stipulated definition used only where decision-bearing.
OPEN	Not established; a real open question, not smoothed over.

Table 1: Claim-tier legend (readout_genesis + readout_universe ladder). Every propositional claim in this paper (H1, H2, H3, Section 5) is DR; the demonstration (Section 7) is FINITE_DIAGNOSTIC where a command was actually executed.

Three acts on the spine are the human's: owning the **problem**, asking the **question**, and **selecting** the hypothesis. Every other node — retrieval, drafting, analysis, checking, rendering — may be performed by any tool. Concretely for this paper: AI (the AI assistant) drafted candidate hypotheses H1, H2, and H3 (`hypotheses.md`); the founder's own act of selection, recorded in `hypothesis_selection.yaml` and confirmed by ruling `BBL-2026-09-04-095` (Thai verbatim in `entries.jsonl`; gloss: "bring all three together"), chose all three — H1 and H3 are carried as untested propositions with stated falsifiers, and H2 is advanced in the paper's own voice as its central conceptual claim (Section 4, `genre_decision.yaml`).

3 THE LENS AND THE FIVE QUESTIONS

3.1 Claim-tier legend

Every claim in this paper carries exactly one tier tag; tiers are never collapsed or silently upgraded (Table 1).

3.2 The readout-not-truth lens

This paper's problem-to-hypothesis translation was performed under *Readout Universe — Yaoharee Lahtee* (lens; DOI 10.5281/zenodo.21529456, 10.5281/zenodo.21665100; repositories `https://github.com/morrocwi/readout_universe`, `https://github.com/morrocwi/readout_genesis`). Naming the lens credits its source and lets a reader replace it; it does not make the lens, or any hypothesis drafted under it, true (disclaimer **D-EXTERNAL-INPUT**). Everything read under this lens is a finite, retained *readout* of a source, never the source itself: a benchmark number, a citation's exact passage, a claim card's own field, this paper's own prior draft — each was produced by some process, at some resolution, under some method, and is read as that readout, not as truth handed down.

3.3 Existence–Attribution–Disclosure and the five questions

The Readout Condition is $E \wedge A \wedge D$: a claim satisfying Existence (some provenance path for every load-bearing distinc-

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#	Founder's question	Mechanically check-able?
Q1	what was actually seen	presence only
Q2	what the data separates	presence only
Q3	what AI filled in	presence only
Q4	what is assumed	presence only
Q5	what independent evidence/objection	**the one genuinely enforce-able check**

Table 2: Founder's five questions ⇔ E-A-D ⇔ claim-card field group (`design/FOUNDATION_v0.5.md` §3.1). A card that answers Q1/Q2/Q4 honestly but misattributes Q3 — crediting the source with a distinction AI actually supplied — fails the Readout Condition even though every field looks filled; `silent_lift_check` is the mechanized test for exactly this. Thai originals: Blackbox Log `BBL-2026-09-04-083` ff.

tion), Attribution (identification of every non-source node), and Disclosure (every AI-supplied distinction named, not silently absorbed into the claim) is what a claim card is built to record, field by field, against the founder's own five questions (Table 2).

Only Q5 — whether an independent check was ever run, and at what independence class — is structurally enforceable by the kernel; Q1–Q4 are presence-checkable, never correctness-checkable, because *mechanical validity ≠ semantic validity*. This is why the independence ladder (Section 6, Section 7) is the licensed route to closing what the other four questions cannot mechanically close on their own.

4 GENRE AND EVIDENCE STANDARD

This paper is genre-routed as **CONCEPTUAL** by a human, adopted-by-human override of the router's own mechanical output (`genre_decision.yaml`). The router (run with `-search-log`) returned `systematic_review` because a frozen `search_log.yaml` exists under each of `h1/h2/h3`; the founder's ruling (`BBL-2026-09-04-097`, Thai verbatim in `entries.jsonl`; gloss: "make this document an academic article, a concept paper") does not adopt that output — a documented search log supporting three candidate hypotheses is not the same fact as the genre of the claim the paper itself advances, and the router has no way to see that distinction (`genre_decision.yaml` note). H2 (the conceptual, portable-property claim) is advanced in the paper's own voice; H1 (design) and H3 (empirical) are carried as propositions with stated falsifiers, neither run as a study this pass. Evidence standard: every citation is a `status: VERIFIED` card (Section 6); every propositional claim stays DR until an independent test says otherwise.

5 THREE PROPOSITIONS

Candidates were drafted by AI from the lens translation above; the founder selected among them (Section 2). All three are carried as propositions, not results.

H1 — Design: the claim-card mechanism ([C1])

Statement. If every claim a standalone scholar produces is recorded as a claim card answering the five questions, tied to E-A-D, and passed through the independence ladder before any public release, then a person with no institutional affiliation *could, as a design proposition,* co-produce knowledge with AI whose rigour a reader can check for themselves, without an institutional credential serving as the certifying step. This is a proposed mechanism, not an established one: this project's own single self-application (n=1, this paper's own production run, Section 7) is offered as one demonstration readout of the mechanism, not as a general test across standalone scholars.

Falsifier. A claim card that passes every field the kernel can mechanically check is nonetheless found by a genuine independent reviewer (I5) to have a false answer at Q2 or Q3 — showing the mechanical check does not track the epistemic property it claims to track. Not yet run: this project has not secured an I5 (outside human, no stake) route, so this falsifier has not been tested in either direction.

What would count as evidence. A corpus of claim cards run through the independence ladder to at least I3, with a subset reaching genuine I5, compared against a matched corpus produced without the discipline, scored on whether an independent reader can recover the evidence/AI-fill boundary the maker recorded.

H2 — Conceptual: rigour as a portable claim-level property ([C2])

Statement. "Rigour" is not a property that institutional machinery (lab, department, review committee) confers on a claim from the outside; it is proposed here, as a conceptual contrast, to be *at least in part* a property a single claim can carry in its own recorded structure — institutions are one historical mechanism that has produced this property, not shown here to be the only possible one. This does not establish that institutions are non-load-bearing: the cited evidence for it is mixed and mostly same-lineage (Table 4).

Falsifier. A claim card that answers all five questions honestly and completely, checked at I3, is shown to still require an institutional credential — not just an independent human reviewer — to be treated as rigorous by its actual intended readers, for at least one class of reader/decision this project cares about.

What would count as evidence. Reader studies on whether a target audience (journal editors, grant reviewers, a practitioner's own community) treats a well-formed claim card as sufficient without also asking who backs it.

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H3 — Empirical: cross-vendor checking raises AI-fill detection ([C3], untested)

Statement. Hypothesis, not yet tested: when claim cards are checked by a different AI vendor from the one that produced them (cross-vendor, independence level I3) instead of being re-checked by the same vendor that produced the card (I0/I1), the rate at which undisclosed AI-fill (a false Q3 field) or undisclosed assumptions (a false Q4 field) is caught *would rise* measurably compared to same-vendor re-checking. No seeded-error benchmark has been run by this project (n=0 own trials); the single most directly on-point source found the effect direction-dependent, not uniform (Table 5).

Falsifier. A controlled comparison (claim cards seeded with a known number of undisclosed AI-fill/assumption instances, same-vendor vs. cross-vendor routes) shows no significant difference, or shows cross-vendor performs worse.

What would count as evidence. A seeded-error benchmark with a pre-registered scoring rubric; tiered `FIT_CALIBRATED` at best until run, DR until then. **This paper does not run that benchmark; H3 is stated, not tested.**

All three propositions were drafted under the same lens (Section 3); none has an evidence relation above independence class I3, and none of the underlying claim cards has yet been approved by the founder for release (`independent_check.status: PENDING` on all three — checked by cross-vendor routes R1–R3 and revised accordingly, not yet re-checked by a new route or founder-approved, Section A).

6 CONVERSATION WITH THE LITERATURE

Each table below is a *conversation with the problem*, not a chronology or priority ranking (`design/S14_literature-review-system.md`): sources are placed by how they talk to the proposition, never by who came first. Every row resolves to a citation card with `status: VERIFIED` in the corresponding `litreview_manifest.yaml` (`records/lit/rigour-without-infrastructure/h{1,2,3}/`), meaning both `metadata_verified` (Crossref/OpenAlex/arXiv lookup) and `claim_match_verified` (a decorrelated cross-vendor AI route, `route:cross-vendor-ai`, I3) hold. **Relation column** follows the repo's comparison rule (same/different/cited, never a priority claim): *same* = the source's own finding is compatible with the proposition's direction; *different* = the source's own finding counts against it; *cited* = the source bears on the topic without a same/different verdict (orthogonal or undetermined in the underlying dialogue table); rows marked *own lineage, not independent* are the founder's own prior published pieces, collided against world literature per instruction `BBL-2026-09-04-098` and never counted as independent support.

Across all three tables, 20 of 48 rows (42%) are the founder's own prior published pieces, collided against world literature per instruction `BBL-2026-09-04-098` rather than left silent; every own-lineage row is marked *own lineage, not independent* and none is counted toward any of the three propositions' evidential support.

7 THE ARTIFACT APPLIED TO ITSELF

This section reports what the 2026-09-04 self-application run — glosa run on glosa's own paper, per instruction `BBL-2026-09-04-091` — actually recorded, as a readout of that run, not as a claim that the method passed cleanly.

Gate results actually executed

Running `./cli/glosa claim validate` directly against the three project claim cards (`GLOSA-CC-20260904-0201/0202/0203`) this session returns `verdict: PASS_WITH_LIMITS` for all three, with zero schema errors, but with one identical warning on every card: *"D-LENS-UNCITED not checked – validate_claim_card was not given citation_cards (lens_translation.lens_ref is set); pass citation_cards=[. . .] to close this gap."* The validator is honest about its own coverage limit here: a schema-level PASS is not, by itself, a claim that the lens-citation gate closed — it closed only once this session additionally cross-checked the lens DOIs against the citation-card apparatus described in Section 3.

Composite quotes caught by the cross-vendor route

The single strongest gate finding this run: an I3 (decorrelated cross-vendor) check on an earlier citation pass caught *composite quotes* — citations assembled from memory rather than a locator taken from the opened source. The founder's ruling in direct response, English gloss (`BBL-2026-09-04-100`):

> *Founder's line (English gloss; verbatim Thai in the Blackbox Log, DOI `10.5281/zenodo.22302518`, entry `BBL-2026-09-04-100`):*
>
> — "should we make it a hard rule: if a source is used it must carry a page and line locator plus a full link, never written from memory — that reduces the problem at the source, before the check."

Adopted immediately (`BBL-2026-09-04-101`, Thai verbatim in `entries.jsonl`; gloss: "update it into the system"), this became the rule under which every one of the 48 citation cards behind Section 6 was subsequently built: each carries `page_or_locator` and `fetched_from_url`, and each `exact_passage` is a source-first quotation, not a recalled paraphrase. This is a real, gate-caught failure that changed the method's own rule (rule 17, `design/FOUNDATION_v0.5.md` §7.8), reported here because it happened, not because it flatters the method.

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Citation card	Source	Relation	Bearing
cite-cstp-boundary-work-001	Mayes 2022, Citizen Science and Scientific Authority	different	Moderate, scoped: credentialed-expert legitimacy-shaping work continues to matter in practice — cross-vendor review narrowed this to bear only on credentialed legitimacy/acceptance pressure, not to be read as evidence the full claim-card + E-A-D + independence-ladder mechanism was tested and found wanting.
cite-disclosure-not-documentation-001	Cwik 2026, Disclosure is not documentation	same	Weak, component-only: supports the H1 component that AI-assisted work needs structured documentation beyond disclosure — cross-vendor review downgraded this from moderate to weak and scoped it to `SUPPORTS_COMPONENT_ONLY`, not evidence the combined mechanism was tested.
cite-llm-hallucination-survey-001	LLM hallucination survey, arXiv:2510.06265	cited	Shows AI cannot be trusted to flag its own errors — exactly why H1 requires an external check rather than AI self-report.
cite-ocsdnet-honf-001	OCSDNET/HONF, Open Hardware and Citizen Science (Indonesia, Thailand, Nepal)	same	Grounds H1's premise that non-institutional knowledge production is a live regional practice, not a hypothetical.
cite-own-blackbox-log-h1-006	Blackbox Log (Lahtee, Zenodo)	own lineage, not independent	Documents the append-only raw-record practice H1 assumes is followed; no claim about rigour by itself.
cite-own-readout-condition-h1-002	The Readout Condition (Lahtee)	own lineage, not independent	Supplies the E-A-D norm H1's claim card operationalizes; does not itself specify the card's fields.
cite-own-readout-genesis-h1-003	Readout Genesis Standalone Synthesis (Lahtee)	own lineage, not independent	Supplies the independence-ladder vocabulary H1 tracks; does not itself define a claim-card schema.
cite-own-standalone-scholar-h1-005	The Standalone Scholar (Lahtee)	own lineage, not independent	Names the dual-track conversion (AI-assisted formation → independent human friction) H1's card is scoped inside, not a replacement for.
cite-own-written-ai-h1-001	Written by AI. Still True. (Lahtee)	own lineage, not independent	Names "knower fetishism"; H1's card targets exactly the constitution-level property this paper argues authority derives from.
cite-own-zero-readout-h1-004	What a Zero Readout Certifies (Lahtee)	own lineage, not independent	Illustrates machine-checked discipline; H1's own mechanical checks (schema validation) are markedly weaker.
cite-pgph-citizen-science-sea-001	Kumar et al. 2025, Community perceptions of citizen science (South/SEA)	cited	Governance-inclusion framing, orthogonal to H1's rigour-checkability framing.
cite-rsos-citizen-epistemology-001	Jaeger et al. 2023, An epistemology for democratic citizen science	same	Weak, general-context only: robust knowledge can arise through plural/citizen epistemology outside narrow institutional credentialing — cross-vendor review confirmed this as `SUPPORTS_GENERAL_CONTEXT` background support, not upgraded to a test of H1's mechanism.
cite-rsos-preprint-credibility-001	Soderberg, Errington & Nosek 2020, Credibility of preprints	same	Researchers rate content/process cues over prestige when prestige is absent — supports the premise, not H1's specific design.
cite-scholarly-kitchen-credentialism-001	Scholarly Kitchen 2026, The Guild and the Gap	cited	A real rejection episode on prior-publication grounds; whether a claim card would change the outcome is untested.
cite-tcij-freelance-th-001	TCIJ (TH) 2020, freelance-academic labour piece	cited	Adjacent labour/credit question, thinner on H1's actual rigour-verification topic than its search match suggested.

Table 3: H1 (design proposition) against 15 verified citation cards, 6 of them own-lineage. Diversity audit (`h1/litreview_manifest.yaml`): 1 Thai / 5 English external sources (own-lineage excluded), stance counts *agrees*:4 *orthogonal*:2 *undetermined*:9 by the manifest's own collapsing convention (Section 7 notes a discrepancy between this convention and the citation cards' own explicit notes). `concentration_flags: []`; `zero_disagree_flag: false`.

A manifest-detected search-mode gap

The genre router's own mechanical output (Section 4) returned `systematic_review` because `context.has_search_log` was true — but every one of `h1/h2/h3`'s own `litreview_manifest.yaml` records `search_mode` as `TARGETED_SEARCH / SCOPING_SEARCH`, never `SYSTEMATIC_REVIEW` — the manifest, read directly, is what let the founder catch that the router's mechanical genre suggestion did not match the actual evidentiary standard the literature runs had met, and overrule it (Section 4). This is the manifest doing exactly the job `design/FOUNDATION_v0.5.md` §7.8's **FC-S8-1** hard block exists for: calling anything less than a full SR protocol a "systematic review" is blocked, here by the human reading the manifest against the router's own claim.

A stance-counting discrepancy this session found and did not paper over

Building Tables 3 and 5 directly from each citation card's own `notes` field ("Dialogue-table stance: NO (disagrees with / challenges H1)" on `cite-cstp-boundary-work-001`) surfaced a discrepancy with the corresponding manifest's own `diversity_audit.counts.by_stance`: the manifests for H1 and H3 both record `disagrees: 0`, while this session confirmed directly (grep against the raw dialogue-table rows, FINITE_DIAGNOSTIC) that at least one row per hypothesis (Mayes 2022 against H1; Xiang et al. 2026 against H3) is an explicit, single-value challenge row that a card's own notes field independently labels a disagreement. The manifest's aggregate counting convention appears to collapse a dual-value "NO/NO" dialogue-table cell into `undetermined` rather than `disagrees`, undercounting real challenges in its own summary statistic by at least one row per hypothesis where this occurs. This paper's own tables (Tables 3 and 5) use the citation card's explicit `notes` field, not the manifest's aggregate count, and this discrepancy is disclosed rather than silently reconciled — a small instance of exactly the "mechanical check does not track the epistemic property it claims to track" failure mode H1's own falsifier names, found here in the tool that mechanically summarizes H1's own supporting literature.

Path leaks caught by CI, not by this run

The repository's public-facing leak gate (`scripts/check_leak.sh`, wired into `.github/workflows/ci.yml`'s `leak-scan` job) is a

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Citation card	Source	Relation	Bearing
cite-rigour-without-infrastructure-001	Wikipedia, Open-notebook science	cited	Treats openness as a property of record-keeping practice, close to but coarser than H2's own five-field structure.
cite-rigour-without-infrastructure-002	Rasmussen 2019, Confronting Research Misconduct in Citizen Science	cited	Calls for institution-like misconduct-detection machinery — cuts against H2's claim that the property needs no such mechanism.
cite-rigour-without-infrastructure-003	Freitag, Meyer & Whiteman 2016, credibility-building strategies	same	Credibility built bottom-up through concrete recorded practices, not top-down certification.
cite-rigour-without-infrastructure-004	Scholarly Kitchen 2026, The Guild and the Gap (+ comments)	different	Close to H2's own falsifier: a named commenter states affiliation/publication history earns *more* scrutiny than any single record could substitute for.
cite-rigour-without-infrastructure-005	Dutta et al. 2021, Decolonizing Open Science	cited	Structural/platform-governance framing, adjacent to but different from H2's single-claim scope.
cite-rigour-without-infrastructure-006	Okafor et al. 2022, Institutionalizing Open Science in Africa	cited	Aims for *more* institutional machinery as the path forward — the opposite emphasis from H2, at a different level.
cite-rigour-without-infrastructure-007	Kumar et al. 2025, community perceptions of citizen science	cited	Access to institutional audiences, not record quality, is the bottleneck this source finds — adjacent friction H2 does not resolve.
cite-rigour-without-infrastructure-008	Srirot & Sohsomboon 2025 (TH), qualitative Trustworthiness framework	same	Record-level techniques (audit trail, triangulation) let a reader judge rigour without institutional standing — the closest structural match to H2 found.
cite-rigour-without-infrastructure-009	THE STANDARD (TH) 2025, Thai Open Science policy piece	cited	Institution-led opening of an existing record — the institution still decides what counts as legitimate, an access question not a rigour-provenance one.
cite-own-blackbox-log-h2-006	Blackbox Log (Lahtee)	own lineage, not independent	No claim about whether rigour is portable at claim level.
cite-own-civilization-of-knowledge-h2-007	The Civilization of Knowledge (Lahtee)	own lineage, not independent	Locates the stabilizing step historically in institutions — would ask whether a claim card is a new non-institutional stabilizer or a mistaken substitute.
cite-own-readout-condition-h2-002	The Readout Condition (Lahtee)	own lineage, not independent	Weak, indirect: supports source-relative warrant architecture, not explicitly "rigour" or institutional non-conferment — cross-vendor review downgraded this from moderate/direct to weak/indirect support; own lineage, not independent confirmation.
cite-own-readout-genesis-h2-003	Readout Genesis Standalone Synthesis (Lahtee)	own lineage, not independent	Institutions are one belief-scale among several, not a precondition for knowledge status.
cite-own-standalone-scholar-h2-005	The Standalone Scholar (Lahtee)	own lineage, not independent	Independent human friction is one specific historical route to stress-tested knowledge, not necessarily the only one — would ask if H2 still needs some friction step.
cite-own-written-ai-h2-001	Written by AI. Still True. (Lahtee)	own lineage, not independent	Institutional authority is derivative, not a source, of epistemic value — the same possession/constitution distinction H2 draws.
cite-own-zero-readout-h2-004	What a Zero Readout Certifies (Lahtee)	own lineage, not independent	Illustrates zero-institutional-apparatus rigour in a formal setting — weak indirect support only.

Table 4: H2 (conceptual proposition, this paper's own advanced claim) against 16 verified citation cards, 7 own-lineage. Diversity audit (`h2/litreview_manifest.yaml`): 3 Thai / 13 English, stance *agrees*:6 *disagrees*:2 *orthogonal*:7 *undetermined*:1. `concentration_flags: []`; `zero_disagree_flag: false` — this is the one proposition of the three whose manifest itself records a nonzero disagree count.

repo-level, not a paper-level, control: it did not need to run specifically against this paper's files this session (none of this paper's content matched a denylisted pattern), but it is the mechanical backstop this paper's own public-facing disclosures rely on, and its presence in CI — rather than only as a manual pre-publish step — is itself part of what this paper's demonstration reports: a leak gate that runs on every push, not only when someone remembers to ask for it.

Independent review of this paper's own claims

The three project claim cards behind this paper (Section A) were themselves put through the independence ladder before this draft was finalized: three cross-vendor AI review routes (R1 Falsifier, R2 HostileReviewer, R3 SourceAuditor; independence class I3), producing nine review reports in total (three routes × three cards, `projects/GLS-2026-001_rigour-without-infrastructure/reviews/*.review_report.json`), read each card independently of the session that drafted it. All nine converged on one defect class: statements written as results when the evidence behind them was one self-application run plus same-lineage or adjacent-domain citations, not a direct test of the mechanism stated. Concretely: H1's "can co-produce knowledge" was rewritten to "could, as a design proposition," with the paper's own n=1 self-application named explicitly as one demonstration readout, not a general test; H2's categorical "rigour is not conferred by institutions" was rewritten to a hedged conceptual proposition ("proposed here … at least in part"), with every same-lineage supporting citation now carrying an explicit not-independent note; H3's asserted "rises measurably" was rewritten to an untested hypothesis, "would rise … no measurement has been made, n=0." Every individual finding from all three routes is logged as a resolved dissent record on its originating claim card (`five_questions.tested.dissent_records`), not silently absorbed into a cleaner-looking rewrite. As of this draft, each card's `independent_check.status` is **PENDING**: checked by cross-vendor routes and revised accordingly, but not yet re-checked by a further independent route, and not yet approved by the founder for release (Section A). This section is offered as a working example of the independence ladder catching over-scope in this paper's own output — one instance, at I3, on one paper — not as proof that the ladder works in general; that broader claim would itself need the same discipline (H1's own falsifier, Section 5) before it could be asserted.

8 LIMITATIONS AND DISCLAIMERS

D-STANDPOINT

See Section 2. This paper is written from a social-enterprise-practitioner/citizen standpoint, not disciplinary authority; a reader who skips to this section should read

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Citation card	Source	Relation	Bearing
cite-selfpreference-judge-001	Wataoka, Takahashi & Ri 2024/25, Self-Preference Bias in LLM-as-a-Judge	cited	Weak, context-only: finds familiarity/self-preference bias, a mechanism relevant to why same-vendor re-checking may under-detect, but does not itself run a cross-vendor-vs-same-vendor comparison — cross-vendor review downgraded this from moderate to weak, scope narrowed to `CONTEXT_ONLY/SUPPORTS_GENERAL_MECHANISM`.
cite-crossmodel-codereview-002	Xiang et al. 2026, Cross-Model LLM Code Review, arXiv:2607.21656	different	The one source that actually runs H3's own comparison: the effect is asymmetric and direction-dependent (one direction helps a lot, the reverse hurts), and same-vendor re-review also improved one condition.
cite-toohconsistent-003	Tan et al. 2025, Too Consistent to Detect (EMNLP)	same	Weak, context-only: same-model repeated sampling cannot catch "self-consistent" errors, relevant to but not a direct test of H3's cross-vendor mechanism — cross-vendor review downgraded this from moderate to weak, scope narrowed to `CONTEXT_ONLY/SUPPORTS_GENERAL_MECHANISM`.
cite-multiagent-debate-004	Du et al. 2023, Multiagent Debate	cited	Multi-instance debate improves factuality without varying vendor identity — the active ingredient could be plurality, not cross-vendor difference.
cite-selfcorrect-reasoning-005	Huang et al. 2023/24, LLMs Cannot Self-Correct Reasoning Yet	cited	Tests zero-external-signal correction, a weaker condition than a same-vendor re-check with a fresh adversarial prompt — an imperfect analogy, flagged as such.
cite-legalai-india-006	John, Singh & Panachakel 2026, Can Legal AI Know When It Is Wrong?	cited	Three vendors' error rates differ nearly 5x on identical inputs, consistent with H3's premise, but the paper's own lever for improved catching is human experience, not cross-vendor AI review.
cite-bangladesh-readiness-007	Sultana et al. 2026, Bangladesh AI Readiness	cited	Names a capacity defeater of H3's practical reach (infrastructure to run any independent review at all), not of H3's empirical claim.
cite-iseas-sea-governance-008	Fong 2026, AI Governance Policies in Southeast Asia	cited	Institutional-capacity constraint on deploying cross-vendor review as policy, orthogonal to H3's mechanism claim.
cite-cofact-thai-detector-009	Cofact Thailand 2025, AI-image-detection reliability	cited	Adjacent domain (detecting whether content is AI-made, not whether AI-authored claims are honest) — genuinely different problem.
cite-mindstudio-crossvendor-010	MindStudio blog 2026, Cross-Vendor AI Agent Review	same	Weak, context-only: states H3's own mechanism in practitioner language, but with no measurement, sample, or method of its own — cross-vendor review scoped this `CONTEXT_ONLY/SUPPORTS_GENERAL_MECHANISM`, flagged as concentration risk, not equivalent-weight evidence.
cite-own-blackbox-log-h3-006	Blackbox Log (Lahtee)	own lineage, not independent	No claim about detection rates.
cite-own-knowledge-stabilized-translation-h3-007	Knowledge as Stabilized Translation (Lahtee)	own lineage, not independent	Stability under variation is the marker of knowing well — conceptually the same move H3 makes; does not specify vendor identity as the variation dimension.
cite-own-readout-condition-h3-002	The Readout Condition (Lahtee)	own lineage, not independent	Disclosure is structural, not optional; does not specify the auditor must be a different AI vendor.
cite-own-readout-genesis-h3-003	Readout Genesis Standalone Synthesis (Lahtee)	own lineage, not independent	Does not separate same-vendor from cross-vendor checking as a variable.
cite-own-standalone-scholar-h3-005	The Standalone Scholar (Lahtee)	own lineage, not independent	Independent validation in this paper's stronger sense requires human friction — cross-vendor AI (I3) is a weaker approximation.
cite-own-written-ai-h3-001	Written by AI. Still True. (Lahtee)	own lineage, not independent	Reinforces that a checker's identity/route matters to what its check can license — no detection-rate comparison.
cite-own-zero-readout-h3-004	What a Zero Readout Certifies (Lahtee)	own lineage, not independent	Machine-checked verification, where available, is a strictly stronger independence route than cross-vendor AI review.

Table 5: H3 (empirical proposition, explicitly untested) against 17 verified citation cards, 7 own-lineage. Diversity audit (`h3/litreview_manifest.yaml`): 1 Thai / 6 English, stance *agrees*:3 *orthogonal*:4 *undetermined*:10 by the manifest's own convention. `concentration_flags: []`; `zero_disagree_flag: false`.

every claim above as offered from that position and access level, never from institutional credential.

D-NONEXPERT
The author is not a substitute for a credentialed expert in citizen-science studies, philosophy of science, library/information science, or empirical AI-evaluation methodology — the disciplines this paper draws on without claiming standing in. Medicine, religious/fiqh authority, law, engineering, anthropology, political science, and formal computer science beyond what is mechanically executed here are explicitly not claimed.

D-SCOPE
Not applicable in the empirical-population sense (this is a CONCEPTUAL genre paper, Section 4). Where a source's own sample size or population matters to a comparison row (Tables 3 to 5), it is stated inline in that row rather than in one blanket population statement.

D-NONCLAIM
This paper does *not* claim: that H1's mechanism has been run as a study; that H2's conceptual-portability claim has been tested against real readers; that H3's seeded-error benchmark has been built or executed; that the 48 citation cards behind Tables 3 to 5 constitute a systematic or exhaustive review (`search_mode: TARGETED_SEARCH / SCOPING_SEARCH`, never `SYSTEMATIC_REVIEW`); that any finding here is a medical, legal, or religious determination; or that cross-vendor AI checking (I3) is equivalent to, or a substitute for, an I5 human review.

D-AIFILL
See Section 9. AI (the AI assistant, "the AI assistant" seat) drafted candidate hypotheses H1/H2/H3, the literature-review runs, the claim cards' `current_evidence`, `retrieved_tool_evidence`, `retained_record_route`, `model_calibration_assumption`, `prompt_system_constraint`, and `decision_policy` fields, and this paper's own prose and LaTeX build. None of this is treated as evidence on its own say-so (Section 9).

D-TIER
The single highest tier claimed anywhere in this paper is DR (declared bridge / human narrative synthesis over a specified mechanism), with one FINITE_DIAGNOSTIC

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exception: the mechanically executed checks reported in Section 7 (schema validation, disclaimer emission, manifest gate reads) and this paper's own LaTeX compile. It cannot go higher because none of H1/H2/H3 has been run as a study, and no claim card behind this paper has passed a class-5 (I5) independent human check.

D-INDEPENDENCE
Every citation card behind Tables 3 to 5 was checked at independence class *I*3 (decorrelated cross-vendor AI route, `claim_match_verified_by: route:cross-vendor-ai`) for `claim_match`, and mechanically (*I*3-adjacent, Crossref/OpenAlex lookup) for `metadata_verified`. The three project claim cards themselves (Section A) have additionally been read by three cross-vendor AI review routes at *I*3 (Section 7) and revised in response, and now sit at `independent_check.status: PENDING` — checked and revised, but not yet re-checked by a route independent of that review pass, and not yet approved by the founder for release. No external or institutional validation is sought or treated as available as a lever that would make any claim here more true; an outside reader's input, if it comes, is optional-level evidence at whatever independence class it actually reaches, never a certification gate.

D-DVP-NOT-K2
A decorrelated-vendor AI review pass (DVP) is at most an *I*3 route. **DVP $\neq$ K2**: no combination of AI-only checks, however many vendors, raises this paper's claims to K2. This paper's K-state is `K0` (Section 8.1); no claim in it is asserted above K0/DR.

D-LEGAL-NEQ-EPISTEMIC
None identified — no license, permit, or regulatory approval is referenced by this paper's claims. The general rule still holds: were one ever cited, it would make an action lawful, never raise epistemic warrant or certify a claim as true.

D-NOT-DIAGNOSTIC
Not applicable — this paper makes no health, clinical, or treatment recommendation of any kind.

D-TRANSLATION
Not applicable: per founder ruling (BBL-2026-09-04-099, Section B), this concept paper is published in English only, with no Thai edition. Every founder line quoted in Section B nonetheless stays Thai, verbatim, untranslated in place, with an English gloss alongside it — the English-only rule governs the paper's own prose, not the Blackbox Note's raw lines. Within the underlying claim cards, the Thai statement fields themselves carry `translation_status: machine` (machine translation, not human-reviewed).

D-DISSENT-PRESERVED
Nine dissent records are logged across the three claim cards (three cross-vendor routes $\times$ three cards, Section 7), each with a `resolved: true` entry naming what was accepted and how the card changed — none is deleted or silently folded into a cleaner rewrite; every one remains readable in the card's own `five_questions.tested.dissent_records` field. The literature-review dialogue tables (Tables 3 and 4) separately carry the closest real dissent this project has found against its own hypotheses (the Mayes 2022 boundary-work finding against H1 [cite-cstp-boundary-work-001]; the Scholarly Kitchen credentialism case against H2 [cite-rigour-without-infrastructure-004]); these are stated as open challenges in Section 6, not resolved or smoothed over.

D-COMPARISON
This paper does not claim priority or precedence. Section 6 states every source relation as *agrees*, *disagrees*, or *orthogonal/undetermined* — never as a priority contest, never as "no other work does X."

D-CITATION-UNVERIFIED
None identified for the 48 citation cards cited in Tables 3 to 5 — every one carries `status: VERIFIED` (`metadata_verified: true`, `claim_match_verified: true`) in its litreview manifest as of 2026-09-04; none is cited from memory (Section 7, BBL-2026-09-04-100/101). Where a source's own full text was not fetched (e.g. `fetch_status: PAYWALLED_ABSTRACT_ONLY`), the citation card's `scope` field names the resulting evidentiary weakness (`PARAPHRASE` rather than `DIRECT_QUOTATION`) inline in the relevant table row rather than silently upgrading it.

D-BLACKBOX-NOTE
This paper carries the mandatory Blackbox Note appendix (Section B). Note id: `BB-2026-09-04-01` (`projects/GLS-2026-001_rigour-without-infrastructure/bl` curated public source: `blackbox/BLACKBOX_NOTE_glosa-paper_2026-09-04.md`; public log entries `BBL-2026-09-04-083, -086, -088, -089, -091, -095, -097, -098, -099, -100, -101` (`blackbox/log/entries.jsonl`, concept DOI `10.5281/zenodo.22302518`).

D-K-STATE
Overall K-state: `K0` (internal working note). Every load-bearing claim in this paper — H1, H2, H3, and every finding in Section 7 — shares this same K0 state; there is no per-claim exception. This matches the three underlying project claim cards' own `k_state: K0` (Section A).

D-AUTHORSHIP
See Section 9. The problem, the research question, and the selection among H1/H2/H3 are the founder's own act (Section 2); AI drafted every candidate, citation search, and this paper's prose, disclosed per claim (`produced_by`, Table 6), and never substitutes for the author's judgment or responsibility (**AIContribution $\neq$ EpistemicResponsibility**).

D-REVISION-LIVE
The three underlying claim cards (`GLOSA-CC-20260904-0201/0202/0203`) are at revision 3 of an actively edited project, the second revision having been driven by the nine cross-vendor review findings described in Section 7; `status: Draft` and `independent_check.status: PENDING` on all three

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reflect that the founder has not yet signed off on any specific card instance, even though the founder has already ruled on genre (BBL-097) and hypothesis selection (BBL-095) at the project level.

D-NO-EPISTEMIC-VETO
No agency — including any AI vendor, reviewer, or institution named in this paper — holds an epistemic veto over H1, H2, or H3 merely by title, credential, or infrastructure ownership; any agency may propose, use, test, challenge, fork, translate, revise, restrict, or withdraw this work without needing permission, provided lineage is preserved (per the repo's own knowledge-validation stance, `AGENTS.md` (identical in the per-vendor gate files)).

D-CANDIDATE-STATUS
H1, H2, and H3 are candidate hypotheses drafted by AI; the act of selecting among them (all three, per BBL-2026-09-04-095) was the founder's alone (Section 2). No claim card here asserts candidate status was itself a form of testing.

D-EXTERNAL-INPUT
The lens this paper's readout-translation and hypothesis-drafting steps were performed under is named, cited, and linked throughout (Section 3); naming it credits its source and lets a reader replace it — it does not make the lens, or any hypothesis drafted under it, true.

8.1 Knowledge state

This paper's overall knowledge state is **K0** (internal working note): every load-bearing claim in it, including Section 7's findings, shares this K0 state. It is not K1 (public-provisional with a named independence class per claim and no class-5 human check yet) because the three underlying claim cards' own `status` field is still `Draft` and `independent_check.status: PENDING` (checked by three cross-vendor routes and revised, not yet re-checked by a further route or founder-approved, Section 7) — the founder has ruled on genre and selection at the project level but has not yet signed off on any specific card instance (Section A).

9 AI-ASSISTANCE DISCLOSURE

This paper is a human–AI co-production, not AI-only automation and not human-only authorship. A generative-AI assistant (vendor recorded in the local cooking log, not named here per gate rule 9) assisted with literature retrieval and card drafting, the three claim cards' field content, the conversation-with-the-literature tables, and this LaTeX build. **AI output is not treated as evidence.** Every claim above is accepted only when it matches a retained source record (a citation card's `exact_passage`, the Blackbox Log's own entries, a mechanically executed command's own output) or a machine-checkable result reported in Section 7. The human holds the standpoint, the falsifier judgment, verification of load-bearing citations at the founder's own discretion, and the final public commitment (**AIContribution $\neq$ EpistemicResponsibility**).

10 WHAT WOULD CHANGE OUR MIND

Per hypothesis, stated plainly rather than left implicit:

- **H1** would be abandoned if a genuine I5 reviewer, working from a schema-passing claim card, found a false answer at Q2 or Q3 — showing the mechanical check gives false reassurance rather than tracking the property it names. It would be strengthened, not proven, by a corpus comparison at I3+ (with a subset at I5) against a matched non-card corpus.
- **H2** would be abandoned if a reader study found that credentialed audiences (editors, grant reviewers, a practitioner's own community) still required an institutional signal even after being shown a complete, well-formed claim card — i.e. the credential doing load-bearing work the card's fields could not substitute for. The Scholarly Kitchen credentialism case (`cite-rigour-without-infrastructure-004`) is already the closest real instance of exactly this challenge and is not explained away.
- **H3** is untested; the one source that actually ran its comparison (`cite-crossmodel-codereview-002`) found a direction-dependent, asymmetric effect rather than a uniform cross-vendor-beats-same-vendor pattern. A seeded-error benchmark that reproduces that asymmetry at the claim-card Q3/Q4 grain would be evidence against H3 as a general, direction-agnostic claim; a benchmark that finds a uniform, direction-independent rise would be evidence for it.

11 CONCLUSION

This paper states three falsifiable propositions about whether a claim-card discipline can substitute for institutional infrastructure in making a standalone scholar's claim reader-checkable, places each against verified literature rather than memory or silence, and then runs the same discipline on its own construction — finding, honestly, a validator warning, a caught set of composite quotes that changed the method's own citation rule mid-project, a router/manifest mismatch the founder had to catch by hand, and a small stance-counting discrepancy in the project's own summary statistics. None of this is smoothed into a clean success story: every claim here stays DR, every underlying card stays K0 and unsigned, and H3 stays explicitly untested. What is offered is not a demonstrated result but a specified mechanism, an honest account of what happened when it was pointed at itself, and a named falsifier for each of the three ways it could be shown wrong.

A CLAIM MATRIX

Per Chair Ruling v1 §B2, this is a print-space compression of the three full claim cards (`projects/GLS-2026-001_rigour-without-infrastructure/clai`
field-presence here is mechanically checkable, correctness is what the independent check (Q5) exists for.

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ID	Informal statement	Evidence / support	Tier	Produced by	Responsible
C1	H1: if every claim is recorded as a claim card tied to E-A-D and passed through the independence ladder before release, then a standalone scholar with no institutional affiliation *could, as a design proposition,* co-produce checkable-rigour knowledge with AI.	15 verified citation cards (Table 3); `GLOSA-CC-20260904-0201`	DR	joint (data→inference); human (inference→claim)	human
C2	H2: rigour is proposed here, as a conceptual contrast, to be *at least in part* a property a single claim can carry in its own recorded structure, not conferred solely from outside by institutional machinery.	16 verified citation cards (Table 4); `GLOSA-CC-20260904-0202`	DR	joint (data→inference); human (inference→claim)	human
C3	H3: cross-vendor (I3) checking of undisclosed AI-fill/assumptions *would rise* measurably compared to same-vendor (I0/I1) re-checking. **Untested; no seeded-error benchmark has been run by this project (n=0).**	17 verified citation cards (Table 5), all analogical, none a direct test of Q3/Q4 detection; `GLOSA-CC-20260904-0203`	DR	joint (data→inference); human (inference→claim)	human

Table 6: Claim Matrix. All three cards: `k_state: K0`, `status: Draft`, `independent_check.status: PENDING` (checked by cross-vendor routes R1–R3 and revised, Section 7; not yet re-checked by a further route or founder-approved), `independence_class: I3` (the citation-verification route, not the card's own sign-off).

A.1 Five-questions detail

B BLACKBOX NOTE: HOW THIS WORK WAS MADE

(Thai project term: "Bantuek Klong Dam".) Curated verbatim lines — typos kept, never polished or translated in place — followed by the cooking log. Full record archived separately (`projects/GLS-2026-001_rigour-without-infrastructure/blackbox_note.yaml`, note id `BB-2026-09-04-01`; public curated source `blackbox/BLACKBOX_NOTE_glosa-paper_2026-09-04.md`; public log entries cited by id below, `blackbox/log/entries.jsonl`, concept DOI `10.5281/zenodo.22302518`).

C HUMAN MASTERY GATE

Before submission the author confirms they can defend, without AI assistance: (1) the problem (Section 1) (2) the major literature families (Section 6) (3) the closest constructs (own-lineage rows, Section 6) (4) the precise inadequacy (the contaminated concept named in Section 1) (5) the proposed mechanism (H1, Section 5) (6) the boundary conditions ($E \wedge A \wedge D$, Section 3) (7) the strongest counterargument (the Scholarly Kitchen credentialism case, Table 4) (8) the strongest counterexample (the direction-dependent cross-model result against H3, Table 5) (9) the falsifying record for each proposition (Section 5) (10) the discriminating empirical test (H3's not-yet-run seeded-error benchmark). **Gate status: PASS WITH NAMED GAPS** — the founder has not yet performed the class-5 (I5) sign-off on any of the three underlying claim cards (Section A); this gap is named, not silently dropped.

D PROVENANCE AND LINEAGE

Status vocabulary: PRESERVE_EXACT, PRESERVE_FUNCTION, EXPAND, SUPERSEDE. This paper PRESERVE_FUNCTIONs the claim-card schema and independence ladder from `design/FOUNDATION_v0.5.md` and EXPANDs it with a three-proposition literature conversation and a self-application demonstration specific to this project (GLS-2026-001). Direct ancestor: the Standalone Scholar dual-track architecture [cite-own-standalone-scholar-h1-005]. Core epistemic engine this paper's disclosure apparatus is built on: the Readout Condition [cite-own-readout-condition-h1-002].

REFERENCES

cite-bangladesh-readiness-007. [cite-bangladesh-readiness-007] ARXIV: 2601.12934. citation card cite-bangladesh-readiness-007.yaml, records/lit/rigour-without-infrastructure/. URL `https://arxiv.org/abs/2601.12934`. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"Bangladesh, ranked 82nd in the 2023 Oxford Insights AI Readiness Index, exhibits significant deficits in technology capacity and research ecosystems, despite strong governmental visions... Findings reveal outdated curri...”.

cite-cofact-thai-detector-009. [cite-cofact-thai-detector-009] OFFICIAL_URL: https://blog.cofact.org/th/ai-detector-20052025/. citation card cite-cofact-thai-detector-009.yaml, records/lit/rigour-without-infrastructure/. URL `https://blog.cofact.org/th/ai-detector-20052025/`. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “[Thai text – see citation card's own exact_passage field] (Artificial Intelligence: AI) [Thai text – see citation card's own exact_passage field]...”.

cite-crossmodel-codereview-002. [cite-crossmodel-codereview-002] ARXIV: 2607.21656. citation card cite-crossmodel-codereview-002.yaml, records/lit/rigour-without-infrastructure/. URL `https://arxiv.org/abs/2607.21656`. scope: DIRECT_QUOTATION; evidence_tier: fit_calibrated; independence_class: I3; fetch_status: FETCHED; citation

ID	seen	ai_filled	assumed	tested (indep.)
C1	Blackbox Note L1; founder's abstract framing	candidate hypothesis text, evidence retrieval, card drafting (all disclosed, `disclaimers_emitted: D-AIFILL`)	no source tests the full combined H1 mechanism as one system	I3 (citation checks); card itself unsigned
C2	same L1; H2's own narrower conceptual framing	same AI-fill fields as C1	the two SUPPORTS sources are same-lineage; cannot alone move H2 past DR	I3; card itself unsigned
C3	same L1; H3's cross-vendor detection-rate framing	same AI-fill fields as C1	no seeded benchmark has been run; the closest analogous study found a direction-dependent effect	I3 (analogical citations only); card itself unsigned

Table 7: Five-questions detail keyed to Table 6. Full cards: `projects/GLS-2026-001_rigour-without-infrastructure/claims/GLOSA-CC-2026090` (and `-0202`, `-0203`).

ID	Kind	English gloss (AI assistant's translation, tier DR; effect on this paper)	Pointer
BBL-083	proposal	"the word 'observation' is abstract, it is a verb; if we started from the problem instead, is there a process like this? We propose focusing on the problem first." — separates "observation" (abstract verb) from "problem state" (concrete pre-existing thing); became FOUNDATION §2.1a (Section 1)	DOI `10.5281/zenodo.22302518`, entry `BBL-2026-09-04-083`
BBL-086/088	finding / correction	"if the problem is still ours, the question is still ours, and the selection of the hypothesis is still ours, and it leads to solving our problem, then use AI, use a cat, use a bear, go ahead — because that is the goal of knowledge: to solve a problem for someone." (088 corrects 086) — fixes which spine nodes must stay human: problem, question, and the *selection* of the hypothesis; became FOUNDATION §2.1c	DOI `10.5281/zenodo.22302518`, entry `BBL-2026-09-04-088` (corrects 086)
BBL-089	ruling	Ratifies the finer-grained spine diagram wording (AI-drafted, human-ratified) — became the spine diagram, FOUNDATION §2.1a	`blackbox/log/entries.jsonl`, entry `BBL-2026-09-04-089`
BBL-091	instruction	Install and use the skill to build this article in full — became task B1/B2 launch	`blackbox/log/entries.jsonl`, entry `BBL-2026-09-04-091`
BBL-095	ruling	"bring all three together" — selects all three hypotheses (H1, H2, H3); became `hypothesis_selection.yaml`	`blackbox/log/entries.jsonl`, entry `BBL-2026-09-04-095`
BBL-097	ruling	"make this document an academic article, a concept paper" — sets genre to CONCEPTUAL, overriding the router's `systematic_review` output; became `genre_decision.yaml` (Section 4)	`blackbox/log/entries.jsonl`, entry `BBL-2026-09-04-097`
BBL-098	instruction	Requires colliding own prior papers against world literature, not silence — became own-lineage rows, Section 6	`blackbox/log/entries.jsonl`, entry `BBL-2026-09-04-098`
BBL-099	ruling	English only, no Thai edition; arXiv two-column theme, polished — became this document's language and theme	`blackbox/log/entries.jsonl`, entry `BBL-2026-09-04-099`
BBL-100	ruling	"should we make it a hard rule: if a source is used it must carry a page and line locator plus a full link, never written from memory — that reduces the problem at the source, before the check." — mandates locator+link, source-first citation, after I3 caught composite quotes; became rule 17, §7.8 (Section 7)	DOI `10.5281/zenodo.22302518`, entry `BBL-2026-09-04-100`
BBL-101	instruction	"update it into the system" — adopts the citation rule immediately; became rule 17 adoption	`blackbox/log/entries.jsonl`, entry `BBL-2026-09-04-101`

Table 8: Blackbox Note — every row is id / English gloss / pointer. Verbatim Thai lines are kept only in the cited records (the Blackbox Log, DOI `10.5281/zenodo.22302518`, or `blackbox/log/entries.jsonl`), by founder ruling `BBL-2026-09-04-103` (English-only paper); the glosses shown here are the AI assistant's translations, not the founder's own words, tier DR.

card status: VERIFIED; exact passage: "Claude review raises Codex drafts from 71.6% to 89.7% ($p\_BH = .001$); Codex self review raises them to 84.5% ($p\_BH = .022$). The reverse direction does not pay off: Codex reviewing Claude drafts drops the pass rate fro...".

cite-cstp-boundary-work-001. [cite-cstp-boundary-work-001] DOI: 10.5334/cstp.519. citation card cite-cstp-boundary-work-001.yaml, records/lit/rigour-without-infrastructure/. URL `https://doi.org/10.5334/cstp.519`. scope: DIRECT_QUOTATION; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: ""the legitimacy of citizen science is powerfully shaped by the perspectives of professional or credentialed experts" – in a 2010-2019 sample of 51 news articles, "perspectives of professional and institutionally affilia...".

cite-disclosure-not-documentation-001. [cite-disclosure-not-documentation-001] DOI: 10.1186/s41073-026-00245-8. citation card cite-disclosure-not-documentation-001.yaml, records/lit/rigour-without-infrastructure/. URL `https://doi.org/10.1186/s41073-026-00245-8`. scope: PARAPHRASE; evidence_tier: Dr; independence_class: I3; fetch_status: PAYWALLED_ABSTRACT_ONLY; citation card status: VERIFIED; exact passage: ""disclosure alone rarely clarifies how AI-assisted work was performed, what information was entered, how outputs were evaluated, or how human responsibility was maintained" – creating a gap between AI-use disclosure as ...".

cite-iseas-sea-governance-008. [cite-iseas-sea-governance-008] OFFICIAL_URL: https://www.iseas.edu.sg/articles-commentaries/iseas-perspective/2025-13-what-is-shaping-artificial-intelligence-ai-governance-policies-in-southeast-asia-by-kristina-fong/. citation card cite-iseas-sea-governance-008.yaml, records/lit/rigour-without-infrastructure/. URL `https://www.iseas.edu.sg/articles-commentaries/iseas-perspective/2025-13-what-is-shaping-artificial-intelligence-ai-governance-policies-in-southeast-asia-by-kristina-fong/`. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status:

10.5281/zenodo.22307841

Step	By	Input lines	Output	What changed
lens_in	ai-assistant	L1	`lens_translation.md`	Drafted `question_readout`, `local_contrast_space_X`, `access_relation_R`, `claim_function_Phi_z0` from L1 under the Readout Universe lens; not yet independently reviewed.
lens_out	ai-assistant	L1–L13	`hypotheses.md`	Drafted H1/H2/H3 as candidate world-language hypotheses; selection reserved to and performed by the founder (BBL-095).
lit_review	ai-assistant	H1/H2/H3	48 citation cards, 3 `litreview_manifest.yaml`	Ran targeted/scoping searches per hypothesis (not systematic), froze manifests, computed diversity/accuracy audits.
claim_card	ai-assistant	hypotheses + literature	`GLOSA-CC-20260904-0201/0202/0203`	Filled full-shape cards; validated `PASS_WITH_LIMITS`; not yet founder-signed.
paper_build	ai-assistant	all of the above	this document	Drafted the concept paper's prose, tables, and LaTeX build; disclosed throughout, never authored.

Table 9: Cooking log (Thai project term: "karn pung") — append-only.

FETCHED; citation card status: VERIFIED; exact passage: “"The capabilities of DPAs across Southeast Asia are very diverse. As such, the use of DPAs as central market surveillance authorities may be viable for some, but not others... existing DPA capacities will need to be bols...”.

cite-legalai-india-006. [cite-legalai-india-006] ARXIV: 2608.21089. citation card cite-legalai-india-006.yaml, records/lit/rigour-without-infrastructure/. URL https://arxiv.org/abs/2608.21089. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"students reporting multiple prior encounters with fabricated citations reported a higher descriptive mean verification score (4.2/5) than students reporting no such encounters (2.8/5)."”.

cite-llm-hallucination-survey-001. [cite-llm-hallucination-survey-001] ARXIV: 2510.06265. citation card cite-llm-hallucination-survey-001.yaml, records/lit/rigour-without-infrastructure/. URL https://arxiv.org/abs/2510.06265. scope: SUPPORTS_GENERAL_CLAIM_ONLY; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"Hallucinations undermine the reliability and trustworthiness of LLMs, especially in domains requiring factual accuracy." Self-consistency detection methods "cannot catch hallucinations that are confidently and consisten...”.

cite-mindstudio-crossvendor-010. [cite-mindstudio-crossvendor-010] OFFICIAL_URL: https://www.mindstudio.ai/blog/cross-vendor-ai-agent-review-claude-codex. citation card cite-mindstudio-crossvendor-010.yaml, records/lit/rigour-without-infrastructure/. URL https://www.mindstudio.ai/blog/cross-vendor-ai-agent-review-claude-codex. scope: SUPPORTS_GENERAL_CLAIM_ONLY; evidence_tier: Open; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"OpenAI's Codex and Anthropic's Claude don't share training data, architectures, or fine-tuning pipelines. When Codex tends to miss a particular class of security issue...Claude often catches it, and vice versa." ... "Wh...”.

cite-multiagent-debate-004. [cite-multiagent-debate-004] ARXIV: 2305.14325. citation card cite-multiagent-debate-004.yaml, records/lit/rigour-without-infrastructure/. URL https://arxiv.org/abs/2305.14325. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: fit_calibrated; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"We present a complementary approach to improve language responses where multiple language model instances propose and debate their individual responses and reasoning processes over multiple rounds to arrive at a common ...”.

cite-ocsdnet-honf-001. [cite-ocsdnet-honf-001] OFFICIAL_URL: https://ocsdnet.org/projects/hita-ordo-natural-fiber-honf-foundation/. citation card cite-ocsdnet-honf-001.yaml, records/lit/rigour-without-infrastructure/. URL https://ocsdnet.org/projects/hita-ordo-natural-fiber-honf-foundation/. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “The main objective is to understand how open science in Indonesia, Thailand and Nepal operates between formal and informal institutions of knowledge and technology transfer and the dynamic of North/South/South networks o...”.

cite-own-blackbox-log-h1-006. [cite-own-blackbox-log-h1-006] DOI: 10.5281/zenodo.22302518. citation card cite-own-blackbox-log-h1-006.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.22302518. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"Blackbox Log – a daily, append-only, verbatim record of the author's raw voice (one pre-release correction is disclosed in BBL-2026-09-04-082), findings and questions (Thai is the source of truth)... Contains the author...”.

cite-own-blackbox-log-h2-006. [cite-own-blackbox-log-h2-006] DOI:

 10.5281/zenodo.22307841

10.5281/zenodo.22302518. citation card cite-own-blackbox-log-h2-006.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.22302518. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"Blackbox Log – a daily, append-only, verbatim record of the author’s raw voice (one pre-release correction is disclosed in BBL-2026-09-04-082), findings and questions (Thai is the source of truth)... Contains the author...”.

cite-own-blackbox-log-h3-006. [cite-own-blackbox-log-h3-006] DOI: 10.5281/zenodo.22302518. citation card cite-own-blackbox-log-h3-006.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.22302518. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"Blackbox Log – a daily, append-only, verbatim record of the author’s raw voice (one pre-release correction is disclosed in BBL-2026-09-04-082), findings and questions (Thai is the source of truth)... Contains the author...”.

cite-own-civilization-of-knowledge-h2-007. [cite-own-civilization-of-knowledge-h2-007] DOI: 10.5281/zenodo.18943971. citation card cite-own-civilization-of-knowledge-h2-007.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.18943971. scope: DIRECT_QUOTATION; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"What changes across civilizations is not only the accumulation of truths, but the succession of agencies authorized to interpret the world’s signals and stabilize those interpretations as social reality... These do not ...”.

cite-own-knowledge-stabilized-translation-h3-007. [cite-own-knowledge-stabilized-translation-h3-007] DOI: 10.5281/zenodo.18925129. citation card cite-own-knowledge-stabilized-translation-h3-007.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.18925129. scope: DIRECT_QUOTATION; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"knowledge is best understood as stabilized translation under bounded observation: a representation that remains stable across admissible variation, calibration, and cross-checking. The result is a framework that preserv...”.

cite-own-readout-condition-h1-002. [cite-own-readout-condition-h1-002] DOI: 10.5281/zenodo.22301318. citation card cite-own-readout-condition-h1-002.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.22301318. scope: DIRECT_QUOTATION; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"This paper develops the Readout Condition as a necessary architecture for source-relative epistemic warrant. Its core is a three-part norm: existence (every claim-level distinction has an epistemic provenance path), att...”.

cite-own-readout-condition-h2-002. [cite-own-readout-condition-h2-002] DOI: 10.5281/zenodo.22301318. citation card cite-own-readout-condition-h2-002.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.22301318. scope: DIRECT_QUOTATION; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"This paper develops the Readout Condition as a necessary architecture for source-relative epistemic warrant. Its core is a three-part norm: existence (every claim-level distinction has an epistemic provenance path), att...”.

cite-own-readout-condition-h3-002. [cite-own-readout-condition-h3-002] DOI: 10.5281/zenodo.22301318. citation card cite-own-readout-condition-h3-002.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.22301318. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"This paper develops the Readout Condition as a necessary architecture for source-relative epistemic warrant. Its core is a three-part norm: existence (every claim-level distinction has an epistemic provenance path), att...”.

cite-own-readout-genesis-h1-003. [cite-own-readout-genesis-h1-003] DOI: 10.5281/zenodo.21529456. citation card cite-own-readout-genesis-h1-003.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.21529456. scope: SUPPORTS_GENERAL_CLAIM_ONLY; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"the paper inserts a typed belief layer between memory and claim evaluation, defines belief as a multidimensional agency-claim relation, and proposes that knowledge be attributed as an auditable status of a claim rather ...”.

cite-own-readout-genesis-h2-003. [cite-own-readout-genesis-h2-003] DOI: 10.5281/zenodo.21529456. citation card cite-own-readout-genesis-h2-003.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.21529456. scope: SUPPORTS_GENERAL_CLAIM_ONLY; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage:

“"the paper inserts a typed belief layer between memory and claim evaluation, defines belief as a multidimensional agency-claim relation, and proposes that knowledge be attributed as an auditable status of a claim rather ...”.

cite-own-readout-genesis-h3-003. [cite-own-readout-genesis-h3-003] DOI: 10.5281/zenodo.21529456. citation card cite-own-readout-genesis-h3-003.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.21529456. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"the paper inserts a typed belief layer between memory and claim evaluation, defines belief as a multidimensional agency-claim relation, and proposes that knowledge be attributed as an auditable status of a claim rather ...”.

cite-own-standalone-scholar-h1-005. [cite-own-standalone-scholar-h1-005] DOI: 10.5281/zenodo.22163849. citation card cite-own-standalone-scholar-h1-005.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.22163849. scope: DIRECT_QUOTATION; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"It develops a dual-track architecture in which AI-assisted formation, open preprint staging, provenance controls, and machine-constrained checks generate defensible provisional assets, while independent human friction, ...”.

cite-own-standalone-scholar-h2-005. [cite-own-standalone-scholar-h2-005] DOI: 10.5281/zenodo.22163849. citation card cite-own-standalone-scholar-h2-005.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.22163849. scope: DIRECT_QUOTATION; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"It develops a dual-track architecture in which AI-assisted formation, open preprint staging, provenance controls, and machine-constrained checks generate defensible provisional assets, while independent human friction, ...”.

cite-own-standalone-scholar-h3-005. [cite-own-standalone-scholar-h3-005] DOI: 10.5281/zenodo.22163849. citation card cite-own-standalone-scholar-h3-005.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.22163849. scope: DIRECT_QUOTATION; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"It develops a dual-track architecture in which AI-assisted formation, open preprint staging, provenance controls, and machine-constrained checks generate defensible provisional assets, while independent human friction, ...”.

cite-own-written-ai-h1-001. [cite-own-written-ai-h1-001] DOI: 10.5281/zenodo.22301202. citation card cite-own-written-ai-h1-001.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.22301202. scope: DIRECT_QUOTATION; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"Institutions remain indispensable, but their authority is derivative from epistemic work, not magical pedigree. AI is philosophically useful because it removes the human face from a claim and reveals how much epistemic ...”.

cite-own-written-ai-h2-001. [cite-own-written-ai-h2-001] DOI: 10.5281/zenodo.22301202. citation card cite-own-written-ai-h2-001.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.22301202. scope: DIRECT_QUOTATION; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"Institutions remain indispensable, but their authority is derivative from epistemic work, not magical pedigree. AI is philosophically useful because it removes the human face from a claim and reveals how much epistemic ...”.

cite-own-written-ai-h3-001. [cite-own-written-ai-h3-001] DOI: 10.5281/zenodo.22301202. citation card cite-own-written-ai-h3-001.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.22301202. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"Institutions remain indispensable, but their authority is derivative from epistemic work, not magical pedigree. AI is philosophically useful because it removes the human face from a claim and reveals how much epistemic ...”.

cite-own-zero-readout-h1-004. [cite-own-zero-readout-h1-004] DOI: 10.5281/zenodo.21665100. citation card cite-own-zero-readout-h1-004.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.21665100. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"We show that a zero readout is not an absence but a classification, and prove which one... The characterisation is machine-checked over the rationals without classical axioms."”.

cite-own-zero-readout-h2-004. [cite-own-zero-readout-h2-004] DOI: 10.5281/zenodo.21665100. citation card cite-own-zero-readout-h2-004.yaml,

10.5281/zenodo.22307841

records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.21665100. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"We show that a zero readout is not an absence but a classification, and prove which one... The characterisation is machine-checked over the rationals without classical axioms."”.

cite-own-zero-readout-h3-004. [cite-own-zero-readout-h3-004] DOI: 10.5281/zenodo.21665100. citation card cite-own-zero-readout-h3-004.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5281/zenodo.21665100. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"We show that a zero readout is not an absence but a classification, and prove which one... The characterisation is machine-checked over the rationals without classical axioms."”.

cite-pgph-citizen-science-sea-001. [cite-pgph-citizen-science-sea-001] DOI: 10.1371/journal.pgph.0003696. citation card cite-pgph-citizen-science-sea-001.yaml, records/lit/rigour-without-infrastructure/. URL https://www.ebi.ac.uk/europepmc/webservices/rest/PMC12370037/fullTextXML. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “Current study showed that people being the recipients of policies and programs, ready to have their voices included.”.

cite-rigour-without-infrastructure-001. [cite-rigour-without-infrastructure-001] OFFICIAL_URL: https://en.wikipedia.org/wiki/Open-notebook_science. citation card cite-rigour-without-infrastructure-001.yaml, records/lit/rigour-without-infrastructure/. URL https://en.wikipedia.org/wiki/Open-notebook_science. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: definition; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"Open-notebook science is the practice of making the entire primary record of a research project publicly available online as it is recorded." (article also notes open-notebook projects have been published in peer-review...”.

cite-rigour-without-infrastructure-002. [cite-rigour-without-infrastructure-002] DOI: 10.5334/cstp.207. citation card cite-rigour-without-infrastructure-002.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5334/cstp.207. scope: SUPPORTS_GENERAL_CLAIM_ONLY; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"the field lacks an overall, widely accepted approach to research integrity, which would at least include training citizen scientists in concepts and methods of research integrity, establishing mechanisms to protect the ...”.

cite-rigour-without-infrastructure-003. [cite-rigour-without-infrastructure-003] DOI: 10.5334/cstp.6. citation card cite-rigour-without-infrastructure-003.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.5334/cstp.6. scope: DIRECT_QUOTATION; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"We surveyed the credibility-building strategies of 30 citizen science programs that monitor environmental aspects of the California coast. We identified a total of twelve strategies: Three that are applied during traini...”.

cite-rigour-without-infrastructure-004. [cite-rigour-without-infrastructure-004] OFFICIAL_URL: https://scholarlykitchen.sspnet.org/2026/08/14/guest-post-the-guild-and-the-gap-what-one-rejection-revealed-about-the-cost-of-credentialism/. citation card cite-rigour-without-infrastructure-004.yaml, records/lit/rigour-without-infrastructure/. URL https://scholarlykitchen.sspnet.org/2026/08/14/guest-post-the-guild-and-the-gap-what-one-rejection-revealed-about-the-cost-of-credentialism/. scope: DIRECT_QUOTATION; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “Preparing manuscripts takes training, practice and experience. If someone has no experience, no affiliation, no supervision or advisory colleagues, etc., then the system should be very careful with that person, especiall...”.

cite-rigour-without-infrastructure-005. [cite-rigour-without-infrastructure-005] DOI: 10.1093/joc/jqab027. citation card cite-rigour-without-infrastructure-005.yaml, records/lit/rigour-without-infrastructure/. URL https://api.crossref.org/works/10.1093/joc/jqab027. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “These decolonizing practices foreground data sovereignty, community ownership, and public ownership of knowledge resources as the bases of resistance to the colonial-capitalist interests of hegemonic Open Science.”.

cite-rigour-without-infrastructure-006. [cite-rigour-without-infrastructure-006] DOI: 10.3389/frma.2022.855198. citation card cite-rigour-without-infrastructure-006.yaml, records/lit/rigour-without-infrastructure/. URL https://www.ebi.ac.uk/europepmc/webservices/rest/PMC9051436/fullTextXML. scope:

10.5281/zenodo.22307841

CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"The advancement of scientific research and raising the next-generation scientists in Africa depend largely on science access. The COVID-19 pandemic has caused discussions around open science (OS) to reemerge globally, e...”.

cite-rigour-without-infrastructure-007. [cite-rigour-without-infrastructure-007] DOI: 10.1371/journal.pgph.0003696. citation card cite-rigour-without-infrastructure-007.yaml, records/lit/rigour-without-infrastructure/. URL https://www.ebi.ac.uk/europepmc/webservices/rest/PMC12370037/fullTextXML. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"Citizen science (CS), a collaboration between people and scientists, is a viable approach utilizing citizens experiences in COVID-19 pandemic to manage future response. This study aimed to understand concepts, experienc...”.

cite-rigour-without-infrastructure-008. [cite-rigour-without-infrastructure-008] OFFICIAL_URL: https://so05.tci-thaijo.org/index.php/JTPLC/article/view/283964. citation card cite-rigour-without-infrastructure-008.yaml, records/lit/rigour-without-infrastructure/. URL https://so05.tci-thaijo.org/index.php/JTPLC/article/view/283964. scope: DIRECT_QUOTATION; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “Confirmability emphasizes neutrality, ensuring that interpretations are derived from actual data rather than the researcher’s imagination, often documented through an audit trail.”.

cite-rigour-without-infrastructure-009. [cite-rigour-without-infrastructure-009] OFFICIAL_URL: https://thestandard.co/thailand-open-science-unlock-ai-big-data-platform/. citation card cite-rigour-without-infrastructure-009.yaml, records/lit/rigour-without-infrastructure/. URL https://thestandard.co/thailand-open-science-unlock-ai-big-data-platform/. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"[Thai text – see citation card’s own exact_passage field] [Thai text – see citation card’s own exact_passage field][Thai text – see citation card’s own exact_passage field]"”.

cite-rsos-citizen-epistemology-001. [cite-rsos-citizen-epistemology-001] DOI: 10.1098/rsos.231100. citation card cite-rsos-citizen-epistemology-001.yaml, records/lit/rigour-without-infrastructure/. URL https://www.ebi.ac.uk/europepmc/webservices/rest/PMC10646465/fullTextXML. scope: SUPPORTS_GENERAL_CLAIM_ONLY; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “Thus, multiple perspectives enhance robust insight, and a multiplicity of perspectives is what democratic citizen science provides.”.

cite-rsos-preprint-credibility-001. [cite-rsos-preprint-credibility-001] DOI: 10.1098/rsos.201520. citation card cite-rsos-preprint-credibility-001.yaml, records/lit/rigour-without-infrastructure/. URL https://www.ebi.ac.uk/europepmc/webservices/rest/PMC7657885/fullTextXML. scope: SUPPORTS_GENERAL_CLAIM_ONLY; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “In general, information related to field perceptions/usage of preprints was rated as less important by respondents, and information related to preprint authors (e.g. institutional information, previous work) received mor...”.

cite-scholarly-kitchen-credentialism-001. [cite-scholarly-kitchen-credentialism-001] OFFICIAL_URL: https://scholarlykitchen.sspnet.org/2026/08/14/guest-post-the-guild-and-the-gap-what-one-rejection-revealed-about-the-cost-of-credentialism/. citation card cite-scholarly-kitchen-credentialism-001.yaml, records/lit/rigour-without-infrastructure/. URL https://scholarlykitchen.sspnet.org/2026/08/14/guest-post-the-guild-and-the-gap-what-one-rejection-revealed-about-the-cost-of-credentialism/. scope: DIRECT_QUOTATION; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “For submissions in this format, we need to look at the author’s expertise via past publications... As we found no record of previous relevant peer-reviewed publications by the author(s) in a brief search, we cannot accep...”.

cite-selfcorrect-reasoning-005. [cite-selfcorrect-reasoning-005] ARXIV: 2310.01798. citation card cite-selfcorrect-reasoning-005.yaml, records/lit/rigour-without-infrastructure/. URL https://arxiv.org/abs/2310.01798. scope: SUPPORTS_GENERAL_CLAIM_ONLY; evidence_tier: fit_calibrated; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"Central to our investigation is the notion of intrinsic self-correction, whereby an LLM attempts to correct its initial responses based solely on its inherent capabilities, without the crutch of external feedback... our...”.

cite-selfpreference-judge-001. [cite-selfpreference-judge-001] ARXIV: 2410.21819. citation card cite-selfpreference-judge-001.yaml, records/lit/rigour-without-infrastructure/. URL https://arxiv.org/abs/2410.21819. scope: SUPPORTS_GENERAL_CLAIM_ONLY; evidence_tier:

 10.5281/zenodo.22307841

fit_calibrated; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"Our findings reveal that LLMs assign significantly higher evaluations to outputs with lower perplexity than human evaluators, regardless of whether the outputs were self-generated. This suggests that the essence of the ...”.

cite-tcij-freelance-th-001. [cite-tcij-freelance-th-001] OFFICIAL_URL: https://www.tcijthai.com/news/2020/1/labour/9775. citation card cite-tcij-freelance-th-001.yaml, records/lit/rigour-without-infrastructure/. URL https://www.tcijthai.com/news/2020/1/labour/9775. scope: CONTEXT_ONLY_NOT_EVIDENCE; evidence_tier: Dr; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “[Thai text – see citation card’s own exact_passage field]”.

cite-toohconsistent-003. [cite-toohconsistent-003] DOI: 10.18653/v1/2025.emnlp-main.238. citation card cite-toohconsistent-003.yaml, records/lit/rigour-without-infrastructure/. URL https://doi.org/10.18653/v1/2025.emnlp-main.238. scope: DIRECT_QUOTATION; evidence_tier: fit_calibrated; independence_class: I3; fetch_status: FETCHED; citation card status: VERIFIED; exact passage: “"Motivated by the observation that self-consistent errors often differ across LLMs, we propose a simple but effective cross-model probe method that fuses hidden state evidence from an external verifier LLM. Our method si...”.

 10.5281/zenodo.22307841

glosa — Rigour Without Infrastructure: A Standalone Scholar Methodology for Human–AI Knowledge Co-Production

v0.4.1 • 5 September 2026 • `10.5281/zenodo.22340255`

What the chapter states. This chapter typesets, as a verbatim excerpt, §2/§3.1/§4.2 of `FOUNDATION_v0.6.md`, this record's design document. "The spine is a round trip: a problem met in the world is translated into the readout vocabulary first (lens-in, per the Lens Law), analysed there, then translated back into world/discipline language to extract the academic hypothesis and its falsifier – with the lineage of every term preserved so nothing is silently promoted."

Status. "K-state unchanged: K0 (no independent check has run on this synthesis pass itself, NC-34)."

Falsifier(s). No explicit falsifier stated in the chapter.

Read before. Ch. 1 *Written by AI. Still True.*; Ch. 2 *The Readout Condition*; Ch. 3 *Readout Genesis Standalone Synthesis*; Ch. 27 *When AI Expands Human Potential*; Ch. 39 *The Standalone Scholar*.

Feeds into. Ch. 8 *Meaning Before Naming*; Ch. 9 *Experience Is Meaning-Giving*; Ch. 17 *Choice Begins Before Choice*; Ch. 18 *Potential as a Readout*; Ch. 37 *Before Meaning, Before Choice*; Ch. 36 *Master Equation River*.

Editorial note (apparatus-fixer pass, 2026-09-06). The Zenodo software record this chapter's headnote cites, `10.5281/zenodo.22340255` (glosa v0.4.1), carries no standalone PDF whose body is the design-document text described below; its own attached files are an internal uplift note, a concept paper already reproduced as Chapter 37 of this book, and a source-code archive. What follows is therefore typeset new for this edition rather than `\includepdf`'d, and consists only of direct quotations from `FOUNDATION_v0.6.md` — the design document that record deposits the software for — cited to the same DOI, plus neutral section labels and two tables reproduced from the source. No summarising prose is added.

§2. The spine: a round trip. *"The spine is a round trip: a problem met in the world is translated into the readout vocabulary first (lens-in, per the Lens Law), analysed there, then translated back into world/discipline language to extract the academic hypothesis and its falsifier — with the lineage of every term preserved so nothing is silently promoted."*

§3.1. Five questions, one crosswalk. Crosswalk reproduced from the source:

Q	Principle	Claim-card field
Q1	Existence (E)	`five_questions.seen`
Q2	What the record alone distinguishes	`five_questions.separates`, `zero_vs_bottom`
Q3	Disclosure (D) of AI's contribution	`five_questions.ai_filled`
Q4	Attribution (A) of every assumption	`five_questions.assumed[]`
Q5	Independent evidence or objection	`five_questions.tested`, `independent_check`

"The Readout Condition is $E \wedge A \wedge D$: a card that answers Q1/Q2/Q4 honestly but misattributes Q3 — crediting the source with a distinction AI actually supplied — fails the condition even though every field looks filled."

§4.2. The Independence Ladder, I0–I5. *"The I0–I5 Independence Ladder and the falsifiable-propositions framing already used in this section are cited to 'Rigour Without Infrastructure' — the ladder's existing five classes and definitions are unchanged."*

Level	Definition	K-state ceiling
I0 — Self	Same session re-reads its own output	K0
I1 — Same model, new session	Fresh conversation, identical model	K0
I2 — Same vendor, other model	Different model, same vendor	K0$\to$K1 only combined with I3+, or the bounded I2+I4 exception
I3 — Different vendor	Materially different model family/vendor; stated as the founder-set minimum bar for K1, but "a minimum, never sufficient on its own"	K1 (floor), never K2
I4 — Mechanical / original-record	Proof kernel, executed test, schema validator, direct original-source lookup	K1 unless combined with I5
I5 — Independent external human (non-founder)	A person outside the authoring session; stated as "the only route to K2"	K2/K3

"the maximum claim tier a route set can support is bounded by the highest independence level actually reached, never by the count of routes at a lower level" (non-collapse row `NC-31`, *ManyModels*$\not\Rightarrow$*Independence*).

Full text and the surrounding schema: `FOUNDATION_v0.6.md` *§2.1, §3.1, §4.2, DOI* `10.5281/zenodo.22340255`.

Appendices

Appendix A — Library Map

Every chapter in this book, in book order, grouped by part, with its DOI, version label, and deposit date as recorded in `manifest.yaml`. A supplementary record, the Human–AI Readout Programme's own library knowledge graph (DOI `10.5281/zenodo.22341671`), lists these same works as machine-readable nodes and edges and is the mechanical source for the "read before" / "feeds into" lines in each chapter's headnote; it is not itself a chapter of this book.

	Ch.	Title	DOI
Part 1 — Root: Before Meaning	1	Written by AI. Still True. Knower Fetishism, Epistemic Pedigree, and the Human Face as a Bad Theory of Truth	`10.5281/zenodo.22301202`
	2	The Readout Condition: Distinguishability, Access, and Epistemic Warrant	`10.5281/zenodo.22301318`
	3	Readout Genesis Standalone Synthesis: Information Epistemic Foundation, Conditioned Agency, and Meta-Readout Governance	`10.5281/zenodo.21529456`
	4	Knowledge as Stabilized Translation: Toward an Observer-Constrained Epistemology	`10.5281/zenodo.18925129`
	5	Constraint-First Epistemology: Normativity, Conditioned Agency, and the Non-Zero Kantian Floor	`10.5281/zenodo.19205869`
	6	Mind as Information Horizon: From Primordial Difference to Expertise Formation on the Discrete Causal Graph	`10.5281/zenodo.19640361`
	7	The Architecture of Mediated Agency: Beyond the Misframing of Free Will and Truth	`10.5281/zenodo.19176260`

	Ch.	Title	DOI
Part 2 — Meaning, Experience, Retention	8	Meaning Before Naming: A Readout-Retention Architecture of Affective-Semantic Reorganization	`10.5281/zenodo.224`
	9	Experience Is Meaning-Giving: A Strong-Form Readout-Retention Theory of Phenomena, Resonance, Rhythm, Accumulation, and Transformative Release	`10.5281/zenodo.223`
	10	Experience Is the Human LoRA: A Readout–Retention Theory of Selective Model Change	`10.5281/zenodo.214`
	11	Human Learning as Epistemic Architecture: A Method for Word Mapping, Life-Concept Graphs, Constraint Testing, and Corrigible Agency in the AI Age	`10.5281/zenodo.223`
	12	Learning Under Generative Abundance: A Structural Law of Epistemic Stabilization	`10.5281/zenodo.187`

	Ch.	Title	DOI
Part 3 — Possibility, Choice, Potential	13	From Problem to Hypothesis: Dynamic Semantic Mobility, Bounded Knowers, and the Readout-Discriminable Hypothesis Space	`10.5281/zenodo.223071`
	14	Knowledge Topology and the First Passage to Usable Hypotheses: A Readout Theory of Discovery Time and Direction	`10.5281/zenodo.223075`
	15	Before Evidence Can Decide: Candidate-Set Formation, Discovery Routing, and Unconceived Alternatives — Questions for a Pre-Evidential Epistemology of Science	`10.5281/zenodo.223075`
	16	The Epistemic Chain Reaction: Human-AI Multiplication from Questions to Readout-Distinguishable Hypotheses — A Controlled Amplification Architecture for Scientific Discovery	`10.5281/zenodo.223080`
	17	Choice Begins Before Choice: Meaning-Shaped Accessibility, Live Possibility, and Effective Agency in Human-AI Systems	`10.5281/zenodo.223577`
	18	Potential as a Readout: Witnessed Envelopes, Two-Layer Agency, and the Measurement of Pseudo-Peace	`10.5281/zenodo.223618`

Part 4 — Society, Power, Justice: Potential Under Compression

Ch.	Title
19	The Causal Grammar of Structured Coexistence: Conflict, Violence, Repair, a Suppressive Peace
20	Violence as a Special Case of Instability in Finite–Memory Causal Systems
21	Causal Ethics: The Mathematics of Regime Choice and Survival (excerpt, pp 73–77)
22	Causal Agency: A Persistence Control Theory of Adaptive Systems (excerpt, p 23–29)
23	The Civilization of Knowledge: Who Has the Authority to Interpret the World

Part 5 — Human–AI Coupling and Return

Ch.	Title	DOI
24	AI–Cognitive Interaction: Activating Youth Potential through Reflective Dialogue and Linguistic Capital	`10.5281/ zenodo.224`
25	Operational Linguistic Wisdom: Elective Connectivity and Linguistic Capital Activation in the Age of LLMs	`10.5281/ zenodo.224`
26	The Dialogue as the Ground of Enlightenment: Religious and Cognitive Frameworks for Understanding Wisdom through Human-AI Conversations	`10.5281/ zenodo.224`
27	When AI Expands Human Potential: Reflective Dissonance, Epistemic Agency, and Constraint	`10.5281/ zenodo.192`
28	The Language Bridge: Expanding Human Potential in the Age of AI	`10.5281/ zenodo.172`
29	AI, Translation, and Access to Event-Specific Context	`10.5281/ zenodo.185`
30	Epistemic Fusion or Epistemic Tunnel: A Phenomenology-Anchored, Global-Literature-Constrained, Problem-First Architecture of Human-AI Collaboration	`10.5281/ zenodo.223`
31	CTSA Human-Return Readout: A Session-Boundary Measurement Architecture for Retained Human Capability After AI	`10.5281/ zenodo.223`
32	How Humans Should Converse with AI: A Problem-First Adaptive Dialogue Conversion Protocol for Durable Human Capability	`10.5281/ zenodo.224`
33	From Assistance to Human Capability: A Readout-Genesis Architecture for Proactive AI, Unequal Life Conditions, and Opportunity Conversion	`10.5281/ zenodo.224`
planned	*Paper C — The Prompt Is Not the Beginning*	—

Licence note, Ch. 28. This chapter's own printed page states CC BY-NC-SA 4.0; the author-held Zenodo record licence (CC BY 4.0, which governs this book) is used here.

Part 6 — Human Position After Labour

Ch.	Title	DOI
34	After Labour: Human Position in an AI-Robotic World System	`10.5281/ zenodo.224819`
35	The Human Conversion Imperative: Machine Acceleration, Human Potential, and the Reversibility Window in an AI-Robotic World	`10.5281/ zenodo.224819`

Part 7 — The River and Its Tests

Ch.	Title	DOI
36	Master Equation River: A Provenance Audit Across the Human–AI Readout Programme	`10.5281/ zenodo.22550491`
37	Before Meaning, Before Choice: A Readout-Native Derivation of Experience, Live Possibility, Agency, and Human-AI Return	`10.5281/ zenodo.22424434`
planned	*Paper B — Testing the Master Equation River*	—
38	State of Evidence for the Readout Hypothesis-Generation Programme: A Shared Evidence Registry for Papers I–IV (with Claim Registry E01–E25)	`10.5281/ zenodo.22308066`

Note on this part's numbering. Chapter numbers in this part, as everywhere in this book, are computed from the current manifest order via the book's `\chnum` macro, not hard-coded, so they move automatically if a chapter is inserted or withheld elsewhere in the book. *Renumbered 2026-09-06 (headnote/bridge pass):* this part's rows previously carried lower position labels, under a book that had no Part 6; inserting the new Part 6, *Human Position After Labour*, shifted every one of this part's chapter numbers upward, which is exactly the drift this macro is designed to absorb. *Included 2026-09-06 (English-edition pass): Master Equation River* was previously withheld pending an English-only edition (DOI `10.5281/zenodo.22414412`, v1.1, Thai-majority body text); its English-only v1.5 edition (DOI `10.5281/zenodo.22550491`, unchanged content through v1.3 plus one new segment, eq. 1–79, with a companion Th_coqc ledger, Appendix F) is now bound as Chapter 36, opening this part, and is no longer withheld.

Part 8 — Method: Rigour Without Infrastructure

Ch.	Title	DOI
39	The Standalone Scholar: A Dual-Track Architecture for AI-Native Scholarship	10.5 zeno
40	Rigour Without Infrastructure: Three Propositions on Claim-Card Discipline as a Substitute for Institutional Certification	10.5 zeno
41	glosa — Rigour Without Infrastructure: A Standalone Scholar Methodology for Human–AI Knowledge Co-Production	10.5 zeno

Appendix B — Independence Ladder, K-State, and the Non-Collapse Rows This Book Cites

Full text, glosa — Rigour Without Infrastructure, `FOUNDATION_v0.6.md` §4.1–§4.4, DOI `10.5281/zenodo.22340255`.

Independence Ladder I0–I5

Level	Definition	Can raise / cannot raise
I0 — Self	Same session re-reads its own output	Nothing; not a `tested:` entry at all / cannot raise tier or `k_state`
I1 — Same model, new session	Fresh conversation, identical model	L1 “2nd reader” only / cannot raise past L2+ DVP minimum, K1 floor
I2 — Same vendor, other model	Different model, same vendor	One DVP route toward L2’s ≥ 3 minimum / cannot raise K1 floor alone (I3 or the bounded exception required)
I3 — Different vendor	Materially different model family/vendor — the founder-set minimum bar for K1 public-provisional, but a minimum, never sufficient on its own	`finite_diagnostic/D` where routes reproduce a mechanical result / never K2, never `Th_coqc` alone
I4 — Mechanical / original-record	Proof kernel, executed test, schema validator, direct original-source lookup	`Th_coqc` or `finite_diagnostic`, paired with I2 under the bounded exception / cannot raise K-state alone
I5 — Independent external human (non-founder)	A person outside this session’s authorship	The only route to K2, and with formal/empirical constraint, K3

K-state: *“K0 Private Candidate → K1 Public Provisional (timestamped, citable, explicitly not peer reviewed) → K2 Socially Stress-Tested (K1 + substantive external human friction) → K3 Reviewed/Empirically Constrained (K2 + certification/replication as appropriate).”*

Non-collapse rows actually cited, by id, somewhere in this book’s chapters. Method: every chapter’s extracted text (via `pdftotext` of the compiled book) was searched mechanically for the pattern `NC-[0-9]+`; the four ids below are every match found (NC-78 and NC-79 in Chapter 18; NC-31 in Chapter 41’s typeset excerpt and NC-34 in Chapter 41’s own headnote, which quotes the source’s status line). This is a lower bound — a chapter may rely on a non-collapse distinction in its own words without citing the id — not a claim that only three of the table’s sixty-seven rows are relevant to this book. *Corrected 2026-09-06 (apparatus-fixer pass):* an earlier pass reported six ids (adding NC-47, NC-58, NC-59, NC-77, attributed to three chapter positions under an earlier numbering, one of them the provenance-audit work, then withheld, and one of them the glosa methodology chapter); those four were artifacts of the glosa methodology chapter previously being bound, wrongly, to an unrelated internal note (see this book’s own build history) rather than to the FOUNDATION_v0.6.md excerpt now printed there, and the provenance-audit work was withheld from that draft of this edition (it is no longer withheld: see Chapter 36) so its own citations did not appear in this book’s text at that time. Source: glosa `methodology/data/non_collapse_table.json`, distributed with the glosa methodology record above; the table itself does not carry a separate DOI.

Id	Pair	Statement
NC-34	No independent check $\not\Rightarrow$ K2 claim	No independent check $\Rightarrow$ no K2 claim; no independent check $\Rightarrow$ no release (MC-05).
NC-31	ManyModels $\not\Rightarrow$ Independence	Calling many AI models is not the same as achieving independence — if they share training data, retrieval corpora, or operator framing, agreement is a weak signal.
NC-78	Potential $\neq$ Exercised $\neq$ Observed	Three readouts of one agent must never be merged: the envelope of what the agent could do over its finite retained option set (potential, p^*), the policy it actually used (exercised, π_{used}), and what an evaluator’s instrument reads of either (observed, O_q). Each is a lossy readout of a different latent (Genesis N2: $M_A = K_A \cdot \theta + \eta$, $\varepsilon_{tot} > 0$, applied twice). Only the witnessed lower bound of potential is ever computable; a sup over unretained options is not a readout.
NC-79	Diagnosis of compression $\neq$ Attribution of responsibility	That an agent’s feasible life-space/capability set has been compressed is diagnosed from an obstruction ledger (Genesis N3/VI.6 obstruction certificates: what was present at the source equals what survived plus what was obstructed at each step) and from cost asymmetry (Causal Ethics CE-30: “only asymmetry,” no intent required). Who is responsible is a separate question answered only by a chooser (CE Axiom II/CE-08: cost attaches to the agency that selected the regime, “not the carrier of consequences”). A diagnosis needs no responsible party; an attribution needs a diagnosis plus a chooser. Neither substitutes for the other.

Appendix C — Symbol Glossary

From *Master Equation River: A Provenance Audit Across the Human–AI Readout Programme* (Ch. 36, v1.5-english, DOI `10.5281/zenodo.22550491`, its own appendix A). This work was withheld from earlier drafts of this edition pending an English-only version (Edition 1 page, prior printings; Appendix A, Part 7); its v1.5 English edition is now bound as Chapter 36 itself, so this appendix's symbol table is a convenience reprint of that chapter's own appendix, cited to its DOI, because several other chapters of this book (Ch. 8, Ch. 9, Ch. 17, Ch. 18, Ch. 31) share the same notation. Per this book's English-only rule, the Thai-script words inside the table's "Meaning" column are mechanically dropped, leaving the English technical vocabulary the source table already carries alongside them — the same mechanical rule this book's own build script applies to bilingual chapter titles. No wording is translated or added. The right-hand column gives the source paper's own internal reference number; its reference list is not reproduced here.

Symbol	**Meaning**	**Ref.**
x_n	phenomenon / world-side encounter	[4],[8]
$r_n = R_H(\cdot)$	finite human readout	[4]
H_n	retained human state (history-bearing operator)	[3],[8]
c_n, Q_n	context; criterion/question	[4],[5]
μ_n, Ψ_H	meaning-giving relation and its operator	[8]
E_n, Φ_E	experience and experience operator	[4] eq. 14
γ_n^{μ}	meaning strength (renamed from an earlier κ_n)	§4.1
ℓ_n, L_H	lexical naming / articulation operator; L_H also used as transport of $H_t \to Q_t$	[7],[8]
U_H, Retain	state-update and selective retention	[3]
$M_{H,n}, I_{H,n}, C_H$	retained memory organization; retrieved internal organization; congruence measure	[8]
T_n, B_n, Ω	encounter history; boundary/structure parameters; rhythm operator	[8]
$m_t, a_{Q,t}, c_{A,t,Q}, \xi_t$	recent-path momentum; attraction; switching cost; residual	[5]
κ_{sem}	semantic-route accessibility	[5] §4.1
$V_{sem}, W_{info}, \Delta V^{\dagger}$	semantic potential; informational work; activation barrier (not physical energy)	[4],[8]
$\Pi_{phys}, \Pi_{feas}, \Pi_{live}, \Pi_{wit}$	physically possible / structurally feasible / experientially live / witnessed policy sets	[10],[9]
$\lambda_{live}, L_{A,t}(g), \tau_{live}$	live-policy accessibility; Live Possibility Field; live threshold	[10]
g, A, z, h, T, B, P	task; agent; conditions; history; horizon; budget; declared model	[9]
$Read_g, D_g, X_g, F_g$	read / actor-directed decision / causal enactment / feedback-objection events	[9]
$p^*_{A,g}, J_{feas}, D_L$	potential envelope; feasible intervention set; live-field distance	[9],[10]
$Y_{obs} = O_q(H_{0:T})$	evaluator's instrument readout of the trajectory	[9],[10]
$Q_t, Y_t, E^{AI}_{H,t}$	prompt; AI output; the human's experience of that output	[6],[8]
$Z_{dlg}, u_H, u_{AI}, \chi_{recip}, D_{recip}, \Sigma$	dialogue state; human/AI inputs; reciprocal-lineage ratio; reciprocal descendants; all descendants	[6]
$\eta_s, \Omega^H_s, B_s, A_s, d_H$	session retention gate; human operator state; rank-restricted candidate update; state dimension	[6] gate; §3.11 proposal
$Y_{return}, RET, AUG, SYN, J^*_s$	unaided return battery; return / augmentation / synergy estimands; three-axis outcome	[6] Repair 7, 9
$K_{like}, K_{validated}$	candidate (knowledge-like) status vs. validated status	[1],[6]
Δ_s, G_s, T_s, P_H	expansion–tunnel balance; gain; tunnel; human performance	[6]
H_{return} (renamed from an earlier R_H in the CTSA chapter), $\langle G_{CTSA}, L, M, P, W, \Delta_{dir} \rangle$	Human Return audit record	[11] §4.4

Appendix D — Falsifier Index, by Part

Each line is quoted or closely paraphrased from the chapter's own headnote, which in turn quotes the chapter's own stated falsifier verbatim; where a chapter states none, that is recorded plainly rather than left blank.

Part 1 — Root: Before Meaning

- **Ch. 1** (`10.5281/zenodo.22301202`): No explicit falsifier stated in the chapter.
- **Ch. 2** (`10.5281/zenodo.22301318`): *"A counterexample would require a target-sensitive route not represented in K^* that nevertheless always passes the access audit without being absorbable into the declared architecture."*
- **Ch. 3** (`10.5281/zenodo.21529456`): No explicit falsifier stated in the chapter.
- **Ch. 4** (`10.5281/zenodo.18925129`): No explicit falsifier stated in the chapter.
- **Ch. 5** (`10.5281/zenodo.19205869`): No explicit falsifier stated in the chapter; it instead lists what "the framework does not claim" (§10).
- **Ch. 6** (`10.5281/zenodo.19640361`): *"the theory's universal falsification condition is stated explicitly: any demonstration that $\varepsilon = 0$ is achievable collapses the entire architecture."*
- **Ch. 7** (`10.5281/zenodo.19176260`): No explicit falsifier stated in the chapter.

Part 2 — Meaning, Experience, Retention

- **Ch. 8** (`10.5281/zenodo.22410666`): Legitimate defeater, verbatim: "pre-nominative measures add no explanatory or predictive information beyond arousal and explicit labeling." Five further hypotheses (H1–H5) are tagged *[Open]* in the chapter's own text.
- **Ch. 9** (`10.5281/zenodo.22357744`): Legitimate defeater, verbatim: "experience is fully predicted by stimulus features and arousal with no added role for reader history, context, or significance."
- **Ch. 10** (`10.5281/zenodo.21425420`): Verbatim, under "9.8 Hard falsifiers": "F1. Durable event-level changes repeatedly have rank above the preregistered active-field bound." "F2. Full-rank or graph-rewiring models consistently outperform low-rank models in bounded learning tasks."
- **Ch. 11** (`10.5281/zenodo.22341297`): Verbatim, under "24.3 What Would Disconfirm the Method": "The method's central claim would be disconfirmed if, in an adequately powered, time-matched design, the method group showed no advantage over the yoked control on protocol-independent transfer and calibration at delay. . . "
- **Ch. 12** (`10.5281/zenodo.18711408`): Verbatim, under "Falsifiable Predictions": "In AI-assisted learning environments, rejection accuracy (correctly discarding implausible explanations) will predict expertise more strongly than recall or production accuracy."

Part 3 — Possibility, Choice, Potential

- **Ch. 13** (`10.5281/zenodo.22307148`): "If increasing attraction never produces measurable alternative suppression once current warrant and knowledge are controlled, P1 weakens."
- **Ch. 14** (`10.5281/zenodo.22307561`): "H1 falsifier. Across well-powered content-matched tasks, organization metrics or manipulations add no out-of-sample predictive value for first-passage time after knowledge amount and generic retrieval/control are modeled."
- **Ch. 15** (`10.5281/zenodo.22307564`): No explicit falsifier stated in the chapter.
- **Ch. 16** (`10.5281/zenodo.22308072`): "If a recursive human-AI system produces thousands of outputs but almost no new usable readout classes, then k_{epi} is low and the claimed 'amplification' is mostly verbosity."
- **Ch. 17** (`10.5281/zenodo.22357788`): "If resource-only models predict equally well, the meaning layer should be narrowed."
- **Ch. 18** (`10.5281/zenodo.22361830`): "P-D (layer 2 dominates the structural case). In cases Violence as Instability classifies as structural (§6.2), the gap closable by layer-2 interventions exceeds the gap closable by layer-1 ones. Defeated if information-only or within-C interventions close as much."

Part 4 — Society, Power, Justice: Potential Under Compression

- **Ch. 19** (`10.5281/zenodo.18925131`): No explicit falsifier stated in the chapter.
- **Ch. 20** (`10.5281/zenodo.18383439`): No explicit falsifier stated in the chapter.

- **Ch. 21** (`10.5281/zenodo.18444260`): "Causal Ethics adopts a strict falsification rule"; the chapter states that once its failure count k reaches the critical threshold k_{crit}, at least one of the regime model, the cost functional, or the assumed proxies must be revised, and that "CE does not permit ad hoc reinterpretation to avoid revision."
- **Ch. 22** (`10.5281/zenodo.18897585`): No explicit falsifier stated in the chapter.
- **Ch. 23** (`10.5281/zenodo.18943971`): No explicit falsifier stated in the chapter.

Part 5 — Human–AI Coupling and Return

- **Ch. 24** (`10.5281/zenodo.22456414`): P1 [Open]: "A longitudinal matched-cohort comparison of youth beginning structured practice-based/income-generating activity at 13, against controls who do not, showing no difference in retained self-regulation competence or in school-to-work mental-health and economic outcomes, refutes the proposal as stated."
- **Ch. 25** (`10.5281/zenodo.22456487`): "An organization's LLM-mediated processing of linguistic traces shows no traceable residue in later organizational readout, policy, or practice… the claim then reduces to ordinary information throughput with no retention-level effect."
- **Ch. 26** (`10.5281/zenodo.22456564`): "If a study finds that reflective-versus-superficial outcome quality is fully explained by a simpler rival model — for instance, prior domain knowledge and time-on-task alone, without any distinct reader-conditioned meaning-giving step — the mirror-of-thought thesis as stated here should be compressed into that simpler model rather than defended as a separate construct."
- **Ch. 27** (`10.5281/zenodo.19215748`): No explicit falsifier stated in the chapter.
- **Ch. 28** (`10.5281/zenodo.17280546`): No explicit falsifier stated in the chapter.
- **Ch. 29** (`10.5281/zenodo.18517054`): No explicit falsifier stated in the chapter.
- **Ch. 30** (`10.5281/zenodo.22331922`): "Human Return is defeated as an augmentation criterion only where durable human capability is outside the declared goal, or where delayed unaided measures add no information beyond in-session performance."
- **Ch. 31** (`10.5281/zenodo.22339909`): "CTSA should be narrowed, merged, or abandoned if (i) independent coders cannot classify the four return classes reliably; [. . .] (iii) delayed CTSA profiles fail to predict or explain anything beyond ordinary performance and transfer; [. . .] or (v) a simpler rival architecture captures the same phenomena with fewer assumptions."
- **Ch. 32** (`10.5281/zenodo.22481928`): "The protocol should be simplified if: (i) ordering the stages adds no value beyond generic 'think critically' instructions; [. . .] or (ix) action/world-feedback closure adds no predictive or calibration value beyond Human Return alone."
- **Ch. 33** (`10.5281/zenodo.22498047`): "barrier categories cannot be coded reliably across independent raters or add no intervention value; [. . .] proactive HCA produces no delayed Return or transfer advantage over reactive DCP at comparable burden."

Part 6 — Human Position After Labour

- **Ch. 34** (`10.5281/zenodo.22481924`): Table 3, "Gate vs concentration" row: "Counterfactual removal/substitution predicts transition failure beyond HHI or top-share measures… Remove Gconv if concentration/interoperability fully explain outcomes."
- **Ch. 35** (`10.5281/zenodo.22481926`): "The Human Conversion architecture should be narrowed or removed under the following results: 1. delayed unaided performance is fully predicted by ordinary skill measures and never dissociates from assisted performance; [. . .] 6. the vector architecture does not improve prediction, diagnosis, or intervention compared with simpler rivals."

Part 7 — The River and Its Tests

- **Ch. 36** (`10.5281/zenodo.22550491`): "It does not claim that the whole river is a universal psychological law or a single physics equation. It functions instead as a typed, multi-level dependency architecture…" Segments it tiers Open (Appendix F) are, in its own words, "proposals that remain open."
- **Ch. 37** (`10.5281/zenodo.22424434`): "H2 — History residual [OPEN]: Prior traversal history will predict later route accessibility after current semantic state, task, and control variables are matched. Failure removes the distinct momentum term."
- **Ch. 38** (`10.5281/zenodo.22308066`): "DECISIVE TEST: a record that would materially confirm, narrow, or falsify the open claim" (applied per claim in the chapter's own registry, Table 1).

Part 8 — Method: Rigour Without Infrastructure

- **Ch. 39** (`10.5281/zenodo.22163849`): "At 36 months, no peer-reviewed anchor, no repeat human node, and no external uptake constitute defeat of the current configuration, not defeat of scholarship itself."
- **Ch. 40** (`10.5281/zenodo.22307841`): H1: "a claim card that passes every field the kernel can mechanically check is nonetheless found by a genuine independent reviewer (I5) to have a false answer at Q2 or Q3." Two further named falsifiers (H2, H3) are stated in the chapter's Section 5.
- **Ch. 41** (`10.5281/zenodo.22340255`): No explicit falsifier stated in the chapter.

Appendix E — Build Record

What was measured for this assembly, and by what, rather than asserted:

- **Chapter count.** 41 chapters, deposited across 8 parts (40 of them `\includepdf`'d from their own source PDF; Chapter 41 is instead a hand-typeset, verbatim-quotation excerpt of its own cited source, since the deposited record carries no matching standalone PDF — see that chapter's headnote and manifest note). Plus 2 named-but-unwritten planned papers (Appendix A, Appendix D). *Corrected 2026-09-06 (apparatus-fixer pass):* an earlier pass counted 37 chapters; that count included, at a chapter position later withheld (then the *Master Equation River* work, see below), and treated the chapter position later occupied by the glosa methodology chapter as a full-PDF chapter when the file actually bound there was an unrelated internal note wrongly matched by the build script's include-type fallback (now fixed). *Further corrected 2026-09-06 (headnote/bridge pass):* a new Part 6, *Human Position After Labour*, was inserted between the part formerly ending at Chapter 31 and the part beginning at Chapter 32, adding three chapters and shifting every later chapter number upward by three. *Further corrected 2026-09-06 (third apparatus-fixer pass):* Chapter 33, From Assistance to Human Capability, was inserted into Part 5, adding one chapter. *Included 2026-09-06 (English-edition pass): Master Equation River*, previously withheld pending an English-only edition, is now bound as Chapter 36 (v1.5, English), opening Part 7; the chapter count below is re-measured against the current 41-chapter, 41-file manifest to include it.
- **Page count.** 41 files were downloaded and measured (`pdfinfo`, per-file counts in `manifest.yaml`), summing to 668 pages; of those, 554 pages across 40 source PDFs are actually `\includepdf`'d into this book. The 114 pages excluded from that figure are: Chapter 41's 15 pages (its downloaded file, kept only for provenance, is superseded by the typeset excerpt in `frontmatter/excerpt_22340255.tex` rather than being bound) and the pages trimmed by this book's two partial-excerpt chapters (Ch. 21, pp. 13–34 and 73–77 of 99, 72 pages excluded; Ch. 22, pp. 13–20 and 23–29 of 42, 27 pages excluded). This figure also excludes the front matter, part openers, appendices, and Chapter 41's typeset excerpt that this build adds, whose combined effect is fixed only by the compiled output below. *Re-measured 2026-09-06 (English-edition pass)* against the current 41-file manifest with *Master Equation River* bound at its v1.5 English page count (24 pages), superseding the earlier withheld-record figures this bullet previously carried.
- **Page size.** A mix of US Letter (612×792 pt) and A4 (595–596×842 pt) source pages; every `\includepdf`'d chapter is included via `pdfpages` at a per-file scale factor computed from its own measured page size (range 0.9727–1.0), so no source page is cropped.
- **Checksums.** A sha256 was computed locally for all 41 downloaded files and compared against the md5 checksum Zenodo's own files API reports for each; all 41 matched. This is an md5 match, not an independent sha256 verification by Zenodo, which does not publish sha256 for these records. One of the 41 (Chapter 41's downloaded file) is retained in `manifest.yaml` as a provenance record only and is not bound into the compiled book (see above).
- **Font embedding.** 40 of the 41 downloaded source PDFs have fully embedded fonts (`pdffonts`). One, Chapter 28, does not; re-embedding before final typesetting is an open item, not yet done as of this front-matter pass.
- **Independent review.** No independent (maker-checker) review pass has yet run on this front matter, on the manifest, or on the assembled book as a whole. Per this workspace's own release rule, no independent check means no release; this collection is not yet released and is not to be treated as such until that pass runs.
- **Zenodo file-list verification.** Record `10.5281/zenodo.22340255`'s live files API was queried directly (2026-09-06) to confirm which files it carries, as part of resolving the Chapter 41 binding correction above; the result (one internal uplift note, one file byte-identical to Chapter 37's, one source tarball) is recorded in that chapter's manifest note rather than only asserted here.
- **Equation registry and Th_coqc status.** Appendix F is generated mechanically at every build from the registry of *Toledo*, the programme's equation library (git tag v1.0.0, DOI `10.5281/zenodo.22548770`): for every bound chapter, each numbered equation with its Toledo code, root, tier and Coq status; the row count printed there is computed at build time. The Master Equation River's own Coq ledger (eq. 1–79 of Chapter 36) is part of that registry.
- **This build.** `build/assemble.py`, run against this version of the front matter and manifest (2026-09-06, English-edition pass, *Master Equation River* included), reported **0 pdflatex errors** and a compiled book of **657 pages**. This page count includes the front matter, part openers, headnotes, appendices (A–F), and both covers this pass adds on top of the 550 measured, actually-bound chapter pages; it is a readout of this one build, not a promised final length, since content still pending (Appendix A's 1 remaining planned paper in Part 5, 1 planned paper in Part 7, license and BBL-id mapping work named in the project's own meeting record) can change it.

Appendix F — Toledo Equation Registry

Every equation cited by this book is registered in *Toledo*, the equation registry spanning the Human–AI Readout Programme (source `https://github.com/morrocwi/toledo`, git tag `v1.0.0`, DOI `10.5281/zenodo.22548770`). **This table replaces the appendix's earlier Master Equation River-only Th_coqc ledger**: every equation of that ledger is already registered here, canonicalised under the record id of the chapter that originally wrote it rather than under the chapter that later audited it, so Ch. 36 (*Master Equation River*) is one of the chapters below for which Toledo lists no directly-keyed row — its equations are counted under their origin chapters instead, not lost. For each bound chapter of this book, every row below is one raw equation occurrence Toledo's own `registry/CANONICAL.json` (field `raw_to_canonical`) keys under that chapter's record id: the label as printed in the source paper, the Toledo code it canonicalises to, that code's root, its tier, and its Coq status, each read directly off Toledo's matching registry entry. Four tiers appear below, unchanged from Toledo's own tagging: **Th_coqc** (a lemma closed under `Print Assumptions` in a stated finite model), **Definition** (typed exactly as the source states it; no proof obligation), **Dr** (a working, author-asserted thesis, not machine-checked at this tier), and **Open** (a stated Coq `Prop`, not proved). A closed Coq status certifies only the internal consistency of the finite model it was checked in; it never certifies that a paper's own empirical or Dr-tier claim is true, and it is not raised or lowered by any neighbouring row. Every count in this appendix, including the row and chapter counts in its closing line, is a readout of `registry/CANONICAL.json` computed at build time, not a fixed constant. A bound chapter Toledo lists no row for is named as such below, never omitted.

Chapter	Label	Toledo code	Root	Tier	Coq status
Ch. 1	`(RA)`	`EQ-002/E.02.v1`	`EQ-002`	Definition	Closed under the global context
Ch. 1	`(RPE)`	`A.8/E.02.v1`	`A.8`	Definition	Closed under the global context
Ch. 1	`(dCr)`	`A.8/E.02.v1`	`A.8`	Definition	Closed under the global context
Ch. 1	`(m-rho)`	`EQ-002/E.02.v1`	`EQ-002`	Definition	Closed under the global context
Ch. 1	`Def-1`	`weld/E.02.v1`	`weld`	Definition	Closed under the global context
Ch. 1	`Def-2`	`weld/E.02.v1`	`weld`	Definition	Closed under the global context
Ch. 1	`Def-3`	`weld/E.02.v1`	`weld`	Definition	Closed under the global context
Ch. 1	`Def-4`	`A.8/E.02.v1`	`A.8`	Definition	Closed under the global context
Ch. 1	`HSC`	`weld/E.02.v1`	`weld`	Definition	Closed under the global context
Ch. 1	`Prin-1`	`A.5/E.04.v1`	`A.5`	Dr	Closed under the global context
Ch. 1	`Prin-2`	`weld/E.10.v1`	`weld`	Dr	Closed under the global context
Ch. 1	`Prin-3`	`EQ-015/E.14.v1`	`EQ-015`	Dr	Closed under the global context
Ch. 1	`Prin-4`	`A.8/E.03.v1`	`A.8`	Dr	Closed under the global context
Ch. 1	`Prin-5`	`EQ-002/E.12.v1`	`EQ-002`	Dr	Closed under the global context
Ch. 1	`Prop-1`	`weld/E.02.v1`	`weld`	Definition	Closed under the global context
Ch. 1	`SC`	`weld/E.02.v1`	`weld`	Definition	Closed under the global context
Ch. 2	`(1)`	`A.8/E.01.v1`	`A.8`	Definition	Closed under the global context
Ch. 2	`(10)`	`EQ-002/M.01.v1`	`EQ-002`	Definition	Closed under the global context
Ch. 2	`(11)`	`EQ-002/M.01.v1`	`EQ-002`	Definition	Closed under the global context
Ch. 2	`(12)`	`EQ-002/M.01.v1`	`EQ-002`	Definition	Closed under the global context
Ch. 2	`(13)`	`EQ-002/M.01.v1`	`EQ-002`	Definition	Closed under the global context
Ch. 2	`(14)`	`EQ-002/M.01.v1`	`EQ-002`	Definition	Closed under the global context
Ch. 2	`(15)`	`weld/M.13.v1`	`weld`	Definition	Closed under the global context
Ch. 2	`(16)`	`weld/M.13.v1`	`weld`	Definition	Closed under the global context
Ch. 2	`(17)`	`A.5/M.05.v1`	`A.5`	Definition	Closed under the global context

Chapter	Label	Toledo code	Root	Tier	Coq status
Ch. 2	(18)	A.5/M.05.v1	A.5	Definition	Closed under the global context
Ch. 2	(19)	EQ-002/M.01.v1	EQ-002	Definition	Closed under the global context
Ch. 2	(2)	A.5/E.05.v1	A.5	Dr	Closed under the global context
Ch. 2	(20)	EQ-002/M.01.v1	EQ-002	Definition	Closed under the global context
Ch. 2	(21)	EQ-002/M.01.v1	EQ-002	Definition	Closed under the global context
Ch. 2	(22)	EQ-002/M.01.v1	EQ-002	Definition	Closed under the global context
Ch. 2	(23)	A.8/M.19.v1	A.8	Definition	Closed under the global context
Ch. 2	(24)	A.8/M.19.v1	A.8	Definition	Closed under the global context
Ch. 2	(25)	weld/M.14.v1	weld	Dr	Closed under the global context
Ch. 2	(26)	A.8/E.01.v1	A.8	Definition	Closed under the global context
Ch. 2	(27)	A.8/E.01.v1	A.8	Definition	Closed under the global context
Ch. 2	(28)	A.8/E.01.v1	A.8	Definition	Closed under the global context
Ch. 2	(29)	A.8/E.01.v1	A.8	Definition	Closed under the global context
Ch. 2	(3)	A.5/E.03.v1	A.5	Definition	Closed under the global context
Ch. 2	(30)	A.8/E.01.v1	A.8	Definition	Closed under the global context
Ch. 2	(31)	EQ-002/M.01.v1	EQ-002	Definition	Closed under the global context
Ch. 2	(32)	EQ-002/M.01.v1	EQ-002	Definition	Closed under the global context
Ch. 2	(4)	A.5/E.03.v1	A.5	Definition	Closed under the global context
Ch. 2	(5)	EQ-002/M.01.v1	EQ-002	Definition	Closed under the global context
Ch. 2	(6)	EQ-002/M.01.v1	EQ-002	Definition	Closed under the global context
Ch. 2	(7)	EQ-002/M.01.v1	EQ-002	Definition	Closed under the global context
Ch. 2	(8)	EQ-002/M.01.v1	EQ-002	Definition	Closed under the global context
Ch. 2	(9)	EQ-002/M.01.v1	EQ-002	Definition	Closed under the global context
Ch. 2	Cor-1	A.5/M.05.v1	A.5	Definition	Closed under the global context
Ch. 2	Def-1	EQ-002/M.01.v1	EQ-002	Definition	Closed under the global context
Ch. 2	Def-2	EQ-015/M.11.v1	EQ-015	Definition	Closed under the global context
Ch. 2	Def-3	EQ-015/M.11.v1	EQ-015	Definition	Closed under the global context
Ch. 2	Def-4	EQ-015/M.11.v1	EQ-015	Definition	Closed under the global context
Ch. 2	Def-5	EQ-015/M.11.v1	EQ-015	Definition	Closed under the global context
Ch. 2	Def-6	A.8/M.18.v1	A.8	Definition	Closed under the global context
Ch. 2	Def-7	A.8/M.18.v1	A.8	Definition	Closed under the global context
Ch. 2	Princ-4	EQ-002/M.01.v1	EQ-002	Definition	Closed under the global context
Ch. 2	Princ-7	A.8/M.17.v1	A.8	Dr	Closed under the global context
Ch. 2	Princ-8	A.8/M.17.v1	A.8	Dr	Closed under the global context
Ch. 2	Princ-9	A.8/M.17.v1	A.8	Dr	Closed under the global context
Ch. 2	Prop-1	EQ-002/M.01.v1	EQ-002	Definition	Closed under the global context
Ch. 2	Prop-2	EQ-002/M.01.v1	EQ-002	Definition	Closed under the global context
Ch. 2	Prop-4	weld/M.14.v1	weld	Dr	Closed under the global context

Chapter	Label	Toledo code	Root	Tier	Coq status
Ch. 3	(1)	weld/M.01.v1	weld	Th_coqc	Closed under the global context
Ch. 3	(10)	A.5/M.06.v1	A.5	finite_diagnostic	Closed under the global context
Ch. 3	(11)	EQ-002/E.09.v1	EQ-002	Definition	Closed under the global context
Ch. 3	(12)	A.5/M.06.v1	A.5	finite_diagnostic	Closed under the global context
Ch. 3	(13)	EQ-015/E.01.v1	EQ-015	Definition	Closed under the global context
Ch. 3	(14)	EQ-015/E.04.v1	EQ-015	Definition	Closed under the global context
Ch. 3	(15)	EQ-015/E.08.v1	EQ-015	Definition	Closed under the global context
Ch. 3	(16)	A.8/M.01.v1	A.8	Dr	Closed under the global context
Ch. 3	(17)	weld/S.43.v1	weld	Definition	Closed under the global context
Ch. 3	(18)	weld/S.44.v1	weld	Definition	Closed under the global context
Ch. 3	(19)	weld/S.45.v1	weld	Definition	Closed under the global context
Ch. 3	(2)	weld/M.02.v1	weld	Definition	Closed under the global context
Ch. 3	(20)	weld/S.46.v1	weld	Definition	Closed under the global context
Ch. 3	(21)	A.5/M.06.v1	A.5	finite_diagnostic	Closed under the global context
Ch. 3	(22)	weld/E.03.v1	weld	Definition	Closed under the global context
Ch. 3	(23)	weld/E.03.v1	weld	Definition	Closed under the global context
Ch. 3	(24)	A.5/M.06.v1	A.5	finite_diagnostic	Closed under the global context
Ch. 3	(25)	weld/E.04.v1	weld	Definition	Closed under the global context
Ch. 3	(26)	A.5/M.06.v1	A.5	finite_diagnostic	Closed under the global context
Ch. 3	(27)	weld/E.05.v1	weld	Definition	Closed under the global context
Ch. 3	(28)	weld/E.06.v1	weld	Definition	Closed under the global context
Ch. 3	(29)	weld/E.06.v1	weld	Definition	Closed under the global context
Ch. 3	(3)	weld/M.01.v1	weld	Th_coqc	Closed under the global context
Ch. 3	(30)	weld/E.07.v1	weld	Dr	Closed under the global context
Ch. 3	(31)	weld/E.03.v1	weld	Definition	Closed under the global context
Ch. 3	(32)	weld/E.03.v1	weld	Definition	Closed under the global context
Ch. 3	(33)	weld/E.08.v1	weld	Definition	Closed under the global context
Ch. 3	(34)	weld/E.08.v1	weld	Definition	Closed under the global context
Ch. 3	(35)	A.5/M.06.v1	A.5	finite_diagnostic	Closed under the global context
Ch. 3	(36)	EQ-002/H.02.v1	EQ-002	Definition	Closed under the global context
Ch. 3	(37)	EQ-002/H.02.v1	EQ-002	Definition	Closed under the global context
Ch. 3	(38)	EQ-002/H.02.v1	EQ-002	Definition	Closed under the global context
Ch. 3	(39)	EQ-002/H.01.v1	EQ-002	Definition	Closed under the global context
Ch. 3	(4)	weld/M.01.v1	weld	Th_coqc	Closed under the global context
Ch. 3	(40)	EQ-002/H.01.v1	EQ-002	Definition	Closed under the global context
Ch. 3	(41)	EQ-002/H.01.v1	EQ-002	Definition	Closed under the global context
Ch. 3	(42)	EQ-002/H.01.v1	EQ-002	Definition	Closed under the global context
Ch. 3	(43)	EQ-002/H.01.v1	EQ-002	Definition	Closed under the global context

Chapter	Label	Toledo code	Root	Tier	Coq status
Ch. 3	(44)	EQ-015/E.02.v1	EQ-015	Definition	Closed under the global context
Ch. 3	(45)	EQ-015/E.02.v1	EQ-015	Definition	Closed under the global context
Ch. 3	(46)	EQ-015/E.02.v1	EQ-015	Definition	Closed under the global context
Ch. 3	(47)	weld/H.01.v1	weld	Definition	Closed under the global context
Ch. 3	(48)	A.5/M.06.v1	A.5	finite_diagnostic	Closed under the global context
Ch. 3	(49)	weld/H.01.v1	weld	Definition	Closed under the global context
Ch. 3	(5)	weld/M.01.v1	weld	Th_coqc	Closed under the global context
Ch. 3	(50)	weld/H.02.v1	weld	Definition	Closed under the global context
Ch. 3	(51)	weld/H.02.v1	weld	Definition	Closed under the global context
Ch. 3	(52)	weld/H.02.v1	weld	Definition	Closed under the global context
Ch. 3	(53)	weld/H.02.v1	weld	Definition	Closed under the global context
Ch. 3	(54)	weld/H.02.v1	weld	Definition	Closed under the global context
Ch. 3	(55)	weld/H.02.v1	weld	Definition	Closed under the global context
Ch. 3	(56)	weld/E.03.v1	weld	Definition	Closed under the global context
Ch. 3	(57)	weld/E.03.v1	weld	Definition	Closed under the global context
Ch. 3	(58)	weld/E.03.v1	weld	Definition	Closed under the global context
Ch. 3	(59)	weld/E.03.v1	weld	Definition	Closed under the global context
Ch. 3	(6)	weld/M.02.v1	weld	Definition	Closed under the global context
Ch. 3	(60)	weld/E.03.v1	weld	Definition	Closed under the global context
Ch. 3	(61)	weld/E.03.v1	weld	Definition	Closed under the global context
Ch. 3	(62)	weld/E.03.v1	weld	Definition	Closed under the global context
Ch. 3	(63)	A.5/H.03.v1	A.5	Definition	Closed under the global context
Ch. 3	(64)	A.5/H.03.v1	A.5	Definition	Closed under the global context
Ch. 3	(65)	A.5/H.03.v1	A.5	Definition	Closed under the global context
Ch. 3	(66)	A.5/H.03.v1	A.5	Definition	Closed under the global context
Ch. 3	(67)	weld/H.03.v1	weld	finite_diagnostic	Closed under the global context
Ch. 3	(68)	weld/H.03.v1	weld	finite_diagnostic	Closed under the global context
Ch. 3	(69)	weld/H.03.v1	weld	finite_diagnostic	Closed under the global context
Ch. 3	(7)	weld/M.02.v1	weld	Definition	Closed under the global context
Ch. 3	(70)	A.5/M.06.v1	A.5	finite_diagnostic	Closed under the global context
Ch. 3	(71)	weld/H.03.v1	weld	finite_diagnostic	Closed under the global context
Ch. 3	(72)	weld/H.03.v1	weld	finite_diagnostic	Closed under the global context
Ch. 3	(73)	weld/H.03.v1	weld	finite_diagnostic	Closed under the global context
Ch. 3	(74)	weld/H.03.v1	weld	finite_diagnostic	Closed under the global context
Ch. 3	(75)	weld/H.03.v1	weld	finite_diagnostic	Closed under the global context
Ch. 3	(76)	weld/H.03.v1	weld	finite_diagnostic	Closed under the global context
Ch. 3	(77)	weld/H.03.v1	weld	finite_diagnostic	Closed under the global context
Ch. 3	(78)	weld/H.03.v1	weld	finite_diagnostic	Closed under the global context

Chapter	Label	Toledo code	Root	Tier	Coq status
Ch. 3	(79)	weld/H.03.v1	weld	finite_diagnostic	Closed under the global context
Ch. 3	(8)	weld/M.03.v1	weld	Definition	Closed under the global context
Ch. 3	(80)	weld/H.03.v1	weld	finite_diagnostic	Closed under the global context
Ch. 3	(81)	weld/H.03.v1	weld	finite_diagnostic	Closed under the global context
Ch. 3	(82)	weld/H.03.v1	weld	finite_diagnostic	Closed under the global context
Ch. 3	(83)	weld/H.03.v1	weld	finite_diagnostic	Closed under the global context
Ch. 3	(84)	weld/H.03.v1	weld	finite_diagnostic	Closed under the global context
Ch. 3	(85)	A.5/M.06.v1	A.5	finite_diagnostic	Closed under the global context
Ch. 3	(86)	weld/H.03.v1	weld	finite_diagnostic	Closed under the global context
Ch. 3	(87)	weld/H.05.v1	weld	Definition	Closed under the global context
Ch. 3	(88)	weld/H.05.v1	weld	Definition	Closed under the global context
Ch. 3	(9)	EQ-015/M.03.v1	EQ-015	Dr	Closed under the global context
Ch. 3	Def-1	weld/E.03.v1	weld	Definition	Closed under the global context
Ch. 3	Prop-1	weld/S.04.v1	weld	Definition	Closed under the global context
Ch. 3	Prop-2	weld/E.07.v1	weld	Dr	Closed under the global context
Ch. 4	(1)	EQ-002/E.03.v1	EQ-002	Definition	Closed under the global context
Ch. 4	(2)	weld/E.01.v1	weld	Definition	Closed under the global context
Ch. 5	(1)	EQ-002/E.10.v1	EQ-002	Definition	Closed under the global context
Ch. 5	(2)	EQ-002/E.10.v1	EQ-002	Definition	Closed under the global context
Ch. 5	(3)	EQ-015/E.08.v1	EQ-015	Definition	Closed under the global context
Ch. 5	(4)	EQ-002/E.07.v1	EQ-002	Ax	Closed under the global context
Ch. 5	(5)	EQ-015/E.09.v1	EQ-015	Definition	Closed under the global context
Ch. 5	(6)	EQ-015/E.09.v1	EQ-015	Definition	Closed under the global context
Ch. 6	(1)	EQ-002/H.01.v1	EQ-002	Definition	Closed under the global context
Ch. 6	(10)	EQ-015/M.09.v1	EQ-015	Definition	Closed under the global context
Ch. 6	(11)	EQ-015/M.09.v1	EQ-015	Definition	Closed under the global context
Ch. 6	(12)	EQ-002/E.07.v1	EQ-002	Ax	Closed under the global context
Ch. 6	(13)	weld/E.01.v1	weld	Definition	Closed under the global context
Ch. 6	(14)	weld/E.03.v1	weld	Definition	Closed under the global context
Ch. 6	(15)	weld/E.03.v1	weld	Definition	Closed under the global context
Ch. 6	(2)	EQ-002/H.01.v1	EQ-002	Definition	Closed under the global context
Ch. 6	(3)	EQ-002/H.01.v1	EQ-002	Definition	Closed under the global context
Ch. 6	(4)	EQ-002/H.01.v1	EQ-002	Definition	Closed under the global context
Ch. 6	(5)	EQ-002/H.01.v1	EQ-002	Definition	Closed under the global context
Ch. 6	(6)	EQ-002/H.01.v1	EQ-002	Definition	Closed under the global context
Ch. 6	(7)	EQ-002/H.01.v1	EQ-002	Definition	Closed under the global context
Ch. 6	(8)	EQ-002/H.01.v1	EQ-002	Definition	Closed under the global context
Ch. 6	(9)	EQ-015/M.09.v1	EQ-015	Definition	Closed under the global context

Chapter	Label	Toledo code	Root	Tier	Coq status
Ch. 6	`MQ.08`	`weld/M.01.v1`	`weld`	Th_coqc	Closed under the global context
Ch. 6	`P3`	`weld/M.01.v1`	`weld`	Th_coqc	Closed under the global context
Ch. 6	`Th-5`	`weld/M.01.v1`	`weld`	Th_coqc	Closed under the global context
Ch. 7	*No equation is registered under this chapter's own record id in Toledo v1.0.0.*				
Ch. 8	`H1`	`EQ-015/E.05.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 8	`H2`	`EQ-002/E.01.v1`	`EQ-002`	Definition	Closed under the global context
Ch. 8	`H3`	`EQ-015/E.05.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 8	`H4`	`EQ-015/E.08.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 8	`H5`	`EQ-015/E.05.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 8	`H6`	`A.5/E.06.v1`	`A.5`	Open	Closed under the global context
Ch. 8	`MBN-articulation`	`EQ-015/H.01.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 8	`MBN-config`	`EQ-015/E.01.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 8	`MBN-experience`	`EQ-015/E.04.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 8	`MBN-naming`	`EQ-015/E.05.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 8	`MBN-readout`	`EQ-002/E.01.v1`	`EQ-002`	Definition	Closed under the global context
Ch. 8	`MBN-resonance-diagnostic`	`EQ-015/E.06.v1`	`EQ-015`	Dr	Closed under the global context
Ch. 8	`MBN-retention`	`EQ-015/E.08.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 8	`§11-non-collapse-laws`	`A.5/E.06.v1`	`A.5`	Open	Closed under the global context
Ch. 8	`§8-non-collapse-chain`	`A.5/E.06.v1`	`A.5`	Open	Closed under the global context
Ch. 9	`EMG-01`	`EQ-002/E.01.v1`	`EQ-002`	Definition	Closed under the global context
Ch. 9	`EMG-02`	`EQ-015/E.01.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 9	`EMG-03`	`EQ-015/E.01.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 9	`EMG-04`	`EQ-015/E.04.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 9	`EMG-05`	`EQ-015/E.04.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 9	`EMG-06`	`EQ-015/E.05.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 9	`EMG-07`	`EQ-015/E.05.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 9	`EMG-08`	`EQ-015/E.08.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 9	`EMG-09`	`EQ-015/E.06.v1`	`EQ-015`	Dr	Closed under the global context
Ch. 9	`EMG-10`	`EQ-015/E.06.v1`	`EQ-015`	Dr	Closed under the global context
Ch. 9	`EMG-11`	`EQ-015/E.06.v1`	`EQ-015`	Dr	Closed under the global context
Ch. 9	`EMG-12`	`EQ-015/E.06.v1`	`EQ-015`	Dr	Closed under the global context
Ch. 9	`EMG-13`	`EQ-015/M.08.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 9	`EMG-14`	`EQ-015/M.08.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 9	`EMG-15`	`A.5/E.07.v1`	`A.5`	Dr	Closed under the global context
Ch. 9	`EMG-16`	`EQ-015/E.07.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 9	`EMG-17`	`EQ-015/E.07.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 9	`EMG-18`	`EQ-015/E.07.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 9	`EMG-19`	`EQ-015/E.07.v1`	`EQ-015`	Definition	Closed under the global context

Chapter	Label	Toledo code	Root	Tier	Coq status
Ch. 9	EMG-20	A.5/E.07.v1	A.5	Dr	Closed under the global context
Ch. 9	EMG-21	EQ-015/H.01.v1	EQ-015	Definition	Closed under the global context
Ch. 9	EMG-22	EQ-015/H.01.v1	EQ-015	Definition	Closed under the global context
Ch. 9	EMG-23	EQ-015/E.08.v1	EQ-015	Definition	Closed under the global context
Ch. 9	EMG-24	EQ-015/H.01.v1	EQ-015	Definition	Closed under the global context
Ch. 9	EMG-25	weld/H.09.v1	weld	Open	Closed under the global context
Ch. 9	EMG-26	weld/H.09.v1	weld	Open	Closed under the global context
Ch. 9	EMG-27	A.5/E.07.v1	A.5	Dr	Closed under the global context
Ch. 9	EMG-28	weld/M.04.v1	weld	Open	Closed under the global context
Ch. 9	EMG-29	A.5/E.07.v1	A.5	Dr	Closed under the global context
Ch. 10	(1)	weld/H.04.v1	weld	Definition	Closed under the global context
Ch. 10	(10)	weld/E.09.v1	weld	Definition	Closed under the global context
Ch. 10	(11)	weld/H.04.v1	weld	Definition	Closed under the global context
Ch. 10	(12)	EQ-015/E.04.v1	EQ-015	Definition	Closed under the global context
Ch. 10	(13)	EQ-015/E.04.v1	EQ-015	Definition	Closed under the global context
Ch. 10	(14)	EQ-015/E.08.v1	EQ-015	Definition	Closed under the global context
Ch. 10	(15)	EQ-015/E.04.v1	EQ-015	Definition	Closed under the global context
Ch. 10	(16)	EQ-015/E.04.v1	EQ-015	Definition	Closed under the global context
Ch. 10	(17)	EQ-015/E.04.v1	EQ-015	Definition	Closed under the global context
Ch. 10	(18)	EQ-015/H.06.v1	EQ-015	Definition	Closed under the global context
Ch. 10	(19)	EQ-015/H.06.v1	EQ-015	Definition	Closed under the global context
Ch. 10	(2)	weld/H.04.v1	weld	Definition	Closed under the global context
Ch. 10	(20)	EQ-015/H.06.v1	EQ-015	Definition	Closed under the global context
Ch. 10	(21)	EQ-015/H.06.v1	EQ-015	Definition	Closed under the global context
Ch. 10	(22)	EQ-015/H.06.v1	EQ-015	Definition	Closed under the global context
Ch. 10	(23)-(24)	EQ-015/H.06.v1	EQ-015	Definition	Closed under the global context
Ch. 10	(25)	EQ-015/H.06.v1	EQ-015	Definition	Closed under the global context
Ch. 10	(26)	EQ-015/H.06.v1	EQ-015	Definition	Closed under the global context
Ch. 10	(27)	EQ-015/H.06.v1	EQ-015	Definition	Closed under the global context
Ch. 10	(28)	EQ-015/H.06.v1	EQ-015	Definition	Closed under the global context
Ch. 10	(29)	EQ-015/H.06.v1	EQ-015	Definition	Closed under the global context
Ch. 10	(3)	weld/H.04.v1	weld	Definition	Closed under the global context
Ch. 10	(30)	weld/H.04.v1	weld	Definition	Closed under the global context
Ch. 10	(31)	weld/H.02.v1	weld	Definition	Closed under the global context
Ch. 10	(32)	EQ-015/H.06.v1	EQ-015	Definition	Closed under the global context
Ch. 10	(33)	EQ-015/H.06.v1	EQ-015	Definition	Closed under the global context
Ch. 10	(34)	EQ-015/H.06.v1	EQ-015	Definition	Closed under the global context
Ch. 10	(35)	A.5/E.08.v1	A.5	Dr	Closed under the global context

Chapter	Label	Toledo code	Root	Tier	Coq status
Ch. 10	(36)	EQ-015/H.37.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 10	(37)	EQ-015/H.37.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 10	(38)	EQ-015/H.06.v1	EQ-015	Definition	Closed under the global context
Ch. 10	(39)	EQ-002/E.11.v1	EQ-002	Dr	Closed under the global context
Ch. 10	(4)	weld/H.04.v1	weld	Definition	Closed under the global context
Ch. 10	(40)	EQ-015/H.06.v1	EQ-015	Definition	Closed under the global context
Ch. 10	(41)	EQ-015/H.06.v1	EQ-015	Definition	Closed under the global context
Ch. 10	(42)	EQ-015/H.06.v1	EQ-015	Definition	Closed under the global context
Ch. 10	(43)	EQ-015/H.36.v1	EQ-015	Open	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 10	(44)	EQ-015/H.36.v1	EQ-015	Open	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 10	(45)	EQ-015/H.36.v1	EQ-015	Open	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 10	(46)	EQ-015/H.36.v1	EQ-015	Open	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 10	(47)	EQ-015/H.36.v1	EQ-015	Open	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 10	(48)	EQ-015/H.36.v1	EQ-015	Open	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 10	(5)	weld/H.04.v1	weld	Definition	Closed under the global context
Ch. 10	(6)	weld/H.04.v1	weld	Definition	Closed under the global context
Ch. 10	(7)	weld/E.09.v1	weld	Definition	Closed under the global context
Ch. 10	(8)	weld/E.09.v1	weld	Definition	Closed under the global context
Ch. 10	(9)	weld/E.09.v1	weld	Definition	Closed under the global context
Ch. 11	*No equation is registered under this chapter's own record id in Toledo v1.0.0.*				
Ch. 12	L1	EQ-015/E.10.v1	EQ-015	Dr	Closed under the global context
Ch. 12	L2	EQ-015/E.10.v1	EQ-015	Dr	Closed under the global context
Ch. 13	(1)	EQ-015/E.12.v1	EQ-015	Definition	Closed under the global context
Ch. 13	(10)	weld/E.05.v1	weld	Definition	Closed under the global context
Ch. 13	(11)-(12)	weld/E.08.v1	weld	Definition	Closed under the global context
Ch. 13	(13)	weld/E.08.v1	weld	Definition	Closed under the global context
Ch. 13	(14)	weld/E.08.v1	weld	Definition	Closed under the global context
Ch. 13	(15)	weld/M.06.v1	weld	Definition	Closed under the global context
Ch. 13	(16)	weld/M.06.v1	weld	Definition	Closed under the global context
Ch. 13	(17)	weld/M.06.v1	weld	Definition	Closed under the global context
Ch. 13	(18)	EQ-015/E.15.v1	EQ-015	Dr	Closed under the global context
Ch. 13	(19)	EQ-015/M.08.v1	EQ-015	Definition	Closed under the global context
Ch. 13	(2)	EQ-015/E.15.v1	EQ-015	Dr	Closed under the global context
Ch. 13	(20)	EQ-015/M.08.v1	EQ-015	Definition	Closed under the global context
Ch. 13	(21)	EQ-015/M.08.v1	EQ-015	Definition	Closed under the global context
Ch. 13	(22)	EQ-015/E.15.v1	EQ-015	Dr	Closed under the global context
Ch. 13	(23)	weld/M.06.v1	weld	Definition	Closed under the global context
Ch. 13	(24)	weld/M.06.v1	weld	Definition	Closed under the global context

Chapter	Label	Toledo code	Root	Tier	Coq status
Ch. 13	`(25)`	`weld/M.06.v1`	`weld`	Definition	Closed under the global context
Ch. 13	`(26)`	`weld/M.06.v1`	`weld`	Definition	Closed under the global context
Ch. 13	`(27)`	`weld/M.06.v1`	`weld`	Definition	Closed under the global context
Ch. 13	`(28)`	`weld/M.06.v1`	`weld`	Definition	Closed under the global context
Ch. 13	`(29)`	`weld/M.06.v1`	`weld`	Definition	Closed under the global context
Ch. 13	`(3)`	`weld/M.05.v1`	`weld`	Definition	Closed under the global context
Ch. 13	`(30)-(32)`	`EQ-015/E.15.v1`	`EQ-015`	Dr	Closed under the global context
Ch. 13	`(33)`	`EQ-015/E.15.v1`	`EQ-015`	Dr	Closed under the global context
Ch. 13	`(34)-(35)`	`weld/M.07.v1`	`weld`	Definition	Closed under the global context
Ch. 13	`(36)`	`weld/M.07.v1`	`weld`	Definition	Closed under the global context
Ch. 13	`(37)`	`weld/M.07.v1`	`weld`	Definition	Closed under the global context
Ch. 13	`(38)`	`EQ-015/E.15.v1`	`EQ-015`	Dr	Closed under the global context
Ch. 13	`(39)`	`weld/E.03.v1`	`weld`	Definition	Closed under the global context
Ch. 13	`(4)`	`weld/M.05.v1`	`weld`	Definition	Closed under the global context
Ch. 13	`(40)`	`weld/E.03.v1`	`weld`	Definition	Closed under the global context
Ch. 13	`(41)`	`weld/E.03.v1`	`weld`	Definition	Closed under the global context
Ch. 13	`(42)`	`weld/E.03.v1`	`weld`	Definition	Closed under the global context
Ch. 13	`(43)`	`weld/S.47.v1`	`weld`	Definition	Closed under the global context
Ch. 13	`(44)`	`weld/S.48.v1`	`weld`	Definition	Closed under the global context
Ch. 13	`(45)`	`weld/S.49.v1`	`weld`	Definition	Closed under the global context
Ch. 13	`(46)`	`weld/S.50.v1`	`weld`	Definition	not yet formalised in Coq
Ch. 13	`(47)`	`EQ-015/E.15.v1`	`EQ-015`	Dr	Closed under the global context
Ch. 13	`(48)`	`weld/M.08.v1`	`weld`	Definition	Closed under the global context
Ch. 13	`(49)`	`weld/M.08.v1`	`weld`	Definition	Closed under the global context
Ch. 13	`(5)`	`weld/M.05.v1`	`weld`	Definition	Closed under the global context
Ch. 13	`(50)`	`weld/M.08.v1`	`weld`	Definition	Closed under the global context
Ch. 13	`(51)`	`weld/M.08.v1`	`weld`	Definition	Closed under the global context
Ch. 13	`(52)`	`weld/M.07.v1`	`weld`	Definition	Closed under the global context
Ch. 13	`(53)`	`weld/M.07.v1`	`weld`	Definition	Closed under the global context
Ch. 13	`(54)`	`weld/M.07.v1`	`weld`	Definition	Closed under the global context
Ch. 13	`(55)`	`EQ-002/H.04.v1`	`EQ-002`	Definition	Closed under the global context
Ch. 13	`(56)`	`EQ-002/H.04.v1`	`EQ-002`	Definition	Closed under the global context
Ch. 13	`(57)`	`EQ-002/H.04.v1`	`EQ-002`	Definition	Closed under the global context
Ch. 13	`(58)`	`EQ-015/E.15.v1`	`EQ-015`	Dr	Closed under the global context
Ch. 13	`(6)`	`EQ-015/E.13.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 13	`(7)`	`EQ-015/E.15.v1`	`EQ-015`	Dr	Closed under the global context
Ch. 13	`(8)`	`weld/M.03.v1`	`weld`	Definition	Closed under the global context
Ch. 13	`(9)`	`weld/E.05.v1`	`weld`	Definition	Closed under the global context

Chapter	Label	Toledo code	Root	Tier	Coq status
Ch. 14	(1)	weld/M.07.v1	weld	Definition	Closed under the global context
Ch. 14	(10)	weld/M.07.v1	weld	Definition	Closed under the global context
Ch. 14	(11)	weld/M.07.v1	weld	Definition	Closed under the global context
Ch. 14	(12)	A.5/E.09.v1	A.5	untagged	Closed under the global context
Ch. 14	(13)	weld/M.07.v1	weld	Definition	Closed under the global context
Ch. 14	(14)	weld/M.07.v1	weld	Definition	Closed under the global context
Ch. 14	(2)-(3)	weld/M.07.v1	weld	Definition	Closed under the global context
Ch. 14	(4)	A.5/E.09.v1	A.5	untagged	Closed under the global context
Ch. 14	(5)	weld/M.07.v1	weld	Definition	Closed under the global context
Ch. 14	(6)	weld/M.07.v1	weld	Definition	Closed under the global context
Ch. 14	(7)	weld/M.07.v1	weld	Definition	Closed under the global context
Ch. 14	(8)	weld/M.07.v1	weld	Definition	Closed under the global context
Ch. 14	(9)	weld/M.07.v1	weld	Definition	Closed under the global context
Ch. 14	H1	weld/M.07.v1	weld	Definition	Closed under the global context
Ch. 14	H2	weld/M.07.v1	weld	Definition	Closed under the global context
Ch. 15	(1)	A.5/E.02.v1	A.5	Definition	Closed under the global context
Ch. 15	(2)	A.5/E.02.v1	A.5	Definition	Closed under the global context
Ch. 15	(3)	A.5/E.02.v1	A.5	Definition	Closed under the global context
Ch. 15	(4)	A.5/E.02.v1	A.5	Definition	Closed under the global context
Ch. 15	(5)	A.5/E.02.v1	A.5	Definition	Closed under the global context
Ch. 15	(6)	A.5/E.02.v1	A.5	Definition	Closed under the global context
Ch. 15	(Q5, unlabeled)	A.5/E.02.v1	A.5	Definition	Closed under the global context
Ch. 16	(1)	A.5/E.02.v1	A.5	Definition	Closed under the global context
Ch. 16	(10)	EQ-002/E.08.v1	EQ-002	Definition	Closed under the global context
Ch. 16	(11)	A.5/H.04.v1	A.5	Definition	Closed under the global context
Ch. 16	(12)	A.5/H.04.v1	A.5	Definition	Closed under the global context
Ch. 16	(13)-(15)	A.5/E.02.v1	A.5	Definition	Closed under the global context
Ch. 16	(16)	A.5/H.04.v1	A.5	Definition	Closed under the global context
Ch. 16	(17)	A.5/H.04.v1	A.5	Definition	Closed under the global context
Ch. 16	(18)	A.5/E.02.v1	A.5	Definition	Closed under the global context
Ch. 16	(19)	A.5/E.02.v1	A.5	Definition	Closed under the global context
Ch. 16	(2)	A.5/E.02.v1	A.5	Definition	Closed under the global context
Ch. 16	(20)	A.5/E.02.v1	A.5	Definition	Closed under the global context
Ch. 16	(21)	A.5/E.02.v1	A.5	Definition	Closed under the global context
Ch. 16	(22)	A.5/E.02.v1	A.5	Definition	Closed under the global context
Ch. 16	(3)	A.5/E.02.v1	A.5	Definition	Closed under the global context
Ch. 16	(4)	A.5/E.02.v1	A.5	Definition	Closed under the global context
Ch. 16	(5)	A.5/E.02.v1	A.5	Definition	Closed under the global context

Chapter	Label	Toledo code	Root	Tier	Coq status
Ch. 16	(6)	A.5/E.02.v1	A.5	Definition	Closed under the global context
Ch. 16	(7)	EQ-002/E.08.v1	EQ-002	Definition	Closed under the global context
Ch. 16	(8)	EQ-002/E.08.v1	EQ-002	Definition	Closed under the global context
Ch. 16	(9)	EQ-002/E.08.v1	EQ-002	Definition	Closed under the global context
Ch. 17	CBC-01	A.5/S.14.v1	A.5	Definition	Closed under the global context
Ch. 17	CBC-02	A.5/S.17.v1	A.5	Definition	not yet formalised in Coq
Ch. 17	CBC-03	A.5/S.18.v1	A.5	Definition	Closed under the global context
Ch. 17	CBC-04	A.5/S.19.v1	A.5	Definition	Closed under the global context
Ch. 17	CBC-05	A.5/S.15.v1	A.5	Definition	Closed under the global context
Ch. 17	CBC-06	A.5/S.16.v1	A.5	Definition	Closed under the global context
Ch. 17	CBC-07	EQ-015/S.04.v1	EQ-015	Definition	Closed under the global context
Ch. 17	CBC-08	EQ-015/S.51.v1	EQ-015	Definition	Closed under the global context
Ch. 17	CBC-09	A.5/S.20.v1	A.5	Definition	not yet formalised in Coq
Ch. 17	CBC-10	EQ-015/S.08.v1	EQ-015	Definition	Closed under the global context
Ch. 17	CBC-11	EQ-015/H.03.v1	EQ-015	Definition	Closed under the global context
Ch. 17	CBC-12	EQ-015/H.03.v1	EQ-015	Definition	Closed under the global context
Ch. 17	CBC-13	EQ-015/H.03.v1	EQ-015	Definition	Closed under the global context
Ch. 17	CBC-14	EQ-015/H.03.v1	EQ-015	Definition	Closed under the global context
Ch. 17	CBC-15	EQ-015/S.04.v1	EQ-015	Definition	Closed under the global context
Ch. 17	CBC-16	EQ-015/S.04.v1	EQ-015	Definition	Closed under the global context
Ch. 17	CBC-H1	weld/S.09.v1	weld	Open	Closed under the global context
Ch. 17	CBC-H2	weld/S.09.v1	weld	Open	Closed under the global context
Ch. 17	CBC-H3	weld/S.09.v1	weld	Open	Closed under the global context
Ch. 17	CBC-H4	weld/S.09.v1	weld	Open	Closed under the global context
Ch. 17	CBC-H5	weld/S.09.v1	weld	Open	Closed under the global context
Ch. 17	CBC-H6	weld/S.09.v1	weld	Open	Closed under the global context
Ch. 17	CBC-H7	weld/S.09.v1	weld	Open	Closed under the global context
Ch. 17	CBC-H8	weld/S.09.v1	weld	Open	Closed under the global context
Ch. 17	CBC-P1	EQ-015/S.05.v1	EQ-015	Dr	Closed under the global context
Ch. 17	CBC-P2	EQ-015/S.05.v1	EQ-015	Dr	Closed under the global context
Ch. 17	CBC-P3	EQ-015/S.05.v1	EQ-015	Dr	Closed under the global context
Ch. 17	CBC-P4	EQ-015/S.05.v1	EQ-015	Dr	Closed under the global context
Ch. 17	CBC-P5	EQ-015/S.05.v1	EQ-015	Dr	Closed under the global context
Ch. 18	(1)	EQ-015/S.52.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 18	(2)	EQ-015/S.53.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 18	(3)	EQ-015/S.54.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 18	(4)	EQ-015/S.56.v1	EQ-015	Definition	Closed under the global context
Ch. 18	(5)	EQ-015/S.57.v1	EQ-015	Definition	Closed under the global context
Ch. 18	(6)	EQ-015/S.09.v1	EQ-015	Definition	Closed under the global context

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Ch. 18	(7)	EQ-015/S.58.v1	EQ-015	Definition	Closed under the global context
Ch. 18	D1 (NC-78)	EQ-015/S.59.v1	EQ-015	Definition	Closed under the global context
Ch. 18	D2	EQ-015/S.60.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 18	D3	EQ-015/S.61.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 18	D4	EQ-015/S.62.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 18	D5	EQ-015/S.63.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 18	D6 (NC-79)	EQ-015/S.64.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 18	D7	EQ-015/S.65.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 18	L1	weld/S.23.v1	weld	Dr	Closed under the global context
Ch. 18	L2	weld/S.24.v1	weld	Dr	Closed under the global context
Ch. 18	L3	weld/S.25.v1	weld	Dr	Closed under the global context
Ch. 18	L4	weld/S.26.v1	weld	Dr	Closed under the global context
Ch. 18	P-A	EQ-015/S.10.v1	EQ-015	Definition	Closed under the global context
Ch. 18	P-B	EQ-015/S.12.v1	EQ-015	Open	Closed under the global context
Ch. 18	P-C	EQ-015/S.12.v1	EQ-015	Open	Closed under the global context
Ch. 18	P-D	EQ-015/S.12.v1	EQ-015	Open	Closed under the global context
Ch. 18	PAR-cert	A.8/S.01.v1	A.8	Dr	Closed under the global context
Ch. 18	PAR-stepper	weld/S.22.v1	weld	Definition	Closed under the global context
Ch. 18	unnumbered	EQ-015/S.55.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 19	*No equation is registered under this chapter's own record id in Toledo v1.0.0.*				
Ch. 20	(1)	weld/S.10.v1	weld	Definition	not yet formalised in Coq
Ch. 20	(10)	weld/S.19.v1	weld	Dr	not yet formalised in Coq
Ch. 20	(11)	weld/S.20.v1	weld	Dr	not yet formalised in Coq
Ch. 20	(12)	weld/S.21.v1	weld	Dr	not yet formalised in Coq
Ch. 20	(2)	weld/S.11.v1	weld	Definition	not yet formalised in Coq
Ch. 20	(3)	weld/S.12.v1	weld	Definition	not yet formalised in Coq
Ch. 20	(4)	weld/S.13.v1	weld	Definition	not yet formalised in Coq
Ch. 20	(5)	weld/S.14.v1	weld	Definition	not yet formalised in Coq
Ch. 20	(6)	weld/S.15.v1	weld	Definition	not yet formalised in Coq
Ch. 20	(7)	weld/S.16.v1	weld	Dr	not yet formalised in Coq
Ch. 20	(8)	weld/S.17.v1	weld	Dr	not yet formalised in Coq
Ch. 20	(9)	weld/S.18.v1	weld	Dr	not yet formalised in Coq
Ch. 20	Theorem (No-Go for Elimination by Suppression)	weld/S.01.v1	weld	Definition	Closed under the global context
Ch. 21	(CE-choice gate)	weld/S.40.v1	weld	Definition	Closed under the global context
Ch. 21	(Tragic)	weld/S.41.v1	weld	Definition	Closed under the global context
Ch. 21	(Tragic-min)	weld/S.42.v1	weld	Definition	not yet formalised in Coq
Ch. 21	CE-01	weld/S.27.v1	weld	Ax	Closed under the global context
Ch. 21	CE-02	weld/S.28.v1	weld	Ax	Closed under the global context
Ch. 21	CE-03	weld/S.29.v1	weld	Ax	Closed under the global context
Ch. 21	CE-04	weld/S.30.v1	weld	Ax	not yet formalised in Coq
Ch. 21	CE-05	EQ-015/S.13.v1	EQ-015	Ax	Closed under the global context
Ch. 21	CE-06	EQ-015/S.14.v1	EQ-015	Definition	Closed under the global context
Ch. 21	CE-07	EQ-015/S.15.v1	EQ-015	Definition	Closed under the global context

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Ch. 21	CE-08	EQ-015/S.16.v1	EQ-015	Definition	Closed under the global context
Ch. 21	CE-09	EQ-015/S.17.v1	EQ-015	Definition	Closed under the global context
Ch. 21	CE-10	EQ-015/S.18.v1	EQ-015	Definition	Closed under the global context
Ch. 21	CE-11	EQ-015/S.19.v1	EQ-015	Definition	Closed under the global context
Ch. 21	CE-12	EQ-015/S.20.v1	EQ-015	Definition	Closed under the global context
Ch. 21	CE-13	EQ-015/S.21.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 21	CE-14	EQ-015/S.23.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 21	CE-15	EQ-015/S.24.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 21	CE-16	EQ-015/S.25.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 21	CE-17	EQ-015/S.26.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 21	CE-18	EQ-015/S.27.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 21	CE-19	EQ-015/S.22.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 21	CE-20	EQ-015/S.28.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 21	CE-21	EQ-015/S.29.v1	EQ-015	Definition	Closed under the global context
Ch. 21	CE-22	EQ-015/S.30.v1	EQ-015	Definition	Closed under the global context
Ch. 21	CE-23	EQ-015/S.31.v1	EQ-015	Definition	Closed under the global context
Ch. 21	CE-24	EQ-015/S.32.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 21	CE-25	weld/S.31.v1	weld	Definition	not yet formalised in Coq
Ch. 21	CE-26	weld/S.32.v1	weld	Definition	not yet formalised in Coq
Ch. 21	CE-27	weld/S.33.v1	weld	Definition	Closed under the global context
Ch. 21	CE-28	weld/S.34.v1	weld	Definition	Closed under the global context
Ch. 21	CE-29	weld/S.35.v1	weld	Definition	not yet formalised in Coq
Ch. 21	CE-30	EQ-015/S.33.v1	EQ-015	Definition	Closed under the global context
Ch. 21	CE-31	weld/S.36.v1	weld	Definition	not yet formalised in Coq
Ch. 21	CE-32	weld/S.37.v1	weld	Dr	Closed under the global context
Ch. 21	CE-33	weld/S.38.v1	weld	Definition	not yet formalised in Coq
Ch. 21	CE-34	weld/S.39.v1	weld	Definition	not yet formalised in Coq
Ch. 22	Constraint-Evolution	A.5/S.07.v1	A.5	Definition	not yet formalised in Coq
Ch. 22	Def-1	A.5/S.04.v1	A.5	Definition	not yet formalised in Coq
Ch. 22	Def-2	A.5/S.05.v1	A.5	Definition	not yet formalised in Coq
Ch. 22	Def-3 (Agency-Condition)	A.5/S.09.v1	A.5	Definition	not yet formalised in Coq
Ch. 22	L3-Constraint-Evolution-History	A.5/S.11.v1	A.5	Definition	Closed under the global context
Ch. 22	L4-Constraint-Evolution-Predictive	A.5/S.12.v1	A.5	Definition	Closed under the global context
Ch. 22	L5-Meta-Regulation	A.5/S.13.v1	A.5	Definition	Closed under the global context
Ch. 22	Persistence-Regulation-Condition	A.5/S.06.v1	A.5	Definition	not yet formalised in Coq
Ch. 22	Proto-Agency-Condition	A.5/S.10.v1	A.5	Definition	Closed under the global context
Ch. 22	State-Constraint-Coupling	A.5/S.08.v1	A.5	Definition	Closed under the global context
Ch. 23	Canonical Formula	EQ-002/E.03.v1	EQ-002	Definition	Closed under the global context
Ch. 24	(1)	EQ-015/H.07.v1	EQ-015	Definition	Closed under the global context
Ch. 24	(2)	EQ-015/E.08.v1	EQ-015	Definition	Closed under the global context
Ch. 25	(1)	A.5/H.01.v1	A.5	Definition	Closed under the global context
Ch. 25	(2)	EQ-015/H.10.v1	EQ-015	Definition	Closed under the global context
Ch. 25	P1	weld/S.08.v1	weld	Open	Closed under the global context

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Ch. 25	`P2`	`weld/S.08.v1`	`weld`	Open	Closed under the global context
Ch. 25	`P3`	`weld/S.08.v1`	`weld`	Open	Closed under the global context
Ch. 25	`P4`	`weld/S.08.v1`	`weld`	Open	Closed under the global context
Ch. 25	`P5`	`weld/S.08.v1`	`weld`	Open	Closed under the global context
Ch. 25	`P6`	`weld/S.08.v1`	`weld`	Open	Closed under the global context
Ch. 25	`P7`	`weld/S.08.v1`	`weld`	Open	Closed under the global context
Ch. 25	`P8`	`weld/S.08.v1`	`weld`	Open	Closed under the global context
Ch. 25	`unlabeled (5, Reflexive Reconfiguration stage)`	`EQ-015/E.08.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 25	`unlabeled (§2, falsifier condition)`	`A.5/S.02.v1`	`A.5`	Open	Closed under the global context
Ch. 25	`unlabeled (§5.1, provenance ledger)`	`A.8/M.02.v1`	`A.8`	Definition	Closed under the global context
Ch. 25	`unlabeled (§5.2, AVRH)`	`EQ-015/H.18.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 26	`(1)`	`EQ-015/E.03.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 26	`(2)`	`EQ-015/H.05.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 26	`(3)`	`EQ-015/H.10.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 26	`(4)`	`A.5/H.08.v1`	`A.5`	Definition	Closed under the global context
Ch. 26	`(5)`	`EQ-015/H.18.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 26	`P1`	`EQ-015/H.35.v1`	`EQ-015`	Open	Closed under the global context
Ch. 26	`P2`	`EQ-015/H.35.v1`	`EQ-015`	Open	Closed under the global context
Ch. 26	`unlabeled (§6, Framing bullet)`	`A.5/H.01.v1`	`A.5`	Definition	Closed under the global context
Ch. 26	`unlabeled (§6, Integration bullet)`	`EQ-015/E.08.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 27	`unlabeled (§7, display 1)`	`EQ-015/E.11.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 27	`unlabeled (§7, display 2)`	`EQ-015/E.11.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 28	*No equation is registered under this chapter's own record id in Toledo v1.0.0.*				
Ch. 29	`(1)`	`EQ-015/H.28.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 29	`(2)`	`weld/H.10.v1`	`weld`	Definition	Closed under the global context
Ch. 29	`(3)`	`weld/H.10.v1`	`weld`	Definition	Closed under the global context
Ch. 29	`Def-1`	`A.5/H.13.v1`	`A.5`	Definition	Closed under the global context
Ch. 29	`Def-2`	`A.5/H.13.v1`	`A.5`	Definition	Closed under the global context
Ch. 29	`Def-3`	`A.5/H.13.v1`	`A.5`	Definition	Closed under the global context
Ch. 29	`Def-4`	`weld/H.10.v1`	`weld`	Definition	Closed under the global context

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Ch. 29	`Def-5`	`weld/H.10.v1`	`weld`	Definition	Closed under the global context
Ch. 29	`Def-6`	`weld/H.10.v1`	`weld`	Definition	Closed under the global context
Ch. 29	`Def-7`	`A.5/H.14.v1`	`A.5`	Definition	Closed under the global context
Ch. 29	`Eq-epoch3`	`A.5/H.13.v1`	`A.5`	Definition	Closed under the global context
Ch. 29	`Eq-unifying`	`A.5/H.13.v1`	`A.5`	Definition	Closed under the global context
Ch. 29	`Finding-1 (H1)`	`EQ-015/H.29.v1`	`EQ-015`	Open	Closed under the global context
Ch. 29	`Finding-2 (H3)`	`EQ-015/H.29.v1`	`EQ-015`	Open	Closed under the global context
Ch. 29	`Finding-3 (H4)`	`EQ-015/H.29.v1`	`EQ-015`	Open	Closed under the global context
Ch. 29	`Prop-1`	`A.5/H.13.v1`	`A.5`	Definition	Closed under the global context
Ch. 29	`Prop-2`	`weld/H.10.v1`	`weld`	Definition	Closed under the global context
Ch. 29	`Prop-3`	`A.5/H.14.v1`	`A.5`	Definition	Closed under the global context
Ch. 29	`Prop-4`	`A.5/H.13.v1`	`A.5`	Definition	Closed under the global context
Ch. 29	`Prop-5 (B1)`	`weld/H.10.v1`	`weld`	Definition	Closed under the global context
Ch. 30	`CN-1`	`EQ-015/H.18.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 30	`EF-01`	`EQ-015/H.02.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 30	`EF-02 (Repair 1)`	`EQ-015/H.02.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 30	`EF-03 (Repair 1)`	`EQ-015/H.02.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 30	`EF-04`	`EQ-015/H.10.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 30	`EF-05`	`EQ-015/H.11.v1`	`EQ-015`	Dr	Closed under the global context
Ch. 30	`EF-06 (Repair 2)`	`EQ-015/H.11.v1`	`EQ-015`	Dr	Closed under the global context
Ch. 30	`EF-07 (Repair 3)`	`weld/H.08.v1`	`weld`	Dr	Closed under the global context
Ch. 30	`EF-08 (Repair 4)`	`EQ-015/H.14.v1`	`EQ-015`	Dr	Closed under the global context
Ch. 30	`EF-09 (Repair 4)`	`EQ-015/H.14.v1`	`EQ-015`	Dr	Closed under the global context
Ch. 30	`EF-10`	`EQ-015/H.14.v1`	`EQ-015`	Dr	Closed under the global context
Ch. 30	`EF-11 (Repair 5)`	`EQ-015/H.15.v1`	`EQ-015`	Dr	Closed under the global context
Ch. 30	`EF-12 (Repair 5)`	`EQ-015/H.15.v1`	`EQ-015`	Dr	Closed under the global context
Ch. 30	`EF-13`	`EQ-015/H.05.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 30	`EF-14`	`EQ-015/H.05.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 30	`EF-15 (Repair 6)`	`EQ-015/H.05.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 30	`EF-16`	`EQ-015/H.05.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 30	`EF-17`	`A.5/H.09.v1`	`A.5`	Definition	Closed under the global context
Ch. 30	`EF-18 (Repair 7)`	`weld/H.07.v1`	`weld`	Definition	Closed under the global context

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Ch. 30	EF-19 (Repair 7)	weld/H.07.v1	weld	Definition	Closed under the global context
Ch. 30	EF-20	EQ-015/H.12.v1	EQ-015	Dr	Closed under the global context
Ch. 30	EF-21	EQ-015/H.12.v1	EQ-015	Dr	Closed under the global context
Ch. 30	EF-22 (Repair 8)	EQ-015/H.12.v1	EQ-015	Dr	Closed under the global context
Ch. 30	EF-23	EQ-015/H.19.v1	EQ-015	Definition	Closed under the global context
Ch. 30	EF-24 (Repair 9)	EQ-015/H.19.v1	EQ-015	Definition	Closed under the global context
Ch. 30	EF-25	EQ-015/H.12.v1	EQ-015	Dr	Closed under the global context
Ch. 30	EF-26 (Repair 10)	EQ-015/H.12.v1	EQ-015	Dr	Closed under the global context
Ch. 30	EF-27	EQ-015/H.13.v1	EQ-015	Dr	Closed under the global context
Ch. 30	NCL-14	A.5/H.07.v1	A.5	Definition	Closed under the global context
Ch. 31	CTSA-01	EQ-015/H.16.v1	EQ-015	Definition	Closed under the global context
Ch. 31	CTSA-02	EQ-015/H.10.v1	EQ-015	Definition	Closed under the global context
Ch. 31	CTSA-03	weld/H.12.v1	weld	Dr	Closed under the global context
Ch. 31	CTSA-04	EQ-015/H.16.v1	EQ-015	Definition	Closed under the global context
Ch. 31	CTSA-05	weld/H.12.v1	weld	Dr	Closed under the global context
Ch. 31	H1	EQ-015/H.20.v1	EQ-015	Open	Closed under the global context
Ch. 31	H2	EQ-015/H.20.v1	EQ-015	Open	Closed under the global context
Ch. 31	H3	EQ-015/H.20.v1	EQ-015	Open	Closed under the global context
Ch. 31	H4	EQ-015/H.20.v1	EQ-015	Open	Closed under the global context
Ch. 31	H5	EQ-015/H.20.v1	EQ-015	Open	Closed under the global context
Ch. 31	H6	EQ-015/H.20.v1	EQ-015	Open	Closed under the global context
Ch. 31	NCL-10	A.5/H.10.v1	A.5	Dr	Closed under the global context
Ch. 32	(1)	A.5/H.01.v1	A.5	Definition	Closed under the global context
Ch. 32	(10)	EQ-015/H.21.v1	EQ-015	Definition	Closed under the global context
Ch. 32	(11)	A.5/H.11.v1	A.5	Definition	Closed under the global context
Ch. 32	(12)	EQ-015/H.02.v1	EQ-015	Definition	Closed under the global context
Ch. 32	(13)	EQ-015/H.02.v1	EQ-015	Definition	Closed under the global context
Ch. 32	(14)	A.5/H.11.v1	A.5	Definition	Closed under the global context
Ch. 32	(15)	A.5/H.11.v1	A.5	Definition	Closed under the global context
Ch. 32	(16)	A.5/H.11.v1	A.5	Definition	Closed under the global context
Ch. 32	(17)	A.5/H.11.v1	A.5	Definition	Closed under the global context
Ch. 32	(18)	weld/H.08.v1	weld	Dr	Closed under the global context
Ch. 32	(19)	A.5/H.12.v1	A.5	Definition	Closed under the global context
Ch. 32	(2)	EQ-015/H.04.v1	EQ-015	Definition	Closed under the global context
Ch. 32	(20)	A.5/H.12.v1	A.5	Definition	Closed under the global context
Ch. 32	(21)	A.5/H.12.v1	A.5	Definition	Closed under the global context

Chapter	Label	Toledo code	Root	Tier	Coq status
Ch. 32	(22)	A.8/H.01.v1	A.8	Definition	Closed under the global context
Ch. 32	(23)	EQ-015/H.22.v1	EQ-015	Definition	Closed under the global context
Ch. 32	(24)	EQ-015/H.22.v1	EQ-015	Definition	Closed under the global context
Ch. 32	(25)	EQ-015/H.17.v1	EQ-015	Definition	Closed under the global context
Ch. 32	(26)	EQ-015/H.23.v1	EQ-015	Definition	Closed under the global context
Ch. 32	(27)	A.5/H.08.v1	A.5	Definition	Closed under the global context
Ch. 32	(28)	EQ-015/H.23.v1	EQ-015	Definition	Closed under the global context
Ch. 32	(29)	EQ-015/H.24.v1	EQ-015	Definition	Closed under the global context
Ch. 32	(3)	EQ-015/E.08.v1	EQ-015	Definition	Closed under the global context
Ch. 32	(30)	EQ-015/H.24.v1	EQ-015	Definition	Closed under the global context
Ch. 32	(31)-(32)	A.8/H.02.v1	A.8	Definition	Closed under the global context
Ch. 32	(33)-(34)	A.8/H.02.v1	A.8	Definition	Closed under the global context
Ch. 32	(35)	EQ-015/H.21.v1	EQ-015	Definition	Closed under the global context
Ch. 32	(36)	weld/S.07.v1	weld	Definition	Closed under the global context
Ch. 32	(37)	EQ-015/M.10.v1	EQ-015	Definition	Closed under the global context
Ch. 32	(38)	EQ-015/M.10.v1	EQ-015	Definition	Closed under the global context
Ch. 32	(39)	A.8/M.03.v1	A.8	Definition	Closed under the global context
Ch. 32	(4)	A.5/H.02.v1	A.5	Definition	Closed under the global context
Ch. 32	(40)-(41)	A.8/H.04.v1	A.8	Definition	Closed under the global context
Ch. 32	(42)	A.8/H.04.v1	A.8	Definition	Closed under the global context
Ch. 32	(43)	A.5/H.02.v1	A.5	Definition	Closed under the global context
Ch. 32	(44)	EQ-015/H.24.v1	EQ-015	Definition	Closed under the global context
Ch. 32	(45)	A.5/H.11.v1	A.5	Definition	Closed under the global context
Ch. 32	(46)	EQ-015/H.27.v1	EQ-015	Dr	Closed under the global context
Ch. 32	(5)	A.5/H.02.v1	A.5	Definition	Closed under the global context
Ch. 32	(6)	A.5/H.02.v1	A.5	Definition	Closed under the global context
Ch. 32	(7)	A.5/H.02.v1	A.5	Definition	Closed under the global context
Ch. 32	(8)	EQ-015/H.21.v1	EQ-015	Definition	Closed under the global context
Ch. 32	(9)	EQ-015/H.21.v1	EQ-015	Definition	Closed under the global context
Ch. 32	AFP	EQ-015/H.21.v1	EQ-015	Definition	Closed under the global context
Ch. 32	AgP	EQ-015/H.25.v1	EQ-015	Open	Closed under the global context
Ch. 32	DCPp	EQ-015/H.25.v1	EQ-015	Open	Closed under the global context
Ch. 32	DEPp	EQ-015/H.25.v1	EQ-015	Open	Closed under the global context
Ch. 32	H1	EQ-015/H.26.v1	EQ-015	Open	Closed under the global context
Ch. 32	H10	EQ-015/H.26.v1	EQ-015	Open	Closed under the global context
Ch. 32	H2	EQ-015/H.26.v1	EQ-015	Open	Closed under the global context
Ch. 32	H3	EQ-015/H.26.v1	EQ-015	Open	Closed under the global context
Ch. 32	H4	EQ-015/H.26.v1	EQ-015	Open	Closed under the global context

Chapter	Label	Toledo code	Root	Tier	Coq status
Ch. 32	H5	EQ-015/H.26.v1	EQ-015	Open	Closed under the global context
Ch. 32	H6	EQ-015/H.26.v1	EQ-015	Open	Closed under the global context
Ch. 32	H7	EQ-015/H.26.v1	EQ-015	Open	Closed under the global context
Ch. 32	H8	EQ-015/H.26.v1	EQ-015	Open	Closed under the global context
Ch. 32	H9	EQ-015/H.26.v1	EQ-015	Open	Closed under the global context
Ch. 32	PFDP	A.5/H.02.v1	A.5	Definition	Closed under the global context
Ch. 32	WCP	EQ-015/H.24.v1	EQ-015	Definition	Closed under the global context
Ch. 33	(1)	EQ-015/M.01.v1	EQ-015	Definition	Closed under the global context
Ch. 33	(10)	A.5/H.15.v1	A.5	Definition	Closed under the global context
Ch. 33	(11)	A.5/H.15.v1	A.5	Definition	Closed under the global context
Ch. 33	(12)	A.5/H.16.v1	A.5	Definition	Closed under the global context
Ch. 33	(13)	A.5/H.16.v1	A.5	Definition	Closed under the global context
Ch. 33	(14)	A.8/H.04.v1	A.8	Definition	Closed under the global context
Ch. 33	(15)	A.5/H.17.v1	A.5	Definition	Closed under the global context
Ch. 33	(16)	A.5/H.17.v1	A.5	Definition	Closed under the global context
Ch. 33	(17)	A.5/H.17.v1	A.5	Definition	Closed under the global context
Ch. 33	(18)	A.5/H.17.v1	A.5	Definition	Closed under the global context
Ch. 33	(19)	A.5/H.17.v1	A.5	Definition	Closed under the global context
Ch. 33	(2)	EQ-015/M.02.v1	EQ-015	untagged	Closed under the global context
Ch. 33	(20)	A.8/H.04.v1	A.8	Definition	Closed under the global context
Ch. 33	(21)	A.8/H.04.v1	A.8	Definition	Closed under the global context
Ch. 33	(22)	EQ-015/H.32.v1	EQ-015	Open	Closed under the global context
Ch. 33	(23)	EQ-015/H.32.v1	EQ-015	Open	Closed under the global context
Ch. 33	(24)	EQ-015/H.17.v1	EQ-015	Definition	Closed under the global context
Ch. 33	(25)	A.5/H.08.v1	A.5	Definition	Closed under the global context
Ch. 33	(26)	EQ-015/H.32.v1	EQ-015	Open	Closed under the global context
Ch. 33	(27)	EQ-015/H.24.v1	EQ-015	Definition	Closed under the global context
Ch. 33	(28)	A.5/H.18.v1	A.5	Definition	Closed under the global context
Ch. 33	(29)	A.5/H.18.v1	A.5	Definition	Closed under the global context
Ch. 33	(3)	weld/M.02.v1	weld	Definition	Closed under the global context
Ch. 33	(30)	A.5/H.18.v1	A.5	Definition	Closed under the global context
Ch. 33	(31)	A.5/H.18.v1	A.5	Definition	Closed under the global context
Ch. 33	(32)	A.5/H.19.v1	A.5	Definition	Closed under the global context
Ch. 33	(33)	A.5/H.19.v1	A.5	Definition	Closed under the global context
Ch. 33	(34)	EQ-015/H.33.v1	EQ-015	Definition	Closed under the global context
Ch. 33	(35)	EQ-015/H.33.v1	EQ-015	Definition	Closed under the global context
Ch. 33	(36)	A.8/H.03.v1	A.8	Definition	Closed under the global context
Ch. 33	(37)	A.8/H.03.v1	A.8	Definition	Closed under the global context

Chapter	Label	Toledo code	Root	Tier	Coq status
Ch. 33	(4)	A.5/M.01.v1	A.5	Definition	Closed under the global context
Ch. 33	(5)	EQ-002/H.03.v1	EQ-002	Definition	Closed under the global context
Ch. 33	(6)	EQ-015/H.30.v1	EQ-015	Definition	Closed under the global context
Ch. 33	(7)	weld/H.06.v1	weld	Definition	Closed under the global context
Ch. 33	(8)	EQ-015/H.31.v1	EQ-015	Definition	Closed under the global context
Ch. 33	(9)	A.5/H.15.v1	A.5	Definition	Closed under the global context
Ch. 33	Human Capability Advancement Principle	EQ-002/H.03.v1	EQ-002	Definition	Closed under the global context
Ch. 34	(1)	EQ-015/W.01.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(10)	EQ-015/W.24.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(11)	EQ-015/W.26.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(12)	EQ-015/W.27.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(13)	EQ-015/W.28.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(14)	EQ-015/W.29.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 34	(15)	EQ-015/W.30.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 34	(16)	EQ-015/W.34.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(17)	EQ-015/W.35.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(18)	EQ-015/W.36.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(19)	A.5/W.04.v1	A.5	Definition	Closed under the global context
Ch. 34	(2)	EQ-002/M.03.v1	EQ-002	Definition	Closed under the global context
Ch. 34	(20)	A.5/W.05.v1	A.5	Definition	Closed under the global context
Ch. 34	(21)	A.5/W.06.v1	A.5	Definition	Closed under the global context
Ch. 34	(22)	EQ-015/W.31.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(23)	EQ-015/W.32.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(24)	EQ-015/W.33.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(25)	EQ-015/W.06.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(26)	EQ-015/W.37.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(27)	EQ-015/W.38.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 34	(28)	EQ-015/W.39.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(29)	EQ-015/W.40.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(3)	EQ-015/W.02.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(30)	EQ-015/W.09.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(31)	weld/W.01.v1	weld	Definition	Closed under the global context
Ch. 34	(32)	EQ-015/H.07.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(33)	A.5/H.06.v1	A.5	Definition	Closed under the global context
Ch. 34	(34)	EQ-015/W.14.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(35)	EQ-015/W.15.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(36)	EQ-015/W.47.v1	EQ-015	Definition	Closed under the global context

Chapter	Label	Toledo code	Root	Tier	Coq status
Ch. 34	(37)	EQ-015/W.48.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(38)	EQ-015/W.49.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(39)	EQ-015/W.50.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(4)	EQ-015/H.10.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(40)	EQ-015/W.42.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(41)	EQ-015/W.43.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 34	(42)	EQ-015/W.44.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 34	(43)	EQ-015/W.45.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(44)	EQ-015/W.46.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 34	(45)	A.5/W.07.v1	A.5	Definition	Closed under the global context
Ch. 34	(46)	A.5/W.08.v1	A.5	Definition	not yet formalised in Coq
Ch. 34	(47)	A.5/W.09.v1	A.5	Definition	not yet formalised in Coq
Ch. 34	(48)	A.5/W.10.v1	A.5	Definition	Closed under the global context
Ch. 34	(49)	A.5/W.02.v1	A.5	Definition	Closed under the global context
Ch. 34	(5)	EQ-015/W.20.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(50)	A.5/W.11.v1	A.5	Definition	Closed under the global context
Ch. 34	(51)	EQ-015/W.01.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(52)	EQ-015/W.13.v1	EQ-015	Dr	Closed under the global context
Ch. 34	(53)	EQ-015/W.41.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 34	(54)	weld/W.02.v1	weld	Definition	Closed under the global context
Ch. 34	(55)	EQ-015/W.06.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(56)	EQ-015/W.12.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(57)	EQ-015/W.10.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(6)	EQ-015/W.21.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 34	(7)	EQ-015/W.22.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(8)	EQ-015/W.25.v1	EQ-015	Definition	Closed under the global context
Ch. 34	(9)	EQ-015/W.23.v1	EQ-015	Definition	Closed under the global context
Ch. 34	Labour-Decentering Proposition	EQ-015/W.04.v1	EQ-015	Definition	Closed under the global context
Ch. 35	(1)	EQ-015/W.51.v1	EQ-015	Definition	Closed under the global context
Ch. 35	(10)	A.5/W.20.v1	A.5	Definition	not yet formalised in Coq
Ch. 35	(11)	EQ-015/W.52.v1	EQ-015	Definition	Closed under the global context
Ch. 35	(12)	EQ-015/W.53.v1	EQ-015	Definition	Closed under the global context
Ch. 35	(13)	EQ-015/W.54.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 35	(14)	EQ-015/W.17.v1	EQ-015	Definition	Closed under the global context
Ch. 35	(15)	EQ-015/W.17.v1	EQ-015	Definition	Closed under the global context
Ch. 35	(16)	EQ-015/W.18.v1	EQ-015	Definition	Closed under the global context
Ch. 35	(17)	EQ-015/W.55.v1	EQ-015	Definition	Closed under the global context
Ch. 35	(18)	EQ-015/W.56.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 35	(19)	EQ-015/W.57.v1	EQ-015	Definition	Closed under the global context

Chapter	Label	Toledo code	Root	Tier	Coq status
Ch. 35	(2)	A.5/W.12.v1	A.5	Definition	Closed under the global context
Ch. 35	(20)	EQ-015/W.58.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 35	(21)	EQ-015/W.59.v1	EQ-015	Definition	mapped to a Coq identifier, not yet Coq-wrapped
Ch. 35	(22)	EQ-002/W.02.v1	EQ-002	Definition	not yet formalised in Coq
Ch. 35	(23)	EQ-002/W.03.v1	EQ-002	Definition	Closed under the global context
Ch. 35	(24)	EQ-015/W.17.v1	EQ-015	Definition	Closed under the global context
Ch. 35	(25)	EQ-015/W.16.v1	EQ-015	Definition	Closed under the global context
Ch. 35	(26)	EQ-015/W.16.v1	EQ-015	Definition	Closed under the global context
Ch. 35	(3)	A.5/W.13.v1	A.5	Definition	not yet formalised in Coq
Ch. 35	(4)	A.5/W.14.v1	A.5	Definition	not yet formalised in Coq
Ch. 35	(5)	A.5/W.15.v1	A.5	Definition	not yet formalised in Coq
Ch. 35	(6)	A.5/W.16.v1	A.5	Definition	not yet formalised in Coq
Ch. 35	(7)	A.5/W.17.v1	A.5	Definition	not yet formalised in Coq
Ch. 35	(8)	A.5/W.18.v1	A.5	Definition	not yet formalised in Coq
Ch. 35	(9)	A.5/W.19.v1	A.5	Definition	not yet formalised in Coq
Ch. 35	Proposition 1	EQ-015/W.16.v1	EQ-015	Definition	Closed under the global context
Ch. 35	Proposition 2	EQ-015/W.17.v1	EQ-015	Definition	Closed under the global context
Ch. 35	Proposition 3	EQ-015/W.19.v1	EQ-015	Definition	Closed under the global context
Ch. 35	Proposition 4	EQ-002/W.01.v1	EQ-002	Definition	Closed under the global context
Ch. 36	*No equation is registered under this chapter's own record id in Toledo v1.0.0.*				
Ch. 37	(1)	EQ-015/M.04.v1	EQ-015	Definition	Closed under the global context
Ch. 37	(10)	EQ-015/E.03.v1	EQ-015	Definition	Closed under the global context
Ch. 37	(11)	EQ-002/E.04.v1	EQ-002	Definition	Closed under the global context
Ch. 37	(12)	EQ-015/E.04.v1	EQ-015	Definition	Closed under the global context
Ch. 37	(13)	EQ-015/E.04.v1	EQ-015	Definition	Closed under the global context
Ch. 37	(14)	EQ-002/E.05.v1	EQ-002	Definition	Closed under the global context
Ch. 37	(15)	EQ-002/E.05.v1	EQ-002	Definition	Closed under the global context
Ch. 37	(16)	EQ-015/E.08.v1	EQ-015	Definition	Closed under the global context
Ch. 37	(17)	EQ-015/E.08.v1	EQ-015	Definition	Closed under the global context
Ch. 37	(18)	EQ-002/E.06.v1	EQ-002	Definition	Closed under the global context
Ch. 37	(19)	EQ-002/E.06.v1	EQ-002	Definition	Closed under the global context
Ch. 37	(2)	EQ-015/M.04.v1	EQ-015	Definition	Closed under the global context
Ch. 37	(20)	EQ-002/E.06.v1	EQ-002	Definition	Closed under the global context
Ch. 37	(21)	EQ-015/H.07.v1	EQ-015	Definition	Closed under the global context
Ch. 37	(22)	EQ-015/H.08.v1	EQ-015	Definition	Closed under the global context
Ch. 37	(23)	EQ-015/H.08.v1	EQ-015	Definition	Closed under the global context
Ch. 37	(24)	A.5/H.05.v1	A.5	Definition	Closed under the global context
Ch. 37	(25)	A.5/H.05.v1	A.5	Definition	Closed under the global context
Ch. 37	(26)	A.5/H.05.v1	A.5	Definition	Closed under the global context
Ch. 37	(27)	EQ-015/H.09.v1	EQ-015	Definition	Closed under the global context
Ch. 37	(28)	weld/S.06.v1	weld	Dr	Closed under the global context
Ch. 37	(29)	A.5/H.01.v1	A.5	Definition	Closed under the global context

Chapter	Label	Toledo code	Root	Tier	Coq status
Ch. 37	(3)	EQ-015/M.01.v1	EQ-015	Definition	Closed under the global context
Ch. 37	(30)	EQ-015/H.04.v1	EQ-015	Definition	Closed under the global context
Ch. 37	(31)	EQ-015/H.10.v1	EQ-015	Definition	Closed under the global context
Ch. 37	(32)	EQ-015/H.04.v1	EQ-015	Definition	Closed under the global context
Ch. 37	(33)	EQ-015/H.04.v1	EQ-015	Definition	Closed under the global context
Ch. 37	(34)	A.5/H.06.v1	A.5	Definition	Closed under the global context
Ch. 37	(35)	EQ-015/H.18.v1	EQ-015	Definition	Closed under the global context
Ch. 37	(36)	weld/H.07.v1	weld	Definition	Closed under the global context
Ch. 37	(37)	weld/H.07.v1	weld	Definition	Closed under the global context
Ch. 37	(38)	EQ-015/H.16.v1	EQ-015	Definition	Closed under the global context
Ch. 37	(39)	A.5/H.09.v1	A.5	Definition	Closed under the global context
Ch. 37	(4)	EQ-015/M.02.v1	EQ-015	untagged	Closed under the global context
Ch. 37	(40)	A.5/H.08.v1	A.5	Definition	Closed under the global context
Ch. 37	(41)	A.8/M.02.v1	A.8	Definition	Closed under the global context
Ch. 37	(42)	A.5/E.01.v1	A.5	Definition	Closed under the global context
Ch. 37	(43)	EQ-015/H.07.v1	EQ-015	Definition	Closed under the global context
Ch. 37	(44)	weld/H.11.v1	weld	Definition	Closed under the global context
Ch. 37	(45)	weld/H.11.v1	weld	Definition	Closed under the global context
Ch. 37	(46)	weld/H.11.v1	weld	Definition	Closed under the global context
Ch. 37	(47)	weld/H.11.v1	weld	Definition	Closed under the global context
Ch. 37	(48)	weld/H.11.v1	weld	Definition	Closed under the global context
Ch. 37	(49)	EQ-015/M.04.v1	EQ-015	Definition	Closed under the global context
Ch. 37	(5)	A.5/M.01.v1	A.5	Definition	Closed under the global context
Ch. 37	(50)	EQ-015/M.04.v1	EQ-015	Definition	Closed under the global context
Ch. 37	(6)	weld/M.02.v1	weld	Definition	Closed under the global context
Ch. 37	(7)	weld/H.06.v1	weld	Definition	Closed under the global context
Ch. 37	(8)	A.5/E.01.v1	A.5	Definition	Closed under the global context
Ch. 37	(9)	EQ-002/E.01.v1	EQ-002	Definition	Closed under the global context
Ch. 37	H1	EQ-015/H.34.v1	EQ-015	Open	Closed under the global context
Ch. 37	H2	EQ-015/H.34.v1	EQ-015	Open	Closed under the global context
Ch. 37	H3	EQ-015/H.34.v1	EQ-015	Open	Closed under the global context
Ch. 37	H4	EQ-015/H.34.v1	EQ-015	Open	Closed under the global context
Ch. 37	H5	EQ-015/H.34.v1	EQ-015	Open	Closed under the global context
Ch. 37	H6	EQ-015/H.34.v1	EQ-015	Open	Closed under the global context
Ch. 38	(1)	A.5/M.03.v1	A.5	untagged	Closed under the global context
Ch. 38	(2)	A.5/M.03.v1	A.5	untagged	Closed under the global context
Ch. 38	(3)	A.5/M.03.v1	A.5	untagged	Closed under the global context
Ch. 38	(4)	A.5/M.04.v1	A.5	Open	Closed under the global context

Chapter	Label	Toledo code	Root	Tier	Coq status
Ch. 38	(unnumbered, Sec.1)	A.5/M.03.v1	A.5	untagged	Closed under the global context
Ch. 38	(unnumbered, Sec.3.4)	A.5/M.03.v1	A.5	untagged	Closed under the global context
Ch. 38	(unnumbered, Sec.4)	A.5/M.03.v1	A.5	untagged	Closed under the global context
Ch. 39	(1)	A.5/M.07.v1	A.5	untagged	Closed under the global context
Ch. 39	(10)	A.8/M.06.v1	A.8	Definition	Closed under the global context
Ch. 39	(101)	A.5/M.12.v1	A.5	untagged	Closed under the global context
Ch. 39	(102)	A.5/M.14.v1	A.5	untagged	Closed under the global context
Ch. 39	(103)	A.5/M.15.v1	A.5	untagged	Closed under the global context
Ch. 39	(104)	A.5/M.29.v1	A.5	untagged	Closed under the global context
Ch. 39	(105)	EQ-015/M.05.v1	EQ-015	finite_diagnostic	Closed under the global context
Ch. 39	(106)	A.8/M.15.v1	A.8	finite_diagnostic	Closed under the global context
Ch. 39	(107)-(108)	A.5/M.30.v1	A.5	finite_diagnostic	Closed under the global context
Ch. 39	(109)	A.8/M.15.v1	A.8	finite_diagnostic	Closed under the global context
Ch. 39	(11)	A.5/M.10.v1	A.5	untagged	Closed under the global context
Ch. 39	(110)	A.8/M.10.v1	A.8	finite_diagnostic	Closed under the global context
Ch. 39	(111)	A.8/M.05.v1	A.8	finite_diagnostic	Closed under the global context
Ch. 39	(112)	A.8/M.05.v1	A.8	finite_diagnostic	Closed under the global context
Ch. 39	(113)	A.8/M.05.v1	A.8	finite_diagnostic	Closed under the global context
Ch. 39	(114)	weld/M.12.v1	weld	finite_diagnostic	Closed under the global context
Ch. 39	(115)	weld/M.12.v1	weld	finite_diagnostic	Closed under the global context
Ch. 39	(116)	weld/M.12.v1	weld	finite_diagnostic	Closed under the global context
Ch. 39	(117)	weld/M.12.v1	weld	finite_diagnostic	Closed under the global context
Ch. 39	(118)	A.8/M.15.v1	A.8	finite_diagnostic	Closed under the global context
Ch. 39	(119)	A.8/M.15.v1	A.8	finite_diagnostic	Closed under the global context
Ch. 39	(12)	A.5/M.11.v1	A.5	untagged	Closed under the global context
Ch. 39	(120)	A.8/M.15.v1	A.8	finite_diagnostic	Closed under the global context
Ch. 39	(121)-(122)	A.5/M.31.v1	A.5	finite_diagnostic	Closed under the global context
Ch. 39	(123)	A.5/M.32.v1	A.5	untagged	Closed under the global context
Ch. 39	(124)	A.8/M.06.v1	A.8	Definition	Closed under the global context
Ch. 39	(125)	A.8/M.16.v1	A.8	finite_diagnostic	Closed under the global context
Ch. 39	(126)	A.8/M.16.v1	A.8	finite_diagnostic	Closed under the global context
Ch. 39	(127)	A.5/M.25.v1	A.5	untagged	Closed under the global context
Ch. 39	(128)	A.8/M.16.v1	A.8	finite_diagnostic	Closed under the global context
Ch. 39	(129)	A.5/M.02.v1	A.5	finite_diagnostic	Closed under the global context
Ch. 39	(13)-(16)	weld/M.11.v1	weld	Definition	Closed under the global context
Ch. 39	(130)	A.5/M.24.v1	A.5	untagged	Closed under the global context
Ch. 39	(131)	A.5/M.22.v1	A.5	untagged	Closed under the global context
Ch. 39	(17)	weld/M.11.v1	weld	Definition	Closed under the global context

Chapter	Label	Toledo code	Root	Tier	Coq status
Ch. 39	(18)	A.5/M.12.v1	A.5	untagged	Closed under the global context
Ch. 39	(19)	A.8/M.07.v1	A.8	finite_diagnostic	Closed under the global context
Ch. 39	(2)	weld/M.10.v1	weld	Definition	Closed under the global context
Ch. 39	(20)	A.8/M.07.v1	A.8	finite_diagnostic	Closed under the global context
Ch. 39	(21)	A.8/M.07.v1	A.8	finite_diagnostic	Closed under the global context
Ch. 39	(22)	A.8/M.08.v1	A.8	finite_diagnostic	Closed under the global context
Ch. 39	(23)	A.8/M.08.v1	A.8	finite_diagnostic	Closed under the global context
Ch. 39	(24)	A.8/M.08.v1	A.8	finite_diagnostic	Closed under the global context
Ch. 39	(25)	A.8/M.08.v1	A.8	finite_diagnostic	Closed under the global context
Ch. 39	(26)	A.5/M.13.v1	A.5	untagged	Closed under the global context
Ch. 39	(27)	EQ-015/M.05.v1	EQ-015	finite_diagnostic	Closed under the global context
Ch. 39	(28)	A.5/M.14.v1	A.5	untagged	Closed under the global context
Ch. 39	(29)	A.5/M.15.v1	A.5	untagged	Closed under the global context
Ch. 39	(3)	A.5/M.08.v1	A.5	untagged	Closed under the global context
Ch. 39	(30)	EQ-015/M.05.v1	EQ-015	finite_diagnostic	Closed under the global context
Ch. 39	(31)	A.5/M.16.v1	A.5	untagged	Closed under the global context
Ch. 39	(32)	A.5/M.17.v1	A.5	untagged	Closed under the global context
Ch. 39	(33)	A.5/M.18.v1	A.5	untagged	Closed under the global context
Ch. 39	(34)	EQ-002/M.02.v1	EQ-002	finite_diagnostic	Closed under the global context
Ch. 39	(35)	EQ-002/M.02.v1	EQ-002	finite_diagnostic	Closed under the global context
Ch. 39	(36)	EQ-002/M.02.v1	EQ-002	finite_diagnostic	Closed under the global context
Ch. 39	(37)	A.8/M.09.v1	A.8	finite_diagnostic	Closed under the global context
Ch. 39	(38)	A.8/M.09.v1	A.8	finite_diagnostic	Closed under the global context
Ch. 39	(39)	A.8/M.09.v1	A.8	finite_diagnostic	Closed under the global context
Ch. 39	(4)	A.8/M.06.v1	A.8	Definition	Closed under the global context
Ch. 39	(40)	weld/M.12.v1	weld	finite_diagnostic	Closed under the global context
Ch. 39	(41)	weld/M.12.v1	weld	finite_diagnostic	Closed under the global context
Ch. 39	(42)	weld/M.12.v1	weld	finite_diagnostic	Closed under the global context
Ch. 39	(43)	weld/M.12.v1	weld	finite_diagnostic	Closed under the global context
Ch. 39	(44)	A.5/M.19.v1	A.5	untagged	Closed under the global context
Ch. 39	(45)	weld/M.12.v1	weld	finite_diagnostic	Closed under the global context
Ch. 39	(46)	weld/M.12.v1	weld	finite_diagnostic	Closed under the global context
Ch. 39	(47)	A.8/M.10.v1	A.8	finite_diagnostic	Closed under the global context
Ch. 39	(48)	A.8/M.10.v1	A.8	finite_diagnostic	Closed under the global context
Ch. 39	(49)	A.8/M.04.v1	A.8	finite_diagnostic	Closed under the global context
Ch. 39	(5)	A.8/M.06.v1	A.8	Definition	Closed under the global context
Ch. 39	(50)	A.8/M.04.v1	A.8	finite_diagnostic	Closed under the global context
Ch. 39	(51)	weld/M.12.v1	weld	finite_diagnostic	Closed under the global context

Chapter	Label	Toledo code	Root	Tier	Coq status
Ch. 39	(52)	weld/M.12.v1	weld	finite_diagnostic	Closed under the global context
Ch. 39	(53)	weld/M.12.v1	weld	finite_diagnostic	Closed under the global context
Ch. 39	(54)	weld/M.12.v1	weld	finite_diagnostic	Closed under the global context
Ch. 39	(55)	weld/M.12.v1	weld	finite_diagnostic	Closed under the global context
Ch. 39	(56)	A.8/M.11.v1	A.8	Definition	Closed under the global context
Ch. 39	(57)	A.8/M.11.v1	A.8	Definition	Closed under the global context
Ch. 39	(58)	A.8/M.11.v1	A.8	Definition	Closed under the global context
Ch. 39	(59)	A.8/M.11.v1	A.8	Definition	Closed under the global context
Ch. 39	(6)	weld/M.10.v1	weld	Definition	Closed under the global context
Ch. 39	(60)	A.5/M.20.v1	A.5	untagged	Closed under the global context
Ch. 39	(61)	A.8/M.11.v1	A.8	Definition	Closed under the global context
Ch. 39	(62)	A.8/M.11.v1	A.8	Definition	Closed under the global context
Ch. 39	(63)	A.5/M.21.v1	A.5	untagged	Closed under the global context
Ch. 39	(64)	EQ-015/M.06.v1	EQ-015	Definition	Closed under the global context
Ch. 39	(65)	EQ-015/E.12.v1	EQ-015	Definition	Closed under the global context
Ch. 39	(66)	EQ-015/M.12.v1	EQ-015	untagged	Closed under the global context
Ch. 39	(67)	A.8/M.12.v1	A.8	Definition	Closed under the global context
Ch. 39	(68)	weld/M.09.v1	weld	finite_diagnostic	Closed under the global context
Ch. 39	(69)	A.5/M.02.v1	A.5	finite_diagnostic	Closed under the global context
Ch. 39	(7)	weld/M.10.v1	weld	Definition	Closed under the global context
Ch. 39	(70)	A.5/M.22.v1	A.5	untagged	Closed under the global context
Ch. 39	(71)	A.5/M.02.v1	A.5	finite_diagnostic	Closed under the global context
Ch. 39	(72)	A.5/M.02.v1	A.5	finite_diagnostic	Closed under the global context
Ch. 39	(73)-(74)	A.5/M.23.v1	A.5	untagged	Closed under the global context
Ch. 39	(75)	A.5/M.02.v1	A.5	finite_diagnostic	Closed under the global context
Ch. 39	(76)	A.5/M.02.v1	A.5	finite_diagnostic	Closed under the global context
Ch. 39	(77)	A.5/M.24.v1	A.5	untagged	Closed under the global context
Ch. 39	(78)	A.5/M.02.v1	A.5	finite_diagnostic	Closed under the global context
Ch. 39	(79)	A.5/M.02.v1	A.5	finite_diagnostic	Closed under the global context
Ch. 39	(8)	weld/M.10.v1	weld	Definition	Closed under the global context
Ch. 39	(80)	A.5/M.02.v1	A.5	finite_diagnostic	Closed under the global context
Ch. 39	(81)	A.8/H.04.v1	A.8	Definition	Closed under the global context
Ch. 39	(82)	A.8/M.13.v1	A.8	finite_diagnostic	Closed under the global context
Ch. 39	(83)	A.8/M.13.v1	A.8	finite_diagnostic	Closed under the global context
Ch. 39	(84)	A.5/M.27.v1	A.5	untagged	Closed under the global context
Ch. 39	(85)	A.8/M.14.v1	A.8	finite_diagnostic	Closed under the global context
Ch. 39	(86)	A.5/M.25.v1	A.5	untagged	Closed under the global context
Ch. 39	(87)	A.5/M.26.v1	A.5	untagged	Closed under the global context

Chapter	Label	Toledo code	Root	Tier	Coq status
Ch. 39	`(88)`	`A.8/M.13.v1`	`A.8`	finite_diagnostic	Closed under the global context
Ch. 39	`(89)`	`EQ-015/M.07.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 39	`(9)`	`A.5/M.09.v1`	`A.5`	untagged	Closed under the global context
Ch. 39	`(90)`	`weld/M.12.v1`	`weld`	finite_diagnostic	Closed under the global context
Ch. 39	`(91)`	`weld/M.12.v1`	`weld`	finite_diagnostic	Closed under the global context
Ch. 39	`(92)`	`A.8/H.04.v1`	`A.8`	Definition	Closed under the global context
Ch. 39	`(93)`	`weld/M.10.v1`	`weld`	Definition	Closed under the global context
Ch. 39	`(94)`	`EQ-015/M.07.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 39	`(95)`	`EQ-015/M.07.v1`	`EQ-015`	Definition	Closed under the global context
Ch. 39	`(96)`	`A.5/M.28.v1`	`A.5`	untagged	Closed under the global context
Ch. 39	`(97)-(100)`	`weld/M.11.v1`	`weld`	Definition	Closed under the global context
Ch. 40	`(unnumbered, Sec.1)`	`EQ-002/E.02.v1`	`EQ-002`	Definition	Closed under the global context
Ch. 40	`(unnumbered, Sec.3.3)`	`A.5/M.14.v1`	`A.5`	untagged	Closed under the global context
Ch. 40	`(unnumbered, Sec.8.1, disclaimer D-DVP-NOT-K2)`	`A.5/M.12.v1`	`A.5`	untagged	Closed under the global context
Ch. 40	`(unnumbered, Sec.9 / Appendix disclaimer D-AUTHORSHIP)`	`A.5/M.32.v1`	`A.5`	untagged	Closed under the global context
Ch. 41	*No equation is registered under this chapter's own record id in Toledo v1.0.0.*				

Rows in this table. 946 equation-occurrence rows across 35 bound chapters, parsed mechanically from Toledo v1.0.0's `registry/CANONICAL.json` at build time; 6 bound chapters list no row (named above, not omitted).

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